Singularities in Hermitian pluripotential theory

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Introduction

The singularities of a plurisubharmonic potential influence its Monge–Ampère measure, the holomorphic functions integrable against its weight, and the geometry of the positive current it represents. On a compact Kähler manifold these relationships are organized by a closed background form. Closedness makes integration by parts compatible with positivity: it conserves the mass of bounded potentials, gives comparison principles, and produces an energy whose derivative is the Monge–Ampère measure. Much of the global theory of singularities depends on this interaction between local analysis and global integration.

This book asks how these relationships change for Hermitian backgrounds. Its starting point is the theory of singularities and $\mathcal{I}$-goodness developed in *Singularities in global pluripotential theory* [Xia]. The aim is to understand which information about a singularity can still be recovered from envelopes, measures and holomorphic sections when closedness is unavailable. Monge–Ampère equations and Gauduchon geometry enter this question together: the equations construct potentials with controlled measures, while Gauduchon metrics supply conserved linear quantities where the nonlinear mass can vary.

The background form and the ambient manifold

Two changes from the Kähler setting must be distinguished. A non-Kähler manifold can carry closed real $(1,1)$-forms and positive closed currents; their local potential theory remains available. Conversely, a nonclosed Hermitian background can be chosen on a Kähler manifold, and its torsion already changes the Monge–Ampère calculus. Many of the difficulties studied here concern the chosen background, rather than the absence of any Kähler metric on the manifold. Global geometric restrictions enter separately, through the existence of closed positive classes, modifications and holomorphic sections.

Let X be a connected compact complex manifold of dimension n. For a smooth real $(1,1)$-form θ on X and a θ-plurisubharmonic potential u, write $\theta_u = \theta + \mathrm{dd^c}u$.

When θ is closed, locally it is a complex Hessian, so θ_u is locally the Hessian of a plurisubharmonic function. For a general θ, a smooth local correction still makes a θ-plurisubharmonic potential plurisubharmonic, but a smooth remainder remains in the current. This is enough to retain compactness, regularization, multiplier ideals and the local construction of bounded products. It does not remove derivatives of the background from products or integration by parts. The general plurisubharmonic tools are therefore developed together in Chapter 1, before their global consequences are separated according to the background.

Mass, comparison and energy

For a Kähler form ω and a smooth real potential u with $\omega + \mathrm{dd^c}u > 0$, Stokes' theorem gives

$$\int_X (\omega + \mathrm{dd^c}u)^n = \int_X \omega^n.$$

The same identity holds for bounded ω-plurisubharmonic potentials. Singular potentials require the non-pluripolar product, and their mass can be smaller; even in the Kähler setting the bounded-potential identity must not be extended to arbitrary singularities. For a general Hermitian form ω, closedness may fail and the mass can already vary among smooth potentials. Its defect is explicit:

$$\int_X (\omega + \mathrm{dd^c}u)^n - \int_X \omega^n = \sum_{k=1}^{n-1} \binom{n}{k} \int_X u(\mathrm{dd^c}u)^{k-1} \wedge \mathrm{dd^c}(\omega^{n-k}).$$

Thus a bounded change of potential can change the mass without changing its ordinary singularity type. Chapter 2 derives this formula and characterizes mass invariance by the vanishing of $\mathrm{dd^c}(\omega^j)$ for $1 \le j \le n-1$.

The consequences go beyond an error term in an identity. For the standard Hopf threefold, masses of smooth admissible potentials are unbounded above, although their infimum is positive. A different compact threefold has a balanced metric with finite upper mass and admits smooth admissible potentials whose masses tend to zero. These constructions, in Chapter 8 and Sections 25.15 and 25.15.5, separate an upper mass bound from a positive lower mass bound. Neither can be imposed silently on the general theory. The global comparison inequality familiar from Kähler geometry would force mass invariance; in its place, local and modified comparison estimates must retain the terms caused by torsion.

Energy imposes a stronger requirement than constant mass. For a smooth real tangent direction v at a smooth admissible potential u, the Monge–Ampère operator defines the one-form

$$\alpha_u(v) = \int_X v(\omega + \mathrm{dd^c}u)^n.$$

For a Kähler background it is the differential of the Monge–Ampère energy. In dimension at least two, the existence of a primitive even on a neighborhood of the zero potential forces $\mathrm{d}\omega = 0$ (Theorem 3.1.1). A Gauduchon metric on a non-Kähler surface conserves total mass but does not supply this primitive. Consequently the Kähler variational argument cannot be recovered from mass conservation alone. The same distinction reappears in interpolation: Perron envelopes and Legendre duality remain meaningful, while the relation between the homogeneous product equation and a metric geodesic equation changes in a nonclosed background (Chapters 16 and 17).

Linear control and nonlinear equations

A Gauduchon metric γ supplies a replacement for one part of the lost conservation law. Its defining equation $\mathrm{dd^c}(\gamma^{n-1}) = 0$ gives

$$\int_X (\theta + \mathrm{dd^c} u) \wedge \gamma^{n-1} = \int_X \theta \wedge \gamma^{n-1}.$$

The trace measure $(\theta + \mathrm{dd^c} u) \wedge \gamma^{n-1}$ determines u up to a constant. Fix a smooth positive volume form $\mathrm{d}V$ on X. In a fixed background, weak convergence of the trace measures is equivalent to L^1 convergence of potentials normalized by $\int_X u \, \mathrm{d}V = 0$ (Chapter 5). The conserved trace mass, however, does not bound the nonlinear Monge–Ampère mass. The examples in Chapter 6 retain strong convergence of currents and traces while their nonlinear masses are infinite. To treat these potentials, the book constructs extended non-pluripolar measures and finite weighted measures. For nonpositive potentials, damping near the pole set restores continuity under convergence in capacity; passing back to undamped measures requires control of the tails (Chapter 7).

The loss of mass conservation also changes what it means to prescribe a volume form. Given a smooth real function F on X, the Hermitian equation has the form

$$(\omega + \mathrm{dd^c} u)^n = c \, \mathrm{e}^F \, \mathrm{d}V, \qquad \omega + \mathrm{dd^c} u > 0, \qquad \sup_X u = 0,$$

and both u and $c > 0$ are unknown. For a Kähler form, integration determines $c = \int_X \omega^n / \int_X \mathrm{e}^F \, \mathrm{d}V$ before the equation is solved. In general the numerator would be the mass of the unknown solution. The equation determines a unique smooth pair, and its adjoint Gauduchon probability describes the variation of $\log c$ with the density. This turns normalization into a geometric functional rather than merely a fixed integral constraint.

Nonnegative, nonzero L^p densities, $p > 1$, still admit continuous solutions with a unique positive scalar. Potential uniqueness is more delicate: a smooth density with zeros can admit distinct normalized smooth solutions in a Hermitian background. For smooth nonnegative data, connectedness and density of the positive set recover

uniqueness among bounded solutions, as proved in Theorem 11.4.1. Scalar uniqueness and selection by exponential approximation remain separate from these potential-uniqueness results. Prescribing a singularity type requires further control of the reference potential, developed in Chapter 13; arbitrary finite non-pluripolar measures are not universally solvable on Hermitian manifolds.

The scalar functional leads back to Gauduchon geometry. Its supporting probabilities extend smooth adjoint measures, and Gauduchon test forms define a finite variational volume whose value on a positive background is its equation constant. This construction remains finite even where upper Monge–Ampère mass is infinite. Its extension to boundary classes, weak dual attainment and degeneration are studied in Chapter 25. The invariant has a precise limit as a description of singularities: distinct divisorial types can have the same entire scalar functional and the same supporting probabilities. Stability of these data does not by itself recover the potential or its ordinary type (Chapter 14).

Envelopes and the information in a singularity

Ordinary singularity type records a potential up to bounded difference. Multiplier ideals record the integrability of holomorphic functions against all positive multiples of its weight. These are different kinds of information, and the corresponding envelopes need not agree. For positive-mass singularities in a closed big class on a compact Kähler manifold, $\mathcal{I}$-goodness links equality of the ordinary and ideal envelopes to a mass identity and to approximation by analytic singularities in the singularity topology. The Hermitian problem is to recover such links with quantities that retain their meaning when mass varies or becomes infinite.

The two envelopes themselves survive for smooth nonclosed backgrounds. In a positive Hermitian background, correcting the curvature error keeps dyadic analytic approximants in the original potential space. Their envelopes decrease to the ideal envelope. Combining convergence of both envelopes with controlled ideal inclusions and divisorial total variation gives a uniform topology whose analytic closure consists exactly of the classes with equal envelopes (Theorem 15.6.1). This applies even to classes with infinite-mass representatives; their canonical envelopes still have finite mass. The topology does not separate all ordinary classes. On curve models it is weaker than the complete polar-variation topology below, since diffuse polar mass can concentrate in an L^1 limit. Keeping the zero obstacle fixed also makes this topology change under smooth changes of the background, by Proposition 15.6.4. This failure already occurs on curves, where the backgrounds are closed; it comes from the lost polar curvature rather than torsion. Controlling all smooth obstacles restores invariance under smooth changes of background. Its analytic closure is characterized by equality of both envelopes for every obstacle. Under bounded mass and positive lower ideal volume, testing the zero obstacle suffices (Proposition 29.8.1). The curve topology is still incomplete and weaker than polar total variation (Theorem 15.6.2 and Proposition 29.12.4).

To extract an ordinary-type invariant from mass, one can take its infimum and supremum over all representatives of the type. The resulting lower and upper volumes need not coincide. For a smooth big background under bounded mass, volume comparison gives a completely metrizable topology on ordinary models of positive lower volume, although the comparison itself need not satisfy the triangle inequality. The complete metric keeps its Cauchy sequences away from zero lower volume. On a compact curve with a fixed positive background, a different complete metric includes the zero-volume boundary: it is exactly total variation of the polar part of the curvature measure (Theorems 29.11.1 and 29.12.1 and Corollary 29.12.1). Weak convergence of the full currents loses information retained by this metric.

Equations for general finite non-pluripolar measures require their own hypotheses. In a semipositive big background, positive lower mass and a model reference admitting a competitor in a smaller background give solutions in the relative envelope class. For an arbitrary smooth big background, the corresponding construction uses bounded mass and positive lower volume of the model reference. These two regimes form Chapters 30 and 31; bounded difference from the reference requires a separate argument. Together with volume comparison they occupy Part IX, while the earlier chapters develop the constructions available without these uniform mass assumptions.

Poles and metric geometry

A logarithmic pole can either be prescribed or forced by the background. In Section 13.20, resolving a prescribed pole leaves a strictly positive residual form, so a bounded Hermitian equation determines the remaining potential. Its bounded residual need not descend, even when the full logarithmic potential does. In the blowup backgrounds of Section 13.21, the exceptional divisor instead occurs in every positive current. Its coefficient belongs to the minimal type, and removing it leaves a current pulled back from the target. The same logarithmic appearance thus conceals different constraints on the residual and its behavior along exceptional fibers.

The metric defined by a positive current off its singular set contains further information. A Hölder potential controls the local trace mass and bounds path distance by the square root of its Hölder modulus; if the current is smooth and strictly positive away from a nonempty analytic set, that set lies at finite completion distance (Theorem 1.18.1). Boundedness alone does not give this conclusion. The prescribed radial densities in Section 27.12 separate the integral controlling the potential from the integral controlling radial length, and produce continuous minimal residuals with complete regular ends. Density zeros also constrain smoothness of exceptional residuals, but their vanishing order alone does not determine regularity or distance (Section 27.11). The radial examples already occur in closed Kähler classes, so their completeness issue is separate from the torsion effects above.

Restriction, sections and geometry

The geometric questions ask how much singularity information survives restriction or passage to another model. Ordinary restriction may lose meaning on a subspace contained in the pole set, whereas restricted ideals and trace envelopes can still encode data there. Divisorial currents give concrete representatives on modifications. Compatibility of numerical classes, compatibility of envelopes and realization by a single ordinary type are separate steps; Chapters 18 and 22 to 24 develops their relationships.

Holomorphic sections introduce another distinction between the background and the ambient manifold. The curvature of a line bundle is closed, even when an auxiliary Hermitian form is not. Thus section counts cannot be expected to measure arbitrary Hermitian Monge–Ampère mass. Gauduchon degrees bound sections and flag valuations, while fibrations explain cases in which their growth and equilibrium live on a lower-dimensional base. Partial kernels connect this algebraic information with the potentials selected by their weights. Closed Bott–Chern classes and locally conformally Kähler structures then provide geometric settings with additional differential constraints, rather than consequences valid for every nonclosed background.

The exposition assumes familiarity with complex manifolds, differential forms and currents. Chapter 1 gives the plurisubharmonic tools used throughout the book. On a first reading, the maximum principle, normalized compactness and exponential integrability provide the starting point; the longer approximation and descent arguments can be consulted when their applications arise. The mass and energy calculations in Chapters 2 and 3 then identify which Kähler arguments need to be replaced. Gauduchon traces, non-pluripolar products and the Hopf example complete the foundations for the equations.

Readers chiefly interested in equations can continue with Chapters 9 to 11 before passing to prescribed singularities in Chapters 12 and 13. For singularity theory, the envelope constructions lead to Chapters 16 to 18: interpolation creates singularity data, and restriction tests what information survives on a submanifold. Holomorphic sections and geometric models then give two ways to realize those data. The later mass-bound part is a separate route for readers seeking volume comparison and general-measure equations under the stated mass hypotheses. The appendix extends descent and resolution-independent measures to unibranch compact spaces and treats equations on projective cones.

The remaining questions are formulated in Chapter 32. The final historical chapter identifies paper sources and the precise dependencies of the arguments; book references also appear where they are used.

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Conventions

Unless otherwise stated, X is a connected compact complex manifold of dimension $n \geq 1$. A Hermitian form is a smooth real strictly positive $(1,1)$-form. We reserve ω for such a form and allow θ to be an arbitrary smooth real $(1,1)$-form. Closedness of θ or ω will always be stated.

We use the normalizations of [Xia]:

$$\mathrm{d}^{\mathrm{c}} = \frac{\mathrm{i}}{4\pi}(\bar{\partial} - \partial), \qquad \mathrm{dd}^{\mathrm{c}} = \frac{\mathrm{i}}{2\pi}\partial\bar{\partial}.$$

On a disk in $\mathbb{C}$ with coordinate z, $\mathrm{dd}^{\mathrm{c}} \log |z|^2$ is the unit Dirac measure at the origin. Multiplier ideals use the weight $\exp(-u)$. Monge–Ampère masses are computed without a factor $1/n!$.

Part I
The Hermitian foundations

Chapter 1
Plurisubharmonic functions and singularities

A potential with a smooth lower Hessian bound becomes plurisubharmonic after a local smooth correction. This brings the local compactness and approximation theorems to a compact complex manifold, provided the correction is retained when potentials are glued or their Hessians multiplied. Normalization controls the resulting families globally. To keep track of singularities under these operations, we compare potentials both by domination up to a constant and by the local integrability of holomorphic functions against their exponential weights.

1.1 Plurisubharmonic potentials

Let X be a connected compact complex manifold of dimension $n \geq 1$, let θ be a smooth real $(1,1)$-form, and let ω be a Hermitian form on X. There are two distinct extensions of the Kähler setting: one can study a closed background on a non-Kähler manifold, or allow the background itself to be nonclosed. In both cases positivity is expressed through quasi-plurisubharmonic functions.

On an open set U in a complex manifold, a *plurisubharmonic function* is a locally integrable upper semicontinuous function satisfying the submean inequality on every complex line in local coordinates. We write $\mathrm{PSH}(U)$ for these functions. A function is *quasi-plurisubharmonic* if it is locally the sum of a plurisubharmonic function and a smooth real function; their space is denoted by $\mathrm{QPSH}(U)$. The following relative positivity condition also makes sense on any such open set.

Definition 1.1.1 A quasi-plurisubharmonic function u on X is *θ-plurisubharmonic* if

$$\theta + \mathrm{dd^c} u \geq 0$$

as a current. We denote the set of these functions by $\mathrm{PSH}(X, \theta)$.

If $\theta = \omega$, every constant belongs to $\mathrm{PSH}(X, \omega)$.

Proposition 1.1.1 *Every real distribution u with $\theta + \mathrm{dd^c}u \geq 0$ has a unique θ-plurisubharmonic representative. On each relatively compact coordinate ball with coordinates $z = (z_1, \ldots, z_n)$, there is a constant $A > 0$ such that $u + A|z|^2$ is plurisubharmonic. Conversely, if u is quasi-plurisubharmonic on the compact manifold X, then there is a constant $C > 0$ such that $u \in \mathrm{PSH}(X, C\omega)$.*

Proof On a coordinate ball with compact closure in a larger chart, smoothness of θ gives

$$\theta \leq A\mathrm{dd^c}|z|^2$$

for a sufficiently large A. Consequently,

$$\mathrm{dd^c}(u + A|z|^2) \geq A\mathrm{dd^c}|z|^2 - \theta \geq 0.$$

The local distributional characterization of plurisubharmonic functions, [Dem12, Chapter I, Theorem 5.8], gives a unique plurisubharmonic representative of $u + A|z|^2$. Subtracting the smooth correction gives a quasi-plurisubharmonic representative of u on the ball. These representatives agree on overlaps: after one common local smooth correction, they are plurisubharmonic functions equal almost everywhere, hence everywhere. They therefore glue to a unique θ-plurisubharmonic representative.

For the converse, choose finitely many relatively compact charts covering X on which $u = p_j + f_j$, with p_j plurisubharmonic and f_j smooth. On the closure of each smaller chart, the smooth form $\mathrm{dd^c} f_j$ is bounded below by $-C_j\omega$. Thus $\mathrm{dd^c}u \geq -C_j\omega$ there. Taking the maximum of finitely many C_j proves the global inequality. □

For $u \in \mathrm{PSH}(X, \theta)$, choose coordinates z and a constant $A > 0$ as in Proposition 1.1.1, and put $p = u + A|z|^2$. Then

$$\theta + \mathrm{dd^c}u = \mathrm{dd^c}p + (\theta - A\mathrm{dd^c}|z|^2).$$

Thus local plurisubharmonic theory applies to p, while the smooth remainder records the original background. It must be retained when forming products or integrating by parts.

Proposition 1.1.2 *If $\theta = \mathrm{dd^c}\rho$ on an open set for a smooth function ρ, then $\mathrm{d}\theta = 0$ on that set. In particular, at a point where $\mathrm{d}\theta \neq 0$, no neighborhood admits such a representation.*

Proof The identity follows from $\mathrm{ddd^c} = 0$. □

For a closed real $(1, 1)$-form, smooth local $\mathrm{dd^c}$-potentials do exist on sufficiently small coordinate balls by the local $\partial\bar{\partial}$-lemma. This local fact does not require the compact manifold to admit a Kähler metric.

A smooth conformal direction also constrains the regularity of a potential. In dimension at least two, the identity below determines the derivative of the conformal factor from smooth coefficients.

Lemma 1.1.1 *Let X be a complex manifold of dimension $n \geq 2$, let θ be a smooth real $(1,1)$-form and let γ be Hermitian. Suppose real-valued $u, a \in L^1_{\mathrm{loc}}(X)$ satisfy*

$$\theta + \mathrm{dd^c} u = a\gamma$$

as currents. Then a and u are smooth.

Proof Taking the exterior derivative and multiplying by γ^{n-2} gives

$$\mathrm{d}a \wedge \gamma^{n-1} = \mathrm{d}\theta \wedge \gamma^{n-2} - a\,\mathrm{d}\gamma \wedge \gamma^{n-2}.$$

At each point, $\xi \mapsto \xi \wedge \gamma^{n-1}$ is an isomorphism from real one-forms to real $(2n-1)$-forms, as one sees in a unitary frame. Its inverse depends smoothly on the point. Let B and C be the images of $\mathrm{d}\theta \wedge \gamma^{n-2}$ and $\mathrm{d}\gamma \wedge \gamma^{n-2}$ under this inverse. They are smooth real one-forms, and

$$\mathrm{d}a = B - aC$$

in distributions. Since $a \in L^1_{\mathrm{loc}}$, this first gives $a \in W^{1,1}_{\mathrm{loc}}$. Differentiating the equation and using smoothness of its coefficients gives $a \in W^{k,1}_{\mathrm{loc}}$ for every integer k. Local Sobolev embedding makes a smooth. Now $\mathrm{dd^c} u = a\gamma - \theta$ is smooth; taking its trace in local coordinates and applying elliptic regularity makes u smooth as well. □

1.2 Changes of background

Let η be another smooth real $(1,1)$-form on X. A positive current can be written as a smooth background plus a complex Hessian in more than one way. A smooth change of decomposition changes the potential by a bounded function; positive linear combinations also allow us to compare potentials belonging to different backgrounds.

Proposition 1.2.1 *Let f be smooth and real, and set $\theta' = \theta + \mathrm{dd^c} f$. Then*

$$\mathrm{PSH}(X, \theta) \longrightarrow \mathrm{PSH}(X, \theta'), \qquad u \longmapsto u - f$$

is a bijection. It preserves the positive current and changes the potential by a bounded function. Moreover, $\mathrm{d}\theta' = \mathrm{d}\theta$.

Proof For every u in the source,

$$\theta' + \mathrm{dd^c}(u - f) = \theta + \mathrm{dd^c} u.$$

The inverse is $v \mapsto v + f$, and f is bounded by compactness. The last assertion follows from $\mathrm{ddd^c} f = 0$. □

The quotient of smooth real $(1,1)$-forms by smooth $\mathrm{dd^c}$-terms thus permits non-closed representatives. The usual Bott–Chern cohomology group is the corresponding quotient of closed forms.

Proposition 1.2.2 *If $u \in \mathrm{PSH}(X, \theta)$ and $v \in \mathrm{PSH}(X, \eta)$, then, for $a, b \geq 0$ with $a + b > 0$,*

$$au + bv \in \mathrm{PSH}(X, a\theta + b\eta).$$

In particular, $\mathrm{PSH}(X, \theta)$ is convex. If $\eta \geq \theta$, then $\mathrm{PSH}(X, \theta) \subseteq \mathrm{PSH}(X, \eta)$.

Proof Sums with nonnegative coefficients preserve quasi-plurisubharmonicity, with zero-coefficient terms omitted. The curvature identity is

$$a\theta + b\eta + \mathrm{dd^c}(au + bv) = a(\theta + \mathrm{dd^c}u) + b(\eta + \mathrm{dd^c}v) \geq 0.$$

For the inclusion, add the positive smooth form $\eta - \theta$ to $\theta + \mathrm{dd^c}u$. □

By Proposition 1.1.1, any two quasi-plurisubharmonic functions belong to $\mathrm{PSH}(X, C\omega)$ for a common sufficiently large C. This lets us compare their singularities within one positive background.

1.3 Maxima and the maximum principle

Maxima allow us to truncate potentials and to replace them on an open set while preserving a lower curvature bound. For a smooth background, the local psh maximum and gluing principles retain that bound by freezing the coefficients and then letting the curvature error tend to zero.

Lemma 1.3.1 *If $u, v \in \mathrm{PSH}(X, \theta)$, then $\max(u, v) \in \mathrm{PSH}(X, \theta)$. Consequently, if $\theta \geq 0$ and $u \in \mathrm{PSH}(X, \theta)$, then $\max(u, -t)$ is θ-plurisubharmonic for every real t.*

Proof The maximum is quasi-plurisubharmonic by a common local smooth correction. To verify its precise lower Hessian bound, fix a positive constant $(1, 1)$-form κ in a coordinate ball. For each $\varepsilon > 0$, freeze θ on smaller neighborhoods to a constant form θ' with $-\varepsilon\kappa \leq \theta - \theta' \leq \varepsilon\kappa$. Write $\theta' + \varepsilon\kappa = \mathrm{dd^c}r$ locally. Then $u + r$ and $v + r$ are plurisubharmonic, hence so is their maximum. It follows that

$$\theta + 2\varepsilon\kappa + \mathrm{dd^c}\max(u, v) \geq 0$$

on each such neighborhood. Let $\varepsilon \to 0$. For the last assertion, constants belong to $\mathrm{PSH}(X, \theta)$ when $\theta \geq 0$. □

Lemma 1.3.2 *Let Y be a complex manifold, let β be a smooth real $(1, 1)$-form on Y, and let $D \subset Y$ be open. Suppose u is β-plurisubharmonic on Y, v is β-plurisubharmonic on D, and*

$$\limsup_{D \ni y \to x} v(y) \leq u(x) \qquad (x \in \partial D).$$

Then the function equal to $\max(u, v)$ on D and to u on $Y \setminus D$ is β-plurisubharmonic.

Proof Call the pasted function w. It is upper semicontinuous by the boundary inequality. Freeze β on a small coordinate ball as in Lemma 1.3.1, and choose a quadratic correction r so that $u + r$ and $v + r$ are plurisubharmonic on their respective domains. Away from ∂D, the function $w + r$ is plurisubharmonic by the maximum theorem. At a point of ∂D it equals $u + r$ and dominates $u + r$ on the ball, so it satisfies the submean inequality on every complex line through that point. Thus $w + r$ is plurisubharmonic on the whole ball. Letting the coefficient-freezing error tend to zero gives $\beta + \mathrm{dd^c} w \geq 0$. □

Almost-everywhere order also determines pointwise order. This permits comparisons proved in a Sobolev class to be applied to the quasi-plurisubharmonic potentials themselves.

Lemma 1.3.3 *Let a, b be quasi-plurisubharmonic functions. If $a \leq b$ almost everywhere, then $a \leq b$ everywhere. In particular, equality almost everywhere implies equality everywhere.*

Proof On a coordinate ball add the same smooth function q so that $a + q$ and $b + q$ are plurisubharmonic. At each point a plurisubharmonic function equals the limit of its ball averages as the radius tends to zero, by the submean inequality and upper semicontinuity. The almost-everywhere inequality orders these averages, so passage to the limit gives $a + q \leq b + q$. Subtract q. □

Lemma 1.3.4 *If a, b are quasi-plurisubharmonic functions and $\{a < b\}$ is nonempty at a point where both are finite, then this set has positive smooth volume.*

Proof Choose a point x with $a(x) < b(x)$ and a number c between them. Upper semicontinuity gives $a < c$ on a neighborhood V of x. If $\{b > c\} \cap V$ had zero volume, then $b \leq c$ almost everywhere on V. By Lemma 1.3.3, this would imply $b(x) \leq c$, a contradiction. Thus $\{a < b\}$ has positive volume. □

Lemma 1.3.5 *A plurisubharmonic function on $\mathbb{C}^m$, $m \geq 1$, that is bounded above is constant.*

Proof First let f be smooth and subharmonic on $\mathbb{C}$, with $f \leq M$. Write $\sigma = (\Delta f)\,\mathrm{d}A$ for its nonnegative Laplacian measure, where Δ and $\mathrm{d}A$ are Euclidean. Green's formula for circular means gives, for $R > 0$,

$$\frac{1}{2\pi}\int_0^{2\pi} f(R\mathrm{e}^{\mathrm{i}t})\,\mathrm{d}t - f(0) = \frac{1}{2\pi}\int_0^R \frac{\sigma(B(0,r))}{r}\,\mathrm{d}r.$$

For fixed $r_0 > 0$ and $R > r_0$, the right side is at least $\sigma(B(0,r_0))\log(R/r_0)/(2\pi)$, while the left side is at most $M - f(0)$. Letting $R \to \infty$ gives $\sigma(B(0,r_0)) = 0$. Thus f is harmonic, and the harmonic Liouville theorem applied to the positive harmonic function $M + 1 - f$ makes f constant.

For the given plurisubharmonic function, convolution with a smooth nonnegative radial probability kernel gives smooth plurisubharmonic functions with the same

upper bound. Their restrictions to every affine complex line are constant by the planar argument, so each convolution is constant on $\mathbb{C}^m$. Local L^1 convergence of the convolutions and the submean representative give the assertion. □

We write $f^*(x) = \limsup_{y\to x} f(y)$ for upper semicontinuous regularization. A subset is *pluripolar* if locally it is contained in $\{p = -\infty\}$ for a plurisubharmonic function p. The local upper-envelope theorem in [Xia] states that a nonempty, locally uniformly upper-bounded family of plurisubharmonic functions has a plurisubharmonic regularized supremum; regularization changes the supremum only on a pluripolar set.

Periodicity also forces rigidity when an entire subharmonic function has only logarithmic upper growth.

Lemma 1.3.6 *Let h be subharmonic on $\mathbb{C}$, with $h(\zeta + 1) = h(\zeta)$. If there are real constants $A \geq 0$ and B such that*

$$h(\zeta) \leq A \log(1 + |\zeta|) + B \qquad (\zeta \in \mathbb{C}),$$

then h is constant.

Proof The locally bounded-above family of real translates has the subharmonic upper envelope

$$H(\zeta) = \big(\sup_{s\in[0,1]} h(\zeta + s)\big)^*.$$

Periodicity makes H invariant under every real translation. Subharmonicity therefore says that H is a convex function of $\operatorname{Im}\zeta$. On the imaginary axis it satisfies $H(\mathrm{i}y) \leq A\log(2 + |y|) + B$. A nonconstant convex function on $\mathbb{R}$ grows at least linearly at one end, so this bound makes H constant. Since $h \leq H$, Lemma 1.3.5 now makes h constant. □

Lemma 1.3.7 *Let U be a complex manifold, let θ be a smooth real $(1,1)$-form on U, and let $Z \subset U$ be closed and locally pluripolar. If u is θ-plurisubharmonic on $U \setminus Z$ and locally bounded above near Z, its upper-limit extension is θ-plurisubharmonic on U. In particular this applies when Z is a proper analytic subset.*

Proof First suppose u is plurisubharmonic. Locally choose a genuine plurisubharmonic $\psi \leq 0$ with $\psi = -\infty$ on Z. For $t > 0$, extend $u + t\psi$ by $-\infty$ on Z. The local upper bound makes this extension upper semicontinuous. It is plurisubharmonic off Z, and its submean inequality on any complex line is automatic at points of Z. It is therefore plurisubharmonic on the whole neighborhood. Let $t \downarrow 0$ and regularize the increasing limit. The local upper-envelope theorem gives a plurisubharmonic function agreeing with u almost everywhere off Z, hence everywhere there. This is the asserted upper-limit extension.

For a smooth background, fix a constant positive form κ in a coordinate chart. On a sufficiently small ball, freeze θ to a constant form θ' with $|\theta - \theta'| \leq \epsilon\kappa$, and choose a quadratic potential q for $\theta' + \epsilon\kappa$. Apply the preceding argument to $u + q$. Its extension gives $\theta + \mathrm{dd^c}u \geq -2\epsilon\kappa$. Let $\epsilon \downarrow 0$. The extensions agree on overlaps

by their upper-limit definition. Analytic subsets are locally pluripolar, as follows by taking the logarithm of the squared norm of local defining functions. □

Lemma 1.3.8 *Let X be a complex manifold of dimension $n \geq 2$, let θ be a smooth real $(1,1)$-form on X, and let $p \in X$. Every θ-plurisubharmonic function on $X \setminus \{p\}$ is locally bounded above near p and extends uniquely to a member of* $\mathrm{PSH}(X, \theta)$. *If X is compact, it is bounded above on $X \setminus \{p\}$.*

Proof Choose coordinates centered at p and write $\Delta_R = \{z \in \mathbb{C} : |z| < R\}$. On the coordinate polydisc Δ_R^n, add a smooth quadratic function to obtain a plurisubharmonic function h on $\Delta_R^n \setminus \{0\}$. Write the coordinates as (z_1, z_2, z'), with z' absent when $n = 2$, and choose $0 < r < R$. The compact set

$$K = \{|z_1| \leq r,\ |z_2| = r,\ |z'_j| \leq r\}$$

avoids the puncture, so $h \leq M$ on K for some finite M. For $z_1 \neq 0$, the maximum principle in the z_2-disc gives the same bound throughout the smaller polydisc. If $z_1 = 0$ and $(z_2, z') \neq 0$, the z_1-disc avoids the puncture, and its boundary is already covered by this bound. Applying the maximum principle once more gives $h \leq M$ there too. Subtracting the smooth correction proves the local upper bound.

Lemma 1.3.7 supplies the extension with its original lower curvature bound. Uniqueness follows from Lemma 1.3.3. On a compact X, upper semicontinuity then gives a global upper bound. □

Lemma 1.3.9 *Let u be θ-plurisubharmonic near a point x. If a C^2 function a touches u from above at x, then $\theta(x) + \mathrm{dd^c} a(x) \geq 0$.*

Proof Freeze $\theta(x)$ on a small coordinate ball and take a quadratic potential for this constant form. After an arbitrarily small positive quadratic correction, its sum with u is plurisubharmonic. An upper test for a plurisubharmonic function has nonnegative complex Hessian, by the submean inequality on complex lines. Let the size of the correction tend to zero. □

1.4 Normalized compactness

Fix a smooth positive probability volume $\mathrm{d}V$ for L^1 norms. The local compactness theorem for plurisubharmonic functions, recalled in [Xia], gives an L^1_{loc}-convergent subsequence or local uniform convergence to $-\infty$ for a locally uniformly upper-bounded sequence on a connected domain. A point at which the global maximum is attained excludes the second alternative, and connectedness propagates the first across X.

The submean inequality also controls upper limits at moving points.

Lemma 1.4.1 *Let u_j, u be quasi-plurisubharmonic on an open set U in a complex manifold. Suppose that $u_j \to u$ in $L^1_{\mathrm{loc}}(U)$ and $\mathrm{dd^c} u_j \geq -\beta$ for one smooth real $(1,1)$-form β on U. If $x_j \to x$ in U, then*

$$\limsup_{j\to\infty} u_j(x_j) \leq u(x).$$

Moreover, the tail envelopes $h_j = (\sup_{k\geq j} u_k)^*$ decrease to u.

Proof On a coordinate ball about x, add a common smooth correction q so that $u_j + q$ and $u + q$ are plurisubharmonic. For a fixed small radius, the submean inequality bounds $(u_j + q)(x_j)$ by its average on the ball centered at x_j. These averages converge to the corresponding average of $u + q$ centered at x: the error from $u_j - u$ tends to zero in L^1, and translations are continuous for the integral of a fixed L^1 function over a ball. Letting the radius decrease to zero proves the inequality.

The same estimate gives local uniform upper bounds for the u_j, so the tail envelopes are quasi-plurisubharmonic with the same lower Hessian bound, by coefficient freezing and the local upper-envelope theorem. If their decreasing limit exceeded $u(x)$ at some x, the definition of regularization would give indices $k_j \geq j$ and points $x_j \to x$ contradicting the first assertion. An almost-everywhere convergent subsequence of (u_j) shows that $h_j \geq u$ almost everywhere, hence everywhere by Lemma 1.3.3. Thus $h_j \downarrow u$. □

Theorem 1.4.1 *For any smooth real $(1,1)$-form θ, the family*

$$\mathcal{K}_\theta = \{u \in \mathrm{PSH}(X,\theta) : \sup_X u = 0\}$$

is compact in $L^1(X, \mathrm{d}V)$. In particular, there is a constant C_θ such that

$$\int_X |u| \, \mathrm{d}V \leq C_\theta \quad (u \in \mathcal{K}_\theta).$$

Proof Let (u_j) be a sequence in this family and choose x_j with $u_j(x_j) = 0$. Passing to a subsequence, suppose $x_j \to x$. Cover X by finitely many connected coordinate balls, and on a slightly larger ball choose a fixed smooth correction r_i for which $u_j + r_i$ is plurisubharmonic. Such corrections exist because θ is smooth. These local functions are uniformly bounded above, since $u_j \leq 0$.

On a ball containing x in its interior, the alternative of locally uniform convergence to $-\infty$ is impossible: the values at x_j stay bounded below. Local compactness therefore gives an L^1-convergent subsequence on this ball. If a second ball overlaps it, local compactness on the second ball cannot yield convergence to $-\infty$ on compact subsets of the overlap, since the first limit is finite almost everywhere there. Thus a further subsequence converges in L^1 on the second ball. The intersection graph of a finite connected open cover of connected X is connected. Repeating along that graph and taking finitely many subsequences gives local L^1 convergence everywhere. A partition of unity then gives convergence in $L^1(X)$ to a function u.

The inequalities $\mathrm{dd^c} u_j \geq -\theta$ pass to distributions. The local canonical representatives of $u + r_i$ are plurisubharmonic and agree on overlaps after subtracting their corrections. They define a global θ-plurisubharmonic representative, still denoted by u. Since $u_j \leq 0$, it satisfies $u \leq 0$.

By Lemma 1.4.1, $0 = \lim_j u_j(x_j) \leq u(x)$. Thus $u(x) = 0$, and the limit belongs to $\mathcal{K}_\theta$. This proves sequential compactness, which is compactness in the metric space $L^1(X)$.

If the displayed norm bound failed, choose a sequence whose L^1 norms tend to infinity. Its convergent subsequence would have bounded norms, a contradiction. □

Corollary 1.4.1 *A decreasing sequence in* $\mathrm{PSH}(X, \theta)$ *with uniformly bounded* L^1 *norms has a genuine* θ*-plurisubharmonic limit and converges to it in* L^1.

Proof Local decreasing-limit theory gives a quasi-plurisubharmonic limit or the function $-\infty$ on each connected coordinate neighborhood. The L^1 bound excludes the latter alternative, by Fatou's lemma applied to the negative parts. The local representatives glue. Finally, the sequence is bounded above by its first member; subtracting the sequence from that member and applying monotone convergence proves L^1 convergence. □

1.5 Uniform exponential integrability

Skoda's local integrability theorem, recalled in [Xia], gives an integrable exponential for each quasi-plurisubharmonic potential v on X. Applying it after local corrections and taking the minimum exponent over a finite cover gives $\mathrm{e}^{-a(v)v} \in L^1(X)$ for some $a(v) > 0$. Normalized compactness makes the exponent and integral bound uniform: otherwise a convergent weighted sum of counterexamples would violate this conclusion.

Write K_ω for the bound in Theorem 1.4.1, so that normalized potentials $v \in \mathrm{PSH}(X, \omega)$ with $\sup_X v = 0$ satisfy

$$\int_X |v| \, \mathrm{d}V \leq K_\omega. \tag{1.1}$$

Lemma 1.5.1 *There are constants* $\alpha > 0$ *and* $C < +\infty$*, depending on* $(X, \omega, \mathrm{d}V)$*, such that*

$$\int_X \mathrm{e}^{-\alpha v} \, \mathrm{d}V \leq C \quad \textit{for all } v \in \mathrm{PSH}(X, \omega), \quad \sup_X v = 0.$$

Proof Suppose no such pair exists. Set $a_j = 2^{-j}$ and choose normalized potentials $v_j \leq 0$ with

$$\int_X \exp\left(-\frac{a_j}{j} v_j\right) \mathrm{d}V \geq 2^j.$$

The partial sums $V_N = \sum_{j \le N} a_j v_j$ decrease, are ω-plurisubharmonic, and have uniformly bounded L^1 norms by (1.1). Their limit V is therefore ω-plurisubharmonic. Since all summands are nonpositive, $V \le a_j v_j$ for every j.

Fix $p > 0$. For every $j \ge 1/p$,

$$\int_X \mathrm{e}^{-pV}\,\mathrm{d}V \ge \int_X \mathrm{e}^{-pa_j v_j}\,\mathrm{d}V \ge \int_X \mathrm{e}^{-(a_j/j)v_j}\,\mathrm{d}V \ge 2^j.$$

Thus e^{-pV} is nonintegrable for every $p > 0$, contradicting the qualitative Skoda input applied to V. □

1.6 Local products and capacity

The local Bedford–Taylor product and capacity measure the exceptional sets of plurisubharmonic functions and control bounded approximation. We recall these classical inputs from [Xia, GZ17]. Fix $1 \le k \le n$. The *plurifine topology* is the coarsest topology making local plurisubharmonic functions continuous. It is finer than the ordinary topology and, in particular, makes strict superlevel sets of quasi-plurisubharmonic functions open.

Theorem 1.6.1 (Bedford–Taylor) *If $p_1, \dots, p_k$ are locally bounded plurisubharmonic functions on a domain in $\mathbb{C}^n$, their mixed product $\mathrm{dd^c} p_1 \wedge \cdots \wedge \mathrm{dd^c} p_k$ is a positive closed current of order zero. It is symmetric, multilinear in its current factors, and continuous under simultaneous locally bounded decreasing approximation. It does not charge pluripolar sets and is local in the plurifine topology.*

Here locality means equality of the restricted currents on a plurifine open Borel set where the potentials agree.

For concentric coordinate balls $B' \Subset B$ and a Borel set $E \subset B'$, let $\mathrm{cap}(E, B) = \sup\{\int_E (\mathrm{dd^c} v)^n : v \in \mathrm{PSH}(B),\ 0 \le v \le 1\}$ be the relative Bedford–Taylor capacity. If v is bounded and plurisubharmonic on B, normalize its range to $[0, 1]$ to obtain

$$(\mathrm{dd^c} v)^n(E) \le (\sup_B v - \inf_B v)^n \,\mathrm{cap}(E, B).$$

The constant case has zero measure. Write $\mathrm{Vol}(E) = \int_E \mathrm{d}V$. We use two standard local estimates on these balls, with constants $C, a > 0$ depending only on the nested balls and the fixed volume form:

(1) If $w \le 0$ is plurisubharmonic on B and v is bounded and plurisubharmonic, then

$$\int_{B'} |w|(\mathrm{dd^c} v)^n \le C\|w\|_{L^1(B)}\|v\|^n_{L^\infty(B)}.$$

An additive constant can first be chosen in v. This is the weighted Chern–Levine–Nirenberg inequality.

(2) For every Borel set $E \subset B'$, the relative Bedford–Taylor capacity satisfies

$$\mathrm{Vol}(E) \le C \exp\bigl(-a \operatorname{cap}(E, B)^{-1/n}\bigr),$$

with value zero when the capacity is zero. This is the classical local volume–capacity estimate.

These are statements of local pluripotential theory, as developed in [GZ17, Chapters 3 and 4]; the bounded product and capacity conventions also agree, up to harmless fixed constants, with [Xia, Chapter 2].

The dependence on the range can be read directly from the cutoff argument. If $M > 0$ and p is plurisubharmonic on B with $0 \le p \le M$, choose a cutoff $0 \le \chi \le 1$ supported in B, equal to one on a smaller ball, with $|\mathrm{dd^c}\chi| \le C\kappa$, where κ is the Euclidean form. For $1 \le j \le n$,

$$\begin{aligned}\int \chi(\mathrm{dd^c}p)^j \wedge \kappa^{n-j} &= \int p\, \mathrm{dd^c}\chi \wedge (\mathrm{dd^c}p)^{j-1} \wedge \kappa^{n-j} \\ &\le CM \int_{\operatorname{supp}\chi} (\mathrm{dd^c}p)^{j-1} \wedge \kappa^{n-j+1}.\end{aligned}$$

Use nested cutoffs between B' and B. Stopping the recursion at degree one, or at degree zero, gives respectively

$$\begin{aligned}\int_{B'} (\mathrm{dd^c}p)^j \wedge \kappa^{n-j} &\le C_j M^{j-1} \int_B \mathrm{dd^c}p \wedge \kappa^{n-1}, \\ \int_{B'} (\mathrm{dd^c}p)^j \wedge \kappa^{n-j} &\le C_j M^j.\end{aligned} \tag{1.2}$$

The constants depend only on the nested balls and j. Local decreasing regularization justifies the integrations by parts for bounded p.

Convergence $f_j \to f$ in local capacity means that $\operatorname{cap}(K \cap \{|f_j - f| > \epsilon\}, B) \to 0$ for every $\epsilon > 0$ and compact $K \Subset B$ in each coordinate ball. We say *quasi-everywhere*, abbreviated q.e., when an assertion holds outside a pluripolar set; differences at common poles are interpreted off that set. A positive measure is *non-pluripolar* if it gives every pluripolar set mass zero.

A function is *quasi-continuous* if it is continuous outside open sets of arbitrarily small capacity. Plurisubharmonic functions have this property for the local Bedford–Taylor capacity, and Borel sets have zero capacity precisely when they are pluripolar. Uniformly bounded psh potentials converging in capacity have weakly convergent local products. The same holds after multiplying the products by uniformly bounded quasi-continuous coefficients converging in capacity. These local results, recalled in [Xia, GZ17], also apply after a common smooth correction to quasi-plurisubharmonic functions.

Theorem 1.6.2 *Let $U \subset \mathbb{C}^n$ be a domain and let β be its Euclidean Hermitian form. The classical local mixed Monge–Ampère inequality says that if a, b are locally bounded plurisubharmonic functions and A, B are nonnegative measurable densities*

on U with

$$(\mathrm{dd^c}a)^n \geq A\beta^n, \qquad (\mathrm{dd^c}b)^n \geq B\beta^n,$$

then, for $0 \leq k \leq n$,

$$(\mathrm{dd^c}a)^k \wedge (\mathrm{dd^c}b)^{n-k} \geq A^{k/n}B^{(n-k)/n}\beta^n.$$

If a, b are locally bounded plurisubharmonic functions, $E = \{a < b\} \Subset U$, and S is a positive closed $(n-1, n-1)$-current obtained as a product of bounded plurisubharmonic Hessians and smooth closed positive forms, then

$$\int_E \mathrm{dd^c}b \wedge S \leq \int_E \mathrm{dd^c}a \wedge S. \tag{1.3}$$

Indeed, apply the Bedford–Taylor integration-by-parts formula to $b_\epsilon = \max\{a, b-\epsilon\}$ and let $\epsilon \downarrow 0$; monotone continuity gives the limit. Replacing one factor at a time yields, for $P = \mathrm{dd^c}a$ and $Q = \mathrm{dd^c}b$,

$$\int_E P^{n-j} \wedge Q^j \leq \int_E P^n, \qquad 0 \leq j \leq n. \tag{1.4}$$

These are the mixed forms of the local Bedford–Taylor comparison principle from [Xia].

The local obstacle theorem gives a corresponding contact statement. Let B be a ball, $h < 0$ a smooth strictly plurisubharmonic defining function, and f a bounded quasi-continuous obstacle with $Ah \leq f \leq 0$ for some $A > 0$. The envelope

$$q = \big(\sup\{v \in \mathrm{PSH}(B) : v \leq f \text{ q.e.}\}\big)^*$$

satisfies $Ah \leq q \leq 0$, tends to zero at the boundary, and obeys

$$\mathbf{1}_{\{q<f\}}(\mathrm{dd^c}q)^n = 0. \tag{1.5}$$

This is the bounded local envelope theorem from [Xia, GZ17].

For a locally bounded quasi-plurisubharmonic function v and a smooth convex increasing function F on a neighborhood of its range, the bounded Bedford–Taylor chain rule is

$$\mathrm{dd^c}F(v) = F'(v)\mathrm{dd^c}v + F''(v)\,\mathrm{d}v \wedge \mathrm{d^c}v. \tag{1.6}$$

The gradient term and its products with positive bounded Bedford–Taylor factors are nonnegative. These local identities follow by decreasing regularization; bounded potentials have the local Sobolev regularity needed for the gradient expression. They are part of the bounded local calculus recalled in [Xia, GZ17]. In particular, inequalities obtained from this formula pass through local smoothing with a lower-curvature error tending to zero.

Lemma 1.6.1 *Let B be a coordinate ball and let τ be a finite non-pluripolar measure with compact support in B. There is a bounded plurisubharmonic function v on B such that $\tau \ll (\mathrm{dd^c}v)^n$.*

Proof Cegrell's decomposition theorem says the following. On a bounded hyperconvex domain in $\mathbb{C}^n$, $n \geq 2$, let $\mathcal{E}_0$ consist of bounded negative plurisubharmonic functions with zero boundary values and finite total Monge–Ampère mass. Every positive compactly supported measure τ has a decomposition

$$\tau = g(\mathrm{dd^c} v)^n + \tau_s, \qquad v \in \mathcal{E}_0, \qquad 0 \leq g \in L^1((\mathrm{dd^c} v)^n),$$

where τ_s is a positive measure carried by a pluripolar set. If τ is non-pluripolar, $\tau_s = 0$, so $\tau \ll (\mathrm{dd^c} v)^n$.

For completeness the required local assertion on a disk also follows from the Green potential. Let $a \leq 0$ be the Green potential of a finite compactly supported non-polar measure τ, normalized by $\mathrm{dd^c} a = \tau$. Then $v = \mathrm{e}^a$ is bounded and subharmonic, and

$$\mathrm{dd^c} v \geq \mathrm{e}^a \tau.$$

To justify the inequality, set $a_M = \max(a, -M)$. The bounded subharmonic chain rule gives $\mathrm{dd^c} \mathrm{e}^{a_M} \geq \mathrm{e}^{a_M} \mathrm{dd^c} a_M$, which on $\{a > -M\}$ is at least $\mathrm{e}^a \tau$. Let $M \to \infty$, using decreasing convergence of the bounded subharmonic functions e^{a_M} and monotone convergence on the right. Since $\{a = -\infty\}$ is polar, $\mathrm{e}^a > 0$ almost everywhere for τ. Thus $\tau \ll \mathrm{dd^c} v$. The chain inequality is (1.6) in dimension one. □

For a signed measure ν, write $|\nu|$ for its total variation measure and $\|\nu\|_{\mathrm{TV}(B)} = |\nu|(B)$. On a curve, the Green representation controls capacity by the total variation of the Laplacian, with an additional L^1 term for the harmonic remainder.

Lemma 1.6.2 *Let $B' \Subset B$ be concentric discs in $\mathbb{C}$. There is a constant C, depending only on these discs, such that for quasi-subharmonic u, v on a neighborhood of $\overline{B}$ and $t > 0$,*

$$\mathrm{cap}(B' \cap \{|u - v| > t\}, B) \leq \frac{C}{t} \left(\|u - v\|_{L^1(B)} + \|\mathrm{dd^c}(u - v)\|_{\mathrm{TV}(B)} \right). \tag{1.7}$$

Pole sets are omitted when forming the difference.

Proof Put $f = u - v$ and write the Jordan decomposition on B as $\mathrm{dd^c} f = \alpha^+ - \alpha^-$. Let $g^\pm \leq 0$ be the Dirichlet Green potentials of $\alpha^\pm$ on B. The $L^1(B)$ norm of the disc Green kernel is bounded uniformly over its pole, so

$$\|g^+\|_{L^1(B)} + \|g^-\|_{L^1(B)} \leq C \|\mathrm{dd^c} f\|_{\mathrm{TV}(B)}.$$

Indeed, on the unit disc its kernel is $\log |(z - \zeta)/(1 - \overline{\zeta} z)|^2$; the logarithmic singularity is uniformly integrable and $|1 - \overline{\zeta} z| \leq 2$. Scaling gives the bound on B.

The distribution $h = f - g^+ + g^-$ is harmonic. The interior mean-value estimate gives

$$\|h\|_{L^\infty(B')} \leq C \left(\|f\|_{L^1(B)} + \|\mathrm{dd^c} f\|_{\mathrm{TV}(B)} \right).$$

The identity $u + g^- = v + g^+ + h$ holds everywhere by almost-everywhere order for quasi-subharmonic functions. Thus $|u - v| \leq -g^+ - g^- + |h|$ off their polar sets.

For every subharmonic test $0 \le a \le 1$ on B, the weighted Chern–Levine–Nirenberg estimate and (1.2) yield

$$\int_{B'} |u - v| \, \mathrm{dd^c} a \le C \left(\|f\|_{L^1(B)} + \|\mathrm{dd^c} f\|_{\mathrm{TV}(B)} \right).$$

The bounded test measure charges no polar set. Restrict to $\{|u - v| > t\}$, divide by t, and take the supremum over a. □

1.6.1 Polar parts on curves

A locally finite signed measure ν on a Riemann surface splits uniquely into its polar part ν^{pol}, carried by a polar Borel set, and a measure charging no polar set. Locally, take a countable union of polar sets whose $|\nu|$-masses approach the supremum over polar sets, and restrict ν to this union. Uniqueness patches these decompositions and makes the operation linear.

For locally quasi-subharmonic u, its polar curvature is the positive measure $\mu_u = (\mathrm{dd^c} u)^{\mathrm{pol}} = \mathbf{1}_{\{u=-\infty\}} \mathrm{dd^c} u$. Indeed, bounded truncations and plurifine locality show that $\mathbf{1}_{\{u>-j\}} \mathrm{dd^c} u$ charges no polar set; smooth corrections do not change polar parts. An upper bound on a difference of potentials forces an order on their polar measures.

Lemma 1.6.3 *Let $B \subset \mathbb{C}$ be a disc and let $f \in L^1_{\mathrm{loc}}(B)$ have a distributional Laplacian that is a signed Radon measure. If $f \le 0$ almost everywhere, then $(\Delta f)^{\mathrm{pol}} \ge 0$.*

Proof It suffices to prove $\Delta f(K) \ge 0$ for every compact polar $K \Subset B$; restriction to smaller discs and inner regularity then give the assertion. Choose smooth open neighborhoods O_j of K with $\overline{O}_{j+1} \subset O_j \Subset B$ and $\bigcap_j \overline{O}_j = K$. Let h_j equal one on $\overline{O}_j$, be harmonic on $B \setminus \overline{O}_j$, and vanish on ∂B. The Dirichlet problem on this smooth domain gives a continuous $0 \le h_j \le 1$. Its outward derivative from O_j is nonpositive, so the distributional boundary jump gives $\Delta h_j \le 0$ on B. The maximum principle gives $h_{j+1} \le h_j$.

The bounded distributional limit h is superharmonic and harmonic off K. Its Laplacian is supported on K, whereas the Laplacian of the bounded subharmonic representative of $-h$ charges no polar set by Theorem 1.6.1. Hence h is harmonic on B. Since $0 \le h \le h_1$ and h_1 tends to zero at ∂B, the maximum principle gives $h = 0$. It follows that $h_j \to 0$ smoothly on compact subsets of $B \setminus K$, while $h_j = 1$ on K.

Choose $0 \le \chi \in C_c^\infty(B)$ equal to one near K. For large j, smooth h_j by convolution on a neighborhood of $\operatorname{supp} \chi$, obtaining $h_{j,\epsilon}$ with $\Delta h_{j,\epsilon} \le 0$. The sign of f gives

$$\begin{aligned} \int_B \chi h_{j,\epsilon} \, \mathrm{d}(\Delta f) &= \int_B f \, \Delta(\chi h_{j,\epsilon}) \, \mathrm{d}x \, \mathrm{d}y \\ &\ge \int_B f \left(2\nabla\chi \cdot \nabla h_{j,\epsilon} + h_{j,\epsilon} \Delta\chi \right) \mathrm{d}x \, \mathrm{d}y. \end{aligned}$$

For fixed j, continuity of h_j gives uniform convergence of its convolutions on $\operatorname{supp}\chi$. On the support of the derivatives of χ, the function h_j is harmonic for all large j. We may therefore first let $\epsilon \downarrow 0$ and then $j \to \infty$. Smooth convergence away from K makes the right-hand side tend to zero. Dominated convergence for the signed measure Δf makes the left-hand side tend to $\Delta f(K)$, proving the claim. □

Corollary 1.6.1 *For locally quasi-subharmonic u, v, the local inequality $u \le v + C$ implies $\mu_u \ge \mu_v$. Moreover,*

$$\mu_{\max(u,v)} = \mu_u \wedge \mu_v, \tag{1.8}$$

where the right-hand side is the greatest measure dominated by both factors.

Proof Apply Lemma 1.6.3 to $u - v - C$. Smooth corrections have zero polar part, so linearity gives the first assertion. Apply it to $u - \max(u, v)$ and $v - \max(u, v)$ to obtain the inequality $\le$ in (1.8).

For the reverse inequality, add a common smooth correction to make u, v subharmonic on a coordinate disc. On a smaller disc let g be the Green potential of $\lambda = \mu_u \wedge \mu_v$ restricted to that disc, so $\mathrm{dd^c} g = \lambda$. The distributions $u - g$ and $v - g$ have positive Hessians and hence subharmonic representatives a, b. Since $\max(u, v) = g + \max(a, b)$ almost everywhere, its Hessian dominates λ. Restriction to a polar carrier of λ proves the reverse inequality. Localizing over the surface completes the proof. □

1.7 Products with smooth backgrounds

The Bedford–Taylor construction also defines products after adding smooth forms to complex Hessians. Independence of the smooth correction makes these products global; positivity requires retaining the original lower curvature bound. Write $\theta_v = \theta + \mathrm{dd^c} v$ and fix $1 \le k \le n$.

Let u be locally bounded and θ-plurisubharmonic. On a relatively compact coordinate ball with coordinates z, choose $A > 0$ and put $q = A|z|^2$ so that $p = u + q$ is plurisubharmonic. Set $\beta = \theta - \mathrm{dd^c} q$ and define

$$(\theta + \mathrm{dd^c} u)^k := \sum_{j=0}^{k} \binom{k}{j} (\mathrm{dd^c} p)^j \wedge \beta^{k-j}. \tag{1.9}$$

Each summand is defined by Theorem 1.6.1 and multiplication by a smooth form. The signed terms in this expansion should not individually be assumed positive.

Proposition 1.7.1 *The construction (1.9), and its mixed version, is independent of the local correction q and of the decomposition of each current into a smooth form and a $\mathrm{dd^c}$-potential. It defines global products of currents with locally bounded quasi-plurisubharmonic potentials. The products are symmetric, multilinear, plurifinely*

local, and give no mass to pluripolar sets. If every factor is positive, their mixed product is positive.

Proof On the overlap of two coordinate balls, the two corrected potentials differ by a smooth function. The multilinear rule $\mathrm{dd^c}(p+f) = \mathrm{dd^c}p + \mathrm{dd^c}f$, valid for bounded local products by smooth approximation, and the binomial formula identify the two expansions. The same argument handles a smooth change of the decomposition of a current. Any two potential decompositions of the same $(1,1)$-current do differ smoothly: their difference has smooth $\mathrm{dd^c}$, and taking its local Euclidean trace gives smoothness by elliptic regularity. Symmetry, multilinearity and locality then follow term by term. Wedges of bounded Bedford–Taylor products with smooth forms have locally finite variation and do not charge pluripolar sets, so these properties also pass to the finite expansion.

It remains to prove positivity; it does not follow from the signs of the summands. Write the factors as $T_i = \theta_i + \mathrm{dd^c}u_i$, $1 \le i \le k$, where each θ_i is smooth and real and each u_i is locally bounded and quasi-plurisubharmonic. Assume $T_i \ge 0$. Fix $\varepsilon > 0$ and a local positive constant form κ. After shrinking a neighborhood of a chosen point, each smooth background θ_i admits a constant, hence closed, form θ_i' such that

$$-\varepsilon\kappa \le \theta_i - \theta_i' \le \varepsilon\kappa.$$

Then $S_i = \theta_i' + \varepsilon\kappa + \mathrm{dd^c}u_i$ is positive and closed there, with a bounded local plurisubharmonic potential. Moreover

$$T_i + 2\varepsilon\kappa = S_i + (\theta_i - \theta_i' + \varepsilon\kappa).$$

All terms on the right are positive, and the second term is smooth. Expanding the mixed product therefore gives a sum of positive currents. Although the neighborhoods depend on ε, they cover the original coordinate ball. The product of the $T_i + 2\varepsilon\kappa$ is thus positive throughout that ball. By multilinearity it converges weakly to the product of the T_i as $\varepsilon \to 0$, proving positivity. □

The local expansion also gives continuity along uniformly locally bounded decreasing sequences, by applying Theorem 1.6.1 to the corrected potentials and then summing the finitely many terms. Each bounded top-degree product has finite mass on compact X. For bounded $u \in \mathrm{PSH}(X,\theta)$, we write $\mathrm{MA}_\theta(u) = (\theta + \mathrm{dd^c}u)^n$.

Decreasing maxima compare the products of ordered potentials on their contact set. The comparison also holds with fixed positive factors, which will be useful in integration by parts.

Lemma 1.7.1 *Let $a \le b$ be locally bounded quasi-plurisubharmonic functions on an n-dimensional complex manifold, and let θ be a smooth real $(1,1)$-form such that $\theta_a, \theta_b \ge 0$. For $1 \le k \le n$, let W be a product of $n-k$ positive currents with smooth backgrounds and locally bounded potentials, with $W = 1$ when $k = n$. Then*

$$\mathbf{1}_{\{a=b\}}\theta_a^k \wedge W \le \mathbf{1}_{\{a=b\}}\theta_b^k \wedge W.$$

Proof The functions $m_\delta = \max(a+\delta, b)$ satisfy $b \le m_\delta \le b+\delta$ and decrease to b as $\delta \downarrow 0$. Their products $\theta^k_{m_\delta} \wedge W$ are positive and converge weakly to $\theta^k_b \wedge W$ by bounded decreasing continuity. Locality on $\{a+\delta > b\}$, which contains $\{a = b\}$, identifies their restrictions there with $\theta^k_a \wedge W$. Thus, for every compact $K \subset \{a = b\}$ in a coordinate chart, the compact-set inequality for weak convergence gives

$$\int_K \theta^k_a \wedge W = \lim_{\delta\downarrow 0} \int_K \theta^k_{m_\delta} \wedge W \le \int_K \theta^k_b \wedge W.$$

The measures are locally finite. Inner regularity extends the inequality to Borel subsets of the contact set, and a countable chart covering proves the assertion. □

The products commute with pullback by a holomorphic submersion. Here the pullback of a current is defined in local product coordinates by pulling back its distributional coefficients.

Lemma 1.7.2 *Let $\pi : Z \to Y$ be a holomorphic submersion of complex manifolds. For $1 \le r \le \dim Z$, let $u_1, \dots, u_r$ be locally bounded quasi-plurisubharmonic functions on Y and let $\theta_1, \dots, \theta_r$ be smooth real $(1,1)$-forms. Then*

$$\bigwedge_{i=1}^{r} \big(\pi^*\theta_i + \mathrm{dd^c}(u_i \circ \pi)\big) = \pi^*\left(\bigwedge_{i=1}^{r} (\theta_i + \mathrm{dd^c} u_i)\right).$$

For $r > \dim Y$, the product on the left vanishes and the right side is interpreted as zero.

Proof Work in product coordinates in which π is a coordinate projection. On a smaller base chart choose smooth quadratics q_i such that $p_i = u_i + q_i$ are plurisubharmonic, and approximate each p_i by a decreasing sequence of smooth plurisubharmonic functions. Their pullbacks have the same properties on the product chart. For smooth forms pullback commutes with wedge products. Apply bounded decreasing continuity to the local expansions (1.9) on both charts to obtain the asserted identity. Pullback of currents by the projection is weakly continuous: pairing with a compactly supported test form amounts to pairing the original current with its integral in the fiber variables. If $r > \dim Y$, every smooth approximating product is zero by degree, so its limit is zero as well. □

For $C^{1,1}$ potentials, these measure-valued products agree with the products of their measurable Hessian coefficients.

Proposition 1.7.2 *Let $u_1, \dots, u_k$ be locally $C^{1,1}$ quasi-plurisubharmonic functions and let $\theta_1, \dots, \theta_k$ be smooth real $(1,1)$-forms. The bounded product*

$$(\theta_1 + \mathrm{dd^c} u_1) \wedge \cdots \wedge (\theta_k + \mathrm{dd^c} u_k)$$

coincides with the almost-everywhere wedge of the locally bounded measurable coefficients of its factors.

Proof On a smaller coordinate ball, add a common smooth quadratic q so that each $u_i + q$ is plurisubharmonic, and convolve with a smooth radial probability kernel. The Hessians of the convolutions are uniformly locally bounded and converge to the weak Hessians in every finite L^s space. Their finite products therefore converge in L^1 on smaller balls. The potentials converge uniformly, so bounded Bedford–Taylor continuity identifies this limit with their pluripotential product. Expanding off the smooth correction, as in (1.9), proves the assertion for the original factors. □

The mixed density inequality survives a smooth change of background. Work on an open set $U \subset \mathbb{C}^n$ and put $\beta = \mathrm{dd^c}|z|^2$.

Lemma 1.7.3 *Let $T_i = \vartheta_i + \mathrm{dd^c} u_i \geq 0$, $i = 1, 2$, where ϑ_i are smooth real $(1,1)$-forms and u_i are locally bounded. If*

$$T_i^n \geq F_i \beta^n, \qquad F_i \geq 0 \text{ measurable},$$

then, for every integer $0 \leq k \leq n$,

$$T_1^k \wedge T_2^{n-k} \geq F_1^{k/n} F_2^{(n-k)/n} \beta^n. \tag{1.10}$$

Consequently, for $a, b \geq 0$,

$$(aT_1 + bT_2)^n \geq \left(aF_1^{1/n} + bF_2^{1/n}\right)^n \beta^n. \tag{1.11}$$

Proof Fix $W \Subset U$. On a ball $B_r(x)$ choose a constant form

$$A_i = \vartheta_i(x) + Cr\beta$$

so that $S_i = A_i - \vartheta_i$ satisfies $0 \leq S_i \leq 2Cr\beta$. If q_i is a quadratic potential for A_i, then $u_i + q_i$ is plurisubharmonic and

$$\left(\mathrm{dd^c}(u_i + q_i)\right)^n = (T_i + S_i)^n \geq F_i \beta^n.$$

Apply Theorem 1.6.2 to the two closed Hessians. After expansion, every term not equal to $T_1^k \wedge T_2^{n-k}$ contains some S_i and is bounded by $C' r\mu$, where

$$\mu = \sum_{a+b \leq n-1} T_1^a \wedge T_2^b \wedge \beta^{n-a-b}$$

is locally finite.

A finite cover and a partition of unity give on W

$$T_1^k \wedge T_2^{n-k} \geq F_1^{k/n} F_2^{(n-k)/n} \beta^n - C' r\mu.$$

Letting $r \downarrow 0$ proves (1.10); the binomial expansion gives (1.11). □

Lemma 1.7.4 *Let ϑ be a smooth real $(1,1)$-form on U. Suppose u is continuous, $T = \vartheta + \mathrm{dd^c}u \geq 0$, and $T^n \geq g^n\beta^n$ with $g \geq 0$ continuous. For a nonnegative convolution kernel χ_δ of mass one,*

$$(T * \chi_\delta)^n \geq (g * \chi_\delta)^n \beta^n \tag{1.12}$$

on every smaller domain on which the convolution is defined.

Proof Approximate the convolution by finite convex combinations of translates and apply (1.11). The background forms converge smoothly and the potentials converge uniformly on compact sets. To pass to the limit, add a common quadratic and telescope the difference of two powers. After pairing with a test form, integration by parts bounds each term by the uniform difference of the potentials times a mixed-current mass. Successive integration against nested cutoffs gives a uniform bound for those masses. Thus the bounded-potential products converge and (1.12) follows. □

For a positive current T, write T_{ac} for the measurable $(1,1)$-form given by the absolutely continuous parts of the coefficients of T, and define $\det_\beta T_{\mathrm{ac}}$ by $(T_{\mathrm{ac}})^n = (\det_\beta T_{\mathrm{ac}})\beta^n$. The convolution estimate identifies this determinant when the top product has a continuous density.

Lemma 1.7.5 *Let u be continuous and $T = \vartheta + \mathrm{dd^c}u \geq 0$, with ϑ smooth. If $T^n = F\beta^n$ for continuous $F \geq 0$, then*

$$\det_\beta T_{\mathrm{ac}} = F \qquad \textit{almost everywhere}.$$

Proof Take a nonnegative smooth radial mollifier χ_δ supported in the ball of radius δ. At Lebesgue points of the coefficient densities, $T * \chi_\delta \to T_{\mathrm{ac}}$ almost everywhere. Indeed, if λ_s is the singular part of a coefficient measure, its convolution is bounded by $C\delta^{-2n}|\lambda_s|(B_\delta)$, which tends to zero almost everywhere by differentiation of measures. Apply Lemma 1.7.4 with $g = F^{1/n}$ to obtain $\det_\beta T_{\mathrm{ac}} \geq F$.

For the reverse inequality, write the convolved current as the Hessian of a locally bounded decreasing psh regularization plus a convergent smooth correction. Bounded product continuity gives $(T * \chi_\delta)^n \rightharpoonup T^n$. Fatou's lemma against a nonnegative compactly supported test function then gives $\det_\beta T_{\mathrm{ac}}\, \beta^n \leq T^n = F\beta^n$. □

Lemma 1.7.6 *Let $h \le a$ be locally bounded θ-plurisubharmonic functions, and let χ be smooth on a neighborhood of $(-\infty, 0]$ with $0 \le \chi' \le 1$ and $\chi'' \ge 0$. Then $w = a + \chi(h - a)$ is θ-plurisubharmonic and*

$$\theta_w = (1 - \chi'(h - a))\theta_a + \chi'(h - a)\theta_h + \chi''(h - a)\, \mathrm{d}(h - a) \wedge \mathrm{d}^{\mathrm{c}}(h - a).$$

In particular, $\theta_w^n \ge \chi'(h - a)^n \theta_h^n$. Positivity of θ_w and this product inequality also hold for convex χ of class $C^{1,1}$ on compact intervals, with $0 \le \chi' \le 1$.

Proof The function $G(x, y) = x + \chi(y - x)$, for $y \le x$, is convex and nondecreasing in each variable: its first derivatives are $1 - \chi'$ and χ', and its Hessian is

$$\chi'' \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \ge 0.$$

For smooth potentials the chain rule gives the identity. For bounded potentials, add a common quadratic correction and convolve. Convolution preserves the order, and $G(x + r, y + r) = r + G(x, y)$ retains the common correction. The bounded chain rule (1.6) and its gradient products give the identity in the limit. Freezing the smooth background as in Proposition 1.7.1 preserves its exact lower bound. Every term on the right is positive; positivity of the bounded mixed products gives the last assertion.

For a locally $C^{1,1}$ weight, extend it convexly across zero with derivative in $[0, 1]$ and smooth it in one variable. The smooth convex weights and their first derivatives converge uniformly on compact intervals, while their derivatives remain in $[0, 1]$. On each smaller coordinate ball the corresponding potentials converge uniformly to w. Bounded product continuity gives convergence of their top powers; the right-hand weights converge uniformly against the fixed measure θ_h^n. Passing to the limit proves positivity and the product inequality with the original smooth background. □

1.8 Hermitian capacity

Local capacity tests can be glued to the constant function on a compact manifold after a fixed rescaling. This compares local Bedford–Taylor capacity with a capacity defined by global potentials and transfers both volume estimates and normalized sublevel estimates to X.

For a Borel set $E \subset X$, put

$$\mathrm{Cap}_\omega(E) = \sup_{v \in \mathrm{PSH}(X,\omega),\ 0 \le v \le 1} \int_E (\omega + \mathrm{dd}^{\mathrm{c}} v)^n.$$

The products are those of Proposition 1.7.1. Their local expansion and (1.2) show that $\mathrm{Cap}_\omega(X) < +\infty$. On relatively compact coordinate balls, a fixed smooth correction bounds the measures of global tests by a constant times the corresponding local capacity. Conversely, the gluing construction below bounds local capacity by global

capacity. Thus the quasi-continuity and capacity-zero characterizations in Section 1.6 hold for Cap_ω. Convergence in this capacity means

$$\mathrm{Cap}_\omega(\{|u_j - u| > \delta\}) \longrightarrow 0 \quad \text{for every } \delta > 0,$$

where pole sets may be omitted. To compare with another Hermitian form ω', choose $C \geq 1$ with $\omega' \leq C\omega$. Every test $0 \leq v \leq 1$ for $\mathrm{Cap}_{\omega'}$ gives a test v/C for Cap_ω, and positivity of mixed products gives

$$(\omega' + \mathrm{dd^c} v)^n \leq C^n (\omega + \mathrm{dd^c}(v/C))^n.$$

Thus $\mathrm{Cap}_{\omega'} \leq C^n \mathrm{Cap}_\omega$. Interchanging the metrics proves equivalence up to fixed multiplicative constants, so convergence in capacity is independent of this choice. The bounded-product continuity statements of Section 1.6 pass to the Hermitian products by their local expansion and a finite cover.

1.8.1 Capacity estimates

For the following estimates, abbreviate $\mathrm{Cap} = \mathrm{Cap}_\omega$ and $\omega_v = \omega + \mathrm{dd^c} v$.

Lemma 1.8.1 *There are constants $a, C > 0$ such that*

$$\mathrm{Vol}(E) \leq C \exp\bigl(-a \,\mathrm{Cap}(E)^{-1/n}\bigr) \tag{1.13}$$

for every Borel $E \subset X$. Also, for $u \in \mathrm{PSH}(X, \omega)$ with $\sup_X u = 0$ and $t > 1$,

$$\mathrm{Cap}(\{u < -t\}) \leq C/t. \tag{1.14}$$

Proof Take a finite cover by smaller coordinate balls $B'_j \Subset B_j$. A local capacity test $0 \leq v \leq 1$ on B_j can be made into a global test as follows. Choose a smooth function χ_j equal to zero near $\overline{B'_j}$, equal to -2 near ∂B_j, and with $-2 \leq \chi_j \leq 0$. Choose $\delta > 0$ small enough that

$$\omega + \delta \mathrm{dd^c} \chi_j \geq 0, \qquad 3\delta/2 \leq 1.$$

The function

$$V = \max\{\delta(v + \chi_j), -\delta/2\} + \delta/2 \quad \text{on } B_j, \qquad V = 0 \quad \text{outside } B_j$$

is globally ω-plurisubharmonic and lies between zero and one. Near the boundary the constant branch is selected. On B'_j we have $\omega_V \geq \delta \mathrm{dd^c} v$. Positivity of bounded products therefore gives

$$\mathrm{cap}(E \cap B'_j, B_j) \leq \delta^{-n} \,\mathrm{Cap}(E).$$

The local volume estimate, followed by summation over the finite cover, proves (1.13).

For the second estimate, choose on each B_j a smooth negative quadratic function r_j with $\mathrm{dd^c} r_j \geq \omega$. For every global test $0 \leq v \leq 1$, the functions $u + r_j \leq 0$ and $v + r_j$ are plurisubharmonic. Thus

$$\begin{aligned}\int_{\{u<-t\}} \omega_v^n &\leq t^{-1} \int_X (-u)\omega_v^n \\ &\leq t^{-1} \sum_j \int_{B'_j} (-u - r_j)\big(\mathrm{dd^c}(v + r_j)\big)^n \\ &\leq \frac{C}{t}\left(1 + \int_X |u| \,\mathrm{d}V\right).\end{aligned}$$

Here the second inequality uses $\omega_v \leq \mathrm{dd^c}(v + r_j)$, and the third is the local weighted estimate; the local test functions have a common bounded range. The normalized L^1 bound in Theorem 1.4.1 finishes the proof after taking the supremum over v. □

Corollary 1.8.1 *Fix $p > 1$. There is C_p such that every nonnegative $F \in L^p(X, \mathrm{d}V)$ satisfies*

$$\int_E F \,\mathrm{d}V \leq C_p \|F\|_p \operatorname{Cap}(E)^2$$

for every Borel set E.

Proof Hölder's inequality and (1.13) bound the integral by

$$C\|F\|_p \exp(-a \operatorname{Cap}(E)^{-1/n}/p'), \qquad p' = p/(p-1).$$

The quotient of this exponential by x^2 is bounded on $0 < x \leq \operatorname{Cap}(X) < \infty$. Capacity-zero sets have zero smooth volume. □

The same proof bounds an L^p density by any fixed positive power of capacity. The choice of the exponent two will be convenient for the sublevel iteration in Lemma 10.3.2.

1.8.2 Convergence of potentials and measures

For quasi-continuous obstacles, a decreasing lower semicontinuous approximation preserves the original bounds and confines the error to sets of vanishing capacity.

Lemma 1.8.2 *Every bounded quasi-continuous function f admits bounded lower semicontinuous functions f_j with common bounds such that $f_j \downarrow f$ quasi-everywhere and $f_j \to f$ in capacity.*

Proof Choose open sets O_j with capacity at most 2^{-j} such that f is continuous on $X \setminus O_j$, and set $G_j = \bigcup_{k \geq j} O_k$. Then G_j decreases, its capacity tends to zero, and f is continuous on its closed complement. Choose $M \geq \sup_X f$ and define

$$f_j = f \text{ on } X \setminus G_j, \qquad f_j = M \text{ on } G_j.$$

At a boundary point of G_j, limits from the closed complement give f, while values from inside are at least that value. Thus f_j is lower semicontinuous. The functions decrease to f outside $\bigcap_j G_j$, a capacity-zero set. Their difference from f is supported in G_j, which also proves convergence in capacity. □

Uniform bounds on potentials give a common capacity bound for their mixed measures.

Lemma 1.8.3 *Fix a Hermitian metric ω, a finite $A \geq 0$ and an integer $0 \leq k \leq n$. All positive mixed measures*

$$\omega_{v_1} \wedge \cdots \wedge \omega_{v_k} \wedge \omega^{n-k}, \qquad \|v_i\|_\infty \leq A, \quad v_i \in \mathrm{PSH}(X, \omega),$$

are bounded above on Borel sets by $C_A \operatorname{Cap}_\omega$. In particular their total masses are bounded. Their integrals against uniformly bounded functions tending to zero in capacity tend uniformly to zero.

Proof Translate the v_i so that $0 \leq v_i \leq 2A$, and put $q = (v_1 + \cdots + v_k)/B$, where $B \geq \max\{2k, 2kA, 1\}$. Then $0 \leq q \leq 1$ and

$$\omega_q = B^{-1} \sum_{i=1}^{k} \omega_{v_i} + (1 - k/B)\omega.$$

All coefficients are positive. Expanding ω_q^n bounds the indicated mixed product by a constant multiple of ω_q^n. The definition of capacity proves the claim. The case $k = 0$ follows by testing with zero. For a bounded function z, split its integral into $\{|z| > t\}$ and its complement; first use capacity convergence for fixed $t > 0$, and then let $t \downarrow 0$.□

The same control applies to the total variation of a product formed with a different smooth background.

Lemma 1.8.4 *Let Ω be Hermitian and θ any smooth real $(1,1)$-form. Suppose v_j are Ω-plurisubharmonic, have common finite upper and lower bounds, and decrease to v. Then the signed measures*

$$\nu_j = (\theta + \mathrm{dd^c} v_j)^n$$

converge to $(\theta + \mathrm{dd^c} v)^n$ against every fixed bounded quasi-continuous function. Their total variations are uniformly controlled by local capacity on a fixed finite cover.

Proof On a relatively compact coordinate ball, choose a fixed quadratic q such that $v_j + q$ are plurisubharmonic. Their common bounds remain uniform. Expand

$$(\theta + \mathrm{dd^c} v_j)^n = \sum_{\ell=0}^{n} \binom{n}{\ell} \big(\mathrm{dd^c}(v_j + q)\big)^\ell \wedge (\theta - \mathrm{dd^c} q)^{n-\ell}.$$

The absolute value of the ℓth summand is bounded by a fixed constant times

$$\left(\mathrm{dd^c}(v_j + q)\right)^\ell \wedge \kappa_0^{n-\ell},$$

where κ_0 is a fixed positive Euclidean form. This mixed positive measure is uniformly bounded by a constant times local capacity: add a fixed strictly plurisubharmonic quadratic to $v_j + q$, expand its top power, and rescale its uniformly bounded range into $[0, 1]$ for the capacity definition. This also handles $\ell = 0$. Summing on a finite cover gives the asserted variation control. Ordinary decreasing Bedford–Taylor continuity gives weak convergence term by term. For a bounded quasi-continuous coefficient, remove an open set of small capacity and extend the coefficient continuously from its compact complement. The variation bound controls the error uniformly for the sequence and its limit. Letting that capacity tend to zero proves the assertion. □

A finite non-pluripolar measure is uniformly absolutely continuous with respect to capacity, although it need not satisfy a power bound.

Lemma 1.8.5 *Let μ be a finite non-pluripolar Borel measure on X. Then*

$$\lim_{\delta \downarrow 0} \sup\{\mu(E) : E \text{ Borel}, \ \mathrm{Cap}_\omega(E) \le \delta\} = 0.$$

Consequently, capacity convergence implies convergence in μ-measure.

Proof If the first assertion failed, there would be $\epsilon > 0$ and Borel sets E_j with capacity at most 2^{-j} and $\mu(E_j) \ge \epsilon$. Countable subadditivity of capacity gives

$$\mathrm{Cap}_\omega\left(\limsup_j E_j\right) = 0.$$

The limsup is therefore pluripolar and has zero μ-measure. On the other hand, finiteness of μ and continuity from above applied to $\bigcup_{j \ge k} E_j$ show that the limsup has measure at least ϵ, a contradiction. Apply the first assertion to the sets on which the difference of two converging potentials exceeds a fixed positive number to get the consequence. □

Lemma 1.8.6 *Let $u_j \in \mathrm{PSH}(X, \theta)$. If u_j decreases to a genuine potential u, or increases with a common upper bound and $u = (\lim_j u_j)^*$, then $u_j \to u$ in capacity.*

Proof In the decreasing case u is a common lower bound; in the increasing case u_1 is one. Choose a Hermitian $\Omega \ge \theta$ and truncate at $-T$. Bounded monotone capacity convergence applies to the truncations. Quasi-continuity of the common lower bound permits removal of an open set of arbitrarily small capacity containing its pole set and all its discontinuities. On the compact complement that lower bound is finite and continuous, hence bounded below. Its deep sublevels therefore have capacity tending to zero. Let $T \to \infty$. □

A fixed bounded reference also controls a mixed trace with one normalized, varying potential.

Lemma 1.8.7 *Let $\theta \geq 0$ and fix a bounded $\psi \in \mathrm{PSH}(X, \theta)$. There is a constant C_ψ such that*

$$\int_X \theta_v \wedge \theta_\psi^{n-1} \leq C_\psi$$

for every bounded $v \in \mathrm{PSH}(X, \theta)$ with $\sup_X v = 0$.

Proof Choose a fixed Hermitian metric $\Omega \geq \theta$. All currents in the left side are positive, and $\Omega_\psi \geq \theta_\psi$. Bounded Stokes gives

$$\int_X \theta_v \wedge \theta_\psi^{n-1} \leq \int_X \theta \wedge \Omega_\psi^{n-1} + \int_X v \, \mathrm{dd^c}(\Omega_\psi^{n-1}).$$

The first term has a fixed bound. For $n \geq 2$, the exact differentiation formula is

$$\mathrm{dd^c}(\Omega_\psi^{n-1}) = (n-1)\mathrm{dd^c}\Omega \wedge \Omega_\psi^{n-2} + (n-1)(n-2)\mathrm{d}\Omega \wedge \mathrm{d^c}\Omega \wedge \Omega_\psi^{n-3},$$

where the second term is absent for $n = 2$. Its absolute value is bounded by fixed multiples of the corresponding positive mixed measures with Ω^2 and Ω^3 in place of the smooth torsion factors. The weighted local Chern–Levine–Nirenberg inequality from Section 1.6 bounds the integrals of $|v|$ against these measures. Indeed, on each larger coordinate ball add fixed smooth corrections to v and ψ, shift the first to be negative, expand in the corrected positive Hessian of ψ, and apply that inequality on a smaller ball. The bound is

$$C(1 + \|\psi\|_\infty)^n (1 + \|v\|_{L^1(X)}).$$

Normalized θ-plurisubharmonic functions have uniformly bounded L^1 norms. Summing the local bounds proves the assertion. For $n = 1$, the second integral is zero by Stokes, and the result is immediate. □

1.8.3 Vanishing area on curves

On a compact curve, the integral of the background is the total mass of every curvature current. When this integral vanishes, positivity forces the current to vanish. Normalized compactness and the local variation-capacity estimate make this uniqueness stable as the area tends to zero.

Let Y be a compact connected Riemann surface and fix a smooth positive form η with $\int_Y \eta = 1$. Use η for L^1 norms and capacity on Y.

Lemma 1.8.8 *For a smooth real $(1,1)$-form σ on Y, put $A = \int_Y \sigma$. There is a smooth real h with $\sigma = A\eta + \mathrm{dd^c}h$. The class $\mathrm{PSH}(Y, \sigma)$ is nonempty exactly when $A \geq 0$. If $A = 0$, it consists of the functions $-h + c$, $c \in \mathbb{R}$, and its unique supremum-zero member is $u_\sigma = -h + \min_Y h$.*

Proof The smooth form $\sigma - A\eta$ has integral zero. The scalar Poisson equation on the compact connected surface therefore has a smooth solution h, unique up to a constant. If $u \in \mathrm{PSH}(Y, \sigma)$, Stokes gives

$$(\sigma + \mathrm{dd^c} u)(Y) = A \geq 0.$$

Conversely, $-h$ has curvature $A\eta$, so is admissible when $A \geq 0$. If $A = 0$, every positive curvature measure has mass zero and vanishes. Thus $\mathrm{dd^c}(u + h) = 0$; elliptic regularity and harmonic uniqueness make $u + h$ constant. Supremum normalization gives the stated formula. □

Theorem 1.8.1 *Let smooth real forms σ_j converge uniformly to a smooth form σ_0, and suppose*

$$A_j = \int_Y \sigma_j \geq 0, \qquad \int_Y \sigma_0 = 0.$$

Let u_0 be the normalized σ_0-plurisubharmonic potential from Lemma 1.8.8. Then, for every $t > 0$,

$$\sup_{u \in \mathrm{PSH}(Y, \sigma_j),\ \sup_Y u = 0} \|u - u_0\|_{L^1(Y)} \longrightarrow 0,$$

$$\sup_{u \in \mathrm{PSH}(Y, \sigma_j),\ \sup_Y u = 0} \mathrm{Cap}_\eta(\{|u - u_0| > t\}) \longrightarrow 0.$$

The full curvature measures tend uniformly to zero in total variation: each has norm A_j.

Proof Choose $C > 0$ with $\sigma_j \leq C\eta$ for every j. Any sequence of normalized σ_j-plurisubharmonic functions belongs to the compact normalized family for $C\eta$. Every L^1 cluster limit u is normalized and satisfies $\sigma_0 + \mathrm{dd^c} u \geq 0$, by passing to distributions. The preceding lemma gives $u = u_0$. A sequence violating the first uniform limit would have a convergent subsequence with this same limit, a contradiction.

For any potential in the jth normalized family, put $T = \sigma_j + \mathrm{dd^c} u$. Positivity and Stokes give $\|T\|_{\mathrm{TV}} = A_j$. Since $\sigma_0 + \mathrm{dd^c} u_0 = 0$,

$$\|\mathrm{dd^c}(u - u_0)\|_{\mathrm{TV}} \leq A_j + \|\sigma_j - \sigma_0\|_{\mathrm{TV}} \longrightarrow 0$$

uniformly over that family. Apply Lemma 1.6.2 on a fixed finite cover by concentric coordinate discs. The first limit and the displayed bound make each local capacity tend uniformly to zero. The local-global capacity comparison then proves the second limit. The current norm identity also proves the final assertion. □

1.9 Comparing local regularizations

To glue local convolutions, we need them to agree up to an error tending uniformly to zero on overlaps. A bounded plurisubharmonic function need not be continuous, so this comparison cannot come from uniform convergence to the original function. Instead, logarithmic convexity of its suprema on balls controls the change of scale and the change of coordinates.

Let $U \subset \mathbb{C}^n$ be a domain and u a bounded plurisubharmonic function on U. On a fixed relatively compact subset, for small $\delta > 0$ put

$$S_\delta u(z) = \sup_{|\zeta - z| \le \delta} u(\zeta), \qquad A_\delta u(z) = \fint_{|\zeta - z| = \delta} u(\zeta)\, \mathrm{d}\sigma(\zeta), \qquad u_\delta = u * \rho_\delta,$$

where $\mathrm{d}\sigma$ is Euclidean surface measure and ρ is a smooth nonnegative radial mollifier of integral one, supported in the unit ball, with $\rho_\delta(\zeta) = \delta^{-2n}\rho(\zeta/\delta)$. A fixed larger neighborhood contains all balls below, and B denotes a bound for the oscillation of u there.

Lemma 1.9.1 *For each fixed $A > 1$,*

$$S_{A\delta} u - S_\delta u \longrightarrow 0, \qquad S_\delta u - u_\delta \longrightarrow 0$$

uniformly on compact subsets as $\delta \downarrow 0$.

Proof The maximum of a plurisubharmonic function on concentric balls is convex as a function of the logarithm of the radius. This follows by restricting to each complex line through the center and comparing on an annulus with the harmonic function affine in $\log|\zeta|$. For a fixed admissible radius $r > A\delta$, convexity gives

$$0 \le S_{A\delta} u - S_\delta u \le \frac{B \log A}{\log(r/\delta)}. \tag{1.15}$$

This proves the first assertion.

We compare the maximum with the spherical mean using the Poisson kernel in real dimension $2n$. For a point at distance at most $\delta/2$ from the center, the normalized Poisson kernel of the ball of radius δ is bounded below by a positive dimensional constant c_n. Since $u \le S_\delta u$, the submean comparison with its harmonic extension gives

$$S_\delta u - S_{\delta/2} u \ge c_n (S_\delta u - A_\delta u).$$

These inequalities are valid for bounded plurisubharmonic functions by local smoothing, followed by a decreasing limit. Equation (1.15) shows that $S_\delta u - A_\delta u$ tends uniformly to zero.

For $0 < t \le 1$, the spherical mean is increasing with radius, and

$$0 \le A_\delta u - A_{t\delta} u \le (S_\delta u - S_{t\delta} u) + (S_{t\delta} u - A_{t\delta} u).$$

The first term is at most $B\log(1/t)/\log(r/(t\delta))$ by logarithmic convexity of S, and the second is at most $C_nB/\log(r/(t\delta))$ by the preceding estimate. Write the radial convolution as an average of $A_{t\delta}u$ with the probability density $\widehat{\rho}(t) = |\mathbb{S}^{2n-1}|t^{2n-1}\rho(t)$. Since $\int_0^1(1+\log(1/t))\widehat{\rho}(t)\,\mathrm{d}t < \infty$, integration gives

$$0 \le A_\delta u - u_\delta \le \frac{C_{n,\rho}B}{\log(r/\delta)}.$$

Combining the two comparisons proves the second assertion. □

Corollary 1.9.1 *Let $U, V \subset \mathbb{C}^n$ be domains and let $F : U \to V$ be biholomorphic and u bounded and plurisubharmonic on U. The difference*

$$u_\delta - \big((u \circ F^{-1}) * \rho_\delta\big) \circ F$$

tends locally uniformly to zero as $\delta \downarrow 0$.

Proof On a compact subset, F and its inverse have uniformly bounded first derivatives. Thus, for a fixed $A > 1$, the inverse image of a radius-δ ball about $F(z)$ lies between balls about z of radii δ/A and $A\delta$. The two supremum regularizations are therefore bounded by one another at these enlarged radii. The first assertion of Lemma 1.9.1 makes their difference uniformly small. Its second assertion compares each of the two convolutions with the corresponding supremum regularization. □

1.10 Gluing with a small curvature error

The overlap estimate permits gluing by a smooth maximum. Small boundary corrections make each local branch irrelevant near the edge of its chart, while convexity of the maximum preserves a common lower curvature bound. Use ω to measure the curvature error.

For an integer $N \ge 1$, choose independent real random variables $Z_1, \dots, Z_N$ with a common smooth even probability density supported in $[-1, 1]$, and define

$$M_\tau(t_1, \dots, t_N) = \mathbb{E}\max_i(t_i + \tau Z_i) \quad (\tau > 0).$$

Convolution makes M_τ smooth. It is convex, increasing in every argument, and, with $\mathbf{1} = (1, \dots, 1)$, satisfies

$$\max_i t_i \le M_\tau(t) \le \max_i t_i + \tau, \qquad M_\tau(t + c\mathbf{1}) = M_\tau(t) + c$$

for $t \in \mathbb{R}^N$ and $c \in \mathbb{R}$. If $t_i < \max_{j\ne i} t_j - 2\tau$, the ith argument can be removed without changing the value. Also $M_\tau(t)$ is nondecreasing in τ: it is a convex even function of τ because the distribution of Z is symmetric.

The second identity expresses translation invariance. If smooth real functions f_i, $1 \le i \le N$, satisfy $\mathrm{dd^c} f_i \ge -\gamma$ for a common smooth real $(1,1)$-form γ, then

$$\mathrm{dd^c} M_\tau(f_1, \dots, f_N) \ge -\gamma. \tag{1.16}$$

Indeed, the first-derivative coefficients are nonnegative and sum to one, by the translation identity, while the real Hessian of the convex function M_τ contributes a semipositive $(1,1)$-form in the chain rule.

Proposition 1.10.1 *Let θ be smooth and real, ω Hermitian, and $u \in \mathrm{PSH}(X,\theta) \cap L^\infty(X)$. There are smooth functions v_j and positive numbers $\varepsilon_j \downarrow 0$ such that*

$$v_j \downarrow u, \qquad \theta + \varepsilon_j \omega + \mathrm{dd^c} v_j \ge 0.$$

Proof Fix $\epsilon > 0$. By continuity of θ, choose finitely many nested coordinate neighborhoods $V_i \Subset W_i \Subset U_i$, with the V_i covering X, and smooth quadratic functions r_i such that

$$0 \le \mathrm{dd^c} r_i - \theta \le \epsilon\omega \quad \text{on } U_i.$$

To make this choice, freeze the coefficients of θ at the center, add a small positive constant form, and shrink the coordinate neighborhood. Put $p_i = u + r_i$, a bounded plurisubharmonic function, and let $p_{i,\delta}$ be its convolution in U_i.

On an overlap, express both p_i and p_j in the jth coordinates. Their difference is the smooth function $r_i - r_j$, so the difference of their convolutions tends uniformly to $r_i - r_j$. By Corollary 1.9.1, changing the coordinates used to convolve p_i introduces another uniformly vanishing error. Thus, on overlaps of the closures of the W_i,

$$(p_{i,\delta} - r_i) - (p_{j,\delta} - r_j) \longrightarrow 0 \quad \text{uniformly}. \tag{1.17}$$

Choose $\eta_i \in C^\infty(U_i, [-1,0])$, zero on V_i and equal to -1 near the boundary of W_i, and $C \ge 1$ with $\mathrm{dd^c}\eta_i \ge -C\omega$. The cover, the functions r_i, η_i and the constant C may depend on ϵ; they are fixed while δ tends to zero. Set

$$f_{i,\delta} = p_{i,\delta} - r_i + (\epsilon/C)\eta_i \quad \text{on } W_i.$$

Each satisfies $\mathrm{dd^c} f_{i,\delta} \ge -\theta - 2\epsilon\omega$. Near a boundary point of W_i, some V_j contains that point. Equation (1.17) implies, for all small δ,

$$f_{i,\delta} < f_{j,\delta} - \epsilon/(2C)$$

on a neighborhood of that boundary point. Finiteness of the cover makes this separation uniform.

Take $\tau(\delta) \downarrow 0$ as $\delta \downarrow 0$, with $\tau(\delta) < \epsilon/(8C)$, and define on each point the smooth maximum of the branches whose W_i contain it. A branch disappears only near the boundary of its W_i, where it is below another branch by more than 2τ. The removal property of M_τ therefore proves that these formulas define a global smooth function $\psi_{\epsilon,\delta}$. Equation (1.16) gives

$$\theta + 2\epsilon\omega + \mathrm{dd^c}\psi_{\epsilon,\delta} \geq 0.$$

At every point there is a branch with $\eta_i = 0$, and $p_{i,\delta} \geq p_i$. Hence $\psi_{\epsilon,\delta} \geq u$. For fixed ϵ, the local convolutions decrease with δ, and so does the smooth maximum when both its arguments and τ decrease. Its pointwise limit is

$$\max_{i:x\in W_i} \big(u(x) + (\epsilon/C)\eta_i(x)\big) = u(x).$$

Thus $\psi_{\epsilon,\delta} \downarrow u$ as $\delta \downarrow 0$, also in L^1.

Choose a decreasing diagonal as follows. Take $\epsilon_j \downarrow 0$ and put $v_j = \psi_{\epsilon_j,\delta_j} + 2^{-j}$. Choose δ_j so small that

$$\|\psi_{\epsilon_j,\delta_j} - u\|_{L^1} \leq 2^{-j}, \qquad v_j < v_{j-1} \quad (j \geq 2).$$

The second choice is possible because $v_{j-1} \geq u + 2^{-j+1}$. More explicitly, the continuous functions $\psi_{\epsilon_j,\delta} + 2^{-j} - v_{j-1}$ decrease pointwise to a function bounded above by -2^{-j}. Their nonnegative level sets are decreasing compact sets with empty intersection, so one of them is empty. The resulting sequence decreases, converges to u in L^1, and stays above u. Its pointwise limit equals u by uniqueness of canonical quasi-plurisubharmonic representatives. Taking $\varepsilon_j = 2\epsilon_j$ proves the proposition. □

1.11 Regularization in a fixed Hermitian background

For $\theta = \omega$, rescaling the approximants absorbs the curvature error into the positive background. Subtracting a constant first makes the approximants negative, so that this rescaling preserves their decreasing order.

Theorem 1.11.1 *For $u \in \mathrm{PSH}(X,\omega) \cap L^\infty(X)$ there are smooth functions $u_j \downarrow u$ such that*

$$\omega + \mathrm{dd^c}u_j > 0.$$

The functions u_j are uniformly bounded.

Proof Subtract a constant so that $u \leq -2$. Apply Proposition 1.10.1 with $\theta = \omega$, obtaining smooth $v_j \downarrow u$ and positive $\varepsilon_j \downarrow 0$ with $(1+\varepsilon_j)\omega + \mathrm{dd^c}v_j \geq 0$. The compact sets $\{v_j \geq -1\}$ decrease to the empty set, since $u \leq -2$. Discarding finitely many terms therefore gives $v_j < 0$ everywhere. Write

$$u_j = \frac{v_j}{1 + 2\varepsilon_j}.$$

Then

$$\omega + \mathrm{dd^c}u_j \geq \frac{\varepsilon_j}{1 + 2\varepsilon_j}\omega > 0.$$

Because v_j is negative and decreases while the positive denominator decreases to one, u_j also decreases and converges to u. It lies between u and its first smooth member, which gives uniform boundedness. Restore the subtracted constant. □

This regularization supplies the approximation used for products and mass in Chapter 6. For an arbitrary smooth θ, Proposition 1.10.1 retains a vanishing curvature error.

1.12 Singularity type and multiplier ideals

We compare quasi-plurisubharmonic functions on X first by domination up to a constant, then through weighted local integrability of holomorphic functions. The first comparison gives ordinary singularity types; the second gives multiplier ideals. A logarithmic pole on a disc will show why the ideals at all positive weights enter the comparison.

Definition 1.12.1 For quasi-plurisubharmonic functions u, v, write $u \preceq v$ if $u \leq v + C$ for some $C \in \mathbb{R}$. Write $u \sim v$ if $u \preceq v$ and $v \preceq u$.

The convention $u \preceq v$ means that u is at least as singular as v. In particular, a logarithmic pole lies below a bounded potential in this order.

Proposition 1.12.1 *The relation $\preceq$ is a preorder, and $\sim$ is an equivalence relation. Adding a bounded smooth function preserves the equivalence class. In particular, all smooth functions on X have the same singularity type.*

Proof Reflexivity follows with $C = 0$. If $u \leq v + C$ and $v \leq w + D$, then $u \leq w + C + D$, proving transitivity. The usual equivalence relation associated to a preorder gives $\sim$. If $|f| \leq A$, then $u - A \leq u + f \leq u + A$. Finally, every smooth function on compact X is bounded. □

A quasi-plurisubharmonic function has *small unbounded locus* if it is locally bounded outside a closed complete pluripolar set.

Definition 1.12.2 Let $\mathcal{O}_X$ denote the sheaf of holomorphic functions. For a quasi-plurisubharmonic function u on X, the multiplier ideal $\mathcal{I}(u) \subseteq \mathcal{O}_X$ is the sheaf whose local sections are holomorphic functions g for which $|g|^2 e^{-u}$ is locally integrable against a smooth positive volume form.

Changing the volume form does not change local integrability: on relatively compact coordinate neighborhoods, the ratio of two smooth positive volume forms is bounded above and below by positive constants. This definition thus requires no chosen Hermitian metric.

Proposition 1.12.2 *If $u \preceq v$, then $\mathcal{I}(ku) \subseteq \mathcal{I}(kv)$ for every real $k > 0$. If $u \sim v$, then these ideals agree for every $k > 0$.*

Proof Let g be a local holomorphic function on X. An inequality $u \le v + C$ implies $e^{-kv} \le e^{kC} e^{-ku}$. Multiplying by $|g|^2$ shows that local integrability for ku implies local integrability for kv. Apply the same argument in both directions when $u \sim v$.□

Example 1.12.1 On a disc in $\mathbb{C}$ centered at the origin, with coordinate z, let $\lambda \ge 0$ and put $u(z) = \lambda \log |z|^2$ for $\lambda > 0$, while $u = 0$ for $\lambda = 0$. Then

$$\mathcal{I}(u)_0 = (z^{\lfloor \lambda \rfloor}).$$

Indeed, if a nonzero holomorphic germ has order m, its squared modulus is comparable to $|z|^{2m}$ near the origin. For a sufficiently small radius $\epsilon > 0$, its weighted integral is finite precisely when

$$\int_0^{\epsilon} r^{2m-2\lambda+1}\, \mathrm{d}r < \infty, \qquad \text{equivalently} \qquad m > \lambda - 1.$$

The smallest nonnegative integer satisfying this strict inequality is $\lfloor \lambda \rfloor$. At an integer value of λ, the excluded borderline exponent is -1, so the strictness of the integrability condition matters.

The example also explains why one multiplier ideal is insufficient to compare singularities: different coefficients λ can have the same integer part. For a real weight $k > 0$, applying the calculation to ku gives $\mathcal{I}(ku)_0 = (z^{\lfloor k\lambda \rfloor})$. The ideals at all $k > 0$ recover λ, since for real coefficients $0 \le \lambda < \mu$ the two integer parts differ as soon as $k(\mu - \lambda) > 1$.

Write $u \preceq_{\mathcal{I}} v$ if $\mathcal{I}(ku) \subseteq \mathcal{I}(kv)$ for every real $k > 0$, and $u \sim_{\mathcal{I}} v$ if equality holds at all these weights. Both definitions make sense on any complex manifold; their relation to envelopes and global masses requires further arguments.

1.13 Analytic types and restriction

Analytic singularities express the poles of a potential through finitely many holomorphic functions. Their logarithmic growth is measured by the Lelong number. For a quasi-plurisubharmonic function u and a point $x \in X$, choose coordinates centered at x and a smooth correction q such that $p = u + q$ is plurisubharmonic. With $B(x, r)$ the coordinate Euclidean ball, set

$$\nu(u, x) = \lim_{r \downarrow 0} \frac{\sup_{B(x,r)} p}{\log r^2}.$$

The limit exists and is independent of the coordinates and the smooth correction, by the local growth theorem recalled in [Xia]. Thus $\log |z|^2$ has Lelong number one at the origin.

We follow the convention of [Xia]: a quasi-plurisubharmonic function u has analytic singularities if locally there are holomorphic functions $f_1, \dots, f_N$, not all identically zero, and a rational $c > 0$ such that

$$u = c\log\bigl(|f_1|^2 + \cdots + |f_N|^2\bigr) + O(1), \qquad c \in \mathbb{Q}_{>0}.$$

The remainder is bounded; smoothness of this remainder is an additional property. A coherent ideal $\mathfrak{a}$ and a common coefficient c specify the *analytic type* $(c, \mathfrak{a})$. Changing local generators changes the logarithm by a bounded amount on smaller relatively compact open sets.

Proposition 1.13.1 *Analytic singularities are preserved by addition, finite maxima, and multiplication by a positive rational number. If $Y \subset X$ is a connected complex submanifold and the ideal of an analytic presentation does not vanish identically on Y, the restriction is genuine and has analytic type*

$$(c, \mathfrak{a}\mathcal{O}_Y).$$

Here $\mathfrak{a}\mathcal{O}_Y$ means the image ideal generated by the restrictions of local sections of $\mathfrak{a}$.

Proof For a rational scaling, multiply the coefficient. For two presentations with coefficients c_1, c_2 and ideals $\mathfrak{a}_1, \mathfrak{a}_2$, choose a common positive rational coefficient d with $c_1 = kd, c_2 = \ell d$, where k, ℓ are positive integers. Products of generators present the sum, since a product of sums of squares is a sum of squares of pairwise products.

For the maximum use

$$\max\{A, B\} \le A + B \le 2\max\{A, B\}, \qquad A, B \ge 0,$$

applied to the appropriate sums of squares of generators of $\mathfrak{a}_1^k$ and $\mathfrak{a}_2^\ell$. Taking logarithms gives the required bounded error. These are local assertions; finite covers give the corresponding ordinary singularity comparisons on a compact manifold.

For restriction, restrict the displayed inequalities defining the bounded remainder. At least one generator remains nonzero on the connected manifold Y, so its common zero set is proper. The restricted function is quasi-plurisubharmonic, is not identically $-\infty$, and has the stated presentation. A smooth local correction gives the same conclusion directly from the usual restriction theorem for psh functions. □

Zero generic Lelong number does not make the restriction of an arbitrary quasi-plurisubharmonic function genuine. For example, on a small polydisc in $\mathbb{C}^n$ with coordinates $z = (z_1, \ldots, z_n)$ and $|z_1| < \mathrm{e}^{-1}$,

$$u(z) = -\sqrt{-\log|z_1|^2}$$

is plurisubharmonic: the function $t \mapsto -\sqrt{-t}$ is increasing and convex for $t < 0$. Along $\{z_1 = 0\}$, its Lelong number is zero but its restriction is identically $-\infty$. The analytic hypothesis in Proposition 1.13.1 will exclude this behavior for the approximants below.

1.13.1 Lelong level sets

Siu's analyticity theorem says that, for every $c > 0$, the set $\{x : \nu(u, x) \geq c\}$ is analytic. A smooth local correction reduces the quasi-plurisubharmonic statement to the theorem for closed positive currents. We use this classical input in the form recalled in [Xia]; its original source is recorded in Section 33.1.

1.14 The local Bergman estimates

Fix a bounded pseudoconvex domain $\Omega \subset \mathbb{C}^n$ and a plurisubharmonic function ψ on Ω. Write $\mathrm{d}\lambda$ for Euclidean volume and $B(x, r)$ for the Euclidean ball of radius $r > 0$ about x. For a positive integer m, define the weighted holomorphic Hilbert space

$$\mathcal{H}_m = \left\{ f \in \mathcal{O}(\Omega) : \|f\|_m^2 := \int_\Omega |f|^2 \mathrm{e}^{-m\psi} \, \mathrm{d}\lambda < \infty \right\}.$$

Choose an orthonormal basis $(g_{m,j})_j$ of $\mathcal{H}_m$ and define $\beta_m = m^{-1} \log \sum_j |g_{m,j}|^2$. The sum is equivalently the squared norm of point evaluation:

$$\sum_j |g_{m,j}(x)|^2 = \sup_{\|f\|_m \leq 1} |f(x)|^2.$$

It is independent of the basis. The standard local finite-generation part of Bergman regularization says that β_m has analytic type $(1/m, \mathcal{I}(m\psi))$ on smaller open sets. This follows from the weighted holomorphic generation of the multiplier ideal and local L^2 division; see [Dem11, Proposition 5.7 and Section 14.A]. We use that local analytic statement as a standard input, together with the point-extension estimate used in the proof below.

Proposition 1.14.1 *There are constants C_1, C_2 independent of m such that, for $B(x, r) \Subset \Omega$,*

$$\psi(x) - \frac{C_1}{m} \leq \beta_m(x) \leq \sup_{B(x,r)} \psi + \frac{1}{m} \log \frac{C_2}{r^{2n}}. \tag{1.18}$$

Consequently $\beta_m \to \psi$ pointwise and in L^1_{loc}, and

$$\max\{\nu(\psi, x) - n/m, 0\} \leq \nu(\beta_m, x) \leq \nu(\psi, x). \tag{1.19}$$

Proof For $\|f\|_m \leq 1$, the submean inequality gives

$$|f(x)|^2 \leq C_2 r^{-2n} \int_{B(x,r)} |f|^2 \, \mathrm{d}\lambda \leq C_2 r^{-2n} \mathrm{e}^{m \sup_{B(x,r)} \psi}.$$

Taking the supremum over the unit ball and then the logarithm proves the upper bound.

The local Ohsawa–Takegoshi point-extension theorem gives a holomorphic f with $f(x) = 1$ and

$$\int_\Omega |f|^2 \mathrm{e}^{-m\psi}\, \mathrm{d}\lambda \le C\mathrm{e}^{-m\psi(x)}$$

when $\psi(x) > -\infty$. The constant is independent of the psh weight and of m. Normalize f in $\mathcal{H}_m$. Point evaluation is then at least $C^{-1}\mathrm{e}^{m\psi(x)}$, proving the lower bound. At a pole the lower bound has no content and remains valid.

Choosing $r = 1/m$ proves pointwise convergence by upper semicontinuity of ψ. On a fixed relatively compact set, the same estimates bound β_m above by a constant and below by $\psi - C_1/m$. Since ψ is locally integrable, dominated convergence also gives convergence in L^1_{loc}.

For the Lelong estimate, take suprema over $B(x, r)$ in the two inequalities and bound the upper one using $B(x, 2r)$. Divide by $\log r^2 < 0$ and let $r \downarrow 0$. The term $-(2n/m)\log r$ contributes $-n/m$ after the division. The remaining constant terms disappear. This proves the two inequalities in (1.19); nonnegativity supplies the maximum with zero. □

In [Dem11], the local weight is $\mathrm{e}^{-2m\phi}$ and Lelong numbers are normalized using $\log|z|$. Putting $\psi = 2\phi$ gives exactly the formulas above. In particular the error is n/m with the present weight $\mathrm{e}^{-m\psi}$ and normalization by $\log|z|^2$.

The same local construction and estimates hold for every real parameter $m > 0$. Its local logarithmic presentation has coefficient $1/m$; this coefficient is rational exactly when m is rational. We use the term analytic type only with the rational-coefficient convention above. The global sequence below is indexed by integers.

The generators also compare multiplier ideals at nearby weights. The loss in the weight compensates for the finite Bergman index.

Lemma 1.14.1 *Let u be quasi-plurisubharmonic, let $m > 0$, and let $a_m = m^{-1}\log\sum_k |g_k|^2 + O(1)$ locally, where the g_k generate $\mathcal{I}(mu)$. For $t > 0$ and $s > 1$,*

$$\mathcal{I}(sta_m) \subseteq \mathcal{I}(tu) \qquad \textit{whenever} \qquad m \ge \frac{st}{s-1}.$$

Also $u \preceq a_m$ locally, and hence $\mathcal{I}(tu) \subseteq \mathcal{I}(ta_m)$ for every $t > 0$.

Proof On a sufficiently small coordinate neighborhood, choose generators $g_1, \dots, g_N$ of $\mathcal{I}(mu)$ and put $B = (\sum_k |g_k|^2)\mathrm{e}^{-mu}$. Shrink so that B is integrable and $a_m = m^{-1}\log\sum_k |g_k|^2 + O(1)$ with a bounded remainder. If $f \in \mathcal{I}(sta_m)$, shrink once more so that f is bounded and $A = |f|^2\mathrm{e}^{-sta_m}$ is integrable. Almost everywhere,

$$|f|^2\mathrm{e}^{-tu} \le C|f|^{2-2/s}A^{1/s}B^{t/m}.$$

Hölder's inequality with exponents s and $s/(s-1)$ applies because $st/(m(s-1)) \le 1$. It proves local integrability of the left side, including at equality in the index bound. The local Bergman lower estimate in Proposition 1.14.1 gives $u \preceq a_m$; bounded smooth corrections do not affect this comparison or any of the ideals. □

1.15 The global curvature bound

Return to the compact manifold X and its fixed Hermitian form ω. The global regularization theorem glues the local Bergman constructions while controlling the curvature error. We retain it as a classical input in the following precise form.

Theorem 1.15.1 (Demailly regularization) *Let u be quasi-plurisubharmonic on X and let γ be a continuous real $(1,1)$-form with $\mathrm{dd^c}u \geq \gamma$. There are quasi-plurisubharmonic u_m, for sufficiently large positive integers m, and numbers $\epsilon_m \downarrow 0$ such that*

(1) *u_m has analytic type $(1/m, \mathcal{I}(mu))$;*
(2) *$u_m \geq u$ and $u_m \to u$ pointwise and in $L^1(X)$;*
(3) *$\mathrm{dd^c}u_m \geq \gamma - \epsilon_m\omega$;*
(4) *$\max\{\nu(u,x) - n/m, 0\} \leq \nu(u_m, x) \leq \nu(u,x)$ for all x.*

No monotonicity of the values u_m is asserted here.

This is the form of [Dem11, Theorem 14.12] used in this book. Its local generators are the weighted Bergman generators of $\mathcal{I}(mu)$; smooth local corrections do not change that ideal. The overlapping sums have comparable norms, so the glued logarithm retains this ordinary analytic type. The global construction freezes the continuous lower bound on sufficiently small coordinate balls. Its curvature error combines the modulus of continuity of that bound with an overlap term tending to zero as the Bergman index increases. It is this global estimate, rather than merely the local convergence in Proposition 1.14.1, which is the standard input in the theorem.

Corollary 1.15.1 *For any smooth real $(1,1)$-form θ on X and $u \in \mathrm{PSH}(X,\theta)$, the approximants can be chosen so that*

$$\theta + \mathrm{dd^c}u_m \geq -\epsilon_m\omega.$$

Their analytic type is $(1/m, \mathcal{I}(mu))$.

Proof Apply the theorem to the closed, almost positive current $\mathrm{dd^c}u$ with lower bound $\gamma = -\theta$. The theorem requires no closedness of γ. Adding θ gives the asserted inequality. In the convention of [Dem11], this application is to the potential $u/2$, since $\frac{\mathrm{i}}{\pi}\partial\bar{\partial}(u/2) = \mathrm{dd^c}u$. □

These approximants retain the curvature lower bound up to a vanishing error. Their multiplier ideals will later identify the limit of their ordinary envelopes in Theorem 15.5.1.

1.15.1 Regularization of closed currents

We also use the closed-current form of regularization recalled in [Xia]. Let $T = \eta + \mathrm{dd^c}v$ be an almost positive closed $(1,1)$-current, where η is smooth and closed and v

is quasi-plurisubharmonic. Its Lelong number is $\nu(T,x) = \nu(v,x)$. If $T \geq \gamma$ for a smooth real $(1,1)$-form γ, there are smooth representatives $T_j = \eta + \mathrm{dd^c} v_j$ with

$$T_j \geq \gamma - C\lambda_j \omega,$$

where $C > 0$ is fixed and the continuous functions λ_j decrease pointwise to $\nu(T,\cdot)$. When all Lelong numbers vanish, Dini's theorem makes this loss uniformly small. Analytic regularization gives representatives S_j with

$$S_j \geq \gamma - \varepsilon_j \omega, \qquad \varepsilon_j \downarrow 0, \qquad \nu(S_j,x) \leq \nu(T,x).$$

Thus it preserves strict positivity after a small loss and introduces no positive Lelong number at a point where the original number is zero.

1.15.2 Smooth remainders after resolution

For arguments on a modification, the bounded remainder in an analytic presentation is sometimes insufficient. The following version of regularization supplies a smooth remainder after resolution.

Theorem 1.15.2 (Demailly regularization) *Let α be any smooth real $(1,1)$-form and let $u \in \mathrm{PSH}(X,\alpha)$. There are $\delta_j \downarrow 0$ and $\widetilde{u}_j \in \mathrm{PSH}(X,\alpha+\delta_j\omega)$ decreasing to u such that each $\widetilde{u}_j$ is smooth off an analytic pole set and, after a modification $\pi_j : Y_j \to X$ given by a finite sequence of blowups with smooth centers, has local expressions as follows. Let s_ℓ be local holomorphic defining functions of divisors, let $a_\ell > 0$ be real numbers, and let b be a smooth real function. Then*

$$\pi_j^* \widetilde{u}_j = \sum_\ell a_\ell \log |s_\ell|^2 + b, \qquad a_\ell > 0, \quad b \in C^\infty.$$

The divisors may be taken to have simple normal crossings.

This classical input applies to the closed current $\mathrm{dd^c} u$ with smooth lower bound $-\alpha$; its source is recorded in Section 33.1. Its lower bound is not required to be closed, and its ambient manifold is only Hermitian. The resolved expression supplies the smooth remainder after resolution; ordinary analytic type with merely a bounded remainder would not suffice here. This version need not be the particular multiplier-ideal sequence in Theorem 1.15.1. We do not assert that the additional conclusions of these two regularization statements hold for the same sequence. Applying this theorem with a sufficiently small curvature error preserves strict positivity.

1.16 Multiplier ideals and logarithmic poles

For a logarithmic pole along a smooth center, weighted integrability is determined by the normal vanishing order of a holomorphic germ. For general singularities, the local valuative characterization and strong openness recalled in [Xia] make the ideal comparison compatible with maxima, addition and increasing limits.

In local coordinates $(z, w) \in \mathbb{C}^d \times \mathbb{C}^k$, with $d \geq 0$ and $k \geq 1$, let $Y = \{w = 0\}$ and denote its ideal by $\mathcal{I}_Y$.

Lemma 1.16.1 *If a quasi-plurisubharmonic function has the form $u = c \log |w|^2 + O(1)$ with $c > 0$ and locally bounded remainder, then for every $t > 0$,*

$$\mathcal{I}(tu) = \mathcal{I}_Y^{\max\{\lfloor tc \rfloor - k + 1, 0\}}.$$

The zeroth power is the full structure sheaf.

Proof The bounded remainder changes integrability by bounded positive factors. Expand a nonzero holomorphic germ $g(z, w)$ in homogeneous terms in w, and let a be the smallest degree whose coefficient is not identically zero as a germ in z. On smaller product neighborhoods, $|g(z, w)| \leq C|w|^a$. Hence $a + k > tc$ suffices for weighted integrability, since its radial bound is

$$\int_0^{\varepsilon} \rho^{2a - 2tc + 2k - 1} \, \mathrm{d}\rho < +\infty.$$

Conversely, at every nearby tangential point where the degree-a term is nonzero, angular integration of $|g|^2$ has a positive leading term of degree $2a$. The radial integral diverges when $a + k \leq tc$, including logarithmic divergence at equality. Such tangential points form a nonempty open set in every sufficiently small neighborhood. Fubini then excludes local weighted integrability of g. For $d = 0$ the same argument is the radial calculation at a point. Thus g is integrable exactly when its order in the normal ideal is at least the integer in the statement. □

Small isolated logarithmic poles can be inserted into any Hermitian background by cutoffs.

Lemma 1.16.2 *Let ω be Hermitian on a compact complex manifold X, and fix distinct points $p_1, \dots, p_N$. There are constants $C_j > 0$ such that, whenever $a_j > 0$ and $\sum_j C_j a_j < 1$, there is $u \in \mathrm{PSH}(X, \omega)$, smooth off these points, with*

$$u = a_j \log |z|^2 + O(1) \quad \textit{near } p_j, \qquad \omega + \mathrm{dd^c} u \geq \left(1 - \sum_j C_j a_j\right) \omega.$$

Here z denotes local coordinates centered at p_j. The function can be chosen to vanish outside arbitrarily small disjoint coordinate neighborhoods, with the constants depending on those neighborhoods.

Proof Choose disjoint coordinate balls and smooth cutoffs χ_j supported there and equal to one near their centers. The function $\chi_j \log |z|^2$, extended by zero, is plurisubharmonic near its pole and smooth elsewhere, so its complex Hessian is bounded below by $-C_j \omega$. Set $u = \sum_j a_j \chi_j \log |z|^2$. The lower bounds give the asserted current inequality, and its remaining properties follow from the disjoint supports. □

Theorem 1.16.1 *For germs of plurisubharmonic functions, the following facts hold.*

(1) *Equality of multiplier ideals at every positive weight is equivalent to equality of the Lelong numbers of the pullbacks at every point on every local modification. Inclusion of the ideals at every positive weight is equivalent to the reverse inequalities for these Lelong numbers.*
(2) *If p_j is an increasing sequence of plurisubharmonic functions converging almost everywhere to a plurisubharmonic function p, then, for each point x and each $\lambda > 0$,*

$$\mathcal{I}(\lambda p_j)_x = \mathcal{I}(\lambda p)_x \quad \textit{for all sufficiently large } j.$$

In the first item one works over a coordinate neighborhood of the germ, allowing smaller neighborhoods as necessary. No assertion that every local modification extends to a modification of the compact manifold is needed. The second item is the increasing-limit form of strong openness used in [Xia]. Smooth additions do not change multiplier ideals or Lelong numbers. Thus these inputs apply to quasi-plurisubharmonic functions after the local correction of Proposition 1.1.1.

Lemma 1.16.3 *Let u, u', v, v' be quasi-plurisubharmonic functions on X. Suppose $u \sim_{\mathcal{I}} v$ and $u' \sim_{\mathcal{I}} v'$. Then*

$$\max(u, u') \sim_{\mathcal{I}} \max(v, v'), \qquad u + u' \sim_{\mathcal{I}} v + v', \qquad au \sim_{\mathcal{I}} av \quad (a > 0).$$

Proof Fix a coordinate neighborhood $U \subset X$ and a smooth local modification $\pi : Y \to U$. At every $y \in Y$, the elementary Lelong identities give

$$\begin{aligned} \nu(\max(u, u') \circ \pi, y) &= \min\{\nu(u \circ \pi, y), \nu(u' \circ \pi, y)\}, \\ \nu((u + u') \circ \pi, y) &= \nu(u \circ \pi, y) + \nu(u' \circ \pi, y), \\ \nu(au \circ \pi, y) &= a\nu(u \circ \pi, y). \end{aligned}$$

The corresponding expressions for v, v' are equal by the first local input. Apply the converse direction of that input on each neighborhood. The Lelong identities are the usual local maximum, addition and scaling identities, unchanged by smooth corrections. □

Let v, w be local quasi-plurisubharmonic functions and let Y be a complex submanifold on which $v|_Y$ is genuine. Restriction and subadditivity give the two inclusions recalled in [Xia] and [Dem11, Section 15]:

$$\mathcal{I}(v|_Y) \subseteq \mathcal{I}(v)\mathcal{O}_Y, \qquad \mathcal{I}(v+w) \subseteq \mathcal{I}(v)\mathcal{I}(w). \tag{1.20}$$

The first follows from local Ohsawa–Takegoshi extension. For the second, weighted holomorphic functions on a product form the Hilbert tensor product of the two weighted spaces; its generators give the product multiplier ideal. Restriction to the diagonal then gives the inclusion. These are local results for psh functions. Smooth corrections extend them to quasi-plurisubharmonic functions with the weight convention of this book.

For a holomorphic submersion, multiplier ideals commute with pullback. Extension of an ideal means the ideal generated by its pullbacks.

Lemma 1.16.4 *Let $p : Z \to Y$ be a holomorphic submersion of complex manifolds and let $u \in \mathrm{QPSH}(Y)$. For every $t > 0$,*

$$\mathcal{I}(tu \circ p) = \mathcal{I}(tu)\mathcal{O}_Z.$$

Proof Work in product coordinates (z, ζ) with $p(z, \zeta) = z$. Write α for a multi-index in the fiber variables and $\mathrm{d}\lambda_\zeta$ for their Euclidean volume. A germ in the ideal on the right is a finite sum of bounded holomorphic multiples of pullbacks of germs integrable against $\mathrm{e}^{-tu(z)}$. Fubini proves its weighted integrability on a sufficiently small product neighborhood.

Conversely, let $F(z, \zeta)$ be a weighted-integrable holomorphic germ and expand it in the fiber variables,

$$F(z, \zeta) = \sum_\alpha f_\alpha(z)\zeta^\alpha.$$

The coefficient estimate for a holomorphic function on a polydisc gives, on smaller product neighborhoods,

$$|f_\alpha(z)|^2 \le C_\alpha \int |F(z, \zeta)|^2 \, \mathrm{d}\lambda_\zeta.$$

Multiply by $\mathrm{e}^{-tu(z)}$ and integrate. Each coefficient germ belongs to $\mathcal{I}(tu)$. Let J be the extension of this ideal to the local analytic ring R of the product. Every finite Taylor sum belongs to J, and the remainder of total fiber degree at least N belongs to $(\zeta)^N$. Thus the class of F in R/J lies in $(\zeta)^N(R/J)$ for every N. The Krull intersection theorem for the finite module R/J over the Noetherian local analytic ring gives intersection zero. Hence $F \in J$. This is the usual local algebra form of

convergence for coherent ideals. Smooth corrections and smooth positive volume densities do not change the integrability ideals. □

1.17 Lelong numbers and divisorial estimates

Lelong numbers measure logarithmic concentration. Divisorial removal turns their lower bounds into inequalities of currents, while the local mass formula controls them under convergence and in a fixed background. For a prime divisor E on a complex manifold, let $[E]$ be its integration current and $\nu(w, E)$ the generic Lelong number of a quasi-plurisubharmonic function w along E.

We use the following precise local consequence of Siu's decomposition, as recalled in [Xia] and the standard current theory of [Dem11]. A closed positive $(1, 1)$-current has, on a prime hypersurface E, coefficient-measure restriction equal to its generic Lelong number times $[E]$. In particular the remaining current after removal of any finite set of its divisorial components is positive. Distinct prime hypersurfaces do not contribute overlapping mass: their intersections have codimension at least two. These are statements about the local closed current.

Lemma 1.17.1 *Let M be a complex manifold, let β be a smooth real $(1, 1)$-form on M, let $w \in \mathrm{PSH}(M, \beta)$, and let $E_1, \ldots, E_N$ be distinct prime divisors on M, with $N \geq 1$. Then*

$$\beta + \mathrm{dd^c} w - \sum_{j=1}^{N} \nu(w, E_j)[E_j] \geq 0.$$

The form β need not be closed.

Proof Work on a coordinate ball and choose a smooth ρ with $\mathrm{dd^c}\rho \geq \beta$. The current

$$T = \mathrm{dd^c}(w + \rho) = (\beta + \mathrm{dd^c} w) + (\mathrm{dd^c}\rho - \beta)$$

is closed and positive. A smooth form, viewed through its coefficient measures, assigns zero mass to every proper analytic set. Thus T and $\beta + \mathrm{dd^c} w$ have the same restriction to $\bigcup_j E_j$. The stated local Siu fact identifies this restriction as $\sum_j \nu(w, E_j)[E_j]$. Subtracting it from the positive current $\beta + \mathrm{dd^c} w$ leaves its coefficient-measure restriction to the complement, which is positive. This proves the inequality locally, hence globally. The argument does not identify $\beta + \mathrm{dd^c} w$ itself as a closed current.□

Order makes the support theorem a rigidity statement for potentials: two ordered quasi-plurisubharmonic functions whose Hessians agree outside an analytic set already differ by a constant.

Lemma 1.17.2 *Let $u \leq v$ be quasi-plurisubharmonic on a connected compact complex manifold X, and let $Z \subsetneq X$ be analytic. If $\mathrm{dd^c} u = \mathrm{dd^c} v$ on $X \setminus Z$, then $v = u + C$ for a real constant C.*

Proof The current $S = \mathrm{dd^c}(v - u)$ is real, closed, of order zero and supported on Z. The support theorem in [Dem12, Chapter III, Corollaries 2.11 and 2.14, pp. 141–143] gives

$$S = \sum_{j=1}^{m} c_j [Z_j], \qquad c_j = \nu(v, Z_j) - \nu(u, Z_j) \leq 0, \tag{1.21}$$

where Z_j are the prime divisorial components of Z. Indeed, near a generic smooth point of Z_j, the difference $v - u - c_j \log |z_1|^2$ has zero complex Hessian and is therefore smooth and pluriharmonic. This identifies c_j with the difference of the generic Lelong numbers. Its sign follows from $u \leq v$. After subtracting the divisorial terms, the remaining closed order-zero current is supported in codimension at least two and vanishes by the same support theorem.

Thus $\mathrm{dd^c}(u - v) = -S \geq 0$. The locally integrable distribution $u - v$ has a global plurisubharmonic representative, which is constant by the maximum principle on compact connected X. Equality almost everywhere between the resulting quasi-plurisubharmonic representatives gives $v = u + C$ everywhere. □

A closed real $(1, 1)$-current admits a smooth residual form and one global distributional potential. Positivity makes that potential quasi-plurisubharmonic. This decomposition lets us attach an ordinary singularity type to a divisorial current.

Lemma 1.17.3 *Let Z be a complex manifold and T a closed real $(1, 1)$-current. There are a smooth closed real $(1, 1)$-form β and a real distribution v such that*

$$T = \beta + \mathrm{dd^c} v.$$

If $T \geq 0$, then v is quasi-plurisubharmonic. Any two distributional potentials in such decompositions differ by a smooth function.

Proof The local $\mathrm{dd^c}$-Poincaré lemma gives real distributional potentials v_i on coordinate neighborhoods, with $T = \mathrm{dd^c} v_i$; for positive T these potentials are plurisubharmonic. See [Dem12, Chapter III, Section 1.18 and Proposition 1.19]. On overlaps $\mathrm{dd^c}(v_i - v_j) = 0$. Taking the Euclidean trace gives a harmonic distribution, so elliptic regularity makes $v_i - v_j$ smooth.

Choose a smooth locally finite partition of unity χ_i subordinate to the cover and put $v = \sum_i \chi_i v_i$, extending each supported term as a distribution. On the jth neighborhood,

$$v - v_j = \sum_i \chi_i (v_i - v_j)$$

is smooth. Thus $\beta = T - \mathrm{dd^c} v = -\mathrm{dd^c}(v - v_j)$ is smooth locally and hence globally. It is real and closed. For positive T, the same identity expresses v locally as a plurisubharmonic function plus a smooth function. Finally, the difference of two distributional potentials has smooth $\mathrm{dd^c}$. Taking a smooth Hermitian trace and applying elliptic regularity shows that the difference is smooth. □

A potential whose curvature is a divisorial current plus a smooth form is determined, up to ordinary domination, by its divisorial coefficients. The comparison extends to countably many components of finite total mass.

Lemma 1.17.4 *Let Y be a connected compact complex manifold. Let D_j be distinct prime divisors, $c_j > 0$ real coefficients such that $S = \sum_j c_j [D_j]$ is a positive current of finite mass, and β a smooth real $(1,1)$-form. Let $v \in \mathrm{QPSH}(Y)$ satisfy*

$$\mathrm{dd^c} v = S - \beta.$$

For $a \in \mathrm{QPSH}(Y)$, the following conditions are equivalent:

$$a \preceq v, \qquad a \preceq_{\mathcal{I}} v, \qquad \nu(a, D_j) \geq c_j \quad \textit{for every } j.$$

Proof The coefficient-measure restriction of S to D_j is $c_j[D_j]$. Distinct divisors have intersections of codimension at least two, which carry no mass of either divisor current. Countable summation preserves this assertion. Smooth forms give every divisor zero coefficient mass. Local Siu decomposition after a smooth correction therefore gives $\nu(v, D_j) = c_j$.

Ordinary domination implies ideal domination by integrability, and ideal domination implies the asserted divisorial Lelong inequalities by Theorem 1.16.1. It remains to prove that these inequalities imply ordinary domination.

Choose a Hermitian form Λ with $a \in \mathrm{PSH}(Y, \Lambda)$. By Lemma 1.17.1, the divisorial inequalities allow removal of each finite partial sum of S from $\Lambda + \mathrm{dd^c} a$ while retaining positivity. Passing to the increasing finite sums gives

$$\Lambda + \mathrm{dd^c} a - S \geq 0.$$

Consequently $\mathrm{dd^c}(a - v) \geq \beta - \Lambda$. The L^1 difference has a quasi-plurisubharmonic representative and is bounded above on compact Y. This yields $a \leq v + C$ almost everywhere. Lemma 1.3.3 extends this inequality everywhere. This proves ordinary domination. □

The preceding lemmas give a canonical ordinary type for a countable divisorial current of finite mass.

Corollary 1.17.1 *Let Z be a connected compact complex manifold, let D_j be distinct prime divisors with integration currents $[D_j]$, and let $c_j > 0$. Suppose $S = \sum_j c_j [D_j]$ is a closed positive current of finite mass. It determines a canonical ordinary quasi-plurisubharmonic type with representatives v satisfying*

$$\mathrm{dd^c} v = S - \beta$$

for a smooth closed real form β. Any two such representatives differ smoothly.

Proof Lemma 1.17.3 supplies the decomposition and shows that its potentials differ smoothly. Thus the ordinary type depends only on S. □

Lemma 1.17.5 *Let u_j, u be quasi-plurisubharmonic on a complex manifold, with a common smooth lower curvature bound. If $u_j \to u$ in L^1_{loc}, then at every point x,*

$$\nu(u, x) \geq \limsup_j \nu(u_j, x).$$

Proof On a coordinate ball about x, add one smooth function to all the potentials to make them plurisubharmonic. Write the resulting potentials as p_j, p, let n be the dimension of the chart, and let κ be its Euclidean form. Distributional convergence gives weak convergence of the positive trace measures $\mathrm{dd^c} p_j \wedge \kappa^{n-1}$ to $\mathrm{dd^c} p \wedge \kappa^{n-1}$. For every sufficiently small radius r whose boundary sphere has zero mass for the latter measure, the masses of the balls converge. The Lelong monotonicity formula bounds each $\nu(p_j, x)$ by its normalized trace mass on that ball. Taking the upper limit in j, and then letting such radii decrease to zero, gives $\limsup_j \nu(p_j, x) \leq \nu(p, x)$. Smooth additions do not change Lelong numbers. □

Lemma 1.17.6 *Let Z be a connected compact complex manifold and let β be a smooth real $(1,1)$-form. There is a finite constant C_β such that*

$$\nu(v, z) \leq C_\beta \qquad \textit{for every } v \in \mathrm{PSH}(Z, \beta) \textit{ and every } z \in Z.$$

In particular all generic divisorial Lelong numbers in this fixed background are uniformly bounded. Additive normalization of the potentials is unnecessary.

Proof Subtract the supremum from v. By Theorem 1.4.1, these normalized potentials have a common L^1 bound. Cover Z by finitely many nested coordinate balls. On each larger ball choose a fixed smooth quadratic function r with $\mathrm{dd^c} r \geq \beta$, so $p = v + r$ is plurisubharmonic there. The functions p have uniformly bounded local L^1 norms.

Write $d = \dim Z$ and let κ be the Euclidean form in the chart. Choose a nonnegative smooth cutoff χ supported in the larger ball and equal to one on an intermediate ball. Positivity and distributional integration by parts give

$$\int_{\text{intermediate ball}} \mathrm{dd^c} p \wedge \kappa^{d-1} \leq \int \chi \, \mathrm{dd^c} p \wedge \kappa^{d-1} = \int p \, \mathrm{dd^c} \chi \wedge \kappa^{d-1} \leq C \|p\|_{L^1}.$$

Every point of the smaller ball admits a surrounding Euclidean ball of a fixed radius inside the intermediate ball. The local Lelong monotonicity formula, recalled in [Xia], bounds the Lelong number at its center by a fixed multiple of this trace mass. The finite cover gives one bound on Z. Smooth additions and additive normalization do not change Lelong numbers, proving the assertion for the original potentials. □

The Bergman estimates also control Lelong numbers on a fixed local modification. A *local divisor over* X means a prime divisor E on a smooth modification $\pi : Z \to U$ of a coordinate neighborhood $U \subset X$. For a quasi-plurisubharmonic function u on X, let $\nu(u \circ \pi, E)$ be its generic Lelong number along E, let ord_E denote divisorial vanishing order (the minimum of the orders of local generators for an ideal), and let $\mathrm{Jac}\,\pi$ be the local holomorphic Jacobian determinant. Write

$$\nu_E(u) = \nu(u \circ \pi, E), \qquad A_U(E) = 1 + \mathrm{ord}_E(\mathrm{Jac}\,\pi).$$

The second number is the log discrepancy. These local modifications are not required to extend over the compact manifold.

Lemma 1.17.7 *For a quasi-plurisubharmonic function u, every fixed local divisor E, and every real $m > 0$, one has*

$$\max\left\{\nu_E(u) - \frac{A_U(E)}{m}, 0\right\} \le \frac{1}{m}\,\mathrm{ord}_E\,\mathcal{I}(mu) \le \nu_E(u).$$

Consequently, a local logarithmic presentation $a_m = m^{-1}\log\sum_k |f_{m,k}|^2 + O(1)$, where the $f_{m,k}$ generate $\mathcal{I}(mu)$, satisfies $\nu_E(a_m) \to \nu_E(u)$ as $m \to \infty$.

Proof The upper estimate follows from the local Bergman lower bound in Proposition 1.14.1. The logarithm of local generators of $\mathcal{I}(mu)$, divided by m, is at least u up to a locally bounded term. Pulling back and taking a generic Lelong number reverses this inequality.

For the lower estimate, take a local generator f of the ideal and set $k = \mathrm{ord}_E(f)$. At a generic smooth point of E, choose a coordinate z_1 defining it. Divisorial removal, after a smooth correction, gives

$$u \circ \pi \le \nu_E(u)\log|z_1|^2 + C.$$

Shrink around a point where the residual factors of $f \circ \pi$ and of the Jacobian are nonvanishing. The change-of-variables formula and local integrability of $|f|^2 e^{-mu}$ then require integrability in z_1 of

$$|z_1|^{2(k+A_U(E)-1-m\nu_E(u))}.$$

This implies $k + A_U(E) > m\nu_E(u)$. In particular the weaker non-strict lower bound in the statement holds. Take the minimum over a finite set of local generators and use nonnegativity of the order. Since $\nu_E(a_m) = m^{-1}\,\mathrm{ord}_E\,\mathcal{I}(mu)$, the last assertion follows by letting $m \to \infty$. □

For rational $m > 0$, the presentation a_m has analytic singularities in the convention of the source book. The real-parameter estimate does not require this restriction.

Blowing up a point converts the divisorial estimate into a pointwise order bound on any local modification.

Lemma 1.17.8 *Let U be a complex manifold of dimension n, let u be quasi-plurisubharmonic on U, and let $\pi : Y \to U$ be a modification with smooth source. Fix $y \in Y$ and an integer $m \geq 1$. Write J_π for the holomorphic Jacobian in local coordinates near y. If a nonzero holomorphic germ g at $\pi(y)$ belongs to $\mathcal{I}(mu)$, then*

$$\operatorname{ord}_y(g \circ \pi) \geq m\nu(u \circ \pi, y) - n - \operatorname{ord}_y J_\pi.$$

Proof If $n \geq 2$, let $\rho : \widetilde{Y} \to Y$ be the blowup of y and let E be its exceptional divisor. For $n = 1$, take ρ to be the identity and $E = \{y\}$. Set $F = \pi \circ \rho$ and write J_F for its local holomorphic Jacobian determinant. Generic orders along E give

$$\nu(u \circ F, E) = \nu(u \circ \pi, y), \qquad \operatorname{ord}_E(g \circ F) = \operatorname{ord}_y(g \circ \pi).$$

On a standard blowup chart the Jacobian of ρ has order $n - 1$ along E. The chain rule therefore gives the log discrepancy over U as

$$A_U(E) = 1 + \operatorname{ord}_E J_F = n + \operatorname{ord}_y J_\pi.$$

Since $g \in \mathcal{I}(mu)$, its order along E is at least the order of this ideal. The lower estimate in Lemma 1.17.7, applied to F, proves the inequality. □

Lemma 1.17.9 *Let v be quasi-plurisubharmonic on X, and let a have analytic singularities on X. If $v \preceq_{\mathcal{I}} a$, then $v \preceq a$. It suffices to assume the finite set of inequalities*

$$\nu(v \circ \pi, E) \geq \nu(a \circ \pi, E)$$

for the prime components E of the resolved ideal on one log resolution π of an analytic presentation of a.

Proof Resolve the coherent ideal of an analytic presentation of a by a smooth modification $\pi : Y \to X$. In local coordinates $z_1, \ldots, z_n$ on Y, its expression has coefficients $b_i \geq 0$, $1 \leq i \leq r \leq n$, and a locally bounded remainder B:

$$a \circ \pi = \sum_{i=1}^{r} b_i \log |z_i|^2 + B, \qquad b_i \geq 0, \qquad B \in L^\infty_{\mathrm{loc}}.$$

The stated finite inequalities give $\nu(v \circ \pi, \{z_i = 0\}) \geq b_i$; under $v \preceq_{\mathcal{I}} a$, they follow from Theorem 1.16.1. Add a smooth local function to $v \circ \pi$ to make it plurisubharmonic. Divisorial removal for this closed positive complex Hessian shows that

$$v \circ \pi - \sum_i b_i \log |z_i|^2$$

has a quasi-plurisubharmonic representative. Here the difference is first interpreted off the divisor, or as an integrable distribution. Such a representative is locally bounded above. Since B is locally bounded, $v \circ \pi \leq a \circ \pi + C$ on each relatively compact resolution chart away from the divisor. A finite cover above a smaller coordinate neighborhood makes C uniform there. Push the inequality down off the analytic exceptional set; the canonical quasi-plurisubharmonic representatives extend it everywhere. Finitely many neighborhoods on the compact manifold give $v \leq a + C$ globally.

The only resolution input is the standard principalization of an analytic ideal. Divisorial removal is applied to a locally closed positive Hessian after smooth correction, not to a purported closed background on the original manifold. □

Lemma 1.17.10 *Let u, v be quasi-plurisubharmonic on X, and let s have analytic singularities on X. If $u + v \leq s + O(1)$, then for analytic potentials U_m, V_m of types $(1/m, \mathcal{I}(mu))$, $(1/m, \mathcal{I}(mv))$, there are numbers $\delta_m \downarrow 0$ such that*

$$U_m + V_m \leq (1 - \delta_m)s + O(1) \quad \textit{for all sufficiently large integers } m.$$

The bounded error may depend on m.

Proof If s is bounded, upper bounds for U_m, V_m give the assertion with $\delta_m = 1/m$ for $m > 1$. Otherwise resolve the ideal of s by a smooth modification $\pi : \widetilde{X} \to X$. Let E_i be the components of its resolved divisor with positive coefficients c_i. In resolution coordinates z_i defining these components, its singular part is $\sum_i c_i \log |z_i|^2$. Set $A_i = 1 + \mathrm{ord}_{E_i}(\mathrm{Jac}\,\pi)$. The hypothesis and Lemma 1.17.7 give

$$\nu(U_m \circ \pi, E_i) + \nu(V_m \circ \pi, E_i) \geq \nu(u \circ \pi, E_i) + \nu(v \circ \pi, E_i) - \frac{2A_i}{m} \geq c_i - \frac{2A_i}{m}.$$

Choose a positive integer $C \geq \max_i 2A_i/c_i$ and put $\delta_m = C/m < 1$ for large m. After a smooth local correction, divisorial removal from the positive Hessian of $(U_m + V_m) \circ \pi$ shows that

$$(U_m + V_m) \circ \pi - (1 - \delta_m) \sum_i c_i \log |z_i|^2$$

has a quasi-plurisubharmonic representative. It is locally bounded above. A finite resolution cover and proper descent therefore give $U_m + V_m \leq (1 - \delta_m)s + O(1)$, just as in Lemma 1.17.9. The bounded constant may depend on m. □

1.18 Hölder potentials and metric distance

A Hölder modulus controls the trace mass of a positive Hessian on small balls. Averaging path lengths against that trace then gives an upper bound for the distance of the regular metric. This converts regularity of the potential into a metric estimate without requiring a pointwise Hessian bound.

Let $U \subset \mathbb{C}^n$ be a coordinate domain, $\omega_{\mathrm{E}} = \mathrm{dd^c}|z|^2$ its Euclidean form, and $\mathrm{d}\lambda$ Euclidean volume. Let $0 < \alpha \le 1$, $v \in C^{0,\alpha}(U)$, θ be a smooth real $(1,1)$-form on U, and $T = \theta + \mathrm{dd^c} v \ge 0$. Write $\mu = T \wedge \omega_{\mathrm{E}}^{n-1}$.

Theorem 1.18.1 *On nested relatively compact charts, for all sufficiently small Euclidean balls,*

$$\mu(B(x,r)) \le C r^{2n-2+\alpha}. \tag{1.22}$$

Every coefficient measure of T has total variation dominated by a multiple of μ. Hence coefficients vanish on sets of zero Hausdorff $(2n-2+\alpha)$-measure, and the local quasi-plurisubharmonic Lelong numbers vanish. This estimate does not imply absolute continuity or a pointwise Hessian bound.

Suppose in addition that T is a smooth strictly positive Hermitian metric η on $U \setminus A$ for a proper analytic subset A. Let d_η^U be the infimum of η-lengths of piecewise smooth paths in $U \setminus A$ joining the given points. On sufficiently small nested charts $V \Subset U$,

$$d_\eta^U(x,y) \le C|x-y|^{\alpha/2}, \qquad x, y \in V \setminus A. \tag{1.23}$$

Each point of $A \cap V$ has a finite completion limit. If these local hypotheses hold in charts covering X, with $A \subset X$ proper analytic and η a smooth Hermitian metric on $X \setminus A$, the completion of $X \setminus A$ for its intrinsic η-distance is a compact continuous metric quotient of X. The metric is incomplete when $A \neq \varnothing$. Both exponents in (1.22) and (1.23) are sharp.

Proof Choose a nonnegative cutoff χ equal to one on $B(x,r)$ and supported on $B(x,2r)$ with $|\mathrm{d}^2\chi| \le Cr^{-2}$. Positivity and integration by parts, subtracting the constant $v(x)$, give

$$\mu(B(x,r)) \le C_1 r^{2n} + C_2 r^{2n-2} \operatorname{osc}_{B(x,2r)} v \le C r^{2n-2+\alpha}.$$

The constant term integrates to zero against $\mathrm{dd^c}\chi \wedge \omega_{\mathrm{E}}^{n-1}$. The Radon–Nikodym matrix of the positive coefficient measures relative to their trace is positive semidefinite almost everywhere, so $|a_{ij}| \le (a_{ii} + a_{jj})/2$. A ball cover proves the Hausdorff-null-set assertion. Dividing (1.22) by r^{2n-2} gives zero Lelong numbers after a smooth local plurisubharmonic shift.

For the distance, let $m = 2n$ and let τ be the Euclidean trace of η. Then $\tau\,\mathrm{d}\lambda \le C\mu$ on the regular locus and $|\xi|_\eta \le C\sqrt{\tau}|\xi|_{\mathrm{E}}$. Polar integration bounds the average length of segments from x to points of $B(x,R)$ by a constant times $\int_{B(x,R)} \sqrt{\tau}(\zeta)|\zeta - x|^{1-m}\,\mathrm{d}\lambda(\zeta)$. On a dyadic annulus of radius s, Cauchy–Schwarz and (1.22) bound this integral by $Cs^{1-m}s^{m/2}s^{(m-2+\alpha)/2} = Cs^{\alpha/2}$. Summing annuli gives $CR^{\alpha/2}$.

For x, y in an inner chart, average broken segments $[x, z] \cup [z, y]$ over a ball of radius $|x - y|/2$ about their midpoint. The preceding bound supplies a path of length at most $C|x - y|^{\alpha/2}$. Almost every such endpoint avoids A along both segments: proper analytic sets have locally countably many smooth strata of real codimension at least two, and the cones from a regular fixed endpoint over these strata have zero volume. Nested charts keep the chosen path inside $U \setminus A$.

The estimate sends all regular sequences approaching one point of A to a common completion point and extends continuously across A. Compactness makes the image the entire completion, but different points of A may be identified. A sequence tending to A cannot converge to a regular point in the metric, since a small closed neighborhood of that regular point has a uniform positive lower metric bound. Thus deletion is incomplete when A is nonempty. The local model $|z|^{\alpha} + |w|^2$, in coordinates $(z, w) \in \mathbb{C} \times \mathbb{C}^{n-1}$, has trace mass of exact ball order $r^{2n-2+\alpha}$ and normal distance $|z|^{\alpha/2}$, proving both sharpness claims. For a logarithmic solution, this theorem is applied to its positive residual current, with the divisor term recorded separately. □

1.19 Obstacle envelopes

An obstacle selects the largest potential lying below it. The local upper-envelope theorem gives the construction once the competitors have a common upper bound; normalized compactness supplies that bound for an obstacle finite almost everywhere or prescribed on a non-pluripolar compact set.

A smooth real $(1, 1)$-form θ is called *big* if there are $\rho \in \mathrm{PSH}(X, \theta)$ and $\delta > 0$ such that

$$\theta + \mathrm{dd^c}\rho \geq \delta\omega.$$

Here ω is any Hermitian metric. Changing that metric changes δ, not the condition. Neither this definition nor the existence of θ-plurisubharmonic functions requires θ to be closed.

For any smooth real $(1, 1)$-form θ and an extended-real obstacle f on X, defined at least outside a pluripolar set, write

$$P_\theta(f) = (\sup\{v \in \mathrm{PSH}(X, \theta) : v \leq f \text{ q.e.}\})^* .$$

For $u, v \in \mathrm{PSH}(X, \theta)$, set $P_\theta(u, v) = P_\theta(\min(u, v))$. The empty supremum is $-\infty$. For an upper semicontinuous obstacle, the pointwise definition gives the same competitors: a quasi-plurisubharmonic function lying below it almost everywhere lies below it everywhere, by the local submean inequality and upper semicontinuity of the obstacle.

To apply the local upper-envelope theorem while retaining the lower bound $-\theta$, fix a constant positive form κ in local coordinates. For each $\epsilon > 0$, shrink the neighborhood and choose a constant form θ' with $-\epsilon\kappa \leq \theta - \theta' \leq \epsilon\kappa$. A smooth potential r for $\theta' + \epsilon\kappa$ makes every competitor plus r plurisubharmonic. The local envelope theorem then gives $\theta + \mathrm{dd^c}P_\theta(f) \geq -2\epsilon\kappa$ there. Letting ϵ tend to zero

proves the exact bound. The same local theorem shows that regularization changes the supremum only on a pluripolar set.

Here is a useful way to check the upper bound for an unbounded obstacle. If $f \le A$ for a real constant A on a Borel set E of positive $\mathrm{d}V$-measure and $h \le f$ is a competitor, normalize $\widetilde{h} = h - \sup_X h$. By Theorem 1.4.1,

$$(\sup_X h - A)\,\mathrm{Vol}(E) \le \int_E |\widetilde{h}|\,\mathrm{d}V \le C_\theta \quad \text{whenever } \sup_X h > A.$$

Thus the competitors have a common upper bound. In particular this applies to any measurable obstacle finite almost everywhere, by taking one of its finite sublevel sets of positive volume.

An obstacle bound on a non-pluripolar compact set also bounds every competitor globally, even when that set has zero smooth volume.

Lemma 1.19.1 *Let θ be a smooth real $(1,1)$-form on the connected compact manifold X, let $K \subseteq X$ be compact and non-pluripolar, and let $w \in C(K,\mathbb{R})$. There is a constant $C = C(\theta, K, w)$ such that every $a \in \mathrm{PSH}(X,\theta)$ satisfying $a \le w$ quasi-everywhere on K has $\sup_X a \le C$.*

Proof Otherwise choose such functions a_j with

$$M_j := \sup_X a_j \ge 2^j + \max_K w.$$

The normalized functions $v_j = a_j - M_j \le 0$ have uniformly bounded L^1 norms by Theorem 1.4.1. Choose $A > 0$ with $\theta \le A\omega$. The partial sums

$$s_N = \sum_{j=1}^{N} 2^{-j} v_j$$

decrease, have uniformly bounded L^1 norms, and satisfy $\mathrm{dd^c} s_N \ge -A\omega$. Thus their limit is a genuine quasi-plurisubharmonic function s; the L^1 bound excludes the identically $-\infty$ alternative in the local decreasing-limit theorem.

Let $N_j \subseteq K$ be a pluripolar exceptional set for the j-th obstacle inequality. On $K \setminus \bigcup_j N_j$ one has $v_j \le -2^j$ for every j, hence $s = -\infty$. The set where a genuine quasi-plurisubharmonic function is $-\infty$ is pluripolar. Together with the countable union of the N_j, it would contain all of K, contradicting the hypothesis on K. □

If $\mathrm{PSH}(X,\theta)$ is nonempty, its normalized envelope

$$V_\theta = P_\theta(0)$$

is θ-plurisubharmonic. Indeed, a normalized θ-plurisubharmonic function is a competitor, and the family is bounded above by zero. Moreover,

$$u - \sup_X u \le V_\theta \qquad (u \in \mathrm{PSH}(X,\theta)).$$

A potential $v \in \mathrm{PSH}(X,\theta)$ has *minimal singularities* if $u \preceq v$ for every $u \in \mathrm{PSH}(X,\theta)$. The displayed inequality shows that V_θ has minimal singularities, and every such v satisfies $v \sim V_\theta$. If $\theta \ge 0$, then $V_\theta = 0$.

Lemma 1.19.2 *Let η be a smooth semipositive real $(1,1)$-form on X. If an obstacle f is finite almost everywhere and bounded below quasi-everywhere by a constant, then $P_\eta(f)$ is a bounded η-plurisubharmonic function.*

Proof The lower constant is a competitor. The preceding finite-sublevel argument gives a common upper bound for all competitors. The local envelope theorem therefore gives a bounded envelope. □

Proposition 1.19.1 *If f is continuous and ω is Hermitian, then $P_\omega(f)$ is continuous. Moreover,*

$$\|P_\omega(f) - P_\omega(g)\|_\infty \le \|f - g\|_\infty \quad (f, g \in C^0(X)).$$

Proof The constant $\min_X f$ is a competitor, so $p = P_\omega(f)$ is bounded. The local envelope theorem gives $p \le f$ quasi-everywhere, and continuity of f makes this inequality pointwise. Choose smooth $v_j \downarrow p$ by Theorem 1.11.1. For every $\epsilon > 0$, the continuous functions $v_j - f$ decrease to a function at most zero. Compactness shows that $v_j \le f + \epsilon$ everywhere for all large j: otherwise their decreasing compact ϵ-superlevel sets would have nonempty intersection. Hence $v_j - \epsilon$ is a competitor and

$$p \le v_j \le p + \epsilon.$$

Thus $v_j \to p$ uniformly, proving continuity. The last inequality follows directly from the competitors and $P_\omega(g + c) = P_\omega(g) + c$ for constants c. □

Definition 1.19.1 For $u \in \mathrm{PSH}(X,\theta)$, set

$$P_\theta[u] = \left(\lim_{C\to+\infty} P_\theta(u + C, 0)\right)^* = (\sup\{h \in \mathrm{PSH}(X,\theta) : h \preceq u,\ h \le 0\})^*.$$

We call u an *envelope fixed point* if $P_\theta[u] = u$. For $F \in C^\infty(X,\mathbb{R})$, replacing the constraint $h \le 0$ by $h \le F$ defines $P_{\theta,F}[u]$.

The two formulas agree by taking the union of the competitor families. They also agree with the definition using V_θ in place of zero: every θ-plurisubharmonic

competitor below zero is below V_θ. The envelope is a genuine potential, since the competitors contain $u - \sup_X u$ and are bounded above by zero. In particular, $u \preceq P_\theta[u]$. The last relation is not an assertion that the two functions differ by a bounded amount.

Directly from the competitors, $P_\theta[u]$ is unchanged when u is replaced by a potential of the same ordinary singularity type, and

$$u \preceq v \quad \Longrightarrow \quad P_\theta[u] \le P_\theta[v].$$

These statements concern ordinary singularities. Replacing $\preceq$ by the multiplier-ideal order would define a different operator, which will require a separate comparison.

For a bounded obstacle in a big background, exceptional sets in the competitor inequality can be removed by a small singular perturbation.

Lemma 1.19.3 *Let θ be big, let f be bounded, and let $u \in \mathrm{PSH}(X,\theta)$ satisfy $u \le f$ quasi-everywhere. There are genuine $u_t \in \mathrm{PSH}(X,\theta)$, $t > 0$, such that $u_t \le f$ everywhere and $u_t \to u$ almost everywhere as $t \downarrow 0$.*

Proof Choose $\rho \in \mathrm{PSH}(X,\theta)$ with $\theta + \mathrm{dd^c}\rho \ge \kappa\omega$ for some $\kappa > 0$. The exceptional set $E = \{u > f\}$ is pluripolar. The global form of Josefson's theorem on compact complex manifolds supplies a genuine quasi-plurisubharmonic χ with $\chi = -\infty$ on E; see [Xia, Theorem 1.1.5]. Rescale and normalize it so that $\chi \le 0$ and $\mathrm{dd^c}\chi \ge -\omega$. Shift ρ so that $\rho \le \inf_X f$. With $s = t/\kappa$, put

$$u_t = \frac{u + t\chi + s\rho}{1+s}.$$

All three potentials are finite almost everywhere, so this is genuine. Moreover

$$\theta + \mathrm{dd^c}u_t = \frac{\theta + \mathrm{dd^c}u + t\mathrm{dd^c}\chi + s(\theta + \mathrm{dd^c}\rho)}{1+s} \ge 0.$$

Outside E, each of u and ρ is at most f, while $\chi \le 0$, so $u_t \le f$. On E the numerator is $-\infty$. Finally $u_t \to u$ wherever all three potentials are finite, which is a set of full measure. □

Minimal singularities separate a potential from the poles of a strict analytic barrier. This gives a proper function on its regular locus.

Lemma 1.19.4 *Let $\rho \in \mathrm{PSH}(X,\theta)$ have analytic singularities with smooth remainder and satisfy $\theta + \mathrm{dd^c}\rho \ge \varepsilon\omega$ for some $\varepsilon > 0$. Set $\Omega = \{\rho > -\infty\}$. If $\varphi \in \mathrm{PSH}(X,\theta)$ has minimal singularities and is continuous on Ω, then $\varphi - \rho$ is continuous, bounded below, and proper on Ω: all its sublevel sets are compact in Ω.*

Proof Translate ρ and φ so that $\rho \le 0$ and $\sup_X \varphi = 0$, and choose C with $\varphi \ge V_\theta - C$. These translations do not affect properness. Choose $M \ge 0$ with $\theta \ge -M\omega$. If $a \in (0,1)$ is sufficiently close to one, then

$$\theta + \mathrm{dd^c}(a\rho) = a(\theta + \mathrm{dd^c}\rho) + (1-a)\theta \ge \big(a\varepsilon - (1-a)M\big)\omega > 0.$$

Thus $a\rho$ is an admissible competitor for V_θ. Minimal singularities and the normalization give

$$0 \geq \varphi \geq V_\theta - C \geq a\rho - C,$$

and hence

$$\varphi - \rho \geq (1-a)(-\rho) - C. \tag{1.24}$$

By hypothesis φ is continuous on Ω; ρ is smooth there because it has analytic singularities. Hence $\varphi - \rho$ is continuous. Since $\rho \to -\infty$ when approaching $X \setminus \Omega$, (1.24) places every sublevel set in a compact superlevel set of ρ. The sublevel set is closed in Ω, so it is compact there. □

1.20 The minimum principle

An infimum of plurisubharmonic functions need not be plurisubharmonic. Kiselman's principle gives a different operation: take an infimum in a real parameter whose complexification is plurisubharmonic and invariant in the imaginary direction. Freezing the background coefficients preserves the precise lower curvature bound.

Lemma 1.20.1 *Let $I \subset \mathbb{R}$ be a nonempty open interval, set $S_I = \{z \in \mathbb{C} : \operatorname{Re} z \in I\}$, and let $\pi : X \times S_I \to X$ be the projection. Suppose $U(x,z) = u_{\operatorname{Re} z}(x)$ is quasi-plurisubharmonic and $\pi^*\theta + \mathrm{dd^c}U \geq 0$. For each x, the function $t \mapsto u_t(x)$ is finite and convex or identically $-\infty$. The finite alternative holds for almost every x.*

Proof A local smooth correction making U plurisubharmonic can be chosen to depend only on x. Restriction to a vertical complex line is therefore subharmonic or identically $-\infty$. In the former case its distributional Laplacian is the second real derivative of $t \mapsto u_t(x)$, which is nonnegative. The resulting convex representative is finite on I and agrees everywhere with the subharmonic function by the submean inequality. Finally, U is locally integrable on the product. Fubini's theorem, applied in coordinate balls times a fixed rectangle in S_I, excludes the identically $-\infty$ alternative for almost every x. □

Corollary 1.20.1 *Let $I \subset \mathbb{R}$ be a nonempty open interval. A radial function $z \mapsto q(\log|z|^2)$ is subharmonic on the annulus $\{z : \log|z|^2 \in I\}$ if and only if q is finite and convex on I. Allowing the identically $-\infty$ function on the annulus adds the corresponding alternative for q.*

Proof The holomorphic covering $z = e^{w/2}$ pulls the function back to $q(\operatorname{Re} w)$. The strip argument in Lemma 1.20.1 proves necessity. Conversely, a finite convex q has nonnegative distributional second derivative. Thus $q(\operatorname{Re} w)$ is subharmonic, and this descends through the local biholomorphisms of the covering. □

Proposition 1.20.1 *Under the hypotheses of Lemma 1.20.1, the function*

$$q(x) = \inf_{t \in I} u_t(x)$$

is either a genuine θ-plurisubharmonic function or identically $-\infty$. No regularization of the infimum is necessary. If I is bounded, the infimum is genuine.

Proof The classical local Kiselman principle says that the infimum in the real variable of a plurisubharmonic function on a product with a strip, invariant in its imaginary variable, is plurisubharmonic or identically $-\infty$. Its parameter interval may be unbounded. We use this local result from [Xia].

Fix $x_0 \in X$, choose local holomorphic coordinates centered at x_0, and let κ be their Euclidean form. Fix $\eta > 0$. On a sufficiently small coordinate ball the constant form $\beta = \theta(x_0)$ satisfies

$$\beta - \eta\kappa \le \theta \le \beta + \eta\kappa.$$

Choose a quadratic function r with $dd^c r = \beta + \eta\kappa$. Then $U + r \circ \pi$ is psh on the ball times S_I. Its infimum is $q + r$, so the local principle gives

$$dd^c q \ge -\beta - \eta\kappa \ge -\theta - 2\eta\kappa$$

where the infimum is genuine. In a fixed atlas the Euclidean forms are bounded by a fixed Hermitian metric. Letting $\eta \downarrow 0$ proves the exact lower bound $dd^c q \ge -\theta$. The local alternative propagates across overlapping balls on connected X: a genuine psh function cannot equal $-\infty$ on an open set. Thus either q is genuine everywhere as a quasi-plurisubharmonic function or $q \equiv -\infty$. The infimum is upper semicontinuous, as an infimum of upper semicontinuous functions.

Suppose I is bounded. By Lemma 1.20.1, the function $t \mapsto u_t(x)$ is finite and convex for almost every x. A supporting affine function at any interior point bounds it from below on I. Hence $q(x) > -\infty$ almost everywhere, excluding the identically $-\infty$ alternative. □

For example, the proposition applies to $u_t - t\tau$ for a fixed real τ. On an unbounded interval the infimum may be $-\infty$; this alternative will encode the cutoff of a test curve.

1.20.1 Pointwise Legendre transforms

The minimum principle can also show that a supremum is already upper semicontinuous. Regularizing the Legendre transform of a concave family gives a potential on the product. The minimum principle identifies its inverse transforms with the original family everywhere; applying real duality once more identifies the regularization with the pointwise supremum. We first specify the one-variable duality, including its endpoints; the general Legendre–Fenchel theorem is recalled in [Xia, appendix on convex analysis].

Lemma 1.20.2 *Let $c \in \mathbb{R}$. Suppose $g : \mathbb{R} \to [-\infty, 0]$ is upper semicontinuous, concave and decreasing, finite somewhere, and equal to $-\infty$ on (c, ∞). Then*

$$f(t) = \sup_{\tau<c}\big(g(\tau) + t\tau\big), \qquad t > 0,$$

is finite and convex, satisfies $f(t) \le ct$, and obeys

$$g(\tau) = \inf_{t>0}\big(f(t) - t\tau\big), \qquad \lim_{t\downarrow 0} f(t) = \sup_{\tau<c} g(\tau).$$

Conversely, every finite convex f on $(0, \infty)$ with $f(t) \le ct$ is recovered by the first formula from $g(\tau) = \inf_{t>0}(f(t) - t\tau)$, and satisfies the same endpoint identity. The formulas also hold for $g \equiv -\infty$, with $f \equiv -\infty$.

Proof Decreasingness makes g finite arbitrarily far to the left. Hence the defining supremum is finite for $t > 0$, bounded above by ct and below by any one finite affine term. Concavity and upper semicontinuity give $g(c) = \lim_{\tau\uparrow c} g(\tau)$, so including c does not change the supremum. The Legendre transform of $-g$ is infinite for $t < 0$. At $t = 0$ it equals $m = \sup_{\tau<c} g(\tau)$; the bounds

$$g(\tau) + t\tau \le f(t) \le m + ct$$

show that $f(t) \to m$ as $t \downarrow 0$. The biconjugacy theorem for the closed convex function $-g$ therefore gives the inverse formula, with $t = 0$ omitted without changing the infimum.

Now start with f. At each $t > 0$, choose a supporting slope p. The upper bound $f(s) \le cs$ as $s \to \infty$ forces $p \le c$. If $p < c$, the supporting line gives $g(p) = f(t) - tp$ and recovers $f(t)$. If $p = c$, then $f(s) - cs$ is constant for $s \ge t$: this function is convex, bounded above, hence decreasing, while its supporting line at t is constant. For every $\tau < c$ we consequently have

$$f(t) - ct \le g(\tau) \le f(t) - t\tau.$$

Letting $\tau \uparrow c$ again recovers $f(t)$. In particular g is finite somewhere. As an infimum of affine functions with slopes $-s < 0$, it is upper semicontinuous, concave and decreasing. Taking $s \downarrow 0$ in the bound $f(s) - s\tau \le s(c - \tau)$ gives $g \le 0$, and taking

$s \to \infty$ gives $g = -\infty$ for $\tau > c$. The endpoint identity now follows from the first part. The identically $-\infty$ case follows directly from the formulas. □

Fix $c \in \mathbb{R}$ and let $\Gamma_\tau \in \mathrm{PSH}(X, \theta)$ for $\tau < c$, with $\Gamma_\tau \le 0$, and suppose $\tau \mapsto \Gamma_\tau$ is pointwise concave and decreasing. Extend the family by its decreasing limit at c and by $-\infty$ for $\tau > c$. Define its forward transform by

$$U_t(x) = \sup_{\tau<c}\big(\Gamma_\tau(x) + t\tau\big), \qquad t > 0. \tag{1.25}$$

For a family $(u_t)_{t>0}$, the inverse transform is

$$\widehat{u}_\tau(x) = \inf_{t>0}\big(u_t(x) - t\tau\big). \tag{1.26}$$

Lemma 1.20.3 *At each x, the extended function $\tau \mapsto \Gamma_\tau(x)$ is upper semicontinuous and concave, or identically $-\infty$. Its forward transform satisfies*

$$\Gamma_\tau(x) = \inf_{t>0}\big(U_t(x) - t\tau\big) \qquad (\tau \in \mathbb{R}).$$

Proof Fix $\tau < c$ and let $\tau_j \uparrow \tau$. The decreasing limit of Γ_{τ_j} is a genuine potential because it dominates Γ_τ. Choose $r \in (\tau, c)$. Outside the pole set of Γ_r, the concave parameter function is finite near τ and hence continuous there. Thus the decreasing limit equals Γ_τ almost everywhere, and everywhere by Lemma 1.3.3. This proves left continuity at every $\tau < c$. The value at c is its left limit by definition. Decreasingness gives upper semicontinuity, also where the value is $-\infty$, and concavity survives the extension. The inverse formula is Lemma 1.20.2 at each x. □

Theorem 1.20.1 *For the preceding family $(\Gamma_\tau)_{\tau<c}$, the pointwise supremum in (1.25) is already upper semicontinuous on $X \times (0, \infty)$. Its complexification $U(x, z) = U_{\mathrm{Re}\, z}(x)$ is quasi-plurisubharmonic on $X \times \{\mathrm{Re}\, z > 0\}$ and satisfies $\pi^*\theta + \mathrm{dd^c}U \ge 0$, where $\pi : X \times \{\mathrm{Re}\, z > 0\} \to X$ is the projection. Each slice belongs to* $\mathrm{PSH}(X, \theta)$.

Proof Every term in the supremum satisfies the product current inequality. The bound $\Gamma_\tau \le 0$ gives $U_t \le ct$, while any fixed $\tau_0 < c$ supplies the lower potential $\Gamma_{\tau_0} + t\tau_0$. By the local envelope theorem in Section 1.19, the upper semicontinuous regularization V on the product is quasi-plurisubharmonic, satisfies $\pi^*\theta + \mathrm{dd^c}V \ge 0$, and agrees with U almost everywhere. It is invariant in the imaginary direction and satisfies $V_t \le ct$. The raw supremum is measurable: left continuity from Lemma 1.20.3 allows the parameters to be restricted to rational numbers.

Outside a set $Z \subset X$ of zero smooth volume containing the pole set of Γ_{τ_0}, Fubini's theorem gives $U_t(x) = V_t(x)$ for almost every $t > 0$. Both sides are finite convex functions of t, by Lemmas 1.20.1 and 1.20.2. Their continuity makes this equality valid for every $t > 0$ and every $x \notin Z$.

For each $\tau < c$, Proposition 1.20.1 gives

$$\chi_\tau = \inf_{t>0}(V_t - t\tau) \in \mathrm{PSH}(X, \theta) \cup \{-\infty\}.$$

Off Z it equals Γ_τ by Lemma 1.20.3. It is therefore genuine, and Lemma 1.3.3 gives $\chi_\tau = \Gamma_\tau$ everywhere. At each x, recover the convex vertical function of V from these inverse transforms using Lemma 1.20.2, also in the identically $-\infty$ case. We obtain

$$V_t(x) = \sup_{\tau<c}\big(\chi_\tau(x) + t\tau\big) = U_t(x).$$

Finally, the lower bound by $\Gamma_{\tau_0} + t\tau_0$ ensures that every slice is genuine. □

1.21 Chern-connection regularization

Fix $C_0 > 0$ and $u \in \mathrm{PSH}(X, \omega)$ with $-C_0 \le u \le 0$. Distances, gradients and tensor norms use the Riemannian metric associated to ω.

Let $R_t u$ be Demailly's regularization formed with the Chern connection of ω. It averages u over the tangent fiber with a fixed smooth nonnegative radial kernel of integral one. The averaging map is a smooth realization of the holomorphic part of the formal fiberwise expansion of the exponential map; it need not be genuinely holomorphic in the fiber. The following classical estimate controls the truncated transform, including its mixed base–fiber terms; its sources are recorded in Section 33.1.

Lemma 1.21.1 *There are constants $A, K > 0$ and $t_* > 0$, depending only on the fixed geometry and bound for u, such that $R_t u$ is smooth for $0 < t < t_*$,*

$$-C_0 \le R_t u \le 0, \qquad R_t u + Kt^2 \downarrow u \quad \text{as } t \downarrow 0.$$

For $0 < \delta < t_*$ and $c > 0$, the bounded function

$$\Phi_{c,\delta} = \inf_{0<t\le\delta}\big\{R_t u + K(t^2 - \delta^2) + K(t - \delta) - c\log(t/\delta)\big\}$$

is quasi-plurisubharmonic and satisfies

$$\omega + \mathrm{dd^c}\Phi_{c,\delta} \ge -(Ac + 2K\delta)\omega.$$

The infimum is attained at some $t > 0$ at every point.

Only this classical curvature estimate is retained as an input. Its linear term $K(t - \delta)$ matters in the Hermitian case: it absorbs the mixed base–fiber terms in the complex Hessian of the regularization. The function $-c\log(t/\delta)$ tends to $+\infty$ as $t \downarrow 0$, while u is bounded, which explains attainment away from zero. We require neither smoothness of the minimizing radius nor smoothness of the transform across the region where that radius equals δ.

We record two elementary estimates for the averaging operators so that their use does not require a Kähler volume identity.

Lemma 1.21.2 *For all sufficiently small $t > 0$,*

$$\|R_t u - u\|_{L^1(\mathrm{d}V)} \le Ct^2, \qquad \|\nabla R_t u\|_\infty \le C/t.$$

The constants are uniform for $-C_0 \le u \le 0$.

Proof Let z be local base coordinates and $\xi \in \mathbb{C}^n$ the fiber variable in a smooth orthonormal tangent frame. Write $\rho(|\xi|^2)$ for the fixed radial averaging kernel, normalized against Lebesgue measure $\mathrm{d}\xi$. In these coordinates the averaging map F gives

$$R_t u(z) = \int u(F(z, t\xi))\rho(|\xi|^2)\,\mathrm{d}\xi, \qquad F(z, 0) = z,$$

where the kernel is supported in $|\xi| \le 1$. The map F is smooth in these variables and its fiber differential at zero is an isomorphism. We first claim that the adjoint with respect to $\mathrm{d}V$ satisfies

$$R_t^* 1 = 1 + O(t^2)$$

uniformly. To see this, insert a smooth partition of unity in the z variable with each term compactly supported in a frame chart. For fixed ξ, change variables $x = F(z, t\xi)$ on a neighborhood of that support. The inverse map and its Jacobian have uniform Taylor expansions in t. The term of order zero in the transformed density is the chosen partition function at x. Its term of order one is linear in ξ. Integrating that term against the even radial kernel gives zero. The remaining density is bounded by $Ct^2\mathrm{d}V(x)$, since the second derivatives of all maps, partition functions and volume coefficients are bounded on these finitely many compact charts. Summing the partition functions proves the claim.

Consequently

$$\left|\int_X (R_t u - u)\,\mathrm{d}V\right| = \left|\int_X u(R_t^* 1 - 1)\,\mathrm{d}V\right| \le Ct^2.$$

The preceding lemma gives $R_t u - u \ge -Kt^2$. For a real function $a \ge -Kt^2$ one has $|a| \le a + 2Kt^2$. Apply this inequality and integrate to obtain the first estimate.

For the derivative bound, change variables in the fiber to express $R_t u(z) = \int_X u(x) k_t(z, x)\,\mathrm{d}V(x)$. In local coordinates the smooth kernel is supported where $|x - z| \le Ct$ and satisfies

$$|\nabla_z k_t(z, x)| \le Ct^{-2n-1}.$$

Indeed, the inverse fiber map and its Jacobian have uniformly bounded derivatives; differentiating the scaled radial argument introduces at most one factor t^{-1} in addition to t^{-2n}. Smooth compact support of the kernel causes no boundary term. Integration over a ball of volume at most Ct^{2n} gives $\int |\nabla_z k_t|\,\mathrm{d}V \le C/t$. Multiplication by the uniform bound for u proves the assertion. □

The first estimate uses cancellation of the odd first-order term of an averaging kernel. It does not use conservation of Hermitian Monge–Ampère mass.

1.21.1 The general curvature modulus

Let θ now be any smooth real $(1,1)$-form and let $u \in \mathrm{PSH}(X,\theta) \cap L^\infty(X)$. Use the same averaging operators and metric.

We first specify the classical input for this background. Apply the classical Chern-connection regularization theorem to the closed current $\mathrm{dd^c}u$ with smooth, possibly nonclosed lower bound $-\theta$. Its tangent-bundle curvature bound is dominated by a fixed multiple of ω. The averaging operators have a constant $K_0 > 0$ such that

$$R_t u + K_0 t^2 \downarrow u.$$

In local base coordinates z and a complex radius variable w, put $t = |w|$. Write $\zeta \in T_z^{1,0}X$ and $\eta \in \mathbb{C}$ for the base and radius components of a tangent vector, and set $\lambda(z,t) = t\partial_t(R_t u(z) + K_0 t^2)$. With fixed constants $A, C > 0$ and an error function $\epsilon_0(t) \geq 0$ tending to zero, the theorem bounds the full base–fiber complex Hessian by the base form $-\theta - A\lambda(z,t)\omega - \epsilon_0(t)\omega$, together with mixed and vertical errors bounded by $C(|\zeta||\eta| + |\eta|^2)$. Here λ is nonnegative and increasing in t, and $\epsilon_0(t) \downarrow 0$ as $t \downarrow 0$. This is the classical regularization estimate for a smooth lower bound.

Lemma 1.21.3 *For K sufficiently large, there is an increasing function $\epsilon(\delta) \geq 0$ tending to zero at zero such that, for every small $\delta > 0$ and $c > 0$,*

$$\Phi_{c,\delta} = \inf_{0<t\leq\delta} \{R_t u + K(t^2 - \delta^2) + K(t - \delta) - c\log(t/\delta)\}$$

is bounded, quasi-plurisubharmonic, and satisfies

$$\theta + \mathrm{dd^c}\Phi_{c,\delta} \geq -(Ac + \epsilon(\delta))\omega.$$

The infimum is attained away from zero. One may also use K in the monotone smoothing inequality above.

Proof Take $K \geq K_0$ and put $H(z,t) = R_t u(z) + Kt^2 + Kt$. The function H is increasing and strictly convex as a function of $\log t$, because $R_t u + K_0 t^2$ has nonnegative increasing logarithmic derivative and the added terms have positive second logarithmic derivative. Its logarithmic derivative $\widetilde{\lambda}$ is at least λ.

The complex Hessian of $K|w|$ has a positive vertical term proportional to $K/|w|$. For sufficiently small $|w| = t$, it absorbs the vertical error and the mixed term, using

$$C|\zeta||\eta| \leq C't|\zeta|^2 + \frac{K}{8t}|\eta|^2.$$

After increasing constants and replacing λ by the larger $\widetilde{\lambda}$, the full base–fiber Hessian of H is therefore bounded below by the base form

$$-\theta - (A\widetilde{\lambda} + \epsilon(t))\omega,$$

where an increasing majorant of $\epsilon_0(t) + C't$ may be used.

Boundedness of u makes the logarithmic penalty tend uniformly to $+\infty$ at zero. Thus the minimum occurs on a compact radial interval away from zero. Strict logarithmic convexity makes the minimizing radius unique. At an interior minimum $\widetilde{\lambda} = c$; at the upper cap $t = \delta$, one has $\widetilde{\lambda} \leq c$. No differentiability of the minimizing radius is needed.

For completeness, localize near any base point and fix $a > 0$. Continuity of compact minimizers and of $\widetilde{\lambda}$ permits a radial interval containing all nearby minimizers on which $\widetilde{\lambda} \leq c + a$. In the cap case keep the original upper radius δ; an open interval approaching the cap has the same infimum. The lower endpoint can be chosen below all minimizers by the uniform logarithmic penalty. Monotonicity of $\widetilde{\lambda}$ ensures its bound throughout the chosen interval. The minimum principle for rotation-invariant plurisubharmonic functions, applied after a local quadratic correction, now gives the lower curvature bound with $A(c + a) + \epsilon(\delta)$ and an arbitrarily small error from freezing the smooth base forms. Let these errors and a tend to zero. This proves the asserted current inequality and quasi-plurisubharmonicity. Finally, $K \geq K_0$ preserves monotonicity and the pointwise limit of $R_t u + Kt^2$. □

The classical theorem supplies a modulus ϵ, not a rate claimed here to be linear in δ. The cancellation and kernel argument of Lemma 1.21.2 still give

$$\|R_t u - u\|_{L^1(\mathrm{d}V)} \leq Ct^2, \qquad \|\nabla R_t u\|_\infty \leq C/t.$$

Indeed the averaging map is unchanged, u is bounded, and the monotone lower bound just proved is all that argument uses.

1.21.2 A linear error for a smooth background

Let θ be a smooth big real $(1, 1)$-form. Choose a strict potential ρ with analytic poles and smooth resolved remainder as in Theorem 1.15.2, and put $\Omega = X \setminus \{\rho = -\infty\}$. The next estimate refines the curvature error for normalized potentials locally bounded on Ω.

Lemma 1.21.4 *Let a be quasi-plurisubharmonic on X, with $\sup_X a = 0$ and $\theta + \mathrm{dd^c} a \geq 0$, and suppose that a is locally bounded on Ω. There are constants $A, K, C > 0$ such that the averaging operators R_t from Lemma 1.21.1 satisfy*

$$R_t a + Kt^2 \downarrow a, \qquad \|R_t a - a\|_{L^1(\mathrm{d}V)} \leq Ct^2.$$

For small $\delta > 0$ and $c > 0$, the transform on Ω

$$\Phi_{c,\delta} = \inf_{0<t\leq\delta} \{R_t a + K(t^2 - \delta^2) + K(t - \delta) - c\log(t/\delta)\}$$

extends canonically to X and satisfies

$$a - K(\delta^2 + \delta) \leq \Phi_{c,\delta} \leq R_\delta a \leq 0, \qquad \theta + \mathrm{dd^c}\Phi_{c,\delta} \geq -(Ac + C\delta)\omega. \tag{1.27}$$

At each point of Ω the infimum is attained at a positive radius. On every compact $K_1 \Subset \Omega$, for sufficiently small t, $\|\nabla R_t a\|_{L^\infty(K_1)} \leq C_{K_1}/t$.

Proof Apply the classical Chern-connection regularization expansion to the closed current $\mathrm{dd^c} a$ and the smooth lower bound $-\theta$. It gives monotonicity and logarithmic convexity: for a constant $K_0 > 0$, set $\lambda(x,t) = t\partial_t(R_t a(x) + K_0 t^2)$. This function is nonnegative and increasing in t. We first justify a linear error for this smooth lower bound, rather than infer a rate from the unspecified modulus in that theorem.

Choose the auxiliary curvature form to be $A\omega$, with A large enough that the Chern curvature plus $A\omega \otimes \mathrm{Id}$ is nonnegative on every decomposable tensor. Equivalently, after absorbing the normalization constants, writing $R_{j\bar{k}\ell\bar{m}}$ for the Chern curvature coefficients and ξ, ζ for tangent vectors in a unitary frame, the coefficients satisfy

$$\sum R_{j\bar{k}\ell\bar{m}} \xi_j \bar{\xi}_k \zeta_\ell \bar{\zeta}_m \geq -A|\xi|^2|\zeta|^2.$$

This choice is available by compactness. Pairing this inequality with the nonnegative eigenvalues of a positive complex Hessian bounds its curvature contraction by $-A$ times its trace, with no further error term.

Fix $x \in X$ and a complex radius variable $w \neq 0$, and put $t = |w|$. Write $\xi \in T_x^{1,0}X$ and $\tau \in \mathbb{C}$ for a base–radius tangent vector. In a uniformly sized coordinate chart about x, write $\beta = \mathrm{dd^c}|z|^2$ and choose a quadratic $q_{x,t}$ with

$$\mathrm{dd^c} q_{x,t} = \theta(x) + Bt\beta,$$

where B is fixed and large enough that $a + q_{x,t}$ is plurisubharmonic on a ball containing the averaging support and its nearby base points. This follows from the smooth variation bound $\|\theta(y) - \theta(x)\| \leq C|y - x|$ on that ball of radius $O(t)$. For the Hessian calculation at (x, w) the quadratic is held fixed. Its coefficients and all needed derivatives are uniformly bounded.

We estimate the errors in the classical regularization expansion. Normalized quasi-plurisubharmonic compactness bounds $\|a\|_1$. On each fixed larger chart, add one fixed strictly plurisubharmonic quadratic Q to a. If $\nu = \mathrm{dd^c}(a + Q) \wedge \beta^{n-1}$, the local Lelong mass inequality and a fixed larger-ball mass bound give $\nu(B_t(x)) \leq Ct^{2n-2}$.

The coefficients of $\partial\bar{\partial}(a+q_{x,t})$ are controlled in total variation by those of $\mathrm{dd^c}(a+Q)$ plus a uniformly bounded smooth form. Write $|\partial\bar{\partial}(a+q_{x,t})|$ for the sum of the total variations of its coefficient measures. The rescaled averaging kernels are supported in $B(x, Ct)$ and bounded by a constant times t^{-2n}. Their integrals against this measure are therefore bounded by

$$Ct^{-2n}|\partial\bar{\partial}(a+q_{x,t})|(B(x,Ct)) \le C't^{-2}.$$

Thus the terms with a coefficient $O(t^3)$ contribute $O(t)$; the order-two expansion has an $O(t)$ remainder as well. The mixed and vertical contributions have bounds $C(|\xi||\tau|+|\tau|^2)$. The curvature term is bounded below by $-A\lambda(x,t)|\xi|^2$, using the preceding curvature choice. The radial term of the same expansion identifies this mass with λ up to $O(t^2)$. Subtracting the regularization of $q_{x,t}$ restores $-\theta(x)$; its base Hessian differs from $\mathrm{dd^c}q_{x,t}$ by $O(t)$, its mixed and vertical Hessians are bounded, and its radial logarithmic derivative is $O(t^2)$. The extra $Bt\beta$ also costs only $O(t)$. For a real $(1,1)$-form α, write $\alpha[v]^2$ for its associated Hermitian quadratic form evaluated on v. Consequently, for a fixed constant $C_1 > 0$, the full base–fiber Hessian satisfies

$$\mathrm{dd^c}_{x,w}R_{|w|}a[\xi,\tau]^2 \ge -\theta_x[\xi]^2 - (A\lambda(x,t)+Ct)|\xi|^2 - C_1(|\xi||\tau|+|\tau|^2).$$

All constants are independent of x, t, c, δ and uniform in the normalized family. In particular, setting $\xi = 0$ in the same local expansion gives a uniform lower bound for the fiber Hessian. Adding a fixed multiple of $|w|^2$ gives its subharmonicity, hence the logarithmic convexity and monotonicity above with a uniform K_0.

Increase $K \ge K_0$ and add $K(t^2+t)$ to the regularization. The positive vertical Hessian of $K|w|$ absorbs the mixed term by

$$C_1|\xi||\tau| \le C't|\xi|^2 + \frac{K}{8t}|\tau|^2,$$

and absorbs the remaining bounded vertical error for small t. The logarithmic derivative of the corrected function is at least λ. The capped Kiselman minimum argument in Lemma 1.21.3 therefore gives the last inequality in (1.27), now with error $C\delta$. On compact subsets of Ω, local boundedness of a makes the logarithmic penalty diverge uniformly at zero, so that the minimum occurs away from zero. The same argument handles minimizers at the upper cap without differentiating their radius.

The other bounds in (1.27) follow from monotonicity and evaluation at $t = \delta$. In particular the transform is bounded above near the analytic complement. The removable singularity theorem extends it, preserving the exact smooth lower Hessian bound by local freezing and passage to the limit. Its lower bound by $a - K(\delta^2+\delta)$ also extends by canonical representatives.

The adjoint computation in Lemma 1.21.2 gives $R_t^*1 = 1 + O(t^2)$. It needs only $a \in L^1$, since

$$\left|\int_X (R_t a - a)\,\mathrm{d}V\right| \le Ct^2\|a\|_1.$$

Together with $R_t a - a \geq -Kt^2$, this proves the stated L^1 estimate. Finally the smoothing kernel near K_1 samples a fixed larger compact subset of Ω, where a is bounded. The same kernel derivative estimate as in Lemma 1.21.2 gives C_{K_1}/t. □

1.22 Pullback and descent

Pullback preserves the local positivity of a potential. Compact fibers force descent when the background has no vertical component; under a modification, a Jacobian factor accounts for the change in multiplier-ideal integrability.

1.22.1 Holomorphic disks

A smooth background need not have local potentials, but its pullback to a disk does. Testing there retains the exact lower curvature bound under a holomorphic map.

Lemma 1.22.1 *Let θ be a smooth real $(1,1)$-form on a complex manifold M, and let u be quasi-plurisubharmonic on M. Then $\theta + \mathrm{dd^c} u \geq 0$ if and only if the following condition holds. For every holomorphic map $f : \Delta \to M$ from the unit disk whose pullback $u \circ f$ is not identically $-\infty$, the function $u \circ f + a$ is subharmonic wherever a is a smooth local potential for $f^*\theta$.*

Proof Suppose first that $\theta + \mathrm{dd^c} u \geq 0$. On a coordinate ball convolve with a smooth nonnegative radial kernel. The smooth functions u_ϵ satisfy

$$\mathrm{dd^c} u_\epsilon \geq -\theta_\epsilon, \qquad \theta_\epsilon \longrightarrow \theta \quad \text{smoothly}.$$

Write $u = p - r$, with p plurisubharmonic and r smooth. The convolutions p_ϵ decrease to p. On a smaller disk where $u \circ f$ is genuine, $p_\epsilon \circ f$ therefore converges in L^1_{loc} to $p \circ f$, while $r_\epsilon \circ f$ converges smoothly. Pull back the smooth inequality and pass to distributions. This gives $\mathrm{dd^c}(u \circ f) + f^*\theta \geq 0$. Since $u \circ f = p \circ f - r \circ f$ is quasi-subharmonic, the distributional inequality makes $u \circ f + a$ subharmonic.

Conversely, test on affine complex lines in a coordinate ball. For any fixed nonzero constant direction ξ, local integrability of u implies that almost every parallel line has a genuine restriction. Fubini's theorem applied to nonnegative smooth test functions gives positivity of the scalar distribution obtained by evaluating the Hermitian coefficient matrix of $\theta + \mathrm{dd^c} u$ on ξ. Positivity for a countable dense set of constant directions proves positivity of the Hermitian matrix of coefficient measures, hence the current inequality. □

Consequently, holomorphic pullback preserves the background inequality wherever the pulled-back potential is genuine: apply the disk criterion to compositions of maps. We also use descent of subharmonicity under a finite disk map $\rho(t) = t^k$, $k \geq 1$.

If $h \circ \rho$ is subharmonic, local inverse branches make h subharmonic off zero. It is bounded above near zero and has the canonical extension value there, since

$$h(0) = (h \circ \rho)(0) = \limsup_{t \to 0,\ t \neq 0} h(t^k) = \limsup_{z \to 0,\ z \neq 0} h(z).$$

The removable-singularity theorem makes h subharmonic at zero as well.

1.22.2 Compact fibers

Proposition 1.22.1 *Let $\pi : Y \to X$ be a surjective holomorphic submersion between connected compact complex manifolds, with connected fibers. Let θ be a smooth real $(1,1)$-form on X. Pullback gives a bijection*

$$\mathrm{PSH}(X, \theta) \longrightarrow \mathrm{PSH}(Y, \pi^*\theta).$$

Every member on the right is constant on each fiber, with the value $-\infty$ allowed on an entire fiber.

Proof In a coordinate ball $V \subset X$, choose a smooth function ρ with $\mathrm{dd^c}\rho \geq \theta$. If $u \in \mathrm{PSH}(Y, \pi^*\theta)$, then $u + \rho \circ \pi$ is plurisubharmonic on $\pi^{-1}(V)$. Its restriction to a fiber is plurisubharmonic or identically $-\infty$. Since a fiber is compact and connected, the maximum principle makes it constant in either case. Thus $u = v \circ \pi$ for a uniquely defined function v on X.

A holomorphic submersion admits local holomorphic sections. Composing $u + \rho \circ \pi$ with such a section shows that $v + \rho$ is plurisubharmonic or identically $-\infty$ on a smaller ball. The latter alternative on a nonempty open ball would make $u = -\infty$ on its open inverse image, contradicting genuineness. Hence v is quasi-plurisubharmonic. Distributional differentiation in local product coordinates gives

$$\pi^*(\theta + \mathrm{dd^c} v) = \pi^*\theta + \mathrm{dd^c} u \geq 0.$$

Positivity of a current of this pulled-back form implies positivity on the base: test in product coordinates and integrate a nonnegative compactly supported volume form of integral one in the fiber variables. It follows that $v \in \mathrm{PSH}(X, \theta)$.

Conversely, Lemma 1.22.1 shows that pullback of a θ-plurisubharmonic function by a submersion is $\pi^*\theta$-plurisubharmonic. It is genuine by surjectivity and local product coordinates. Uniqueness follows from surjectivity. □

1.22.3 Modifications

Proposition 1.22.2 *Let $\pi : Y \to X$ be a modification of connected compact complex manifolds, and let θ be a smooth real $(1,1)$-form on X. Pullback gives a bijection*

$$\mathrm{PSH}(X,\theta) \longrightarrow \mathrm{PSH}(Y,\pi^*\theta).$$

Proof Let $S \subset X$ be a proper analytic set over whose complement π is an isomorphism, and put $D = \pi^{-1}(S)$. Pullback preserves the curvature inequality by Lemma 1.22.1 and is genuine on the isomorphism locus. It is injective because π is surjective.

To prove surjectivity, let $b \in \mathrm{PSH}(Y,\pi^*\theta)$ and define u off S by the inverse map. On a coordinate neighborhood choose a smooth ρ with $\mathrm{dd^c}\rho \geq \theta$. Then $b + \rho \circ \pi$ is plurisubharmonic. Properness and its local upper bounds give a local upper bound for $u + \rho$ near S, so the removable-singularity theorem extends $u + \rho$ plurisubharmonically across S. The resulting quasi-plurisubharmonic extensions of u agree almost everywhere on overlaps and hence glue as canonical representatives.

The current $\theta + \mathrm{dd^c}u$ is positive off S. Locally it is the sum of the positive current $\mathrm{dd^c}(u + \rho)$ and the smooth form $\theta - \mathrm{dd^c}\rho$. The latter has zero coefficient-measure mass on S, so the restriction of $\theta + \mathrm{dd^c}u$ to S is positive as well. Thus $u \in \mathrm{PSH}(X,\theta)$. The plurisubharmonic functions $b + \rho \circ \pi$ and $(u + \rho) \circ \pi$ agree off D, hence almost everywhere and everywhere. Therefore $b = u \circ \pi$. □

A modification preserves L^1 convergence within a family with one lower curvature bound. Compactness controls the pullbacks near the exceptional locus, where a direct change-of-variables estimate would involve a vanishing Jacobian.

Lemma 1.22.2 *Let $\pi : Y \to X$ be a modification between connected complex manifolds, and let θ be a smooth real $(1,1)$-form on X. If $v_j \to v$ in $L^1_{\mathrm{loc}}(X)$ and all v_j, v belong to $\mathrm{PSH}(X,\theta)$, then*

$$v_j \circ \pi \longrightarrow v \circ \pi \quad \text{in } L^1_{\mathrm{loc}}(Y).$$

In particular, on compact manifolds this is convergence in L^1.

Proof Lemma 1.4.1 bounds v_j uniformly above on relatively compact subsets of X. The pullbacks are therefore locally uniformly bounded above and have the common smooth lower curvature bound $-\pi^*\theta$. After a smooth local correction, plurisubharmonic compactness gives a convergent subsequence on each connected coordinate region in Y, or convergence to $-\infty$ locally uniformly.

Every such region meets the complement of the exceptional locus. There π is a biholomorphism, and change of variables gives L^1_{loc} convergence to $v \circ \pi$. Thus collapse is excluded. Each subsequential limit agrees with $v \circ \pi$ off the exceptional locus, hence almost everywhere, since that analytic locus has smooth volume zero. The limit is unique. Applying this argument to every subsequence and a countable coordinate cover proves convergence of the whole sequence in $L^1_{\mathrm{loc}}(Y)$. □

Capacity convergence is also preserved by a modification. The comparison takes place away from small neighborhoods of the exceptional sets; a uniform comparison of capacities on the whole manifolds is unnecessary.

Lemma 1.22.3 *Let $\pi : Y \to X$ be a modification between connected compact complex manifolds, equipped with Hermitian forms ω_Y, ω_X. Write* $\mathrm{Cap}_Y = \mathrm{Cap}_{\omega_Y}$ *and* $\mathrm{Cap}_X = \mathrm{Cap}_{\omega_X}$. *For quasi-plurisubharmonic functions v_j, v on X,*

$$v_j \longrightarrow v \text{ in capacity on } X \quad \Longleftrightarrow \quad v_j \circ \pi \longrightarrow v \circ \pi \text{ in capacity on } Y.$$

Proof Choose a proper analytic set $S \subset X$ such that π is an isomorphism outside $D = \pi^{-1}(S)$. Both sets have capacity zero. For $\epsilon > 0$, choose an open neighborhood V of D of capacity less than ϵ. Properness gives a neighborhood U_0 of S with $\pi^{-1}(U_0) \subset V$. Shrink it to an open U with $\mathrm{Cap}_X(U) < \epsilon$. The inverse map is biholomorphic near the compact set $X \setminus U$. A finite cover by nested coordinate charts and the local-global capacity comparison of Section 1.8 give a constant C_ϵ such that, for every Borel set $E \subset X$,

$$\begin{aligned} \mathrm{Cap}_Y(\pi^{-1}E) &\le \epsilon + C_\epsilon \, \mathrm{Cap}_X(E), \\ \mathrm{Cap}_X(E) &\le \epsilon + C_\epsilon \, \mathrm{Cap}_Y(\pi^{-1}E). \end{aligned}$$

Apply these inequalities to $E = \{|v_j - v| > \delta\}$ for each fixed $\delta > 0$, omitting the pluripolar pole sets. First let $j \to \infty$, then $\epsilon \downarrow 0$. □

1.22.4 Multiplier ideals under modifications

Let $\pi : Y \to X$ be a modification of connected compact complex manifolds of dimension n. There is an analytic subset $S \subset X$ of codimension at least two such that π is an isomorphism over $X \setminus S$. Let K_X and K_Y be the canonical holomorphic line bundles. Set $D = \pi^{-1}(S)$ and

$$R_\pi = K_Y \otimes \pi^* K_X^{-1}.$$

The determinant of the differential is a canonical holomorphic section J_π of R_π. In local coordinates its coefficient is the complex Jacobian determinant. Its zeros are contained in D. These standard facts about modifications of smooth manifolds include the case $n = 1$, when the modification is an isomorphism.

Lemma 1.22.4 *Let a be quasi-plurisubharmonic on X. Under the natural identification away from S,*

$$\pi_*\big(R_\pi \otimes \mathcal{I}(a \circ \pi)\big) = \mathcal{I}(a).$$

More precisely, for any holomorphic line bundle M on X, the map

$$s \longmapsto J_\pi \, \pi^* s$$

identifies

$$H^0\big(X, M \otimes \mathcal{I}(a)\big) \simeq H^0\big(Y, \pi^* M \otimes R_\pi \otimes \mathcal{I}(a \circ \pi)\big).$$

Proof The assertion is local on X. In coordinates and local frames the proposed map sends a holomorphic coefficient f to $(f \circ \pi)\operatorname{Jac}\pi$. Change of variables gives the local integrability equivalence between

$$|f|^2 \mathrm{e}^{-a} \quad \text{and} \quad |f \circ \pi|^2 |\operatorname{Jac}\pi|^2 \mathrm{e}^{-a\circ\pi}.$$

For the implication from integrability upstairs to integrability near a downstairs point, take a relatively compact smaller neighborhood. Its inverse image has compact closure by properness, so finitely many integrability charts suffice. Smooth positive coordinate densities are comparable on these smaller charts. Analytic exceptional sets have smooth volume zero and do not change the integrals.

It remains to justify that every section upstairs has the required form. Divide it by J_π off D and use the inverse of π there. This gives a holomorphic section of M off S. Hartogs extension across the codimension-at-least-two analytic set S supplies a holomorphic section s on X. Its image $J_\pi \pi^* s$ agrees with the given section off D, hence everywhere by the identity theorem. The integrability equivalence then places s in the required multiplier ideal. The same argument on each open set proves the sheaf identity and the global identification. □

In particular, the formula does not identify $\pi^*\mathcal{I}(a)$ with $\mathcal{I}(a \circ \pi)$. The Jacobian is part of the holomorphic coefficient upstairs, rather than a factor that may be deleted from the integrability test.

We also need invariance of the ideal *order*, which asks for all positive weights. We give the local argument to avoid assuming that a modification of a small open subset of Y extends over X.

Lemma 1.22.5 *For quasi-plurisubharmonic a, b on X,*

$$a \preceq_{\mathcal{I}} b \quad \Longleftrightarrow \quad a \circ \pi \preceq_{\mathcal{I}} b \circ \pi.$$

The assertion is local on X and also holds for a proper bimeromorphic map between noncompact smooth complex manifolds.

Proof The reverse implication follows at every positive weight from the sheaf identity in Lemma 1.22.4: tensor the upstairs ideal inclusion with R_π and apply the left-exact direct image functor. The proof of that sheaf identity is local on the target; properness makes each fiber compact and supplies the required finite integrability cover. Compactness of the entire manifolds is unnecessary.

For the forward implication, fix a coordinate neighborhood $W \subset Y$, a smooth local modification $\sigma : Z \to W$, and a prime divisor E on Z. Shrinking W as necessary near the center under consideration, its image is contained in a coordinate neighborhood $U \subset X$. Put $F = \pi \circ \sigma$ and

$$A_F(E) = 1 + \operatorname{ord}_E \operatorname{Jac} F.$$

The map F is holomorphic and is one-to-one and biholomorphic off a proper analytic set. It need not be a proper modification of U.

For a positive integer m, let ν_E denote the generic Lelong number along E, and let ord_E denote the generic order of an ideal there. The proof of Lemma 1.17.7 applies to this map on relatively compact coordinate patches and gives

$$\max\left\{\nu_E(a \circ F) - \frac{A_F(E)}{m}, 0\right\} \le \frac{1}{m} \operatorname{ord}_E\big(\mathcal{I}(ma)\mathcal{O}_Z\big) \le \nu_E(a \circ F).$$

For clarity, the upper inequality pulls back the local Bergman lower bound on U. For the lower one, a germ $f \in \mathcal{I}(ma)$ pulls back to an integrable coefficient $(f \circ F) \operatorname{Jac} F$: on a relatively compact upstairs patch the change-of-variables integral is bounded by the integral over its downstairs image. Generic divisorial removal and the one-variable integral give

$$\operatorname{ord}_E(f \circ F) + A_F(E) > m\nu_E(a \circ F).$$

Only local injectivity off the exceptional set and the compact patch are needed for this estimate, not properness of F over U. The same estimates hold for b.

The hypothesis implies $\mathcal{I}(ma) \subseteq \mathcal{I}(mb)$ for every $m > 0$. Orders of the pulled-back ideals therefore satisfy the reverse inequality. Letting $m \to \infty$ in the two estimates proves

$$\nu_E(a \circ F) \ge \nu_E(b \circ F).$$

This holds for every local divisor over Y. Blowing up a point on any local modification recovers its pointwise Lelong number; in dimension one the point already is a divisor. The exact local criterion in Theorem 1.16.1 now proves the upstairs ideal order. No extension of σ outside its local domain was used. □

Lemma 1.22.6 *For a proper bimeromorphic map $\rho : Z \to Y$ between smooth complex manifolds and genuine quasi-plurisubharmonic functions a, b on Y,*

$$a \sim_{\mathcal{I}} b \quad \Longleftrightarrow \quad a \circ \rho \sim_{\mathcal{I}} b \circ \rho.$$

The statement is local on Y.

Proof Apply the local order equivalence of Lemma 1.22.5 to both (a, b) and (b, a). The two order inequalities are exactly equality of the multiplier ideals at every positive weight. □

1.22.5 Restriction of analytic approximants

An arbitrary quasi-plurisubharmonic function with zero generic Lelong number along a submanifold may still restrict identically to $-\infty$. Analytic regularization removes this obstruction: the approximating ideals have genuine restrictions, and a fixed lower curvature bound controls their divisorial orders on every fixed modification.

Corollary 1.22.1 *Fix a smooth real $(1,1)$-form θ, a Hermitian form ω, and $u \in \mathrm{PSH}(X, \theta)$. Let u_m with curvature errors ϵ_m be the approximants of Corollary 1.15.1. Let $Y \subset X$ be a connected compact complex submanifold and suppose*

$$\nu(u, Y) := \inf_{y \in Y} \nu(u, y) = 0.$$

Each $u_m|_Y$ is genuine. If $\pi : Z \to Y$ is a smooth compact modification and $E \subset Z$ is a prime divisor, then

$$\sup_m \nu\big((u_m|_Y) \circ \pi, E\big) < \infty.$$

Proof For each m, choose $y \in Y$ with $\nu(u, y) < 1/m$. The local Skoda integrability criterion gives $\mathcal{I}(mu)_y = \mathcal{O}_{X,y}$. Thus the image ideal $\mathcal{I}(mu)\mathcal{O}_Y$ is not identically zero. Analytic restriction in Proposition 1.13.1 makes $u_m|_Y$ genuine. Its pullback to Z is also genuine.

The restricted and pulled-back positivity inequality gives

$$\pi^*(\theta|_Y) + \mathrm{dd^c}\big((u_m|_Y) \circ \pi\big) \geq -\epsilon_m \pi^*(\omega|_Y).$$

A single smooth Hermitian form on Z, multiplied by a large enough constant, dominates these lower bounds for all indices. Lemma 1.17.6 therefore applies on Z and bounds the generic Lelong number along E. □

1.23 Integration against non-pluripolar measures

A common bound on the potentials allows L^1 convergence for smooth volume to be transferred to a fixed finite measure that does not charge pluripolar sets. Bounded exponentials extend this statement to upper-bounded families.

1.23.1 A fixed measure

Cegrell's fixed-measure convergence lemma says that a uniformly bounded sequence of plurisubharmonic functions converging in L^1_{loc} converges in $L^1(\nu)$ on compact subsets for each finite non-pluripolar measure ν there. The usual formulation allowing extraction gives the same conclusion: apply it to every subsequence, extract almost-everywhere convergence for smooth volume, and use the subsequence criterion for convergence. This is the classical fixed-measure argument used in [GZ17, Section 11.1.2]. We retain this local lemma as a standard input.

Lemma 1.23.1 *Let ν be a finite non-pluripolar measure on X. If uniformly bounded quasi-plurisubharmonic functions a_j, a have a common smooth lower curvature bound and $a_j \to a$ in $L^1(X)$, then*

$$\int_X |a_j - a| \, \mathrm{d}\nu \longrightarrow 0.$$

If instead $u_j, u \in \mathrm{PSH}(X, \theta)$ have a common upper bound and $u_j \to u$ in $L^1(X)$, then, for every $b > 0$,

$$\int_X |\mathrm{e}^{bu_j} - \mathrm{e}^{bu}| \, \mathrm{d}\nu \longrightarrow 0.$$

Proof For the first statement add a fixed smooth strictly plurisubharmonic correction on each of finitely many coordinate balls. The resulting functions are uniformly bounded and plurisubharmonic. Apply the local input on smaller compact balls covering X and sum the integrals; the corrections cancel in the difference.

For the second statement choose $A\omega \geq \theta$. If $u_j \leq C$, the convex chain rule gives $\mathrm{dd^c}\mathrm{e}^{bu_j} \geq -bA\mathrm{e}^{bC}\omega$, also for unbounded u_j by bounded truncation. Thus e^{bu_j} are uniformly bounded quasi-plurisubharmonic functions in one lower curvature class. Their $L^1(\mathrm{d}V)$ convergence follows from the Lipschitz bound $b\mathrm{e}^{bC}$ for the exponential on $(-\infty, C]$. Apply the first statement. We assign the value zero to the exponential at a pole. □

1.23.2 Vanishing exponents

Proposition 1.23.1 *Let θ be smooth and let $\varphi \in \mathrm{PSH}(X, \theta)$. Suppose φ is unbounded below and $u_\lambda \sim \varphi$ for every $\lambda > 0$. Then λu_λ cannot converge uniformly on X to a finite constant as $\lambda \downarrow 0$. Its distance to any finite constant is also infinite in $L^\infty(\mathrm{d}V)$ for each fixed $\lambda > 0$.*

Proof Bounded equivalence preserves unboundedness below. Every nonempty strict sublevel set of the upper semicontinuous function u_λ is open and has positive smooth volume. Thus u_λ is essentially unbounded below, as is λu_λ. The stated conclusions follow. □

A vanishing exponent gives convergence to one even when the normalized potentials vary.

Lemma 1.23.2 *Let $v_j \in \mathrm{PSH}(X, \theta)$ satisfy $\sup_X v_j = 0$, and let $\lambda_j > 0$ decrease to zero. For every finite non-pluripolar measure μ,*

$$\int_X \left|\mathrm{e}^{\lambda_j v_j} - 1\right| \mathrm{d}\mu \longrightarrow 0.$$

If $f \in L^p(\mathrm{d}V)$ for some $p > 1$, then more precisely

$$\int_X \left|\mathrm{e}^{\lambda_j v_j} - 1\right| |f| \,\mathrm{d}V \leq C_p \lambda_j \|f\|_{L^p(\mathrm{d}V)}.$$

The constant depends on the fixed background, p and $\mathrm{d}V$.

Proof Choose a Hermitian form $\Omega \geq \theta$. By Lemma 1.8.1, for every $T > 1$,

$$\mathrm{Cap}_\Omega(\{v_j < -T\}) \leq C/T$$

uniformly in j. Since $v_j \leq 0$, splitting at this sublevel gives

$$\int_X (1 - \mathrm{e}^{\lambda_j v_j}) \,\mathrm{d}\mu \leq \lambda_j T \mu(X) + \mu(\{v_j < -T\}).$$

Lemma 1.8.5 makes the second term uniformly small when T is large. For this fixed T, the first term tends to zero as $j \to \infty$. This proves the general assertion. The value of the exponential is zero at a pole; pole sets have zero μ-measure.

For the quantitative assertion, Lemma 1.5.1, applied in background Ω, gives a uniform L^q bound for $-v_j$, where $q = p/(p-1)$. Hölder's inequality and

$$0 \leq 1 - \mathrm{e}^{\lambda_j v_j} \leq \lambda_j |v_j|$$

give the claimed estimate for every j. □

1.23.3 Measures with bounded potentials

A moving bounded-product measure can also be used if its potentials converge in capacity. The proof estimates the torsion terms in integration by parts.

Lemma 1.23.3 *Let $a_j, a, w_j, w \in \mathrm{PSH}(X, \omega)$ be uniformly bounded. If $a_j \to a$ in $L^1(X)$ and $w_j \to w$ in capacity, then, for $0 \le k \le n$,*

$$\int_X |a_j - a|\, \omega_{w_j}^k \wedge \omega^{n-k} \longrightarrow 0.$$

The same conclusion holds when the a_j have any fixed smooth lower curvature bound.

Proof By Lemma 1.4.1, the tail envelopes $A_j = (\sup_{\ell \ge j} a_\ell)^*$ decrease to a. Since $a \le \max(a_j, a) \le A_j$, bounded monotone capacity convergence implies

$$\alpha_j := \max(a_j, a) \longrightarrow a \quad \text{in capacity.}$$

The part $\alpha_j - a$ of $|a_j - a| = (\alpha_j - a) + (\alpha_j - a_j)$ tends to zero against the moving products by Lemma 1.8.3.

Put $h_j = \alpha_j - a_j \ge 0$. We compare its integrals for two indices j, ℓ. The case $k = 0$ is the assumed L^1 convergence. For $k \ge 1$, put $z = w_j - w_\ell$ and

$$T = \sum_{s=0}^{k-1} \omega_{w_j}^s \wedge \omega_{w_\ell}^{k-1-s} \wedge \omega^{n-k}.$$

The sum includes $s = 0$. Factorization and compact Stokes give

$$\int_X h_j \big(\omega_{w_j}^k - \omega_{w_\ell}^k\big) \wedge \omega^{n-k} = \int_X h_j \mathrm{dd^c} z \wedge T = \int_X z\, \mathrm{dd^c}(h_j T).$$

Here the full product rule is

$$\mathrm{dd^c}(h_j T) = \mathrm{dd^c} h_j \wedge T + \mathrm{d}h_j \wedge \mathrm{d^c} T - \mathrm{d^c} h_j \wedge \mathrm{d}T + h_j \mathrm{dd^c} T. \tag{1.28}$$

The last term is retained. It need not vanish on a Hermitian manifold.

We estimate the four terms. Since $\mathrm{dd^c} h_j = \omega_{\alpha_j} - \omega_{a_j}$, the absolute contribution of the first is at most

$$\int_X |z| (\omega_{\alpha_j} + \omega_{a_j}) \wedge T.$$

It tends to zero uniformly as $j, \ell \to \infty$ by Lemma 1.8.3, since $w_j - w_\ell \to 0$ in capacity. Differentiating each factor shows $\mathrm{d}T = \mathrm{d}\omega \wedge R$ and $\mathrm{d^c} T = \mathrm{d^c}\omega \wedge R$, where R is a positive sum of mixed products of degree $(n-2, n-2)$. Moreover $\mathrm{dd^c} T$ is a sum of terms containing $\mathrm{dd^c}\omega$ and mixed products of degree $n-2$, or $\mathrm{d}\omega \wedge \mathrm{d^c}\omega$ and mixed products of degree $n-3$. Terms with negative degree are absent. These

smooth torsion factors are bounded in absolute value by fixed multiples of ω^2 and ω^3. As h_j is uniformly bounded, the contribution of $h_j \mathrm{dd^c} T$ also tends uniformly to zero by the same capacity estimate.

For the two gradient terms, the pointwise Cauchy–Schwarz inequality for a positive mixed product gives, for example,

$$\left| \int_X z \, \mathrm{d}h_j \wedge \mathrm{d^c}\omega \wedge R \right|^2 \leq C \left(\int_X |z| \, \mathrm{d}h_j \wedge \mathrm{d^c} h_j \wedge \omega \wedge R \right) \left(\int_X |z| \, \omega^2 \wedge R \right).$$

The second factor tends uniformly to zero. The first is uniformly bounded. For completeness, if p is any of the uniformly bounded potentials and S any positive mixed product of degree $n-1$ occurring here, then

$$\int_X \mathrm{d}p \wedge \mathrm{d^c} p \wedge S = \frac{1}{2} \int_X p^2 \mathrm{dd^c} S - \int_X p \, \mathrm{dd^c} p \wedge S.$$

Expand $\mathrm{dd^c} S$ with all its torsion terms as above, and write $\mathrm{dd^c} p = \omega_p - \omega$. Boundedness of p and Lemma 1.8.3 bound both integrals uniformly. Finally, $\mathrm{d}h_j \wedge \mathrm{d^c} h_j \leq 2\mathrm{d}\alpha_j \wedge \mathrm{d^c}\alpha_j + 2\mathrm{d}a_j \wedge \mathrm{d^c} a_j$ as positive quadratic forms. This proves the asserted bound. The other gradient term has the same estimate. In dimension one $T = 1$ and all torsion terms in (1.28) vanish.

These bounded integration-by-parts and energy identities may first be written for smooth decreasing regularizations in the exact Hermitian background. The local Bedford–Taylor chain rule, its gradient products and quasi-continuity give the displayed identities for bounded potentials. Equivalently they follow from the bounded current calculus of Section 1.7. Only the common bound on the potentials enters the estimates.

We have shown that the difference of the two h_j integrals tends uniformly to zero when j, ℓ are large. For a fixed sufficiently large ℓ, apply Lemma 1.23.1 separately to $\alpha_j \to a$ and $a_j \to a$ in $L^1(X)$. The triangle inequality gives

$$\int_X h_j \, \omega_{w_\ell}^k \wedge \omega^{n-k} \longrightarrow 0.$$

Then let $\ell \to \infty$. Together with the estimate for $\alpha_j - a$, this proves the result. If the a_j have a larger fixed lower curvature bound, enlarge ω to Ω so that they are Ω-plurisubharmonic. Capacity convergence is unchanged by this replacement, by Section 1.8. Apply the proved statement with Ω; positivity bounds the original mixed measures by their Ω counterparts. □

1.23.4 Smoothing the measure

Coordinate convolution gives smooth approximations of a finite measure. For a varying quasi-plurisubharmonic integrand, the submean inequality transfers the

error to the original measure. This is the additional control needed beyond weak convergence of the measures themselves.

For a smooth background θ with $\mathrm{PSH}(X,\theta) \neq \varnothing$ and a non-pluripolar probability measure ν, put

$$A_\theta(\nu) = \sup\left\{\int_X (-w)\,\mathrm{d}\nu : w \in \mathrm{PSH}(X,\theta),\ \sup_X w = 0\right\}.$$

This number may be infinite. Bounded Bedford–Taylor measures satisfy the following estimate.

Lemma 1.23.4 *Let ω be Hermitian, let $\psi \in \mathrm{PSH}(X,\omega)$ be bounded, and put $\nu = (\omega + \mathrm{dd^c}\psi)^n$. There is a finite constant C, depending on the fixed backgrounds and $\|\psi\|_\infty$, such that*

$$\int_X (-w)\,\mathrm{d}\nu \le C \qquad (w \in \mathrm{PSH}(X,\theta),\ \sup_X w = 0).$$

Proof On finitely many nested coordinate balls $B'_a \Subset B_a$ covering X, choose fixed smooth corrections q_a, r_a with $\mathrm{dd^c} q_a \ge \omega$ and $\mathrm{dd^c} r_a \ge \theta$, and shift r_a to be nonpositive. Then $\psi + q_a$ is bounded plurisubharmonic, $w + r_a$ is negative plurisubharmonic, and positivity of bounded products gives

$$\int_{B'_a} (-w)\,\mathrm{d}\nu \le \int_{B'_a} (-w - r_a)\big(\mathrm{dd^c}(\psi + q_a)\big)^n \le C_a\big(1 + \|w\|_{L^1(B_a)}\big).$$

The last step is the weighted Chern–Levine–Nirenberg estimate in Section 1.6. Sum over the cover and use the normalized L^1 bound of Theorem 1.4.1. □

Choose finitely many coordinate balls B_a, smooth functions $\eta_a \ge 0$ with compact support in these balls and $\sum_a \eta_a = 1$, and inner neighborhoods of their supports whose closures lie in B_a. Let $\chi \ge 0$ be a smooth radial convolution kernel on the unit ball of $\mathbb{C}^n$, of integral one. Fix $\epsilon_j \downarrow 0$ small enough for every convolution below to stay in the corresponding coordinate ball. Denote convolution of a function in that chart by $S_{a,j}$. For a Borel test function F and a probability measure ν, define a smooth probability measure ν_j by convolving each coordinate measure $\eta_a \nu$ and adding its extensions by zero. Its defining dual identity is

$$\int_X F\,\mathrm{d}\nu_j = \sum_a \int_X \eta_a S_{a,j} F\,\mathrm{d}\nu. \tag{1.29}$$

The kernel is even, so the same convolution occurs on the right. This identity first holds for nonnegative Borel functions, and then for integrable signed functions. In particular $\nu_j \rightharpoonup \nu$.

Lemma 1.23.5 *If ν is non-pluripolar and $A_\theta(\nu) < \infty$, then*

$$\sup_j A_\theta(\nu_j) < \infty.$$

More generally, let z_j, z be quasi-plurisubharmonic functions with a common upper bound and a common smooth lower curvature bound. Suppose they converge both in $L^1(X)$ and in $L^1(\nu)$. Then

$$\int_X |z_j - z| \, \mathrm{d}\nu_j \longrightarrow 0. \tag{1.30}$$

Consequently, for every $\lambda > 0$,

$$\mathrm{e}^{\lambda z_j} \nu_j \rightharpoonup \mathrm{e}^{\lambda z} \nu.$$

Proof On each coordinate ball choose a fixed smooth function q_a whose Levi form dominates θ. If $w \in \mathrm{PSH}(X, \theta)$ and $\sup_X w = 0$, then $w + q_a$ is plurisubharmonic. The submean inequality gives

$$-S_{a,j} w \le -w + S_{a,j} q_a - q_a \le -w + C.$$

The constant is independent of w and j. Integrating (1.29) proves $A_\theta(\nu_j) \le A_\theta(\nu) + C$.

For the second assertion the tail envelopes

$$H_j = \left(\sup_{k \ge j} z_k \right)^*$$

decrease to z by Lemma 1.4.1. Enlarge the fixed corrections q_a so that $H_j + q_a, z_j + q_a, z + q_a$ are all plurisubharmonic. Since $H_j \ge z_j, z$, the dual identity and the submean inequality imply

$$\begin{aligned} \int_X |z_j - z| \, \mathrm{d}\nu_j &\le \sum_a \int_X \eta_a S_{a,j}(2H_j - z_j - z) \, \mathrm{d}\nu \\ &\le 2 \sum_a \int_X \eta_a \big(S_{a,j}(H_j + q_a) - (z + q_a) \big) \, \mathrm{d}\nu + \int_X (z - z_j) \, \mathrm{d}\nu. \end{aligned}$$

All integrals are well defined: each function on the left of a submean inequality is bounded above, and its negative part is controlled by that of its integrable unsmoothed counterpart plus a constant.

Radial averages of a plurisubharmonic function decrease when the radius decreases. Therefore $S_{a,j}(H_j + q_a)$ decreases to $z + q_a$ on the support of η_a. To identify the limit, fix an index k, bound $H_j \le H_k$ for $j \ge k$, let $j \to \infty$ in the radius, and then let $k \to \infty$. The reverse inequality is the submean inequality. The nonnegative difference in the last display is dominated by an integrable constant minus z on this compact

support. Thus its integral tends to zero. The final integral tends to zero by $L^1(\nu)$ convergence. This proves (1.30).

The exponential is Lipschitz on the common upper range, so its moving L^1 error also tends to zero. If $\zeta > 0$ is smooth, then $\zeta e^{\lambda z} = e^{\lambda z + \log \zeta}$ is bounded and quasi-plurisubharmonic. For any fixed bounded quasi-plurisubharmonic function F, the same local submean argument and dominated convergence in (1.29) give

$$\int_X F \, d\nu_j \longrightarrow \int_X F \, d\nu.$$

Apply this to $F = \zeta e^{\lambda z}$. Subtracting two positive smooth tests proves the required convergence for every smooth test function. Uniformly bounded masses and uniform approximation of continuous functions by smooth ones prove weak convergence. □

1.24 Curvature of line bundles

Let M be a complex manifold and let θ be a smooth real $(1, 1)$-form on M. To connect potentials with holomorphic sections, we ask when a current $\theta + dd^c u$, with u locally integrable, can be the curvature of a line-bundle metric. The transition law for its local weights forces a closedness condition on the background.

Proposition 1.24.1 *The curvature current of a singular Hermitian metric with locally integrable weights on a holomorphic line bundle is closed. In particular, a non-closed form θ cannot itself be the curvature of such a metric, nor can $\theta + dd^c u$ be closed at a point where $d\theta \neq 0$.*

Proof Let h be the singular Hermitian metric on the holomorphic line bundle $L \to M$ in the statement. Choose a local holomorphic frame e of L and write its squared norm as $|e|_h^2 = e^{-\varphi}$. The curvature current is locally $dd^c\varphi$. If $e' = ge$ is another frame, with g holomorphic and nowhere zero, the new weight is $\varphi - \log |g|^2$. Since $\log |g|^2$ is pluriharmonic, these local currents glue. Each is closed because $ddd^c = 0$. The final assertion follows from

$$d(\theta + dd^c u) = d\theta$$

in the sense of currents. □

When studying sections with multiplier-ideal conditions, we shall therefore specify both the line bundle with its closed curvature and any auxiliary Hermitian form on M. An algebraic interpretation then requires an argument relating the line bundle's invariants to that analytic background.

Chapter 2
Torsion and the variation of mass

Let X be a connected compact complex manifold of dimension $n \geq 1$, and fix a Hermitian form ω on X.

The total Monge–Ampère mass need not be preserved by adding a complex Hessian to a Hermitian form. Integration by parts expresses its variation through derivatives of the background powers; polarization shows that the vanishing of these derivatives is also necessary for invariance. When they do not vanish, even smooth potentials of the same singularity type can have different masses, and the ordinary global comparison principle fails. Uniform upper bounds then provide a weaker property that depends only on the manifold.

2.1 Integration by parts with a non-closed form

Integration by parts transfers the complex Hessian of a function to the form against which it is integrated. For powers of a nonclosed background, the resulting derivative contains both its complex Hessian and a quadratic torsion term.

Lemma 2.1.1 *For a smooth real function f and a smooth form T of bidegree $(n-1, n-1)$,*

$$\int_X \mathrm{dd^c} f \wedge T = \int_X f \mathrm{dd^c} T.$$

Proof Stokes' theorem applied to $\mathrm{d}(\mathrm{d^c} f \wedge T)$ gives the first equality in

$$\int_X \mathrm{dd^c} f \wedge T = \int_X \mathrm{d^c} f \wedge \mathrm{d}T = \int_X f \mathrm{dd^c} T.$$

For the second, apply Stokes' theorem for $\mathrm{d^c}$ to $f\,\mathrm{d}T$ and use $\mathrm{d^c d} = -\mathrm{dd^c}$. The integral of $\mathrm{d^c}$ on forms of total degree $2n-1$ vanishes by Stokes' theorem for ∂ and $\bar{\partial}$. □

Lemma 2.1.2 *For a smooth real* $(1,1)$*-form* θ *and an integer* $q \geq 1$*,*

$$\mathrm{dd^c}(\theta^q) = q\,\theta^{q-1} \wedge \mathrm{dd^c}\theta + q(q-1)\,\theta^{q-2} \wedge \mathrm{d}\theta \wedge \mathrm{d^c}\theta,$$

where the second term is omitted when $q = 1$*.*

Proof For $q \geq 2$, since θ has even degree, $\mathrm{d^c}(\theta^q) = q\,\theta^{q-1} \wedge \mathrm{d^c}\theta$. Applying d and again using the even degree of θ^{q-1} gives

$$\mathrm{dd^c}(\theta^q) = q(q-1)\,\theta^{q-2} \wedge \mathrm{d}\theta \wedge \mathrm{d^c}\theta + q\,\theta^{q-1} \wedge \mathrm{dd^c}\theta.$$

For the Hermitian form ω, the term $\mathrm{d}\omega \wedge \mathrm{d^c}\omega$ can contribute to derivatives of higher powers even when $\mathrm{dd^c}\omega = 0$. The mass expansion below involves these powers separately.

2.2 The mass polynomial

For a smooth real function u on X, write $\omega_u = \omega + \mathrm{dd^c}u$ and set

$$\mathcal{H}_\omega = \{u \in C^\infty(X,\mathbb{R}) : \omega + \mathrm{dd^c}u > 0\}, \qquad \mathrm{M}_\omega(u) = \int_X (\omega + \mathrm{dd^c}u)^n.$$

The set $\mathcal{H}_\omega$ is convex and open in the C^2 topology. Expanding the mass along a smooth direction gives a polynomial whose nonconstant coefficients detect the derivatives of the background.

Theorem 2.2.1 *For every smooth real function u,*

$$\mathrm{M}_\omega(u) = \int_X \omega^n + \sum_{k=1}^{n-1} \binom{n}{k} \int_X u(\mathrm{dd^c}u)^{k-1} \wedge \mathrm{dd^c}(\omega^{n-k}). \tag{2.1}$$

Proof Expand $(\omega + \mathrm{dd^c}u)^n$ by the binomial theorem. The term with $k = n$ integrates to zero, since

$$(\mathrm{dd^c}u)^n = \mathrm{d}\big(\mathrm{d^c}u \wedge (\mathrm{dd^c}u)^{n-1}\big).$$

For $1 \leq k \leq n-1$, apply Lemma 2.1.1 with $T = (\mathrm{dd^c}u)^{k-1} \wedge \omega^{n-k}$. Both d and $\mathrm{d^c}$ annihilate $\mathrm{dd^c}u$, so

$$\mathrm{dd^c}T = (\mathrm{dd^c}u)^{k-1} \wedge \mathrm{dd^c}(\omega^{n-k}).$$

This gives each of the terms in (2.1). When $n = 1$, the sum is empty and the mass is constant. □

If every $\mathrm{dd^c}(\omega^j)$ in (2.1) vanishes, the mass is constant. To prove necessity, vary the potential in several directions and recover each of these forms by polarization.

Theorem 2.2.2 (Chiose) *Assume $n \geq 2$. The following conditions are equivalent:*

(1) $\int_X (\omega + \mathrm{dd^c} u)^n$ *is independent of* $u \in \mathcal{H}_\omega$*;*
(2) $\mathrm{dd^c}(\omega^j) = 0$ *for* $1 \leq j \leq n-1$*;*
(3) $\mathrm{dd^c}\omega = 0$ *and* $\mathrm{d}\omega \wedge \mathrm{d^c}\omega = 0$.

They are also equivalent to constancy of the mass over all bounded ω-plurisubharmonic functions. Under these conditions the common mass is $\int_X \omega^n$.

Proof Assume the first condition. For every smooth real u, both signs of tu are admissible when $|t|$ is small. The polynomial $\int_X (\omega + t\mathrm{dd^c} u)^n$ is therefore constant on an interval. Its nonconstant coefficients vanish. For $1 \leq j \leq n-1$ this gives

$$\int_X \omega^j \wedge (\mathrm{dd^c} u)^{n-j} = 0.$$

Replace u by a linear combination of $n-j$ smooth real functions and compare the coefficient containing every parameter once. We obtain

$$\int_X \omega^j \wedge \mathrm{dd^c} u_1 \wedge \cdots \wedge \mathrm{dd^c} u_{n-j} = 0.$$

Integration by parts in the first factor, followed by testing arbitrary u_1, yields the pointwise identity

$$\mathrm{dd^c}(\omega^j) \wedge \mathrm{dd^c} u_2 \wedge \cdots \wedge \mathrm{dd^c} u_{n-j} = 0$$

for every choice of the remaining functions. At a given point their complex Hessians can be prescribed arbitrarily: multiply a quadratic polynomial in a coordinate chart by a cutoff equal to one near that point. Complexifying the multilinear identity permits the quadratic monomials $z_a \bar{z}_b$. Products of their Hessians give all coordinate forms of complementary bidegree. Pairing with those forms shows that every coefficient of $\mathrm{dd^c}(\omega^j)$ vanishes. When $j = n-1$, there are no remaining factors, and the same conclusion follows directly from testing u_1. This proves the second condition.

The second condition implies the first by Theorem 2.2.1. For $n \geq 3$, the identities for $j = 1, 2$ give

$$0 = \mathrm{dd^c}(\omega^2) = 2\omega \wedge \mathrm{dd^c}\omega + 2\mathrm{d}\omega \wedge \mathrm{d^c}\omega,$$

so the second condition implies the third. For $n = 2$, the last torsion form vanishes for degree reasons, and the implication is immediate. Conversely Lemma 2.1.2 proves all the power identities from the third condition.

For a bounded potential, Theorem 1.11.1 gives uniformly bounded smooth strictly positive approximants decreasing to it. The bounded-product continuity of Section 1.7 passes their common mass to the limit. Conversely, constancy for bounded potentials implies constancy on $\mathcal{H}_\omega$. The zero potential gives the value of the common mass.□

In particular, a closed background has invariant mass. More generally, under the equivalent conditions above, (2.1) gives the same identity for every smooth real u. In dimension one the mass is always $\int_X \omega$.

The first derivative of the mass singles out the highest power of the perturbed background.

Proposition 2.2.1 *For smooth real functions u, v on X,*

$$\frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} \mathrm{M}_\omega(u+tv) = n\int_X v\mathrm{dd^c}(\omega_u^{n-1}). \tag{2.2}$$

If M_ω is constant on $\mathcal{H}_\omega$, then $\mathrm{dd^c}(\omega_u^{n-1}) = 0$ for every $u \in \mathcal{H}_\omega$.

Proof Differentiating the finite polynomial gives

$$n\int_X \mathrm{dd^c}v \wedge \omega_u^{n-1}.$$

Apply Lemma 2.1.1. For the second assertion, every smooth direction v gives a line segment in $\mathcal{H}_\omega$ for sufficiently small $|t|$, so the right side vanishes for all test functions v. The smooth top-degree form $\mathrm{dd^c}(\omega_u^{n-1})$ must therefore vanish. □

Corollary 2.2.1 *If $n = 2$, then for every smooth real function u on X,*

$$\mathrm{M}_\omega(u) = \int_X \omega^2 + 2\int_X u\mathrm{dd^c}\omega.$$

Consequently, M_ω is constant on $\mathcal{H}_\omega$ if and only if $\mathrm{dd^c}\omega = 0$.

Proof The formula is (2.1) for $n = 2$. It proves one implication directly; the other follows from Proposition 2.2.1 at $u = 0$. □

The identity $\mathrm{dd^c}(\omega^{n-1}) = 0$ makes the first variation vanish at $u = 0$; it will be studied in Chapter 4. On a surface this already gives mass invariance. In dimension three, however,

$$\mathrm{M}_\omega(u) = \int_X \omega^3 + 3\int_X u\mathrm{dd^c}(\omega^2) + 3\int_X u\mathrm{dd^c}u \wedge \mathrm{dd^c}\omega.$$

Vanishing of $\mathrm{dd^c}(\omega^2)$ removes the linear term, while $\mathrm{dd^c}\omega$ still governs the quadratic term. Thus stationarity at one potential must be distinguished from invariance on $\mathcal{H}_\omega$.

2.3 Mass variation on a complex torus

The surface formula suggests choosing a background whose complex Hessian is nonzero and a potential that pairs nontrivially with it. A product of elliptic curves gives an explicit example in which the mass varies along a family of smooth potentials.

Example 2.3.1 Let $X = (\mathbb{C}/(\mathbb{Z}+\mathrm{i}\mathbb{Z}))^2$. For $j = 1, 2$, let x_j, y_j be the real coordinates on its universal cover and put

$$z_j = x_j + \mathrm{i}y_j, \qquad \beta_j = \frac{\mathrm{i}}{2\pi}\,\mathrm{d}z_j \wedge \mathrm{d}\bar{z}_j.$$

The forms β_j descend to X. For a real parameter t, define

$$b(x_1) = 2 + \cos(2\pi x_1), \qquad \omega = \beta_1 + b(x_1)\beta_2, \qquad u_t = t\cos(2\pi x_1).$$

Since $1 \le b \le 3$, the form ω is Hermitian. It is not closed: $\mathrm{d}\omega = \mathrm{d}b \wedge \beta_2$ is not identically zero.

For a function f of x_1 alone, with derivatives taken in x_1, one has $\mathrm{dd^c} f = (f''/4)\beta_1$. Hence

$$\omega_{u_t} = \big(1 - \pi^2 t\cos(2\pi x_1)\big)\beta_1 + b\beta_2 > 0 \quad \text{if} \quad |t| < \pi^{-2}.$$

Writing $B = \int_X \beta_1 \wedge \beta_2 > 0$, direct expansion and integration give

$$\mathrm{M}_\omega(u_t) = (4 - \pi^2 t)B. \tag{2.3}$$

Indeed, the average of $\cos(2\pi x_1)$ is zero and that of its square is $1/2$, while

$$\omega_{u_t}^2 = 2\big(1 - \pi^2 t\cos(2\pi x_1)\big)\big(2 + \cos(2\pi x_1)\big)\beta_1 \wedge \beta_2.$$

The complex torus admits Kähler metrics, but the chosen equation form is nonclosed. Mass variation already occurs in this setting. The Hopf example in Chapter 8 will exhibit the further failure of an upper mass bound on a non-Kähler manifold.

Corollary 2.3.1 *In general, the total Monge–Ampère mass is not determined by the ordinary singularity type or by the multiplier ideals, even among smooth strictly ω-plurisubharmonic potentials.*

Proof All the potentials u_t in Example 2.3.1 are smooth on compact X, so they have the same singularity type. Their multiplier ideals are the structure sheaf for every positive multiple of the weight. Their masses differ by (2.3). □

2.4 Comparison and mass invariance

Adding a constant to a potential leaves its measure unchanged. Consequently, a comparison inequality on strict sublevel sets would compare total masses after a suitable constant shift. The varying-mass example therefore obstructs the usual global inequality.

Proposition 2.4.1 *Suppose that ω has the following property for every pair $u, v \in \mathcal{H}_\omega$:*

$$\int_{\{u<v\}} \omega_v^n \leq \int_{\{u<v\}} \omega_u^n. \tag{2.4}$$

Then M_ω is constant on $\mathcal{H}_\omega$.

Proof Fix $u, v \in \mathcal{H}_\omega$. By compactness choose a constant C such that $u < v + C$ everywhere. Since $\omega_{v+C} = \omega_v$, applying (2.4) to u and $v + C$ gives $\mathrm{M}_\omega(v) \leq \mathrm{M}_\omega(u)$. Exchange u and v and repeat the argument to obtain the opposite inequality. □

Corollary 2.4.1 *The inequality (2.4) is false for general compact Hermitian backgrounds.*

Proof Use the torus X, form ω, potentials u_t and constant $B = \int_X \beta_1 \wedge \beta_2$ of Example 2.3.1. For $-\pi^{-2} < t < 0$, set $u = 0$ and $v = u_t + C$ with $C > |t|$. Then $\{u < v\} = X$, but

$$\int_X \omega_v^2 = (4 - \pi^2 t)B > 4B = \int_X \omega_u^2.$$

The modified comparison theorem in Theorem 10.2.1 accounts for the derivatives of the background. Conversely, the mass identity in Theorem 2.2.2 yields ordinary comparison for bounded potentials, as proved in Theorem 11.5.2.

2.5 Uniform mass bounds

Masses can vary while remaining uniformly bounded. Comparing two Hermitian forms and rescaling their potentials shows that finiteness of the upper bound depends only on X. Define

$$\overline{\mathrm{v}}_{\mathrm{sm}}(\omega) = \sup_{u \in \mathcal{H}_\omega} \mathrm{M}_\omega(u) \in (0, +\infty].$$

Proposition 2.5.1 *If η is another Hermitian form and $\omega \leq C\eta$ for a constant $C > 0$, then*

$$\overline{\mathrm{v}}_{\mathrm{sm}}(\omega) \leq C^n \overline{\mathrm{v}}_{\mathrm{sm}}(\eta).$$

In particular, finiteness of $\overline{\mathrm{v}}_{\mathrm{sm}}$ is independent of the Hermitian form.

Proof For $u \in \mathcal{H}_\omega$, the function u/C belongs to $\mathcal{H}_\eta$, because $C\eta + \mathrm{dd^c}u$ is positive and dominates $\omega + \mathrm{dd^c}u$. For positive Hermitian matrices $A \leq B$, all eigenvalues of $A^{-1/2}BA^{-1/2}$ are at least one, so $\det A \leq \det B$. Applying this pointwise gives monotonicity of the top exterior power. Therefore

$$\int_X (\omega + \mathrm{dd^c}u)^n \leq C^n \int_X \big(\eta + \mathrm{dd^c}(u/C)\big)^n \leq C^n \overline{\mathrm{v}}_{\mathrm{sm}}(\eta).$$

Take the supremum over u. Any two Hermitian forms on compact X are comparable in both directions, proving the last assertion. □

Chapter 3
Monge–Ampère energy and torsion

Let X be a connected compact complex manifold of dimension $n \geq 1$, and fix a Hermitian form ω on X. We use ω_u and $\mathcal{H}_\omega$ as in Chapter 2.

An energy for the Monge–Ampère operator would have directional derivative $\int_X v\,\omega_u^n$ at $u \in \mathcal{H}_\omega$ in every smooth real direction v. Such a primitive exists exactly when ω is closed: symmetry of mixed derivatives first detects $\mathrm{d}(\omega_u^{n-1})$, and varying the potential then detects $\mathrm{d}\omega$. This is stronger than the mass-invariance condition of Chapter 2. Integrating the one-form along radial segments still defines a functional for a nonclosed background; its first variation displays the torsion terms that change the Euler equation.

3.1 Primitives of the Monge–Ampère one-form

On $\mathcal{H}_\omega$ consider the Monge–Ampère one-form

$$\alpha_u(v) = \int_X v\,\omega_u^n, \qquad v \in C^\infty(X, \mathbb{R}).$$

A primitive on a subset $U \subseteq \mathcal{H}_\omega$ open in the C^2 topology on smooth functions is a functional E satisfying $DE_u(v) = \alpha_u(v)$ for every $u \in U$ and every smooth real direction v. Restrict E to the intersection of U with a finite-dimensional affine plane. Its first derivatives depend smoothly on the parameters, so its mixed derivatives agree. The following integration-by-parts criterion identifies the differential condition imposed by that symmetry.

Lemma 3.1.1 *For a smooth real $(n-1, n-1)$-form Ω, the pairing $B(v, w) = \int_X w\mathrm{dd^c}v \wedge \Omega$ is symmetric on smooth real functions if and only if $\mathrm{d}\Omega = 0$.*

Proof By integration by parts and testing against every smooth real function v, symmetry is equivalent to the pointwise identity

$$\mathrm{dd^c}(w\Omega) - \mathrm{dd^c}w \wedge \Omega = w\mathrm{dd^c}\Omega + \mathrm{d}w \wedge \mathrm{d^c}\Omega - \mathrm{d^c}w \wedge \mathrm{d}\Omega = 0$$

for every smooth w. Taking $w = 1$ gives $\mathrm{dd^c}\Omega = 0$. The remaining expression is linear in the first jet of w, so we can complexify the test functions. At a chosen point take a cutoff holomorphic linear function with $\partial w = \lambda$ arbitrary and $\bar\partial w = 0$. The first-order expression is

$$\frac{\mathrm{i}}{2\pi}\big(\partial w \wedge \bar\partial\Omega - \bar\partial w \wedge \partial\Omega\big).$$

It follows that $\lambda \wedge \bar\partial\Omega = 0$ for every $(1,0)$-covector λ. The pairing with $(n-1,n)$-forms is nondegenerate, hence $\bar\partial\Omega = 0$. Reality gives $\partial\Omega = 0$ as well. Conversely, if $\mathrm{d}\Omega = 0$, its pure type implies $\mathrm{d^c}\Omega = 0$, and the same product formula proves symmetry. □

Lemma 3.1.2 *Let T be a form of type $(2,1)+(1,2)$ on a complex vector space of dimension $n \geq 2$. If $T \wedge A^{n-2} = 0$ for every real $(1,1)$-form A in an open neighborhood of a positive form, then $T = 0$.*

Proof For $n = 2$ the assertion is immediate. Otherwise the expression is polynomial in the real coefficients of A, so it vanishes for every real $(1,1)$-form. Polarization and then complexification give $T \wedge A_1 \wedge \cdots \wedge A_{n-2} = 0$ for arbitrary complex $(1,1)$-forms A_j. Such products span all forms of bidegree $(n-2, n-2)$.

The two type components of T vanish separately after these wedges. Wedge the identity for $T^{2,1}$ with arbitrary $(0,1)$-covectors. The resulting test forms span bidegree $(n-2, n-1)$, which pairs perfectly with bidegree $(2,1)$ to top degree. Thus $T^{2,1} = 0$. The other component is treated with $(1,0)$-covectors. □

Theorem 3.1.1 *For $n \geq 2$, the one-form α has a primitive on $\mathcal{H}_\omega$ if and only if $\mathrm{d}\omega = 0$. In fact, a primitive on a C^2-open neighborhood of 0 already forces $\mathrm{d}\omega = 0$.*

Proof Suppose a primitive E exists on such a neighborhood U. For $u \in U$ and smooth real directions v, w, differentiate $E(u + sv + tw)$ in both orders. Equality of its mixed derivatives, after division by n, gives

$$\int_X w\mathrm{dd^c}v \wedge \omega_u^{n-1} = \int_X v\mathrm{dd^c}w \wedge \omega_u^{n-1}. \tag{3.1}$$

By Lemma 3.1.1,

$$0 = \mathrm{d}(\omega_u^{n-1}) = (n-1)\mathrm{d}\omega \wedge (\omega + \mathrm{dd^c}u)^{n-2}.$$

Fix a point x. A real quadratic polynomial in local coordinates, multiplied by a cutoff equal to one near x, realizes any prescribed small real $(1,1)$-form $H = \mathrm{dd^c}u(x)$. These functions can be chosen linearly in H; their global C^2 norms are bounded by a fixed constant times the coefficient norm of H. Thus sufficiently small H are realized by potentials in U. The preceding identity holds with $\omega(x) + H$ ranging over an open set. Apply Lemma 3.1.2 with $T = \mathrm{d}\omega(x)$ to conclude that $\mathrm{d}\omega(x) = 0$.

Since x was arbitrary, $\mathrm{d}\omega = 0$. Conversely, Proposition 3.2.1 constructs a primitive for a closed background. □

In dimension one every real $(1,1)$-form is closed, and Proposition 3.2.1 always supplies a primitive. For $n \geq 2$, the symmetry condition at a single potential is $\mathrm{d}(\omega_u^{n-1}) = 0$; the theorem uses it on an open set. Mass invariance alone does not suffice. Indeed, on a compact complex surface admitting no Kähler metric, choose a Gauduchon form ω by Theorem 4.1.1. Its mass is constant by Theorem 2.2.2, but its Monge–Ampère one-form has no primitive.

3.2 Radial energy and closed backgrounds

Integrating along the segment from 0 to u gives a functional for an arbitrary background. Let θ be a smooth real $(1,1)$-form on X. For $u \in C^\infty(X,\mathbb{R})$, put $\theta_u = \theta + \mathrm{dd^c}u$ and define

$$\begin{aligned} E_\theta(u) &= \int_0^1 \int_X u(\theta + t\mathrm{dd^c}u)^n \,\mathrm{d}t \\ &= \frac{1}{n+1} \sum_{j=0}^{n} \int_X u\, \theta_u^j \wedge \theta^{n-j}. \end{aligned} \tag{3.2}$$

To obtain the second expression, write $\theta + t\mathrm{dd^c}u = (1-t)\theta + t\theta_u$, expand, and use

$$\binom{n}{j} \int_0^1 t^j (1-t)^{n-j} \,\mathrm{d}t = \frac{1}{n+1}.$$

When θ is closed, integration by parts makes the derivatives of the mixed terms telescope.

Proposition 3.2.1 *If* $\mathrm{d}\theta = 0$*, then* $DE_{\theta,u}(v) = \int_X v\theta_u^n$ *for every pair of smooth real functions* u, v*.*

Proof Differentiate the finite sum in (3.2). The term indexed by j contributes

$$\int_X v\theta_u^j \wedge \theta^{n-j} + j \int_X u\mathrm{dd^c}v \wedge \theta_u^{j-1} \wedge \theta^{n-j},$$

with the second summand absent for $j = 0$. For $j \geq 1$, the form $\theta_u^{j-1} \wedge \theta^{n-j}$ is closed. By Lemma 3.1.1, u and v may therefore be interchanged in the second summand. Using $\mathrm{dd^c}u = \theta_u - \theta$, the contribution for $j \geq 1$ becomes

$$(j+1) \int_X v\theta_u^j \wedge \theta^{n-j} - j \int_X v\theta_u^{j-1} \wedge \theta^{n-j+1}.$$

Together with the $j = 0$ term $\int_X v\theta^n$, these terms telescope to $(n+1)\int_X v\theta_u^n$. Division by $n+1$ completes the proof. □

Taking $\theta = \omega$ when $\mathrm{d}\omega = 0$, the restriction of E_ω to $\mathcal{H}_\omega$ is the primitive required in Theorem 3.1.1.

3.3 First variation in a nonclosed background

For the fixed Hermitian background $\theta = \omega$, the functional (3.2) integrates α along the segment from 0 to $u \in \mathcal{H}_\omega$. Its derivative differs from α by the terms that remain when integration by parts differentiates the background powers. Computing this difference gives the Euler equation even when no primitive exists.

Proposition 3.3.1 *For $u \in \mathcal{H}_\omega$ and $v \in C^\infty(X, \mathbb{R})$, put $\Omega_t = (\omega + t\mathrm{dd^c}u)^{n-1}$ for $0 \le t \le 1$. Then*

$$DE_{\omega,u}(v) - \int_X v\omega_u^n = n\int_0^1 t\int_X v\big(u\mathrm{dd^c}\Omega_t + \mathrm{d}u \wedge \mathrm{d^c}\Omega_t - \mathrm{d^c}u \wedge \mathrm{d}\Omega_t\big)\,\mathrm{d}t. \tag{3.3}$$

Proof Differentiate the radial integral in (3.2) and use

$$\int_X v\omega_u^n = \int_0^1 \frac{\mathrm{d}}{\mathrm{d}t}\left(t\int_X v(\omega + t\mathrm{dd^c}u)^n\right)\mathrm{d}t.$$

The terms without a factor nt cancel, leaving

$$\begin{aligned} DE_{\omega,u}(v) - \int_X v\omega_u^n &= n\int_0^1 t\int_X (u\mathrm{dd^c}v - v\mathrm{dd^c}u) \wedge \Omega_t\,\mathrm{d}t \\ &= n\int_0^1 t\int_X v\big(\mathrm{dd^c}(u\Omega_t) - \mathrm{dd^c}u \wedge \Omega_t\big)\,\mathrm{d}t. \end{aligned}$$

The second equality is Lemma 2.1.1. The product rule in the proof of Lemma 3.1.1 expands the last parenthesis and gives (3.3). When $n = 1$, one has $\Omega_t = 1$ and the defect vanishes. □

Proposition 3.3.2 *If $n = 2$, then every $u \in \mathcal{H}_\omega$ satisfies*

$$E_\omega(u) = \int_X u\omega^2 + \int_X u\omega \wedge \mathrm{dd^c}u + \frac{1}{3}\int_X u(\mathrm{dd^c}u)^2.$$

For every smooth real direction v, its differential satisfies

$$DE_{\omega,u}(v) - \int_X v\omega_u^2 = \int_X v\big(u\mathrm{dd^c}\omega + \mathrm{d}u \wedge \mathrm{d^c}\omega - \mathrm{d^c}u \wedge \mathrm{d}\omega\big). \tag{3.4}$$

Proof Expanding the radial integral gives the first identity. Here $\Omega_t = \omega + t\mathrm{dd^c}u$, whose d- and $\mathrm{d^c}$-derivatives are those of ω. Formula (3.3) and $2\int_0^1 t\,\mathrm{d}t = 1$ give the second identity. □

For a fixed smooth real $(2,2)$-form μ on X with $n = 2$, a critical point of $E_\omega(u) - \int_X u\,\mu$ in $\mathcal{H}_\omega$ therefore satisfies

$$\omega_u^2 + u\mathrm{dd^c}\omega + \mathrm{d}u \wedge \mathrm{d^c}\omega - \mathrm{d^c}u \wedge \mathrm{d}\omega = \mu.$$

Chapter 4
Gauduchon metrics and linear mass

Let X be a connected compact complex manifold of dimension $n \geq 2$, and fix a Hermitian form ω on X.

If θ is a smooth real $(1,1)$-form and u is θ-plurisubharmonic, pairing $\theta + \mathrm{dd^c} u$ with a smooth $\mathrm{dd^c}$-closed $(n-1, n-1)$-form depends only on θ. Gauduchon's conformal theorem supplies positive test forms of this kind, and their pairings detect whether a smooth background admits a positive current. The weak closure of the normalized tests turns the largest attainable lower curvature bound into a minimum. Applied to a solution metric, the same conformal construction gives the probability measure that characterizes the range of the linearized Monge–Ampère operator.

4.1 Conformal Gauduchon metrics

The condition needed for linear mass conservation can be achieved in every conformal class. Its construction reduces to finding a positive generator of the adjoint kernel of a scalar elliptic operator.

Definition 4.1.1 A Hermitian form γ is *Gauduchon* if

$$\mathrm{dd^c}(\gamma^{n-1}) = 0.$$

Theorem 4.1.1 (Gauduchon's theorem) *Every Hermitian form η has a conformal Gauduchon representative. More precisely, there is a smooth strictly positive function ρ, unique up to multiplication by a positive constant, such that*

$$\mathrm{dd^c}(\rho\eta^{n-1}) = 0.$$

The associated Gauduchon form is $\gamma = \rho^{1/(n-1)}\eta$.

Proof For a smooth real function v on X, define

$$Lv = n\frac{\mathrm{dd^c}v \wedge \eta^{n-1}}{\eta^n}.$$

This real second-order operator has positive definite principal symbol and no zeroth-order term. The maximum principle gives $\ker L = \mathbb{R}$. Its index is zero, since the linear path joining it to a Laplace operator is elliptic. The Fredholm theorem and elliptic regularity therefore give a one-dimensional kernel for its formal adjoint L^* with respect to η^n, generated by a nonzero smooth real function ρ.

We first prove that $\int_X \rho\eta^n \neq 0$. Otherwise the Fredholm alternative would solve $Lv = 1$, contrary to the maximum principle at a maximum of v. Rescale so that $\int_X \rho\eta^n = 1$.

Suppose that ρ is negative somewhere. Choose a nonzero smooth $h \geq 0$ supported where $\rho < 0$, and put $c = \int_X h\rho\eta^n < 0$. The function $h - c$ is strictly positive and orthogonal to the adjoint kernel. The Fredholm alternative would therefore solve $Lv = h - c > 0$, again impossible. Thus $\rho \geq 0$.

In real local coordinates $(x_1, \ldots, x_{2n})$, write $L^* = A^{ij}\partial_i\partial_j + B^i\partial_i + C$, where $\partial_i = \partial/\partial x_i$, the coefficients are smooth, and repeated indices are summed from 1 to $2n$. Its leading symbol is positive definite. Since

$$(L^* - \max(C, 0))\rho = -\max(C, 0)\rho \leq 0,$$

the strong minimum principle for an operator with nonpositive zeroth-order term shows that a zero of ρ would force ρ to vanish identically on the connected manifold. Hence $\rho > 0$.

Finally, integration by parts gives

$$\int_X \rho Lv\,\eta^n = n\int_X v\,\mathrm{dd^c}(\rho\eta^{n-1}).$$

Thus $L^*\rho = 0$ is precisely the required equation. The one-dimensional adjoint kernel proves uniqueness up to positive scaling. □

The Kähler condition $\mathrm{d}\gamma = 0$ implies the balanced condition $\mathrm{d}(\gamma^{n-1}) = 0$, which in turn implies that γ is Gauduchon. On surfaces the first two conditions coincide.

4.2 Conservation of the trace mass

The power γ^{n-1} of a Gauduchon metric annihilates every $\mathrm{dd^c}$-potential under integration. It therefore measures a positive current by a number depending only on its smooth background, including when the potential is unbounded.

Proposition 4.2.1 *Let θ be a smooth real $(1,1)$-form, let u be a θ-plurisubharmonic function, and let γ be a Gauduchon metric. Then*

$$\int_X (\theta + \mathrm{dd^c} u) \wedge \gamma^{n-1} = \int_X \theta \wedge \gamma^{n-1}. \tag{4.1}$$

Proof Distributional integration by parts gives

$$\langle \mathrm{dd^c} u, \gamma^{n-1} \rangle = \langle u, \mathrm{dd^c}(\gamma^{n-1}) \rangle = 0.$$

The product on the left of (4.1) is a positive measure, whose mass is the finite number on the right. □

Here $\theta + \mathrm{dd^c} u$ is the full distributional current, including any mass on the pole set of u. The non-pluripolar construction will discard this pole mass; Example 7.5.1 illustrates the difference on a curve.

Corollary 4.2.1 *There are constants $0 < a \leq b < +\infty$, depending only on ω and a chosen Gauduchon metric, such that every $u \in \mathrm{PSH}(X, \omega)$ satisfies*

$$a \leq \int_X (\omega + \mathrm{dd^c} u) \wedge \omega^{n-1} \leq b.$$

Proof Fix a Gauduchon form γ on X and choose positive constants c, C with $c\gamma^{n-1} \leq \omega^{n-1} \leq C\gamma^{n-1}$. Wedge these inequalities with the positive current $\omega + \mathrm{dd^c} u$ and integrate. By Proposition 4.2.1, the resulting bounds are cA and CA, where $A = \int_X \omega \wedge \gamma^{n-1} > 0$. □

The trace bound gives control of positive currents in the weak topology. By itself, it gives no bound for the Monge–Ampère mass: a bound on the integral of a trace does not bound the integral of its nth power.

The pointwise arithmetic–geometric mean inequality relates this conserved trace to the density in a smooth Monge–Ampère equation. Fix a smooth positive volume form $\mathrm{d}V$.

Proposition 4.2.2 *For any two Hermitian forms η, γ,*

$$\eta \wedge \gamma^{n-1} \geq \left(\frac{\eta^n}{\gamma^n}\right)^{1/n} \gamma^n \tag{4.2}$$

pointwise. Consequently, let θ be a smooth real $(1,1)$-form and f a smooth strictly positive function. If $u \in C^\infty(X, \mathbb{R})$ and $c > 0$ satisfy $(\theta + \mathrm{dd^c}u)^n = cf\,\mathrm{d}V$ with $\theta + \mathrm{dd^c}u > 0$, then

$$c^{1/n} \int_X \left(\frac{f\,\mathrm{d}V}{\gamma^n}\right)^{1/n} \gamma^n \leq \int_X \theta \wedge \gamma^{n-1} \tag{4.3}$$

for every Gauduchon metric γ.

Proof In a frame where γ is the identity, let $\lambda_1, \ldots, \lambda_n > 0$ be the eigenvalues of η. The two ratios in (4.2) are respectively $n^{-1} \sum_j \lambda_j$ and $(\prod_j \lambda_j)^{1/n}$. Their comparison is the arithmetic–geometric mean inequality.

Apply it with $\eta = \theta + \mathrm{dd^c}u$, integrate, and use (4.1). □

The corresponding density inequality for unbounded potentials in a Hermitian background is proved in Lemma 6.4.2.

4.3 The Gauduchon criterion for positivity

The trace identity gives a necessary condition for $\mathrm{PSH}(X, \theta)$ to be nonempty: every Gauduchon degree of θ must be nonnegative. It is also sufficient. Separation produces a positive current with these prescribed pairings, and a scalar elliptic equation shows that its difference from θ is $\mathrm{dd^c}$-exact.

Theorem 4.3.1 (The generalized Lamari criterion) *For a smooth real $(1,1)$-form θ, the following conditions are equivalent:*

(1) *There exists $u \in \mathrm{PSH}(X, \theta)$.*
(2) $\int_X \theta \wedge \gamma^{n-1} \geq 0$ *for every Gauduchon metric γ.*

To apply separation, we express the test in terms of smooth strictly positive $\mathrm{dd^c}$-closed $(n-1, n-1)$-forms Γ. Every strictly positive $(n-1, n-1)$-form has a unique strictly positive $(1,1)$-root $\Gamma = \gamma^{n-1}$. In a diagonal frame, after using the same wedge normalization for both sides, the coefficients g_i of Γ are the products $\prod_{j\neq i} \lambda_j$ of the eigenvalues of γ. Hence

$$\lambda_i = \frac{(\prod_j g_j)^{1/(n-1)}}{g_i}.$$

The spectral formula uses only a positive determinant and a matrix inverse, so the root depends smoothly on the coefficients, including at repeated eigenvalues. Pointwise uniqueness makes the local roots agree on overlaps. Thus $\mathrm{dd^c}\Gamma = 0$ says exactly that this root is Gauduchon.

Proof (of Theorem 4.3.1) Necessity is Proposition 4.2.1. For sufficiency, let F be the real vector space of smooth real $(n-1, n-1)$-forms and let

$$E = \{\Gamma \in F : \mathrm{dd^c}\Gamma = 0\}, \qquad \ell(\Gamma) = \int_X \theta \wedge \Gamma \quad (\Gamma \in E).$$

Fix a strictly positive $\Gamma_0 \in E$, supplied by Theorem 4.1.1. The hypothesis says that ℓ is nonnegative on strictly positive elements of E. By adding $\epsilon\Gamma_0$ and letting $\epsilon \downarrow 0$, the same holds on semipositive elements of E.

For $\Phi \in F$ put

$$p(\Phi) = \inf\{\ell(\Gamma) : \Gamma \in E,\ \Gamma \geq \Phi\}.$$

This is a finite sublinear function. To see finiteness, choose C with $-C\Gamma_0 \leq \Phi \leq C\Gamma_0$. The admissible element $C\Gamma_0$ gives an upper bound, while any admissible Γ satisfies $\ell(\Gamma) \geq -C\ell(\Gamma_0)$. Addition and positive scaling of competitors prove sublinearity. For $\Phi \in E$, positivity of $\ell(\Gamma - \Phi)$ also shows that $p(\Phi) = \ell(\Phi)$.

Hahn–Banach gives a real linear extension T of ℓ to F with $T(\Phi) \leq p(\Phi)$. If $\Phi \geq 0$, the zero form is admissible for $p(-\Phi)$, so $T(\Phi) \geq 0$. Moreover, the bounds by $C\Gamma_0$ show that T is continuous for the uniform norm. It is therefore a positive $(1,1)$-current. The current $S = T - \theta$ annihilates E.

To recover a potential from S, consider the scalar operator

$$Dv = \frac{\mathrm{dd^c}(v\Gamma_0)}{\mathrm{d}V}.$$

This is an elliptic scalar operator with no zeroth-order term, since $\mathrm{dd^c}\Gamma_0 = 0$. Its kernel consists of the constants and its index is zero. Stokes' theorem gives $\int_X Dv\,\mathrm{d}V = 0$. Its range is consequently exactly the smooth functions of integral zero. The inverse on functions normalized by $\int_X v\,\mathrm{d}V = 0$ is continuous in the smooth topology, by elliptic estimates.

For a smooth top-degree form ζ with $\int_X \zeta = 0$, solve $Dv = \zeta/\mathrm{d}V$ with this normalization and set $R\zeta = v\Gamma_0$. Then $\mathrm{dd^c}R\zeta = \zeta$ and R is continuous. Define a distribution on such top-degree forms by

$$\langle u, \zeta\rangle = S(R\zeta),$$

and extend it to all top-degree forms by assigning any value on one fixed volume form of integral one. For every $\Phi \in F$, the form $\Phi - R(\mathrm{dd^c}\Phi)$ belongs to E. It follows that

$$\langle \mathrm{dd^c}u, \Phi\rangle = S(R(\mathrm{dd^c}\Phi)) = S(\Phi).$$

Thus $\theta + \mathrm{dd^c}u = T \geq 0$, and Proposition 1.1.1 gives a representative in $\mathrm{PSH}(X, \theta)$.□

Applying the criterion to $\theta - a\omega$, for $a \in \mathbb{R}$, determines the largest lower curvature bound attainable in the background θ.

Corollary 4.3.1 *For a smooth real* $(1,1)$*-form* θ*, set*

$$\kappa_\omega(\theta) = \inf_{\gamma \text{ Gauduchon}} \frac{\int_X \theta \wedge \gamma^{n-1}}{\int_X \omega \wedge \gamma^{n-1}}.$$

This is a finite real number and equals the largest real a *for which there is a quasi-plurisubharmonic* u *satisfying*

$$\theta + \mathrm{dd^c}u \geq a\omega.$$

The largest value is attained. In particular, θ *is big precisely when* $\kappa_\omega(\theta) > 0$*.*

Proof Smoothness gives $-C\omega \leq \theta \leq C\omega$, so the infimum lies in $[-C, C]$. The trace identity shows that any admissible a is at most every displayed ratio. Conversely, with $a = \kappa_\omega(\theta)$, all Gauduchon integrals of $\theta - a\omega$ are nonnegative. Apply Theorem 4.3.1 to this smooth background. The last assertion is the definition of bigness as existence of a strictly positive representative. □

Vanishing of one Gauduchon degree forces a positive current to vanish.

Lemma 4.3.1 *Let* θ *be a smooth real* $(1,1)$*-form with* $\mathrm{PSH}(X, \theta) \neq \varnothing$*. If*

$$\int_X \theta \wedge \gamma^{n-1} = 0$$

for one Gauduchon metric γ*, then* $\theta = \mathrm{dd^c} f$ *for a smooth real function* f*. Conversely, a smoothly* $\mathrm{dd^c}$*-exact form has zero degree against every Gauduchon metric. The conclusion also holds for* $n = 1$*, with* $\gamma^{n-1} = 1$*.*

Proof Choose $u \in \mathrm{PSH}(X, \theta)$ and put $S = \theta + \mathrm{dd^c} u \geq 0$. Stokes' theorem gives $\int_X S \wedge \gamma^{n-1} = 0$. Since the trace measure controls all coefficient measures of S, one has $S = 0$. Taking the trace of $\mathrm{dd^c} u = -\theta$ with respect to ω gives an elliptic equation with smooth right-hand side. Elliptic regularity makes u smooth, so $f = -u$ works. The converse is Stokes' theorem. For $n = 1$, the trace measure is S itself and the same argument applies. □

For a smooth real $(1, 1)$-form θ, put $a = \kappa_\omega(\theta)$. By Corollary 4.3.1, $\theta - a\omega$ admits a positive current. Hence Lemma 4.3.1 shows that the infimum defining $\kappa_\omega(\theta)$ is attained by a smooth Gauduchon metric if and only if

$$\theta - a\omega = \mathrm{dd^c} f \quad \text{for some } f \in C^\infty(X, \mathbb{R}).$$

In that case every Gauduchon metric attains it. The positive current at the threshold always exists; a smooth minimizing test is much more restrictive.

4.4 Compactness and duality of Gauduchon currents

The weak closure of normalized smooth Gauduchon forms supplies a minimizing test even when no smooth form attains the threshold. The normalization gives compactness by bounding the coefficient measures. Let $\mathscr{G}_\omega$ be the set of positive $(n-1, n-1)$-currents T on X satisfying

$$\mathrm{dd^c} T = 0, \qquad \int_X \omega \wedge T = 1.$$

These are the normalized Gauduchon currents. Let $\mathscr{C}_\omega$ be the weak closure of the smooth strictly positive members of $\mathscr{G}_\omega$. The root calculation for Theorem 4.3.1 identifies those members with normalized powers of Gauduchon metrics.

Proposition 4.4.1 *The sets $\mathscr{G}_\omega$ and $\mathscr{C}_\omega$ are nonempty, compact and convex. For every smooth real $(1, 1)$-form θ,*

$$\kappa_\omega(\theta) = \min_{T \in \mathscr{C}_\omega} \int_X \theta \wedge T. \tag{4.4}$$

Moreover, for $T \in \mathscr{G}_\omega$, membership in $\mathscr{C}_\omega$ is equivalent to

$$\int_X \theta \wedge T \geq 0 \quad \textit{for every smooth real } \theta \textit{ with } \mathrm{PSH}(X, \theta) \neq \varnothing. \tag{4.5}$$

Proof Gauduchon's theorem supplies a smooth positive member after normalization. The fixed trace mass bounds the total variation of all coefficient measures in finitely many local frames. Weak compactness of finite measures therefore gives compactness of the family of positive currents with this trace. Positivity, the trace normalization and $\mathrm{dd^c} T = 0$ pass to limits. On this bounded set, testing against smooth forms or against

continuous coefficient forms gives the same topology, by uniform approximation. It is metrizable since the continuous test space is separable. Thus the compactness can also be used sequentially. The defining conditions of $\mathscr{G}_\omega$ are convex. Its smooth strictly positive members form a convex set, so their closure $\mathscr{C}_\omega$ is compact and convex as well.

Pairing with a fixed smooth θ is continuous. Its minimum on the closure of the smooth tests is therefore their infimum, proving (4.4). The Lamari criterion shows that (4.5) holds for every smooth positive test, and hence for every member of its weak closure.

Conversely, if $T \notin \mathscr{C}_\omega$, separate this point from the compact convex set $\mathscr{C}_\omega$ in the weak space of currents. A continuous linear functional for that weak topology is evaluation against a smooth form. After changing its sign if necessary, there is a smooth real $(1,1)$-form β such that

$$\int_X \beta \wedge T < \min_{R \in \mathscr{C}_\omega} \int_X \beta \wedge R = \kappa_\omega(\beta).$$

By Corollary 4.3.1, the smooth form $\beta - \kappa_\omega(\beta)\omega$ admits a positive current. Its pairing with T is negative because $\int_X \omega \wedge T = 1$. This contradicts (4.5) and proves the converse. □

Corollary 4.4.1 *The function κ_ω is concave, positively homogeneous and invariant under addition of $\mathrm{dd^c} f$ for $f \in C^\infty(X, \mathbb{R})$. Let θ and η be smooth real $(1,1)$-forms and let $\varepsilon \geq 0$. If $-\varepsilon\omega \leq \theta - \eta \leq \varepsilon\omega$, then*

$$|\kappa_\omega(\theta) - \kappa_\omega(\eta)| \leq \varepsilon.$$

If smooth real $(1,1)$-forms θ_j converge uniformly to a smooth form θ and $T_j \in \mathscr{C}_\omega$ minimize their pairings, every weak cluster point of (T_j) minimizes the pairing with θ.

Proof Formula (4.4) is an infimum of linear functionals, which proves concavity and positive homogeneity. Gauge invariance follows from $\mathrm{dd^c} T = 0$. The normalized trace gives

$$\left| \int_X (\theta - \eta) \wedge T \right| \leq \varepsilon \quad (T \in \mathscr{C}_\omega),$$

and taking infima proves the estimate. For the last assertion, pass to a subsequence $T_j \rightharpoonup T \in \mathscr{C}_\omega$. The estimate and uniform convergence of the backgrounds give

$$\int_X \theta \wedge T = \lim_j \int_X \theta_j \wedge T_j = \lim_j \kappa_\omega(\theta_j) = \kappa_\omega(\theta).$$

Thus T is a minimizer. □

The closure $\mathscr{C}_\omega$ cannot in general be replaced by all of $\mathscr{G}_\omega$. The exceptional-curve example in Example 27.8.1 separates the two sets by a smooth closed form. Testing against all of $\mathscr{G}_\omega$ will instead detect smooth positivity and nefness in Theorem 25.17.1.

4.5 The adjoint measure of the linearization

The derivative of the logarithmic Monge–Ampère operator has constants in its kernel, so solving its linearized equation requires a compatibility condition. The Gauduchon multiplier identifies that condition. Fix $u \in \mathcal{H}_\omega$ and write $\eta = \omega_u$. For a smooth real direction v, logarithmic differentiation gives

$$L_u v = n \frac{\mathrm{dd^c} v \wedge \eta^{n-1}}{\eta^n}. \tag{4.6}$$

Choose the Gauduchon multiplier ρ_u from Theorem 4.1.1, normalized by

$$\int_X \rho_u \eta^n = 1, \qquad \mathrm{d}\mu_u := \rho_u \eta^n.$$

The measure μ_u is a smooth positive probability measure.

Proposition 4.5.1 *For every smooth real v,*

$$\int_X L_u v \, \mathrm{d}\mu_u = 0. \tag{4.7}$$

Moreover, the equation $L_u v = h$, for smooth real h, has a smooth solution if and only if $\int_X h \, \mathrm{d}\mu_u = 0$. When it exists, the solution is unique modulo constants.

Proof The integral in (4.7) is

$$n \int_X \mathrm{dd^c} v \wedge \rho_u \eta^{n-1} = n \int_X v \, \mathrm{dd^c}(\rho_u \eta^{n-1}) = 0.$$

This proves the necessary compatibility condition.

By the Fredholm argument in Theorem 4.1.1, the range of $L_u : C^{k+2,\alpha}(X) \to C^{k,\alpha}(X)$ has codimension one for a nonnegative integer k and $0 < \alpha < 1$. The nonzero functional $h \mapsto \int_X h \, \mathrm{d}\mu_u$ annihilates that range, so its kernel is the range. Elliptic regularity gives a smooth solution; the identity $\ker L_u = \mathbb{R}$ gives uniqueness modulo constants. □

The multiplier ρ_u is constant precisely when η is Gauduchon, by its defining equation and uniqueness in Theorem 4.1.1. Thus

$$\mu_u = \frac{\eta^n}{\int_X \eta^n} \quad \text{if } \mathrm{dd^c}(\eta^{n-1}) = 0.$$

On a surface, $\mathrm{dd^c}\omega_u = \mathrm{dd^c}\omega$, so every positive potential perturbation of a Gauduchon form is still Gauduchon. In higher dimension this persistence requires more than the Gauduchon condition at $u = 0$. Write $M = \int_X \omega^n$.

Proposition 4.5.2 *Every form ω_u, for $u \in \mathcal{H}_\omega$, is Gauduchon if and only if*

$$\int_X \omega_u^n = M \quad \text{for every } u \in \mathcal{H}_\omega.$$

In that case the normalized multiplier is $\rho_u = M^{-1}$ and $\mu_u = M^{-1}\omega_u^n$ for every $u \in \mathcal{H}_\omega$.

Proof If all ω_u are Gauduchon, the mass derivative (2.2) vanishes in every direction. Integration along the segment from 0 to u gives the mass identity. Conversely, Proposition 2.2.1 shows that constant mass forces $\mathrm{dd^c}(\omega_u^{n-1}) = 0$ for every $u \in \mathcal{H}_\omega$. The normalized multiplier is the constant M^{-1} by Theorem 4.1.1. □

The differential characterization is given by Theorem 2.2.2: persistence is equivalent to $\mathrm{dd^c}(\omega^j) = 0$ for all integers $1 \le j \le n-1$. The following example distinguishes it from the single Gauduchon identity in dimension three.

Example 4.5.1 Let $E = \mathbb{C}/(\mathbb{Z} + \mathrm{i}\mathbb{Z})$ and $X = E^3$. Write $z_j = x_j + \mathrm{i}y_j$ for the coordinates on the covering space, for $1 \le j \le 3$, and let $\eta_j = (\mathrm{i}/2)\mathrm{d}z_j \wedge \mathrm{d}\bar{z}_j$ be the descended forms. The periodic functions

$$b = \exp(\cos(2\pi x_1)), \qquad c = b^{-1}$$

define a Hermitian form

$$\omega = \eta_1 + b\eta_2 + c\eta_3.$$

Since $bc = 1$,

$$\omega^2 = 2b\eta_1 \wedge \eta_2 + 2c\eta_1 \wedge \eta_3 + 2\eta_2 \wedge \eta_3.$$

The derivatives of b, c involve only x_1, so $\mathrm{dd^c}(\omega^2) = 0$.

Put $v = \cos(2\pi x_2)$ and $h = -\pi\cos(2\pi x_2)$, so that $\mathrm{dd^c}v = h\eta_2$. For a real s with $0 < |s| < 1/(\pi\mathrm{e})$, the form

$$\omega_{sv} = \eta_1 + (b + sh)\eta_2 + c\eta_3$$

is positive, since $b \ge \mathrm{e}^{-1}$ and $|h| \le \pi$. However,

$$\mathrm{dd^c}(\omega_{sv}^2) = 2s\mathrm{dd^c}\omega \wedge \mathrm{dd^c}v = 2sh\mathrm{dd^c}c \wedge \eta_2 \wedge \eta_3 \ne 0.$$

Indeed, $\mathrm{dd^c}c = (c''/(4\pi))\eta_1$, where the prime differentiates in x_1, and $c''(0) = 4\pi^2\mathrm{e}^{-1}$ while $h = -\pi$ at $x_2 = 0$. Thus a Gauduchon form can have arbitrarily small positive potential perturbations which are not Gauduchon.

When persistence holds, the one-form $v \mapsto \int_X v\,\mathrm{d}\mu_u$ has a primitive exactly when ω is closed: it is the raw Monge–Ampère one-form divided by M, so Theorem 3.1.1 applies.

4.6 Limits of adjoint measures

When every positive potential perturbation of ω is Gauduchon, its adjoint probability is determined by its Monge–Ampère measure and the common mass. Mass invariance and bounded-product continuity extend this formula to bounded potentials.

Proposition 4.6.1 *Assume that ω_v is Gauduchon for every $v \in \mathcal{H}_\omega$. For every $u \in \mathrm{PSH}(X, \omega) \cap L^\infty(X)$, the measure*

$$\mu_u := M^{-1}(\omega + \mathrm{dd^c} u)^n$$

is a probability measure and agrees with the preceding definition when $u \in \mathcal{H}_\omega$. If $u_j, u \in \mathrm{PSH}(X, \omega) \cap L^\infty(X)$ are uniformly bounded and $u_j \to u$ in capacity, then $\mu_{u_j} \rightharpoonup \mu_u$.

Proof By Proposition 4.5.2 and Theorem 2.2.2, the positive bounded product $(\omega + \mathrm{dd^c} u)^n$ has mass M. Its normalization agrees with the smooth adjoint measure by Proposition 4.5.2. The bounded-product capacity continuity in Section 1.8 gives weak convergence; division by M preserves it. □

The proposition applies in particular to Gauduchon surfaces. For a general Hermitian background, passing from smooth $u_j \in \mathcal{H}_\omega$ to $u \in \mathrm{PSH}(X, \omega)$ requires control of the multipliers ρ_{u_j} as well as the potentials. The supporting probabilities of Chapter 14 recover μ_u in the smooth case and give a conditional approximation theorem. General approximation independence and an adjoint identity for a weak linearization remain separate questions.

Chapter 5
Gauduchon traces and the topology of potentials

Let X be a connected compact complex manifold of dimension $n \geq 1$, and let θ be a smooth real $(1,1)$-form with $\mathrm{PSH}(X,\theta) \neq \varnothing$. Fix a Gauduchon metric γ and a smooth positive probability volume form $\mathrm{d}V$ for L^1 norms. In dimension one, γ may be any Hermitian metric.

Taking the Gauduchon trace reduces recovery of a potential, modulo constants, to a scalar elliptic equation. On normalized potentials, weak trace convergence is equivalent to L^1 convergence. The elliptic inverse and smoothing of measure tests give sharp quantitative estimates for this equivalence.

5.1 The trace map

Adding a constant leaves the trace unchanged. We therefore use the mean normalization

$$\mathcal{K}^0_{\theta,\mathrm{d}V} = \left\{ u \in \mathrm{PSH}(X,\theta) : \int_X u \,\mathrm{d}V = 0 \right\}.$$

For such a potential put

$$\mathcal{T}_\gamma(u) = (\theta + \mathrm{dd^c} u) \wedge \gamma^{n-1}, \qquad M_{\theta,\gamma} = \int_X \theta \wedge \gamma^{n-1}.$$

By Proposition 4.2.1, the measure $\mathcal{T}_\gamma(u)$ is positive and has mass $M_{\theta,\gamma}$. In dimension one this follows from Stokes' theorem. For $f \in C^\infty(X,\mathbb{R})$, set $Af = (\mathrm{dd^c} f \wedge \gamma^{n-1})/\mathrm{d}V$.

Theorem 5.1.1 *The map $\mathcal{T}_\gamma$ is an affine homeomorphism from the compact convex space $\mathcal{K}^0_{\theta,\mathrm{d}V}$, with its L^1 topology, onto a compact convex subset of the positive measures of mass $M_{\theta,\gamma}$, with their weak topology. For $u_j, u \in \mathcal{K}^0_{\theta,\mathrm{d}V}$,*

$$u_j \longrightarrow u \text{ in } L^1 \quad \Longleftrightarrow \quad \mathcal{T}_\gamma(u_j) \rightharpoonup \mathcal{T}_\gamma(u).$$

If $M_{\theta,\gamma} = 0$, both spaces consist of a single point.

Proof The mean-normalized space is convex. It is the image of the compact space in Theorem 1.4.1 under the continuous surjection $v \mapsto v - \int_X v \,\mathrm{d}V$, hence compact.

If $u_j \to u$ in L^1, their trace measures converge in distributions. Their common finite mass upgrades this to weak convergence, by uniform approximation of continuous tests by smooth functions.

For injectivity, equality of traces gives $A(u - v) = 0$. Elliptic regularity for A makes $u - v$ smooth, and the maximum principle makes it constant. Its mean is zero, so it vanishes. The potentials agree everywhere by Lemma 1.3.3.

A continuous injection from a compact space to the Hausdorff space of finite measures is a homeomorphism onto its image. Affineness gives convexity of the image. If its common mass is zero, every trace is the zero measure, and injectivity makes the potential space a singleton. In this case Lemma 4.3.1 also gives $\theta + \mathrm{dd^c}u = 0$ and smoothness of the unique normalized potential. □

Admissibility for the scalar inverse requires positivity of the full current. For a finite signed measure σ of total mass zero, let $G_\gamma\sigma$ denote the mean-zero distribution solving

$$\mathrm{dd^c}(G_\gamma\sigma) \wedge \gamma^{n-1} = \sigma. \tag{5.1}$$

Indeed, $\ker A = \mathbb{R}$ by the maximum principle, and A has index zero because its linear homotopy to a Laplace operator stays elliptic. Let A^* be its formal adjoint with respect to $\mathrm{d}V$. Stokes' theorem gives $A^*1 = 0$, so $\ker A^* = \mathbb{R}$ as well. The Fredholm alternative solves the equation for smooth data exactly when $\int_X \sigma = 0$. Duality with the continuous inverse of A^* on smooth functions of mean zero extends G_γ to top-degree distributions of total mass zero, including finite signed measures; imposing zero mean fixes the additive constant.

Proposition 5.1.1 *A positive measure ν of mass $M_{\theta,\gamma}$ belongs to the image of $\mathcal{T}_\gamma$ if and only if the distribution*

$$w = G_\gamma\big(\nu - \theta \wedge \gamma^{n-1}\big)$$

satisfies $\theta + \mathrm{dd^c}w \geq 0$. In that case its canonical representative is the corresponding member of $\mathcal{K}^0_{\theta,\mathrm{d}V}$.

Proof If $\nu = \mathcal{T}_\gamma(u)$, uniqueness in (5.1) gives $w = u$ as distributions. Conversely, Proposition 1.1.1 gives a θ-plurisubharmonic representative of w. Its mean is zero by construction, and (5.1) identifies its trace with ν. □

In dimension one the positivity condition is automatic: $\theta + \mathrm{dd^c} w = \nu$. Thus every positive measure of the specified mass occurs. In higher dimension, the scalar trace condition must be supplemented by positivity of the full current.

5.2 A metric on normalized potentials

Testing trace differences against bounded Lipschitz functions gives a complete metric, which the elliptic inverse compares quantitatively with the L^1 norm. Fix a distance d induced by a smooth Riemannian metric on X. For a real Lipschitz function f on X, write $\mathrm{Lip}_d(f) = \sup_{x \neq y} |f(x) - f(y)|/d(x, y)$.

Corollary 5.2.1 *For $u, v \in \mathcal{K}^0_{\theta,\mathrm{d}V}$, the formula*

$$d_{\gamma,d}(u, v) = \sup_{\substack{\|f\|_\infty \le 1 \\ \mathrm{Lip}_d(f) \le 1}} \left| \int_X f \, \mathrm{d}\big(\mathcal{T}_\gamma(u) - \mathcal{T}_\gamma(v)\big) \right|$$

is a bounded complete metric on $\mathcal{K}^0_{\theta,\mathrm{d}V}$, inducing its L^1 topology. Changing γ or d gives the same topology and a uniformly equivalent metric on this compact space.

Proof Lipschitz functions are uniformly dense in continuous functions, and the test family in the supremum is uniformly bounded and equicontinuous. The first fact separates measures and gives weak convergence from metric convergence. The Arzelà–Ascoli theorem supplies finite uniform nets for the test family and gives the converse. Symmetry and the triangle inequality follow directly from the formula, and its value is at most $2M_{\theta,\gamma}$.

Pull back this metric by the homeomorphism in Theorem 5.1.1. The resulting metric space is compact, hence complete. For two choices of the auxiliary data, both metrics induce the same compact L^1 topology, so the identity maps in both directions are uniformly continuous. □

Proposition 5.2.1 *There is a constant $C \ge 1$, depending only on $X, \theta, \gamma, \mathrm{d}V$ and d, such that*

$$C^{-1}\|u - v\|_{L^1} \le d_{\gamma,d}(u, v) \le C\|u - v\|_{L^1}^{1/2} \qquad (u, v \in \mathcal{K}^0_{\theta,\mathrm{d}V}).$$

Proof Put $w = u - v$, $\sigma = \mathcal{T}_\gamma(u) - \mathcal{T}_\gamma(v)$ and $M = M_{\theta,\gamma}$, and write $\|\sigma\|_{\mathrm{TV}}$ for the total variation of σ. Then $\int_X w \, \mathrm{d}V = 0$, $\sigma = \mathrm{dd^c} w \wedge \gamma^{n-1}$ and $\|\sigma\|_{\mathrm{TV}} \le 2M$. If $M = 0$, Theorem 5.1.1 gives $u = v$, so both estimates hold.

Since $\ker A = \ker A^* = \mathbb{R}$, for a smooth real function g, the equation

$$A^*\phi = g - \int_X g \, \mathrm{d}V, \qquad \int_X \phi \, \mathrm{d}V = 0$$

has a unique smooth solution. For $p > 2n$, elliptic $W^{2,p}$ estimates on mean-zero functions and Sobolev embedding give $\|\phi\|_\infty + \mathrm{Lip}_d(\phi) \le C\|g\|_\infty$.

Pairing the distributional equation $Aw\,\mathrm{d}V = \sigma$ with ϕ yields

$$\left|\int_X wg\,\mathrm{d}V\right| = \left|\int_X \phi\,\mathrm{d}\sigma\right| \le C\|g\|_\infty d_{\gamma,d}(u,v).$$

Taking the supremum over smooth real g with $\|g\|_\infty \le 1$ gives $\|w\|_{L^1}$ and proves the first estimate.

For the second estimate, let f be any test function in the definition of $d_{\gamma,d}$. A fixed finite coordinate cover, smooth partition of unity and convolution at scale $0 < \epsilon < \epsilon_0$ give smooth functions f_ϵ satisfying

$$\|f - f_\epsilon\|_\infty \le C\epsilon, \qquad \|A^* f_\epsilon\|_\infty \le C\epsilon^{-1}.$$

The constants are uniform over this test family. To see the second bound, localize f by the fixed partition. The localized functions have uniformly bounded values and first weak derivatives. In each second derivative of their convolution, move one derivative onto the mollifier; its L^1 norm is $O(\epsilon^{-1})$. The lower-order terms of A^* have fixed smooth coefficients and satisfy the same bound. Hence

$$\left|\int_X f\,\mathrm{d}\sigma\right| \le C\big(M\epsilon + \epsilon^{-1}\|w\|_{L^1}\big).$$

If $0 < \|w\|_{L^1} < \epsilon_0^2$, choose $\epsilon = \|w\|_{L^1}^{1/2}$. For larger norms use $d_{\gamma,d}(u,v) \le 2M$, and for $w = 0$ use injectivity. Taking the supremum over f proves the asserted bound. □

Example 5.2.1 Let $X = \mathbb{C}/(\mathbb{Z} + \mathrm{i}\mathbb{Z})$, with $z = x + \mathrm{i}y$, $\theta = \gamma = \eta = (\mathrm{i}/2)\mathrm{d}z \wedge \mathrm{d}\bar{z}$, $\mathrm{d}V = \eta$ and the flat distance d. For integers $k \ge 1$, set

$$u_k = \frac{\cos(2\pi kx)}{2\pi k^2}.$$

These smooth mean-zero potentials satisfy

$$\theta + \mathrm{dd^c}u_k = \left(1 - \frac{1}{2}\cos(2\pi kx)\right)\eta > 0, \qquad \|u_k\|_{L^1} = \frac{1}{\pi^2 k^2}.$$

The function $f_k = \cos(2\pi kx)/(2\pi k)$ has supremum norm at most one and Lipschitz constant at most one. It follows that

$$d_{\gamma,d}(u_k, 0) \ge \left|\int_X f_k\,\mathrm{dd^c}u_k\right| = \frac{1}{8\pi k}.$$

Consequently, the exponent $1/2$ in Proposition 5.2.1 cannot be replaced by any larger exponent, even for smooth strictly positive potentials on a fixed curve. The power one in the lower bound cannot be replaced by a smaller positive power: for $0 < t \le 1$, both $\|tu_1\|_{L^1}$ and $d_{\gamma,d}(tu_1, 0)$ equal t times their positive values at $t = 1$.

Chapter 6
Non-pluripolar Monge–Ampère measures

Let X be a connected compact complex manifold of dimension $n \geq 1$. Fix a Hermitian form ω and a smooth positive probability volume form $\mathrm{d}V$ for L^1 norms and integrals.

A Monge–Ampère measure can be recovered from bounded products on the sets where its potential stays above a finite level. For a general smooth background, the truncation may lose positivity, but its product remains positive on these sets. For a Hermitian background, finiteness of the resulting measure for every potential is equivalent to a uniform bound on the masses of bounded potentials. If that bound fails, a potential with trivial multiplier ideals can have an integrable Hessian whose determinant is not integrable.

6.1 Truncation and non-pluripolar measures

Let θ be a smooth real $(1,1)$-form and $u \in \mathrm{PSH}(X,\theta)$. For $t > 0$, put

$$u_t = \max(u, -t), \qquad D_t = \{u > -t\}.$$

Choose a Hermitian form $\Omega \geq \theta$. Then u_t is bounded and Ω-plurisubharmonic, so Proposition 1.7.1 defines the signed product $(\theta + \mathrm{dd^c} u_t)^n$. On the plurifinely open set D_t, it agrees with local truncations that preserve positivity up to an arbitrarily small smooth error.

Lemma 6.1.1 *The restricted product satisfies*

$$0 \leq \mathbf{1}_{D_t}(\theta + \mathrm{dd^c} u_t)^n \leq \mathbf{1}_{D_t}(\Omega + \mathrm{dd^c} u_t)^n.$$

The same local positivity assertion holds for mixed products, on a common set where all the truncations leave their potentials unchanged.

Proof Work in a coordinate ball with a fixed positive constant form κ, and fix $\epsilon > 0$. Around any point choose a smaller neighborhood and a constant real form θ' such that

$$-\epsilon\kappa \le \theta - \theta' \le \epsilon\kappa.$$

Take a smooth local potential r for $\theta' + \epsilon\kappa$. Then $u + r$ is plurisubharmonic. On a relatively compact smaller neighborhood choose $k > t + \sup|r|$ and set

$$h_k = \max(u + r, -k) - r.$$

It is locally bounded and equals $u_t = u$ on D_t. Moreover,

$$\theta + \mathrm{dd^c} h_k + 2\epsilon\kappa = \mathrm{dd^c} \max(u + r, -k) + (\theta - \theta' + \epsilon\kappa)$$

is a sum of a positive closed bounded-potential current and a smooth positive form. Its bounded products are positive. Plurifine locality transfers this positivity to the restriction of $(\theta + \mathrm{dd^c} u_t + 2\epsilon\kappa)^n$ to D_t.

These neighborhoods cover the coordinate ball. Let $\epsilon \downarrow 0$. The product with the fixed bounded potential u_t is polynomial in ϵ, with coefficients of locally finite variation. Consequently convergence holds in local total variation, also after restriction to D_t. This proves the first inequality. Apply the same construction to each factor to obtain mixed positivity.

For the second inequality, expand the difference of the two powers by the telescoping identity, using the smooth positive difference $\Omega - \theta$. Every summand is a mixed product of positive factors on D_t, as just proved. The difference is therefore nonnegative there. □

Theorem 6.1.1 *The measures $\mathbf{1}_{D_t}(\theta + \mathrm{dd^c} u_t)^n$ increase with t. Their limit*

$$\mathrm{MA}^{\mathrm{ext}}_\theta(u) := \lim_{t \to +\infty} \mathbf{1}_{\{u > -t\}} (\theta + \mathrm{dd^c} \max(u, -t))^n \tag{6.1}$$

is a positive non-pluripolar Borel measure, finite on each D_t and sigma-finite on X. It depends only on the current $\theta + \mathrm{dd^c} u$, is unchanged by adding a constant to u, and satisfies

$$\mathrm{MA}^{\mathrm{ext}}_\theta(u) \le \mathrm{MA}^{\mathrm{ext}}_\Omega(u) \quad \textit{if } \Omega \textit{ is Hermitian and } \theta \le \Omega.$$

Its total mass may be infinite. When finite, it is the usual non-pluripolar Monge–Ampère measure.

Proof For $s > t$, locality on $D_t \subset D_s$ gives

$$\mathbf{1}_{D_t}(\theta + \mathrm{dd^c} u_s)^n = \mathbf{1}_{D_t}(\theta + \mathrm{dd^c} u_t)^n.$$

Together with Lemma 6.1.1, this proves monotonicity. The increasing limit is countably additive by monotone convergence, and vanishes on pluripolar sets because each bounded product does. The comparison with Ω passes to the limit. Compatibility gives

$$\mathbf{1}_{D_t} \mathrm{MA}^{\mathrm{ext}}_\theta(u) = \mathbf{1}_{D_t}(\theta + \mathrm{dd^c} u_t)^n.$$

Its mass is bounded by that of the bounded Ω-product. The sets D_t exhaust the complement of $\{u = -\infty\}$, which has zero mass, proving sigma-finiteness.

For a smooth real function f, consider $\theta' = \theta + \mathrm{dd^c} f$ and $u' = u - f$. On a common superlevel set of u and u', sufficiently deep truncations give the same bounded-potential current. Their restricted products therefore agree. These sets exhaust the complement of the common pole set, proving invariance under this change of decomposition. By the elliptic-regularity argument in Proposition 1.7.1, any two smooth decompositions of the same current differ in this way. Constants are a special case. If the limit has finite mass, this construction is exactly the extension of the local bounded products by zero across the pole set. □

The notation $\mathrm{MA}^{\mathrm{ext}}$ permits infinite mass. On compact X, the measure defines a current of order zero precisely when its total mass is finite.

6.2 Contact sets

The bounded contact inequality of Lemma 1.7.1 passes to unbounded potentials on common finite levels. Local positive corrections are needed because the global truncations need not remain θ-plurisubharmonic.

Proposition 6.2.1 *Let θ be any smooth real $(1,1)$-form and let $u, v \in \mathrm{PSH}(X, \theta)$ satisfy $u \le v$. Then*

$$\mathbf{1}_{\{u=v\}} \mathrm{MA}^{\mathrm{ext}}_{\theta}(u) \le \mathbf{1}_{\{u=v\}} \mathrm{MA}^{\mathrm{ext}}_{\theta}(v).$$

Proof Fix a coordinate ball with a constant positive form κ, a level $t > 0$, and $\epsilon > 0$. On smaller neighborhoods freeze θ to a constant form θ' with $-\epsilon\kappa \le \theta - \theta' \le \epsilon\kappa$. Choose a quadratic function r with $\mathrm{dd^c} r = \theta' + \epsilon\kappa$. Then $u + r$ and $v + r$ are plurisubharmonic. For a large integer k put

$$a_k = \max(u + r, -k) - r, \qquad b_k = \max(v + r, -k) - r.$$

They are bounded on relatively compact neighborhoods, satisfy $a_k \le b_k$, and

$$\theta + 2\epsilon\kappa + \mathrm{dd^c} a_k = \mathrm{dd^c} \max(u + r, -k) + (\theta - \theta' + \epsilon\kappa) \ge 0,$$

with the same inequality for b_k. Apply the top-degree case of Lemma 1.7.1 with background $\theta + 2\epsilon\kappa$.

On $D_t = \{u > -t\}$, both u and v are above $-t$. Taking $k > t + \sup|r|$ makes $a_k = u$ and $b_k = v$ on D_t. Restrict the bounded inequality to $\{u = v\} \cap D_t$ and use locality. It gives the desired contact inequality for background $\theta + 2\epsilon\kappa$ on that set. The smaller neighborhoods cover the original ball, so the restricted inequality holds throughout it.

Choose a Hermitian form $\Omega \ge \theta$. The fixed truncations $u_t = \max(u, -t)$ and $v_t = \max(v, -t)$ are bounded and Ω-plurisubharmonic, so their signed products

with background θ are defined. On D_t, the products just used equal the restrictions of $(\theta + \mathrm{dd^c} u_t + 2\epsilon\kappa)^n$ and $(\theta + \mathrm{dd^c} v_t + 2\epsilon\kappa)^n$. These are polynomials in ϵ whose coefficients have locally finite variation. They converge in local total variation as $\epsilon \downarrow 0$, including after restriction to $\{u = v\} \cap D_t$. This proves the inequality for background θ on each finite level. Finally let $t \to +\infty$. The common pole set has zero extended mass, and the levels exhaust its complement in the equality set. □

6.3 Weighted lower semicontinuity

Capacity convergence identifies the bounded products after a cutoff in the potential. Increasing the cutoff recovers a lower bound for the extended measure, even when its mass is infinite. The weights may vary with the potentials.

Lemma 6.3.1 *Let θ be any smooth real $(1,1)$-form. Suppose $u_j, u \in \mathrm{PSH}(X,\theta)$ have a common upper bound and $u_j \to u$ in capacity. If b_j, b are nonnegative uniformly bounded quasi-continuous functions and $b_j \to b$ in capacity, then*

$$\int_X b\,\mathrm{MA}_\theta^{\mathrm{ext}}(u) \le \liminf_j \int_X b_j\,\mathrm{MA}_\theta^{\mathrm{ext}}(u_j).$$

Both sides may be infinite.

Proof Translate the potentials below zero and choose a Hermitian metric $\Omega \ge \theta$. For $T > 0$, let $g_T : [-\infty, 0] \to [0,1]$ be a continuous nondecreasing cutoff, zero on $[-\infty, -2T]$, one on $[-T, 0]$, increasing to one at finite points. Put $v_j = \max(u_j, -2T)$ and $v = \max(u, -2T)$. These are uniformly bounded Ω-plurisubharmonic functions and converge in capacity. Expand $(\theta+\mathrm{dd^c} v_j)^n$ using $\theta+\mathrm{dd^c} v_j = (\Omega+\mathrm{dd^c} v_j)+(\theta-\Omega)$. Each smooth factor is bounded by a fixed power of Ω up to a constant. Thus the signed terms have total variations controlled by positive bounded mixed measures, and Lemma 1.8.3 gives a common capacity bound for fixed T. Bounded Bedford–Taylor continuity gives weak convergence term by term. It also permits the weights $b_j g_T(v_j)$: their differences from the limiting weight tend to zero in capacity, and the common capacity bound makes their integrals tend to zero. For the fixed limiting weight, remove a set of arbitrarily small capacity, extend continuously from its complement, and control the discarded part by the same bound. Locality and the vanishing cutoff on the truncation level give

$$\begin{aligned}\int_X b_j g_T(u_j)\,\mathrm{MA}_\theta^{\mathrm{ext}}(u_j) &= \int_X b_j g_T(v_j)(\theta + \mathrm{dd^c} v_j)^n \\ &\longrightarrow \int_X b g_T(u)\,\mathrm{MA}_\theta^{\mathrm{ext}}(u).\end{aligned}$$

The first integral is nonnegative and bounded above by the corresponding uncut integral, although the bounded product before multiplication may be signed. Let $T \to \infty$ and use monotone convergence. □

6.4 Density and trace inequalities

The mixed density inequality of Lemma 1.7.3 passes to unbounded potentials on common finite levels. Convex combinations then give a lower bound for their extended Monge–Ampère measures; comparison with a smooth metric gives a lower bound for the full linear trace.

Lemma 6.4.1 *Let θ be a smooth semipositive $(1,1)$-form and let $f, g \geq 0$ be measurable functions on X. Suppose $u, v \in \mathrm{PSH}(X, \theta)$ and*

$$\mathrm{MA}_\theta^{\mathrm{ext}}(u) \geq f \, \mathrm{d}V, \qquad \mathrm{MA}_\theta^{\mathrm{ext}}(v) \geq g \, \mathrm{d}V.$$

For $0 \leq t \leq 1$ we have

$$\mathrm{MA}_\theta^{\mathrm{ext}}(tu + (1-t)v) \geq \left(t f^{1/n} + (1-t) g^{1/n}\right)^n \mathrm{d}V. \tag{6.2}$$

Proof The endpoints are immediate. For $0 < t < 1$, translate u, v to be nonpositive and put $u_k = \max(u, -k)$ and $v_k = \max(v, -k)$. Semipositivity of θ makes these bounded θ-plurisubharmonic functions. By locality,

$$\mathrm{MA}_\theta(u_k) \geq \mathbf{1}_{\{u > -k\}} f \, \mathrm{d}V, \qquad \mathrm{MA}_\theta(v_k) \geq \mathbf{1}_{\{v > -k\}} g \, \mathrm{d}V.$$

Apply Lemma 1.7.3 to each mixed term in the positive expansion of $\theta^n_{tu_k + (1-t)v_k}$. On $E_k = \{u > -k\} \cap \{v > -k\}$ this gives exactly the density on the right of (6.2). On the same set, plurifine locality identifies the left side with the extended measure of $tu + (1-t)v$. Let $k \to \infty$. The sets E_k exhaust the complement of the two pole sets, which have zero smooth volume. Monotone convergence on Borel sets proves the inequality. □

The same mixed inequality gives a linear trace bound for unbounded potentials. The linear current retains its mass on the pole set, whereas the extended measure discards that set.

Lemma 6.4.2 *Let γ be a Hermitian form. If $u \in \mathrm{PSH}(X, \omega)$ and $0 \leq g \in L^1(\mathrm{d}V)$ satisfy $\mathrm{MA}_\omega^{\mathrm{ext}}(u) \geq g \, \mathrm{d}V$, then*

$$(\omega + \mathrm{dd^c} u) \wedge \gamma^{n-1} \geq (g \, \mathrm{d}V / \gamma^n)^{1/n} \gamma^n.$$

Proof For bounded u, Lemma 1.7.3 with $T_1 = \omega + \mathrm{dd^c} u$, $T_2 = \gamma$ and $k = 1$ gives the displayed inequality.

In general, subtract a constant so that $u \leq 0$ and put $u_k = \max(u, -k)$. Positivity and locality give

$$\mathrm{MA}_\omega(u_k) \geq \mathbf{1}_{\{u > -k\}} g \, \mathrm{d}V.$$

Apply the bounded inequality to u_k. Since $u_k \to u$ in L^1, their full traces converge weakly. The right-hand densities converge in L^1: $g^{1/n}$ is integrable and the pole set has zero smooth volume. Passing to the limit proves the assertion. □

6.5 The bounded mass property

The restricted products have uniformly bounded mass whenever the bounded potentials do. Comparison of Hermitian forms makes this a property of the manifold, and smooth decreasing regularization reduces its verification to smooth potentials.

Definition 6.5.1 The lower and upper masses of ω are

$$v_-(\omega) = \inf_{v \in \mathrm{PSH}(X,\omega)\cap L^\infty(X)} \int_X (\omega + \mathrm{dd^c} v)^n,$$
$$v_+(\omega) = \sup_{v \in \mathrm{PSH}(X,\omega)\cap L^\infty(X)} \int_X (\omega + \mathrm{dd^c} v)^n.$$

The manifold has the *bounded mass property* if $v_+(\omega) < +\infty$. The form ω has *positive lower mass* if $v_-(\omega) > 0$.

Proposition 6.5.1 *The bounded mass property is independent of the Hermitian metric. Moreover, the supremum in Definition 6.5.1 is unchanged if only smooth potentials are used.*

Proof For another Hermitian form η, choose $C > 0$ such that $\eta \le C\omega$. If v is a bounded η-plurisubharmonic function, then v/C is ω-plurisubharmonic and

$$0 \le \eta + \mathrm{dd^c} v \le C\big(\omega + \mathrm{dd^c}(v/C)\big).$$

By positivity of mixed bounded products, $\int_X (\eta + \mathrm{dd^c} v)^n \le C^n v_+(\omega)$. Comparison in the other direction proves metric independence.

For the last assertion apply Theorem 1.11.1. The smooth approximants decrease and remain uniformly bounded, so continuity of bounded products gives convergence of their total masses. The reverse inequality between the two suprema is immediate. □

Proposition 6.5.2 *Assume $v_+(\omega) < +\infty$. If θ is a smooth real $(1,1)$-form with $\theta \le C\omega$, $C > 0$, and $u \in \mathrm{PSH}(X,\theta)$, then*

$$\mathrm{MA}^{\mathrm{ext}}_\theta(u)(X) \le C^n v_+(\omega).$$

Proof The comparison in Theorem 6.1.1 with $C\omega$ gives

$$\mathbf{1}_{D_t}\mathrm{MA}^{\mathrm{ext}}_\theta(u) \le \mathbf{1}_{D_t}(C\omega + \mathrm{dd^c} u_t)^n.$$

Since u_t/C is bounded and ω-plurisubharmonic, the right-hand side has mass at most $C^n v_+(\omega)$. Let $t \to \infty$. □

6.6 Infinite mass with trivial multiplier ideals

A summable convex combination retains a fixed fraction of the mass of each smooth summand. Choosing their masses to outgrow the reciprocal nth powers of the weights forces infinite mass in the limit. The same weights make the tail small enough for the uniform exponential estimate of Lemma 1.5.1 to apply at every exponent.

Theorem 6.6.1 *If $v_+(\omega) = +\infty$, there is $U \in \mathrm{PSH}(X, \omega)$, $U \leq 0$, such that*

$$\mathrm{MA}^{\mathrm{ext}}_{\omega}(U)(X) = +\infty, \qquad \int_X \mathrm{e}^{-pU}\,\mathrm{d}V < +\infty \quad \textit{for every } p > 0.$$

In particular, $\mathcal{I}(pU) = \mathcal{O}_X$ for every $p > 0$.

Proof Set $a_j = 2^{-j}$. By Proposition 6.5.1, choose smooth ω-plurisubharmonic functions u_j with $\sup_X u_j = 0$ and

$$M_j := \int_X \omega^n_{u_j} \geq 2^{j(n+1)}.$$

The partial sums $U_N = \sum_{j \leq N} a_j u_j$ decrease and satisfy

$$\omega + \mathrm{dd^c} U_N = \sum_{j \leq N} a_j \omega_{u_j} + \left(1 - \sum_{j \leq N} a_j\right)\omega \geq 0.$$

Their L^1 norms are bounded by (1.1), so their limit U is ω-plurisubharmonic and nonpositive.

Fix j and write $v_j = U - a_j u_j$. Taking limits in the remaining partial sums gives $(1 - a_j)\omega + \mathrm{dd^c} v_j \geq 0$. For $t > 0$, put

$$v_{j,t} = \max(v_j, -t), \qquad W_{j,t} = a_j u_j + v_{j,t}.$$

The first is a bounded potential for the positive background $(1 - a_j)\omega$, so $W_{j,t}$ is bounded and ω-plurisubharmonic.

On $D_t = \{U > -t\}$, the inequality $u_j \leq 0$ gives $v_j = U - a_j u_j \geq U > -t$. Thus $W_{j,t} = U = \max(U, -t)$ there. Plurifine locality and positivity of bounded mixed products yield

$$\begin{aligned} \mathbf{1}_{D_t} \omega^n_{\max(U,-t)} &= \mathbf{1}_{D_t} \left(a_j \omega_{u_j} + (1 - a_j)\omega + \mathrm{dd^c} v_{j,t}\right)^n \\ &\geq a_j^n \mathbf{1}_{D_t} \omega^n_{u_j}. \end{aligned}$$

The sets D_t exhaust $X \setminus \{U = -\infty\}$. This pole set is pluripolar, hence has zero smooth volume; the smooth measure $\omega^n_{u_j}$ gives it zero mass. Letting $t \to +\infty$, we obtain

$$\mathrm{MA}^{\mathrm{ext}}_{\omega}(U)(X) \geq a_j^n M_j \geq 2^j.$$

Since j is arbitrary, the mass is infinite.

For integrability, fix $p > 0$ and take N so large that $p\delta \le \alpha$, where $\delta = \sum_{j>N} a_j = 2^{-N}$ and α is from Lemma 1.5.1. Write $U = A_N + V_N$ with $A_N = \sum_{j\le N} a_j u_j$ smooth. For a finite tail ending at $m > N$, Jensen's inequality, with the unused weight assigned to zero, gives

$$\exp\left(-p \sum_{N<j\le m} a_j u_j\right) \le \sum_{N<j\le m} \frac{a_j}{\delta} \mathrm{e}^{-p\delta u_j} + 1 - \sum_{N<j\le m} \frac{a_j}{\delta}.$$

Because $u_j \le 0$ and $p\delta \le \alpha$, the integrals of the exponentials on the right are uniformly bounded. Integrate and let $m \to +\infty$ by monotone convergence to obtain $\int_X \mathrm{e}^{-pV_N} \mathrm{d}V < +\infty$. Multiplication by the bounded smooth function e^{-pA_N} proves $\int_X \mathrm{e}^{-pU} \mathrm{d}V < +\infty$.

The constant germ 1 is consequently integrable against each weight e^{-pU}, so every multiplier ideal in the statement is the whole structure sheaf. □

Corollary 6.6.1 *The bounded mass property is equivalent to finiteness of* $\mathrm{MA}_\omega^{\mathrm{ext}}(u)(X)$ *for every* $u \in \mathrm{PSH}(X, \omega)$. *If bounded mass fails, potentials of infinite extended mass with all multiplier ideals trivial can approach zero in* L^1.

Proof The first assertion follows from Proposition 6.5.2 and Theorem 6.6.1. For the second, use εU, $0 < \varepsilon \le 1$, with U as above. It is ω-plurisubharmonic and has all the required exponential integrability. On $D_t = \{U > -t\}$ its truncation at level εt is $\varepsilon \max(U, -t)$, and

$$\omega + \mathrm{dd^c}\big(\varepsilon \max(U, -t)\big) = \varepsilon\omega_{\max(U,-t)} + (1-\varepsilon)\omega.$$

Positivity of mixed products gives a lower bound of ε^n times the truncation measure for U. Its limiting mass is therefore infinite. Finally, $\|\varepsilon U\|_{L^1} \to 0$. □

6.7 Integrable Hessians and infinite mass

The Gauduchon identity fixes the trace integral of each positive summand in the preceding construction. Their weighted series therefore converges in the coefficient L^1 norm, although its determinant is not integrable. This strengthens the distinction between linear convergence and nonlinear mass: even total-variation convergence of the trace measures does not control the Monge–Ampère masses.

Fix a Gauduchon form γ and set $M = \int_X \omega \wedge \gamma^{n-1}$. For a real $(1,1)$-form T with measurable coefficients, let A_T be its Hermitian matrix relative to γ, and let $|A_T| = (A_T^2)^{1/2}$ be its positive matrix absolute value. Use the norm

$$\|T\|_{L^1(\gamma)} = \frac{1}{n}\int_X \operatorname{tr}|A_T|\,\gamma^n.$$

For $T \geq 0$ this equals $\int_X T \wedge \gamma^{n-1}$. Any smooth background norm gives the same space of integrable coefficients on compact X. We write $\det_\gamma T = \det A_T$, defined almost everywhere when the coefficients are integrable.

Proposition 6.7.1 *Suppose $v_+(\omega) = +\infty$. For the potential U and smooth decreasing approximants U_N constructed in Theorem 6.6.1, the current $T = \omega + \mathrm{dd^c}U$ has integrable coefficients and*

$$\|\omega + \mathrm{dd^c}U_N - T\|_{L^1(\gamma)} \leq 2^{1-N}M.$$

Its pointwise determinant satisfies

$$\mathrm{MA}_\omega^{\mathrm{ext}}(U) \geq (\det_\gamma T)\gamma^n, \qquad \int_X (\det_\gamma T)\gamma^n = +\infty.$$

Moreover,

$$\int_X (\omega + \mathrm{dd^c}U_N)^n \longrightarrow +\infty.$$

Proof Write $T_j = \omega + \mathrm{dd^c}u_j$ for the smooth positive currents in that construction. Put

$$S_N = \sum_{j\leq N} a_j T_j, \qquad \delta_N = \sum_{j>N} a_j = 2^{-N}.$$

The smooth positive matrices S_N increase pointwise. The Gauduchon identity gives

$$\int_X S_N \wedge \gamma^{n-1} = (1-\delta_N)M.$$

Monotone convergence of their traces shows that the matrix series has a finite positive limit S almost everywhere. Its coefficients are integrable: in a γ-unitary frame the absolute value of every coefficient is bounded by the trace. Positivity and the same trace identity give

$$\|S - S_N\|_{L^1(\gamma)} = \delta_N M.$$

Thus $S_N \to S$ also in distributions. Since

$$\omega + \mathrm{dd^c}U_N = S_N + \delta_N\omega \longrightarrow \omega + \mathrm{dd^c}U$$

in distributions by L^1 convergence of the potentials, we obtain $T = S$. The triangle inequality now gives

$$\|\omega + \mathrm{dd^c}U_N - T\|_{L^1(\gamma)} \leq \|S_N - S\|_{L^1(\gamma)} + \delta_N\|\omega\|_{L^1(\gamma)} = 2\delta_N M.$$

Fix N and set $V_N = U - U_N$. The remaining partial sums give $\delta_N\omega + \mathrm{dd^c}V_N \geq 0$ and $V_N \leq 0$. For $t > 0$, put $W_{N,t} = U_N + \max(V_N, -t)$. It is bounded, and its current is the sum of two positive factors:

$$\omega + \mathrm{dd^c}W_{N,t} = S_N + \big(\delta_N\omega + \mathrm{dd^c}\max(V_N, -t)\big).$$

On $D_t = \{U > -t\}$, the inequality $U_N \leq 0$ gives $V_N \geq U > -t$. Consequently $W_{N,t} = U = \max(U, -t)$ there. Locality and positivity of bounded products imply

$$\mathbf{1}_{D_t}\mathrm{MA}^{\mathrm{ext}}_\omega(U) \geq \mathbf{1}_{D_t}S_N^n.$$

The smooth measure S_N^n ignores the pluripolar pole set of U. Letting $t \to \infty$ proves $\mathrm{MA}^{\mathrm{ext}}_\omega(U) \geq S_N^n$. The positive matrices S_N increase to T almost everywhere, so their determinants increase to $\det_\gamma T$. Monotone convergence on every Borel set gives the asserted determinant-measure inequality.

For every $N \geq j$, positivity gives $S_N \geq a_jT_j$. Hence both the measurable limit and the smooth approximating current satisfy

$$\int_X (\det_\gamma T)\gamma^n \geq a_j^n \int_X T_j^n \geq 2^j, \qquad \int_X (\omega + \mathrm{dd^c}U_N)^n \geq a_j^n \int_X T_j^n \geq 2^j.$$

The first bound holds for every j and the second for every $N \geq j$. They give the two assertions about infinite mass. □

Corollary 6.7.1 *If bounded mass fails, there are nonpositive potentials* $w_\varepsilon \in \mathrm{PSH}(X, \omega)$, $0 < \varepsilon < 1$, *such that*

$$\omega + \mathrm{dd^c}w_\varepsilon \geq (1-\varepsilon)\omega, \qquad \mathcal{I}(pw_\varepsilon) = \mathcal{O}_X \quad (p > 0),$$

and both the pointwise determinant integral and the extended Monge–Ampère mass are infinite. As $\varepsilon \downarrow 0$, *the potentials converge to zero in* L^1 *and their currents converge to* ω *in the coefficient* L^1 *norm. Their full Gauduchon trace measures converge to* $\omega \wedge \gamma^{n-1}$ *in total variation.*

Proof Take $w_\varepsilon = \varepsilon U$. Its current is

$$T_\varepsilon = (1-\varepsilon)\omega + \varepsilon T.$$

The lower bound and the two L^1 convergence assertions follow immediately. The full trace measures have integrable densities, and their total-variation distance is at most

$$\varepsilon\|T - \omega\|_{L^1(\gamma)}.$$

Every multiplier ideal is trivial by the exponential integrability already proved for U. Since $T_\varepsilon \geq \varepsilon T$ almost everywhere, its determinant integral is at least ε^n times that of T, hence infinite. Its extended mass is infinite by Corollary 6.6.1. □

For smooth potentials, the total masses can be specified in advance while both the currents and the exponential weights converge.

Corollary 6.7.2 *Suppose $v_+(\omega) = +\infty$. Let $(\varepsilon_j)_{j\geq 1}$ and $(b_j)_{j\geq 1}$ be real sequences with $0 < \varepsilon_j < 1$, $\varepsilon_j \to 0$, and $b_j > \int_X \omega^n$. There are smooth potentials $w_j \in \mathrm{PSH}(X,\omega)$ with $\sup_X w_j = 0$ such that*

$$\omega + \mathrm{dd^c} w_j \geq (1-\varepsilon_j)\omega, \qquad \int_X (\omega + \mathrm{dd^c} w_j)^n = b_j,$$
$$\|\mathrm{dd^c} w_j\|_{L^1(\gamma)} \leq 2M\varepsilon_j.$$

Moreover, $w_j \to 0$ in $L^1(\mathrm{d}V)$ and $\mathrm{e}^{-pw_j} \to 1$ in $L^1(\mathrm{d}V)$ for every fixed $p > 0$. The full Gauduchon trace measures converge to $\omega \wedge \gamma^{n-1}$ in total variation.

Proof By Proposition 6.5.1, choose a smooth $u_j \in \mathrm{PSH}(X,\omega)$ with $\sup_X u_j = 0$ and

$$\int_X (\omega + \mathrm{dd^c} u_j)^n > \varepsilon_j^{-n} b_j.$$

Put $T_j = \omega + \mathrm{dd^c} u_j$. The polynomial

$$F_j(s) = \int_X \big((1-s)\omega + sT_j\big)^n, \qquad 0 \leq s \leq \varepsilon_j,$$

satisfies $F_j(0) < b_j$ and $F_j(\varepsilon_j) > b_j$, the latter by positivity of the mixed terms. The intermediate value theorem gives $s_j \in (0,\varepsilon_j)$ with $F_j(s_j) = b_j$. Set $w_j = s_j u_j$. This proves the normalization, strict positivity and prescribed mass.

The Gauduchon identity gives $\|T_j\|_{L^1(\gamma)} = \|\omega\|_{L^1(\gamma)} = M$. Hence

$$\|\mathrm{dd^c} w_j\|_{L^1(\gamma)} = s_j \|T_j - \omega\|_{L^1(\gamma)} \leq 2M\varepsilon_j.$$

The absolute value of the trace is bounded by the trace of the matrix absolute value, so the same bound holds for the total variation of $\mathrm{dd^c} w_j \wedge \gamma^{n-1}$. With K_ω as in (1.1), we also have $\|w_j\|_{L^1(\mathrm{d}V)} \leq K_\omega \varepsilon_j$.

Let α, C be the constants in Lemma 1.5.1 for $(X,\omega,\mathrm{d}V)$, and fix $p > 0$. For all sufficiently large j, one has $ps_j \leq \alpha$. Convexity of the exponential function, applied with weight ps_j/α, gives

$$0 \leq \mathrm{e}^{-pw_j} - 1 \leq \frac{ps_j}{\alpha}\big(\mathrm{e}^{-\alpha u_j} - 1\big).$$

Integration bounds its $L^1(\mathrm{d}V)$ norm by $pC\varepsilon_j/\alpha$, proving the last convergence. □

Chapter 7
Weighted Monge–Ampère measures

Let X be a connected compact complex manifold of dimension $n \geq 1$, and fix a Hermitian form ω. Capacity refers to Cap_ω from Section 1.8.

An extended Monge–Ampère measure can have infinite mass, while its restriction away from deep sublevels admits a polynomial mass bound. A sufficiently fast decreasing function of the depth compensates for this growth and gives a finite measure that varies continuously under capacity convergence of the potentials. Removing the weight requires uniform control of the remaining mass on deep sublevels. The examples below show how this mass can survive even when the potentials converge in capacity.

For nonpositive $u \in \mathrm{PSH}(X, \omega)$, write

$$\mu_u = \mathrm{MA}_\omega^{\mathrm{ext}}(u), \qquad r_n = \max(n-2, 0).$$

7.1 Range and restricted mass

The first mixed mass has a uniform bound from the Gauduchon trace. Local cutoff estimates turn this into a bound of degree $j-1$ for the jth mixed mass. Stokes' theorem then expresses the top mass in terms of the lower ones.

Proposition 7.1.1 *There is a constant C_ω such that, for every $A \geq 0$,*

$$\int_X (\omega + \mathrm{dd^c} v)^n \leq C_\omega (1+A)^{r_n} \quad \textit{if } v \in \mathrm{PSH}(X, \omega),\ 0 \leq v \leq A.$$

Proof Choose finitely many coordinate balls with smaller relatively compact balls covering X. On each larger ball choose a fixed quadratic q with $\mathrm{dd^c}q \geq \omega$ and shift it so that $p = v + q$ is plurisubharmonic with $0 \leq p \leq A + C$. If κ is the Euclidean form, the identity

$$\mathrm{dd^c}p = (\omega + \mathrm{dd^c}v) + (\mathrm{dd^c}q - \omega)$$

and Corollary 4.2.1 bound $\int \mathrm{dd^c}p \wedge \kappa^{n-1}$ uniformly in v and A. Indeed, the first term is bounded by a fixed multiple of the global trace mass, and the second is smooth and fixed. In dimension one use Stokes' theorem instead of the trace bound. By (1.2), on each smaller ball,

$$\int (\mathrm{dd^c}p)^j \wedge \kappa^{n-j} \leq C_j(1 + A)^{j-1} \quad (1 \leq j \leq n).$$

Expand $(\omega + \mathrm{dd^c}v)^j = (\mathrm{dd^c}p + \omega - \mathrm{dd^c}q)^j$ and sum over the finite cover. The smooth factors have bounded coefficients, so the positive mixed masses satisfy

$$0 \leq M_j(v) := \int_X (\omega + \mathrm{dd^c}v)^j \wedge \omega^{n-j} \leq C_j(1 + A)^{\max(j-1,0)} \quad (0 \leq j \leq n).$$

For smooth v, the exact top wedge integrates to zero, as in the proof of Theorem 2.2.1. The regularization of Theorem 1.11.1 and bounded-product continuity give the same identity for bounded v. Thus

$$0 = \int_X (\mathrm{dd^c}v)^n = \sum_{j=0}^{n} (-1)^{n-j} \binom{n}{j} M_j(v).$$

Since the mixed masses are nonnegative,

$$M_n(v) \leq \sum_{j=0}^{n-1} \binom{n}{j} M_j(v) \leq C_\omega(1 + A)^{r_n}.$$

For $n \leq 2$ this gives bounded mass. In higher dimensions the bound grows with the range; truncation makes it a bound for the mass at finite depth.

Lemma 7.1.1 *For $t \geq 0$ and every Borel set $E \subset X$,*

$$\mu_u(\{u \geq -t\}) \leq C_\omega(1 + t)^{r_n}, \qquad \mu_u(E \cap \{u \geq -t\}) \leq (1 + t)^n \operatorname{Cap}_\omega(E). \quad (7.1)$$

Proof Put $v_t = \max(u, -t - 1) + t + 1$. It is ω-plurisubharmonic and has range $[0, t + 1]$. Its measure agrees with μ_u on $\{u \geq -t\} \subset \{u > -t - 1\}$ by plurifine locality. The first bound follows from Proposition 7.1.1, after increasing the constant. For the second, $v_t/(1 + t)$ is an admissible capacity test, and positivity gives

$$(\omega + \mathrm{dd^c}v_t)^n \leq (1 + t)^n \big(\omega + \mathrm{dd^c}(v_t/(1 + t))\big)^n.$$

7.2 Exponential damping

An exponential transform has bounded range and preserves positivity. Its Hessian dominates an exponential multiple of the original positive current, so its top product controls the weighted measure.

For $a > 0$ put $F_a(t) = a^{-1}\mathrm{e}^{at}$ on $t \le 0$, with $F_a(-\infty) = 0$. Then $0 \le F_a \le a^{-1}$, while $0 < F_a' \le 1$ and $F_a'' \ge 0$ at finite arguments.

Proposition 7.2.1 *If $u \in \mathrm{PSH}(X, \omega)$ and $u \le 0$, then $F_a(u)$ is a bounded ω-plurisubharmonic function, and*

$$\mathrm{e}^{nau}\mu_u \le \mathrm{MA}_\omega(F_a(u)).$$

Proof First suppose u is bounded. The chain rule (1.6) gives

$$\omega + \mathrm{dd^c} F_a(u) = \mathrm{e}^{au}(\omega + \mathrm{dd^c} u) + (1 - \mathrm{e}^{au})\omega + a\mathrm{e}^{au}\,\mathrm{d}u \wedge \mathrm{d^c} u.$$

All three summands are positive. Positivity of their bounded products yields the claimed measure inequality.

For arbitrary u, use $u_t = \max(u, -t)$. The bounded potentials $F_a(u_t)$ decrease and lie between zero and a^{-1}, so their limit $F_a(u)$ is a bounded ω-plurisubharmonic function. On the plurifine open set $D_t = \{u > -t\}$, it agrees with $F_a(u_t)$. Locality and the bounded case therefore give

$$\mathrm{e}^{nau}\mathbf{1}_{D_t}\mu_u \le \mathbf{1}_{D_t}\mathrm{MA}_\omega(F_a(u)).$$

Let $t \to +\infty$. The pole set is discarded by μ_u and has zero mass for the bounded product on the right. □

Theorem 7.2.1 *For $p > 0$, the non-pluripolar measures $\nu_{p,u} = \mathrm{e}^{pu}\mu_u$ satisfy*

$$\nu_{p,u}(X) \le C_\omega(1 + p^{-1})^{r_n}, \qquad \nu_{p,u}(E) \le \max(1, n/p)^n \,\mathrm{Cap}_\omega(E)$$

for every Borel set E, uniformly over the nonpositive potentials u.

Proof Take $a = p/n$ in Proposition 7.2.1. The range estimate with $A = n/p$ gives the total-mass bound. For the capacity bound put $B = \max(1, n/p)$ and $v = F_{p/n}(u)$. Then v/B is an admissible capacity test, and

$$B\big(\omega + \mathrm{dd^c}(v/B)\big) = (\omega + \mathrm{dd^c} v) + (B - 1)\omega \ge \omega + \mathrm{dd^c} v.$$

Consequently $\nu_{p,u} \le \mathrm{MA}_\omega(v) \le B^n \mathrm{MA}_\omega(v/B)$. The measures vanish on pluripolar sets because μ_u does. □

Since μ_u vanishes on the pole set, multiplication by e^{-pu} recovers it from $\nu_{p,u}$, with infinite mass allowed. An additive shift $u \mapsto u + c$ multiplies the weighted measure by e^{pc}.

7.3 Decreasing weights and capacity continuity

For a decreasing weight, integration over its level parameter converts (7.1) into a mass and tail estimate. At any fixed depth, capacity convergence passes through the bounded products. The tail estimate then removes this restriction.

Theorem 7.3.1 *Let $w : [0, +\infty) \to (0, +\infty)$ be continuously differentiable and nonincreasing, with $w(t) \to 0$. For $T \geq 0$ define*

$$B_w(T) = \int_T^\infty (1+t)^{r_n}(-w'(t))\,\mathrm{d}t.$$

Assume $B_w(0) < +\infty$. For nonpositive $u \in \mathrm{PSH}(X, \omega)$, the measures $\nu_{w,u} = w(-u)\mu_u$, with weight zero on the pole set, satisfy

$$\nu_{w,u}(X) \leq C_\omega B_w(0), \qquad \nu_{w,u}(\{u < -T\}) \leq C_\omega B_w(T).$$

For every Borel set E and $T \geq 0$,

$$\nu_{w,u}(E) \leq w(0)(1+T)^n \operatorname{Cap}_\omega(E) + C_\omega B_w(T).$$

In particular they are uniformly absolutely continuous with respect to capacity. If nonpositive $u_j, u \in \mathrm{PSH}(X, \omega)$ satisfy $u_j \to u$ in capacity, then $\nu_{w,u_j} \rightharpoonup \nu_{w,u}$.

Proof Write $s = -u$. Since $w(s) = \int_s^\infty (-w'(t))\,\mathrm{d}t$, Tonelli's theorem and Lemma 7.1.1 give

$$\int_X w(s)\,\mathrm{d}\mu_u = \int_0^\infty (-w'(t))\mu_u(\{s \leq t\})\,\mathrm{d}t \leq C_\omega B_w(0).$$

This identity includes the set $\{s = 0\}$; the pole set has zero μ_u-mass. Restricting the integral to $\{s > T\}$ gives the same bound with $B_w(T)$, since the inner set is then $\{T < s \leq t\}$ and $t \geq T$.

The second bound in (7.1) gives

$$\int_{E\cap\{s\leq T\}} w(s)\,\mathrm{d}\mu_u \leq w(0)(1+T)^n \operatorname{Cap}_\omega(E).$$

Add the tail estimate. Since $B_w(T) \to 0$, first taking T large and then the capacity small gives uniform absolute continuity.

For continuity, fix a continuous nonincreasing $\eta : [0, +\infty) \to [0, 1]$, equal to one on $[0, 1]$ and zero on $[2, +\infty)$, and put $\eta_T(s) = \eta(s/T)$. For fixed $T > 0$, set $v_{j,T} = \max(u_j, -2T)$ and $g_T(t) = \eta_T(-t)w(-t)$ on $[-2T, 0]$. Since $g_T(-2T) = 0$, locality gives

$$\eta_T(-u_j)w(-u_j)\mu_{u_j} = g_T(v_{j,T})\,\mathrm{MA}_\omega(v_{j,T}). \tag{7.2}$$

Truncation preserves capacity convergence. The functions $g_T(v_{j,T})$ are uniformly bounded and quasi-continuous, and converge in capacity to the corresponding

coefficient for u. The weighted bounded-product continuity in Sections 1.6 and 1.8 therefore gives weak convergence in (7.2) at each fixed T. The tail bounds control the error by $C_\omega B_w(T)$ times the supremum norm of a continuous test function, uniformly in j and for u. Let $T \to \infty$ to conclude. □

For $w(t) = \mathrm{e}^{-pt}$, $p > 0$, the integral $B_w(0)$ is finite because the exponential dominates the polynomial factor. Thus the exponential measures of Theorem 7.2.1 are continuous under capacity convergence. Their tails satisfy the useful bound

$$\nu_{p,u}(\{u < -T\}) \le \mathrm{e}^{-pT/2}\nu_{p/2,u}(X) \le C_p \mathrm{e}^{-pT/2}.$$

Polynomial weights give a more explicit form of the general estimate.

Theorem 7.3.2 *For every $q > r_n$, the measures*

$$\sigma_{q,u} := (1-u)^{-q}\mu_u$$

have uniformly bounded mass for $u \in \mathrm{PSH}(X,\omega)$, $u \le 0$. More precisely,

$$\sigma_{q,u}(\{u < -T\}) \le C\frac{q}{q-r_n}(1+T)^{r_n-q} \quad (T \ge 0).$$

There is also C_q such that, for every Borel set $E \subset X$,

$$\sigma_{q,u}(E) \le C_q \operatorname{Cap}_\omega(E)^{(q-r_n)/(q+n-r_n)}.$$

Proof For $w(t) = (1+t)^{-q}$, direct integration gives

$$B_w(T) = q\int_T^\infty (1+t)^{r_n-q-1}\,\mathrm{d}t = \frac{q}{q-r_n}(1+T)^{r_n-q}.$$

Thus Theorem 7.3.1 gives the mass and tail estimates. For the capacity bound put $\delta = \operatorname{Cap}_\omega(E)$. If $0 < \delta \le 1$, choose $T = \delta^{-1/(q+n-r_n)} \ge 1$ in that theorem. Both $T^n\delta$ and T^{r_n-q} equal $\delta^{(q-r_n)/(q+n-r_n)}$, so its two terms have the required bound. If $\delta = 0$, let $T \to \infty$; if $\delta > 1$, use the total-mass bound. □

For $n \ge 2$, the capacity exponent is $1 - n/(q+2)$. The threshold $q > n-2$ comes from the mixed-mass estimate in Proposition 7.1.1.

For $n \ge 3$ and $a > 1$, take

$$w(t) = (1+t)^{-(n-2)}[\log(\mathrm{e}+t)]^{-a}.$$

Indeed, writing $r = n-2 > 0$, differentiation gives

$$(1+t)^r(-w'(t)) = \frac{r}{(1+t)[\log(\mathrm{e}+t)]^a} + \frac{a}{(\mathrm{e}+t)[\log(\mathrm{e}+t)]^{a+1}}.$$

The ratios $(\mathrm{e}+t)/(1+t)$ are bounded. Substitution $y = \log(\mathrm{e}+t)$ therefore bounds the two tail integrals by fixed multiples of $[\log(\mathrm{e}+T)]^{1-a}$ and $[\log(\mathrm{e}+T)]^{-a}$,

respectively. Since $a > 1$ and $\log(\mathrm{e} + T) \geq 1$, $B_w(T) \leq C_{n,a}[\log(\mathrm{e} + T)]^{1-a}$. Thus $(1-u)^{-(n-2)}[\log(\mathrm{e} - u)]^{-a}\mu_u$ has uniformly bounded mass, the displayed logarithmic tail bound and capacity continuity. In dimensions one and two, $r_n = 0$ and $B_w(T) = w(T)$, so every weight satisfying the stated monotonicity and vanishing conditions works.

When bounded mass fails, choose the potential U used in Corollary 6.7.1 and normalize $\sup_X U = 0$. For $0 < \varepsilon < 1$, the potentials εU approach zero in capacity as $\varepsilon \downarrow 0$: by (1.14), the capacity of $\{\varepsilon U < -\delta\}$ is at most $C\varepsilon/\delta$ for each fixed $\delta > 0$ and small ε. Consequently, for $p > 0$ and $q > r_n$,

$$\mathrm{e}^{p\varepsilon U}\mu_{\varepsilon U} \rightharpoonup \omega^n, \qquad (1-\varepsilon U)^{-q}\mu_{\varepsilon U} \rightharpoonup \omega^n,$$

although $\mu_{\varepsilon U}(X) = +\infty$ for every $0 < \varepsilon < 1$. The full currents also converge to ω in the coefficient L^1 norm, and their Gauduchon traces converge in total variation.

7.4 Uniform tails and undamped convergence

Removing the weight requires uniform control of the mass on deep sublevels of the approximating potentials. The following condition supplies both a finite limit measure and weak convergence. It can equivalently be expressed by an increasing integrability weight.

Theorem 7.4.1 *Let $u_j, u \in \mathrm{PSH}(X, \omega)$ be nonpositive and suppose $u_j \to u$ in capacity. Assume*

$$\lim_{T\to\infty} \sup_j \mu_{u_j}(\{u_j < -T\}) = 0.$$

Then μ_u is finite and $\mu_{u_j} \rightharpoonup \mu_u$.

Proof Choose T_0 so that every tail beyond T_0 has mass at most one. By Lemma 7.1.1, $\mu_{u_j}(X) \leq C(1+T_0)^{r_n} + 1$. Thus the tail condition already gives a uniform bound for the total masses.

With the cutoffs η_T from (7.2), the fixed-truncation argument with $w = 1$ gives

$$\eta_T(-u_j)\mu_{u_j} \rightharpoonup \eta_T(-u)\mu_u \quad \text{for every fixed } T > 0.$$

The integrals on the left are bounded by this uniform total mass. Since $\eta_T(-u)$ increases to one off the pole set as $T \to \infty$, monotone convergence proves $\mu_u(X) < +\infty$.

For a continuous test function $f : X \to \mathbb{R}$, the error made by inserting $\eta_T(-u_j)$ is at most $\|f\|_\infty \mu_{u_j}(\{u_j < -T\})$, uniformly small for large T. For the limit potential the corresponding error tends to zero because μ_u is finite and its pole set has zero mass. Pass first $j \to \infty$ with T fixed, and then $T \to \infty$. □

Let $\chi : [0, +\infty) \to [1, +\infty)$ be continuous, increasing and unbounded. For a nonpositive potential define

$$J_{\chi,\omega}(u) = \int_X \chi(-u)\, \mathrm{d}\mu_u,$$

with the value $+\infty$ allowed. A uniform bound for this functional forces the sublevel tails to vanish because $\chi(T) \to +\infty$.

Corollary 7.4.1 *Let $u_j, u \in \mathrm{PSH}(X, \omega)$ be nonpositive. If $u_j \to u$ in capacity and $\sup_j J_{\chi,\omega}(u_j) \le C$, then*

$$J_{\chi,\omega}(u) \le C, \qquad \mu_{u_j} \rightharpoonup \mu_u.$$

Proof Since $\chi \ge 1$, the total masses are at most C, and

$$\mu_{u_j}(\{u_j < -T\}) \le C/\chi(T).$$

Thus Theorem 7.4.1 applies. For the integrability assertion, use (7.2) with the continuous function $s \mapsto \chi(s)\eta_T(s)$, which has compact support. The corresponding weighted integrals converge at each fixed T and are at most C. Let $T \to \infty$ by monotone convergence. □

Proposition 7.4.1 *For a family of nonpositive potentials in $\mathrm{PSH}(X, \omega)$, the uniform tail condition in Theorem 7.4.1 is equivalent to a uniform bound on $J_{\chi,\omega}$ for some continuous increasing unbounded $\chi \ge 1$.*

Proof One implication was proved in the preceding corollary. Conversely, write the family as $(u_j)_{j\in J}$, where J is its index set. The tail condition and Lemma 7.1.1 give a uniform bound for the total masses, as in Theorem 7.4.1. Choose numbers $T_k \to +\infty$, with $T_{k+1} > T_k + 1$, such that the mass of $\{-u_j > T_k\}$ is at most 2^{-k} for every member of the family. Let r_k be a continuous nondecreasing function, zero on $[0, T_k]$ and one on $[T_k + 1, +\infty)$. Define

$$\chi(s) = 1 + \sum_{k\ge 1} r_k(s).$$

The sum is locally finite, so χ is continuous, increasing and unbounded. Tonelli's theorem gives

$$J_{\chi,\omega}(u_j) \le \mu_{u_j}(X) + \sum_{k\ge 1} \mu_{u_j}(\{-u_j > T_k\}) \le \sup_j \mu_{u_j}(X) + 1.$$

7.5 Mass concentration

The tail condition excludes mass carried by levels that disappear in the capacity limit. On a curve this mass can concentrate at a pole. When bounded mass fails, it can also persist along smooth potentials converging to zero.

Example 7.5.1 On $X = \mathbb{P}^1$, let z be the standard affine coordinate and let δ_0 denote the unit Dirac measure at its origin. Take the Fubini–Study form

$$\omega = \mathrm{dd^c} \log(1 + |z|^2), \qquad \int_X \omega = 1,$$

and the global nonpositive potential

$$u(z) = \log \frac{|z|^2}{1 + |z|^2}, \qquad u_j = \max(u, -j).$$

At infinity, in the coordinate $\zeta = 1/z$, the function u is $-\log(1 + |\zeta|^2)$, so it is smooth there. The Poincaré–Lelong formula gives

$$\omega + \mathrm{dd^c} u = \delta_0, \qquad \mathrm{MA}^{\mathrm{ext}}_{\omega}(u) = 0.$$

The second identity follows because the extended measure discards the pole set and the current vanishes on its complement.

Each u_j is bounded and ω-plurisubharmonic. Its measure $\mu_j = \omega + \mathrm{dd^c} u_j$ has mass one by Stokes' theorem. On the open set $\{u > -j\}$, locality gives $\mu_j = 0$. Hence

$$\operatorname{supp} \mu_j \subset \{u \le -j\} = \left\{ |z|^2 \le \frac{\mathrm{e}^{-j}}{1 - \mathrm{e}^{-j}} \right\}.$$

These discs shrink to the origin, so $\mu_j \rightharpoonup \delta_0$.

On their supports, $u_j = -j$. Consequently, for $j > T$,

$$\mu_j(\{u_j < -T\}) = 1, \qquad \mathrm{e}^{p u_j} \mu_j = \mathrm{e}^{-pj} \mu_j \rightharpoonup 0 \quad (p > 0).$$

Nevertheless, $u_j \to u$ in capacity. Indeed, outside any neighborhood of the origin the functions eventually agree, and neighborhoods of this capacity-zero point can be chosen with arbitrarily small capacity. Thus bounded mass and capacity convergence do not prevent the undamped measures from acquiring mass on a pole of the limit. The damped limit agrees with the extended measure, as predicted by Theorem 7.3.1.

We return to the fixed compact Hermitian manifold (X, ω).

Corollary 7.5.1 *Assume $v_+(\omega) = +\infty$ and put $V_\omega = \int_X \omega^n$. Fix a real number $b > V_\omega$ and a sequence $0 < \varepsilon_j < 1$ tending to zero. The smooth potentials w_j in Corollary 6.7.2, chosen with mass b, converge to zero in capacity. For every $T > 0$,*

$$\left\|\mathbf{1}_{\{w_j \geq -T\}}(\omega + \mathrm{dd^c} w_j)^n - \omega^n\right\|_{\mathrm{TV}} \longrightarrow 0, \qquad \int_{\{w_j < -T\}} (\omega + \mathrm{dd^c} w_j)^n \longrightarrow b - V_\omega.$$

After passage to a subsequence, there is a positive measure σ of mass $b - V_\omega$ such that

$$(\omega + \mathrm{dd^c} w_j)^n \rightharpoonup \omega^n + \sigma, \qquad \mathbf{1}_{\{w_j < -T\}}(\omega + \mathrm{dd^c} w_j)^n \rightharpoonup \sigma \quad \textit{for every } T > 0.$$

Proof The construction in Corollary 6.7.2 gives $w_j = s_j u_j$, where u_j is smooth and ω-plurisubharmonic, $\sup_X u_j = 0$ and $0 < s_j < \varepsilon_j$. The normalized sublevel estimate (1.14) gives, for every fixed $\delta > 0$ and all sufficiently large j,

$$\mathrm{Cap}_\omega(\{|w_j| > \delta\}) = \mathrm{Cap}_\omega(\{u_j < -\delta/s_j\}) \leq C s_j/\delta \longrightarrow 0.$$

Put $\mu_j = (\omega + \mathrm{dd^c} w_j)^n$. Fix $T > 0$ and write $E_{j,T} = \{w_j \geq -T\}$. The lower bound in Corollary 6.7.2 gives

$$\mathbf{1}_{E_{j,T}} \mu_j \geq \lambda_{j,T} := (1 - \varepsilon_j)^n \mathbf{1}_{E_{j,T}} \omega^n.$$

Since $w_j \to 0$ in L^1, the smooth volume of $X \setminus E_{j,T}$ tends to zero. Hence $\lambda_{j,T} \to \omega^n$ in total variation. For every $p > 0$, Theorem 7.3.1 gives $\int_X \mathrm{e}^{p w_j}\, \mathrm{d}\mu_j \to V_\omega$, and therefore

$$\limsup_j \mu_j(E_{j,T}) \leq \mathrm{e}^{pT} \lim_j \int_X \mathrm{e}^{p w_j}\, \mathrm{d}\mu_j = \mathrm{e}^{pT} V_\omega.$$

Let $p \downarrow 0$. Together with the lower bound this proves $\mu_j(E_{j,T}) \to V_\omega$. Positivity now yields

$$\begin{aligned} \|\mathbf{1}_{E_{j,T}} \mu_j - \omega^n\|_{\mathrm{TV}} &\leq \mu_j(E_{j,T}) - \lambda_{j,T}(X) + \|\lambda_{j,T} - \omega^n\|_{\mathrm{TV}} \\ &\longrightarrow 0. \end{aligned}$$

Subtracting this shallow mass from $\mu_j(X) = b$ proves the tail identity.

The positive measures μ_j have fixed mass on compact X, so a subsequence converges weakly to a measure ν of mass b. The inequality $\mu_j \geq (1 - \varepsilon_j)^n \omega^n$ gives $\nu \geq \omega^n$. Set $\sigma = \nu - \omega^n$. For each $T > 0$, subtract the already established total-variation limit of the shallow restrictions from $\mu_j \rightharpoonup \nu$. This gives the last assertion along the same subsequence for every T. □

The currents in Corollary 7.5.1 also converge in coefficient L^1, and their Gauduchon traces converge in total variation. The argument identifies the mass of σ, but does not determine its support or whether it charges pluripolar sets.

7.6 Damping for a smooth background

For a general smooth background θ, constant truncation may fail to preserve θ-plurisubharmonicity. A larger Hermitian form makes the truncations positive and dominates the measure; the local expansion of the θ-product then gives continuity at each fixed truncation level.

Theorem 7.6.1 *Let θ be any smooth real $(1,1)$-form. For nonpositive $u \in \mathrm{PSH}(X,\theta)$ put $\mu_{\theta,u} = \mathrm{MA}_\theta^{\mathrm{ext}}(u)$. The finite damping, capacity estimates and capacity-continuity statements of this chapter, including Theorem 7.3.1, hold for these measures, with constants depending on θ and a fixed Hermitian metric. In particular, for $p > 0$ and $q > r_n$,*

$$\int_X \mathrm{e}^{pu}\,\mathrm{d}\mu_{\theta,u} \le C(1+p^{-1})^{r_n}, \qquad \int_X (1-u)^{-q}\,\mathrm{d}\mu_{\theta,u} \le C_q.$$

The undamped tail criterion and the integrability criterion also hold with $\mu_{\theta,u}$ in place of μ_u.

Proof Choose $C_0 \ge 1$ such that $\theta \le \Omega := C_0\omega$. Then $u \in \mathrm{PSH}(X,\Omega)$ and Theorem 6.1.1 gives

$$0 \le \mu_{\theta,u} \le \mathrm{MA}_\Omega^{\mathrm{ext}}(u).$$

All the mass, tail and capacity bounds follow from the already proved estimates for Ω. The metric comparison in Section 1.8 gives, for every Borel set $E \subset X$,

$$\mathrm{Cap}_\omega(E) \le \mathrm{Cap}_\Omega(E) \le C_0^n\,\mathrm{Cap}_\omega(E).$$

For continuity, let nonpositive $u_j, u \in \mathrm{PSH}(X,\theta)$ converge in capacity. Use (7.2) with the θ-product. Its truncations are bounded and Ω-plurisubharmonic. The weighted capacity continuity in Sections 1.6 and 1.8 applies term by term to the local expansion of their signed θ-products, with the same bounded quasi-continuous coefficients. The cutoff vanishes at the truncation level, giving exactly the restricted positive measure. The uniform tails supplied by domination remove the cutoff. Domination also supplies the sublevel estimate, so the preceding proofs of the undamped tail and integrability criteria apply verbatim. □

Chapter 8
Unbounded mass on a Gauduchon threefold

The Gauduchon condition cancels the linear term in the mass polynomial. On the Hopf threefold below, the quadratic term is a positive multiple of the Dirichlet energy along the elliptic fibers. We construct smooth potentials with arbitrarily large vertical energy while keeping the perturbed form positive. The heat kernel supplies the concentration; varying its time parameter across the base controls the mixed derivatives, and a function of the base alone restores positivity while preserving the vertical energy. The same formula identifies all bounded potentials of minimum mass through their fiber averages.

Theorem 8.0.1 (Xia–Zhang) *The compact threefold*

$$X = (\mathbb{C}^3 \setminus \{0\})/\langle z \mapsto \mathrm{e}^{-1} z\rangle$$

admits a Gauduchon form ω and smooth real functions V_j such that

$$\omega + \mathrm{dd^c} V_j > 0, \qquad \int_X (\omega + \mathrm{dd^c} V_j)^3 \longrightarrow +\infty.$$

In particular, X does not have the bounded mass property.

8.1 The Hopf metric and its mass

The projective fibration separates the horizontal and vertical parts of the metric. Normalizing the fiber measure to have mass one fixes the constants in the total-mass formula.

On the covering space $\mathbb{C}^3 \setminus \{0\}$, with coordinates $z = (z_0, z_1, z_2)$, put

$$s = \log |z|^2, \qquad \alpha = \mathrm{dd^c} s, \qquad \beta = \mathrm{d}s \wedge \mathrm{d^c} s = \frac{\mathrm{i}}{2\pi} \partial s \wedge \bar{\partial} s, \qquad \omega = \alpha + \beta.$$

The contraction changes s by the constant -2, so these forms descend to X. A closed annulus between two successive radii is a compact fundamental region; this also proves compactness of the quotient. Let ω_{FS} be the Fubini–Study form on $\mathbb{P}^2$, given by $\mathrm{dd^c} \log(1 + |\xi|^2)$ in affine coordinates $\xi = (\xi_1, \xi_2)$. Projectivization gives a holomorphic map

$$\pi : X \longrightarrow \mathbb{P}^2, \qquad \alpha = \pi^* \omega_{\mathrm{FS}}, \qquad \int_{\mathbb{P}^2} \omega_{\mathrm{FS}}^2 = 1.$$

The form α has rank two and is positive transverse to the fibers. The rank-one form β is positive along the vertical line. Their kernels have zero intersection, and therefore $\omega > 0$.

The identities needed below are

$$\alpha^3 = 0, \quad \beta^2 = 0, \quad \mathrm{d}\alpha = 0, \quad \mathrm{dd^c}\beta = -\alpha^2, \quad \mathrm{dd^c}(\omega^2) = 0. \tag{8.1}$$

The first three follow from the definitions and the dimension of the base. For the fourth, differentiate $\beta = \mathrm{i}\partial s \wedge \bar{\partial} s/(2\pi)$; the result is $(\partial\bar{\partial} s)^2/(4\pi^2) = -\alpha^2$. Finally, $\omega^2 = \alpha^2 + 2\alpha \wedge \beta$, so its $\mathrm{dd^c}$ is $-2\alpha^3 = 0$. Thus ω is Gauduchon, although $\mathrm{dd^c}\omega = -\alpha^2 \neq 0$.

On the affine base chart $U_0 = \{[z_0 : z_1 : z_2] \in \mathbb{P}^2 : z_0 \neq 0\} \simeq \mathbb{C}^2$ write

$$\xi_j = z_j/z_0 \quad (j = 1, 2), \qquad w = \log z_0.$$

The logarithm and the contraction change w by elements of $\Lambda = \mathbb{Z} + 2\pi\mathrm{i}\mathbb{Z}$. Hence

$$\pi^{-1}(U_0) \simeq U_0 \times E, \qquad E = \mathbb{C}/\Lambda, \qquad \mathrm{d}A_E = \frac{\mathrm{i}}{4\pi} \mathrm{d}w \wedge \mathrm{d}\bar{w}, \quad \int_E \mathrm{d}A_E = 1.$$

On another base chart the fiber coordinate changes by a holomorphic function of the base. Consequently, the vertical derivative ∂_w and the fiber integral below are globally defined. For a smooth function V on X, write $V_w = \partial V/\partial w$.

Proposition 8.1.1 *For every $V \in C^\infty(X, \mathbb{R})$,*

$$\int_X (\omega + \mathrm{dd^c} V)^3 = 6 + 6 \int_{\mathbb{P}^2} \left(\int_{\pi^{-1}(y)} |V_w|^2 \, \mathrm{d}A_E \right) \omega_{\mathrm{FS}}^2(y). \tag{8.2}$$

This is an identity of integrals; positivity is not required.

Proof Since β restricts to $2\mathrm{d}A_E$ on each fiber,

$$\int_X \omega^3 = 3 \int_X \alpha^2 \wedge \beta = 6.$$

Apply (2.1) and (8.1), then integrate by parts against the closed form α^2:

$$\int_X (\omega + \mathrm{dd^c}V)^3 - \int_X \omega^3 = 3\int_X V\mathrm{dd^c}\omega \wedge \mathrm{dd^c}V$$
$$= -3\int_X V\alpha^2 \wedge \mathrm{dd^c}V = 3\int_X \mathrm{d}V \wedge \mathrm{d^c}V \wedge \alpha^2.$$

Wedging with α^2 removes every horizontal component of $\mathrm{d}V \wedge \mathrm{d^c}V$. Its vertical component is $\mathrm{i}|V_w|^2\mathrm{d}w \wedge \mathrm{d}\bar{w}/(2\pi) = 2|V_w|^2\mathrm{d}A_E$. Fiber integration proves the formula. □

8.2 The radial–vertical block

A potential depending on the base only through its radius has one tangential horizontal coefficient and a two-by-two radial–vertical block. The Schur complement of that block isolates the mixed derivative that must be controlled. Subscripts denote partial derivatives in the displayed variables.

For $\xi \neq 0$ introduce

$$x = \log|\xi|^2, \qquad \mu = \frac{\mathrm{e}^x}{1+\mathrm{e}^x}, \qquad \mu_x = \mu(1-\mu).$$

We use smooth real functions $V : \mathbb{R} \times E \to \mathbb{R}$, written $V = V(x, w)$, that are independent of x near the negative end and constant in both variables near the positive end. These conditions give smooth extension across $\xi = 0$ and over the complement of the affine chart, using the same constant at the positive end. The nonconstant base region may depend on V.

Lemma 8.2.1 *For such a function, at $\xi \neq 0$ the inequality $\omega + \mathrm{dd^c}V > 0$ is equivalent to*

$$1 + V_{w\bar{w}} > 0, \qquad \mu + V_x > 0, \qquad Q[V] := V_{xx} + \mu - \frac{|\mu + V_{x\bar{w}}|^2}{1 + V_{w\bar{w}}} > 0. \tag{8.3}$$

At $\xi = 0$, it is enough and necessary that $1 + V_{w\bar{w}} > 0$.

Proof In these coordinates,

$$s = w + \bar{w} + \log(1 + \mathrm{e}^x), \qquad \partial s = \mathrm{d}w + \mu\partial x,$$

and

$$2\pi\alpha = \mathrm{i}\mu\partial\bar{\partial}x + \mathrm{i}\mu(1-\mu)\partial x \wedge \bar{\partial}x.$$

Choose horizontal vectors of unit length for $2\pi\alpha$, one orthogonal to the radial direction and the other

$$e_r = \frac{\xi_1\partial_{\xi_1} + \xi_2\partial_{\xi_2}}{\sqrt{\mu(1-\mu)}}.$$

Adjoin the vertical vector ∂_w. In this frame the matrix of $2\pi(\omega + \mathrm{dd^c}V)$ is

$$\begin{pmatrix} (\mu + V_x)/\mu & 0 & 0 \\ 0 & (\mu + V_{xx})/[\mu(1-\mu)] & (\mu + V_{x\bar{w}})/\sqrt{\mu(1-\mu)} \\ 0 & (\mu + V_{wx})/\sqrt{\mu(1-\mu)} & 1 + V_{w\bar{w}} \end{pmatrix}.$$

The Schur complement of the last diagonal entry gives exactly (8.3). At $\xi = 0$, the derivatives of V in the base vanish and $\partial s = \mathrm{d}w$, which gives the last assertion. □

A rapid cutoff of a concentrated fiber potential can make the mixed derivative $V_{x\bar{w}}$ too large for this criterion. We therefore spread the concentration by the heat equation before cutting off, so that the fiber derivatives are already exponentially small in the cutoff region.

8.3 Heat potentials and the mixed derivative

A heat potential gives large vertical energy at small times and decays at large times, where it can be cut off. Its lower Hessian bound controls vertical positivity. To control the radial–vertical block as the time changes, we also need an estimate relating the kernel to its logarithmic gradient.

For $t > 0$ define

$$k_t(w) = \frac{2}{t} \sum_{\lambda \in \Lambda} \exp\left(-\frac{|w - \lambda|^2}{t}\right). \tag{8.4}$$

This is a smooth positive periodic function of integral one. Indeed, unfolding the integral reduces it to the Gaussian integral on $\mathbb{C}$, since $\mathrm{d}A_E = \mathrm{d}a\,\mathrm{d}b/(2\pi)$ for $w = a + \mathrm{i}b$. For $(p, q) \in \mathbb{Z}^2$ set

$$\nu_{p,q} = p + \frac{\mathrm{i}q}{2\pi}, \qquad \chi_{p,q}(w) = \mathrm{e}^{2\pi \mathrm{i} p \operatorname{Re} w + \mathrm{i} q \operatorname{Im} w}.$$

These characters form an orthonormal basis of $L^2(E, \mathrm{d}A_E)$. Computing the Fourier coefficients by the same unfolding gives

$$k_t = \sum_{p,q} \mathrm{e}^{-\pi^2 |\nu_{p,q}|^2 t} \chi_{p,q}, \qquad \partial_t k_t = \partial_w \partial_{\bar{w}} k_t. \tag{8.5}$$

All differentiated series converge locally uniformly for $t > 0$.

Lemma 8.3.1 *Let* $0 < \eta < 1/10$ *and define*

$$F_\eta(t,w) = -\eta \int_t^\infty (k_r(w) - 1)\,\mathrm{d}r.$$

Then F_η *is smooth and real, and*

$$(F_\eta)_t = (F_\eta)_{w\bar{w}} = \eta(k_t - 1), \tag{8.6}$$

$$(\log k_t)_{w\bar{w}} \geq -1/t, \tag{8.7}$$

$$\int_E |(F_\eta)_w(t,w)|^2 \mathrm{d}A_E = 2\eta^2 \log(1/t) + O(\eta^2) \quad (0 < t \leq 1). \tag{8.8}$$

The last error is uniform in t *and* η*. For fixed nonnegative integers* a, m *there is also a uniform estimate*

$$\|\partial_t^a F_\eta(t,\cdot)\|_{C^m(E)} \leq C_{a,m}\eta \mathrm{e}^{-t/4} \quad (t \geq 1). \tag{8.9}$$

Proof Integrating the nonconstant Fourier modes gives

$$F_\eta(t,w) = -\frac{\eta}{\pi^2} \sum_{(p,q)\neq(0,0)} \frac{\mathrm{e}^{-\pi^2|\nu_{p,q}|^2 t}}{|\nu_{p,q}|^2} \chi_{p,q}(w).$$

This proves smoothness, reality and (8.6). The least nonzero value of $\pi^2|\nu_{p,q}|^2$ is $1/4$. After any fixed number of differentiations, the additional coefficients have polynomial growth in $|\nu_{p,q}|$. Factoring out $\mathrm{e}^{-t/4}$ for $t \geq 1$ leaves a convergent Gaussian series, proving (8.9).

To prove (8.7), fix a lift of w and give Λ the probability weights

$$P_\lambda = \frac{\mathrm{e}^{-|w-\lambda|^2/t}}{\sum_{\lambda'} \mathrm{e}^{-|w-\lambda'|^2/t}}.$$

Logarithmic differentiation of the Gaussian sum yields

$$t\partial_{\bar{w}} \log k_t = -\sum_\lambda (w-\lambda)P_\lambda,$$

and one more derivative gives

$$(\log k_t)_{w\bar{w}} = -\frac{1}{t} + \frac{1}{t^2}\left(\sum_\lambda |w-\lambda|^2 P_\lambda - \left|\sum_\lambda (w-\lambda)P_\lambda\right|^2\right).$$

The expression in parentheses is a variance and is nonnegative.

Finally, Parseval's identity gives, with $\mathcal{E}_\eta(t)$ denoting the energy in (8.8),

$$\mathcal{E}_\eta(t) = \frac{\eta^2}{\pi^2} \sum_{(p,q)\neq(0,0)} \frac{\mathrm{e}^{-2\pi^2|\nu_{p,q}|^2 t}}{|\nu_{p,q}|^2}, \qquad \mathcal{E}'_\eta(t) = -2\eta^2(k_{2t}(0) - 1).$$

The Gaussian formula at the origin implies

$$k_{2t}(0) = \frac{1}{t} + O\left(t^{-1}\mathrm{e}^{-1/(4t)}\right) \quad (0 < t \leq 1).$$

For this bound, isolate the zero lattice vector and use $\mathrm{e}^{-|\lambda|^2/(2t)} \leq \mathrm{e}^{-1/(4t)}\mathrm{e}^{-|\lambda|^2/4}$ for $\lambda \neq 0$. The error is integrable at zero, and $\mathcal{E}_\eta(1) = O(\eta^2)$. Integrating the derivative between t and 1 proves (8.8). □

The Schur complement subtracts a square involving the mixed derivative. The heat equation produces a compensating positive square after the time parameter is allowed to vary. A bound relating the heat kernel to its logarithmic gradient controls the region where that positive square is small.

Lemma 8.3.2 *For a constant $C_0 \geq 1$ depending only on E,*

$$tk_t(w) \leq C_0 \exp\left(-\frac{|t\partial_{\bar{w}} \log k_t(w)|^2}{C_0 t}\right) \quad (0 < t \leq 1). \tag{8.10}$$

Proof Choose a nearest lattice point λ_0, and put $d = |w-\lambda_0|$ and $s_\lambda = |w-\lambda|^2 - d^2 \geq 0$. The distance d is bounded by the diameter of a fundamental rectangle. The triangle inequality shows that $s_\lambda \geq |\lambda - \lambda_0|^2/4 - C$; outside a fixed finite set it is at least $|\lambda - \lambda_0|^2/8$. Consequently,

$$\sum_\lambda (1 + |w - \lambda|)\mathrm{e}^{-s_\lambda/t} \leq C, \qquad tk_t(w) \leq C\mathrm{e}^{-d^2/t}.$$

The probability formula in the previous proof gives, with $G = |t\partial_{\bar{w}} \log k_t|$,

$$G \leq d + \sum_{\lambda\neq\lambda_0} |w - \lambda|\mathrm{e}^{-s_\lambda/t}.$$

If $d \geq 1/4$, this is at most Cd. If $d < 1/4$, the nonzero lattice distances are at least one and

$$s_\lambda \geq |\lambda - \lambda_0|(|\lambda - \lambda_0| - 2d) \geq \tfrac{1}{2}|\lambda - \lambda_0|^2.$$

Summing the resulting Gaussian bound gives $G \leq C(d + \mathrm{e}^{-c/t})$ in all cases. When $G > 2C\mathrm{e}^{-c/t}$, we have $d \geq G/(2C)$ and can use $tk_t \leq C\mathrm{e}^{-d^2/t}$. In the remaining case G^2/t is bounded, and the same conclusion follows from $tk_t \leq C$, after increasing C_0. □

Lemma 8.3.3 *There is a constant C depending only on E with the following property. For $0 < \kappa < 1/(200C_0)$, $0 < \eta < 1/10$, $0 < t \le 1$ and $1/2 \le \mu \le 1$, define $\sigma > 0$ and the functions h, ζ of $w \in E$ by*

$$\sigma^2 = \frac{2\kappa}{\eta} \log \frac{1}{\eta t}, \qquad h = \eta k_t, \qquad \zeta = \frac{\sigma}{\mu}\, t\partial_{\bar{w}} \log k_t.$$

Then

$$\kappa t k_t - \frac{\mu^2(1-\eta)h}{1-\eta+h}\left|\zeta - \frac{1}{1-\eta}\right|^2 \le C\kappa \exp\left(-\frac{1}{C\sigma^2 t}\right). \tag{8.11}$$

Proof If $\sigma^2 t \ge 1/100$, use $tk_t \le C_0$ and enlarge C. Assume $\sigma^2 t < 1/100$. If $|\zeta - (1-\eta)^{-1}| \ge 1/4$, the subtracted expression is at least $h/[100(1+h)]$. Since $\log(1/(\eta t)) > 2$, the definition of σ gives

$$\kappa t(1+h) \le \frac{\eta}{400} + C_0\kappa\eta < \frac{\eta}{100}.$$

Multiplication by $k_t/(1+h)$ shows that the left side of (8.11) is nonpositive.

Otherwise $|\zeta| \ge 3/4$, so

$$|t\partial_{\bar{w}} \log k_t| = \frac{\mu|\zeta|}{\sigma} \ge \frac{3}{8\sigma}.$$

Apply Lemma 8.3.2 and discard the nonnegative square. This gives (8.11) in the last case as well. □

8.4 Variation of the heat time

The heat time will remain small on a fixed base ball and increase until the fiber potential can be cut off. We choose its rate so that the negative part of the Schur complement has small integral in the radial variable. Fix κ as in Lemma 8.3.3.

All estimates below are for $\eta \downarrow 0$, with κ fixed. Write

$$L = \log(1/\eta), \quad \varepsilon = \eta^{1/\eta^2}, \quad q_0 = \sqrt{\log \frac{1}{\eta\varepsilon}}, \quad a = \sqrt{\frac{\kappa}{2\eta}}.$$

Fix a smooth nondecreasing $\chi : \mathbb{R} \to [0,1]$, independent of η, equal to zero on $(-\infty, 0]$ and one on $[1, +\infty)$, with $\int_0^1 \chi = 1/2$. Define

$$q(x) = q_0 - \int_0^{ax} \chi(s)\,\mathrm{d}s, \qquad t(x) = \eta^{-1}\mathrm{e}^{-q(x)^2}. \tag{8.12}$$

Thus $t = \varepsilon$ on $x \le 0$, and $q = q_0 + 1/2 - ax$ on $x \ge a^{-1}$. Put

$$b = \frac{q_0 - 1/2}{a}, \qquad x_* = \frac{q_0 + 1/2}{a}.$$

For small η, we have $0 < a^{-1} < b < x_*$, with

$$q(b) = 1, \quad q(x_*) = 0, \quad b, x_* = O\big(\sqrt{L/\eta}\big), \quad \eta b = o(1). \tag{8.13}$$

Fix also a smooth cutoff $\vartheta : \mathbb{R} \to [0,1]$, independent of η, zero on $(-\infty, 0]$ and one on $[1, +\infty)$, and define

$$W(x,w) = \vartheta(q(x)) F_\eta(t(x), w).$$

This is $F_\eta(\varepsilon, w)$ on $x \le 0$ and zero on $x \ge x_*$. Moreover,

$$1 + W_{w\bar{w}} = 1 + \vartheta(q)\eta(k_t - 1) > 1 - \eta. \tag{8.14}$$

Lemma 8.4.1 *For the functions just constructed, let*

$$d_\eta(x) = \max_{w \in E} \max\{-Q[W](x,w), 0\}.$$

For all sufficiently small η, the function d_η is continuous, supported in $[0, b]$, and

$$\int_{\mathbb{R}} d_\eta(x)\,\mathrm{d}x = o(1), \qquad \sup_{\mathbb{R}\times E} \max\{-W_x, 0\} = o(1). \tag{8.15}$$

On $[-2,-1] \times E$, one also has $Q[W] \ge 2/27$.

Proof Write $F = F_\eta$ for the heat potential. We first work on $[0, b]$, where $\vartheta(q) = 1$. Set

$$h = \eta k_t, \qquad \sigma = (\log t)_x, \qquad \zeta = \frac{\sigma t}{\mu} \partial_{\bar{w}} \log k_t.$$

The heat equation gives $F_t = h - \eta$, and the chain rule gives

$$W_x = \sigma t F_t, \quad W_{xx} = (\sigma_x + \sigma^2) t F_t + \sigma^2 t^2 F_{tt}, \quad W_{x\bar{w}} = \sigma t F_{t\bar{w}}.$$

Substitute these in $Q[W]$, use $h_t = h((\log k_t)_{w\bar{w}} + |\partial_{\bar{w}} \log k_t|^2)$, and complete the square. The exact result is

$$\begin{aligned} Q[W] = {} & \sigma^2 t\{h[1 + t(\log k_t)_{w\bar{w}}] - \eta\} + \sigma_x t F_t + \mu(1-\mu) - \frac{\mu^2 \eta}{1-\eta} \\ & + \frac{\mu^2(1-\eta)h}{1-\eta+h} \left| \zeta - \frac{1}{1-\eta} \right|^2. \end{aligned} \tag{8.16}$$

In particular, the part $\sigma^2 t h[1 + t(\log k_t)_{w\bar{w}}]$ is nonnegative.

On $[0, a^{-1}]$ we have

$$\sigma = 2aq\chi(ax), \qquad \sigma_x = -\frac{\kappa}{\eta}\chi(ax)^2 + \frac{\kappa}{\eta}q\chi'(ax).$$

Here $q_0 - 1/2 \le q \le q_0$ and $t \le \varepsilon e^{q_0}$. The latter tends to zero faster than any power of η. In particular,

$$\kappa q\chi'(ax)t \le \eta, \qquad \sigma^2 t \le \eta^2.$$

At each point where $\chi(ax) > 0$, apply Lemma 8.3.3 with $\kappa\chi(ax)^2$ in place of κ; its constant is uniform in x. It absorbs the negative term $-\kappa\chi^2 t k_t$ coming from $\sigma_x t F_t$. The remaining part involving χ' is at least $-\eta$, because $F_t \ge -\eta$. The other negative terms in (8.16) are $O(\eta)$; the exponential error is also $O(\eta)$ since $\sigma^2 t \le \eta^2$. Thus $d_\eta \le C\eta$ on this interval. If $\chi = 0$, smooth nonnegativity implies $\chi' = 0$, and directly

$$Q[W] = \mu - \frac{\mu^2}{1 - \eta + h} > 0,$$

since $\mu \le \mu(a^{-1}) < 1 - \eta$ for small η.

On $[a^{-1}, b]$ the formulas simplify to

$$\sigma = 2aq, \qquad \sigma_x = -\kappa/\eta, \qquad \mathrm{d}x = -\frac{\mathrm{d}q}{a} = \sqrt{\frac{\eta}{2\kappa}}\frac{\mathrm{d}t}{tq}. \tag{8.17}$$

For $t \le 1$, use Lemma 8.3.3 in (8.16) to obtain

$$d_\eta(x) \le 2\eta + 2\kappa q^2 t + C\kappa \exp\left(-\frac{\eta}{C\kappa q^2 t}\right). \tag{8.18}$$

Indeed, $-\eta\sigma^2 t = -2\kappa q^2 t$ and $\sigma_x t F_t = -\kappa t k_t + \kappa t$; these are exactly the two terms for which the scale was chosen. When $t \ge 1$, use the exponential derivative estimates (8.9) directly in the definition of Q. The denominator stays above $1 - \eta$, and the chain rule gives

$$d_\eta(x) \le C\kappa q^2 t^2 e^{-t/4}. \tag{8.19}$$

The perturbing derivatives are bounded by constants times $\eta e^{-t/4}$, $\eta\sigma t e^{-t/4}$ and $\eta[(|\sigma_x| + \sigma^2)t + \sigma^2 t^2]e^{-t/4}$. Since $q, t \ge 1$ here, each is bounded by a constant times the right side of (8.19). Since $1 + W_{w\bar{w}} \ge 1 - \eta$, subtracting the two quotients gives

$$|Q[W] - Q[0]| \le |W_{xx}| + C\big(|W_{x\bar{w}}| + |W_{x\bar{w}}|^2 + |W_{w\bar{w}}|\big).$$

The square of the mixed derivative is at most $C\kappa\eta q^2 t^2 e^{-t/2}$, so it satisfies the same bound. Since $Q[0] > 0$, this proves (8.19).

We now integrate the bounds for the negative part. The constant 2η and the initial interval contribute $O(\eta b) = o(1)$. In the small-time region $q \ge \sqrt{L}$ and $t = \eta^{-1}e^{-q^2}$. Therefore

$$\int 2\kappa q^2 t \,\mathrm{d}x \le \frac{2\kappa}{a\eta}\int_{\sqrt{L}}^{\infty} q^2 e^{-q^2}\,\mathrm{d}q \le C\sqrt{\eta L} = o(1).$$

For the exponential term the integral is at most

$$\frac{C\kappa}{a}\int_{\sqrt{L}}^{q_0}\exp\left(-\frac{\eta^2 e^{q^2}}{C\kappa q^2}\right)\mathrm{d}q.$$

The interval $[\sqrt{L}, 2\sqrt{L}]$ has length $O(\sqrt{L})$. For $q \geq 2\sqrt{L}$ and sufficiently small η, $\eta^2 e^{q^2}/q^2 \geq e^{q^2/4}$. The integrand on this remaining interval is bounded by $\exp(-e^{q^2/4}/(C\kappa))$, which is integrable on $[1, +\infty)$. The prefactor is $O(\sqrt{\eta})$, and the initial interval contributes $O(\sqrt{\eta L})$, so the full expression is $o(1)$. In the large-time region, (8.17) gives

$$\int d_\eta(x)\,\mathrm{d}x \leq C\sqrt{\eta}\int_1^{1/(e\eta)} qte^{-t/4}\,\mathrm{d}t \leq C\sqrt{\eta L}\int_1^{\infty} te^{-t/4}\,\mathrm{d}t = o(1).$$

This proves the integral assertion on $[0, b]$.

The negative slope has a uniform small bound as well. For $t \leq 1$, $F_t \geq -\eta$ and $0 \leq \sigma \leq 2aq$, so

$$\max\{-W_x, 0\} \leq \sqrt{2\kappa\eta}\,t\sqrt{\log\frac{1}{\eta t}} \leq C\sqrt{\eta L}.$$

The last inequality follows because $t\sqrt{L + \log(1/t)}$ is increasing on $(0, 1]$ when $L > 1$. For $t \geq 1$ and $q \geq 1$, use $|W_x| \leq C\sqrt{\eta}\,qte^{-t/4}$ and $q \leq \sqrt{L}$.

It remains to check the ends, including the cutoff region. If $b \leq x \leq x_*$, then $0 \leq q \leq 1$ and $1/(e\eta) \leq t \leq 1/\eta$. The derivatives of the cutoffs and scale have at most polynomial growth in η^{-1}. Thus (8.9) implies

$$\sup_w\big(|W_x| + |Q[W] - Q[0]|\big) \leq Ce^{-1/(5e\eta)}.$$

But $Q[0] = \mu(1-\mu) \geq e^{-x_*}/4$ on this interval, and (8.13) shows that the error is $o(e^{-x_*})$. Hence $Q[W] > 0$ there. For $x \geq x_*$, $W = 0$ and positivity is immediate. For $x \leq 0$, $W_x = 0$ and

$$Q[W] \geq \mu\left(1 - \frac{\mu}{1-\eta}\right) > 0,$$

since $\mu \leq 1/2$. On $[-2, -1]$, use $1/9 \leq \mu \leq 3/10$ and $1-\eta \geq 9/10$ to obtain the lower bound $2/27$. Finally $Q[W]$ is continuous with positive denominator; its negative part has a continuous maximum over compact E. This proves the support, continuity and all remaining assertions. □

8.5 A base correction and unbounded mass

Adding a function of x changes $Q[W]$ by its second derivative and leaves the vertical energy unchanged. The integral bound for d_η lets us choose this derivative above the defect, then balance its integral in an interval where $Q[W]$ is uniformly positive. The correction has small first derivative, so the remaining horizontal coefficient stays positive as well.

Proof (of Theorem 8.0.1) Choose a smooth function $r_\eta \geq 0$ supported in $(-1/2, b+1/2)$ with

$$r_\eta > d_\eta \text{ on } [0,b], \qquad m_\eta := \int_{\mathbb{R}} r_\eta \,\mathrm{d}x \leq \int_{\mathbb{R}} d_\eta \,\mathrm{d}x + \eta = o(1).$$

Such a majorant exists by uniform approximation of the compactly supported continuous function d_η, followed by adding a positive bump on $[0,b]$; for each fixed η the approximation error can be chosen as small as the displayed integral requires. Fix a smooth $\psi \geq 0$ supported in $(-2,-1)$ with integral one and set

$$g_\eta(x) = \int_{-\infty}^{x} (x-s)(r_\eta(s) - m_\eta \psi(s))\,\mathrm{d}s, \qquad V_\eta = W + g_\eta.$$

The negative part of g_η'' is placed to the left of the positive part, and their integrals agree. Consequently,

$$-m_\eta \leq g_\eta' \leq 0, \qquad g_\eta' = 0 \text{ for } x \leq -2 \text{ and for } x \geq b + 1/2.$$

Thus g_η is zero at the negative end and constant at the positive end. The function V_η has the endpoint properties required for smooth extension to X.

Since g_η is independent of w, the Schur complement changes by exactly its second derivative:

$$Q[V_\eta] = Q[W] + r_\eta - m_\eta \psi.$$

This is positive on $[0,b]$ by the strict majorant property. On $(-2,-1)$ it is at least $2/27 - m_\eta \|\psi\|_\infty > 0$. Elsewhere $Q[W] > 0$, $\psi = 0$ and $r_\eta \geq 0$. The vertical coefficient remains above $1-\eta$ by (8.14). For the remaining coefficient, if $x \leq -2$ both W_x and g_η' vanish; if $-2 \leq x \leq 0$, then $W_x = 0$ and $\mu \geq \mu(-2) > 0$; and if $x \geq 0$, then $\mu \geq 1/2$ and $W_x \geq -o(1)$. In all cases $\mu + (V_\eta)_x > 0$ for sufficiently small η. The positivity lemma therefore gives $\omega + \mathrm{dd^c} V_\eta > 0$ on X.

On the fixed ball $B = \{|\xi|^2 < \mathrm{e}^{-3}\} \Subset U_0$, one has $V_\eta = F_\eta(\varepsilon, w)$, including at $\xi = 0$. Thus

$$\int_{\pi^{-1}(y)} |(V_\eta)_w|^2 \mathrm{d}A_E = 2\eta^2 \log(1/\varepsilon) + O(\eta^2) = 2\log(1/\eta) + O(\eta^2) \longrightarrow +\infty$$

uniformly for $y \in B$. Since $\int_B \omega_{\mathrm{FS}}^2 > 0$, the mass identity (8.2) proves the theorem along any sequence $\eta \downarrow 0$. □

8.6 Minimum mass and fiber averaging

Averaging along the elliptic fibers removes the variation measured by (8.2). The fiber Poincaré inequality bounds the distance to this average by the excess mass. Bounded regularization preserves that inequality, so it identifies all bounded minimizers as well as their stability.

Let E act on X by scalar multiplication: $\tau_a[z] = [\mathrm{e}^a z]$ for $a \in E$. The action preserves α, β and ω, and $\mathrm{d}A_E$ is its probability Haar measure. For a bounded Borel function $u : X \to \mathbb{R}$, set

$$\mathrm{d}\nu = \frac{\omega^3}{6} = \frac{1}{2}\alpha^2 \wedge \beta, \qquad Au(p) = \int_E u(\tau_a(p))\,\mathrm{d}A_E(a) \quad (p \in X).$$

Thus Au is constant on each fiber, and ν is a probability measure whose fiber measures are $\mathrm{d}A_E$ and whose base measure is ω_{FS}^2.

Proposition 8.6.1 *For every $u \in \mathrm{PSH}(X,\omega) \cap L^\infty(X)$, the average Au is the pullback of a bounded ω_{FS}-plurisubharmonic function on $\mathbb{P}^2$, and*

$$\|u - Au\|^2_{L^2(X,\nu)} \le \frac{2}{3}\left(\int_X (\omega + \mathrm{dd^c}u)^3 - 6\right). \tag{8.20}$$

The coefficient $2/3$ is optimal, even for smooth u with $\omega + \mathrm{dd^c}u > 0$. Moreover,

$$\int_X (\omega + \mathrm{dd^c}u)^3 = 6 \quad \Longleftrightarrow \quad u = \pi^* v \quad \textit{for some } v \in \mathrm{PSH}(\mathbb{P}^2, \omega_{\mathrm{FS}}) \cap L^\infty(\mathbb{P}^2).$$

Proof First suppose that u is smooth and $\omega + \mathrm{dd^c}u > 0$. Averaging the positive forms $\tau_a^*(\omega + \mathrm{dd^c}u)$ gives $\omega + \mathrm{dd^c}Au > 0$. Write $Au = \pi^* v$. At each point, the kernel of ∂s maps isomorphically to the tangent space of the base. Restricting $\pi^*(\omega_{\mathrm{FS}} + \mathrm{dd^c}v) + \beta$ to this kernel shows that $\omega_{\mathrm{FS}} + \mathrm{dd^c}v > 0$.

For a smooth function $f : E \to \mathbb{R}$, the nonconstant Fourier characters in (8.5), indexed by $(p,q) \in \mathbb{Z}^2 \setminus \{(0,0)\}$, have eigenvalues $\pi^2 p^2 + q^2/4 \ge 1/4$ for $-\partial_w \partial_{\bar{w}}$. Parseval's identity therefore gives

$$\int_E \left| f - \int_E f\,\mathrm{d}A_E \right|^2 \mathrm{d}A_E \le 4 \int_E |f_w|^2\,\mathrm{d}A_E.$$

Apply this inequality on every fiber and integrate against ω_{FS}^2. The mass identity (8.2) gives (8.20).

For bounded u, choose smooth $u_j \downarrow u$ with $\omega + \mathrm{dd^c}u_j > 0$ by Theorem 1.11.1. The sequence is uniformly bounded. Its averages decrease pointwise to Au by dominated convergence. The corresponding smooth base potentials v_j decrease to a bounded ω_{FS}-plurisubharmonic function v, so $Au = \pi^* v$. Dominated convergence gives $u_j \to u$ and $Au_j \to Au$ in $L^2(X,\nu)$, while bounded Bedford–Taylor continuity gives convergence of the total masses. Thus (8.20) passes to u.

If the mass is 6, the inequality gives $u = Au$ almost everywhere. Since both functions are bounded and quasi-plurisubharmonic, Lemma 1.3.4 gives equality everywhere. Conversely, if $u = \pi^* v$ with bounded $v \in \mathrm{PSH}(\mathbb{P}^2, \omega_{\mathrm{FS}})$, choose smooth $v_j \downarrow v$ with $\omega_{\mathrm{FS}} + \mathrm{dd^c} v_j > 0$ by Theorem 1.11.1. The pullbacks decrease to u and are uniformly bounded. They are strictly ω-plurisubharmonic because the base forms are positive and β is positive on the vertical line. Their vertical derivatives vanish, so (8.2) gives mass 6 for each of them. Bounded Bedford–Taylor continuity gives the same mass for u.

To check the constant, choose a nonzero real $\rho \in C_c^\infty(U_0)$. For $\delta \in \mathbb{R}$, the function

$$u(\xi, w) = \delta \rho(\xi) \cos(\operatorname{Im} w)$$

extends by zero to a smooth function on X. For sufficiently small nonzero real δ, it satisfies $\omega + \mathrm{dd^c} u > 0$. Its fiber average is zero, and its only fiber frequencies are $(p, q) = (0, \pm 1)$, with eigenvalue $1/4$. Equality holds in the fiber inequality and hence in (8.20), with nonzero left side. No smaller coefficient can hold. □

Corollary 8.6.1 *The Hopf metric satisfies*

$$v_-(\omega) = 6, \qquad v_+(\omega) = +\infty.$$

The lower bound follows from (8.20) and is attained by constants; Theorem 8.0.1 gives the infinite supremum.

In particular, for any sequence of bounded potentials whose masses tend to 6, the differences from their fiber averages tend to zero in $L^2(X, \nu)$. This conclusion controls fiber variation; the averaged potentials still vary in the bounded potential space of the base.

For nonpositive $u \in \mathrm{PSH}(X, \omega)$ on this threefold, Theorem 7.3.1 gives finite measures for the critical weight $(1 - u)^{-1}[\log(\mathrm{e} - u)]^{-a}$, $a > 1$, with uniform tails and capacity continuity. The optimal damping rate remains undetermined here.

Part II
Monge–Ampère equations

Chapter 9
Smooth Monge–Ampère equations

Let X be a connected compact complex manifold of dimension $n \geq 1$. Fix a Hermitian form ω and a smooth positive volume form $\mathrm{d}V$ on X. Given $F \in C^\infty(X, \mathbb{R})$, we seek a smooth real potential u and a scalar $b \in \mathbb{R}$ satisfying

$$\omega_u^n = \mathrm{e}^{F+b}\mathrm{d}V, \qquad \omega_u > 0. \tag{9.1}$$

Here $\omega_u = \omega + \mathrm{dd^c}u$, and adding a constant to u does not change the equation. For a closed background, integration determines b from the prescribed density. For a Hermitian background, the total mass can depend on u, so the potential and the scalar must be found together. The adjoint probability supplies their linearized compatibility condition. Weighted integration and a trace estimate give the compactness needed to continue solutions from the background density; the same adjoint probability then governs the variation of b.

For $n \geq 2$, prescribing the volume of a Gauduchon metric requires a different potential variation: we deform its $(n-1)$st power within an Aeppli class. The resulting first-order torsion terms have a specific matrix structure. Their coefficients make the derivative estimates possible, and rescaling reduces the gradient bound to a Liouville theorem on $\mathbb{C}^n$.

9.1 Scalar bounds and uniqueness

Extrema of the potential bound the normalization scalar in terms of the prescribed data. Applied to the difference of two solutions, the same argument identifies their scalars; ellipticity then determines the potential up to a constant. The scalar bounds also make the continuity path uniform.

Theorem 9.1.1 (Tosatti–Weinkove) *For every smooth real F, (9.1) has a unique pair $(u, b) \in C^\infty(X, \mathbb{R}) \times \mathbb{R}$ with $\sup_X u = 0$.*

In dimension one, integration determines $\mathrm{e}^b = \int_X \omega / \int_X \mathrm{e}^F \mathrm{d}V$, and the remaining equation for u is the usual Poisson equation. Thus the statement holds in that dimension as well.

Proposition 9.1.1 *Every solution of* (9.1) *satisfies*

$$\inf_X \left(\log \frac{\omega^n}{\mathrm{d}V} - F \right) \le b \le \sup_X \left(\log \frac{\omega^n}{\mathrm{d}V} - F \right).$$

Proof At a maximum point x_+ of u, the Hermitian form $\mathrm{dd^c}u$ is nonpositive. Hence $\omega_u^n(x_+) \le \omega^n(x_+)$, and $b \le \log(\omega^n/\mathrm{d}V)(x_+) - F(x_+)$. At a minimum point the reverse argument gives the lower bound. □

When $\mathrm{d}\omega = 0$, integration gives the familiar explicit formula

$$b = \log \frac{\int_X \omega^n}{\int_X \mathrm{e}^F \mathrm{d}V}. \tag{9.2}$$

The same formula holds whenever the mass-invariance criterion of Theorem 2.2.2 is satisfied. In general, integrating the equation gives only

$$\mathrm{e}^b = \frac{\mathrm{M}_\omega(u)}{\int_X \mathrm{e}^F \mathrm{d}V},$$

whose numerator depends on the unknown potential.

Proposition 9.1.2 *Suppose* (u, b) *and* (v, c) *are smooth solutions of* (9.1) *with the same* F *and* $\mathrm{d}V$. *Then* $b = c$ *and* $u - v$ *is constant.*

Proof At a maximum of $u - v$, one has $\omega_u \le \omega_v$, so $\mathrm{e}^b \le \mathrm{e}^c$. At a minimum one obtains $\mathrm{e}^b \ge \mathrm{e}^c$. Therefore $b = c$.

Put $h = u - v$ and $\eta_t = \omega_v + t\mathrm{dd^c}h$ for $0 \le t \le 1$. This is a family of strictly positive forms. Subtracting the logarithms of the two equations and integrating the derivative along the segment gives

$$0 = \int_0^1 n \, \frac{\mathrm{dd^c}h \wedge \eta_t^{n-1}}{\eta_t^n} \, \mathrm{d}t.$$

The right side is a linear elliptic operator applied to h. Its coefficients are smooth and uniformly positive on the compact manifold, and it has no zeroth-order term. The strong maximum principle implies that h is constant. □

9.2 The potential bound from weighted integration

An upper bound for the density controls the normalized potential. Integrating against e^{-pu} produces a gradient term and mixed weighted masses; a finite recurrence

absorbs the latter together with the torsion error. Sobolev iteration then gives a set of uniformly positive volume near the minimum, and normalized L^1 compactness bounds that minimum. Norms below use the fixed metric ω.

Lemma 9.2.1 *There is a constant C depending on ω such that, for every positive $(1,1)$-form η, every smooth real f, every $\delta > 0$, and $0 \le k \le n-2$,*

$$|\mathrm{d}^{\mathrm{c}} f \wedge \mathrm{d}\omega \wedge \eta^k \wedge \omega^{n-k-2}| \le \frac{C}{\delta}\, \mathrm{d}f \wedge \mathrm{d}^{\mathrm{c}} f \wedge \eta^k \wedge \omega^{n-k-1} + C\delta\, \eta^k \wedge \omega^{n-k}.$$

The absolute value means that of the coefficient relative to the positive volume form ω^n.

Proof At the point in question choose an ω-unitary frame in which η is diagonal, with positive eigenvalues $\lambda_1, \dots, \lambda_n$. Expand η^k as a sum over k-element subsets I, with coefficient a positive dimensional multiple of $\lambda_I = \prod_{i \in I} \lambda_i$. A surviving coefficient from $\mathrm{d}^{\mathrm{c}} f \wedge \mathrm{d}\omega$ has the form of a bounded coefficient of $\mathrm{d}\omega$ times one first derivative f_a, where $a \notin I$. Young's inequality bounds its absolute value by

$$C\lambda_I |f_a| \le (C/\delta)\lambda_I |f_a|^2 + C\delta\lambda_I.$$

The sum of the first terms is bounded by a fixed multiple of the first positive form in the assertion: its diagonal expansion contains precisely the terms $\lambda_I |f_a|^2$ with $a \notin I$. The second positive form contains all the λ_I. Finitely many indices and the uniform bound for $\mathrm{d}\omega$ give a uniform C. □

Lemma 9.2.2 *Suppose u is smooth, $\eta = \omega + \mathrm{dd}^{\mathrm{c}} u > 0$, and $\eta^n \le \Lambda\omega^n$ for a fixed $\Lambda > 0$. There are constants $p_0 \ge 1$ and C depending only on ω and Λ such that*

$$\int_X \mathrm{d}(\mathrm{e}^{-pu/2}) \wedge \mathrm{d}^{\mathrm{c}}(\mathrm{e}^{-pu/2}) \wedge \omega^{n-1} \le Cp \int_X \mathrm{e}^{-pu}\omega^n \quad (p \ge p_0).$$

Proof For a fixed real $p \ge 1$, define M_k for $0 \le k \le n$ and J_k for $0 \le k \le n-1$ by

$$M_k = \int_X \mathrm{e}^{-pu}\eta^k \wedge \omega^{n-k}, \qquad J_k = \int_X \mathrm{e}^{-pu}\mathrm{d}u \wedge \mathrm{d}^{\mathrm{c}} u \wedge \eta^k \wedge \omega^{n-k-1}, \quad J = \sum_{k=0}^{n-1} J_k.$$

All these numbers are nonnegative. Let $T = \sum_{k=0}^{n-1} \eta^k \wedge \omega^{n-k-1}$. Since $\mathrm{d}\eta = \mathrm{d}\omega$, differentiation gives

$$\mathrm{d}T = n \sum_{k=0}^{n-2} \eta^k \wedge \omega^{n-k-2} \wedge \mathrm{d}\omega.$$

The sum is empty when $n = 1$. Stokes' theorem gives the exact identity

$$\int_X e^{-pu}(\eta^n - \omega^n) = \int_X e^{-pu} dd^c u \wedge T$$
$$= pJ + \int_X e^{-pu} d^c u \wedge dT.$$

The density assumption bounds the left side above by $(\Lambda + 1)M_0$. Apply Lemma 9.2.1 to the last integral, and choose $\delta = K/p$ for a sufficiently large fixed K. The contribution proportional to J/δ is then at most $pJ/2$. After absorption we obtain

$$pJ \le CM_0 + \frac{C}{p}\sum_{k=0}^{n-2} M_k. \tag{9.3}$$

This estimate alone is not enough, because the mixed weighted masses on the right have not been bounded.

For $1 \le k \le n-2$, a second integration by parts gives

$$M_k - M_{k-1} = pJ_{k-1} + R_k, \qquad R_k = \int_X e^{-pu} d^c u \wedge d(\eta^{k-1} \wedge \omega^{n-k}). \tag{9.4}$$

The derivative in R_k has two terms, with coefficients $k-1$ and $n-k$. Apply the pointwise estimate with $\delta = 1/p$ to each of them. Omitting nonexistent negative indices yields

$$|R_k| \le Cp(J_{k-2} + J_{k-1}) + \frac{C}{p}(M_{k-2} + M_{k-1}), \qquad M_{-1} = J_{-1} = 0.$$

Consequently, for $p \ge 1$,

$$M_k \le C(M_{k-1} + M_{k-2} + pJ).$$

Induction over the finitely many values of k proves

$$\sum_{k=0}^{n-2} M_k \le C(M_0 + pJ).$$

For $n = 2$ this is immediate, and for $n = 1$ the sum is empty. Inserting this bound in (9.3) gives $pJ \le CM_0 + C'J$. Taking $p \ge 2C'$ absorbs the last term and proves $pJ \le CM_0$.

Finally,

$$\int_X d(e^{-pu/2}) \wedge d^c(e^{-pu/2}) \wedge \omega^{n-1} = \frac{p^2}{4} J_0 \le \frac{p^2}{4} J \le CpM_0.$$

Theorem 9.2.1 *For each $\Lambda > 0$ there is C_Λ such that every smooth u with*

$$\omega + \mathrm{dd^c} u > 0, \qquad \sup_X u = 0, \qquad (\omega + \mathrm{dd^c} u)^n \le \Lambda \omega^n$$

satisfies $\|u\|_\infty \le C_\Lambda$.

Proof Let $\mathrm{d}\mu = \omega^n / \int_X \omega^n$. The usual Sobolev inequality for the fixed Riemannian metric of ω gives, for some $\beta > 1$ and every smooth real test function w,

$$\left(\int_X |w|^{2\beta} \, \mathrm{d}\mu \right)^{1/\beta} \le C \left(\int_X \mathrm{d}w \wedge \mathrm{d^c} w \wedge \omega^{n-1} + \int_X |w|^2 \, \mathrm{d}\mu \right).$$

We may take $\beta = n/(n-1)$ if $n \ge 2$, and for example $\beta = 2$ if $n = 1$. Choose p_0 as in Lemma 9.2.2. Apply the Sobolev inequality to $w = \mathrm{e}^{-pu/2}$ and use that lemma. With $Z = \mathrm{e}^{-u}$ we get

$$\|Z\|_{L^{p\beta}(\mu)} \le (Cp)^{1/p} \|Z\|_{L^p(\mu)} \quad (p \ge p_0).$$

Iterate at $p = p_0 \beta^j$. The product of the constants is finite because $\sum_j (1 + \log(p_0 \beta^j))/(p_0 \beta^j) < \infty$. Hence, for a uniform $C_1 \ge 1$ and $m = \inf_X u$,

$$\mathrm{e}^{-p_0 m} \le C_1 \int_X \mathrm{e}^{-p_0 u} \, \mathrm{d}\mu. \tag{9.5}$$

This still involves the unknown potential on the right. Divide by $\mathrm{e}^{-p_0 m}$ and set $A = p_0^{-1} \log(2C_1)$. The integrand $\mathrm{e}^{-p_0(u-m)}$ is at most one everywhere and at most $(2C_1)^{-1}$ outside $S = \{u \le m + A\}$. Therefore (9.5) implies

$$\mu(S) \ge \frac{1}{2C_1} =: \delta > 0.$$

By Theorem 1.4.1, $\int_X (-u) \, \mathrm{d}\mu \le C_0$. On S we have $-u \ge -m - A$, so

$$C_0 \ge \int_X (-u) \, \mathrm{d}\mu \ge \delta(-m - A)$$

when $-m \ge A$; otherwise the desired bound is immediate. Thus $-m \le A + C_0/\delta$, proving the theorem. □

9.3 The linearized equation and smooth dependence

The linearized operator L_u of (4.6) annihilates constants, and its range is the space of functions with zero mean under the adjoint probability μ_u of Proposition 4.5.1. Including the scalar as an unknown removes this obstruction. In dimension one, $\mu_u = \omega_u / \int_X \omega_u$; the compatibility condition follows from Stokes' theorem.

The normalization $\sup_X u = 0$ is convenient for compactness, but need not vary differentiably with the data. For differentiating solutions, fix instead the linear normalization

$$\int_X u \, \mathrm{d}V = 0.$$

Changing between these normalizations only adds a constant to u and leaves b unchanged.

Proposition 9.3.1 *With this linear normalization, the solution (u, b) of (9.1) depends smoothly on F in Hölder spaces, and hence along every smooth finite-dimensional family of smooth data.*

Proof Fix an integer $k \geq 0$ and a real exponent $0 < \alpha < 1$. Consider the map

$$(u, b, F) \longmapsto \log \frac{(\omega + \mathrm{dd^c} u)^n}{\mathrm{d}V} - F - b$$

on the open set of potentials in $C^{k+2,\alpha}$ with $\omega_u > 0$, restricted to $\int_X u \, \mathrm{d}V = 0$. Its derivative in (u, b) is

$$(v, a) \longmapsto L_u v - a.$$

Given $h \in C^{k,\alpha}$, integrating $L_u v - a = h$ against μ_u forces $a = -\int_X h \, \mathrm{d}\mu_u$. For this value, the equation for v satisfies the compatibility condition of Proposition 4.5.1, and has a unique solution with zero $\mathrm{d}V$-mean. Elliptic estimates make this inverse bounded. The Banach-space implicit function theorem gives a local solution map around the given positive solution; this step does not require the global existence theorem. Uniqueness identifies the local maps wherever they overlap, and elliptic regularity allows arbitrary k for smooth data. □

9.4 Second derivatives and ellipticity

The determinant equation controls the product of the eigenvalues, so an upper bound for their sum gives uniform ellipticity. We obtain that trace bound by completing a square in the differentiated equation. The remaining torsion term is first order in the trace and can be absorbed at the maximum of a suitable function of the trace and the potential.

In holomorphic coordinates $(z^1, \ldots, z^n)$, subscripts denote coordinate derivatives, inverse matrices carry upper indices, and repeated indices run from 1 to n. Write

$$\omega = \frac{\mathrm{i}}{2\pi} g_{i\bar{j}} \, \mathrm{d}z^i \wedge \mathrm{d}\bar{z}^j, \qquad \omega_u = \frac{\mathrm{i}}{2\pi} h_{i\bar{j}} \, \mathrm{d}z^i \wedge \mathrm{d}\bar{z}^j, \qquad h_{i\bar{j}} = g_{i\bar{j}} + u_{i\bar{j}}.$$

For a smooth real function v, set $\Delta_g v = g^{i\bar{j}} v_{i\bar{j}}$ and $\Delta_h v = h^{i\bar{j}} v_{i\bar{j}}$. Thus $\Delta_h = L_u$ and $\Delta_h u = n - \mathrm{tr}_h \, g$. Write also $|\partial v|_h^2 = h^{i\bar{j}} v_i v_{\bar{j}}$. Let ∇ be the Chern connection of the fixed metric g, with

$$\Gamma^p_{ik} = g^{p\bar q}\partial_i g_{k\bar q}, \qquad T^p_{ik} = \Gamma^p_{ik} - \Gamma^p_{ki}, \qquad \tau_k = \sum_i T^i_{ik}.$$

All bounds for this connection and its curvature depend only on g.

Lemma 9.4.1 *Suppose $h = g + (u_{i\bar j}) > 0$, and put*

$$S = \operatorname{tr}_g h, \qquad B = \operatorname{tr}_h g, \qquad G = \log \frac{\det h}{\det g}.$$

There is C_g such that, at every point where $S \geq 1$,

$$\Delta_h \log S \geq \frac{2}{S^2}\operatorname{Re}\bigl(h^{k\bar q}\tau_k S_{\bar q}\bigr) + \frac{\Delta_g G}{S} - C_g B. \tag{9.6}$$

Proof The identity $\mathrm{d}\omega_u = \mathrm{d}\omega$ is equivalent to

$$\nabla_i h_{k\bar j} - \nabla_k h_{i\bar j} = T^p_{ik}(g_{p\bar j} - h_{p\bar j}), \tag{9.7}$$

and its complex conjugate. This also follows directly by cancelling the third derivatives of u.

For tensors with two holomorphic and one antiholomorphic lower indices, use the positive norm

$$\|A\|^2 = g^{i\bar\ell}h^{k\bar q}h^{p\bar j}A_{ik\bar j}\overline{A_{\ell q\bar p}}.$$

Set $A_{ik\bar j} = \nabla_i h_{k\bar j}$ and $D_{ik\bar j} = T^p_{ik}h_{p\bar j}$. We first claim

$$\Delta_h S = \|A + D\|^2 + \Delta_g G + E, \qquad |E| \leq C_g(1 + S)B. \tag{9.8}$$

Here is the calculation and the bound for its remainder. Start with $\Delta_h S = h^{i\bar j}g^{k\bar\ell}\nabla_i\nabla_{\bar j}h_{k\bar\ell}$. Switch $\bar j, \bar\ell$ by the conjugate of (9.7), commute $\nabla_i, \nabla_{\bar\ell}$, and then switch i, k by (9.7). The terms with a derivative of h are

$$h^{i\bar j}g^{k\bar\ell}\nabla_{\bar\ell}\nabla_k h_{i\bar j} - 2\operatorname{Re}\bigl(h^{i\bar j}g^{k\bar\ell}T^p_{ik}\nabla_{\bar\ell}h_{p\bar j}\bigr).$$

In making the first switch there is also a term $-h^{i\bar j}g^{k\bar\ell}\overline{T^q_{j\ell}}\nabla_i h_{k\bar q}$. Using (9.7) on it is what gives one of the two conjugate derivative terms just displayed, together with

$$h^{i\bar j}g^{k\bar\ell}T^p_{ik}\overline{T^q_{j\ell}}(h_{p\bar q} - g_{p\bar q}).$$

The part containing h is $\|D\|^2$. Differentiating the logarithmic determinant gives

$$\Delta_g G = h^{i\bar j}g^{k\bar\ell}\nabla_{\bar\ell}\nabla_k h_{i\bar j} - \|A\|^2.$$

The two derivative terms left over are $2\operatorname{Re}\langle A, D\rangle$, using the antisymmetry of T. These identities give (9.8).

More explicitly, let

$$C_{i\bar{\ell}k\bar{j}} = (\nabla_i \nabla_{\bar{\ell}} - \nabla_{\bar{\ell}} \nabla_i) h_{k\bar{j}}.$$

The remainder is the contraction by $h^{i\bar{j}} g^{k\bar{\ell}}$ of

$$\begin{aligned}(\nabla_i \overline{T^q_{j\ell}})(g_{k\bar{q}} - h_{k\bar{q}}) + (\nabla_{\bar{\ell}} T^p_{ik})(g_{p\bar{j}} - h_{p\bar{j}}) \\ + C_{i\bar{\ell}k\bar{j}} - T^p_{ik} \overline{T^q_{j\ell}} g_{p\bar{q}}.\end{aligned}$$

The commutator C is curvature acting linearly on h; this follows by inserting $\nabla = \partial + \Gamma$ and cancelling the derivatives of h. In a g-unitary frame diagonalizing h, every coefficient inside this remainder is bounded by $C_g(1 + S)$. The contraction is consequently bounded by $C_g(1 + S) \sum_i h^{i\bar{i}}$, as asserted. No derivative of h remains in E.

It remains to control the gradient introduced by the logarithm. Define

$$H_{ik\bar{j}} = h_{i\bar{j}} \frac{S_k}{S}, \qquad K = \|A + D - H\|^2 \geq 0.$$

By (9.7),

$$g^{i\bar{\ell}} (A + D)_{ik\bar{\ell}} = S_k + \tau_k.$$

Also $\|H\|^2 = |\partial S|^2_h / S$. Expanding the square therefore gives

$$K = \|A + D\|^2 - \frac{|\partial S|^2_h}{S} - \frac{2}{S} \operatorname{Re}(h^{k\bar{q}} \tau_k S_{\bar{q}}).$$

Combine this with (9.8) to obtain the exact identity

$$\Delta_h \log S = \frac{K}{S} + \frac{2}{S^2} \operatorname{Re}(h^{k\bar{q}} \tau_k S_{\bar{q}}) + \frac{\Delta_g G}{S} + \frac{E}{S}.$$

Since $S \geq 1$, the remainder is at least $-2C_g B$. Enlarging the constant proves the lemma. □

Theorem 9.4.1 *Fix $D, K > 0$ and a smooth real function G on X. Suppose u is smooth, $\omega_u > 0$, and*

$$\operatorname{osc}_X u \leq D, \qquad \omega_u^n = e^G \omega^n, \qquad |G| \leq K, \qquad \Delta_g G \geq -K.$$

There is a constant $C = C(\omega, D, K)$ such that

$$C^{-1} \omega \leq \omega_u \leq C\omega.$$

Proof Use $h = g + (u_{i\bar{j}})$, $S = \operatorname{tr}_g h$, and $B = \operatorname{tr}_h g$ as in Lemma 9.4.1. Write $s = u - \inf_X u + 1$, so $1 \leq s \leq D + 1$, and maximize

$$Q = \log S - Au + s^{-1}$$

on X, with $A \geq 1$ to be fixed. At a maximum point x, if $S(x) < 1$ the upper bound for S everywhere already follows from the oscillation bound and the definition of Q. We may therefore assume $S(x) \geq 1$ and use Lemma 9.4.1.

The positive eigenvalues of h relative to g give $SB \geq n^2$. Hence $\Delta_g G/S \geq -KB/n^2$. Moreover, $\partial Q = 0$ gives

$$\frac{S_k}{S} = (A + s^{-2})u_k.$$

Because $|\tau|_h^2 \leq C_g B$, the absolute value of the torsion term is bounded by

$$\begin{aligned} \left| \frac{2}{S^2} \operatorname{Re}(h^{k\bar{q}} \tau_k S_{\bar{q}}) \right| &\leq \frac{C_g(A+1)}{S} \sqrt{B}\, |\partial u|_h \\ &\leq \frac{|\partial u|_h^2}{s^3} + \frac{C_g(A+1)^2 s^3}{S^2} B. \end{aligned} \tag{9.9}$$

If $S(x)^2 \leq (A+1)^2(D+1)^3$, then $S(x)$ is bounded once A is fixed, and again so is S everywhere. In the remaining case the last coefficient of B is at most C_g.

On the other hand, direct differentiation gives

$$\Delta_h Q = \Delta_h \log S + (A + s^{-2})(B - n) + 2s^{-3}|\partial u|_h^2.$$

The trace inequality and (9.9) imply at x

$$0 \geq \Delta_h Q \geq (A - C_g - K/n^2)B - n(A+1),$$

after a fixed enlargement of C_g. Choose $A > C_g + K/n^2 + 1$. It follows that $B(x)$ is bounded. If λ_i are the eigenvalues, then for $n \geq 2$

$$S = \sum_i \lambda_i = \mathrm{e}^G \sum_i \prod_{j \neq i} \lambda_j^{-1} \leq n\mathrm{e}^K B^{n-1}.$$

For $n = 1$, $S = \mathrm{e}^G$ directly. Thus $S(x)$ is bounded in all cases. For every $y \in X$, the maximum property gives

$$\log S(y) \leq \log S(x) + A(u(y) - u(x)) + s(x)^{-1} - s(y)^{-1} \leq \log S(x) + AD + 1.$$

This proves $S \leq C$ on X. The determinant lower bound then gives $\lambda_i \geq \mathrm{e}^{-K} C^{-(n-1)}$, proving uniform equivalence. □

Corollary 9.4.1 *Fix ω. For smooth solutions of*

$$(\omega + \mathrm{dd^c} u)^n = \mathrm{e}^G \omega^n, \qquad \omega + \mathrm{dd^c} u > 0,$$

a uniform C^0 bound for u and uniform bounds for the smooth real logarithmic density G in every C^k norm imply uniform bounds for u in every C^k norm.

Proof Theorem 9.4.1 gives uniform ellipticity and a bound for $\Delta_g u = S - n$. The standard linear elliptic $W^{2,p}$ estimate for this fixed operator first bounds u in $C^{1,\alpha}$

for every $0 < \alpha < 1$. The local complex Evans–Krylov estimate for the concave logarithmic determinant then gives a uniform $C^{2,\beta}$ bound for some $0 < \beta < 1$. One can use its real uniformly elliptic formulation: project real symmetric Hessians to their complex Hermitian part and extend the logarithmic determinant concavely from the compact set of positive matrices supplied by the preceding theorem. The extension is uniformly elliptic on real symmetric matrices because that projection preserves positivity and its trace. The usual Evans–Krylov theorem applies to this extension. Finally, differentiation of the equation gives linear uniformly elliptic equations with Hölder coefficients; Schauder estimates and induction give all higher bounds. Only these standard local elliptic regularity theorems are used here. □

The same conclusion holds for equations $\omega_u^n = \mathrm{e}^{au+F}\omega^n$ with a in a fixed bounded interval of nonnegative numbers, u uniformly bounded, and F uniformly bounded in every C^k. Indeed,

$$\Delta_g(au + F) = a(S - n) + \Delta_g F \geq -an + \Delta_g F,$$

so the second-derivative estimate applies without a prior bound for derivatives of u. The fixed-operator estimate first gives $C^{1,\alpha}$, after which the same local regularity and bootstrapping apply.

9.5 The continuity method

The local inverse gives openness of the solvable set. The extrema bound for the scalar, the potential estimate and the derivative estimates give compactness along a path from the background density to the prescribed one. Uniform ellipticity preserves positivity at a limiting parameter.

Proof (of Theorem 9.1.1) The dimension-one argument was given at the start of the chapter. In higher dimension put $G = F + \log(\mathrm{d}V/\omega^n)$ and consider

$$(\omega + \mathrm{dd^c}u_t)^n = \mathrm{e}^{tG+b_t}\omega^n, \qquad \sup_X u_t = 0, \qquad 0 \leq t \leq 1.$$

At $t = 0$, take $u_0 = 0$ and $b_0 = 0$. The local implicit-function argument in Proposition 9.3.1, with the additive constant temporarily fixed by a linear mean, shows that the set of solvable parameters is open.

At every solvable parameter, Proposition 9.1.1 gives

$$-\sup_X(tG) \leq b_t \leq -\inf_X(tG), \qquad \mathrm{e}^{tG+b_t} \leq \mathrm{e}^{\operatorname{osc}_X G}.$$

The constants b_t are uniformly bounded, and Theorem 9.2.1 bounds u_t in C^0. Apply Corollary 9.4.1. Along any sequence of solvable parameters converging in $[0, 1]$, a subsequence of (u_t, b_t) therefore converges smoothly to a solution; uniform equivalence of the metrics preserves strict positivity. The supremum normalization

also passes to the limit. Thus the solvable set is closed. Connectedness of $[0, 1]$ gives a solution at $t = 1$, and Proposition 9.1.2 gives uniqueness. □

9.6 The smooth exponential equation

When the logarithmic density contains a positive multiple of the potential, extrema bound the potential itself and the linearization acquires a negative zeroth-order term. The continuity method therefore has neither an additive normalization nor an extra scalar unknown.

Proposition 9.6.1 *Let $\lambda > 0$, let F be smooth, and let v, w be smooth strictly ω-plurisubharmonic functions. If*

$$\omega_v^n \geq e^{\lambda v + F}\omega^n, \qquad \omega_w^n \leq e^{\lambda w + F}\omega^n,$$

then $v \leq w$. In particular, two smooth solutions coincide.

Proof If $v - w$ had a positive maximum at x, then $\omega_v(x) \leq \omega_w(x)$. Taking determinants and applying the two assumptions would give $e^{\lambda v(x)+F(x)} \leq e^{\lambda w(x)+F(x)}$, contradicting $v(x) > w(x)$. For two solutions, apply the comparison in both orders.□

Theorem 9.6.1 *For every $\lambda > 0$ and every smooth real F, there is a unique smooth u with $\omega_u > 0$ such that*

$$\omega_u^n = e^{\lambda u + F}\omega^n.$$

The solution satisfies

$$-\lambda^{-1} \sup_X F \leq u \leq -\lambda^{-1} \inf_X F.$$

Proof At a minimum of u, the equation gives $\lambda u + F \geq 0$; at a maximum it gives $\lambda u + F \leq 0$. These inequalities prove the stated bounds.

Consider the equations

$$\omega_{u_t}^n = e^{\lambda u_t + tF}\omega^n, \qquad 0 \leq t \leq 1.$$

At $t = 0$, $u_0 = 0$ is a solution. The linearization in u is $L_u - \lambda$. Its kernel is zero by the maximum principle. Its Fredholm index is zero: deform its positive principal coefficients to those of a fixed Laplacian while keeping the negative zeroth-order term. The latter operator is invertible, and elliptic Fredholm index is constant under this deformation. Thus $L_u - \lambda$ is an isomorphism between the usual Hölder spaces, with bounded inverse by Schauder theory. The implicit function theorem proves openness.

The extrema bounds give $\|u_t\|_\infty \leq \lambda^{-1}\|F\|_\infty$. The semilinear consequence of Theorem 9.4.1 gives uniform ellipticity and all higher bounds. A subsequence along any converging sequence of solvable parameters therefore converges smoothly to a strictly positive solution at the limit. This proves closedness, and hence existence at $t = 1$. Uniqueness is Proposition 9.6.1. □

For a fixed smooth real obstacle f on X and a parameter $\beta > 0$, the choice $\lambda = \beta$ and $F = -\beta f$ gives

$$\omega_{u_\beta}^n = \mathrm{e}^{\beta(u_\beta - f)}\omega^n.$$

These equations penalize values above the obstacle. In Proposition 12.1.1 we use them to construct its envelope and prove contact support. The necessary convergence will use only bounded regularization and the comparison just proved.

9.7 Variation of the normalization scalar

Integration against the adjoint probability eliminates the derivatives of the potential from the first variation. In the second variation, the matrix logarithm leaves a nonnegative square with a minus sign. This gives concavity and identifies its equality case.

Write $b(F)$ for the scalar in (9.1), and use the linear normalization $\int_X u \,\mathrm{d}V = 0$. For a smooth family F_t, with t in an open interval $I \subset \mathbb{R}$, write (u_t, b_t) for its solutions and put $\eta_t = \omega_{u_t}$. A dot denotes differentiation in t.

Theorem 9.7.1 *Put $A_t = \eta_t^{-1}\mathrm{dd^c}\dot{u}_t$, the Hermitian endomorphism determined by the indicated forms. The normalization constant satisfies*

$$\dot{b}_t = -\int_X \dot{F}_t \,\mathrm{d}\mu_{u_t}, \tag{9.10}$$

$$\ddot{b}_t = -\int_X \ddot{F}_t \,\mathrm{d}\mu_{u_t} - \int_X \mathrm{tr}(A_t^2)\,\mathrm{d}\mu_{u_t}, \tag{9.11}$$

In particular, for smooth real functions F_0, G on X and the affine family $F_t = F_0 + tG$, the function b_t is concave. Its second derivative vanishes at a parameter value if and only if G is constant on X.

Proof Differentiating the logarithm of (9.1) gives

$$L_{u_t}\dot{u}_t = \dot{F}_t + \dot{b}_t.$$

Integration against μ_{u_t} proves (9.10).

For positive matrices H_t, direct differentiation of the inverse gives

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}\log\det H_t = \mathrm{tr}(H_t^{-1}\ddot{H}_t) - \mathrm{tr}\big((H_t^{-1}\dot{H}_t)^2\big).$$

Apply this pointwise to η_t. The second derivative of the equation is

$$L_{u_t}\ddot{u}_t - \mathrm{tr}(A_t^2) = \ddot{F}_t + \ddot{b}_t.$$

At this fixed parameter, integrate against μ_{u_t} once more. This proves (9.11); there is no need to differentiate the measure.

The eigenvalues of the Hermitian endomorphism A_t are real, so $\operatorname{tr}(A_t^2) \geq 0$. For affine F_t the second derivative is therefore nonpositive. If it vanishes, strict positivity of the smooth density of μ_{u_t} implies $A_t = 0$ everywhere. Thus $\mathrm{dd^c}\dot{u}_t = 0$; the maximum principle gives that $\dot{u}_t$ is constant. The first differentiated equation then gives $G = -\dot{b}_t$, a constant. Conversely, if G is constant, the solution metric is independent of t and $b_t = b_0 - tG$. □

Corollary 9.7.1 *For two smooth functions F, G, the normalization constants satisfy*

$$-\sup_X(G - F) \leq b(G) - b(F) \leq -\inf_X(G - F).$$

In particular, $|b(G) - b(F)| \leq \|G - F\|_\infty$. If $G - F$ is nonconstant, both displayed inequalities are strict.

Proof Apply (9.10) along $F_t = F + t(G - F)$ and integrate over $0 \leq t \leq 1$. Each μ_{u_t} is a probability measure with strictly positive smooth density. The integral of a nonconstant continuous function against it lies strictly between that function's minimum and maximum. □

Thus b is a decreasing, concave, Lipschitz functional of the logarithmic density, with the translation rule $b(F + a) = b(F) - a$ for every real constant a. It extends uniquely as a Lipschitz functional from smooth to continuous F by uniform approximation. This last statement concerns the scalar functional only; existence or regularity of limiting potentials requires further estimates.

9.8 Normalization on the Hopf threefold

The Hopf metric of Chapter 8 makes the discrepancy from the closed-background normalization explicit: it is the energy of the solution along the elliptic fibers. In this section use that threefold, its projection $\pi : X \to \mathbb{P}^2$, the probability measure $\nu = \omega^3/6$ and the averaging operator A of Proposition 8.6.1. Write $\beta = \omega - \pi^*\omega_{\mathrm{FS}}$ for the rank-one form in that construction. The fiber derivative u_w is independent of the base trivialization.

For $F \in C^\infty(X, \mathbb{R})$, let $(u_F, b(F))$ be the solution of

$$(\omega + \mathrm{dd^c}u_F)^3 = \mathrm{e}^{F+b(F)}\omega^3, \qquad \omega + \mathrm{dd^c}u_F > 0, \qquad \int_X u_F \,\mathrm{d}\nu = 0.$$

Define

$$D(F) = b(F) + \log \int_X \mathrm{e}^F \,\mathrm{d}\nu.$$

The corresponding expression vanishes for a closed background by (9.2).

Proposition 9.8.1 *For every smooth real F on this threefold,*

$$\mathrm{e}^{D(F)} = 1 + \int_X |(u_F)_w|^2 \,\mathrm{d}\nu, \tag{9.12}$$

$$\|u_F - Au_F\|^2_{L^2(X,\nu)} \le 4\big(\mathrm{e}^{D(F)} - 1\big). \tag{9.13}$$

Thus $D(F) \ge 0$, with equality if and only if $F = \pi^* H$ for some $H \in C^\infty(\mathbb{P}^2, \mathbb{R})$.

Proof Integration of the equation and (8.2) give

$$\mathrm{e}^{D(F)} = \frac{1}{6}\int_X (\omega + \mathrm{dd^c} u_F)^3 = 1 + \int_X |(u_F)_w|^2 \,\mathrm{d}\nu.$$

Substituting the first equality into (8.20) proves (9.13).

If $D(F) = 0$, smoothness gives $(u_F)_w = 0$ everywhere. Since u_F is real, it is constant on each connected fiber. It descends to a smooth real function q on $\mathbb{P}^2$, with $\omega_{\mathrm{FS}} + \mathrm{dd^c} q > 0$. Since $\beta^2 = 0$ and $\omega = \pi^*\omega_{\mathrm{FS}} + \beta$, we have

$$(\omega + \mathrm{dd^c}\pi^* q)^3 = 3\pi^*(\omega_{\mathrm{FS}} + \mathrm{dd^c} q)^2 \wedge \beta.$$

Dividing by $\omega^3 = 3\pi^*\omega_{\mathrm{FS}}^2 \wedge \beta$ and taking logarithms proves that F is a smooth base pullback.

Conversely, suppose $F = \pi^* H$. Apply Theorem 9.1.1 on $(\mathbb{P}^2, \omega_{\mathrm{FS}})$ to obtain a smooth real q and a scalar c with

$$\omega_{\mathrm{FS}} + \mathrm{dd^c} q > 0, \qquad (\omega_{\mathrm{FS}} + \mathrm{dd^c} q)^2 = \mathrm{e}^{H+c}\omega_{\mathrm{FS}}^2.$$

Closedness of the base form and Stokes' theorem give $\mathrm{e}^c \int_{\mathbb{P}^2} \mathrm{e}^H \omega_{\mathrm{FS}}^2 = 1$. The preceding cube identity shows that $\pi^* q$, after subtraction of its ν-mean, solves the equation on X with scalar c. Uniqueness gives $b(F) = c$. Disintegration of ν now yields $D(F) = 0$. □

In particular, if a smooth positive density f satisfies $\int_X f \,\mathrm{d}\nu = 1$, the equation $(\omega + \mathrm{dd^c} u)^3 = \mathrm{e}^b f \omega^3$ has $b \ge 0$, with equality precisely when f comes from the base. Fixing the total integral of the density therefore does not determine the scalar on this non-closed background.

Corollary 9.8.1 *Let $G \in C^\infty(X, \mathbb{R})$, and let v be the unique smooth real function satisfying*

$$L_0 v = G - \int_X G \,\mathrm{d}\nu, \qquad \int_X v \,\mathrm{d}\nu = 0.$$

Then

$$\frac{\mathrm{d}}{\mathrm{d}t} D(tG)\bigg|_{t=0} = 0, \qquad \frac{\mathrm{d}^2}{\mathrm{d}t^2} D(tG)\bigg|_{t=0} = 2\int_X |v_w|^2 \,\mathrm{d}\nu.$$

The second derivative vanishes if and only if G is a smooth base pullback.

Proof For real t near zero put $u_t = u_{tG}$ and $b_t = b(tG)$. By Proposition 9.3.1, this is a smooth family, with $u_0 = b_0 = 0$. Differentiating the logarithmic equation at zero gives $L_0\dot{u}_0 = G + \dot{b}_0$. Since ω is Gauduchon, ν is the adjoint probability of L_0. Integration gives $\dot{b}_0 = -\int_X G\,\mathrm{d}\nu$, so $\dot{u}_0 = \nu$. This also proves existence and uniqueness of the displayed mean-zero solution.

Smooth dependence gives $(u_t)_w = t\nu_w + O(t^2)$ uniformly on X. Equation (9.12) therefore gives

$$\mathrm{e}^{D(tG)} = 1 + t^2 \int_X |\nu_w|^2\,\mathrm{d}\nu + O(t^3).$$

Taking logarithms proves the derivative formulas.

If the second derivative vanishes, then $\nu = \pi^* q$ for a smooth real q on the base. Direct wedge expansion gives

$$L_0(\pi^* q) = \pi^*\left(\frac{2\mathrm{dd^c}q \wedge \omega_{\mathrm{FS}}}{\omega_{\mathrm{FS}}^2}\right).$$

Hence $G = L_0\nu + \int_X G\,\mathrm{d}\nu$ is a base pullback. Conversely, when G is a base pullback, the proposition gives $D(tG) = 0$ for every real t, proving the converse. □

This is a second-variation formula at the background density. It does not determine global convexity of D: the second derivatives of b and of the logarithmic integral have opposite signs.

9.9 Prescribing a Gauduchon volume form

We now return to a general connected compact complex manifold X of dimension $n \geq 2$. To preserve a Gauduchon Aeppli class, we vary the $(n-1)$st power of the metric by a ∂- and $\bar{\partial}$-exact term. The resulting scalar ansatz contains first derivatives of the potential as well as its complex Hessian.

Theorem 9.9.1 (Székelyhidi–Tosatti–Weinkove) *Fix a Hermitian form γ_0, a Gauduchon form γ, and smooth real F. There is a unique pair $(u, b) \in C^\infty(X, \mathbb{R}) \times \mathbb{R}$ with $\sup_X u = 0$ such that the positive $(n-1, n-1)$-form*

$$\Gamma_u = \gamma_0^{n-1} + \mathrm{dd^c} u \wedge \gamma^{n-2} + \mathrm{Re}\left(\frac{\mathrm{i}}{2\pi}\partial u \wedge \bar{\partial}(\gamma^{n-2})\right)$$

has a positive root $\eta_u^{n-1} = \Gamma_u$ satisfying

$$\eta_u^n = \mathrm{e}^{F+b}\gamma^n.$$

Keep γ_0 and γ fixed for the remainder of the chapter, and write $\mathcal{D}_\gamma u = \Gamma_u - \gamma_0^{n-1}$ for the linear part of the ansatz.

To see the preserved geometry, set $\xi_u = \mathrm{i}\,\bar{\partial} u \wedge \gamma^{n-2}/(4\pi)$. The Leibniz rule gives $\Gamma_u - \gamma_0^{n-1} = \partial\xi_u + \bar{\partial}\overline{\xi_u}$. If γ_0 is Gauduchon, so is η_u, and their $(n-1)$st powers have the same Aeppli class: the difference has the displayed exact form. Uniqueness here concerns the stated scalar ansatz; it does not assert uniqueness among all Gauduchon metrics in that Aeppli class.

9.10 Bounds for the Gauduchon potential

Taking the trace of the positive ansatz gives a lower bound for a fixed linear elliptic operator applied to the potential. Its Green kernel gives an L^1 bound. Near a minimum, supporting affine functions select a contact set on which the gradient is small and the real Hessian has a lower bound; the determinant equation then forces this set to have uniformly positive volume. Combining the two estimates bounds the minimum.

Proposition 9.10.1 *Every smooth normalized solution (u, b) of the equation in Theorem 9.9.1 satisfies*

$$|b| + \|u\|_{L^\infty(X)} \le C.$$

The constant depends only on the fixed geometry and an upper bound for $\|F\|_\infty$. It does not depend on derivatives of F.

Proof We first bound the scalar. At a maximum of u, its first derivatives vanish and $\mathrm{dd^c} u \le 0$, so $\Gamma_u \le \gamma_0^{n-1}$. At a minimum the reverse inequality holds. The determinant of the positive cofactor matrix is monotone under the order of positive matrices; the positive-root volume is its $1/(n-1)$ power. Evaluating the equation at these two points gives

$$\inf_X \left(\log \frac{\gamma_0^n}{\gamma^n} - F\right) \le b \le \sup_X \left(\log \frac{\gamma_0^n}{\gamma^n} - F\right).$$

This proves the scalar bound.

Consider the fixed real linear elliptic operator

$$Lu = \frac{\gamma \wedge \mathcal{D}_\gamma u}{\gamma^n}.$$

Its principal part is $\mathrm{dd^c}u \wedge \gamma^{n-1}/\gamma^n$, its other term is first order, and it has no zeroth-order term. Positivity of Γ_u implies $Lu \geq -C_0$ for a fixed constant C_0.

The fixed operator L has a smooth strictly positive adjoint probability ν, by the Fredholm and maximum-principle argument in the proof of Theorem 4.1.1. With the sign for which L is nonpositive at a maximum, its normalized Green kernel can be written

$$v(x) - \int_X v \,\mathrm{d}\nu = -\int_X K(x,y) Lv(y) \,\mathrm{d}\nu(y),$$
$$K \geq 0, \qquad \sup_x \int_X K(x,y) \,\mathrm{d}\nu(y) < +\infty.$$

Here the kernel is smooth off the diagonal and locally integrable. To obtain this version, construct the inverse on functions of zero ν-mean by the Fredholm alternative. The local fundamental solution has a positive leading singularity after the displayed minus sign, so compactness bounds its negative part. Adding a constant to K makes it nonnegative and does not change the formula, since $\int_X Lv \,\mathrm{d}\nu = 0$. The local Green estimates give the uniform integral bound.

Apply the formula to u and use $Lu \geq -C_0$. At a maximum point, where $u = 0$, it gives

$$-\int_X u \,\mathrm{d}\nu \leq C_1.$$

Because $u \leq 0$ and ν has a fixed smooth positive density, this is a uniform L^1 bound in any fixed smooth volume form.

Let $M = \min_X u$ and choose holomorphic coordinates on a ball centered at a minimum point. Compactness permits a fixed coordinate radius and uniform bounds for the metric coefficients and their derivatives; rescale the ball to the unit ball $B \subset \mathbb{R}^{2n}$. Write x for its real coordinates, D and D^2 for the Euclidean gradient and Hessian, $\mathrm{d}x$ for Euclidean volume, and I for an identity matrix of the indicated size. Fix a small number $\varepsilon > 0$, depending only on those bounds, and set

$$v(x) = u(x) + \varepsilon |x|^2.$$

For any real vector p with $|p| < \varepsilon/2$, the function $v(x) - p \cdot x$ has an interior minimum: its boundary values are at least $M + \varepsilon/2$, whereas its value at the center is M. At such a point,

$$Dv = p, \qquad D^2 v \geq 0, \qquad v \leq M + \varepsilon/2.$$

Let $A \subset B$ be the set of all points satisfying $|Dv| \le \varepsilon/2$, $D^2 v \ge 0$, and $v \le M + \varepsilon/2$. Since $u \ge M$, this set lies in $\{|x| \le 1/\sqrt{2}\}$. It is closed there, hence compact in B, and its gradient image contains the ball of radius $\varepsilon/2$. The area formula for the smooth gradient map therefore gives

$$|B_{\varepsilon/2}| \le \int_A \det D^2 v \, \mathrm{d}x. \tag{9.14}$$

The Hessian is nonnegative on A, so no absolute value is needed in the determinant.

We bound this determinant pointwise on A. There,

$$|Du| \le \frac{5}{2}\varepsilon, \qquad D^2 u \ge -2\varepsilon I.$$

The complex Hessian has the corresponding uniform lower bound. Wedge with the fixed positive γ^{n-2} and bound the first-order term using the gradient estimate. Let Q_u be the matrix of Γ_u in a fixed local frame of $(n-1,0)$-forms. Since γ_0^{n-1} is strictly positive, choosing ε small gives

$$Q_u \ge cI$$

with a uniform $c > 0$. The equation and the already proved bounds for $F + b$ bound $\det Q_u$ from above, because the determinant of this cofactor matrix is, up to a fixed positive frame factor, the $(n-1)$st power of η_u^n. All eigenvalues of Q_u are therefore bounded above as well: their product is bounded and each of the other $n-1$ eigenvalues is at least c.

In particular $\gamma \wedge \Gamma_u / \gamma^n$ is uniformly bounded on A. Subtracting its fixed background term and its bounded first-order term bounds

$$\frac{\mathrm{dd^c} u \wedge \gamma^{n-1}}{\gamma^n}$$

from above. This is a positive fixed multiple of the γ-trace of the complex Hessian of u. The complex Hessian of v is nonnegative on A, since the real Hessian is nonnegative. Metric comparison thus bounds its Euclidean complex trace from above. The real trace of $D^2 v$ is four times that complex trace. It follows that

$$0 \le \det D^2 v \le \left(\frac{\mathrm{tr}_{\mathbb{R}} D^2 v}{2n} \right)^{2n} \le C_2 \quad \text{on } A.$$

Equation (9.14) now gives $|A| \ge c_2 > 0$, where c_2 is independent of u. On A we also have $u \le v \le M + \varepsilon/2$. Consequently the uniform L^1 bound gives

$$C_3 \ge \int_A (-u) \, \mathrm{d}x \ge c_2(-M - \varepsilon/2).$$

This bounds $-M$, and completes the proof. □

9.11 The adjoint equation for Gauduchon volume

The positive-root operation changes the trace linearization by a factor $n/(n-1)$, while the first-order term of the ansatz introduces drift. Its adjoint probability gives the compatibility condition for varying both the potential and the scalar. We compute this operator at any potential with $\Gamma_u > 0$ and then express its adjoint as an equation for a positive $(1,1)$-form. Put $K = \gamma^{n-2}$.

Proposition 9.11.1 *On the open set $\Gamma_u > 0$, the derivative of*

$$\mathcal{G}(u) = \log \frac{\eta_u^n}{\gamma^n}$$

is the real elliptic operator

$$\mathcal{L}_u v = \frac{n}{n-1} \frac{\eta_u \wedge \mathcal{D}_\gamma v}{\eta_u^n}. \tag{9.15}$$

It has no zeroth-order term and its kernel consists of constants. There is a unique smooth positive function ρ_u, normalized by $\int_X \rho_u \eta_u^n = 1$, such that the probability measure

$$\mathrm{d}\mu_u^\gamma = \rho_u \eta_u^n$$

satisfies $\int_X \mathcal{L}_u v \, \mathrm{d}\mu_u^\gamma = 0$ for every smooth real v.

Proof For a smooth path of potentials, write $\eta = \eta_{u(t)}$ and $\Gamma = \Gamma_{u(t)}$. Differentiating $\eta^{n-1} = \Gamma$ gives $(n-1)\dot{\eta} \wedge \eta^{n-2} = \dot{\Gamma}$. Wedge with η and differentiate η^n. The result is

$$\frac{\mathrm{d}}{\mathrm{d}t} \log \eta^n = \frac{n}{n-1} \frac{\eta \wedge \dot{\Gamma}}{\eta^n},$$

which proves (9.15).

Only the first term of $\mathcal{D}_\gamma$ contributes to its principal symbol. For a nonzero complex covector ζ, the form

$$\eta_u \wedge \mathrm{i}\zeta \wedge \overline{\zeta} \wedge \gamma^{n-2}$$

is strictly positive. Hence the principal symbol is positive definite, and compactness makes the smooth operator uniformly elliptic at each fixed u. The remaining term is first order, and constants are annihilated. The strong maximum principle gives $\ker \mathcal{L}_u = \mathbb{R}$.

The Fredholm and positivity arguments in the proof of Theorem 4.1.1 apply to this real elliptic operator with first-order terms and no zeroth-order term. Its index is zero by deformation to a Laplacian, so the adjoint for η_u^n has a one-dimensional smooth kernel with a strictly positive generator. Normalizing its integral to one gives ρ_u and uniqueness. □

The adjoint can be expressed as an equation for a positive $(1,1)$-form. For a smooth real $(1,1)$-form A, set

$$\mathcal{A}_\gamma(A) = \mathrm{dd^c}A \wedge \gamma^{n-2} + \mathrm{Re}\left(\frac{\mathrm{i}}{2\pi}\partial A \wedge \bar\partial(\gamma^{n-2})\right).$$

This is a real top-degree form.

Proposition 9.11.2 *The multiplier in Proposition 9.11.1 is characterized by*

$$\mathcal{A}_\gamma(\rho_u\eta_u) = 0, \qquad \int_X \rho_u\eta_u^n = 1. \tag{9.16}$$

When $n = 2$, its first equation is $\mathrm{dd^c}(\rho_u\eta_u) = 0$. When $\eta_u = \gamma$, the multiplier is constant and $\mathrm{d}\mu_u^\gamma = \gamma^n/\int_X \gamma^n$.

Proof For a smooth real $(1,1)$-form A and a smooth real function v on X, we prove the integration formula

$$\int_X A \wedge \mathcal{D}_\gamma v = \int_X v\,\mathcal{A}_\gamma(A). \tag{9.17}$$

Write $c = \mathrm{i}/(2\pi)$. Since A, K, v are real, expanding the real part gives

$$\mathcal{D}_\gamma v = c\,\partial\bar\partial v \wedge K + \frac{c}{2}\big(\partial v \wedge \bar\partial K - \bar\partial v \wedge \partial K\big).$$

Twice integrating the first term by parts gives $\int_X v\,c\partial\bar\partial(A \wedge K)$. Integrating each first-order term once gives the correction

$$\int_X v\,\frac{c}{2}\big(-\partial A \wedge \bar\partial K + \bar\partial A \wedge \partial K - 2A \wedge \partial\bar\partial K\big).$$

Finally,

$$\partial\bar\partial(A \wedge K) = \partial\bar\partial A \wedge K - \bar\partial A \wedge \partial K + \partial A \wedge \bar\partial K + A \wedge \partial\bar\partial K.$$

Adding cancels the terms containing $A \wedge \partial\bar\partial K$ and leaves precisely $\mathcal{A}_\gamma(A)$. This proves (9.17). Apply it to $A = \rho_u\eta_u$ in (9.15). It identifies the adjoint condition with (9.16).

For $n = 2$, $K = 1$, proving the first specialization. For the second, the product rule gives

$$\mathcal{A}_\gamma(\gamma) = \frac{1}{n-1}\mathrm{dd^c}(\gamma^{n-1}) = 0.$$

The unique normalized multiplier is consequently the constant $(\int_X \gamma^n)^{-1}$. □

For $n > 2$ the equation for $\rho_u\eta_u$ contains the first derivatives of γ^{n-2}. The Gauduchon condition on η_u alone is not the displayed adjoint equation. The fixed auxiliary metric γ remains part of it.

9.12 The matrix structure of the Gauduchon equation

The Hodge star converts the positive $(n-1, n-1)$-form into a matrix whose logarithmic determinant is concave. Two features will control its largest eigenvalue: the positive background gives coercivity, and the torsion coefficients omit precisely the gradient indices that would otherwise carry the largest inverse weight. We derive both properties before differentiating the equation.

For the matrix and derivative calculations set $v = u/(2\pi)$, so that $\mathrm{dd^c}u = \mathrm{i}\partial\bar{\partial}v$. All covariant derivatives are for the Chern connection ∇ of γ. In holomorphic coordinates $z^1, \dots, z^n$, write $v_i = \partial_i v$ and $v_{\bar{j}} = \partial_{\bar{j}} v$; indices range from 1 to n, and δ_{ij} is the Kronecker symbol. Write $\gamma = \mathrm{i}\gamma_{a\bar{b}}\, \mathrm{d}z^a \wedge \mathrm{d}\bar{z}^b$, with repeated indices summed. The Hodge star of this metric is denoted by $\star_\gamma$, and $\det_\gamma$ is the determinant relative to it. Write its torsion as T^c_{ab} and lower the last index with γ. Thus

$$T_{ab\bar{c}} = \partial_a \gamma_{b\bar{c}} - \partial_b \gamma_{a\bar{c}}, \qquad T_{ab\bar{c}} = -T_{ba\bar{c}}.$$

Let $Q_0 = \star_\gamma \gamma_0^{n-1}/(n-1)!$ and

$$Z = \frac{1}{(n-1)!} \star_\gamma \mathrm{Re}\big(\mathrm{i}\partial v \wedge \bar{\partial}\gamma^{n-2}\big).$$

Here the real $(1,1)$-forms on the right are identified with their Hermitian coefficient matrices. Set

$$\chi = (\mathrm{tr}_\gamma Q_0)\gamma - (n-1)Q_0, \quad W = (\mathrm{tr}_\gamma Z)\gamma - (n-1)Z, \quad g = \chi + \mathrm{i}\partial\bar{\partial}v + W.$$

The linear map $P_\gamma(A) = ((\mathrm{tr}_\gamma A)\gamma - A)/(n-1)$ satisfies

$$\begin{aligned} Q_u &:= P_\gamma(g) = Q_0 + P_\gamma(\mathrm{i}\partial\bar{\partial}v) + Z > 0, \\ \log \det_\gamma Q_u &= h = (n-1)(F + b). \end{aligned} \tag{9.18}$$

The tensor g need not be positive. To check the Hessian term, diagonalize it in a unitary frame: its complementary wedge coefficient is the sum of the other diagonal entries divided by $n-1$. If η_u has eigenvalues a_i relative to γ, then Q_u has eigenvalues $\prod_{j\neq i} a_j$, whose product is $(\prod_i a_i)^{n-1}$. This proves the determinant identity.

Lemma 9.12.1 *At a point in a unitary frame,*

$$\begin{aligned} 2(n-1)Z_{i\bar{j}} = \delta_{ij} \sum_{p,k} v_p \overline{T_{pk\bar{k}}} - v_i \sum_k \overline{T_{jk\bar{k}}} - \sum_k v_k \overline{T_{kj\bar{i}}} \\ + \delta_{ij} \sum_{p,k} v_{\bar{p}} T_{pk\bar{k}} - v_{\bar{j}} \sum_k T_{ik\bar{k}} - \sum_k v_{\bar{k}} T_{ki\bar{j}}. \end{aligned} \tag{9.19}$$

In particular $Z_{i\bar{j}}$ contains no v_j or $v_{\bar{i}}$. Moreover

$$2(n-1)Z_{i\bar{i}} = \sum_{p\neq i,\ k\neq i} \big(v_p \overline{T_{pk\bar{k}}} + v_{\bar{p}} T_{pk\bar{k}}\big). \tag{9.20}$$

Covariant differentiation preserves these coefficient identities. For example $\nabla_1 Z_{1\bar{1}}$ contains no pure Hessian v_{11} or its conjugate, and $\nabla_{\bar{1}}\nabla_1 Z_{1\bar{1}}$ contains no $v_{11\bar{1}}$ or its conjugate, nor a pure v_{11} or its conjugate.

Proof For $n = 2$, the defining expression is zero. The formulas below also vanish because the alternating symbol with three indices vanishes in two dimensions. Suppose $n \geq 3$ and use coordinates unitary at the point. Direct antisymmetrization gives

$$\bar{\partial}\gamma = -\frac{\mathrm{i}}{2} \sum_{a,b,c} \overline{T_{bc\bar{a}}}\, \mathrm{d}z^a \wedge \mathrm{d}\bar{z}^b \wedge \mathrm{d}\bar{z}^c.$$

Use $\bar{\partial}\gamma^{n-2} = (n-2)\bar{\partial}\gamma \wedge \gamma^{n-3}$. To exhibit the wedge contraction explicitly, put

$$\delta^{paj}_{bci} = \det\begin{pmatrix} \delta_{pb} & \delta_{pc} & \delta_{pi} \\ \delta_{ab} & \delta_{ac} & \delta_{ai} \\ \delta_{jb} & \delta_{jc} & \delta_{ji} \end{pmatrix}.$$

If $A = \star_\gamma(\mathrm{i}\partial v \wedge \bar{\partial}\gamma^{n-2})/(n-1)!$, expansion of the $n-3$ remaining copies of γ leaves the complementary three indices and gives

$$A_{i\bar{j}} = \frac{1}{2(n-1)} \sum_{p,a,b,c} v_p \overline{T_{bc\bar{a}}}\, \delta^{paj}_{bci}.$$

Indeed the factor is $(n-2)(n-3)!/[2(n-1)!]$; the determinant records the signs of the remaining holomorphic and antiholomorphic indices. Expanding its six terms and using $T_{bc\bar{a}} = -T_{cb\bar{a}}$ gives

$$A_{i\bar{j}} = \frac{1}{n-1}\left(\delta_{ij} \sum_{p,k} v_p \overline{T_{pk\bar{k}}} - v_i \sum_k \overline{T_{jk\bar{k}}} - \sum_k v_k \overline{T_{kj\bar{i}}}\right).$$

Here A^* denotes the Hermitian adjoint matrix. Since $Z = (A + A^*)/2$, this is (9.19).

For $i \neq j$, the coefficient of v_j in this expression vanishes by $T_{jj\bar{i}} = 0$; the coefficient of $v_{\bar{i}}$ vanishes in the conjugate expression. For $i = j$, cancel the terms with $p = i$ and $k = i$ to obtain (9.20). These are tensor identities valid in every unitary frame. The Chern connection preserves the metric, so one can differentiate them in a unitary frame parallel at the point. For instance the first derivative of a summand in (9.20) is

$$(\nabla_1 v_p)\overline{T_{pk\bar{k}}} + v_p \nabla_1 \overline{T_{pk\bar{k}}}, \qquad p, k \neq 1,$$

and its conjugate counterpart. A further $\bar{1}$ derivative still has $p \neq 1$. This gives exactly the stated exclusions, including the terms arising when a derivative falls on a coefficient. □

The explicit contraction explains why an arbitrary first-order perturbation cannot be substituted for Z in the estimate.

Lemma 9.12.2 *Diagonalize g at a point and order its eigenvalues as $\lambda_1 \geq \cdots \geq \lambda_n$. Put*

$$\mu_i = \frac{\sum_{j\neq i} \lambda_j}{n-1} > 0, \quad t_i = \mu_i^{-1}, \quad f_i = \frac{1}{n-1}\sum_{j\neq i} t_j, \quad \mathcal{F} = \sum_i f_i = \sum_i t_i.$$

Then $\mu_1 \leq \cdots \leq \mu_n$, $f_1 \leq \cdots \leq f_n$, and

$$\sum_i f_i\lambda_i = n, \qquad \frac{\mathcal{F}}{n(n-1)} \leq f_k \leq \frac{\mathcal{F}}{n-1} \quad (k>1), \qquad t_i \leq (n-1)f_1 \quad (i>1), \tag{9.21}$$
$$f_n \geq \mathcal{F}/n.$$

Fix $\tau, D > 0$ and assume $Q_0 \geq \tau\gamma$ and $\det P_\gamma(g) \leq D$. For λ_1 larger than a constant depending only on n, τ, D,

$$\sum_i f_i(\chi_{i\bar{i}} - \lambda_i) \geq \frac{\tau}{2}\mathcal{F}. \tag{9.22}$$

There is also a uniform positive lower bound for $\mathcal{F}$ depending only on n, D.

Proof The ordering follows from the formula for μ_i and then for f_i. Euler's identity here can be checked without differentiation:

$$\sum_i f_i\lambda_i = \sum_j \frac{t_j}{n-1}\sum_{i\neq j}\lambda_i = \sum_j t_j\mu_j = n.$$

For $k > 1$, the sum defining f_k contains $t_1 \geq \mathcal{F}/n$. Its upper bound and the estimate for t_i are immediate. The largest f_n is at least the average.

The map P_γ is self-adjoint for the matrix trace pairing. Consequently

$$\sum_i f_i\chi_{i\bar{i}} = \sum_i t_i(Q_0)_{i\bar{i}} \geq \tau\mathcal{F}.$$

Moreover $\sum_i \lambda_i = \sum_i \mu_i > \lambda_1 > 0$; the strict inequality follows from $\mu_1 > 0$. Hence $\mu_n > \lambda_1/n$. Since the other $n-1$ positive numbers are at least μ_1,

$$\mu_1^{n-1}\mu_n \leq \prod_i \mu_i \leq D, \qquad \mathcal{F} \geq \mu_1^{-1} \geq \left(\frac{\lambda_1}{nD}\right)^{1/(n-1)}.$$

For λ_1 sufficiently large this is at least $2n/\tau$. Subtracting $\sum_i f_i\lambda_i = n$ proves (9.22). Finally arithmetic–geometric mean applied to t_i gives $\mathcal{F} \geq nD^{-1/n}$. □

The concavity term also has a direct formula. For the matrix function $\Phi(g) = \log\det P_\gamma(g)$, the negative second differential in a complex matrix direction B and its adjoint is

$$-D^2\Phi_g(B, B^*) = \operatorname{tr}\bigl(P_\gamma(g)^{-1}P_\gamma(B)P_\gamma(g)^{-1}P_\gamma(B)^*\bigr) \geq 0. \tag{9.23}$$

This follows by differentiating the inverse matrix. In a diagonal unitary frame its off-diagonal terms include

$$\frac{|B_{k\bar{1}}|^2}{(n-1)^2\mu_k\mu_1}.$$

When $\lambda_k < \lambda_1$, that coefficient is exactly

$$\frac{f_k - f_1}{\lambda_1 - \lambda_k} = \frac{1}{(n-1)^2\mu_k\mu_1}.$$

The expression on the right also makes sense when the two eigenvalues coincide. The positive square and the weights f_i enter the largest-eigenvalue estimate below.

9.13 The largest-eigenvalue estimate

The matrix structure now gives a bound for the complex Hessian in terms of the squared gradient. We apply the maximum principle to the largest eigenvalue, adding functions of the gradient and the potential. The missing torsion indices let the concavity square absorb the third derivatives. If the smallest eigenvalue is not too negative, the positive background closes the estimate; otherwise that eigenvalue supplies a positive Hessian square.

Plain subscripts on scalar functions and comma subscripts on matrix coefficients denote coordinate derivatives. The coefficients of the Chern connection are written Γ^q_{ik}. Choose $\tau > 0$ with $Q_0 \geq \tau\gamma$.

Theorem 9.13.1 *Fix smooth Hermitian forms γ_0, γ on a compact complex manifold X of dimension $n \geq 2$. For a smooth real function h on X, suppose that v is a smooth solution of*

$$Q = Q_0 + P_\gamma(\mathrm{i}\partial\bar{\partial}v) + Z(\partial v) > 0, \qquad \log \det_\gamma Q = h,$$

with the tensors of Section 9.12. If $\|v\|_{C^0}$ and $\|h\|_{C^2}$ are bounded, then

$$\sup_X |\mathrm{i}\partial\bar{\partial}v|_\gamma \leq C\big(1 + \sup_X |\partial v|_\gamma^2\big),$$

where C depends only on these bounds and the fixed smooth data.

Proof Adding a constant, arrange $\sup v = 0$ and write $-M \leq v \leq 0$. Set

$$K = 1 + \sup_X |\partial v|_\gamma^2, \qquad \phi(s) = -\tfrac{1}{2}\log(1 - s/(2K)), \qquad \psi(t) = D_1 \mathrm{e}^{-D_2 t}.$$

The positive constants D_1, D_2 will be chosen below using only the fixed data. On $0 \leq s \leq K$,

$$\frac{1}{4K} \leq \phi' \leq \frac{1}{2K}, \qquad \phi'' = 2(\phi')^2, \qquad 0 \leq \phi \leq \tfrac{1}{2}\log 2.$$

Let λ_1 be the largest eigenvalue of $g = \chi + \mathrm{i}\partial\bar{\partial}v + W$ relative to γ. It is positive because $\mathrm{tr}_\gamma\, g = \mathrm{tr}_\gamma\, Q > 0$. Take a maximum of

$$H = \log \lambda_1 + \phi(|\partial v|_\gamma^2) + \psi(v).$$

It suffices to bound λ_1/K at this point: the other two terms are uniformly bounded once D_1, D_2 are fixed. We may assume that this ratio exceeds any fixed constant specified in the course of the argument.

Choose holomorphic coordinates unitary at the maximum, with g diagonal there and eigenvalues ordered $\lambda_1 \geq \cdots \geq \lambda_n$. All estimates below are at this point. Repeated indices are summed only when a sum is displayed. Put $\lambda = \lambda_1$ and use the weights t_i, f_k and their sum $\mathcal{F}$ from Lemma 9.12.2. Since $\sum_i \lambda_i > 0$ and the eigenvalues are ordered,

$$|\lambda_i| \leq (n-1)\lambda, \qquad |v_{i\bar{j}}| \leq C\lambda \quad \text{if } \lambda \geq K.$$

The constant C may change and is independent of K. Until D_1, D_2 are fixed below, it is also independent of them. Pure derivatives v_{ij} are not included in this bound.

Differentiating the largest eigenvalue.

We make its use at a multiple eigenvalue precise. Choose a smooth local unitary frame parallel for the Chern connection at the point, with first vector an eigenvector for λ. In this frame let B have eigenvalues $0, \epsilon_0, \ldots, \epsilon_0$, where $\epsilon_0 > 0$ is fixed and small. Then $B \geq 0$, $\nabla B = 0$ at the point, and its second derivatives are bounded by

fixed background constants. The largest eigenvalue $\widetilde{\lambda}$ of $g - B$ is simple near the point, is at most $\lambda_1(g)$, and equals λ there. Hence the corresponding smooth function $\widetilde{H}$ has a local maximum there. Only the largest eigenvalue needs to be simple.

For a self-adjoint matrix family with a simple top eigenvalue, differentiate its normalized eigenvector equation. Its component in the pth eigenspace is the corresponding derivative of the matrix divided by the eigenvalue gap. Differentiating the eigenvalue once more therefore gives

$$\begin{aligned} a_k &:= \widetilde{\lambda}_k = \nabla_k g_{1\bar{1}}, \\ \widetilde{\lambda}_{k\bar{k}} &= \nabla_{\bar{k}} \nabla_k g_{1\bar{1}} + O(1) + \sum_{p>1} \frac{|\nabla_k g_{1\bar{p}}|^2 + |\nabla_k g_{p\bar{1}}|^2}{\lambda - \lambda_p + \epsilon_0}. \end{aligned}$$

This formula remains valid if some lower eigenvalues coincide: sum over any orthonormal basis of their eigenspaces. The gaps are positive and at most $n\lambda + \epsilon_0$. In holomorphic coordinates, $a_k = g_{1\bar{1},k} + O(\lambda)$, while converting the second covariant derivative to partial derivatives introduces at most $C \sum_p |g_{1\bar{p},k}| + C\lambda$. Convert the off-diagonal covariant derivatives in the positive squares as well; their errors are $O(\lambda)$. Young's inequality absorbs the linear off-diagonal terms. Thus, after decreasing a fixed positive coefficient,

$$\widetilde{\lambda}_{k\bar{k}} \geq g_{1\bar{1},k\bar{k}} + \frac{1}{8n\lambda} \sum_{p>1} (|g_{1\bar{p},k}|^2 + |g_{p\bar{1},k}|^2) - C(|g_{1\bar{1},k}| + \lambda). \tag{9.24}$$

The precise positive constant will not be used.

Write $V_k = \partial_k(|\partial v|^2_\gamma)$. The first derivative of $\widetilde{H}$ gives

$$\frac{a_k}{\lambda} = -\phi' V_k - \psi' v_k, \qquad |V_k| \leq C\sqrt{K} \sum_p (|v_{pk}| + |v_{p\bar{k}}|) + CK. \tag{9.25}$$

For the second derivative, the derivatives of the fixed metric in $|\partial v|^2$ cost only a small fraction of its positive Hessian squares and a constant. Therefore

$$\begin{aligned} \widetilde{H}_{k\bar{k}} \geq \frac{\widetilde{\lambda}_{k\bar{k}}}{\lambda} - \frac{|a_k|^2}{\lambda^2} + \phi' \sum_p (v_p v_{\bar{p}k\bar{k}} + v_{\bar{p}} v_{pk\bar{k}}) + \frac{1}{5K} \sum_p (|v_{pk}|^2 + |v_{p\bar{k}}|^2) \\ + \phi''|V_k|^2 + \psi''|v_k|^2 + \psi' v_{k\bar{k}} - C. \end{aligned} \tag{9.26}$$

The third term follows directly by differentiating $\sum \gamma^{p\bar{q}} v_p v_{\bar{q}}$ twice and applying $ab \leq \varepsilon a^2 + C_\varepsilon b^2$ to each metric-derivative term.

The third derivatives and the torsion term.

We provide the estimates that remove third derivatives from this inequality. Define

$$E = \sum_{k,p} f_k(|v_{pk}|^2 + |v_{p\bar{k}}|^2), \qquad S = \sum_{k,p} f_k(|v_{pk}| + |v_{p\bar{k}}|),$$
$$C = -D^2\Phi_g(\nabla_1 g, (\nabla_1 g)^*) \geq 0.$$

The first differentiated equation is

$$\sum_k f_k \nabla_p g_{k\bar{k}} = h_p. \tag{9.27}$$

Since $P_\gamma(W) = Z$ and the connection preserves γ, one has the exact contractions

$$\sum_k f_k \nabla_p W_{k\bar{k}} = \sum_i t_i \nabla_p Z_{i\bar{i}}, \qquad \sum_k f_k \nabla_{\bar{1}} \nabla_1 W_{k\bar{k}} = \sum_i t_i \nabla_{\bar{1}} \nabla_1 Z_{i\bar{i}}. \tag{9.28}$$

The weights in these identities are evaluated at the point; they are not differentiated.

Here is how the large weights are controlled. The only weight t_i not bounded by $(n-1)f_1$ is t_1. In $Z_{1\bar{1}}$, every gradient index differs from 1. Thus a pure Hessian in $\nabla_p Z_{1\bar{1}}$ has an index $q > 1$, and can be weighted by f_q, which is comparable to $\mathcal{F}$. Pure derivatives commute in these coordinates. A mixed Hessian can similarly be placed in column q using conjugate symmetry. All other t_i are bounded by a constant times every f_k. Derivatives of coefficients cost $C\sqrt{K}\mathcal{F}$. Applying these observations to the first identity, and then to (9.27), gives, for each p,

$$\left|\sum_k f_k v_{k\bar{k}p}\right| + \left|\sum_k f_k v_{k\bar{k}\bar{p}}\right| \leq C(S + \sqrt{K}\mathcal{F}). \tag{9.29}$$

The connection terms in the differentiated equation contribute weighted mixed Hessians and $C\sqrt{K}\mathcal{F}$, already on the right. Bounded derivatives of h are absorbed using the positive lower bound for $\mathcal{F}$.

For clarity we also track the second derivative of the equation through the eigenvalue estimate. Partial derivatives of v commute, so

$$g_{1\bar{1},k\bar{k}} = g_{k\bar{k},1\bar{1}} + W_{1\bar{1},k\bar{k}} - W_{k\bar{k},1\bar{1}} + O(1).$$

Twice differentiating the equation covariantly gives

$$\sum_k f_k \nabla_{\bar{1}} \nabla_1 g_{k\bar{k}} = h_{1\bar{1}} + C.$$

The conversion to partial derivatives uses

$$\nabla_{\bar{1}} \nabla_1 g_{k\bar{k}} = g_{k\bar{k},1\bar{1}} - 2\,\mathrm{Re} \sum_q \overline{\Gamma^q_{1k}} g_{k\bar{q},1} + O(\lambda).$$

Rewrite the derivative of g in this term using $g_{k\bar{q},1} = g_{1\bar{q},k} + W_{k\bar{q},1} - W_{1\bar{q},k} + O(1)$ and its conjugate. The first part is absorbed by the off-diagonal squares in (9.24), with cost $C\lambda\mathcal{F} + C\sum_k f_k|g_{1\bar{1},k}|$. The $W_{1\bar{q},k}$ part is bounded by $C\lambda\mathcal{F} + C\sum_{k,p} f_k|v_{pk}|$. For the remaining part use

$$2\,\mathrm{Re}\sum_q \overline{\Gamma^q_{1k}} W_{k\bar{q},1} = W_{k\bar{k},1\bar{1}} - \nabla_{\bar{1}}\nabla_1 W_{k\bar{k}} + O(\sqrt{K}).$$

It cancels the partial second derivative $-W_{k\bar{k},1\bar{1}}$ above. We obtain

$$\begin{aligned}\sum_k f_k\widetilde{\lambda}_{k\bar{k}} \geq C + \sum_k f_k(W_{1\bar{1},k\bar{k}} - \nabla_{\bar{1}}\nabla_1 W_{k\bar{k}}) \\ - C\left(\sum_k f_k|g_{1\bar{1},k}| + \sum_{k,p} f_k|v_{pk}| + \lambda\mathcal{F}\right).\end{aligned} \tag{9.30}$$

The second contraction in (9.28) bounds its negative torsion term. In the t_1 part the possible third derivatives are $v_{q1\bar{1}}$ and their conjugates with $q > 1$; all other parts have weights bounded by Cf_1. Commute partial derivatives and use

$$v_{1\bar{1},k} = g_{1\bar{1},k} - \chi_{1\bar{1},k} - W_{1\bar{1},k}.$$

The exclusions in Lemma 9.12.1 ensure that any remaining pure v_{11} carries only the small weight Cf_1. Derivatives of the connection contribute second derivatives with the same weighted bounds. It follows that

$$\left|\sum_k f_k\nabla_{\bar{1}}\nabla_1 W_{k\bar{k}}\right| \leq C\left(\sum_k f_k|g_{1\bar{1},k}| + \sum_{k,p} f_k|v_{pk}| + \lambda\mathcal{F}\right).$$

For the positive torsion term write $W_{1\bar{1}} = \sum_p(c^p v_p + \overline{c^p} v_{\bar{p}})$ with fixed smooth coefficients. Two derivatives give $c^p\sum_k f_k v_{pk\bar{k}}$ and its conjugate, plus terms bounded by $C(S + \sqrt{K}\mathcal{F})$. Equation (9.29) bounds the former. In this estimate the mixed Hessians can be bounded by $C\lambda$, giving

$$\left|\sum_k f_k W_{1\bar{1},k\bar{k}}\right| \leq C\left(\sum_{k,p} f_k|v_{pk}| + \lambda\mathcal{F}\right).$$

Substitute these estimates in (9.30), then in (9.26). The third derivatives in the latter are bounded by (9.29) and $\phi'|\partial v| \le C/\sqrt{K}$. The resulting linear Hessian terms are absorbed by a small part of E/K, with cost $C\mathcal{F}$. Thus

$$0 \ge \frac{C}{\lambda} - \sum_k f_k \frac{|a_k|^2}{\lambda^2} + \frac{E}{6K} + \phi'' \sum_k f_k |V_k|^2 + \psi'' \sum_k f_k |v_k|^2 + \psi' \sum_k f_k v_{k\bar{k}} - C\left(\sum_k f_k \frac{|g_{1\bar{1},k}|}{\lambda} + \mathcal{F}\right). \tag{9.31}$$

Cancellation when the smallest eigenvalue is controlled.

Fix $0 < \delta < 1/2$, to be chosen after D_1, D_2. First suppose $\lambda_n \ge -\delta\lambda$. Put $I = \{k : f_k > \delta^{-1} f_1\}$, so $1 \notin I$. Equation (9.25) and $\phi'' = 2(\phi')^2$ give

$$-\sum_{k\notin I} f_k |a_k|^2/\lambda^2 \ge -\phi'' \sum_{k\notin I} f_k |V_k|^2 - 2(\psi')^2 \delta^{-1} f_1 K,$$
$$-2\delta \sum_{k\in I} f_k |a_k|^2/\lambda^2 \ge -2\delta\phi'' \sum_{k\in I} f_k |V_k|^2 - 4\delta(\psi')^2 \sum_{k\in I} f_k |v_k|^2.$$

Choose δ small enough that $4\delta(\psi')^2 \le \psi''/2$ throughout $[-M, 0]$. By (9.23),

$$\frac{C}{\lambda} \ge \sum_{k\in I} \frac{f_k - f_1}{\lambda(\lambda - \lambda_k)} |\nabla_1 g_{k\bar{1}}|^2 \ge (1 - 2\delta) \sum_{k\in I} \frac{f_k |\nabla_1 g_{k\bar{1}}|^2}{\lambda^2}.$$

Here $(1-\delta)/(1+\delta) \ge 1 - 2\delta$.

We need the following quantitative comparison of the two derivatives. For every $\varepsilon > 0$, provided λ/K is sufficiently large depending on the fixed function ψ,

$$\sum_{k\in I} f_k \frac{|\nabla_1 g_{k\bar{1}}|^2 - |a_k|^2}{\lambda^2} \ge -\frac{E}{12K} - C\mathcal{F} - \varepsilon C|\psi'|\mathcal{F} - C_\varepsilon |\psi'| \sum_k f_k |v_k|^2. \tag{9.32}$$

Here is a proof keeping the weights explicit. Let $U = \sum_{p>1,q} |v_{pq}|$. Commuting partial derivatives in g gives, for $k > 1$,

$$\nabla_1 g_{k\bar{1}} - a_k = W_{k\bar{1},1} - W_{1\bar{1},k} + O(\lambda), \qquad |W_{k\bar{1},1}| + |W_{1\bar{1},k}| \le C(\lambda + U).$$

The second inequality uses the absence of v_1 in $Z_{k\bar{1}}$: differentiating in direction 1 introduces no pure v_{11}. All mixed Hessians are bounded by $C\lambda$, and every other pure Hessian belongs to U.

Use $|a+e|^2 - |a|^2 \geq -2|a||e|$, then (9.25). For $k > 1$, the pure Hessians in V_k all belong to U. If $H_k = \sum_p |v_{p\bar{k}}|$, the resulting negative terms, divided by λ^2, are bounded by a constant times

$$K^{-1/2}H_k + K^{-1/2}U + \frac{U^2}{\lambda\sqrt{K}} + 1 + |\psi'||v_k| + \frac{|\psi'|}{\lambda}|v_k|U.$$

We used $H_k \leq C\lambda$ and $\lambda \geq K$ to simplify the cross terms. The weight comparison (9.21), symmetry of pure derivatives and Cauchy–Schwarz give $\mathcal{F}U^2 \leq CE$. Consequently, after multiplication by f_k and summation, the first two terms are bounded by any fixed small fraction of E/K plus $C\mathcal{F}$. The third is at most $CE/(\lambda\sqrt{K})$ and is absorbed by another such fraction for large λ/K. For the last two terms use

$$|v_k| \leq \varepsilon + C_\varepsilon |v_k|^2, \qquad |v_k|U \leq |v_k|^2 + U^2.$$

The U^2 contribution is at most $C|\psi'|E/\lambda$, again absorbable for large λ/K. Allocating the fractions to sum to $1/12$ proves (9.32).

Insert this cancellation and the preceding inequalities into (9.31). The remaining derivative error satisfies, by (9.25),

$$C\sum_k f_k|g_{1\bar{1},k}|/\lambda \leq \frac{E}{60K} + C\mathcal{F} + \varepsilon C|\psi'|\mathcal{F} + C_\varepsilon|\psi'|\sum_k f_k|v_k|^2,$$

after Young's inequality and adjustment of constants. Also the torsion contraction satisfies

$$\left|\sum_k f_k W_{k\bar{k}}\right| = \left|\sum_i t_i Z_{i\bar{i}}\right| \leq C\sum_k f_k|v_k| \leq \varepsilon C\mathcal{F} + C_\varepsilon \sum_k f_k|v_k|^2.$$

The first inequality uses the same missing gradient index in $Z_{1\bar{1}}$. Since $\psi' < 0$ and $v_{k\bar{k}} = \lambda_k - \chi_{k\bar{k}} - W_{k\bar{k}}$, we conclude, decreasing the remaining coefficient of E if necessary, that

$$\begin{aligned} 0 \geq f_1\left(\frac{\lambda^2}{40K} - 2(\psi')^2\delta^{-1}K\right) + \left(\frac{\psi''}{2} - C_\varepsilon|\psi'|\right)\sum_k f_k|v_k|^2 \\ - C_0\mathcal{F} - \varepsilon C_0|\psi'|\mathcal{F} + |\psi'|\sum_k f_k(\chi_{k\bar{k}} - \lambda_k). \end{aligned} \tag{9.33}$$

In obtaining the first term we used $|v_{1\bar{1}}|^2 \geq \lambda^2/2 - CK$, which follows from $\chi + W = O(\sqrt{K})$.

For large λ, Lemma 9.12.2 bounds the last sum below by $\tau\mathcal{F}/2$. Choose first ε with $\varepsilon C_0 < \tau/4$. Next choose D_2 so that $\psi''/2 > C_\varepsilon|\psi'|$, and then choose D_1 so that $|\psi'|\tau/4 > C_0$ on $[-M, 0]$. Finally choose δ as above. All these choices depend only on the fixed data. Equation (9.33) then forces $\lambda^2 \leq 80\,\delta^{-1}\sup_{[-M,0]}|\psi'|^2K^2$, proving the desired bound in the first case.

The remaining eigenvalue configuration.

Suppose now $\lambda_n < -\delta\lambda$, keeping the same constants. Since $f_n \geq \mathcal{F}/n$ and $|\chi + W| \leq C\sqrt{K}$,

$$\frac{E}{6K} \geq \frac{f_n}{6K}|v_{n\bar{n}}|^2 \geq \frac{\delta^2\lambda^2}{12nK}\mathcal{F} - C\mathcal{F}.$$

In (9.31) discard C and the ψ'' term. Since ψ is now fixed, $|\psi' \sum_k f_k v_{k\bar{k}}| \leq C\lambda\mathcal{F}$. Use $g_{1\bar{1},k} = a_k + O(\lambda)$ and Young's inequality to bound its error by $\frac{1}{2}\sum_k f_k|a_k|^2/\lambda^2 + C\mathcal{F}$. This leaves

$$0 \geq -\tfrac{3}{2}\sum_k f_k|a_k|^2/\lambda^2 + \frac{\delta^2\lambda^2}{12nK}\mathcal{F} + \phi''\sum_k f_k|V_k|^2 - C\lambda\mathcal{F}.$$

By (9.25) and the scalar inequality $\frac{3}{2}|a+b|^2 \leq 2|a|^2 + 6|b|^2$,

$$\tfrac{3}{2}\sum_k f_k|a_k|^2/\lambda^2 \leq \phi''\sum_k f_k|V_k|^2 + CK\mathcal{F}.$$

As $\lambda \geq K$, cancellation and division by $\lambda\mathcal{F} > 0$ give $\lambda/K \leq C$. This finishes both cases. The maximum property of H propagates the bound to all points, and $|\lambda_i| \leq (n-1)\lambda_1$ then bounds every mixed Hessian entry, as asserted. □

9.14 A Liouville theorem for the limiting cone

Rescaling at a point of large gradient removes the background and first-order terms, leaving a degenerate equation on $\mathbb{C}^n$. Its bounded Lipschitz solutions are constant. The proof uses dimension reduction when a directional energy becomes small, and a strict admissible comparison function when all directional energies stay positive.

Use Euclidean coordinates $z = (z_1, \ldots, z_n)$, the identity matrix I, the balls $B(x, r)$ and Euclidean volume $\mathrm{d}y$. In this section ∇ and Δ denote the Euclidean real gradient and Laplacian, and ∂_ξ denotes differentiation in a complex direction ξ.

For $n \geq 2$ and a Hermitian matrix H, set

$$P_n(H) = \frac{(\operatorname{tr} H)I - H}{n-1}.$$

This linear map preserves nonnegativity and satisfies $P_n(I) = I$ and $\operatorname{tr} P_n(H) = \operatorname{tr} H$. For a real function v write $H(v) = (v_{i\bar{j}})$ in Euclidean coordinates. We say that a continuous v is *admissible* if $P_n(H(v)) \geq 0$ in the sense of matrix-valued distributions. We say that it is *maximal for this cone* if it is admissible and every smooth function h touching v from below at a point satisfies

$$P_n(H(h)) \not> 0$$

at that point. Touching means that $v - h$ has a local minimum zero; it need not be a strict minimum. This is the viscosity supersolution condition for the degenerate boundary equation, together with distributional admissibility.

Admissible functions are subharmonic, by taking the trace. Convolution preserves admissibility because P_n is linear. Nonnegative linear combinations of smooth admissible Hessians remain admissible, as do their integrals. Translation and unitary change of coordinates preserve both conditions, as does multiplication of a function by a positive constant and addition of a constant.

Lemma 9.14.1 *Local uniform limits preserve admissibility and maximality. If v is maximal on a bounded open set $D \subset \mathbb{C}^n$, h is smooth and admissible on D, both functions are continuous on $\overline{D}$, and $h \le v$ on ∂D, then $h \le v$ on D.*

Proof Let v_j be a sequence of admissible maximal functions converging locally uniformly to v. Distributional positivity is closed under this convergence. For maximality, suppose a smooth h touches a limit v from below at z_0 and $P_n(H(h)(z_0)) > 0$. Choose a small ball on which $v - h \ge 0$. Replacing h by $h - \delta|z - z_0|^2$, with $\delta > 0$ sufficiently small, keeps its matrix strictly positive near z_0 and makes the minimum strict. Uniform convergence produces interior minimum points of $v_j - h + \delta|z - z_0|^2$ tending to z_0. Adding the minimum value to the test function produces a lower touching function for v_j with strictly positive matrix, a contradiction.

For comparison, choose R with $\overline{D} \subset \{|z| \le R\}$. If $h > v$ somewhere, then for small $\delta > 0$ the smooth function

$$h_\delta = h + \delta(|z|^2 - R^2)$$

still exceeds v at some interior point and is at most v on the boundary. The negative minimum of $v - h_\delta$ lies in the interior. But $P_n(H(h_\delta)) = P_n(H(h)) + \delta I > 0$ there, contradicting maximality after adding the minimum value to h_δ. No boundary regularity of D is needed for this argument. □

Theorem 9.14.1 *A bounded globally Lipschitz function on $\mathbb{C}^n$, $n \ge 2$, that is maximal for this cone is constant.*

Proof Let v be a bounded globally Lipschitz maximal function on $\mathbb{C}^n$. We argue by induction on n. For $n = 2$, $P_2(H) \ge 0$ is equivalent to $H \ge 0$, so v is plurisubharmonic. By Lemma 1.3.5, it is constant.

We record the reduction in dimension for $n \ge 3$. If a function is independent of z_n, its Hessian has the block form $H \oplus 0$. The first block of $P_n(H \oplus 0)$ is

$$\frac{n-2}{n-1} P_{n-1}(H),$$

and the last entry is $\operatorname{tr} H/(n-1)$. Positivity of $P_{n-1}(H)$ implies $\operatorname{tr} H \ge 0$, and strict positivity implies strict positivity of this trace. Thus nonnegative, and respectively positive definite, matrices correspond in these two dimensions. The distributional version follows by the same linear identities. A lower test function downstairs extends

independently of z_n, so maximality also descends. Consequently the induction hypothesis applies to a bounded Lipschitz limit independent of one complex variable.

Suppose now that v is nonconstant. Translate and rescale its values so that $\inf v = 0$, $\sup v = 1$, and $0 \le v \le 1$. Let its Lipschitz constant be L. Choose a smooth radial nonnegative probability kernel η, supported on the unit ball and strictly positive in its interior. For a bounded measurable function f and $r > 0$, write

$$[f]_r(x) = \int_{\mathbb{C}^n} f(x+y)\eta_r(y)\,\mathrm{d}y, \qquad \eta_r(y) = r^{-2n}\eta(y/r), \qquad v_\epsilon = [v]_\epsilon.$$

Both v and v^2 are subharmonic. For the latter, convolve v, use $v_\epsilon \ge 0$ and $\Delta(v_\epsilon^2) = 2v_\epsilon \Delta v_\epsilon + 2|\nabla v_\epsilon|^2 \ge 0$, and pass to the uniform limit.

For either $f = v$ or $f = v^2$, the radial averages $[f]_r(x)$ increase with r by the subharmonic spherical-mean inequality. Their limit is independent of x, since

$$|[f]_r(x) - [f]_r(y)| \le \|f\|_\infty \|\eta_r(\cdot - x) - \eta_r(\cdot - y)\|_{L^1} \le C\frac{|x-y|}{r}.$$

The submean inequality makes that limit at least $f(x)$ for every x, while it is at most $\sup f = 1$. Therefore

$$[v]_r(x) \longrightarrow 1, \qquad [v^2]_r(x) \longrightarrow 1. \tag{9.34}$$

First suppose that there exist $\rho > 0$, sequences $\epsilon_k \downarrow 0$, $r_k \to \infty$, points x_k and unit complex directions ξ_k such that

$$[v^2]_{r_k}(x_k) + [v]_\rho(x_k) - 2v(x_k) \ge \frac{4}{3}, \qquad \int_{B(x_k,r_k)} |\partial_{\xi_k} v_{\epsilon_k}|^2\,\mathrm{d}y \longrightarrow 0. \tag{9.35}$$

Translate x_k to the origin and make ξ_k the last complex direction by a unitary rotation. The resulting functions have the same uniform bound and Lipschitz constant. A subsequence converges locally uniformly to a maximal function a, by the lemma and compactness of Lipschitz functions. The corresponding mollifications at ϵ_k have the same local uniform limit. Their derivatives in the last complex direction tend to zero in L^2 on every fixed ball, by the integral condition. Hence that derivative of a is zero as a distribution. Since a is real and continuous, it is independent of z_n.

The dimension reduction and the induction hypothesis make a a constant $c \in [0,1]$. In the first inequality of (9.35), the first term is at most one, the fixed-radius average tends to c, and the value at the origin tends to c. It would follow that $4/3 \le 1 - c \le 1$, a contradiction.

We are left with the negation of this case. For every fixed $\rho > 0$, there is $C_\rho \geq 1$ such that, whenever $0 < \epsilon < C_\rho^{-1}$, $r > C_\rho$, and

$$[v^2]_r(x) + [v]_\rho(x) - 2v(x) \geq \frac{4}{3},$$

one has

$$\int_{B(x,r)} |\partial_\xi v_\epsilon|^2 \, \mathrm{d}y \geq C_\rho^{-1} \quad \text{for every unit complex direction } \xi. \tag{9.36}$$

Indeed, failure for every integer bound would provide the sequences in (9.35) by a diagonal choice.

Choose the origin with $v(0) < 1/12$. By (9.34), choose ρ with $[v]_\rho(0) > 3/4$, and then $r > C_\rho$ with $[v^2]_r(0) > 3/4$. The open set

$$D = \{x : 2v(x) < [v^2]_r(x) + [v]_\rho(x) - 4/3\}$$

contains the origin. We claim that, for some $c > 0$,

$$[v^2]_r - c|z|^2$$

is smooth and admissible on D.

To see this, differentiate the square of the smooth mollification:

$$H(v_\epsilon^2) = 2v_\epsilon H(v_\epsilon) + 2(\partial_i v_\epsilon \, \overline{\partial_j v_\epsilon})_{ij}.$$

After convolution at the fixed radius r, call these two terms A_ϵ and B_ϵ. The first has $P_n(A_\epsilon) \geq 0$, because $v_\epsilon \geq 0$, v_ϵ is admissible, and P_n is linear.

The second satisfies $B_\epsilon \geq c_0 I$ uniformly on D for small ϵ. Here the passage from the unweighted integral in (9.36) to the convolution needs care. The bound $|\nabla v_\epsilon| \leq L$ makes the energy of a sufficiently thin boundary annulus of $B(x,r)$ at most $C_\rho^{-1}/2$, uniformly in x, ϵ, ξ. On the remaining smaller closed ball, the fixed kernel η_r has a positive minimum. Equation (9.36) then gives a uniform positive lower bound for the weighted integral in every direction. This is exactly the asserted matrix inequality for B_ϵ.

Choose $0 < c \leq c_0$. Since P_n preserves nonnegative matrices,

$$P_n\big(H([v_\epsilon^2]_r) - cI\big) \geq 0 \quad \text{on } D.$$

As $\epsilon \downarrow 0$, $v_\epsilon^2 \to v^2$ uniformly, and convolution at the fixed positive radius gives convergence of every derivative. This proves the claim.

The smooth function

$$h = \tfrac{1}{2}\big([v^2]_r - c|z|^2 + [v]_\rho - 4/3\big)$$

is therefore admissible on D. The set $D' = \{v < h\}$ contains the origin, is bounded because v and its averages are bounded, and lies in D. Moreover $v = h$ on $\partial D'$. Apply

Lemma 9.14.1 on D' to obtain $h \le v$, contradicting its definition. This excludes the second case and proves the theorem. □

The fixed Lipschitz bound is used both for compactness of translates and for the annulus estimate. The conclusion does not follow merely from bounded subharmonicity on $\mathbb{C}^n$ when $n > 1$.

9.15 The gradient bound by rescaling

The Hessian estimate makes a sequence rescaled by its maximum gradient compact in C^1 on every fixed Euclidean ball. A hypothetical gradient blowup would therefore give a bounded Lipschitz function with a nonzero derivative at the origin. The rescaled determinant tends to zero, so the limit is maximal for the cone of Section 9.14, contradicting its Liouville theorem.

Consider a family of normalized solutions of Theorem 9.9.1 with $\|F\|_{C^0} \le M$ for a fixed $M > 0$. The potential estimate bounds $\|u\|_{C^0}$ and $|b|$ uniformly. We isolate the consequence of a Hessian estimate for this family.

Proposition 9.15.1 *Suppose there is a constant K such that every solution in this family satisfies*

$$\sup_X |\mathrm{dd^c} u|_\gamma \le K\big(1 + \sup_X |\partial u|_\gamma^2\big). \tag{9.37}$$

Then $\sup_X |\partial u|_\gamma$ *and* $\sup_X |\mathrm{dd^c} u|_\gamma$ *are bounded by a constant depending only on* X, γ_0, γ, M *and* K.

Proof Use the cofactor matrices Q_u of (9.18). Their background term is fixed, and Z is linear in the first derivatives of $u/(2\pi)$.

Suppose the gradient bound fails. Choose solutions (u_j, b_j) with prescribed functions F_j such that

$$A_j = \sup_X |\partial u_j|_\gamma = |\partial u_j(x_j)|_\gamma \longrightarrow \infty.$$

After extraction the points lie in one fixed coordinate neighborhood. Translate their coordinate centers to zero and make a uniformly controlled complex linear change so that $\gamma(x_j)$ is the identity. A fixed-radius coordinate ball remains available at every center. On balls whose radii tend to infinity define

$$v_j(z) = u_j(x_j + z/A_j),$$

where the preceding linear changes are understood. The coefficient matrices $g_j(z)$ of γ evaluated at these scaled points converge smoothly on every fixed ball to the Euclidean identity. In this notation

$$|v_j| \le C, \qquad |\partial v_j|_{g_j} \le 1, \qquad |\partial v_j(0)| = 1.$$

The assumed estimate gives a uniform bound for all complex Hessian entries of v_j on every fixed ball, since two derivatives contribute A_j^{-2}. Hence their Euclidean Laplacians are uniformly bounded. The standard interior estimates for the scalar Laplacian give local $W^{2,p}$ bounds for every finite p. Choose $p > 2n$ and use Sobolev embedding and compactness to extract a diagonal subsequence converging in C^1 on compact subsets to a function v on $\mathbb{C}^n$. It is bounded and globally Lipschitz, and

$$|\partial v(0)| = 1.$$

Divide the cofactor matrix in (9.18) by A_j^2 and evaluate at the scaled points. The background term is $O(A_j^{-2})$. The first-order term is $O(A_j^{-1})$, since one derivative of u_j is A_j times a derivative of v_j. The uniformly bounded complex Hessians and $g_j \to I$ therefore give, in fixed Euclidean coordinates on any fixed ball,

$$A_j^{-2} Q_{u_j} = P_n\left(\frac{H(v_j)}{2\pi}\right) + o(1), \qquad \det(A_j^{-2} Q_{u_j}) = A_j^{-2n} \mathrm{e}^{(n-1)(F_j+b_j)} \longrightarrow 0. \tag{9.38}$$

For precision, take the compatible unitary frames obtained from the positive square roots $S_j = g_j^{-1/2}$. The principal matrix in these frames is $P_n(S_j H(v_j) S_j/(2\pi))$, which differs from $P_n(H(v_j)/(2\pi))$ by a uniformly vanishing error. No derivatives of these frames are taken, and the determinant identity is exact in the unitary frames. The positive factor $1/(2\pi)$ does not change the cone. The positive matrices on the left imply, by passage to distributions, $P_n(H(v)) \geq 0$.

We check maximality. Suppose a smooth h touches v from below at z_0 and $P_n(H(h)(z_0)) > 0$. Subtract $\delta|z - z_0|^2$ with $\delta > 0$ small. On a small ball this makes the contact strict and preserves strict positivity of the matrix. Local uniform convergence supplies interior lower contact points $z_j \to z_0$ for v_j and this perturbed test function, after adding constants to the latter. The complex Hessian inequality at a minimum gives

$$H(v_j)(z_j) \geq H(h)(z_j) - \delta I.$$

Because P_n preserves nonnegative matrices, (9.38) now implies $A_j^{-2} Q_{u_j}(z_j) \geq cI$ for a fixed $c > 0$ and all sufficiently large j. Its determinant is at least c^n, contradicting the limit zero in that equation. Thus v is maximal for the cone of Section 9.14.

The Liouville theorem makes v constant, contradicting $|\partial v(0)| = 1$. This proves the uniform gradient bound. Substitution in (9.37) gives the uniform complex Hessian bound. □

For prescribed functions bounded in C^2, the potential estimate bounds u and b, and Theorem 9.13.1, after the fixed conversion $v = u/(2\pi)$, proves (9.37). Thus both the gradient and the complex Hessian are uniformly bounded.

9.16 Existence for the Gauduchon ansatz

The potential, gradient and Hessian bounds put the cofactor matrices in a compact set of positive matrices. This yields uniform ellipticity after the real Hessian is projected to its complex part and transformed by P_n. Local elliptic regularity gives the compactness needed for closedness; the adjoint probability gives openness. The maximum principle then proves uniqueness within the scalar ansatz.

Lemma 9.16.1 *For the fixed forms in Theorem 9.9.1, the preceding potential, gradient and second-order estimates give uniform C^k bounds for smooth normalized solutions whenever the smooth prescribed functions F range over a set bounded in every C^k norm.*

Proof Work first with $v = u/(2\pi)$ and write $Q = Q_u$ for the matrix in (9.18). The second-order theorem and the rescaling proposition give uniform bounds for v, its gradient and its complex Hessian. The constants b are bounded by the potential estimate. Hence Q has uniformly bounded eigenvalues above, and its determinant is uniformly bounded below by the equation. Every eigenvalue is therefore bounded below by a positive constant as well. Choose fixed $0 < a < b_0$ such that $aI \le Q \le b_0 I$ in unitary frames for every solution.

The fixed scalar operator $\Delta_\gamma v = \mathrm{tr}_\gamma(\mathrm{i}\partial\bar\partial v)$ is uniformly elliptic and has bounded right-hand side. Its standard interior $W^{2,p}$ estimates, applied on a finite cover, bound v in $W^{2,p}$ for every finite p. Choose $p > 2n$ and $0 < \alpha < 1 - 2n/p$. Sobolev embedding gives a uniform $C^{1,\alpha}$ bound. Thus the matrix $R(x) = Q_0(x) + Z_x(\partial v(x))$ is uniformly C^α.

On a coordinate ball, let $\mathcal{H}_x(A)$ denote the Hermitian matrix of $\mathrm{i}\partial\bar\partial v$ in a fixed smooth unitary frame when $A = D^2_{\mathbb{R}} v$. It is a real linear map in A, smooth in x. If a real symmetric matrix N is nonnegative, then $\mathcal{H}_x(N) \ge 0$, and its trace is comparable to $\mathrm{tr}_{\mathbb{R}} N$ with uniform positive constants on the ball. This follows either by evaluating on a vector and its complex rotation, or by the coordinate formula for $v_{i\bar j}$.

Freeze the already obtained function $\partial v(x)$ in $R(x)$ and define on all real symmetric A

$$\widetilde{\Phi}(A,x) = \inf_{aI \le B \le b_0 I} \left\{ \log\det B + \mathrm{tr}\left[B^{-1}\big(R(x) + P_n(\mathcal{H}_x(A)) - B\big)\right]\right\}.$$

The infimum is over a fixed compact set of positive Hermitian matrices. This function is concave in A. If $R(x) + P_n(\mathcal{H}_x(A))$ lies in that set, it agrees with its logarithmic determinant: each expression is a supporting affine function of the concave logarithmic determinant, and the tangent at that very matrix attains equality.

For $N \ge 0$, each affine function increases by

$$\mathrm{tr}\big(B^{-1} P_n(\mathcal{H}_x(N))\big).$$

Since P_n preserves positivity and trace, this lies between fixed positive multiples of $\mathrm{tr}_{\mathbb{R}} N$, uniformly in B, x. Taking infima preserves both inequalities. Thus $\widetilde{\Phi}$ is uniformly elliptic on all real symmetric matrices.

Smoothness of $\mathcal{H}_x$ and the uniform C^α bound for R also give

$$|\widetilde{\Phi}(A,x) - \widetilde{\Phi}(A,y)| \le C|x-y|^\alpha(1+\|A\|).$$

These are the hypotheses of the standard local concave, uniformly elliptic Evans–Krylov estimate with Hölder coefficients, in the formulation recorded in Section 33.6. The value $\widetilde{\Phi}(0,x)$ is uniformly bounded by the compact tangent set and the bound for R. The given smooth function solves

$$\widetilde{\Phi}(D^2_{\mathbb{R}} v, x) = (n-1)(F+b)$$

on the ball. That estimate, on smaller balls of a finite cover, yields a uniform $C^{2,\beta}$ bound for some $\beta > 0$.

Differentiating the original smooth matrix equation gives linear uniformly elliptic equations for first derivatives of v. Their coefficients, including the first-order coefficients from Z, are now Hölder continuous. Local Schauder estimates give $C^{3,\beta}$ bounds, and differentiation and induction give every higher bound. The normalizations and the finite coordinate cover make the bounds global. The only regularity inputs here are the standard scalar $W^{2,p}$, concave Evans–Krylov and Schauder estimates.□

Proof (of Theorem 9.9.1) Let

$$F_0 = \log \frac{\gamma_0^n}{\gamma^n}, \qquad F_t = (1-t)F_0 + tF, \quad 0 \le t \le 1.$$

At $t = 0$, $u = 0, b = 0$ solves the equation. For the continuity argument normalize by $\int_X u\,\gamma^n = 0$; adding a constant has no effect on the equation, and at the end we impose $\sup u = 0$ instead.

For openness, the derivative in the mean-zero potential and the scalar is $(w,a) \mapsto \mathcal{L}_u w - a$. Given a right-hand side q, integration against the adjoint probability of Proposition 9.11.1 forces $a = -\int_X q\,\mathrm{d}\mu_u^\gamma$. The Fredholm alternative then gives a unique mean-zero w, and Schauder estimates bound this inverse between Hölder spaces. Since positivity of Γ_u is open, the implicit function theorem gives solutions for all nearby parameters.

For closedness, the prescribed path has uniform bounds in every smooth norm. The potential estimate bounds u after supremum normalization and bounds b. The second-order theorem and the rescaling argument give the gradient and complex Hessian bounds, and the preceding lemma gives every higher bound. Subtracting the mean preserves all derivative bounds and changes the uniformly bounded potentials by uniformly bounded constants. A convergent subsequence of mean-zero solutions therefore gives a smooth solution at a limiting parameter. The determinant lower bound and matrix upper bound preserve strict positivity in the limit. Thus the nonempty open and closed set of parameters is all of $[0,1]$.

Finally consider two solutions (u, b) and (w, c) for the same F. At a maximum of $u - w$ their gradients agree and their complex Hessians are ordered. The first-order terms are equal there, and P_n preserves the Hessian ordering. Consequently $Q_u \leq Q_w$, so $b \leq c$ by the equation. At a minimum the reverse inequality gives $b \geq c$. Hence $b = c$. Along the segment between u and w, the matrices Q remain positive because the ansatz is affine. Integrating the linearized logarithmic determinant along that segment produces a scalar uniformly elliptic operator with drift and no zeroth-order term annihilating $u - w$. The maximum principle makes $u - w$ constant. Supremum normalization makes the constant zero. □

9.17 Normalization for the Gauduchon equation

The existence theorem now defines a scalar $b_\gamma(F)$ for every smooth real F. Its variation is governed by the adjoint probability μ_u^γ and the concavity of the logarithmic determinant of Γ_u. To differentiate solutions, fix a smooth positive volume form $\mathrm{d}V_0$ and use $\int_X u \, \mathrm{d}V_0 = 0$ in place of the supremum normalization.

Lemma 9.17.1 *The mean-normalized pair $(u, b_\gamma(F))$ depends smoothly on smooth finite-dimensional families of F.*

Proof The positive-root map is smooth, since in a frame it is the inverse of the positive cofactor-matrix map. The openness argument in the proof of Theorem 9.9.1 gives a bounded inverse for the derivative in the potential and scalar between the relevant Hölder spaces, also with the chosen $\mathrm{d}V_0$-mean. The implicit function theorem gives local smooth solution maps; uniqueness joins them. Applying this for every order of Hölder regularity gives the assertion for smooth data. □

For a smooth family of prescribed functions $F_t \in C^\infty(X, \mathbb{R})$, with t in a real open interval, let u_t be the mean-normalized solutions and set $Q_t = Q_{u_t}$ as in (9.18). The expression

$$\mathrm{tr}\big((Q_t^{-1}\dot{Q}_t)^2\big)$$

is independent of the chosen frame and is nonnegative: the endomorphism is similar to the Hermitian matrix $Q_t^{-1/2}\dot{Q}_t Q_t^{-1/2}$.

Theorem 9.17.1 *For a smooth family F_t and its solutions,*

$$\dot b_\gamma(F_t) = -\int_X \dot F_t \,\mathrm{d}\mu^\gamma_{u_t},$$
$$\ddot b_\gamma(F_t) = -\int_X \ddot F_t \,\mathrm{d}\mu^\gamma_{u_t} - \frac{1}{n-1}\int_X \operatorname{tr}\big((Q_t^{-1}\dot Q_t)^2\big)\,\mathrm{d}\mu^\gamma_{u_t}.$$

In particular b_γ is concave, and along $F_t = F_0 + tG$ its second derivative is strictly negative unless G is constant. For smooth real functions F, G on X and a real constant a, moreover,

$$b_\gamma(F + a) = b_\gamma(F) - a, \qquad |b_\gamma(G) - b_\gamma(F)| \le \|G - F\|_\infty.$$

Proof By (9.18),

$$\mathcal{G}(u) = \frac{1}{n-1}\log\det_\gamma Q_u.$$

Differentiate $\mathcal{G}(u_t) = F_t + b_\gamma(F_t)$ once. Integration against the adjoint probability gives the first formula.

Because Γ_u is affine in u, its second derivative along the family is $\mathcal{D}_\gamma \ddot u_t$. The matrix derivative identity for log det therefore gives

$$\mathcal{L}_{u_t}\ddot u_t - \frac{1}{n-1}\operatorname{tr}\big((Q_t^{-1}\dot Q_t)^2\big) = \ddot F_t + \ddot b_\gamma(F_t).$$

Integrate at this fixed parameter against $\mu^\gamma_{u_t}$ to obtain the second formula. No derivative of the measure is needed.

For affine F_t, positivity of the density of $\mu^\gamma_{u_t}$ proves concavity. Equality in the second derivative forces $\dot\Gamma_{u_t} = 0$, hence $\mathcal{L}_{u_t}\dot u_t = 0$. The kernel statement makes $\dot u_t$ constant, so the first differentiated equation forces G to be constant. Conversely a constant change of F changes only b by the opposite constant. Finally integrate the first derivative along the line from F to G; it lies between $-\sup_X(G - F)$ and $-\inf_X(G - F)$, proving the stated Lipschitz bound. □

The Lipschitz bound extends b_γ uniquely to continuous F. Constructing corresponding weak solutions would additionally require a root-volume measure for singular forms in the Aeppli ansatz, compactness of their potentials and stability of the adjoint equation.

On a surface with Gauduchon background, conservation of total mass makes the scalar and its adjoint probability explicit.

Example 9.17.1 If $n = 2$ and γ_0 is Gauduchon, then $\eta_u = \gamma_0 + \mathrm{dd^c} u$ is also Gauduchon. The mass identity of Theorem 2.2.2 gives $\int_X \eta_u^2 = \int_X \gamma_0^2$. Therefore

$$b_\gamma(F) = \log \int_X \gamma_0^2 - \log \int_X \mathrm{e}^F \gamma^2, \qquad \mathrm{d}\mu_u^\gamma = \frac{\mathrm{e}^F \gamma^2}{\int_X \mathrm{e}^F \gamma^2}.$$

The adjoint equation gives the same measure because ρ_u is constant. Along $F + tG$, the second derivative is minus the variance of G under this probability.

Chapter 10
Continuous solutions and normalization

Let X be a connected compact complex manifold of dimension $n \geq 1$, let ω be a Hermitian form, and fix a smooth positive volume form $\mathrm{d}V$. Fix $p > 1$ and a nonnegative $f \in L^p(X, \mathrm{d}V)$ with $\int_X f \, \mathrm{d}V > 0$. Norms and almost-everywhere statements use $\mathrm{d}V$ unless another measure is specified.

Smooth approximations of f give solutions by the preceding chapter, but normalized L^1 compactness alone does not identify the limit of their Monge–Ampère measures. A Gauduchon trace bounds the normalization scalars, giving uniform L^p control of the equation densities. Modified comparison then strengthens normalized L^1 convergence to uniform convergence. For the exponential equation, the same trace controls the additive constants of the potentials. Letting the exponent tend to zero determines a unique scalar for the prescribed density, even when the normalized potential is not unique.

10.1 Continuous solvability

The prescribed density can be normalized by an unknown scalar or by an exponential of the potential:

Theorem 10.1.1 (Kołodziej–Nguyen) *There are $c > 0$ and $u \in \mathrm{PSH}(X, \omega) \cap C(X)$ such that*

$$(\omega + \mathrm{dd^c} u)^n = c f \, \mathrm{d}V.$$

One may normalize $\sup_X u = 0$.

The scalar is unique by Proposition 10.5.1. The normalized potential need not be unique, even for a smooth density; Theorem 11.6.1 gives an example.

Theorem 10.1.2 (Nguyen) *For every $\lambda > 0$, there is a unique continuous ω-plurisubharmonic function u_λ satisfying*

$$(\omega + \mathrm{dd^c} u_\lambda)^n = \mathrm{e}^{\lambda u_\lambda} f \, \mathrm{d}V. \tag{10.1}$$

10.2 The modified comparison principle

The variation of total mass prevents the ordinary global comparison principle. Replacing a comparison potential v by $(1-\varepsilon)v$ reserves strict positivity that absorbs the torsion terms. On a sublevel of height s, integration by parts makes the error proportional to s; a finite induction controls the intervening mixed products. The multiplicative correction tends to one as the height tends to zero with ε fixed.

Choose $B_0 \geq 1$ such that

$$-B_0\omega^2 \leq \mathrm{dd^c}\omega \leq B_0\omega^2, \qquad -B_0\omega^3 \leq \mathrm{d}\omega \wedge \mathrm{d^c}\omega \leq B_0\omega^3.$$

Only inequalities after wedging with positive complementary forms are needed. Such a constant exists because ω is Hermitian and X is compact. Terms whose degree exceeds the dimension are omitted.

Theorem 10.2.1 *Let $u, v \in \mathrm{PSH}(X,\omega) \cap L^\infty(X)$ and $0 < \varepsilon < 1$. For $s > 0$, put*

$$m_\varepsilon = \inf_X \big(u - (1-\varepsilon)v\big), \qquad U_{\varepsilon,s} = \{u < (1-\varepsilon)v + m_\varepsilon + s\}.$$

If $0 < s < \varepsilon^3/(32B_0n^2)$, then, with $a = B_0n^2s\varepsilon^{-3}$,

$$(1-4a)^n \int_{U_{\varepsilon,s}} \omega^n_{(1-\varepsilon)v} \leq \int_{U_{\varepsilon,s}} \omega^n_u. \tag{10.2}$$

In particular, the reverse factor $(1-4a)^{-n}$ tends to one as $s \downarrow 0$ for fixed ε.

Proof Write $v_\varepsilon = (1-\varepsilon)v$, $U = U_{\varepsilon,s}$ and

$$\phi = \max(u, v_\varepsilon + m_\varepsilon + s), \qquad T_k = \omega_u^k \wedge \omega_\phi^{n-k}, \qquad I_k = \int_U T_k \quad (0 \leq k \leq n).$$

Set $I_{-1} = I_{-2} = 0$. Then $0 \leq \phi - u \leq s$, this difference vanishes outside U, and on U locality gives $\omega_\phi = \omega_{v_\varepsilon} \geq \varepsilon\omega$ in all bounded products.

By Lemma 1.7.1, for every fixed positive mixed product W of $n-1$ bounded-potential factors,

$$\mathbf{1}_{\{u=\phi\}}\omega_u \wedge W \leq \mathbf{1}_{\{u=\phi\}}\omega_\phi \wedge W.$$

For $0 \leq k \leq n-1$, apply it with $W = \omega_u^k \wedge \omega_\phi^{n-k-1}$. On $X \setminus U$ the difference $T_k - T_{k+1}$ is nonnegative. Consequently

$$I_k - I_{k+1} \leq \int_X \mathrm{dd^c}(\phi - u) \wedge W = \int_X (\phi - u)\mathrm{dd^c}W. \tag{10.3}$$

This Stokes identity is valid for the bounded potentials in use. To check it, approximate u and ϕ by bounded smooth decreasing ω-plurisubharmonic functions using Theorem 1.11.1. For smooth functions it is ordinary integration by parts. The derivative $\mathrm{dd^c}W$ expands as below in bounded Hessian products and smooth forms; bounded Bedford–Taylor convergence, also against bounded quasi-continuous coefficients converging in capacity, passes each term to the limit.

Set $r = n - k - 1$. Since $\mathrm{d}\omega_u = \mathrm{d}\omega_\phi = \mathrm{d}\omega$, expansion gives

$$\begin{aligned}\mathrm{dd^c}W = {} & k\,\mathrm{dd^c}\omega \wedge \omega_u^{k-1} \wedge \omega_\phi^r + r\,\mathrm{dd^c}\omega \wedge \omega_u^k \wedge \omega_\phi^{r-1} \\ & + k(k-1)\mathrm{d}\omega \wedge \mathrm{d^c}\omega \wedge \omega_u^{k-2} \wedge \omega_\phi^r \\ & + 2kr\,\mathrm{d}\omega \wedge \mathrm{d^c}\omega \wedge \omega_u^{k-1} \wedge \omega_\phi^{r-1} \\ & + r(r-1)\mathrm{d}\omega \wedge \mathrm{d^c}\omega \wedge \omega_u^k \wedge \omega_\phi^{r-2}.\end{aligned}$$

Factors with a zero coefficient and a negative power are absent. On U we can replace each ω^j in an upper bound by $\varepsilon^{-j}\omega_\phi^j$. Using $\phi - u \le s$ and $\varepsilon^{-2} \le \varepsilon^{-3}$ in (10.3), we obtain

$$I_k - I_{k+1} \le B_0 s \varepsilon^{-3}\big(k(k-1)I_{k-2} + k(1+2r)I_{k-1} + r^2 I_k\big).$$

Each coefficient is at most n^2. Hence

$$I_k - I_{k+1} \le a(I_{k-2} + I_{k-1} + I_k). \tag{10.4}$$

Put $q = 1 - 4a > 7/8$. Inductively suppose $qI_j \le I_{j+1}$ for $j < k$. Then $I_{k-1} \le q^{-1}I_k$ and $I_{k-2} \le q^{-2}I_k$ whenever the indices exist. Since

$$1 + q^{-1} + q^{-2} \le 1 + 8/7 + 64/49 < 4,$$

(10.4) gives $qI_k \le I_{k+1}$. The case $k = 0$ follows from the same formula with the two absent terms set to zero. Induction proves $q^n I_0 \le I_n$. On U, $T_0 = \omega_{v_\varepsilon}^n$, while $T_n = \omega_u^n$, giving the theorem. In dimension one all torsion terms vanish and the same argument applies. □

A strict multiplicative inequality between the measures absorbs the comparison error. Choose the positivity reserve first, and then a sublevel thin enough to retain this strict inequality.

Theorem 10.2.2 *Let $u, v \in \mathrm{PSH}(X,\omega) \cap L^\infty(X)$, and let $0 \le c < 1$. If*

$$\mathbf{1}_{\{u<v\}}\omega_u^n \le c\,\mathbf{1}_{\{u<v\}}\omega_v^n,$$

then $u \ge v$. In particular, $\omega_u^n(\{u < v\}) = 0$ implies $u \ge v$.

Proof Suppose $m = \inf_X(u - v) < 0$. Choose $\varepsilon > 0$ so small that

$$(1-\varepsilon)^n > c, \qquad 2\varepsilon\|v\|_\infty < -m/4,$$

omitting the second restriction if $v = 0$. Use m_ε and $U_{\varepsilon,s}$ defined in Theorem 10.2.1 for this pair u, v. Since $|m_\varepsilon - m| \le \varepsilon\|v\|_\infty$, for all sufficiently small $s > 0$ the set

$U = U_{\varepsilon,s}$ is contained in $\{u < v\}$: there

$$u - v < m_\varepsilon + s - \varepsilon v \le m + 2\varepsilon\|v\|_\infty + s < 0.$$

The set U is nonempty by the definition of the infimum and has positive volume by Lemma 1.3.4.

Make s smaller so that Theorem 10.2.1 applies and $(1-4B_0n^2s\varepsilon^{-3})^n(1-\varepsilon)^n > c$. Write the first factor as q^n. Positivity of every mixed term gives

$$\omega^n_{(1-\varepsilon)v} \ge (1-\varepsilon)^n\omega_v^n + \varepsilon^n\omega^n.$$

Comparison and the assumption would therefore imply

$$q^n(1-\varepsilon)^n\int_U \omega_v^n + q^n\varepsilon^n\int_U \omega^n \le \int_U \omega_u^n \le c\int_U \omega_v^n.$$

All integrals are finite. The chosen strict inequality and the positive volume of U make this impossible. □

Corollary 10.2.1 *Every bounded ω-plurisubharmonic function has strictly positive Monge–Ampère mass.*

Proof Let u be a bounded ω-plurisubharmonic function. If $\omega_u^n = 0$, bounded domination with the constant potential $v = \sup_X u + 1$ would give $u \ge v$, a contradiction.□

The lower mass bound here depends on the potential.

Corollary 10.2.2 *If $u, v \in \mathrm{PSH}(X,\omega) \cap L^\infty(X)$ and $\lambda > 0$, then*

$$\mathrm{e}^{-\lambda v}\omega_v^n \ge \mathrm{e}^{-\lambda u}\omega_u^n \quad\Longrightarrow\quad v \le u.$$

Proof For every $t > 0$, on $\{u < v - t\}$ the assumption gives $\omega_u^n \le \mathrm{e}^{-\lambda t}\omega_v^n$. Apply Theorem 10.2.2 to $u, v - t$, and then let $t \downarrow 0$. □

10.3 Capacity estimates and uniform compactness

Modified comparison bounds the capacity of a thin sublevel by the Monge–Ampère mass of a slightly larger one. For an L^p density, Corollary 1.8.1 bounds this mass by the square of the capacity. Iterating the two inequalities gives a lower capacity bound for every nonempty sublevel of fixed height. The tail estimate (1.14) then bounds the depth of a normalized potential. Write $\mathrm{Cap} = \mathrm{Cap}_\omega$ for the capacity of Section 1.8.

Lemma 10.3.1 *Let $u, v \in \mathrm{PSH}(X, \omega) \cap L^\infty(X)$, with $-H \le v \le 0$ and $H \ge 1$. Fix $0 < \varepsilon < 1$, and put*

$$m = \inf_X (u - (1-\varepsilon)v), \qquad U_s = \{u < (1-\varepsilon)v + m + s\}.$$

There are $\delta_\varepsilon > 0$, depending only on (X, ω, ε), and $C_H > 0$, depending only on (n, H), such that

$$t^n \operatorname{Cap}(U_s) \le C_H \int_{U_{s+t}} \omega_u^n \qquad (0 < s, t,\ s + t < \delta_\varepsilon). \tag{10.5}$$

Every U_s, $s > 0$, has positive capacity.

Proof Use the constant B_0 from Theorem 10.2.1 and set

$$\delta_\varepsilon = \min\{1 - \varepsilon, \varepsilon^3/(128 B_0 n^2)\}.$$

For a capacity test $0 \le \rho \le 1$, put

$$\tau = \frac{t}{2(1-\varepsilon)(H+1)}, \quad w = (1-\tau)v + \tau\rho, \quad m' = \inf_X (u - (1-\varepsilon)w).$$

Then $0 < \tau < 1$ and $0 \le (1-\varepsilon)(w - v) \le t/2$, so $m - t/2 \le m' \le m$. Consequently

$$U_s \subset V := \{u < (1-\varepsilon)w + m' + s + t/2\} \subset U_{s+t}.$$

Apply modified comparison to u, w at height $s + t/2$. Its inverse factor is at most $(8/7)^n$. Moreover,

$$\omega_{(1-\varepsilon)w} \ge (1-\varepsilon)\tau\omega_\rho = \frac{t}{2(H+1)}\omega_\rho.$$

Integration over $U_s \subset V$ and comparison over V give (10.5), with $C_H = [2(H+1)]^n (8/7)^n$, after taking the supremum over ρ. Each U_s is nonempty by the definition of the infimum. Lemma 1.3.4 gives positive smooth volume, hence positive capacity by the constant test. □

Lemma 10.3.2 *In the preceding setting, suppose*

$$\int_E \omega_u^n \le A \operatorname{Cap}(E)^2 \qquad (E \subset X \text{ Borel}), \qquad A > 0.$$

For every $0 < D < \delta_\varepsilon$,

$$\operatorname{Cap}(U_D) \ge \frac{D^n}{4^n C_H A}. \tag{10.6}$$

Proof Write $a(s) = \operatorname{Cap}(U_s)^{1/n} > 0$ and $K = (C_H A)^{1/n}$. The preceding lemma gives

$$t\, a(s) \le K a(s+t)^2 \qquad (s + t < \delta_\varepsilon).$$

Suppose $4Ka(D) < D$. Starting with $s_0 = D$, define

$$t_j = 2Ka(s_j), \qquad s_{j+1} = s_j - t_j.$$

We check positivity before using the sublevel inequality. If $a(s_i) \le 2^{-i}a(D)$ for $0 \le i \le j$, then

$$\sum_{i=0}^{j} t_i \le 2Ka(D) \sum_{i=0}^{j} 2^{-i} < 4Ka(D) < D.$$

Hence $s_{j+1} > 0$, and the displayed sublevel inequality gives $a(s_{j+1}) \le a(s_j)/2$. Induction justifies every step, and $s_j \downarrow s_\infty \ge D - 4Ka(D) > 0$. Monotonicity gives $0 < a(s_\infty) \le a(s_j) \le 2^{-j}a(D)$, a contradiction. Therefore $D \le 4Ka(D)$, which is (10.6). □

Theorem 10.3.1 *There are constants $C_p, c_p > 0$, depending only on $(X, \omega, \mathrm{d}V, p)$, such that, whenever $u \in \mathrm{PSH}(X, \omega) \cap L^\infty(X)$, $\sup_X u = 0$, and*

$$\omega_u^n = F\,\mathrm{d}V, \qquad F \in L^p(X, \mathrm{d}V), \quad F \ge 0,$$

we have

$$\|u\|_\infty \le C_p(1 + \|F\|_p), \qquad \|F\|_p \ge c_p. \tag{10.7}$$

Proof The zero density is excluded by Corollary 10.2.1. Apply Corollary 1.8.1 and Lemma 10.3.2 with $v = 0$, $H = 1$, $\varepsilon = 1/2$, and $A = C_p\|F\|_p > 0$. Fix $D = \delta_{1/2}/2$. If $M = -\inf_X u$, the resulting inequality is

$$\mathrm{Cap}(\{u < -M + D\}) \ge c/\|F\|_p.$$

If $M > D + 1$, (1.14) bounds this capacity by $C/(M - D)$, giving the first assertion; the remaining case is immediate. Bounding the same capacity by $\mathrm{Cap}(X)$ gives the second assertion. □

If two uniformly bounded solutions remain separated at some point, the preceding estimate gives a fixed positive capacity to a sublevel of their difference. Small L^1 distance makes its volume small, while the uniform L^p density bound makes its Monge–Ampère mass small. This contradicts the sublevel estimate and upgrades L^1 compactness to uniform compactness. Put $p' = p/(p-1)$.

Theorem 10.3.2 *Suppose $u_j \in \mathrm{PSH}(X, \omega) \cap C(X)$, $\sup_X u_j = 0$, and*

$$\omega_{u_j}^n = F_j\,\mathrm{d}V, \qquad F_j \ge 0, \qquad \sup_j \|F_j\|_p < \infty$$

for the fixed $p > 1$. If (u_j) is Cauchy in $L^1(X, \mathrm{d}V)$, then it is Cauchy uniformly. In particular, every such family has a subsequence converging uniformly to a continuous ω-plurisubharmonic function.

Proof By Theorem 10.3.1, there is a common $H \geq 1$ with $-H \leq u_j \leq 0$. Suppose uniform Cauchy convergence fails. After exchanging indices when necessary, there are a number $a > 0$ and arbitrarily large j, k with

$$S := \inf_X (u_k - u_j) \leq -a.$$

Choose $0 < \varepsilon < 1$ such that $\varepsilon H < a/4$, and choose a fixed $D > 0$ with $2D < \delta_\varepsilon$ and $2D < a/4$. Apply the preceding sublevel lemmas with $u = u_k$, $v = u_j$. Since $v \leq 0$, $m = \inf(u - (1-\varepsilon)v) \leq S$. Thus

$$U_{2D} \subset \{u - v < S + 2D + \varepsilon H\} \subset \{|u - v| > a/2\}.$$

The capacity lower bound (10.6), with a common A supplied by Corollary 1.8.1, gives $\mathrm{Cap}(U_D) \geq c > 0$. On the other hand, (10.5), Hölder and Chebyshev give

$$\begin{aligned} D^n \,\mathrm{Cap}(U_D) &\leq C_H \int_{U_{2D}} F_k \,\mathrm{d}V \\ &\leq C_H \|F_k\|_p \left(\frac{2\|u_k - u_j\|_1}{a} \right)^{1/p'} \longrightarrow 0. \end{aligned}$$

This contradiction proves uniform Cauchy convergence. The normalized compactness theorem, Theorem 1.4.1, supplies an L^1-Cauchy subsequence of any such family. Its uniform limit is continuous, and distributional convergence preserves ω-plurisubharmonicity. □

10.4 Construction of continuous solutions

Smooth approximation gives solutions with unknown normalization scalars. A Gauduchon trace bounds those scalars above, and the lower bound for the density norm bounds them away from zero. The resulting uniform L^p bound puts the potentials in the compact family just constructed, so their limit solves the prescribed equation.

Proof (of Theorem 10.1.1) Choose smooth strictly positive $f_j \to f$ in $L^p(X, \mathrm{d}V)$. By Theorem 9.1.1, smooth existence gives

$$\omega_{u_j}^n = c_j f_j \,\mathrm{d}V, \qquad \sup_X u_j = 0, \qquad c_j > 0.$$

We first control the constants. For $n \geq 2$, fix a Gauduchon metric γ. By Proposition 4.2.2,

$$c_j^{1/n} A_j \leq \int_X \omega \wedge \gamma^{n-1}, \qquad A_j = \int_X (f_j \mathrm{d}V/\gamma^n)^{1/n} \gamma^n.$$

Since $|x^{1/n} - y^{1/n}| \le |x - y|^{1/n}$ for $x, y \ge 0$, Hölder's inequality gives $A_j \to A > 0$, where $A = \int_X (f\,\mathrm{d}V/\gamma^n)^{1/n}\gamma^n$. Thus c_j is bounded above. In dimension one the same estimate is Stokes' identity and $A_j = \int_X f_j\,\mathrm{d}V$.

The lower-density estimate in Theorem 10.3.1 gives $c_j\|f_j\|_p \ge c_p$, so the constants are bounded away from zero. The densities $c_j f_j$ are now uniformly bounded in L^p, and the same theorem gives a common bound for u_j. Pass to a subsequence with $c_j \to c > 0$ and u_j Cauchy in L^1. By Theorem 10.3.2, $u_j \to u$ uniformly, with u continuous and $\sup_X u = 0$.

Bounded Bedford–Taylor continuity gives $\omega_{u_j}^n \rightharpoonup \omega_u^n$, while $c_j f_j \to c f$ in L^p. Hence $\omega_u^n = c f\,\mathrm{d}V$, as required. □

For the exponential equation, the corresponding scalar is $\mathrm{e}^{\lambda \sup_X u}$. Its upper bound comes from Jensen's inequality: weighting a Gauduchon probability by the root of the density relates the conserved trace to $\sup_X u$. The lower-density estimate again gives the reverse bound.

Lemma 10.4.1 *Fix $\lambda > 0$. Suppose smooth u_j with $\omega_{u_j} > 0$ solve*

$$\omega_{u_j}^n = \mathrm{e}^{\lambda u_j} f_j\,\mathrm{d}V,$$

where smooth strictly positive f_j converge in L^p to a nonnegative nonzero f. Then $M_j = \sup_X u_j$ is bounded above and below, uniformly in j.

Proof Absorb the fixed smooth positive ratio $\mathrm{d}V/\omega^n$ into both f_j and f. This preserves their L^p convergence, so we may use $\mathrm{d}V = \omega^n$ in the calculation. Choose $\rho > 0$ smooth with $\mathrm{dd^c}(\rho\omega^{n-1}) = 0$, using Theorem 4.1.1; for $n = 1$, take $\rho = 1$. Put

$$\alpha = \int_X \rho\omega^n, \quad A_j = \int_X f_j^{1/n}\rho\omega^n, \quad \psi_j = u_j - M_j \le 0.$$

Set $A = \int_X f^{1/n}\rho\omega^n$. The root inequality used in the preceding existence proof gives $A_j \to A > 0$. Arithmetic–geometric mean and the conserved trace imply

$$\alpha = \int_X \omega_{u_j} \wedge \rho\omega^{n-1} \ge \mathrm{e}^{\lambda M_j/n} \int_X \mathrm{e}^{\lambda\psi_j/n} f_j^{1/n}\rho\omega^n.$$

Apply Jensen's inequality to the probability measure $A_j^{-1} f_j^{1/n}\rho\omega^n$. Taking logarithms yields

$$\lambda M_j \le n\log(\alpha/A_j) + \frac{\lambda}{A_j}\int_X (-\psi_j) f_j^{1/n}\rho\omega^n. \tag{10.8}$$

The normalized potentials ψ_j have uniform L^q bounds for every finite q, by Lemma 1.5.1. With $q = np/(np-1)$, Hölder gives a uniform bound for the last integral, since $\|f_j^{1/n}\|_{np} = \|f_j\|_p^{1/n}$. This proves the upper bound for M_j.

Finally, the density $F_j = \mathrm{e}^{\lambda u_j} f_j$ satisfies $\|F_j\|_p \le \mathrm{e}^{\lambda M_j}\|f_j\|_p$. The lower-density bound of Theorem 10.3.1, applied to ψ_j, gives $c_p \le \mathrm{e}^{\lambda M_j}\|f_j\|_p$, which bounds M_j below. □

Proof (of Theorem 10.1.2) Choose smooth strictly positive $f_j \to f$ in L^p. The smooth proper equation, Theorem 9.6.1, gives smooth u_j satisfying the required equations. The preceding lemma bounds M_j and consequently bounds $F_j = \mathrm{e}^{\lambda u_j} f_j$ in L^p. The normalized potentials $\psi_j = u_j - M_j$ are uniformly bounded by Theorem 10.3.1. Pass to a subsequence with $M_j \to M$ and ψ_j Cauchy in L^1. Theorem 10.3.2 gives uniform convergence $u_j \to u \in \mathrm{PSH}(X, \omega) \cap C(X)$.

Uniform convergence and $f_j \to f$ in L^p give $\mathrm{e}^{\lambda u_j} f_j \to \mathrm{e}^{\lambda u} f$ in L^p. Bounded-product continuity therefore gives $\omega_u^n = \mathrm{e}^{\lambda u} f \, \mathrm{d}V$. Bounded exponential comparison, Corollary 10.2.2, gives uniqueness, in fact among all bounded solutions. □

Increasing the prescribed density decreases the solution: the exponential factor compensates for the change. Bounded exponential comparison gives this order even where the densities vanish.

Lemma 10.4.2 *Let $a, b \in \mathrm{PSH}(X, \omega) \cap L^\infty(X)$ solve*

$$\omega_a^n = \mathrm{e}^{\lambda a} f_a \, \mathrm{d}V, \qquad \omega_b^n = \mathrm{e}^{\lambda b} f_b \, \mathrm{d}V,$$

where $\lambda > 0$ and f_a, f_b are nonnegative nonzero L^p densities. If $f_a \le f_b$ almost everywhere, then $b \le a$ on X.

Proof The equations and $f_a \le f_b$ give $\mathrm{e}^{-\lambda b} \omega_b^n \ge \mathrm{e}^{-\lambda a} \omega_a^n$. Apply Corollary 10.2.2 with $u = a$ and $v = b$. □

Corollary 10.4.1 *Every bounded ω-plurisubharmonic solution of $\omega_u^n = f \, \mathrm{d}V$ is continuous.*

Proof The density $\mathrm{e}^{-u} f$ is nonnegative, nonzero and belongs to L^p. By Theorem 10.1.2, there is a continuous solution r of

$$\omega_r^n = \mathrm{e}^{r-u} f \, \mathrm{d}V.$$

The bounded potential u satisfies this same proper equation. Applying Lemma 10.4.2 in both directions gives $u = r$. □

10.5 The vanishing exponent and scalar uniqueness

For the fixed density f, compare u_λ from (10.1) with constant translates of any solution of the normalized equation. The maximum and minimum of that solution give two barriers whose separation is its oscillation. After multiplication by λ, both converge to the logarithm of the same scalar.

Proposition 10.5.1 *Suppose $u \in \mathrm{PSH}(X, \omega) \cap C(X)$ and $c > 0$ satisfy $\omega_u^n = c f \, \mathrm{d}V$. Set $m = \inf_X u$ and $M = \sup_X u$. Then the solution of (10.1) obeys*

$$u + \frac{\log c}{\lambda} - M \le u_\lambda \le u + \frac{\log c}{\lambda} - m. \tag{10.9}$$

In particular,

$$\|\lambda u_\lambda - \log c\|_\infty \le \lambda(M - m), \qquad \mathrm{e}^{\lambda u_\lambda} \longrightarrow c \quad \textit{uniformly on } X.$$

The scaling constant c is unique.

Proof Put $a = u + \lambda^{-1} \log c - M$ and $b = u + \lambda^{-1} \log c - m$. Both have the same Monge–Ampère measure as u, and

$$\mathrm{e}^{\lambda a} f \, \mathrm{d}V \le c f \, \mathrm{d}V = \omega_a^n = \omega_b^n \le \mathrm{e}^{\lambda b} f \, \mathrm{d}V.$$

Apply Corollary 10.2.2 first to a, u_λ and then to u_λ, b. This gives $a \le u_\lambda \le b$, proving (10.9). Multiplication by λ, together with $m \le u \le M$, gives the norm bound and the uniform limit.

If another pair (v, c') solves the equation with the same f, the same unique functions u_λ satisfy $\lambda u_\lambda \to \log c'$ as well. Hence $c' = c$. □

The barriers bound $\operatorname{osc}_X u_\lambda$ by $2(M - m)$. Moreover, the densities $\mathrm{e}^{\lambda u_\lambda} f$ converge to $c f$ in L^p. By Theorem 10.3.2, the normalized potentials $u_\lambda - \sup_X u_\lambda$ are uniformly precompact as $\lambda \downarrow 0$. Every cluster limit solves the equation with scalar c, but scalar uniqueness does not identify a unique cluster potential. Proposition 13.18.1 describes the symmetries preserved by these limits.

For classical potentials, a strict positivity reserve at an extremum gives a direct proof of scalar uniqueness.

Proposition 10.5.2 *Let $u, v \in C^2(X, \mathbb{R})$ and set $A = \omega + \mathrm{dd^c} u$, $B = \omega + \mathrm{dd^c} v$. If $A, B \ge 0$ and $A^n = c B^n$ for a constant $c > 0$, then $c = 1$.*

Proof Fix $\varepsilon > 0$. At a maximum of $u - (1 + \varepsilon) v$,

$$A + \varepsilon \omega \le (1 + \varepsilon) B.$$

At this maximum, $B > 0$ because $A + \varepsilon\omega > 0$. Determinant monotonicity and $A^n = c B^n$ give $c \le (1 + \varepsilon)^n$. Let $\varepsilon \downarrow 0$ and interchange the two potentials to obtain $c \le 1$ and $c^{-1} \le 1$. □

10.6 Normalization on the Hopf threefold

In this section, use the Hopf threefold X, Hermitian form ω and projection $\pi : X \to \mathbb{P}^2$ of Chapter 8. Recall $\alpha = \pi^*\omega_{\mathrm{FS}}$ and $\beta = \omega - \alpha$, and put $\nu = \omega^3/6$. Write E for the elliptic fiber, τ_a for translation by $a \in E$, and $\mathrm{d}A_E$ for its probability Haar measure. The fiber-averaging operator is denoted by A, as in Proposition 8.6.1.

The minimum-mass characterization in that proposition gives a lower bound for the normalization scalar. Equality forces every solution to descend to the projective base. Uniform L^p control makes this equality stable under limits of data, although the densities themselves need only converge weakly.

Here take $\mathrm{d}V = \omega^3$ and retain the nonnegative nonzero density f. By Theorem 10.1.1 and Proposition 10.5.1, there is a unique positive scalar c_f for which

$$(\omega + \mathrm{dd^c}u)^3 = c_f f\, \omega^3$$

has a continuous ω-plurisubharmonic solution. Put $I_f = \int_X f\, \mathrm{d}\nu$.

Proposition 10.6.1 *One has $c_f I_f \geq 1$, and every continuous solution u satisfies*

$$\|u - Au\|^2_{L^2(X,\nu)} \leq 4(c_f I_f - 1). \tag{10.10}$$

Moreover, $c_f I_f = 1$ if and only if $f = \pi^* g$ almost everywhere with respect to ν, for some nonnegative $g \in L^p(\mathbb{P}^2, \omega^2_{\mathrm{FS}})$. In this case every continuous solution is the pullback of a continuous ω_{FS}-plurisubharmonic function.

Proof Integrating the equation gives $\int_X \omega_u^3 = 6c_f I_f$. The lower mass bound and (8.20) give the first two assertions.

Suppose $c_f I_f = 1$. By Proposition 8.6.1, $u = \pi^* q$ for a bounded ω_{FS}-plurisubharmonic function q. Local holomorphic sections of π show that q is continuous. Thus $u \circ \tau_a = u$ for every $a \in E$. Naturality of bounded products and invariance of ω give

$$\tau_a^*(f\nu) = f\nu \qquad (a \in E).$$

We use this invariance as an identity of measures.

Fubini's theorem defines the fiber average Af almost everywhere. It equals $\pi^* g$ for a measurable function g on the base: local trivializations differ by translations of the Haar variable. Jensen's inequality gives

$$\int_{\mathbb{P}^2} |g|^p \omega^2_{\mathrm{FS}} \leq \int_X |f|^p\, \mathrm{d}\nu.$$

For every continuous real test function χ on X, measure invariance gives $\int_X \chi f\, \mathrm{d}\nu = \int_X (\chi \circ \tau_a) f\, \mathrm{d}\nu$. Integrate in a and use Fubini and invariance of Haar measure under $a \mapsto -a$. The result is $\int_X \chi f\, \mathrm{d}\nu = \int_X \chi Af\, \mathrm{d}\nu$. Hence $f = Af = \pi^* g$ almost everywhere, with $g \geq 0$.

Conversely, suppose $f = \pi^* g$ almost everywhere. Disintegration gives $I_f = \int_{\mathbb{P}^2} g\omega^2_{\mathrm{FS}}$. Apply Theorem 10.1.1 on the base. There are a continuous ω_{FS}-

plurisubharmonic function q and a scalar $k > 0$ with $(\omega_{\mathrm{FS}} + \mathrm{dd^c} q)^2 = k g\, \omega_{\mathrm{FS}}^2$. Bounded decreasing regularization and Stokes give $\int_{\mathbb{P}^2} (\omega_{\mathrm{FS}} + \mathrm{dd^c} q)^2 = 1$, so $k = I_f^{-1}$.

Put $u = \pi^* q$. To compute its bounded product, take smooth strictly ω_{FS}-plurisubharmonic $q_j \downarrow q$ by Theorem 1.11.1. Their pullbacks decrease to u and are strictly ω-plurisubharmonic. For each j, the smooth cube identity gives

$$(\omega + \mathrm{dd^c} \pi^* q_j)^3 = 3\pi^* (\omega_{\mathrm{FS}} + \mathrm{dd^c} q_j)^2 \wedge \beta.$$

On a local trivialization, the right side is $6\mathrm{d}A_E$ times the base measure $(\omega_{\mathrm{FS}} + \mathrm{dd^c} q_j)^2$: horizontal components of β vanish against the base top-degree form, and its vertical restriction is $2\mathrm{d}A_E$. Bounded-product continuity on the base and on X, tested in these trivializations and patched by a partition of unity, therefore gives

$$(\omega + \mathrm{dd^c} u)^3 = \frac{f}{I_f}\, \omega^3.$$

Uniqueness of the scalar implies $c_f = I_f^{-1}$. Every continuous solution then has mass 6, so the same mass characterization makes it a base pullback. Its descended potential is continuous by local sections, as above. □

Near equality, compactness gives uniform convergence of subsequences of potentials and weak convergence of their densities. The exact equality case identifies the limits.

Proposition 10.6.2 *Fix $1 < p < \infty$ and $M \geq 1$. Suppose $f_j \geq 0$ satisfy*

$$\int_X f_j \, \mathrm{d}\nu = 1, \qquad \|f_j\|_{L^p(X,\nu)} \leq M, \qquad c_{f_j} \longrightarrow 1.$$

For each j, choose any continuous solution u_j of $\omega_{u_j}^3 = c_{f_j} f_j \omega^3$, normalized by $\sup_X u_j = 0$. Then

$$\|u_j - Au_j\|_{C^0(X)} \longrightarrow 0, \qquad f_j - Af_j \rightharpoonup 0 \quad \text{in } L^p(X, \nu).$$

Every subsequence has a further subsequence with $u_j \to \pi^* q$ uniformly and $f_j \rightharpoonup \pi^* g$ weakly in L^p, where q is continuous and ω_{FS}-plurisubharmonic, $g \geq 0$, $\int_{\mathbb{P}^2} g\omega_{\mathrm{FS}}^2 = 1$, and

$$(\omega_{\mathrm{FS}} + \mathrm{dd^c} q)^2 = g\omega_{\mathrm{FS}}^2.$$

Proof The equation densities $c_{f_j} f_j$ have a common L^p bound because the scalars converge. By Theorem 10.3.2, every subsequence has a further subsequence on which $u_j \to u$ uniformly. Reflexivity of L^p permits a further extraction with $f_j \rightharpoonup f$ weakly in L^p. Positivity, unit integral and the bound $\|f\|_p \leq M$ pass to the limit. Bounded-product continuity under uniform convergence identifies

$$(\omega + \mathrm{dd^c} u)^3 = f\omega^3.$$

Thus $c_f = 1$, and Proposition 10.6.1 gives $u = \pi^* q$ and $f = \pi^* g$. The bounded cube identity proved there identifies the displayed base equation by testing against pullbacks of continuous base functions. In particular, both limits are fixed by A.

Haar averaging is a contraction on $C^0(X)$ and on $L^p(X, \nu)$. It is also self-adjoint for integration against ν: Fubini and invariance under the fiber translations give $\int \chi A f \, \mathrm{d}\nu = \int A\chi f \, \mathrm{d}\nu$ for $\chi \in L^{p'}$. Along the extracted subsequence, therefore, $u_j - Au_j \to 0$ uniformly and $f_j - Af_j \rightharpoonup 0$ in L^p. If either asserted convergence failed for the original sequence, a subsequence witnessing that failure, with one fixed $L^{p'}$ test in the weak case, would have a further subsequence of this kind. This contradiction proves both whole-sequence conclusions. □

Example 10.6.1 Choose $0 < a < 1/2$ and write $z \in \mathbb{C}^3 \setminus \{0\}$ for a point of the covering space. Set $s = \log |z|^2$. Since s changes by -2 under the defining contraction, every 2-periodic function of s descends to X. For an integer $m \geq 1$, define the smooth normalized potential

$$u_m = \frac{a}{\pi^2 m^2}\big(\cos(\pi m s) - 1\big), \qquad \|u_m - Au_m\|_{C^0} = \frac{a}{\pi^2 m^2} \longrightarrow 0.$$

Primes below denote derivatives with respect to s. Since $\mathrm{dd^c} s = \alpha$ and $\mathrm{d}s \wedge \mathrm{d^c} s = \beta$, the chain rule gives

$$\omega + \mathrm{dd^c} u_m = (1 + u_m')\alpha + (1 + u_m'')\beta, \quad u_m' = -\frac{a}{\pi m}\sin(\pi m s), \quad u_m'' = -a\cos(\pi m s).$$

Both coefficients are bounded below by positive constants independent of m. The rank identities $\alpha^3 = \beta^2 = 0$ imply

$$(\omega + \mathrm{dd^c} u_m)^3 = H_m \omega^3, \qquad H_m = \Big(1 - \frac{a}{\pi m}\sin(\pi m s)\Big)^2 \big(1 - a\cos(\pi m s)\big).$$

On each fiber s is twice the real Haar coordinate plus a base-dependent constant. Its distribution modulo 2 is uniform. Averaging the periodic expression, the terms containing u_m'', $u_m' u_m''$ and $(u_m')^2 u_m''$ integrate to zero, as derivatives of periodic functions. Hence

$$I_m := \int_X H_m \, \mathrm{d}\nu = AH_m = 1 + \frac{a^2}{2\pi^2 m^2}.$$

Set $f_m = H_m / I_m$. These smooth densities have unit integral and common strictly positive lower and finite upper bounds. Scalar uniqueness gives $c_{f_m} = I_m \to 1$, and $Af_m = 1$. Nevertheless,

$$\|f_m - 1 + a\cos(\pi m s)\|_{C^0(X)} = O(m^{-1}), \qquad \lim_{m\to\infty} \|f_m - Af_m\|_{L^1(X,\nu)} = \frac{2a}{\pi} > 0.$$

Here the constant in $O(m^{-1})$ depends only on a, and the last identity uses the period average of $|\cos|$, which is $2/\pi$. Thus near equality does not imply strong L^1 convergence to the fiber average, even for uniformly positive bounded smooth densities. For any base pullback g, the contraction property gives $\|f_m - Af_m\|_1 \leq 2\|f_m - g\|_1$, so approximation by base pullbacks also fails in L^1.

Chapter 11
Uniqueness for degenerate equations

Let X be a connected compact complex manifold of dimension $n \geq 1$, let ω be a Hermitian form, and let $\mathrm{d}V$ be a smooth positive volume form. The preceding chapter gives uniqueness of the normalization constant. We now ask when the corresponding potential is unique up to an additive constant.

For real potentials u, v, of the regularity stated in each result, set

$$A = \omega + \mathrm{dd^c} u, \qquad B = \omega + \mathrm{dd^c} v, \qquad w = u - v, \qquad T = \sum_{j=0}^{n-1} A^j \wedge B^{n-1-j}.$$

Then

$$\mathrm{dd^c} w \wedge T = A^n - B^n. \tag{11.1}$$

When $A, B \geq 0$, the polarized cofactor T is semipositive. It may be degenerate and need not be closed. For Sobolev or $C^{1,1}$ potentials, coefficient products in these identities are understood almost everywhere.

For $C^{1,1}$ solutions with the same continuous density, strict positivity of the density supplies local uniform ellipticity. The main issue is to locate the extrema of w in these elliptic regions and to decide whether their values propagate across the zero set. A positive weight making T $\mathrm{dd^c}$-closed gives another route: a global energy identity forces the derivative of w to vanish in the cofactor directions. For bounded potentials, an energy measure replaces this pointwise calculation.

If the background has invariant mass, ordinary comparison proves bounded uniqueness for L^p densities, $p > 1$, without a connectedness condition on the positive-density set. For a general Hermitian background, however, two zero hypersurfaces can separate regions carrying different extremal values. The projective examples below first place this freedom in an open zero-density band, then retain it with only quadratic zeros, even for a real analytic density. They show why smallness of the zero set alone does not restore uniqueness.

11.1 Extrema and density sublevels

To use the strong maximum principle, we first locate a maximum and a minimum inside regions of uniform ellipticity. A small strictly positive perturbation does this even for measurable Hessians. Uniqueness then follows if the closures of the extremal components meet.

For $u, v \in C^{1,1}(X)$ with $A, B \geq 0$ almost everywhere, consider

$$A^n = B^n = f\,\mathrm{d}V, \qquad f \in C^0(X), \qquad f \geq 0. \tag{11.2}$$

Fix a smooth positive $(1,1)$-form β, put $H = A + B$, and set

$$E_{\mathrm{unif}} = \left\{ x : \begin{array}{l} \text{there are a neighborhood } V \ni x \text{ and } c > 0 \\ \text{with } H \geq c\beta \text{ almost everywhere on } V \end{array} \right\}. \tag{11.3}$$

This open set is independent of β and of the representatives of the measurable coefficients.

Theorem 11.1.1 *Assume* (11.2). *If $\chi \in C^2(X, \mathbb{R})$ satisfies $C := \omega + \mathrm{dd^c}\chi > 0$, put*

$$P_\chi = \{x \in X : f(x)\mathrm{d}V_x \geq C_x^n\}, \qquad U = \{f > 0\}.$$

Then P_χ is a nonempty compact subset of U, and $U \subset E_{\mathrm{unif}}$. There are components G_+, G_- of E_{unif}, each meeting P_χ, such that

$$w \equiv \max_X w \quad \text{on } \overline{G_+}, \qquad w \equiv \min_X w \quad \text{on } \overline{G_-}. \tag{11.4}$$

If w is nonconstant, these closures are disjoint. Consequently w is constant if any one of the following holds:

(1) *every two components of E_{unif} meeting P_χ have intersecting closures;*
(2) *E_{unif} is connected;*
(3) *E_{unif} is dense and has at most two components.*

The same three conclusions hold with U in place of E_{unif}. In particular, uniqueness holds when $\{f = 0\}$ is a nonempty connected embedded C^1 real hypersurface. No orientability or coorientability assumption is needed.

The contact argument must use points where the Hessians exist. The following lemma finds such points near a maximum; it also gives the strict comparison step in the measurable strong maximum principle.

Lemma 11.1.1 *In Euclidean space $\mathbb{R}^m$, let $B(p,r)$ be the ball of center p and radius $r > 0$, and let I be the identity matrix. Let $g \in C^{1,1}$ near $\overline{B(p,r)} \subset \mathbb{R}^m$, suppose $g \le g(p)$ on the ball, and let E be any full-measure subset on which g has a second-order Taylor expansion. For every $\delta > 0$ and $0 < \rho < r$, there is $x \in E \cap B(p,\rho)$ such that*

$$D^2 g(x) \le 2\delta I.$$

Proof Put $g_\delta(x) = g(x) - \delta|x-p|^2$. The gradient map $F = \nabla g_\delta$ is Lipschitz on the ball, so it maps the null set $N = \overline{B(p,r)} \setminus E$ to a null set. Choose $s \in B(0,\delta\rho) \setminus F(N)$ and maximize $g_\delta(x) - s\cdot(x-p)$ on the closed ball. At any point with $|x-p| \ge \rho$, its value is strictly smaller than its value at p, because

$$g(x) - g(p) - \delta|x-p|^2 - s\cdot(x-p) \le -\delta|x-p|^2 + |s|\,|x-p| < 0.$$

Thus a maximizer x lies in $B(p,\rho)$. Its first derivative gives $F(x) = s$, so $x \notin N$. The second-order Taylor expansion at this upper contact point gives $D^2 g(x) \le 2\delta I$. The null-set assertion for a locally Lipschitz gradient follows by a countable cover by sets on which it is Lipschitz. The full-measure set of second-order Taylor points exists since the first derivatives of a $C^{1,1}$ function are Lipschitz. □

Lemma 11.1.2 *Let $a = (a^{ij})$ be a measurable symmetric matrix field on a connected domain $\Omega \subset \mathbb{R}^m$. For each compact $K \Subset \Omega$, suppose there are constants $0 < \lambda_K \le \Lambda_K < \infty$ such that*

$$\lambda_K I \le a(x) \le \Lambda_K I$$

almost everywhere on K, where I is the identity matrix. Repeated indices $i, j = 1,\dots,m$ are summed, and $z_{ij} = \partial_i\partial_j z$. If $z \in W^{2,\infty}_{\mathrm{loc}}(\Omega)$, $a^{ij}z_{ij} \ge 0$ almost everywhere, and z attains its supremum at an interior point, then z is constant.

Proof We first prove strict weak comparison on a relatively compact domain: if $b \in C^{1,1}$, $Lb := a^{ij}b_{ij} \ge \kappa > 0$ almost everywhere, and $b \le M$ on the boundary, then $b \le M$ inside. Otherwise an interior maximum lies in a closed coordinate ball on which $a \le \Lambda I$. Apply Lemma 11.1.1 to the full-measure set where both the differential inequality and the Taylor expansion hold. Choosing $2\delta m\Lambda < \kappa$ gives the contradiction $\kappa \le Lb \le 2\delta \operatorname{tr} a \le 2\delta m\Lambda$.

Take the $C^{1,1}$ representative of z, write $M = \sup_\Omega z$, and suppose $K = \{z = M\}$ is nonempty and proper. Near a relative boundary point of K, choose $y \notin K$ so that the ball of radius $r = \operatorname{dist}(y,K) > 0$ is compactly contained in Ω and touches K at x_0. On the closed annulus $r/2 \le |x-y| \le r$, use fixed bounds $\lambda I \le a \le \Lambda I$ and set

$$b(x) = \mathrm{e}^{-\alpha|x-y|^2} - \mathrm{e}^{-\alpha r^2}.$$

For sufficiently large α,

$$Lb = \mathrm{e}^{-\alpha|x-y|^2}\big(4\alpha^2 a^{ij}(x_i-y_i)(x_j-y_j) - 2\alpha \operatorname{tr} a\big) \ge \kappa > 0$$

almost everywhere on the annulus. On its inner sphere $z < M$ uniformly; on its outer sphere $b = 0$. Choose $\eta > 0$ small enough that $z + \eta b \le M$ on both spheres. Strict

weak comparison gives this inequality throughout the annulus. At x_0, however, $\nabla z = 0$ and b increases to first order in the inward radial direction, which contradicts that inequality. Thus K has no relative boundary and connectedness proves the assertion. This argument uses no derivative of the measurable coefficients. □

At a point where A, B are semipositive, define the cofactor quadratic form on real cotangent vectors

$$Q_x(\xi)\mathrm{d}V_x = \mathrm{i}\,\xi^{1,0} \wedge \overline{\xi^{1,0}} \wedge T_x. \tag{11.5}$$

Mixed products of semipositive $(1,1)$-forms are semipositive in this covector-testing order. Expanding H^{n-1}, $H = A + B$, gives

$$2^{-(n-1)} H^{n-1} \le T \le H^{n-1}. \tag{11.6}$$

At a point, choose a β-unitary coframe that diagonalizes H, and write $\lambda_1, \dots, \lambda_n$ for its eigenvalues. In that coframe the coefficient of $\mathrm{i}\alpha \wedge \bar{\alpha} \wedge H^{n-1}$, for $\alpha = \sum_{k=1}^n b_k\,\mathrm{d}z_k$, with complex coefficients b_k, is a positive dimensional constant times

$$\sum_{k=1}^n |b_k|^2 \prod_{\ell \ne k} \lambda_\ell. \tag{11.7}$$

For $n \ge 2$, (11.6)–(11.7) show that Q is locally uniformly elliptic exactly where H has a local essential positive lower bound. The reverse implication uses the local upper bound for H: testing each eigendirection bounds every $\prod_{\ell \ne k} \lambda_\ell$ below, and choosing k different from an index of a smallest eigenvalue bounds that eigenvalue below. For $n = 1$ one has $T = 1$, so the linearized equation is elliptic everywhere.

Proof (of Theorem 11.1.1) We first locate the extrema. Fix $s > 0$ and maximize

$$w - s(v - \chi). \tag{11.8}$$

For smooth data, the complex Hessian inequality at a maximizer is

$$(1 + s)B \ge A + sC. \tag{11.9}$$

The positive term sC first gives $B > 0$, and determinant equality then gives $A > 0$. In a B-unitary frame put $J = B^{-1/2}AB^{-1/2}$, $K = B^{-1/2}CB^{-1/2}$. Since $\det J = 1$, arithmetic–geometric mean and (11.9) give

$$n + s\,\mathrm{tr}\,K \le \mathrm{tr}\,J + s\,\mathrm{tr}\,K \le n(1 + s).$$

Consequently $\mathrm{tr}\,K \le n$, $B \ge C/n$, and $\det K \le (\mathrm{tr}\,K/n)^n \le 1$. In particular $B^n \ge C^n$.

For $C^{1,1}$ potentials, fix a maximizer p of (11.8) and apply Lemma 11.1.1 on a coordinate ball, choosing radii tending to zero and errors $\delta_j \downarrow 0$. Include in its full-measure set the points where both Hessians, their semipositivity, and their common determinant are defined. With $\beta_0 = \mathrm{dd^c}|z|^2$, the resulting points $x_j \to p$ satisfy

$$A(x_j) + sC_j(x_j) \le (1+s)B(x_j), \qquad C_j = C - (\delta_j/s)\beta_0 > 0$$

for large j. The matrix argument just given yields $f(x_j)\mathrm{d}V_{x_j} \ge C_j(x_j)^n$. Continuity of $f, \mathrm{d}V, C$, not continuity of the Hessians, gives $p \in P_\chi$. Thus every maximizer of the perturbed function lies in P_χ. Letting $s \downarrow 0$ places an actual maximizer of w in P_χ. Applying the same argument to $-w$ places a minimizer there as well. Compactness and positivity of C show that P_χ is compact, nonempty, and contained in U.

On a relatively compact ball in U, the determinants of A and B have a positive lower bound, while their coefficients have an essential upper bound. Every eigenvalue is therefore bounded below by the determinant lower bound divided by a fixed power of the upper bound. Thus $U \subset E_{\mathrm{unif}}$.

By (11.1), w solves a measurable nondivergence equation whose principal quadratic form is a fixed positive multiple of Q. On each relatively compact coordinate ball in E_{unif}, its coefficients are measurable and uniformly elliptic. Lemma 11.1.2 makes w constant on the component containing its maximum found above, and makes $-w$ constant on the component containing its minimum found above. Continuity extends the two constant values to the closures, proving (11.4). Distinct extrema force disjoint closures.

The same argument applies on U itself: it contains the extremal points in P_χ, and the cofactor is locally uniformly elliptic there. This proves the assertions with U in place of E_{unif}.

The component criteria are immediate except for the dense two-component case. If an open dense subset of connected X has at most two components with disjoint closures, those closures are disjoint nonempty closed sets whose union is X, contradicting connectedness. Finally, when $Z = \{f = 0\}$ is a connected embedded real hypersurface, flattening charts assign to each point of Z the unordered pair of components containing its two local sides. This pair is locally constant along Z, so connectedness makes it independent of the point; the two labels may coincide or be exchanged along a loop. Every component of $X \setminus Z$ has a nonempty boundary in Z, since otherwise it would be both open and closed in connected X. Near such a boundary point it contains a local side. Thus the pair accounts for every component, and U has at most two components. It is dense because a hypersurface has empty interior. This proves the last assertion without choosing a global side. □

The $C^{1,1}$ bound controls the measurable coefficients; continuity of the density places the limiting contact points in $\{f > 0\}$. The next criterion gives a quantitative condition for that set to be connected.

11.1.1 A square-root threshold

A nonnegative $C^{1,1}$ density grows at most quadratically away from a zero. If its zero set separates a small ball, a family of transverse segments therefore contributes at least a constant times $\sqrt{t}$ to the volume of $\{f \le t\}$ for small $t > 0$. Excluding this lower bound gives connectedness and hence uniqueness.

Write $\mathrm{Vol}(E) = \int_E \mathrm{d}V$ for Borel sets E.

Theorem 11.1.2 *Let $u, v \in C^2(X)$, $A, B \ge 0$, and $A^n = B^n = f\,\mathrm{d}V$, where $f \ge 0$ is $C^{1,1}$. If*

$$\liminf_{t \downarrow 0} \frac{\mathrm{Vol}\{f \le t\}}{\sqrt{t}} = 0, \tag{11.10}$$

then $\{f > 0\}$ is connected and $u - v$ is constant. In particular this holds when $f^{-1/2} \in L^1(\mathrm{d}V)$, with $f^{-1/2} = +\infty$ on $\{f = 0\}$. The exponent $1/2$ is sharp in every dimension $n \ge 3$.

Proof Write $Z = \{f = 0\}$. Condition (11.10) first implies $\mathrm{Vol}(Z) = 0$. We show that the complement of Z in every sufficiently small coordinate ball is connected. Fix such a ball with convex Euclidean image. If its complement were disconnected, choose points a, b in different components and small closed endpoint disks perpendicular to the segment $[a, b]$, contained in those respective components. By making the disks small, all parallel segments between corresponding points stay in the ball. Each segment meets Z, since otherwise it would connect the two components.

On a slightly smaller fixed coordinate region the gradient of f is Lipschitz with some constant $K > 0$. Nonnegativity gives $\nabla f(z) = 0$ at every zero, hence

$$f(x) \le \frac{K}{2}|x - z|^2 \tag{11.11}$$

whenever the segment from z to x stays in that region. The endpoint disks are compact and disjoint from Z, so their nearby segment portions are uniformly free of zeros. Thus every parallel segment contains a zero a fixed positive distance from its endpoints. For small $t > 0$, (11.11) supplies on that segment an interval of length $2\sqrt{2t/K}$ belonging to $\{f \le t\}$. Integration over the positive-measure transverse disk, and comparability of $\mathrm{d}V$ with coordinate volume, give

$$\mathrm{Vol}\{f \le t\} \ge c\sqrt{t}$$

for all sufficiently small t, contradicting (11.10). This proves the local connectedness assertion.

Each such ball has a nonempty complement of Z, since Z has zero volume, and that complement belongs to one component of $U = X \setminus Z$. Two overlapping balls have the same component label: their open overlap contains a point outside Z. These locally constant labels on connected X are therefore all the same. Hence U is connected, and Theorem 11.1.1 gives uniqueness.

If $f^{-1/2} \in L^1$, then, outside the null zero set,

$$\frac{\mathrm{Vol}\{f \le t\}}{\sqrt{t}} \le \int_{\{f \le t\}} f^{-1/2}\,\mathrm{d}V \longrightarrow 0$$

by dominated convergence. This proves the stated sufficient integrability condition. The sharpness assertion follows from the examples in Section 11.9, as quantified at the end of that section. □

11.2 Polarized cofactors and propagation

Uniform ellipticity is sufficient for propagation of an extremum, but it is not necessary. The range of the cofactor may already connect enough points to force constancy. For C^2 potentials, the quadratic form Q in (11.5) is continuous. Its polarization defines $Q^\sharp : T^*X \to TX$ by $\eta(Q^\sharp\xi) = Q(\xi,\eta)$. A reachable set from a point consists of endpoints of absolutely continuous curves starting there with velocity in $\operatorname{im} Q^\sharp$ almost everywhere. A control orbit consists of endpoints of finite concatenations of flows of locally Lipschitz vector fields taking values in that range.

A positive weight h satisfying $\mathrm{dd^c}(hT) = 0$ gives an energy identity controlling $\mathrm{d}w$ everywhere. Without such a weight, a degenerate maximum principle gives propagation only from the extremal sets. This distinction accounts for the different density assumptions on reachable sets and orbits in the theorem.

Theorem 11.2.1 *Let $u, v \in C^2(X)$, $A, B \ge 0$, and $A^n = B^n$.*

(1) *If there is a continuous $h > 0$ such that $\mathrm{dd^c}(hT) = 0$ in distributions, then*

$$\mathrm{i}\,\partial w \wedge \bar\partial w \wedge T \equiv 0. \tag{11.12}$$

It follows that w is constant if the strict positivity locus of the cofactor is dense, or if one cofactor reachable set is dense.

(2) *Without the weight hypothesis, the global maximum and minimum sets of w are invariant under every locally Lipschitz vector field taking values in the range of the cofactor symbol. If every such control orbit is dense, then w is constant. For smooth data it is enough that smooth range fields, for example the cofactor columns, and their iterated Lie brackets span TX at every point.*

11.2.1 Weighted energy

Testing the weighted cofactor against w^2 produces a nonnegative gradient integral. Only distributional $\mathrm{dd^c}$-closedness is needed: for a continuous $h > 0$, the condition is

$$\int_X \mathrm{dd^c}\varphi \wedge hT = 0 \qquad (\varphi \in C^\infty(X,\mathbb{R})). \tag{11.13}$$

Proof (of Theorem 11.2.1(1)) The product rule gives

$$\mathrm{dd^c}(w^2) = 2w\mathrm{dd^c}w + 2\mathrm{d}w \wedge \mathrm{d^c}w = 2w\mathrm{dd^c}w + \frac{\mathrm{i}}{\pi}\partial w \wedge \bar{\partial} w. \tag{11.14}$$

Approximate w^2 in C^2 by smooth functions in (11.13). Since hT is continuous and $\mathrm{dd^c}w \wedge T = 0$, this gives

$$0 = \frac{1}{\pi}\int_X h\,\mathrm{i}\partial w \wedge \bar{\partial} w \wedge T. \tag{11.15}$$

The continuous integrand is nonnegative, so it vanishes pointwise.

For the nonnegative quadratic form Q, $Q(\mathrm{d}w) = 0$ implies $Q(\mathrm{d}w, \eta) = 0$ for every η. Therefore

$$\mathrm{d}w_x\big(Q_x^\sharp \eta\big) = 0 \qquad (x \in X,\ \eta \in T_x^* X). \tag{11.16}$$

The chain rule and (11.16) make w constant along every absolutely continuous curve whose velocity lies in $\operatorname{im} Q^\sharp$ almost everywhere. One dense reachable set suffices by continuity. If instead Q is positive definite on a dense set, (11.15) gives $\mathrm{d}w = 0$ there and hence everywhere. □

The integration uses only the test function w^2. For general tests, integration by parts introduces derivatives of hT; $\mathrm{dd^c}(hT) = 0$ alone does not make $L\varphi = (\mathrm{dd^c}\varphi \wedge T)/\mathrm{d}V$ a symmetric divergence operator.

11.2.2 The real symbol and exterior-ball propagation

To propagate an extremum without the weight, we write the linearized operator in real coordinates. Its symbol is continuous and semipositive. An exterior tangent ball to the maximum set can have its normal only in the kernel of that symbol; this forces the maximum set to be invariant under the range fields.

In a holomorphic chart $z_j = x_j + \mathrm{i}y_j$, let C_{cf} be the Hermitian matrix characterized by

$$\frac{\big(\mathrm{i}\sum H_{j\bar{k}}\mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k\big) \wedge T}{\mathrm{d}V} = \operatorname{Tr}(C_{\mathrm{cf}}H) \tag{11.17}$$

for every Hermitian matrix H. Positivity of mixed wedges makes C_{cf} positive semidefinite. Write $C_{\mathrm{cf}} = P_{\mathrm{cf}} + \mathrm{i}Q_{\mathrm{cf}}$, where P_{cf} is real symmetric and Q_{cf} is real skew-symmetric. In the real coordinate order $(x_1, \dots, x_n, y_1, \dots, y_n)$,

$$L\varphi := \frac{\mathrm{dd^c}\varphi \wedge T}{\mathrm{d}V} = \operatorname{Tr}(aD^2\varphi), \qquad a = \frac{1}{8\pi}\begin{pmatrix} P_{\mathrm{cf}} & Q_{\mathrm{cf}} \\ -Q_{\mathrm{cf}} & P_{\mathrm{cf}} \end{pmatrix}. \tag{11.18}$$

Indeed $\varphi_{j\bar{k}} = (\varphi_{x_jx_k} + \varphi_{y_jy_k} + \mathrm{i}\varphi_{x_jy_k} - \mathrm{i}\varphi_{y_jx_k})/4$; inserting this expression in (11.17) proves (11.18). For real coordinate vectors $\xi, \eta \in \mathbb{R}^n$, moreover,

$$(\xi, \eta)^T a(\xi, \eta) = \frac{1}{8\pi}(\xi - \mathrm{i}\eta)^* C_{\mathrm{cf}}(\xi - \mathrm{i}\eta) \geq 0.$$

Thus a is a continuous principal symbol, with no first-order term in holomorphic coordinates, and its range is that of $Q^\sharp$.

An exterior-ball argument propagates the maximum for this operator. Let $W \in C^2$ satisfy $LW \geq 0$ in a chart, suppose $W \leq M$, and put $K = \{W = M\}$. If $z \in K$ is a nearest point to $y \notin K$, then

$$(y - z)^T a(z)(y - z) = 0. \tag{11.19}$$

To prove this, suppose the left side is positive, set $r_0 = |y - z|$, and shrink the tangent ball toward z: $y' = z + \theta(y - z)$, $R = \theta r_0$, with $0 < \theta < 1$ small. The closed ball $\overline{B(y', R)}$ lies in the chart and meets K only at z. This follows from the triangle inequality; equality in $|k - y| \leq |k - y'| + |y' - y| \leq r_0$, for $k \in K$, forces $k = z$. For large γ, the function

$$\psi(x) = \mathrm{e}^{-\gamma R^2} - \mathrm{e}^{-\gamma|x-y'|^2}$$

satisfies

$$L\psi = 2\gamma \mathrm{e}^{-\gamma|x-y'|^2}\big(\operatorname{tr} a - 2\gamma(x - y')^T a(x)(x - y')\big) < 0$$

in a neighborhood of z. Choose a sufficiently small $r < R$, and consider the lens $D = B(z, r) \cap B(y', R)$. On its boundary lying on the outer tangent sphere, $\psi = 0$ and $W - M \leq 0$. On the remaining compact boundary portion, $W - M < 0$, since the tangent ball meets K only at z. Hence, for sufficiently small $\varepsilon > 0$, $W - M \leq \varepsilon\psi$ on ∂D. A positive interior maximum of $W - M - \varepsilon\psi$ would have nonpositive Hessian but strictly positive image under L, which is impossible. The same inequality therefore holds in D. Outside the tangent ball $\psi \geq 0$, so it also holds in a full neighborhood of z. Second-order contact at z now yields

$$0 \leq LW(z) \leq \varepsilon L\psi(z) < 0,$$

proving (11.19).

Since $a \geq 0$, its range is the orthogonal complement of its kernel. Equation (11.19) consequently implies $(y - z) \cdot Z(z) = 0$ for every range field Z. If Z is locally Lipschitz and γ is one of its flow curves starting in K, let $d(t)$ denote its squared distance to K in a sufficiently small compact chart neighborhood. Shortening the time interval ensures that nearest points z_t lie in the chart interior. Write $D^+d(t) = \limsup_{h\downarrow 0}(d(t + h) - d(t))/h$ for the upper right derivative. Holding such a nearest point fixed when differentiating the distance from above gives

$$\begin{aligned} D^+d(t) &\leq 2(\gamma(t) - z_t) \cdot Z(\gamma(t)) \\ &= 2(\gamma(t) - z_t) \cdot \big(Z(\gamma(t)) - Z(z_t)\big) \leq 2\operatorname{Lip}(Z)d(t). \end{aligned}$$

As $d(0) = 0$, Gronwall's inequality gives $d(t) = 0$. The same argument for $-Z$, followed by continuation, proves invariance under both signs of time and under finite concatenations of such flows.

Proof (of Theorem 11.2.1(2)) Apply the preceding argument to $W = w$ and to $W = -w$ in each chart. Both are solutions of $LW = 0$. The maximum set of w is nonempty, closed, and contains every orbit starting in it. If every orbit is dense, this maximum set is all of X, proving constancy. The argument for the minimum set is identical.

For completeness, let a smooth family of range fields be bracket generating. At p, let $\mathcal{D}_p$ be the span of the values of all original fields pushed forward by finite compositions of their local flows and inverse flows. This distribution contains the original fields and is invariant under their flows. Differentiation of pushed-forward fields along a flow shows that it contains their nested Lie brackets; finite-dimensional subspaces are closed, so the derivative remains in $\mathcal{D}_p$. Induction and the Jacobi identity give every iterated bracket. The assumed rank condition therefore gives $\mathcal{D}_p = T_pX$.

Choose $2n$ such pushed-forward fields independent at p. The flow of each is a conjugate of an original flow and hence preserves its orbit. The map obtained by successively flowing for $2n$ small times has invertible derivative at zero. The inverse function theorem puts a neighborhood of p in its orbit. All orbits are open, and their partition of connected X has only one member. This proves the stated Hörmander-type sufficient condition, without a constant-rank hypothesis or a smooth matrix square root. In particular, the columns of a smooth symbol a may be used as the original range fields. □

For $C^{2,1}$ potentials and a locally Lipschitz background, the cofactor columns are locally Lipschitz and the orbit criterion still applies. The bracket criterion uses smooth fields.

11.2.3 The unweighted energy defect

Integration against w^2 also measures the obstruction to using the unweighted cofactor. For smooth potentials, the error is exactly the pairing of w^2 with $\mathrm{dd^c}T$. On a surface the Gauduchon condition makes this error vanish for every pair; in dimension three an additional term depends on the potentials.

Proposition 11.2.1 *Let $n \geq 2$ and let u, v be smooth ω-plurisubharmonic functions with $A^n = B^n = f\,\mathrm{d}V$ for a smooth $f \geq 0$. Then*

$$\int_X \mathrm{d}w \wedge \mathrm{d^c}w \wedge T = \frac{1}{2}\int_X w^2 \mathrm{dd^c}T. \tag{11.20}$$

If $\mathrm{dd^c}T = 0$ and $\{f > 0\}$ is dense in X, then $u - v$ is constant.

Proof By (11.1), $\mathrm{dd^c}w \wedge T = 0$. Integrating (11.14) against T and applying Stokes twice yields

$$\int_X \mathrm{dd^c}(w^2) \wedge T = \int_X w^2 \mathrm{dd^c}T.$$

Substitution proves (11.20).

For the uniqueness conclusion, apply Theorem 11.2.1(1) with $h = 1$. At every point with $f > 0$, the form B is positive definite, so the term B^{n-1} makes the cofactor strictly positive. Its positivity locus is therefore dense, as required. □

Corollary 11.2.1 *Assume $n = 2$ and $\mathrm{dd^c}\omega = 0$. If $f \geq 0$ is smooth and its zero set has empty interior, then two smooth ω-plurisubharmonic solutions, both normalized to have supremum zero, of*

$$(\omega + \mathrm{dd^c}u)^2 = c f \, \mathrm{d}V$$

with the same $c > 0$ coincide.

Proof Here $T = A + B$ and $\mathrm{dd^c}T = 2\mathrm{dd^c}\omega = 0$. The set $\{f > 0\}$ is dense, so the proposition makes the difference constant. Normalization removes that constant. □

In dimension three, direct expansion gives

$$\mathrm{dd^c}T = 3\mathrm{dd^c}(\omega^2) + 3\mathrm{dd^c}\omega \wedge \mathrm{dd^c}(u + v).$$

The Gauduchon condition cancels the first term but need not cancel the second. Take both potentials equal to the perturbing function in Example 4.5.1. Their common solution form $A = B$ is positive and satisfies $\mathrm{dd^c}(A^2) \neq 0$, so $\mathrm{dd^c}T = 3\mathrm{dd^c}(A^2) \neq 0$. Thus even an equal pair of positive solution forms can have a cofactor that is not $\mathrm{dd^c}$-closed.

11.3 Integrated rigidity in dimensions two and three

In dimensions two and three, equality of positive determinants gives a quadratic difference with a strict sign unless the two forms agree. On a surface its integral is zero by exactness. On a threefold the integral contains $\mathrm{dd^c}\omega$, so a sign condition on that form gives the same conclusion. Strong Sobolev approximation justifies the identities for $W^{2,n}$ potentials.

Assume $n \in \{2, 3\}$. All coefficient norms below use a fixed smooth metric and $\mathrm{d}V$.

Theorem 11.3.1 *Let $u, v \in W^{2,n}(X, \mathbb{R})$. Assume $A, B \geq 0$ almost everywhere and*

$$A^n = B^n = f\,\mathrm{d}V, \qquad f > 0 \quad \text{almost everywhere}.$$

For $n = 2$ impose no differential condition on ω. For $n = 3$, assume

$$\mathrm{dd^c}\omega \wedge \mathrm{i}\eta \wedge \bar{\eta} \leq 0 \tag{11.21}$$

for every $(1,0)$-covector η. Then $\mathrm{dd^c}(u - v) = 0$ distributionally and $u - v$ is almost everywhere constant. For ω-plurisubharmonic representatives the equality holds pointwise.

Proof Put

$$s = (u + v)/2, \qquad D = \mathrm{dd^c}w = A - B,$$
$$C = (A + B)/2 = \omega + \mathrm{dd^c}s.$$

The coefficients of C, D lie in L^n; their top products are integrable by Hölder's inequality. At almost every point the common determinant is positive, so A, B are positive definite. Diagonalize them simultaneously by congruence. For a coframe of $(1,0)$-covectors $\theta_1, \ldots, \theta_n$, write $A = \sum e_j$, $B = \sum \lambda_j e_j$, where $e_j = \mathrm{i}\theta_j \wedge \bar{\theta}_j$, $\lambda_j > 0$, and $\prod \lambda_j = 1$.

In dimension two,

$$D^2 = A^2 + B^2 - 2A \wedge B = 2(2 - \lambda_1 - \lambda_2)e_1 \wedge e_2 \leq 0, \tag{11.22}$$

with equality precisely when $A = B$. In dimension three,

$$2C \wedge D^2 = A^3 + B^3 - A^2 \wedge B - A \wedge B^2 \leq 0. \tag{11.23}$$

Indeed, with $E = e_1 \wedge e_2 \wedge e_3$, the four terms on the right are

$$6E, \qquad 6E, \qquad 2\Big(\sum_j \lambda_j\Big)E, \qquad 2\Big(\sum_{i<j} \lambda_i\lambda_j\Big)E.$$

Each sum is at least three, and equality again forces all $\lambda_j = 1$.

Choose smooth real $w_k \to w$ in $W^{2,n}$ and set $D_k = \mathrm{dd^c}w_k$. Such approximations follow by a partition of unity, local extension of compactly supported pieces, and convolution. They are not required to preserve semipositivity. For $n = 2$, Stokes gives $\int_X D_k^2 = 0$, since $D_k^2 = \mathrm{d}(\mathrm{d^c}w_k \wedge D_k)$. For a fixed norm-comparison constant $C_0 > 0$,

$$\|D_k^2 - D^2\|_{L^1} \leq C_0\|D_k - D\|_{L^2}(\|D_k\|_{L^2} + \|D\|_{L^2}) \longrightarrow 0.$$

Thus $\int_X D^2 = 0$, and the nonpositive integrand in (11.22) vanishes almost everywhere. Its strict equality case gives $D = 0$ almost everywhere.

For $n = 3$, also choose smooth $s_k \to s$ in $W^{2,3}$, and put $C_k = \omega + \mathrm{dd^c} s_k$. The needed identity is

$$\int_X C_k \wedge D_k^2 = \int_X \omega \wedge D_k^2 = -\int_X \mathrm{d}w_k \wedge \mathrm{d^c} w_k \wedge \mathrm{dd^c}\omega. \tag{11.24}$$

Here the first equality follows because $\mathrm{dd^c} s_k \wedge D_k^2 = \mathrm{d}(\mathrm{d^c} s_k \wedge D_k^2)$. To check the sign in the second equality, put $J_k = \mathrm{d}w_k \wedge \mathrm{d^c} w_k$. Using $\mathrm{d^c d} = -\mathrm{dd^c}$ and $(\mathrm{d^c})^2 = 0$,

$$\mathrm{d^c} J_k = -D_k \wedge \mathrm{d^c} w_k, \qquad \mathrm{dd^c} J_k = -D_k^2.$$

Stokes, first for $\mathrm{d}(\omega \wedge \mathrm{d^c} J_k)$ and then for $\mathrm{d^c}(\mathrm{d}\omega \wedge J_k)$, gives

$$\int_X \omega \wedge \mathrm{dd^c} J_k = -\int_X \mathrm{d}\omega \wedge \mathrm{d^c} J_k,$$
$$0 = -\int_X \mathrm{dd^c}\omega \wedge J_k - \int_X \mathrm{d}\omega \wedge \mathrm{d^c} J_k.$$

Combining them yields (11.24).

Using a fixed norm-comparison constant $C_0 > 0$, strong L^3 convergence implies convergence of all three-factor products in L^1. For example,

$$\|(C_k - C) \wedge D_k^2\|_1 \le C_0 \|C_k - C\|_3 \|D_k\|_3^2 \longrightarrow 0,$$

and

$$\|C \wedge (D_k^2 - D^2)\|_1 \le C_0 \|C\|_3 \|D_k - D\|_3 (\|D_k\|_3 + \|D\|_3) \longrightarrow 0.$$

Also $\mathrm{d}w_k \wedge \mathrm{d^c} w_k \to \mathrm{d}w \wedge \mathrm{d^c} w$ in $L^{3/2}$, so the right side of (11.24) converges. Consequently

$$\int_X C \wedge D^2 = -\int_X \mathrm{d}w \wedge \mathrm{d^c} w \wedge \mathrm{dd^c}\omega \ge 0. \tag{11.25}$$

The last sign uses $\mathrm{d}w \wedge \mathrm{d^c} w = (\mathrm{i}/(2\pi))\partial w \wedge \bar{\partial} w$ and (11.21). The left integrand is nonpositive by (11.23), so it vanishes almost everywhere. The positive-density equality case gives $D = 0$ almost everywhere.

In both dimensions $\mathrm{dd^c} w = 0$ distributionally. By Proposition 1.1.1 with background zero, w has a global plurisubharmonic representative. It attains a maximum on compact X, so the maximum principle and connectedness make it constant.

For ω-plurisubharmonic representatives, Lemma 1.3.3 upgrades equality almost everywhere to equality everywhere. Equal mean normalizations then give equality in the Sobolev class, and equal supremum normalizations give equality for the plurisubharmonic representatives. □

Corollary 11.3.1 *For C^2 potentials in dimensions two and three, the corresponding conclusion remains valid if the common continuous density is positive merely on a dense set. In dimension three retain* (11.21)*; in particular* $\mathrm{dd^c}\omega = 0$ *is sufficient.*

Proof Put $D = A - B$ and $C = (A + B)/2$ for the two C^2 potentials in the corollary. At a point where the common top power vanishes, the algebraic expressions

are still nonpositive: in dimension two $D^2 = -2A \wedge B$, and in dimension three $2C \wedge D^2 = -A^2 \wedge B - A \wedge B^2$. The same integral identities force these continuous expressions to vanish everywhere. Their strict equality cases give $D = 0$ on $\{f > 0\}$. Density and continuity then give $D = 0$ globally. □

For the Hopf threefold X of Chapter 8, recall the projection $\pi : X \to \mathbb{P}^2$ and the semipositive form $\alpha = \pi^*\omega_{\mathrm{FS}}$, where ω_{FS} is the Fubini–Study form. The Hermitian metric ω used there satisfies $\mathrm{dd^c}\omega = -\alpha^2$, hence (11.21). The Sobolev uniqueness theorem therefore applies to this metric even though its upper Monge–Ampère volume is infinite. The following sections return to the general compact Hermitian setting.

11.4 Bounded uniqueness and connectedness

Connectedness of the positive-density region also gives uniqueness for bounded potentials. The missing differentiability is replaced by an energy measure of the difference. Its pushforward to the real line has a Lipschitz density vanishing at the extreme levels. This forces the extreme sets to have zero equation mass; smooth test potentials concentrated near such a set then contradict modified comparison.

Theorem 11.4.1 *Let $F \in C^\infty(X)$ satisfy $F \geq 0$, $F \not\equiv 0$, with $U = \{F > 0\}$ connected and dense. If bounded ω-plurisubharmonic functions u, v satisfy*

$$\omega_u^n = \omega_v^n = F\omega^n,$$

then $u - v$ is constant. In particular, $\sup_X u = \sup_X v$ implies $u = v$.

Bounded solutions are continuous by Corollary 10.4.1. We may therefore use pointwise extrema throughout the proof. If b is continuous and $\omega_b \geq \varepsilon\omega$, with $0 < \varepsilon < 1$, apply Theorem 10.2.1 to $b/(1-\varepsilon)$. Thus its thin-sublevel estimate applies directly to any such strictly positive comparison potential. The correction factor tends to one with the sublevel height, while ε stays fixed.

Proposition 11.4.1 *Let continuous ω-plurisubharmonic u, v satisfy $\omega_u^n = \omega_v^n = F\omega^n$, where $F \geq 0$ is bounded measurable. Let q be smooth and strictly ω-plurisubharmonic, and write $\omega_q^n = G\omega^n$. For every $0 < \kappa < 1$ and $\eta > 0$,*

$$F\omega^n\big(\{u - v < \min_X(u - v) + \eta\} \cap \{F > \kappa G\}\big) > 0. \tag{11.26}$$

The analogous assertion holds for the near-maximum set.

Proof Choose $\delta \in (0, 1]$ with $\omega_q \geq \delta\omega$. For $0 < t < 1$, put $b_t = (1-t)v + tq$, $m_t = \min_X(u - b_t)$, and $E_s = \{u < b_t + m_t + s\}$. Then $\omega_{b_t} \geq t\delta\omega$. Write $\mu = F\omega^n$, $d = \kappa^{-1/n} - 1$, and $\gamma = nd$. By (1.11),

$$\omega_{b_t}^n \geq \big((1-t)F^{1/n} + tG^{1/n}\big)^n\omega^n.$$

On $L = \{F \le \kappa G\}$ this is at least $(1+\gamma t)\mu$, and on $R = \{F > \kappa G\}$ it is at least $(1-nt)\mu$. Choose $s > 0$ so small that the reverse comparison factor in (10.2), with $\varepsilon = t\delta$, is at most $1 + \gamma t/2$. Setting $A = \mu(E_s \cap L)$ and $B = \mu(E_s \cap R)$ gives

$$(1+\gamma t)A + (1-nt)B \le (1+\gamma t/2)(A+B), \qquad B \ge \frac{1-\kappa^{1/n}}{2}\,\mu(E_s).$$

The open set E_s is nonempty by the definition of m_t. As $\omega_{b_t}^n \ge (t\delta)^n\omega^n$, (10.2) forces $\mu(E_s) > 0$, hence $B > 0$.

Let $a = \min_X(u-v)$ and $\Lambda = \|v-q\|_\infty$. The perturbation satisfies $|m_t - a| \le t\Lambda$, and $E_s \subset \{u - v < a + 2t\Lambda + s\}$. Given η, choose t and then s with $2t\Lambda + s < \eta$. This proves (11.26). Interchanging u and v gives the near-maximum assertion. □

11.4.1 The energy of the difference

Assume u, v are continuous ω-plurisubharmonic functions and $A^n = B^n = F\omega^n$ for bounded measurable $F \ge 0$. Although the cofactor is now a current, its pairing with the gradient square still defines a positive measure. The identity $\mathrm{dd^c}w \wedge T = 0$ determines the distribution of this energy over the levels of w.

Proposition 11.4.2 *There is a finite positive measure $E = \mathrm{d}w \wedge \mathrm{d^c}w \wedge T$ with the following properties.*

For every smooth real function χ on $\mathbb{R}$,

$$\int_X \chi''(w)\,\mathrm{d}E = \int_X \chi(w)\,\mathrm{dd^c}T. \tag{11.27}$$

The signed current $\mathrm{dd^c}T$ is a finite measure. If $a = \min_X w$, $b = \max_X w$, and

$$k(t) = \int_X (w-t)_+\,\mathrm{dd^c}T,$$

then k is nonnegative, globally Lipschitz, supported on $[a,b]$, and

$$w_*E = k(t)\,\mathrm{d}t. \tag{11.28}$$

In particular, E charges no level set of w.

On every relatively compact coordinate domain V on whose neighborhood F has a positive essential lower bound, there is $C_V < \infty$ such that every piecewise C^1 Lipschitz function p on $[a,b]$ satisfies

$$\int_V |\mathrm{d}(p(w))|\,\mathrm{d}\lambda \le C_V \left(\int_V |p'(w)|^2\,\mathrm{d}E\right)^{1/2}, \tag{11.29}$$

where $\mathrm{d}\lambda$ is Euclidean coordinate volume.

Proof Expand T locally into smooth powers of ω multiplied by closed Bedford–Taylor products of $\mathrm{dd^c}u$ and $\mathrm{dd^c}v$. Differentiating twice acts only on those smooth powers. The local mass bounds in Section 1.6 show that $\mathrm{dd^c}T$ is a finite signed measure.

We first construct E and justify (11.27). On a coordinate ball, convolve u, v, ω with a common nonnegative radial mollifier. Write the resulting smooth positive forms as $A_\rho = \omega_\rho + \mathrm{dd^c}u_\rho$ and $B_\rho = \omega_\rho + \mathrm{dd^c}v_\rho$, and let T_ρ be their mixed sum. On a slightly larger ball, add the same fixed multiple of $|z|^2$ to each potential to obtain plurisubharmonic functions. Their radial regularizations decrease to the originals, so Bedford–Taylor continuity applies to all closed Hessian products. Since u, v are continuous, u_ρ, v_ρ, w_ρ converge uniformly on compact subsets.

For a closed smooth product S_ρ appearing in T_ρ, the identity

$$\mathrm{d}w_\rho \wedge \mathrm{d^c}w_\rho \wedge S_\rho = \tfrac{1}{2}\mathrm{dd^c}(w_\rho^2 S_\rho) - w_\rho \mathrm{dd^c}w_\rho \wedge S_\rho \tag{11.30}$$

has a distributional limit. Both Hessian products on its right converge weakly with uniformly bounded local variation, and their weights converge uniformly. The same holds after multiplying by the convergent smooth factors from ω_ρ. Thus the positive measures $E_\rho = \mathrm{d}w_\rho \wedge \mathrm{d^c}w_\rho \wedge T_\rho$ converge to a positive finite measure E. Formula (11.30) makes this limit intrinsic on overlapping charts.

For smooth regularizations, the chain rule gives

$$\mathrm{dd^c}\chi(w_\rho) \wedge T_\rho = \chi''(w_\rho)E_\rho + \chi'(w_\rho)\mathrm{dd^c}w_\rho \wedge T_\rho.$$

The last current tends to zero, because $\mathrm{dd^c}w_\rho \wedge T_\rho = A_\rho^n - B_\rho^n$ tends weakly to $A^n - B^n = 0$ and has uniformly bounded variation. Uniform convergence of $\chi'(w_\rho)$ permits the weight. Passing to the limit and integrating by parts on compact X gives (11.27). A partition of unity justifies this passage without a global regularization; the derivative terms cancel after summation.

Let $\sigma = w_*(\mathrm{dd^c}T)$ and $\nu = w_*E$. Equation (11.27) says $\nu'' = \sigma$ as distributions on $\mathbb{R}$. Stokes and $\mathrm{dd^c}w \wedge T = 0$ give

$$\sigma(\mathbb{R}) = \int_X \mathrm{dd^c}T = 0, \qquad \int_{\mathbb{R}} s\,\mathrm{d}\sigma(s) = \int_X w\,\mathrm{dd^c}T = \int_X \mathrm{dd^c}w \wedge T = 0.$$

Consequently $k(t) = \int (s-t)_+\,\mathrm{d}\sigma(s)$ is globally Lipschitz and vanishes outside $[a, b]$. It has $k'' = \sigma$; hence the compactly supported distribution $\nu - k(t)\,\mathrm{d}t$ has second derivative zero and must vanish. Positivity of ν and continuity of k imply $k \geq 0$. This proves (11.28) and the assertion about level sets.

We establish (11.29). On a coordinate neighborhood of V, set $\beta = \mathrm{dd^c}|z|^2$ and choose $c_V > 0$ with $F\omega^n \geq c_V\beta^n$. Apply Lemma 1.7.4 with the constant function $g = c_V^{1/n}$. Differentiation of coefficient measures then gives $\det_\beta A_{\mathrm{ac}} \geq c_V$ almost everywhere. The local Sobolev regularity in Section 1.6 gives $w_\rho \to w$ strongly in local $W^{1,2}$; a subsequence of gradients converges almost everywhere. Along the same subsequence $A_\rho \to A_{\mathrm{ac}}$ almost everywhere by differentiation of its coefficient measures.

Since T_ρ contains A_ρ^{n-1} and all its terms are positive, Fatou's lemma applied to E_ρ yields the measure inequality

$$E \geq \mathrm{d}w \wedge \mathrm{d}^{\mathrm{c}}w \wedge A_{\mathrm{ac}}^{n-1}. \tag{11.31}$$

The cofactor matrix of A_{ac} has smallest eigenvalue at least $\det A_{\mathrm{ac}}/\operatorname{tr} A_{\mathrm{ac}} \geq c_V/\operatorname{tr} A_{\mathrm{ac}}$. The trace is locally integrable because A has finite mass. The Sobolev chain rule, (11.31), and Cauchy–Schwarz give

$$\int_V |\mathrm{d}(p(w))|\,\mathrm{d}\lambda \leq C\left(\int_V |p'(w)|^2\,\mathrm{d}E\right)^{1/2}\left(\int_V \frac{\operatorname{tr} A_{\mathrm{ac}}}{c_V}\,\mathrm{d}\lambda\right)^{1/2}.$$

The last factor is finite and is absorbed into C_V. Values assigned to p' at its finitely many corners are immaterial because E charges no level of w. □

Lemma 11.4.1 *Assume in addition that F is continuous and $U = \{F > 0\}$ is connected and dense. If $w = u - v$ is nonconstant, then*

$$F\omega^n(\{w = \min_X w\}) = F\omega^n(\{w = \max_X w\}) = 0.$$

Proof Put $a = \min_X w < b = \max_X w$ and let k be given by Proposition 11.4.2. It is Lipschitz, nonnegative, and $k(a) = 0$. Choose $\epsilon_j \downarrow 0$ with $a + \epsilon_j < b$. For $\delta > 0$, set

$$J_j(\delta) = \int_a^{a+\epsilon_j} \frac{\mathrm{d}t}{k(t) + \delta}.$$

If L is a Lipschitz constant for k, then $k(t) \leq L(t - a)$ on this interval, so $J_j(\delta) \to \infty$ as $\delta \downarrow 0$. Choose $\delta_j > 0$ with $J_j = J_j(\delta_j) \geq j$. Let $p_j = 0$ on $(-\infty, a]$, $p_j = 1$ on $[a + \epsilon_j, \infty)$, and

$$p_j'(t) = \frac{1}{J_j(k(t) + \delta_j)} \qquad (a < t < a + \epsilon_j).$$

The pushforward identity (11.28) gives

$$\int_X |p_j'(w)|^2\,\mathrm{d}E = \frac{1}{J_j^2}\int_a^{a+\epsilon_j} \frac{k(t)}{(k(t) + \delta_j)^2}\,\mathrm{d}t \leq \frac{1}{J_j} \longrightarrow 0.$$

By (11.29), $\mathrm{d}(p_j(w)) \to 0$ in local L^1 on U. The bounded functions $p_j(w)$ converge pointwise and in local L^1 to $\mathbf{1}_{\{w>a\}}$, whose distributional gradient therefore vanishes on U. Connectedness makes this indicator constant almost everywhere on U. Since w is nonconstant, $\{w > a\}$ is a nonempty open set; density of U makes its intersection with U nonempty and open. The constant is one. Thus $\{w = a\} \cap U$ has zero ordinary volume, and $F = 0$ off U gives the first assertion. Apply the argument to $-w$ for the maximum set. □

11.4.2 Concentrating smooth test potentials

The following construction has a constant independent of the height and derivatives of the test density. This uniformity permits concentration near a null set.

Proposition 11.4.3 *For every $B_0 > \|1\|_{L^2(\omega^n)}$ there is $\varepsilon = \varepsilon(X, \omega, B_0) > 0$ such that every smooth function $g \geq 1$ with $\|g\|_{L^2(\omega^n)} \leq B_0$ admits a smooth strictly ω-plurisubharmonic function q satisfying*

$$\omega_q^n \geq \varepsilon g \omega^n. \tag{11.32}$$

Proof Choose finitely many coordinate balls $B_j = \{|z_j| < R_j\}$, with closures in their charts, whose concentric smaller balls $\{|z_j| < r_j\}$ cover X. On B_j, let ψ_j solve

$$(\mathrm{dd^c}\psi_j)^n = g\omega^n, \qquad \psi_j|_{\partial B_j} = 0. \tag{11.33}$$

The coordinate density is smooth and strictly positive. The classical smooth Dirichlet theorem applies: $C_j(|z_j|^2 - R_j^2)$ is a smooth strict subsolution with zero boundary values for sufficiently large $C_j > 0$. Thus ψ_j is smooth and strictly plurisubharmonic on $\overline{B}_j$. The maximum principle gives $\psi_j \leq 0$.

We need a bound independent of $\sup g$. Write the coordinate equation as $\det(\partial_a \bar{\partial}_b \psi_j) = D_j$, and put $M_j = -\min_{B_j} \psi_j$. At a minimum point x_j, each real slope p with $|p| < M_j/(2R_j)$ has an interior contact point for $\psi_j - p \cdot (x - x_j)$: its value at x_j is $-M_j$, while its boundary values are larger than $-M_j$. At these contact points the real Hessian S is nonnegative, and the gradient map covers the slope ball. The real area formula and the matrix inequality

$$\det_{\mathbb{R}} S \leq 4^n \big(\det_{\mathbb{C}}(\partial_a \bar{\partial}_b \psi_j)\big)^2 = 4^n D_j^2$$

therefore give $M_j \leq C(n, R_j)\|D_j\|_{L^2(B_j)}^{1/n}$. For the matrix inequality, let J be the standard complex structure on the real coordinate space and average S with $J^{\mathsf{T}} S J$. Concavity of $\log\det$ and the paired eigenvalues of the J-invariant average give the displayed factor, with singular matrices handled by approximation. The ratio of ω^n to coordinate volume is fixed and bounded, so $\|D_j\|_2$ is controlled by B_0. Consequently

$$-M_0 \leq \psi_j \leq 0 \quad \text{on every } B_j, \tag{11.34}$$

where M_0 depends only on X, ω, B_0 and the chosen finite cover.

Put $\beta_j = \mathrm{dd^c}|z_j|^2$. Choose $a_j > 0$ such that $a_j \beta_j \leq \omega/2$ on B_j, and let $t_j = (R_j^2 + r_j^2)/2$. Fix $\tau > 0$ so small that $\tau M_0 < a_j(R_j^2 - r_j^2)/4$ for every j. The smooth function

$$s_j = \tau\psi_j + a_j(t_j - |z_j|^2)$$

is positive by a fixed margin on the smaller ball, negative on a collar of ∂B_j, and satisfies $\omega_{s_j} \geq \omega/2 + \tau \mathrm{dd^c}\psi_j$. Choose a smooth convex regularization M_j^{sm} of $s \mapsto \max(s, 0)$ that is zero below a small negative threshold and equal to s above

the corresponding positive threshold, with $0 \le (M_j^{\mathrm{sm}})' \le 1$ and $(M_j^{\mathrm{sm}})'' \ge 0$. The thresholds can be chosen below both margins. Set $Q_j = M_j^{\mathrm{sm}}(s_j)$ on B_j and extend it by zero. This extension is smooth, and

$$\omega_{Q_j} = (M_j^{\mathrm{sm}})'(s_j)\omega_{s_j} + \big(1 - (M_j^{\mathrm{sm}})'(s_j)\big)\omega + (M_j^{\mathrm{sm}})''(s_j)\,\mathrm{d}s_j \wedge \mathrm{d}^{\mathrm{c}} s_j \ge \omega/2.$$

On the smaller ball, $Q_j = s_j$ and $\omega_{Q_j} \ge \tau \mathrm{dd}^{\mathrm{c}}\psi_j$. If N is the number of balls, take $q = N^{-1}\sum_j Q_j$. Then $\omega_q \ge \omega/2$ everywhere. Each point belongs to at least one smaller ball, so positivity of the other summands and (11.33) give

$$\omega_q^n \ge (\tau/N)^n (\mathrm{dd}^{\mathrm{c}}\psi_j)^n = (\tau/N)^n g\omega^n.$$

Thus (11.32) holds with $\varepsilon = (\tau/N)^n$, independent of $\sup g$ and of all derivatives of g. □

Proof (Proof of Theorem 11.4.1) By Corollary 10.4.1, u, v are continuous. In dimension one, $\mathrm{dd}^{\mathrm{c}} w = 0$, so the maximum principle makes w constant. Assume $n \ge 2$.

Suppose w is nonconstant. Let $a = \min_X w$, $K = \{w = a\}$, and $\mu = F\omega^n$. By Lemma 11.4.1, $\mu(K) = 0$. Choose $B_0 > \|1\|_{L^2(\omega^n)}$ and let $\varepsilon > 0$ be the constant from Proposition 11.4.3. The compact set $K_0 = K \cap \{F \ge \varepsilon/2\}$ has zero ω^n-volume. Fix $A > 1$ with $\varepsilon A > \max_X F$. By regularity of the smooth volume measure, choose an open neighborhood O of K_0 so small that

$$\int_X \omega^n + (A^2 - 1)\int_O \omega^n < B_0^2.$$

Take a smooth $0 \le \chi \le 1$ equal to one near K_0 and supported in O, and set $g = 1 + (A-1)\chi$. If $K_0 = \varnothing$, take $\chi = 0$. Then $g \ge 1$, $g = A$ on K_0, and $\|g\|_{L^2(\omega^n)} < B_0$.

By Proposition 11.4.3, there is a smooth strictly ω-plurisubharmonic q with $\omega_q^n = G\omega^n \ge \varepsilon g\omega^n$. On K_0 we have $G \ge \varepsilon A > F$; on $K \setminus K_0$ we have $F < \varepsilon/2 < \varepsilon \le G$. Thus $G > F$ throughout the compact set K. Choose $0 < \kappa < 1$ such that $F < \kappa G$ on K, and put $W = \{F < \kappa G\}$. The complement of this open neighborhood of K is compact, and $w - a$ has a strictly positive minimum there if it is nonempty. Hence some $\eta > 0$ satisfies $\{w < a + \eta\} \subset W$.

Proposition 11.4.1 asserts that $\mu(\{w < a+\eta\} \cap \{F > \kappa G\}) > 0$. The set inside this measure is empty by the preceding inclusion. This contradiction makes w constant. Equality of the suprema makes the constant zero. □

A positive lower bound on a bounded measurable density gives the same conclusion. In this case coercivity holds on all of X, and the concentrating test can dominate the essential upper bound on a whole neighborhood of the extreme set.

Corollary 11.4.1 *If $0 < c_0 \le F \le C_0 < \infty$ almost everywhere and bounded ω-plurisubharmonic functions satisfy $\omega_u^n = \omega_v^n = F\omega^n$, then $u - v$ is constant.*

Proof Continuity follows from Corollary 10.4.1; dimension one follows from $\mathrm{dd^c}(u - v) = 0$. In higher dimension, suppose $w = u - v$ is nonconstant and put $K = \{w = \min_X w\}$. The proof of Lemma 11.4.1 applies with $U = X$: Proposition 11.4.2 gives its pushforward identity and local coercivity on every chart. Thus K has zero volume.

Fix $B_0 > \|1\|_{L^2(\omega^n)}$ and the corresponding ε in Proposition 11.4.3. Choose $A > 1$ with $\varepsilon A > C_0$ and a smooth cutoff χ equal to one on a neighborhood W of K, supported in an open set of sufficiently small volume that $g = 1 + (A - 1)\chi$ has $\|g\|_2 < B_0$. The test potential has density $G \ge \varepsilon g$. Choose $\kappa < 1$ with $C_0 < \kappa\varepsilon A$. Then $F < \kappa G$ almost everywhere on W. Continuity of w puts a sufficiently thin near-minimum set inside W, contradicting Proposition 11.4.1. Hence w is constant.□

11.5 Comparison and invariant mass

The preceding argument controls the extreme levels through connectedness of the positive-density region. When the background satisfies the Guan–Li condition, invariance of total mass gives ordinary comparison instead. An auxiliary solution with larger density on a strict sublevel set then forces that set to have zero equation mass, and domination proves uniqueness.

Set $V = \int_X \omega^n$. The Guan–Li condition

$$\mathrm{dd^c}\omega = 0, \qquad \mathrm{dd^c}(\omega^2) = 0 \tag{11.35}$$

is equivalent to mass invariance by Lemma 2.1.2 and Theorem 2.2.2. In dimension one it is automatic.

Theorem 11.5.1 *Assume* (11.35). *If $p > 1$ and $f \geq 0$ belongs to $L^p(X, \omega^n)$ with positive integral, there is exactly one normalized pair (c, u), with $c > 0$, $u \in \mathrm{PSH}(X, \omega) \cap L^\infty$, $\sup_X u = 0$, satisfying*

$$(\omega + \mathrm{dd^c} u)^n = c f \omega^n.$$

Moreover

$$c = \frac{\int_X \omega^n}{\int f \omega^n},$$

and u is continuous. Thus two bounded solutions with the same L^p density differ by a constant; no positive lower bound on f is required.

We first establish the comparison principle through its equivalence with mass invariance.

Theorem 11.5.2 *Condition* (11.35) *holds if and only if every pair of bounded ω-plurisubharmonic functions a, b satisfies*

$$\int_{\{a<b\}} \mathrm{MA}_\omega(b) \leq \int_{\{a<b\}} \mathrm{MA}_\omega(a). \tag{11.36}$$

It suffices to require this inequality for smooth potentials.

Proof Under (11.35), Theorem 2.2.2 gives $\int_X \mathrm{MA}_\omega(a) = V$ for every bounded potential a; for $n = 1$ this is Stokes' theorem.

Set $q = \max(a, b)$. On $\{a < b\}$, locality gives $\mathrm{MA}_\omega(q) = \mathrm{MA}_\omega(b)$. On $\{a \geq b\}$, one has $a = q$ and $a \leq q$, so bounded ordered contact (Lemma 1.7.1) gives

$$\mathbf{1}_{\{a \geq b\}} \mathrm{MA}_\omega(a) \leq \mathbf{1}_{\{a \geq b\}} \mathrm{MA}_\omega(q).$$

All three total masses equal V. Subtracting the last inequality from this finite common mass proves (11.36).

Conversely, comparison for smooth potentials implies smooth mass invariance: shift one member of a smooth pair by a sufficiently large constant to make its strict sublevel set all of X, apply (11.36), and reverse the pair. Comparison with the zero potential gives the common mass V. By Theorem 2.2.2 and Lemma 2.1.2, this implies (11.35) when $n \geq 2$; for $n = 1$ that condition holds by degree. □

For the uniqueness argument we need an auxiliary solution and the mixed inequality. Continuous existence supplies the first. If $G \geq 0$ lies in $L^p(X, \omega^n)$ and $\int_X G\omega^n = V$, Theorem 10.1.1 gives a continuous bounded solution with a positive normalizing constant. The mass identity makes that constant equal to one. Thus

$$\mathrm{MA}_\omega(\rho) = G\omega^n \tag{11.37}$$

is solvable, including for the indicator-truncated density below.

Proof (of Theorem 11.5.1) The domination statement needed here is the case $c = 0$ of Theorem 10.2.2: if $\mathrm{MA}_\omega(a)(\{a < b\}) = 0$ for bounded ω-plurisubharmonic functions a, b, then $a \geq b$.

Let $\mathrm{MA}_\omega(u) = \mathrm{MA}_\omega(v) = \mu = F\omega^n$, with $F \in L^p$, $p > 1$. The mass identity gives $\mu(X) = V$. If $u - v$ is nonconstant, shift u so that its difference from v has supremum $s > 0$ and infimum $-s$. Interchanging the pair if necessary, arrange that $S = \{v < u\}$ satisfies $\mu(S) \leq V/2$. Define

$$b_0 = \frac{V - 2\mu(S)}{V} \geq 0, \qquad G = 2F\mathbf{1}_S + b_0.$$

This nonnegative L^p function has total integral V, so choose the continuous bounded solution ρ of (11.37). Take C_0 with $\rho - C_0 \leq u$, and set

$$q_\delta = (1-\delta)u + \delta(\rho - C_0), \qquad E_\delta = \{v < q_\delta\} \subset S, \qquad 0 < \delta < 1.$$

On S, the mixed inequality (1.11) gives

$$\mathrm{MA}_\omega(q_\delta) \geq \left((1-\delta) + \delta 2^{1/n}\right)^n \mu =: A_\delta \mu, \qquad A_\delta > 1. \tag{11.38}$$

Ordinary comparison on E_δ yields

$$A_\delta \mu(E_\delta) \leq \int_{E_\delta} \mathrm{MA}_\omega(q_\delta) \leq \int_{E_\delta} \mathrm{MA}_\omega(v) = \mu(E_\delta).$$

Thus $\mu(E_\delta) = 0$. Bounded domination gives $v \geq q_\delta$ everywhere; letting $\delta \downarrow 0$ gives $v \geq u$, contradicting $\sup_X(u - v) = s > 0$. Therefore the difference is constant. The argument never divides by F and permits it to vanish on an open set.

Finally, for the density f in the theorem put $I = \int_X f\omega^n > 0$ and $c = V/I$. Equation (11.37) supplies a continuous bounded solution for $G = cf$, normalized by subtracting its supremum. Any competing bounded solution with constant c' satisfies $V = c'I$, hence $c' = c$; the just-proved uniqueness and the common normalization identify the potentials. In particular every bounded solution with this L^p density is continuous, even though continuity was not assumed. □

Remark 11.5.1 Theorem 11.5.2 is an equivalence of differential, comparison, and mass-invariance properties. It is not an equivalence between the Guan–Li condition and uniqueness. The other criteria in this chapter apply without that background condition. Conversely, imposing it on an auxiliary metric rather than on ω does not justify (11.36) for MA_ω.

11.6 Smooth nonuniqueness for a degenerate density

The first example puts the freedom in the potential inside an open region where the density is zero. Three independent rank-one directions reduce equality of the Monge–Ampère measures to equality of products of three scalar coefficients. We construct two semipositive forms with the same cube, then add an exact correction to obtain a strictly positive equation background.

Let $X = \mathbb{P}(\mathbb{C}^2 \oplus \mathbb{C}^2) = \mathbb{P}^3$, with block homogeneous coordinates $[A : B]$, where $A, B \in \mathbb{C}^2$, and let ω_X be its Fubini–Study form.

Theorem 11.6.1 *On $X = \mathbb{P}^3$ there exist a smooth Hermitian form ω with $\mathrm{d}\omega \neq 0$, a smooth nonnegative nonzero function f, and distinct smooth functions $u, v \in \mathrm{PSH}(X, \omega)$ such that*

$$\sup_X u = \sup_X v = 0, \qquad (\omega + \mathrm{dd^c} u)^3 = f\omega_X^3 = (\omega + \mathrm{dd^c} v)^3.$$

The density vanishes on a nonempty open set. Both potentials have minimal singularities.

11.6.1 Three rank-one directions

On $U = \{A \neq 0, B \neq 0\}$ put

$$r = \log \frac{|A|^2}{|B|^2}, \qquad P = \mathrm{dd^c} \log |A|^2, \quad Q = \mathrm{dd^c} \log |B|^2, \quad R = \mathrm{d}r \wedge \mathrm{d^c} r.$$

The logarithms individually use a local homogeneous lift; their Hessians are independent of that lift. The forms P, Q are pullbacks of Fubini–Study forms from the two projective lines. All three forms are semipositive of rank one, and

$$P^2 = Q^2 = R^2 = 0, \qquad P \wedge Q \wedge R > 0. \tag{11.39}$$

Indeed, in local coordinates $A = (1, a)$ and $B = \tau(1, b)$, P and Q are positive multiples of $\mathrm{i}\,\mathrm{d}a \wedge \mathrm{d}\bar{a}$ and $\mathrm{i}\,\mathrm{d}b \wedge \mathrm{d}\bar{b}$, while

$$\partial r = \frac{\bar{a}\,\mathrm{d}a}{1 + |a|^2} - \frac{\mathrm{d}\tau}{\tau} - \frac{\bar{b}\,\mathrm{d}b}{1 + |b|^2}.$$

Its $\mathrm{d}\tau$ component is nonzero. The three rank-one directions are therefore independent. In particular, for nonnegative x, y, z,

$$(xP + yQ + zR)^3 = 6xyz\, P \wedge Q \wedge R, \tag{11.40}$$

and the form is positive exactly when all three coefficients are positive.

For a smooth function $G : \mathbb{R} \to \mathbb{R}$ the chain rule is

$$\mathrm{dd^c} G(r) = G'(r)(P - Q) + G''(r)R. \tag{11.41}$$

Let $\sigma(r) = \mathrm{e}^r/(1 + \mathrm{e}^r)$. Since $\log(|A|^2 + |B|^2) = \log |B|^2 + \log(1 + \mathrm{e}^r)$, we have

$$\omega_X = \sigma(r)P + \sigma(-r)Q + \sigma'(r)R. \tag{11.42}$$

A function of r constant at each end of $\mathbb{R}$ extends smoothly over $X \setminus U$. A form whose coefficients equal those of (11.42) for $|r| \geq 2$ extends smoothly by ω_X there.

11.6.2 The scalar coefficients

The central coefficients will make a different rank-one factor vanish in each solution form. Outside the central interval they will return to the Fubini–Study coefficients, so the resulting forms extend across the omitted projective lines. Primes denote derivatives with respect to r.

Choose an even smooth bump β supported on $[-1, 1]$, strictly positive on $(-1, 1)$ and with $\beta' > 0$ on $(-1, 0)$, normalized by $\int_{-1}^{1} \beta = 2$. A positive multiple of $\exp(-1/(1 - r^2))$ on $(-1, 1)$, extended by zero, does this. Define

$$F(r) = -1 + \int_{-1}^{r} \beta(t)\,\mathrm{d}t \quad (-1 < r < 1), \qquad F = -1 \ (r \leq -1), \quad F = 1 \ (r \geq 1),$$

and put $p = \beta/2$, $q = \beta'/2$. Then F is smooth and odd, p is even, and q is odd, positive on $(-1, 0)$ and negative on $(0, 1)$. Both p and q vanish to infinite order at the endpoints.

Take smooth $a \geq 0$ with $a = 1$ on $(-\infty, -2/3]$ and $a = 0$ on $[-1/3, \infty)$, and put $b(r) = a(-r)$. Choose an even nonnegative $\chi \in C_c^\infty((-1/3, 1/3))$ with $\chi(0) > 0$. On $[-1, 1]$ set

$$z_0 = (q^2 + \chi^2)^{1/2}.$$

It is smooth: the expression under the square root is positive in the interior; near each endpoint $z_0 = |q|$ with fixed sign of q. It is flat at the endpoints and positive in the interior. Also,

$$z_0 \geq |q|, \qquad a(z_0 - q) = 0, \qquad b(z_0 + q) = 0. \tag{11.43}$$

The cancellations follow from the disjoint supports of a and χ, and from the sign of q on the support of a; the other equality follows by reflection.

On $(-\infty, -1]$ choose smooth functions ℓ, s such that $0 \le \ell \le 1$, $\ell = 0$ on $(-\infty, -2]$, $\ell = 1$ on $[-3/2, -1]$, while $s > 0$ for $r < -1$, $s = 1$ on $(-\infty, -3/2]$, and s is flat with value zero at -1. Such functions are obtained from ordinary smooth cutoffs and a flat positive endpoint function. Define

$$\begin{aligned} &x = (1-\ell)\sigma + \ell, \quad y = (1-\ell)\sigma(-r) + \ell s, \quad z = (1-\ell)\sigma' + \ell s \quad (r \le -1),\\ &x = p + a, \qquad\qquad y = p + b, \qquad\qquad z = z_0 \qquad\qquad (-1 \le r \le 1), \end{aligned}$$

and for $r \ge 1$ use $x(r) = y(-r)$, $y(r) = x(-r)$, $z(r) = z(-r)$.

The definitions join smoothly. At -1 their values are $(1, 0, 0)$, with all discrepancies flat; near that endpoint $a = 1$ and $b = 0$. The central definitions are symmetric under reflection, so the same holds at 1. We have

$$x, y, z > 0 \quad \text{for } r \ne \pm 1, \qquad (x, y, z)(-1) = (1, 0, 0), \quad (x, y, z)(1) = (0, 1, 0).$$

For $|r| \ge 2$ their values are $(\sigma(r), \sigma(-r), \sigma'(r))$.

11.6.3 Two forms with the same cube

The cancellation identities now turn the scalar coefficients into two semipositive forms with the same cube. Their average will be strictly positive except at the two ends of the central interval.

The globally smooth form and function

$$\omega_0 = x(r)P + y(r)Q + z(r)R, \qquad w = F(r)$$

are understood by the extensions above. The form is semipositive, and positive except on $M_- = \{r = -1\}$ and $M_+ = \{r = 1\}$. Put

$$H = \tfrac{1}{2}\mathrm{dd^c} w = p(P - Q) + qR, \qquad A_0 = \omega_0 - H, \quad B_0 = \omega_0 + H.$$

For $|r| \ge 1$ we have $H = 0$. On $|r| < 1$,

$$A_0 = aP + (2p + b)Q + (z_0 - q)R, \qquad B_0 = (2p + a)P + bQ + (z_0 + q)R.$$

All coefficients are nonnegative. The cancellations in (11.43) give $A_0^3 = B_0^3 = 0$ on this band. Outside it both cubes equal ω_0^3. Thus A_0, B_0 are globally smooth semipositive forms with equal cubes. That common top-degree form is smooth, vanishes on the closed band $|r| \le 1$, and is positive on its complement, including near $X \setminus U$.

11.6.4 Restoring strict positivity of the background

The forms A_0, B_0 already have the required common measure. We obtain a Hermitian background by adding the same exact correction to their average and subtracting its potential from both solutions. The correction must restore positivity at $M_\pm$ while preserving it in the transition regions.

Take $\kappa \in C_c^\infty((-2,2))$, equal to one on $[-3/2, 3/2]$. For a parameter $\varepsilon > 0$ to be chosen, set

$$\rho(r) = \varepsilon(r^2 - 1)\kappa(r), \qquad \omega = \omega_0 + \mathrm{dd^c}\rho.$$

Its coefficients on U are $x + \rho'$, $y - \rho'$ and $z + \rho''$. On $[-3/2, 3/2]$ one has $\rho' = 2\varepsilon r$ and $\rho'' = 2\varepsilon$. The last coefficient is positive. For $r \geq 0$, $x + 2\varepsilon r$ is positive, including at $r = 0$ since $x(0) = p(0) > 0$. On $[-3/2, 0]$, the function x has a positive minimum m, so $x + 2\varepsilon r \geq m - 3\varepsilon > 0$ for small ε. Reflection gives the corresponding estimate for $y - 2\varepsilon r$. On the compact region $3/2 \leq |r| \leq 2$, all three original coefficients have a positive minimum and the perturbations are $O(\varepsilon)$. Outside that region and the central interval the form is exactly ω_X. This proves strict positivity globally.

For completeness, nonclosedness can be checked directly. Since $\mathrm{d}R = -\mathrm{d}r \wedge (P - Q)$,

$$\mathrm{d}\omega = \mathrm{d}\omega_0 = \mathrm{d}r \wedge \big((x' - z)P + (y' + z)Q\big). \tag{11.44}$$

At $r = 0$, $x' = y' = 0$ and $z = \chi(0) > 0$. Hence $\mathrm{d}\omega = z\,\mathrm{d}r \wedge (Q - P) \neq 0$ there. The nonzero $\mathrm{d}\tau$ component used in (11.39) also shows that $\mathrm{d}r \wedge P$ and $\mathrm{d}r \wedge Q$ are linearly independent.

Proof (of Theorem 11.6.1) Set $\widetilde{u} = -w/2 - \rho$ and $\widetilde{v} = w/2 - \rho$. The construction gives

$$\omega + \mathrm{dd^c}\widetilde{u} = A_0 \geq 0, \qquad \omega + \mathrm{dd^c}\widetilde{v} = B_0 \geq 0.$$

Subtract their respective suprema to obtain u, v. Their Hessians are unchanged, and $v - u = w +$ constant is nonconstant. The function $f = A_0^3/\omega_X^3 = B_0^3/\omega_X^3$ is smooth, nonnegative and nonzero. It vanishes on the nonempty open band $|r| < 1$. Both smooth potentials have minimal singularities because $V_\omega = 0$ for a Hermitian background. □

Every Hermitian background on $\mathbb{P}^3$ has bounded mass: the Fubini–Study form has invariant mass by closed bounded Stokes, and Proposition 6.5.1 makes finiteness independent of the metric. This applies to all the projective backgrounds constructed below.

The linearized difference also explains the failure of the smooth maximum argument. The tensor

$$T = A_0^2 + A_0 \wedge B_0 + B_0^2$$

satisfies $\mathrm{dd^c}w \wedge T = B_0^3 - A_0^3 = 0$, but $T = 0$ on $M_\pm$, where $A_0 = B_0 = \omega_0$ has rank one. The resulting equation is not globally elliptic, even though both potentials and their common density are smooth.

11.6.5 A continuum of radial solutions

The open zero-density band gives more freedom than the two smooth solutions exhibit. Writing a further potential as $G(r) - \rho$, with G a function of one real variable, reduces the equation to choosing a slope G' for which one of the three factors vanishes. Switching the vanishing factor at two suitable levels produces a continuum of $C^{1,1}$ solutions for the same background ω and density f.

The strict inequality $z_0 > |p'|$ near zero gives

$$\int_0^1 z_0(r)\,\mathrm{d}r > \int_0^1 |p'(r)|\,\mathrm{d}r = p(0). \tag{11.45}$$

Proposition 11.6.1 *The same smooth background and smooth density in Theorem 11.6.1 admit uncountably many distinct normalized $C^{1,1}$ solutions of*

$$\mathrm{MA}_\omega(u) = f\omega_X^3.$$

They include a directly constructed solution invariant under the block exchange $\iota[A : B] = [B : A]$, if the cutoff defining ρ is chosen even.

Proof Choose $-1 < I < 0 < J < 1$ satisfying

$$\int_I^J z_0(r)\,\mathrm{d}r = p(I) + p(J). \tag{11.46}$$

We first show there are an interval's worth of choices. For fixed I, define

$$K(I, J) = \int_I^J z_0(r)\,\mathrm{d}r - p(I) - p(J).$$

At $I = 0$ we have $K(0, 0) = -2p(0) < 0$ and $K(0, 1) > 0$ by (11.45). Continuity gives $\epsilon > 0$ such that these opposite signs persist for every $I \in [-\epsilon, 0]$. For $0 < J < 1$,

$$\partial_J K(I, J) = z_0(J) - p'(J) > 0.$$

The intermediate value theorem therefore supplies a unique $J = J(I) \in (0, 1)$ for every $I \in [-\epsilon, 0)$. Shrink ϵ below one.

For any pair satisfying (11.46), define the slope on $[-1, 1]$ by

$$h_{I,J}(r) = \begin{cases} p(r), & -1 \le r \le I, \\ p(I) - \displaystyle\int_I^r z_0(s)\,\mathrm{d}s, & I \le r \le J, \\ -p(r), & J \le r \le 1, \end{cases}$$

and extend it by zero outside that interval. The balance identity makes it continuous at J, and continuity at I is immediate. Each piece has bounded derivative. Thus $h_{I,J}$ is globally Lipschitz; it is flat at ± 1. Let $G_{I,J}$ be any primitive. It is $C^{1,1}$ and constant

at each end of the real line. Consequently $G_{I,J}(r)$ extends to a global $C^{1,1}$ function on X, smooth near $X \setminus U$.

On the middle interval the slope lies between $-p$ and p. Indeed,

$$(h_{I,J} - p)' = -z_0 - p' \le 0, \qquad (h_{I,J} + p)' = -z_0 + p' \le 0.$$

The first difference starts at zero at I; the second ends at zero at J. These two inequalities give $h_{I,J} \le p$ and $h_{I,J} \ge -p$, respectively.

Put $\widetilde{u}_{I,J} = G_{I,J}(r) - \rho$. Its solution form is, almost everywhere on U,

$$\omega + \mathrm{dd^c}\widetilde{u}_{I,J} = \omega_0 + \mathrm{dd^c}G_{I,J}(r) = (x + h_{I,J})P + (y - h_{I,J})Q + (z + h'_{I,J})R. \quad (11.47)$$

On $I < r < J$, the first two coefficients are nonnegative because $x = p + a$, $y = p + b$ and $-p \le h_{I,J} \le p$. The third is zero. On $-1 < r < I$, we have $h_{I,J} = p$ and $b = 0$, since $I < 0$; the second coefficient is zero and the others are nonnegative, using $z_0 + p' \ge 0$. On $J < r < 1$, we have $h_{I,J} = -p$ and $a = 0$; the first coefficient is zero and the others are nonnegative, using $z_0 - p' \ge 0$. Outside the band, $h_{I,J} = h'_{I,J} = 0$, so the form equals ω_0.

There is no additional hypersurface term in (11.47): $G_{I,J}$ is $C^{1,1}$, and its first derivative is continuous at both transition points. Its distributional Hessian is the bounded almost-everywhere Hessian. The preceding coefficient inequalities therefore prove $\omega + \mathrm{dd^c}\widetilde{u}_{I,J} \ge 0$ as a current.

By Proposition 1.7.2, its bounded product agrees with the almost-everywhere wedge of its Hessian. The rank-one formula (11.39) now shows that the cube of (11.47) is zero almost everywhere on the band, and equals ω_0^3 off it. These are exactly the defining values of $f\omega_X^3$ in the preceding construction. Thus $\widetilde{u}_{I,J}$ solves the equation. Subtracting its supremum gives a normalized solution $u_{I,J}$.

Different I in the interval above give different normalized solutions. If $I_1 < I_2 < 0$ and $I_1 < r < I_2$, then the slope belonging to I_2 is $p(r)$, whereas

$$p(r) - h_{I_1,J(I_1)}(r) = \int_{I_1}^{r} (p'(s) + z_0(s))\,\mathrm{d}s > 0,$$

because $p' > 0$ on $(-1, 0)$. The two primitives therefore cannot differ by a constant. An interval contains uncountably many parameters, proving the first assertion. None of these slopes is identically p or identically $-p$, so the constructed solutions are also distinct from the original two smooth solutions.

Finally, the function

$$L(t) = \int_0^t z_0(r)\,\mathrm{d}r - p(t), \qquad 0 \le t \le 1,$$

is negative at zero, positive at one, and strictly increasing on $(0, 1)$. Its unique zero t_* gives an admissible symmetric pair $I = -t_*$, $J = t_*$. Evenness of z_0 and p then makes $h_{I,J}$ odd: on the middle interval it is $-\int_0^r z_0(s)\,\mathrm{d}s$. Choose its primitive even. If ρ is

even, $G_{I,J}(r) - \rho$ and its normalized version are fixed by ι. This proves the explicit invariant-solution assertion. □

For the symmetric data, positive-exponent approximation selects the symmetric member just constructed; this will be proved in Theorem 13.18.1.

11.7 Nonuniqueness with a zero set of empty interior

We now keep the Hermitian background of Theorem 11.6.1 and change its density to be positive off $M_- \cup M_+$, where $M_\pm = \{r = \pm 1\}$. On the middle band, the determinant equation becomes a scalar initial-value problem. Barriers keep its solutions between the admissible slopes; infinite-order vanishing at the endpoints then makes their extensions smooth.

Theorem 11.7.1 *There is a smooth nonclosed Hermitian form ω on $\mathbb{P}^3$ and a smooth nonnegative density g, positive off two disjoint smooth real hypersurfaces, such that*

$$\mathrm{MA}_\omega(u) = g\omega_X^3$$

has uncountably many distinct smooth solutions normalized by $\sup_X u = 0$. *Thus a zero set of empty interior, even of smooth-volume measure zero, does not ensure uniqueness of smooth normalized solutions. The background has bounded mass.*

Use the coefficients of Section 11.6.2 with the explicit bump

$$p(r) = C\exp\left(-\frac{1}{1-r^2}\right), \qquad -1 < r < 1, \qquad \int_{-1}^{1} p(r)\,\mathrm{d}r = 1,$$

extended by zero; the integral fixes $C > 0$ and $q = p'$. Fix a scalar $0 < s < 1$, and set

$$k_s(r) = (x(r) + sp(r))(y(r) - sp(r))(z(r) + sq(r)).$$

For $|r| > 1$ this is xyz. For $G' = sp$, the potential $G(r) - \rho$ has solution form $\omega_0 + sH$. It is strictly positive for $-1 < r < 1$, since

$$x + sp = a + (1+s)p > 0, \quad y - sp = b + (1-s)p > 0, \quad z_0 + sq \geq (1-s)z_0 > 0.$$

It is also strictly positive outside the closed band. At $r = \pm 1$ it equals the rank-one form ω_0. Consequently

$$g\omega_X^3 = (\omega_0 + sH)^3 \tag{11.48}$$

defines a smooth density positive exactly off $M_- \cup M_+$. Those hypersurfaces have measure zero and empty interior.

Lemma 11.7.1 *There exists $\eta_0 > 0$ such that for every $0 < \eta < \eta_0$ the initial-value problem*

$$h' = \frac{k_s(r)}{(x(r)+h)(y(r)-h)} - z(r), \qquad h(0) = sp(0) - \eta, \tag{11.49}$$

has a smooth solution on $(-1,1)$ with $-x < h < y$ and $h < sp$. Moreover, on some fixed endpoint neighborhoods there are constants $0 < c_L < s < t < 1$, independent of η, such that

$$c_L p < h < sp \quad \textit{near } -1, \qquad -tp < h < sp \quad \textit{near } 1.$$

Proof The function $h_s = sp$ is a solution of (11.49) and lies strictly between $-x$ and y throughout $(-1,1)$. Write $F_s(r,h)$ for that equation's smooth right side on the open domain $-x(r) < h < y(r)$.

Near -1 the coefficients are exactly

$$x = 1 + p, \qquad y = p, \qquad z = q > 0.$$

For a constant $c \in (-1,1)$ the measure coefficient of the trial slope cp is

$$k_c = (1-c^2)\big(1 + (1+c)p\big)pq.$$

Fix $c_L \in (0,s)$. At $p = 0$ the difference of the two factors preceding pq for $c = c_L$ and $c = s$ is $s^2 - c_L^2 > 0$. Thus, on a sufficiently small fixed left endpoint neighborhood, $k_{c_L} > k_s$ and

$$F_s(r, c_L p) - (c_L p)' = \frac{k_s - k_{c_L}}{(x + c_L p)(y - c_L p)} < 0. \tag{11.50}$$

This is the lower barrier when the equation is followed backwards.

Near 1 the exact coefficients are

$$x = p, \qquad y = 1 + p, \qquad z = -q > 0,$$

and the trial slope cp has coefficient

$$k_c = (1-c^2)\big(1 + (1-c)p\big)p(-q).$$

Fix $s < t < 1$. The difference of the factors for $c = s$ and $c = -t$ is positive at $p = 0$, namely $t^2 - s^2$. On a sufficiently small fixed right endpoint neighborhood, $k_s > k_{-t}$, so

$$F_s(r, -tp) - (-tp)' = \frac{k_s - k_{-t}}{(x - tp)(y + tp)} > 0. \tag{11.51}$$

This is a lower barrier for forward evolution.

Choose $r_L < 0 < r_R$ within these respective neighborhoods. The graph of h_s on $[r_L, r_R]$ is a compact subset of the smooth domain of F_s. Existence and continuous dependence for smooth ordinary differential equations therefore give $\eta_0 > 0$ such

that every initial value $sp(0) - \eta$, $0 < \eta < \eta_0$, has a solution throughout that interval, still in the domain, and

$$h(r_L) > c_L p(r_L), \qquad h(r_R) > -tp(r_R).$$

Two solutions of the scalar equation cannot cross, by local uniqueness. Since $h(0) < h_s(0)$, we have $h < h_s$ wherever the solution exists.

Follow the solution backwards from r_L. It cannot cross $c_L p$. At a first crossing, the difference $h - c_L p$ would be zero and positive immediately to its right; its right derivative there would be nonnegative. Formula (11.50) gives a strictly negative derivative instead. Thus $c_L p < h < sp$ on its leftward interval of existence. Likewise (11.51) prevents a first downward crossing of $-tp$ in forward time from r_R, giving $-tp < h < sp$ to the right.

These bounds also prevent a finite endpoint of either interval of existence inside $(-1, 1)$. On any compact subinterval approaching such a proposed endpoint, both horizontal factors are bounded away from zero, the slope remains bounded, and F_s and its derivatives are bounded on a compact neighborhood of the graph. The ordinary continuation theorem extends the solution across that endpoint. It therefore exists on the entire interval $(-1, 1)$ with the claimed bounds. □

Lemma 11.7.2 *Every solution furnished by Lemma 11.7.1, extended by zero outside* $(-1, 1)$*, is smooth on* $\mathbb{R}$ *and vanishes to infinite order at* ± 1.

Proof Let h be one of the solutions in Lemma 11.7.1. On either endpoint neighborhood put $\xi = h/p$. The trapping bounds place ξ in a fixed compact subinterval of $(-1, 1)$. Since $p > 0$ there, this quotient is smooth in the open interval. Define the rational functions

$$R_L(p, \xi) = \frac{(1 - s^2)(1 + (1 + s)p)}{(1 + (1 + \xi)p)(1 - \xi)} - 1,$$

$$R_R(p, \xi) = 1 - \frac{(1 - s^2)(1 + (1 - s)p)}{(1 + \xi)(1 + (1 - \xi)p)}.$$

The explicit endpoint formulas for the equation give

$$h' = q\, R_L(p, \xi) \quad \text{on the left,} \qquad h' = q\, R_R(p, \xi) \quad \text{on the right.}$$

Both rational functions are smooth, with all derivatives bounded, on a fixed neighborhood of the relevant compact set with $p \geq 0$. Writing $R_{L/R}$ for the corresponding function at either endpoint, we obtain

$$\xi' = \frac{p'}{p}\big(R_{L/R}(p, \xi) - \xi\big). \tag{11.52}$$

For the chosen bump,

$$\frac{p'}{p} = -\frac{2r}{(1 - r^2)^2}.$$

This rational function and each of its derivatives grow at most as a power of the inverse distance to the endpoint. The same holds for every derivative of p, and more precisely each derivative of p is p times such a rational function.

An induction using (11.52) shows that every derivative $\xi^{(j)}$ grows at most polynomially in the inverse distance to the endpoint. The starting bound is the trapping bound on ξ. For the induction step, differentiate the full right side, including its $-\xi$ term, j times: every term is a product of a derivative of p'/p, a bounded partial derivative of $R_{L/R}$, and derivatives of p and of ξ of order at most j. All of these have the required polynomial bound by the induction hypothesis. This proves it for $\xi^{(j+1)}$.

Leibniz' rule for $h = p\xi$ now gives, for each j, constants C_j, N_j such that

$$|h^{(j)}(r)| \le C_j p(r)(1 - |r|)^{-N_j}$$

near the endpoints. The right side tends to zero because of exponential flatness of p. Thus all one-sided derivatives tend to zero. Extend each derivative by zero at the endpoint. The mean value theorem shows that its endpoint difference quotient tends to the zero limit of the next derivative. These extensions are therefore successive derivatives of the extended function, which is smooth and flat. □

Proof (of Theorem 11.7.1) Let η_0 be as in Lemma 11.7.1, and for $0 < \eta < \eta_0$ denote by h_η its solution with initial value $sp(0) - \eta$. Extend h_η by zero using the preceding lemma, and take a smooth primitive G_η. It is constant on each end of the real line, so $G_\eta(r)$ extends smoothly to all of $\mathbb{P}^3$, including the two omitted projective lines. Set

$$\widetilde{u}_\eta = G_\eta(r) - \rho.$$

On the open band its solution form is

$$\omega + \mathrm{dd^c}\widetilde{u}_\eta = (x + h_\eta)P + (y - h_\eta)Q + (z + h'_\eta)R.$$

The horizontal coefficients are strictly positive by trapping and continuation, while the equation gives

$$z + h'_\eta = \frac{k_s}{(x + h_\eta)(y - h_\eta)} > 0.$$

Outside the band the form is ω_0. At the two boundary hypersurfaces, flatness gives the same rank-one form ω_0. It is therefore globally semipositive, and its smooth cube equals the measure in (11.48) everywhere. Normalize by subtracting $\sup_X \widetilde{u}_\eta$.

Distinct values of η have different slopes at zero. The corresponding potentials cannot differ by a constant, so their normalized versions remain distinct. This gives an uncountable family. The Hermitian background is unchanged, so it retains bounded mass. □

The density degenerates to infinite order on the two hypersurfaces. The complement has three components, and different solutions can be constant translates of one another on the two exterior components with different translation constants. The scalar equation on the middle component joins these constants smoothly across the degenerate hypersurfaces. This is why ellipticity at every point of the positive-density set does not give a global maximum-principle argument across its boundary.

11.8 Quadratic density zeros

Infinite-order vanishing of the density is not needed for smooth nonuniqueness. We change the coefficients of ω_0 so that its radial coefficient stays positive and one horizontal coefficient vanishes quadratically at each interface. The resulting singular scalar equation forces the slopes of potential differences to decay exponentially in the inverse distance to the interface. The slopes therefore extend smoothly by zero, despite the finite vanishing order of the density.

Theorem 11.8.1 *On $\mathbb{P}^3$ there exist a smooth nonclosed Hermitian form ω and a smooth nonnegative density g such that:*

(1) *The zero set of g consists of two disjoint smooth real hypersurfaces. Near either hypersurface, g is the square of a smooth defining function times a smooth strictly positive function.*
(2) *The equation $(\omega + \mathrm{dd^c}u)^3 = g\omega_X^3$ has uncountably many distinct smooth ω-plurisubharmonic solutions normalized by* $\sup_X u = 0$.

The fixed background has bounded mass.

We retain the rank-one forms P, Q, R and the real function r, with new scalar coefficients.

Choose a smooth function a valued in $[0, 1]$, equal to one for $r \le -2/3$ and zero for $r \ge -1/3$, and put $b(r) = a(-r)$. Choose a smooth even cutoff χ valued in $[0, 1]$, zero for $|r| \le 5/4$ and one for $|r| \ge 3/2$. Set $p(r) = (1 - r^2)^2$ and

$$\begin{aligned} x &= (1-\chi)(p+a) + \chi\sigma, \\ y &= (1-\chi)(p+b) + \chi(1-\sigma), \\ z &= 1 - \chi + \chi\sigma', \qquad \omega_0 = xP + yQ + zR. \end{aligned} \tag{11.53}$$

These forms extend by ω_X across the two omitted projective lines. The coefficient z is strictly positive. Both x, y are strictly positive except that $y(-1) = 0$ and $x(1) = 0$. Thus ω_0 is positive off $M_\pm = \{r = \pm 1\}$ and has rank two there.

Choose a smooth cutoff $\kappa \in C_c^\infty((-2, 2))$ equal to one on $[-3/2, 3/2]$. For a parameter $\varepsilon > 0$ to be chosen, define the correction

$$\rho = \varepsilon(r^2 - 1)\kappa(r), \qquad \omega = \omega_0 + \mathrm{dd^c}\rho.$$

For sufficiently small $\varepsilon > 0$, ω is strictly positive. Indeed, on the central interval its coefficients are $x + 2\varepsilon r$, $y - 2\varepsilon r$, $z + 2\varepsilon$. For $r \geq 0$ the first is positive; on $[-3/2, 0]$, x has a positive minimum, so it remains positive for small ε. Reflection proves the second assertion, and the third is immediate. On $3/2 \leq |r| \leq 2$, the original form is strictly positive on a compact set and the correction is $O(\varepsilon)$. Near the projective ends there is no correction. At zero, $x' = y' = 0$ and $z = 1$, so

$$\mathrm{d}\omega = \mathrm{d}r \wedge \big((x' - z)P + (y' + z)Q\big) \neq 0.$$

Define the density before constructing further solutions by

$$g\omega_X^3 = \omega_0^3.$$

It is smooth and positive exactly off $M_- \cup M_+$. Near either hypersurface, $xyz = (1+p)p$ and

$$g(r) = \frac{(1 + p(r))(1 - r^2)^2}{\sigma(r)(1 - \sigma(r))\sigma'(r)}.$$

The denominator is smooth and strictly positive at ± 1. This proves the asserted quadratic vanishing. The potential $-\rho$ is one smooth solution.

Lemma 11.8.1 *For every sufficiently small $\eta > 0$, the equation*

$$h' = \frac{xyz}{(x + h)(y - h)} - z, \qquad h(0) = -\eta \tag{11.54}$$

has a smooth solution on $(-1, 1)$ with $-x < h < y$ and $h < 0$. For some fixed $0 < c < 1$, on fixed neighborhoods of both endpoints one has

$$-cp < h < 0.$$

Every such solution extends by zero to a smooth function on $\mathbb{R}$, flat at ± 1.

Proof The right side $F(r, h)$ is smooth on $-x < h < y$, and $h = 0$ is an exact solution. Near -1, $(x, y, z) = (1 + p, p, 1)$; near 1, $(x, y, z) = (p, 1 + p, 1)$. Fix $0 < c < 1$. For the trial function $-cp$, at the left endpoint

$$F(r, -cp) + cp' = \frac{1 + p}{(1 + (1 - c)p)(1 + c)} - 1 + cp' \longrightarrow -\frac{c}{1 + c} < 0,$$

whereas at the right endpoint

$$F(r, -cp) + cp' = \frac{1 + p}{(1 - c)(1 + (1 + c)p)} - 1 + cp' \longrightarrow \frac{c}{1 - c} > 0.$$

Here $p, p' \to 0$ at both endpoints. The strict inequalities hold on fixed endpoint neighborhoods. They give a lower barrier backwards on the left and forwards on the right.

Choose a compact middle interval with endpoints in those neighborhoods. Continuous dependence around the exact solution zero supplies one $\eta_0 > 0$ such that all

$0 < \eta < \eta_0$ give a solution on that interval, in the smooth domain, and above $-cp$ at its endpoints. Scalar uniqueness gives $h < 0$. The strict crossing arguments then give $-cp < h < 0$ at both ends. On each compact subinterval of $(-1, 1)$ the trapped graph stays a positive distance from both horizontal boundaries; hence ordinary continuation excludes any finite interior endpoint. This proves existence throughout $(-1, 1)$.

The equation factors as

$$h' = h\,L(r, h), \qquad L(r, h) = \frac{z(x - y + h)}{(x + h)(y - h)}. \tag{11.55}$$

Since $h < 0$, $(\log(-h))' = L(r, h)$. Near the left endpoint, the trapping bounds and the exact coefficients imply

$$L(r, h) = \frac{1 + h}{(1 + p + h)(p - h)} \geq \frac{C_1}{p}$$

after shrinking that neighborhood, with $C_1 > 0$. Near the right endpoint they give

$$L(r, h) = \frac{-1 + h}{(p + h)(1 + p - h)} \leq -\frac{C_2}{p}, \qquad C_2 > 0.$$

For example on the left $1 + h \geq 1 - cp \geq 1/2$ and the denominator is at most $(1 + p)(1 + c)p$. On the right the numerator is at most -1, and the denominator is at most $p(1 + (1 + c)p)$. Because p is comparable to the square of the endpoint distance $d = 1 - |r|$, integrating these inequalities backwards on the left and forwards on the right gives, for some $A > 0$,

$$|h(r)| \leq C\mathrm{e}^{-A/d} \tag{11.56}$$

near either endpoint. Constants may depend on the solution; uniformity in η is not needed for smooth extension.

It remains to control every derivative. On the trapped graph, all partial derivatives in (r, h) of the rational function L have at most polynomial growth in d^{-1}. Indeed, near an endpoint x, y are polynomials in r, $z = 1$, h is bounded, and the two denominator factors are bounded below by a positive constant and by $(1 - c)p$, in the appropriate order. Repeated quotient differentiation therefore only introduces finite powers of p^{-1}.

Inductively, for every $j \geq 0$ there are constants C_j, N_j with

$$|h^{(j)}(r)| \leq C_j \mathrm{e}^{-A/d} d^{-N_j}.$$

The case $j = 0$ is (11.56). Differentiate $h' = hL(r, h(r))$ j times. Every term has a derivative of the explicit factor h, of order at most j, and a derivative of the composite L. The latter uses only partial derivatives of L and derivatives of h of order at most j, all polynomially bounded by the induction hypothesis. The explicit factor retains the exponential bound, proving the next case. All derivatives therefore tend to zero.

Extending them by zero and applying the mean value theorem to endpoint difference quotients proves that the zero extension is smooth and flat. □

Proof (of Theorem 11.8.1) Let $\eta_0 > 0$ be the initial-value threshold obtained in Lemma 11.8.1, and let h_η denote the solution with $h_\eta(0) = -\eta$. For each $0 < \eta < \eta_0$, take a primitive G_η of its smooth zero extension. It is constant near each projective end and therefore defines a global smooth function. The potential $u_\eta = G_\eta(r) - \rho$ has solution form

$$\omega + \mathrm{dd^c} u_\eta = (x + h_\eta)P + (y - h_\eta)Q + (z + h'_\eta)R.$$

Inside the band the first two coefficients are positive and the equation gives

$$z + h'_\eta = \frac{xyz}{(x + h_\eta)(y - h_\eta)} > 0.$$

Outside the band the form is ω_0, and at the two boundary hypersurfaces flatness gives the same rank-two form ω_0. Thus it is globally semipositive and has cube exactly $\omega_0^3 = g\omega_X^3$. Different initial slopes give potentials which do not differ by constants, so supremum normalization preserves an uncountable family. Bounded mass follows from Proposition 6.5.1, as above. □

At each interface a difference of two solutions agrees to infinite order with a constant, because its derivative is a difference of flat slopes. The constants at the two interfaces need not agree. The singular scalar equation thus produces flat differences even though the density has only quadratic zeros.

11.9 A real analytic density

The last construction prescribes a globally real analytic density before choosing the background. We express it as a polynomial in a real analytic function on projective space, then choose a symmetric pair of radial perturbations with that determinant. Patching their coefficients while preserving the determinant extends the construction across the projective ends. The two quadratic zero hypersurfaces retain the separation that allows different extremal values.

Theorem 11.9.1 *Put*

$$t = \frac{|A|^2}{|A|^2 + |B|^2}, \qquad a = \frac{1}{1+\mathrm{e}}, \qquad b = \frac{\mathrm{e}}{1+\mathrm{e}}, \qquad f_*(t) = (t-a)^2(t-b)^2.$$

There exist a smooth positive nonclosed ω and distinct smooth normalized ω-plurisubharmonic functions u, v with

$$(\omega + \mathrm{dd^c}u)^3 = (\omega + \mathrm{dd^c}v)^3 = f_*(t)\omega_X^3. \tag{11.57}$$

Both solution forms are positive away from the two connected real analytic hypersurfaces $\{t = a\}$, $\{t = b\}$, and have complex rank two on them. The density is globally real analytic and has precisely quadratic transverse zeros. For every $n \geq 4$ the same assertions hold on $\mathbb{P}^3 \times \mathbb{P}^{n-3}$, with the pulled-back density and the product Fubini–Study volume, and with rank $n-1$ on the two product hypersurfaces.

Proof The rank-one formula (11.40) reduces the construction to scalar coefficients with a prescribed product. By (11.42),

$$\omega_X^3 = 6(\sigma')^2 P \wedge Q \wedge R, \qquad \sigma' = \sigma(r)\sigma(-r) > 0. \tag{11.58}$$

11.9.1 Prescribing the determinant

On U, one has $t = \sigma(r)$. We first construct two positive triples with product $f_*(\sigma)(\sigma')^2$ on the middle band. Their horizontal factors will lose exactly one direction at each endpoint.

Put $d_*(r) = f_*(\sigma(r))(\sigma'(r))^2$. Since $a = \sigma(-1)$, $b = \sigma(1)$, define the positive smooth divided differences

$$k_-(r) = \int_0^1 \sigma'(-1 + s(r+1))\,\mathrm{d}s, \qquad k_+(r) = \int_0^1 \sigma'(1 + s(r-1))\,\mathrm{d}s,$$

and $H_*(r) = k_-(r)k_+(r)\sigma'(r) > 0$. The fundamental theorem of calculus gives

$$\sigma(r) - a = (r+1)k_-(r), \quad \sigma(r) - b = (r-1)k_+(r), \quad d_* = (1-r^2)^2 H_*^2. \tag{11.59}$$

These formulas supply the smooth positive extension at both endpoints, without a division by a vanishing function. The symmetry $\sigma(-r) = 1 - \sigma(r)$ makes d_*, and hence H_*, even.

Choose the flat bump

$$p(r) = \begin{cases} \mathrm{e}^{-1/(1-r^2)}, & |r| < 1, \\ 0, & |r| \geq 1, \end{cases} \qquad q = p'.$$

All derivatives vanish at $r = \pm 1$: differentiating gives a finite sum of powers of $(1-r^2)^{-1}$ multiplied by the exponential, and the exponential dominates every such

power. On $|r| < 1$, $q = -2rp/(1-r^2)^2$. Define

$$z_b = \sqrt{q^2 + 2H_*^2}, \qquad S_* = \sqrt{r^2 + 4p^2 + 2z_b(1-r^2)^2}, \tag{11.60}$$

$$x_b = (S_* - r)/2, \qquad y_b = (S_* + r)/2. \tag{11.61}$$

Both radicands are strictly positive everywhere, so these are single smooth formulas, including at the endpoints. They satisfy

$$z_b > |q|, \qquad x_b - y_b = -r, \qquad x_b y_b = p^2 + \frac{z_b}{2}(1-r^2)^2. \tag{11.62}$$

The coefficients x_b, y_b are nonnegative, positive away from the endpoints, with values $(1,0)$ at -1 and $(0,1)$ at 1. On the open band,

$$S_*^2 - (2p + |r|)^2 = 2(1-r^2)^2(z_b - |q|) > 0,$$

so $x_b > p$, $y_b > p$. Consequently both triples

$$(x_b - p, y_b + p, z_b - q), \qquad (x_b + p, y_b - p, z_b + q) \tag{11.63}$$

are strictly positive there.

Here is the determinant cancellation. Set $D_* = x_b y_b - p^2$, $E_* = p(x_b - y_b) = -rp$. Equations (11.62) and the formula for q give

$$D_* = (z_b/2)(1-r^2)^2, \qquad qD_* = z_b E_*.$$

The two products in (11.63) therefore equal

$$(D_* + E_*)(z_b - q) = (D_* - E_*)(z_b + q) = \frac{D_*}{z_b}(z_b^2 - q^2) = d_*. \tag{11.64}$$

For $|r| \geq 1$, $p = q = 0$ and the same identity reads $x_b y_b z_b = d_*$. Both perturbed forms have exactly the prescribed determinant, and rank two at the endpoints.

11.9.2 Extension over the projective ends

The coefficients just constructed have the right determinant but need not extend over the projective ends. We interpolate the two horizontal coefficients to those of $f_*(t)^{1/3}\omega_X$ away from the zero levels, and determine the radial coefficient by the prescribed determinant.

Choose a smooth even cutoff $0 \le \eta \le 1$, equal to one on $|r| \le 3/2$ and zero on $|r| \ge 2$. On $|r| < 3/2$ set $(x, y, z) = (x_b, y_b, z_b)$. On $|r| > 5/4$, where $f_*(\sigma(r)) > 0$, put

$$\begin{aligned} \kappa &= f_*(\sigma(r))^{1/3}, \\ x &= \eta x_b + (1-\eta)\kappa\sigma(r), \\ y &= \eta y_b + (1-\eta)\kappa\sigma(-r), \qquad z = d_*/(xy). \end{aligned}$$

These coefficients are smooth and positive on the outer region. On the overlap $5/4 < |r| < 3/2$, $\eta = 1$, $p = q = 0$, and $x_b y_b z_b = d_*$; hence the two definitions agree identically. For $|r| \ge 2$ they equal $\kappa(\sigma(r), \sigma(-r), \sigma'(r))$. Thus

$$\omega_0 = xP + yQ + zR$$

extends smoothly across both omitted projective lines, being equal there to $f_*(t)^{1/3}\omega_X$. This is smooth positive near those lines since $f_*(0) = f_*(1) = a^2b^2 > 0$. Globally ω_0 is semipositive, positive away from $r = \pm 1$, and rank two on these levels.

The function

$$F(r) = 2\int_0^r p(s)\,\mathrm{d}s$$

is constant on each tail and extends to a smooth nonconstant function on X. Its half-Hessian is $H_0 = \mathrm{dd^c}F/2 = p(P-Q) + qR$. By (11.64), the forms $A_0 = \omega_0 - H_0$, $B_0 = \omega_0 + H_0$ satisfy

$$A_0^3 = B_0^3 = f_*(t)\omega_X^3$$

on the open block set and hence globally. Their ranks and positivity are those already checked in (11.63). The construction is symmetric: $x(-r) = y(r)$ and z is even.

11.9.3 A positive equation background

As in the first example, an exact correction makes the average form strictly positive without changing the two solution forms. Here the radial coefficient is already positive; the correction restores the missing horizontal direction at each endpoint. Choose a parameter $\varepsilon > 0$ as restricted below.

Let $h(r) = (r^2 - 1)\eta(r)$, extend it by zero on the tails, and set $\rho = \varepsilon h$. The coefficients of $\omega = \omega_0 + \mathrm{dd^c}\rho$ are

$$(x + \varepsilon h',\ y - \varepsilon h',\ z + \varepsilon h''). \tag{11.65}$$

On $|r| \le 3/2$, $h' = 2r$, $h'' = 2$. For $0 \le r \le 3/2$, $x + 2\varepsilon r > 0$: the only zero of x is at $r = 1$, where the correction is positive. On $[-3/2, 0]$, x has a positive minimum m_0, so positivity holds if $3\varepsilon < m_0$. Reflection proves positivity of the second coefficient, and the third is $z + 2\varepsilon > 0$. On the compact transition region $3/2 \le |r| \le 2$, all original coefficients have a common positive lower bound m_1. If M_1 bounds $|h'| + |h''|$, taking

also $\varepsilon < m_1/(M_1 + 1)$ preserves their positivity there. For $|r| \geq 2$, the background equals $f_*(t)^{1/3}\omega_X > 0$. This proves global strict positivity.

The correction is exact, so $\mathrm{d}\omega = \mathrm{d}\omega_0$. At $r = 0$, evenness of S_* gives $x'(0) = -1/2$, $y'(0) = 1/2$. Hence (11.44) gives

$$\mathrm{d}\omega_0 = (z_b(0) + 1/2)\,\mathrm{d}r \wedge (Q - P) \neq 0.$$

Indeed, if $\mathrm{d}r \wedge (Q - P) = 0$, wedging with P and $\mathrm{d}^{\mathrm{c}}r$ would contradict $P \wedge Q \wedge R > 0$. Thus the positive background is nonclosed.

Now put $\widetilde{u} = -F/2 - \rho$, $\widetilde{v} = F/2 - \rho$. Then $\omega + \mathrm{dd}^{\mathrm{c}}\widetilde{u} = A_0$, $\omega + \mathrm{dd}^{\mathrm{c}}\widetilde{v} = B_0$. Subtract their respective suprema to obtain u, v with common normalization zero. Their difference remains nonconstant because F is nonconstant.

11.9.4 Zeros and higher dimensions

It remains to check the global regularity of the prescribed density and the geometry of its zero levels. Taking a product with projective space then preserves the determinant equality and supplies examples in every higher dimension.

The function t is real analytic in every projective chart, being a quotient of real polynomials with positive denominator. Hence $f_*(t)$ is globally real analytic. Its two zero levels are regular because $t = \sigma(r)$, $\sigma' > 0$, and $\mathrm{d}r \neq 0$. Each level is connected: it is the quotient of the product of two three-spheres of fixed nonzero radii by simultaneous circle multiplication. It is compact, and a hypersurface has zero smooth volume. The divided differences in (11.59) show near the two levels that

$$f_*(\sigma(r)) = (r + 1)^2 g_-(r), \qquad f_*(\sigma(r)) = (r - 1)^2 g_+(r), \tag{11.66}$$

respectively, with smooth strictly positive transverse factors.

For $n \geq 4$, let θ be the Fubini–Study form on $Y = \mathbb{P}^{n-3}$, let $\pi_1 : X \times Y \to X$ and $\pi_2 : X \times Y \to Y$ be the projections, and set on $X \times Y$

$$\omega_n = \pi_1^*\omega + \pi_2^*\theta, \qquad \Omega_n = \pi_1^*\omega_X + \pi_2^*\theta, \qquad u_n = \pi_1^*u, \quad v_n = \pi_1^*v.$$

Only the term with three factors from X survives in an n-fold product. Consequently

$$(\omega_n + \mathrm{dd}^{\mathrm{c}}u_n)^n = \binom{n}{3}\pi_1^*(A_0^3) \wedge \pi_2^*(\theta^{n-3}) = f_*(\pi_1^*t)\Omega_n^n,$$

and the same holds for v_n. The solution ranks on the two product levels are $2+(n-3) = n - 1$. Restriction to a slice recovers $\mathrm{d}\omega \neq 0$, so ω_n is nonclosed. Normalization, distinctness, and real analyticity are preserved by pullback. □

Write Vol for volume with respect to ω_X^3, or Ω_n^n in dimension n. The quadratic zeros complete the sharpness assertion of Theorem 11.1.2. In finitely many tubular charts, the density is bounded above and below by positive multiples of the square

of a real transverse coordinate, while the smooth volume has a positive bounded Jacobian. Away from these charts the density is bounded below. Thus, with $s > 0$ a sublevel parameter and $\alpha > 0$ an integrability exponent,

$$\mathrm{Vol}\{f_* \le s\} \asymp \sqrt{s} \quad (s \downarrow 0), \qquad f_*^{-\alpha} \in L^1 \quad \Longleftrightarrow \quad \alpha < \tfrac{1}{2}. \tag{11.67}$$

Thus every weaker inverse-power hypothesis with positive exponent below $1/2$ admits smooth nonunique examples in every $n \ge 3$.

Remark 11.9.1 For this example, the positive-density set $\{f_*(t) > 0\}$ has three components: the middle band and the two outer regions. The solution forms have the same missing direction on each zero level, so $E_{\mathrm{unif}} = \{f_*(t) > 0\}$. The maximum and minimum of their potential difference occur on the two outer components, whose closures are disjoint, in accordance with Theorem 11.1.1. Although the closure of the middle component meets both outer closures, this adjacency does not propagate an extremal value through it.

Part III
Relative singularities and equations

Chapter 12
Envelopes and volumes of singularities

Let X be a connected compact complex manifold of dimension $n \geq 1$. Fix an auxiliary Hermitian metric ω and a smooth positive volume form $\mathrm{d}V$.

A singularity type can contain potentials with different Monge–Ampère masses. Its lower and upper volumes record this variation. Stretching a potential away from a reference gives a different test of its singularities: requiring a competitor for every finite stretch defines a relative envelope class without a mass condition.

Contact support supplies the measure estimates needed for domination. For a semipositive big background, a competitor in a smaller background then forces positive mass. The argument uses individual positivity and allows the infimum over bounded potentials to be zero. Conversely, the relative envelope class can contain potentials of infinite mass. The distinction between envelope conditions and mass bounds is essential for the prescribed-singularity equations of the next chapter.

12.1 Contact sets

An envelope's measure is concentrated where it meets its obstacle. For a smooth obstacle, exponential equations approximate the envelope and their densities vanish in the limit on every region with a strict gap. Bounded-product continuity carries this property through quasi-continuous approximation and truncation.

Proposition 12.1.1 *Let ω be Hermitian and $f \in C^\infty(X, \mathbb{R})$. For $\beta > 0$, denote by u_β the smooth strictly ω-plurisubharmonic solution of*

$$\omega_{u_\beta}^n = \mathrm{e}^{\beta(u_\beta - f)}\omega^n.$$

These solutions converge uniformly as $\beta \to +\infty$ to $p = P_\omega(f)$. Moreover,

$$\mathbf{1}_{\{p<f\}}\omega_p^n = 0.$$

Proof Existence is Theorem 9.6.1. Choose $A \geq 0$ with $\mathrm{dd^c} f \leq A\omega$. At a maximum of $u_\beta - f$,

$$\omega_{u_\beta} \leq \omega + \mathrm{dd^c} f \leq (1 + A)\omega.$$

The equation gives

$$u_\beta \leq f + \frac{C_f}{\beta}, \qquad C_f = n \log(1 + A).$$

Thus $u_\beta - C_f/\beta$ is an envelope competitor, and $u_\beta \leq p + C_f/\beta$.

The function p is continuous by Proposition 1.19.1. Choose smooth strictly ω-plurisubharmonic $v_j \downarrow p$ by Theorem 1.11.1; convergence is uniform by compactness. For any $\delta > 0$, choose j so that

$$p \leq v_j \leq p + \delta/2.$$

Then $v_j - \delta \leq f$. Positivity and compactness give a number $c_j > 0$ such that $\omega_{v_j}^n \geq c_j\omega^n$. Put $C_j = \max(0, -\log c_j)$. The function

$$w_{j,\beta} = v_j - \delta - C_j/\beta$$

satisfies

$$\omega_{w_{j,\beta}}^n = \omega_{v_j}^n \geq \mathrm{e}^{-C_j}\omega^n \geq \mathrm{e}^{\beta(w_{j,\beta} - f)}\omega^n.$$

Smooth exponential comparison gives $w_{j,\beta} \leq u_\beta$. Consequently

$$p - \delta - C_j/\beta \leq u_\beta \leq p + C_f/\beta.$$

First let β tend to infinity with j fixed, so no uniform lower bound on c_j is needed, and then let $\delta \downarrow 0$. This proves uniform convergence. Bounded Bedford–Taylor continuity gives weak convergence of the measures.

On a compact subset of the open set $\{p < f\}$, uniform convergence gives $u_\beta - f \leq -\epsilon$ for all large β, with some $\epsilon > 0$. There the measures have densities at most $\mathrm{e}^{-\beta\epsilon}$. Testing with continuous functions of compact support in that open set and passing to the bounded-product limit gives $\omega_p^n = 0$ there. □

Proposition 12.1.2 *If ω is Hermitian and f is bounded and lower semicontinuous, then $P_\omega(f)$ is bounded, lies below f quasi-everywhere, and its measure is concentrated on $\{P_\omega(f) = f\}$.*

Proof Suppose first that f is continuous. Choose smooth $f_j \to f$ uniformly. The envelope Lipschitz bound gives $p_j = P_\omega(f_j) \to p = P_\omega(f)$ uniformly. On every compact subset of $\{p < f\}$, the inequalities $p_j < f_j$ hold on a neighborhood for all large j. The preceding proposition and bounded-product convergence then give $\omega_p^n = 0$ on $\{p < f\}$.

For lower semicontinuous f, choose continuous $f_j \uparrow f$, with common upper and lower bounds. For example, for a fixed compatible distance d one may take

$$f_j(x) = \inf_{y \in X} \big(f(y) + j d(x, y)\big).$$

These functions are Lipschitz and increase to f: if convergence failed at x, approximate minimizers would approach x, contradicting lower semicontinuity.

The functions $p_j = P_\omega(f_j)$ increase, stay bounded, and have a bounded ω-plurisubharmonic regularized limit $p_\infty = (\lim_j p_j)^*$. A countable union of pluripolar exceptional sets shows that $p_\infty \le f$ quasi-everywhere. Thus $p_\infty \le P_\omega(f)$.

For fixed k, the set $\{p_\infty < f_k\}$ is open. If $j \ge k$, then $p_j \le p_\infty < f_k \le f_j$ there, so the continuous-obstacle case gives zero measure on that open set. Increasing bounded Bedford–Taylor convergence applies after adding a fixed smooth local correction whose Hessian dominates ω and expanding against the resulting smooth background. Together with the weak-limit inequality for open sets, it yields

$$\omega_{p_\infty}^n(\{p_\infty < f_k\}) = 0.$$

Taking the union over k gives zero measure on $\{p_\infty < f\}$, hence also on $\{p_\infty < P_\omega(f)\}$ outside a pluripolar set. By bounded domination, Theorem 10.2.2, $p_\infty \ge P_\omega(f)$. They are therefore equal, and the assertion follows. □

Theorem 12.1.1 *Let $\theta \ge 0$ be smooth and let f be bounded and quasi-continuous. Then $p = P_\theta(f)$ is bounded, $p \le f$ quasi-everywhere, and*

$$\mathbf{1}_{\{p<f\}} \mathrm{MA}_\theta(p) = 0.$$

Proof First suppose $\theta = \omega$ is Hermitian. Choose $f_j \downarrow f$ quasi-everywhere as in Lemma 1.8.2. The bounded functions $p_j = P_\omega(f_j)$ decrease to p. Indeed their limit is a competitor for p, since it lies below f quasi-everywhere; the reverse inequality follows from monotonicity.

For fixed k, put $H_k = \min\{1, (f - p_k)_+\}$. It is bounded and quasi-continuous. Where $H_k > 0$ and $j \ge k$, we have $p_j \le p_k < f \le f_j$, outside a pluripolar set. The lower semicontinuous case therefore gives

$$\int_X H_k \, \omega_{p_j}^n = 0.$$

The weighted bounded-product convergence in Section 1.8 gives $\int_X H_k \, \omega_p^n = 0$. The sets $\{p_k < f\}$ increase to $\{p < f\}$, proving support.

For general semipositive θ, set $\theta_j = \theta + j^{-1}\omega$ with ω Hermitian. The functions $p_j = P_{\theta_j}(f)$ have the common bounds $\inf_X f \le p_j \le \sup_X f$ and decrease.

Their limit is θ-plurisubharmonic by distributional convergence, lies below f quasi-everywhere, and hence equals $p = P_\theta(f)$. For each fixed k, the same coefficient $H_k = \min\{1, (f - p_k)_+\}$ gives

$$\int_X H_k \, \mathrm{MA}_{\theta_j}(p_j) = 0 \qquad (j \geq k).$$

All p_j are bounded potentials for the fixed Hermitian form $\theta + \omega$. Expand the products with θ_j against that fixed background. Bounded-product convergence and uniform capacity control pass the displayed identity to $\mathrm{MA}_\theta(p)$, including its quasi-continuous coefficient. Exhausting $\{p < f\}$ as before completes the proof. □

Theorem 12.1.2 *Let $\theta \geq 0$ be smooth. If f is quasi-continuous and $p = P_\theta(f)$ is a genuine potential, then $p \leq f$ quasi-everywhere and*

$$\mathbf{1}_{\{p<f\}} \mathrm{MA}_\theta(p) = 0.$$

The total extended mass is allowed to be infinite.

Proof The local envelope theorem and a countable Choquet family give $p \leq f$ outside a pluripolar set. Choose $M > \sup_X p$ and replace f by $\min(f, M)$. This does not change its envelope: p itself is a competitor for the smaller obstacle. It suffices to prove support on the contact set for this smaller obstacle, because $p < M$ and therefore it has the same contact set as the original one.

Set $f_j = \max(f, -j)$ and $p_j = P_\theta(f_j)$. The constants $-j$ are admissible, so p_j is bounded. Theorem 12.1.1 gives $p_j \leq f_j$ quasi-everywhere and

$$\mathbf{1}_{\{p_j<f_j\}} \mathrm{MA}_\theta(p_j) = 0.$$

The sequence p_j decreases and stays above p. Its limit is a genuine θ-plurisubharmonic function, lies below f quasi-everywhere, and is therefore a competitor for p. Hence $p_j \downarrow p$.

Fix an integer $k \geq 1$ and $C > 0$. Define the bounded nonnegative quasi-continuous function

$$H_{k,C} = \min\{1, (f - p_k)_+\} \, \min\{1, (p + C)_+\}.$$

Values on pluripolar exceptional sets are immaterial. Where this coefficient is positive, $p_k < f$ and $p > -C$. For $j \geq k$, we then have $p_j \leq p_k < f \leq f_j$ and $p_j \geq p > -C$. Locality and Theorem 12.1.1 imply

$$\int_X H_{k,C} \, \mathrm{MA}_\theta(\max(p_j, -C)) = 0.$$

The truncated potentials are uniformly bounded and decrease to $\max(p, -C)$. Their measures converge weakly by bounded Bedford–Taylor continuity. The uniform capacity bounds and weighted convergence in Section 1.8 also allow testing against $H_{k,C}$. Hence the displayed identity holds with p in place of p_j. By locality this says that the extended measure of p vanishes on $\{p_k < f\} \cap \{p > -C\}$. These sets

increase to $\{p < f\}$ off the pole set when first k and then C tend to infinity. Continuity from below of positive measures proves the assertion, even if the total extended mass is infinite. □

For a smooth big background, a strict singular barrier replaces the constant lower truncations. The same contact assertion for quasi-continuous obstacles finite almost everywhere is proved in Theorem 13.7.2, using local homogeneous replacements. Combining that result, or Theorem 12.1.2 in a semipositive background, with ordered contact controls rooftop measures. The strict inequality in the first term below assigns the common contact set to the second term.

Proposition 12.1.3 *Let θ be a smooth real $(1,1)$-form that is semipositive or big, and let $u, v \in \mathrm{PSH}(X,\theta)$. If $p = P_\theta(u,v)$ is genuine, then*

$$\mathrm{MA}_\theta(p) \le \mathbf{1}_{\{p=u<v\}}\mathrm{MA}_\theta(u) + \mathbf{1}_{\{p=v\}}\mathrm{MA}_\theta(v).$$

Proof The envelope is below both obstacles; their upper semicontinuity gives this inequality everywhere. Its measure is supported on $\{p = \min(u,v)\}$ by the contact theorem. Partition this set into $\{p = u < v\}$ and $\{p = v\}$. On each part apply Proposition 6.2.1. All measures vanish on the pluripolar pole sets, which can be discarded throughout. □

Proposition 12.1.4 *Let $\theta \ge 0$ be smooth. Every envelope fixed point φ satisfies*

$$\mathrm{MA}_\theta(\varphi) \le \mathbf{1}_{\{\varphi=0\}}\theta^n.$$

In particular, its mass is finite.

Proof Since $\varphi \le 0$, it is a competitor for $P_\theta(\varphi+1,0)$. This rooftop is also at most $P_\theta[\varphi] = \varphi$, so it equals φ. Apply Proposition 12.1.3. The contact set $\{\varphi = \varphi+1 < 0\}$ is contained in the pole set, which the extended measure does not charge. Only contact with zero contributes, giving the asserted measure bound. □

12.2 Bounded domination in a semipositive background

The Hermitian domination theorem used above extends to a smooth semipositive background θ when every bounded θ-plurisubharmonic potential has positive mass. For bounded potentials u, v satisfying the measure inequality below, contact support bounds $\sup_X P_\theta(bu - (b-1)v)$ from below as the stretching parameter $b > 1$ increases. A fixed gap between u and v would then force the normalized L^1 norms of these envelopes to diverge, contradicting compactness.

Call a smooth semipositive form θ *non-collapsing on bounded potentials* if

$$\int_X \mathrm{MA}_\theta(w) > 0 \quad \text{for every } w \in \mathrm{PSH}(X,\theta) \cap L^\infty(X).$$

Every Hermitian form has this property by Corollary 10.2.1. Bigness also supplies it, by an envelope argument.

Proposition 12.2.1 *If a smooth real background θ is big, every bounded $u \in \mathrm{PSH}(X,\theta)$ has strictly positive Monge–Ampère mass.*

Proof Choose $\rho \in \mathrm{PSH}(X,\theta)$ and $a > 0$ with $\theta_\rho \geq a\omega$. Normalize $\rho \leq 0$ and put

$$p = P_{a\omega}(u - \rho).$$

The obstacle is finite almost everywhere and bounded below, so Lemma 1.19.2 gives bounded p. The obstacle is quasi-continuous off a pluripolar set. Contact support, applied in the Hermitian background $a\omega$, gives $\mathrm{MA}_{a\omega}(p)$ carried by $D = \{p+\rho = u\}$, outside the pole set of ρ.

By Lemma 1.3.3, $p + \rho \leq u$ everywhere, and

$$\theta + \mathrm{dd^c}(p + \rho) = (a\omega + \mathrm{dd^c}p) + (\theta_\rho - a\omega) \geq a\omega + \mathrm{dd^c}p \geq 0.$$

On common finite truncation levels, positivity of the mixed products and ordered contact give

$$\mathbf{1}_D \mathrm{MA}_{a\omega}(p) \leq \mathbf{1}_D \mathrm{MA}_\theta(p + \rho) \leq \mathbf{1}_D \mathrm{MA}_\theta(u).$$

Exhausting these levels proves the same inequalities for the extended measures. Since p is bounded, its Hermitian Monge–Ampère mass is positive by Corollary 10.2.1. Integrating proves the assertion. □

Theorem 12.2.1 (Guedj–Lu) *Let $\theta \geq 0$ be non-collapsing on bounded potentials. Suppose $u, v \in \mathrm{PSH}(X,\theta) \cap L^\infty(X)$ and $0 \leq c < 1$. If*

$$\mathbf{1}_{\{u<v\}} \mathrm{MA}_\theta(u) \leq c\, \mathbf{1}_{\{u<v\}} \mathrm{MA}_\theta(v),$$

then $u \geq v$.

Proof If the conclusion failed, choose $a > 0$ such that $E = \{u < v - a\}$ is nonempty. This set has positive volume by Lemma 1.3.4. Fix C with $|u|, |v| \leq C$.

For $b > 1$ define the bounded envelope

$$w_b = P_\theta\big(bu - (b-1)v\big), \qquad D_b = \{w_b = bu - (b-1)v\}.$$

Constants bounded below are competitors, since $\theta \geq 0$. Bounded contact support gives $\mathrm{MA}_\theta(w_b)$ concentrated on D_b. The potential $b^{-1}w_b + (1 - b^{-1})v$ lies below u quasi-everywhere, hence everywhere by Lemma 1.3.3, and equals u on D_b outside a pluripolar set. Ordered contact and positivity of the mixed terms give

$$\mathbf{1}_{D_b}\left[b^{-n}\mathrm{MA}_\theta(w_b) + (1-b^{-1})^n\mathrm{MA}_\theta(v)\right] \le \mathbf{1}_{D_b}\mathrm{MA}_\theta(u).$$

Choose b sufficiently large that $(1-b^{-1})^n > c$. On $D_b \cap \{u < v\}$ this forces $\mathrm{MA}_\theta(w_b) = 0$. Non-collapsing gives positive total mass for w_b, so some mass is carried by $D_b \cap \{u \ge v\}$. At a point of that set outside the exceptional pluripolar set,

$$w_b = bu - (b-1)v = u + (b-1)(u-v) \ge -C.$$

Consequently $\sup_X w_b \ge -C$, uniformly as $b \to \infty$.

On the fixed set E, the obstacle bound gives $w_b \le C - (b-1)a$ quasi-everywhere. Hence

$$\int_X |w_b - \sup_X w_b|\,\mathrm{d}V \ge \big((b-1)a - 2C\big)\,\mathrm{vol}_{\mathrm{d}V}(E)$$

for all sufficiently large b. This contradicts the normalized L^1 bound of Theorem 1.4.1. □

Exponential comparison follows as in Corollary 10.2.2. For unbounded potentials, the stretched envelopes must also be kept in the required singularity class; this is the role of the relative classes introduced below.

12.3 Upper and lower volumes

Let θ be any smooth real $(1,1)$-form. Taking the infimum and supremum of mass over an ordinary singularity type gives two invariants with values in $[0,+\infty]$. They can differ because a bounded change of potential need not preserve mass.

Definition 12.3.1 For $u \in \mathrm{PSH}(X,\theta)$, define

$$\underline{\mathrm{vol}}_\theta(u) = \inf_{v\in\mathrm{PSH}(X,\theta),\ v\sim u}\int_X \mathrm{MA}_\theta(v),$$
$$\overline{\mathrm{vol}}_\theta(u) = \sup_{v\in\mathrm{PSH}(X,\theta),\ v\sim u}\int_X \mathrm{MA}_\theta(v).$$

These are the lower and upper volumes of the singularity type of u in the fixed background. We write $\underline{\mathrm{vol}}(\theta)$ and $\overline{\mathrm{vol}}(\theta)$ for their values at the potential V_θ with minimal singularities.

For semipositive θ, the type of zero consists of the bounded potentials. Extending the lower mass in Definition 6.5.1 to this background gives

$$\nu_-(\theta) = \underline{\mathrm{vol}}_\theta(0) = \inf_{v\in\mathrm{PSH}(X,\theta)\cap L^\infty(X)}\int_X \mathrm{MA}_\theta(v).$$

The condition $\nu_-(\theta) > 0$ is stronger than the non-collapsing property used in bounded domination: it requires one positive lower bound for every bounded potential. Comparison of singularity volumes under bounded mass is developed in Chapter 29.

The two volumes are invariant under bounded changes of u by definition. They are also invariant under the smooth gauge change $(\theta, u) \mapsto (\theta + \mathrm{dd^c} f, u - f)$, for $f \in C^\infty(X, \mathbb{R})$: the map $v \mapsto v - f$ identifies the competitor currents and their measures. Scaling gives

$$\underline{\mathrm{vol}}_{t\theta}(tu) = t^n \underline{\mathrm{vol}}_\theta(u), \qquad \overline{\mathrm{vol}}_{t\theta}(tu) = t^n \overline{\mathrm{vol}}_\theta(u) \quad (t > 0).$$

12.4 Relative envelope classes

Fix a reference $\varphi \in \mathrm{PSH}(X, \theta)$. Stretching a potential away from φ makes the obstacle more singular when the potential lies below the reference. Requiring a competitor after every finite stretch gives a relative class stable under convex combinations and finite rooftops.

For $u \in \mathrm{PSH}(X, \theta)$ and $b \geq 1$, define the competitor family

$$\mathcal{A}_b(u, \varphi) = \left\{ h \in \mathrm{PSH}(X, \theta) : b^{-1} h + (1 - b^{-1})\varphi \leq u \right\}.$$

For $b = 1$ the inequality means $h \leq u$. For $b > 1$ the displayed sum of quasi-plurisubharmonic functions is defined also at poles. This notation avoids assigning a value to $bu - (b-1)\varphi$ at a common pole. When the family is nonempty, its upper envelope is denoted

$$Q_b(u, \varphi) = P_\theta(bu - (b-1)\varphi).$$

The finite-sublevel bound in Section 1.19 makes this a genuine potential whenever the competitor family is nonempty. The upper-envelope theorem gives the convex inequality almost everywhere, and Lemma 1.3.3 makes it pointwise.

Definition 12.4.1 Define

$$\mathcal{E}_{\mathrm{env}}(X, \theta, \varphi) = \{u \in \mathrm{PSH}(X, \theta) : u \preceq \varphi,\ \mathcal{A}_b(u, \varphi) \neq \varnothing \text{ for every } b \geq 1\}.$$

Write $\mathcal{E}_\varphi = \mathcal{E}_{\mathrm{env}}(X, \theta, \varphi)$.

Proposition 12.4.1 *The class $\mathcal{E}_{\mathrm{env}}(X, \theta, \varphi)$ is convex, contains all potentials equivalent to φ, and is stable under bounded changes of its members. If u belongs to it, $v \in \mathrm{PSH}(X, \theta)$, $u \leq v$ and $v \preceq \varphi$, then v belongs to it. In particular, it is stable under taking maxima of two members.*

Proof If $u \geq \varphi - C$, use $h = \varphi - bC$ for each b. More generally, if $u' \geq u - C$ and $h \in \mathcal{A}_b(u, \varphi)$, then $h - bC \in \mathcal{A}_b(u', \varphi)$. The singularity-order condition persists under bounded changes. Convex combinations of two competitors give competitors for the same convex combination of their target functions. Finally a competitor for u is a competitor for any larger v; the assumed relation $v \preceq \varphi$ supplies the other defining condition. □

Lemma 12.4.1 *If $u \in \mathcal{E}_\varphi$ and $u \leq \varphi$, then $Q_b(u, \varphi) \in \mathcal{E}_\varphi$ for every $b \geq 1$. More generally, if $u, w \in \mathcal{E}_\varphi$ and $u \leq w$, then $Q_b(u, w) \in \mathcal{E}_\varphi$.*

Proof Fix $b, s \geq 1$ and put $h = Q_{bs}(u, \varphi)$. Its defining inequality gives, off the pole sets,

$$s^{-1}h + (1 - s^{-1})\varphi \leq bu - (b - 1)\varphi.$$

The left side is θ-plurisubharmonic, so it is at most $Q_b(u, \varphi)$. Hence $h \in \mathcal{A}_s(Q_b(u, \varphi), \varphi)$ for every s. Also $Q_b(u, \varphi) \leq u \leq \varphi$. The inequalities between convex combinations extend across the poles by Lemma 1.3.3, proving the first assertion.

For the second assertion choose a constant C such that $u - C \leq \varphi$ and $w - C \leq \varphi$. Translation reduces to $u, w \leq \varphi$. The obstacle for $Q_b(u, w)$ lies above that for $Q_b(u, \varphi)$ and below u, because $u \leq w \leq \varphi$. Thus

$$Q_b(u, \varphi) \leq Q_b(u, w) \leq u \leq \varphi.$$

Upward stability of the relative class, proved in Proposition 12.4.1, gives the conclusion. Restoring the translation does not change class membership. □

Proposition 12.4.2 *The rooftop of any nonempty finite collection of members of $\mathcal{E}_\varphi$ is genuine and belongs to $\mathcal{E}_\varphi$.*

Proof Let $u_1, \ldots, u_N \in \mathcal{E}_\varphi$, with $N \geq 1$. A common constant translation reduces to $u_i \leq \varphi$. By Lemma 12.4.1 and Proposition 12.4.1,

$$h = \frac{1}{N} \sum_{i=1}^{N} Q_N(u_i, \varphi) \in \mathcal{E}_\varphi.$$

Each summand is at most φ. For every i, its defining inequality therefore gives

$$h \leq N^{-1} Q_N(u_i, \varphi) + (1 - N^{-1})\varphi \leq u_i.$$

These convex-combination inequalities are meaningful also at common poles. Thus h is a competitor for $p = P_\theta(\min_i u_i)$, which is genuine and satisfies $h \leq p \leq u_1 \preceq \varphi$. Upward stability gives $p \in \mathcal{E}_\varphi$. Restoring the translation preserves this conclusion.□

The finiteness of the family is essential: every $\varphi - j$ belongs to $\mathcal{E}_\varphi$, whereas $P_\theta(\inf_{j \geq 1}(\varphi - j)) \equiv -\infty$. The common-envelope constructions for sequences in Chapter 30 require additional estimates.

Proposition 12.4.3 *If $u \in \mathcal{E}_\varphi$, then for every smooth real obstacle F,*

$$P_{\theta,F}[u] = P_{\theta,F}[\varphi].$$

In particular, $P_\theta[u] = P_\theta[\varphi]$.

Proof The relation $u \preceq \varphi$ gives one inequality. For $a \in (0, 1)$ choose $h_a \in \mathrm{PSH}(X, \theta)$ with $a\varphi + (1 - a)h_a \leq u$. Combining competitors below F gives

$$aP_{\theta,F}[\varphi] + (1 - a)P_{\theta,F}[h_a] \leq P_{\theta,F}[u].$$

Indeed, their convex combination lies below F and is more singular than u; taking the two upper envelopes preserves the inequality by local regularization. Write $m = \min_X F$ and $M = \max_X F$. Translation of competitors gives

$$P_\theta[h_a] + m \le P_{\theta,F}[h_a] \le P_\theta[h_a] + M.$$

Every $P_\theta[h_a]$ has supremum zero, since it lies above $h_a - \sup_X h_a$ and below zero. Normalized compactness therefore bounds the L^1 norms of $P_{\theta,F}[h_a]$ uniformly in a. Letting $a \uparrow 1$ in the first inequality gives $P_{\theta,F}[\varphi] \le P_{\theta,F}[u]$ almost everywhere. By Lemma 1.3.3 the inequality is pointwise. □

12.5 Model references and positive mass

Assume now that θ is semipositive and big. A competitor for u in background $a\theta$, with $0 < a < 1$, leaves the big form $(1-a)\theta$ available. An envelope in that remaining background has positive mass and transfers it to u on the contact set. To obtain such competitors throughout a relative class, take a reference satisfying

$$P_\theta[\varphi] = \varphi, \qquad P_{\varepsilon\theta}(\varphi) \not\equiv -\infty \quad \text{for some } 0 < \varepsilon < 1. \tag{12.1}$$

Fix one such ε. The difference envelope separates a member of $\mathcal{E}_\varphi$ from the reference, and stretching realizes that difference in an arbitrarily small background. Write $\mathcal{E}_0 = \mathcal{E}_{\mathrm{env}}(X, \theta, 0)$. Under bounded mass, (29.4) uses positive lower volume in place of the smaller-background condition; the comparison of these conditions belongs to that regime.

Lemma 12.5.1 *If $u \in \mathcal{E}_\varphi$ and $u \le \varphi$, then*

$$w = P_\theta(u - \varphi) \in \mathcal{E}_0, \qquad w \le 0, \qquad w + \varphi \le u.$$

Proof Since $\varphi \le 0$, the potential u is a competitor for the obstacle $u - \varphi$ away from its pole set. The envelope is genuine, bounded above by zero, and satisfies the asserted sum inequality by Lemma 1.3.3. For $b \ge 1$, let $h_b = Q_b(u, \varphi)$. Then

$$h_b \le bu - (b-1)\varphi \le b(u - \varphi).$$

Semipositivity makes $b^{-1}h_b$ a θ-plurisubharmonic competitor for w. Hence $h_b \le bw$, supplying a competitor for $P_\theta(bw)$ for every b. This is exactly $w \in \mathcal{E}_0$. □

Lemma 12.5.2 *If $0 < a < 1$ and $q = P_{a\theta}(u)$ is genuine for $u \in \mathrm{PSH}(X, \theta)$, then*

$$\int_X \mathrm{MA}_\theta(u) \ge (1-a)^n v_-(\theta).$$

The mass on the left is also strictly positive individually, even if $v_-(\theta) = 0$.

Proof The obstacle $u - q$ is nonnegative away from the poles. The bounded-below envelope lemma makes $p = P_{(1-a)\theta}(u - q)$ bounded. Its measure is carried by $D = \{p + q = u\}$, and $p + q \le u$ everywhere. Positivity of mixed products and ordered contact imply

$$\mathrm{MA}_{(1-a)\theta}(p) \le \mathbf{1}_D \mathrm{MA}_\theta(p + q) \le \mathbf{1}_D \mathrm{MA}_\theta(u).$$

All assertions are first restricted away from the poles. Integrating and writing $p = (1 - a)\widetilde{p}$ proves the lower bound from the definition of $v_-(\theta)$. The bounded potential p has strictly positive mass by Proposition 12.2.1, since $(1 - a)\theta$ is big. This proves the individual assertion as well. □

Theorem 12.5.1 *Every $u \in \mathcal{E}_\varphi$ admits a genuine $P_{(\varepsilon+\delta)\theta}(u)$ whenever $0 < \delta < 1-\varepsilon$. Consequently*

$$\int_X \mathrm{MA}_\theta(u) \ge (1 - \varepsilon)^n v_-(\theta), \qquad \int_X \mathrm{MA}_\theta(u) > 0.$$

The mass may be infinite.

Proof Translate u below φ and use $w \in \mathcal{E}_0$ from Lemma 12.5.1. The potential

$$P_{\varepsilon\theta}(\varphi) + \delta P_\theta(w/\delta) \le \varphi + w \le u$$

has background $(\varepsilon + \delta)\theta$. It supplies the asserted competitor. Apply Lemma 12.5.2 and let $\delta \downarrow 0$ for the uniform estimate. The individual conclusion follows by taking any one such δ. □

12.6 Infinite mass inside the envelope class

Return to the Hermitian background ω and reference zero. The infinite-mass series of Theorem 6.6.1 splits into a smooth finite sum and a tail whose negative curvature bound tends to zero. Multiplying a sufficiently late tail supplies a competitor for any prescribed stretch, placing the series in the relative envelope class.

Proposition 12.6.1 *If $v_+(\omega) = +\infty$, the potential U in Theorem 6.6.1 can be chosen so that, in addition,*

$$U \in \mathcal{E}_{\mathrm{env}}(X, \omega, 0), \qquad P_\omega[U] = 0.$$

The same conclusions hold for εU, $0 < \varepsilon \le 1$. Each of these potentials has infinite extended mass and trivial multiplier ideals at every positive weight.

Proof Use the smooth nonpositive ω-plurisubharmonic functions u_j, $j \geq 1$, of Theorem 6.6.1 and their convergent series $U = \sum_{j\geq 1} 2^{-j} u_j$. For each integer $N \geq 1$, write

$$U = A_N + V_N, \qquad A_N = \sum_{j\leq N} 2^{-j} u_j, \qquad V_N = \sum_{j>N} 2^{-j} u_j, \qquad \delta_N = 2^{-N}.$$

The head A_N is smooth, and the decreasing finite sums of the tail give $V_N \in \mathrm{PSH}(X, \delta_N \omega)$. The identity $U = A_N + V_N$ makes each tail genuine.

Fix $b > 1$ and choose N so large that $b\delta_N \leq 1$. Then

$$h_b = bV_N + \min_X (bA_N)$$

is ω-plurisubharmonic: its curvature is bounded below by $-b\delta_N\omega \geq -\omega$. Moreover $h_b \leq bU$. This gives a member of $\mathcal{A}_b(U, 0)$ for every b, proving the relative-class assertion. For εU, use the tails εV_N and choose N with $b\varepsilon\delta_N \leq 1$. By Proposition 12.4.3, the ordinary envelope of each of these potentials is $P_\omega[0] = 0$. Their infinite mass and trivial multiplier ideals are supplied by Corollary 6.6.1. □

The finite-mass bound of Proposition 12.1.4 concerns the envelope fixed point itself. This example shows that equality of ordinary envelopes, even together with trivial multiplier ideals at every positive weight, does not give finite mass to its representatives.

Corollary 12.6.1 *The following are equivalent for a Hermitian metric ω:*

(1) $v_+(\omega) < +\infty$;
(2) *every member of* $\mathcal{E}_{\mathrm{env}}(X, \omega, 0)$ *has finite extended mass;*
(3) *every* $u \in \mathrm{PSH}(X, \omega)$ *with* $P_\omega[u] = 0$ *has finite extended mass.*

Proof Bounded mass gives finiteness for all ω-plurisubharmonic functions by Proposition 6.5.2. If it fails, Proposition 12.6.1 contradicts each of the other two assertions. □

Chapter 13
Equations with prescribed singularities

Let X be a connected compact complex manifold of dimension $n \geq 1$. Fix a Hermitian metric ω and a smooth positive probability volume form $\mathrm{d}V$. Unqualified function norms use this volume. Write MA_θ for the extended measure $\mathrm{MA}_\theta^{\mathrm{ext}}$ and $s_+ = \max(s, 0)$ for a real s.

Solving a Monge–Ampère equation in a prescribed ordinary type requires a uniform bound for the difference from its reference potential. In a semipositive big background, smaller-background envelopes supply comparison, while subsolutions give relative estimates. Weighted addition then removes a possible deficit in the limiting measure without changing the ordinary type.

For an arbitrary smooth big background, a strict analytic barrier replaces the smaller-background construction. Resolution supplies approximating equations, and stability preserves minimal type and gives Hölder regularity off the poles. A smooth strictly positive density on the whole regular locus of this barrier then gives interior smoothness. Projective and local examples explain the roles of positivity and global control. Resolved logarithmic poles and exceptional minimal types give concrete applications in which the residual form and the Jacobian determine the equation.

Positive-exponent comparison gives uniqueness in the prescribed type; at exponent zero it determines only the scalar. Symmetry can identify a selected potential. For an individual L^1 density, bounded truncations and Gauduchon tests instead determine the largest coefficient admitting a subsolution. Passing from this threshold to an exact zero-exponent equation requires control of the potentials as well as their measures.

13.1 Domination with a smaller background

Assume in this section that θ is semipositive and big. A potential for $\delta\theta$, with $0 < \delta < 1$, leaves the semipositive big background $(1 - \delta)\theta$ for an auxiliary envelope. Each bounded potential in this remaining background has positive mass by Proposition 12.2.1. Bounded domination therefore supplies a lower bound for the

envelope, and a family of rooftops converts a strict measure inequality into pointwise order.

Lemma 13.1.1 *Suppose* $u, v \in \mathrm{PSH}(X, \theta)$*,* $v \preceq u$*, and* $\mathrm{MA}_\theta^{\mathrm{ext}}(u)(\{u < v\}) = 0$*. If* $0 < \delta < 1$ *and* $v_\delta = P_{\delta\theta}(v)$ *is genuine, then* $u \geq v_\delta$*.*

Proof Choose C with $v - C \leq u$. Since $v_\delta \leq v$, the obstacle $u - v_\delta$ is bounded below by $-C$ off the pole sets and is finite almost everywhere. Hence

$$p = P_{(1-\delta)\theta}(u - v_\delta)$$

is bounded by Lemma 1.19.2. Its measure is carried by $D = \{p + v_\delta = u\}$ by contact support. Also $p + v_\delta \leq u$ everywhere and

$$\theta + \mathrm{dd^c}(p + v_\delta) = ((1-\delta)\theta + \mathrm{dd^c} p) + (\delta\theta + \mathrm{dd^c} v_\delta).$$

Positivity of the two summands and ordered contact imply

$$\mathbf{1}_D \mathrm{MA}_{(1-\delta)\theta}(p) \leq \mathbf{1}_D \mathrm{MA}_\theta^{\mathrm{ext}}(u).$$

These measure inequalities can first be taken on common finite truncation levels and then exhausted. On $D \cap \{p < 0\}$ outside the pole sets, we have $u = p + v_\delta < v_\delta \leq v$. The hypothesis therefore gives $\mathrm{MA}_{(1-\delta)\theta}(p)(\{p < 0\}) = 0$. Bounded domination in the non-collapsing semipositive background $(1-\delta)\theta$, applied to p and zero, yields $p \geq 0$. Hence $u \geq p + v_\delta \geq v_\delta$. □

Theorem 13.1.1 *Let* $u \in \mathrm{PSH}(X, \theta)$ *satisfy* $P_{\delta\theta}(u) \not\equiv -\infty$ *for some* $0 < \delta < 1$*. Suppose* $v \in \mathrm{PSH}(X, \theta)$*,* $v \preceq u$*, and* $0 \leq c < 1$*. If*

$$\mathbf{1}_{\{u<v\}} \mathrm{MA}_\theta^{\mathrm{ext}}(u) \leq c\, \mathbf{1}_{\{u<v\}} \mathrm{MA}_\theta^{\mathrm{ext}}(v),$$

then $u \geq v$*.*

Proof Replace v by $\max(u, v)$. Plurifine locality preserves the displayed inequality on the strict set, and there is $C \geq 0$ with $u \leq v \leq u + C$. For $b > 1$ define

$$p_b = P_\theta(bu - (b-1)v), \qquad D_b = \{p_b = bu - (b-1)v\}.$$

The competitor $u - (b-1)C$ gives a genuine envelope, and $u - (b-1)C \leq p_b \leq u$. Contact support concentrates its measure on D_b. The potential $b^{-1}p_b + (1 - b^{-1})v$ lies below u and agrees with it on D_b. Ordered contact and positivity give

$$\mathbf{1}_{D_b}\left[b^{-n}\mathrm{MA}_\theta^{\mathrm{ext}}(p_b) + (1-b^{-1})^n \mathrm{MA}_\theta^{\mathrm{ext}}(v)\right] \leq \mathbf{1}_{D_b}\mathrm{MA}_\theta^{\mathrm{ext}}(u).$$

Choose b large enough that $(1 - b^{-1})^n > c$. On $D_b \cap \{u < v\}$ the hypothesis forces $\mathrm{MA}_\theta^{\mathrm{ext}}(p_b)$ to vanish. To justify this when masses are infinite, first restrict to the sets where $u, v, p_b > -k$. Every measure there is finite, being bounded by the mass of its fixed bounded truncation. Subtract the finite terms on each such set and then take the union over k. No subtraction of infinite masses occurs.

It follows that the measure of p_b is carried by $D_b \cap \{u = v\} \subset \{p_b = u\}$. Apply Lemma 13.1.1 to the pair (p_b, u); their singularities are equivalent and the smaller-background envelope of u is genuine. We obtain

$$p_b \geq P_{\delta\theta}(u)$$

for all sufficiently large b. If $u \not\geq v$, Lemma 1.3.3 shows that $\{u < v\}$ has positive smooth volume. Since both potentials are finite almost everywhere, some set $E = \{u < v - a\}$, $a > 0$, has positive volume with both functions finite there. On E, the obstacle inequality gives

$$P_{\delta\theta}(u) \leq p_b \leq v - ba$$

quasi-everywhere. Let b tend to infinity through integers. This forces the genuine quasi-plurisubharmonic function $P_{\delta\theta}(u)$ to be $-\infty$ on a set of positive volume, a contradiction. Thus $u \geq v$ off the pole sets, and the canonical representatives give the inequality everywhere. □

Corollary 13.1.1 *Under the smaller-background and singularity-order assumptions of the theorem, if $\lambda > 0$ and*

$$\mathrm{e}^{-\lambda u}\mathrm{MA}_\theta^{\mathrm{ext}}(u) \leq \mathrm{e}^{-\lambda v}\mathrm{MA}_\theta^{\mathrm{ext}}(v)$$

off the pole sets, then $u \geq v$. In particular the exponential equation has at most one solution in a fixed ordinary singularity type admitting a smaller-background envelope.

Proof Fix $t > 0$. On $\{u < v - t\}$, the assumed inequality gives $\mathrm{MA}_\theta^{\mathrm{ext}}(u) \leq \mathrm{e}^{-\lambda t}\mathrm{MA}_\theta^{\mathrm{ext}}(v - t)$. Apply the theorem and let $t \downarrow 0$. For two solutions of one equation in the stated type, apply the result in both directions. Weighted measures are interpreted off the common pluripolar exceptional sets; no value of an infinite weight on such a set is needed. □

13.2 Bounded equations in a semipositive big background

Keep θ semipositive and big. A convex function of a strict singular potential turns a Hermitian solution into a bounded subsolution for θ. Its derivative decays polynomially; multiplying the auxiliary density by the reciprocal weight compensates for this decay.

Fix $\rho \in \mathrm{PSH}(X, \theta)$ and $0 < a \leq 1$ such that

$$\theta + \mathrm{dd^c}\rho \geq a\omega, \qquad \rho \leq -1.$$

The construction uses only this curvature inequality.

Lemma 13.2.1 *Fix $p > 1$ and $L > 0$. There is $b > 0$ such that, for every $f \geq 0$ with $\|f\|_{L^p(\mathrm{d}V)} \leq L$, one can find $s \in \mathrm{PSH}(X, \theta)$ satisfying*

$$-1 \leq s \leq 0, \qquad \mathrm{MA}_\theta(s) \geq b f \, \mathrm{d}V.$$

The constant depends on the fixed geometry, ρ, a, p, L.

Proof Choose $1 < q < p$. Since a fixed quasi-plurisubharmonic function belongs to every finite L^r space, Hölder's inequality gives a uniform L^q bound for

$$g = (-\rho)^{2n} f.$$

By Theorem 10.1.1 there are $c > 0$ and a continuous ω-plurisubharmonic v, normalized by $\sup_X v = -1$, such that

$$\omega_v^n = c(1 + g) \, \mathrm{d}V.$$

The construction of that theorem gives a uniform upper bound for c: in its Gauduchon trace estimate the root integral of $1 + g$ is bounded below by the root integral of 1. The lower-density estimate gives $c\|1 + g\|_q \geq c_q > 0$. Thus c is also bounded below, and the potential estimate gives $-C \leq v \leq -1$ with a uniform C.

Put $h = \rho + av \leq -1$. Then $\theta_h \geq a\omega_v$. On $(-\infty, -1]$ define

$$\chi(t) = -t^{-1} - 1, \qquad \chi'(t) = t^{-2} \in (0, 1], \qquad \chi''(t) = -2t^{-3} > 0.$$

Set $s = \chi(h)$, with value -1 at poles of h. To justify both positivity and the product calculation, first use $h_k = \max(h, -k)$, $k \geq 1$. The bounded chain rule and $\theta \geq 0$ give

$$\theta + \mathrm{dd^c}\chi(h_k) = (1 - \chi'(h_k))\theta + \chi'(h_k)\theta_{h_k} + \chi''(h_k) \, \mathrm{d}h_k \wedge \mathrm{d^c}h_k \geq 0.$$

These bounded potentials decrease to s, so $s \in \mathrm{PSH}(X, \theta)$ and $-1 \leq s \leq 0$. On $\{h > -k\}$, locality and positivity give

$$\mathrm{MA}_\theta(s) \geq a^n |h|^{-2n} \omega_v^n \geq a^n c |h|^{-2n} (-\rho)^{2n} f \, \mathrm{d}V.$$

Exhaust the finite locus of h. Its complement is pluripolar and has zero $\mathrm{d}V$ measure. Since $|h| \leq (-\rho) + aC \leq (1 + aC)(-\rho)$, the assertion follows with any common lower bound for $a^n c(1 + aC)^{-2n}$. □

The lemma gives a subsolution for each prescribed density in an L^p bounded family. It is not a lower bound for the masses of all bounded θ-plurisubharmonic functions.

13.2.1 The exponential equation

Approximate θ from above by Hermitian forms. The bounded subsolution supplies a common lower barrier, while exponential comparison orders the resulting solutions. Their decreasing limit solves the equation in the original semipositive big background.

Theorem 13.2.1 *Let $0 \le f \in L^p(\mathrm{d}V)$, $p > 1$, be nonzero, and let $\lambda > 0$. There is a unique bounded $v \in \mathrm{PSH}(X, \theta)$ such that*

$$\mathrm{MA}_\theta(v) = \mathrm{e}^{\lambda v} f \, \mathrm{d}V.$$

For fixed λ, these solutions are uniformly bounded when $\|f\|_p$ is bounded above and $\int_X f^{1/n} \, \mathrm{d}V$ is bounded away from zero.

Proof For $0 < \varepsilon \le 1$, set $\theta_\varepsilon = \theta + \varepsilon\omega$. The Hermitian existence theorem gives a continuous solution v_ε with

$$\mathrm{MA}_{\theta_\varepsilon}(v_\varepsilon) = \mathrm{e}^{\lambda v_\varepsilon} f \, \mathrm{d}V.$$

Take s, b from Lemma 13.2.1, and put $w = s + \lambda^{-1} \log b$. Positivity of bounded products gives

$$\mathrm{MA}_{\theta_\varepsilon}(w) \ge \mathrm{MA}_\theta(s) \ge b f \, \mathrm{d}V \ge \mathrm{e}^{\lambda w} f \, \mathrm{d}V.$$

Bounded exponential comparison yields $w \le v_\varepsilon$. If $0 < \varepsilon' < \varepsilon$, then $v_{\varepsilon'}$ is also θ_ε-plurisubharmonic and

$$\mathrm{MA}_{\theta_\varepsilon}(v_{\varepsilon'}) \ge \mathrm{e}^{\lambda v_{\varepsilon'}} f \, \mathrm{d}V.$$

The same comparison gives $v_{\varepsilon'} \le v_\varepsilon$. Hence v_ε decreases to a bounded θ-plurisubharmonic v, between w and v_1.

To pass to the equation, use the fixed Hermitian background $\Omega = \theta + \omega$. Expand

$$(\theta_\varepsilon + \mathrm{dd^c} v_\varepsilon)^n = \sum_{k=0}^{n} \binom{n}{k} (\Omega + \mathrm{dd^c} v_\varepsilon)^k \wedge \big(-(1-\varepsilon)\omega\big)^{n-k}.$$

Bounded monotone continuity applies to every term, and the smooth coefficients converge. Dominated convergence applies to the right-hand sides. This proves the equation. Uniqueness follows from semipositive bounded domination and Proposition 12.2.1, by the exponential comparison argument.

For the uniform assertion the lower barrier w is uniform by the subsolution lemma. The upper barrier v_1 is an equation in the fixed Hermitian background Ω. The proof of Lemma 10.4.1 bounds its supremum using only an upper density norm and a positive lower root integral. Indeed those two bounds control, respectively, the Hölder term and the denominator in (10.8); smooth approximation retains these bounds. Thus v_1, and hence v, has a uniform upper bound. □

Local continuity of this bounded solution will follow from Theorem 13.5.1.

13.2.2 The normalized equation

For the normalized equation, the approximating potentials need not be monotone. Upper envelopes and lower rooftop envelopes supply monotone sequences around an L^1 cluster limit. Comparison identifies the two limits and hence the equation, while the scalar is controlled by the Gauduchon trace and the bounded subsolution.

Theorem 13.2.2 *For every nonzero $0 \le f \in L^p(\mathrm{d}V)$, $p > 1$, there are a bounded $\varphi \in \mathrm{PSH}(X, \theta)$ and a uniquely determined $c > 0$ such that*

$$\mathrm{MA}_\theta(\varphi) = c f \,\mathrm{d}V, \qquad \sup_X \varphi = 0.$$

The construction gives uniform bounds for $\|\varphi\|_\infty, c, c^{-1}$ when $\|f\|_p$ is bounded above and $\int_X f^{1/n}\,\mathrm{d}V$ is bounded away from zero. Uniqueness of the normalized potential is not asserted here.

Proof For integers $j \ge 1$, put $\omega_j = \theta + 2^{-j}\omega$. Choose continuous normalized solutions

$$\mathrm{MA}_{\omega_j}(\varphi_j) = c_j f \,\mathrm{d}V, \qquad \sup_X \varphi_j = 0.$$

First we bound c_j. Fix a Hermitian form γ_0, chosen Gauduchon when $n \ge 2$; in dimension one, the same calculation is Stokes. The smooth approximation proof of Theorem 10.1.1 and the Gauduchon trace give a common upper bound: the numerator $\int_X \omega_j \wedge \gamma_0^{n-1}$ is bounded above by $\int_X (\theta + \omega) \wedge \gamma_0^{n-1}$, and the root integral of f is positive and fixed. This uses the smooth approximating equations before passage to their continuous limits.

For a lower bound take s, b from Lemma 13.2.1. If $c_j < b$, then

$$\mathrm{MA}_{\omega_j}(\varphi_j) \le (c_j/b)\mathrm{MA}_{\omega_j}(s + A)$$

for every constant A. Strict domination would give $\varphi_j \ge s + A$ for every A, a contradiction. Thus $c_j \ge b$. In fact the constants decrease with j: if $k > j$, positivity gives $\mathrm{MA}_{\omega_j}(\varphi_k) \ge c_k f \,\mathrm{d}V$, and the same contradiction excludes $c_j < c_k$. Hence $c_j \downarrow c > 0$.

We next bound the potentials uniformly. All φ_j are normalized Ω-plurisubharmonic for $\Omega = \theta + \omega$. Choose $\gamma > 0$ small and $1 < q < p$ such that

$$g_j = \mathrm{e}^{-\gamma\varphi_j} f$$

is bounded in L^q, using uniform exponential integrability and Hölder's inequality. Since $\varphi_j \le 0$, we have $g_j \ge f$. The preceding exponential theorem therefore gives uniformly bounded $w_j \in \mathrm{PSH}(X, \theta)$ with

$$\mathrm{MA}_\theta(w_j) = \mathrm{e}^{\gamma w_j} g_j \,\mathrm{d}V.$$

Choose $C_0 \ge \max\{1, \sup_j c_j\}$ and set $z_j = w_j - \gamma^{-1}\log C_0$. Then

$$\mathrm{MA}_{\omega_j}(z_j) \geq \mathrm{e}^{\gamma z_j} c_j g_j \,\mathrm{d}V, \qquad \mathrm{MA}_{\omega_j}(\varphi_j) = \mathrm{e}^{\gamma \varphi_j} c_j g_j \,\mathrm{d}V.$$

Exponential comparison gives $z_j \leq \varphi_j \leq 0$. This is the desired common bounded range.

Extract a subsequence converging in L^1 and almost everywhere to the canonical θ-plurisubharmonic representative φ. We keep the indices j; the positive coefficients in ω_j still decrease to zero. Define

$$\Phi_j^+ = \left(\sup_{\ell \geq j} \varphi_\ell\right)^*, \qquad h_j = \inf_{\ell \geq j} \varphi_\ell, \qquad p_j = P_\theta(h_j).$$

All potentials here have a common bounded range. The upper envelopes are ω_j-plurisubharmonic and decrease to φ; the usual upper-envelope representative agrees almost everywhere with the L^1 limit. This follows also from the local Hartogs lemma applied to the bounded tails.

For $\ell \geq j$, positivity gives $\mathrm{MA}_{\omega_j}(\varphi_\ell) \geq c_\ell f \,\mathrm{d}V \geq c f \,\mathrm{d}V$. By ordered contact, Lemma 1.7.1, the finite maxima retain this lower bound. Bounded increasing continuity, followed by the decreasing limit in j and the smooth background expansion, gives

$$\mathrm{MA}_{\omega_j}(\Phi_j^+) \geq c f \,\mathrm{d}V, \qquad \mathrm{MA}_\theta(\varphi) \geq c f \,\mathrm{d}V. \tag{13.1}$$

We obtain the reverse inequality by lower envelopes, retaining a useful exponential weight. For $N \geq j$ put

$$p_{j,N} = P_\theta\left(\min_{j \leq \ell \leq N} \varphi_\ell\right).$$

The obstacle is continuous and bounded, so the measure of $p_{j,N}$ is carried by its contact set. Partition that set into disjoint Borel pieces by choosing the first index ℓ at which $p_{j,N} = \varphi_\ell$. On this piece, ordered contact in background ω_ℓ and positivity give

$$\mathrm{MA}_\theta(p_{j,N}) \leq \mathrm{MA}_{\omega_\ell}(p_{j,N}) \leq c_\ell f \,\mathrm{d}V.$$

There $p_{j,N} = \varphi_\ell \geq h_j$. Summing the restrictions therefore yields the global inequality

$$\mathrm{MA}_\theta(p_{j,N}) \leq \mathrm{e}^{p_{j,N} - h_j} c_j f \,\mathrm{d}V.$$

As $N \to \infty$, the envelopes decrease to p_j: the limit lies below each tail obstacle quasi-everywhere, hence is a competitor for $P_\theta(h_j)$, and the reverse inequality is immediate. Bounded monotone continuity and dominated convergence give

$$\mathrm{MA}_\theta(p_j) \leq \mathrm{e}^{p_j - h_j} c_j f \,\mathrm{d}V.$$

The functions p_j increase. Let $\varphi' = (\lim_j p_j)^*$; since $h_j \to \varphi$ almost everywhere, we have $\varphi' \leq \varphi$ by canonical representatives. Passing to the increasing limit gives

$$\mathrm{MA}_\theta(\varphi') \leq \mathrm{e}^{\varphi' - \varphi} c f \,\mathrm{d}V \leq \mathrm{e}^{\varphi' - \varphi} \mathrm{MA}_\theta(\varphi).$$

Bounded exponential comparison forces $\varphi' \geq \varphi$. Thus they coincide, and this last inequality, together with (13.1), proves the equation. Subtracting $\sup_X \varphi$ gives the required normalization if needed.

All bounds used above depend only on the two stated density bounds and the fixed data; this proves the uniform assertion for the constructed solution. Finally, two different positive constants for the same f would contradict strict domination after arbitrary translation of one bounded solution, exactly as in the lower-bound argument. This proves uniqueness of c. □

For $\lambda > 0$, let v_λ be the bounded solution of $\mathrm{MA}_\theta(v_\lambda) = \mathrm{e}^{\lambda v_\lambda} f \, \mathrm{d}V$, and let (φ, c) be a normalized solution furnished by Theorem 13.2.2. The same two exponential barriers as in Proposition 10.5.1 give

$$\|\lambda v_\lambda - \log c\|_\infty \leq \lambda \operatorname{osc}_X \varphi$$

for these bounded degenerate solutions.

13.3 A quantitative stability estimate

A small L^1 excess of a comparison potential over a bounded solution gives a small uniform excess. To prove this, solve an auxiliary subsolution problem for the density restricted to a strict superlevel set. Domination bounds the height of that set, and Hölder's inequality converts its volume into the desired power of the L^1 error.

Keep θ semipositive and big, and fix $p > 1$.

Theorem 13.3.1 *Fix $L, C_0 > 0$. Let $u \in \mathrm{PSH}(X, \theta)$ satisfy*

$$-C_0 \leq u \leq 0, \qquad \mathrm{MA}_\theta(u) = f \, \mathrm{d}V, \qquad 0 \leq f \in L^p(\mathrm{d}V), \quad \|f\|_p \leq L.$$

For every

$$0 < \eta < \frac{p-1}{np + p - 1}$$

there is a constant C, depending only on the fixed data and η, such that every $v \in \mathrm{PSH}(X, \theta)$ with $v \leq 0$ satisfies

$$\sup_X (v - u)_+ \leq C \left(\int_X (v - u)_+ \, \mathrm{d}V \right)^\eta .$$

The comparison potential v may be unbounded.

Proof Replace v by $\max(v, u)$. This leaves both sides unchanged and reduces to $-C_0 \leq u \leq v \leq 0$. Put $I = \int_X (v - u) \, \mathrm{d}V$. If $I = 0$, the canonical potentials agree everywhere and the conclusion follows. We may assume $I > 0$.

Choose $q \in (1, p)$, to be specified at the end. For $\delta > 0$ put

$$E_\delta = \{u < v - \delta\}, \qquad g_\delta = \mathbf{1}_{E_\delta} f, \qquad m_\delta = \|g_\delta\|_q.$$

If $m_\delta > 0$, apply Lemma 13.2.1 to g_δ/m_δ. There are $h \in \mathrm{PSH}(X, \theta)$ and $b > 0$ such that

$$-1 \le h \le 0, \qquad \mathrm{MA}_\theta(h) \ge b\, g_\delta/m_\delta \,\mathrm{d}V.$$

The constant b is independent of δ, f, u, v, since the auxiliary density has L^q norm one.

First suppose $0 < m_\delta < b/2$, and take

$$t = (2m_\delta/b)^{1/n} \in (0, 1), \qquad w = (1 - t)v + th - \delta - C_0 t.$$

Because $h - v \le C_0$, we have $w \le v - \delta$. Hence $\{u < w\} \subset E_\delta$. Positivity of bounded mixed products gives

$$\mathrm{MA}_\theta(w) \ge t^n \mathrm{MA}_\theta(h) \ge 2g_\delta \,\mathrm{d}V.$$

On $\{u < w\}$ the last measure is $2\mathrm{MA}_\theta(u)$. Bounded strict domination in the semipositive big background, proved in Theorem 12.2.1, implies $u \ge w$. Consequently

$$v - u \le v - w = \delta + C_0 t + t(v - h) \le \delta + (C_0 + 1)t.$$

If $m_\delta = 0$, the measure of u vanishes on E_δ. Apply the same strict domination theorem with coefficient zero to u and $v - \delta$; it gives $v - u \le \delta$. If $m_\delta \ge b/2$, the bound $v - u \le C_0$ proves the following inequality after increasing its constant. Thus in every case,

$$\sup_X (v - u) \le \delta + C_1 m_\delta^{1/n}. \tag{13.2}$$

Hölder's inequality and Chebyshev's inequality give

$$m_\delta \le \|f\|_p \,\mathrm{d}V(E_\delta)^{1/q - 1/p} \le L(I/\delta)^{(p-q)/(pq)}.$$

Set $a = (p - q)/(npq) > 0$. For $0 < I \le 1$, choose $\delta = I^{a/(1+a)}$. Substitution in (13.2) yields

$$\sup_X (v - u) \le C_2 I^{\eta_q}, \qquad \eta_q = \frac{p - q}{npq + p - q}.$$

As $q \downarrow 1$, the number η_q tends to $(p - 1)/(np + p - 1)$. For the prescribed η, choose $q > 1$ close enough that $\eta_q > \eta$. Then $I^{\eta_q} \le I^\eta$ on $(0, 1]$. When $I > 1$, the bound by C_0 gives the same conclusion after increasing C. This proves the theorem. □

Corollary 13.3.1 *Fix $C_0 > 0$. Let $u_j, u \in \mathrm{PSH}(X, \theta)$ satisfy $-C_0 \le u_j, u \le 0$ and*

$$\mathrm{MA}_\theta(u_j) = f_j \, \mathrm{d}V, \qquad \mathrm{MA}_\theta(u) = f \, \mathrm{d}V, \qquad \sup_j \|f_j\|_p + \|f\|_p < +\infty.$$

If $u_j \to u$ in L^1, then $u_j \to u$ uniformly. More precisely, for every exponent in the theorem,

$$\|u_j - u\|_\infty \le C \|u_j - u\|^\eta_{L^1(\mathrm{d}V)}.$$

Proof Apply the theorem with solution u and comparison u_j, and then with solution u_j and comparison u. The constants are uniform by the common range and density bound. The larger of the two suprema is the uniform norm, and each positive-part integral is at most the full L^1 difference. □

In an existence argument, the equation and density bound for the limit must be established before applying this corollary.

13.4 Hölder continuity from bounded stability

Apply bounded stability to the Chern-connection regularizations of a solution. Their L^1 error controls the pointwise smoothing error; a gradient estimate for the smooth averages then gives a Hölder modulus. We use the operators R_t and transforms $\Phi_{c,\delta}$ from Lemmas 1.21.1 and 1.21.2. The final interpolation step yields the explicit exponent below.

Theorem 13.4.1 *Let $p > 1$ and let $u \in \mathrm{PSH}(X, \omega) \cap L^\infty(X)$ solve*

$$\mathrm{MA}_\omega(u) = f \, \mathrm{d}V, \qquad 0 \le f \in L^p(\mathrm{d}V).$$

Then u is Hölder continuous with every exponent

$$0 < \beta < \frac{2(p-1)}{np + 3(p-1)}.$$

For a fixed bound $-C_0 \le u \le 0$ and a fixed bound for $\|f\|_p$, the Hölder constants are uniform. In particular the continuous solutions constructed in Theorem 10.1.1 have this regularity.

Proof Normalize by a constant so that $-C_0 \le u \le 0$. Choose

$$0 < \eta < \frac{p-1}{np + p - 1}, \qquad \alpha = 2\eta < 1.$$

For small $\delta > 0$ set

$$c = \delta^\alpha, \qquad e = Ac + 2K\delta, \qquad r = -C_0 - 1, \qquad v_\delta = \frac{\Phi_{c,\delta} + er}{1 + e}.$$

The curvature estimate gives $v_\delta \in \mathrm{PSH}(X, \omega)$. Evaluation of the infimum at $t = \delta$ gives $\Phi_{c,\delta} \le R_\delta u \le 0$. Its lower bound is

$$\Phi_{c,\delta} \ge u - K(\delta^2 + \delta),$$

since $R_t u + Kt^2 \ge u$, $Kt \ge 0$ and $-c\log(t/\delta) \ge 0$. For δ small this is at least r. Thus

$$v_\delta \le \Phi_{c,\delta} \le R_\delta u \le 0.$$

Apply Theorem 13.3.1 and the integral estimate:

$$\sup_X (v_\delta - u)_+ \le C\left(\int_X (v_\delta - u)_+ \,\mathrm{d}V\right)^\eta \le C\|R_\delta u - u\|_1^\eta \le C\delta^\alpha.$$

As $e \le C\delta^\alpha$ and $u - r \le C_0 + 1$, it follows that

$$\Phi_{c,\delta} - u = (1 + e)(v_\delta - u) + e(u - r) \le C\delta^\alpha. \tag{13.3}$$

Fix z and a minimizing radius $t = t(z) \in (0, \delta]$. The lower estimate at this radius and (13.3) give

$$-c\log(t/\delta) \le \Phi_{c,\delta}(z) - u(z) + K(\delta^2 + \delta) \le C\delta^\alpha = Cc.$$

Thus $t(z) \ge a\delta$ with a constant $a = \mathrm{e}^{-C} > 0$ independent of z and δ. Monotonicity of $R_s u + Ks^2$ yields

$$\begin{aligned} R_{a\delta}u + K(a\delta)^2 &\le R_t u + Kt^2 \\ &= \Phi_{c,\delta} + K\delta^2 + K\delta - Kt + c\log(t/\delta) \\ &\le \Phi_{c,\delta} + K\delta^2 + K\delta. \end{aligned}$$

Together with (13.3) and the lower bound $R_s u - u \ge -Ks^2$, this proves, after setting $s = a\delta$,

$$\|R_s u - u\|_\infty \le Cs^\alpha$$

for every sufficiently small $s > 0$.

For points x, y at distance d small enough, the derivative estimate gives

$$|u(x) - u(y)| \le 2Cs^\alpha + Cd/s.$$

Choose $s = d^{1/(1+\alpha)}$. Both terms are then bounded by a constant times $d^{\alpha/(1+\alpha)}$. Larger distances are controlled by the oscillation bound. This proves Hölder continuity with exponent $\alpha/(1 + \alpha)$. Letting η approach its permitted upper endpoint gives

every β in the theorem; any smaller exponent follows on compact X by changing the constant. □

13.5 Local continuity in a semipositive big background

Return to a semipositive big background θ. A strict singular barrier repairs the curvature error of regularization where the background itself is degenerate. Bounded stability then gives continuity of a bounded solution on every open set where the barrier is locally bounded.

Choose $\rho \in \mathrm{PSH}(X, \theta)$ with $\theta + \mathrm{dd^c}\rho \geq \kappa\omega$ for some $\kappa > 0$, and let U be such an open set. Use the averages $R_t u$, transforms $\Phi_{c,\delta}$ and curvature modulus from Section 1.21.1 and Lemma 1.21.3.

Theorem 13.5.1 *Let $p > 1$ and $u \in \mathrm{PSH}(X, \theta) \cap L^\infty(X)$ satisfy*

$$\mathrm{MA}_\theta(u) = f\,\mathrm{d}V, \qquad 0 \leq f \in L^p(\mathrm{d}V).$$

Then u is continuous on every open set where the fixed strict barrier ρ is locally bounded.

Proof Normalize u so that $-C_0 \leq u \leq 0$, and subtract a constant from ρ so that $\rho \leq -C_0 - 1$. Choose

$$0 < \eta < \frac{p-1}{np+p-1}, \qquad \alpha = 2\eta < 1,$$

and set, for small $\delta > 0$,

$$c = \delta^\alpha + \epsilon(\delta), \qquad e = Ac + \epsilon(\delta), \qquad s = e/\kappa, \qquad v_\delta = \frac{\Phi_{c,\delta} + s\rho}{1+s}.$$

The curvature bounds give $v_\delta \in \mathrm{PSH}(X, \theta)$, since $s\kappa = e$. Moreover

$$\rho \leq -C_0 - 1 \leq \Phi_{c,\delta} \leq R_\delta u \leq 0$$

for small δ. The lower bound follows from $\Phi_{c,\delta} \geq u - K(\delta^2 + \delta)$ as in the strictly positive case. Hence $v_\delta \leq \Phi_{c,\delta} \leq R_\delta u \leq 0$.

Apply Theorem 13.3.1, whose comparison potential may be unbounded. The preceding smoothing estimate gives

$$\sup_X (v_\delta - u)_+ \leq C\left(\int_X (v_\delta - u)_+\,\mathrm{d}V\right)^\eta \leq C\delta^\alpha.$$

Fix M and restrict to points where $\rho \geq -M$. As $s \leq Cc$ and $c \to 0$, the exact identity

$$\Phi_{c,\delta} - u = (1+s)(v_\delta - u) + s(u - \rho)$$

gives $\Phi_{c,\delta} - u \le C_M c$ there.

At a minimizing radius t, the same lower estimate in its defining expression yields

$$-c\log(t/\delta) \le \Phi_{c,\delta} - u + K(\delta^2 + \delta) \le C_M c.$$

Thus $t \ge a_M\delta$ with $a_M > 0$ independent of the point and δ. Monotonicity of smoothing now gives

$$R_{a_M\delta}u + K(a_M\delta)^2 \le R_t u + Kt^2 \le \Phi_{c,\delta} + K\delta^2 + K\delta.$$

Its upper bound and the monotone lower bound imply

$$\sup_{\{\rho\ge -M\}} |R_{a_M\delta}u - u| \le C_M\big(\delta^\alpha + \epsilon(\delta)\big) \longrightarrow 0.$$

Every compact subset of U has a lower bound for ρ. The smooth functions on the left therefore converge uniformly to u on that compact subset. This proves continuity on U. □

The proof also gives a modulus on each fixed compact subset: combine its uniform smoothing error with the derivative bound C/t. A Hölder exponent would require a proved power bound on the curvature error modulus. For a smooth big background, Lemma 1.21.4 supplies a linear curvature error by refining the classical expansion. Together with stability above a strict barrier, it gives local Hölder continuity even for minimal-singularity solutions in Theorem 13.16.1. Neither argument asserts continuity across the pole set.

13.6 Homogeneous replacement on a coordinate ball

To prove that an envelope measure is supported on the contact set, we need a local replacement with zero Monge–Ampère measure and prescribed boundary values. We construct it by letting a positive density tend to zero in the smooth Dirichlet problem, then approximate continuous boundary values.

Work on $B = B_R = \{|z| < R\} \subset \mathbb{C}^n$, put $\beta = \mathrm{dd^c}|z|^2$, and let θ be any smooth real $(1,1)$-form near $\overline{B}$. Distances and the Laplacian are Euclidean.

Lemma 13.6.1 *For $\varphi \in C^\infty(\partial B)$ and $\varepsilon > 0$, there is a unique smooth function v_ε on $\overline{B}$ such that*

$$\theta + \mathrm{dd^c}v_\varepsilon > 0, \qquad (\theta + \mathrm{dd^c}v_\varepsilon)^n = \varepsilon\beta^n, \qquad v_\varepsilon|_{\partial B} = \varphi.$$

Proof Extend φ smoothly to $\widetilde{\varphi}$ on $\overline{B}$. For sufficiently large A, the function $\underline{v} = \widetilde{\varphi} + A(|z|^2 - R^2)$ is a smooth strict subsolution with the prescribed boundary values: its background form is positive and its determinant is at least $\varepsilon\beta^n$. For $n \ge 2$ the classical smooth Dirichlet theorem of Guan–Li applies to the compact ball, with arbitrary smooth background $\chi = \theta$ and positive constant right-hand side. Their

operator is $(\mathrm{i}/2)\partial\bar{\partial}$, so our potential is π times theirs. No closedness condition from their separate compact-manifold theorems is used here. For $n = 1$ this is the ordinary Poisson Dirichlet problem. Uniqueness follows from the maximum principle applied to the difference: the difference of the two log determinants is a smooth uniformly elliptic linear operator on that difference. □

We will also use smooth comparison on domains inside the ball. If a, b are smooth and strictly admissible, $(\theta + \mathrm{dd^c}a)^n \geq (\theta + \mathrm{dd^c}b)^n$, and $a \leq b$ on the boundary, then $a \leq b$. Integrating the derivative of the log determinant along the segment between their positive matrices gives a linear elliptic inequality for $a - b$. The maximum principle proves the assertion; a boundary need not be smooth for this maximum argument.

Theorem 13.6.1 *For every $\varphi \in C(\partial B)$ there exists a continuous θ-plurisubharmonic function H_φ on $\overline{B}$ such that*

$$(\theta + \mathrm{dd^c}H_\varphi)^n = 0 \quad \text{on } B, \qquad H_\varphi|_{\partial B} = \varphi.$$

It dominates every continuous θ-plurisubharmonic function p on $\overline{B}$ with $p|_{\partial B} \leq \varphi$. For $\varphi, \psi \in C(\partial B)$, the construction satisfies

$$\|H_\varphi - H_\psi\|_{C(\overline{B})} \leq \|\varphi - \psi\|_{C(\partial B)}.$$

For smooth boundary data the constructed function is Lipschitz on $\overline{B}$.

Proof We first take φ smooth and $0 < \varepsilon \leq 1$. Fix the strict subsolution above for $\varepsilon = 1$; it is below every v_ε by smooth comparison. Choose a smooth function q with $q = \varphi$ on the boundary and $\Delta q = -C$, where C is large enough that $\Delta v_\varepsilon \geq -C$ follows from $\theta + \mathrm{dd^c}v_\varepsilon \geq 0$. The scalar maximum principle gives $v_\varepsilon \leq q$. Both barriers are smooth and equal $\widetilde{\varphi}$ on the boundary. Consequently

$$|v_\varepsilon(z) - \widetilde{\varphi}(z)| \leq C_1 \operatorname{dist}(z, \partial B), \tag{13.4}$$

with C_1 independent of ε.

Here is a uniform modulus inside the ball. Let $h \in \mathbb{C}^n$ be small, and set $D_h = \{z \in B : z + h \in B\}$. There is $C_2 > 0$, independent of the translation h, such that smoothness of θ gives

$$\theta(z) - \theta(z + h) + C_2|h|\beta \geq 0.$$

For a constant $C_3 > 0$ to be fixed, define on D_h

$$w(z) = v_\varepsilon(z + h) + C_2|h|(|z|^2 - R^2) - C_3|h|.$$

Then $\theta + \mathrm{dd^c}w \geq \theta(z + h) + \mathrm{dd^c}v_\varepsilon(z + h) > 0$ and its determinant is at least $\varepsilon\beta^n$; the Euclidean volume form is translation invariant. On ∂D_h, either z or $z + h$ is on ∂B, and the other point has boundary distance at most $|h|$. Estimate (13.4) and the

Lipschitz bound for $\widetilde{\varphi}$ therefore give $w \le v_\varepsilon$ if C_3 is fixed sufficiently large. Smooth comparison yields this inequality throughout D_h. Reversing the translation gives

$$|v_\varepsilon(z+h) - v_\varepsilon(z)| \le C_4|h|.$$

The constants are independent of ε. The same bound for arbitrary pairs follows by subdividing their segment inside the convex ball, and extends to $\overline{B}$ because each v_ε is continuous there.

Smooth comparison also shows that v_ε increases as ε decreases. The barriers and common Lipschitz bound imply that its limit H_φ is Lipschitz and that convergence is uniform on $\overline{B}$. The current inequality passes to the limit. Continuity of the local bounded products under uniform convergence gives $(\theta + \mathrm{dd^c} H_\varphi)^n = 0$. For smooth boundary values φ, ψ, denote the positive-density solutions by $v_{\varepsilon,\varphi}$ and $v_{\varepsilon,\psi}$. Smooth comparison gives $\|v_{\varepsilon,\varphi} - v_{\varepsilon,\psi}\|_\infty \le \|\varphi - \psi\|_\infty$; passing to the limit proves the asserted contraction for such data.

For continuous φ, choose smooth $\varphi_j \to \varphi$ uniformly on ∂B. The contraction makes H_{φ_j} uniformly Cauchy. Its limit is continuous, has the prescribed boundary values, and is θ-plurisubharmonic. Uniform convergence of the bounded products again proves the homogeneous equation. Passing to the limit also proves the contraction for all continuous boundary data.

It remains to prove the domination property; it does not follow from a comparison of total masses. First take smooth φ and continuous p as in the statement. For $\delta > 0$, put $p_\delta = p + \delta(|z|^2 - R^2)$, so $\theta + \mathrm{dd^c} p_\delta \ge \delta\beta$. By Lemma 1.3.9, any C^2 function a touching p_δ from above at an interior point z_0 satisfies

$$\theta(z_0) + \mathrm{dd^c} a(z_0) \ge \delta\beta.$$

If $p_\delta - v_\varepsilon$ had a positive maximum, it would occur inside B, since the boundary difference is nonpositive. The smooth function v_ε plus the maximum would be an upper test there. Its determinant would thus be at least $\delta^n\beta^n$, contradicting its equation when $\varepsilon < \delta^n$. Hence $p_\delta \le v_\varepsilon$. Let ε decrease to zero, then let δ decrease to zero, obtaining $p \le H_\varphi$.

Finally, for continuous φ, choose the smooth approximants above and set $a_j = \|\varphi_j - \varphi\|_\infty$. The boundary values of $p - a_j$ are at most φ_j. The smooth-data result gives $p - a_j \le H_{\varphi_j}$; uniform convergence proves $p \le H_\varphi$. □

The domination statement identifies H_φ as the largest continuous admissible function with boundary values at most φ. This is the property needed for local envelope replacement.

13.7 Contact support in a big background

Let θ now be an arbitrary smooth big background. Regularization makes the envelope of a continuous obstacle continuous off the poles of a strict analytic barrier. A homogeneous replacement below the obstacle then forces its Monge–Ampère measure to vanish away from contact.

Choose ρ with analytic singularities and smooth remainder so that

$$\theta + \mathrm{dd^c}\rho \geq \kappa\omega, \qquad \kappa > 0,$$

and set $Z = \{\rho = -\infty\}$. Such a barrier exists by Theorem 1.15.2.

Theorem 13.7.1 *For $h \in C(X)$, the envelope $p = P_\theta(h)$ is genuine and continuous on $X \setminus Z$. Moreover its extended Monge–Ampère measure satisfies*

$$\mathbf{1}_{\{p<h\}}\mathrm{MA}_\theta(p) = 0.$$

The measure in this formula is allowed to have infinite total mass.

Proof Subtract a constant from ρ so that $\rho + 2 \leq h$. The elementary envelope construction and upper semicontinuity of the obstacle give

$$\rho + 2 \leq p \leq h.$$

In particular p is locally bounded off Z. We first prove continuity there, without using the contact assertion.

Apply Lemma 1.21.4 to $p - \sup_X p$ and restore the constant. The transforms $\Phi_{c,\delta}$ from that lemma are defined on X and, with fixed $A, K, C > 0$ and $\epsilon(\delta) = C\delta$, satisfy

$$\theta + \mathrm{dd^c}\Phi_{c,\delta} \geq -(Ac + \epsilon(\delta))\omega. \tag{13.5}$$

Their infimum is attained at a positive radius off Z, and

$$p - K(\delta^2 + \delta) \leq \Phi_{c,\delta} \leq R_\delta p \leq R_\delta h.$$

For small δ the first bound gives $\Phi_{c,\delta} \geq \rho + 1$, hence $\rho \leq \Phi_{c,\delta}$ on X.

Set

$$m(\delta) = \sup_{0<t\leq\delta} \|R_t h - h\|_\infty, \qquad c = m(\delta) + \epsilon(\delta) + \delta, \qquad s = \frac{Ac + \epsilon(\delta)}{\kappa}.$$

Positivity, normalization, and localization of the averaging kernels imply $m(\delta) \to 0$ by uniform continuity of h. Define

$$v_\delta = \frac{\Phi_{c,\delta} + s\rho}{1 + s}.$$

The strict barrier and (13.5) show that v_δ is a genuine θ-plurisubharmonic function. Since $\rho \leq \Phi_{c,\delta}$,

$$v_\delta \le \Phi_{c,\delta} \le R_\delta h \le h + m(\delta).$$

Thus $v_\delta - m(\delta)$ is an envelope competitor, and $v_\delta \le p + m(\delta)$.

On a fixed compact subset $L \Subset X \setminus Z$, the difference $p - \rho$ is bounded above. As $s \le Cc$, the exact identity

$$\Phi_{c,\delta} - p = (1+s)(v_\delta - p) + s(p - \rho)$$

therefore gives $\Phi_{c,\delta} - p \le C_L c$. At a minimizing radius t,

$$-c\log(t/\delta) \le \Phi_{c,\delta} - p + K(\delta^2 + \delta) \le C_L c.$$

Since $c \ge \delta$, this yields $t \ge a_L \delta$ with $a_L > 0$ fixed. Monotonicity of smoothing now implies

$$p - K(a_L\delta)^2 \le R_{a_L\delta} p \le \Phi_{c,\delta} + K\delta^2 + K\delta.$$

Consequently $R_{a_L\delta} p \to p$ uniformly on L. These averages are smooth for positive radius even though p is globally unbounded. Applying this to compact neighborhoods proves continuity on $X \setminus Z$.

Now take $x \in \{p < h\} \setminus Z$. By the continuity just proved this is an ordinary open set near x. Choose a sufficiently small coordinate ball B_r centered at x, with closure in that set. Let H be the homogeneous replacement of Theorem 13.6.1 with boundary data $p|_{\partial B_r}$. Its domination property gives $H \ge p$ on the ball.

We can make the ball small enough that $H \le h$. Indeed, with z denoting the chosen Euclidean coordinates and Δ their scalar Laplacian, $\theta + \mathrm{dd^c} H \ge 0$ implies a fixed bound $\Delta H \ge -C$. Comparison with the function $\sup_{\partial B_r} p + C'(r^2 - |z - x|^2)$, with C' fixed large enough, gives

$$H \le \sup_{\partial B_r} p + C' r^2.$$

As $r \to 0$, the right-hand side tends to $p(x)$, whereas $\inf_{B_r} h \to h(x) > p(x)$. This proves the claim for sufficiently small r.

Paste H into p on the ball. The result is θ-plurisubharmonic: for each $a > 0$, paste $\max\{p, H - a\}$ inside the ball and p outside. Continuity and equality of the boundary values ensure that $H - a < p$ on a collar, so this is an ordinary local maximum construction. Letting $a \downarrow 0$ gives the desired pasted function, with convergence locally uniform near the boundary. It is bounded above by h on all of X. The defining property of the envelope therefore forces it to equal p. Thus $p = H$ on B_r and $\mathrm{MA}_\theta(p)$ vanishes there.

These balls cover $\{p < h\} \setminus Z$. The local bounded product there agrees with the extended measure, and the latter does not charge the analytic, hence pluripolar, set Z. This proves the contact assertion on X, without taking or bounding its total mass. □

13.7.1 Quasi-continuous obstacles

Continuous approximations of an obstacle need not preserve every envelope competitor pointwise. The approximation in Lemma 1.19.3 repairs this issue for lower semicontinuous obstacles; quasi-continuity then permits weighted measure limits. For an unbounded obstacle, finite truncation in an arbitrary smooth background introduces signed products. Lemma 1.8.4 controls their limits before restriction to the region where the product is positive.

Proposition 13.7.1 *If f is bounded and lower semicontinuous, then $p = P_\theta(f)$ is locally bounded on $X \setminus Z$, and*

$$\mathbf{1}_{\{p<f\}}\mathrm{MA}_\theta(p) = 0.$$

Proof Take continuous $f_j \uparrow f$ with common bounds $m \le f_j \le M$, using the infimum construction in Proposition 12.1.2. Shift ρ below m. The envelopes $p_j = P_\theta(f_j)$ increase and satisfy

$$\rho \le p_j \le M.$$

Their regularized limit $q = (\lim_j p_j)^*$ is genuine. Outside a pluripolar set it equals the pointwise limit and is at most f, so $q \le p = P_\theta(f)$.

We prove the reverse inequality without a domination theorem. Let u be any envelope competitor. Apply Lemma 1.19.3 to obtain $u_t \le f$ everywhere. For fixed $t, \eta > 0$, the sets

$$\{u_t < f_j + \eta\}, \qquad j \ge 1,$$

are open, increase with j, and cover X. Compactness gives an index after which $u_t \le f_j + \eta$ everywhere. Thus $u_t - \eta$ is a competitor for p_j, and $q \ge u_t - \eta$. Let $\eta \downarrow 0$ and then $t \downarrow 0$. We obtain $q \ge u$ almost everywhere, hence everywhere by the canonical representatives of quasi-plurisubharmonic functions. Taking the envelope over u gives $q \ge p$.

For fixed k, the set $U_k = \{p < f_k\}$ is open. If $j \ge k$, then $p_j \le p < f_k \le f_j$ on U_k. The preceding continuous-obstacle theorem gives $\mathrm{MA}_\theta(p_j) = 0$ there. On every relatively compact open subset of $X \setminus Z$, the functions p_j have common bounds. Increasing local Bedford–Taylor continuity, applied after a common smooth correction and finite expansion, gives convergence of their measures to $\mathrm{MA}_\theta(p)$. The weak-limit inequality for open sets therefore gives zero measure on $U_k \setminus Z$. The union of these sets is $\{p < f\} \setminus Z$. The extended measure charges no pluripolar set, so it also charges no part of Z. □

Proposition 13.7.2 *The conclusion of Proposition 13.7.1 holds for every bounded quasi-continuous obstacle f.*

Proof By Lemma 1.8.2, there are bounded lower semicontinuous f_j with common bounds and $f_j \downarrow f$ quasi-everywhere. The envelopes $p_j = P_\theta(f_j)$ decrease to $p = P_\theta(f)$. Indeed, the common lower barrier makes the limit genuine; the limit lies below f quasi-everywhere and is thus an envelope competitor, while monotonicity gives the other inequality. All these functions have common local bounds off Z.

For fixed k, put $H_k = \min\{1, (f - p_k)_+\}$ on $X \setminus Z$. This is a bounded quasi-continuous function there. If $j \geq k$, then on $\{H_k > 0\}$ we have $p_j \leq p_k < f \leq f_j$, outside a pluripolar set. The preceding proposition gives $H_k \mathrm{MA}_\theta(p_j) = 0$ off Z.

Let g be continuous with compact support in $X \setminus Z$. Weighted bounded-product continuity from Section 1.6 applies to the fixed quasi-continuous coefficient gH_k. Consequently $H_k \mathrm{MA}_\theta(p) = 0$ off Z. Letting $k \to \infty$ exhausts $\{p < f\} \setminus Z$. Non-pluripolarity removes Z. □

For an unbounded obstacle, we use the signed-product continuity of Lemma 1.8.4 after truncation.

Theorem 13.7.2 *Let θ be smooth and big. Let f be a quasi-continuous obstacle, finite almost everywhere, and suppose $p = P_\theta(f)$ is genuine. Then $p \leq f$ quasi-everywhere and*

$$\mathbf{1}_{\{p<f\}} \mathrm{MA}_\theta(p) = 0.$$

The measure may have infinite total mass.

Proof Set $f_j = \max(f, -j)$ and $p_j = P_\theta(f_j)$. The genuine potential p is a competitor for every obstacle. Since $f_j \leq f_1$ and f_1 is finite almost everywhere, the finite-sublevel argument in Section 1.19 gives a common upper bound for all competitors. Thus the envelopes p_j are genuine and uniformly bounded above. Choose M strictly larger than this bound.

The obstacle $g_j = \min(f_j, M)$ is bounded and quasi-continuous. Since $p_j < M$ and $p_j \leq f_j$ quasi-everywhere, p_j is a competitor for $P_\theta(g_j)$; the reverse inequality follows from $g_j \leq f_j$. Thus $p_j = P_\theta(g_j)$. The preceding bounded result gives contact support for p_j at g_j. The strict inequality $p_j < M$ excludes contact at the artificial upper cutoff, so

$$\mathbf{1}_{\{p_j<f_j\}} \mathrm{MA}_\theta(p_j) = 0. \tag{13.6}$$

Each p_j is locally bounded off Z, with a lower barrier of the form $\rho - C_j$. The sequence decreases to p: its limit is genuine since it dominates p, lies below f quasi-everywhere, and is therefore itself an envelope competitor.

Fix a Hermitian form $\Omega \geq \theta$. For integers $C, k \geq 1$, define

$$v_{j,C} = \max(p_j, -C), \qquad v_C = \max(p, -C),$$

which are bounded Ω-plurisubharmonic functions, and set

$$H_{k,C} = \min\{1, (f - p_k)_+\} \min\{1, (p + C)_+\} \quad \text{on } X \setminus Z.$$

Extend $H_{k,C}$ by zero on Z. It is bounded and quasi-continuous: off Z the first difference has a finite quasi-plurisubharmonic second term; bounded continuous truncations preserve quasi-continuity, and the pluripolar set Z can be removed with arbitrarily small capacity. At a pole of p the second factor is zero.

For $j \geq k$, on $\{H_{k,C} > 0\}$ we have, outside irrelevant pluripolar sets,

$$p_j \geq p > -C, \qquad p_j \leq p_k < f \leq f_j.$$

The signed measure $\nu_{j,C} = (\theta + \mathrm{dd^c}\nu_{j,C})^n$ therefore agrees there with $\mathrm{MA}_\theta(p_j)$ by plurifine locality. Equation (13.6) gives

$$H_{k,C}\nu_{j,C} = 0.$$

For fixed C, the potentials $\nu_{j,C}$ have common global bounds and decrease to ν_C. The signed convergence lemma yields

$$\int_X H_{k,C}(\theta + \mathrm{dd^c}\nu_C)^n = 0.$$

On the set where $H_{k,C}$ is positive, $p > -C$. The restricted product in Theorem 6.1.1 identifies the last integrand with $H_{k,C}\mathrm{MA}_\theta(p)$, a positive measure; on the complement the coefficient is zero. It follows that $\mathrm{MA}_\theta(p)$ vanishes on $\{H_{k,C} > 0\}$.

If $p(x)$ is finite and $p(x) < f(x)$, then $p_k(x) < f(x)$ for large k, and $p(x) > -C$ for large C. The countable union of these positive sets therefore covers $\{p < f\}$ outside $Z \cup \{p = -\infty\}$. Both excluded sets are pluripolar. This proves the theorem.□

13.8 Compactness with a common lower barrier

A common lower barrier makes all finite and infinite rooftops genuine. An error integral against each approximating measure bounds the rooftop error by a summable tail. Weighted lower semicontinuity passes this bound to the limit, and singular domination identifies the lower envelope with the L^1 limit. Sandwiching by upper envelopes then gives capacity convergence; suitable domination also controls the measure tails.

Theorem 13.8.1 *Let $\theta \geq 0$ be big. Suppose $\varphi \in \mathrm{PSH}(X, \theta)$ admits a genuine $P_{\delta\theta}(\varphi)$ for some $0 < \delta < 1$. Let $u_j \in \mathrm{PSH}(X, \theta)$ satisfy, for a fixed C,*

$$\varphi - C \leq u_j \leq \varphi + C, \qquad u_j \longrightarrow u \text{ in } L^1(X).$$

If

$$I_j := \int_X |\mathrm{e}^{u_j} - \mathrm{e}^u|\,\mathrm{MA}_\theta^{\mathrm{ext}}(u_j) \longrightarrow 0, \tag{13.7}$$

then $u_j \to u$ in capacity.

Proof The canonical limit lies between $\varphi - C$ and $\varphi + C$. It suffices to prove capacity convergence after further extraction from an arbitrary subsequence. Choose an extraction with $I_j \leq 2^{-j}$ and almost-everywhere convergence for smooth volume. For $N \geq j$ define

$$p_{j,N} = P_\theta(u_j, \ldots, u_N), \qquad p_j = P_\theta\left(\inf_{\ell \geq j} u_\ell\right).$$

All these envelopes lie between $\varphi - C$ and $\varphi + C$. The contact theorem and ordered contact give a finite-rooftop inequality: partition the contact set according to the first index ℓ for which $p_{j,N} = u_\ell$. On that piece the measure of $p_{j,N}$ is bounded by that of u_ℓ, and $e^{p_{j,N}} = e^{u_\ell}$. Therefore

$$\int_X |e^{p_{j,N}} - e^u| \, \mathrm{MA}_\theta^{\mathrm{ext}}(p_{j,N}) \le \sum_{\ell=j}^{N} I_\ell \le 2^{1-j}.$$

This uses only positive measures, including when their total masses are infinite.

As $N \to \infty$, $p_{j,N} \downarrow p_j$; the limit is the envelope of the infimum because every competitor is below every finite rooftop. The sequence p_j increases to a genuine canonical limit p. Both passages hold in capacity by Lemma 1.8.6. Apply Lemma 6.3.1 to the bounded nonnegative weights $|e^{p_{j,N}} - e^u|$, first in N and then in j. We obtain

$$\int_X |e^p - e^u| \, \mathrm{MA}_\theta^{\mathrm{ext}}(p) = 0.$$

The inequalities $p_j \le u_\ell$ for all $\ell \ge j$ imply $p \le u$, first almost everywhere and then for canonical representatives. Also $p \sim u \sim \varphi$. Off their poles, $e^p < e^u$ exactly on $\{p < u\}$, so $\mathrm{MA}_\theta^{\mathrm{ext}}(p)(\{p < u\}) = 0$. The competitor $P_{\delta\theta}(\varphi) - C$ makes $P_{\delta\theta}(p)$ genuine. Apply Theorem 13.1.1 with coefficient zero to obtain $p \ge u$, and hence $p = u$.

Finally $U_j = (\sup_{\ell \ge j} u_\ell)^*$ decreases to u by Hartogs' lemma and the almost-everywhere extraction. The inequalities $p_j \le u_j \le U_j$, together with the monotone capacity convergence of both outer sequences, prove the claim. □

Proposition 13.8.1 *In Theorem 13.8.1, one may replace the two bounds on u_j by*

$$\varphi - C \le u_j \le C$$

provided the L^1 limit satisfies $u \preceq \varphi$. In particular the approximants may be bounded while φ is unbounded.

Proof The L^1 limit satisfies $u \ge \varphi - C$, so the additional hypothesis implies $u \sim \varphi$. Repeat the finite rooftop construction in the theorem. Every $p_{j,N}$ remains genuine and lies between $\varphi - C$ and C, which is enough for contact support and for the bounded exponential weights. The resulting inequality is still

$$\int_X |e^{p_{j,N}} - e^u| \, \mathrm{MA}_\theta^{\mathrm{ext}}(p_{j,N}) \le 2^{1-j}.$$

The limits p_j exist by the common lower bound, and $p_j \le u$ follows from $p_j \le u_\ell$ for every $\ell \ge j$ and almost-everywhere convergence. Thus $\varphi - C \le p_j \le u$, and all p_j and their increasing limit p have the same ordinary type as φ. Monotone capacity convergence and weighted lower semicontinuity give $\mathrm{MA}_\theta^{\mathrm{ext}}(p)(\{p < u\}) = 0$. Singular domination applies to (p, u) exactly as before, so $p = u$. The upper tail envelopes decrease to u, and the same sandwich gives capacity convergence. No

singularity-type assertion was needed for the finite rooftops or for the individual approximants. □

Corollary 13.8.1 *Under the geometric and common-barrier hypotheses of Theorem 13.8.1, suppose either*

(1) $\mathrm{MA}_\theta^{\mathrm{ext}}(u_j) \le \nu$ *for a fixed finite non-pluripolar measure* ν*; or*
(2) $\mathrm{MA}_\theta^{\mathrm{ext}}(u_j) \le A\omega_{w_j}^n$*, where* A *is fixed and* $w_j \in \mathrm{PSH}(X,\omega)$ *are uniformly bounded and converge in capacity to a bounded* w*.*

Then $u_j \to u$ *in capacity, all the measures have finite mass, and*

$$\mathrm{MA}_\theta^{\mathrm{ext}}(u_j) \rightharpoonup \mathrm{MA}_\theta^{\mathrm{ext}}(u).$$

Proof In the first case Lemma 1.23.1 applied to exponentials proves (13.7). In the second case use Lemma 1.23.3, since the exponentials are bounded and have a common lower curvature bound. The criterion gives capacity convergence in both cases.

The masses are uniformly bounded by the corresponding dominating measures. For the deep tails use $u_j \ge \varphi - C$. In the first case their masses are at most $\nu(\{\varphi < C - T\})$, which tends to zero because ν ignores the pole set of φ. In the second case Lemma 1.8.3 bounds them by $C' \operatorname{Cap}_\omega(\{\varphi < C - T\})$, which also tends to zero. Apply the raw-tail continuity theorem Theorem 7.4.1 in its general-background form. It gives finiteness and weak convergence for the limiting measure as well. □

Its reference need not be an envelope fixed point; the smaller-background condition is enough. A model reference becomes useful when an a priori estimate supplies the two uniform barriers. If such a lower barrier is unavailable, constructing one in a relative envelope class is an additional problem, treated in Section 30.6.

13.9 A bounded subsolution theorem

A bounded density relative to the Monge–Ampère measure of one bounded potential also admits a bounded exponential solution. This permits right-hand sides singular with respect to smooth volume. Regularizing the reference gives a moving dominating measure; the mixed density inequality and fixed-reference trace estimate of Lemmas 1.7.3 and 1.8.7 bound the normalizations when the density is allowed to vanish.

Theorem 13.9.1 *Let θ be semipositive and big. Suppose*

$$\mu = f\,\mathrm{MA}_\theta(\psi), \qquad \psi \in \mathrm{PSH}(X,\theta) \cap L^\infty(X), \qquad 0 \le f \in L^\infty(\mathrm{MA}_\theta(\psi)),$$

and $\mu(X) > 0$. For every $\lambda > 0$ there is a unique bounded $u \in \mathrm{PSH}(X,\theta)$ satisfying

$$\mathrm{MA}_\theta(u) = \mathrm{e}^{\lambda u}\mu.$$

Proof Write $\nu = \mathrm{MA}_\theta(\psi)$, a finite non-pluripolar measure.

Smooth strictly positive density. First suppose f is smooth and strictly positive. Choose a Hermitian $\Omega \ge \theta$ and smooth Ω-plurisubharmonic $\psi_j \downarrow \psi$, using the bounded regularization theorem. The rooftops

$$q_j = P_\theta(\psi_j)$$

are uniformly bounded and decrease to ψ: they lie between ψ and ψ_j. Contact support and ordered contact in background Ω give

$$\mathrm{MA}_\theta(q_j) \le \mathbf{1}_{\{q_j=\psi_j\}}\Omega^n_{\psi_j}.$$

Each right side has a bounded density with respect to smooth volume. Moreover $\mathrm{MA}_\theta(q_j)(X) > 0$ by individual noncollapse. The bounded big-background theorem therefore supplies u_j with

$$\mathrm{MA}_\theta(u_j) = \mathrm{e}^{\lambda u_j} f\,\mathrm{MA}_\theta(q_j).$$

Since f is bounded above and away from zero, $q_j - C$ and $q_j + C$ are lower and upper barriers for a fixed C. More explicitly choose C so that $\mathrm{e}^{\lambda(q_j-C)} f \le 1 \le \mathrm{e}^{\lambda(q_j+C)} f$ for all j. Exponential order then gives $q_j - C \le u_j \le q_j + C$.

Extract an L^1 limit u, still uniformly bounded. The moving domination

$$\mathrm{MA}_\theta(u_j) \le C'\Omega^n_{\psi_j}$$

and Corollary 13.8.1, with reference zero, give convergence in capacity. Bounded capacity continuity with the quasi-continuous weights $\mathrm{e}^{\lambda u_j} f$ gives

$$\mathrm{MA}_\theta(u) = \mathrm{e}^{\lambda u} f\nu.$$

Bounded measurable density. Choose positive smooth f_j with a common upper bound $L > 0$, converging to f in $L^1(\nu)$. Such an approximation follows from regularity of the finite measure, continuous approximation on large compact sets, and smooth uniform approximation of the continuous functions; truncation to a fixed bounded interval and addition of a positive constant tending to zero preserve the bounds and convergence. The first step gives bounded u_j with

$$\mathrm{MA}_\theta(u_j) = \mathrm{e}^{\lambda u_j} f_j\nu.$$

A fixed $\psi - C$ is a lower barrier, since $e^{\lambda(\psi - C)} f_j \le 1$ for all j. Hence $u_j \ge \psi - C$. The upper bound requires a separate argument because $\inf_X f_j$ can tend to zero.

The construction in the first step also gives the mixed inequality

$$\theta_{u_j} \wedge \theta_\psi^{n-1} \ge e^{\lambda u_j/n} f_j^{1/n} \nu. \tag{13.8}$$

To see this without assuming a mixed inequality for arbitrary singular measures, use that step's approximants $u_{j,k}$ and q_k. Their top measures have smooth-volume densities and

$$\mathrm{MA}_\theta(u_{j,k}) = e^{\lambda u_{j,k}} f_j \mathrm{MA}_\theta(q_k).$$

Apply Lemma 1.7.3 to $(u_{j,k}, q_k)$. For this fixed j both sequences are uniformly bounded and converge in capacity to (u_j, ψ). Bounded weighted continuity passes the mixed inequality to the limit and yields (13.8).

Put $M_j = \sup_X u_j$, $v_j = u_j - M_j$ and $c_j = e^{\lambda M_j}$. If the M_j were unbounded above, extract $M_j \to +\infty$ and then $v_j \to v$ in $L^1(X)$, with v a genuine normalized potential. By Lemma 1.23.1, $e^{\lambda v_j/n} \to e^{\lambda v/n}$ in $L^1(\nu)$. The inequalities

$$|f_j^{1/n} - f^{1/n}|^n \le |f_j - f|, \qquad 0 \le f_j \le L,$$

then give

$$\int_X e^{\lambda v_j/n} f_j^{1/n} \, d\nu \longrightarrow \int_X e^{\lambda v/n} f^{1/n} \, d\nu > 0.$$

The last integral is positive because $\mu(X) > 0$ and ν gives no mass to the pole set of v. Integrate (13.8) and apply Lemma 1.8.7. For some $a > 0$ along the extraction, this gives

$$a c_j^{1/n} \le \int_X \theta_{v_j} \wedge \theta_\psi^{n-1} \le C_\psi,$$

contradicting $M_j \to +\infty$. Thus u_j is uniformly bounded above as well as below.

Extract an L^1 limit. Now $\mathrm{MA}_\theta(u_j) \le C'' \nu$, so Corollary 13.8.1 gives capacity and weak Monge–Ampère convergence. The right sides converge in total variation: use the fixed-measure exponential convergence and

$$\int_X |e^{\lambda u_j} f_j - e^{\lambda u} f| \, d\nu \le C \|f_j - f\|_{L^1(\nu)} + \|f\|_\infty \int_X |e^{\lambda u_j} - e^{\lambda u}| \, d\nu.$$

Consequently $\mathrm{MA}_\theta(u) = e^{\lambda u} \mu$. Uniqueness is bounded exponential order. □

13.10 Adding weighted Monge–Ampère measures

Weighted measures of two solutions in the same ordinary type can be added while retaining that type. Truncation gives bounded equations whose solutions have a common lower barrier. Their exponential error tends to zero even if the approximating masses grow, so the lower-barrier criterion identifies the limit. This addition step will remove a measure deficit in the prescribed-type existence proof.

Theorem 13.10.1 *Let $\theta \geq 0$ be big, and suppose $\varphi \in \mathrm{PSH}(X, \theta)$ admits $P_{\delta\theta}(\varphi) \not\equiv -\infty$ for some $0 < \delta < 1$. Let $u, v \in \mathrm{PSH}(X, \theta)$ satisfy $u, v \sim \varphi$, and let $\lambda > 0$. If*

$$\mu = \mathrm{e}^{-\lambda u}\mathrm{MA}_\theta^{\mathrm{ext}}(u) + \mathrm{e}^{-\lambda v}\mathrm{MA}_\theta^{\mathrm{ext}}(v)$$

is a finite measure, then there exists $a \in \mathrm{PSH}(X, \theta)$, with $a \sim \varphi$, such that

$$\mathrm{MA}_\theta^{\mathrm{ext}}(a) = \mathrm{e}^{\lambda a}\mu.$$

Weights are interpreted off the pluripolar pole sets. The solution is unique in this ordinary singularity type.

Proof Choose $S \geq \max(\sup_X u, \sup_X v)$. A common translation reduces to $u, v \leq 0$: translating by $-S$ multiplies μ by $\mathrm{e}^{\lambda S}$ and translates the resulting solution back by S.

For integers $j \geq 1$, set $u_j = \max(u, -j)$, $v_j = \max(v, -j)$ and

$$\mu_j = \mathrm{e}^{-\lambda u_j}\mathrm{MA}_\theta(u_j) + \mathrm{e}^{-\lambda v_j}\mathrm{MA}_\theta(v_j).$$

For $s_j = (u_j + v_j)/2$, positivity gives

$$\mu_j \leq 2^n \mathrm{e}^{\lambda j}\mathrm{MA}_\theta(s_j).$$

Each measure is finite and nonzero by individual bounded noncollapse. The bounded subsolution theorem gives a_j with $\mathrm{MA}_\theta(a_j) = \mathrm{e}^{\lambda a_j}\mu_j$. Bounded exponential order yields

$$a_j \leq \min(u_j, v_j) \leq 0.$$

Choose C with $|u - v| \leq C$ off the common poles; then $|u_j - v_j| \leq C$. Put

$$D = C/2 + n\lambda^{-1}\log 2, \qquad b_j = s_j - D.$$

The inequality

$$\mathrm{e}^{-\lambda b_j}\mathrm{MA}_\theta(b_j) \geq 2^{-n}\mathrm{e}^{\lambda D - \lambda s_j}\big(\mathrm{MA}_\theta(u_j) + \mathrm{MA}_\theta(v_j)\big) \geq \mu_j$$

follows from $|u_j - v_j| \leq C$. Hence $a_j \geq b_j \geq w$, where

$$w = (u + v)/2 - D \sim \varphi$$

is a fixed genuine potential. Extract $a_j \to a$ in $L^1(X)$. The upper barriers decrease to $\min(u, v)$, so $w \leq a \leq \min(u, v)$ and $a \sim \varphi$.

To prove convergence in capacity, write $E_j = \{\min(u, v) > -j\}$. Locality gives $\mu_j = \mu$ there. Thus

$$\int_{E_j} |\mathrm{e}^{a_j} - \mathrm{e}^{a}|\,\mathrm{MA}_\theta(a_j) \le \int_X |\mathrm{e}^{a_j} - \mathrm{e}^{a}|\,\mathrm{d}\mu \longrightarrow 0$$

by Lemma 1.23.1. On $X \setminus E_j$ we have $a_j \le -j$ and $a \le -j$, whence $|\mathrm{e}^{a_j} - \mathrm{e}^{a}| \le \mathrm{e}^{-j}$. Also

$$\mathrm{MA}_\theta(a_j) = \mathrm{e}^{\lambda(a_j - u_j)}\mathrm{MA}_\theta(u_j) + \mathrm{e}^{\lambda(a_j - v_j)}\mathrm{MA}_\theta(v_j) \le \mathrm{MA}_\theta(u_j) + \mathrm{MA}_\theta(v_j).$$

Choose a Hermitian $\Omega \ge \theta$. Write $r_n = \max(n-2, 0)$. The bounded-range estimate gives

$$\int_{X\setminus E_j} |\mathrm{e}^{a_j} - \mathrm{e}^{a}|\,\mathrm{MA}_\theta(a_j) \le C'\mathrm{e}^{-j}(1+j)^{r_n} \longrightarrow 0.$$

The lower-barrier criterion Proposition 13.8.1 now gives $a_j \to a$ in capacity. Only polynomial control of the approximating masses was needed.

It remains to identify the equation. This can be done on bounded levels without claiming raw weak convergence prematurely. For $T > 0$ choose the continuous cutoff g_T from Lemma 6.3.1 and put $\zeta_T = g_T(w)$. Where $\zeta_T > 0$, we have $w > -2T$, hence every a_j and a is above $-2T$. If $j > 2T$ and $x \notin E_j$, the inequalities $w(x) \le a_j(x) \le -j$ force $\zeta_T(x) = 0$. Consequently, for all $j > 2T$,

$$\zeta_T \mathrm{MA}_\theta(a_j) = \zeta_T \mathrm{e}^{\lambda a_j}\mu.$$

Test this identity against any continuous function. On the left, replace a_j by $\max(a_j, -2T)$ and use bounded capacity continuity with the fixed bounded quasi-continuous weight ζ_T. On the right, fixed-measure exponential convergence gives convergence in total variation. Therefore

$$\zeta_T \mathrm{MA}_\theta^{\mathrm{ext}}(a) = \zeta_T \mathrm{e}^{\lambda a}\mu.$$

Let $T \to \infty$. The cutoffs increase to one off the pole set of the genuine w; all measures ignore that set. This proves the equation and, in particular, finiteness of the limiting raw measure. Uniqueness follows from singular exponential order. □

13.11 Equations in a prescribed ordinary type

Return to a semipositive big θ. We now require the solution to have the ordinary type of an envelope fixed point φ satisfying (12.1). Fix $\varepsilon \in (0,1)$ for which $P_{\varepsilon\theta}(\varphi)$ is genuine. The smaller-background envelope yields a uniform bound for the difference from φ. A supersolution and the addition theorem then prove existence in this type.

Theorem 13.11.1 *Let $\mu = f\,\mathrm{d}V$, where $0 \le f \in L^p(\mathrm{d}V)$, $p > 1$, and $\mu(X) > 0$. Then:*

(1) *For every $\lambda > 0$ there is a unique $u \in \mathrm{PSH}(X,\theta)$ with $u \sim \varphi$ such that*

$$\mathrm{MA}_\theta(u) = \mathrm{e}^{\lambda u}\mu.$$

(2) *There are $u \in \mathrm{PSH}(X,\theta)$ with $u \sim \varphi$ and a uniquely determined constant $c > 0$ such that*

$$\mathrm{MA}_\theta(u) = c\mu.$$

The counterexample in Theorem 11.6.1 shows that the potential in part (2) need not be unique modulo constants, even when the background is Hermitian and the reference is zero.

We first bound the density norm from below in the prescribed ordinary type.

Lemma 13.11.1 *Fix $q > 1$. There is $b_q > 0$ such that every $u \in \mathrm{PSH}(X,\theta)$ of ordinary type φ with*

$$\mathrm{MA}_\theta^{\mathrm{ext}}(u) = F\,\mathrm{d}V, \qquad 0 \le F \in L^q(\mathrm{d}V),$$

satisfies $\|F\|_q \ge b_q$. For fixed θ and smaller-background parameter ε, this constant is independent of φ.

Proof Bounded equivalence with φ gives a genuine smaller-background envelope for u. If $F = 0$, strict domination applied to $u, u+1$ is impossible, so $N = \|F\|_q > 0$.

Apply Lemma 13.2.1 in background $(1-\varepsilon)\theta$ to F/N, whose L^q norm is one. It gives a bounded s, with $-1 \le s \le 0$, and a constant $b_q > 0$ depending only on this fixed background and q, such that

$$\mathrm{MA}_{(1-\varepsilon)\theta}(s) \ge (b_q/N)F\,\mathrm{d}V.$$

Put $h = P_{\varepsilon\theta}(\varphi) + s$. This is a genuine θ-plurisubharmonic potential with $h \preceq u$. The current splitting is positive, so the same density bound holds for $\mathrm{MA}_\theta^{\mathrm{ext}}(h)$. To justify it, first truncate $P_{\varepsilon\theta}(\varphi)$ on finite levels, use positivity of bounded mixed products, and then exhaust its finite locus. Its pole set has zero smooth volume.

If $N < b_q$, we would have

$$\mathrm{MA}_\theta^{\mathrm{ext}}(u) \le (N/b_q)\mathrm{MA}_\theta^{\mathrm{ext}}(h+t) \quad \text{for every } t \in \mathbb{R}.$$

Since $N/b_q < 1$, Theorem 13.1.1 would imply $u \geq h + t$ for every t. At a point where both potentials are finite this is impossible. Thus $N \geq b_q$. The construction of s used no property of φ, proving the uniform assertion. □

This is a lower bound for the density norm in a prescribed ordinary type. It does not impose a positive lower bound on its total mass.

The next estimate explains how the same singularity type is retained in the limit.

Theorem 13.11.2 *Under the hypotheses on (X, θ, φ) above, suppose $u \in \mathrm{PSH}(X, \theta)$, $u \sim \varphi$, $\sup_X u = 0$, and*

$$\mathrm{MA}_\theta(u) \leq f \, \mathrm{d}V, \qquad 0 \leq f \in L^p(\mathrm{d}V), \quad p > 1.$$

There is a constant C, depending on the fixed geometry, φ, p, and an upper bound for $\|f\|_{L^p}$, such that

$$\varphi - C \leq u \leq \varphi.$$

Proof The upper bound follows directly from the fixed-point property: $u \leq 0$ and $u \preceq \varphi$, so u is a competitor for $P_\theta[\varphi]$.

Put $\varphi_\varepsilon = P_{\varepsilon\theta}(\varphi)$. Since $\varphi_\varepsilon \leq \varphi \leq 0$, uniform exponential integrability for normalized u and exponential integrability for this fixed φ_ε allow a choice of $\gamma > 0$ and $1 < q < p$ such that

$$g = \mathrm{e}^{-\gamma(\varphi_\varepsilon + u)} f$$

has a uniform L^q bound. More explicitly, choose $q < p$ first, apply Hölder with exponents p/q and $p/(p-q)$, and then choose γ small enough to control the exponential factor by Cauchy–Schwarz and the two exponential integrability estimates.

Apply Theorem 13.2.2 in background $(1-\varepsilon)\theta$ to the density $1 + g$. We obtain a bounded v, with $\sup_X v = 0$, and $c > 0$ such that

$$((1-\varepsilon)\theta + \mathrm{dd^c} v)^n = c(1+g)\,\mathrm{d}V.$$

Both $\|v\|_\infty$ and c^{-1} are uniformly bounded. The added constant one ensures a uniform positive root integral, even if no lower bound for the integral of f was specified.

Put $h = \varphi_\varepsilon + v \leq \varphi$ and define

$$\chi(t) = \frac{1}{1-t} \quad (t \leq 0), \qquad \psi = \varphi + \chi(h - \varphi).$$

Then $0 \leq \chi \leq 1$, $0 < \chi' \leq 1$, and $\chi'' > 0$. Interpret the formula off the pole sets. For bounded ordered potentials, Lemma 1.7.6, with $G(x, y) = x + \chi(y - x)$, gives

$$\theta + \mathrm{dd^c} G(\varphi, h) \geq (1 - \chi'(h - \varphi))\theta_\varphi + \chi'(h - \varphi)\theta_h \geq 0.$$

For unbounded potentials apply the bounded assertion to $\varphi_k = \max(\varphi, -k)$ and $h_k = \max(h, -k) \leq \varphi_k$. Monotonicity of G makes $G(\varphi_k, h_k)$ decrease. Its limit is θ-plurisubharmonic and lies between φ and $\varphi + 1$, so it is genuine and equals ψ as defined above. Locality on common finite levels and bounded product positivity give

$$\mathrm{MA}_\theta^{\mathrm{ext}}(\psi) \geq (\varphi - h + 1)^{-2n}\mathrm{MA}_\theta^{\mathrm{ext}}(h) \geq c(\varphi - h + 1)^{-2n}\mathrm{e}^{-\gamma\varphi_\varepsilon}\mathrm{e}^{-\gamma u} f\,\mathrm{d}V.$$

Pole sets have zero smooth volume, so the last inequality extends by exhaustion to all Borel sets.

Write $T = -\varphi_\varepsilon \geq 0$. If $C \geq 1 + \|v\|_\infty$, then $\varphi - h + 1 \leq T + C$. Since

$$A := \inf_{T\geq 0} \frac{\mathrm{e}^{\gamma T}}{(T + C)^{2n}} > 0,$$

we obtain, with a uniform $d > 0$,

$$\mathrm{MA}_\theta^{\mathrm{ext}}(\psi) \geq d\mathrm{e}^{-\gamma u}\mathrm{MA}_\theta^{\mathrm{ext}}(u).$$

Choose $K = 2 + \gamma^{-1}|\log d|$. On $\{u < \psi - K\}$ we have $u < 1 - K$, hence

$$\mathrm{MA}_\theta^{\mathrm{ext}}(u) \leq d^{-1}\mathrm{e}^{\gamma(1-K)}\mathrm{MA}_\theta^{\mathrm{ext}}(\psi - K) \leq \mathrm{e}^{-\gamma}\mathrm{MA}_\theta^{\mathrm{ext}}(\psi - K).$$

The two potentials have the same ordinary singularity type. Moreover $P_{\varepsilon\theta}(u)$ is genuine, since $u \geq \varphi - C_u$ for some constant C_u. Singular strict domination yields $u \geq \psi - K \geq \varphi - K$. All constants used to choose K are uniform in the asserted family. The possibly nonuniform C_u is used only to provide a genuine envelope, not in any estimate. □

Lemma 13.11.2 *For nonzero* $0 \leq f \in L^p(\mathrm{d}V)$ *and* $\lambda > 0$*, there exists* $w \in \mathrm{PSH}(X, \theta)$ *with* $w \sim \varphi$ *such that*

$$\mathrm{MA}_\theta^{\mathrm{ext}}(w) \leq \mathrm{e}^{\lambda w} f\,\mathrm{d}V.$$

Proof Let v be the bounded solution of $\mathrm{MA}_\theta(v) = \mathrm{e}^{\lambda v} f\,\mathrm{d}V$ supplied by Theorem 13.2.1, and write $M = \sup_X v$, $a = v - M \leq 0$. The rooftop $p = P_\theta(a, \varphi)$ satisfies $\varphi - \|a\|_\infty \leq p \leq \varphi$, so $p \sim \varphi$. By the rooftop inequality,

$$\mathrm{MA}_\theta^{\mathrm{ext}}(p) \leq \mathbf{1}_{\{p=a\leq\varphi\}}\mathrm{MA}_\theta(a) + \mathbf{1}_{\{p=\varphi<a\}}\mathrm{MA}_\theta^{\mathrm{ext}}(\varphi).$$

The second term vanishes: Proposition 12.1.4 carries the measure of φ on $\{\varphi = 0\}$, while $\varphi < a \leq 0$ excludes that set. On the first contact set, $a = p$ and $\mathrm{MA}_\theta(a) = \mathrm{e}^{\lambda(a+M)} f\,\mathrm{d}V$. Thus $\mathrm{MA}_\theta^{\mathrm{ext}}(p) \leq \mathrm{e}^{\lambda(p+M)} f\,\mathrm{d}V$. Take $w = p + M$. □

13.11.1 Existence for a positive exponent

We prove part (1) of Theorem 13.11.1 by maximizing the weighted mass below $\mu = f\,\mathrm{d}V$. The relative estimate supplies compactness of the potentials. If the maximizing measure leaves a nonzero deficit, the addition theorem increases its mass, giving a contradiction.

Proof (of Theorem 13.11.1, part (1)) Fix $\lambda > 0$. By Lemma 13.11.2, choose $w \sim \varphi$ such that

$$\nu_0 := \mathrm{e}^{-\lambda w}\mathrm{MA}_\theta^{\mathrm{ext}}(w) \le \mu = f\,\mathrm{d}V.$$

Consider all potentials $z \sim \varphi$ for which

$$\nu_0 \le \nu_z := \mathrm{e}^{-\lambda z}\mathrm{MA}_\theta^{\mathrm{ext}}(z) \le \mu,$$

and let B be the supremum of their weighted masses $\nu_z(X)$. This is a nonempty family, since it contains w, and $B \le \mu(X) < +\infty$. Choose u_j in the family with $\nu_{u_j}(X) \to B$. Singular exponential order gives $u_j \le w$.

We need a lower bound for their suprema. Put $\varphi_\varepsilon = P_{\varepsilon\theta}(\varphi) \le 0$ and solve the bounded equation

$$((1-\varepsilon)\theta + \mathrm{dd^c}\psi)^n = \mathrm{e}^{\lambda\psi} f\,\mathrm{d}V$$

using Theorem 13.2.1. For $h = \psi + \varphi_\varepsilon$, positivity of the current splitting gives

$$\mathrm{e}^{-\lambda h}\mathrm{MA}_\theta^{\mathrm{ext}}(h) \ge \mathrm{e}^{-\lambda\varphi_\varepsilon} f\,\mathrm{d}V \ge \mu \ge \nu_{u_j}.$$

Here $h \preceq u_j$ because $h \preceq \varphi$ and $u_j \sim \varphi$. Each u_j admits a smaller-background envelope by bounded equivalence with φ. Singular exponential order therefore gives $u_j \ge h$. Consequently

$$\sup_X h \le M_j := \sup_X u_j \le \sup_X w.$$

The lower bound is finite because h is genuine.

Apply the relative uniform estimate to $v_j = u_j - M_j$. Their measures satisfy

$$\mathrm{MA}_\theta^{\mathrm{ext}}(v_j) = \mathrm{MA}_\theta^{\mathrm{ext}}(u_j) \le \mathrm{e}^{\lambda \sup_X w} f\,\mathrm{d}V.$$

Thus $\varphi - C \le v_j \le \varphi$. The bounds for M_j give a fixed bounded distance between all u_j and φ. Extract an L^1 limit u. The common-barrier dominated continuity theorem gives convergence in capacity and

$$\mathrm{MA}_\theta^{\mathrm{ext}}(u_j) \rightharpoonup \mathrm{MA}_\theta^{\mathrm{ext}}(u), \qquad u \sim \varphi.$$

Extract also $\nu_{u_j} \rightharpoonup \nu$. Since $\nu_0 \le \nu_{u_j} \le \mu$, the limit satisfies the same inequalities. We claim that

$$\mathrm{e}^{\lambda u_j}\nu_{u_j} \rightharpoonup \mathrm{e}^{\lambda u}\nu.$$

Indeed $\mathrm{e}^{\lambda u_j} \to \mathrm{e}^{\lambda u}$ in $L^1(\mu)$ by Lemma 1.23.1, and the measures ν_{u_j} are all at most μ. This controls the error from changing the weight. For the fixed bounded weight $\mathrm{e}^{\lambda u}$, approximate it in $L^1(\mu)$ by continuous functions; the domination of both ν_{u_j} and ν makes this approximation uniform in the integrals. Their weak convergence then gives the claim. The same argument works after multiplication by any continuous test function.

It follows that

$$\mathrm{MA}_\theta^{\mathrm{ext}}(u) = \mathrm{e}^{\lambda u}\nu, \qquad \mathrm{e}^{-\lambda u}\mathrm{MA}_\theta^{\mathrm{ext}}(u) = \nu, \qquad \nu(X) = B.$$

The second identity is taken off the pole set, which has zero μ-measure. Hence u belongs to the maximizing family. Write $\nu = g\,\mathrm{d}V$, with $0 \le g \le f$.

Suppose $g \ne f$ in $L^1(\mathrm{d}V)$. Apply the supersolution lemma to $f - g$ to find $v \sim \varphi$ with

$$\nu_v := \mathrm{e}^{-\lambda v}\mathrm{MA}_\theta^{\mathrm{ext}}(v) \le (f - g)\,\mathrm{d}V.$$

This measure is nonzero. More generally, a potential v with a genuine smaller-background envelope cannot have zero Monge–Ampère measure: applying strict domination with coefficient zero to v and $v + 1$ would give $v \ge v + 1$. Thus $\nu_v(X) > 0$.

The sum $\nu + \nu_v$ is finite, so Theorem 13.10.1 supplies $a \sim \varphi$ with

$$\mathrm{e}^{-\lambda a}\mathrm{MA}_\theta^{\mathrm{ext}}(a) = \nu + \nu_v.$$

This measure still lies between ν_0 and μ, but has mass $B + \nu_v(X) > B$, a contradiction. Therefore $g = f$ and u solves the required equation. Uniqueness is singular exponential order in the prescribed ordinary type. □

13.11.2 The normalized equation

Let the positive exponent tend to zero. Bounded comparison controls the normalization from above, while the relative lower-density estimate keeps it positive. The common barriers then give convergence of the Monge–Ampère measures and preserve the prescribed type.

Proof (of Theorem 13.11.1, part (2)) For each $j \ge 1$, use part (1) to solve

$$\mathrm{MA}_\theta^{\mathrm{ext}}(u_j) = \mathrm{e}^{u_j/j} f\,\mathrm{d}V, \qquad u_j \sim \varphi.$$

Write $M_j = \sup_X u_j$, $v_j = u_j - M_j$ and $c_j = \mathrm{e}^{M_j/j}$. Choose a bounded normalized solution h and $\alpha > 0$ of $\mathrm{MA}_\theta(h) = \alpha f\,\mathrm{d}V$, supplied by Theorem 13.2.2.

For $H_j = h + \|h\|_\infty + j\log\alpha$, we have

$$\mathrm{e}^{-H_j/j}\mathrm{MA}_\theta(H_j) = \mathrm{e}^{-(h+\|h\|_\infty)/j} f\,\mathrm{d}V \le f\,\mathrm{d}V = \mathrm{e}^{-u_j/j}\mathrm{MA}_\theta^{\mathrm{ext}}(u_j).$$

The bounded H_j admits a smaller-background envelope, and $u_j \preceq H_j$. Exponential order gives $u_j \le H_j$. In particular

$$0 < c_j \le \alpha\exp(2\|h\|_\infty/j).$$

The lower-density bound of Lemma 13.11.1, applied to v_j, gives

$$b_p \le \|c_j\mathrm{e}^{v_j/j} f\|_p \le c_j\|f\|_p,$$

so c_j is also bounded away from zero.

Since $\sup_X v_j = 0$ and $\mathrm{MA}_\theta^{\mathrm{ext}}(v_j) = c_j e^{v_j/j} f\,\mathrm{d}V \le Cf\,\mathrm{d}V$, the relative uniform estimate gives $\varphi - C' \le v_j \le \varphi$. Extract $v_j \to v$ in $L^1(X)$ and $c_j \to c > 0$. The common-barrier dominated continuity theorem yields capacity convergence, $v \sim \varphi$, and raw weak convergence. Meanwhile Lemma 1.23.2 gives

$$\int_X |e^{v_j/j} - 1| f\,\mathrm{d}V \longrightarrow 0.$$

Thus the right sides converge in total variation to $cf\,\mathrm{d}V$, and $\mathrm{MA}_\theta^{\mathrm{ext}}(v) = cf\,\mathrm{d}V$.

For uniqueness of the scalar, suppose $a, b \sim \varphi$ solve the normalized equation with constants $c_a < c_b$. For every real t,

$$\mathrm{MA}_\theta^{\mathrm{ext}}(a) = \frac{c_a}{c_b}\mathrm{MA}_\theta^{\mathrm{ext}}(b+t).$$

Strict domination gives $a \ge b + t$. At a point where both are finite this is impossible as $t \to +\infty$. Interchanging the two constants proves uniqueness. This argument does not assert uniqueness of the potential after the constants have been shown equal. □

13.12 Comparison above a strict singular barrier

Let θ now be any smooth big real $(1,1)$-form. A strict analytic barrier replaces the smaller-background construction used above. It confines a prospective negative minimum away from its pole set; a local comparison then rules out that minimum.

Fix $\rho \le 0$ with analytic singularities and smooth remainder, and decrease the Hermitian metric if necessary so that $\theta_\rho \ge \omega$. Put $Z = \{\rho = -\infty\}$ and $\Omega = X \setminus Z$. These data remain fixed through Section 13.17. A potential bounded below by $\rho - C$ is locally bounded on Ω, where its extended measure agrees with its bounded product.

Lemma 13.12.1 *Let $B \Subset \Omega$ be a coordinate ball, and let u, v be bounded θ-plurisubharmonic functions on B. Assume*

$$\liminf_{x\to\partial B}(u - v)(x) \ge 0, \qquad \mathbf{1}_{\{u<v\}}\mathrm{MA}_\theta(u) \le \mathbf{1}_{\{u<v\}}\mathrm{MA}_\theta(v).$$

Then $u \ge v$ on B.

Proof Fix $\varepsilon > 0$ and set $v_\varepsilon = \max(u, v - \varepsilon)$. It equals u near the boundary. Locality on $\{u > v - \varepsilon\}$ and ordered contact on $\{u \le v - \varepsilon\}$ show that

$$\mathrm{MA}_\theta(u) \le \mathrm{MA}_\theta(v_\varepsilon)$$

throughout B: the latter set is contained in $\{u < v\}$, where the assumed inequality applies.

Let q be the ordinary plurisubharmonic envelope on B of $u - v_\varepsilon$. This bounded quasi-continuous obstacle is zero near the boundary and lies between Ah and zero

for a sufficiently large A, where $h < 0$ is a smooth strictly plurisubharmonic defining function for the ball. Thus q tends to zero at the boundary, and (1.5) shows that $(\mathrm{dd^c}q)^n$ is carried by $D = \{q = u - v_\varepsilon\}$.

Canonical representatives give $q + v_\varepsilon \le u$. Ordered contact, followed by the preceding measure inequality, yields

$$\mathbf{1}_D \mathrm{MA}_\theta(q + v_\varepsilon) \le \mathbf{1}_D \mathrm{MA}_\theta(u) \le \mathbf{1}_D \mathrm{MA}_\theta(v_\varepsilon).$$

The positive expansion of $(\theta_{v_\varepsilon} + \mathrm{dd^c}q)^n$ contains both $\mathrm{MA}_\theta(v_\varepsilon)$ and $(\mathrm{dd^c}q)^n$. Thus $\mathbf{1}_D(\mathrm{dd^c}q)^n = 0$, and contact support gives $(\mathrm{dd^c}q)^n = 0$. The classical bounded Bedford–Taylor comparison principle with zero boundary values implies $q = 0$. Consequently $u \ge v_\varepsilon$, first quasi-everywhere and then everywhere. Let $\varepsilon \downarrow 0$. □

Theorem 13.12.1 *Suppose $u, v \in \mathrm{PSH}(X, \theta)$ satisfy, for constants $C_0, C_1 \ge 0$,*

$$\min(u, v) \ge \rho - C_0, \qquad v \le u + C_1.$$

If, for some $0 \le c < 1$,

$$\mathbf{1}_{\{u<v\}} \mathrm{MA}_\theta(u) \le c\, \mathbf{1}_{\{u<v\}} \mathrm{MA}_\theta(v),$$

then $u \ge v$. The measures need not have finite total mass.

Proof Replace ρ by $\widehat{\rho} = a\rho - A$, where $a > 1$ is sufficiently close to one and A is sufficiently large. Since

$$\theta + \mathrm{dd^c}(a\rho) = a\theta_\rho + (1 - a)\theta,$$

we can arrange $\theta_{\widehat{\rho}} \ge \omega/2$, $\widehat{\rho} - v \le -1$, and $u - \widehat{\rho} \to +\infty$ when approaching Z. When Z is empty, the last boundary condition is unnecessary. In the proof below write $r = \widehat{\rho}$.

Choose $0 < \delta < 1$ with $(1 - \delta)^n > c$, and put $v_\delta = (1 - \delta)v + \delta r$. Then

$$u - v_\delta \ge \delta(u - r) - (1 - \delta)C_1.$$

If its infimum m on Ω were negative, a sequence x_j approaching that infimum would have a subsequence converging to a point $x_0 \in \Omega$. Choose a coordinate ball $B \Subset \Omega$ centered at x_0, identify it with $\{|z| < 1\}$, and take

$$\eta(z) = b(1 - |z|^2), \qquad 0 < b < 1, \qquad \omega/2 + \mathrm{dd^c}\eta \ge 0.$$

The function $w = v_\delta + \delta\eta + m$ is θ-plurisubharmonic on B, and $\liminf_{\partial B}(u - w) \ge 0$. Also

$$w - v = \delta(r - v + \eta) + m < 0.$$

Thus $\{u < w\} \subset \{u < v\}$, and positivity gives on this set

$$\mathrm{MA}_\theta(u) \le c\mathrm{MA}_\theta(v) \le (1 - \delta)^n \mathrm{MA}_\theta(v) \le \mathrm{MA}_\theta(w).$$

The local comparison lemma implies $u \geq w$ on B. Evaluating at x_j gives $u(x_j) - v_\delta(x_j) \geq m + \delta\eta(x_j)$, contrary to $\eta(x_0) > 0$. We have proved $u \geq v_\delta$. Let $\delta \downarrow 0$ along values with $(1-\delta)^n > c$, and extend the resulting inequality from Ω by canonical representatives. □

Corollary 13.12.1 *Let $u, v \in \mathrm{PSH}(X, \theta)$ satisfy $u, v \geq \rho - C$ for some $C \geq 0$ and $u \sim v$.*

(1) *If $\mathrm{MA}_\theta(u) \leq c\mathrm{MA}_\theta(v)$ for $c \geq 0$, then $c \geq 1$.*
(2) *If $\mathrm{MA}_\theta(u) \leq \mathrm{e}^{\lambda(u-v)}\mathrm{MA}_\theta(v)$ for $\lambda > 0$, then $u \geq v$.*

In particular the positive-exponent equation has at most one solution in each such ordinary singularity type, and the normalized equation has at most one positive scalar in that type.

Proof If $c < 1$, apply the theorem to $u, v + A$ for every constant $A > 0$. Its lower-barrier constant may depend on A, which causes no difficulty. The conclusion $u \geq v + A$ is impossible for all A. For the second assertion, fix $a > 0$ and restrict the measure inequality to $\{u < v - a\}$. There the factor is at most $\mathrm{e}^{-\lambda a} < 1$, so the theorem gives $u \geq v - a$. Let $a \downarrow 0$. Apply this assertion in both orders for two exponential solutions. Apply the first assertion in both orders for two normalized solutions. □

13.13 Subsolutions and the Gauduchon trace

The strict barrier also lets us construct a subsolution within a bounded distance of any reference between ρ and zero. A convex combination of positive currents gives the curvature estimate. Pairing the equation with a Gauduchon metric then bounds its normalization through a linear trace.

Lemma 13.13.1 *Fix $p > 1$ and $L > 0$. There is $m > 0$ such that, whenever*

$$\rho \leq \eta \leq 0, \quad \eta \in \mathrm{PSH}(X, \theta), \qquad g \geq 0, \quad \|g\|_{L^p(\mathrm{d}V)} \leq L,$$

there exists $v \in \mathrm{PSH}(X, \theta)$ with

$$\eta \leq v \leq \eta + 1, \qquad \mathrm{MA}_\theta(v) \geq mg\,\mathrm{d}V.$$

The constant depends only on the fixed geometry, ρ, p, L.

Proof Choose $1 < q < p$. The functions $h = g(2-\rho)^{2n}$ have uniformly bounded L^q norms, by Hölder's inequality and the finite polynomial moments of a fixed quasi-plurisubharmonic function. Apply Lemma 13.2.1 with the Hermitian background ω. It gives $-1 \leq s \leq 0$, $s \in \mathrm{PSH}(X, \omega)$, and

$$\omega_s^n \geq mh\,\mathrm{d}V$$

for a fixed $m > 0$. Put $u_1 = \rho + s$. Then $\theta_{u_1} \geq \omega_s$. On Ω define

$$t = 1 + \eta - u_1, \qquad v = \eta + t^{-1}.$$

Here $1 \leq t \leq 2 - \rho$. Apply Lemma 1.7.6 on relatively compact subsets of Ω, with $a = \eta$, $h = u_1$ and $\chi(s) = (1 - s)^{-1}$. It gives

$$\theta_v = (1 - t^{-2})\theta_\eta + t^{-2}\theta_{u_1} + 2t^{-3}\,\mathrm{d}(\eta - u_1) \wedge \mathrm{d}^{\mathrm{c}}(\eta - u_1) \geq t^{-2}\theta_{u_1}.$$

Since $\eta \leq v \leq \eta + 1$, Lemma 1.3.7 extends v across Z, and the bounds extend by Lemma 1.3.3. On Ω, positivity of the local products gives

$$\mathrm{MA}_\theta(v) \geq t^{-2n}\theta_{u_1}^n \geq (2 - \rho)^{-2n}\omega_s^n \geq mg\,\mathrm{d}V.$$

Both sides give zero mass to Z, proving the global inequality. □

Proposition 13.13.1 *Let γ be Gauduchon, write $\gamma^n = G\,\mathrm{d}V$, and put*

$$T_\gamma = \int_X \theta \wedge \gamma^{n-1} > 0.$$

If $u \geq \rho - C$ is θ-plurisubharmonic and $\mathrm{MA}_\theta(u) \geq F\,\mathrm{d}V$, $F \geq 0$, then

$$\int_X F^{1/n} G^{1-1/n}\,\mathrm{d}V \leq T_\gamma. \tag{13.9}$$

Consequently, for a fixed nonzero $0 \leq f \in L^p(\mathrm{d}V)$, $p > 1$, solutions above this barrier of $\mathrm{MA}_\theta(u) = \mathrm{e}^{\lambda u} f\,\mathrm{d}V$ have uniformly bounded suprema when $\lambda > 0$ is fixed. If instead $v_j \in \mathrm{PSH}(X, \theta)$, $j \geq 1$, and constants $C_j \geq 0$ satisfy

$$\sup_X v_j = 0, \qquad \mathrm{MA}_\theta(v_j) = c_j \mathrm{e}^{v_j/j} f\,\mathrm{d}V, \qquad v_j \geq \rho - C_j,$$

with $c_j > 0$, then the constants c_j are uniformly bounded above.

Proof The linear current $\theta_u \wedge \gamma^{n-1}$ is positive and globally defined, even at Z. Distributional Stokes gives

$$\int_X \theta_u \wedge \gamma^{n-1} = \int_X \theta \wedge \gamma^{n-1},$$

since $\mathrm{dd}^{\mathrm{c}}\gamma^{n-1} = 0$. Applying the identity to ρ proves $T_\gamma > 0$.

On Ω, apply Lemma 1.7.3 to θ_u and γ to obtain

$$\theta_u \wedge \gamma^{n-1} \geq F^{1/n} G^{1-1/n}\,\mathrm{d}V.$$

Exhaust Ω and use positivity of the global trace measure. The complement has zero smooth volume, giving (13.9).

For the exponential assertion write $a = \sup_X u$ and $w = u - a$. The estimate becomes

$$\mathrm{e}^{\lambda a/n} \int_X \mathrm{e}^{\lambda w/n} f^{1/n} G^{1-1/n} \, \mathrm{d}V \le T_\gamma.$$

The integral has a positive lower bound over all normalized θ-plurisubharmonic w. Otherwise normalized compactness supplies a subsequence converging almost everywhere to a genuine potential, and dominated convergence gives a strictly positive limit integral, a contradiction. We used $w \le 0$ and the integrability of $f^{1/n}$. This bounds a above.

For the last assertion the same estimate contains $c_j^{1/n}$ and $\mathrm{e}^{v_j/(jn)}$. If those integrals approached zero along a subsequence with $j \to \infty$, normalized compactness and almost-everywhere convergence would give $v_j/j \to 0$ almost everywhere. Dominated convergence then gives the positive limit $\int_X f^{1/n} G^{1-1/n} \, \mathrm{d}V$. This contradiction bounds c_j; finitely many remaining indices do not matter. No total top-degree mass was used in either argument. □

13.14 Stability above a strict barrier

The reference-preserving subsolution transfers the bounded stability argument to unbounded potentials above ρ. The estimate is uniform over all references between ρ and zero. For a fixed reference, it upgrades L^1 compactness to uniform convergence of differences on Ω.

Theorem 13.14.1 *Fix $C_0, L > 0$, $p > 1$, and*

$$0 < \gamma < \frac{p-1}{np + p - 1}.$$

Let $\eta \in \mathrm{PSH}(X, \theta)$ satisfy $\rho \le \eta \le 0$. Suppose $u \in \mathrm{PSH}(X, \theta)$ satisfies

$$\eta - C_0 \le u \le \eta, \qquad \mathrm{MA}_\theta(u) = f \, \mathrm{d}V, \qquad 0 \le f \in L^p(\mathrm{d}V), \quad \|f\|_p \le L.$$

There is a constant C, depending only on the fixed geometry, ρ, C_0, L, p, γ, such that every $v \in \mathrm{PSH}(X, \theta)$ with $v \le \eta$ satisfies

$$\sup_\Omega (v - u)_+ \le C \left(\int_\Omega (v - u)_+ \, \mathrm{d}V \right)^\gamma. \tag{13.10}$$

The constant is independent of η in the stated range. The comparison potential v need not have the ordinary singularity type of η.

Differences in this statement are taken on Ω, where u and η are locally bounded; their integrals are unchanged by extending them by zero on the analytic complement. The bound also means the corresponding order inequality for the canonical potentials on all of X.

Proof Replace v by $\max(v, u)$, so that $\eta - C_0 \le u \le v \le \eta$ and $0 \le v - u \le C_0$ on Ω.

Fix $q \in (1, p)$ and $\delta > 0$, and put

$$E_\delta = \{u < v - \delta\} \cap \Omega, \qquad g_\delta = \mathbf{1}_{E_\delta} f, \qquad m_\delta = \|g_\delta\|_q.$$

When $m_\delta > 0$, Lemma 13.13.1, followed by subtracting one from its potential, gives $h \in \mathrm{PSH}(X, \theta)$ and a fixed $b > 0$ such that

$$\eta - 1 \le h \le \eta, \qquad \mathrm{MA}_\theta(h) \ge b\, g_\delta / m_\delta \, \mathrm{d}V.$$

The constant is uniform in η because the auxiliary density has L^q norm one and the strict barrier is fixed.

If $0 < m_\delta < b/2$, set

$$t = (2m_\delta/b)^{1/n}, \qquad w = (1 - t)v + th - \delta - C_0 t.$$

Since $h - v \le C_0$, the set $\{u < w\} \cap \Omega$ is contained in E_δ. On this set the local positive-product expansion gives $\mathrm{MA}_\theta(w) \ge t^n \mathrm{MA}_\theta(h) \ge 2\mathrm{MA}_\theta(u)$. Both potentials are bounded below by ρ minus a constant, and $w \le u + C_0$. Thus Theorem 13.12.1 applies and gives $u \ge w$. It follows that

$$v - u \le \delta + C_0 t + t(v - h) \le \delta + (C_0 + 1)t.$$

If $m_\delta = 0$, apply the same domination theorem with coefficient zero to u and $v - \delta$. If $m_\delta \ge b/2$, use $v - u \le C_0$. In all cases,

$$\sup_\Omega (v - u) \le \delta + C_1 m_\delta^{1/n}.$$

Write $I = \int_\Omega (v - u)\, \mathrm{d}V$. For $I = 0$, the canonical potentials agree; otherwise Hölder and Chebyshev give

$$m_\delta \le L(I/\delta)^{(p-q)/(pq)}.$$

For $0 < I \le 1$, choosing $\delta = I^{(p-q)/(npq+p-q)}$ yields the exponent $\gamma_q = (p - q)/(npq + p - q)$. Choose $q > 1$ close enough to one that $\gamma_q > \gamma$. The case $I > 1$ follows from the bound by C_0. This proves (13.10). □

Corollary 13.14.1 *Fix $\eta \in \mathrm{PSH}(X,\theta)$ with $\rho \le \eta \le 0$, constants $C_0, L > 0$, and $p > 1$. Consider the potentials $u \in \mathrm{PSH}(X,\theta)$ satisfying*

$$\eta - C_0 \le u \le \eta, \qquad \mathrm{MA}_\theta(u) = f_u \,\mathrm{d}V, \qquad 0 \le f_u \in L^p(\mathrm{d}V), \quad \|f_u\|_p \le L.$$

This family is compact for uniform convergence of differences on Ω. If $u_j \to u$ in L^1, then u belongs to the same family and

$$\sup_\Omega |u_j - u| \le C\|u_j - u\|^\gamma_{L^1(\mathrm{d}V)} \longrightarrow 0$$

for every exponent in Theorem 13.14.1. Moreover f_{u_j} converges weakly to f_u in L^p.

Proof Apply the theorem in both orders to u_j, u_k, with the common reference η. It follows that an L^1-Cauchy sequence is uniformly Cauchy in differences on Ω. The uniform limit there is locally θ-plurisubharmonic and agrees almost everywhere, hence everywhere in canonical representatives, with the L^1 limit u. The bounds $\eta - C_0 \le u \le \eta$ hold on all of X. Letting $k \to \infty$ in the pairwise estimate proves the displayed quantitative bound, before using any equation for u.

The densities form a bounded set in the reflexive space L^p. For a weakly convergent subsequence, say $f_{u_{j_k}} \rightharpoonup f$, local uniform convergence of the bounded potentials gives $\mathrm{MA}_\theta(u) = f\,\mathrm{d}V$ on Ω, by Bedford–Taylor continuity for the smooth nonclosed background. Both measures give zero mass to the analytic complement, so this equality holds on X. Positivity and weak lower semicontinuity give $f \ge 0$ and $\|f\|_p \le L$. Every weak cluster point is therefore the same density f_u, proving weak convergence of the whole sequence. Finally, the common bounds by η and $\eta - C_0$ give L^1 compactness by quasi-plurisubharmonic compactness. The preceding argument upgrades each convergent subsequence to uniform convergence in the stated family.□

For a normalized solution with minimal singularities, take $\eta = V_\theta$. Every normalized comparison potential lies below this reference.

13.15 Minimal singularities in an arbitrary big background

We solve the equation in the type of V_θ. Resolve strict analytic approximants to reduce first to a semipositive big background. Their solutions have uniform bounds relative to the approximating singularities; upper and lower envelopes then recover a solution of minimal type. For the normalized equation, a common interval below V_θ allows us to apply the preceding compactness theorem directly.

The fixed barrier satisfies $\rho \le V_\theta \le 0$. We use the resolution form of analytic regularization in Theorem 1.15.2.

Lemma 13.15.1 *Let $\psi \in \mathrm{PSH}(X,\theta)$ satisfy $\theta_\psi \geq a\omega$ for some $a > 0$ and have the resolved smooth-remainder form of Theorem 1.15.2. For every nonzero $0 \leq f \in L^p(\mathrm{d}V)$, $p > 1$, and $\lambda > 0$, there exists $u \in \mathrm{PSH}(X,\theta)$ with $u \sim \psi$ such that*

$$\mathrm{MA}_\theta(u) = \mathrm{e}^{\lambda u} f \,\mathrm{d}V.$$

Proof Resolve ψ by $\pi : Y \to X$. The pulled-back current has the form

$$\pi^*\theta + \mathrm{dd^c}\pi^*\psi = \beta + [D],$$

where D is an effective real divisor, $[D]$ is its current of integration, and β is a smooth real $(1,1)$-form on Y. Off D one has $\beta \geq a\pi^*\omega$, so this inequality holds everywhere by smoothness. Thus β is semipositive.

We check its bigness. For a blowup with smooth center and exceptional divisor E, choose a metric on $\mathcal{O}(-E)$ with curvature positive in the directions tangent to the projective fibers. The pullback of a Hermitian metric is positive in the complementary directions along E. Consequently its sum with a sufficiently small positive multiple of this curvature is a Hermitian metric: first check positivity along E, then in a neighborhood, and finally on the compact complement, where the pullback is already positive. Iterating over the blowups gives an effective real exceptional divisor F, with logarithmic potential ζ, and a smooth Hermitian form κ such that

$$\pi^*\omega + \mathrm{dd^c}\zeta = \kappa + [F].$$

For the iteration, pull back the previous smooth positive form, apply the one-blowup construction, and add the pulled-back previous divisor to the new positive multiple of E. Hence $\beta + a\mathrm{dd^c}\zeta \geq a\kappa$, proving bigness of β. This argument concerns Hermitian metrics and never requires a closed positive form on Y.

Fix a smooth positive volume form $\mathrm{d}V_Y$, and write $\pi^*\mathrm{d}V = J\,\mathrm{d}V_Y$. The function J is smooth, nonnegative and bounded, and is positive off the exceptional locus. Define

$$H = \mathrm{e}^{\lambda\pi^*\psi}(f \circ \pi)J.$$

It is nonzero and belongs to $L^p(\mathrm{d}V_Y)$. Indeed ψ is bounded above, and change of variables gives

$$\int_Y |(f \circ \pi)J|^p \,\mathrm{d}V_Y \leq \|J\|_\infty^{p-1} \int_X |f|^p \,\mathrm{d}V.$$

Values on the analytic exceptional sets do not affect these integrals. The semipositive big-background theorem Theorem 13.2.1 produces a bounded $s \in \mathrm{PSH}(Y,\beta)$ with

$$\mathrm{MA}_\beta(s) = \mathrm{e}^{\lambda s} H \,\mathrm{d}V_Y.$$

The function $\pi^*\psi + s$ is $\pi^*\theta$-plurisubharmonic, because its current is $\beta_s + [D] \geq 0$, where $\beta_s = \beta + \mathrm{dd^c}s$. By Theorem A.2.1 it descends to $u \in \mathrm{PSH}(X,\theta)$. Boundedness of s gives $u \sim \psi$. Off the exceptional and polar sets, change of variables proves the

required equation. Both its extended left-hand side and its smooth-volume right-hand side ignore those analytic sets. The equation therefore holds on all of X. □

Theorem 13.15.1 *Let θ be any smooth big real $(1,1)$-form and let $0 \le f \in L^p(\mathrm{d}V)$, $p > 1$, be nonzero. For every $\lambda > 0$ there is a unique $u \in \mathrm{PSH}(X,\theta)$ such that*

$$u \sim V_\theta, \qquad \mathrm{MA}_\theta(u) = \mathrm{e}^{\lambda u} f \,\mathrm{d}V.$$

Proof Apply Theorem 1.15.2 to V_θ. Set $\delta_j = 2^{-j}$ and take a subsequence whose curvature errors are at most δ_j. Denote its potentials by $\widetilde{\psi}_j \downarrow V_\theta$; then $\theta_{\widetilde{\psi}_j} \ge -\delta_j \omega$. Set

$$a_j = (1-\delta_j) \sup_X \widetilde{\psi}_j \ge 0, \qquad \psi_j = (1-\delta_j)\widetilde{\psi}_j + \delta_j \rho - a_j.$$

Normalized compactness, or monotonicity and Hartogs' lemma, gives $a_j \to 0$. Resolving the singularities of $\widetilde{\psi}_j$ and ρ together shows that ψ_j has a smooth remainder after resolution. Moreover,

$$\theta_{\psi_j} \ge \delta_j^2 \omega, \qquad (1-\delta_j)V_\theta + \delta_j \rho - a_j \le \psi_j \le V_\theta. \tag{13.11}$$

The upper inequality follows because ψ_j is θ-plurisubharmonic and at most zero. The explicit subtraction of a_j keeps the normalization separate from the fixed pole barrier.

By the resolved equation lemma there are $u_j \sim \psi_j$ satisfying

$$\mathrm{MA}_\theta(u_j) = \mathrm{e}^{\lambda u_j} f \,\mathrm{d}V.$$

Each u_j is bounded below by ρ up to a constant depending on j. The Gauduchon trace proposition gives a uniform upper bound for $\sup_X u_j$.

There is also a uniform lower estimate. Translate the fixed ρ down by $1 + \sup_j a_j$ when applying Lemma 13.13.1, and take its reference to be $\psi_j - 1$. The lemma gives $\psi_j - 1 \le v_j \le \psi_j \le 0$ and $\mathrm{MA}_\theta(v_j) \ge m f \,\mathrm{d}V$, with $m > 0$ independent of j. Put $w_j = v_j + \lambda^{-1} \log m$. Then

$$\mathrm{MA}_\theta(w_j) \ge \mathrm{e}^{\lambda w_j} f \,\mathrm{d}V, \qquad u_j \sim w_j \sim \psi_j.$$

The exponential order in Corollary 13.12.1 gives $u_j \ge w_j$. Thus for a uniform C,

$$\psi_j - C \le u_j \le V_\theta + C. \tag{13.12}$$

After extraction, u_j converges in L^1 and almost everywhere to a genuine u. The preceding bounds give $V_\theta - C \le u \le V_\theta + C$.

We verify its equation using envelopes. Put $A_j = \sup_{k \geq j} a_k$ and, for $N \geq j$, define

$$p_{j,N} = P_\theta(u_j, \dots, u_N), \qquad t_{j,N} = \max_{j \leq k \leq N} u_k.$$

Both functions lie above the genuine competitor

$$b_j = (1 - \delta_j)V_\theta + \delta_j \rho - C - A_j.$$

In particular every envelope is genuine. All these functions are uniformly locally bounded on Ω, also as j varies. Contact support from Theorem 13.7.2 and a disjoint partition by the first contact index give

$$\mathrm{MA}_\theta(p_{j,N}) \leq \mathrm{e}^{\lambda p_{j,N}} f \, \mathrm{d}V.$$

There is no factor N here: on each piece the envelope equals one u_k, and ordered contact bounds its measure by that single measure. Locality and ordered contact for a finite maximum similarly give

$$\mathrm{MA}_\theta(t_{j,N}) \geq \mathrm{e}^{\lambda t_{j,N}} f \, \mathrm{d}V.$$

Let $N \to \infty$, setting $p_j = \lim_N p_{j,N}$ and $t_j = (\lim_N t_{j,N})^*$. Local bounded monotone continuity passes both inequalities to these limits. Next p_j increases to a regularized limit p, while t_j decreases to u by Hartogs' lemma. The bound by b_j gives $p \sim V_\theta$, and $p \leq u$. A second use of local bounded monotone continuity gives

$$\mathrm{MA}_\theta(p) \leq \mathrm{e}^{\lambda p} f \, \mathrm{d}V, \qquad \mathrm{MA}_\theta(u) \geq \mathrm{e}^{\lambda u} f \, \mathrm{d}V.$$

These inequalities hold on Ω and extend across Z because all measures involved ignore it. Thus $\mathrm{MA}_\theta(p) \leq \mathrm{e}^{\lambda(p-u)} \mathrm{MA}_\theta(u)$. The exponential order gives $p \geq u$, so $p = u$ and the equation follows. Applying that order in both directions also proves uniqueness. Every passage of the Monge–Ampère operator above was along a locally bounded monotone sequence on Ω. □

Lemma 13.15.2 *Under the hypotheses of Theorem 13.15.1, let u_λ be its solution and put*

$$v_\lambda = u_\lambda - \sup_X u_\lambda, \qquad c_\lambda = \mathrm{e}^{\lambda \sup_X u_\lambda}.$$

There are constants $B, C, m > 0$, independent of $0 < \lambda \leq 1$, such that

$$V_\theta - B \leq v_\lambda \leq V_\theta, \qquad m \leq c_\lambda \leq C. \tag{13.13}$$

Proof The equation for v_λ is

$$\mathrm{MA}_\theta(v_\lambda) = c_\lambda \mathrm{e}^{\lambda v_\lambda} f \, \mathrm{d}V.$$

Fix a Gauduchon metric γ and write $\gamma^n = G\,\mathrm{d}V$. Since $v_\lambda \le 0$ and $0 < \lambda \le 1$, (13.9) gives

$$c_\lambda^{1/n} \int_X \mathrm{e}^{v_\lambda/n} f^{1/n} G^{1-1/n}\,\mathrm{d}V \le \int_X \theta \wedge \gamma^{n-1}.$$

The integrals on the left have a positive uniform lower bound. Otherwise quasi-plurisubharmonic compactness would give an almost everywhere convergent subsequence of the normalized v_λ with a genuine limit, and dominated convergence would give a positive limit integral, since f is nonzero. Thus $c_\lambda \le C$.

To control the singularity type, uniform Skoda integrability gives fixed $\varepsilon > 0$ and $1 < q < p$ with $\|\mathrm{e}^{-\varepsilon v_\lambda} f\|_q$ uniformly bounded. Apply Lemma 13.13.1 with reference $V_\theta - 1$ and barrier $\rho - 1$. It supplies

$$V_\theta - 1 \le \eta_\lambda \le V_\theta, \qquad \mathrm{MA}_\theta(\eta_\lambda) \ge m\mathrm{e}^{-\varepsilon v_\lambda} f\,\mathrm{d}V \ge m f\,\mathrm{d}V$$

for a fixed $m > 0$. Scalar order, together with $\mathrm{MA}_\theta(v_\lambda) \le c_\lambda f\,\mathrm{d}V$, gives $c_\lambda \ge m$. Set $D = \varepsilon^{-1}\max\{0, \log(C/m)\}$ and $w_\lambda = \eta_\lambda - D$. Then

$$\mathrm{e}^{\varepsilon(w_\lambda - v_\lambda)}\mathrm{MA}_\theta(v_\lambda) \le m\mathrm{e}^{-\varepsilon v_\lambda} f\,\mathrm{d}V \le \mathrm{MA}_\theta(w_\lambda),$$

since $\lambda v_\lambda \le 0$ and $\eta_\lambda \le 0$. Exponential order gives $v_\lambda \ge w_\lambda$, proving the uniform bounds with $B = 1 + D$. □

Theorem 13.15.2 *Under the assumptions of Theorem 13.15.1, there are* $v \in \mathrm{PSH}(X, \theta)$*, a constant* $C > 0$ *and a uniquely determined* $c > 0$ *such that*

$$\sup_X v = 0, \qquad V_\theta - C \le v \le V_\theta, \qquad \mathrm{MA}_\theta(v) = c f\,\mathrm{d}V.$$

Uniqueness of the normalized potential is not asserted.

Proof Apply Lemma 13.15.2 with $\lambda = 1/j$ and write $v_j = v_{1/j}$, $c_j = c_{1/j}$. Extract a subsequence with $v_j \to v$ in L^1 and almost everywhere, and $c_j \to c \in [m, C]$. The limit is normalized, genuine and of minimal ordinary type. The densities $c_j \mathrm{e}^{v_j/j} f$ have a common L^p bound. Hence Corollary 13.14.1, with reference V_θ, upgrades the convergence to uniform convergence of differences on Ω and identifies the weak L^p limit of these densities with the density of $\mathrm{MA}_\theta(v)$.

At almost every point v_j converges to a finite value, so $v_j/j \to 0$. Since $v_j \le 0$ and $c_j \le C$, dominated convergence gives $c_j \mathrm{e}^{v_j/j} f \to c f$ in L^p. Therefore $\mathrm{MA}_\theta(v) = c f\,\mathrm{d}V$. Scalar uniqueness is Corollary 13.12.1. □

13.16 Local Hölder continuity in a big background

The stability estimate above ρ gives a local modulus for solutions of minimal type. Combine it with the quantitative regularization of Lemma 1.21.4: the strict barrier absorbs the curvature error, and a lower bound for the minimizing radius converts the envelope estimate into a smoothing estimate on compact subsets of Ω.

Theorem 13.16.1 *Fix $C_0 > 0$. Let $u \in \mathrm{PSH}(X, \theta)$ satisfy*

$$\sup_X u = 0, \qquad V_\theta - C_0 \le u \le V_\theta, \qquad \mathrm{MA}_\theta(u) = f\,\mathrm{d}V,$$

where $0 \le f \in L^p(\mathrm{d}V)$ and $p > 1$. Then u is locally Hölder continuous on Ω, with every exponent

$$0 < \beta < \frac{2(p-1)}{np + 3(p-1)}.$$

For a fixed bound on C_0 and $\|f\|_p$, the constants are uniform on each fixed compact subset of Ω.

Proof Subtract a constant from the strict barrier so that $\rho \le u - 1$. It still satisfies $\theta + \mathrm{dd^c}\rho \ge \kappa\omega$ for a fixed $\kappa > 0$. Choose $0 < \gamma < (p-1)/(np + p - 1)$ and write $\alpha = 2\gamma < 1$. Apply Lemma 1.21.4 with $a = u$ and use its constants A, K, C and transform $\Phi_{c,\delta}$. For small $\delta > 0$ set

$$c = \delta^\alpha, \qquad e = Ac + C\delta, \qquad s = e/\kappa, \qquad v_\delta = \frac{\Phi_{c,\delta} + s\rho}{1 + s}.$$

Lemma 1.21.4 gives $v_\delta \in \mathrm{PSH}(X, \theta)$. Its lower bound for $\Phi_{c,\delta}$ implies $\rho \le \Phi_{c,\delta}$ when δ is small. Thus

$$v_\delta \le \Phi_{c,\delta} \le R_\delta u \le 0.$$

Apply Theorem 13.14.1 with reference V_θ and comparison v_δ. The L^1 smoothing estimate gives

$$\sup_\Omega (v_\delta - u)_+ \le C\|(R_\delta u - u)_+\|_1^\gamma \le C\delta^\alpha.$$

Fix a compact subset on which $\rho \ge -M$. Since $s \le C\delta^\alpha$ and $u \le 0$, the identity

$$\Phi_{c,\delta} - u = (1 + s)(v_\delta - u) + s(u - \rho)$$

gives $\Phi_{c,\delta} - u \le C_M\delta^\alpha$ there.

At a minimizing radius t the lower monotone bound yields

$$-c\log(t/\delta) \le \Phi_{c,\delta} - u + K(\delta^2 + \delta) \le C_M c.$$

Hence $t \ge a_M\delta$ for a fixed $a_M > 0$. As in the proof of Theorem 13.4.1, monotonicity now gives

$$R_{a_M\delta}u - u \le C_M\delta^\alpha, \qquad R_{a_M\delta}u - u \ge -K(a_M\delta)^2.$$

The constants are uniform on the chosen compact subset. Apply this to a slightly larger compact neighborhood, where also $\|\nabla R_t u\| \le C_M/t$. For nearby points x, y at distance $d > 0$, smooth approximation and a short path in that neighborhood give

$$|u(x) - u(y)| \le C_M(t^\alpha + d/t).$$

Take $t = d^{1/(1+\alpha)}$. This gives exponent $\alpha/(1+\alpha)$; letting γ approach its upper endpoint gives the asserted range. Boundedness on the compact neighborhood handles pairs farther apart. □

The formulation with a scalar and an equation only on Ω follows as well. Suppose $\mathrm{MA}_\theta(u) = cf\,\mathrm{d}V$ on Ω, where $c > 0$ and $0 \le f \in L^p(X, \mathrm{d}V)$, and retain the same minimal bounds. The lower bound $u \ge V_\theta - C_0 \ge \rho - C_0$ makes u locally bounded on Ω, so its bounded-potential product is $\mathrm{MA}_\theta(u)$ there. The analytic complement has zero smooth volume and zero non-pluripolar mass. Thus the equation holds on X, and the theorem applies with cf in place of f. For uniform estimates the density bound is $\|cf\|_p$, including the scalar.

13.16.1 Limits of normalized exponential solutions

For the normalized exponential family (v_λ, c_λ) of Lemma 13.15.2, the reference interval is uniform in $0 < \lambda \le 1$. Stability therefore controls differences on all of Ω, and the local Hölder estimates strengthen the convergence on compact subsets. Fix the nonzero density $0 \le f \in L^p(\mathrm{d}V)$, $p > 1$, and use the constants B, C, m of that lemma.

Proposition 13.16.1 *For the family (v_λ, c_λ) of Lemma 13.15.2, the normalized potentials are relatively compact for uniform convergence of differences on Ω. As $\lambda \downarrow 0$, every cluster limit v is normalized, has minimal singularities, and satisfies*

$$\mathrm{MA}_\theta(v) = cf\,\mathrm{d}V,$$

where $c > 0$ is the unique scalar in Theorem 13.15.2. Moreover,

$$c_\lambda \longrightarrow c, \qquad c_\lambda \mathrm{e}^{\lambda v_\lambda} f \longrightarrow cf \quad \text{in } L^p(\mathrm{d}V).$$

Every sequence of parameters in $(0, 1]$ has a subsequence for which the potentials converge in C^β on each compact subset of Ω, for all

$$0 < \beta < \frac{2(p-1)}{np + 3(p-1)}.$$

Proof Lemma 13.15.2 gives a common reference interval, and the densities are bounded in L^p by $C\|f\|_p$. Apply Corollary 13.14.1 with $\eta = V_\theta$. Given any sequence $\lambda_j \downarrow 0$, pass to a subsequence with $v_{\lambda_j} \to v$ uniformly in differences on Ω and in

$L^1(X)$, and with $c_{\lambda_j} \to c_* \in [m, C]$. The limit has the same reference bounds and is normalized. On the complement of the pole set,

$$0 \le 1 - \mathrm{e}^{\lambda v_\lambda} \le 1 - \mathrm{e}^{\lambda(V_\theta - B)} \longrightarrow 0.$$

Dominated convergence after multiplication by f therefore gives $\mathrm{e}^{\lambda v_\lambda} f \to f$ in L^p for the whole family. The compactness corollary identifies the limit equation as $\mathrm{MA}_\theta(v) = c_* f \, \mathrm{d}V$. Scalar uniqueness gives $c_* = c$. Every subsequential scalar limit as $\lambda \downarrow 0$ is therefore c, so $c_\lambda \to c$. The stated density convergence follows.

Finally Theorem 13.16.1 gives uniform local C^α bounds for every α below its endpoint. Their uniform limit inherits those bounds by pointwise convergence. For $0 < \beta < \alpha$, uniform convergence and the elementary interpolation inequality on smaller coordinate balls give

$$[v_{\lambda_j} - v]_{C^\beta} \le C \|v_{\lambda_j} - v\|_\infty^{1-\beta/\alpha} [v_{\lambda_j} - v]_{C^\alpha}^{\beta/\alpha} \longrightarrow 0.$$

Use slightly larger compact neighborhoods for these estimates. The uniformly convergent subsequence already supplied by compactness works on every compact subset and for each sub-endpoint exponent; no further choice depending on that exponent is needed. □

Corollary 13.16.1 *Suppose a group G of biholomorphisms preserves θ and the measure $f \, \mathrm{d}V$. Every v_λ and every cluster limit as $\lambda \downarrow 0$ is G-invariant. If the normalized zero-exponent equation has only one G-invariant solution of minimal singularities, then the whole family v_λ converges to it, uniformly in differences on Ω and in each local C^β topology from the proposition.*

Proof For $g \in G$, pullback preserves θ-plurisubharmonicity and $V_\theta \circ g = V_\theta$, by the envelope definition. It also preserves the positive-exponent equation. Its uniqueness gives $u_\lambda \circ g = u_\lambda$, and normalization preserves this equality. Pullback by a fixed smooth biholomorphism is continuous on $L^1(X)$, since its Jacobian relative to the fixed smooth volume is bounded. Every cluster limit is therefore invariant almost everywhere, hence in canonical representatives. This argument does not require g to preserve the chosen barrier or Ω.

Under the stated uniqueness within the invariant class, every cluster limit is the same potential. Relative compactness then implies convergence of the whole family in the uniform-difference metric: a sequence staying a fixed positive distance from that potential would have a convergent subsequence, a contradiction. The interpolation argument above upgrades it to the asserted local C^β convergence. □

The proposition gives compactness and identifies every cluster equation. The symmetry criterion supplies one way to identify the potential. For the full minimal-type family, Theorem 14.11.1 will identify the selected limit as the greatest critical subsolution normalized by the scalar's supporting probabilities, even when the critical equation has several normalized solutions.

13.17 Interior smoothness on the regular locus

A smooth strictly positive density on the whole regular locus Ω of the fixed strict barrier gives interior smoothness of a minimal-type solution once local Hölder continuity is known there. The proper gap from Lemma 1.19.4 forces the difference between the solution and the barrier to attain a minimum on any nonregular set. Near that minimum, smooth Dirichlet approximants have polynomial trace growth; localization improves this growth until a uniform trace bound yields elliptic regularity.

Theorem 13.17.1 *Suppose $\varphi \in \mathrm{PSH}(X, \theta)$ has minimal singularities and is locally Hölder continuous on Ω. Let $c > 0$ and let f be smooth and strictly positive on Ω. Assume*

$$(\theta + \mathrm{dd^c}\varphi)^n = c f \omega^n \quad \text{on } \Omega.$$

Then $\varphi \in C^\infty(\Omega)$ and $\theta + \mathrm{dd^c}\varphi > 0$ there.

Subtracting a constant, we may assume $\sup_X \varphi = 0$ and choose $C > 0$ such that $V_\theta - C \le \varphi \le V_\theta$. A sufficient condition for the Hölder premise is $0 \le f \in L^p(X, \omega^n)$ for some $p > 1$, by Theorem 13.16.1. The theorem also applies when local Hölder continuity is obtained by another argument.

All coordinate balls have compact closure in a larger chart.

13.17.1 Local comparison

Local comparison identifies the given continuous potential with limits of smooth Dirichlet solutions. Its proof uses the mixed inequalities of Lemmas 1.7.3 and 1.7.4 and the closed-current comparison formula (1.4).

In the local arguments, put $\beta = \mathrm{dd^c}|z|^2$ in Euclidean coordinates and take coefficient matrices relative to β. Write $\partial_i = \partial/\partial z_i$, $\partial_{\bar{i}} = \partial/\partial \bar{z}_i$ and $\Delta_\beta = \mathrm{tr}_\beta \mathrm{dd^c}$. Repeated indices run from 1 to n; I denotes the identity matrix. The smooth real background is denoted by ϑ.

Proposition 13.17.1 *Let $D \Subset \mathbb{C}^n$ be a bounded domain and let ϑ be a smooth real $(1,1)$-form near $\overline{D}$. Suppose $u, v \in C(\overline{D})$ are ϑ-plurisubharmonic,*

$$(\vartheta + \mathrm{dd^c}u)^n \ge F\beta^n, \qquad (\vartheta + \mathrm{dd^c}v)^n \le F\beta^n,$$

where $F > 0$ is continuous on D. If $u \le v$ on ∂D, then $u \le v$ in D.

Proof We first prove the lower-test property for v. Suppose that a C^2 function q touches v from below at p and, at p,

$$\vartheta + \mathrm{dd^c}q > 0, \qquad \det_\beta(\vartheta + \mathrm{dd^c}q) > F.$$

Choose a sufficiently small $a > 0$. After replacing q by $q - a|z-p|^2$ and shrinking a ball $B_r(p)$, there are fixed $\lambda, \eta > 0$ such that

$$A := \vartheta + \mathrm{dd^c} q \geq \lambda\beta, \qquad A^n \geq (F+\eta)\beta^n.$$

Put $\kappa = \sup_{B_r} \|\vartheta - \vartheta(p)\|_\beta$ and $B_0 = \vartheta(p) + \kappa\beta$. Thus $S = B_0 - \vartheta$ satisfies $0 \leq S \leq 2\kappa\beta$. For small $s > 0$ the set $E = \{v < q + s\}$ is nonempty and relatively compact in B_r. If Q_0 is a quadratic potential for B_0, then

$$P = \mathrm{dd^c}(v + Q_0) = \vartheta + \mathrm{dd^c} v + S, \qquad Q = \mathrm{dd^c}(q + s + Q_0) = A + S \geq \lambda\beta$$

are closed positive Hessians. Applying (1.4), expanding P, and using $S \leq 2\kappa\beta \leq 2\kappa\lambda^{-1}Q$, we obtain

$$\int_E P^n \leq \int_E F\beta^n + e_\kappa \int_E P^n, \qquad e_\kappa = \left(1 + \frac{2\kappa}{\lambda}\right)^n - 1.$$

On the other hand, comparison gives

$$\int_E P^n \geq \int_E Q^n \geq \int_E (F+\eta)\beta^n.$$

If M bounds F on the ball, then $(1 - e_\kappa)\eta \leq e_\kappa M$, which is impossible after r is decreased because $e_\kappa \to 0$. Thus no such lower test exists.

Suppose now that $\max_D(u - v) > 0$. Let χ_δ be a nonnegative smooth mollifier of mass one and radius $\delta > 0$. For small $t > 0$ and $C \geq \sup_D |z|^2$, set

$$u_t = u + t(|z|^2 - C).$$

Then $u_t \leq v$ on ∂D, the positive maximum persists, and $T_t = \vartheta + \mathrm{dd^c} u_t \geq t\beta$. Lemma 1.7.3 gives

$$T_t^n \geq (F^{1/n} + t)^n \beta^n.$$

Choose a relatively compact neighborhood W of the positive maximum set. By Lemma 1.7.4,

$$(T_t * \chi_\delta)^n \geq \left((F^{1/n} + t) * \chi_\delta\right)^n \beta^n.$$

Since

$$\vartheta + \mathrm{dd^c}(u_t * \chi_\delta) = T_t * \chi_\delta + (\vartheta - \vartheta * \chi_\delta),$$

the convolution satisfies $T_t * \chi_\delta \geq t\beta$. If $\epsilon_\delta = \|\vartheta - \vartheta * \chi_\delta\|_{C^0(W)}$, then

$$\vartheta + \mathrm{dd^c}(u_t * \chi_\delta) \geq (1 - \epsilon_\delta/t)(T_t * \chi_\delta).$$

For fixed $t > 0$, uniform convergence of the convolved density and $\epsilon_\delta \to 0$ show that the left-hand side is positive and has determinant strictly larger than F for small δ. Uniform convergence preserves an interior maximum of $u_t * \chi_\delta - v$. Subtracting its value produces a forbidden smooth lower test for v, a contradiction. □

13.17.2 Bounded traces and smoothness

A bounded trace of a positive current bounds all its coefficient measures. The equation then gives a lower eigenvalue bound, so it becomes uniformly elliptic. We record this criterion before constructing approximants whose traces can be controlled.

Proposition 13.17.2 *Let $B \subset \mathbb{C}^n$ be a coordinate ball, let ϑ be a smooth real $(1,1)$-form there and let $F \in C^\infty(B)$ be strictly positive. Suppose u is continuous, $T = \vartheta + \mathrm{dd^c} u \geq 0$, and*

$$T^n = F\beta^n. \tag{13.14}$$

At a point p, the following are locally equivalent:

(1) *u is smooth;*
(2) *$T \geq \delta\beta$ for some $\delta > 0$;*
(3) *$\mathrm{tr}_\beta T$ has a locally bounded measure density.*

Proof Choose C so that $v = u + C|z|^2$ is plurisubharmonic, and use nonnegative radial mollifiers for which $v_\eta \downarrow v$. The smooth matrix convolutions $T_\eta = T * \chi_\eta$ can be written as

$$T_\eta = (\vartheta * \chi_\eta - C\beta) + \mathrm{dd^c} v_\eta.$$

Bedford–Taylor monotone continuity and smooth convergence of the background imply $T_\eta^k \to T^k$ for every k.

Assume (2). For every Hermitian matrix $M \geq \delta I$,

$$\det M \geq \delta^{n-1}\big(\mathrm{tr}\, M - (n-1)\delta\big).$$

Apply this to the regularizations and pass to currents. Equation (13.14) gives

$$\mathrm{tr}_\beta T \leq \delta^{1-n} F + (n-1)\delta,$$

which is (3).

Conversely assume (3), say $\mathrm{tr}_\beta T \leq L$. Positivity makes every coefficient of the matrix-valued current T a bounded density. Hence the Euclidean Laplacian of u is locally bounded, and interior Poisson estimates give $u \in W^{2,q}_{\mathrm{loc}}$ for every finite q. Mollification converges in $W^{2,q}$; taking $q > 2n$ identifies the bounded-potential determinant with the ordinary almost-everywhere determinant. Thus $\det_\beta T = F$ almost everywhere. On a smaller ball with $F \geq m > 0$, all eigenvalues of T are at most L and their product is at least m, so

$$\frac{m}{L^{n-1}} I \leq T \leq LI$$

almost everywhere.

It remains to regularize a strong solution. Let P be the real linear map from real symmetric $2n \times 2n$ Hessian matrices to Hermitian $n \times n$ matrices, sending D^2u to the coefficient matrix of $\mathrm{dd^c}u$ relative to β, and let $\Theta(x)$ be the matrix of ϑ. With $\delta = mL^{1-n}$, define the concave extension

$$G(x, M) = \inf_{\delta I \le A \le LI} \left\{ \log\det A + \operatorname{tr}\left[A^{-1}\big(\Theta(x) + P(M) - A\big)\right]\right\} - \log F(x).$$

For fixed A, write $\mathcal{E}_A(x, M)$ for the expression in braces minus $\log F(x)$. The tangent-plane formula for log det gives

$$G(x, D^2u) = 0 \quad \text{almost everywhere.}$$

The operator G is concave and uniformly elliptic on real symmetric matrices. The affine operators $\mathcal{E}_A$ have constant principal matrices with eigenvalues bounded above and below in terms of δ and L. Their x-dependence is smooth uniformly in A, so G has a uniform local Lipschitz modulus in x.

We verify that the strong solution is a viscosity solution. Suppose that a smooth function p touches u from above at x_0 and that $G(x_0, D^2p) < 0$. Some fixed matrix A then satisfies $\mathcal{E}_A(x, D^2p) < -\eta$ on a small ball. Since $G(x, D^2u) = 0$ almost everywhere, $\mathcal{E}_A(x, D^2u) \ge 0$ there. Thus $u - p$ has an interior maximum and satisfies

$$\operatorname{tr}\Big(A^{-1}P(D^2(u-p))\Big) \ge \eta \quad \text{almost everywhere.}$$

After a small quadratic perturbation makes the contact strict, this contradicts the Aleksandrov maximum principle [GT01, Chapter 9, Theorem 9.1]. If instead p touches u from below and $G(x_0, D^2p) > 0$, then $\mathcal{E}_A(x, D^2p) > \eta$ for every admissible A on a small ball. At almost every point choose an admissible $A(x)$ with $\mathcal{E}_{A(x)}(x, D^2u) < \eta/2$. A measurable choice is obtained by taking the first qualifying element of a countable dense family. The function $p - u$ then violates the same maximum principle for the measurable uniformly elliptic coefficients $A(x)^{-1}$. Hence u is a viscosity solution of $G(x, D^2u) = 0$.

The classical variable-coefficient Evans–Krylov theorem gives $u \in C^{2,\alpha}_{\mathrm{loc}}$ for some $0 < \alpha < 1$. The original equation is now uniformly elliptic with smooth data. Difference quotients satisfy linear equations whose principal coefficients are averages of inverse matrices along positive definite segments. Interior Schauder estimates first give $u \in C^{3,\alpha}_{\mathrm{loc}}$ and then, by iteration, $u \in C^\infty$. This proves (3)⇒(1). Finally, (1) and $F > 0$ imply that the smooth semipositive matrix T is positive definite, giving (2) and (3). □

13.17.3 Smooth exact approximants

The smooth Dirichlet theorem in Lemma 13.6.1 also applies to a smooth positive density F: a smooth extension of the boundary value plus a large quadratic function is a strict subsolution. We need its dependence on the boundary data explicitly. The next estimates make that dependence polynomial, so Hölder approximation of the boundary values gives a polynomial rate in both the uniform error and the trace.

The balls are centered at the origin. On their boundaries, ν is the outward Euclidean unit normal and $\nabla_{\tan}$ denotes tangential differentiation. For a positive form T, covector inner products with subscript T use its inverse matrix.

Lemma 13.17.1 *Let B be a coordinate ball, let ϑ and $F > 0$ be smooth near $\overline{B}$, and let q be smooth with $R = \vartheta + \mathrm{dd^c} q \geq \varepsilon\beta$ for a fixed $\varepsilon > 0$. Fix a finite $L \geq 0$. If $u \in C^\infty(B)$ satisfies*

$$T = \vartheta + \mathrm{dd^c} u > 0, \qquad T^n = F\beta^n, \qquad |u - q| \leq L,$$

put $S = \mathrm{tr}_\beta T$ and $v = u - q$. For $A > 0$, define

$$Q_A = \log S - Av + \mathrm{e}^{-v}.$$

There are A_0, C, depending only on the fixed data and L, such that at every interior local maximum of Q_A, with $A \geq A_0$,

$$S \leq C\sqrt{A}. \tag{13.15}$$

No gradient bound for u is assumed.

Proof At the point in question make a constant unitary change of coordinates so that the matrix of T is diagonal, with eigenvalues λ_i. Write

$$\Delta_T = \sum_i \lambda_i^{-1} \partial_i \partial_{\bar{i}}, \qquad H = \mathrm{tr}_T\, \beta.$$

Differentiating $\log\det T = \log F$ and commuting derivatives gives

$$\Delta_T S = \Delta_\beta \log F + P + E_0, \quad P = \sum_{k,i,j} \frac{|\partial_k T_{i\bar{j}}|^2}{\lambda_i \lambda_j}, \quad |E_0| \leq C_0 H. \tag{13.16}$$

The non-closedness of ϑ enters through the bounded vector

$$E_p = \sum_k (\partial_p \vartheta_{k\bar{k}} - \partial_k \vartheta_{p\bar{k}}).$$

Since $\partial_p S - E_p = \sum_k \partial_k T_{p\bar{k}}$, weighted Cauchy–Schwarz gives $|\partial S - E|_T^2 \le SP$. From (13.16),

$$\Delta_T \log S \ge -\frac{C_0(1+H)}{S} - \frac{2}{S}\operatorname{Re}\langle \partial \log S, E\rangle_T. \tag{13.17}$$

Set $g_A(s) = -As + \mathrm{e}^{-s}$. At a local maximum of Q_A, $\partial \log S = -g_A'(v)\partial v$ and $\Delta_T v = n - \operatorname{tr}_T R$. Combining these identities with (13.17), then completing the square in ∂v, yields

$$0 \ge \left[\varepsilon A - \frac{C_2}{S} - \frac{C_2(A+1)^2}{S^2}\right] H - C_3(A+1) - \frac{C_2}{S}. \tag{13.18}$$

Here $|E|_T^2 \le CH$ and the bounds for v have been used. If $S \ge K\sqrt{A}$ and K is large, the bracket is at least $\varepsilon A/2$. Thus $H \le C$ independently of A and u. The determinant upper bound then gives $S \le C'$. Enlarging K makes this incompatible with $S \ge K\sqrt{A}$, proving (13.15). □

We shall also use the following gradient-controlled variant. If $|\nabla(u-q)|_\beta$ is bounded, then at an interior maximum of $\log S - A(u-q)$ the trace is bounded independently of A for all large A. Indeed, at the maximum $\partial \log S = A\partial(u-q)$, and (13.17) gives

$$0 \ge \left[A\varepsilon - \frac{C}{S} - \frac{CA}{S}\right] H - nA - \frac{C}{S}.$$

For S larger than a fixed constant the bracket is at least $A\varepsilon/2$; the determinant again controls S.

A polynomial boundary trace estimate

Proposition 13.17.3 *Let B be the unit ball. Fix smooth ϑ, smooth $F > 0$ near $\overline{B}$, and $M < \infty$. There are $C, N < \infty$, depending only on these fixed data, such that every smooth solution u with boundary value $g \in C^\infty(\partial B)$ of*

$$T = \vartheta + \mathrm{dd^c} u > 0, \qquad T^n = F\beta^n, \qquad \|u\|_{L^\infty(B)} \le M, \qquad u|_{\partial B} = g,$$

satisfies

$$\sup_{\overline{B}} \operatorname{tr}_\beta T \le C\big(1 + \|g\|_{C^4(\partial B)}\big)^N. \tag{13.19}$$

Proof Fix a sufficiently large constant $K > 0$, depending only on the fixed data. Put $L = 1 + \|g\|_{C^4}$. Extend g to G on $\overline{B}$ with $\|G\|_{C^4} \le CL$. For $A = CL$ large enough,

$$\underline{u} = G + A(|z|^2 - 1), \qquad \underline{T} = \vartheta + \mathrm{dd^c}\underline{u} \ge K\beta, \qquad \underline{T}^n \ge F\beta^n.$$

Proposition 13.17.1 gives $\underline{u} \le u$. If h solves $\Delta_\beta h = -C_\vartheta$ with $h = g$ on ∂B and $C_\vartheta \ge \sup \operatorname{tr}_\beta \vartheta$, then $u \le h$. Boundary differentiation of these barriers gives

$$|\partial_\nu u| + |\nabla_{\tan} u| \le CL \quad \text{on } \partial B. \tag{13.20}$$

Let $H = \operatorname{tr}_T \beta$, $\mathcal{L} = \operatorname{tr}_T \mathrm{dd^c}$, and $w = u - \underline{u} \ge 0$. Taking K fixed and sufficiently large gives

$$\mathcal{L}w = n - \operatorname{tr}_T \underline{T} \le -(1 + H). \tag{13.21}$$

Let $E = \sum_k (z_k \partial_{z_k} + \bar{z}_k \partial_{\bar{z}_k})$ be the real radial field. Write Θ for the coefficient matrix of ϑ. Differentiating $\log \det T = \log F$ and commuting E with complex second derivatives yields

$$\mathcal{L}(Eu) = E(\log F) + 2n - \operatorname{tr}_T(E\Theta + 2\Theta),$$

Hence $|\mathcal{L}(Eu)| \le C(1+H)$. The same estimate holds for Zu when Z is an infinitesimal unitary rotation. Applying the minimum principle to $CL + C'w \pm Yu$, with $Y = E$ or Z, gives

$$|Eu| + |Zu| \le CL \quad \text{on } B. \tag{13.22}$$

For a rotation Z, the function $R_Z = Z(u - G)$ vanishes on ∂B and satisfies $|\mathcal{L}R_Z| \le CL(1 + H)$. The barriers $CLw \pm R_Z$ and (13.21) imply $|\partial_\nu R_Z| \le CL^2$ on the boundary. Rotations span its real tangent space, so all real mixed normal–tangential second derivatives are $O(L^2)$. Pure tangential second derivatives are $O(L)$ by differentiating $u = g$ twice and using (13.20). Thus all complex Hessian entries except the complex normal–normal entry are $O(L^2)$.

It remains to bound the tangential block from below. For $n \ge 2$, let

$$\lambda_0 = \min\{T_p(\xi, \bar{\xi}) : p \in \partial B,\ \xi \in T_p^{1,0}(\partial B),\ |\xi|_\beta = 1\}.$$

Choose a minimizing pair (p, ξ) and extend ξ to a smooth unit tangential field near p, with uniformly bounded derivatives. On the sphere, any smooth function vanishing there satisfies

$$\mathrm{dd^c}(u - G)(\xi, \bar{\xi}) = \frac{1}{2} E(u - G).$$

Define

$$a(z) = (\vartheta + \mathrm{dd^c} G)_z(\xi(z), \overline{\xi(z)}) - \frac{1}{2} EG(z), \qquad P = a + \frac{1}{2} Eu - \lambda_0.$$

Then $P \ge 0$ on the nearby spherical boundary and $P(p) = 0$. The boundary first-derivative bound also gives $\lambda_0 \le CL$ by the displayed tangential identity. Consequently

$$|P| \le CL, \qquad |\mathcal{L}P| \le CL(1 + H).$$

On a fixed cap $D = B \cap B_r(p)$ consider

$$\Psi = P + C_1 Lw + C_2 L|z - p|^2.$$

Choose C_2 so that $\Psi \ge 0$ on the interior spherical part of ∂D, and then C_1 so that $\mathcal{L}\Psi \le 0$ by (13.21). The minimum principle yields $\Psi \ge 0$. At p the normal derivative

of $|z-p|^2$ vanishes, and therefore

$$\partial_\nu P + C_1 L \partial_\nu w \le 0.$$

Equations (13.20)–(13.22) now give $D^2u(\nu,\nu)(p) \le CL^2$, hence $\operatorname{tr}_\beta T(p) \le CL^2$. Positivity and $\det_\beta T = F$ imply

$$\lambda_0 \ge cL^{-2(n-1)}. \tag{13.23}$$

At any boundary point write

$$T = \begin{pmatrix} B_0 & b \\ b^* & d \end{pmatrix}$$

in tangential–normal blocks, with B_0 an $(n-1)\times(n-1)$ Hermitian matrix, $b \in \mathbb{C}^{n-1}$ and $d \in \mathbb{R}$. We have $|b| \le CL^2$ and $B_0 \ge \lambda_0 I$. The determinant equation and the Schur complement give

$$d = \frac{F}{\det B_0} + b^* B_0^{-1} b \le (\sup F)\lambda_0^{-(n-1)} + CL^4\lambda_0^{-1},$$

which is polynomial in L. For $n = 1$, the trace is F and this argument is unnecessary.

Finally, choose a fixed quadratic q with $\vartheta + \mathrm{dd^c} q \ge \varepsilon\beta$. Apply Lemma 13.17.1 with one fixed A. At an interior maximum of its auxiliary function the trace is uniformly bounded, while at a boundary maximum the preceding estimate applies. Since the remaining terms are uniformly bounded, (13.19) follows. □

Exact approximation

Apply these estimates to the boundary values of φ on a ball in Ω.

Proposition 13.17.4 *Let $B \Subset \Omega$ be a coordinate ball, identified with $\{|z| < r_0\}$, with closure in a larger coordinate chart, and put*

$$F = \frac{c f \omega^n}{\beta^n}.$$

There are $a > 0$, $b, C < \infty$, and $u_j \in C^\infty(\overline{B})$ such that

$$\theta + \mathrm{dd^c} u_j > 0, \qquad (\theta + \mathrm{dd^c} u_j)^n = F\beta^n, \tag{13.24}$$

and

$$\|u_j - \varphi\|_{L^\infty(B)} \le C j^{-a}, \qquad \sup_{\overline{B}} \mathrm{tr}_\beta(\theta + \mathrm{dd^c} u_j) \le C j^b. \tag{13.25}$$

Proof Local Hölder continuity of φ gives smooth boundary approximations g_j with

$$\|g_j - \varphi\|_{L^\infty(\partial B)} \le C j^{-a}, \qquad \|g_j\|_{C^4(\partial B)} \le C j^4.$$

Choose a fixed quadratic q with $\eta = \theta + \mathrm{dd^c} q > 0$ near $\overline{B}$. Extend g_j smoothly to G_j. For a sufficiently large, polynomially bounded A_j, the function

$$\underline{w}_j = G_j - q + A_j(|z|^2 - r_0^2)$$

has boundary value $g_j - q$ and satisfies

$$\eta + \mathrm{dd^c} \underline{w}_j = \theta + \mathrm{dd^c} G_j + A_j \beta > 0, \qquad (\eta + \mathrm{dd^c} \underline{w}_j)^n \ge F\beta^n.$$

Apply the smooth Dirichlet theorem just recalled with density $H = F\beta^n/\eta^n$ to obtain a solution w_j for the Hermitian background η; put $u_j = w_j + q$. Comparing u_j with $\varphi \pm \|g_j - \varphi\|_\infty$ by Proposition 13.17.1 proves the first bound in (13.25). The functions u_j are therefore uniformly bounded. Proposition 13.17.3 and the polynomial C^4 boundary norm give the second bound. □

13.17.4 Localization near the nonregular set

A quadratic gap along the nonregular set suppresses the boundary contribution to a maximum-principle estimate. Choosing the coefficient of that gap proportional to $\log j$ improves polynomial trace growth to growth smaller than every positive power. Combining this improvement with the approximation rate gives uniform $C^{1,\alpha}$ bounds.

Distances and norms below are Euclidean. A smooth exact solution for (ϑ, F) satisfies $\vartheta + \mathrm{dd^c} u_j > 0$ and $(\vartheta + \mathrm{dd^c} u_j)^n = F\beta^n$.

Proposition 13.17.5 *For $R > 0$, let $B = B_R(0)$, let ϑ and $F > 0$ be smooth near $\overline{B}$, and let $u \in C(\overline{B})$. Suppose q is smooth and $\vartheta + \mathrm{dd^c} q \geq \varepsilon\beta$ for some $\varepsilon > 0$. Put $v = u - q$ and $m = v(0)$, and fix $\kappa > 0$. Let $\mathcal{S} \subset \overline{B}$ be closed, contain 0, and satisfy*

$$v(0) = m, \qquad v(z) \geq m + \kappa|z|^2 \quad (z \in \mathcal{S}). \tag{13.26}$$

Assume that smooth exact solutions u_j, $j \geq 2$, satisfy, for fixed constants $a > 0$, $b \geq 0$ and $C > 0$,

$$\|u_j - u\|_\infty \leq C j^{-a}, \tag{13.27}$$

$$\sup_{\overline{B}} \mathrm{tr}_\beta(\vartheta + \mathrm{dd^c} u_j) \leq C j^b, \tag{13.28}$$

$$\sup_j \sup_K \mathrm{tr}_\beta(\vartheta + \mathrm{dd^c} u_j) < \infty \quad \textit{for every } K \Subset \overline{B} \setminus \mathcal{S}. \tag{13.29}$$

Then for every $\gamma > 0$ there are a neighborhood U_γ of 0 and C_γ such that

$$\mathrm{tr}_\beta(\vartheta + \mathrm{dd^c} u_j) \leq C_\gamma j^\gamma \quad \textit{on } U_\gamma. \tag{13.30}$$

For every $0 < \alpha < 1$, there is also a neighborhood U of 0 on which $u_j \to u$ in $C^{1,\alpha}$. In particular the approximants are uniformly bounded in that norm and $u \in C^{1,\alpha}(U)$.

Proof Write $e_j = C j^{-a}$, fix $d = \kappa R^2/2$, and choose $D > (b+1)/d$. Choose $\eta > 0$ so that $2D\eta < \gamma/2$. For small $\delta > 0$, the neighborhood

$$W_\delta = \{z \in B : \mathrm{dist}(z, \mathcal{S}) < \delta\}$$

contains 0, has $v \geq m - \eta$ on its closure, and satisfies $v \geq m + d$ on $\partial B \cap \overline{W}_\delta$ with the fixed d above. This follows from (13.26), continuity, and compactness. The other boundary portion is a compact subset of $\overline{B} \setminus \mathcal{S}$, where (13.29) applies.

Set $A_j = D \log j$. Apply the maximum principle on $\overline{W}_\delta$ to

$$Q_j = \log \mathrm{tr}_\beta(\vartheta + \mathrm{dd^c} u_j) - A_j(u_j - q) + \mathrm{e}^{-(u_j - q)}.$$

At an interior maximum Lemma 13.17.1 gives $\log \mathrm{tr}_\beta(\vartheta + \mathrm{dd^c} u_j) \leq C + \frac{1}{2}\log A_j$. On the boundary away from $\mathcal{S}$ there is a fixed trace bound. On the outer boundary, (13.28) and the gap d give

$$Q_j \leq C + b \log j - A_j(m + d - e_j),$$

which is suppressed by the choice of D. Hence throughout W_δ,

$$Q_j \leq C_\delta + \tfrac{1}{2}\log A_j - A_j(m - \eta - e_j).$$

On a small ball about 0 where $v \leq m + \eta$, this implies

$$\log \mathrm{tr}_\beta(\vartheta + \mathrm{dd^c} u_j) \leq C_\delta + \tfrac{1}{2}\log A_j + A_j(2\eta + 2e_j).$$

By the choice of η, the logarithmic factor and $A_j e_j$ are absorbed into the remaining power. This proves (13.30).

On a smaller ball, the real Laplacian of u_j is bounded by Cj^γ. Interior Poisson estimates and Sobolev embedding give, for every finite q_0 and every $\sigma < 1$ obtainable by taking q_0 large,

$$\|u_j\|_{W^{2,q_0}} + \|u_j\|_{C^{1,\sigma}} \le Cj^\gamma.$$

For $d_k = u_{2^{k+1}} - u_{2^k}$,

$$\|d_k\|_\infty \le C2^{-ka}, \qquad \|d_k\|_{C^{1,\sigma}} \le C2^{k\gamma}.$$

If $\alpha < \sigma$ and $t = (1+\alpha)/(1+\sigma)$, Hölder interpolation gives

$$\|d_k\|_{C^{1,\alpha}} \le C2^{k[-a(1-t)+\gamma t]}.$$

Since γ may be arbitrarily small, the exponent can be made negative. The dyadic series converges in $C^{1,\alpha}$ to its uniform limit u. If $2^k \le j < 2^{k+1}$, interpolation applied to $u_j - u_{2^k}$ gives the same decaying bound. Hence the full sequence converges to u in $C^{1,\alpha}$ on a smaller ball. □

Proposition 13.17.6 *Suppose u_j solve the same smooth equation as in Proposition 13.17.5, satisfy (13.27)–(13.28), and u is smooth with $\vartheta + \mathrm{dd^c} u > 0$ on an open set $O \subset B$. Then, for every $K \Subset O$,*

$$\sup_j \sup_K \mathrm{tr}_\beta(\vartheta + \mathrm{dd^c} u_j) < \infty. \tag{13.31}$$

Proof For p in a fixed compact subset of O, take a uniformly sized ball $B_r(p) \Subset O$ and set

$$q_p(z) = u(z) - \kappa|z-p|^2$$

with $\kappa > 0$ small, so $\vartheta + \mathrm{dd^c} q_p \ge \varepsilon\beta$. If $e_j = Cj^{-a}$ and $v_{j,p} = u_j - q_p$, then

$$v_{j,p} \ge -e_j, \quad v_{j,p}(p) \le e_j, \quad v_{j,p} \ge \kappa r^2 - e_j \text{ on } \partial B_r(p).$$

With $A_j = D\log j$ and $D\kappa r^2 > b+1$, the same maximum argument as above gives

$$\mathrm{tr}_\beta(\vartheta + \mathrm{dd^c} u_j)(p) \le C\sqrt{\log j}. \tag{13.32}$$

The constants are uniform on compact subsets of O.

Put $T_j = \vartheta + \mathrm{dd^c} u_j$ and $w_j = u_j - u$. On slightly larger compact sets, (13.32) bounds $|\Delta w_j|$ by $C\sqrt{\log j}$. Poisson estimates and Sobolev embedding give

$$\|w_j\|_{C^{1,\sigma}} \le C_\sigma\sqrt{\log j}, \qquad \sigma < 1.$$

Interpolating with $\|w_j\|_\infty \le Cj^{-a}$ shows $\nabla w_j \to 0$ uniformly on smaller compact sets. The gradients of $v_{j,p}$ are therefore uniformly bounded. Repeat the maximum argument with $\log \mathrm{tr}_\beta T_j - A_j v_{j,p}$ and use the gradient-controlled version of

Lemma 13.17.1. Its interior trace bound is now independent of A_j, while the boundary term is suppressed as before. Evaluation at p gives

$$\operatorname{tr}_\beta T_j(p) \le C \exp(2A_j e_j) \le C,$$

proving (13.31). □

13.17.5 Derivative estimates and bounded traces

To turn the $C^{1,\alpha}$ bounds into bounded traces, we need a derivative estimate whose dependence on ellipticity is polynomial. The Chern-connection calculation below retains the torsion of the background. An averaging estimate for the trace then closes a nonlinear iteration when α is sufficiently close to one.

Here B is the unit ball, $B_r(p)$ is a Euclidean ball, and ∂g denotes ordinary coordinate derivatives of its coefficient matrix. Tensor derivatives in the proof use the specified Chern connection.

Proposition 13.17.7 *Fix smooth ϑ and smooth $F > 0$ near the closed unit ball. There are $C, N < \infty$, depending only on finitely many derivatives of the fixed data, such that any smooth solution*

$$g = \vartheta + \mathrm{dd^c} u, \qquad g^n = F\beta^n, \qquad M^{-1}\beta \le g \le M\beta, \qquad M \ge 1,$$

satisfies

$$\sup_{B_{1/2}} |\partial g|_\beta \le C M^N. \tag{13.33}$$

If only $\sup_B \operatorname{tr}_\beta g \le L$ is given, the same bound holds with a fixed power of $1 + L$. On a ball of radius $r \le 1$,

$$\sup_{B_{r/2}} |\nabla \operatorname{tr}_\beta g|_\beta \le C r^{-1} (1 + L)^N. \tag{13.34}$$

Proof Use the flat connection of β and the Chern connection ∇ of g. Write $g^{k\bar{\ell}}$ for inverse matrix coefficients, and $\bar{\nabla}$ for covariant differentiation in $(0, 1)$ directions. Unsubscripted tensor norms and inner products in this calculation use g. Their connection difference is

$$\Psi^k_{ij} = g^{k\bar{\ell}} \partial_i g_{j\bar{\ell}}, \qquad \mathcal{R} = |\Psi|^2_g.$$

The lowered torsion tensor is determined by the fixed background. Put

$$C_{ij\bar{\ell}} = \partial_i \vartheta_{j\bar{\ell}} - \partial_j \vartheta_{i\bar{\ell}}, \qquad \mathcal{T}^k_{ij} = g^{k\bar{\ell}} C_{ij\bar{\ell}}. \tag{13.35}$$

Consequently, for a fixed integer N,

$$\begin{aligned} |\mathcal{T}| &\le CM^N, \\ |\nabla\mathcal{T}| + |\bar\nabla\mathcal{T}| &\le CM^N(1+\sqrt{\mathcal{R}}), \\ |\nabla^2\mathcal{T}| + |\nabla\bar\nabla\mathcal{T}| + |\bar\nabla^2\mathcal{T}| &\le CM^N(1+\mathcal{R}+|\nabla\Psi|+|\bar\nabla\Psi|). \end{aligned} \tag{13.36}$$

Every factor here is either a fixed derivative of ϑ, a connection factor Ψ, or a factor of $g^{\pm1}$; this explains the polynomial dependence.

For tensor fields, use $\Delta_g = g^{p\bar q}\nabla_p\nabla_{\bar q}$. We record the curvature identities used in the estimate. With

$$\mathrm{Rm}^k{}_{i\bar jl} = -\partial_{\bar j}\Psi^k_{il}, \qquad \mathrm{Rm}_{i\bar jl\bar k} = g_{m\bar k}\,\mathrm{Rm}^m{}_{i\bar jl},$$

set

$$\mathrm{Ric}^{(2)}_{l\bar k} = g^{i\bar j}\,\mathrm{Rm}_{i\bar jl\bar k}, \qquad \mathrm{Ric}^{(1)}_{l\bar k} = g^{i\bar j}\,\mathrm{Rm}_{l\bar ki\bar j} = -\partial_l\partial_{\bar k}\log F.$$

The two curvature interchanges are

$$\begin{aligned} \mathrm{Rm}_{i\bar jl\bar k} - \mathrm{Rm}_{l\bar ji\bar k} &= -\nabla_{\bar j}C_{il\bar k}, \\ \mathrm{Rm}_{i\bar jl\bar k} - \mathrm{Rm}_{i\bar kl\bar j} &= -\nabla_i\overline{C}_{\bar jkl}. \end{aligned}$$

They follow by differentiating the torsion identity and its conjugate. Successive interchange and contraction give

$$\mathrm{Ric}^{(2)}_{l\bar k} = \mathrm{Ric}^{(1)}_{l\bar k} - g^{i\bar j}\big(\nabla_l\overline{C}_{\bar jki} + \nabla_{\bar j}C_{il\bar k}\big). \tag{13.37}$$

Metric compatibility, (13.36), and the fixed smooth formula for $\mathrm{Ric}^{(1)}$ imply

$$|\,\mathrm{Ric}^{(2)}\,| \le CM^N(1+\sqrt{\mathcal{R}}), \qquad |\nabla\,\mathrm{Ric}^{(2)}\,| \le CM^N(1+\mathcal{R}+|\nabla\Psi|+|\bar\nabla\Psi|). \tag{13.38}$$

Moreover,

$$\nabla_{\bar q}\Psi^k_{ij} = -\,\mathrm{Rm}^k{}_{i\bar qj},$$

and the Chern–Bianchi identity, including its torsion term, is

$$\nabla_p\,\mathrm{Rm}^k{}_{i\bar qj} - \nabla_i\,\mathrm{Rm}^k{}_{p\bar qj} = -\mathcal{T}^r_{pi}\,\mathrm{Rm}^k{}_{r\bar qj}.$$

Consequently,

$$\Delta_g\Psi^k_{ij} = -\nabla_i(\mathrm{Ric}^{(2)})^k{}_j + g^{p\bar q}\mathcal{T}^r_{pi}\,\mathrm{Rm}^k{}_{r\bar qj}. \tag{13.39}$$

If $A^k{}_l = (\mathrm{Ric}^{(2)})^k{}_l$, define its action on a tensor of type $(1,2)$ by

$$(A\cdot\Psi)^k_{ij} = A^k{}_l\Psi^l_{ij} - A^l{}_i\Psi^k_{lj} - A^l{}_j\Psi^k_{il}.$$

Differentiating the squared norm and using metric compatibility gives

$$\Delta_g\mathcal{R} = |\nabla\Psi|^2 + |\bar\nabla\Psi|^2 + 2\,\mathrm{Re}\langle\Delta_g\Psi,\Psi\rangle - \mathrm{Re}\langle A\cdot\Psi,\Psi\rangle. \tag{13.40}$$

There is no mixed torsion term because the Chern torsion has type $(2,0)$. Using (13.36), (13.38), and (13.39) in (13.40), Young's inequality absorbs the products of $\sqrt{\mathcal{R}}$ with $|\nabla\Psi| + |\bar{\nabla}\Psi|$. We obtain

$$\Delta_g \mathcal{R} \geq -CM^N(\mathcal{R}^{3/2} + 1). \tag{13.41}$$

Let $s = \operatorname{tr}_\beta g$. Direct differentiation as in (13.16) gives

$$\Delta_g s \geq M^{-1}\mathcal{R} - CM, \qquad |\partial s|_g^2 \leq nM^2\mathcal{R}. \tag{13.42}$$

Take a cutoff ζ supported in B, equal to one on $B_{1/2}$, set $\eta = \zeta^2$ and $h = 2nM - s$, so $nM \leq h \leq 2nM$, and maximize

$$Q = \frac{\eta\mathcal{R}}{h}.$$

At a positive interior maximum of Q, the identity $\partial\mathcal{R}/\mathcal{R} = \partial h/h - \partial\eta/\eta$ gives

$$0 \geq \frac{\Delta_g Q}{Q} = \frac{\Delta_g \mathcal{R}}{\mathcal{R}} + \frac{\Delta_g \eta}{\eta} - \frac{\Delta_g h}{h} - \frac{2|\partial\eta|_g^2}{\eta^2} + \frac{2\operatorname{Re}\langle\partial\eta, \partial h\rangle_g}{\eta h}.$$

We have $|\partial\eta|_g^2 \leq CM\eta$, $|\Delta_g\eta| \leq CM$ and $|\partial h|_g \leq \sqrt{n}M\sqrt{\mathcal{R}}$. Insert (13.41) and (13.42) and multiply by ηQ. The positive term is $M^{-1}Q^2$. The two terms involving $\sqrt{\mathcal{R}}$ are bounded in absolute value by a fixed power of M times $Q^{3/2}$, since $\mathcal{R} = hQ/\eta$ and $h \leq 2nM$. The remaining cutoff terms are bounded by $CM^N(Q+1)$. Therefore

$$0 \geq M^{-1}Q^2 - CM^N(Q^{3/2} + Q + 1).$$

Thus Q, and then $\mathcal{R}$ on $B_{1/2}$, are bounded by a fixed polynomial in M. Since $|\partial g|_\beta^2 \leq M^3\mathcal{R}$, this proves (13.33).

If $s \leq L$, the determinant lower bound gives $\lambda_{\min}(g) \geq (\inf F)L^{1-n}$, so M is a fixed power of $1 + L$. For a ball $B_r(p)$, set $u_r(z) = r^{-2}u(p + rz)$ and use the coefficient matrices $\vartheta_r(z) = \vartheta(p + rz)$, $g_r(z) = g(p + rz)$ and $F_r(z) = F(p + rz)$. Then $g_r = \vartheta_r + \mathrm{dd^c}u_r$ in the unit coordinates and $\det g_r = F_r$. For $r \leq 1$ the derivatives of the fixed data remain uniformly bounded. The unit-ball estimate for the ordinary coefficient derivatives ∂g_r and their conjugates gives the factor r^{-1} in (13.34). □

Proposition 13.17.8 *There is $\alpha_* < 1$, depending only on n and the exponent N in Proposition 13.17.7, with the following property. Let B_0 be a Euclidean ball, fix a smooth real $(1,1)$-form ϑ and a smooth function $F > 0$ near $\overline{B_0}$, and let u_j be smooth exact solutions of $(\vartheta + \mathrm{dd^c} u_j)^n = F\beta^n$ with $\vartheta + \mathrm{dd^c} u_j > 0$. If*

$$\sup_j \|u_j\|_{C^{1,\alpha}(B_0)} < \infty \qquad \textit{for some } \alpha_* < \alpha < 1,$$

then $\mathrm{tr}_\beta(\vartheta + \mathrm{dd^c} u_j)$ is uniformly bounded on every smaller ball.

Proof Fix the center of the smaller concentric balls $B_r \Subset B_0$. Put $s_j = \mathrm{tr}_\beta(\vartheta + \mathrm{dd^c} u_j)$ and, for their radii r,

$$M_j(r) = 1 + \sup_{B_r} s_j.$$

For concentric balls $B_r \Subset B_R \Subset B_0$, $d = R - r \le 1$, Proposition 13.17.7 gives

$$|\nabla s_j| \le C d^{-1} M_j(R)^N \quad \text{on } B_{r+d/2}. \tag{13.43}$$

Let χ_h be a nonnegative mollifier of radius $h \le d/4$. Since

$$s_j = \mathrm{tr}_\beta \vartheta + \Delta_\beta u_j,$$

integration by parts, after subtracting the constant vector $\nabla u_j(p)$, gives

$$(s_j * \chi_h)(p) \le C(1 + h^{\alpha-1}).$$

Equation (13.43) gives

$$|s_j(p) - (s_j * \chi_h)(p)| \le C d^{-1} M_j(R)^N h.$$

Choose

$$h = \frac{d}{4} M_j(R)^{-N/(2-\alpha)}.$$

After absorbing fixed powers of d^{-1}, we obtain

$$M_j(r) \le C d^{-1} M_j(R)^q, \qquad q = \frac{N(1-\alpha)}{2-\alpha}. \tag{13.44}$$

Choose α_* so close to one that $q < 1$ for $\alpha > \alpha_*$. Choose $0 < r_0 < R_0$ with $B_{R_0} \Subset B_0$. Let $r_k \uparrow R_0$ with $r_{k+1} - r_k = (R_0 - r_0)2^{-k-1}$. Iterating (13.44) gives

$$M_j(r_0) \le \prod_{k=0}^{m-1} \left[C(r_{k+1} - r_k)^{-1}\right]^{q^k} M_j(r_m)^{q^m}.$$

The infinite product converges because $\sum q^k(1+k) < \infty$. For each fixed j, the final factor tends to one as $m \to \infty$. The resulting bound is independent of j. □

13.17.6 Proof of the main theorem

If a nonregular point exists, minimize $\varphi - \rho$ over the nonregular set. The strict barrier gives quadratic separation at that point. The preceding estimates then supply a bounded trace in a neighborhood, contradicting its choice.

Proof (of Theorem 13.17.1) If $n = 1$, the equation is the linear identity

$$\mathrm{dd^c}\varphi = c f \omega - \theta \quad \text{on } \Omega.$$

Its right-hand side is smooth, so interior elliptic regularity gives $\varphi \in C^\infty(\Omega)$, and the original equation gives $\theta + \mathrm{dd^c}\varphi = c f \omega > 0$. We henceforth assume $n \geq 2$.

Let

$$\mathcal{S} = \{p \in \Omega : \varphi \text{ is not smooth on any neighborhood of } p\}.$$

This is closed in Ω. Suppose $\mathcal{S} \neq \varnothing$. By Lemma 1.19.4, the restriction of $\varphi - \rho$ to $\mathcal{S}$ attains its minimum m at some $p \in \mathcal{S}$.

Choose nested coordinate balls

$$B \Subset B_0 \Subset \Omega$$

centered at p. On B_0, Proposition 13.17.4 gives smooth exact Dirichlet solutions u_j converging uniformly to φ at a polynomial rate and having polynomial global trace growth.

On the regular set $\mathcal{R} = \Omega \setminus \mathcal{S}$, the function φ is smooth. There $\theta + \mathrm{dd^c}\varphi$ is semi-positive and has positive determinant, hence is positive definite. Proposition 13.17.6 therefore gives a uniform trace bound for the u_j on every compact subset of $\mathcal{R} \cap B_0$.

In coordinates centered at p, take $\kappa > 0$ small and set

$$q(z) = \rho(z) - \kappa|z|^2.$$

After shrinking B, the strict lower bound for θ_ρ gives $\theta + \mathrm{dd^c}q > 0$. If $v = \varphi - q$, then the minimizing property of p gives

$$v(p) = m, \qquad v(z) = (\varphi - \rho)(z) + \kappa|z|^2 \geq m + \kappa|z|^2 \quad (z \in \mathcal{S} \cap \overline{B}).$$

Thus Proposition 13.17.5 applies on B with the closed set $\mathcal{S} \cap \overline{B}$: the approximation rate and global polynomial trace bound come from Proposition 13.17.4, and the bound away from $\mathcal{S}$ comes from Proposition 13.17.6. The same proposition gives, for every $0 < \alpha < 1$, a neighborhood of p on which the approximants have a uniform $C^{1,\alpha}$ bound.

Choose $\alpha > \alpha_*$ as in Proposition 13.17.8. That proposition yields

$$\sup_j \mathrm{tr}_\beta(\theta + \mathrm{dd^c}u_j) < \infty$$

on a still smaller neighborhood of p. Since $u_j \to \varphi$ uniformly, $\mathrm{dd^c} u_j \to \mathrm{dd^c}\varphi$ as currents. The trace bound passes to the limit, so $\mathrm{tr}_\beta(\theta + \mathrm{dd^c}\varphi)$ has a bounded density near p. Proposition 13.17.2 now implies that φ is smooth near p, contradicting $p \in \mathcal{S}$.

Hence $\mathcal{S} = \varnothing$ and $\varphi \in C^\infty(\Omega)$. The form $\theta + \mathrm{dd^c}\varphi$ is smooth and semipositive there, while

$$(\theta + \mathrm{dd^c}\varphi)^n = c f \omega^n > 0.$$

Its nonnegative eigenvalues have positive product, so every eigenvalue is positive. Therefore $\theta + \mathrm{dd^c}\varphi > 0$ on Ω. □

Corollary 13.17.1 *Let $0 \le f \in L^p(X, \omega^n)$, $p > 1$, be smooth and strictly positive on Ω. Every minimal-singularity solution of $(\theta + \mathrm{dd^c}\varphi)^n = c f \omega^n$, $c > 0$, is smooth and strictly θ-plurisubharmonic on Ω.*

Proof The analytic complement of Ω has zero mass for both measures, so it suffices to know the equation in the bounded-potential sense on Ω. Apply Theorem 13.16.1 with density $c f$ to obtain local Hölder continuity. All hypotheses of Theorem 13.17.1 are now satisfied. The existence of a normalized minimal-singularity solution is already proved in Theorem 13.15.2. □

For a smooth closed big representative on a compact Kähler manifold, Theorem 27.10.1 applies this conclusion to the classical full-mass solution and identifies the regular open set with the ample locus.

13.18 Symmetry and the selected projective limit

The density in the projective example of Theorem 11.6.1 vanishes on an open band. Its symmetries identify the zero-exponent limit and locate a failure of second differentiability even at positive exponent. The selection starts with a general observation: uniqueness of the positive-exponent equation preserves the symmetries of the data, and uniform compactness carries those symmetries to every normalized cluster limit. For the projective data this already produces a solution distinct from the two exchanged by the involution.

Proposition 13.18.1 *Let ω be Hermitian, let $\mathrm{d}V$ be a smooth positive volume form, and let $0 \le f \in L^p(\mathrm{d}V)$, $p > 1$, be nonzero. Let $c > 0$ be the unique scalar for the bounded equation $\mathrm{MA}_\omega(u) = c f \,\mathrm{d}V$. For $\lambda > 0$ denote by u_λ the unique bounded solution of*

$$\mathrm{MA}_\omega(u_\lambda) = \mathrm{e}^{\lambda u_\lambda} f \,\mathrm{d}V, \qquad a_\lambda = u_\lambda - \sup_X u_\lambda.$$

The family $\{a_\lambda : 0 < \lambda \le 1\}$ is relatively compact in $C(X)$. Every cluster limit as $\lambda \downarrow 0$ is a normalized bounded solution of $\mathrm{MA}_\omega(a) = c f \,\mathrm{d}V$.

Suppose a group G of biholomorphisms of X preserves ω and the measure $f \,\mathrm{d}V$. Then every a_λ and every such cluster limit is G-invariant. In particular, the zero-exponent equation has a G-invariant normalized solution.

Proof The barriers of Proposition 10.5.1 give a uniform oscillation bound for u_λ and uniform convergence $e^{\lambda u_\lambda} \to c$. Thus the equation densities are uniformly bounded in L^p for $0 < \lambda \le 1$ and tend to cf in L^p as $\lambda \downarrow 0$. Apply Theorem 10.3.2 to obtain relative compactness of a_λ in $C(X)$ and the asserted equation for every cluster limit.

For each $g \in G$, pullback of the positive-exponent equation shows that $u_\lambda \circ g$ solves the same equation. Its uniqueness gives $u_\lambda \circ g = u_\lambda$. Surjectivity of g preserves the supremum, so $a_\lambda \circ g = a_\lambda$. Uniform convergence preserves these equalities for every g. Compactness supplies at least one cluster limit as $\lambda \downarrow 0$, yielding the invariant solution. □

Corollary 13.18.1 *On $\mathbb{P}(\mathbb{C}^2 \oplus \mathbb{C}^2)$, with block homogeneous coordinates $[A : B]$, the data in Theorem 11.6.1 can be chosen so that the same zero-exponent equation has at least three distinct normalized bounded solutions. Two are smooth, and a third is Hölder continuous and fixed by the involution $\iota[A : B] = [B : A]$.*

Proof Use the radial coordinate r, rank-one forms P, Q, R, Fubini–Study form ω_X, and scalar functions F, x, y, z from Sections 11.6.1 and 11.6.2. Set $w = F(r)$, $\omega_0 = xP + yQ + zR$, $H = \mathrm{dd^c} w/2$, $A_0 = \omega_0 - H$ and $B_0 = \omega_0 + H$ as in Section 11.6.3. The Hermitian form is $\omega = \omega_0 + \mathrm{dd^c}\rho$, with $\rho = \varepsilon(r^2 - 1)\kappa(r)$ from Section 11.6.4. Choose the cutoff κ even; all its required support and plateau properties are preserved by this choice. Then ρ is even, while $w = F(r)$ is odd. The involution sends r to $-r$, interchanges P, Q, and fixes R. The coefficient identities $x(-r) = y(r)$ and $z(-r) = z(r)$ show that $\iota^*\omega_0 = \omega_0$. Thus $\iota^*\omega = \omega$, and $\iota^*\omega_X = \omega_X$.

Since $\iota^* H = -H$, the involution interchanges A_0 and B_0. Their cubes are equal, so $f\omega_X^3$ is invariant. It also interchanges $\widetilde{u} = -w/2 - \rho$ and $\widetilde{v} = w/2 - \rho$. These functions therefore have the same supremum, and the normalized solutions u, v are interchanged. They are distinct, so neither is invariant.

Apply Proposition 13.18.1 to the group generated by ι. Here $c = 1$, as the two constructed solutions already have that scalar. An invariant cluster solution a cannot equal either u or v. Its bounded equation with smooth density gives Hölder regularity by Theorem 13.4.1. This gives a third solution; smoothness of that third solution is not asserted. □

To identify the whole normalized limit for these symmetric data, we now use the radial equation. It determines the invariant solution even among bounded potentials whose differentiability is not known beforehand.

13.18.1 The invariant projective solution

Retain the even correction ρ and the coefficients of Section 11.6.2. Let K be generated by the block unitary group $U(2) \times U(2)$ and the exchange $\iota[A : B] = [B : A]$. Both preserve ω and $f\omega_X^3$. Write $U = \{A \neq 0,\ B \neq 0\}$, and denote by u_* the normalized even solution with the symmetric transition pair from Proposition 11.6.1.

Theorem 13.18.1 *For the symmetric projective data, u_* is the unique normalized bounded K-invariant solution of*

$$\mathrm{MA}_\omega(u) = f\omega_X^3.$$

If u_λ is the unique solution of

$$\mathrm{MA}_\omega(u_\lambda) = \mathrm{e}^{\lambda u_\lambda} f\omega_X^3, \qquad \lambda > 0,$$

then

$$u_\lambda - \sup_X u_\lambda \longrightarrow u_* \quad \textit{uniformly as } \lambda \downarrow 0.$$

The selected limit is $C^{1,1}$. This is a selection theorem for these data; it does not assert uniqueness of all normalized solutions of the zero-exponent equation.

We give the radial measure calculation needed to treat bounded solutions, rather than assume differentiability of a cluster limit. Use the notation x, y, z, P, Q, R of Section 11.6. For a real scalar s, define the coefficient polynomial $A(r, s)$ and its primitive $\Phi(r, s)$ by

$$A(r, s) = (x(r) + s)(y(r) - s), \qquad \Phi(r, s) = \int_0^s A(r, t)\,\mathrm{d}t.$$

Partial derivatives of Φ with respect to r below keep s fixed.

Lemma 13.18.1 *Suppose u is a bounded block-unitary-invariant solution of the zero-exponent equation. On U write $u + \rho = G(r)$. Then G is locally Lipschitz, its derivative $h = G'$ has locally bounded variation, and its one-sided representatives satisfy*

$$-x(r) \le h(r\pm) \le y(r), \qquad \mathrm{d}h + z\,\mathrm{d}r \ge 0.$$

In fact h is continuous. With

$$k(r) = \begin{cases} 0, & |r| \le 1, \\ x(r)y(r)z(r), & |r| > 1, \end{cases}$$

it satisfies the distributional identity on $\mathbb{R}$

$$\mathrm{d}\big(\Phi(r, h(r))\big) = \big(k(r) - A(r, h(r))z(r) + \partial_r \Phi(r, h(r))\big)\,\mathrm{d}r. \tag{13.45}$$

On every open interval where $A(r, h(r)) > 0$, h is smooth and

$$h' = \frac{k(r)}{A(r, h)} - z.$$

Proof Block-unitary orbits in U are exactly the level sets of r: normalize the sum of the squared norms of the two coordinate blocks, and map the two vectors of the same respective lengths by their unitary groups. Thus invariance gives a scalar G.

On a compact r-interval, the radial direction annihilates P, Q, while R is strictly positive in that direction. Testing positivity of $\omega_0 + \mathrm{dd^c}G(r)$ in this direction gives $G'' + z \ge 0$ as a one-dimensional distribution. One may obtain this first by smooth approximation of the scalar distribution and then by testing with nonnegative functions. Adding a twice integrated primitive of z makes G convex. A finite bounded convex function on an interval is locally Lipschitz, and its derivative is locally of bounded variation. The quasi-plurisubharmonic representative is the corresponding continuous representative. The horizontal directions give $x + h \ge 0$, $y - h \ge 0$ almost everywhere. Continuity of x, y and the one-sided limits of a function of bounded variation give these bounds for both one-sided representatives. The radial inequality is precisely $\mathrm{d}h + z\,\mathrm{d}r \ge 0$; in particular jumps of h can only be upward.

We next justify the formula for the measure, including possible jumps. Identify U smoothly with $\mathbb{R} \times Y$, where $Y = \{r = 0\}$, by real scaling of one block. The forms P, Q are independent of r in this identification. So is $\mathrm{d^c}r$: in local coordinates it is the angular differential of the relative complex scale plus fixed base terms. Therefore

$$P \wedge Q \wedge R = \mathrm{d}r \wedge \nu,$$

where ν is a fixed smooth positive volume form on the compact orbit Y, with the orientation chosen by this equality.

For smooth G, the rank-one wedge formula gives the radial coefficient of its Monge–Ampère measure, after dividing by 6ν, as

$$A(r, h)(z + h')\,\mathrm{d}r = A(r, h)z\,\mathrm{d}r + \mathrm{d}\big(\Phi(r, h)\big) - \partial_r \Phi(r, h)\,\mathrm{d}r.$$

This identity passes to a locally Lipschitz G with the preceding lower Hessian bound. Indeed, convolve G on a slightly larger compact interval. The derivatives h_ϵ are uniformly bounded and converge to h almost everywhere and in every finite L^q space. Their second derivatives have a uniform lower bound. Consequently the functions $G_\epsilon(r)$ have a common quasi-plurisubharmonic lower Hessian bound on compact subsets of U, and converge uniformly to $G(r)$. A fixed smooth local correction and bounded Bedford–Taylor continuity give convergence of their signed products. On the displayed right side, the two ordinary terms converge in L^1, and $\Phi(r, h_\epsilon) \to \Phi(r, h)$ in L^1, giving distributional convergence of its derivative. This proves the same formula for the bounded product of G.

The specified smooth data measure has radial coefficient $k(r)\,\mathrm{d}r$, giving (13.45) initially for the almost-everywhere representative of h. If h jumped at r_0, the atom of its left side would be

$$\Phi(r_0, h(r_0+)) - \Phi(r_0, h(r_0-)) = \int_{h(r_0-)}^{h(r_0+)} (x(r_0) + s)(y(r_0) - s)\,\mathrm{d}s.$$

The right side has no atom. An upward jump of positive size lies in $[-x(r_0), y(r_0)]$, where the integrand is strictly positive in the interior. Since $x + y > 0$ everywhere, such an integral is strictly positive. Thus there are no jumps. A locally bounded-variation function whose jumps all vanish has a continuous representative here: its distributional derivative consists of an absolutely continuous measure and a nonnegative singular measure, so its one-sided representatives coincide at every point. We henceforth use this representative. In particular G is C^1 and $G' = h$ everywhere.

All coefficients on the right of (13.45) are now continuous, so $\Phi(r, h(r))$ is C^1. Where $\partial_s \Phi = A > 0$, the implicit function theorem gives $h \in C^1$ and the stated ordinary differential equation. Smoothness then follows by the smooth coefficients of that equation. The function k is smooth, since the original density and its flat joins are smooth. □

Proof (of Theorem 13.18.1) Let u be a normalized bounded K-invariant solution. Apply the radial lemma. Because ρ and u are invariant under block exchange, G is even, so h is odd and $h(0) = 0$.

On $0 < r < 1$ we have $x = p > 0$ and $y = p + b \geq p > 0$. Starting from zero, as long as $-p < h < y$, the equation is $h' = -z_0$. Thus $h(r) = -\int_0^r z_0(s)\,\mathrm{d}s$. This decreasing function cannot meet the positive upper boundary y. Its first meeting with the lower boundary $-p$ occurs at the unique $t_* \in (0, 1)$ satisfying

$$\int_0^{t_*} z_0(s)\,\mathrm{d}s = p(t_*),$$

whose existence and uniqueness were proved in Proposition 11.6.1.

After that meeting, h must stay on $-p$ until $r = 1$. To prove it, suppose a connected component (a_0, b_0) of $\{h + p > 0\}$ lies to its right. Its left endpoint has $a_0 \geq t_*$ and $h(a_0) = -p(a_0) < 0$. Near that endpoint, $h < 0 < y$; hence on the component the interior equation applies for as long as $h < 0$. It forces $h' = -z_0 < 0$, so h cannot first

reach zero. The equation therefore holds throughout the component. But then

$$(h + p)' = p' - z_0 < 0,$$

contradicting positivity of $h + p$ immediately to the right of its zero at a_0. Thus $h = -p$ on $[t_*, 1]$. Oddness determines the entire negative half of the band. The slope on $[-1, 1]$ is exactly that of the symmetric explicit solution.

We also identify the slope outside the band; boundedness at the projective ends matters here. On $r > 1$, $k = xyz > 0$. Neither horizontal coefficient can vanish at a finite point of this interval. For the lower one define

$$\Psi(r) = \int_{-x(r)}^{h(r)} A(r, s)\,\mathrm{d}s \geq 0.$$

Equation (13.45) and the smooth lower endpoint show that Ψ is C^1. If $h(r_0) = -x(r_0)$, its derivative at that point is $k(r_0) > 0$: the Az term vanishes, and the two partial-r terms agree at the common endpoint. This contradicts that the nonnegative C^1 function Ψ has an interior minimum zero there. The upper-endpoint integral $\int_{h(r)}^{y(r)} A(r, s)\,\mathrm{d}s$ gives the same contradiction, with derivative $-k(r_0)$, when $h = y$. Thus $-x < h < y$ everywhere on $r > 1$, and the smooth ordinary differential equation of the lemma applies.

For $r \geq 2$ the background is Fubini–Study:

$$x = \sigma, \qquad y = 1 - \sigma, \qquad z = \sigma', \qquad \sigma = \frac{\mathrm{e}^r}{1 + \mathrm{e}^r}.$$

Set $a = \sigma + h \in (0, 1)$. Its equation is

$$a(1 - a)a' = \sigma(1 - \sigma)\sigma' > 0.$$

Hence a is increasing and has a limit in $(0, 1]$ at $+\infty$. If that limit were less than one, $h = a - \sigma$ would approach a negative constant and its primitive G would be unbounded below, contrary to boundedness of u and ρ. The limit is therefore one. Integrating the equation gives

$$\frac{a^2}{2} - \frac{a^3}{3} = \frac{\sigma^2}{2} - \frac{\sigma^3}{3} + C.$$

Taking the limit at infinity gives $C = 0$. The scalar function $s^2/2 - s^3/3$ is strictly increasing on $(0, 1)$, so $a = \sigma$ and $h = 0$ for $r \geq 2$. Ordinary uniqueness propagates this zero solution throughout $r > 1$, since the coefficients are smooth and the two horizontal coefficients stay positive there. Oddness gives $h = 0$ on $r < -1$ as well.

We have proved that G' agrees everywhere with the symmetric explicit slope. Thus u differs from u_* by a constant on U. The two quasi-plurisubharmonic functions then agree up to that constant on X by their canonical representatives. Their supremum

normalization removes the constant. This proves uniqueness within the bounded K-invariant class.

Finally Proposition 13.18.1 gives uniform precompactness of the normalized positive-exponent family. Every cluster limit solves the zero-exponent equation and is K-invariant, because the data are K-invariant. The scalar is one for these data. The uniqueness just proved identifies every cluster limit with u_*. If convergence of the whole family failed, a sequence staying a fixed positive uniform distance from u_* would have a convergent subsequence, contradicting that identification. This proves uniform convergence and the asserted $C^{1,1}$ regularity of the limit. □

13.18.2 Second derivatives at positive exponent

The zero-density band in the same projective example fixes the radial derivative at every positive exponent. Consequently the exponential solutions have the same failure of second differentiability as the selected limit u_*. Retain A, k and the transition radius t_* from the radial calculation above.

Proposition 13.18.2 *For the symmetric projective data of Theorem 13.18.1, let u_λ solve*

$$\mathrm{MA}_\omega(u_\lambda) = \mathrm{e}^{\lambda u_\lambda} f \omega_X^3, \qquad \lambda > 0.$$

On the open band $\{-1 < r < 1\}$, $u_\lambda - u_*$ is constant. Both potentials are $C^{1,1}$ there and fail to be C^2 at every point of the two hypersurfaces $r = \pm t_*$. In particular, none of these positive-exponent solutions is a globally C^2 potential.

Proof Uniqueness and invariance of the data imply that u_λ is K-invariant. Write $u_\lambda + \rho = G_\lambda(r)$ on U. The positivity argument in Lemma 13.18.1 uses only boundedness and invariance, so G_λ is locally Lipschitz and $h_\lambda = G'_\lambda$ is locally of bounded variation, with

$$-x \le h_\lambda(r\pm) \le y, \qquad \mathrm{d}h_\lambda + z\,\mathrm{d}r \ge 0.$$

Its weak flux equation is the same one, with the prescribed coefficient replaced by

$$k_\lambda(r) = \mathrm{e}^{\lambda(G_\lambda(r) - \rho(r))} k(r).$$

Indeed, the scalar convolution and signed-product argument of that lemma is unchanged, and this is exactly the radial coefficient of the right-hand measure. At the initial bounded stage k_λ is locally bounded. Hence the right side of the flux equation has no atoms. The same strictly positive jump integral eliminates every jump of h_λ. Thus h_λ is continuous and G_λ is C^1. The coefficient k_λ is consequently continuous. Where $A(r, h_\lambda) > 0$, inversion of the C^1 flux gives

$$h'_\lambda = \frac{k_\lambda(r)}{A(r, h_\lambda)} - z.$$

On $-1 < r < 1$ the coefficient k_λ is identically zero. Exchange invariance gives $h_\lambda(0) = 0$ and oddness. On the positive half-band, the first-contact and no-departure arguments in the proof of Theorem 13.18.1 apply verbatim: the first branch solves $h' = -z_0$, reaches $-p$ at t_*, and cannot leave that lower boundary to its right because $(h + p)' = p' - z_0 < 0$ on every proposed departure interval. Therefore

$$h_\lambda(r) = \begin{cases} p(r), & -1 < r \le -t_*, \\ -\displaystyle\int_0^r z_0(s)\,\mathrm{d}s, & -t_* \le r \le t_*, \\ -p(r), & t_* \le r < 1. \end{cases}$$

This is the derivative of $u_* + \rho$. Its primitive differs from G_λ by one constant on the connected band. The claimed local $C^{1,1}$ regularity follows from the explicit slope.

At $t_* \in (0, 1)$ the two one-sided derivatives of this slope are

$$h'_\lambda(t_*-) = -z_0(t_*) < 0, \qquad h'_\lambda(t_*+) = -p'(t_*) > 0.$$

These unequal derivatives prevent the scalar primitive from having a second derivative at t_*. Through every point of that level set, the submersion r supplies a smooth curve parametrized by r, transferring the failure to u_λ. Smooth ρ cannot cancel the jump. Oddness gives the conclusion at $-t_*$, and the identical slope proves it for u_*. □

The zero set of f contains the whole band. This example therefore does not concern the smooth strictly positive densities in the Hermitian Calabi–Yau theorem. It does show why smoothness of a nonnegative density alone is insufficient for second-order regularity, even when the equation includes a strictly increasing exponential factor.

13.19 Local density and boundary control

Return to the compact complex manifold X and its auxiliary Hermitian metric ω. Let θ be smooth, and let $\rho \in \mathrm{PSH}(X, \theta)$ satisfy $\theta_\rho \ge \varepsilon\omega$ for some $\varepsilon > 0$. In this section Ω denotes an open set where ρ is smooth and $\mathrm{d}\theta = 0$.

Theorem 13.17.1 uses the equation on the whole regular locus of a global strict barrier. On the present open set, smooth positive density alone does not give the same conclusion. A smooth boundary trace provides a different route: Dirichlet uniqueness identifies the bounded solution with a smooth one. The examples below show why both the domain and its boundary data matter.

13.19.1 Smooth boundary traces

For a coordinate ball $B \Subset \Omega$, a boundary trace g means that $u(z) \to g(\xi)$ as $B \ni z \to \xi$, for every $\xi \in \partial B$.

Theorem 13.19.1 *Suppose that $u \in \mathrm{PSH}(X, \theta)$ has minimal ordinary singularities and*

$$\mathrm{MA}_\theta(u) = f\,\mathrm{d}V \quad \text{on } \Omega, \qquad f \in C^\infty(\Omega), \quad f > 0.$$

Assume that Ω is covered by coordinate Euclidean balls $B_j \Subset \Omega$ such that θ has a smooth local potential near $\overline{B_j}$ and u has a smooth boundary trace on ∂B_j. Then $u \in C^\infty(\Omega)$ and $\theta + \mathrm{dd^c}u > 0$ there.

Proof Minimality gives $u \geq \rho - C$ for a constant C. Thus u is locally bounded on Ω. Fix one of the balls, write it as $B_R(z_0)$, and choose a smooth function h near its closure with $\mathrm{dd^c}h = \theta$. The bounded plurisubharmonic function $v = u + h$ has, by Proposition 1.7.1, Bedford–Taylor measure $f\,\mathrm{d}V$ and smooth boundary trace $\varphi = g + h|_{\partial B}$.

For smooth V, Euclidean volume $\mathrm{d}\Lambda$ gives

$$(\mathrm{dd^c}V)^n = \frac{n!}{\pi^n}\det(V_{j\bar{k}})\,\mathrm{d}\Lambda. \tag{13.46}$$

The smooth Dirichlet input of Section 13.17.3, with zero background and density $f\,\mathrm{d}V$, gives a smooth strictly plurisubharmonic V on $\overline{B}$ with $(\mathrm{dd^c}V)^n = f\,\mathrm{d}V$ and boundary value φ. A smooth extension Φ of φ gives the required strict subsolution $\Phi + A(|z - z_0|^2 - R^2)$ for large A. In dimension one this is the Poisson Dirichlet problem.

The difference $v - V$ tends to zero at every boundary point. Compactness of ∂B makes this convergence uniform near the boundary: otherwise a sequence approaching the boundary with difference bounded away from zero would have a subsequence approaching one fixed boundary point. Bounded Dirichlet uniqueness gives $v = V$. Therefore $u = V - h$ is smooth and its background form is positive on B. The balls cover Ω. □

The proof uses minimality only to obtain the local lower bound from ρ. Consequently the same conclusion holds for any solution already known to be locally bounded on Ω and satisfying the same trace condition. Local closedness supplies the potentials after the coordinate balls are made small enough; the background may be nonclosed elsewhere.

Example 13.19.1 For every $n \geq 1$, the trace criterion admits a bounded continuous solution whose global density belongs to L^1 and to no L^p with $p > 1$. The solution need not be Hölder at the complement of Ω.

Let $E = \mathbb{C}/(\mathbb{Z} + \mathrm{i}\mathbb{Z})$, $X = E^n$, and

$$\omega_j = \frac{\mathrm{i}}{2\pi}\,\mathrm{d}z_j \wedge \mathrm{d}\bar{z}_j, \qquad \theta = \sum_{j=1}^n \omega_j, \qquad D = \{0\} \times E^{n-1}.$$

Take $\mathrm{d}V = \theta^n$ and $\rho = 0$. Choose a nonnegative smooth cutoff χ supported in a coordinate disc $|z| < r_0 < 1$ of E and equal to one near zero. For $\beta > 0$, define

$$\psi(z) = \chi(z)(-\log|z|^2)^{-\beta} \quad (z \neq 0), \qquad \psi(0) = 0,$$

and extend by zero. On the smaller punctured disc put $t = |z|^2$ and $L = -\log t$. Differentiation gives

$$\psi_{z\bar{z}} = \frac{\beta(\beta+1)}{tL^{\beta+2}}, \qquad t\frac{\mathrm{d}}{\mathrm{d}t}L^{-\beta} = \beta L^{-\beta-1} \longrightarrow 0. \tag{13.47}$$

The first expression is integrable and the second is the radial flux. Integration by parts on punctured discs, using also $\psi \to 0$, shows that the distributional Hessian has no atom at zero. On the cutoff annulus its negative part is bounded. Hence, for a sufficiently small fixed $\eta > 0$, $u = \eta\psi(z_1)$ satisfies $\theta + \mathrm{dd^c}u \geq \theta/2$.

Only one coordinate contributes to $\mathrm{dd^c}u$. Locally approximate the bounded subharmonic potential in that coordinate from above and use Theorem 1.6.1; the smooth diagonal determinant identity passes to the limit and yields

$$\mathrm{MA}_\theta(u) = f\theta^n, \qquad f = 1 + \eta\psi_{z\bar{z}}(z_1).$$

Give f any positive value on the null set D. It is smooth and positive on $\Omega = X \setminus D$, where u is smooth and has smooth traces on all relatively compact coordinate balls. Near D its singular term is comparable to $|z_1|^{-2}(-\log|z_1|^2)^{-\beta-2}$. Its radial L^1 integral converges, whereas

$$\int_0^{r_0} s^{1-2p}(-\log s^2)^{-p(\beta+2)}\,\mathrm{d}s = +\infty \qquad (p > 1).$$

Also $(-\log|z|^2)^{-\beta}/|z|^\alpha \to +\infty$ for every $\alpha > 0$, so u is not Hölder at D. Finally, boundedness gives minimal ordinary singularities, since every global θ-plurisubharmonic competitor is bounded above.

13.19.2 A bounded discontinuous potential

Let $U = W = \{|z| < 1/4\} \subset \mathbb{C}$, with coordinates z and w on the two factors. Write $\mathrm{d}A$ for planar Lebesgue measure and set

$$a_k = \alpha_k = 2^{-k-4}, \qquad A = \sum_{k\geq 1}\alpha_k = \frac{1}{16}, \qquad S = \{0\} \cup \{a_k : k \geq 1\}.$$

Define on U

$$h(z) = \sum_{k\geq 1}\alpha_k \log|z - a_k|^2, \qquad b(z) = \mathrm{e}^{h(z)}, \tag{13.48}$$

with $b = 0$ where $h = -\infty$.

Lemma 13.19.1 *The function*

$$a(z) = \frac{1}{\pi} \int_U \log|z-\zeta|^2 \, b(\zeta)^{-1} \, \mathrm{d}A(\zeta)$$

is bounded and continuous on U. The bounded function

$$P(z,w) = a(z) + b(z)|w|^2$$

is plurisubharmonic on $U \times W$, is smooth off $S \times W$, and satisfies

$$(\mathrm{dd^c} P)^2 = \frac{2}{\pi^2} \, \mathrm{d}A(z) \, \mathrm{d}A(w). \tag{13.49}$$

It is discontinuous at every $(0, w_0)$ with $w_0 \neq 0$.

Proof The summands defining h are negative on U, so their partial sums decrease. Since $h(0) = 2\sum_k \alpha_k \log a_k > -\infty$, the limit is subharmonic. On compact subsets of $U \setminus S$, the series and its derivatives converge uniformly, making h harmonic there. Moreover $0 \le b \le 1$, $b(a_k) = 0$, and $b(0) > 0$.

Weighted arithmetic–geometric mean gives, almost everywhere,

$$b(z)^{-2} \le \sum_{k \ge 1} \frac{\alpha_k}{A} |z - a_k|^{-4A}. \tag{13.50}$$

For the countable inequality, first use the first N weights and assign the remaining weight to the factor 1; then let N increase. Since $4A = 1/4 < 2$, the integrals of the terms $|z - a_k|^{-4A}$ on U are bounded uniformly in k. Thus $b^{-1} \in L^2(U)$.

The logarithmic kernel has uniformly bounded L^2 norm on the translates of U that occur in the definition of a. Cauchy–Schwarz bounds a uniformly. Choose a compactly supported $L^2(\mathbb{C})$ kernel agreeing with $\log|z|^2$ on $U - U$. Translation continuity in L^2, paired with $b^{-1}\mathbf{1}_U \in L^2$, proves continuity of a. The distribution identity $\partial_z \partial_{\bar z} \log|z|^2 = \pi\delta_0$ gives $a_{z\bar z} = b^{-1}$. In particular a is subharmonic and, by local elliptic regularity, smooth off S.

The second summand of P is $\exp(h(z) + \log|w|^2)$, a convex increasing composition of a plurisubharmonic function, extended by zero at its poles. Thus P is bounded and plurisubharmonic. At smooth points the harmonicity of $h = \log b$ gives $b b_{z\bar z} = |b_z|^2$, and hence

$$\det(P_{j\bar k}) = b a_{z\bar z} + |w|^2 \big(b b_{z\bar z} - |b_z|^2\big) = 1.$$

The set $S \times W$ is contained in the pole set of $h(z) + \log|z|$, a plurisubharmonic function not identically $-\infty$. It is also Lebesgue-null. Pluripolar nullity in Theorem 1.6.1 and (13.46) prove (13.49) throughout the bidisc. Finally,

$$P(a_k, w_0) = a(a_k) \longrightarrow a(0) < a(0) + b(0)|w_0|^2 = P(0, w_0)$$

when $w_0 \neq 0$. □

13.19.3 Compact gluing

We work on the flat complex torus $X = (\mathbb{C}/(\mathbb{Z}+\mathrm{i}\mathbb{Z}))^2$, identifying a small coordinate ball with a ball in $U \times W$. For a smooth function v on the ball write $H_v = (v_{j\bar{k}})$ for its complex Hessian and I for the two-dimensional identity matrix.

Theorem 13.19.2 *There are a positive flat Kähler form θ on X, a bounded $u \in \mathrm{PSH}(X,\theta)$ with minimal ordinary singularities, and a density $F > 0$ almost everywhere such that*

$$\mathrm{MA}_\theta(u) = F\theta^2, \qquad F \in L^1(X,\theta^2), \qquad F \notin L^p(X,\theta^2) \quad (p > 1).$$

On an open coordinate ball Ω, F is a positive constant and u is discontinuous. The constant function $\rho = 0$ is a smooth strict barrier on X.

Proof Set $R = 1/8$ and choose M so that $P_0 = P + M$ satisfies $0 \le P_0 \le L$ on $B_R \Subset U \times W$ for some finite L. Choose $C > 0$ with

$$\frac{3CR^2}{16} \ge 2, \qquad \frac{5CR^2}{16} > L + 2, \tag{13.51}$$

and put $Q(Z) = C(|Z|^2 - R^2/4)$ for $Z = (z,w)$. Let χ be the convolution of $|t|$ with an even nonnegative smooth probability kernel supported in $(-1,1)$. Then χ is smooth and convex, $|\chi'| \le 1$, $\chi'(0) = 0$, and $\chi(t) = |t|$ for $|t| \ge 1$. The function

$$m(s,t) = \frac{s + t + \chi(s-t)}{2}, \qquad v = m(P_0, Q)$$

is bounded and plurisubharmonic, because m is convex and nondecreasing in each variable. The collar inequalities imply

$$v = P_0 \quad \text{on } B_{R/4}, \qquad v = Q \quad \text{when } 3R/4 \le |Z| < R.$$

Indeed, $Q \le -2 \le P_0 - 2$ on the first ball and $Q > L + 2 \ge P_0 + 2$ on the collar. Define

$$\theta = \frac{\mathrm{i}C}{2\pi}(\mathrm{d}z \wedge \mathrm{d}\bar{z} + \mathrm{d}w \wedge \mathrm{d}\bar{w}), \qquad u = \begin{cases} v - Q & \text{on } B_R, \\ 0 & \text{outside } B_R. \end{cases}$$

The collar identity makes u identically zero near ∂B_R and gives a global bounded potential with $\theta + \mathrm{dd^c}u = \mathrm{dd^c}v$ on the ball and $\theta + \mathrm{dd^c}u = \theta$ outside. Thus u is admissible and has minimal ordinary singularities.

Outside the pluripolar set $(S \times W) \cap B_R$, set $\lambda = (1 + \chi'(P_0 - Q))/2$. Differentiating the regularized maximum gives

$$H_v = \lambda H_P + (1-\lambda)CI + \frac{\chi''(P_0 - Q)}{2}\partial(P_0 - Q) \otimes \overline{\partial(P_0 - Q)}. \tag{13.52}$$

Both H_P and CI are positive definite there; the final term is positive semidefinite. Hence the classical density is positive. Bounded products charge no pluripolar sets, so there is no further measure on the exceptional set. Their mass on compact X is finite by Proposition 1.7.1. This proves $\mathrm{MA}_\theta(u) = F\theta^2$ for an almost-everywhere positive $F \in L^1$. On $\Omega = B_{R/4}$, (13.49) gives the representative $F = C^{-2}$. There $u = P + M - Q$ is discontinuous at $(0, w_0)$ for $0 < |w_0| < R/4$.

We prove failure of every global L^p directly in the transition region. For large k, set

$$m_k = a(a_k)+M \in [0, L], \qquad w_k = \big(R^2/4+m_k/C-|a_k|^2\big)^{1/2}, \qquad Z_k = (a_k, w_k).$$

Then $w_k > 0$, $P_0(Z_k) = Q(Z_k) = m_k$, and (13.51) gives $|Z_k|^2 < 9R^2/16$. Near the isolated pole a_k,

$$h(z) = \alpha_k \log|z - a_k|^2 + H_k(z),$$

where H_k is smooth and harmonic. Consequently b and P_0 are continuous near Z_k. There is a product neighborhood $V_k \Subset B_R$ of Z_k, with $|w|$ bounded below by a positive number, on which $1/4 \le \lambda \le 3/4$.

At smooth points of V_k, positivity and the two-dimensional determinant expansion give

$$\begin{aligned}\det H_v &\ge \det\big(\lambda H_P + (1-\lambda)CI\big)\\ &\ge \lambda(1-\lambda)C \operatorname{tr} H_P \ge c_k b_{z\bar z},\end{aligned}$$

where $c_k > 0$ may depend on k and C. Here $P_{z\bar z} = b^{-1} + |w|^2 b_{z\bar z}$. With $\zeta = z - a_k$, shrink the first factor of V_k so that

$$b_{z\bar z} = |\zeta|^{2\alpha_k} \mathrm{e}^{H_k} \left|\frac{\alpha_k}{\zeta} + \partial_z H_k\right|^2 \ge c_k' |\zeta|^{2\alpha_k - 2}$$

for some $c_k' > 0$. Since $F = C^{-2} \det H_v$, this lower bound implies $F \notin L^p$ whenever $p(1-\alpha_k) \ge 1$: the w-disc has positive area and the radial integral in z contains

$$\int_0^{\delta_k} r^{1-2p(1-\alpha_k)}\, \mathrm{d}r = +\infty.$$

For each $p > 1$ such a k exists because $\alpha_k \downarrow 0$. □

The bounded-product capacity estimate in Section 1.6 also holds for the discontinuous potential P. Thus a first-order capacity bound, even together with a constant positive density, cannot supply the missing boundary control.

Example 13.19.2 On the bidisc in $\mathbb{C}^2$ with coordinates (z, w), the function

$$H(z, w) = 2|z|(1 + |w|^2)$$

is locally Lipschitz, plurisubharmonic and not C^1 along $\{z = 0\}$. Plurisubharmonicity follows by writing its two summands as convex exponential compositions of $\log|z|$

and $\log|z| + \log|w|^2$. At $z \neq 0$,

$$H_{z\bar{z}} = \frac{1+|w|^2}{2|z|}, \qquad H_{w\bar{w}} = 2|z|, \qquad |H_{z\bar{w}}|^2 = |w|^2.$$

Hence $\det(H_{j\bar{k}}) = 1$, and pluripolar nullity gives $(\mathrm{dd^c}H)^2 = 2\pi^{-2}\,\mathrm{d}\Lambda$ on the entire bidisc.

This last example rules out a purely local implication from Hölder continuity and a smooth positive density to smoothness. It does not supply the global hypotheses on the regular locus in Theorem 13.17.1. In Theorem 13.19.2, the solution lacks local Hölder continuity on the indicated ball and the density is rough elsewhere on X, which is the regular locus of the chosen barrier $\rho = 0$. The trace criterion instead identifies the given bounded solution with a smooth Dirichlet solution on each ball.

13.20 Equations with resolved logarithmic poles

Let X be a connected compact complex manifold of dimension $n \geq 2$. A prescribed logarithmic singularity can be removed on a resolution. When its residual form is strictly positive, the bounded Hermitian equation supplies the remaining potential.

Let $h > 0$ be a smooth Hermitian form on X, and let $\psi \leq 0$ be quasi-plurisubharmonic, smooth outside a nonempty analytic pole set A. Assume there is a modification $\pi : Y \to X$, biholomorphic over $X \setminus A$, such that $\pi^{-1}(A) = |D|$ is a simple normal crossing divisor and

$$\psi \circ \pi = \sum_j c_j \log|s_j|^2 + b, \qquad \beta = \pi^* h + \mathrm{dd^c} b - \sum_j c_j \Theta_j > 0, \tag{13.53}$$

where $c_j > 0$, b is smooth, and $\mathrm{dd^c}\log|s_j|^2 = [D_j] - \Theta_j$.

Theorem 13.20.1 *For every nonzero nonnegative $f \in L^p(X, h^n)$, $p > 1$, there are $\kappa > 0$ and a normalized $u \in \mathrm{PSH}(X, h)$ such that*

$$\mathrm{MA}_h^{\mathrm{ext}}(u) = \kappa f h^n, \qquad u = \psi + O(1). \tag{13.54}$$

The lifted remainder $v = (u - \psi) \circ \pi$ extends Hölder continuously over Y, so u is locally Hölder on $X \setminus A$. Its current is

$$\pi^* h + \mathrm{dd^c}(u \circ \pi) = \sum_j c_j [D_j] + \beta + \mathrm{dd^c} v. \tag{13.55}$$

The positive residual current $\beta + \mathrm{dd^c} v$ has zero Lelong numbers. If h is Kähler, $\kappa = \int_Y \beta^n / \int_X f h^n$, and the normalized solution is unique among all exact-ordinary-type competitors for (13.54), including discontinuous competitors. If the transported density in the proof is smooth and strictly positive, the residual may be chosen smooth and strictly positive.

Proof Define $J = \pi^* h^n/\beta^n$ and $g = (f \circ \pi)J$ off the exceptional set, assigning any value to g on that volume-zero set. The smooth nonnegative J is bounded. Change of variables on the biholomorphic locus gives

$$\int_Y g\beta^n = \int_X f h^n, \qquad \int_Y g^p \beta^n \le \|J\|_\infty^{p-1} \int_X f^p h^n. \tag{13.56}$$

Thus g is a nonzero L^p density. Theorems 10.1.1 and 13.4.1 give a bounded Hölder β-plurisubharmonic v and $\kappa > 0$ with $(\beta + \mathrm{dd^c} v)^n = \kappa g \beta^n$. In the closed case, β is Kähler and the mass identity fixes the stated scalar. For smooth positive g, use the smooth Hermitian theorem instead by Theorem 9.1.1.

Set $u = \psi + v \circ \pi^{-1}$ on $X \setminus A$ and $u = -\infty$ on A. Push forward the positive current $\sum c_j [D_j] + \beta + \mathrm{dd^c} v$. Pushforward commutes with $\mathrm{dd^c}$, $\pi_* \pi^* h = h$, and the pushed locally integrable potential is u almost everywhere. Its quasi-plurisubharmonic representative equals the displayed function off A and tends to $-\infty$ on A because v is bounded. Hence $h + \mathrm{dd^c} u \ge 0$ and $u = \psi + O(1)$ in the two-sided ordinary sense. The pulled-back identity is (13.55); bounded residuals have zero Lelong number. The equation holds off A by (13.56), and Theorem 6.1.1 extends it as (13.54). Subtracting $\sup_X u$ changes neither the equation nor the ordinary type.

For closed-case uniqueness, let u' be an exact-type competitor. On $Y \setminus D$, $w = (u' - \psi) \circ \pi$ is bounded and β-plurisubharmonic. Local bounded removal extends it to a bounded β-plurisubharmonic function on Y by Lemma 1.3.7. On each relatively compact subset of $Y \setminus D$, sufficiently deep truncations of u' equal u'. Thus locality in Theorem 6.1.1 gives the bounded equation $\kappa g \beta^n$ there; both sides give D zero mass. The global equation follows, its mass fixes κ, and closed full-mass uniqueness gives $w = v$+constant as recalled in Section 33.17. Normalization fixes that constant. □

The Hölder constants depend on the fixed geometry and divisor metrics, the c_j, a lower positivity bound for β, p, and the actual normalized L^p density κg. No estimate uniform under degenerating β or uncontrolled κ follows. In simple normal crossing coordinates a density comparable to $\prod_j |z_j|^{-2a_j}$ belongs locally to L^p exactly when

$$p a_j < 1 \quad \text{for every } j, \tag{13.57}$$

by the factors $\int_0^\epsilon r_j^{1-2pa_j}\, \mathrm{d}r_j$. Smooth positive f downstairs need not make g strictly positive upstairs, since J can vanish on the exceptional set.

Example 13.20.1 Take $X = \mathbb{P}^n$, $h = \lambda \omega_{\mathrm{FS}}$, $\int_X \omega_{\mathrm{FS}}^n = 1$, and $0 < c < \lambda$. Put

$$\psi_c = c \log \frac{\sum_{j=1}^n |Z_j|^2}{\sum_{j=0}^n |Z_j|^2}.$$

Its pole is $P = [1 : 0 : \cdots : 0]$. On the point blowup $\pi : Y \to X$, let $q : Y \to \mathbb{P}^{n-1}$ be the direction map. In affine blowup coordinates $x = (z, zw)$, Poincaré–Lelong gives

$$\pi^* h + \mathrm{dd^c}(\psi_c \circ \pi) = c[E] + (\lambda - c)\pi^* \omega_{\mathrm{FS}} + c q^* \omega_{\mathrm{FS}}.$$

The residual form is positive: π detects normal directions on E, and q detects its tangent directions. Write $H = \pi^*[\omega_{\mathrm{FS}}]$. Since $H|_E = 0$ and $E^n = (-1)^{n-1}$, its mass is

$$\int_Y \big((\lambda - c)\pi^*\omega_{\mathrm{FS}} + cq^*\omega_{\mathrm{FS}}\big)^n = \lambda^n - c^n. \tag{13.58}$$

This yields the exact nonminimal type for every nonnegative L^p density of the prescribed residual mass. Choose a small smooth v with $v|_E$ nonconstant and positive smooth residual $S = \beta + \mathrm{dd^c}v$. Its pushforward density lies in every L^p with $1 < p < n/(n-1)$: π^*h^n vanishes as $|z|^{2(n-1)}$, so the local pth-power integral is finite precisely when $(n-1)(p-1) < 1$. Different directions towards P give different limits of $u - \psi_c$. The downstairs current has no divisor supported at a point. The regular metric is $(Y \setminus E, S)$ and completes by restoring the whole divisor E; deletion of a smooth real-codimension-two subset does not alter the smooth ambient distance infimum. A sufficiently small nonclosed perturbation of h preserving both positive forms retains the Hermitian existence conclusion.

13.20.1 Residual mass

The power condition of Theorem 2.2.2 fixes the mass of every bounded residual solution. It persists under pullback and addition of a smooth closed form: expand each power and apply the original identities to its terms. On a Gauduchon surface the condition is simply $\mathrm{dd^c}h = 0$, so it includes the nonclosed Hopf metric of Proposition 19.6.1.

Proposition 13.20.1 *Under* (13.53)*, assume* $\mathrm{dd^c}(h^k) = 0$ *for* $1 \le k \le n-1$*, and put* $H = \pi^*h$*. Then*

$$\int_X h^n - \int_Y \beta^n = \sum_j c_j \sum_{i=0}^{n-1} \int_{D_j} H^i \wedge \beta^{n-1-i} > 0. \tag{13.59}$$

Thus the constructed solution of Theorem 13.20.1 has $\kappa = \int_Y \beta^n / \int_X f h^n$*. If* $\int_X f h^n = \int_X h^n$*, its scalar is strictly less than one.*

Proof Let $C = \sum_j c_j \Theta_j - \mathrm{dd^c}b$, so $\beta = H - C$ with C closed. The mixed forms $Q_i = H^i \wedge \beta^{n-1-i}$ satisfy $\mathrm{dd^c}Q_i = 0$: expand $\beta = H - C$ and apply the power condition to each H^k. With $\Psi = \psi \circ \pi$, $\mathrm{dd^c}\Psi = \sum_j c_j [D_j] - C$. Distributional integration by parts against the smooth $\mathrm{dd^c}$-closed Q_i gives

$$\int_Y (H^n - \beta^n) = \sum_i \int_Y C \wedge Q_i = \sum_j c_j \sum_i \int_{D_j} Q_i.$$

Change of variables gives $\int_Y H^n = \int_X h^n$. Every integrand on the right is nonnegative, since H is semipositive and β positive; the $i = 0$ term on each nonempty D_j is strictly

positive. This includes exceptional divisors. By Theorem 2.2.2, the residual solution has mass $\int_Y \beta^n$, giving the scalar. □

13.21 Minimal singularities on blowups

An exceptional divisor can be forced by the background itself. Its coefficient is then part of the minimal ordinary type, while the remainder descends to the target. The density changes by the Jacobian, which determines the loss of integrability.

Let (Y, h) be a connected compact Hermitian n-fold, $n \geq 2$, let $Z \subset Y$ be a connected smooth center of codimension $r \geq 2$, and write $\pi : X = \mathrm{Bl}_Z Y \to Y$ with exceptional divisor E. Choose a smooth metric on $\mathcal{O}(E)$, put $L = \log |s_E|^2$, and fix $c > 0$. The smooth background

$$\theta = \pi^* h + c\Theta_E \tag{13.60}$$

can be indefinite, and it is nonclosed when h is nonclosed. Positivity is imposed on $\theta + \mathrm{dd^c} u$, not on θ.

Proposition 13.21.1 *Every $u \in \mathrm{PSH}(X, \theta)$ has, up to an additive constant, a unique h-plurisubharmonic target potential q such that*

$$u = cL + q \circ \pi + C, \qquad \theta + \mathrm{dd^c} u = \pi^*(h + \mathrm{dd^c} q) + c[E]. \tag{13.61}$$

The exact minimal ordinary type in this class is $cL + O(1)$, and u has this type if and only if q is bounded. There is a strict θ-plurisubharmonic barrier with logarithmic coefficient $c + \delta$ for some $\delta > 0$.

Proof Put $T = \theta + \mathrm{dd^c} u$ and $q = \pi_*(u - cL)$ as a scalar distribution. The latter is locally integrable: off Z it is the inverse-coordinate function, and a smooth target volume pulls back with bounded Jacobian, so its absolute integral is controlled by that of $u - cL$ on X. Since $\pi_*[E] = 0$,

$$S = \pi_* T = h + \mathrm{dd^c} q \geq 0.$$

A smooth local strictly plurisubharmonic shift gives a quasi-plurisubharmonic representative of q. Its pullback is genuine and positive by Lemma 1.22.1: the modification is dominant, so a local plurisubharmonic correction remains genuine after composition.

Off E, $T = \pi^* S$. The difference is an order-zero current supported on E, and it is closed: both derivatives equal $\pi^* \mathrm{d}h$. To identify it, write $R = T - \pi^* S$. In a chart where $E = \{z = 0\}$, order zero gives $zR = \bar{z}R = 0$. Closedness then gives $\mathrm{d}z \wedge R = \mathrm{d}\bar{z} \wedge R = 0$. Thus only its normal $(1, 1)$ coefficient remains, a measure on E; the tangential equations in $\mathrm{d}R = 0$ make that measure a constant multiple of integration on E. Connectedness of E identifies one constant a globally, so $R = a[E]$. Its cohomology class is $c[E]$. On a line in an exceptional fiber, $\mathcal{O}(E)$ has degree -1,

so $[E]$ is nonzero and $a = c$. The difference between the two scalar potentials in (13.61) has $\mathrm{dd^c}$ zero. Local elliptic regularity makes it pluriharmonic; compactness and connectedness make it constant.

When $h > 0$, $q = 0$ is admissible and every h-plurisubharmonic q is bounded above on compact Y. Hence cL is minimal and two-sided bounded comparison with it is exactly boundedness of q. For the barrier choose the divisor metric so that $-\Theta_E$ is positive on the vertical tangent bundle of $E \to Z$, using the tautological $\mathcal{O}(-1)$. In adapted coordinates

$$\pi(y, z, w) = (y, z, zw), \tag{13.62}$$

$\pi^* h$ controls the nonvertical directions and $|z|^2|\mathrm{d}w|^2$. For small $\delta > 0$, the vertical positivity of $-\delta\Theta_E$ dominates its mixed errors by completing the square; away from E, compactness allows the same δ. Thus $\pi^* h - \delta\Theta_E > 0$, and the potential $(c + \delta)L$ has current $(c+\delta)[E] + \pi^* h - \delta\Theta_E$, dominating a positive Hermitian form. Changing the divisor metric changes this potential only by a smooth correction. □

Theorem 13.21.1 *Fix a smooth positive volume form Ω on X and a nonzero nonnegative $f \in L^p(X, \Omega)$, $p > 1$. There exist $\kappa > 0$ and a normalized minimal-type solution of*

$$\mathrm{MA}_\theta^{\mathrm{ext}}(u) = \kappa f \Omega, \qquad u = cL + q \circ \pi + C, \tag{13.63}$$

where q is Hölder on Y. The divisor coefficient is exactly c. In (13.62), $u - cL = q(y, 0) + C + O(|z|^\alpha)$ on smaller charts for some $\alpha > 0$, so its limit is constant along each exceptional fiber.

If $\pi_*(f\Omega) = F h^n$, then $F \in L^s(Y, h^n)$ for every

$$1 < s < \frac{rp}{(r-1)p + 1}. \tag{13.64}$$

For a point center, data bounded above and below by positive constants near E attain the strict threshold $s < n/(n-1)$, and for finite p no strictly larger universal exponent than $np/((n-1)p + 1)$ can be guaranteed. If h is Kähler, the class $[\theta]$ is big and non-nef, has volume $\int_Y h^n$, and $\kappa = \int_Y h^n / \int_X f\Omega$; the solution is unique among normalized minimal ordinary-type competitors for (13.63). Under $\mathrm{dd^c}(h^j) = 0$, $1 \le j \le n-1$, the scalar of the constructed solution has the same integral formula.

Proof In (13.62) the complex Jacobian is z^{r-1}. Put $J = \pi^* h^n / \Omega$, so $J \asymp |z|^{2(r-1)}$ on nested charts and $F \circ \pi = f/J$ off E. For $1 < s < p$, change of variables and Hölder's inequality give

$$\int_Y F^s h^n = \int_X f^s J^{1-s} \Omega \tag{13.65}$$

$$\le \|f\|_{L^p(\Omega)}^s \left(\int_X J^{-(s-1)p/(p-s)} \Omega \right)^{(p-s)/p}. \tag{13.66}$$

The normal complex integral is finite precisely if $(r-1)(s-1)p/(p-s) < 1$, which is (13.64). Theorems 10.1.1 and 13.4.1 on (Y, h) supply q and κ with $(h + \mathrm{dd^c} q)^n =$

$\kappa F h^n$. Use (13.61) off E and Theorem 6.1.1 across it. Hölder continuity gives $q(\pi(y,z,w)) = q(y,0) + O(|z|^\alpha)$ uniformly on smaller charts.

For f bounded above and below by positive constants near a point exceptional divisor, F is comparable to $|y|^{-2(n-1)}$, giving the strict threshold. More generally, $f \asymp |s_E|^{-2a}$ with $0 \le a < 1/p$ belongs to L^p upstairs, while its pushforward has threshold $s < n/(n-1+a)$. Letting $a \uparrow 1/p$ proves the stated limitation; it makes no finite-p endpoint assertion.

When h is Kähler, the positive form $\pi^* h - \delta\Theta_E$ makes $[\theta]$ big. Its degree on an exceptional-fiber line is $-c$, so it is not nef. The minimal current $c[E] + \pi^* h$ has non-pluripolar top mass $\int_Y h^n$, and the minimal mass theorem identifies this with $\mathrm{vol}([\theta])$ as recalled in Section 33.14. Full-mass uniqueness and mass integration fix the scalar and normalized current as recalled in Section 33.17. A possibly discontinuous exact-type competitor still has bounded q by Proposition 13.21.1; on every relatively compact set off E, deep truncations equal that competitor, so locality in Theorem 6.1.1 gives the bounded target equation off Z. Both target measures give Z zero mass, so closed uniqueness applies globally. If only the power conditions hold, Theorem 2.2.2 fixes the scalar of the constructed solution. Quantitative Hölder bounds depend on κF, not on an unnormalized F alone. □

For the prescribed pullback measure $\pi^*(h^n)$, choose $q = 0$ and $\kappa = 1$. The current is $c[E] + \pi^* h$; its residual form has exactly the exceptional vertical directions as kernel on E, and its density has order $|z|^{2(r-1)}$ times a smooth positive factor. The regular metric is isometric to $(Y \setminus Z, h)$ and its completion restores Z, collapsing each exceptional fiber. The distance to E is comparable to $|z|$ on bounded adapted charts. Neither the density order nor this distance exponent depends on c.

Example 13.21.1 If $n = 2$ and Z is a point on a compact Kähler surface, $[\theta] = \pi^*[h] + c[E]$ has Zariski positive part $\pi^*[h]$ and negative part cE: $E^2 = -1$ and $\pi^*[h] \cdot E = 0$. Its volume is $\int_Y h^2$, although it is non-nef. The scalar-one solution for $\pi^*(h^2)$ has exact logarithmic coefficient c, a quadratic density zero upstairs, and a regular metric whose completion collapses E to the target point. The stronger barrier from Proposition 13.21.1 has coefficient $c + \delta$.

Theorem 13.21.2 *Let $\pi : Y_m \to Y_0$ be a finite nonempty sequence of blowups of connected smooth centers of codimensions $r_j \ge 2$ over a compact Hermitian (Y_0, h_0). Let E_j be the exceptional divisor at stage j, let $\pi_{>j}$ denote the subsequent map, choose $a_j > 0$, and set*

$$\Delta = \sum_j a_j \pi_{>j}^* E_j, \qquad \mathrm{dd^c} L_\Delta = [\Delta] - \Theta_\Delta, \qquad \theta = \pi^* h_0 + \Theta_\Delta.$$

Every θ-positive current is $\pi^*(h_0 + \mathrm{dd^c} q) + [\Delta]$ and every potential is $L_\Delta + q \circ \pi + C$, with $q \in \mathrm{PSH}(Y_0, h_0)$. Minimal ordinary type is exactly $L_\Delta + O(1)$, equivalent to bounded q. For potentials of this type, every prime exceptional coefficient is the one prescribed by the effective total transform Δ. There is a strict barrier with divisor $\Delta + B$ for an effective exceptional B positive on every exceptional component. For

nonzero nonnegative upstairs L^p data, $p > 1$, a minimal solution has Hölder q, with pushed density in every L^s satisfying

$$1 < s < \frac{Rp}{(R-1)p+1}, \qquad R = \prod_j r_j. \tag{13.67}$$

For Kähler h_0 the mass, scalar, and closed full-mass uniqueness are those of the target; the power conditions on h_0 fix the scalar for the constructed nonclosed solution. For the pullback measure $\pi^*(h_0^n)$, the residual metric completes to Y_0, identifying fibers, and the density-zero divisor is fixed by the squared holomorphic Jacobian independently of the a_j.

Proof Apply Proposition 13.21.1 successively, allowing an intermediate smooth background to be indefinite. The difference between each current and the pullback of its pushforward is closed of order zero, and exceptional-fiber degree fixes its coefficient. Pullback of earlier divisor currents produces the total transforms, proving the representation without an SNC requirement. The initial $q = 0$ is admissible; upper bounds on compact targets prove minimality and the bounded-q criterion. At SNC charts this is the usual sum of logarithms.

At each stage the positive vertical curvature construction in Proposition 13.21.1 gives $h_j = \pi_j^* h_{j-1} - \delta_j \Theta_{E_j} > 0$ for a sufficiently small $\delta_j > 0$. Expanding yields $h_m = \pi^* h_0 - \Theta_B$ for $B = \sum_j \delta_j \pi_{>j}^* E_j$, effective with a positive coefficient on every exceptional component. Hence $h_m + [\Delta + B]$ is the strict barrier current. The Jacobian calculation and Hölder estimate apply stagewise. If $s = rp/((r-1)p+1)$, then $1 - 1/s = (1 - 1/p)/r$; iteration over the stages gives (13.67), with strictness retained. The target Hölder equation, mass and uniqueness are obtained as in Theorem 13.21.1. For the pullback measure take $q = 0$ and use the target path comparison away from the exceptional set. The Jacobian fixes the density, while $[\Delta]$ fixes the divisor current; these are independent pieces of the equation. □

13.22 Representing measures in an ordinary type

Return to a connected compact complex manifold X of dimension $n \geq 1$, with auxiliary Hermitian metric ω and smooth positive probability volume form $\mathrm{d}V$. Local measure representations can be patched with a strict barrier and truncated relative to a prescribed potential. A decreasing convex sum then gives a single Monge–Ampère measure dominating every local piece in the sense of absolute continuity. Its potential has the prescribed ordinary type, and the given measure is an integrable density against it.

Proposition 13.22.1 *Let $\theta \geq 0$ be smooth and big, and let $\varphi \in \mathrm{PSH}(X, \theta)$ admit a genuine $P_{\epsilon\theta}(\varphi)$ for some $0 < \epsilon < 1$. For every finite non-pluripolar measure μ there are*

$$\psi \in \mathrm{PSH}(X, \theta), \qquad \psi \sim \varphi, \qquad 0 \leq f \in L^1\big(\mathrm{MA}_\theta^{\mathrm{ext}}(\psi)\big)$$

such that

$$\mu = f\,\mathrm{MA}_\theta^{\mathrm{ext}}(\psi).$$

The reference measure may have infinite mass; its density f is integrable.

Proof Translate φ below zero. Put $q = P_{\epsilon\theta}(\varphi) \leq \varphi$, and choose a strict big potential $\rho \leq 0$ with $\theta + \mathrm{dd^c}\rho \geq a\omega$, where $a > 0$. Then

$$s = q + (1-\epsilon)\rho \leq \varphi, \qquad \theta + \mathrm{dd^c} s \geq (1-\epsilon)a\omega.$$

In particular s is genuine. Its pole set is pluripolar, as is that of φ, so both are negligible for μ.

Choose finitely many compact inner coordinate balls K_i covering X, with larger balls B_i. Apply Lemma 1.6.1 to $\mu_i = \mathbf{1}_{K_i}\mu$. This gives bounded local plurisubharmonic v_i, normalized by $1 \leq v_i \leq 2$, with

$$\mu_i \ll \nu_i := (\mathrm{dd^c} v_i)^n.$$

Take a smooth cutoff χ_i which is zero near K_i and -3 on a boundary collar of B_i. Choose $\delta_i > 0$ so small that

$$\theta + \mathrm{dd^c}\big(s + \delta_i(v_i + \chi_i)\big) \geq \tfrac{1}{2}(1-\epsilon)a\omega + \delta_i \mathrm{dd^c} v_i.$$

Let S_i equal $\max(s, s + \delta_i(v_i + \chi_i))$ on B_i and s elsewhere. The boundary collar makes this a global θ-plurisubharmonic function. Since $\chi_i \leq 0$ may be arranged,

$$s \leq S_i \leq s + 2\delta_i \leq \varphi + 2\delta_i.$$

For every positive integer m, set

$$q_{i,m} = \max(\varphi - m, S_i) - 2\delta_i.$$

These potentials have the same ordinary type as φ, and

$$\varphi - m - 2\delta_i \leq q_{i,m} \leq \varphi. \tag{13.68}$$

On a neighborhood of K_i, $S_i = s + \delta_i v_i$ away from the pole set of s. Let

$$D_{i,m} = \{s + \delta_i v_i > \varphi - m\}$$

in that neighborhood, restricting to points where the potentials are finite. This is a plurifinely open set there. Locality and positive mixed products give

$$\mathbf{1}_{D_{i,m}}\mathrm{MA}_\theta^{\mathrm{ext}}(q_{i,m}) \geq \delta_i^n \mathbf{1}_{D_{i,m}}\nu_i. \tag{13.69}$$

Here no boundedness of s is being assumed. After a local smooth correction, the positive product inequality holds at every common finite truncation level of s and the bounded potential v_i. Exhaust the complement of the pole set of s. The measure ν_i is non-pluripolar, so it loses no mass in that exhaustion. Then restrict the resulting inequality by plurifine locality. This proves (13.69) for the extended measures.

For μ_i-almost every point of K_i, both s and φ are finite, and hence that point belongs to $D_{i,m}$ for some m. Consequently, a Borel set which is null for every $\mathrm{MA}_\theta^{\mathrm{ext}}(q_{i,m})$ is null for every μ_i, and therefore for μ.

Enumerate the countable family $q_{i,m}$ as q_k, and write its bounds as $\varphi - C_k \le q_k \le \varphi$ with $C_k < \infty$. Choose positive t_k such that

$$\sum_k t_k < 1, \qquad B := \sum_k t_k C_k < \infty.$$

The finite convex combinations

$$\psi_N = \sum_{k=1}^{N} t_k q_k + \left(1 - \sum_{k=1}^{N} t_k\right)\varphi$$

decrease and lie between $\varphi - B$ and φ. Their genuine limit ψ is θ-plurisubharmonic and satisfies $\psi \sim \varphi$. For any fixed k, separate its summand to write

$$\psi = t_k q_k + (1 - t_k) w_k, \qquad w_k \in \mathrm{PSH}(X, \theta), \qquad w_k \sim \varphi.$$

The remainder is obtained by the same decreasing convex-sum construction. Positive non-pluripolar mixed products give

$$\mathrm{MA}_\theta^{\mathrm{ext}}(\psi) \ge t_k^n \mathrm{MA}_\theta^{\mathrm{ext}}(q_k).$$

This is valid also for infinite measures: first restrict away from the common pole set, which is the pole set of φ by the ordinary-type bounds, and then exhaust by finite truncations. All measures ignore that pole set. The preceding null-set conclusion now gives $\mu \ll \mathrm{MA}_\theta^{\mathrm{ext}}(\psi)$.

By Theorem 6.1.1, the reference measure is σ-finite. The Radon–Nikodym theorem applies to this σ-finite measure and the finite μ. It gives a nonnegative density f with

$$\int_X f \, \mathrm{MA}_\theta^{\mathrm{ext}}(\psi) = \mu(X) < \infty,$$

as claimed. If $\mu = 0$, one may simply take $\psi = \varphi$ and $f = 0$. □

Lemma 13.22.1 *Let $\theta \ge 0$ be smooth and big, and let μ be a finite non-pluripolar measure on X. There are a bounded $\psi \in \mathrm{PSH}(X, \theta)$ and a nonnegative $f \in L^1(\mathrm{MA}_\theta(\psi))$ such that*

$$\mu = f \, \mathrm{MA}_\theta(\psi).$$

Proof Apply Proposition 13.22.1 with $\varphi = 0$. Semipositivity gives $P_{\epsilon\theta}(0) = 0$ for every $0 < \epsilon < 1$, and $\psi \sim 0$ means that ψ is bounded. □

These representations give an L^1 density, whereas Theorem 13.9.1 requires a bounded density against the measure of a bounded potential. For an unbounded prescribed type, the representing potential is itself unbounded. The general-measure existence argument in Section 30.6 uses a bounded Hermitian reference and relative compactness to pass to the equation.

13.23 Lower mass and solvability for all measures

For a Hermitian background h, solvability for every finite non-pluripolar measure is equivalent to positive lower mass. When $\nu_-(h) = 0$, smooth densities with rapidly decreasing normalizers can be summed to produce a density admitting no positive scaled subsolution. Strict domination supplies the bound that makes this construction work. In the remaining sections, h denotes a Hermitian form on X.

For nonzero $0 \le q \in L^p(\mathrm{d}V)$ with $p > 1$, let $c_h(q) > 0$ be the scalar in its continuous equation, supplied by Theorem 10.1.1 and Proposition 10.5.1. For the measure $\nu = q\,\mathrm{d}V$ we also write $c_h(\nu) = c_h(q)$.

Lemma 13.23.1 *If $u \in \mathrm{PSH}(X, h)$ and $a > 0$ satisfy*

$$\mathrm{MA}_h^{\mathrm{ext}}(u) \ge aq\,\mathrm{d}V,$$

then $a \le c_h(q)$. Consequently, for nonzero nonnegative bounded densities q, r,

$$q \le r \implies c_h(q) \ge c_h(r), \qquad c_h(tq) = t^{-1}c_h(q) \quad (t > 0).$$

Proof Let v be a continuous solution for q. If $a > c_h(q)$, put $\tau = c_h(q)/a < 1$. Every translate $v + A$ admits a genuine envelope in background $h/2$, and $u \preceq v + A$. The inequality

$$\mathrm{MA}_h(v + A) = c_h(q)q\,\mathrm{d}V \le \tau\,\mathrm{MA}_h^{\mathrm{ext}}(u)$$

and Theorem 13.1.1 give $v + A \ge u$ for every real A. Letting $A \to -\infty$ at a point where u is finite is a contradiction.

If $q \le r$, a continuous solution for r is a subsolution for $c_h(r)q\,\mathrm{d}V$, giving the order inequality. Scalar uniqueness gives the homogeneity. The comparison used only boundedness of v, so its first conclusion also holds for any nonnegative density that has a bounded solution with scalar $c_h(q)$. □

Lemma 13.23.2 *Fix $p > 1$. If nonzero nonnegative densities $q_j, q \in L^p(\mathrm{d}V)$ satisfy $q_j \to q$ in L^p, then $c_h(q_j) \to c_h(q)$.*

Proof The Gauduchon trace inequality bounds $c_h(q_j)$ above, since $q_j^{1/n} \to q^{1/n}$ in L^n and every fixed positive trace test has a positive integral at q. The lower-density estimate of Theorem 10.3.1 bounds the scalars away from zero. Normalized continuous solutions consequently have uniformly bounded L^p densities. By Theorem 10.3.2, every subsequence has a further subsequence on which the potentials converge

uniformly and the scalars converge to some $c > 0$. Bounded product continuity identifies the limit equation as $\mathrm{MA}_h(u) = cq\,\mathrm{d}V$. Scalar uniqueness gives $c = c_h(q)$, proving convergence of the original sequence. □

Theorem 13.23.1 *Suppose $v_-(h) = 0$. There is a probability measure $\mu = f\,\mathrm{d}V$ with $f \in L^1(\mathrm{d}V)$ and $f \geq m > 0$ almost everywhere such that no $u \in \mathrm{PSH}(X, h)$ and $a > 0$ satisfy*

$$\mathrm{MA}_h^{\mathrm{ext}}(u) \geq a\mu. \tag{13.70}$$

In particular, the equation $\mathrm{MA}_h^{\mathrm{ext}}(u) = c\mu$ has no solution for any $c > 0$, even without a prescribed singularity type or a finite-mass requirement on the candidate potential.

Proof By Theorem 25.12.1, there are smooth strictly positive probability measures $\nu_j = f_j\,\mathrm{d}V$ such that

$$c_h(\nu_j) \leq 2^{-2j} \qquad (j \geq 1).$$

Set

$$\mu = \sum_{j=1}^{\infty} 2^{-j}\nu_j, \qquad f = \sum_{j=1}^{\infty} 2^{-j} f_j.$$

Monotone convergence gives $\int_X f\,\mathrm{d}V = 1$, so $f \in L^1$. Also $f \geq \frac{1}{2}\min_X f_1 =: m > 0$ almost everywhere. Absolute continuity with respect to smooth volume makes μ non-pluripolar.

If (13.70) held, then for every j

$$\mathrm{MA}_h^{\mathrm{ext}}(u) \geq a2^{-j}\nu_j.$$

Lemma 13.23.1 would imply $a2^{-j} \leq c_h(\nu_j) \leq 2^{-2j}$. Thus $a \leq 2^{-j}$ for all j, contradicting $a > 0$. □

Corollary 13.23.1 *The following conditions are equivalent:*

(1) $v_-(h) > 0$.
(2) *Every nonzero finite positive non-pluripolar measure μ admits $c > 0$ and $u \in \mathcal{E}_{\mathrm{env}}(X, h, 0)$ such that $\mathrm{MA}_h^{\mathrm{ext}}(u) = c\mu$.*
(3) *For every probability density $f \in L^1(\mathrm{d}V)$ bounded below by a positive constant, there are $u \in \mathrm{PSH}(X, h)$ and $a > 0$ such that $\mathrm{MA}_h^{\mathrm{ext}}(u) \geq af\,\mathrm{d}V$.*

Proof For the reference zero, $P_h[0] = 0$ and $P_{\varepsilon h}(0) = 0$ for every $0 < \varepsilon < 1$. The form h is semipositive and big. Thus (1) implies (2) by Theorem 30.1.1. Condition (2) implies (3), since an absolutely continuous measure is non-pluripolar. The preceding theorem proves the contrapositive of (3) implies (1). □

Corollary 13.23.2 *Assume $v_-(h) = 0$, and let $\mu = f\,\mathrm{d}V$ be the measure constructed above. Then $f \notin L^p(\mathrm{d}V)$ for every $p > 1$. Moreover, for every nonnegative probability density $g \in L^1(\mathrm{d}V)$ and every $0 < t < 1$, the probability density*

$$g_t = (1-t)g + tf$$

admits no positive scaled subsolution as in (13.70). Nevertheless $\|g_t - g\|_{L^1} \le 2t$. In particular, densities with this obstruction are dense among probability densities in L^1.

Proof If $f \in L^p$ for some $p > 1$, Theorem 10.1.1 would give a continuous solution of $\mathrm{MA}_h(u) = c f\,\mathrm{d}V$ with $c > 0$, contradicting the theorem. Since $g_t\,\mathrm{d}V \ge t\mu$, a positive scaled subsolution for $g_t\,\mathrm{d}V$ would give one for μ. Finally $\|g_t - g\|_1 = t\|f - g\|_1 \le 2t$.□

Smooth strictly positive probability densities are also dense in L^1 and always admit a smooth normalized solution. Thus ordinary L^1 approximation alone cannot preserve solvability when the lower mass vanishes. The obstruction concerns the concentration of the density; it occurs even when that density is bounded below by a positive constant. It is realized on the compact threefold of Theorem 25.15.1.

This argument does not exclude unbounded solutions of the exponential equation. If $\mathrm{MA}_h^{\mathrm{ext}}(u) = \mathrm{e}^{\lambda u}\mu$ with $\lambda > 0$, the factor $\mathrm{e}^{\lambda u}$ need not have a positive lower bound. It does exclude bounded solutions for the measure above. Determining whether the exponential equation can absorb this concentration, and characterizing individual measures that admit a normalized solution when $v_-(h) = 0$, are distinct questions.

13.24 Subsolutions for an individual density

For one fixed density, the normalizers of its bounded truncations give an exact test for the existence of a positive scaled subsolution. A cluster limit attains the threshold because density lower bounds survive L^1 convergence of potentials through upper envelopes.

Let $0 \le f \in L^1(\mathrm{d}V)$ have positive integral, and put

$$f_k = \min(f, k), \qquad b_h(f) = \lim_{k\to\infty} c_h(f_k). \tag{13.71}$$

The limit exists in $[0, +\infty)$ because the constants decrease and $c_h(f_1)$ is finite. The order lemma also gives

$$b_h(f) = \inf\{c_h(q) : 0 \le q \le f,\ q \in L^\infty,\ \textstyle\int_X q\,\mathrm{d}V > 0\}. \tag{13.72}$$

In fact every such q is bounded above by f_k for sufficiently large k, and the f_k themselves are admissible tests.

A lower bound for a Monge–Ampère density survives L^1 convergence of potentials. This uses upper envelopes and damped continuity, rather than continuity of the original Monge–Ampère measures.

Lemma 13.24.1 *Let $s_j, s \in \mathrm{PSH}(X,h)$ have a common upper bound and satisfy $s_j \to s$ in $L^1(X)$. Suppose $g_j, g \geq 0$, $g_j \to g$ in $L^1(\mathrm{d}V)$, and*

$$\mathrm{MA}_h^{\mathrm{ext}}(s_j) \geq g_j \,\mathrm{d}V.$$

Then $\mathrm{MA}_h^{\mathrm{ext}}(s) \geq g \,\mathrm{d}V$.

Proof Translate the potentials to be nonpositive and extract a subsequence for which both convergences hold almost everywhere. Set

$$a_j = \inf_{\ell \geq j} g_\ell, \qquad w_j = \left(\sup_{\ell \geq j} s_\ell\right)^*.$$

Then $0 \leq a_j \uparrow g$ almost everywhere and $w_j \downarrow s$. For fixed j, the finite maxima $\max_{j \leq \ell \leq N} s_\ell$ have measures at least $a_j \,\mathrm{d}V$ by ordered contact. They increase to w_j after regularization, with the genuine lower bound s_j and upper bound zero. The convergence holds in capacity by Lemma 1.8.6. Damped continuity therefore gives

$$\mathrm{e}^{w_j} \mathrm{MA}_h^{\mathrm{ext}}(w_j) \geq \mathrm{e}^{w_j} a_j \,\mathrm{d}V.$$

Indeed the right sides for the finite maxima converge by dominated convergence, since $a_j \leq g$ almost everywhere. Divide by the positive weight off the pole set to obtain $\mathrm{MA}_h^{\mathrm{ext}}(w_j) \geq a_j \,\mathrm{d}V$.

Now $w_j \downarrow s$ is again convergence in capacity with a genuine lower bound. Apply damped continuity once more. Since $0 \leq \mathrm{e}^{w_j} a_j \leq g$, dominated convergence gives

$$\mathrm{e}^{s} \mathrm{MA}_h^{\mathrm{ext}}(s) \geq \mathrm{e}^{s} g \,\mathrm{d}V.$$

Divide on $\{s > -N\}$ and let $N \to \infty$. Both sides of the desired inequality give zero mass to the pole set. Restore the common translation. □

Theorem 13.24.1 *For every such f,*

$$b_h(f) = \max\{a \geq 0 : \textit{there exists } u \in \mathrm{PSH}(X,h) \textit{ with } \mathrm{MA}_h^{\mathrm{ext}}(u) \geq a f \,\mathrm{d}V\}.$$

In particular, a positive scaled subsolution exists if and only if $b_h(f) > 0$.

Proof If $\mathrm{MA}_h^{\mathrm{ext}}(u) \geq a f \,\mathrm{d}V$ with $a > 0$, then it dominates $a f_k \,\mathrm{d}V$. Lemma 13.23.1 gives $a \leq c_h(f_k)$ for every k, hence $a \leq b_h(f)$. If $b_h(f) = 0$, the zero potential realizes the coefficient zero.

Suppose $b = b_h(f) > 0$. Choose continuous normalized solutions

$$\mathrm{MA}_h(u_k) = c_h(f_k) f_k \,\mathrm{d}V, \qquad \sup_X u_k = 0.$$

After extraction, $u_{k_j} \to u$ in L^1, where u is a genuine normalized h-plurisubharmonic function. The densities $c_h(f_{k_j}) f_{k_j}$ converge to bf in L^1: they are dominated by $c_h(f_1) f$ and converge pointwise almost everywhere. Lemma 13.24.1 gives $\mathrm{MA}_h^{\mathrm{ext}}(u) \geq b f \,\mathrm{d}V$, attaining the maximum. □

The argument preserves a density lower bound. It does not claim that the original normalized solutions converge in capacity, that their limit has finite mass, or that the limiting inequality is an equality. These are additional requirements in the individual equation problem. The definition is unchanged if truncation is replaced by any increasing sequence of nonzero bounded densities tending to f: necessity bounds its limiting constants below by $b_h(f)$, and the same upper-envelope construction bounds them above by $b_h(f)$.

13.24.1 The greatest normalized threshold subsolution

At a positive threshold, the normalized subsolutions are stable under maxima and L^1 limits. Their upper envelope is therefore itself a subsolution and dominates every other normalized one.

Proposition 13.24.1 *For $b = b_h(f) > 0$, the family*

$$\mathcal{S}_f = \{s \in \mathrm{PSH}(X, h) : \sup_X s = 0,\ \mathrm{MA}_h^{\mathrm{ext}}(s) \geq b f \, \mathrm{d}V\}$$

is nonempty and compact in $L^1(X)$. It has a greatest member

$$\Phi_f = \left(\sup_{s \in \mathcal{S}_f} s\right)^* .$$

Every normalized L^1 cluster limit of the greatest exponential solutions from Corollary 13.28.1 belongs to $\mathcal{S}_f$ and hence lies below Φ_f.

Proof The family is nonempty by Theorem 13.24.1. Its compactness follows from normalized compactness and Lemma 13.24.1. Ordered contact makes it stable under finite maxima. Choose a countable Choquet family for its supremum and take successive finite maxima, including one fixed member of $\mathcal{S}_f$. This increasing sequence has a genuine lower bound and upper bound zero. Its canonical limit is Φ_f and its convergence holds in L^1. The same lemma gives the required measure inequality. Its supremum is zero because it lies between a normalized member and zero. Thus $\Phi_f \in \mathcal{S}_f$ and it dominates every member.

For an exponential cluster limit, the densities of the measures converge in L^1 to bf by Corollary 13.28.1. Apply the lemma to these densities and the normalized potentials. □

This proposition asserts a greatest normalized subsolution, not equality of its measure with $bf \, \mathrm{d}V$. In particular it does not assert that Φ_f has finite mass.

Proposition 13.24.2 *If $f \in L^p(\mathrm{d}V)$ for some $p > 1$, then $b_h(f)$ equals the normalization constant of the continuous equation for f. In general,*

$$b_h(f) \geq \frac{v_-(h)}{\int_X f \,\mathrm{d}V}.$$

If h is Kähler, then $b_h(f) = \int_X h^n / \int_X f \,\mathrm{d}V$ for every nonzero nonnegative L^1 density.

Proof Let v be a continuous solution for $f \in L^p$, with scalar $c > 0$. It is a subsolution for $c f \,\mathrm{d}V$, so the maximal-subsolution characterization gives $c \leq b_h(f)$. Conversely, the bounded-solution comparison in Lemma 13.23.1, applied to the attained threshold subsolution and v, gives $b_h(f) \leq c$.

Each bounded solution has total mass at least $v_-(h)$, so $c_h(f_k) \int_X f_k \,\mathrm{d}V \geq v_-(h)$. Take the limit. In the Kähler case, bounded Stokes instead gives the exact identity $c_h(f_k) \int_X f_k \,\mathrm{d}V = \int_X h^n$, which proves the last assertion. □

13.24.2 Smooth comparison tests

Bounded measurable tests in (13.72) detect concentrations that smooth tests can miss. To compare them, let f be a probability density bounded below by a positive constant and define

$$C_h^{\mathrm{sm}}(f) = \inf_g c_h(g) \|g/f\|_{L^\infty},$$

where g ranges over smooth strictly positive probability densities. The subsolution bound gives $b_h(f) \leq C_h^{\mathrm{sm}}(f)$, with equality for smooth f. The next construction separates even their positivity when lower mass vanishes.

Theorem 13.24.2 *If $v_-(h) = 0$, there is a probability density $f \in L^1(\mathrm{d}V)$ bounded below by a positive constant such that*

$$b_h(f) = 0 < C_h^{\mathrm{sm}}(f).$$

Thus no positive multiple of $f \,\mathrm{d}V$ is dominated by an extended Monge–Ampère measure, although every smooth comparison test gives a common positive bound.

Proof Let $(U_i)_{i \geq 1}$ be a countable basis of nonempty open subsets of X. We construct numbers $\alpha_j > 0$, bounded densities $q_j \not\equiv 0$, and closed nowhere dense sets K_j such that

$$\int_X q_j \,\mathrm{d}V \leq 1, \qquad \operatorname{supp} q_j \subset K_j, \qquad c_h(q_j) \leq \alpha_j 2^{-j}.$$

Start with $\alpha_1 = 1/4$. Once α_j is chosen, the identity between lower mass and the infimum of smooth normalization constants gives a smooth strictly positive probability density g_j with $c_h(g_j) < \alpha_j 2^{-j}$. Choose a closed nowhere dense K_j so that $q_j = g_j \mathbf{1}_{K_j}$ is sufficiently close to g_j in L^2 for the displayed scalar bound to hold, using Lemma 13.23.2. Such sets exist: remove an open dense set of arbitrarily

small measure from X; boundedness of the fixed g_j makes the L^2 error arbitrarily small. Choose it small enough also that q_j is nonzero.

After choosing $K_1, \dots, K_N$, put

$$m_N = \min_{1 \le i \le N} \mathrm{d}V\left(U_i \setminus \bigcup_{j=1}^{N} K_j\right) > 0.$$

Each set inside the minimum is nonempty and open. Choose

$$0 < \alpha_{N+1} \le \min\{\alpha_N/2,\ 2^{-N-2} m_N\},$$

and continue. In particular,

$$\sum_{j>N} \alpha_j \le 2\alpha_{N+1} \le 2^{-N-1} m_N.$$

The function $Q = \sum_j \alpha_j q_j$ is nonnegative and integrable. Its essential infimum on every nonempty open set is zero. To see this on U_i, take $N \ge i$ and remove the first N sets K_j. On the remaining set the first N summands vanish. Markov's inequality bounds the measure where the tail exceeds 2^{-N} by

$$2^N \sum_{j>N} \alpha_j \int_X q_j \,\mathrm{d}V \le m_N/2.$$

The remaining set originally has measure at least m_N. A positive-measure subset therefore satisfies $Q \le 2^{-N}$. Let $N \to \infty$ and use the basis property.

Set $M = \int_X (1+Q)\,\mathrm{d}V$ and $f = (1+Q)/M$. If $\mathrm{MA}_h^{\mathrm{ext}}(u) \ge a f \,\mathrm{d}V$ with $a > 0$, then the bounded comparison lemma gives

$$\frac{a\alpha_j}{M} \le c_h(q_j) \le \alpha_j 2^{-j} \quad \text{for every } j.$$

This is impossible. Hence $b_h(f) = 0$.

Finally let g be any smooth strictly positive probability density and put $R = \|g/f\|_\infty$. The inequality $g \le R(1+Q)/M$ almost everywhere forces $g \le R/M$ everywhere. Otherwise continuity would give a nonempty open set on which $g > R(1+\varepsilon)/M$ for some $\varepsilon > 0$, contradicting the essential infimum property of Q on that set. Order and homogeneity of the scalar now give

$$R c_h(g) \ge M c_h(1).$$

The constant test $g = 1$ attains this value, since $\operatorname{ess\,inf}_X Q = 0$. Thus $C_h^{\mathrm{sm}}(f) = M c_h(1) > 0$, as required. □

The distinction is caused by the order restriction on a smooth test: it cannot use a measurable increase of a density whose essential lower value is unchanged in every open set. Bounded measurable tests retain this information. The theorem

concerns subsolution existence for individual L^1 densities. A characterization of exact normalized equations at $b_h(f)$, including their natural singularity class and finite-mass compactness, remains a separate problem.

13.25 Gauduchon tests for integrable densities

The dual formula for smooth normalization constants extends to the threshold of an integrable density. First bound all Gauduchon tests in one norm to pass to bounded measurable densities. Monotone truncation then gives the formula at the L^1 endpoint, including a zero threshold.

For γ Gauduchon and a nonnegative measurable function p, put

$$L_{h,\gamma}(p) = \frac{\displaystyle\int_X p(\mathrm{d}V/\gamma^n)^{1/n}\gamma^n}{\displaystyle\int_X h \wedge \gamma^{n-1}}.$$

For real $p \in L^n(\mathrm{d}V)$, define

$$\rho_h(p) = b_h(|p|^n)^{-1/n} \quad (p \neq 0), \qquad \rho_h(0) = 0,$$

with $0^{-1/n} = +\infty$. Each individual test is finite on L^n, since its coefficient relative to $\mathrm{d}V$ is smooth. The supremum over γ may be infinite. In dimension one, the test is $\int_X p\,\mathrm{d}V / \int_X h$.

Theorem 13.25.1 *For every real $p \in L^n(\mathrm{d}V)$,*

$$\rho_h(p) = \sup_{\gamma \text{ Gauduchon}} L_{h,\gamma}(|p|). \tag{13.73}$$

Equivalently, for every nonzero nonnegative $f \in L^1(\mathrm{d}V)$,

$$b_h(f)^{1/n} = \inf_{\gamma \text{ Gauduchon}} \frac{\displaystyle\int_X h \wedge \gamma^{n-1}}{\displaystyle\int_X (f\,\mathrm{d}V/\gamma^n)^{1/n}\gamma^n}. \tag{13.74}$$

The formulas allow $\rho_h(p) = +\infty$ and $b_h(f) = 0$. No attainment of the infimum for a nonsmooth density is asserted.

Proof It suffices to prove (13.73) for $p \geq 0$, since both sides depend only on $|p|$; the case $p = 0$ is immediate. If p is smooth and strictly positive, Theorem 25.1.1, with the density p^n divided by its integral, gives

$$c_h(p^n)^{-1/n} = \sup_{\gamma} L_{h,\gamma}(p).$$

The integral normalization cancels by homogeneity. Also $b_h(p^n) = c_h(p^n)$.

We need uniformity in γ before approximating a measurable density. The lower-density estimate in Theorem 10.3.1, with exponent two, gives a constant $a_h > 0$ such that

$$c_h(p^n)\|p^n\|_{L^2} \geq a_h$$

for smooth strictly positive p. Therefore

$$L_{h,\gamma}(p) \leq a_h^{-1/n}\|p\|_{L^{2n}} \quad \text{for every } \gamma. \tag{13.75}$$

Positive smooth approximation and duality show that the smooth coefficient of each $L_{h,\gamma}$ has $L^{2n/(2n-1)}$ norm at most $a_h^{-1/n}$. In particular, the supremum of these functionals is Lipschitz on the positive cone of L^{2n}, with that same bound.

Now let $p \geq 0$ be bounded and nonzero. Approximate it in L^{2n} by smooth strictly positive p_j. Then $p_j^n \to p^n$ in L^2. By Lemma 13.23.2,

$$c_h(p_j^n) \longrightarrow c_h(p^n) > 0.$$

The uniform bound (13.75) also gives

$$\sup_\gamma L_{h,\gamma}(p_j) \longrightarrow \sup_\gamma L_{h,\gamma}(p).$$

The formula follows for bounded p.

For general nonnegative $p \in L^n$, put $p_k = \min(p, k)$. The definition of b_h gives

$$\rho_h(p) = \lim_{k\to\infty} c_h(p_k^n)^{-1/n} = \sup_k \sup_\gamma L_{h,\gamma}(p_k).$$

For each fixed γ, monotone convergence gives $\sup_k L_{h,\gamma}(p_k) = L_{h,\gamma}(p)$. The two suprema commute, proving (13.73). Every denominator in (13.74) is positive; taking reciprocals proves that formula, including the zero endpoint. □

For $n \geq 2$, write $L_{h,\gamma}(p) = \int_X p\, a_\gamma \,\mathrm{d}V$. Its coefficient satisfies

$$\|a_\gamma\|_{L^{n/(n-1)}} = \frac{\left(\int_X \gamma^n\right)^{(n-1)/n}}{\int_X h \wedge \gamma^{n-1}}.$$

Thus Theorem 25.12.1 identifies positive lower mass with a uniform bound for these endpoint norms. The weaker bound used above is available on every compact Hermitian manifold. It suffices to pass from smooth tests to bounded densities; monotone truncation then treats all integrable densities.

Proposition 13.25.1 *Let f be a nonzero nonnegative function in $L^1(\mathrm{d}V)$ and let $p \in L^n(\mathrm{d}V)$ be real. If h_1, h_2 are Hermitian forms on X and $mh_1 \le h_2 \le Mh_1$ for positive constants m, M, then*

$$m^n b_{h_1}(f) \le b_{h_2}(f) \le M^n b_{h_1}(f), \qquad M^{-1}\rho_{h_1}(p) \le \rho_{h_2}(p) \le m^{-1}\rho_{h_1}(p).$$

If $h_2 - h_1 = \mathrm{dd^c} a$ for a smooth real function a, their thresholds agree.

Proof For every Gauduchon metric the numerators in (13.74) satisfy the same bounds with factors m, M; the denominators agree. Taking the infimum and then the nth power proves the threshold bounds. Taking reciprocals gives the bounds for ρ, with the stated conventions at zero and infinity. In the last assertion, distributional Stokes makes the numerators equal. □

All Hermitian metrics on compact X are comparable. Thus positivity of the individual threshold and the set of admissible density roots are independent of the chosen Hermitian metric. If the reference volume is replaced by $\mathrm{d}V' = a\,\mathrm{d}V$ with $a > 0$ smooth, the same measure has density $f' = f/a$. The maximal-subsolution characterization gives $b'_h(f') = b_h(f)$, and $p \mapsto a^{-1/n}p$ preserves both the root functional and the L^n norm. This identifies the construction under a change of reference volume.

13.26 Stability of the subsolution threshold

The Gauduchon formula expresses $b_h^{1/n}$ as an infimum of continuous functions of the density. It therefore gives upper semicontinuity in L^1. For density roots, the reciprocal functional ρ_h is a supremum of positive linear tests; this gives a complete normed space on which the threshold has stronger stability.

Theorem 13.26.1 *Suppose f_j, f are nonzero nonnegative L^1 densities and $f_j \to f$ in $L^1(\mathrm{d}V)$. Then*

$$\limsup_{j\to\infty} b_h(f_j) \le b_h(f).$$

Moreover,

$$b_h(f) = \lim_{\delta\downarrow 0} \sup\{c_h(q) : q \in L^\infty,\ q \ge 0,\ q \not\equiv 0,\ \|q - f\|_1 < \delta\}. \tag{13.76}$$

The suprema are finite for sufficiently small δ.

Proof For a Gauduchon metric γ on X, write $A_\gamma(f) = \int_X (f\,\mathrm{d}V/\gamma^n)^{1/n}\gamma^n$. For each fixed γ, Hölder's inequality and $|s^{1/n} - t^{1/n}| \le |s - t|^{1/n}$ show that A_γ is continuous in L^1. It is positive at every nonzero nonnegative density. By Theorem 13.25.1, $b_h^{1/n}$ is the infimum of the continuous positive quotients $\int_X h \wedge \gamma^{n-1}/A_\gamma(f)$. It is therefore upper semicontinuous. Testing one fixed γ also bounds it above on an L^1 neighborhood of f.

For bounded q, $b_h(q) = c_h(q)$. The first assertion therefore bounds the upper limit of $c_h(q_j)$ along every bounded-density approximation $q_j \to f$. Conversely the canonical truncations of f attain $b_h(f)$ in the limit. Local boundedness and these two observations prove (13.76). □

Consequently b_h is continuous at each density where it vanishes. If $v_-(h) = 0$, it is discontinuous at every density where it is positive: normalize the density, use the obstructing mixtures of Corollary 13.23.2, and restore its integral. Thus the zero set is exactly the continuity set in that case. This statement concerns the scalar functional; it supplies neither capacity convergence of the original equation potentials nor equality of their limiting measures.

13.26.1 Finite sums of densities

The test formula makes ρ_h homogeneous and monotone. Its triangle inequality proves that adding finitely many admissible densities preserves a positive threshold. In the statements below, f, g are nonzero nonnegative L^1 densities, and p, q are real functions in L^n.

Proposition 13.26.1 *The functional ρ_h is homogeneous, satisfies $\rho_h(p) \le \rho_h(q)$ whenever $|p| \le |q|$, and obeys*

$$\rho_h(p + q) \le \rho_h(p) + \rho_h(q).$$

In particular, if $b_h(f), b_h(g) > 0$, then

$$b_h(f + g) \ge \left(b_h(f)^{-1/n} + b_h(g)^{-1/n}\right)^{-n} > 0.$$

Proof The representation (13.73) is a supremum of positive linear tests applied to $|p|$. It gives homogeneity, order, and the triangle inequality by $|p + q| \le |p| + |q|$. Finally $(f + g)^{1/n} \le f^{1/n} + g^{1/n}$ gives the displayed density inequality. □

Proposition 13.26.2 *The real vector space*

$$\mathcal{D}_h = \{p \in L^n(\mathrm{d}V) : \rho_h(p) < +\infty\}$$

is complete for the norm $\|p\|_{\mathcal{D}_h} = \|p\|_{L^n} + \rho_h(p)$. It is an order ideal in L^n: if $q \in \mathcal{D}_h$ and $|p| \le |q|$, then $p \in \mathcal{D}_h$. It contains L^∞ and is therefore dense in L^n for the L^n norm.

Moreover, $\mathcal{D}_h = L^n$ if and only if $v_-(h) > 0$. In that case its norm is equivalent to the L^n norm. If h is Kähler, then

$$\rho_h(p) = \left(\int_X h^n\right)^{-1/n} \|p\|_{L^n}.$$

Proof The preceding proposition proves that $\mathcal{D}_h$ is a vector space and that the displayed expression is a norm. It also proves the order-ideal assertion. Bounded functions belong to $\mathcal{D}_h$ by continuous equation existence.

If $p_j \to p$ in L^n, then $|p_j|^n \to |p|^n$ in L^1. Upper semicontinuity of b_h gives

$$\rho_h(p) \le \liminf_j \rho_h(p_j).$$

This includes $b_h(|p|^n) = 0$, when the right side must be infinite; for $p = 0$ the inequality follows from nonnegativity. Now let (p_j) be Cauchy in $\mathcal{D}_h$. It converges in L^n to some p. Lower semicontinuity, applied first to p_j and then to $p_j - p_k$ as $k \to \infty$, gives $p \in \mathcal{D}_h$ and

$$\rho_h(p_j - p) \le \liminf_{k\to\infty} \rho_h(p_j - p_k) \longrightarrow 0.$$

Together with L^n convergence, this proves completeness.

If $v_-(h) > 0$, Proposition 13.24.2 gives $\rho_h(p) \le v_-(h)^{-1/n}\|p\|_{L^n}$, proving equality of the spaces and equivalence of the norms. If $v_-(h) = 0$, take the obstructing density f from Theorem 13.23.1. Its root $f^{1/n}$ belongs to L^n but not to $\mathcal{D}_h$. The Kähler formula follows from the same normalization proposition. □

Thus, when lower mass vanishes, the roots of densities admitting positive scaled subsolutions form a proper dense subspace of L^n, complete for the stronger displayed norm. Finite addition preserves admissibility. An infinite sum need not: the positive smooth summands in Theorem 13.23.1 already give an L^1 counterexample. Density of bounded functions in the $\mathcal{D}_h$ norm has not been asserted. The metric comparisons in Proposition 13.25.1 show that this space is independent of h and that its norms for different Hermitian metrics are equivalent.

Proposition 13.26.3 *The norm ρ_h alone is complete on $\mathcal{D}_h$ if and only if $v_+(h) < +\infty$. When upper mass is finite,*

$$\|p\|_{L^n} \le v_+(h)^{1/n}\rho_h(p),$$

so ρ_h and $\|\cdot\|_{\mathcal{D}_h}$ are equivalent norms.

Proof If $v_+(h) < +\infty$, every continuous solution for a bounded density q satisfies $c_h(q)\int_X q\,\mathrm{d}V \le v_+(h)$. Apply this to the truncations of $|p|^n$ and let the truncation level increase. It gives the displayed bound. Completeness follows from Proposition 13.26.2.

Suppose instead that $v_+(h) = +\infty$. By Theorem 25.12.1, the constants of smooth positive probability densities are unbounded above. For each j, choose such a density f_j with

$$c_h(f_j) \ge (j2^j)^n, \qquad p_j = j f_j^{1/n}.$$

Then $\|p_j\|_{L^n} = j$ and $\rho_h(p_j) \le 2^{-j}$. The positive partial sums $P_N = \sum_{j\le N} p_j$ are Cauchy for ρ_h.

A fixed Gauduchon test has a smooth strictly positive coefficient, so (13.73) gives $\rho_h(p) \ge c\|p\|_{L^1}$ with a constant $c > 0$. Thus $\sum_j p_j$ converges in L^1 to a nonnegative

function P. Were P_N to converge for ρ_h to an element of $\mathcal{D}_h$, that element would equal P almost everywhere. But $P \geq p_j$ for all j, so $\|P\|_{L^n} \geq j$ for all j. This contradicts membership in L^n. Therefore ρ_h is not complete. □

The two mass properties have separate roles: positive lower mass makes every L^n root admissible, whereas finite upper mass makes the root norm control the L^n norm. The norm $\|\cdot\|_{\mathcal{D}_h}$ is complete without either assumption.

The geometric examples realize all four possibilities. The non-Kähler entries below follow from Corollaries 8.6.1 and 25.15.2 and Proposition 25.15.1.

Manifold	Domain $\mathcal{D}_h$	Is ρ_h complete?
Kähler manifold	L^n	Yes
Hopf threefold	L^n	No
Balanced elliptic quotient	$\mathcal{D}_h \subsetneq L^n$	Yes
Product example	$\mathcal{D}_h \subsetneq L^n$	No

13.26.2 Approximation by bounded density roots

Bounded approximation in the norm of $\mathcal{D}_h$ requires the tails to tend to zero uniformly over all Gauduchon tests. Denote the closure of bounded functions by $\mathcal{D}_h^0$. For every $r > n$, the lower-density estimate in Theorem 10.3.1, with exponent r/n, and Proposition 13.24.2 give

$$\rho_h(p) \leq C_r \|p\|_{L^r} \qquad (p \in L^r). \tag{13.77}$$

The constant depends on h, the reference volume and r, but not on p.

Proposition 13.26.4 *For $p \in \mathcal{D}_h$, the following conditions are equivalent:*

(1) $p \in \mathcal{D}_h^0$;
(2) $\rho_h(|p|\mathbf{1}_{\{|p|>K\}}) \to 0$ *as* $K \to \infty$;
(3) *whenever* $|q_j| \leq |p|$ *and* $q_j \to 0$ *in measure, one has* $\|q_j\|_{\mathcal{D}_h} \to 0$.

Equivalently, the second condition is

$$\sup_{\gamma \text{ Gauduchon}} L_{h,\gamma}(|p|\mathbf{1}_{\{|p|>K\}}) \longrightarrow 0.$$

The space $\mathcal{D}_h^0$ is a closed order ideal. It contains L^r for every $r > n$, and smooth functions are dense in it. If $v_-(h) > 0$, then $\mathcal{D}_h^0 = \mathcal{D}_h = L^n$.

Proof For $K > 0$, set $p_K = \operatorname{sgn}(p)\min(|p|, K)$. Then $|p - p_K| = (|p| - K)_+$ and

$$(|p| - K)_+ \leq |p|\mathbf{1}_{\{|p|>K\}}, \qquad |p|\mathbf{1}_{\{|p|>2K\}} \leq 2(|p| - K)_+.$$

The L^n tails tend to zero. These inequalities show that the second condition is equivalent to $p_K \to p$ in $\mathcal{D}_h$. Conversely, if a is bounded and $K \geq \|a\|_\infty$, then

$(|p| - K)_+ \le |p - a|$. Approximation by bounded a therefore proves the equivalence of the first two conditions.

Assume the second condition and let q_j be as in the third. Fix K. On $\{|p| \le K\}$, the functions q_j are uniformly bounded and tend to zero in measure, hence in L^{2n}. Equation (13.77) makes the root norm of this part tend to zero. On the complementary set it is at most $\rho_h(|p|\mathbf{1}_{\{|p|>K\}})$. Let first j and then K tend to infinity. This proves $\rho_h(q_j) \to 0$. The same split, or dominated convergence in measure, gives $\|q_j\|_{L^n} \to 0$. Conversely, apply the third condition to $q_j = |p|\mathbf{1}_{\{|p|>j\}}$.

The Gauduchon version is (13.73). Solidity transfers the tail condition from p to every function dominated by $|p|$, proving the order-ideal assertion. For $p \in L^r$, its bounded truncations converge in L^r, hence in $\mathcal{D}_h$ by (13.77). Bounded functions admit smooth L^{2n} approximations, which converge in $\mathcal{D}_h$ by the same estimate; smooth functions are thus dense in the closure. Under positive lower mass, Proposition 13.26.2 identifies the $\mathcal{D}_h$ and L^n norms up to equivalence, proving the last assertion. □

The metric comparisons in Proposition 13.25.1 identify $\mathcal{D}_h^0$ for different Hermitian metrics. Its defining closure is therefore intrinsic to the compact complex manifold and the identification of density roots. The tail condition controls the scalar threshold under approximation: if $p_j \to p \ne 0$ in $\mathcal{D}_h$, then

$$|\rho_h(p_j) - \rho_h(p)| \le \rho_h(p_j - p) \longrightarrow 0, \qquad b_h(|p_j|^n) \longrightarrow b_h(|p|^n) > 0.$$

The same tail estimate gives a uniform exponential-damping modulus and scalar error bounds in Proposition 13.28.1. It does not, by itself, give capacity convergence of equation potentials. When lower mass vanishes, equality of $\mathcal{D}_h^0$ and $\mathcal{D}_h$ remains a separate question.

13.27 Limits of exponential equations

For an unbounded prescribed type, the zero-exponent limit must be formulated through normalized potentials and their measures. Even multiplying an individual unbounded potential by a positive exponent leaves its uniform distance from every finite constant infinite. The useful convergence comes instead from integration against the fixed right-hand-side measure.

Proposition 1.23.1 makes this obstruction precise for every unbounded ordinary type.

For a fixed finite non-pluripolar measure μ and solutions of $\mathrm{MA}_\theta(u_\lambda) = \mathrm{e}^{\lambda u_\lambda}\mu$, write

$$v_\lambda = u_\lambda - \sup_X u_\lambda, \qquad c_\lambda = \mathrm{e}^{\lambda \sup_X u_\lambda}.$$

By Lemma 1.23.2, if $c_\lambda \to c < +\infty$, then the right-hand sides converge in total variation to $c\mu$. For an L^p density, Lemma 1.23.2 gives a linear error in the exponential factor. Identifying the measure of a limit of v_λ additionally requires a continuity theorem for the Monge–Ampère operator.

For L^1-convergent potentials, Corollary 13.8.1 identifies the limiting measure under its ordinary-type, common-barrier and domination hypotheses. In a relative envelope class, Theorem 30.3.1 instead constructs the lower envelope under positive lower mass: for a semipositive big θ and a reference satisfying (12.1), an L^1-convergent sequence in that class has capacity and weak measure convergence if one finite non-pluripolar measure dominates its measures. The latter argument is developed with its mass hypothesis in Chapter 30.

13.27.1 A positive limiting exponent

At a strictly positive limiting exponent, exponential decay controls the mass on deep sublevels uniformly. Once the potentials converge in capacity, the tail criterion identifies their limiting Monge–Ampère measure. Convergence of the right-hand-side densities then identifies the equation.

Theorem 13.27.1 *Let θ be any smooth real $(1,1)$-form and μ a finite non-pluripolar measure. Suppose $f_j \geq 0$, $f_j \to f$ in $L^1(\mu)$, and $\lambda_j > 0$ with $\lambda_j \to \lambda > 0$. Let $u_j, u \in \mathrm{PSH}(X,\theta)$ be nonpositive, suppose $u_j \to u$ in capacity, and assume*

$$\mathrm{MA}_\theta^{\mathrm{ext}}(u_j) = \mathrm{e}^{\lambda_j u_j} f_j\, \mu.$$

Then $\mathrm{MA}_\theta^{\mathrm{ext}}(u)$ is finite and

$$\mathrm{MA}_\theta^{\mathrm{ext}}(u) = \mathrm{e}^{\lambda u} f\, \mu.$$

The measures on the left converge weakly; the right-hand sides converge in total variation.

Proof Discard finitely many indices so that $\lambda_j \geq \lambda/2$. Their masses are at most $\|f_j\|_{L^1(\mu)}$, hence uniformly bounded. On a deep sublevel,

$$\mathrm{MA}_\theta^{\mathrm{ext}}(u_j)(\{u_j < -T\}) \leq \mathrm{e}^{-\lambda T/2}\|f_j\|_{L^1(\mu)}.$$

The tail criterion Theorem 7.4.1, in its general-background form Theorem 7.6.1, shows that the limiting Monge–Ampère measure is finite and that the left-hand sides converge weakly to it.

It remains to identify this weak limit. By Lemma 1.8.5, $u_j \to u$ in μ-measure. The functions $\mathrm{e}^{\lambda u_j}$ are bounded by one and converge in μ-measure to $\mathrm{e}^{\lambda u}$. The change of exponent is uniform for nonpositive arguments, because the mean-value theorem gives

$$\sup_{t\leq 0} |\mathrm{e}^{\lambda_j t} - \mathrm{e}^{\lambda t}| \leq |\lambda_j - \lambda| \sup_{s\geq 0} s\mathrm{e}^{-(\lambda/2)s} \longrightarrow 0.$$

Thus $\mathrm{e}^{\lambda_j u_j} \to \mathrm{e}^{\lambda u}$ in μ-measure and, by absolute continuity, in $f\mu$-measure as well. Boundedness by one and finiteness of $f\mu$ give convergence in $L^1(f\mu)$. Therefore

$$\int_X |\mathrm{e}^{\lambda_j u_j} f_j - \mathrm{e}^{\lambda u} f| \, \mathrm{d}\mu \le \|f_j - f\|_{L^1(\mu)} + \int_X |\mathrm{e}^{\lambda_j u_j} - \mathrm{e}^{\lambda u}| f \, \mathrm{d}\mu \longrightarrow 0.$$

This proves total-variation convergence and identifies the equation for u. □

Normalizing constants can be absorbed into f_j whenever the resulting densities converge in $L^1(\mu)$. The theorem does not assert that an arbitrary L^1-convergent sequence converges in capacity or stays in a prescribed envelope class.

This argument also pinpoints the difficulty as $\lambda_j \downarrow 0$: the tail bound $\mathrm{e}^{-\lambda_j T}\|f_j\|_1$ no longer tends to zero uniformly in j as $T \to \infty$. A zero-exponent approximation needs another source of uniform tail control, such as the integrability condition in Corollary 7.4.1. Global bounded mass would not by itself supply that missing tail estimate.

13.28 Exponential equations for an individual density

Fix a nonzero nonnegative $f \in L^1(\mathrm{d}V)$. An exponential subsolution is a common lower bound for the continuous solutions of the truncated equations. Those solutions decrease, so Theorem 13.27.1 gives an exact equation at the limit. The limiting solution dominates every subsolution. A positive threshold $b_h(f)$ supplies the required lower bound for each exponent.

Theorem 13.28.1 *Fix $\lambda > 0$. Suppose there is $s \in \mathrm{PSH}(X, h)$ such that*

$$\mathrm{MA}_h^{\mathrm{ext}}(s) \ge \mathrm{e}^{\lambda s} f \, \mathrm{d}V.$$

Then the equation

$$\mathrm{MA}_h^{\mathrm{ext}}(u_\lambda) = \mathrm{e}^{\lambda u_\lambda} f \, \mathrm{d}V$$

has a greatest solution. More precisely, u_λ is the greatest genuine h-plurisubharmonic subsolution of this equation, and its Monge–Ampère measure is finite. It is the decreasing limit of the continuous solutions for $f_k = \min(f, k)$.

Proof Let u_k solve the continuous exponential equation for f_k, using Theorem 10.1.2. By Lemma 10.4.2, $u_{k+1} \le u_k$.

Every genuine subsolution v for f lies below each u_k. Indeed, for $t > 0$, on $\{u_k < v - t\}$ we have

$$\mathrm{MA}_h(u_k) = \mathrm{e}^{\lambda u_k} f_k \, \mathrm{d}V \le \mathrm{e}^{-\lambda t} \mathrm{e}^{\lambda v} f \, \mathrm{d}V \le \mathrm{e}^{-\lambda t} \mathrm{MA}_h^{\mathrm{ext}}(v - t).$$

The first potential u_k is bounded, so $P_{h/2}(u_k)$ is genuine and $v - t \preceq u_k$. Strict domination, Theorem 13.1.1, gives $u_k \ge v - t$. Let $t \downarrow 0$.

In particular, $s \le u_k \le u_1$. Thus the decreasing limit u_λ is genuine. By Lemma 1.8.6, the convergence holds in capacity. Choose a common constant $A \ge \sup_X u_1$. Apply

Theorem 13.27.1 to $u_k - A$, with densities $e^{\lambda A} f_k \to e^{\lambda A} f$ in L^1. It proves the equation and finiteness of its left side. Finally every subsolution v lies below all u_k, hence below u_λ, which is itself a subsolution. □

Every subsolution lies below this solution, so its extremal property is independent of the truncation used to construct it. Smaller solutions with other singularities are not excluded. For $f \in L^p$, $p > 1$, the continuous solution gives a bounded lower barrier. The greatest solution is then bounded and agrees with it by bounded exponential comparison.

Corollary 13.28.1 *If $b = b_h(f) > 0$, the preceding greatest solution exists for every $\lambda > 0$. If $v \in \mathrm{PSH}(X, h)$ has $\sup_X v = 0$ and $\mathrm{MA}_h^{\mathrm{ext}}(v) \geq b f \, \mathrm{d}V$, then*

$$u_\lambda \geq v + \lambda^{-1} \log b.$$

Writing

$$M_\lambda = \sup_X u_\lambda, \qquad v_\lambda = u_\lambda - M_\lambda, \qquad c_\lambda = e^{\lambda M_\lambda},$$

we have

$$c_\lambda \longrightarrow b_h(f), \qquad e^{\lambda u_\lambda} f \longrightarrow b_h(f) f \quad \text{in } L^1(\mathrm{d}V) \quad (\lambda \downarrow 0).$$

Proof The attained-threshold theorem supplies v. Since $v \leq 0$, the function $s = v + \lambda^{-1} \log b$ satisfies

$$\mathrm{MA}_h^{\mathrm{ext}}(s) \geq b f \, \mathrm{d}V \geq e^{\lambda s} f \, \mathrm{d}V.$$

The theorem gives $u_\lambda \geq s$, hence $c_\lambda \geq b$.

Set $g_\lambda = e^{\lambda v_\lambda} f$. By Lemma 1.23.2, applied to the fixed finite measure $f \, \mathrm{d}V$, we have $g_\lambda \to f$ in L^1. Each g_λ is nonzero, since a genuine quasi-plurisubharmonic function is finite almost everywhere. The equation for u_λ implies

$$c_\lambda \leq b_h(g_\lambda)$$

by the maximal subsolution characterization. Upper semicontinuity therefore gives

$$b \leq \liminf_{\lambda \downarrow 0} c_\lambda \leq \limsup_{\lambda \downarrow 0} c_\lambda \leq b_h(f) = b.$$

This proves the scalar limit. Since $g_\lambda \to f$ in L^1, it also proves $c_\lambda g_\lambda \to b f$ in L^1. □

The right-hand-side measures therefore converge in total variation. This answers the scalar and measure questions in Section 13.27. Under positive lower mass, Corollary 30.7.1 also gives capacity convergence along every L^1-convergent normalized subsequence and identifies an exact solution at $b_h(f)$. For L^p data, $p > 1$, Proposition 13.18.1 gives uniform precompactness. In either case, convergence of the whole normalized family requires identification of all cluster limits. Smooth strictly positive dual attainment gives whole-family uniform convergence in Corollary 25.9.2. If the

smooth maximizing form is only semipositive, but strictly positive outside a proper analytic set, Proposition 25.9.5 gives whole-family L^1 convergence and a fixed lower barrier. For L^p data, $p > 1$, Corollary 25.9.5 strengthens this to uniform convergence.

For a general positive individual threshold, the corollary gives

$$v_\lambda \geq v - \lambda^{-1} \log(c_\lambda / b_h(f)).$$

The next estimate quantifies this scalar error when the density root is approximable by bounded roots. Whether an exponential subsolution exists when $b_h(f) = 0$ is a separate question; Theorem 13.28.1 applies whenever such a subsolution is available.

13.28.1 An error bound for the scalar limit

When $f^{1/n} \in \mathcal{D}_h^0$, the uniform tail condition quantifies the scalar limit. Split the root at height K: its bounded part has a linear exponential error, while its tail is small in the $\mathcal{D}_h$ norm. Optimizing K gives a modulus valid for every normalized potential.

Proposition 13.28.1 *Let $0 \leq p \in \mathcal{D}_h^0$ be nonzero, put $f = p^n$, $R = \rho_h(p)$ and $b = R^{-n}$. There is a constant C_h such that the function*

$$\eta_p(\lambda) = \inf_{K>0} \left(\|p \mathbf{1}_{\{p>K\}}\|_{\mathcal{D}_h} + C_h K \lambda \right)$$

tends to zero as $\lambda \downarrow 0$ and satisfies

$$\left\| p(1 - \mathrm{e}^{\lambda v/n}) \right\|_{\mathcal{D}_h} \leq \eta_p(\lambda) \qquad \left(v \in \mathrm{PSH}(X, h),\ \sup_X v = 0 \right). \tag{13.78}$$

For the greatest exponential solution and the normalizations of Corollary 13.28.1, if $\eta_p(\lambda) < R$, then

$$b \leq c_\lambda \leq \left(R - \eta_p(\lambda) \right)^{-n}$$

and

$$\left\| \mathrm{MA}_h^{\mathrm{ext}}(v_\lambda) - b f \, \mathrm{d}V \right\|_{\mathrm{TV}} \leq (c_\lambda - b) \|p\|_{L^n}^n + nb \|p\|_{L^n}^{n-1} \eta_p(\lambda).$$

If $p \in L^r$ for some finite $r > n$, the same bounds hold with $\eta_p(\lambda)$ replaced by $C_{h,r} \|p\|_{L^r} \lambda$. In particular the scalar and total-variation errors are $O(\lambda)$ for each such fixed density.

Proof By Lemma 1.5.1, normalized h-plurisubharmonic functions have uniformly bounded norms in every finite L^s. Equation (13.77), with $s = 2n$, therefore gives a uniform bound for $\|v\|_{\mathcal{D}_h}$. On $\{p \le K\}$ use

$$0 \le p(1 - \mathrm{e}^{\lambda v/n}) \le K\lambda(-v)/n;$$

on its complement use the upper bound p. Solidity and the triangle inequality give (13.78) for a fixed constant C_h, after taking the infimum over K. The root-tail criterion and the L^n tail estimate make the first term tend to zero as $K \to \infty$. Choose K first and then let $\lambda \downarrow 0$ to obtain $\eta_p(\lambda) \to 0$.

Set $q_\lambda = p\mathrm{e}^{\lambda v_\lambda/n}$, so that $q_\lambda^n = g_\lambda = \mathrm{e}^{\lambda v_\lambda} f$. The triangle inequality gives

$$R - \eta_p(\lambda) \le \rho_h(q_\lambda) \le R.$$

The previously proved inequalities $b \le c_\lambda \le b_h(g_\lambda) = \rho_h(q_\lambda)^{-n}$ give the scalar bound. Since $0 \le q_\lambda \le p$, the elementary factorization of $p^n - q_\lambda^n$ and Hölder's inequality imply

$$\|f - g_\lambda\|_{L^1} \le n\|p\|_{L^n}^{n-1}\|p - q_\lambda\|_{L^n} \le n\|p\|_{L^n}^{n-1}\eta_p(\lambda).$$

The identity $c_\lambda g_\lambda - bf = (c_\lambda - b)g_\lambda - b(f - g_\lambda)$, together with $\|g_\lambda\|_{L^1} \le \|f\|_{L^1}$, proves the total-variation bound.

Finally suppose $p \in L^r$ with $n < r < +\infty$, and choose $n < s < r$. By the same subcritical estimate and Hölder,

$$\|p(-v)\|_{\mathcal{D}_h} \le C_s\|p(-v)\|_{L^s} \le C_s\|p\|_{L^r}\|v\|_{L^{rs/(r-s)}} \le C_{h,r}\|p\|_{L^r}.$$

Thus the bound $1 - \mathrm{e}^{\lambda v/n} \le \lambda(-v)/n$ gives the asserted linear replacement in (13.78). Repeat the two estimates with this replacement. For small λ, the function $t \mapsto (R - t)^{-n}$ has bounded derivative on $0 \le t \le R/2$, proving the stated error orders. □

The general modulus tends to zero but has not been shown to be $O(\lambda)$. In particular,

$$0 \le \lambda^{-1}\log(c_\lambda/b) \le -\frac{n}{\lambda}\log\bigl(1 - \eta_p(\lambda)/R\bigr)$$

does not provide a uniform bound from root tails alone. Even when a uniform lower barrier is obtained, a capacity argument must check the singularity-type or lower-mass hypotheses of its continuity theorem. The proposition estimates scalars and measures; it does not supply exact threshold existence when lower mass vanishes.

Chapter 14
Normalization and weak adjoint measures

Let X be a connected compact complex manifold of dimension $n \geq 1$, with a smooth positive volume form $\mathrm{d}V$. Until Section 14.8, take θ smooth, semipositive and big, and a model reference φ satisfying (12.1). Function spaces are real, and their Lebesgue norms use $\mathrm{d}V$.

The prescribed equation determines a unique scalar even when it has several normalized potentials. Varying the logarithmic density makes this scalar a concave functional. Its supporting probabilities measure the response to that variation and extend the smooth Gauduchon adjoint; the convex dual controls their concentration through entropy. These measures give compactness for changing densities, references and positive backgrounds. They also provide the integral normalizations that single out the limit of the exponential equations. The examples with moving divisors show why the scalar functional alone cannot recover the singularities of a potential.

14.1 The scalar functional

Comparison makes the scalar decrease as the density increases. Mixed products strengthen this order to concavity in the logarithmic density; compactness of solutions then gives continuity under uniformly bounded L^1 perturbations.

For a bounded real measurable function F, Theorem 13.11.1 gives $u \sim \varphi$ and a unique $c_\varphi(F) > 0$ such that

$$\mathrm{MA}_\theta^{\mathrm{ext}}(u) = c_\varphi(F)\mathrm{e}^F \,\mathrm{d}V. \tag{14.1}$$

Define

$$B_\varphi(F) = \log c_\varphi(F).$$

Scalar uniqueness attaches B_φ to the ordinary type of φ.

Proposition 14.1.1 *Let $F, G \in L^\infty(\mathrm{d}V)$ be real. The functional B_φ is decreasing, and for every constant $a \in \mathbb{R}$,*

$$B_\varphi(F + a) = B_\varphi(F) - a.$$

Consequently

$$-\operatorname*{ess\,sup}_X (G - F) \le B_\varphi(G) - B_\varphi(F) \le -\operatorname*{ess\,inf}_X (G - F), \tag{14.2}$$

and $|B_\varphi(G) - B_\varphi(F)| \le \|G - F\|_\infty$.

Proof Suppose $F \le G$ almost everywhere and, to obtain a contradiction, $c_\varphi(F) < c_\varphi(G)$. Let $u, v \sim \varphi$ solve the two equations. Then, for every constant a,

$$\mathrm{MA}_\theta^{\mathrm{ext}}(u) \le \frac{c_\varphi(F)}{c_\varphi(G)} \mathrm{MA}_\theta^{\mathrm{ext}}(v + a).$$

The ratio is strictly less than one. Bounded equivalence transfers the genuine smaller-background envelope from φ to u, and $v + a \preceq u$. Singular strict domination gives $u \ge v + a$ for every a. This is impossible at any point where both potentials are finite. Thus $B_\varphi(F) \ge B_\varphi(G)$.

Replacing F by $F + a$ and the constant by $\mathrm{e}^{-a} c_\varphi(F)$ leaves the right side of (14.1) unchanged. Scalar uniqueness proves the translation rule. Apply monotonicity to the two constant translates of F bounding G to obtain (14.2). □

For a smooth real h, replacing θ by $\theta + \mathrm{dd^c} h$ and each potential by $u - h$ preserves the current and the prescribed ordinary type. Transporting the solution problem in this way leaves its scalar unchanged.

A convex combination of two solutions stays in the prescribed ordinary type. Its mixed density lower bound gives concavity.

Theorem 14.1.1 *The functional B_φ is concave on the real space $L^\infty(\mathrm{d}V)$: for F, G in that space,*

$$B_\varphi(tF + (1 - t)G) \ge tB_\varphi(F) + (1 - t)B_\varphi(G), \qquad 0 \le t \le 1.$$

Proof Let $u, v \sim \varphi$ solve the equations for F, G. Put $H = tF + (1 - t)G$ and $D = c_\varphi(F)^t c_\varphi(G)^{1-t}$. Lemma 6.4.1 and the arithmetic–geometric mean inequality give

$$\begin{aligned} \mathrm{MA}_\theta^{\mathrm{ext}}(tu + (1 - t)v) &\ge \big(t(c_\varphi(F)\mathrm{e}^F)^{1/n} + (1 - t)(c_\varphi(G)\mathrm{e}^G)^{1/n}\big)^n \,\mathrm{d}V \\ &\ge D\mathrm{e}^H \,\mathrm{d}V. \end{aligned}$$

The averaged potential has ordinary type φ. If $c_\varphi(H) < D$, its measure dominates that of a solution for H by a factor strictly greater than one. Apply singular strict domination to that solution and every constant translate of the averaged potential, as in the proof of Proposition 14.1.1. The same contradiction gives $c_\varphi(H) \ge D$. Taking logarithms proves concavity. □

For smooth Hermitian data, this concavity agrees with the differential calculation of Theorem 9.7.1.

Uniformly bounded logarithmic densities give common barriers for the solutions. Their compactness identifies the limiting scalar even when the logarithmic densities converge only in L^1.

Theorem 14.1.2 *If F_j, F are uniformly bounded and $F_j \to F$ in $L^1(\mathrm{d}V)$, then*

$$B_\varphi(F_j) \longrightarrow B_\varphi(F).$$

Proof Choose solutions $u_j \sim \varphi$ normalized by $\sup_X u_j = 0$, and write $c_j = c_\varphi(F_j)$. If $|F_j| \le M$, order and translation give

$$\mathrm{e}^{-M} c_\varphi(0) \le c_j \le \mathrm{e}^{M} c_\varphi(0).$$

In particular their Monge–Ampère measures are dominated by a fixed multiple of $\mathrm{d}V$. Theorem 13.11.2 gives a constant C such that

$$\varphi - C \le u_j \le \varphi.$$

From any subsequence extract $c_j \to c > 0$ and $u_j \to u$ in $L^1(X)$. Corollary 13.8.1 gives convergence in capacity, $u \sim \varphi$, and raw weak convergence of their measures. On the other hand, boundedness makes the exponential Lipschitz on the common range, so $c_j \mathrm{e}^{F_j} \to c\mathrm{e}^{F}$ in $L^1(\mathrm{d}V)$. Therefore

$$\mathrm{MA}_\theta^{\mathrm{ext}}(u) = c\mathrm{e}^{F}\,\mathrm{d}V.$$

Scalar uniqueness gives $c = c_\varphi(F)$. Every subsequence has a further subsequence with this scalar limit, proving convergence of the full sequence and then of its logarithms. □

14.2 Supporting probability measures

A supporting functional records the first-order response of the scalar to a change of logarithmic density. Monotonicity and the translation rule force its negative to be integration against a probability. Work first on $C(X)$ with the uniform norm. For $F \in C(X)$, let $\mathcal{A}_\varphi(F)$ be the set of Borel probability measures ν satisfying

$$B_\varphi(G) \le B_\varphi(F) - \int_X (G - F)\,\mathrm{d}\nu \qquad \text{for every } G \in C(X). \tag{14.3}$$

Thus $-\nu$ is a supergradient of B_φ at F.

Theorem 14.2.1 *For every $F \in C(X)$, the set $\mathcal{A}_\varphi(F)$ is nonempty, convex and weakly compact. Every one of its measures is absolutely continuous with respect to* $\mathrm{d}V$.

Proof Write $C_\varphi = -B_\varphi$, a convex Lipschitz function. For fixed $F \in C(X, \mathbb{R})$ and a direction $h \in C(X, \mathbb{R})$, the limit

$$p(h) = \lim_{t \downarrow 0} \frac{C_\varphi(F + th) - C_\varphi(F)}{t}$$

exists and is finite: convexity makes the difference quotients monotone in $t > 0$, and their absolute values are at most $\|h\|_\infty$. Positive homogeneity is immediate. Applying convexity with another direction $k \in C(X, \mathbb{R})$ at the midpoint of $F + 2th$ and $F + 2tk$ and then letting $t \downarrow 0$ gives $p(h + k) \le p(h) + p(k)$. Thus p is sublinear.

Hahn–Banach extends the zero functional on $\{0\}$ to a linear functional $\ell \le p$. Since $p(h) \le \|h\|_\infty$, application to h and $-h$ makes ℓ continuous. The monotonicity of the convex difference quotients gives

$$C_\varphi(F + h) \ge C_\varphi(F) + \ell(h).$$

The translation identity, tested with positive and negative constants, implies $\ell(1) = 1$. If $h \ge 0$, monotonicity gives $C_\varphi(F - th) \le C_\varphi(F)$ for $t > 0$, whereas the supporting inequality gives the reverse lower bound $C_\varphi(F) - t\ell(h)$. Hence $\ell(h) \ge 0$. By the Riesz representation theorem, $\ell(h) = \int_X h \, \mathrm{d}\nu$ for a positive probability measure ν. This is exactly (14.3), proving nonemptiness. Each inequality in that definition specifies a weakly closed half-space in the compact space of probability measures. Their intersection is closed and convex, hence weakly compact.

It remains to prove absolute continuity. Let K be compact with $\mathrm{d}V(K) = 0$. Choose continuous functions $0 \le \chi_j \le 1$ equal to one on K, whose supports have volumes tending to zero. For any $\nu \in \mathcal{A}_\varphi(F)$, the supporting inequality with $G = F + \chi_j$ gives

$$0 \le \nu(K) \le \int_X \chi_j \, \mathrm{d}\nu \le B_\varphi(F) - B_\varphi(F + \chi_j).$$

The right side tends to zero by Theorem 14.1.2. Thus $\nu(K) = 0$. Inner regularity of a finite Borel measure on the compact manifold proves the same for every Borel set of zero volume. The Radon–Nikodym theorem now gives $\mathrm{d}\nu = h_\nu \mathrm{d}V$ with $h_\nu \ge 0$ and $\int_X h_\nu \mathrm{d}V = 1$. □

Proposition 14.2.1 *If $F_j \to F$ uniformly and $\nu_j \in \mathcal{A}_\varphi(F_j)$, every weak cluster point of ν_j belongs to $\mathcal{A}_\varphi(F)$. In particular, if the latter set consists of a single measure, ν_j converges to it.*

Proof Take a weakly convergent subsequence and fix $G \in C(X)$. In

$$B_\varphi(G) \le B_\varphi(F_j) - \int_X (G - F_j) \, \mathrm{d}\nu_j,$$

uniform continuity of B_φ, uniform convergence of F_j and weak convergence of the probabilities allow passage to the limit. The resulting inequality holds for every G, so the limiting measure is in $\mathcal{A}_\varphi(F)$. If that set is a singleton, every convergent subsequence has the same limit; compactness proves the final assertion. □

14.2.1 Directional derivatives

The range of integrals against the supporting probabilities is exactly the gap between the two directional derivatives. Thus differentiability is equivalent to uniqueness of the supporting probability. For $F, G \in C(X)$, write $D^+B_\varphi(F)[G]$ and $D^-B_\varphi(F)[G]$ for the limits of the difference quotient as $t \downarrow 0$ and $t \uparrow 0$, respectively.

Proposition 14.2.2 *The one-sided derivatives exist, and*

$$D^+B_\varphi(F)[G] = -\max_{\nu\in\mathcal{A}_\varphi(F)} \int_X G \,\mathrm{d}\nu,$$
$$D^-B_\varphi(F)[G] = -\min_{\nu\in\mathcal{A}_\varphi(F)} \int_X G \,\mathrm{d}\nu.$$

Consequently B_φ is Gâteaux differentiable at F if and only if $\mathcal{A}_\varphi(F)$ is a singleton. Its derivative is then $G \mapsto -\int_X G \,\mathrm{d}\nu$.

Proof Existence and finiteness of the one-sided limits follow from concavity and the Lipschitz bound. For $t > 0$, every supporting measure at F gives

$$\frac{B_\varphi(F + tG) - B_\varphi(F)}{t} \le -\int_X G \,\mathrm{d}\nu.$$

Conversely choose $\nu_t \in \mathcal{A}_\varphi(F + tG)$ and apply its supporting inequality to F. This gives the lower bound $-\int_X G \,\mathrm{d}\nu_t$ for the same quotient. Along a sequence $t \downarrow 0$, extract a weak limit of ν_t. It belongs to $\mathcal{A}_\varphi(F)$ by Proposition 14.2.1. The two bounds identify the right derivative with the negative maximum. Apply this identity to $-G$ to obtain the left derivative.

If the set is a singleton, the two derivatives coincide for every G and equal the displayed continuous linear functional. Conversely, equality of the two one-sided derivatives for every G forces all supporting probabilities to have identical integrals against every continuous function. The Riesz theorem makes them identical. □

14.2.2 Generic uniqueness

The gap between the two directional derivatives vanishes densely along every affine line. Upper semicontinuity of that gap and separability of $C(X)$ turn this one-dimensional fact into uniqueness of the supporting probability on a dense G_δ in the uniform norm.

Theorem 14.2.2 *The set*

$$\mathcal{U}_\varphi = \{F \in C(X, \mathbb{R}) : \mathcal{A}_\varphi(F) \text{ is a singleton}\}$$

is a dense G_δ. At every $F \in \mathcal{U}_\varphi$, the functional B_φ is Gâteaux differentiable and its derivative is integration against minus the unique supporting probability. If $F_j \to F$ uniformly and $\nu_j \in \mathcal{A}_\varphi(F_j)$, then ν_j converge weakly to that probability.

Proof For $h \in C(X, \mathbb{R})$ define the width

$$w_h(F) = \max_{\nu \in \mathcal{A}_\varphi(F)} \int_X h \, \mathrm{d}\nu - \min_{\nu \in \mathcal{A}_\varphi(F)} \int_X h \, \mathrm{d}\nu \geq 0.$$

The extrema exist by weak compactness. This width is upper semicontinuous in F. Indeed, if $F_j \to F$, choose maximizing and minimizing probabilities at F_j. Along a subsequence realizing the upper limit of the widths, compactness of the space of probabilities gives weak limits of both choices. By Proposition 14.2.1, both limits belong to $\mathcal{A}_\varphi(F)$. Their difference of integrals is at most $w_h(F)$.

For every $\epsilon > 0$ the set $\{F : w_h(F) < \epsilon\}$ is therefore open. It is dense as well. If $h = 0$ there is nothing to prove. Otherwise fix F and any open ball about it. The function

$$b(t) = B_\varphi(F + th)$$

is finite and concave on the real line. Its nondifferentiability set is at most countable: its finite one-sided derivatives are monotone, and the nonempty open intervals between unequal derivatives at distinct points are disjoint, so each can be assigned a distinct rational number. Choose a differentiability point t sufficiently close to zero that $F + th$ remains in the given ball. The one-sided derivative formula of Proposition 14.2.2 gives $w_h(F + th) = 0$. This proves density.

The compact manifold X is metrizable, so $C(X, \mathbb{R})$ is separable. Choose a countable uniformly dense family (h_j). The Baire theorem shows that

$$\bigcap_{j,k \geq 1} \{F : w_{h_j}(F) < 1/k\}$$

is a dense G_δ. At every point of this intersection, all supporting probabilities have the same integral against every h_j. Uniform approximation and their unit masses then give equality of their integrals against every continuous function. The Riesz representation theorem makes the probabilities identical. Conversely, a singleton has zero width in every direction, so the displayed intersection is exactly $\mathcal{U}_\varphi$. The

derivative follows from Proposition 14.2.2, and the approximation assertion follows from Proposition 14.2.1. □

14.2.3 Gauduchon adjoints

For a smooth Hermitian equation, differentiating the scalar identifies the supporting probability with the adjoint measure of the linearized operator.

Theorem 14.2.3 *Let $\theta = \omega$ be Hermitian, let $\varphi = 0$, and let F be smooth. Let u be the smooth normalized solution of* (14.1)*, and let μ_u be its Gauduchon adjoint probability from Proposition 4.5.1. Then*

$$\mathcal{A}_0(F) = \{\mu_u\}.$$

For $n = 1$, put $\mu_u = (\omega + \mathrm{dd^c}u)/\int_X \omega$.

Proof In dimensions at least two, Theorem 9.7.1 shows that for every smooth real G,

$$\frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} B_0(F + tG) = -\int_X G \,\mathrm{d}\mu_u.$$

Scalar uniqueness identifies its normalizing constant with the present B_0. For any $\nu \in \mathcal{A}_0(F)$, testing the supporting inequality at $F + tG$ for both signs of t gives equality of the integrals of G against ν and μ_u. Smooth functions are uniformly dense in $C(X)$, so $\nu = \mu_u$. Nonemptiness proves the assertion. In dimension one Stokes gives $c_0(F) = \int_X \omega / \int_X \mathrm{e}^F \,\mathrm{d}V$, whose logarithmic derivative gives the stated measure directly. □

Thus for a continuous F in the Hermitian case, smooth approximations $F_j \to F$ uniformly produce Gauduchon adjoint probabilities whose weak cluster points all belong to $\mathcal{A}_0(F)$ and are absolutely continuous. If $\mathcal{A}_0(F)$ is a singleton, the weak limit is independent of this approximation. In particular this holds at every $F \in \mathcal{U}_0$. Uniqueness at every continuous density remains unresolved in this Hermitian setting, and likewise for the general singular references considered above.

In dimension $n \geq 2$, the Hermitian zero-reference case has a quantitative strengthening: Corollary 25.8.1 gives uniform L^q bounds for all these densities when $\|F\|_\infty$ is bounded and $1 < q < n/(n-1)$. It includes $q = n/(n-1)$ under the positive lower-volume hypothesis. Its proof uses the finite Gauduchon-current formulation of Chapter 25; no such bound for an arbitrary singular reference is inferred here.

A weak linearized operator requires more than a supporting probability. In this Hermitian zero-reference setting, Theorem 25.7.1 represents each supporting probability by a determinant-weighted positive $\mathrm{dd^c}$-closed current. Identifying that current with a scalar multiple of the weak power ω_u^{n-1} requires a justified product and compatibility with approximation.

For nonzero $0 \le f \in L^1(\mathrm{d}V)$ with $f^{1/n} \in \mathcal{D}^0_\omega$ as in Section 13.26.2, Proposition 25.9.3 gives a similar representation for $F \mapsto \log b_\omega(\mathrm{e}^F f)$. This functional comes from the subsolution threshold rather than an equation in a prescribed singularity type.

14.3 A convex functional on probability measures

To pass to limits in the supporting inequality, we need control of the mass that supporting probabilities can put on small sets. The convex dual of B_φ supplies this control: its sublevels are uniformly absolutely continuous. This will allow the logarithmic density and the supporting measure to vary together.

Let $\mathcal{P}(X)$ denote the Borel probability measures on X. Define

$$H_\varphi(\nu) = \sup_{G \in C(X)} \left(B_\varphi(G) + \int_X G \, \mathrm{d}\nu \right), \qquad \nu \in \mathcal{P}(X). \tag{14.4}$$

The value $+\infty$ is allowed. Taking $G = 0$ shows that $H_\varphi \ge B_\varphi(0)$, so the value $-\infty$ never occurs.

Proposition 14.3.1 *The functional H_φ is convex and weakly lower semicontinuous. For every $F \in C(X)$,*

$$B_\varphi(F) = \min_{\nu \in \mathcal{P}(X)} \left(H_\varphi(\nu) - \int_X F \, \mathrm{d}\nu \right). \tag{14.5}$$

The minimizers are exactly $\mathcal{A}_\varphi(F)$.

Proof Every function of ν in the supremum defining H_φ is affine and weakly continuous. Their supremum is convex and lower semicontinuous. The definition gives $H_\varphi(\nu) - \int_X F \, \mathrm{d}\nu \ge B_\varphi(F)$. Equality holds precisely when

$$B_\varphi(G) + \int_X G \, \mathrm{d}\nu \le B_\varphi(F) + \int_X F \, \mathrm{d}\nu \quad \text{for every } G \in C(X).$$

This is the defining inequality for $\mathcal{A}_\varphi(F)$. Its nonemptiness, proved in Theorem 14.2.1, proves attainment. □

Theorem 14.3.1 *For every finite real L, the set*

$$\mathcal{K}_L = \{\nu \in \mathcal{P}(X) : H_\varphi(\nu) \le L\}$$

is weakly compact. Its measures are uniformly absolutely continuous with respect to $\mathrm{d}V$*: for every $\epsilon > 0$ there is a $\delta > 0$ such that*

$$\mathrm{d}V(E) < \delta \quad \Longrightarrow \quad \sup_{\nu \in \mathcal{K}_L} \nu(E) < \epsilon \tag{14.6}$$

for every Borel set E. In particular, writing $\mathrm{d}\nu = h_\nu \mathrm{d}V$, the densities in this set are uniformly integrable in $L^1(\mathrm{d}V)$. On $\mathcal{K}_L$, weak convergence of measures also implies weak convergence of their densities in $L^1(\mathrm{d}V)$.

Proof Weak compactness follows from lower semicontinuity and compactness of $\mathcal{P}(X)$. We may assume that the sublevel is nonempty, so $L \ge B_\varphi(0)$. Fix $t > 0$. By Theorem 14.1.2 there is $\delta_t > 0$ such that

$$0 \le \chi \le 1, \quad \chi \in C(X), \quad \int_X \chi \,\mathrm{d}V < \delta_t \quad \Longrightarrow \quad B_\varphi(t\chi) > B_\varphi(0) - 1.$$

Indeed, failure would give a sequence of such χ tending to zero in L^1, contrary to that theorem. For $\nu \in \mathcal{K}_L$, the defining supremum gives

$$t \int_X \chi \,\mathrm{d}\nu \le L - B_\varphi(t\chi) < L - B_\varphi(0) + 1.$$

Choose t so large that the last numerator divided by t is less than $\epsilon/2$. If $\mathrm{d}V(E) < \delta_t/2$, outer regularity supplies an open neighborhood O of E with $\mathrm{d}V(O) < \delta_t$. For any compact $K \subset E$, choose a continuous cutoff equal to one on K, between zero and one, and supported in O. The last inequality gives $\nu(K) < \epsilon/2$, uniformly in ν. Inner regularity yields $\nu(E) \le \epsilon/2 < \epsilon$, proving (14.6).

This proves absolute continuity of every measure in the sublevel, and the uniform small-set criterion for the densities, whose integrals all equal one. These two facts are exactly uniform integrability on a finite measure space.

Finally suppose $\nu_j \to \nu$ weakly within the sublevel. For a bounded measurable G, choose continuous G_k with $|G_k| \le \|G\|_\infty$ and $G_k \to G$ in $\mathrm{d}V$-measure. Such approximations follow, for example, by approximating simple functions using regularity and continuous cutoffs. Uniform absolute continuity gives

$$\sup_{\sigma \in \mathcal{K}_L} \int_X |G_k - G| \,\mathrm{d}\sigma \longrightarrow 0.$$

To see this, split into $\{|G_k - G| \le a\}$ and its complement; the first integral is at most a, and the second is bounded by $2\|G\|_\infty$ times a uniformly small measure. For fixed k, weak convergence applies to G_k. Passing first $j \to \infty$ and then $k \to \infty$ proves convergence against G. This is precisely weak $L^1(\mathrm{d}V)$ convergence of the

densities. In particular every sequence in the sublevel has a weakly convergent L^1 subsequence. □

14.4 Bounded measurable densities and stability

Continuous approximations of a bounded measurable logarithmic density have supporting probabilities in one dual sublevel. Uniform integrability then permits both the density and the supporting measure to vary in the limiting inequality.

Theorem 14.4.1 *For every $F \in L^\infty(\mathrm{d}V)$ there is a nonempty set of absolutely continuous probability measures, again denoted $\mathcal{A}_\varphi(F)$, characterized by*

$$B_\varphi(G) \le B_\varphi(F) - \int_X (G - F)\,\mathrm{d}\nu \qquad \text{for every } G \in L^\infty(\mathrm{d}V). \tag{14.7}$$

For continuous F, this agrees with the preceding definition. Formula (14.5) holds for bounded measurable F as well, with these measures as its minimizers. In that formula the objective is declared to be $+\infty$ whenever $H_\varphi(\nu) = +\infty$; otherwise ν is absolutely continuous and the integral of the L^∞ class F is well defined.

Suppose F_j, F are uniformly bounded and $F_j \to F$ in $L^1(\mathrm{d}V)$. If $\nu_j \in \mathcal{A}_\varphi(F_j)$, their densities are uniformly integrable. Every weak L^1 cluster point belongs to $\mathcal{A}_\varphi(F)$. If the latter is a singleton, the full sequence converges to it weakly in $L^1(\mathrm{d}V)$.

Proof First, if $H_\varphi(\nu) < \infty$, the measure ν is absolutely continuous by Theorem 14.3.1. A bounded measurable G admits uniformly bounded continuous approximations $G_k \to G$ in $L^1(\mathrm{d}V)$. Absolute continuity and the common bound imply $\int_X G_k\,\mathrm{d}\nu \to \int_X G\,\mathrm{d}\nu$. Together with Theorem 14.1.2, this proves

$$H_\varphi(\nu) = \sup_{G \in L^\infty(\mathrm{d}V)} \left(B_\varphi(G) + \int_X G\,\mathrm{d}\nu \right) \tag{14.8}$$

for such ν.

Choose continuous F_j with $|F_j| \le M = \|F\|_\infty$ and $F_j \to F$ in $L^1(\mathrm{d}V)$, and choose $\nu_j \in \mathcal{A}_\varphi(F_j)$. The preceding duality and Lipschitz bound give

$$H_\varphi(\nu_j) = B_\varphi(F_j) + \int_X F_j\,\mathrm{d}\nu_j \le B_\varphi(0) + 2M.$$

Thus a subsequence converges weakly in L^1 to a probability measure ν. Uniform absolute continuity gives

$$\int_X |F_j - F|\,\mathrm{d}\nu_j \longrightarrow 0.$$

Here again one splits at a fixed small positive threshold; the exceptional sets have $\mathrm{d}V$-measure tending to zero. Since F is bounded measurable, weak L^1 convergence applies to its integral. Lower semicontinuity and continuity of B_φ yield

$$H_\varphi(\nu) \le \liminf_j H_\varphi(\nu_j) = B_\varphi(F) + \int_X F \, \mathrm{d}\nu.$$

The reverse inequality follows from (14.8). Equality and that same supremum formula prove (14.7) and the minimum formula.

For continuous F, every measure in the old supporting set has finite dual value by Proposition 14.3.1, so (14.8) extends all its inequalities to bounded measurable tests. This proves agreement of the definitions. Conversely, an absolutely continuous probability satisfying (14.7) has finite dual value and achieves the minimum. Hence the characterization is exact.

For the stability assertion, begin with the stated F_j, ν_j rather than with an approximating construction. The common uniform bound again places all ν_j in a single finite sublevel of H_φ. The same compactness and limiting argument show that every weak L^1 cluster point achieves the minimum for F. If the minimizing set is a singleton, subsequence compactness forces convergence of the full sequence to that measure. □

14.5 Entropy and the closed case

When closedness fixes the mass of an ordinary singularity type, integration determines the scalar explicitly. Its convex dual is then relative entropy. The variational identity below also supplies the estimates for nonclosed backgrounds.

For a Borel probability measure ν absolutely continuous with respect to $\mathrm{d}V$, let h be its nonnegative density and define

$$\mathrm{Ent}_{\mathrm{d}V}(\nu) = \int_X h \log h \, \mathrm{d}V, \qquad 0 \log 0 = 0,$$

and set its entropy to $+\infty$ if it is not absolutely continuous. This entropy may be negative when $\mathrm{d}V(X) \neq 1$, but its negative part is finite since $h \log h \ge -\mathrm{e}^{-1}$.

Lemma 14.5.1 *For every $\nu \in \mathcal{P}(X)$,*

$$\mathrm{Ent}_{\mathrm{d}V}(\nu) = \sup_{F \in C(X)} \left(\int_X F \, \mathrm{d}\nu - \log \int_X \mathrm{e}^F \, \mathrm{d}V \right).$$

Proof If ν is not absolutely continuous, choose a compact set K with $\mathrm{d}V(K) = 0$ and $\nu(K) > 0$. For $t > 0$, regularity and a continuous cutoff give $0 \le \chi_t \le 1$, equal to one on K, with support of volume at most e^{-2t}. With $F = t\chi_t$, the expression in the supremum is at least

$$t\nu(K) - \log\bigl(\mathrm{d}V(X) + \mathrm{e}^{-t}\bigr),$$

which tends to infinity.

Suppose $\nu = h \, \mathrm{d}V$. When its entropy is finite, Jensen's inequality for the logarithm gives

$$\int_X F\,\mathrm{d}\nu - \mathrm{Ent}_{\mathrm{d}V}(\nu) \le \log \int_{\{h>0\}} \mathrm{e}^F\,\mathrm{d}V \le \log \int_X \mathrm{e}^F\,\mathrm{d}V.$$

For infinite entropy the resulting upper bound is automatic. For the opposite inequality set

$$F_k = \log\bigl(\max\{k^{-1}, \min\{h, k\}\}\bigr), \qquad k \ge 1.$$

Then $\int_X \mathrm{e}^{F_k}\,\mathrm{d}V \to 1$, and $\int_X F_k\,\mathrm{d}\nu \to \mathrm{Ent}_{\mathrm{d}V}(\nu)$, with infinity allowed. The positive parts converge increasingly, and the negative parts are dominated by the integrable negative part of $h \log h$. For each fixed k, approximate F_k in $L^1(\mathrm{d}V + \nu)$ by continuous functions with the same bounds. Their two integrals converge to those of F_k, including the exponential integral. The continuous supremum is therefore at least its value at F_k. Let $k \to \infty$. □

If θ is Kähler, the ordinary type φ has a fixed non-pluripolar mass $m_\varphi > 0$ under the standing hypotheses, by the closed-background theory of [Xia]. For $F \in L^\infty(\mathrm{d}V)$, integrating the equation gives

$$B_\varphi(F) = \log m_\varphi - \log \int_X \mathrm{e}^F\,\mathrm{d}V, \qquad \mathcal{A}_\varphi(F) = \left\{ \frac{\mathrm{e}^F\,\mathrm{d}V}{\int_X \mathrm{e}^F\,\mathrm{d}V} \right\}. \tag{14.9}$$

The second identity follows by differentiating in bounded directions and testing the supporting inequality with both signs of the parameter.

Proposition 14.5.1 *If θ is Kähler, then*

$$H_\varphi(\nu) = \log m_\varphi + \mathrm{Ent}_{\mathrm{d}V}(\nu).$$

Thus the minimum formula becomes the usual entropy variational formula for the logarithmic integral in (14.9).

Proof Substitute (14.9) into the defining supremum and apply Lemma 14.5.1. □

The reference volume is part of this dual functional. If it is replaced by $\mathrm{d}V' = \mathrm{e}^a \mathrm{d}V$ for a continuous real a, then

$$B'_\varphi(F) = B_\varphi(F + a), \qquad H'_\varphi(\nu) = H_\varphi(\nu) - \int_X a\,\mathrm{d}\nu.$$

The first identity follows from the equation; the second follows by the change of variable $G = F + a$ in the defining supremum. The displayed identity for the dual is valid with infinite values as well. In a nonclosed background, H_φ is still defined by the same supremum, but total mass need not be fixed on an ordinary type, so the entropy computation does not determine it.

14.6 Entropy bounds

For a Hermitian background and zero reference, a lower bound for the equation density and an upper Gauduchon trace bound place B_0 between two logarithmic integrals. Dualizing gives two-sided entropy estimates, and hence identifies the finite domain of H_0. For the entropy estimates in this section, normalize $\mathrm{d}V(X) = 1$.

Proposition 14.6.1 *Let $\theta = \omega$ be Hermitian, and let $\varphi = 0$. For every $q > 1$ there is a finite real constant a_q, and there is a finite real constant A, such that every Borel probability measure ν satisfies*

$$a_q + \frac{1}{q}\operatorname{Ent}_{\mathrm{d}V}(\nu) \le H_0(\nu) \le A + n\operatorname{Ent}_{\mathrm{d}V}(\nu).$$

The inequalities include the value $+\infty$. Thus $H_0(\nu) < \infty$ if and only if ν has finite entropy. A family of probabilities has uniformly bounded H_0 if and only if its entropies are uniformly bounded.

Proof In dimension one the closed formula (14.9) gives $H_0 = \log \int_X \omega + \operatorname{Ent}_{\mathrm{d}V}$, so the assertions follow. Suppose $n \ge 2$.

For $G \in C(X)$, apply Theorem 10.3.1 to a bounded solution with density $\mathrm{e}^{B_0(G)+G}$. It gives a constant $c_q > 0$, independent of G, such that

$$B_0(G) \ge \log c_q - \frac{1}{q}\log \int_X \mathrm{e}^{qG}\,\mathrm{d}V.$$

For the other bound, fix a smooth Gauduchon metric γ, rescaled so that $\int_X \omega\wedge\gamma^{n-1} = 1$, and write

$$k_\gamma = \left(\frac{\gamma^n}{\mathrm{d}V}\right)^{(n-1)/n}, \qquad m = \min_X k_\gamma > 0.$$

For smooth G, the trace bound (4.3) gives

$$\mathrm{e}^{B_0(G)/n}\int_X \mathrm{e}^{G/n}k_\gamma\,\mathrm{d}V \le 1.$$

Uniform approximation and continuity of B_0 extend this inequality to continuous G. Consequently

$$B_0(G) \le -n\log m - n\log \int_X \mathrm{e}^{G/n}\,\mathrm{d}V.$$

Take the defining suprema in the two scalar bounds and apply Lemma 14.5.1, with $F = qG$ and $F = G/n$, respectively. This proves both inequalities for every probability measure, including infinite values, with $a_q = \log c_q$ and $A = -n\log m$. Fix $q = 2$ for the finite-domain and bounded-family assertions. □

14.6.1 The mass bounds as entropy differences

The difference between H_0 and entropy measures the variation of total Monge–Ampère mass. Its two extrema recover $v_-(\omega)$ and $v_+(\omega)$.

Let $\mathcal{P}_{\mathrm{ent}}(X)$ denote the probabilities of finite entropy relative to the fixed smooth probability volume $\mathrm{d}V$. By Proposition 14.6.1, the difference $H_0(\nu) - \mathrm{Ent}_{\mathrm{d}V}(\nu)$ is a finite real number for every ν in this set.

Proposition 14.6.2 *For every Hermitian background ω,*

$$\inf_{\nu\in\mathcal{P}_{\mathrm{ent}}(X)} \big(H_0(\nu) - \mathrm{Ent}_{\mathrm{d}V}(\nu)\big) = \log v_-(\omega),$$
$$\sup_{\nu\in\mathcal{P}_{\mathrm{ent}}(X)} \big(H_0(\nu) - \mathrm{Ent}_{\mathrm{d}V}(\nu)\big) = \log v_+(\omega).$$

The same extrema are obtained by restricting to smooth strictly positive probability measures. Here $\log 0 = -\infty$ and $\log(+\infty) = +\infty$. Thus boundedness of this difference above is equivalent to bounded mass, and boundedness below is equivalent to positive lower mass.

Proof Write $Z_G = \int_X \mathrm{e}^G \,\mathrm{d}V$. The bounded solution for $G \in C(X)$ has mass $\mathrm{e}^{B_0(G)} Z_G$. If $v_-(\omega) > 0$, its lower mass bound gives

$$B_0(G) \geq \log v_-(\omega) - \log Z_G.$$

Taking the defining supremum for H_0 and using the entropy variational formula gives $H_0 - \mathrm{Ent}_{\mathrm{d}V} \geq \log v_-(\omega)$ on $\mathcal{P}_{\mathrm{ent}}(X)$. If $v_+(\omega) < \infty$, the upper mass bound gives the reverse estimate $H_0 - \mathrm{Ent}_{\mathrm{d}V} \leq \log v_+(\omega)$ in the same way. The corresponding bounds are automatic when their endpoints are infinite.

To prove sharpness, first take smooth real G and its Gibbs probability $\nu_G = Z_G^{-1}\mathrm{e}^G \,\mathrm{d}V$. This has finite entropy. Testing the supremum defining H_0 with G gives

$$H_0(\nu_G) - \mathrm{Ent}_{\mathrm{d}V}(\nu_G) \geq B_0(G) + \log Z_G.$$

For the other direction, let σ_G be the smooth strictly positive Gauduchon adjoint probability of Theorem 14.2.3. It lies in $\mathcal{A}_0(G)$, so its dual value is $B_0(G) + \int_X G \,\mathrm{d}\sigma_G$. The entropy variational inequality yields

$$H_0(\sigma_G) - \mathrm{Ent}_{\mathrm{d}V}(\sigma_G) \leq B_0(G) + \log Z_G.$$

As smooth G varies, the infimum and supremum of the solution masses $\mathrm{e}^{B_0(G)} Z_G$ are $v_-(\omega)$ and $v_+(\omega)$ because these are all smooth strictly positive potential masses. By Theorem 1.11.1, every bounded potential mass is a limit of such masses. Both chosen probabilities are smooth and strictly positive. The last two inequalities prove sharpness on this smaller domain, including zero lower mass and infinite upper mass. The two equivalences follow. □

The difference is independent of the smooth positive reference volume used to define it. For smooth real a, under $\mathrm{d}V' = \mathrm{e}^a \mathrm{d}V$, both H_0 and Ent change by $-\int_X a \,\mathrm{d}\nu$; see the transformation formula preceding Proposition 14.6.1. Finite entropy is unchanged because a is bounded. If h is a smooth real function and $\omega + \mathrm{dd^c} h$ is also Hermitian, transporting bounded solutions by $u \mapsto u - h$ preserves B_0 and hence this difference. Finally it is constant on the finite-entropy domain exactly when $\nu_-(\omega) = \nu_+(\omega)$, that is, when the masses of all bounded potentials agree.

14.6.2 Entropy for a prescribed ordinary type

Return to a semipositive big θ and a model reference φ satisfying (12.1). Fix a corresponding $\varepsilon \in (0, 1)$. The unused part of the background supplies a uniform lower density bound; comparison in a larger Hermitian background supplies the upper bound. Together they extend the entropy estimates to the prescribed type.

Corollary 14.6.1 *For every $q > 1$ there is a finite real a_q, and there is a finite real A, such that*

$$a_q + \frac{1}{q} \operatorname{Ent}_{\mathrm{d}V}(\nu) \le H_\varphi(\nu) \le A + n \operatorname{Ent}_{\mathrm{d}V}(\nu)$$

for every Borel probability ν, including infinite values. Thus finite dual value and finite entropy have the same domain, and uniformly bounded dual values are equivalent to uniformly bounded entropies. For fixed θ and ε in (12.1), the constants can be chosen independently of the model reference φ.

Proof For $G \in C(X)$, let $u \sim \varphi$ solve the equation with density $c\mathrm{e}^G$, where $c = \mathrm{e}^{B_\varphi(G)}$. Applying Lemma 13.11.1 gives $c\|\mathrm{e}^G\|_q \ge b_q$, hence

$$B_\varphi(G) \ge \log b_q - \frac{1}{q} \log \int_X \mathrm{e}^{qG} \,\mathrm{d}V.$$

For the upper bound choose a Hermitian form $\Omega \ge \theta$, and let $B_{\Omega,0}$ denote its bounded-type normalization. A bounded solution v for G in this background has a genuine smaller-background envelope. Moreover $u \in \mathrm{PSH}(X, \Omega)$, $u \preceq v$, and positivity gives

$$\mathrm{MA}^{\mathrm{ext}}_\Omega(u) \ge \mathrm{MA}^{\mathrm{ext}}_\theta(u) = c\mathrm{e}^G \,\mathrm{d}V.$$

If $c > \mathrm{e}^{B_{\Omega,0}(G)}$, the same strict domination argument, now applied to v and $u + t$ in background Ω, is impossible. Hence $B_\varphi \le B_{\Omega,0}$ and $H_\varphi \le H_{\Omega,0}$. Proposition 14.6.1 gives the required upper bound.

Taking the defining supremum in the lower scalar estimate and applying Lemma 14.5.1 with $F = qG$ gives the lower bound for every probability, including infinite values, with $a_q = \log b_q$. The finite-domain and bounded-family assertions follow by taking $q = 2$. The lower-density lemma is uniform in the reference for fixed ε, and the upper bound uses only Ω. This proves the final assertion. □

For example, let the model references φ_j satisfy the same smaller-background condition with fixed ε, let $G_j \in L^\infty(\mathrm{d}V)$ satisfy $\|G_j\|_\infty \le M$ for a fixed constant M, and choose $\nu_j \in \mathcal{A}_{\varphi_j}(G_j)$. Choose a Hermitian form $\Omega \ge \theta$ and use its zero-reference functionals $B_{\Omega,0}, H_{\Omega,0}$. Then

$$H_{\varphi_j}(\nu_j) = B_{\varphi_j}(G_j) + \int_X G_j \,\mathrm{d}\nu_j \le B_{\Omega,0}(0) + 2M.$$

The uniform lower entropy estimate bounds their entropies. The Hermitian upper estimate places all ν_j in a single finite sublevel of $H_{\Omega,0}$. Hence Theorem 14.3.1 gives weak compactness and uniform integrability of their densities. Every weak limit has finite entropy. This gives compactness as the reference varies; identifying the limit as a supporting probability for a limiting reference requires further information.

14.7 Changing the prescribed singularities

More singular references have smaller normalizing scalars. The smaller-background competitor also gives a quantitative comparison with reference zero, uniform over all logarithmic densities. In this section the references vary among the model potentials satisfying (12.1); B_0, H_0 refer to the same semipositive big background.

Proposition 14.7.1 *Let φ, ψ satisfy (12.1). If $\varphi \preceq \psi$, then $B_\varphi \le B_\psi$ and $H_\varphi \le H_\psi$. Moreover, if $P_{\varepsilon\theta}(\varphi)$ is genuine for $0 < \varepsilon < 1$, then*

$$B_0(G) + n\log(1-\varepsilon) \le B_\varphi(G) \le B_0(G) \qquad (G \in C(X)),$$

and

$$H_0(\nu) + n\log(1-\varepsilon) \le H_\varphi(\nu) \le H_0(\nu) \qquad (\nu \in \mathcal{P}(X)).$$

The dual inequalities include infinite values.

Proof Fix G, and let u_φ, u_ψ solve the two ordinary-type equations, with scalars c_φ, c_ψ. Then $u_\varphi \preceq u_\psi$. If $c_\varphi > c_\psi$, singular strict domination applied to u_ψ and $u_\varphi + t$ would give $u_\psi \ge u_\varphi + t$ for every real t, a contradiction at a common finite point. This proves the monotonicity of B, and its defining supremum gives the monotonicity of H. Since $\varphi \preceq 0$, it also proves both upper bounds.

For the lower bound, let v be the bounded solution for G, with scalar c_0, and put

$$h = P_{\varepsilon\theta}(\varphi) + (1-\varepsilon)v.$$

The two currents in this sum have backgrounds $\varepsilon\theta$ and $(1-\varepsilon)\theta$, respectively. Positivity and homogeneity give

$$\mathrm{MA}_\theta^{\mathrm{ext}}(h) \ge (1-\varepsilon)^n \mathrm{MA}_\theta(v) = (1-\varepsilon)^n c_0 \mathrm{e}^G \,\mathrm{d}V.$$

As in the lower-density lemma, the inequality is justified first on finite levels and then off the pole set of $P_{\varepsilon\theta}(\varphi)$, which has zero smooth volume. Also $h \preceq u_\varphi$. If $c_\varphi < (1-\varepsilon)^n c_0$, strict domination would force $u_\varphi \geq h + t$ for every t. Thus $c_\varphi \geq (1-\varepsilon)^n c_0$. Taking logarithms proves the scalar bound; taking the defining suprema proves the dual bound. □

14.7.1 Uniform comparison of the scalar functionals

Uniform distance between the scalar functionals gives a stability criterion for their supporting probabilities. For two model references φ, ψ in the preceding setting, define

$$\Delta(\varphi, \psi) = \sup_{G \in C(X)} |B_\varphi(G) - B_\psi(G)|. \tag{14.10}$$

The comparison with B_0 makes this number finite for every such pair. It is a pseudometric: symmetry, vanishing on the diagonal and the triangle inequality follow from the uniform norm. Distinct singularity types can have distance zero, as Example 14.10.1 shows.

Proposition 14.7.2 *For these model references,*

$$\Delta(\varphi, \psi) = \sup_{\mathrm{Ent}_{\mathrm{d}V}(\nu) < \infty} |H_\varphi(\nu) - H_\psi(\nu)|,$$

where the supremum is over probability measures. The same value is obtained in (14.10) *using all bounded measurable G. It is independent of the smooth positive reference volume.*

Proof By Corollary 14.6.1, finite entropy is the common finite domain of the two duals; rescaling the reference volume to have mass one does not change finiteness. A uniform bound for $|B_\varphi - B_\psi|$ passes to their defining suprema, so the displayed supremum is at most $\Delta(\varphi, \psi)$. Conversely, denote that supremum by D. If $\nu \in \mathcal{A}_\psi(G)$, it has finite entropy and the dual minimum formula gives

$$B_\varphi(G) \leq H_\varphi(\nu) - \int_X G \, \mathrm{d}\nu \leq B_\psi(G) + D.$$

Interchanging the references proves the opposite inequality. Thus $\Delta \leq D$.

For bounded measurable G, take continuous approximations with the same uniform bound converging in $L^1(\mathrm{d}V)$. The continuity theorem Theorem 14.1.2, applied separately to the two fixed references, extends the scalar bound to G. Finally a volume change $\mathrm{d}V' = \mathrm{e}^a \mathrm{d}V$ replaces each $B(G)$ by $B(G + a)$. Translation by $a \in C(X)$ is a bijection of the test space, so the supremum is unchanged. □

Proposition 14.7.3 *Let φ_j, φ be model references in this same regime and suppose $\Delta(\varphi_j, \varphi) \to 0$. Let G_j, G be uniformly bounded measurable functions with $G_j \to G$ in $L^1(\mathrm{d}V)$, and choose $\nu_j \in \mathcal{A}_{\varphi_j}(G_j)$. Their densities are uniformly integrable. Every weak cluster point belongs to $\mathcal{A}_\varphi(G)$, and convergence of a subsequence also holds weakly in L^1. If $\mathcal{A}_\varphi(G)$ is a singleton, the full sequence converges to it in these two senses.*

Proof Write $\delta_j = \Delta(\varphi_j, \varphi)$. The scalar and dual bounds, together with the supporting identity, give

$$0 \le H_\varphi(\nu_j) - \int_X G_j \,\mathrm{d}\nu_j - B_\varphi(G_j) \le 2\delta_j.$$

If $\|G_j\|_\infty \le M$, then $H_\varphi(\nu_j) \le B_\varphi(0) + 2M + 2\delta_j$. Thus all measures lie in one finite dual sublevel. By Theorem 14.3.1, their densities are uniformly integrable, and a weakly convergent subsequence also converges weakly in L^1.

Let its limit be ν. Uniform integrability, the common bound and convergence in measure give $\int_X |G_j - G| \,\mathrm{d}\nu_j \to 0$. Weak L^1 convergence applies to the bounded function G, and Theorem 14.1.2 gives $B_\varphi(G_j) \to B_\varphi(G)$. Lower semicontinuity in the displayed inequality yields

$$H_\varphi(\nu) - \int_X G \,\mathrm{d}\nu - B_\varphi(G) \le 0.$$

The measurable dual minimum formula gives the reverse inequality and identifies equality with $\nu \in \mathcal{A}_\varphi(G)$. Compactness proves whole-sequence convergence when this set is a singleton. □

The smaller-background comparison gives an explicit family converging to reference zero.

Corollary 14.7.1 *Let the model references φ_j satisfy $P_{\varepsilon_j\theta}(\varphi_j) \not\equiv -\infty$, where $0 < \varepsilon_j < 1$ and $\varepsilon_j \to 0$. Then $B_{\varphi_j} \to B_0$ uniformly on $C(X)$. If $G_j, G \in C(X, \mathbb{R})$ and $G_j \to G$ uniformly and $\nu_j \in \mathcal{A}_{\varphi_j}(G_j)$, every weak cluster point belongs to $\mathcal{A}_0(G)$. The densities of ν_j are uniformly integrable, and weak convergence of a subsequence also holds weakly in $L^1(\mathrm{d}V)$. If $\mathcal{A}_0(G)$ is a singleton, the full sequence converges to it in these two senses.*

Proof Put $\delta_j = -n\log(1-\varepsilon_j) \to 0$. The comparison proposition gives $\Delta(\varphi_j, 0) \le \delta_j$, hence uniform convergence of B_{φ_j} to B_0 on all of $C(X)$. Apply Proposition 14.7.3 with limiting reference zero for the assertions about supporting probabilities. □

For ordinary divisorial types, a relative comparison of positive residual forms supplies a geometric criterion for this functional convergence, as proved next. General convergence criteria for references with degenerate residuals or arbitrary singularities remain open.

14.8 Varying positive backgrounds

A relative inequality between two Hermitian forms compares their scalars independently of the size of the logarithmic density. This converts geometric convergence of backgrounds into the uniform functional comparison used above.

We now vary a smooth positive real $(1,1)$-form β on X, with reference zero. Write B_β, H_β and $\mathcal{A}_\beta$ for its functionals and supporting sets. For two such backgrounds set

$$d(B_\beta, B_{\beta'}) = \sup_{F \in L^\infty(\mathrm{d}V)} |B_\beta(F) - B_{\beta'}(F)|.$$

The supremum ranges over all bounded tests, with no common bound on their norms.

Theorem 14.8.1 *Suppose $0 < \varepsilon < 1$ and $(1-\varepsilon)\beta \le \beta' \le (1+\varepsilon)\beta$. For every bounded measurable F,*

$$n\log(1-\varepsilon) \le B_{\beta'}(F) - B_\beta(F) \le n\log(1+\varepsilon). \tag{14.11}$$

In particular $d(B_\beta, B_{\beta'}) \le -n\log(1-\varepsilon)$. The same conclusion holds if the form comparison is satisfied by positive smooth representatives $\beta + \mathrm{dd^c}h$ and $\beta' + \mathrm{dd^c}h'$.

Proof First let F be smooth, with smooth solutions v, v' in the two backgrounds. At a maximum of $v' - (1+\varepsilon)v$,

$$0 < \beta' + \mathrm{dd^c}v' \le (1+\varepsilon)(\beta + \mathrm{dd^c}v).$$

Taking determinants and canceling $\mathrm{e}^F\mathrm{d}V$ gives the upper bound. At a minimum of $v' - (1-\varepsilon)v$, the reverse form inequality gives the lower bound. For bounded measurable F, choose smooth uniformly bounded approximations converging in L^1. Apply Theorem 14.1.2 separately in the two fixed backgrounds. Finally the correspondence $v \mapsto v - h$ preserves both boundedness and the current, so $B_{\beta+\mathrm{dd^c}h} = B_\beta$ whenever both backgrounds are positive. This proves the representative assertion.□

Corollary 14.8.1 *Let β_j satisfy the form comparison with β and $\varepsilon_j \to 0$. Suppose $F_j \to F$ in $L^1(\mathrm{d}V)$ with a common L^∞ bound. Then $B_{\beta_j}(F_j) \to B_\beta(F)$. For any $\nu_j \in \mathcal{A}_{\beta_j}(F_j)$, their densities are uniformly integrable and every subsequence has a further subsequence converging weakly in L^1 to an element of $\mathcal{A}_\beta(F)$. If this set is a singleton, the full sequence converges to it weakly in L^1 and as probability measures. This holds in particular when F is smooth.*

Proof Put $\delta_j = -n\log(1-\varepsilon_j)$. The scalar assertion follows from (14.11) and Theorem 14.1.2 in the fixed background β. Taking suprema over tests gives the extended-valued inequalities

$$H_{\beta_j} \le H_\beta + \delta_j, \qquad H_\beta \le H_{\beta_j} + \delta_j.$$

Their finite domains coincide. The support identity therefore gives

$$0 \le H_\beta(\nu_j) - B_\beta(F_j) - \int_X F_j \, \mathrm{d}\nu_j \le 2\delta_j.$$

If $\|F_j\|_\infty \le M$, this puts the measures in a fixed sublevel $H_\beta \le B_\beta(0) + 2M + 2\sup_j \delta_j$. Apply Theorem 14.3.1. For a weak L^1 cluster limit, uniform integrability gives $\int |F_j - F| \, \mathrm{d}\nu_j \to 0$. Lower semicontinuity in the last display and the measurable dual minimum formula identify the limit as a support at F. Compactness gives full convergence when that support is unique; Theorem 14.2.3 supplies uniqueness at smooth F. □

Smooth changes of the background also control the potentials. The inverse of the linearized equation used in Proposition 9.3.1 remains available when the form itself is included among the parameters.

Proposition 14.8.1 *If positive $\beta_j \to \beta$ and $F_j \to F$ in C^∞, the solutions normalized by $\int_X v_j \, \mathrm{d}V = 0$ converge in C^∞ to the similarly normalized solution for (β, F). Their scalars converge as well.*

Proof Fix $k \ge 0$ and $0 < \alpha < 1$. On the open positivity set, consider the smooth map

$$(\eta, G, w, a) \longmapsto \log \frac{(\eta + \mathrm{dd^c} w)^n}{\mathrm{d}V} - a - G$$

from $C^{k,\alpha}$ forms and functions, mean-zero $C^{k+2,\alpha}$ potentials, and $\mathbb{R}$, into $C^{k,\alpha}$. At the limiting solution its derivative in (w, a) is $(h, q) \mapsto \Delta_\Omega h - q$, where $\Omega = \beta + \mathrm{dd^c} v$ and $\Delta_\Omega h = n \mathrm{dd^c} h \wedge \Omega^{n-1}/\Omega^n$. For a target g, choose $q = -\int g \, \mathrm{d}\mu_v$, with the adjoint probability of Proposition 4.5.1. The Fredholm alternative then gives the unique mean-zero h solving $\Delta_\Omega h = g + q$, with a bounded inverse as in Proposition 9.3.1. In dimension one this is the Poisson alternative.

The implicit function theorem gives a local branch continuous in these spaces. For smooth parameters, bounded uniqueness from Corollary 11.4.1 identifies this branch with the global smooth solution. Thus $v_j \to v$ in $C^{k+2,\alpha}$ for every k, proving the assertion. □

The smooth adjoint probabilities determine the functional once its additive constant is fixed. If $\nu_{\beta,F}$ denotes the unique support at a smooth test, integration of the derivative in Theorem 14.2.3 along sF gives

$$B_\beta(F) = B_\beta(0) - \int_0^1 \int_X F \, \mathrm{d}\nu_{\beta,sF} \, \mathrm{d}s. \tag{14.12}$$

The implicit function theorem makes this derivative continuous along the path. Consequently two positive backgrounds on the same $X, \mathrm{d}V$ have identical bounded-test functionals if and only if their values at zero and all their smooth-test adjoint probabilities agree. One direction follows by differentiation; the other follows from (14.12) for smooth tests and bounded L^1 approximation for all remaining tests.

14.9 Divisorial types and moving sections

Removing an actual divisorial part identifies an ordinary singularity class with bounded potentials in its residual background. The scalar therefore sees the residual form. Convergence of full potentials also needs control of the sections defining the divisors.

Let θ be a smooth real $(1,1)$-form on X. Fix finitely many holomorphic line bundles L_r on X with smooth metrics h_r, nonzero sections s_r, and weights $t_r > 0$. Put

$$\varphi = \sum_r t_r \log |s_r|^2_{h_r}, \qquad \Gamma = \sum_r t_r \operatorname{div}(s_r), \qquad \beta = \theta - \sum_r t_r c_1(L_r, h_r),$$

and assume $\beta > 0$. The empty family is allowed, with $\varphi = 0$ and $\beta = \theta$.

Theorem 14.9.1 *There is an exact correspondence*

$$\{u \in \mathrm{PSH}(X,\theta) : u \sim \varphi\} = \{\varphi + w : w \in \mathrm{PSH}(X,\beta) \cap L^\infty(X)\},$$

and for each corresponding pair,

$$\mathrm{MA}^{\mathrm{ext}}_\theta(\varphi + w) = (\beta + \mathrm{dd^c} w)^n. \tag{14.13}$$

For every bounded measurable F, the equation in this ordinary type has a unique scalar and a unique solution modulo constants. Its scalar functional satisfies

$$B_{\theta,[\varphi]}(F) = B_\beta(F).$$

Its dual and supporting sets consequently coincide with H_β and $\mathcal{A}_\beta$.

Proof Poincaré–Lelong gives $\theta + \mathrm{dd^c}\varphi = [\Gamma] + \beta$, so every potential on the right is admissible and has type φ. Conversely, an ordinary-type competitor u has exactly the divisorial coefficients of Γ. By Lemma 1.17.1,

$$\theta + \mathrm{dd^c} u - [\Gamma] \geq 0.$$

The locally integrable difference $w = u - \varphi$ therefore has a β-plurisubharmonic representative. It is bounded off the divisor by ordinary equivalence; the local submean property preserves both bounds across the divisor. Equality almost everywhere gives $u = \varphi + w$ for the quasi-plurisubharmonic representatives.

Off the divisor the two currents agree. Locality identifies their products there. The extended measure on the left gives the divisor zero mass, and the bounded product on the right is finite and gives analytic sets zero mass by Section 1.7. This proves (14.13). The bounded equation in background β has a continuous solution and a unique scalar by Theorem 10.1.1 and Proposition 10.5.1. Its density is bounded above and below by positive constants, so Corollary 11.4.1 gives uniqueness modulo constants. The correspondence transfers these assertions to the entire ordinary type and identifies the functionals. □

In particular the background comparison and adjoint stability of Theorem 14.8.1 and Corollary 14.8.1 apply whenever the positive residuals satisfy the relative form inequalities. The divisor locations do not enter those estimates. This assertion concerns ordinary types; a model-envelope interpretation additionally requires bounded equivalence of the envelope and the displayed reference.

For a fixed positive residual β, uniformly bounded $F_j \to F$ in L^1 give uniform convergence of the normalized residual solutions. Indeed, scalar continuity gives uniformly bounded equation densities converging in every finite L^p. The compactness theorem Theorem 10.3.2 gives uniform subsequential convergence, bounded-product continuity identifies the limiting equation, and Corollary 11.4.1 identifies every limit. Subtracting the means gives the same conclusion for mean-zero normalization.

Theorem 14.9.2 *Keep the line bundles and metrics fixed. Let nonzero sections s_{jr} converge to $s_r \neq 0$ in fixed finite-dimensional spaces of global sections, and let the positive weights satisfy $t_{jr} \to t_r > 0$. Define φ_j, θ_j from these data and positive residuals β_j by the preceding formulas. Suppose either that $\beta_j \to \beta > 0$ and $F_j \to F$ in C^∞, or that $\beta_j = \beta > 0$ and the tests converge in L^1 with a common L^∞ bound. Then $\varphi_j \to \varphi$ in L^1, and the ordinary-type solutions normalized by $\sup_X u_j = 0$ converge in L^1 to the similarly normalized limiting solution. For mean-zero residual solutions v_j, one also has*

$$\sup_X(\varphi_j + v_j) \longrightarrow \sup_X(\varphi + v).$$

The residuals converge smoothly in the first regime and uniformly in the second.

Proof The residual convergence follows from Proposition 14.8.1 and the preceding compactness argument. The fixed metrics and bounded weights give a common smooth lower bound for $\mathrm{dd^c}\varphi_j$. The convergent sections give a common upper bound for φ_j and convergence at a point outside all limiting divisors, so their suprema are bounded above and below. By Theorem 1.4.1, every subsequence has a further subsequence converging in L^1. Off the limiting divisors, convergence of the sections and weights gives local uniform convergence to φ. Since those divisors have zero volume, every L^1 cluster equals φ. Thus $\varphi_j \to \varphi$ in $L^1(X)$.

Put $C_j = \sup_X(\varphi_j + v_j)$ and $C = \sup_X(\varphi + v)$. The sums converge in L^1 and have a common smooth lower Hessian bound, since $\mathrm{dd^c}v_j \geq -\beta_j$ and the residual backgrounds are uniformly bounded. Applying Lemma 1.4.1 at maximizing points gives $\limsup C_j \leq C$. A maximizing point of $\varphi + v$ lies off its poles. Evaluation there, using local uniform convergence, gives $\liminf C_j \geq C$.

It follows that $u_j = \varphi_j + v_j - C_j \to \varphi + v - C$ in L^1. For an empty divisor family the same assertion is just residual convergence and convergence of its supremum. □

14.10 The mass determined by the functional

For closed backgrounds, the functional has a precise loss of information: on the model domain below, equality of functionals is exactly equality of mass. This also gives examples in which supporting probabilities stay fixed while the full potentials have different limits.

In this section X is compact Kähler. Allow θ to be any smooth closed big form, and let $\varphi = P_\theta[\varphi]$ have small unbounded locus in the sense of Section 1.12 and positive mass

$$m_\varphi = \int_X \mathrm{MA}_\theta(\varphi) > 0.$$

Write $B_{\theta,\varphi}$ for the scalar functional of its ordinary type, and put $Z_F = \int_X \mathrm{e}^F \,\mathrm{d}V$.

Theorem 14.10.1 *For every such pair and every bounded measurable F, the ordinary-type equation has a unique scalar and a solution unique modulo constants. Its functional and supporting set are*

$$B_{\theta,\varphi}(F) = \log m_\varphi - \log Z_F, \qquad \mathcal{A}_{\theta,\varphi}(F) = \{\mathrm{e}^F \mathrm{d}V / Z_F\}.$$

For two such pairs on the same $X, \mathrm{d}V$, possibly in different classes,

$$\sup_{F \in L^\infty} |B_{\theta,\varphi}(F) - B_{\theta',\psi}(F)| = |\log m_\varphi - \log m_\psi|. \tag{14.14}$$

Thus an invariant of these pairs is determined by the entire functional if and only if it is constant on every fiber of the mass map.

Proof Existence follows from Corollary 31.8.1. Indeed, a Kähler manifold has bounded mass by Lemma 20.4.1, and closed ordinary-type mass invariance gives the positive lower reference volume m_φ in (29.4). The classical closed prescribed-type theorem, recorded in Section 33.10, gives uniqueness modulo constants under the stated model and small-unbounded-locus hypotheses.

Every ordinary-type competitor has mass m_φ. Integration of its equation forces $cZ_F = m_\varphi$, proving the scalar formula. As in (14.9), Jensen's inequality gives the stated support, and differentiation in every bounded direction makes it unique. Subtracting the scalar formulas gives (14.14). Finally $m_\varphi = \mathrm{d}V(X) \exp B_{\theta,\varphi}(0)$, so the functional and the mass determine each other. An invariant factors through either one exactly when it is constant on these fibers. □

For an ordinary divisorial type with a positive closed residual β, Theorem 14.9.1 gives the same formula with mass $\int_X \beta^n$: bounded closed-product Stokes fixes this mass. This computation does not require a model representative for a real weighted sum of divisors. If $\beta_j \to \beta > 0$ uniformly, continuity of the wedge polynomial gives convergence of the masses; the relative form bounds give the sharper inequalities $(1-\varepsilon_j)^n \int \beta^n \le \int \beta_j^n \le (1+\varepsilon_j)^n \int \beta^n$.

Example 14.10.1 Put $X = \mathbb{P}^n$ for an integer $n \geq 1$ and let θ be its Fubini–Study form, normalized by $\int_X \theta^n = 1$, and let D be a hyperplane. Let s_D be a defining section of $\mathcal{O}(1)$ with its Fubini–Study metric, and choose a real $t \in (0,1)$. Then $a_D = \log|s_D|^2 \leq 0$ after rescaling, with

$$\mathrm{dd^c} a_D = [D] - \theta, \qquad \theta + \mathrm{dd^c}(ta_D) = (1-t)\theta + t[D].$$

Put $\varphi_{t,D} = P_\theta[ta_D]$. By Proposition 15.2.3, this is a model reference of ordinary type ta_D. Also $ta_D \in \mathrm{PSH}(X, t\theta)$ and $ta_D \leq \varphi_{t,D}$, so it satisfies (12.1). Off D its original representative has Monge–Ampère measure $(1-t)^n\theta^n$, and its non-pluripolar measure puts no mass on D.

Invariance of mass under bounded changes in the closed background gives

$$m_{\varphi_{t,D}} = (1-t)^n, \qquad B_{\varphi_{t,D}}(G) - B_0(G) = n\log(1-t) \quad (G \in C(X,\mathbb{R})).$$

Thus the constant in Proposition 14.7.1 is sharp.

On $\mathbb{P}^1$, choose two distinct points p, q for D. The corresponding references have distance zero, although one is unbounded at p and bounded at q, while the other has the reverse behavior. They are not ordinarily equivalent. For every positive integer m,

$$\mathcal{I}(m\varphi_{t,p}) = \mathcal{O}(-\lfloor mt \rfloor p), \qquad \mathcal{I}(m\varphi_{t,q}) = \mathcal{O}(-\lfloor mt \rfloor q).$$

These identities follow from Lemma 1.16.1. For $mt \geq 1$ the ideals differ. Thus Δ does not separate ideal types either, and cannot determine the ordinary or ideal singularity type.

Fix a smooth test F on $\mathbb{P}^1$, and let w_F be the mean-zero solution of $\theta + \mathrm{dd^c} w_F = \mathrm{e}^F \mathrm{d}V/Z_F$. The full solutions for $D = p, q$ are

$$u_{t,D} = ta_D + (1-t)w_F - \sup_X \big(ta_D + (1-t)w_F\big).$$

Their distributional difference satisfies $\mathrm{dd^c}(u_{t,p} - u_{t,q}) = t([p] - [q]) \neq 0$. Alternating the two divisors therefore gives two different L^1 subsequential limits, although the entire scalar functional and each of its supporting sets are constant. Within this fixed one-weight family, the functional does recover t, since $t \mapsto (1-t)^n$ is injective.

Example 14.10.2 On $\mathbb{P}^n$, $n \geq 2$, let ω be the Fubini–Study form and let f be a nonconstant smooth real function. Fix $0 < t < 1$ and put

$$\beta = \mathrm{e}^f \omega, \qquad \theta = \beta + t\omega, \qquad \varphi_D = t\log|s_D|^2.$$

Every hyperplane D has the same positive residual β. This form is nonclosed: $\mathrm{d}\beta = \mathrm{e}^f \mathrm{d}f \wedge \omega \not\equiv 0$, since wedging a one-form with ω^{n-1} is injective and $\mathrm{d}f$ is nonzero somewhere. By Theorem 14.9.1, every ordinary type has the same functional B_β on all bounded tests, hence the same supporting sets. At a smooth test the common support is the unique Gauduchon adjoint probability.

If v_F is the common smooth residual solution, the full solutions are the supremum normalizations of $\varphi_D + v_F$. Two distinct hyperplanes give different poles, divisorial

coefficients and multiplier ideals $\mathcal{O}(-\lfloor kt \rfloor D)$ for sufficiently large k; the ideal formula is the normal-coordinate calculation of Lemma 1.16.1. Their potentials differ by more than a constant because their distributional Hessian difference is $t([D_1] - [D_2])$. Alternation again has two distinct L^1 limits. These conclusions follow from a common residual background and make no mass-only assertion for general nonclosed backgrounds.

14.11 Selection by a vanishing exponential parameter

The zero-exponent equation can have several normalized solutions. The positive-exponent family nevertheless selects one: its scalar functional supplies a family of integral normalizations, and the selected potential is the greatest critical subsolution satisfying all of them. To prove greatestness, we use supporting probabilities for a second scalar functional, obtained by varying the forcing in the subsolution inequality.

Return to a connected compact complex manifold X of dimension $n \geq 1$, with smooth positive volume form $\mathrm{d}V$. Let θ be an arbitrary smooth big real $(1,1)$-form. Fix a strict analytic barrier $\rho \leq 0$ as in Section 13.12, and put $\Omega = X \setminus \{\rho = -\infty\}$. Fix a nonzero $0 \leq f \in L^p(\mathrm{d}V)$, with $1 < p < \infty$. For a nonzero nonnegative density g in some L^q, $q > 1$, let $c(g) > 0$ be the scalar in Theorem 13.15.2. Its solutions have ordinary type V_θ and are continuous on Ω by Theorem 13.16.1. Write MA_θ for the extended measure. On Ω this is the locally bounded product for every potential of minimal type, and its extension gives zero mass to the analytic complement.

Put $c = c(f)$ and define

$$\begin{aligned} \kappa(r) &= \log c - \log c(\mathrm{e}^r f), \\ \mathcal{P}_f &= \left\{ \nu \in \mathcal{P}(X) : \int_X r \, \mathrm{d}\nu \leq \kappa(r) \text{ for every } r \in C(X) \right\}. \end{aligned} \tag{14.15}$$

When θ is semipositive and $f = \mathrm{e}^F$ with $F \in C(X)$, this is the supporting set $\mathcal{A}_0(F)$. The definition above also allows f to vanish and θ to be nonsemipositive.

Theorem 14.11.1 *Let u_λ be the solution of minimal type of*

$$\mathrm{MA}_\theta(u_\lambda) = \mathrm{e}^{\lambda u_\lambda} f \, \mathrm{d}V, \qquad \lambda > 0.$$

The set $\mathcal{P}_f$ is nonempty and weakly compact, and every potential of minimal type is integrable against all its measures. As $\lambda \downarrow 0$,

$$w_\lambda = u_\lambda - \lambda^{-1} \log c$$

converges locally uniformly on Ω and in $L^1(X)$ to a potential w of minimal type, continuous on Ω. It is the greatest potential s of minimal type, continuous on Ω, satisfying

$$\mathrm{MA}_\theta(s) \geq c f \, \mathrm{d}V, \qquad \int_X s \, \mathrm{d}\nu \leq 0 \quad (\nu \in \mathcal{P}_f). \tag{14.16}$$

Moreover,

$$\mathrm{MA}_\theta(w) = c f \, \mathrm{d}V, \qquad \max_{\nu \in \mathcal{P}_f} \int_X w \, \mathrm{d}\nu = 0.$$

Thus $u_\lambda - \sup_X u_\lambda$ *converges to* $w - \sup_X w$ *in the same two topologies. For a Hermitian* θ*, one may take* $\Omega = X$*, and both potential convergences are uniform.*

14.11.1 A norm defined by the critical scalar

Mixed products turn scalar comparison into a triangle inequality for density roots. The strict-barrier subsolution estimate gives its continuity in the corresponding Lebesgue norm.

Fix $1 < q < p$ and set $\ell = nq$. On real $L^\ell(\mathrm{d}V)$ define

$$N(a) = \begin{cases} c(|a|^n)^{-1/n}, & a \neq 0, \\ 0, & a = 0. \end{cases}$$

Proposition 14.11.1 *The function* N *is a lattice norm. There is* $C_q < \infty$*, depending only on* $X, \theta, \mathrm{d}V, q$*, such that*

$$N(a) \leq C_q \|a\|_\ell, \qquad |N(a) - N(b)| \leq C_q \|a - b\|_\ell. \tag{14.17}$$

Consequently, for nonzero $g, h \geq 0$ *in* L^q*,*

$$|c(g)^{-1/n} - c(h)^{-1/n}| \leq C_q \|g - h\|_q^{1/n}.$$

Proof If a potential v of minimal type satisfies $\mathrm{MA}_\theta(v) \geq dg \, \mathrm{d}V$, then $d \leq c(g)$. Otherwise comparison with a critical solution for g contradicts Corollary 13.12.1. Hence $g \leq h$ implies $c(g) \geq c(h)$. Scalar uniqueness also gives $c(tg) = c(g)/t$ for $t > 0$. These facts prove lattice monotonicity and absolute homogeneity of N; positivity follows from $0 < c(g) < \infty$.

For nonzero $a, b \geq 0$, put $A = N(a)$ and $B = N(b)$. Choose critical potentials v_a, v_b with measures $(a/A)^n \mathrm{d}V$ and $(b/B)^n \mathrm{d}V$. On every relatively compact subset of Ω, (1.11) gives

$$\mathrm{MA}_\theta\left(\frac{A v_a + B v_b}{A + B}\right) \geq \left(\frac{a + b}{A + B}\right)^n \mathrm{d}V.$$

Both potentials are locally bounded there. Exhaustion of Ω gives the global inequality, since its complement has zero smooth volume. The averaged potential has minimal type, so scalar order gives $N(a+b) \leq A + B$. The zero cases follow from homogeneity. For real functions, use $|a + b| \leq |a| + |b|$.

Apply Lemma 13.13.1 with reference V_θ and nonnegative densities of L^q norm at most one. It gives a uniform $m > 0$ and subsolutions of minimal type, hence $c(g) \geq m$ for each such nonzero density. For $a \neq 0$, take $g = |a|^n / \|a\|_\ell^n$. Homogeneity gives

$N(a) \le m^{-1/n}\|a\|_\ell$, proving the first bound in (14.17). The reverse triangle inequality gives the second bound. Apply it to $g^{1/n}, h^{1/n}$ and use $|g^{1/n} - h^{1/n}|^n \le |g - h|$ to obtain the last assertion. □

14.11.2 Norming probabilities and integral bounds

Set $a = f^{1/n}$ and $A = N(a) = c^{-1/n}$. The probabilities in (14.15) come from positive functionals norming N at a. Their representation will control integrals of unbounded potentials as well as bounded logarithmic perturbations.

With $\ell' = \ell/(\ell - 1)$, define

$$\mathcal{H}_a = \left\{\eta \in L^{\ell'}(\mathrm{d}V) : \eta \ge 0,\ \int_X a\eta\,\mathrm{d}V = A,\ \int_X b\eta\,\mathrm{d}V \le N(b) \text{ for every } b \in L^\ell\right\}. \tag{14.18}$$

Proposition 14.11.2 *The set $\mathcal{H}_a$ is nonempty and weakly compact in $L^{\ell'}$. The function*

$$\kappa(r) = n \log \frac{N(a\mathrm{e}^{r/n})}{A}$$

is convex, increasing and 1-Lipschitz on $C(X)$, with $\kappa(0) = 0$ and $\kappa(r + b) = \kappa(r) + b$ for constants b. Its subdifferential at zero is

$$\mathcal{P}_f = \{\nu_\eta : \eta \in \mathcal{H}_a\}, \qquad \mathrm{d}\nu_\eta = A^{-1}a\eta\,\mathrm{d}V. \tag{14.19}$$

If z is measurable and $az \in L^\ell$, then

$$\sup_{\nu \in \mathcal{P}_f} \int_X |z|\,\mathrm{d}\nu \le C_q A^{-1}\|az\|_\ell. \tag{14.20}$$

If also $a\mathrm{e}^{tz/n} \in L^\ell$ and $N(a\mathrm{e}^{tz/n}) = A$ for some $t > 0$, then $\int_X z\,\mathrm{d}\nu \le 0$ for every $\nu \in \mathcal{P}_f$.

Proof Hahn–Banach gives a linear functional norming N at a and dominated by N. By (14.17) it is continuous in L^ℓ, and hence is represented by $\eta_0 \in L^{\ell'}$. For $b \ge 0$, testing on $\mathrm{sgn}(\eta_0)b$ gives

$$\int_X b|\eta_0|\,\mathrm{d}V \le N(b).$$

Since $a \ge 0$ and $\int_X a\eta_0\,\mathrm{d}V = A = N(a)$, equality holds also with $|\eta_0|$. Positivity extends domination to real b, so $|\eta_0| \in \mathcal{H}_a$. The defining conditions are weakly closed and give $\|\eta\|_{\ell'} \le C_q$. Reflexivity proves weak compactness.

For nonzero nonnegative b_0, b_1, apply weighted arithmetic–geometric mean to $b_j/N(b_j)$, then use monotonicity and the triangle inequality:

$$N(b_0^{1-t}b_1^t) \le N(b_0)^{1-t}N(b_1)^t, \qquad 0 \le t \le 1.$$

Taking $b_j = a\mathrm{e}^{r_j/n}$ proves convexity of κ. Order and the constant translation rule follow from those of N, and together imply its Lipschitz bound.

Every ν_η is a probability and satisfies

$$\frac{N(a\mathrm{e}^{r/n})}{A} \geq \int_X \mathrm{e}^{r/n}\,\mathrm{d}\nu_\eta.$$

Jensen's inequality gives $\kappa(r) \geq \int_X r\,\mathrm{d}\nu_\eta$, proving one inclusion in (14.19). The map $\eta \mapsto \nu_\eta$ is weakly continuous, since $ar \in L^\ell$ for $r \in C(X)$.

Conversely, let $\nu \in \mathcal{P}_f$ and put $L(r) = \int_X r\,\mathrm{d}\nu$. Differentiating the exponential curve at zero in L^ℓ gives

$$N(a\mathrm{e}^{tr/n}) = N(a) + \frac{t}{n}N'_+(a;ar) + o(t), \qquad t \downarrow 0.$$

Here the one-sided derivative of the continuous convex norm exists, and the error follows from its Lipschitz bound. The supporting inequality, used with r and $-r$, therefore gives

$$|AL(r)| \leq C_q\|ar\|_\ell, \qquad AL(r) \leq N'_+(a;ar) \leq N(a+ar) - A. \tag{14.21}$$

Thus $T(ar) = AL(r)$ is well defined and continuous on these products. They are dense in

$$S_a = \{b \in L^\ell : b = 0 \text{ almost everywhere on } \{a = 0\}\} :$$

approximate b/a in $L^\ell(a^\ell \mathrm{d}V)$ by continuous functions. If $ar \geq 0$, then $ar = ar_+$ almost everywhere. The first inequality in (14.21) gives $L(r - r_+) = 0$, so $T(ar) \geq 0$. Positive continuous approximations in the weighted space show that the extension of T to S_a is positive.

Represent this extension by $\eta \geq 0$ in $L^{\ell'}$, chosen zero on $\{a = 0\}$. Since $T(a) = A$, extending the second inequality in (14.21) and replacing its argument b by $b - a$ gives $T(b) \leq N(b)$ on S_a. For general $b \in L^\ell$,

$$\int_X b\eta\,\mathrm{d}V \leq N(b\mathbf{1}_{\{a>0\}}) \leq N(b).$$

Thus $\eta \in \mathcal{H}_a$, and its probability agrees with ν on all continuous tests. This proves the representation and weak compactness of $\mathcal{P}_f$.

The bound (14.20) is Hölder's inequality. Under the last hypothesis, the norming inequality gives $\int_X \mathrm{e}^{tz/n}\,\mathrm{d}\nu \leq 1$. Since z is integrable by that bound, Jensen's inequality implies $\int_X z\,\mathrm{d}\nu \leq 0$. □

These estimates apply uniformly to supremum-normalized θ-plurisubharmonic potentials v. Indeed Lemma 1.5.1 gives uniform finite moments, after enlarging θ to a Hermitian background, and Hölder's inequality gives

$$\int_X |v|^{nq} f^q\,\mathrm{d}V \leq \|f\|_p^q \left(\int_X |v|^{nqp/(p-q)}\,\mathrm{d}V\right)^{(p-q)/p}. \tag{14.22}$$

Consequently (14.20) gives a uniform bound for $\int_X |v|\,\mathrm{d}\nu$ over $\nu \in \mathcal{P}_f$. Constant translates are integrable as well. The representation also shows that these probabilities charge neither $\{f = 0\}$ nor $X \setminus \Omega$.

14.11.3 Bounds for the centered potentials

Choose a critical potential $\psi \le 0$, with $\sup_X \psi = 0$ and $\mathrm{MA}_\theta(\psi) = c f\,\mathrm{d}V$. The integral normalizations prevent the centered family from drifting upward by constants.

Lemma 14.11.1 *There is $K < \infty$, independent of $0 < \lambda \le 1$, such that*

$$\psi \le w_\lambda, \qquad 0 \le k_\lambda := \sup_X w_\lambda \le K, \qquad \psi - K \le v_\lambda := w_\lambda - k_\lambda \le V_\theta. \tag{14.23}$$

Moreover $\int_X w_\lambda\,\mathrm{d}\nu \le 0$ for every $\nu \in \mathcal{P}_f$. The family w_λ is relatively compact for locally uniform convergence on Ω; every such cluster also converges in $L^1(X)$ to a potential of minimal type.

Proof Since $\psi \le 0$, its measure dominates $c\mathrm{e}^{\lambda\psi} f\,\mathrm{d}V$. Exponential order in Corollary 13.12.1 gives $\psi \le w_\lambda$, hence $k_\lambda \ge 0$. For each fixed λ, the density $\mathrm{e}^{\lambda w_\lambda} f$ lies in L^p, and its critical scalar is c. Therefore

$$N(a\mathrm{e}^{\lambda w_\lambda/n}) = A.$$

The moment estimate above makes $a w_\lambda \in L^\ell$, so Proposition 14.11.2 gives all the asserted integral inequalities. Fix one $\nu \in \mathcal{P}_f$. Then

$$k_\lambda \le -\int_X v_\lambda\,\mathrm{d}\nu \le C_q A^{-1} \|a v_\lambda\|_\ell \le K$$

by (14.22). Normalization gives $v_\lambda \le V_\theta$, and subtracting k_λ from the lower bound proves (14.23).

The potentials $v_\lambda = u_\lambda - \sup_X u_\lambda$ are the normalized exponential family of Proposition 13.16.1. That proposition gives subsequences converging locally uniformly on Ω and in $L^1(X)$ to potentials of minimal type. Extracting a convergent subsequence of $k_\lambda \in [0, K]$ gives the same compactness for $w_\lambda = v_\lambda + k_\lambda$. The bound $|w_\lambda| \le |\psi| + K$ also gives L^1 convergence along every locally uniformly convergent subsequence, since $X \setminus \Omega$ has zero volume. □

In particular,

$$\mathrm{e}^{\lambda \sup_X u_\lambda} = c\mathrm{e}^{\lambda k_\lambda} \longrightarrow c.$$

The actual equation densities satisfy $\|ce^{\lambda w_\lambda} f\|_p \le ce^K \|f\|_p$. This bound includes the scalar. An upper L^p bound on arbitrarily rescaled data alone would not bound the normalized potentials: on $\mathbb{P}^1$, with $\omega_{\mathrm{FS}} = \mathrm{dd^c} \log(1 + |z|^2)$ and $0 < \epsilon < 1$,

$$\psi_\epsilon = \log \frac{|z|^2 + \epsilon^2}{1 + |z|^2}$$

is smooth and ω_{FS}-plurisubharmonic, with supremum zero and infimum $\log \epsilon^2$. Write $\mathrm{MA}_{\omega_{\mathrm{FS}}}(\psi_\epsilon) = g_\epsilon \omega_{\mathrm{FS}}$, and take L^p norms with respect to ω_{FS} in this example. The data $g_\epsilon / \|g_\epsilon\|_p$ have norm one but scalar $\|g_\epsilon\|_p$ and the same potentials. This rescaling does not bound the actual equation densities g_ϵ.

14.11.4 Scalar envelopes of subsolutions

To compare a cluster potential with every member of (14.16), fix a point $x \in \Omega$ and vary a continuous forcing. Let $\mathcal{S}$ be the convex class of potentials of minimal type that are continuous on Ω. Define

$$\mathcal{F}_{\lambda,x}(r) = \sup\{s(x) : s \in \mathcal{S},\ \mathrm{MA}_\theta(s) \ge ce^{\lambda s + r} f \, \mathrm{d}V\}. \tag{14.24}$$

Lemma 14.11.2 *The functional $\mathcal{F}_{\lambda,x}$ is finite, concave, decreasing and λ^{-1}-Lipschitz on $C(X)$. It satisfies*

$$\mathcal{F}_{\lambda,x}(0) = w_\lambda(x), \qquad \mathcal{F}_{\lambda,x}(r + b) = \mathcal{F}_{\lambda,x}(r) - b/\lambda.$$

There is a probability $\nu_{\lambda,x}$ such that, for every s admissible in (14.24)*,*

$$\lambda s(x) + \int_X r \, \mathrm{d}\nu_{\lambda,x} \le \lambda w_\lambda(x). \tag{14.25}$$

Proof Exponential order makes w_λ the greatest admissible potential for $r = 0$. If $m = \min_X r$ and $M = \max_X r$, then $w_\lambda - M/\lambda$ is admissible for r, and translating any admissible s by m/λ makes it admissible for zero forcing. Hence

$$w_\lambda(x) - M/\lambda \le \mathcal{F}_{\lambda,x}(r) \le w_\lambda(x) - m/\lambda.$$

The local mixed inequality on Ω and arithmetic–geometric mean show that a convex combination of potentials admissible for r_0, r_1 is admissible for the same combination of the forcings. Approximating the two suprema at x proves concavity. Inclusion of the admissible classes proves order; constant translation proves the displayed covariance and the Lipschitz bound.

Apply the Hahn–Banach supporting-functional construction in the proof of Theorem 14.2.1 to $-\lambda\mathcal{F}_{\lambda,x}$ at zero. This is a finite continuous convex functional, increasing and with constant translation slope one. The same order and constant tests

make its subgradient a probability. The supporting inequality and the definition of the supremum give (14.25). Attainment of the supremum is unnecessary. □

Proof (Theorem 14.11.1) The representation and moment bound prove the assertions about $\mathcal{P}_f$. The centered bounds give precompactness. We identify every cluster potential.

Fix $x \in \Omega$. For $r \in C(X)$, choose a critical potential ψ_r with supremum zero for the density $\mathrm{e}^r f$. Then

$$\mathrm{MA}_\theta(\psi_r) = c\mathrm{e}^{-\kappa(r)+r} f \,\mathrm{d}V.$$

Since $\psi_r \le 0$, the potential $\psi_r - \kappa(r)/\lambda$ is admissible in (14.24). Thus

$$\int_X r \,\mathrm{d}\nu_{\lambda,x} \le \kappa(r) + \lambda\big(w_\lambda(x) - \psi_r(x)\big). \tag{14.26}$$

For fixed x, r, the values $w_\lambda(x)$ are uniformly bounded and $\psi_r(x)$ is finite. Every weak cluster of $\nu_{\lambda,x}$ as $\lambda \downarrow 0$ therefore belongs to $\mathcal{P}_f$.

Let s satisfy (14.16). The function $-\lambda s$ is lower semicontinuous and bounded below on X, with the value $+\infty$ allowed. Choose finite continuous $r_j \uparrow -\lambda s$, by metric inf-convolution and truncation. Since $r_j \le -\lambda s$, the potential s is admissible for every r_j. Inequality (14.25) bounds their integrals from above. Monotone convergence after subtracting a common lower bound proves that s is integrable against $\nu_{\lambda,x}$ and gives

$$w_\lambda(x) \ge s(x) - \int_X s \,\mathrm{d}\nu_{\lambda,x}. \tag{14.27}$$

Take a sequence $\lambda_j \downarrow 0$ with $w_{\lambda_j} \to z$ locally uniformly on Ω and in $L^1(X)$. The relative bounds preserve minimal type and local continuity. Local bounded-product continuity identifies the limit equation, since $c\mathrm{e}^{\lambda_j w_{\lambda_j}} f \to cf$ in L^p by dominated convergence and the common bound $w_\lambda \le K$. Thus $\mathrm{MA}_\theta(z) = cf \,\mathrm{d}V$ on Ω, and then on X because both sides ignore its analytic complement.

For each $\nu \in \mathcal{P}_f$, the bound $|w_\lambda| \le |\psi| + K$ has a ν-integrable right side. These measures charge only Ω. Dominated convergence in $\int_X w_\lambda \,\mathrm{d}\nu \le 0$ gives $\int_X z \,\mathrm{d}\nu \le 0$. Hence z belongs to the selection class.

For any other s in that class and fixed $x \in \Omega$, extract a further subsequence with $\nu_{\lambda_j,x} \rightharpoonup \nu \in \mathcal{P}_f$. The potential s is upper semicontinuous and bounded above on X, so

$$\limsup_j \int_X s \,\mathrm{d}\nu_{\lambda_j,x} \le \int_X s \,\mathrm{d}\nu \le 0.$$

For an unbounded s, this is Portmanteau applied first to $\max(s, -M)$, followed by $M \to \infty$; the final integral is finite by (14.20). Passing to the limit in (14.27) gives $z(x) \ge s(x)$. This holds for every $x \in \Omega$ and extends to all of X by Lemma 1.3.3. Every cluster is therefore the same greatest element w. Precompactness proves convergence of the whole family.

The maximum of $\int_X w \,\mathrm{d}\nu$ over $\mathcal{P}_f$ is attained: under (14.19) this is the weakly continuous functional $\eta \mapsto A^{-1} \int_X aw\eta \,\mathrm{d}V$ on $\mathcal{H}_a$, since $aw \in L^\ell$. If the maximum

were negative, a small positive constant translate of w would still satisfy (14.16), contrary to greatestness. Thus it is zero.

Finally $k_\lambda \to \sup_X w$. Indeed, from any subsequence choose a further one with $k_\lambda \to k$. Then $v_\lambda = w_\lambda - k_\lambda$ converges in L^1 to $w - k$; Theorem 1.4.1 gives $\sup_X(w - k) = 0$. Thus $k = \sup_X w$ for every scalar cluster. Subtracting the constants proves the claimed normalized convergence. If θ is Hermitian, take $\rho = 0$ and $\Omega = X$, so local uniform convergence is uniform. □

Together with Proposition 13.16.1, the theorem also gives convergence of the normalized potentials uniformly in differences on Ω and in every local C^β topology stated there: compactness in those topologies and uniqueness of the L^1 limit force every subsequence to have the same limit. Selection of this limit does not imply uniqueness of all normalized solutions of the critical equation.

Chapter 15
Multiplier-ideal envelopes and $\mathcal{I}$-goodness

Let X be a connected compact complex manifold of dimension $n \geq 1$, and let θ be a smooth real $(1,1)$-form. Fix a smooth positive probability volume $\mathrm{d}V$ for L^1 norms. We use the extended non-pluripolar measure.

Multiplier ideals record weighted holomorphic integrability. They can agree for two potentials whose ordinary envelopes differ, so passing from an ordinary type to an ideal class loses information. Dyadic analytic approximation makes this loss visible: the associated envelopes converge to the greatest representative of the ideal class below zero, which may exceed the ordinary envelope of the original potential.

In the Kähler theory, equality of these envelopes also has a mass characterization. For a nonclosed background, their equality remains meaningful even when the original potential has infinite mass. We therefore track both envelopes when studying analytic approximation. Curve examples show why this still does not ensure continuity of mass or completeness: diffuse polar curvature can disappear in the limit. They also explain the role of the obstacle when the background changes.

15.1 The Kähler characterization

First suppose X is Kähler, θ is a smooth closed big real $(1,1)$-form, and $u \in \mathrm{PSH}(X,\theta)$ has positive non-pluripolar mass. Use $P_\theta[u]_{\mathcal{I}}$ for the ideal envelope defined in Definition 15.2.1, and let d_S be the singularity pseudometric of [Xia, Section 6.2]. The characterization of $\mathcal{I}$-good singularities in [Xia, Theorem 7.1.1] states the equivalence of

$$u \text{ lies in the } d_S\text{-closure of analytic singularities,}$$
$$\int_X \mathrm{MA}_\theta(u) = \int_X \mathrm{MA}_\theta(P_\theta[u]_{\mathcal{I}}),$$
$$P_\theta[u] = P_\theta[u]_{\mathcal{I}}.$$

The last condition involves envelopes alone. It therefore gives a starting point for the Hermitian problem even when the metric and mass characterizations are unavailable.

The integral on the right in the middle condition is the *multiplier-ideal volume* in [Xia]. The upper and lower volumes of Definition 12.3.1 instead vary the potential within its ordinary type. This distinction matters in a nonclosed background, where even bounded changes can alter total mass.

15.2 Ideal projection

Return to the general compact complex setting at the chapter's opening. Taking the supremum over an ideal class could enlarge its multiplier ideals. Increasing-limit stability prevents this and makes the envelope a projection.

Definition 15.2.1 For $u \in \mathrm{PSH}(X, \theta)$ define

$$P_\theta[u]_{\mathcal{I}} := (\sup\{v \in \mathrm{PSH}(X, \theta) : v \le 0,\ v \sim_{\mathcal{I}} u\})^* .$$

The family is nonempty because $u - \sup_X u$ belongs to it; bounded perturbations do not change multiplier ideals. It is bounded above by zero, so its regularized envelope is a genuine θ-plurisubharmonic function. It depends only on the multiplier-ideal equivalence class of u.

Theorem 15.2.1 *For every $u \in \mathrm{PSH}(X, \theta)$,*

$$P_\theta[u]_{\mathcal{I}} \sim_{\mathcal{I}} u, \qquad P_\theta[P_\theta[u]_{\mathcal{I}}]_{\mathcal{I}} = P_\theta[u]_{\mathcal{I}}.$$

The regularization in Definition 15.2.1 does not change the pointwise supremum. Moreover,

$$\sup_X P_\theta[u]_{\mathcal{I}} = 0.$$

Proof Write $q = P_\theta[u]_{\mathcal{I}}$. By Choquet's lemma choose a sequence from the defining family whose regularized supremum is q. Replace it by successive finite maxima. By Lemma 1.16.3, the resulting sequence v_j remains in the family, is increasing, and converges to q almost everywhere.

Fix $x \in X$ and $\lambda > 0$. On a coordinate ball about x, choose one smooth function r such that every $v_j + r$ and $q + r$ is plurisubharmonic. This is possible because they all have the same smooth curvature lower bound $-\theta$. The increasing-limit assertion of Theorem 1.16.1 gives

$$\mathcal{I}(\lambda(q + r))_x = \mathcal{I}(\lambda(v_j + r))_x = \mathcal{I}(\lambda u)_x$$

for large j; smooth additions give the second equality. Hence $q \sim_{\mathcal{I}} u$.

Since $q \le 0$, it now belongs to its own defining family. The pointwise supremum is at least q and at most its regularization q, proving the assertion about regularization. The defining families for $P_\theta[q]_{\mathcal{I}}$ and $P_\theta[u]_{\mathcal{I}}$ coincide, which proves idempotence.

Finally, if $\sup_X q < 0$, the function $q - \sup_X q$ would be a larger competitor, a contradiction. □

Proposition 15.2.1 *For every $u \in \mathrm{PSH}(X, \theta)$,*

$$u - \sup_X u \le P_\theta[u] \le P_\theta[u]_{\mathcal{I}} \le V_\theta.$$

Moreover,

$$P_\theta[u] \sim_{\mathcal{I}} u, \qquad P_\theta[P_\theta[u]]_{\mathcal{I}} = P_\theta[u]_{\mathcal{I}},$$
$$P_\theta[P_\theta[u]_{\mathcal{I}}] = P_\theta[u]_{\mathcal{I}}.$$

Proof For a competitor $h \le 0$ with $h \preceq u$ in the ordinary envelope, the function $\max(h, u - \sup_X u)$ is below zero, ordinarily equivalent to u, and dominates h. It is therefore admissible for the ideal envelope. Taking regularized suprema gives $P_\theta[u] \le P_\theta[u]_{\mathcal{I}}$; the two outer bounds follow from the defining families.

Projection and monotonicity of local integrability now give, for every $\lambda > 0$,

$$\mathcal{I}(\lambda u) \subseteq \mathcal{I}(\lambda P_\theta[u]) \subseteq \mathcal{I}(\lambda P_\theta[u]_{\mathcal{I}}) = \mathcal{I}(\lambda u).$$

Thus the ordinary envelope preserves all multiplier ideals and has the same ideal envelope as u. For $q = P_\theta[u]_{\mathcal{I}}$, the already proved bounds and idempotence give $q \le P_\theta[q] \le P_\theta[q]_{\mathcal{I}} = q$. □

Corollary 15.2.1 *If θ is semipositive and big, then $P_\theta[u]_{\mathcal{I}}$ has finite extended Monge–Ampère mass for every $u \in \mathrm{PSH}(X, \theta)$.*

Proof The last identity in Proposition 15.2.1 makes $P_\theta[u]_{\mathcal{I}}$ an ordinary envelope fixed point. Apply the fixed-point mass estimate in Proposition 12.1.4. □

Proposition 15.2.2 *For $u, v \in \mathrm{PSH}(X, \theta)$,*

$$u \preceq_{\mathcal{I}} v \quad \Longrightarrow \quad P_\theta[u]_{\mathcal{I}} \le P_\theta[v]_{\mathcal{I}}.$$

For $a > 0$ one has $P_{a\theta}[au]_{\mathcal{I}} = aP_\theta[u]_{\mathcal{I}}$. For smooth real $(1,1)$-forms θ_i and potentials $u_i \in \mathrm{PSH}(X, \theta_i)$, $i = 1, 2$,

$$P_{\theta_1}[u_1]_{\mathcal{I}} + P_{\theta_2}[u_2]_{\mathcal{I}} \le P_{\theta_1+\theta_2}[u_1 + u_2]_{\mathcal{I}}.$$

Proof If h is a competitor for $P_\theta[u]_{\mathcal{I}}$, the reverse Lelong inequalities in Theorem 1.16.1 show that $\max(h, v - \sup_X v)$ is multiplier-ideal equivalent to v. It is below zero and dominates h. Taking the supremum proves order preservation. Scaling gives a bijection of the two defining families. For addition, the sum of the two envelopes is below zero, is $(\theta_1 + \theta_2)$-plurisubharmonic, and is multiplier-ideal equivalent to $u_1 + u_2$ by projection and Lemma 1.16.3. It is therefore an admissible competitor on the right. □

15.2.1 Analytic and divisorial singularities

For an analytic singularity, the ideal bound forces ordinary domination by Lemma 1.17.9. The ideal envelope is then a competitor for the ordinary envelope, giving their equality. This extends [Xia, Proposition 3.2.10] to smooth backgrounds.

Proposition 15.2.3 *Let $a \in \mathrm{PSH}(X, \theta)$ have analytic singularities. Then*

$$a \sim P_\theta[a] = P_\theta[a]_{\mathcal{I}}.$$

In particular the common envelope has analytic singularities with the same coefficient and ideal as a, up to a bounded remainder.

Proof Put $q = P_\theta[a]_{\mathcal{I}}$. Projection gives $q \sim_{\mathcal{I}} a$, so Lemma 1.17.9 gives $q \preceq a$. Since $q \leq 0$, it is a competitor for $P_\theta[a]$. Combine this with Proposition 15.2.1 to obtain

$$a - \sup_X a \leq P_\theta[a] \leq q \leq P_\theta[a].$$

The two envelopes are equal and ordinarily equivalent to a. Their bounded difference from a preserves the analytic presentation. □

The same argument applies to a countable divisorial current of finite mass, using Lemma 1.17.4.

Corollary 15.2.2 *Let S be a closed positive divisorial current of finite mass on a connected compact complex manifold Z, and choose a smooth residual decomposition $S = \beta + \mathrm{dd^c} v$ as in Corollary 1.17.1. For every Hermitian form Ω with $v \in \mathrm{PSH}(Z, \Omega)$,*

$$P_\Omega[v] = P_\Omega[v]_{\mathcal{I}} \sim v.$$

Proof Put $q = P_\Omega[v]_{\mathcal{I}}$. Projection gives $q \sim_{\mathcal{I}} v$, and Lemma 1.17.4 gives $q \preceq v$. Thus q is admissible for $P_\Omega[v]$. Together with Proposition 15.2.1, this yields $v - \sup_Z v \leq P_\Omega[v] = q \preceq v$. □

15.3 Changes of background and envelope mass

A smooth change of representative transports the current but changes the zero obstacle. The resulting envelopes remain ordinarily equivalent after transport. For a nonclosed background, their masses can nevertheless differ.

Proposition 15.3.1 *Let $u \in \mathrm{PSH}(X,\theta)$ and $f \in C^\infty(X,\mathbb{R})$. Set $\theta' = \theta + \mathrm{dd^c} f$, $u' = u - f$, $m = \min_X f$ and $M = \max_X f$. Then*

$$P_\theta[u]_{\mathcal{I}} + m \le P_{\theta'}[u']_{\mathcal{I}} + f \le P_\theta[u]_{\mathcal{I}} + M.$$

The identical inequalities hold for ordinary envelopes.

Proof If v is a competitor for $P_\theta[u]_{\mathcal{I}}$, then $v - f + m$ is θ'-plurisubharmonic, below zero, and multiplier-ideal equivalent to u'. Therefore $v \le P_{\theta'}[u']_{\mathcal{I}} + f - m$. Taking the envelope gives the first inequality. Conversely, a competitor v' in the new background gives $v' + f - M$ in the old one, which proves the second inequality. For P, use the ordinary singularity order instead of multiplier-ideal equivalence in the same calculation. □

For $u \in \mathrm{PSH}(X,\theta)$, put

$$W_\theta(u) = \int_X \mathrm{MA}_\theta(P_\theta[u]_{\mathcal{I}}) \in [0,+\infty].$$

The following example shows that this quantity depends on the decomposition of the current, even for smooth potentials.

Proposition 15.3.2 *There are two smooth decompositions of the same positive current for which the corresponding values of W are different.*

Proof Use the Hermitian form ω and smooth function $g = u_t$ from Example 2.3.1, choosing $t \neq 0$ sufficiently small. Both ω and $\omega' = \omega + \mathrm{dd^c} g$ are Hermitian, and

$$\omega + \mathrm{dd^c} 0 = \omega' + \mathrm{dd^c}(-g).$$

For a smooth potential in a Hermitian background, the multiplier ideals at all weights are trivial and zero is an admissible competitor for its multiplier-ideal envelope. That envelope is therefore zero. Consequently

$$W_\omega(0) = \int_X \omega^2, \qquad W_{\omega'}(-g) = \int_X (\omega + \mathrm{dd^c} g)^2.$$

The two integrals differ by the explicit formula (2.3). □

The underlying torus is Kähler; it is the chosen nonclosed background that causes this dependence. In the closed theory, invariance of mass under bounded changes supplies the further step after Proposition 15.3.1.

Singularity volumes instead range over the entire ordinary type, so the gauge bijection of competitors preserves them. Under bounded mass, Proposition 29.9.1 consequently transports positive-volume models while preserving their upper-volume comparison.

15.3.1 Envelope equality and infinite mass

The equality

$$P_\theta[u] = P_\theta[u]_{\mathcal{I}} \tag{15.1}$$

is meaningful for each $u \in \mathrm{PSH}(X, \theta)$. We call it *envelope goodness in the fixed background*. Its relation to mass differs from the Kähler characterization: it can hold even when the potential has infinite mass and its envelope has finite mass.

Proposition 15.3.3 *If a Hermitian metric ω fails bounded mass, there exists $U \in \mathrm{PSH}(X, \omega)$ such that*

$$P_\omega[U] = P_\omega[U]_{\mathcal{I}} = 0, \qquad U \in \mathcal{E}_{\mathrm{env}}(X, \omega, 0), \qquad \mathrm{MA}_\omega^{\mathrm{ext}}(U)(X) = +\infty.$$

Proof Take U from Proposition 12.6.1, which supplies its ordinary envelope, class membership and infinite mass. Since $P_\omega[U] = V_\omega = 0$, Proposition 15.2.1 gives $P_\omega[U]_{\mathcal{I}} = 0$. □

15.4 L^1 geometry of ideal classes

The canonical ideal representatives let us measure the distance between ideal classes by an L^1 norm. Normalized compactness gives a compact completion, but the new points need not remain canonical representatives of their own ideal classes. On a curve, finite atomic approximation produces every normalized potential in this completion.

Let

$$\mathfrak{S}_{\mathcal{I}}(X, \theta) = \mathrm{PSH}(X, \theta)/\sim_{\mathcal{I}}, \qquad \mathfrak{M}_{\mathcal{I}}(X, \theta) = \{q \in \mathrm{PSH}(X, \theta) : P_\theta[q]_{\mathcal{I}} = q\}.$$

Write $[u]_{\mathcal{I}}$ for the multiplier-ideal class of u. By Theorem 15.2.1, $[u]_{\mathcal{I}} \mapsto P_\theta[u]_{\mathcal{I}}$ identifies these two sets, and every ideal envelope has supremum zero.

Proposition 15.4.1 *The formula*

$$d_{\mathcal{I},\theta,\mathrm{d}V}([u]_{\mathcal{I}},[v]_{\mathcal{I}}) = \int_X |P_\theta[u]_{\mathcal{I}} - P_\theta[v]_{\mathcal{I}}|\,\mathrm{d}V \tag{15.2}$$

defines a finite metric on $\mathfrak{S}_{\mathcal{I}}(X,\theta)$. This metric space is totally bounded. Its metric completion is canonically the L^1 closure of $\mathfrak{M}_{\mathcal{I}}(X,\theta)$ inside the compact space $\mathcal{K}_\theta$ of Theorem 1.4.1. Changing $\mathrm{d}V$ *to another smooth positive probability volume gives a bi-Lipschitz equivalent metric.*

Proof The envelopes are integrable and depend only on the ideal classes. Symmetry and the triangle inequality are those of the L^1 norm. If the distance is zero, Lemma 1.3.3 identifies the two envelopes everywhere. Projection then identifies their ideal classes. Conversely, equal ideal classes have the same envelope.

The normalized compactness theorem places $\mathfrak{M}_{\mathcal{I}}$ inside compact $\mathcal{K}_\theta$. Every subset of a compact metric space is totally bounded, and its completion is its closure in that space. Finally the ratio of two smooth positive volume forms on compact X is bounded above and below by positive constants. Multiplying the integral by these bounds proves the last assertion. □

15.4.1 Completion on a curve

On a curve, the curvature measure determines a normalized potential. Discretizing that measure produces ideal envelopes even when the limiting potential is not an ideal envelope.

Theorem 15.4.1 *Let Y be a compact connected Riemann surface and θ a smooth positive form of total mass one. Fix a smooth positive probability volume on Y, again denoted by* $\mathrm{d}V$*, and use $\mathcal{K}_\theta$, the envelopes and the ideal metric on Y in place of X. Then the normalized ideal envelopes with finitely many logarithmic poles and zero non-pluripolar mass are dense in $\mathcal{K}_\theta$ for L^1. The logarithmic coefficients can be chosen rational, so these approximants have analytic singularities in the convention of [Xia]. Consequently the completion in Proposition 15.4.1 is all of $\mathcal{K}_\theta$. The metric on ideal classes is not complete.*

Proof Let δ_p denote the Dirac mass at p. For distinct points $p_1,\dots,p_N \in Y$, consider a finitely supported probability measure

$$\nu = \sum_{j=1}^{N} a_j\delta_{p_j}, \qquad a_j > 0, \quad \sum_j a_j = 1.$$

The Green-kernel construction gives a potential q_ν with

$$\theta + \mathrm{dd^c}q_\nu = \nu, \qquad \sup_Y q_\nu = 0.$$

Indeed, choose a kernel with $\mathrm{dd}^{\mathrm{c}}_x G(x,p) = \delta_p - \theta$. The sum $\sum_j a_j G(x,p_j)$ has this curvature, and subtracting its finite supremum gives q_ν. In a coordinate z centered at p_j, it is $a_j \log|z|^2$ plus a smooth function; it is smooth away from the finite support. All of its curvature lies on its pole set, so its non-pluripolar measure is zero.

We claim $P_\theta[q_\nu]_{\mathcal{I}} = q_\nu$. If $w \in \mathrm{PSH}(Y,\theta)$ and $w \sim_{\mathcal{I}} q_\nu$, the local ideal criterion gives $\nu(w,p_j) = a_j$, where $\nu(w,p_j)$ is the Lelong number. On a curve, this Lelong number is exactly the atomic mass of the positive measure $\theta + \mathrm{dd}^{\mathrm{c}} w$. That measure has total mass one by distributional Stokes, so its prescribed atoms already exhaust the mass. It follows that $\theta + \mathrm{dd}^{\mathrm{c}} w = \nu$. Thus $\mathrm{dd}^{\mathrm{c}}(w - q_\nu) = 0$. The difference is a distributional harmonic function, hence smooth and constant on the compact connected curve. Among these translates the greatest one below zero is q_ν. This proves the claim. It also shows that every such q_ν is an ordinary model potential by Proposition 15.2.1.

Now take any $u \in \mathcal{K}_\theta$. Its curvature $\nu = \theta + \mathrm{dd}^{\mathrm{c}} u$ is a positive probability measure. Choose finitely supported probability measures ν_j converging weakly to ν. One elementary construction partitions a compact metric space into finitely many Borel sets of diameter tending to zero, places their masses at chosen points, and omits the sets of mass zero. Uniform continuity of each continuous test function proves weak convergence.

We may make these weights rational. For a partition with weights $a_1, \dots, a_N$, choose a large integer M, replace a_i by $b_i = \lfloor M a_i \rfloor / M$ for $i < N$, and put $b_N = 1 - \sum_{i<N} b_i$. Discard zero weights. The total absolute change is at most $2(N-1)/M$, which can be made to tend to zero along the partitions. This preserves weak convergence. Writing $b_i = m_i/M$ gives the analytic type

$$\left(\frac{1}{M}, \mathcal{O}_Y\left(-\sum_i m_i p_i\right)\right)$$

with smooth local remainders.

Let $q_j = q_{\nu_j}$. By normalized compactness, every subsequence has an L^1-convergent further subsequence with limit $q \in \mathcal{K}_\theta$. Passing to distributions in the curvature identity gives

$$\theta + \mathrm{dd}^{\mathrm{c}} q = \nu = \theta + \mathrm{dd}^{\mathrm{c}} u.$$

The difference is constant by the same harmonic argument. Since both suprema are zero, it is zero. Every cluster limit is therefore u, so the full sequence converges to u in L^1. This proves the density assertion and identifies the completion.

To see incompleteness concretely, choose a nonconstant smooth real function f. For sufficiently small $\varepsilon > 0$, $\theta + \varepsilon \mathrm{dd}^{\mathrm{c}} f > 0$. Put $u = \varepsilon f - \sup_Y(\varepsilon f)$. It is smooth, normalized and nonconstant. Its multiplier ideals are all trivial, so $P_\theta[u]_{\mathcal{I}} = 0$, because zero is an admissible competitor. Thus $u \notin \mathfrak{M}_{\mathcal{I}}(Y,\theta)$. This set is not closed in its completion, proving incompleteness. □

15.4.2 Mass and capacity convergence

For the nonconstant smooth u in the proof, its approximants q_j have zero non-pluripolar measure, whereas $\mathrm{MA}_\theta(u) = \theta + \mathrm{dd^c}u > 0$.

After mean normalization, the same sequence converges in the Gauduchon trace metric of Corollary 5.2.1. The original sequence q_j cannot converge to u in capacity. Indeed, $q_j, u \le 0$, so Theorem 7.3.1 would give

$$0 = \mathrm{e}^{q_j}\mathrm{MA}_\theta(q_j) \rightharpoonup \mathrm{e}^u(\theta + \mathrm{dd^c}u),$$

contrary to strict positivity of the limiting measure. Thus weak trace convergence and L^1 convergence do not imply capacity convergence even on a curve.

The same sequence shows that ideal projection is discontinuous for L^1: $P_\theta[q_j]_{\mathcal{I}} = q_j \to u$, whereas $P_\theta[u]_{\mathcal{I}} = 0$. The completion distinguishes smooth normalized potentials that the original ideal quotient identifies.

The mass defect can also persist at a fixed positive mass.

Proposition 15.4.2 *In the curve setting above, for every rational* $0 < \tau < 1$ *there are normalized analytic ideal envelopes* r_j *such that*

$$d_{\mathcal{I},\theta,\mathrm{d}V}([r_j]_{\mathcal{I}}, [0]_{\mathcal{I}}) \longrightarrow 0,$$
$$\mathrm{MA}_\theta(r_j)(Y) = \tau, \qquad \mathrm{MA}_\theta(r_j) \rightharpoonup \tau\theta.$$

In particular non-pluripolar mass is not continuous for this metric, even on a family whose masses have a fixed positive lower bound.

Proof Choose finitely supported rational probability measures ν_j converging weakly to θ, and let q_j be their normalized Green potentials. The preceding proof gives $q_j \to 0$ in L^1. Put $a = 1 - \tau$ and $r_j = P_\theta[aq_j]_{\mathcal{I}}$. The scaling is admissible in the same background, since

$$\theta + \mathrm{dd^c}(aq_j) = (1-a)\theta + a\nu_j \ge 0.$$

The function aq_j is a competitor, so $aq_j \le r_j \le 0$ and $r_j \to 0$ in L^1. Projection makes r_j an ideal envelope. By Proposition 15.2.3, it has exactly the logarithmic coefficients of aq_j, with bounded local remainders. These coefficients are rational. Away from the finite pole set it is locally bounded. Thus, on the curve, removing the polar mass gives precisely

$$\mathrm{MA}_\theta(r_j) = \theta + \mathrm{dd^c}r_j - a\nu_j.$$

Distributional Stokes gives total mass $1 - a = \tau$. The positive curvature measures $\theta + \mathrm{dd^c}r_j$ have mass one and converge distributionally, hence weakly, to θ. Since $\nu_j \rightharpoonup \theta$, the displayed identity proves the claimed measure limit. Finally $P_\theta[0]_{\mathcal{I}} = 0$, so the metric distance is $\int_Y |r_j|\,\mathrm{d}V$. The limiting potential has non-pluripolar measure θ, of mass one. □

For comparison, let $d_{S,\theta}$ denote the source singularity pseudometric in background θ. The ordered-distance formula in [Xia, Lemma 6.2.2] gives

$$d_{S,\theta}(r_j, 0) = \frac{1}{2}\left(\int_Y \theta - \int_Y \mathrm{MA}_\theta(r_j)\right) = \frac{1-\tau}{2} > 0.$$

Here $r_j \leq 0$, so the ordered formula applies. The source distance retains the fixed mass gap that disappears in the elementary ideal metric.

15.4.3 Comparison with scalar normalization

Return to X in arbitrary dimension and the fixed volume $\mathrm{d}V$. The scalar pseudometric Δ of (14.10) compares normalizing constants over all densities. Shrinking an analytic singularity makes this scalar distance tend to zero through the smaller-background estimate. At a fixed mass, however, the closed-background functional loses the locations of poles.

Example 15.4.1 Let θ be semipositive and big, and let $a \in \mathrm{PSH}(X, \theta)$ have analytic singularities with $a \leq 0$. Use B_φ and $\mathcal{A}_\varphi$ from Sections 14.1 and 14.2 with the fixed volume $\mathrm{d}V$. Set

$$\varphi_t = P_\theta[ta], \qquad t \in \mathbb{Q} \cap (0, 1).$$

Proposition 15.2.3 gives $\varphi_t \sim ta$, so invariance under bounded changes gives $P_\theta[\varphi_t] = \varphi_t$. Moreover $ta \in \mathrm{PSH}(X, t\theta)$ and $ta \leq \varphi_t$, hence $P_{t\theta}(\varphi_t)$ is genuine. By Proposition 14.7.1,

$$\sup_{G \in C(X)} |B_{\varphi_t}(G) - B_0(G)| \leq -n \log(1-t) \longrightarrow 0 \qquad (t \downarrow 0).$$

When θ is Hermitian and G is fixed and smooth, every choice in $\mathcal{A}_{\varphi_t}(G)$ converges weakly in L^1 to the smooth Gauduchon adjoint probability, by Corollary 14.7.1 and Theorem 14.2.3.

The projective references in Example 14.10.1 have equal mass and identical normalization functionals, although their poles and multiplier ideals differ.

15.5 Limits of analytic envelopes

The curve construction shows that convergence of potentials alone need not identify the limit of their ideal envelopes. A common lower bound in the ideal order prevents this failure. Together with convergence of divisorial Lelong numbers, it forces every cluster limit to have the prescribed ideals. The maximality of the envelope then identifies the limit. The dyadic analytic approximants satisfy these two conditions.

Here θ is again an arbitrary smooth real $(1,1)$-form. Fix $u \in \mathrm{PSH}(X,\theta)$ and a Hermitian form h on X.

Proposition 15.5.1 *Let $\epsilon_j \geq 0$ tend to zero, and let $a_j \in \mathrm{PSH}(X, \theta + \epsilon_j h)$ satisfy*

$$u \preceq_{\mathcal{I}} a_j, \qquad \nu_E(a_j) \longrightarrow \nu_E(u) \quad \textit{for every local divisor } E \textit{ over } X.$$

Then

$$P_{\theta+\epsilon_j h}[a_j]_{\mathcal{I}} \longrightarrow P_\theta[u]_{\mathcal{I}} \quad \textit{in } L^1.$$

If the a_j have analytic singularities, their ordinary envelopes converge to $P_\theta[u]$ in L^1 if and only if $P_\theta[u] = P_\theta[u]_{\mathcal{I}}$.

Proof Put $p = P_\theta[u]_{\mathcal{I}}$ and $q_j = P_{\theta+\epsilon_j h}[a_j]_{\mathcal{I}}$. Projection and ideal order give

$$p \leq P_{\theta+\epsilon_j h}[p]_{\mathcal{I}} \leq q_j \leq 0.$$

The first inequality uses p as a competitor in the larger background. All q_j have supremum zero and belong to $\mathrm{PSH}(X, \theta + Ch)$ for a fixed $C \geq \sup_j \epsilon_j$. Normalized compactness therefore gives an L^1-convergent further subsequence of every subsequence. Let q be such a limit. Passing the current inequalities to the limit gives $q \in \mathrm{PSH}(X,\theta)$, and Lemma 1.3.3 gives $p \leq q \leq 0$.

Fix a local modification $\pi : Z \to U$ and a prime divisor $E \subset Z$. By Lemma 1.22.2, the chosen subsequence converges after pullback in L^1_{loc}. The pointwise semicontinuity in Lemma 1.17.5 yields

$$\nu_E(q) \geq \limsup_j \nu_E(q_j) = \lim_j \nu_E(a_j) = \nu_E(u).$$

Indeed, at every point of E each pullback has Lelong number at least its generic number on E; apply the pointwise inequality and then take the generic value of the limit. On the other hand, $p \leq q$ gives $\nu_E(q) \leq \nu_E(p) = \nu_E(u)$. Thus q and u have the same Lelong numbers on all local divisors. Blowing up a point on any local modification recovers its pointwise Lelong number as a generic divisorial number; on a curve the point is already a divisor. By Theorem 1.16.1, $q \sim_{\mathcal{I}} u$.

Consequently q is a competitor for p, so $q \leq p$. Every cluster limit is therefore p, which proves convergence of the entire sequence. For analytic a_j, Proposition 15.2.3 identifies the ordinary and ideal envelopes. The last assertion follows from uniqueness of the L^1 limit and Lemma 1.3.3. □

Multiplier-ideal inclusions at fixed weights imply the required divisorial convergence. The local Bergman estimate controls the error from log discrepancy; after division by the weight, this error tends to zero.

Corollary 15.5.1 *Let $\epsilon_j \geq 0$ tend to zero, and let $a_j \in \mathrm{PSH}(X, \theta + \epsilon_j h)$ satisfy $u \preceq_{\mathcal{I}} a_j$. Suppose that, for every integer $m \geq 1$ and every $\delta > 0$,*

$$\mathcal{I}((1+\delta)ma_j) \subseteq \mathcal{I}(mu) \quad \textit{for all sufficiently large } j.$$

Then the conclusions of Proposition 15.5.1 hold.

Proof Fix a local divisor E over a coordinate neighborhood U. Write $A = A_U(E)$ for its log discrepancy. By Lemma 1.17.7, the asserted inclusion implies

$$(1+\delta)m\nu_E(a_j) \geq \mathrm{ord}_E\, \mathcal{I}((1+\delta)ma_j) \geq \mathrm{ord}_E\, \mathcal{I}(mu) \geq m\nu_E(u) - A.$$

Take the lower limit in j first, then let $m \to \infty$ and $\delta \downarrow 0$. It follows that $\liminf_j \nu_E(a_j) \geq \nu_E(u)$. The opposite inequality holds term by term because $u \preceq_{\mathcal{I}} a_j$. The preceding proposition applies. □

Divisorial convergence cannot replace the common ideal bound. In the atomic approximation above, every individual atom can tend to zero while the limiting potential still fails to be an envelope.

Example 15.5.1 On a compact connected curve Y with a positive form θ of total mass one, there are normalized analytic $a_j \in \mathrm{PSH}(Y, \theta)$ and a smooth $u \in \mathrm{PSH}(Y, \theta)$ such that

$$a_j \longrightarrow u \text{ in } L^1, \qquad \nu_E(a_j) \longrightarrow \nu_E(u) \text{ for every local divisor } E,$$

but $P_\theta[a_j] = P_\theta[a_j]_{\mathcal{I}} \to u \neq P_\theta[u] = P_\theta[u]_{\mathcal{I}}$ in L^1.

Proof Choose u smooth, nonconstant and normalized, with $\theta + \mathrm{dd^c}u > 0$. Apply the construction in Theorem 15.4.1 to the smooth probability measure $\nu = \theta + \mathrm{dd^c}u$. In partitions of vanishing diameter, the largest cell mass tends to zero because ν has no atoms. Choose the rational approximations of the weights so that their total absolute error tends to zero. The resulting atomic probability measures ν_j converge weakly to ν, and their largest atoms tend to zero.

Their normalized Green potentials a_j are analytic, satisfy $P_\theta[a_j] = P_\theta[a_j]_{\mathcal{I}} = a_j$, and converge to u in L^1, by the same theorem. At every point $y \in Y$,

$$0 \leq \nu(a_j, y) = \nu_j(\{y\}) \leq \max_{z \in Y} \nu_j(\{z\}) \longrightarrow 0 = \nu(u, y).$$

Local modifications of a smooth curve are isomorphisms, so this gives all the required divisorial limits. Since u is bounded and $\theta > 0$, both its envelopes are zero. Its nonconstancy proves the failure of envelope convergence. Each a_j has a logarithmic pole, so $u \preceq_{\mathcal{I}} a_j$ fails. □

15.5.1 Dyadic regularization

Demailly regularization supplies the common ideal bound, and the Bergman estimate gives convergence of each divisorial order. Dyadic indices add an order on ordinary

singularity types through subadditivity, making the envelopes decrease pointwise. This is the mechanism used in [Xia, Theorem 7.1.1, proof of (3) ⇒ (1), Step 1].

Choose the regularization from Theorem 1.15.1 along $m_j = 2^j$, starting at a sufficiently large index. Denote it by a_j and choose numbers $\epsilon_j > 0$ decreasing to zero such that

$$a_j \in \mathrm{PSH}(X, \theta + \epsilon_j h), \qquad a_j \geq u, \qquad a_j \text{ has type } (1/m_j, \mathcal{I}(m_j u)). \tag{15.3}$$

The chosen regularization satisfies $a_j \to u$ in L^1. Its values need not decrease. Its ordinary singularity types do: subadditivity in (1.20) gives

$$\mathcal{I}(2m_j u) \subseteq \mathcal{I}(m_j u)^2, \qquad a_{j+1} \preceq a_j.$$

Taking logarithms of generators proves the ordinary comparison, with a constant allowed to depend on j.

Theorem 15.5.1 *For every $u \in \mathrm{PSH}(X, \theta)$ and every choice in (15.3), put*

$$q_j = P_{\theta+\epsilon_j h}[a_j].$$

Then q_j has analytic singularities, and

$$q_j = P_{\theta+\epsilon_j h}[a_j]_{\mathcal{I}} \searrow P_\theta[u]_{\mathcal{I}}$$

pointwise and in L^1. The limit is independent of the regularizing representatives, the admissible error sequence and the metric h.

Proof By Proposition 15.2.3, $q_j = P_{\theta+\epsilon_j h}[a_j]_{\mathcal{I}} \sim a_j$. Since the backgrounds decrease and $a_{j+1} \preceq a_j$, the function q_{j+1} is a competitor for $P_{\theta+\epsilon_j h}[a_j]$. Hence $q_{j+1} \leq q_j$.

The inequality $a_j \geq u$ gives $u \preceq_{\mathcal{I}} a_j$. For every local divisor E, Lemma 1.17.7 gives

$$\nu_E(a_j) = \frac{1}{m_j} \operatorname{ord}_E \mathcal{I}(m_j u) \longrightarrow \nu_E(u).$$

By Proposition 15.5.1, $q_j \to p := P_\theta[u]_{\mathcal{I}}$ in L^1 and $p \leq q_j \leq 0$. The decreasing limit is a genuine θ-plurisubharmonic function and agrees almost everywhere with p, hence everywhere. This identifies the limit solely from θ and the ideal class of u, proving every asserted independence. □

Corollary 15.5.2 *For the sequence above, the following are equivalent:*

$$P_\theta[u] = P_\theta[u]_{\mathcal{I}}, \qquad q_j \longrightarrow P_\theta[u] \text{ in } L^1, \qquad q_j \searrow P_\theta[u] \text{ pointwise}.$$

They hold for one admissible choice of approximation if and only if they hold for every such choice. For the fixed volume $\mathrm{d}V$*,*

$$\lim_j \int_X (q_j - P_\theta[u])\,\mathrm{d}V = \int_X (P_\theta[u]_{\mathcal{I}} - P_\theta[u])\,\mathrm{d}V \geq 0,$$

and equality to zero is another equivalent condition.

Proof The theorem identifies both limits, and its L^1 convergence gives the integral formula. By Proposition 15.2.1, the limiting difference is nonnegative. Its integral vanishes precisely when the two envelopes agree almost everywhere, hence everywhere by Lemma 1.3.3. This proves all the equivalences. □

15.5.2 Absorbing the curvature error

When $\theta = \omega$ is Hermitian, choose $h = \omega$. Dividing the approximants by $1 + \epsilon_j$ then absorbs the curvature error. Rational error bounds preserve the analytic coefficients, so the entire approximation lies in $\mathrm{PSH}(X, \omega)$.

Corollary 15.5.3 *Let ω be Hermitian and let $u \in \mathrm{PSH}(X, \omega)$. There are analytic $b_j \in \mathrm{PSH}(X, \omega)$ with $b_j \to u$ in L^1, with decreasing ordinary singularity types, such that*

$$P_\omega[b_j] = P_\omega[b_j]_{\mathcal{I}} \searrow P_\omega[u]_{\mathcal{I}}.$$

Consequently analytic singularity classes are dense in the entire ideal-type metric space of Proposition 15.4.1. The equality $P_\omega[u] = P_\omega[u]_{\mathcal{I}}$ is equivalent to convergence of these ordinary envelopes to $P_\omega[u]$ in L^1.

Proof Use $h = \omega$ in (15.3). Enlarge the error bounds to positive rational numbers $\epsilon_j \downarrow 0$, and set

$$c_j = \frac{1}{1 + \epsilon_j}, \qquad b_j = c_j a_j.$$

Then $c_j \uparrow 1$ is rational and

$$\omega + \mathrm{dd^c} b_j = c_j\big((1 + \epsilon_j)\omega + \mathrm{dd^c} a_j\big) \geq 0.$$

The analytic coefficient is c_j/m_j. Also $b_j \to u$ in L^1, since $a_j \to u$ there and the sequence has bounded L^1 norm. Because $a_{j+1} \preceq a_j$, $c_{j+1} \geq c_j$, and each a_j is bounded above, one has $b_{j+1} \preceq b_j$.

Put $q_j = P_{(1+\epsilon_j)\omega}[a_j] = P_{(1+\epsilon_j)\omega}[a_j]_{\mathcal{I}}$. The preceding theorem gives $q_j \searrow P_\omega[u]_{\mathcal{I}}$ pointwise and in L^1. Since b_j is analytic, Proposition 15.2.3 and the homogeneity in Proposition 15.2.2 give

$$r_j := P_\omega[b_j] = P_\omega[b_j]_{\mathcal{I}} = c_j q_j.$$

Here $q_j \le 0$ decreases and $c_j \uparrow 1$, so r_j also decreases, with pointwise limit $P_\omega[u]_{\mathcal{I}}$. Moreover

$$P_\omega[u]_{\mathcal{I}} \le q_j \le r_j \le 0.$$

The genuine lower bound proves L^1 convergence by dominated convergence.

The metric distance from $[b_j]_{\mathcal{I}}$ to $[u]_{\mathcal{I}}$ is exactly the L^1 distance from r_j to $P_\omega[u]_{\mathcal{I}}$, so it tends to zero. The last assertion follows from the identified limit. □

15.6 A topology on ordinary singularity types

The two envelopes retain different parts of a singularity. Requiring both to converge makes their equality a closed condition. Controlled ideal inclusions and total variation of divisorial orders supply additional coordinates that the dyadic approximation can satisfy in a fixed Hermitian background. Together they give an analytic-closure criterion on the whole space of ordinary types.

Fix a Hermitian form ω on X and write

$$\mathfrak{S}(X,\omega) = \mathrm{PSH}(X,\omega)/\sim.$$

In this section $[u]$ denotes the ordinary bounded-difference class. For each smooth compact modification $\pi : Y \to X$, fix a Hermitian form γ_π on Y. For a prime divisor $E \subset Y$, set $w_E = \int_E \gamma_\pi^{n-1}$ and use the generic number $\nu_E(u) = \nu(u \circ \pi, E)$ of Section 1.17.

Lemma 15.6.1 *For $u, v \in \mathrm{PSH}(X,\omega)$, the sum*

$$d_\pi([u],[v]) = \sum_{E \subset Y \text{ prime}} w_E |\nu_E(u) - \nu_E(v)| \tag{15.4}$$

is finite and defines a pseudometric on $\mathfrak{S}(X,\omega)$. Changing γ_π gives an equivalent pseudometric on this fixed model.

Proof The current $T = \pi^*\omega + \mathrm{dd^c}(u \circ \pi)$ is positive. Divisorial removal in Lemma 1.17.1 gives, for every finite family of prime divisors,

$$\sum_E w_E \nu_E(u) \le \int_Y T \wedge \gamma_\pi^{n-1} < \infty.$$

Taking the supremum over finite families proves summability and countability of the positive coefficients. Apply the same bound to v. Bounded changes preserve

the coefficients, and the triangle inequality holds term by term. On compact Y, two Hermitian forms bound one another by fixed positive multiples, so their divisor weights do also. □

For $t > 0$ and $a > 1$, let

$$\mathcal{U}(t,a) = \{([u],[v]) : \mathcal{I}(atu) \subseteq \mathcal{I}(tv), \quad \mathcal{I}(atv) \subseteq \mathcal{I}(tu)\}. \tag{15.5}$$

These are relations between ordinary classes, since a bounded change leaves each ideal unchanged. Let d_I be the pullback of the metric in (15.2) under $[u] \mapsto [u]_{\mathcal{I}}$, with background ω, and put

$$d_O([u],[v]) = \|P_\omega[u] - P_\omega[v]\|_{L^1(\mathrm{d}V)}.$$

Both are pseudometrics on ordinary classes. The distance d_I retains the ideal envelope, while d_O also detects ordinary-envelope differences within a single ideal class.

Proposition 15.6.1 *Finite intersections of the relations $d_O < r$, $d_I < r$, $d_\pi < r$ and $\mathcal{U}(t,a)$, with $r, t > 0$ and $a > 1$, form a base of entourages for a uniformity on $\mathfrak{S}(X,\omega)$. Denote its topology by $\tau_{\omega,0}$. A net $[u_j]$ converges to $[u]$ precisely when both envelopes converge in L^1, $d_\pi([u_j],[u]) \to 0$ for every fixed π, and $([u_j],[u]) \in \mathcal{U}(t,a)$ eventually for every fixed t, a. Smooth positive choices of* $\mathrm{d}V$ *and of the γ_π do not change the uniformity.*

Proof All generators are symmetric and contain the diagonal. Multiplier ideals decrease with the weight: locally subtract an upper bound from a potential to see this from integrability. For $b = \sqrt{a}$, put $\mathcal{V} = \mathcal{U}(t,b) \cap \mathcal{U}(bt,b)$. If $([x],[y]), ([y],[z]) \in \mathcal{V}$, then

$$\mathcal{I}(atx) \subseteq \mathcal{I}(bty) \subseteq \mathcal{I}(tz),$$

and the same argument with x, z exchanged proves $\mathcal{V} \circ \mathcal{V} \subseteq \mathcal{U}(t,a)$. Halving a pseudometric radius gives the corresponding composition refinement. A finite intersection of these refinements proves the uniformity axiom. Its convergence criterion is exactly eventual membership in each generating entourage. Bounded ratios of smooth positive volumes and Lemma 15.6.1 prove independence of the auxiliary choices. □

Each entourage involves finitely many fixed models. The definition does not require a countable cofinal family of modifications or simultaneous uniform convergence over all models.

15.6.1 Analytic closure

Let $\mathcal{A}_\omega$ be the ordinary classes with analytic singularities in the rational-coefficient convention of Section 1.13. The curvature correction in Corollary 15.5.3 preserves this convention. The next theorem also retains the ideal weights affected by that correction.

Theorem 15.6.1 *The closure and sequential closure of $\mathcal{A}_\omega$ in $\tau_{\omega,0}$ are both*

$$\mathcal{G}_\omega = \{[u] : P_\omega[u] = P_\omega[u]_{\mathcal{I}}\}. \tag{15.6}$$

For every $u \in \mathrm{PSH}(X, \omega)$, the dyadic sequence b_j of Corollary 15.5.3 satisfies

$$[b_j] \longrightarrow [u] \text{ in } \tau_{\omega,0} \quad \Longleftrightarrow \quad [u] \in \mathcal{G}_\omega.$$

The approximants may be chosen with one positive rational analytic coefficient on all charts.

Proof Use the notation there: $m_j = 2^j$, a_j has type $(1/m_j, \mathcal{I}(m_j u))$, $c_j = (1 + \epsilon_j)^{-1} \uparrow 1$ is rational, and $b_j = c_j a_j$. Its ordinary and ideal envelopes agree and decrease to $P_\omega[u]_{\mathcal{I}}$. Thus the two envelope coordinates converge to those of u precisely when $[u] \in \mathcal{G}_\omega$.

Fix $t > 0$, $a > 1$, and choose $1 < s < a$. For large j, $ac_j > s$ and $m_j \geq st/(s-1)$. By Lemma 1.14.1,

$$\mathcal{I}(atb_j) = \mathcal{I}(atc_j a_j) \subseteq \mathcal{I}(sta_j) \subseteq \mathcal{I}(tu).$$

Since $u \preceq a_j \preceq b_j$, the reverse relation $\mathcal{I}(atu) \subseteq \mathcal{I}(tb_j)$ holds for every j. The scale c_j belongs inside the first ideal weight.

For a divisor E on a fixed π, the Bergman divisorial estimate gives

$$c_j \max\left\{\nu_E(u) - \frac{A(E)}{m_j}, 0\right\} \leq \nu_E(b_j) \leq c_j \nu_E(u) \leq \nu_E(u).$$

Here $A(E)$ is the log discrepancy, computed over a coordinate neighborhood of its center. Each coefficient converges to $\nu_E(u)$. The summability in Lemma 15.6.1 therefore gives $d_\pi([b_j], [u]) \to 0$ by dominated convergence. This proves every remaining coordinate of convergence.

Finally analytic classes have equal envelopes by Proposition 15.2.3. The map $[u] \mapsto (P_\omega[u], P_\omega[u]_{\mathcal{I}})$ is continuous into $L^1 \times L^1$, whose diagonal is closed. Thus any limit of analytic nets lies in $\mathcal{G}_\omega$. The constructed sequence gives the converse and the sequential assertion. □

15.6.2 Separation and mass

The topology is defined before identifying classes with their ordinary envelopes. Those envelopes can coincide for distinct ordinary types.

Proposition 15.6.2 *Two classes have the same $\tau_{\omega,0}$ neighborhoods if and only if their ordinary envelopes coincide. This topology is not T_0 and cannot be induced by a metric on $\mathfrak{S}(X,\omega)$.*

Proof The d_O coordinate distinguishes different ordinary envelopes. Conversely, equal ordinary envelopes have equal multiplier ideals at every weight by Proposition 15.2.1. Their ideal envelopes and all divisorial coefficients therefore agree, so every generating entourage has identical sections at the two classes.

To produce distinct such classes, choose a cutoff logarithmic pole $w \le 0$ supported in a coordinate ball, with w plurisubharmonic near its pole and smooth on the cutoff annulus. A small positive multiple makes $w \in \mathrm{PSH}(X,\omega)$. Put $\chi(s) = 1 - \sqrt{1-s}$ for $s \le 0$ and $u = \chi(w)$. The inequalities $0 < \chi' \le 1/2$ and $\chi'' \ge 0$, applied first to bounded truncations, give $u \in \mathrm{PSH}(X,\omega)$. In local coordinates z centered at the pole, u has magnitude comparable to $\sqrt{-\log|z|^2}$, so it is unbounded and e^{-tu} is integrable for every $t > 0$. All its multiplier ideals are trivial and $P_\omega[u]_{\mathcal{I}} = 0$.

For every $0 < \delta \le 1$, the function $\delta s - \chi(s)$ is bounded above on $(-\infty, 0]$. Hence $\delta w \preceq u$ and δw is an ordinary-envelope competitor. Letting $\delta \downarrow 0$ off the pole gives $P_\omega[u] = 0$. Thus $[u] \ne [0]$ have the same neighborhoods. □

The extended measures of Theorem 6.1.1 give the two envelope masses

$$\int_X \mathrm{MA}_\omega(P_\omega[u]), \qquad \int_X \mathrm{MA}_\omega(P_\omega[u]_{\mathcal{I}}).$$

In the present positive background, both are at most $\int_X \omega^n$. For the ideal envelope this follows from Propositions 12.1.4 and 15.2.1. For the ordinary envelope, normalize $u \le 0$ and put $p_j = P_\omega(u + j, 0)$ and $p = P_\omega[u]$. Then $p_j \to p$ in capacity by Lemma 1.8.6. For $k > 0$, set $\eta_k = \min\{1, (u+k)_+\}$. If $j > k$, the contact with $u + j$ lies outside $\{\eta_k > 0\}$, so Proposition 12.1.3 gives $\eta_k \mathrm{MA}_\omega(p_j) \le \eta_k \omega^n$. Test against a nonnegative continuous function and apply Lemma 6.3.1 to obtain the same inequality for p. Letting $k \to \infty$ proves $\mathrm{MA}_\omega(p) \le \omega^n$, since the remaining set $\{u = -\infty\}$ is pluripolar.

The original representative can nevertheless have infinite mass: Proposition 15.3.3 gives such a u with both envelopes equal to zero. The topology and its analytic-closure criterion include this ordinary class. A mass condition on the representative is therefore unnecessary, while infinite canonical mass does not arise for these two envelopes in a positive Hermitian background.

15.6.3 Transport with an obstacle

For a smooth real obstacle F, use the ordinary envelope $P_{\omega,F}[u]$ of Definition 1.19.1, and write $P_{\omega,F}[u]_{\mathcal{I}}$ for the ideal envelope with $v \le F$ in place of $v \le 0$. Its competitor family contains a translate of u. Replacing a competitor by its maximum with this translate shows that the ideal envelope can equivalently use $v \preceq_{\mathcal{I}} u$. The same notation

applies to any smooth real background admitting u. Define $\tau_{\omega,F}$ by replacing the two envelope coordinates in Proposition 15.6.1 by these obstacle envelopes.

Proposition 15.6.3 *Let $\chi \in C^\infty(X,\mathbb{R})$ and $\omega' = \omega + \mathrm{dd^c}\chi > 0$. The map $[u] \mapsto [u-\chi]$ is a uniform isomorphism*

$$(\mathfrak{S}(X,\omega),\tau_{\omega,F}) \longrightarrow (\mathfrak{S}(X,\omega'),\tau_{\omega',F-\chi}).$$

For $s > 0$, $[u] \mapsto [su]$ is a uniform isomorphism from $\tau_{\omega,F}$ to $\tau_{s\omega,sF}$. Both assertions transport the ordinary and ideal envelopes exactly.

Proof For either envelope operator P, its competitors give

$$P_{\omega',F-\chi}[u-\chi] = P_{\omega,F}[u] - \chi, \qquad P_{s\omega,sF}[su] = sP_{\omega,F}[u].$$

Smooth additions leave all ideals and divisorial numbers unchanged. Scaling multiplies the divisorial and envelope distances by s and reindexes the ideal weights by $t \mapsto st$. Each map and its inverse therefore transport the generating entourages. □

For two smooth obstacles, translation of competitors gives

$$P_{\omega,F}[u] + \min_X(G-F) \le P_{\omega,G}[u] \le P_{\omega,F}[u] + \max_X(G-F),$$

with the same inequalities for ideal envelopes. The associated extended measures satisfy $\mathrm{MA}_{\omega'}(u-\chi) = \mathrm{MA}_\omega(u)$ and $\mathrm{MA}_{s\omega}(su) = s^n\mathrm{MA}_\omega(u)$ by the local construction in Theorem 6.1.1, including at infinite total mass. Keeping the zero obstacle fixed in both backgrounds instead gives $P_{\omega',0}[u-\chi] = P_{\omega,\chi}[u] - \chi$. The bounded comparison of the obstacles does not give continuity of this change.

Proposition 15.6.4 *Let Y be a compact connected Riemann surface and $\theta > 0$ a smooth form of mass one. If F is a nonconstant smooth function with $\theta' = \theta + \mathrm{dd^c}F > 0$, the bijection*

$$\Gamma_F : (\mathfrak{S}(Y,\theta),\tau_{\theta,0}) \longrightarrow (\mathfrak{S}(Y,\theta'),\tau_{\theta',0}), \qquad [u] \longmapsto [u-F],$$

and its inverse are both discontinuous. Equivalently, $\tau_{\theta,0}$ and $\tau_{\theta,F}$ are incomparable topologies on the same ordinary-type space.

Proof The smooth function θ'/θ has mean one and is not constant, since a harmonic function on Y is constant. Choose $\min_Y(\theta'/\theta) < c < 1$, so $\theta' \not\ge c\theta$.

Choose nonatomic polar probability measures $\mu_j \rightharpoonup \theta$. To construct them, start with finite atomic probabilities converging weakly to θ as in Theorem 15.4.1. Replace each atom by a probability supported on a polar Cantor set in an arbitrarily small coordinate disc about that atom, preserving its weight. The construction in Example 29.12.2 supplies such nonatomic probabilities in every disc. Choose the discs with diameters tending uniformly to zero. Uniform continuity of continuous test functions preserves weak convergence, and a finite union of their polar carriers is polar.

Let q_j be the normalized Green potentials with $\theta + \mathrm{dd^c} q_j = \mu_j$ and $\sup_Y q_j = 0$. Normalized compactness gives L^1 cluster limits, and every such limit has curvature θ and supremum zero. It is therefore zero, so $q_j \to 0$ in L^1. Put $u_j = cq_j$. Then

$$\theta + \mathrm{dd^c} u_j = (1-c)\theta + c\mu_j, \qquad u_j \le P_\theta[u_j] \le 0.$$

All point Lelong numbers vanish. Since modifications of a smooth curve are isomorphisms, Theorem 1.16.1 gives trivial multiplier ideals at every weight. Thus $P_\theta[u_j]_{\mathcal{I}} = 0$, all controlled ideal relations hold, and every divisorial distance to zero vanishes. The displayed bounds give ordinary-envelope convergence in L^1. Consequently $[u_j] \to [0]$ in $\tau_{\theta,0}$.

Set $v_j = P_{\theta,F}[u_j]$. For every competitor $h \preceq u_j$, Corollary 1.6.1 gives $\theta + \mathrm{dd^c} h \ge c\mu_j$. The family below F contains a translate of u_j and is stable under finite maxima. Choquet selection and distributional convergence of its increasing maxima retain this inequality at v_j. If $[u_j] \to [0]$ in $\tau_{\theta,F}$, its ordinary-envelope coordinate would give $v_j \to P_{\theta,F}[0] = F$ in L^1, because F is a bounded θ-plurisubharmonic competitor. Passing the current inequality to the limit would give $\theta' \ge c\theta$, a contradiction.

The exact paired transport in Proposition 15.6.3 identifies this failure with discontinuity of Γ_F. Apply the same argument to θ' and $-F$ to prove discontinuity of its inverse. Under paired transport these are precisely the two directions of the identity between $\tau_{\theta,0}$ and $\tau_{\theta,F}$, proving incomparability. □

The sequence has trivial ideals and polar curvature $c\mu_j$. By Lemma 29.12.1 and the bounded obstacle comparison, its ordinary envelopes in both backgrounds have non-pluripolar mass $1-c$. The discontinuity therefore persists inside a positive-mass cutoff. On the same curve, the complete polar-variation metric of Theorem 29.12.1 is invariant under smooth background changes after ordinary projection; it retains the diffuse curvature that the sequence above loses.

15.6.4 All smooth obstacles

Return to the Hermitian background ω on X. The preceding curve example shows that a smooth change of background need not preserve convergence when the obstacle is kept at zero. Requiring convergence for every smooth obstacle makes the collection stable under such changes. The same dyadic approximation still determines its analytic closure, now by testing envelope equality at every obstacle.

Let $\tau_\omega^{\mathrm{all}}$ be the uniform topology generated by all the uniformities $\tau_{\omega,F}$, with $F \in C^\infty(X,\mathbb{R})$. Thus an entourage uses finitely many obstacle-envelope distances, together with finitely many of the ideal relations and divisorial distances already defined.

Proposition 15.6.5 *A countable uniformly dense family $\{F_k\}_{k\geq 1} \subset C^\infty(X,\mathbb{R})$ suffices for the obstacle coordinates of $\tau^{\mathrm{all}}_\omega$. For χ smooth with $\omega' = \omega + \mathrm{dd^c}\chi > 0$, the map $[u] \mapsto [u-\chi]$ is a uniform isomorphism between $\tau^{\mathrm{all}}_\omega$ and $\tau^{\mathrm{all}}_{\omega'}$. For $s > 0$, positive scaling gives the corresponding uniform isomorphism to $\tau^{\mathrm{all}}_{s\omega}$. This topology is not T_0 on ordinary classes.*

Proof For either envelope operator, the obstacle comparison gives

$$\|P_{\omega,F}[u] - P_{\omega,G}[u]\|_{L^1(\mathrm{d}V)} \leq \|F-G\|_\infty.$$

The difference between the two envelope distances for a pair $[u],[v]$ is therefore at most $2\|F-G\|_\infty$, uniformly in the pair. Smooth functions contain a countable dense subset for the uniform norm: approximate each member of a countable dense subset of $C(X)$ by smooth functions with errors $1/m$, $m \geq 1$. Approximating each of the finitely many obstacles in an entourage and then shrinking its radii proves equality of the uniformities. The divisorial models are still allowed to range over every fixed compact modification.

The exact transport in Proposition 15.6.3 reindexes the entire smooth-obstacle family by $F \mapsto F-\chi$, or by $F \mapsto sF$. These are bijections of that family, proving both isomorphisms.

For nonseparation, use the unbounded $u = 1-\sqrt{1-w}$ in Proposition 15.6.2. It has trivial ideals and zero divisorial numbers, and $\delta w \preceq u$ for $0 < \delta \leq 1$. Fix F and put $q = P_\omega(F)$. Since $\omega > 0$, the constant $\min_X F$ is a competitor, so q is bounded and $q \leq F$. The function

$$h_\delta = (1-\delta)q + \delta w - \delta\|F\|_\infty$$

is ω-plurisubharmonic, lies below F, and satisfies $h_\delta \preceq u$. Letting $\delta \downarrow 0$ off the pole of w gives $P_{\omega,F}[u] = q = P_{\omega,F}[0]$. Trivial ideals give the same equality for the ideal envelopes. Every generating entourage therefore has identical sections at $[u]$ and $[0]$, although these ordinary classes differ. □

The ideal envelope below F is the ordinary envelope below F of its zero-obstacle ideal representative. This identity isolates the remaining ordinary-projection question.

Proposition 15.6.6 *Let θ be a smooth real $(1,1)$-form, let $u \in \mathrm{PSH}(X,\theta)$ and put $q = P_\theta[u]_{\mathcal{I}}$. For every smooth real F,*

$$P_{\theta,F}[q] = P_{\theta,F}[u]_{\mathcal{I}} \sim q.$$

If $u \in \mathcal{E}_{\mathrm{env}}(X,\theta,q)$, then also

$$P_{\theta,F}[u] = P_{\theta,F}[u]_{\mathcal{I}} \quad \textit{for every smooth real } F.$$

Proof Set $q_F = P_{\theta,F}[u]_{\mathcal{I}}$. Translation of competitors gives

$$q + \min_X F \leq q_F \leq q + \max_X F,$$

so $q_F \sim q$. The identity $q_F - F = P_{\theta+\mathrm{dd^c}F}[u - F]_{\mathcal{I}}$ and Theorem 15.2.1 give $q_F \sim_{\mathcal{I}} u$. Since $q_F \le F$ and $q_F \sim q$, it is an ordinary competitor for $P_{\theta,F}[q]$. Conversely, an ordinary competitor for q is admissible for its ideal envelope after taking its maximum with a sufficiently low translate of q. As $q \sim_{\mathcal{I}} u$, these bounds give

$$q_F \le P_{\theta,F}[q] \le P_{\theta,F}[q]_{\mathcal{I}} = q_F.$$

The last assertion follows from Proposition 12.4.3. □

Theorem 15.6.2 *The closure and sequential closure of $\mathcal{A}_\omega$ in $\tau_\omega^{\mathrm{all}}$ both equal*

$$\mathcal{G}_\omega^{\mathrm{all}} = \{[u] : P_{\omega,F}[u] = P_{\omega,F}[u]_{\mathcal{I}} \text{ for every smooth } F\}.$$

It suffices to test equality on the countable dense family above. The dyadic sequence b_j of Corollary 15.5.3 converges to $[u]$ in this topology if and only if $[u] \in \mathcal{G}_\omega^{\mathrm{all}}$.

Proof Fix a smooth obstacle F. Obstacle transport and Theorem 15.2.1, as in the preceding proof, give $P_{\omega,F}[u]_{\mathcal{I}} \sim_{\mathcal{I}} u$. For an analytic a, Lemma 1.17.9 makes this ideal envelope an ordinary competitor, so $P_{\omega,F}[a] = P_{\omega,F}[a]_{\mathcal{I}}$.

Write $p = P_{\omega,F}[u]_{\mathcal{I}}$ and $q_j = P_{\omega,F}[b_j]_{\mathcal{I}}$. The dyadic construction gives $u \preceq_{\mathcal{I}} b_j$, decreasing ordinary types, and convergence of every local divisorial Lelong number to that of u. Ideal order, proved by taking maxima with a translate of b_j below F, gives $p \le q_j \le F$. Analytic envelope equality and the decreasing types make q_j decrease. Its limit q is genuine and ω-plurisubharmonic, with $p \le q \le F$; the same bounds give convergence in L^1.

Pullback convergence and Lelong semicontinuity, as in Proposition 15.5.1, give $\nu_E(q) \ge \nu_E(u)$ for every local divisor E. The inequality $p \le q$ gives the reverse bound. The local ideal criterion yields $q \sim_{\mathcal{I}} u$. Since $q \le F$, it is admissible for p, proving

$$P_{\omega,F}[b_j] = P_{\omega,F}[b_j]_{\mathcal{I}} \searrow P_{\omega,F}[u]_{\mathcal{I}}.$$

The controlled ideal relations and all fixed-model divisorial distances converge by the proof of Theorem 15.6.1; they do not depend on F. This one dyadic sequence consequently converges in $\tau_\omega^{\mathrm{all}}$ exactly under the stated envelope equalities.

For each F, the equality of the two L^1 envelope coordinates is closed. Analytic classes belong to every such equality set, so their net closure lies in the displayed intersection. The constructed sequence proves the converse. Finally the uniform obstacle bound in the proof of Proposition 15.6.5 extends equality from a dense family to every smooth obstacle. □

The inclusion $\mathcal{G}_\omega^{\mathrm{all}} \subseteq \mathcal{G}_\omega$ follows by taking $F = 0$. For $q = P_\omega[u]_{\mathcal{I}}$, Proposition 15.6.6 reduces the reverse inclusion to $P_{\omega,F}[u] = P_{\omega,F}[q]$ for every F. Membership in $\mathcal{E}_{\mathrm{env}}(X, \omega, q)$ suffices. Under bounded mass and positive lower volume of q, this membership is equivalent to zero-obstacle equality, by Proposition 29.8.1. On curves the two closure sets coincide even at zero mass, by Proposition 29.12.3. The implication outside these regimes remains unresolved here. Although $\tau_\omega^{\mathrm{all}}$ removes the fixed-obstacle gauge dependence, it still has incomplete curve families

and remains strictly weaker than polar total variation on ordinary curve models, by Proposition 29.12.4.

15.6.5 Complete model topologies

On a curve, total variation of polar curvature gives a complete metric on ordinary models, and ideal projection takes the atomic part. The diffuse polar mass is exactly the distance to the analytic closure, by Corollary 29.12.1 and Proposition 29.12.1. This topology is stronger than $\tau_{\omega,0}$ on ordinary models: Proposition 29.12.2 exhibits the diffuse concentration lost by the latter.

For a smooth big background under bounded mass, Theorem 29.11.1 gives a complete topology on models of positive lower volume. Its ideal-model locus is the completion of the positive-volume analytic models by Corollary 29.11.1.

Problem 15.6.1 Extend the complete topology of Theorem 29.11.1 to ordinary models of zero lower volume and beyond bounded mass, retaining analytic closure exactly at the ideal models. Recover the curve topology of Corollary 29.12.1 and the background compatibility of Sections 29.9 and 29.10, including higher-dimensional limits that lose bigness or positive lower volume.

Part IV
Interpolation and restriction

Chapter 16
Perron interpolation and the geodesic equation

Let X be a connected compact complex manifold of dimension $n \geq 1$, and let θ be a smooth real $(1,1)$-form. Fix a smooth positive volume form $\mathrm{d}V$ for the $L^1(X)$ topology.

In Kähler geometry, the smooth geodesic equation for the L^2 metric on potentials is a homogeneous Monge–Ampère equation on a product. Product positivity also defines Perron interpolation for a nonclosed background, but the two constructions can differ: integration by parts in the action produces torsion terms. Even a smooth Perron segment given by constant translation can fail to be stationary for that action.

The Perron construction itself rests on convexity and barriers. Convexity bounds every admissible path by the endpoint chord, while a lower barrier ensures that the envelope attains the endpoints. For potentials of the same ordinary singularity type, two affine barriers suffice and give uniform Lipschitz control in time. Maximality on each finite subinterval will supply the link with Legendre duality in the next chapter.

16.1 Subgeodesics and endpoint bounds

For a nonempty open interval $I \subset \mathbb{R}$, write

$$S_I = \{z \in \mathbb{C} : \operatorname{Re} z \in I\}, \qquad \pi : X \times S_I \longrightarrow X.$$

A family $u_t \in \mathrm{PSH}(X,\theta)$, $t \in I$, is a *subgeodesic* if its complexification $U(x,z) = u_{\operatorname{Re} z}(x)$ is quasi-plurisubharmonic and satisfies $\pi^*\theta + \mathrm{dd^c}U \geq 0$ on the product. For finite $a < b$ and $u_a, u_b \in \mathrm{PSH}(X,\theta)$, a subgeodesic on (a,b) is said to run from u_a to u_b if it converges to them in $L^1(X)$ at the respective endpoints.

The elementary operations below construct lower barriers. Vertical convexity and the minimum principle from Section 1.20 then give the continuity and upper bounds needed for the Perron envelope.

Proposition 16.1.1 *Subgeodesics are preserved by adding an affine function of t, by a positive affine reparametrization of time, by finite maxima, and by convex combinations. For $h \in C^\infty(X, \mathbb{R})$, replacing θ by $\theta + \mathrm{dd^c} h$ and u_t by $u_t - h$ also preserves them.*

Proof Affine functions of $\operatorname{Re} z$ have zero complex Hessian, and a positive affine change of time is induced by a holomorphic change of strip coordinates. Convex combinations preserve the current inequality, and smooth gauge changes preserve the current itself. Maxima are covered by Lemma 1.3.1. □

Proposition 16.1.2 *A subgeodesic is continuous in $L^1(X)$ on the interior of its parameter interval. If $a < b$ are finite endpoints and its L^1 endpoint limits are $u_a, u_b \in \mathrm{PSH}(X, \theta)$, then*

$$u_t \le \frac{b-t}{b-a} u_a + \frac{t-a}{b-a} u_b, \qquad a < t < b. \tag{16.1}$$

Moreover the boundary upper limits hold jointly in space and time:

$$\limsup_{\substack{y \to x \\ t \downarrow a}} u_t(y) \le u_a(x), \qquad \limsup_{\substack{y \to x \\ t \uparrow b}} u_t(y) \le u_b(x).$$

Proof For a compact interval $J \Subset I$, choose $a' < \inf J \le \sup J < b'$ in I. By Lemma 1.20.1, the path lies below the chord through $u_{a'}, u_{b'}$, hence below a constant on J. The function $\inf_{a'<t<b'} u_t$ is an integrable lower bound by Proposition 1.20.1. For almost every x, the function $t \mapsto u_t(x)$ is finite and convex, hence continuous. Dominated convergence proves continuity in L^1 on J.

For the endpoint inequality, choose sequences approaching a, b along which the stipulated L^1 convergence holds almost everywhere. Apply convexity between these times and pass to the limit. For each t, this proves (16.1) almost everywhere in x, hence everywhere by Lemma 1.3.3.

For the joint limit at a, the coefficient of u_a tends to one and the term involving u_b has a vanishing coefficient and a uniform upper bound. Upper semicontinuity of u_a gives the assertion when $u_a(x)$ is finite. At a pole, bound u_a above near x by an arbitrary real number, take the joint limit, and let that number tend to $-\infty$. The argument at b is the same. □

16.2 Perron segments

The chord bound survives taking a regularized supremum. A single competitor attaining the endpoints will bound that supremum from below and force the same endpoint limits. Fix $u_0, u_1 \in \mathrm{PSH}(X, \theta)$ and call a subgeodesic on $(0, 1)$ *admissible* if its joint boundary upper limits are at most u_0, u_1, respectively. Define

$$U = (\sup\{V : V \text{ is an admissible complexification}\})^* .$$

The regularization is taken on $X \times S_{(0,1)}$. Denote its slices by u_t^{P}. When these form a subgeodesic attaining the prescribed endpoints in L^1, they define the *Perron segment*, with $u_0^{\mathrm{P}} = u_0$ and $u_1^{\mathrm{P}} = u_1$. Affine reparametrization gives the same construction on any finite interval.

Proposition 16.2.1 *If an admissible competitor exists, the Perron envelope is the largest admissible subgeodesic. If one competitor attains u_0, u_1 in L^1, so does the envelope. In that case, its restriction to each $[a, b] \subset [0, 1]$, $a < b$, is the Perron segment between its own endpoints.*

Proof Vertical convexity and the boundary inequalities bound every competitor by $(1 - t)u_0 + tu_1$. This bound is jointly upper semicontinuous on the interior and locally bounded above, so it holds after regularization. The local envelope theorem of Section 1.19 gives a product potential, still invariant in $\Im z$. It dominates an admissible competitor, so each slice belongs to $\mathrm{PSH}(X, \theta)$. The chord bound gives the joint boundary inequalities by the endpoint argument in Proposition 16.1.2. The envelope therefore belongs to its defining family and is its largest member.

If a competitor w_t has the required L^1 endpoints, then

$$w_t \le u_t^{\mathrm{P}} \le (1 - t)u_0 + tu_1.$$

In particular, almost everywhere,

$$|u_t^{\mathrm{P}} - u_0| \le |w_t - u_0| + t|u_1 - u_0|,$$

which proves $u_t^{\mathrm{P}} \to u_0$ in L^1 as $t \downarrow 0$. The argument at 1 is identical.

For $0 \le a < b \le 1$, let v_t be the Perron envelope between u_a^{P} and u_b^{P}. The original path restricted to (a, b) attains these endpoints and is admissible by Proposition 16.1.2. Thus v_t attains them and dominates u_t^{P}. Its joint boundary bounds permit gluing its complexification to U outside the smaller strip by Lemma 1.3.2. At an original endpoint, the same bounds preserve admissibility for u_0, u_1. The resulting competitor cannot exceed U, proving equality on (a, b). □

A bounded difference between the endpoints supplies the required lower barrier explicitly.

Theorem 16.2.1 *Suppose $u_0 \sim u_1$ and choose $C \geq 0$ with $u_0 - C \leq u_1 \leq u_0 + C$. The Perron segment exists and attains both endpoints. Every slice has their ordinary singularity type, and*

$$u_0 - Ct \leq u_t^{\mathrm{P}} \leq u_0 + Ct, \qquad u_1 - C(1-t) \leq u_t^{\mathrm{P}} \leq u_1 + C(1-t).$$

Away from the common pole set,

$$|u_t^{\mathrm{P}} - u_s^{\mathrm{P}}| \leq C|t-s|, \qquad 0 \leq s, t \leq 1.$$

Proof Consider the barrier

$$w_t = \max\{u_0 - Ct,\ u_1 - C(1-t)\}.$$

It is a subgeodesic by Proposition 16.1.1, and the bounded difference gives $w_0 = u_0, w_1 = u_1$. Upper semicontinuity gives the joint boundary upper limits. Off the common poles, $|w_t - u_0| \leq Ct$ and $|w_t - u_1| \leq C(1-t)$, so w_t attains both endpoints in L^1. Apply Proposition 16.2.1.

The lower bounds follow from $w_t \leq u_t^{\mathrm{P}}$ and the upper bounds from the chord estimate. They also give the asserted singularity type. Off the common poles, put $f(t) = u_t^{\mathrm{P}}(x)$. For $0 < s < t < 1$, monotonicity of convex secant slopes gives

$$-C \leq \frac{f(s) - f(0)}{s} \leq \frac{f(t) - f(s)}{t - s} \leq \frac{f(1) - f(t)}{1 - t} \leq C.$$

The endpoint estimates cover $s = 0$ and $t = 1$, proving the Lipschitz bound on $[0, 1]$.□

For endpoints of different types, a common lower potential still gives an admissible constant path. Endpoint attainment requires a barrier approaching the prescribed values; Proposition 16.2.1 separates this requirement from existence of the envelope.

16.3 The smooth product equation

For a smooth path, product positivity compares acceleration with the spatial gradient of the velocity. Let u_t be smooth in $(x,t) \in X \times I$, suppose $\eta_t = \theta + \mathrm{dd^c} u_t > 0$, and write, in local holomorphic coordinates $(x_1, \dots, x_n)$,

$$\theta = \frac{\mathrm{i}}{2\pi} \sum_{i,j=1}^{n} g_{i\bar{j}} \, \mathrm{d}x_i \wedge \mathrm{d}\bar{x}_j, \qquad H = (g_{i\bar{j}} + \partial_i \partial_{\bar{j}} u), \qquad \xi = (\partial_i \dot{u})_{i=1}^{n}.$$

Dots denote time derivatives, ξ is a column vector, and ξ^* is its conjugate transpose. Since $t = \operatorname{Re} z$, the product form $\pi^*\theta + \mathrm{dd^c} U$ has coefficient matrix

$$\begin{pmatrix} H & \frac{1}{2}\xi \\ \frac{1}{2}\xi^* & \frac{1}{4}\ddot{u} \end{pmatrix}.$$

Proposition 16.3.1 *The product form is semipositive exactly when*

$$\ddot{u} \geq \xi^* H^{-1} \xi.$$

Its $(n+1)$st power vanishes exactly when equality holds. Equivalently, the homogeneous product equation is

$$\ddot{u}\, \eta_t^n = n \, \mathrm{d}\dot{u} \wedge \mathrm{d^c}\dot{u} \wedge \eta_t^{n-1}. \tag{16.2}$$

Proof The Schur complement of H is $(\ddot{u} - \xi^* H^{-1} \xi)/4$. Since $H > 0$, the block matrix is semipositive exactly when this complement is nonnegative, and its determinant vanishes exactly when the complement vanishes.

The invariant equation (16.2) follows from

$$\mathrm{d}\dot{u} \wedge \mathrm{d^c}\dot{u} = \frac{\mathrm{i}}{2\pi} \partial \dot{u} \wedge \bar{\partial} \dot{u}, \qquad n \frac{\mathrm{d}\dot{u} \wedge \mathrm{d^c}\dot{u} \wedge \eta_t^{n-1}}{\eta_t^n} = \operatorname{tr}(H^{-1} \xi \xi^*) = \xi^* H^{-1} \xi.$$

This calculation identifies the smooth homogeneous product equation. A weak formulation and proof for nonsmooth Perron segments require a separate argument.

16.4 The L^2 action and torsion

The product equation uses the spatial form pointwise. In the L^2 action, integration by parts also differentiates its powers, producing terms absent from that equation. Fix a Hermitian form ω. On the space $\mathcal{H}_\omega$ of Chapter 2, consider the metric

$$G_u(v, w) = \int_X vw\, \omega_u^n, \qquad v, w \in C^\infty(X, \mathbb{R}).$$

For a smooth path $u_t \in \mathcal{H}_\omega$, $0 \le t \le 1$, its action is

$$\mathcal{S}(u) = \frac{1}{2} \int_0^1 \int_X \dot{u}_t^2\, \omega_{u_t}^n\, \mathrm{d}t.$$

Stationarity under smooth variations fixing both endpoints gives the formal geodesic equation for this metric.

Theorem 16.4.1 *Put $\eta_t = \omega_{u_t}$ and $T_t = \eta_t^{n-1}$. The path is stationary for $\mathcal{S}$ if and only if*

$$\begin{aligned} \ddot{u}_t\, \eta_t^n &= n\, \mathrm{d}\dot{u}_t \wedge \mathrm{d}^c \dot{u}_t \wedge T_t \\ &\quad + n\dot{u}_t \big(\mathrm{d}\dot{u}_t \wedge \mathrm{d}^c T_t - \mathrm{d}^c \dot{u}_t \wedge \mathrm{d}T_t\big) + \frac{n}{2} \dot{u}_t^2 \mathrm{dd}^c T_t. \end{aligned} \tag{16.3}$$

All differentials on the right act only in the X variable.

Proof Vary the path as $u_t + s v_t$, where v_t is smooth in (x, t) and $v_0 = v_1 = 0$, and put $f = \dot{u}_t$. Differentiating the action at $s = 0$ gives

$$\delta \mathcal{S} = \int_0^1 \left(\int_X f \dot{v}_t\, \eta_t^n + \frac{n}{2} \int_X f^2 \mathrm{dd}^c v_t \wedge T_t \right) \mathrm{d}t.$$

Integration in time uses $\partial_t(\eta_t^n) = n\mathrm{dd}^c f \wedge T_t$. Integration on X uses Lemma 2.1.1 with the entire form $f^2 T_t$. Thus

$$\delta \mathcal{S} = \int_0^1 \int_X v_t \left(-\ddot{u}_t \eta_t^n - n f \mathrm{dd}^c f \wedge T_t + \frac{n}{2} \mathrm{dd}^c (f^2 T_t) \right) \mathrm{d}t.$$

The full product rule is

$$\begin{aligned} \mathrm{dd}^c (f^2 T_t) &= 2 f \mathrm{dd}^c f \wedge T_t + 2 \mathrm{d}f \wedge \mathrm{d}^c f \wedge T_t \\ &\quad + 2 f \big(\mathrm{d}f \wedge \mathrm{d}^c T_t - \mathrm{d}^c f \wedge \mathrm{d}T_t\big) + f^2 \mathrm{dd}^c T_t. \end{aligned}$$

After cancellation with the volume derivative, the integrand is v_t times the difference between the right and left sides of (16.3). Positivity of η_t on $X \times [0, 1]$ permits every sufficiently small variation. The fundamental lemma of the calculus of variations gives both implications. □

For a Kähler background, $dT_t = d^c T_t = 0$ and the equation reduces to (16.2). The Gauduchon condition on η_t removes only the last term. Constant translations isolate that term and provide an exact test for stationarity.

Corollary 16.4.1 *Fix $u \in \mathcal{H}_\omega$ and a nonzero constant a. The path $u_t = u + at$ is a Perron segment and solves the smooth homogeneous product equation. It is stationary for the L^2 action if and only if*

$$dd^c(\omega_u^{n-1}) = 0.$$

For $n \geq 2$, this says precisely that ω_u is Gauduchon.

Proof The path is admissible and equals the chord between its endpoints, so no admissible competitor can be larger. It is therefore the Perron segment. Its time acceleration and the spatial derivatives of its velocity vanish, which proves the homogeneous product equation. In (16.3), the sole remaining term is $\frac{n}{2}a^2 dd^c(\omega_u^{n-1})$. This vanishes exactly under the displayed condition. □

The three-dimensional torus in Example 4.5.1 carries a Gauduchon form ω and a smooth potential $u \in \mathcal{H}_\omega$ for which ω_u is not Gauduchon. Thus $u_t = u + t$ is a smooth Perron segment that is not stationary for the L^2 action. On a surface, the Gauduchon condition persists under potential perturbations by Proposition 4.5.2, but it does not cancel the first-derivative terms in (16.3). Relating Perron interpolation to a complete metric on Hermitian singularity classes remains a separate problem.

Chapter 17
Test curves and Legendre duality

Let X be a connected compact complex manifold of dimension $n \geq 1$, and let θ be a smooth real $(1,1)$-form. Fix $\varphi \in \mathrm{PSH}(X,\theta)$ with $P_\theta[\varphi] = \varphi$; thus $\varphi \leq 0$ and $\sup_X \varphi = 0$. Use a fixed smooth positive volume form for the $L^1(X)$ topology.

Legendre duality exchanges two maximality conditions. Each member of a test curve is fixed by its singularity envelope, while every finite segment of the corresponding ray is a Perron envelope. Replacing such a segment changes each inverse transform only by a bounded amount, so the envelope condition forces Perron maximality. Affine rooftop comparisons prove the converse. This argument uses convexity, the minimum principle and the Perron construction of Chapter 16, and applies to nonclosed backgrounds as well as to Kähler forms.

We use the pointwise transforms of Section 1.20.1; the proofs follow the envelope arguments in [Xia, section on the Ross–Witt Nyström correspondence].

17.1 Envelope test curves and rays

The parameter of a test curve measures increasing singularity. Its cutoff becomes the slope of the transformed ray, while its limit towards $-\infty$ gives the initial potential.

Definition 17.1.1 An *envelope test curve* with initial potential φ consists of a real number c and potentials $\Gamma_\tau \in \mathrm{PSH}(X,\theta)$ for $\tau < c$ such that

(1) $\tau \mapsto \Gamma_\tau$ is pointwise concave and decreasing;
(2) $P_\theta[\Gamma_\tau] = \Gamma_\tau$ for every $\tau < c$;
(3) $(\sup_{\tau<c} \Gamma_\tau)^* = \varphi$.

Its cutoff is c. We extend it by

$$\Gamma_c = \lim_{\tau \uparrow c} \Gamma_\tau, \qquad \Gamma_\tau \equiv -\infty \quad (\tau > c).$$

Each Γ_τ has supremum zero, so Theorem 1.4.1 and Corollary 1.4.1 give $\Gamma_c \in \mathrm{PSH}(X,\theta)$.

Definition 17.1.2 An *envelope ray* with initial potential φ and slope c is a subgeodesic u_t, $t > 0$, such that

(1) $u_t \to \varphi$ in $L^1(X)$ as $t \downarrow 0$, and every finite restriction, with $u_0 = \varphi$, is a Perron segment;
(2) $P_\theta[u_t] = \varphi$ for every $t > 0$;
(3) $\sup_X u_t = ct$ for every $t > 0$.

Write U_t for the forward transform (1.25) and $\widehat{u}_\tau$ for the inverse transform (1.26). The forward transform is already a subgeodesic by Theorem 1.20.1; its finite restrictions must still be shown to be Perron segments.

17.2 Perron maximality

The envelope normalization gives the transformed ray's slope and singularity envelope directly. To prove Perron maximality, replace the transform on a finite interval by its Perron segment. The estimate (17.1) keeps each inverse transform in its original ordinary singularity type, where the envelope condition forces equality.

Theorem 17.2.1 *The forward transform of an envelope test curve is an envelope ray with the same initial potential and cutoff slope. Moreover $U_t - ct$ decreases as t increases.*

Proof By Theorem 1.20.1, U_t is a subgeodesic. The inequalities

$$\Gamma_\tau + t\tau \le U_t \le \varphi + ct$$

give $\sup_X U_t \ge t\tau$ for each $\tau < c$ and $\sup_X U_t \le ct$. Letting $\tau \uparrow c$ proves equality. The formula

$$U_t - ct = \sup_{\tau<c}\big(\Gamma_\tau - t(c-\tau)\big)$$

proves decreasingness.

The order property of the singularity envelope gives

$$\Gamma_\tau = P_\theta[\Gamma_\tau] \le P_\theta[U_t] \le P_\theta[\varphi] = \varphi.$$

Taking the regularized supremum in τ proves $P_\theta[U_t] = \varphi$.

The local envelope theorem in Section 1.19 gives $\sup_{\tau<c}\Gamma_\tau = \varphi$ almost everywhere. At those points, the preceding bounds imply $U_t \to \varphi$ as $t \downarrow 0$. For any fixed $\tau_0 < c$ and $0 < t \le 1$, they also give

$$\Gamma_{\tau_0} - |\tau_0| \le U_t \le \varphi + |c|.$$

Both bounds are integrable, so dominated convergence proves the initial L^1 limit.

Fix $0 \le a < b$ and replace U_t on (a, b) by the Perron segment between U_a, U_b, using $U_0 = \varphi$ when $a = 0$. The restricted original path attains these endpoints, so

Proposition 16.2.1 gives the replacement. Its joint boundary bounds permit gluing by Lemma 1.3.2 and Proposition 16.1.2. The resulting subgeodesic v_t dominates U_t and equals it outside (a, b). The endpoint chord on the replaced interval gives $v_t \le ct$ everywhere.

For $\tau < c$, put $\chi_\tau = \inf_{t>0}(v_t - t\tau)$. The lower bound by Γ_τ and Proposition 1.20.1 give $\chi_\tau \in \mathrm{PSH}(X, \theta)$. We claim that

$$\Gamma_\tau \le \chi_\tau \le \Gamma_\tau + (b - a)(c - \tau). \tag{17.1}$$

Indeed for $s > 0$ with $s \in [a, b]$, the decreasingness of $U_t - ct$ gives

$$\chi_\tau \le U_b - b\tau \le U_s - s\tau + (b - s)(c - \tau).$$

For s outside the interval, use $v_s = U_s$ to obtain the same bound with $(b - a)(c - \tau)$. Taking the infimum over $s > 0$ and using Lemma 1.20.3 proves (17.1).

Moreover $\chi_\tau \le t(c - \tau)$ for every $t > 0$, hence $\chi_\tau \le 0$. By (17.1), it has the ordinary type of Γ_τ and is therefore a competitor for $P_\theta[\Gamma_\tau] = \Gamma_\tau$. Thus $\chi_\tau = \Gamma_\tau$. Recovering v_t by Lemma 1.20.2 gives $v_t = U_t$, proving Perron maximality on every finite interval. □

17.3 Recovering the singularity envelopes

The minimum principle produces the inverse potentials, but their envelope property uses the whole ray. A rooftop under a translated ray is an admissible affine path on every interval $[0, M]$. Letting the interval length vary, and then shrinking the relative time, removes the rooftop's additive constant.

Theorem 17.3.1 *If u_t is an envelope ray of slope c, its inverse transform is an envelope test curve with cutoff c and initial potential φ. The forward and inverse transforms are mutually inverse bijections between the two classes in Definitions 17.1.1 and 17.1.2.*

Proof By Lemma 1.20.1, each vertical function $u_t(x) - ct$ is finite and convex, or identically $-\infty$. The upper bound by zero makes it decreasing. Since $\sup_X(u_t - ct) = 0$, Theorem 1.4.1 and Corollary 1.4.1 give

$$\widehat{u}_c = \inf_{t>0}(u_t - ct) \in \mathrm{PSH}(X, \theta).$$

For $\tau < c$, we have $\widehat{u}_c \le \widehat{u}_\tau \le u_1 - \tau$, so Proposition 1.20.1 gives $\widehat{u}_\tau \in \mathrm{PSH}(X, \theta)$. Also

$$\widehat{u}_\tau \le \inf_{t>0} t(c - \tau) = 0 \quad (\tau \le c),$$

while the same upper bound with $t \to \infty$ shows $\widehat{u}_\tau \equiv -\infty$ for $\tau > c$. Concavity and decreasingness in τ follow by taking an infimum of affine functions with slopes $-t$.

Fix $\tau < c$ and write $g = \widehat{u}_\tau$. The inequality $g \le u_M - M\tau$, valid for every $M > 0$, gives

$$P_\theta[g] \le P_\theta[u_M] = \varphi.$$

For $C > 0$ put $q_C = P_\theta(g + C, 0)$. It is genuine since $g \le 0$ is a competitor, and

$$g \le q_C \le P_\theta[g] \le \varphi, \qquad q_C - C \le g \le u_M - M\tau.$$

The affine path $q_C - Ct/M$ is admissible for $\varphi, u_M - M\tau$ on $[0, M]$. Adding $-t\tau$ gives a bijection of Perron competitor families by Proposition 16.1.1, so the Perron segment for these endpoints is $u_t - t\tau$. For $0 < s \le 1$ this yields

$$q_C - Cs \le u_{sM} - sM\tau.$$

Take the infimum in $M > 0$ to get $q_C - Cs \le g$, and then let $s \downarrow 0$. Thus $q_C = g$ for every $C > 0$. Taking its increasing envelope limit proves $P_\theta[g] = g$.

Let $\psi = (\sup_{\tau<c} \widehat{u}_\tau)^* \le \varphi$. Monotonicity of $u_t - ct$ gives a pointwise limit at zero; an almost-everywhere convergent sequence from the L^1 endpoint condition identifies it with φ almost everywhere. At these points, the endpoint identity in Lemma 1.20.2 gives

$$\sup_{\tau<c} \inf_{t>0} (u_t(x) - t\tau) = \varphi(x).$$

Therefore $\psi = \varphi$ by Lemma 1.3.3.

As $\tau \uparrow c$, the transforms decrease to $\widehat{u}_c$: the infima over $t > 0$ and $\tau < c$ commute. This gives the required cutoff convention. Finally, Lemma 1.20.2 recovers u_t from its inverse transforms, including at identically $-\infty$ vertical functions, and Lemma 1.20.3 recovers each test curve from its forward transform. Hence the correspondence is a bijection. □

17.4 A logarithmic pole on the projective line

Set $X = \mathbb{P}^1$ and $\varphi = 0$. The envelope of a logarithmic pole is obtained by replacing part of a convex function by a tangent line. Its transform is a smooth ray whose bounded finite slices retain the pole through their long-time behavior. In the affine coordinate z, take

$$\theta = \mathrm{dd^c} \log(1 + |z|^2), \qquad g(z) = \log \frac{|z|^2}{1 + |z|^2}.$$

The form extends to $\mathbb{P}^1$ and has integral one. The potential g has a logarithmic pole at zero and extends smoothly across infinity, with value zero there.

Put $H(\lambda) = -\lambda \log \lambda - (1-\lambda)\log(1-\lambda)$ for $0 < \lambda < 1$, with $H(0) = H(1) = 0$. For $0 < \lambda < 1$, define

$$\Gamma_\lambda(z) = \begin{cases} \lambda \log |z|^2 - \log(1+|z|^2) + H(\lambda), & |z|^2 \le \lambda/(1-\lambda), \\ 0, & |z|^2 \ge \lambda/(1-\lambda). \end{cases} \tag{17.2}$$

Set $\Gamma_\lambda = 0$ for $\lambda \le 0$, $\Gamma_1 = g$, and $\Gamma_\lambda = -\infty$ for $\lambda > 1$.

Proposition 17.4.1 *This is an envelope test curve with initial potential zero and cutoff one. Its forward transform is*

$$U_t(z) = \log \frac{1 + e^t |z|^2}{1 + |z|^2}, \qquad t > 0. \tag{17.3}$$

At $z = 0$ its value is zero, and at infinity its value is t.

Proof Fix $0 < \lambda < 1$. For $s = \log |z|^2$ let $q_\lambda(s) = \Gamma_\lambda(z) + \log(1 + e^s)$. In the first region of (17.2) it is the affine function $\lambda s + H(\lambda)$, tangent to $\log(1 + e^s)$ at $s_\lambda = \log(\lambda/(1-\lambda))$. In the second region it is $\log(1+e^s)$. Its derivative is continuous and nondecreasing at the joining point, so q_λ is convex. By Corollary 1.20.1, Γ_λ is θ-plurisubharmonic away from zero. It extends across its logarithmic pole by Lemma 1.3.7. It is zero near infinity and has ordinary type λg.

To prove $\Gamma_\lambda = P_\theta[\lambda g]$, let $v \le 0$ be a competitor with $v \preceq \lambda g$. On the punctured disc $0 < |z|^2 < \lambda/(1-\lambda)$, the function

$$v - \Gamma_\lambda = v + \log(1 + |z|^2) - \lambda \log |z|^2 - H(\lambda)$$

is subharmonic, since $v + \log(1 + |z|^2)$ is subharmonic and $\log |z|^2$ is harmonic there. The singularity bound on v makes this difference bounded above near zero, so it extends subharmonically across zero by Lemma 1.3.7. Its upper boundary limit is at most zero, since $v \le 0$ and $\Gamma_\lambda = 0$ on the boundary circle. The maximum principle gives $v \le \Gamma_\lambda$ on the disc; at zero both have value $-\infty$. Outside the disc, the same inequality follows from $\Gamma_\lambda = 0$. Thus every competitor lies below Γ_λ, which is itself a competitor. Ordinary-type invariance gives $P_\theta[\Gamma_\lambda] = \Gamma_\lambda$. Zero is also an envelope fixed point.

For a fixed $0 < |z| < \infty$, put $p = |z|^2/(1 + |z|^2)$. The function $\lambda \mapsto \Gamma_\lambda(z)$ is zero for $\lambda \le p$. For $p < \lambda < 1$ its derivative is

$$\log |z|^2 + \log \frac{1-\lambda}{\lambda}.$$

This starts at zero and decreases, proving concavity and decreasingness. At zero the curve is zero for $\lambda \le 0$ and $-\infty$ thereafter; at infinity it is zero for $\lambda \le 1$. These also satisfy the asserted conditions. Its limit towards $-\infty$ is zero, and its limit at one is g, proving that it is a test curve.

For finite nonzero z and $t > 0$, the maximum in the transform occurs in the second parameter region at

$$\lambda = \frac{e^t |z|^2}{1 + e^t |z|^2}.$$

Substitution gives (17.3). At zero only parameters $\lambda \le 0$ have finite values, giving zero; at infinity the supremum of $t\lambda$ for $\lambda < 1$ is t. In the coordinate $w = 1/z$, the formula is $\log((|w|^2 + e^t)/(|w|^2 + 1))$, so the ray is smooth in both charts. □

Each finite slice of this ray is bounded and has envelope zero, while its inverse transforms for $0 < \lambda \le 1$ have logarithmic poles. At $z = 0$, the parameters $\lambda \le 0$ determine the transform: omitting them would incorrectly give $U_t(0) = -\infty$.

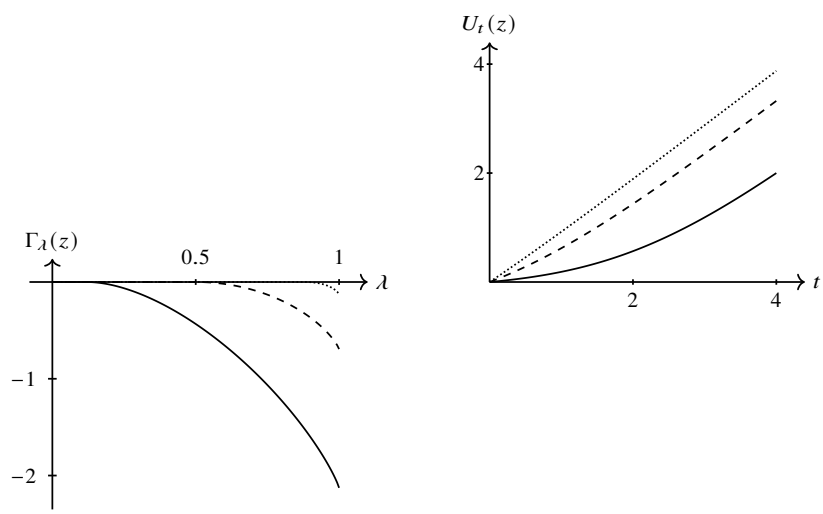

Fig. 17.1 The explicit test curve and its ray, at points with $\log |z|^2 = -2$ (solid), 0 (dashed), and 2 (dotted). The parameter functions on the left are concave; their transforms on the right are convex.

Chapter 18
Restriction and intrinsic trace data

Let X be a connected compact complex manifold, let $Y \subset X$ be a connected compact complex submanifold of positive dimension, and fix $u \in \mathrm{QPSH}(X)$ with $\nu(u, Y) = 0$.

Even when all multiplier ideals of u are trivial, its restriction to Y can have logarithmic poles. To retain information determined by the ambient ideal class, we restrict the ideals themselves and take their asymptotic vanishing orders on modifications of Y. These orders form the divisorial trace data. Subadditivity and analytic approximation make this data additive under sums of potentials.

The next question is whether a potential on Y realizes these numbers. On a curve, their atomic measure has a Green potential with exactly the prescribed Lelong numbers. In higher dimension, restricted analytic envelopes converge to the largest potential satisfying the trace inequalities. Lelong semicontinuity gives only a lower bound for its divisorial numbers. When Y is Kähler, convergence in a sufficiently large closed background supplies the reverse bound; pullback from a curve gives another exact realization, also for non-Kähler Y. The product examples at the end show why exact trace data still does not determine Monge–Ampère mass.

Choose a smooth real $(1, 1)$-form θ with $u \in \mathrm{PSH}(X, \theta)$ and a Hermitian form ω. We use the analytic approximants u_m and errors ϵ_m of Corollary 1.15.1. Their restrictions are genuine by Corollary 1.22.1.

18.1 Restricted ideals and trace data

The image of a multiplier ideal on Y records the vanishing of its local generators along Y. Its order on a modification can grow with the weight, even when the ideal is generically trivial on Y.

For $t > 0$ put

$$\mathfrak{a}_t(u; Y) = \mathcal{I}(tu)\mathcal{O}_Y.$$

Here we take the image of the restriction map in $\mathcal{O}_Y$. It is coherent and nonzero. Indeed, choose $y \in Y$ with $\nu(u, y) < 1/t$; Skoda integrability gives $\mathcal{I}(tu)_y = \mathcal{O}_{X,y}$.

A *divisor over* Y in this chapter means a prime divisor E on a connected smooth compact modification $\pi : Z \to Y$. For a nonzero coherent ideal $\mathfrak{a}$ on Y, let $\operatorname{ord}_E(\mathfrak{a})$ be the generic vanishing order of its image on Z. It is a finite nonnegative integer.

Theorem 18.1.1 *For every divisor E over Y, the limit*

$$\tau_{Y,u}(E) := \lim_{t\to\infty} \frac{\operatorname{ord}_E \mathfrak{a}_t(u;Y)}{t} = \sup_{t>0} \frac{\operatorname{ord}_E \mathfrak{a}_t(u;Y)}{t} \tag{18.1}$$

exists and is finite. It also equals the limit along positive integers. The numbers depend only on the multiplier-ideal equivalence class of u on X.

Proof Fix the modification $\pi : Z \to Y$ carrying E and write $b(t) = \operatorname{ord}_E \mathfrak{a}_t(u;Y)$ for $t > 0$. For $s, t > 0$, the second inclusion in (1.20), followed by restriction, gives

$$\mathfrak{a}_{s+t} \subseteq \mathfrak{a}_s\mathfrak{a}_t, \qquad b(s+t) \geq b(s) + b(t).$$

Moreover $b(t) \geq 0$. For fixed $s > 0$ and $t = ks + r$, $0 \leq r < s$, repeated superadditivity gives $b(t) \geq kb(s)$; for $r = 0$ the same conclusion follows without a remainder. Hence

$$\liminf_{t\to\infty} \frac{b(t)}{t} \geq \frac{b(s)}{s}.$$

Taking the supremum over s proves existence and the supremum formula, initially allowing $+\infty$. In particular the positive-integer subsequence has the same limit.

For integer m sufficiently large, Corollary 1.15.1 and analytic restriction give

$$\frac{b(m)}{m} = \nu\big((u_m|_Y) \circ \pi, E\big).$$

These numbers are uniformly bounded by Corollary 1.22.1. The integer limit, and therefore the full supremum, is finite. Finally all the image ideals in the definition are unchanged when u is replaced by a multiplier-ideal equivalent potential. This proves the last assertion. □

We call $\tau_{Y,u}$ the *divisorial trace data*.

Ideal inclusions give order and scaling directly. For addition, subadditivity gives one inequality; an analytic approximant to the sum controls sufficiently fine approximants to both summands and gives the reverse inequality.

Proposition 18.1.1 *Let $v \in \mathrm{QPSH}(X)$ also have zero generic Lelong number along Y. Then, pointwise on divisors over Y,*

(1) $u \preceq_{\mathcal{I}} v$ *implies* $\tau_{Y,u} \geq \tau_{Y,v}$*;*
(2) $\tau_{Y,au} = a\tau_{Y,u}$ *for every* $a > 0$*;*
(3) $\tau_{Y,u+v} = \tau_{Y,u} + \tau_{Y,v}$*;*
(4) $\tau_{Y,\max(u,v)} \leq \min\{\tau_{Y,u}, \tau_{Y,v}\}$.

Proof The ideal inclusion defining $\preceq_{\mathcal{I}}$ reverses vanishing orders after restriction. Replacing t by at in (18.1) proves homogeneity. Moreover,

$$\mathcal{I}(tu) + \mathcal{I}(tv) \subseteq \mathcal{I}(t\max(u,v)).$$

The order of a sum of ideals is the minimum of their orders; restriction and passage to the limit prove (4). Generic Lelong numbers of sums add, so $u + v$ still has zero generic Lelong number along Y.

Subadditivity of multiplier ideals gives $\tau_{Y,u+v} \geq \tau_{Y,u} + \tau_{Y,v}$. For the reverse inequality, fix a sufficiently large integer k and an analytic approximant S_k to $u + v$, of type $(1/k, \mathcal{I}(k(u+v)))$. We have $u + v \leq S_k$ by Theorem 1.15.1. Let U_m, V_m be the corresponding approximants to u, v. The analytic comparison in Lemma 1.17.10 gives

$$U_m + V_m \leq (1 - \delta_m)S_k + O(1), \qquad \delta_m \downarrow 0.$$

Both δ_m and the bounded error may depend on the fixed k. All three analytic restrictions are genuine by Corollary 1.22.1. Restrict, pull back to a fixed modification, and take the generic Lelong number along its divisor E:

$$\frac{\mathrm{ord}_E\, \mathfrak{a}_m(u;Y)}{m} + \frac{\mathrm{ord}_E\, \mathfrak{a}_m(v;Y)}{m} \geq (1 - \delta_m)\frac{\mathrm{ord}_E\, \mathfrak{a}_k(u+v;Y)}{k}.$$

First let $m \to \infty$, with k fixed, and then let $k \to \infty$. The finite limits in Theorem 18.1.1 give the reverse inequality. □

18.2 Comparison with ordinary restriction

When the ordinary restriction exists, the restriction theorem for multiplier ideals bounds the trace data by its Lelong numbers. For analytic singularities, ordinary comparison with the approximants supplies the reverse inequality.

Proposition 18.2.1 *If $v = u|_Y$ is genuine, then*

$$\tau_{Y,u}(E) \leq \nu(v \circ \pi, E)$$

for every divisor E over Y. If u has analytic singularities, equality holds.

Proof The restriction theorem and Lemma 1.17.7, applied on Y, give for every $t > 0$

$$\operatorname{ord}_E \mathfrak{a}_t(u; Y) \le \operatorname{ord}_E \mathcal{I}(tv) \le t\nu(v \circ \pi, E).$$

Divide by t and use Theorem 18.1.1.

If u has analytic singularities, apply Lemma 1.17.10 with $v = 0$ and $s = u$, taking $V_m = 0$ and $U_m = u_m$. It gives

$$u_m \le (1 - \delta_m)u + O(1), \qquad \delta_m \downarrow 0.$$

Restrict to Y, pull back by π, and take the generic Lelong number along E. Since u_m has type $(1/m, \mathcal{I}(mu))$,

$$\frac{1}{m} \operatorname{ord}_E \mathfrak{a}_m(u; Y) \ge (1 - \delta_m)\nu((u|_Y) \circ \pi, E).$$

Let $m \to \infty$. This proves the reverse inequality and hence equality. All restrictions are genuine because $\nu(u, Y) = 0$. □

The inequality can be strict, as the following local example shows. On a small bidisc in $\mathbb{C}^2$ with coordinates (z, w), write

$$u(z, w) = \max\{-\log(-\log|z|^2),\ \log|w|^2\}, \qquad Y = \{z = 0\}.$$

The bidisc is chosen with $|z| < \mathrm{e}^{-1}$, and the first function is extended by $-\infty$ at $z = 0$. It is psh: the function $s \mapsto -\log(-s)$ is increasing and convex for $s < 0$. The maximum is psh as well. For every $t > 0$,

$$\mathrm{e}^{-tu} \le (-\log|z|^2)^t,$$

which is locally integrable. Thus all ambient multiplier ideals are trivial and all the trace data is zero. But $u|_Y = \log|w|^2$ has Lelong number one at the origin. This example is from [Xia, Example 8.1.2].

18.3 Realization on a compact curve

Return to the compact setting, and assume until Section 18.6 that Y is a curve. Its divisors are points, so the trace data prescribes an atomic curvature measure. Stokes bounds the total atomic mass by the area of the background. Filling the remaining mass with a smooth measure then gives a positive current whose Green potential realizes the data.

With the sum initially understood through finite subsums, set

$$D_u = \sum_{p \in Y} \tau_{Y,u}(p)\, \delta_p.$$

Theorem 18.3.1 *Put $M = \int_Y \theta|_Y$. Then*

$$0 \le D_u(Y) \le M. \tag{18.2}$$

The finite atomic measure D_u defines a canonical ordinary quasi-plurisubharmonic singularity type on Y. For every smooth positive form η on Y with $\int_Y \eta = 1$, that type has a representative $v \in \mathrm{PSH}(Y, \theta|_Y)$ with

$$\theta|_Y + \mathrm{dd^c} v = D_u + (M - D_u(Y))\eta. \tag{18.3}$$

Its extended Monge–Ampère measure is $(M - D_u(Y))\eta$. The ordinary type is independent of η, of the chosen smooth background admitting the construction, and of the ambient representative within its multiplier-ideal class.

Proof For integer m let

$$D_m = \frac{1}{m} \sum_{p \in Y} \mathrm{ord}_p\, \mathfrak{a}_m(u; Y)\, \delta_p.$$

A nonzero coherent ideal on a compact smooth curve has finitely many zeros, so this is a finite sum. It is the atomic part of the positive current $(\theta + \epsilon_m \omega)|_Y + \mathrm{dd^c}(u_m|_Y)$. Stokes on the curve gives

$$D_m(Y) \le M + \epsilon_m \int_Y \omega|_Y.$$

For any finite subset of Y, pass to the limit in the sum of its coefficients using Theorem 18.1.1. Then take the supremum over finite subsets. This gives (18.2). In particular the positive coefficients have at most countable support, and they define the claimed finite measure. Write $d = D_u(Y)$.

Choose a Green kernel for $\mathrm{dd^c}$ on the curve, normalized so that

$$\mathrm{dd^c_x} G(x, p) = \delta_p - \eta, \qquad G(x, p) \le 0.$$

The Green kernel of the scalar Laplacian on the compact Riemann surface has the following properties: its singularity is $\log |z(x) - z(p)|^2$, it is bounded above uniformly, and $\sup_p \int_Y |G(x, p)|\eta(x) < \infty$. Subtracting one uniform constant gives the indicated sign. Fubini therefore shows that

$$w(x) = \int_Y G(x, p)\, \mathrm{d}D_u(p)$$

is an L^1 function, with its canonical quasi-plurisubharmonic representative satisfying $\mathrm{dd^c} w = D_u - d\,\eta$. It has Lelong number $\tau_{Y,u}(p)$ at every point p.

Since $M\eta - \theta|_Y$ is smooth and has integral zero, the scalar Poisson equation gives a smooth s with $\mathrm{dd^c} s = M\eta - \theta|_Y$. Then $v = w + s$ satisfies (18.3). Positivity follows from $M \ge d$. At every atom of D_u, the positive Lelong number forces $v = -\infty$. The remaining pole set has zero smooth volume. Removing the pole set from the current

consequently leaves exactly $(M - d)\eta$, proving the assertion about the extended measure.

If another smooth volume form or another admissible smooth background is used, the difference of the two constructed representatives has smooth $\mathrm{dd^c}$: their atomic parts are the same, and all other terms in their equations are smooth. Elliptic regularity makes their difference smooth, hence bounded. This proves independence of these choices at the level of ordinary type. Independence of the ambient multiplier-ideal representative follows from equality of the numbers defining D_u. □

We denote this ordinary type by $\mathrm{Tr}_Y^{\mathrm{at}}(u)$, the *atomic trace on a curve*. All potentials on a curve having these Lelong numbers have the same multiplier ideals, but they need not have the same ordinary type. The construction selects the type with a smooth residual curvature measure. Additivity of the trace data and linearity of the Green operator give

$$\mathrm{Tr}_Y^{\mathrm{at}}(u + v) \sim \mathrm{Tr}_Y^{\mathrm{at}}(u) + \mathrm{Tr}_Y^{\mathrm{at}}(v), \qquad \mathrm{Tr}_Y^{\mathrm{at}}(au) \sim a\,\mathrm{Tr}_Y^{\mathrm{at}}(u) \quad (a > 0).$$

Here $v \in \mathrm{QPSH}(X)$ also has zero generic Lelong number along Y.

Keep D_m, G, w from the proof and put $w_m = \int_Y G(\cdot, p)\,\mathrm{d}D_m(p)$.

Proposition 18.3.1 *The measures D_m above satisfy $D_m \le D_u$ and converge to D_u in total variation. Along dyadic indices they increase. With a fixed nonpositive Green kernel, the corresponding potentials w_{2^j} decrease to w and converge in $L^1(Y)$. Each w_m has the ordinary singularity type of $u_m|_Y$.*

If the ordinary restriction $u|_Y$ is genuine, it is more singular than the atomic trace:

$$u|_Y \preceq \mathrm{Tr}_Y^{\mathrm{at}}(u).$$

For analytic u, ordinary equivalence holds.

Proof The supremum formula (18.1) gives $D_m \le D_u$ coefficient by coefficient. Since D_u is a finite atomic measure, dominated convergence of its coefficients proves $D_m(Y) \to D_u(Y)$ and hence total-variation convergence. Superadditivity at $m + m$ gives $D_{2m} \ge D_m$. The sign of G then makes w_{2^j} decrease. Monotone convergence for $-G$ identifies its pointwise limit with w, including possible poles. The uniform L^1 bound for the kernel also gives

$$\|w_m - w\|_{L^1(\eta)} \le C\|D_m - D_u\|_{\mathrm{TV}} \longrightarrow 0.$$

Each D_m is finitely supported. Locally both w_m and $u_m|_Y$ are its logarithmic singular part plus a bounded remainder. Thus their ordinary types agree.

If $a = u|_Y$ is genuine, Proposition 18.2.1 gives $\nu(a, p) \ge \tau_{Y,u}(p)$ at every point. Since $\mathrm{dd^c}w = D_u - D_u(Y)\eta$, Lemma 1.17.4 implies $a \preceq w$. If u is analytic, Proposition 18.2.1 identifies its finitely many restricted logarithmic coefficients with those of D_u. Its bounded local remainders then give ordinary equivalence. □

18.4 The Poisson equation in the atomic trace type

Write $d = D_u(Y)$ and let w be the Green representative above. Every $\theta|_Y$-plurisubharmonic potential in the ordinary type of w has polar curvature D_u. Its extended measure is the remaining curvature, of total mass $M - d$. Prescribing it amounts to solving a Poisson equation for the bounded correction.

Theorem 18.4.1 *Let* $\mathrm{d}V$ *be a smooth positive volume form on* Y*, and let* $0 \le f \in L^p(\mathrm{d}V)$*,* $p > 1$*, with* $\int_Y f\,\mathrm{d}V > 0$*. There is a solution in the atomic trace type to*

$$\mathrm{MA}^{\mathrm{ext}}_{\theta|_Y}(v) = c f\,\mathrm{d}V, \qquad c > 0,$$

if and only if $M > d$*. In that case*

$$c = \frac{M - d}{\int_Y f\,\mathrm{d}V},$$

and v *is unique up to an additive constant. If* $M = d$*, there is instead a unique such ordinary-type solution up to constants with extended measure zero.*

Proof Assume $M > d$ and use the Green potential w above. Choose the displayed c and solve

$$\mathrm{dd^c} s = d\,\eta + c f\,\mathrm{d}V - \theta|_Y.$$

The right side has integral zero. The scalar elliptic estimate gives $s \in W^{2,p}(Y)$, unique up to constants. In real dimension two the Sobolev embedding for $p > 1$ makes s continuous and bounded. The potential $v = w + s$ has the atomic trace type and satisfies

$$\theta|_Y + \mathrm{dd^c} v = D_u + c f\,\mathrm{d}V \ge 0.$$

Its extended measure is $c f\,\mathrm{d}V$, since this density does not charge the pole set while the atoms do. For $M = d$ the identical construction with $c = 0$ has a smooth right side and produces the asserted zero-measure solution.

To prove necessity and uniqueness, let a be any $\theta|_Y$-plurisubharmonic potential in the ordinary type of w, and put $\mu = \theta|_Y + \mathrm{dd^c} a$. By Corollary 1.6.1, ordinary equivalence with w makes the polar part of μ exactly D_u. Consequently

$$\mathrm{MA}^{\mathrm{ext}}_{\theta|_Y}(a) = \mu - D_u, \qquad \int_Y \mathrm{MA}^{\mathrm{ext}}_{\theta|_Y}(a) = M - d.$$

Integrating any proposed equation forces the displayed value of c, so a positive c requires $M > d$. If two potentials solve with the same density and scalar, their full currents both equal $D_u + c f\,\mathrm{d}V$. Their difference has zero $\mathrm{dd^c}$ and is therefore constant on the compact connected curve. The argument also proves uniqueness in the zero-measure case. □

18.5 Singularity distance on a curve

The singularity pseudometric d_S of [Xia, Definition 6.2.1], applied on the curve Y, measures the mass lost between ordered Green representatives. For w and w_m, it is half the missing atomic mass. This identifies the atomic trace with the P-equivalence class of the Kähler trace in [Xia, Section 8.1] when X is Kähler; the distance calculation itself uses only the curve geometry.

Proposition 18.5.1 *Choose a smooth positive form η of integral one on Y and choose $A > D_u(Y)$ sufficiently large that every restricted approximant $u_m|_Y$ is $A\eta$-plurisubharmonic. Use a fixed nonpositive Green kernel to define w, w_m as above. Then*

$$d_{S,A\eta}(w, u_m|_Y) = d_{S,A\eta}(w, w_m) = \frac{1}{2}\big(D_u(Y) - D_m(Y)\big) \longrightarrow 0. \tag{18.4}$$

In particular the atomic trace has $\mathcal{I}$-good singularities on the curve. If the ambient manifold is Kähler, it represents the P-equivalence class of the Kähler trace.

Proof The restricted curvature bounds of Corollary 1.15.1 are dominated by a single $A\eta$, after increasing A. Since $D_m \le D_u$ and $G \le 0$, we have $w \le w_m$. Also

$$A\eta + \mathrm{dd^c} w = D_u + (A - D_u(Y))\eta, \qquad A\eta + \mathrm{dd^c} w_m = D_m + (A - D_m(Y))\eta.$$

Their extended masses are $A - D_u(Y)$ and $A - D_m(Y)$, both positive. For $a, b \in \mathrm{PSH}(Y, A\eta)$ with $a \le b$, the ordered-distance formula [Xia, Lemma 6.2.2] reads

$$d_{S,A\eta}(a, b) = \frac{1}{2}\left(\int_Y \mathrm{MA}_{A\eta}(b) - \int_Y \mathrm{MA}_{A\eta}(a)\right).$$

Substituting $a = w$ and $b = w_m$ proves the second equality in (18.4). Ordinary equivalence of $u_m|_Y$ and w_m gives distance zero between them for $d_{S,A\eta}$, proving the first equality. Total-variation convergence of D_m proves the limit.

Each w_m has finitely many logarithmic poles with rational coefficients, hence analytic singularities. The characterization in [Xia, Theorem 7.1.1], now applied on the Kähler curve Y, shows that their d_S limit w is $\mathcal{I}$-good.

Finally suppose X is Kähler. The source book supplies a quasi-equisingular approximation with analytic type $(2^{-j}, \mathcal{I}(2^j u))$. Its restriction to Y has the ordinary type of w_{2^j}. The same distance calculation therefore proves that these restrictions converge to w for $d_{S,A\eta}$. By [Xia, Definition 8.1.1 and Proposition 8.1.2], w therefore represents that trace modulo P-equivalence. □

18.5.1 Total variation of the atomic data

The trace calculation extends to arbitrary ideal envelopes on Y. Their atomic curvature determines them, and taking a maximum replaces two atomic measures by their common part. The ordered mass formula can therefore compare envelopes whose poles are unrelated. For this subsection, fix a smooth positive form β on Y, put $A = \int_Y \beta$, and choose a nonpositive Green kernel G for $\eta = \beta/A$. All atomic measures below have countable support and finite total mass.

For a positive atomic measure D with $D(Y) \le A$, set

$$w_D(x) = \int_Y G(x,p)\,\mathrm{d}D(p), \qquad q_D = P_\beta[w_D]_{\mathcal{I}}.$$

The Green construction gives a genuine quasi-plurisubharmonic function with $\mathrm{dd^c} w_D = D - D(Y)\eta$.

Lemma 18.5.1 *The function w_D is β-plurisubharmonic, and*

$$q_D \sim w_D, \qquad \int_Y \mathrm{MA}_\beta(q_D) = \int_Y \mathrm{MA}_\beta(w_D) = A - D(Y).$$

The map $D \mapsto q_D$ is a bijection onto all normalized ideal envelopes. The positive-mass locus corresponds to $D(Y) < A$. The measure D is the atomic part of $\beta + \mathrm{dd^c} q_D$.

Proof One has

$$\beta + \mathrm{dd^c} w_D = D + (A - D(Y))\eta \ge 0.$$

Every atom of D is a pole of w_D. The smooth measure η charges no polar set, including any additional poles caused by accumulation of the atoms. Hence $\mathrm{MA}_\beta(w_D) = (A - D(Y))\eta$.

Put $q = q_D$. Ideal projection preserves the point Lelong numbers. Applied to $\mathrm{dd^c} w_D = D - D(Y)\eta$, Lemma 1.17.4 gives $q \preceq w_D$. The envelope inequality $q \ge w_D - \sup_Y w_D$ proves $q \sim w_D$. By Corollary 1.6.1, the polar part of $\beta + \mathrm{dd^c} q$ is therefore also D. Stokes gives its non-pluripolar mass $A - D(Y)$.

For any $u \in \mathrm{PSH}(Y,\beta)$, its point Lelong numbers give an atomic measure D_u of mass at most A. On a smooth curve these numbers determine all multiplier ideals, by Theorem 1.16.1. Thus $u \sim_{\mathcal{I}} w_{D_u}$, so $P_\beta[u]_{\mathcal{I}} = q_{D_u}$. If $D_u(Y) = A$, its atoms exhaust the full curvature measure and its non-pluripolar mass is zero. This proves surjectivity and the description of the positive-mass locus. Distinct atomic measures have distinct Lelong data, proving injectivity. □

Theorem 18.5.1 *For positive atomic measures D, E with masses less than A,*

$$\frac{1}{96}\|D - E\|_{\mathrm{TV}} \le d_{S,\beta}(q_D, q_E) \le \frac{1}{2}\|D - E\|_{\mathrm{TV}}.$$

Thus the singularity distance is a metric on these ideal envelopes and induces exactly total-variation convergence of their atomic data.

Proof Write $d_S = d_{S,\beta}$ and $a \vee b = \max(a, b)$. For positive-mass potentials $u, v \in \mathrm{PSH}(Y, \beta)$, besides the ordered-distance formula we use the inequality [Xia, Lemma 6.2.1]

$$d_S(u, v) \le d_S(u, u \vee v) + d_S(v, u \vee v) \le 48 d_S(u, v)$$

on a curve. Here $48 = 3(1 + 1)2^{1+2}$ is the constant in that lemma for dimension one.

Put $z = \max(w_D, w_E)$. The lemma identifies the polar curvature measures of w_D, w_E as D, E. By (1.8), the polar curvature of z is $D \wedge E$, the atomic measure with the smaller coefficient at each point. Stokes therefore gives

$$\mathrm{MA}_\beta(z)(Y) = A - (D \wedge E)(Y).$$

The ordered-distance formula yields

$$\begin{aligned} d_S(w_D, z) + d_S(w_E, z) &= \frac{1}{2}\big(D(Y) + E(Y) - 2(D \wedge E)(Y)\big) \\ &= \frac{1}{2}\|D - E\|_{\mathrm{TV}}. \end{aligned}$$

Combining this equality with the two inequalities above proves both bounds. Ordinary equivalence with q_D, q_E leaves the singularity distance unchanged. The lower bound and the lemma's injectivity make the pseudometric a metric on these representatives.□

Corollary 18.5.1 *For $0 < \varepsilon < A$, the normalized ideal envelopes satisfying $\mathrm{MA}_\beta(q)(Y) \ge \varepsilon$ form a complete space for $d_{S,\beta}$. Analytic ideal envelopes are dense in this space. The space is not totally bounded.*

Proof By the lemma, the space corresponds to the positive atomic measures of mass at most $A - \varepsilon$. This set is complete for total variation. Indeed a Cauchy sequence has a positive finite-measure limit with the same mass bound. The union of the sequence's countable supports is countable, and total-variation convergence gives zero limit mass off that union. The limit is therefore still atomic. The two metric estimates transfer completeness to the ideal envelopes.

Finite truncation followed by rational approximation of each coefficient from below preserves the mass bound and converges in total variation. The resulting Green potentials have finitely many rational logarithmic poles; their ideal envelopes have the same ordinary types by the lemma. They are analytic, and the upper metric estimate proves density.

Finally fix $0 < a < A - \varepsilon$. The measures $a\delta_p$, for distinct $p \in Y$, have pairwise total-variation distance $2a$. Their ideal envelopes have pairwise d_S distance at least $a/48$. This infinite separated family rules out total boundedness. □

This topology differs from the L^1 ideal metric of Proposition 15.4.1: that metric is totally bounded. The sequence in Proposition 15.4.2 makes the distinction quantitative, since it converges in the ideal metric while remaining at a fixed positive d_S distance from its limit.

18.6 The trace envelope

Return to the original setting with $Y \subset X$ of arbitrary positive dimension. The restricted approximants have vanishing curvature error, so normalized compactness gives a potential in the original background $\theta|_Y$. Its divisorial numbers dominate the trace data. These inequalities define an envelope, and comparison with the resolved analytic ideals identifies it as the limit of the restricted analytic envelopes.

Proposition 18.6.1 *There is a potential $v \in \mathrm{PSH}(Y, \theta|_Y)$ such that*

$$\nu(v \circ \pi, E) \geq \tau_{Y,u}(E) \quad \textit{for every divisor } E \textit{ over } Y.$$

In particular, $v \in \mathrm{PSH}(Y, \Omega_0)$ for every Hermitian form $\Omega_0 \geq \theta|_Y$.

Proof The restrictions $u_m|_Y$ have a common smooth curvature lower bound. Subtract their suprema and extract an L^1 limit v by normalized compactness. Their current inequalities pass to the limit as $\epsilon_m \to 0$, giving $\theta|_Y + \mathrm{dd^c} v \geq 0$.

For each fixed compact modification, Lemma 1.22.2 gives convergence of the pullbacks along that same subsequence. Lemma 1.17.5 gives, at every point z of E,

$$\nu(v \circ \pi, z) \geq \limsup_m \nu((u_m|_Y) \circ \pi, z) \geq \lim_m \frac{\mathrm{ord}_E\, \mathfrak{a}_m(u; Y)}{m} = \tau_{Y,u}(E).$$

Taking the generic value proves the assertion. No simultaneous extraction over divisors is required: the pullback lemma applies to each modification after the subsequence on Y has been fixed. □

For a smooth real $(1,1)$-form Ω on Y for which the following family is nonempty, define

$$\mathcal{T}_\Omega(u;Y) := \Big(\sup\{a \in \mathrm{PSH}(Y,\Omega) : a \le 0, \\ \nu(a\circ\pi, E) \ge \tau_{Y,u}(E) \ \text{ for every } E\}\Big)^*. \tag{18.5}$$

The preceding proposition supplies a competitor for $\Omega = \theta|_Y$ and for every larger background. The envelope depends on Ω and the ambient multiplier-ideal class of u.

18.6.1 Limits of restricted analytic envelopes

On a log resolution of $\mathfrak{a}_m(u;Y)$, the trace inequalities dominate every coefficient of the restricted analytic singularity. Each trace competitor is consequently more singular than $u_m|_Y$. This gives a common lower barrier for the analytic envelopes. Semicontinuity puts every cluster limit back in the trace family, where maximality identifies it.

Theorem 18.6.1 *Let Ω be a smooth real $(1,1)$-form on Y, let h be Hermitian, and choose $\delta_m \ge 0$ tending to zero such that, for all sufficiently large m,*

$$a_m := u_m|_Y \in \mathrm{PSH}(Y, \Omega + \delta_m h).$$

Then $T = \mathcal{T}_\Omega(u;Y)$ is genuine, and

$$q_m := P_{\Omega+\delta_m h}[a_m] = P_{\Omega+\delta_m h}[a_m]_{\mathcal{I}} \longrightarrow T \quad \text{in } L^1(Y). \tag{18.6}$$

If δ_{2^j} decreases, then $q_{2^j} \searrow T$ pointwise. The limit depends only on Ω and the ambient multiplier-ideal class of u. Moreover,

$$P_\Omega[T]_{\mathcal{I}} = P_\Omega[T] = T.$$

Proof The argument of Proposition 18.6.1, applied to the assumed curvature bounds, gives a normalized Ω-plurisubharmonic limit v of a subsequence of the a_m. Its divisorial numbers dominate $\tau_{Y,u}$. Hence the family defining T is nonempty, and its supremum lies between v and zero. It is stable under finite maxima, since Lelong numbers of a maximum are the minima of the two Lelong numbers, also after pullback. Choquet extraction and finite maxima give an increasing sequence of competitors with canonical limit T. Pullback convergence and Lelong semicontinuity show that T retains every defining inequality. Thus T is itself a competitor and is the largest one.

Fix m and resolve $\mathfrak{a}_m(u;Y)$ on $\pi : Z \to Y$. For each prime component E of the resolved ideal, (18.1) gives

$$\nu(T\circ\pi, E) \ge \tau_{Y,u}(E) \ge \frac{1}{m}\,\mathrm{ord}_E\,\mathfrak{a}_m(u;Y) = \nu(a_m\circ\pi, E).$$

The finite-divisor criterion in Lemma 1.17.9 gives $T \preceq a_m$. Since $T \leq 0$ and $\delta_m \geq 0$, it is a competitor for q_m. Also Proposition 15.2.3 identifies the two envelopes in (18.6) and gives $q_m \sim a_m$. Therefore

$$T \leq q_m \leq 0, \qquad \nu(q_m \circ \pi, E) = \frac{1}{m} \operatorname{ord}_E \mathfrak{a}_m(u; Y)$$

for every divisor over Y.

All q_m have supremum zero and a common smooth curvature lower bound. Consider any L^1 cluster limit q. It belongs to $\mathrm{PSH}(Y, \Omega)$, lies between T and zero, and has

$$\nu(q \circ \pi, E) \geq \limsup_m \nu(q_m \circ \pi, E) = \tau_{Y,u}(E)$$

along the convergent subsequence. Here the pullback and semicontinuity argument of Proposition 18.6.1 applies to each fixed compact modification. Thus q is a competitor for T, which proves $q = T$. Normalized compactness now gives convergence of the full sequence in L^1.

Subadditivity gives $\mathfrak{a}_{2m}(u; Y) \subseteq \mathfrak{a}_m(u; Y)^2$, hence $a_{2m} \preceq a_m$. When the dyadic backgrounds decrease, this implies $q_{2^{j+1}} \leq q_{2^j}$. The lower bound by T makes the pointwise limit genuine, and its L^1 identification gives equality with T everywhere. Finally the defining family for T depends only on Ω and the ideals of u, proving the asserted independence.

Ideal projection preserves all divisorial Lelong numbers, so $P_\Omega[T]_{\mathcal{I}}$ is a trace competitor. Maximality and the envelope inequality give $P_\Omega[T]_{\mathcal{I}} = T$. The ordinary fixed-point assertion follows from Proposition 15.2.1. □

Taking $\Omega = \theta|_Y$, $h = \omega|_Y$ and $\delta_m = \epsilon_m$ gives the limit in the original restricted background. Along dyadic indices the errors can be enlarged to a decreasing sequence with limit zero. A fixed sufficiently large Hermitian background permits $\delta_m = 0$.

Inclusion of the competitor families also orders the trace envelopes: increasing the prescribed data decreases $\mathcal{T}_\Omega$, whereas increasing the background increases it whenever both envelopes are defined.

18.6.2 Envelopes on modifications

Let $\pi : Z \to Y$ be a modification of connected compact complex manifolds. Extend the restricted ideals to Z by taking their image in $\mathcal{O}_Z$. For a divisor F on a further modification $\rho : W \to Z$, their asymptotic order is $\tau_{Y,u}(F)$, computed on $\pi \circ \rho : W \to Y$. Write $\mathcal{T}_\Gamma^\pi(u; Y)$ for the envelope on Z defined by (18.5) with background Γ and these inherited inequalities. Thus the trace data still comes from the original ambient ideals.

The pulled-back background forces every competitor to descend. A common refinement then identifies its divisorial inequalities on the two models.

Theorem 18.6.2 *Suppose Ω is a smooth real $(1,1)$-form on Y for which $T = \mathcal{T}_\Omega(u;Y)$ is defined, and let h be a Hermitian form on Z. Then*

$$\mathcal{T}^{\pi}_{\pi^*\Omega}(u;Y) = T \circ \pi, \qquad \mathcal{T}^{\pi}_{\pi^*\Omega+\varepsilon h}(u;Y) \searrow T \circ \pi \quad (\varepsilon \downarrow 0). \tag{18.7}$$

The convergence is pointwise and in $L^1(Z)$, and the limit is independent of h.

Let $\delta_m \geq 0$ tend to zero and suppose, for all sufficiently large m, that

$$b_m := (u_m|_Y) \circ \pi \in \mathrm{PSH}(Z, \pi^*\Omega + \delta_m h).$$

Then

$$P_{\pi^*\Omega+\delta_m h}[b_m] = P_{\pi^*\Omega+\delta_m h}[b_m]_{\mathcal{I}} \longrightarrow T \circ \pi \quad \text{in } L^1(Z).$$

If δ_{2^j} decreases, these envelopes decrease pointwise to $T \circ \pi$ along the dyadic indices.

Proof For any nonempty trace family, the finite-maxima and semicontinuity argument of Theorem 18.6.1 shows that the envelope is its largest member. By Proposition 1.22.2, every $\pi^*\Omega$-plurisubharmonic function is uniquely $a \circ \pi$ with $a \in \mathrm{PSH}(Y, \Omega)$, and surjectivity preserves the bound by zero. If a satisfies all trace inequalities over Y, its pullback satisfies those over Z, since every modification of Z composes to one of Y.

Conversely, suppose the pullback satisfies the inequalities over Z. For a prime divisor E on any smooth compact modification $V \to Y$, resolve the graph of the bimeromorphic map between V and Z. This gives a smooth compact common refinement W. The map $W \to V$ is an isomorphism at the generic point of E, so the strict transform F has the same ideal orders and generic Lelong numbers as E. Its inequality over Z is therefore exactly the required inequality at E. The competitor families correspond in both directions. Their largest member on Z is consequently $T \circ \pi$, proving the first identity.

For $\varepsilon > 0$, the pullback $T \circ \pi$ remains a competitor. The perturbed envelopes lie between $T \circ \pi$ and zero and decrease as $\varepsilon \downarrow 0$. Their limit S is genuine and $\pi^*\Omega$-plurisubharmonic. Pullback L^1 convergence and Lelong semicontinuity retain every trace inequality, so S belongs to the unperturbed family. Its maximality gives $S \leq T \circ \pi$, and the lower bound gives equality. The same lower bound, which is integrable on Z, gives L^1 convergence by dominated convergence.

For the final assertion, denote the displayed analytic envelope by q_m. The proof of Theorem 18.6.1 applies to the global coherent ideals $\mathfrak{a}_m(u;Y)\mathcal{O}_Z$: resolving each ideal and applying Lemma 1.17.9 gives $T \circ \pi \preceq b_m$. Hence

$$T \circ \pi \leq q_m \leq 0, \qquad \nu(q_m \circ \rho, F) = \frac{1}{m} \operatorname{ord}_F \mathfrak{a}_m(u;Y)$$

for every divisor over Z, where ordinary and ideal envelopes agree by Proposition 15.2.3. Every L^1 cluster limit is $\pi^*\Omega$-plurisubharmonic and retains all trace inequalities. The first identity in (18.7) identifies it with $T \circ \pi$. Compactness gives convergence of the full sequence. Subadditivity orders the dyadic analytic types, and decreasing dyadic backgrounds then order the envelopes; their pointwise limit is the same L^1 limit. □

For $\Omega = \theta|_Y$, the final hypothesis holds with $\delta_m = C\epsilon_m$, where the fixed constant C satisfies $\pi^*(\omega|_Y) \le Ch$. Thus the restricted analytic envelopes converge to the same pulled-back trace on every compact model. Formula (18.7) also gives compatibility under successive modifications.

For arbitrary changes of background, inclusion of the competitor families gives an order on the envelopes. Equality of their ordinary types would require a reverse singularity bound. For curve fibrations, the fixed divisorial potential in Theorem 18.7.1 supplies this bound in every sufficiently large background.

18.6.3 Exact realization on a Kähler submanifold

Lelong semicontinuity in the envelope limit gives $\nu(\mathcal{T}_\Omega(u;Y)\circ\pi, E) \ge \tau_{Y,u}(E)$. When Y is Kähler, positive mass in a sufficiently large closed background strengthens this to equality: the decreasing-model theorem gives convergence of the divisorial numbers. An exact representative then supplies the reverse inequality in every background that admits it.

Theorem 18.6.3 *Suppose Y is Kähler. Choose a Hermitian form $\Omega_0 \ge \theta|_Y$ and a Kähler form $\kappa \ge 2\Omega_0$ on Y. For every Hermitian form $\Omega \ge \kappa$,*

$$\nu\big(\mathcal{T}_\Omega(u;Y) \circ \pi, E\big) = \tau_{Y,u}(E)$$

for every prime divisor E on every smooth compact modification $\pi : Z \to Y$.

Proof The restricted curvature errors tend to zero, so $u_m|_Y \in \mathrm{PSH}(Y,\kappa)$ for all sufficiently large m. By Theorem 18.6.1, the ordinary model potentials

$$q_j = P_\kappa[u_{2^j}|_Y]$$

decrease to $T = \mathcal{T}_\kappa(u;Y)$ after discarding finitely many indices. The normalized majorant v of Proposition 18.6.1 satisfies $v \le T$ and belongs to $\mathrm{PSH}(Y,\kappa/2)$. Thus Lemma 12.5.2 and Stokes give, with $r = \dim Y$,

$$\int_Y \mathrm{MA}_\kappa(T) \ge 2^{-r} \int_Y \kappa^r > 0.$$

For decreasing ordinary model potentials in a closed background, positive mass of the limit gives convergence of every divisorial Lelong number [Xia, Proposition 3.1.12]. Applied on Y, this yields

$$\nu(T \circ \pi, E) = \lim_j \nu(q_j \circ \pi, E) = \lim_j 2^{-j} \operatorname{ord}_E \mathfrak{a}_{2^j}(u;Y) = \tau_{Y,u}(E).$$

Here the middle equality uses the ordinary equivalence of each analytic envelope with its restricted approximant.

If $\Omega \geq \kappa$, then T is a competitor for $T_\Omega = \mathcal{T}_\Omega(u;Y)$, so $T \leq T_\Omega$. Its defining trace inequalities give the other bound:

$$\tau_{Y,u}(E) \leq \nu(T_\Omega \circ \pi, E) \leq \nu(T \circ \pi, E) = \tau_{Y,u}(E).$$

This proves the assertion for every such Ω. □

The closed theory is applied on Y. Under bounded mass on Y, Corollary 29.5.2 gives the same exactness in a sufficiently large Hermitian background with positive lower mass. For general non-Kähler Y, the envelope limit still supplies only the lower bound on its divisorial numbers.

18.6.4 Positive mass and prescribed equations

Doubling a Hermitian background that admits a trace competitor leaves that competitor in the half-sized background. This gives positive mass to the trace envelope and the model condition for equations in its ordinary type.

Theorem 18.6.4 *Choose Ω_0, v as in Proposition 18.6.1 and set $\Omega = 2\Omega_0$. Then $T = \mathcal{T}_\Omega(u;Y)$ is genuine and satisfies all the divisorial inequalities in (18.5). Moreover,*

$$P_\Omega[T]_{\mathcal{I}} = T, \qquad P_\Omega[T] = T, \qquad P_{\Omega/2}(T) \not\equiv -\infty.$$

Writing $r = \dim Y$, its extended measure satisfies

$$0 < \int_Y \mathrm{MA}^{\mathrm{ext}}_\Omega(T) < +\infty, \qquad \mathrm{MA}^{\mathrm{ext}}_\Omega(T) \leq \mathbf{1}_{\{T=0\}}\Omega^r.$$

Proof Since $\theta|_Y + \epsilon_m\omega|_Y \leq \Omega$ for all sufficiently large m, the restricted approximants belong to $\mathrm{PSH}(Y,\Omega)$. Apply Theorem 18.6.1 with zero curvature error to obtain T, its defining inequalities and both fixed-point identities.

Normalize $v \leq 0$. Since $v \in \mathrm{PSH}(Y,\Omega/2)$ is a trace competitor, it lies below T and supplies a competitor for $P_{\Omega/2}(T)$. The fixed-point contact estimate Proposition 12.1.4, applied on Y, gives the displayed measure bound and finiteness. The smaller-background individual positivity in Lemma 12.5.2 gives strict positivity. □

Corollary 18.6.1 *For the chosen Ω and T, let $\mathrm{d}V$ be a smooth positive volume form on Y and let $0 \le f \in L^p(\mathrm{d}V)$, $p > 1$, with $\int_Y f\,\mathrm{d}V > 0$. For every $\lambda > 0$ there is a unique $a \sim T$ in $\mathrm{PSH}(Y, \Omega)$ with*

$$\mathrm{MA}^{\mathrm{ext}}_{\Omega}(a) = \mathrm{e}^{\lambda a} f\,\mathrm{d}V.$$

There is also a positive scalar c and $a \sim T$ with $\mathrm{MA}^{\mathrm{ext}}_{\Omega}(a) = c f\,\mathrm{d}V$; the scalar is unique.

Proof The preceding theorem checks both parts of the model condition (12.1) on the compact Hermitian manifold Y. Apply the prescribed ordinary-type theorem Theorem 13.11.1. □

On a curve, Lemma 1.17.4 identifies the trace inequalities with ordinary domination by the Green representative w. Thus

$$\mathcal{T}_{A\eta}(u; Y) = P_{A\eta}[w] = P_{A\eta}[w]_{\mathcal{I}}$$

for every $A > 0$ with $A \ge D_u(Y)$. The last equality and exactness of the prescribed Lelong numbers follow from Lemma 18.5.1, including when the residual mass is zero.

18.7 Exact traces over a curve

Let $p : X \to M$ be a surjective holomorphic submersion of connected compact complex manifolds with connected fibers. Let $C \subset M$ be a connected compact smooth curve, put $Y = p^{-1}(C)$, and write $q = p|_Y : Y \to C$. Then Y is a connected smooth compact submanifold. Take $U \in \mathrm{QPSH}(M)$ with $\nu(U, C) = 0$, and set $u = U \circ p$. The generic Lelong number of u along Y is zero, since in product coordinates it equals that of U along C.

An exact representative on a curve also gives higher-dimensional examples by pullback. Multiplier ideals commute with a submersion, so the trace coefficients on a modification are the curve coefficients multiplied by the multiplicities of the pulled-back fibers. The pulled-back Green potential has precisely these divisorial coefficients. Its domination property then identifies the ordinary type of the trace envelope.

Let

$$D = \sum_{b \in C} \tau_{C,U}(b)\,\delta_b, \qquad \delta = D(C) < +\infty.$$

Choose a smooth positive probability form η on C and its nonpositive Green kernel. The curve construction gives a genuine $w \in \mathrm{QPSH}(C)$ with

$$\mathrm{dd^c} w = D - \delta\eta.$$

For $b \in C$, put $Y_b = q^{-1}(b)$ and let $[Y_b]$ be its current of integration. Set $v = w \circ q$. Pullback of distributions by a smooth submersion is defined by integration along the fibers. It commutes with differentiation and with positive sums here, so

$$\mathrm{dd^c} v = \sum_{b \in C} \tau_{C,U}(b)[Y_b] - \delta q^* \eta. \tag{18.8}$$

Each fiber is a smooth connected prime divisor. The sum is a positive closed current of finite mass, as also follows locally from the positive current $\mathrm{dd^c}(w \circ q) + \delta q^* \eta$.

Theorem 18.7.1 *In this setting, v realizes the entire divisorial trace data of u along Y:*

$$\nu(v \circ \pi, E) = \tau_{Y,u}(E)$$

for every prime divisor E on every smooth compact modification $\pi : Z \to Y$. More explicitly, write $h = q \circ \pi$. If $h(E) = C$, both sides are zero. If $h(E) = \{b\}$, choose a local coordinate t_b on C vanishing at b. Both sides equal

$$\mathrm{ord}_E(t_b \circ h)\, \tau_{C,U}(b).$$

Choose any Hermitian form Ω_0 with $v \in \mathrm{PSH}(Y, \Omega_0)$, and set $\Omega = 2\Omega_0$. Then

$$\mathcal{T}_\Omega(u; Y) = P_\Omega[v] = P_\Omega[v]_{\mathcal{I}} \sim v. \tag{18.9}$$

The envelope has positive finite extended mass and satisfies the prescribed-type equation conclusions of Corollary 18.6.1. Its ordinary type is independent of the chosen Green normalization, smooth probability form, and sufficiently large background.

Proof Lemma 1.16.4 and restriction to Y give

$$\mathfrak{a}_t(u; Y) = \mathfrak{a}_t(U; C)\mathcal{O}_Y.$$

The nonzero coherent ideal on the smooth curve is locally generated at b by $t_b^{k_t(b)}$, where $k_t(b) = \mathrm{ord}_b\, \mathfrak{a}_t(U; C)$. For a vertical divisor E the order of its extension is therefore $k_t(b)\, \mathrm{ord}_E(t_b \circ h)$. For a divisor dominating C, the generic order is zero, since this curve ideal has only finitely many zeros. Divide by t and use the defining trace limit. This gives exactly the stated formula for $\tau_{Y,u}(E)$.

For a vertical E, at a general point its local equation $z_1 = 0$ satisfies $t_b \circ h = z_1^k g$ with g holomorphic and nonvanishing. A local holomorphic kth root of g changes this coordinate so that $t_b \circ h = z_1^k$. The one-variable Lelong number of $w(z_1^k)$ is $k\nu(w, b) = k\tau_{C,U}(b)$, by the definition through circular means. This is also its generic divisorial Lelong number in the product coordinates. For a horizontal E, its regular locus contains a point mapped to a point where w is finite; the nonconstant map from that locus to C has open image. The pullback is finite there, so its generic Lelong number on E is zero. This proves exact realization on every compact model.

By Lemma 1.17.4 and (18.8), a quasi-plurisubharmonic a is more singular than v if and only if $\nu(a, Y_b) \geq \tau_{C,U}(b)$ for each positive coefficient. These fiber inequalities are among the defining trace constraints, and ordinary domination implies all the

constraints on further modifications, by the exact formula just proved. Thus the trace competitor family equals the ordinary-envelope family, and $\mathcal{T}_{\overline{\Omega}}(u;Y) = P_\Omega[v]$.

For $\psi = P_\Omega[v]_{\mathcal{I}}$, ideal projection preserves the fiber Lelong numbers and the same domination lemma gives $\psi \preceq v$. Hence

$$v - \sup_Y v \le P_\Omega[v] \le \psi \preceq v.$$

Since $\psi \le 0$, it belongs to the ordinary-envelope family and is at most $P_\Omega[v]$. This proves (18.9).

The potential $v - \sup_Y v$ supplies a competitor in the half-sized background. The fixed-point contact estimate Proposition 12.1.4 gives finite mass, and Lemma 12.5.2 gives strict positivity. The two envelope properties satisfy (12.1), so Theorem 13.11.1 supplies the stated equations.

Two Green constructions on C differ smoothly, by the scalar Poisson argument in Theorem 18.3.1. Their pullbacks therefore have the same ordinary type. Every background in the statement gives an envelope ordinarily equivalent to this pullback. This proves the claimed independence, at the level of ordinary type rather than equality of normalized potentials. □

The same construction respects sums and positive real scalars at the level of ordinary types. Indeed the curve trace measures add by Proposition 18.1.1; their Green representatives add with one fixed kernel, and pullback preserves the sum. Each envelope in a sufficiently large background represents that same ordinary type.

18.8 Product realizations and their measures

For a sum of potentials on a product with a curve, dimension forces the Monge–Ampère expansion to contain one curve factor. The resulting product formula computes the measure of the pulled-back Green trace and solves equations with separated densities. A transverse potential with trivial multiplier ideals can preserve every trace coefficient while changing finite mass to infinite mass.

Let C be a compact connected curve with a smooth positive probability form η, and let (Z, ω_Z) be a compact connected Hermitian manifold of dimension $k \ge 1$. On $Y = C \times Z$, let $q : Y \to C$ and $r : Y \to Z$ be the projections. Choose $A > 0$ and set

$$\Omega = q^*(A\eta) + r^*\omega_Z.$$

For positive Borel measures μ on C and ν on Z, $\mu \otimes \nu$ denotes their usual product measure. The extended measures below are sigma-finite by Theorem 6.1.1, so this notation remains meaningful when a total mass is infinite.

Proposition 18.8.1 *For $w \in \mathrm{PSH}(C, A\eta)$ and $h \in \mathrm{PSH}(Z, \omega_Z)$, the potential $a = w \circ q + h \circ r$ is Ω-plurisubharmonic and*

$$\mathrm{MA}^{\mathrm{ext}}_\Omega(a) = (k+1)\,\mathrm{MA}^{\mathrm{ext}}_{A\eta}(w) \otimes \mathrm{MA}^{\mathrm{ext}}_{\omega_Z}(h). \tag{18.10}$$

Proof The curvature is the sum of the two positive pullback currents, proving positivity. Suppose first that both potentials are bounded. Multilinearity of the bounded products leaves only the term with one curve factor and k factors from Z: two curve factors vanish by dimension, as do $k+1$ factors from Z. The binomial coefficient is $k+1$. This gives the formula for bounded potentials. To justify the exterior product identity at this regularity, use local psh corrections on each factor and decreasing convolution on smaller product charts. For smooth approximants it is the ordinary exterior product of forms. Local Bedford–Taylor convergence and the finite smooth-background expansion give weak convergence of the factor measures with uniformly bounded variation on compact subcharts. Products then converge weakly: first test against products of continuous functions and then use uniform approximation by finite sums of such products. This proves the identity in the limit.

For general potentials subtract constants so that $w, h \le 0$. Set $w_j = \max(w, -j)$ and $h_j = \max(h, -j)$. On

$$R_j = \{w > -j\} \times \{h > -j\}$$

the sum agrees with the bounded potential $w_j \circ q + h_j \circ r$. This is a plurifine open Borel set, and it lies in $\{a > -2j\}$. Locality of the bounded truncation products therefore identifies the extended measure of a restricted to R_j with that of this bounded sum. The bounded formula identifies it with $k+1$ times the product of the two restricted truncation measures. These are exactly the restrictions of the two extended factor measures.

The sets R_j increase to the product of the complements of the two pole sets. Both sides of the proposed identity give the remaining set zero mass. For the left side it is the pole set of the sum; for the right side this follows from the non-pluripolar property of each factor and sigma-finiteness of the product measures. Monotone convergence of positive measures proves the formula, even when the resulting mass is infinite. □

Suppose now that D is the finite atomic trace measure of a curve problem as above, put $\delta = D(C)$, and let w have $\mathrm{dd^c} w = D - \delta\eta$. For $A > \delta$, the canonical product representative $v = w \circ q$ satisfies

$$\mathrm{MA}^{\mathrm{ext}}_{\Omega}(v) = (k+1)(A-\delta)\,\eta \otimes \omega_Z^k. \tag{18.11}$$

Thus its measure is smooth and positive, despite the possibly infinite family of singular fibers. This identity concerns the extended measure: its definition discards the divisorial part of the full curvature current.

The product formula also gives a separated-density equation in this ordinary trace type. Fix a smooth positive volume form $\mathrm{d}V_Z$ on Z and $p > 1$. Let $0 \le f \in L^p(C, \eta)$ and $0 \le g \in L^p(Z, \mathrm{d}V_Z)$ have positive integrals. The curve equation gives $w_f \sim w$ with

$$\mathrm{MA}^{\mathrm{ext}}_{A\eta}(w_f) = c_C f\eta, \qquad c_C = \frac{A-\delta}{\int_C f\eta}.$$

The bounded Hermitian equation on Z gives a bounded h_g and $c_Z > 0$ with

$$\mathrm{MA}_{\omega_Z}(h_g) = c_Z g \, \mathrm{d}V_Z.$$

Here we use Theorem 10.1.1. The sum $a = w_f \circ q + h_g \circ r$ is ordinarily equivalent to v, and

$$\mathrm{MA}^{\mathrm{ext}}_{\Omega}(a) = (k+1)c_C c_Z \, (f\eta) \otimes (g \, \mathrm{d}V_Z).$$

The tensor density belongs to L^p by Fubini. This gives an explicit separated construction; existence for nonseparated L^p data in a sufficiently large trace background follows from Corollary 18.6.1.

Proposition 18.8.2 *Suppose $v_+(\omega_Z) = +\infty$ and $A > \delta$. In the same product background there is $a \in \mathrm{PSH}(Y, \Omega)$ such that*

$$a \sim_{\mathcal{I}} v, \qquad \int_Y \mathrm{MA}^{\mathrm{ext}}_{\Omega}(v) < +\infty, \qquad \int_Y \mathrm{MA}^{\mathrm{ext}}_{\Omega}(a) = +\infty.$$

Both potentials have the same divisorial Lelong numbers on every modification. When D comes from the ambient curve problem, these are the exact trace numbers in Theorem 18.7.1. The product manifold itself fails BMP.

Proof Choose the nonpositive potential H of Theorem 6.6.1 on Z. It has infinite extended mass and e^{-tH} is integrable for every $t > 0$. Put $a = w \circ q + H \circ r$. The product formula and $A - \delta > 0$ show that its mass is infinite. Formula (18.11) gives the finite positive mass of v.

We verify ideal equivalence locally for every $t > 0$. Since $H \leq 0$, integrability against e^{-ta} implies integrability against e^{-tv}, so $\mathcal{I}(ta) \subseteq \mathcal{I}(tv)$. Conversely the submersion identity gives $\mathcal{I}(tv) = \mathcal{I}(tw)\mathcal{O}_Y$. Local generators pulled back from C are integrable against e^{-tw}; their products with e^{-tH} are integrable by Fubini. Bounded holomorphic multipliers and finite sums preserve local integrability. Thus $\mathcal{I}(tv) \subseteq \mathcal{I}(ta)$, proving equality at every positive weight. Theorem 1.16.1 then gives equality of all divisorial Lelong numbers after modification.

The infinite mass of a rules out BMP on Y by Proposition 6.5.2. □

For a concrete ambient realization, take a compact manifold M containing the curve C, a potential $U \in \mathrm{QPSH}(M)$ with $\nu(U, C) = 0$, and $X = M \times Z$. Then $Y = C \times Z$ and $u = U \circ \mathrm{pr}_M$, where $\mathrm{pr}_M : X \to M$ is the projection, satisfy the preceding fibration theorem. The factor Z can be the Hopf threefold of Theorem 8.0.1. Small logarithmic poles of U at finitely many points of C already give nonzero trace data; the curve construction also allows any countable trace data produced by the ambient ideal sequence.

The mass formula in Section 27.7 uses stronger information: an actual restriction in a closed background, with a finite divisorial part and bounded remainder on a Kähler model. Its residual class is nef and computes that specified restriction's mass. Divisorial trace data alone does not supply such a remainder, as the preceding example demonstrates.

Part V
Holomorphic sections and geometric models

Chapter 19
Holomorphic sections and Hermitian mass

Let X be a connected compact complex manifold of dimension $n \geq 1$, let L be a holomorphic line bundle with smooth Hermitian metric h, and fix $u \in \mathrm{QPSH}(X)$. Choose a smooth positive volume form $\mathrm{d}V$.

In a local holomorphic frame e, write $|e|_h^2 = \mathrm{e}^{-\chi}$ and set $\beta = \mathrm{dd^c}\chi$. This defines the smooth closed curvature form of h.

The weight selects global sections whose vanishing compensates for its singularities. Their Bergman kernels produce positive currents in the closed curvature class of L. A Gauduchon degree bounds how much vanishing a nonzero section can carry; on a submanifold, this gives a necessary condition for realizing the intrinsic trace data by restricted sections.

When $\beta + \mathrm{dd^c}u \geq 0$ on a curve and u has finitely many logarithmic poles, Riemann–Roch identifies the leading section growth with the residual curvature mass. Higher-dimensional examples explain why the choice of background matters. Homogeneous sections on a Hopf manifold detect its projective base, while an infinite-mass weight with trivial multiplier ideals leaves every section space unchanged. Section counts alone therefore cannot recover a mass computed in an unrelated Hermitian background.

19.1 Weighted sections and Bergman currents

The multiplier ideal records which sections have finite weighted norm. Evaluation on the resulting Hilbert space gives the kernel; its extremal formula turns comparisons of norms into comparisons of potentials.

For every positive integer m, set

$$\mathcal{H}_m(u) = H^0(X, L^m \otimes \mathcal{I}(mu)), \qquad \|s\|_{m,u}^2 = \int_X |s|_{h^m}^2 \mathrm{e}^{-mu}\,\mathrm{d}V.$$

The integral is taken off the pole set of u, a set of smooth volume zero. Membership in $\mathcal{H}_m(u)$ is equivalent to finiteness of this norm: the definition of the multiplier ideal gives local integrability, and compactness gives a finite covering. The converse follows by restriction to coordinate balls. Thus $\mathcal{H}_m(u)$ is a finite-dimensional Hilbert space, as a subspace of $H^0(X, L^m)$. Write $N_m = \dim \mathcal{H}_m(u)$.

For $v \in \mathrm{QPSH}(X)$, the ideal order $u \preceq_{\mathcal{I}} v$ gives $\mathcal{H}_m(u) \subseteq \mathcal{H}_m(v)$ for every m. Ordinary equivalence of weights gives the same section spaces. Their Hilbert norms satisfy the following comparison.

Proposition 19.1.1 *Let $v \in \mathrm{QPSH}(X)$, let $D, E \geq 0$, and let $F \in C^\infty(X, \mathbb{R})$. Suppose $|u - v| \leq D$ off the poles and $\mathrm{d}V' = \mathrm{e}^F \mathrm{d}V$ with $|F| \leq E$. The section spaces for $(u, \mathrm{d}V)$ and $(v, \mathrm{d}V')$ coincide. For every s in this common space and every positive integer m,*

$$\mathrm{e}^{-mD-E} \|s\|^2_{m,u} \leq \|s\|^2_{m,v,\mathrm{d}V'} \leq \mathrm{e}^{mD+E} \|s\|^2_{m,u}.$$

Proof The two integrands differ by the factor $\mathrm{e}^{-m(v-u)+F}$, which lies between the displayed constants. Integrability in either norm is therefore equivalent to integrability in the other. Integrating proves both estimates. □

Fix $m \geq 1$ and choose an orthonormal basis $s_1, \ldots, s_{N_m}$ of $\mathcal{H}_m(u)$. Define $K_{m,u}$ on X and $B_{m,u}$ off the pole set by

$$K_{m,u}(x) = \sum_{j=1}^{N_m} |s_j(x)|^2_{h^m}, \qquad B_{m,u}(x) = K_{m,u}(x)\mathrm{e}^{-mu(x)}.$$

Set $B_{m,u} = 0$ on the pole set. When $N_m = 0$, both kernels are zero. Unitary invariance makes the definition independent of the orthonormal basis.

Proposition 19.1.2 *If $N_m > 0$, then*

$$K_{m,u}(x) = \sup_{0 \neq s \in \mathcal{H}_m(u)} \frac{|s(x)|^2_{h^m}}{\|s\|^2_{m,u}}, \qquad \int_X B_{m,u}\, \mathrm{d}V = N_m.$$

The potential $\Phi_m = m^{-1} \log K_{m,u}$ is genuine, has analytic singularities, and satisfies

$$\beta + \mathrm{dd^c}\Phi_m \geq 0.$$

This positive current is closed. If $N_m = 0$, then $\Phi_m \equiv -\infty$.

Proof Write a section in the orthonormal basis. The Cauchy–Schwarz inequality bounds its evaluation by the sum of squared basis evaluations, with equality for a coefficient vector proportional to their complex conjugates after trivializing the one-dimensional fiber. At a common zero both sides vanish. This proves the extremal formula. Integrating the defining finite sum for $B_{m,u}$ sums the squared norms of the basis vectors and gives N_m.

Locally $s_j = f_j e^{\otimes m}$, so

$$\Phi_m = \frac{1}{m} \log \sum_j |f_j|^2 - \chi, \qquad \beta + \mathrm{dd^c}\Phi_m = \frac{1}{m}\mathrm{dd^c} \log \sum_j |f_j|^2 \geq 0.$$

A nonzero section prevents the sum from vanishing identically. The local expression therefore defines a genuine potential of analytic singularity type. It also proves closedness directly. □

For the weights and volume forms in Proposition 19.1.1, the extremal formula gives, whenever the common section space is nonzero,

$$|\Phi_m(u, \mathrm{d}V) - \Phi_m(v, \mathrm{d}V')| \leq D + E/m$$

off the common base locus. The two potentials have the same ordinary singularity type. In particular the smooth volume form affects these potentials by $O(m^{-1})$ when the weight is fixed.

A simultaneous change of metric and weight preserves the weighted norm exactly. Let $a \in C^\infty(X, \mathbb{R})$. If the squared metric is changed to $h' = h\mathrm{e}^{-a}$ and the weight to $u' = u - a$, their product $h'^m \mathrm{e}^{-mu'}$ is unchanged. Hence the Hilbert norm and the density $B_{m,u}\,\mathrm{d}V$ are unchanged, while

$$\beta' = \beta + \mathrm{dd^c}a, \qquad \Phi'_m = \Phi_m - a.$$

The current $\beta + \mathrm{dd^c}\Phi_m$ is consequently unchanged.

19.2 Gauduchon degree and vanishing

The zero divisor of a section is positive. Pairing its Poincaré–Lelong formula with a Gauduchon metric therefore bounds its size by the degree of the line bundle. Write $h^0(X, L^m) = \dim H^0(X, L^m)$.

For a Gauduchon metric γ on X, define the degree

$$\deg_\gamma L = \int_X \beta \wedge \gamma^{n-1}.$$

Changing h changes β by $\mathrm{dd^c}a$ for a smooth real function a. Its integral against γ^{n-1} vanishes by Stokes, so the degree is independent of the smooth metric on L.

Theorem 19.2.1 *If* $\deg_\gamma L < 0$*, then* $H^0(X, L^m) = 0$ *for every* $m \geq 1$*. If* $\deg_\gamma L = 0$ *and* $H^0(X, L^m) \neq 0$*, then* L^m *is holomorphically trivial and* $h^0(X, L^m) = 1$*. At degree zero, each nonzero weighted subspace* $\mathcal{H}_m(u)$ *is also one-dimensional.*

Proof Let s be a nonzero section of L^m with zero divisor D. The Poincaré–Lelong formula gives

$$\mathrm{dd^c} \log |s|^2_{h^m} = [D] - m\beta.$$

The logarithm is locally integrable. Testing against γ^{n-1} gives

$$m \deg_\gamma L = \int_D \gamma^{n-1} \geq 0.$$

Every nonempty effective divisor has strictly positive integral because γ is positive on its regular locus. Negative degree is therefore impossible. At degree zero the divisor is empty, so s is nowhere zero and trivializes L^m. All holomorphic functions on the connected compact manifold are constant, giving dimension one. □

The same degree also constrains positive curvature currents. If $T = \beta + \mathrm{dd^c} u \geq 0$, the trace identity of Proposition 4.2.1 gives

$$\deg_\gamma L = \int_X T \wedge \gamma^{n-1} \geq 0.$$

If the degree is zero, positivity and the strict positivity of γ imply $T = 0$.

19.3 Restricted sections and trace data

Let $Y \subset X$ be smooth, compact and connected, of positive dimension r, with $\nu(u, Y) = 0$. Use the image ideals $\mathfrak{a}_m = \mathfrak{a}_m(u; Y)$ of Section 18.1 and put

$$V_m = \mathrm{im}\big(\mathcal{H}_m(u) \longrightarrow H^0(Y, L^m|_Y)\big).$$

Thus $V_m \subseteq H^0(Y, L^m|_Y \otimes \mathfrak{a}_m)$; the right-hand space can contain sections that do not extend to X.

A nonzero restricted section has the vanishing orders imposed by the image ideal. Removing finitely many of these orders leaves an effective divisor. If such sections exist in arbitrarily large degrees, normalized compactness gives a positive current in the residual class determined by the trace data. Pairing this current with a Gauduchon metric yields the degree obstruction. Write c_1 for the real Bott–Chern first Chern class.

Theorem 19.3.1 *Suppose $V_m \neq 0$ for arbitrarily large integers m. Let $\pi : Z \to Y$ be any smooth compact modification, let $E_1, \dots, E_q$ be distinct prime divisors on Z, and let γ be a Gauduchon metric on Z. Then*

$$\deg_\gamma(\pi^*(L|_Y)) \geq \sum_{i=1}^{q} \tau_{Y,u}(E_i) \int_{E_i} \gamma^{r-1}.$$

Moreover the real closed $(1,1)$ class

$$c_1(\pi^*(L|_Y)) - \sum_{i=1}^{q} \tau_{Y,u}(E_i) c_1(\mathcal{O}_Z(E_i))$$

is pseudoeffective: it admits a closed positive current.

Proof Choose a nonzero $s_m \in V_m$ along an unbounded sequence of degrees, and put $b_i(m) = \operatorname{ord}_{E_i} \mathfrak{a}_m$. Its pullback is nonzero, and its local coefficients belong to the pulled-back image ideal. Thus

$$D_m := \operatorname{div}(\pi^* s_m) - \sum_{i=1}^{q} b_i(m) E_i \geq 0.$$

Choose smooth metrics on $\mathcal{O}_Z(E_i)$ with curvatures β_i, and set

$$\Theta_m = \pi^*(\beta|_Y) - \sum_{i=1}^{q} \frac{b_i(m)}{m} \beta_i, \qquad \Theta = \pi^*(\beta|_Y) - \sum_{i=1}^{q} \tau_{Y,u}(E_i) \beta_i.$$

Dividing by the canonical divisor sections gives a nonzero holomorphic section of $\pi^*(L^m|_Y) \otimes \mathcal{O}_Z(-\sum_i b_i(m) E_i)$ with divisor D_m. Equip this bundle with the induced quotient metric. The logarithm of the squared norm of this section, divided by m, is a global quasi-plurisubharmonic function w_m satisfying the Poincaré–Lelong identity

$$\Theta_m + \mathrm{dd^c} w_m = \frac{1}{m}[D_m] \geq 0.$$

The full integer trace limit in Theorem 18.1.1 gives $\Theta_m \to \Theta$ smoothly, also along the selected degrees. One Hermitian form Λ on Z dominates all Θ_m. Normalize $\sup_Z w_m = 0$ and apply Theorem 1.4.1 in $\mathrm{PSH}(Z, \Lambda)$ to obtain an L^1-convergent subsequence. Its limit w satisfies

$$\Theta + \mathrm{dd^c} w \geq 0.$$

Since Θ is closed, this positive current proves the asserted pseudoeffectivity in every dimension.

Finally, pair it with γ^{r-1}. Stokes removes $\mathrm{dd^c} w$, while Poincaré–Lelong gives $\int_Z \beta_i \wedge \gamma^{r-1} = \int_{E_i} \gamma^{r-1}$. The resulting nonnegative pairing is exactly the stated degree inequality. □

Equivalently, a strict violation of one of these finite Gauduchon inequalities forces $V_m = 0$ for all sufficiently large m. Otherwise an unbounded sequence of nonzero restricted spaces would contradict the theorem.

The necessary condition has no general converse. On the elliptic curve $E = \mathbb{C}/(\mathbb{Z}+\mathrm{i}\mathbb{Z})$, choose an irrational real number a and the flat line bundle with multipliers $\mathrm{e}^{2\pi\mathrm{i}a}$ and 1 on the two generators. Its constant metric has curvature zero. Take $u = 0$ and $Y = X = E$, so every trace number is zero and the displayed pseudoeffectivity condition holds. Nevertheless every positive tensor power has no nonzero holomorphic section. To see this, a hypothetical section of a power has a zero divisor of degree zero and hence is nowhere zero. Its logarithmic norm is harmonic on E, and is constant. The lifted holomorphic function therefore has constant modulus and is constant on $\mathbb{C}$. The nontrivial multiplier $\mathrm{e}^{2\pi\mathrm{i}ma}$ forces that constant to be zero, a contradiction.

19.4 Bounded weighted sections

Integrability of two weighted squared norms does not ensure integrability of their product. The logarithmic example below shows how this prevents multiplication of the selected section spaces. Bounded weighted norms are preserved under multiplication and give the graded algebra used in the next chapter.

Example 19.4.1 Put $X = \mathbb{P}^1$, let z be its affine coordinate, and let δ_0 be the Dirac mass at the origin. Take $L = \mathcal{O}(1)$ with Fubini–Study curvature β, and let

$$u(z) = \frac{3}{5}\log\frac{|z|^2}{1+|z|^2}.$$

Then $\beta + \mathrm{dd^c}u = \frac{2}{5}\beta + \frac{3}{5}\delta_0 \geq 0$, $\mathcal{I}(u) = \mathcal{O}$ and $\mathcal{I}(2u) = \mathcal{O}(-[0])$. A section of L nonzero at 0 belongs to $\mathcal{H}_1(u)$, but its square does not belong to $\mathcal{H}_2(u)$.

Proof The weight has only one singularity, $\frac{3}{5}\log|z|^2 + O(1)$ at zero. The one-variable integrability calculation gives the two stated ideals. The curvature follows from Poincaré–Lelong and the Fubini–Study potential. The section assertion is precisely the failure to vanish at zero. □

Return to the original data (X, L, h, u). Define $\mathcal{W}_0 = \mathbb{C}$ and, for $m \geq 1$,

$$\mathcal{W}_m(u) = \{s \in H^0(X, L^m) : |s|^2_{h^m}\mathrm{e}^{-mu} \text{ is bounded off } \{u = -\infty\}\}.$$

These finite-dimensional vector spaces are unchanged when the smooth metric is changed, since the ratio of two smooth metrics is bounded above and below on X. They lie in $\mathcal{H}_m(u)$, because X has finite smooth volume. For positive integers m, k, $s \in \mathcal{W}_m(u)$ and $t \in \mathcal{W}_k(u)$, the identity

$$|st|^2_{h^{m+k}}\mathrm{e}^{-(m+k)u} = \left(|s|^2_{h^m}\mathrm{e}^{-mu}\right)\left(|t|^2_{h^k}\mathrm{e}^{-ku}\right)$$

proves $\mathcal{W}_m(u)\mathcal{W}_k(u) \subseteq \mathcal{W}_{m+k}(u)$. Closure under addition follows from $|s+t|^2 \le 2|s|^2 + 2|t|^2$. Thus $\bigoplus_m \mathcal{W}_m(u)$ is a graded subalgebra of the section ring.

On a smooth projective variety, the almost-semigroup construction of [Xia, Section 10.3] treats metrics with positive curvature current on big line bundles with positive multiplier-ideal volume. It uses the integrable section spaces themselves.

19.5 Section growth on a curve

At a logarithmic pole on a curve, integrability rounds the required vanishing order down, while boundedness rounds it up. The resulting divisor degrees differ by a bounded amount. Riemann–Roch therefore gives the same leading growth for both section spaces.

Assume in this section that X is a Riemann surface and $\beta + \mathrm{dd^c}u \ge 0$. Let $p_1, \dots, p_r$ be distinct points such that u is locally bounded on $X \setminus \{p_1, \dots, p_r\}$ and, in coordinates z_i centered at p_i,

$$u = c_i \log |z_i|^2 + O(1), \qquad c_i > 0.$$

Set $\ell = \deg L = \int_X \beta$ and $d = \sum_i c_i$.

Theorem 19.5.1 *In this situation* $0 \le d \le \ell$, *and*

$$\mathcal{H}_m(u) = H^0\left(X, L^m\Big(-\sum_i \lfloor mc_i \rfloor p_i\Big)\right),$$
$$\mathcal{W}_m(u) = H^0\left(X, L^m\Big(-\sum_i \lceil mc_i \rceil p_i\Big)\right).$$

Moreover,

$$\lim_{m\to\infty} \frac{\dim \mathcal{H}_m(u)}{m} = \lim_{m\to\infty} \frac{\dim \mathcal{W}_m(u)}{m} = \ell - d = \int_X \mathrm{MA}_\beta^{\mathrm{ext}}(u).$$

Proof A germ vanishing to order k has weighted squared norm comparable to $|z_i|^{2(k-mc_i)}$. Local integrability means $k - mc_i > -1$, or $k \ge \lfloor mc_i \rfloor$; boundedness means $k \ge mc_i$, or $k \ge \lceil mc_i \rceil$. Compactness gives the displayed global identifications.

The positive curvature measure has atom c_i at p_i and total mass ℓ. Subtracting these atoms leaves a positive measure, proving $d \le \ell$. The remainder does not charge polar sets: after subtracting the logarithmic singularity, the local potential has a bounded quasi-plurisubharmonic representative. By Proposition 1.7.1, its curvature gives polar sets zero mass. Thus this remaining measure is exactly the extended measure, of mass $\ell - d$.

If $\ell > d$, the degrees of both line bundles in the display equal $m(\ell - d) + O(1)$ and tend to infinity. Let g be the genus of X. Riemann–Roch and Serre duality give $h^0 = \deg + 1 - g$ once the degree exceeds $2g - 2$. Dividing by m proves both limits.

If $\ell = d$, both degrees remain bounded. A line bundle of negative degree has no nonzero section, and one of nonnegative degree k has at most $k + 1$ sections: impose vanishing at $k + 1$ distinct points and use the negative-degree case for the kernel. Hence both dimensions are uniformly bounded, and both limits are zero. These arguments include $r = 0$. □

19.6 Homogeneous sections on a Hopf manifold

Let $n \geq 2$ and let $z = (z_1, \ldots, z_n)$ be the coordinates on $\mathbb{C}^n$. Set

$$X_n = (\mathbb{C}^n \setminus \{0\})/\langle z \mapsto \mathrm{e}^{-1} z\rangle.$$

The scalar contraction imposes an exact homogeneity condition on lifted sections. It reduces their dimension to a monomial count, while the invariant metric makes the kernel constant. The resulting growth is at most m^{n-1}, reflecting the dimension of the projective base.

On the covering space put

$$s = \log |z|^2, \qquad \alpha = \mathrm{dd^c} s, \qquad \zeta = \mathrm{d}s \wedge \mathrm{d^c} s, \qquad \omega = \alpha + \zeta.$$

The contraction changes s by -2, so the displayed forms descend to X_n. The calculation in Chapter 8, with ζ in place of its symbol β, gives

$$\alpha^n = 0, \qquad \zeta^2 = 0, \qquad \mathrm{dd^c}\zeta = -\alpha^2, \qquad \mathrm{dd^c}(\omega^{n-1}) = -(n-1)\alpha^n = 0.$$

Thus ω is Gauduchon. Under projectivization $\pi : X_n \to \mathbb{P}^{n-1}$, the form α is the pullback of the normalized Fubini–Study form. The elliptic fiber calculation in Chapter 8 gives

$$\int_{X_n} \alpha^{n-1} \wedge \zeta = 2, \qquad \int_{X_n} \omega^n = 2n.$$

Set $\mathrm{d}V = \omega^n/(2n)$, a probability measure.

For an integer $k \geq 0$ let L_k be the line bundle defined by

$$(z, t) \sim (\mathrm{e}^{-1} z, \mathrm{e}^{-k} t)$$

on $(\mathbb{C}^n \setminus \{0\}) \times \mathbb{C}$. Give it the metric $|t|^2_{h_k} = |t|^2 |z|^{-2k}$. Both factors transform inversely, so this metric descends. Its curvature is $k\alpha$. Apply the section and kernel constructions above with $L = L_k$, $h = h_k$ and this probability volume.

Proposition 19.6.1 *For $m \geq 1$, with weight $u = 0$, we have*

$$N_m := h^0(X_n, L_k^m) = \binom{km + n - 1}{n - 1}, \qquad B_{m,0} = N_m.$$

The Bergman potential is the constant $m^{-1} \log N_m$, and its curvature current is $k\alpha$. Moreover,

$$\deg_\omega L_k = 2k(n-1), \qquad \lim_{m\to\infty} m^{-n} N_m = 0, \qquad (k\alpha)^n = 0.$$

In contrast, $\int_{X_n} \omega^n = 2n > 0$.

Proof A section lifts to a holomorphic function f on $\mathbb{C}^n \setminus \{0\}$ satisfying

$$f(\mathrm{e}^{-1} z) = \mathrm{e}^{-km} f(z).$$

Since $n \geq 2$, Hartogs' removable-singularity theorem extends f to all of $\mathbb{C}^n$. In its Taylor series, a monomial of total degree j transforms by e^{-j}. Comparing coefficients shows that only degree km occurs. Conversely every homogeneous polynomial of that degree has the required equivariance. Counting monomials gives the binomial coefficient, including the constant polynomial when $k = 0$.

The inner product defined by h_k^m and $\mathrm{d}V$ is invariant under $U(n)$, so the basis-independent function $K_{m,0}$ is $U(n)$-invariant. It is also unchanged under any positive radial dilation: every section is homogeneous of degree km, and the factor $|z|^{-2km}$ in its squared norm cancels that dilation. Unitary transformations and positive radial dilations act transitively on $\mathbb{C}^n \setminus \{0\}$. Consequently $K_{m,0}$ is constant. Its integral is N_m by the orthonormal-basis identity, and $\mathrm{d}V$ has mass one. Thus $K_{m,0} = B_{m,0} = N_m$.

The potential and curvature statements follow immediately. Finally,

$$\int_{X_n} k\alpha \wedge \omega^{n-1} = k(n-1) \int_{X_n} \alpha^{n-1} \wedge \zeta = 2k(n-1).$$

The binomial coefficient grows at most as m^{n-1}, proving the normalized limit, and $\alpha^n = 0$ proves the vanishing curvature mass. □

For $n = 3$ and $k > 0$, these positive-degree line bundles have quadratic section growth on the Gauduchon threefold of Theorem 8.0.1. The unrestricted Hermitian masses on that same manifold are unbounded.

19.7 Section counts and infinite Hermitian mass

Return to the initial compact setting. A weight with trivial multiplier ideals imposes no integrability restriction on global sections. The infinite-mass witness of Theorem 6.6.1 therefore leaves every section count unchanged. For the trivial bundle, all Bergman currents vanish while the Hermitian mass is infinite.

Proposition 19.7.1 *Let ω be a Hermitian metric on X with $v_+(\omega) = +\infty$. There is a potential $U \in \mathrm{PSH}(X, \omega)$ such that, with $L = \mathcal{O}_X$ and its constant metric,*

$$\dim \mathcal{H}_m(U) = 1 \quad \textit{for every } m \geq 1, \qquad \int_X \mathrm{MA}^{\mathrm{ext}}_\omega(U) = +\infty.$$

The currents produced by its global Bergman potentials are identically zero.

Proof Take U from Theorem 6.6.1, which has $\mathcal{I}(tU) = \mathcal{O}_X$ at every positive weight and infinite extended mass. Global holomorphic functions on X are constant, so every weighted section space is one-dimensional. The constant section is integrable by the ideal condition and compactness. Its normalized basis consists of a nonzero constant; therefore each Φ_m is constant. Since the trivial metric has curvature zero, its Bergman current is zero. □

In particular $m^{-n} \dim \mathcal{H}_m(U) \to 0$ while the Hermitian mass is infinite. Here the line-bundle curvature is zero and the nonlinear operator uses ω; the two quantities arise from different backgrounds.

The bounded section algebra now supplies the multiplicative structure for the flag bodies of Chapter 20. Quantitative kernel limits require further relations between the selecting weight and line-bundle curvature; these are developed in Chapter 26.

Chapter 20
Flags, convex bodies and section growth

Let X be a connected compact complex manifold of dimension $n \geq 1$, and let L be a holomorphic line bundle on X.

A complete compact flag turns the vanishing of sections into lattice points. The Gauduchon degree bounds from the preceding chapter keep their normalized values in a compact convex body. For a graded section algebra with nonzero positive-degree pieces, counting these points gives a growth exponent and a leading coefficient: the exponent is the transcendence degree of the field of section ratios, and the coefficient is the body's intrinsic lattice volume.

The flag is initially assumed to lie on X. A point blowup later supplies one on a modification while preserving the section algebra, so the count applies to every compact complex manifold. Full-dimensional growth forces a projective modification and hence bounded Hermitian Monge–Ampère mass. The Hopf threefold with unbounded mass still carries positive-degree line bundles whose bodies are triangles: their area detects quadratic section growth, although their three-dimensional volume is zero.

20.1 Flag valuations and degree bounds

Successive vanishing orders along a flag give the lexicographically first exponent in a local Taylor series. Its coefficient identifies the number of distinct values with the dimension of a section space. The effective divisor remaining at each restriction then gives a Gauduchon bound for the next order.

First suppose there are a point $p \in X$ and a flag

$$X = Y_0 \supset Y_1 \supset \cdots \supset Y_n = \{p\},$$

where Y_i is a smooth connected compact prime divisor in Y_{i-1} for $1 \leq i \leq n$. For a nonnegative integer m and a nonzero section $s \in H^0(X, L^m)$, define $\nu_1(s)$ to be its vanishing order along Y_1. Divide by the corresponding power of the canonical section

of $\mathcal{O}_X(Y_1)$ and restrict to Y_1. The result is a nonzero section of $L^m(-\nu_1 Y_1)|_{Y_1}$. Repeat the procedure down the flag. The resulting vector $\nu(s) = (\nu_1(s), \dots, \nu_n(s)) \in \mathbb{N}^n$ is its flag valuation.

Near p, choose coordinates adapted to the flag and a nonvanishing frame of L. This construction is the lexicographically first exponent of the Taylor expansion of the local coefficient of s. In particular, for nonzero sections of the indicated degrees,

$$\nu(st) = \nu(s) + \nu(t), \qquad \nu(s+t) \geq \min\{\nu(s), \nu(t)\}$$

when s, t have the same degree and $s + t \neq 0$. If their valuations are different, equality holds in the second formula. The inequality uses lexicographic order.

Lemma 20.1.1 *For every finite-dimensional vector subspace $V \subset H^0(X, L^m)$,*

$$\#\{\nu(s) : 0 \neq s \in V\} = \dim V.$$

Proof Sections with distinct valuations are linearly independent: in a nontrivial finite linear combination, the least valuation occurs in exactly one summand and cannot cancel. Thus there are at most $\dim V$ values.

For the reverse bound, proceed by induction on $\dim V$. The nonempty set of values has a least element in the lexicographic order on $\mathbb{N}^n$. The coefficient of that monomial is a nonzero linear functional on V. Its kernel has codimension one, and every nonzero member of the kernel has strictly larger valuation. By induction the kernel contributes $\dim V - 1$ distinct values, in addition to the least one. The zero-dimensional case is immediate. □

The values are uniformly bounded in terms of the tensor power. The bound follows by pairing the residual divisor at each stage with a Gauduchon metric.

Theorem 20.1.1 *There are constants $C_1, \dots, C_n \geq 0$, depending only on L and the chosen flag, such that*

$$0 \leq \nu_i(s) \leq C_i m \quad (1 \leq i \leq n)$$

for every $m \geq 1$ and every nonzero $s \in H^0(X, L^m)$.

Proof For each i, equip Y_{i-1} with a Gauduchon metric γ_i; when it is a curve, any Hermitian metric will do. Put $d_i = n - i + 1$ and

$$a_i = \int_{Y_i} \gamma_i^{d_i - 1} > 0, \qquad A_i = \deg_{\gamma_i}(L|_{Y_{i-1}}).$$

For $i = n$ the integral defining a_i is the integral of 1 over the point p, so $a_n = 1$.

For $j < i$, also put

$$B_{ij} = \deg_{\gamma_i}\left(\mathcal{O}_{Y_{j-1}}(Y_j)|_{Y_{i-1}}\right).$$

All these are fixed real numbers.

At stage i of the valuation algorithm, the residual section belongs to the line bundle

$$M_{i-1} = L^m|_{Y_{i-1}} \otimes \bigotimes_{j<i} \mathcal{O}_{Y_{j-1}}(-\nu_j(s)Y_j)|_{Y_{i-1}}.$$

Its zero divisor contains $\nu_i(s)Y_i$. Poincaré–Lelong, tested against $\gamma_i^{d_i-1}$, therefore gives

$$a_i\nu_i(s) \le mA_i - \sum_{j<i} B_{ij}\nu_j(s). \tag{20.1}$$

Define recursively

$$C_i = \frac{|A_i| + \sum_{j<i}|B_{ij}|C_j}{a_i}.$$

Induction in (20.1) gives $\nu_i(s) \le C_i m$. All choices depend only on the line bundle and flag. □

20.2 The body of a graded section algebra

The valuation semigroup records each graded dimension exactly. Its linear bound gives a compact slice of the closed cone, and the semigroup growth theorem relates the asymptotic count to the volume of this slice.

Let $W_\bullet = \bigoplus_{m\ge 0} W_m$ be a graded subalgebra of the section ring of L, with $W_0 = \mathbb{C}$. For a weight $u \in \mathrm{QPSH}(X)$, the bounded weighted section algebra $\mathcal{W}_\bullet(u)$ of Section 19.4 is one example.

Set

$$S = \{(0,0)\} \cup \{(\nu(s), m) : m \ge 1,\ 0 \ne s \in W_m\}.$$

It is an additive semigroup. Indeed the product of two nonzero sections is nonzero on the connected manifold, and valuations add. Choose constants C_i from Theorem 20.1.1. Then S lies in the closed cone

$$\{(x,t) \in \mathbb{R}^n \times \mathbb{R} : t \ge 0,\ 0 \le x_i \le C_i t\}.$$

The only point of this cone at height zero is the origin. Define

$$\Delta(W_\bullet) = \overline{\mathrm{conv}}\{m^{-1}\nu(s) : m \ge 1,\ 0 \ne s \in W_m\}.$$

This is a compact convex subset of $\mathbb{R}^n$ when the set is nonempty. If every positive-degree piece is zero, set $\Delta(W_\bullet) = \varnothing$. Equivalently it is the height-one slice of the closed cone generated by S. Write $\mathrm{vol}_{\mathbb{R}^n}$ for Euclidean volume.

Proposition 20.2.1 *For every such algebra,*

$$\limsup_{m\to\infty} m^{-n} \dim W_m \le \mathrm{vol}_{\mathbb{R}^n} \Delta(W_\bullet).$$

In particular a body with empty interior gives $\dim W_m = o(m^n)$.

Proof Let $S_m = \{v(s) : 0 \neq s \in W_m\}$. Its cardinality is $\dim W_m$ by Lemma 20.1.1, and $m^{-1}S_m \subseteq \Delta(W_\bullet)$. Attach the half-open cube $m^{-1}(a + [0,1)^n)$ to each $a \in S_m$. The cubes are disjoint, their total volume is $m^{-n} \dim W_m$, and their union is contained in

$$\Delta(W_\bullet) + [0, m^{-1}]^n.$$

These compact neighborhoods decrease to $\Delta(W_\bullet)$, so their volumes converge to its volume. This proves the inequality. A compact convex set has positive n-dimensional volume exactly when it has nonempty interior, proving the last statement. The empty-body case is immediate. □

The semigroup growth theorem recalled in [Xia, Theorem C.2.1] applies in every dimension $r \ge 0$: if an additive semigroup in $\mathbb{Z}^{r+1}$ lies in a closed convex cone in $\mathbb{R}^r \times \mathbb{R}_{\ge 0}$ with compact height-one slice and with only the origin at height zero, and generates $\mathbb{Z}^{r+1}$ as a group, then its degree-m cardinalities divided by m^r converge to the Euclidean volume of the height-one slice of its own closed cone.

Corollary 20.2.1 *If the subgroup generated by S is all of $\mathbb{Z}^{n+1}$, then*

$$\lim_{m\to\infty} m^{-n} \dim W_m = \mathrm{vol}_{\mathbb{R}^n} \Delta(W_\bullet).$$

Proof The cone bound and its height-zero condition were proved above, and the subgroup hypothesis is precisely the one in the stated semigroup theorem. Apply it and use Lemma 20.1.1. □

20.3 Intrinsic dimension, lattice and growth

A zero ambient volume can conceal growth in fewer than n directions. The group generated by the valuation semigroup determines the degree period and the lattice in which the body has full dimension. Counting in this lattice gives the growth exponent and coefficient. Section ratios identify the exponent with a transcendence degree; the limit of graded dimensions then makes the intrinsic volume independent of the flag.

Assume in this section that some positive-degree piece of $W_\bullet$ is nonzero. Let $G \subset \mathbb{Z}^{n+1}$ be the subgroup generated by S, and let pr_{n+1} be projection to the last coordinate. Define $e \ge 1$ by

$$\mathrm{pr}_{n+1}(G) = e\mathbb{Z}.$$

Thus e is the greatest common divisor of the degrees with nonzero sections. Set

$$\Lambda = G \cap (\mathbb{R}^n \times \{0\}), \qquad d = \operatorname{rank} \Lambda.$$

We identify Λ with a lattice in its real horizontal span. Its lattice measure assigns volume one to a fundamental parallelepiped. On a translate of this span use the translated measure, denoted by vol_Λ; in dimension zero a point has volume one.

Theorem 20.3.1 *The affine dimension of $\Delta(W_\bullet)$ is d. Its lattice volume is finite and strictly positive, and*

$$\lim_{q\to\infty} \frac{\dim W_{eq}}{(eq)^d} = \mathrm{vol}_\Lambda\, \Delta(W_\bullet).$$

If e does not divide m, then $W_m = 0$. In particular, all sufficiently large multiples of e have nonzero sections.

Proof Choose $g_0 = (b, e) \in G$ and an integral basis $\lambda_1, \dots, \lambda_d$ of Λ. Every element of G has a unique expression

$$\sum_{i=1}^{d} a_i\lambda_i + qg_0, \qquad (a, q) \in \mathbb{Z}^d \times \mathbb{Z}.$$

These coordinates identify G with $\mathbb{Z}^{d+1}$. Let S' be the image of S. It generates that entire lattice, and its height is $q = m/e \geq 0$.

The points $(m^{-1}x, 1)$, for $(x, m) \in S$ with $m > 0$, linearly span the $(d + 1)$-dimensional space $\mathrm{span}_{\mathbb{R}}\, G$. Since they all have height one, their affine span has dimension d. Taking the convex hull and closure does not change this affine span, proving the dimension assertion.

The linear change of coordinates carries the cone bound already proved for S to a closed cone with compact height-one slice and with only the origin at height zero. Indeed a vector (x, m) with $|x| \leq Cm$ has horizontal coordinates bounded by a constant times m, and hence by a constant times q. The height-one body for S' is

$$\Delta' = e\big(\Delta(W_\bullet) - b/e\big),$$

where the right side is expressed in the basis λ_i. It is full-dimensional in $\mathbb{R}^d$. Therefore

$$\mathrm{vol}_{\mathbb{R}^d}\, \Delta' = e^d\, \mathrm{vol}_\Lambda\, \Delta(W_\bullet) > 0.$$

Apply the semigroup growth theorem cited in Section 20.2 to S' with its full lattice. Its degree-q slice is in bijection with S_{eq}, so

$$\lim_{q\to\infty} \frac{\dim W_{eq}}{q^d} = \mathrm{vol}_{\mathbb{R}^d}\, \Delta'.$$

Dividing by e^d gives the asserted limit. Its positivity gives eventual nonvanishing. For $d = 0$ the body is a point of volume one, so the integer dimensions converge to one and are eventually equal to one. Every nonzero degree lies in the projection $e\mathbb{Z}$ of G, proving vanishing outside these multiples. □

The growth dimension also has a description independent of the flag. Write $\mathcal{M}(X)$ for the field of meromorphic functions on X and define the section-ratio field

$$K(W_\bullet) = \{s/t : s, t \in W_m \text{ for some } m \geq 0,\ t \neq 0\} \subset \mathcal{M}(X).$$

It is a field: addition and multiplication use the common denominator $t_1 t_2$ in degree $m_1 + m_2$, and inversion exchanges numerator and denominator. It contains $\mathbb{C}$ because $W_0 = \mathbb{C}$.

Proposition 20.3.1 *The transcendence degree of $K(W_\bullet)$ over $\mathbb{C}$ is exactly d. If $\Delta(W_\bullet)$ has nonempty interior in $\mathbb{R}^n$, then X has n algebraically independent meromorphic functions. Moreover d, the period e and the intrinsic volume in Theorem 20.3.1 are independent of the chosen complete compact flag whenever two such flags are available.*

Proof For $d > 0$, choose sections $s_0, \ldots, s_d$ of positive degrees $m_0, \ldots, m_d$ with affinely independent normalized valuation vectors. Let M be a common multiple of their degrees and put $t_i = s_i^{M/m_i} \in W_M$. The ratios $f_i = t_i/t_0$ belong to $K(W_\bullet)$. The flag valuation extends to quotients by subtraction, giving

$$\nu(f_i) = M\left(\frac{\nu(s_i)}{m_i} - \frac{\nu(s_0)}{m_0}\right), \qquad 1 \leq i \leq d.$$

These vectors are linearly independent. Distinct monomials in the f_i therefore have distinct valuations, so the least term in a nonzero polynomial cannot cancel. The ratios are algebraically independent. Thus the transcendence degree is at least d, including the immediate case $d = 0$.

Conversely let $f_1, \ldots, f_r$, with $r \geq 1$, be algebraically independent members of the field. Taking the product of their denominators, write $f_i = A_i/B$ with $A_i, B \in W_M$ and $B \neq 0$ for a common positive degree M. For every integer $q \geq 1$ and multi-index $a = (a_1, \ldots, a_r) \in \mathbb{N}^r$ with $|a| = a_1 + \cdots + a_r \leq q$, the sections

$$B^{q-|a|} A_1^{a_1} \cdots A_r^{a_r} \in W_{Mq}$$

are linearly independent: division by B^q turns any linear relation into a polynomial relation in the f_i. There are $\binom{q+r}{r}$ such sections. The integer M is a multiple of e, so Theorem 20.3.1 implies $\dim W_{Mq} = O(q^d)$. The binomial lower bound then forces $r \leq d$. This proves equality of transcendence degrees. A body with nonempty interior has $d = n$, giving the stated consequence for X.

The field and the set of nonzero degrees do not depend on the flag, so neither d nor e does. The lattice volume is the limit of the same dimensions with the same denominator for either flag, by Theorem 20.3.1. It is therefore independent as well. The bodies and their ambient lattices may still differ. □

20.4 Moishezon manifolds and bounded mass

When the section-ratio field has full transcendence degree, a projective modification bounds the masses of Hermitian potentials on X. The bound comes from dominating the pulled-back background by a fixed Kähler form, whose Monge–Ampère mass is unchanged by adding a smooth potential.

We use the classical projective-modification theorem: a compact irreducible complex space of dimension n with n algebraically independent meromorphic functions admits a modification from a smooth projective manifold. This form of Moishezon's theorem is recorded in Section 33.12.

Lemma 20.4.1 *If a compact complex manifold X admits a modification $\pi : Z \to X$ from a compact Kähler manifold, then X has the bounded mass property.*

Proof Fix a Hermitian form ω on X and a Kähler form κ on Z. Choose $C > 0$ with $\pi^*\omega \le C\kappa$. By Proposition 6.5.1, it suffices to bound the masses of smooth $u \in \mathrm{PSH}(X, \omega)$. Their pullbacks satisfy

$$0 \le \pi^*\omega + \mathrm{dd^c}(u \circ \pi) \le C\kappa + \mathrm{dd^c}(u \circ \pi).$$

Taking top powers and integrating gives

$$\begin{aligned}\int_X (\omega + \mathrm{dd^c} u)^n &= \int_Z (\pi^*\omega + \mathrm{dd^c}(u \circ \pi))^n \\ &\le \int_Z (C\kappa + \mathrm{dd^c}(u \circ \pi))^n = C^n \int_Z \kappa^n.\end{aligned}$$

The first equality follows by change of variables on the biholomorphic locus: smooth top-degree forms give the exceptional analytic sets zero mass. The inequality follows by expanding in the smooth semipositive difference $C\kappa - \pi^*\omega$. Since κ is closed, Stokes' theorem gives the last equality. The bound is independent of u, proving bounded mass. □

Theorem 20.4.1 *If X fails the bounded mass property, then for every complete smooth compact flag, every holomorphic line bundle L and every graded section subalgebra $W_\bullet$ as above,*

$$\mathrm{vol}_{\mathbb{R}^n} \Delta(W_\bullet) = 0, \qquad \dim W_m = o(m^n).$$

Proof A body of nonempty interior would give n algebraically independent meromorphic functions by Proposition 20.3.1. The stated Moishezon theorem would give a smooth projective, hence Kähler, modification of X. This contradicts Lemma 20.4.1. The body therefore has empty interior, and Proposition 20.2.1 gives both conclusions. □

20.5 Flags on a point blowup

We now remove the assumption that X itself carries a complete compact flag. In dimension at least two, the exceptional divisor of a point blowup carries a complete projective flag. Hartogs extension preserves holomorphic sections, so the growth formula on the modification counts the original section spaces. On a curve, a point already completes the flag.

Lemma 20.5.1 *Every connected compact complex manifold X of dimension $n \geq 1$ admits a smooth compact modification $\pi : \widetilde{X} \to X$ with a complete smooth compact flag. It may be chosen so that, for every holomorphic line bundle L and every $m \geq 0$, pullback gives*

$$H^0(X, L^m) \simeq H^0(\widetilde{X}, \pi^* L^m).$$

These identifications respect products.

Proof If $n = 1$, take π to be the identity and choose any point as the last member of the flag. If $n \geq 2$, blow up a point $p \in X$. The usual local construction of the blowup of $\mathbb{C}^n$ at the origin shows that $\widetilde{X}$ is smooth and its exceptional divisor is $E \simeq \mathbb{P}^{n-1}$. Choose a nested linear flag in E. Together with $\widetilde{X}$ it gives

$$\widetilde{X} \supset E \supset \mathbb{P}^{n-2} \supset \cdots \supset \mathbb{P}^0.$$

Every member is smooth and compact, and is a prime divisor in the preceding member. The modification is an isomorphism off E.

Let s be a section of $\pi^* L^m$. Off E, it gives a section of L^m on $X \setminus \{p\}$. In a trivialization of L^m on a coordinate ball at p, its coefficient is holomorphic on the punctured ball. The classical Hartogs extension theorem in dimension $n \geq 2$ extends that coefficient holomorphically across p. It glues with the section away from p. Its pullback agrees with s off E, and hence everywhere by the identity theorem. This proves surjectivity; injectivity and product compatibility follow from equality on the same dense open set. The identity-map case is immediate. □

On the modification in Lemma 20.5.1, apply the flag construction to $\widetilde{W}_m = \pi^* W_m \subseteq H^0(\widetilde{X}, \pi^* L^m)$. The resulting body's shape may depend on the point and flag; its dimension and normalized count are determined by the original section algebra.

Theorem 20.5.1 *Assume that some positive-degree piece of $W_\bullet$ is nonzero. There are an integer $0 \leq d \leq n$, a degree period $e \geq 1$, and a number $V(W_\bullet) > 0$ such that*

$$\lim_{q \to \infty} \frac{\dim W_{eq}}{(eq)^d} = V(W_\bullet), \qquad W_m = 0 \quad \text{if } e \nmid m.$$

Here d is the transcendence degree of the section-ratio field $K(W_\bullet)$, e is the greatest common divisor of the nonzero degrees, and $V(W_\bullet)$ is the intrinsic lattice volume of every auxiliary body constructed above. In particular these three quantities are independent of the auxiliary modification and flag.

Proof Pullback preserves the graded dimensions and the nonzero degrees. Apply Theorems 20.1.1 and 20.3.1 on $\widetilde{X}$ to obtain the displayed limit and positive lattice volume. Pullback also identifies the section-ratio fields: their elements are the corresponding ratios, and injectivity follows on the dense isomorphism locus. Thus Proposition 20.3.1 identifies d with the original field's transcendence degree. The unchanged nonzero degrees determine e, and the limit of the unchanged dimensions determines $V(W_\bullet)$. All three are independent of the auxiliary choices. □

Corollary 20.5.1 *For every holomorphic line bundle L on a connected compact complex manifold X of dimension n,*

$$h^0(X, L^m) = O(m^n).$$

If X is not Moishezon, then, for each L, either all its positive-degree section spaces vanish, or there is an integer $d < n$ for which

$$h^0(X, L^m) = O(m^d).$$

In particular $m^{-n} h^0(X, L^m) \to 0$. These conclusions apply when X fails the bounded mass property.

Proof Apply the theorem to the full section ring, taking the zero-ring-in-positive-degrees case separately. The periodic count gives $O(m^d)$ for all degrees, with $d \le n$. If $d = n$, the section-ratio field supplies n algebraically independent meromorphic functions on X. By Moishezon's theorem, X admits a projective modification and is Moishezon. Thus a non-Moishezon manifold has $d < n$. Finally, a Moishezon manifold has the bounded mass property by Lemma 20.4.1. Its contrapositive proves the last assertion. □

The same upper bounds apply to multiplier-ideal section spaces by their inclusion in $H^0(X, L^m)$. Identifying weighted spaces on a modification additionally requires the Jacobian correction; Lemma 1.22.4 gives that formula with the relative canonical bundle.

20.6 The Hopf flag and its lower-dimensional body

The coordinate divisors on a scalar Hopf manifold reduce a homogeneous section to a nowhere-vanishing section on an elliptic fiber. The final valuation coordinate is therefore zero, and the remaining coordinates fill a simplex whose intrinsic volume gives the section growth.

Use X_n and L_k from Proposition 19.6.1, with $n \ge 2$ and $k \ge 0$. In the covering coordinates $z_1, \dots, z_n$, choose a point p on the elliptic curve $z_1 = \cdots = z_{n-1} = 0$. The flag is $Y_0 = X_n$ and

$$Y_i = \{z_1 = \cdots = z_i = 0\}/\langle \mathrm{e}^{-1} \rangle \quad (1 \le i \le n-1), \qquad Y_n = \{p\} \subset Y_{n-1}.$$

Each member is smooth and compact and is a prime divisor in the preceding member. Take $W_m = H^0(X_n, L_k^m)$.

Proposition 20.6.1 *For this flag,*

$$\{\nu(s) : 0 \neq s \in W_m\} = \{(a_1, \dots, a_{n-1}, 0) : a_i \in \mathbb{N},\ \textstyle\sum_i a_i \leq km\}.$$

Consequently

$$\Delta(W_\bullet) = \{(x_1, \dots, x_{n-1}, 0) : x_i \geq 0,\ \textstyle\sum_i x_i \leq k\}.$$

Its n-dimensional volume is zero. When $k > 0$, its volume in the coordinate hyperplane $x_n = 0$, with the Euclidean measure for which the coordinate lattice has covolume one, is

$$\frac{k^{n-1}}{(n-1)!} = \lim_{m\to\infty} \frac{\dim W_m}{m^{n-1}}.$$

For $k = 0$ the body is a point and $\dim W_m = 1$.

Proof By Proposition 19.6.1, sections are homogeneous polynomials of degree km. Dividing successively by the least powers of $z_1, \dots, z_{n-1}$ and restricting leaves a nonzero multiple of $z_n^{km-a_1-\cdots-a_{n-1}}$. This residual section has no zeros on Y_{n-1}, so the final valuation is zero. The preceding coordinates are nonnegative and sum to at most km.

Conversely every such tuple is realized by the monomial

$$z_1^{a_1} \cdots z_{n-1}^{a_{n-1}} z_n^{km-a_1-\cdots-a_{n-1}}.$$

The normalized values lie in the displayed simplex and include all its vertices in degree one. Their closed convex hull is therefore that simplex. For $k > 0$, its coordinate-lattice volume is $k^{n-1}/(n-1)!$, the leading coefficient of the same proposition's binomial count. For $k = 0$, only constant polynomials remain.

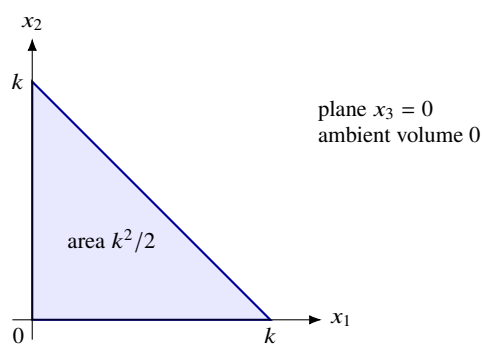

Fig. 20.1 For the Hopf threefold and $k > 0$, the body is a triangle in a coordinate plane. Its area counts quadratic section growth; its ambient volume vanishes.

Chapter 21
Hopf fibrations and toric potentials

Fix $n \geq 2$ and use the scalar Hopf manifold X_n, projection $\pi : X_n \to \mathbb{P}^{n-1}$ and forms α, ζ, ω of Proposition 19.6.1. Write σ for the Fubini–Study form normalized by $\int_{\mathbb{P}^{n-1}} \sigma^{n-1} = 1$.

Potentials in the pulled-back closed classes descend to the projective base because the elliptic fibers are compact. Fiber integration identifies the weighted section norms on both spaces. For descended ω-plurisubharmonic potentials, it also computes the Monge–Ampère measure and reduces the equation to the base. There closedness fixes the normalization and gives uniqueness, although unrestricted masses on the Hopf threefold are unbounded. For smooth base potentials invariant under separate coordinate rotations, logarithmic coordinates turn curvature positivity into convexity; strict positivity on the base makes the gradient parametrize the relative interior of a simplex.

The fiber calculations use the identities, for $y \in \mathbb{P}^{n-1}$,

$$\alpha = \pi^*\sigma, \qquad \omega = \alpha + \zeta, \qquad \zeta^2 = 0, \qquad \mathrm{d}V = \tfrac{1}{2}\alpha^{n-1} \wedge \zeta, \qquad \int_{\pi^{-1}(y)} \zeta = 2.$$

21.1 Weighted sections on the projective base

The line bundles L_k from Proposition 19.6.1, for integers $k \geq 0$, are the pullbacks of $\mathcal{O}(k)$ with its Fubini–Study metric. Represent a section by a homogeneous polynomial f of degree k in the covering coordinates $z \in \mathbb{C}^n \setminus \{0\}$. Its squared norm $|f(z)|^2|z|^{-2k}$ is invariant under every nonzero complex scalar. Descent of the weight therefore makes the weighted integrand a function on the base, and the fiber integral identifies the norms and kernels of Section 19.1.

Theorem 21.1.1 *Let $k \geq 0$ be an integer and $u \in \mathrm{PSH}(X_n, k\alpha)$. Then $u = \pi^* v$ for a unique $v \in \mathrm{PSH}(\mathbb{P}^{n-1}, k\sigma)$. For every positive integer m, pullback identifies the weighted section spaces isometrically; for a section s in the displayed space on the base,*

$$H^0\big(\mathbb{P}^{n-1}, \mathcal{O}(km) \otimes \mathcal{I}(mv)\big) \simeq H^0\big(X_n, L_k^m \otimes \mathcal{I}(mu)\big),$$
$$\|\pi^* s\|^2_{X_n,m,u} = \|s\|^2_{\mathbb{P}^{n-1},m,v}.$$

The norm on the base uses σ^{n-1}. The kernels K and, off their pole sets, B are pullbacks of the corresponding base kernels. Pullback also identifies the bounded weighted graded section algebras.

Proof Descent of the potential follows from Proposition 1.22.1. The unweighted section spaces on both manifolds consist of the same homogeneous polynomials of degree km, by the proof of Proposition 19.6.1. Pointwise norms of corresponding sections agree by the metric identification. For every nonnegative Borel function F on the base, fiber integration gives

$$\int_{X_n} (F \circ \pi)\, \mathrm{d}V = \frac{1}{2} \int_{\mathbb{P}^{n-1}} F\sigma^{n-1} \int_{\pi^{-1}(y)} \zeta = \int_{\mathbb{P}^{n-1}} F\sigma^{n-1}.$$

Apply this formula to $F = |s|^2 \mathrm{e}^{-mv}$, with the value on the pole set immaterial. It proves equality of the possibly infinite integrals. Finiteness characterizes the multiplier-ideal section spaces, giving the asserted isometry. An orthonormal basis therefore pulls back to an orthonormal basis, which proves the kernel identities. Finally, a nonnegative function on the base is bounded if and only if its pullback is bounded, proving the statement about bounded weighted sections and their multiplication. □

21.2 Descended measures and equations

The form ζ supplies the fiber direction missing from the pulled-back curvature. Its square vanishes, so the Hermitian top power of a descended potential has a single mixed term. Integrating this term determines the mass and reduces the equation within this class to the base.

If ν is a positive Borel measure on $\mathbb{P}^{n-1}$, for nonnegative Borel test functions F on X_n we write $\pi^*\nu \wedge \zeta$ for the measure defined by

$$\int_{X_n} F\, \pi^*\nu \wedge \zeta = \int_{\mathbb{P}^{n-1}} \left(\int_{\pi^{-1}(y)} F\zeta \right) \mathrm{d}\nu(y).$$

For a smooth top-degree form ν, this agrees with its pullback wedged with ζ.

Theorem 21.2.1 *For a quasi-plurisubharmonic function v on $\mathbb{P}^{n-1}$, the pullback $u = \pi^* v$ belongs to* $\mathrm{PSH}(X_n, \omega)$ *if and only if* $v \in \mathrm{PSH}(\mathbb{P}^{n-1}, \sigma)$. *In this case*

$$\mathrm{MA}^{\mathrm{ext}}_{\omega}(u) = n\,\pi^* \mathrm{MA}^{\mathrm{ext}}_{\sigma}(v) \wedge \zeta.$$

Consequently

$$\int_{X_n} \mathrm{MA}^{\mathrm{ext}}_{\omega}(u) = 2n \int_{\mathbb{P}^{n-1}} \mathrm{MA}^{\mathrm{ext}}_{\sigma}(v) \le 2n.$$

Equality holds for bounded v. For an integer $k \ge 0$ and $v \in \mathrm{PSH}(\mathbb{P}^{n-1}, k\sigma)$,

$$\mathrm{MA}^{\mathrm{ext}}_{k\alpha}(\pi^* v) = 0$$

as an n-dimensional measure on X_n.

Proof If v is σ-plurisubharmonic, then

$$\omega + \mathrm{dd^c} u = \pi^*(\sigma + \mathrm{dd^c} v) + \zeta \ge 0.$$

Conversely, if the left side is positive, wedge it with the smooth semipositive form ζ and push forward. Since $\zeta^2 = 0$ and its fiber integral is two,

$$\pi_*\big((\omega + \mathrm{dd^c} u) \wedge \zeta\big) = 2(\sigma + \mathrm{dd^c} v) \ge 0.$$

This identity of currents follows by testing against a smooth form on the base and integrating in the fiber variables. It proves the converse.

For the measure formula, first take bounded $v \in \mathrm{PSH}(\mathbb{P}^{n-1}, \sigma)$. By Lemma 1.7.2, its bounded products commute with π^*. Since $\zeta^2 = 0$ and the base has dimension $n-1$, expansion gives

$$(\omega + \mathrm{dd^c} u)^n = n\,\pi^*(\sigma + \mathrm{dd^c} v)^{n-1} \wedge \zeta.$$

The definition of the measure on the right agrees with this local product by fiber integration.

For general v, normalize $v \le 0$ and use $v_j = \max(v, -j)$. Because σ is positive, these are bounded σ-plurisubharmonic functions. Their pullbacks are exactly $u_j = \max(u, -j)$. Multiply the bounded identity by the common indicator $\mathbf{1}_{\{u > -j\}} = \pi^* \mathbf{1}_{\{v > -j\}}$ and pass to the increasing limits defining the extended measures. Fiber integration for nonnegative functions justifies the limit on the right, including at poles.

The mass formula follows because each fiber has ζ-mass two. The base extended mass is at most one: its truncation measure restricted to $\{v > -j\}$ is bounded above in total mass by the full bounded product, whose mass is $\int \sigma^{n-1} = 1$ by closed Stokes. For bounded v no part is removed and equality holds.

For the final assertion, take bounded truncations of $v \in \mathrm{PSH}(\mathbb{P}^{n-1}, k\sigma)$. Their pulled-back currents have zero nth power by Lemma 1.7.2. Restricting to the truncation levels and taking the increasing limit gives zero extended measure, also when $k = 0$.□

If an ω-plurisubharmonic potential is constant on the fibers, the local-section argument in Proposition 1.22.1 gives a quasi-plurisubharmonic base potential. The theorem therefore bounds its Hermitian mass by $2n$. On the Hopf threefold this bound is six, so the unbounded masses in Chapter 8 require potentials that vary along the fibers.

For the equation, let $f \geq 0$ be in $L^p(\mathbb{P}^{n-1}, \sigma^{n-1})$, $p > 1$, with $\int f\sigma^{n-1} > 0$. Among potentials $u = \pi^* v$ with bounded $v \in \mathrm{PSH}(\mathbb{P}^{n-1}, \sigma)$,

$$(\omega + \mathrm{dd^c} u)^n = c(f \circ \pi)\,\mathrm{d}V, \qquad c > 0, \tag{21.1}$$

is equivalent to the base equation

$$(\sigma + \mathrm{dd^c} v)^{n-1} = \frac{c}{2n} f\sigma^{n-1}.$$

The base equation implies (21.1) by Theorem 21.2.1 and the formula for $\mathrm{d}V$. Conversely, pushing (21.1) forward gives $2n(\sigma + \mathrm{dd^c} v)^{n-1} = cf\sigma^{n-1}$.

Integrating the base equation fixes

$$c = \frac{2n}{\int_{\mathbb{P}^{n-1}} f\sigma^{n-1}}.$$

Since σ is closed, it satisfies the Guan–Li condition. Apply Theorem 11.5.1 to $(\mathbb{P}^{n-1}, \sigma)$. It gives a continuous base solution with the scalar just computed, unique up to constants among bounded σ-plurisubharmonic functions. Consequently (21.1) has a continuous solution unique up to constants within the descended class. For smooth positive f, Theorem 9.1.1 on the same base gives smooth v, with the same scalar fixed by Stokes.

For a real parameter λ, the same calculation identifies an equation with right side $c\mathrm{e}^{\lambda u}(f \circ \pi)\mathrm{d}V$ with the base equation having right side $(c/(2n))\mathrm{e}^{\lambda v} f\sigma^{n-1}$.

21.3 Logarithmic coordinates and the gradient simplex

Projective scaling leaves one null direction in the logarithmic Hessian. On its quotient, positivity becomes convexity, and boundedness of the potential forces the gradient into a simplex. The logarithmic growth gives every interior gradient value; strict positivity gives uniqueness on the quotient.

Here allow $k > 0$ to be real. Let v be a smooth real function on $\mathbb{P}^{n-1}$ invariant under separate rotations of the homogeneous coordinates, and put $u = \pi^* v$. In the covering coordinates $z_1, \ldots, z_n$, on the locus where all coordinates are nonzero, set $x_i = \log |z_i|^2$ and

$$F(x) = k \log \left(\sum_{i=1}^{n} \mathrm{e}^{x_i} \right) + u(x).$$

Write $\nabla F = (F_i)$ and $D^2F = (F_{ij})$ for its Euclidean gradient and Hessian, and let $\mathbf{1} = (1, \dots, 1)$. This invariance makes u a function of x, while projective scaling gives, for every $t \in \mathbb{R}$,

$$F(x + t\mathbf{1}) = F(x) + kt. \tag{21.2}$$

Proposition 21.3.1 *On this locus, positivity of $k\alpha + \mathrm{dd^c}u$ is equivalent to convexity of F. When F is convex, its gradient lies in*

$$\Sigma_k = \{p \in \mathbb{R}^n : p_i \geq 0,\ \textstyle\sum_i p_i = k\},$$

and $D^2F\,\mathbf{1} = 0$. If $k\sigma + \mathrm{dd^c}v$ is positive definite on the base, the gradient induces a bijection from $\mathbb{R}^n/\mathbb{R}\mathbf{1}$ onto the relative interior of Σ_k.

Proof On the covering space, $k\alpha + \mathrm{dd^c}u = \mathrm{dd^c}F$. In the coordinates z_i, its Hermitian coefficient matrix is a positive constant times

$$\left(\frac{F_{ij}}{z_i\bar{z}_j}\right)_{i,j}.$$

Congruence by the invertible diagonal matrix with entries $1/z_i$ proves the equivalence with the nonnegative real Hessian of F. Differentiating (21.2) gives $\sum_i F_i = k$ and $D^2F\,\mathbf{1} = 0$.

Boundedness of v on the compact base gives a constant C such that

$$\left|F(x) - k\log\sum_i \mathrm{e}^{x_i}\right| \leq C.$$

Fix x and a coordinate vector e_i. The convex function $t \mapsto F(x + te_i)$ is bounded above as $t \to -\infty$, since $n \geq 2$. Its derivative cannot be negative: the supporting line at such a point would tend to $+\infty$ as $t \to -\infty$. Thus $F_i \geq 0$, proving $\nabla F \in \Sigma_k$.

Assume the base form is positive definite. The Hessian is positive definite transverse to $\mathbf{1}$, because the differential of projectivization has kernel exactly the complex scalar direction. In particular F is strictly convex on $H = \{x : \sum_i x_i = 0\}$. Along each line $x + te_i$, its derivative is strictly increasing and nonnegative everywhere, hence positive. Thus every gradient lies in the relative interior of Σ_k.

Conversely take $p_i > 0$ with $\sum_i p_i = k$. The function $F(x) - p \cdot x$ is translation invariant along $\mathbf{1}$. On H it tends to $+\infty$ as $|x| \to \infty$. To see this, note that

$$k\max_i x_i - p \cdot x = \sum_i p_i(\max_j x_j - x_i)$$

is strictly positive on every nonzero vector of H, and has a positive minimum on its unit sphere. The bounded difference between F and $k\log\sum \mathrm{e}^{x_i}$, together with $\log\sum \mathrm{e}^{x_i} \geq \max_i x_i$, gives the claimed coercivity. There is thus a unique minimizer on H. Its critical-point equation says that $\nabla F - p$ is a multiple of $\mathbf{1}$; both vectors have coordinate sum k, so this multiple is zero. This proves surjectivity. Strict convexity on H proves injectivity. □

Chapter 22
Divisorial currents on compact modifications

Let X be a connected compact complex manifold and $Y \subset X$ a connected compact complex submanifold of dimension $r \geq 1$. Fix a smooth real $(1,1)$-form θ and a Hermitian form ω on X, and let $u \in \mathrm{PSH}(X,\theta)$ satisfy $\nu(u,Y) = 0$. We study the divisorial trace data $\tau_{Y,u}$ of Chapter 18 on smooth compact modifications of Y.

A model carries only part of this data: blowing up can reveal vanishing that had higher codimension on the original manifold. Summing the visible coefficients gives a closed positive current on each model, controlled by its Gauduchon mass. These currents agree under pushforward, while pullback acquires an effective exceptional correction. The full trace envelope, which imposes the inequalities on every model at once, pulls back exactly. The difference explains why the ordinary type of a model's divisorial part need not realize the full trace data.

The divisorial currents are closed even when θ is not. With a closed restricted background, subtracting them gives residual Bott–Chern classes compatible between models. To identify the exceptional correction for an infinite divisorial sum, we need continuity of pullback for positive closed currents. A scalar Gauduchon equation supplies this continuity. Residual classes with a nonzero prescribed exterior derivative belong to the affine spaces of Chapter 23.

Use the analytic approximants u_m and errors ϵ_m of Corollary 1.15.1. For a smooth compact modification $\pi : Z \to Y$, put

$$\theta_Z = \pi^*(\theta|_Y), \qquad \omega_Z^0 = \pi^*(\omega|_Y), \qquad b_m(E) = \mathrm{ord}_E\big(\mathcal{I}(mu)\mathcal{O}_Y\big) \quad (m \in \mathbb{Z}_{\geq 1}).$$

Here E is a prime divisor on Z, and the order is that of the image ideal pulled back to Z. Write $[E]$ for its integration current. By Theorem 18.1.1,

$$\frac{b_m(E)}{m} \longrightarrow \tau_{Y,u}(E), \qquad \tau_{Y,u}(E) = \sup_{m\geq 1} \frac{b_m(E)}{m}.$$

The analytic approximants satisfy $\mathrm{dd^c}((u_m|_Y)\circ\pi) \geq -\theta_Z - \epsilon_m\omega_Z^0$ by Corollary 1.22.1.

22.1 Divisorial trace currents

The restricted analytic approximants dominate the finite divisors of their image ideals. Pairing with a Gauduchon metric removes the potential term from this inequality. The resulting bound survives the limit of each coefficient and controls their entire sum.

For a positive integer m, with E ranging over prime divisors on Z, define the finite divisorial current

$$D_{m,Z} = \frac{1}{m} \sum_E b_m(E)[E].$$

The sum is finite because the image ideal on the smooth compact manifold Z is coherent and nonzero. Its codimension-one zero components form a finite family. Define initially through finite subsums

$$S_Z(u;Y) = \sum_E \tau_{Y,u}(E)[E].$$

Theorem 22.1.1 *The last sum defines a closed positive* $(1,1)$*-current with finite mass on* Z*. For every Gauduchon metric* γ *on* Z*,*

$$\int_Z S_Z(u;Y) \wedge \gamma^{r-1} \le \int_Z \theta_Z \wedge \gamma^{r-1}. \tag{22.1}$$

For $r = 1$ *take* $\gamma^{r-1} = 1$*. Moreover*

$$0 \le D_{m,Z} \le S_Z(u;Y), \qquad \int_Z \big(S_Z(u;Y) - D_{m,Z}\big) \wedge \gamma^{r-1} \longrightarrow 0.$$

In particular $D_{m,Z}$ *converges weakly as a current to* $S_Z(u;Y)$*. The dyadic sequence* $D_{2^j,Z}$ *increases.*

Proof Set $w_m = (u_m|_Y) \circ \pi$. By Corollary 1.22.1, these pullbacks are genuine, and

$$T_m = \theta_Z + \epsilon_m \omega_Z^0 + \mathrm{dd^c} w_m \ge 0, \qquad \nu(w_m, E) = b_m(E)/m.$$

Lemma 1.17.1 gives $D_{m,Z} \le T_m$. Current Stokes and the Gauduchon identity therefore give

$$\int_Z D_{m,Z} \wedge \gamma^{r-1} \le \int_Z \theta_Z \wedge \gamma^{r-1} + \epsilon_m \int_Z \omega_Z^0 \wedge \gamma^{r-1}.$$

Pass to the limit in the coefficients of any fixed finite set of prime divisors, using Theorem 18.1.1. Then take the supremum over finite sets. This proves (22.1) as a finite-subsum bound.

Every prime divisor has strictly positive γ-area. Hence there are at most countably many positive coefficients: for each positive integer k, at most finitely many terms can have weighted mass at least $1/k$, and their union includes every positive term. Enumerate these divisors. Their finite partial sums are increasing positive closed

currents with a uniform trace-mass bound. They therefore converge, in coefficient measures and weakly as currents, to a positive current of finite mass. Distributional differentiation is continuous on currents, so the limit is closed. This constructs $S_Z(u;Y)$ and proves the integral inequality.

The supremum formula for $\tau_{Y,u}(E)$ gives $b_m(E)/m \le \tau_{Y,u}(E)$ for every m and E. The full integer limit gives coefficientwise convergence. Apply dominated convergence to the countable series with summable weights $\tau_{Y,u}(E)\int_E \gamma^{r-1}$. It yields the asserted vanishing trace mass of $S_Z - D_{m,Z}$. A fixed smooth test form is bounded in absolute value on positive $(1,1)$-currents by a constant times this trace mass, proving weak convergence. Finally $b_{2m}(E) \ge 2b_m(E)$ gives $D_{2m,Z} \ge D_{m,Z}$. □

22.2 Compatibility between models

Under a modification, a divisor and its strict transform define the same order on the image ideals. Exceptional divisors disappear under pushforward but can carry additional vanishing. Divisibility of the pulled-back ideals measures that additional vanishing.

Theorem 22.2.1 *Let $g : Z' \to Z$ be a smooth compact modification, and view both manifolds as models over Y. Then*

$$g_* S_{Z'}(u;Y) = S_Z(u;Y).$$

The assignment is additive and positively homogeneous in u. For every finite set $\mathcal{F}$ of prime divisors on Z, the finite effective real divisor

$$A_{\mathcal{F}} = \sum_{E\in\mathcal{F}} \tau_{Y,u}(E)E$$

satisfies

$$[g^* A_{\mathcal{F}}] \le S_{Z'}(u;Y).$$

Proof If E' is the strict transform of a prime divisor $E \subset Z$, the map is biholomorphic near their generic points. They define the same vanishing order on the image ideals from Y, and hence $\tau_{Y,u}(E') = \tau_{Y,u}(E)$. Also $g_*[E'] = [E]$.

For an exceptional prime divisor F, its image has codimension at least two. The pullback of any smooth $(r-1, r-1)$ test form restricts to zero on its regular locus, because the map there has complex rank at most $r-2$. Thus $g_*[F] = 0$. In dimension one a modification between smooth compact curves is an isomorphism, so there are no such exceptional divisors. Pushforward is continuous on currents for this proper map. Applying it to increasing finite sums, or equivalently testing their summable positive measures, proves the displayed compatibility. Additivity and homogeneity follow coefficientwise from Proposition 18.1.1.

For the final inequality, let $\mathfrak{a}_{m,Z} = \mathcal{I}(mu)\mathcal{O}_Z$ be the image ideal on Z, for a positive integer m. Its divisorial vanishing gives

$$\mathfrak{a}_{m,Z} \subseteq \mathcal{O}_Z\left(-\sum_{E\in\mathcal{F}} b_m(E)E\right).$$

Indeed on a smooth manifold a holomorphic function with the indicated generic orders is divisible by the corresponding local Cartier equations, with multiplicity. Pull back this inclusion and take the order at any prime divisor $F \subset Z'$. It gives

$$b_m(F) \geq \sum_{E\in\mathcal{F}} b_m(E)\,\mathrm{ord}_F(g^*E). \tag{22.2}$$

Divide by m and pass to the limit; the sum is finite. Thus every coefficient of $g^*A_{\mathcal{F}}$ is at most $\tau_{Y,u}(F)$. Both sides are divisorial positive currents, which proves the asserted current inequality. □

22.3 The trace envelope as a common majorant

The trace envelope imposes the inequalities for all divisors over Y, including those absent from any one model. It therefore supplies a single potential whose pullbacks dominate every divisorial trace current. Its background is the original restriction $\theta|_Y$.

Proposition 22.3.1 *There exists $v \in \mathrm{PSH}(Y, \theta|_Y)$ such that for every smooth compact modification $\pi : Z \to Y$,*

$$\nu(v \circ \pi, E) \geq \tau_{Y,u}(E) \quad \textit{for every prime } E \subset Z,$$

and

$$\pi^*(\theta|_Y) + \mathrm{dd^c}(v \circ \pi) \geq S_Z(u; Y).$$

Proof Take $v = \mathcal{T}_{\theta|_Y}(u; Y)$ from (18.5). By Theorem 18.6.1, it belongs to $\mathrm{PSH}(Y, \theta|_Y)$ and satisfies every defining trace inequality. On a fixed model Z, Lemma 1.17.1 therefore allows removal of every finite partial sum of $S_Z(u; Y)$ from $\theta_Z + \mathrm{dd^c}(v \circ \pi)$. Passing to the current limit of these sums, constructed in Theorem 22.1.1, preserves positivity and proves the second assertion. □

The choice in the proof is canonical: Theorem 18.6.2 identifies $v \circ \pi$ with the trace envelope in background θ_Z for the inherited data. Its Lelong numbers may still be larger than those prescribed. Exact realization requires the reverse inequality, as established in Chapter 18 for analytic singularities, Kähler targets in sufficiently large backgrounds, and curve fibrations. The ordinary type attached to S_Z records only the divisorial part on that particular model.

22.4 Bott–Chern classes and closed currents

Bott–Chern cohomology separates the smooth background of a closed current from its global potential. Subtracting the trace current from a closed background then gives a residual class on each model. Pushforward compatibility organizes these classes into a b-divisor class, as in [Xia, section *The notion of b-divisors*].

For a complex manifold Z, let $\Lambda^{1,1}_{\mathbb{R}}$ denote the bundle of real $(1,1)$-forms and define

$$H^{1,1}_{\mathrm{BC}}(Z,\mathbb{R}) = \frac{\{\beta \in C^\infty(Z,\Lambda^{1,1}_{\mathbb{R}}) : \mathrm{d}\beta = 0\}}{\{\mathrm{dd^c} f : f \in C^\infty(Z,\mathbb{R})\}}.$$

Lemma 22.4.1 *Replacing smooth forms and smooth functions in this quotient by real currents and real distributions gives the same space. In particular every closed positive* $(1,1)$*-current* T *on* Z *can be written*

$$T = \beta + \mathrm{dd^c} w$$

with β *smooth and closed and* w *a global quasi-plurisubharmonic function.*

Proof The decomposition in Lemma 1.17.3 proves surjectivity of the smooth-to-current map and gives a quasi-plurisubharmonic potential for positive currents. Its smooth-difference assertion also proves injectivity: a distribution whose $\mathrm{dd^c}$ is smooth is smooth, so two smooth representatives identified in the current quotient are already identified in the smooth quotient. □

For global pairings, take Z connected and compact. The class of a closed current T is written $\{T\}_{\mathrm{BC}}$. It is called pseudoeffective when it has a positive closed representative. Pullback of classes uses smooth representatives. Pushforward under a proper modification uses currents; it is well-defined because pushforward commutes with $\mathrm{dd^c}$. The lemma identifies its result with a smooth Bott–Chern class. For a modification g, we have $g_* g^* \beta = \beta$ as currents for every smooth form β of the indicated degree, by degree-one change of variables.

A Gauduchon metric γ defines a linear functional on this cohomology:

$$\ell_\gamma(\{T\}_{\mathrm{BC}}) = \int_Z T \wedge \gamma^{\dim Z - 1}.$$

Current Stokes makes it independent of the representative. For a positive representative it is nonnegative, and equality forces that representative to vanish, by positivity and the strict positivity of the metric. In particular the pseudoeffective cone contains no nonzero line: if $T_1, T_2 \geq 0$ represent opposite classes, their sum has zero Gauduchon pairing, hence $T_1 = T_2 = 0$.

Use the Hopf manifold X_n, $n \geq 2$, and its Gauduchon metric ω from Proposition 19.6.1, with covering coordinates $z \in \mathbb{C}^n \setminus \{0\}$. The form $\alpha = \mathrm{dd^c} \log |z|^2$ is exact as a de Rham form. Indeed $\mathrm{d^c} \log |z|^2$ descends to the quotient because the logarithm changes by a constant under the contraction. Nevertheless

$$\ell_\omega(\{\alpha\}_{\mathrm{BC}}) = \int_{X_n} \alpha \wedge \omega^{n-1} = 2(n-1) > 0.$$

Thus its Bott–Chern class is nonzero. Gauduchon integration detects closed-curvature information that the de Rham class loses in this example.

Return now to the smooth compact modifications of the original manifold Y.

Definition 22.4.1 A *Bott–Chern b-divisor class* over Y is a family

$$\mathbb{A} = (\mathbb{A}_Z)_{\pi: Z \to Y}, \qquad \mathbb{A}_Z \in H^{1,1}_{\mathrm{BC}}(Z, \mathbb{R}),$$

over smooth compact modifications, compatible with pushforward. It is pseudoeffective if every $\mathbb{A}_Z$ is pseudoeffective. It is *Cartier* if there are one model Z and one class a_Z such that its class on every model Z' dominating Z is the pullback of a_Z.

When the models are Kähler, the standard $\partial\bar{\partial}$ lemma identifies this cohomology with the real $(1,1)$-cohomology used in [Xia, section *The notion of b-divisors*]. The definition then agrees with the source definition.

For the trace data fixed at the beginning of the chapter, the common majorant supplies positive representatives of the residual classes.

Proposition 22.4.1 *If the original ambient background θ is closed, the classes*

$$\mathbb{A}_Z(u; Y, \theta) = \{\pi^*(\theta|_Y) - S_Z(u; Y)\}_{\mathrm{BC}}$$

form a pseudoeffective Bott–Chern b-divisor class. They depend only on the ambient multiplier-ideal class of u and the Bott–Chern class of $\theta|_Y$.

Proof By Proposition 22.3.1, there is one v on Y such that

$$R_Z = \pi^*(\theta|_Y) + \mathrm{dd^c}(v \circ \pi) - S_Z(u; Y) \geq 0$$

on every model. Since θ and S_Z are closed, R_Z is closed. Its class is exactly the displayed one, proving pseudoeffectivity. Compatibility follows from Theorem 22.2.1 and $g_* g^*(\theta_Z) = \theta_Z$. The trace data is unchanged by ambient multiplier-ideal equivalence, and replacing $\theta|_Y$ by a smooth $\mathrm{dd^c}$ perturbation changes no class. These facts prove independence. □

If θ is not closed, the same expression R_Z is still a positive current, but

$$\mathrm{d}R_Z = \pi^*(\mathrm{d}\theta|_Y).$$

It therefore need not define a class in ordinary Bott–Chern cohomology. The larger classes with a specified non-closed derivative will be treated in Chapter 23.

22.5 The b-divisor class of a closed positive current

For a closed positive current, the same construction uses its own local potentials in place of restricted ambient ideals. The trace coefficients are then its divisorial Lelong numbers, and divisorial removal gives the positive residual currents directly.

Let T be a closed positive $(1,1)$-current on a connected compact complex manifold Y, and write $T = \beta + \mathrm{dd^c} w$ as above. On a compact modification $\pi : Z \to Y$, local psh potentials define the positive pullback π^*T: dominance prevents their compositions from being identically $-\infty$. With E ranging over prime divisors on Z, set

$$N_Z(T) = \sum_E \nu(w \circ \pi, E)[E], \qquad \mathrm{Reg}_Z(T) = \pi^*T - N_Z(T).$$

Divisorial removal makes the second current positive and closed. The following lemma identifies the coefficients with the trace construction.

Lemma 22.5.1 *For every quasi-plurisubharmonic function w on Y and every prime divisor E on a smooth compact modification $\pi : Z \to Y$, the self-trace in* (18.1) *satisfies*

$$\tau_{Y,w}(E) = \nu(w \circ \pi, E).$$

In particular, for the potential of the closed positive current T, one has $N_Z(T) = S_Z(w; Y)$.

Proof A genuine quasi-plurisubharmonic function has zero Lelong number almost everywhere, so its self-trace satisfies the required generic hypothesis. For a fixed divisor E on a compact modification, apply Lemma 1.17.7 on a coordinate neighborhood of its center. With $\nu_E = \nu(w \circ \pi, E)$ and A_E the log discrepancy, it gives

$$\max\{\nu_E - A_E/m, 0\} \le \frac{1}{m} \operatorname{ord}_E \mathcal{I}(mw) \le \nu_E.$$

The integer limit defining $\tau_{Y,w}(E)$ is therefore ν_E. Equality of all coefficients identifies the two divisorial currents. □

Theorem 22.5.1 *The family*

$$\mathbb{D}(T)_Z = \{\mathrm{Reg}_Z(T)\}_{\mathrm{BC}}$$

is a pseudoeffective Bott–Chern b-divisor class. For closed positive currents T_1, T_2 and $a \geq 0$,

$$\mathbb{D}(T_1 + T_2) = \mathbb{D}(T_1) + \mathbb{D}(T_2), \qquad \mathbb{D}(aT_1) = a\mathbb{D}(T_1).$$

On a Kähler manifold, with Kähler models, it agrees with the b-divisor class of the same current in [Xia, subsection The b-divisors of currents*].*

Proof The preceding lemma identifies $N_Z(T)$ with the already constructed compatible divisorial currents. For $g : Z' \to Z$, we have $g_* N_{Z'}(T) = N_Z(T)$ and $g_* g^*(\pi^* T) = \pi^* T$. For the latter identity, write the current as a smooth form plus $\mathrm{dd^c}$ of a potential; degree-one change of variables applies both to the smooth form and to the locally integrable pulled-back potential, and pushforward commutes with $\mathrm{dd^c}$. Therefore

$$g_* \mathrm{Reg}_{Z'}(T) = \mathrm{Reg}_Z(T).$$

Taking classes proves compatibility, and positivity proves pseudoeffectivity. Generic Lelong numbers add under addition of positive currents and scale by nonnegative constants. Hence their divisorial parts, and then their regular parts, have the same properties. This proves the two identities. In the Kähler case both the regular-current construction and the identified cohomology classes are exactly those in the source definition. □

22.6 Nef Cartier realizations

If T has analytic singularities, a resolution removes them in a finite divisor and leaves a current with bounded potentials. This remainder acquires no new divisorial part on higher models, so its class determines the entire family there. Smooth regularization of its bounded potential makes the realizing class nef.

A class $a \in H^{1,1}_{\mathrm{BC}}(Z, \mathbb{R})$ is *nef* if, for one Hermitian metric γ, it has a smooth representative $\beta_\epsilon \geq -\epsilon\gamma$ for every $\epsilon > 0$. Comparison of two fixed Hermitian metrics shows that this definition is independent of the metric. A Cartier b-divisor class is called nef here if it has a realization by a nef class.

Lemma 22.6.1 *On a connected compact complex manifold Z, let β be a smooth real $(1,1)$-form and v a bounded quasi-plurisubharmonic function. If $R = \beta + \mathrm{dd^c} v \geq 0$ is closed, then $\{R\}_{\mathrm{BC}}$ is nef.*

Proof Since R is closed, β is closed. Fix a Hermitian metric γ. By Proposition 1.10.1, there are smooth v_j and $\epsilon_j \downarrow 0$ such that

$$\beta + \mathrm{dd^c} v_j \geq -\epsilon_j \gamma.$$

These are closed representatives of $\{R\}_{\mathrm{BC}}$. Given any $\epsilon > 0$, choose j with $\epsilon_j \leq \epsilon$. This is the defining nef estimate. □

Theorem 22.6.1 *If the closed positive current T has analytic singularities, then $\mathbb{D}(T)$ is Cartier and has a nef realization on a smooth compact modification.*

Proof Resolve the analytic singularity ideal, using the log-resolution theorem in [Xia, section on analytic singularities, theorem on existence of log resolutions]. On one smooth compact model $\pi : Z \to Y$,

$$\pi^* T = [D] + R,$$

where D is a finite effective rational divisor and R is positive, closed, and has locally bounded potentials. To obtain a global bounded potential, choose smooth metrics on the divisor bundles. Subtract the corresponding logarithmic squared norms from $w \circ \pi$. The remainder v is globally bounded, and

$$R = \beta_Z + \mathrm{dd^c} v$$

for a smooth closed β_Z. The lemma makes its class nef.

On every further compact modification $g : Z' \to Z$, the current $g^* R$ still has bounded local potentials and consequently zero Lelong numbers. The entire divisorial part of $g^* \pi^* T$ is therefore $[g^* D]$, and its regular part is $g^* R$. It follows that $\mathbb{D}(T)_{Z'} = g^* \{R\}_{\mathrm{BC}}$. Thus the pair $(Z, \{R\}_{\mathrm{BC}})$ is the required nef Cartier realization.□

For a general current, analytic regularization gives closed approximants bounded below by $-\epsilon_m \gamma$. Adding $\epsilon_m \gamma$ makes them positive but changes their exterior derivative when γ is not closed. The approximation then belongs to the larger spaces of Chapter 23; it does not by itself approximate the original Bott–Chern b-divisor class by nef Cartier classes.

22.7 Continuous representatives and pullback

A finite divisorial sum pulls back by its local equations. Passing to an infinite sum requires continuity for closed positive currents whose Bott–Chern classes may vary. A scalar Gauduchon equation selects a smooth representative in each class; the complementary normalized potentials then converge in L^1, where the pullback theorem of Lemma 1.22.2 applies.

Fix a Gauduchon metric γ on a connected compact complex manifold Y of dimension $n \geq 1$. In dimension one any Hermitian metric will do. Fix smooth norms on tensor fields induced by γ. Put $V_\gamma = \int_Y \gamma^n$ and, for a real $(1,1)$-form or current B, define the scalar trace by

$$\sigma_\gamma(B)\,\gamma^n = B \wedge \gamma^{n-1}.$$

For a current this is an identity of distributions. We use the classical elliptic Fredholm theorem, regularity, and the continuity on distributions of the normalized elliptic inverse, as in the proof of Theorem 4.1.1.

Lemma 22.7.1 *Every closed real $(1,1)$-current T has a unique decomposition*

$$T = B_\gamma(T) + \mathrm{dd^c} v_\gamma(T), \qquad \int_Y v_\gamma(T)\,\gamma^n = 0,$$

where $v_\gamma(T)$ is a real distribution and $B_\gamma(T)$ is a smooth closed real form with constant scalar trace. Both assignments are linear and continuous from weak convergence of currents to convergence of distributions. The forms $B_\gamma(T)$ lie in a finite-dimensional space, and their convergence is in C^∞.

That space maps isomorphically to $H^{1,1}_{\mathrm{BC}}(Y,\mathbb{R})$. If $T \geq 0$, then $v_\gamma(T)$ is quasi-plurisubharmonic. On the cone of positive closed currents its dependence on T is continuous in L^1, and for every nonnegative integer l,

$$\|B_\gamma(T)\|_{C^l} + \|v_\gamma(T)\|_{L^1(\gamma^n)} \leq C_l \int_Y T \wedge \gamma^{n-1}. \tag{22.3}$$

The constants depend on the fixed metric and norms.

Proof Set $L_\gamma f = \sigma_\gamma(\mathrm{dd^c} f)$. Use the inverse G_γ in (5.1), normalized with volume form γ^n/V_γ. It acts on top-degree distributions of total mass zero and is continuous by the elliptic estimates for the inverse and its transpose.

Define

$$c_T = \frac{\int_Y T \wedge \gamma^{n-1}}{V_\gamma}, \qquad v_\gamma(T) = G_\gamma\big((\sigma_\gamma(T) - c_T)\gamma^n\big), \qquad B_\gamma(T) = T - \mathrm{dd^c} v_\gamma(T).$$

The last current is closed and has scalar trace c_T. We verify its smoothness and the finite-dimensional assertion without a Kähler identity.

Let J be the complex structure acting on real covectors. On real $(1,1)$-forms consider the first-order operator

$$\mathscr{P}B = (\mathrm{d}B, \mathrm{d}\sigma_\gamma(B)).$$

Its principal symbol is injective at every nonzero real covector ξ. Indeed, $\xi \wedge B = 0$ and the $(1,1)$ condition force B to be a real multiple of $\xi \wedge J\xi$. To check this, choose unitary real coordinates with $\xi = \mathrm{d}x_1$. Every term of B contains $\mathrm{d}x_1$, and invariance under J forces every term to contain $\mathrm{d}y_1$ as well. The scalar trace of this remaining term is nonzero unless its coefficient is zero. The other symbol equation is $\xi\,\sigma_\gamma(B) = 0$, so it removes that coefficient.

Let $\mathscr{P}^*$ be the formal adjoint for the fixed metric and volume. Consequently $\mathscr{P}^*\mathscr{P}$ is elliptic. Its distributional kernel is a finite-dimensional space of smooth forms, by elliptic regularity and compactness. Since $\mathscr{P}B_\gamma(T) = 0$, the representative belongs to this space. Conversely its smooth kernel consists exactly of closed forms with constant scalar trace: integration of $\langle \mathscr{P}^*\mathscr{P}B, B\rangle$ gives $\mathscr{P}B = 0$, and connectedness makes the trace constant. Denote this kernel by $\mathscr{H}_\gamma$.

The construction is linear and continuous on distributions. Convergence of its B components therefore takes place in the finite-dimensional space $\mathscr{H}_\gamma$, hence in every smooth norm. If $T = \mathrm{dd^c} f$ for a distribution f, then $c_T = 0$ and the inverse gives $v_\gamma(T) = f - V_\gamma^{-1}\int_Y f\gamma^n$, so $B_\gamma(T) = 0$. Applied to smooth closed forms, this proves that $\mathscr{H}_\gamma$ is a system of representatives for Bott–Chern cohomology, with exactly one representative in each class. For uniqueness of the displayed decomposition directly, the difference of the two potentials has L_γ equal to a constant. Integration forces that constant to be zero, and the normalized kernel is zero.

Suppose now that $T \geq 0$. The inequality $\mathrm{dd^c} v_\gamma(T) \geq -B_\gamma(T)$ gives a quasi-plurisubharmonic representative of this distribution. If positive closed currents T_j converge weakly to T, their B components converge smoothly, so the potentials have one common lower curvature bound. By Theorem 1.4.1, subtracting their suprema gives an L^1 precompact family. If $a_j = \sup_Y v_\gamma(T_j)$, the zero-mean normalization yields

$$0 \leq a_j V_\gamma = \int_Y \big(a_j - v_\gamma(T_j)\big)\,\gamma^n.$$

The right side is uniformly bounded, so the original potentials are also L^1 precompact. Every cluster limit equals their distributional limit $v_\gamma(T)$. Thus the whole sequence converges in L^1.

Finally the positive closed currents of trace mass one form a weakly compact set. Positivity bounds their coefficient measures, weak limits remain positive and closed, and testing with γ^{n-1} preserves their mass. The smooth and L^1 continuity just proved bounds both norms on this set. Positive homogeneity gives (22.3). A positive current of mass zero is zero, so that case is included. □

The smooth representatives give a common curvature bound for converging normalized potentials. This is precisely the hypothesis needed to pull back their L^1 convergence.

Proposition 22.7.1 *Let $g : Z \to Y$ be a smooth compact modification, and fix a Gauduchon metric γ' on Z. Pullback of positive closed $(1,1)$-currents by their local potentials is weakly continuous. There is a constant C_g such that*

$$\int_Z g^*T \wedge (\gamma')^{n-1} \le C_g \int_Y T \wedge \gamma^{n-1} \qquad (T \ge 0,\ \mathrm{d}T = 0). \tag{22.4}$$

The constant depends on the fixed modification and metrics.

Proof Write $T = B_\gamma(T) + \mathrm{dd^c} v_\gamma(T)$. The pullback is

$$g^*T = g^*B_\gamma(T) + \mathrm{dd^c}(v_\gamma(T) \circ g).$$

It is positive and closed by local psh potentials, and the composed potential is genuine because the modification is dominant. This formula agrees with the local definition, so it is independent of the decomposition.

For $T_j \to T$, the smooth parts converge in C^∞ and the potentials converge in $L^1(Y)$ with a common lower curvature bound. Lemma 1.22.2 gives convergence of their compositions in $L^1(Z)$. Taking $\mathrm{dd^c}$ proves weak convergence of the pullbacks.

Current Stokes on Z gives

$$\int_Z g^*T \wedge (\gamma')^{n-1} = \int_Z g^*B_\gamma(T) \wedge (\gamma')^{n-1}.$$

The absolute value of the right side is bounded by a fixed constant times $\|B_\gamma(T)\|_{C^0}$. Apply (22.3). The left side is nonnegative, giving (22.4). □

Corollary 22.7.1 *Fix the modification $g : Z \to Y$ and metrics of Proposition 22.7.1. Let D_j be prime divisors on Y, $[D_j]$ their integration currents, and $c_j > 0$. If $S = \sum_j c_j [D_j]$ is a positive closed current of finite mass, then*

$$g^*S = \sum_j c_j [g^*D_j]$$

as positive currents of finite mass. The partial sums converge with vanishing trace mass of their positive tails on Z. In particular g^*S is again purely divisorial.

Proof Apply the proposition to the increasing finite partial sums S_N and their positive closed tails $S - S_N$. Pullback is additive on positive closed currents by addition of local potentials. Thus $g^*S - g^*S_N = g^*(S - S_N) \ge 0$, and its trace mass is bounded by C_g times the trace mass of $S - S_N$, which tends to zero. Pullback of a finite divisor is the usual Cartier pullback by its local equation. This identifies the series and its limit. Its terms are integration currents of prime divisors with nonnegative coefficients; regrouping this countable positive finite-mass sum preserves equality. □

22.8 Exceptional divisors between trace models

Return to the ambient trace data fixed at the beginning of the chapter, and write $S_Z = S_Z(u; Y)$. Pullback of the full current is now available. Under a modification, it is bounded above by the trace current on the higher model, with equality of coefficients along strict transforms. The difference is therefore a finite effective exceptional divisor. Its logarithmic potential describes the change in the ordinary types of the divisorial parts.

Theorem 22.8.1 *For a smooth compact modification $g : Z' \to Z$, there is a unique finite effective real divisor E_g supported on its exceptional locus such that*

$$S_{Z'} = g^* S_Z + [E_g]. \tag{22.5}$$

For $h : Z'' \to Z'$, the divisors satisfy

$$E_{g\circ h} = E_h + h^* E_g. \tag{22.6}$$

Their coefficients are additive and positively homogeneous in the ambient trace potential u.

The analytic-ideal approximants on the two models give finite effective exceptional divisors

$$[E_{m,g}] = D_{m,Z'} - g^* D_{m,Z}.$$

For any Gauduchon metric γ' on Z', the measures

$$[E_{m,g}] \wedge (\gamma')^{r-1} \longrightarrow [E_g] \wedge (\gamma')^{r-1}, \qquad r = \dim Y,$$

converge in total variation.

Proof Enumerate the positive divisorial components of S_Z. Every finite partial sum has pullback at most $S_{Z'}$ by Theorem 22.2.1. Corollary 22.7.1 identifies the increasing limit of those pullbacks with $g^* S_Z$. Thus $S_{Z'} - g^* S_Z$ is a positive closed divisorial current. The two currents have equal coefficients along every strict transform of a divisor on Z, because these orders describe the same valuation. Every other prime divisor of Z' is exceptional for g. The exceptional analytic set has only finitely many irreducible components on compact Z'. Its divisorial components therefore support a finite effective real divisor E_g. Coefficients of integration currents identify it uniquely.

Pullback of positive closed $(1, 1)$-currents is functorial, by composition of local psh potentials. Using (22.5) for h and g gives

$$S_{Z''} = h^* S_{Z'} + [E_h] = (g \circ h)^* S_Z + [h^* E_g] + [E_h].$$

Uniqueness gives (22.6). Additivity and positive homogeneity follow from the same properties of every trace coefficient and linearity of pullback.

For a fixed m, apply (22.2) to the finite divisorial support of the image ideal on Z. After division by m, it gives $D_{m,Z'} - g^* D_{m,Z} \geq 0$. The coefficients on strict transforms are zero, so this difference is the integration current of an effective exceptional real divisor $E_{m,g}$.

Choose also a Gauduchon metric γ on Z. By Theorem 22.1.1, the currents $S_Z - D_{m,Z}$ and $S_{Z'} - D_{m,Z'}$ are positive and their trace masses tend to zero. Subtraction of the two identities for the exceptional currents gives

$$[E_{m,g}] - [E_g] = -(S_{Z'} - D_{m,Z'}) + g^*(S_Z - D_{m,Z}).$$

Both terms on the right, before the minus sign, are positive. The total variation of the signed trace measure is consequently at most

$$\int_{Z'} (S_{Z'} - D_{m,Z'}) \wedge (\gamma')^{r-1} + C_g \int_Z (S_Z - D_{m,Z}) \wedge \gamma^{r-1},$$

by (22.4). Both terms tend to zero, proving total variation convergence. □

Fix a modification $g : Z' \to Z$ as in the theorem and its divisor E_g. The coefficient at an exceptional prime $F \subset Z'$ is explicitly

$$\tau_{Y,u}(F) - \sum_{D \subset Z} \tau_{Y,u}(D) \operatorname{ord}_F(g^* D). \tag{22.7}$$

Here D ranges over prime divisors on Z with positive trace coefficient. The sum is countable and convergent. Indeed it is the coefficient of $g^* S_Z$ in the positive divisorial series of Corollary 22.7.1; it is bounded above by $\tau_{Y,u}(F)$. The same current identity compares the ordinary types supplied by Corollary 1.17.1.

Corollary 22.8.1 *Choose decompositions $S_Z = B_Z + \mathrm{dd^c} v_Z$ and $S_{Z'} = B_{Z'} + \mathrm{dd^c} v_{Z'}$ with smooth closed forms $B_Z, B_{Z'}$, as in Corollary 1.17.1. If ρ_{E_g} is a global logarithmic potential of the finite real divisor E_g, then*

$$v_{Z'} - (v_Z \circ g) - \rho_{E_g}$$

is represented by a smooth function. In particular

$$v_{Z'} \preceq v_Z \circ g,$$

and the two ordinary types are equal if and only if $E_g = 0$.

Proof Choose smooth Hermitian metrics on the line bundles of the prime components of E_g. Let α_{E_g} be the weighted sum of the smooth closed curvature forms of these metrics. The weighted sum of the logarithms of the squared canonical-section norms gives ρ_{E_g} with

$$\mathrm{dd^c} \rho_{E_g} = [E_g] - \alpha_{E_g}.$$

The function ρ_{E_g} is quasi-plurisubharmonic and bounded above on compact Z'.

The chosen decompositions and (22.5) give

$$\mathrm{dd^c}\big(v_{Z'} - v_Z \circ g - \rho_{E_g}\big) = g^* B_Z + \alpha_{E_g} - B_{Z'},$$

which is smooth. By Lemma 1.17.3, the L^1 difference is represented by a smooth function. The corresponding equality of quasi-plurisubharmonic functions extends from almost everywhere to everywhere by Lemma 1.3.3. The bounded-above logarithmic potential gives the ordinary domination.

If $E_g = 0$, the smooth difference gives ordinary equivalence. Conversely ordinary equivalence would give equal generic Lelong numbers on every prime divisor of Z'. The smooth residual terms show that their difference on a prime component of E_g is precisely its positive coefficient, a contradiction. Hence ordinary equivalence forces $E_g = 0$. □

When θ is closed, the positive residual currents R_Z constructed in the proof of Proposition 22.4.1 and their Bott–Chern classes satisfy

$$\begin{aligned} R_{Z'} &= g^* R_Z - [E_g], \\ \mathbb{A}_{Z'}(u; Y, \theta) &= g^* \mathbb{A}_Z(u; Y, \theta) - \{[E_g]\}_{\mathrm{BC}}. \end{aligned}$$

Indeed the smooth background and the common trace potential pull back, while (22.5) gives the change in the subtracted divisorial current. The exceptional term removes the additional prescribed divisorial vanishing on the higher model. It is compatible with exact pullback of the trace envelope, which retains the inequalities on all models. The following example also shows that this term can be nonzero when the trace data has an exact potential.

Example 22.8.1 Let $Y = \mathbb{P}^2$ with homogeneous coordinates $[Z_0 : Z_1 : Z_2]$ and choose $a \in \mathbb{Q}_{>0}$. Put

$$u = a \log \frac{|Z_1|^2 + |Z_2|^2}{|Z_0|^2 + |Z_1|^2 + |Z_2|^2}.$$

Take the ambient manifold to be Y itself. If σ is the Fubini–Study form, then $u \in \mathrm{PSH}(Y, a\sigma)$. It has analytic singularities only at $p = [1 : 0 : 0]$, and has zero generic Lelong number on Y. Its trace is exactly the ordinary singularity data of u, by Proposition 18.2.1. There are no divisorial poles on Y, so $S_Y = 0$ and its canonical ordinary type is smooth. If $\pi : \widetilde{Y} \to Y$ is the blowup at p with exceptional curve F, then

$$S_{\widetilde{Y}} = a[F], \qquad E_\pi = aF.$$

Indeed on either standard blowup chart the numerator has one exceptional factor of squared modulus and a smooth positive remaining factor. The pole order is a along F and zero along every other prime divisor. Thus a nonzero exceptional difference does not obstruct the existence of an exact trace potential. It obstructs identifying that potential with the canonical type of the divisorial part on the original model.

The same distinction occurs on a manifold failing BMP. Take a compact Hermitian manifold (W, ω_W) of dimension k failing BMP, such as the Hopf threefold in Theorem 8.0.1, and pull this potential back to $Y \times W$. Its pole set is $\{p\} \times W$, of codimension two, and the modification $\widetilde{Y} \times W \to Y \times W$ has exceptional coefficient a along $F \times W$. Analytic restriction again gives an exact trace. Let pr_Y and pr_W be the two product projections.

Choose smooth ω_W-plurisubharmonic functions H_j whose masses tend to infinity. For the Hermitian form $\Omega = \mathrm{pr}_Y^* \sigma + \mathrm{pr}_W^* \omega_W$, their pullbacks have masses

$$\binom{k+2}{2} \left(\int_Y \sigma^2 \right) \left(\int_W (\omega_W + \mathrm{dd^c} H_j)^k \right) \longrightarrow +\infty,$$

by the smooth binomial expansion and the dimensions of the two factors. Hence the product also fails BMP. The pulled-back potential belongs to $\mathrm{PSH}(Y \times W, C\Omega)$ for every $C \geq \max\{a, 1\}$, since its curvature is bounded below by $-a\, \mathrm{pr}_Y^* \sigma$. Scaling the smooth witnesses by C shows that $C\Omega$ also has unbounded mass.

Chapter 23
Classes with a non-closed background

Let Y be a connected compact complex manifold of dimension $n \geq 1$, let θ be a smooth real $(1,1)$-form on Y, and write $\pi : Z \to Y$ for a smooth compact modification.

Subtracting a divisorial current from $\pi^*\theta$ leaves its exterior derivative unchanged. When θ is closed, the result defines a Bott–Chern class, as in the preceding chapter. For a general background, it belongs instead to an affine space over Bott–Chern cohomology: differences of representatives are closed, although the representatives themselves need not be. The trace envelope still supplies positive residual representatives, and exceptional vanishing describes their change between models.

Analytic singularities give nef Cartier realizations in these affine spaces. For general potentials, analytic regularization produces such realizations in nearby backgrounds, with weak convergence of the positive residual currents on each fixed model. The background error tends smoothly to zero, but removing it can destroy nefness. Nor does this linear convergence control Monge–Ampère mass: when bounded mass fails, even the family realized by a smooth positive form has an infinite-mass representative.

23.1 Affine classes with prescribed derivative

Fixing the exterior derivative leaves closed currents as the differences between representatives. After quotienting by global potential terms, these differences are Bott–Chern classes. Write $\mathcal{D}'(Z,\mathbb{R})$ for real distributions and set

$$\mathscr{A}_Z(\theta) = \frac{\{\pi^*\theta + C : C \text{ is a closed real } (1,1)\text{-current on } Z\}}{\mathrm{dd^c}\mathcal{D}'(Z,\mathbb{R})}.$$

Write $\{T\}$ for the class of T. By Lemma 22.4.1, every class has a smooth representative, and translation by Bott–Chern classes gives

$$\mathscr{A}_Z(\theta) = \{\pi^*\theta\} + H^{1,1}_{\mathrm{BC}}(Z,\mathbb{R}).$$

A smooth real $(1,1)$-form η represents an element of $\mathscr{A}_Z(\theta)$ exactly when $\mathrm{d}\eta = \pi^*\mathrm{d}\theta$. Pullback of a class uses a smooth representative; pushforward uses currents, just as for Bott–Chern classes.

Proposition 23.1.1 *If $g : Z' \to Z$ is a compact modification over Y, pushforward and pullback induce maps*

$$g_* : \mathscr{A}_{Z'}(\theta) \to \mathscr{A}_Z(\theta), \qquad g^* : \mathscr{A}_Z(\theta) \to \mathscr{A}_{Z'}(\theta), \qquad g_* g^* = \mathrm{id}.$$

Pushforward preserves classes with positive current representatives.

Proof Write $\pi' = \pi \circ g$. Translation by the pulled-back background identifies the required maps with the Bott–Chern maps of Section 22.4. Indeed, for a closed current C' on Z' and a smooth closed form β on Z,

$$g_*(\pi'^*\theta + C') = \pi^*\theta + g_*C', \qquad g^*(\pi^*\theta + \beta) = \pi'^*\theta + g^*\beta.$$

The first equality uses degree-one change of variables on the smooth background. The Bott–Chern maps are well-defined on the closed terms, so these identities define the affine maps independently of representatives. Their composition is the identity because g_*g^* is the identity on the background and on Bott–Chern classes. Proper pushforward preserves positive currents, proving the final assertion. □

The prescribed derivative is essential to this pushforward construction: $g_*\pi'^*\mathrm{d}\theta = \pi^*\mathrm{d}\theta$ is smooth, whereas pushing forward an arbitrary smooth exterior derivative can produce a singular current.

A family with entries in $\mathscr{A}_Z(\theta)$ compatible with pushforward is a *b-divisor class over the background* θ. Pseudoeffectivity and Cartier realization have the meanings of Definition 22.4.1, with affine classes in place of Bott–Chern classes. Nefness uses the almost positive smooth-representative condition of Section 22.6, now within $\mathscr{A}_Z(\theta)$.

23.2 Residual trace classes

Suppose Y is a smooth submanifold of a connected compact complex manifold X. Let Θ be a smooth real $(1,1)$-form on X with $\Theta|_Y = \theta$, and let $u \in \mathrm{PSH}(X,\Theta)$ satisfy $\nu(u,Y) = 0$. Use the trace currents $S_Z(u;Y)$ of Section 22.1.

The trace envelope dominates these currents on every model. Subtracting them from its curvature gives positive residual representatives with exterior derivative $\pi^*\mathrm{d}\theta$. The prescribed trace currents determine the classes even when the envelope has larger Lelong numbers.

Theorem 23.2.1 *For the trace data just fixed, the family defined by*

$$\mathbb{B}_Z(u;Y,\theta) = \{\pi^*\theta - S_Z(u;Y)\} \in \mathscr{A}_Z(\theta)$$

is a pseudoeffective b-divisor class over θ. It is unchanged by ambient multiplier-ideal equivalence and by a smooth $\mathrm{dd^c}$ *change of the restricted background, after identifying the resulting affine spaces.*

Proof The currents S_Z are closed, so the displayed classes belong to the indicated affine spaces. Their compatibility follows from Theorem 22.2.1 and Proposition 23.1.1. Take the canonical $v = \mathcal{T}_\theta(u;Y)$ of (18.5). By Proposition 22.3.1,

$$\pi^*\theta + \mathrm{dd^c}(v \circ \pi) - S_Z(u;Y) \geq 0$$

on every model, proving pseudoeffectivity in the original restricted background. The currents S_Z depend only on the ambient multiplier ideals. For a smooth real function f on Y, a change $\theta \mapsto \theta + \mathrm{dd^c} f$ adds $\mathrm{dd^c}(f \circ \pi)$ to every representative, which is zero in the quotient. This proves both invariances. □

When θ is closed, this family is the Bott–Chern family of Proposition 22.4.1.

On a higher model, the trace current can exceed the pullback of the trace current on the original model. The effective exceptional difference from Theorem 22.8.1 therefore translates the residual class, while preserving the prescribed exterior derivative.

Corollary 23.2.1 *For the trace classes in Theorem 23.2.1 and a compact modification $g : Z' \to Z$ over Y, let E_g be the effective exceptional divisor in Theorem 22.8.1. Then*

$$\mathbb{B}_{Z'}(u;Y,\theta) = g^*\mathbb{B}_Z(u;Y,\theta) - \{[E_g]\}_{\mathrm{BC}}.$$

Here the Bott–Chern class on the right acts by translation on the affine space $\mathscr{A}_{Z'}(\theta)$.

Proof Let $\pi : Z \to Y$ be the structure map. Abbreviate $S_Z = S_Z(u;Y)$ and use Lemma 1.17.3 to write $S_Z = B_Z + \mathrm{dd^c} v_Z$ with B_Z smooth and closed and v_Z quasi-plurisubharmonic. The affine class $\mathbb{B}_Z$ has smooth representative $\pi^*\theta - B_Z$. Its pullback therefore equals the class of $\pi'^*\theta - g^*S_Z$, where $\pi' = \pi \circ g$; the two representatives differ by $\mathrm{dd^c}(v_Z \circ g)$. Substituting $S_{Z'} = g^*S_Z + [E_g]$ gives the displayed identity. The exterior derivative of both representatives is $\pi'^*\mathrm{d}\theta$, since the subtracted currents are closed. Thus the subtraction takes place within the stated affine space. □

The cocycle identity (22.6) makes these translations compatible with composition, and Theorem 22.8.1 gives total variation convergence of their analytic-ideal approximants in trace mass. Meanwhile, Theorem 18.6.2 identifies the envelopes on the models as pullbacks of the original trace envelope. The exceptional translation records the additional divisorial subtraction; it is compatible with exact pullback of this envelope and can be nonzero even for exactly realized trace data, as in Example 22.8.1.

For the rest of the chapter, take the ambient manifold to be Y itself and let $u \in \mathrm{PSH}(Y, \theta)$. The self-trace identity Lemma 22.5.1 gives

$$N_Z(u) = \sum_E \nu(u \circ \pi, E)[E] = S_Z(u; Y), \qquad \mathbb{D}_\theta(u)_Z = \{\pi^*\theta - N_Z(u)\}.$$

Here E ranges over prime divisors on Z. Theorem 23.2.1 makes $\mathbb{D}_\theta(u)$ pseudoeffective. Divisorial removal gives the representative $\pi^*\theta + \mathrm{dd^c}(u \circ \pi) - N_Z(u) \geq 0$ directly, by Lemma 1.17.1 and passage to increasing finite sums.

23.3 Nef realizations and residual-current limits

On a resolution of analytic singularities, the bounded residual potential has smooth approximants with arbitrarily small curvature error. They prove nefness within the same affine class. Boundedness also prevents new divisorial terms on further modifications, giving the Cartier realization.

Proposition 23.3.1 *If $u \in \mathrm{PSH}(Y, \theta)$ has analytic singularities, then $\mathbb{D}_\theta(u)$ has a nef Cartier realization on a smooth compact modification.*

Proof Resolve the analytic singularity ideal as in the proof of Theorem 22.6.1. On one model $\pi : Z \to Y$ we have

$$\pi^*\theta + \mathrm{dd^c}(u \circ \pi) = [D] + R, \qquad R = \beta + \mathrm{dd^c} v \geq 0,$$

where D is a finite effective rational divisor, β is smooth, v is globally bounded after choosing divisor metrics, and $\mathrm{d}\beta = \pi^*\mathrm{d}\theta$. Positivity follows from divisorial removal.

Fix a Hermitian form γ on Z and apply Proposition 1.10.1 to v in the smooth background β. It gives smooth v_j with $\beta + \mathrm{dd^c} v_j \geq -\epsilon_j \gamma$, $\epsilon_j \downarrow 0$. These forms represent the same affine class and have exterior derivative $\pi^*\mathrm{d}\theta$, proving nefness.

On every further modification $g : Z' \to Z$, the current $g^*\beta + \mathrm{dd^c}(v \circ g)$ is positive by Lemma 1.22.1. Its bounded potential has zero Lelong numbers, so the divisorial part of the full pullback is precisely g^*D. The residual affine class is consequently $g^*\{R\}$, proving the Cartier realization. □

For general u, add twice the curvature error of analytic regularization to the background. This gives positive analytic currents to which the proposition applies, while the ideal orders identify the limiting residual family.

Theorem 23.3.1 *Let $u \in \mathrm{PSH}(Y, \theta)$ and fix a Hermitian form ω on Y. There are analytic potentials $u_m \to u$ in $L^1(Y)$ and positive numbers $\epsilon_m \downarrow 0$, indexed by sufficiently large positive integers m, such that*

$$\theta_m = \theta + 2\epsilon_m\omega, \qquad \theta_m + \mathrm{dd^c} u_m \geq \epsilon_m\omega.$$

For each m, the family $\mathbb{D}_{\theta_m}(u_m)$ has a nef Cartier realization. On every fixed compact model $\pi : Z \to Y$, its class has the current representative

$$C_{m,Z} = \pi^*\theta + 2\epsilon_m\pi^*\omega - D_{m,Z}, \qquad D_{m,Z} = \frac{1}{m}\sum_E \mathrm{ord}_E\, \mathcal{I}(mu)[E].$$

Here E ranges over prime divisors on Z, and the order is computed after pulling the ideal back to Z. These representatives satisfy

$$C_{m,Z} \longrightarrow \pi^*\theta - N_Z(u) \quad \textit{weakly as currents.}$$

The positive residual representatives satisfy

$$R_{m,Z} := C_{m,Z} + \mathrm{dd^c}(u_m \circ \pi) \longrightarrow \pi^*\theta + \mathrm{dd^c}(u \circ \pi) - N_Z(u) \geq 0 \quad \textit{weakly as currents.}$$

The exterior derivatives of both $C_{m,Z}$ and $R_{m,Z}$ converge smoothly to $\pi^*\mathrm{d}\theta$.

Proof Take u_m from Corollary 1.15.1, retaining its L^1 convergence. If $e_m \to 0$ are its curvature errors, put $\epsilon_m = \sup_{\ell \geq m} e_\ell + 1/m$. This gives positive decreasing errors while keeping the original integer indices. Adding $2\epsilon_m\omega$ changes the lower bound $-\epsilon_m\omega$ into $\epsilon_m\omega$. Each u_m has analytic type $(1/m, \mathcal{I}(mu))$, so Proposition 23.3.1 gives a nef Cartier realization in background θ_m.

On a fixed model the generic divisorial Lelong numbers of $u_m \circ \pi$ are exactly $m^{-1}\,\mathrm{ord}_E\, \mathcal{I}(mu)$. Dropping the $\mathrm{dd^c}$ of that potential from the residual-current representative leaves precisely $C_{m,Z}$ in its affine class. The self-trace identity and Theorem 22.1.1 give $D_{m,Z} \to N_Z(u)$ in trace mass, hence weakly as currents. The smooth error $2\epsilon_m\pi^*\omega$ tends to zero in every derivative, proving the convergence of $C_{m,Z}$.

Divisorial removal gives $R_{m,Z} \geq 0$. All the u_m and u have one smooth lower curvature bound: choose $A > 0$ with $\theta + \epsilon_m\omega \leq A\omega$ for every retained index. By Lemma 1.22.2, their $L^1(Y)$ convergence implies $u_m \circ \pi \to u \circ \pi$ in $L^1(Z)$. Applying $\mathrm{dd^c}$ and adding the already convergent $C_{m,Z}$ proves the residual-current limit. Positivity persists under weak convergence. Since each $D_{m,Z}$ is closed,

$$\mathrm{d}R_{m,Z} = \mathrm{d}C_{m,Z} = \pi^*\mathrm{d}\theta + 2\epsilon_m\pi^*\mathrm{d}\omega,$$

which proves the stated smooth convergence. □

The classes lie in $\mathcal{A}_Z(\theta_m)$. Translation by $-2\epsilon_m\pi^*\omega$ identifies this space with $\mathcal{A}_Z(\theta)$, but subtracting that form need not preserve nefness. Thus the current convergence above does not give nef Cartier approximation with exactly fixed exterior derivative.

23.4 Divisorial classes and infinite mass

Now take the background to be a Hermitian form ω on Y, with upper mass $v_+(\omega)$ and extended measure $\mathrm{MA}_\omega^{\mathrm{ext}}$ as in Chapter 6. When $v_+(\omega) = +\infty$, the potential of Theorem 6.6.1 has the same multiplier ideals as a smooth potential. Their residual classes agree on every model, while their extended masses differ.

Proposition 23.4.1 *If $v_+(\omega) = +\infty$, there is $U \in \mathrm{PSH}(Y, \omega)$ such that*

$$\mathbb{D}_\omega(U) = \mathbb{D}_\omega(0), \qquad \int_Y \mathrm{MA}_\omega^{\mathrm{ext}}(U) = +\infty, \qquad \int_Y \omega^n < +\infty.$$

The common family has the nef Cartier realization $(Y, \{\omega\})$.

Proof Take U from Theorem 6.6.1. Every positive-weight multiplier ideal is the unit ideal. Hence every divisorial order in $D_{m,Z}$ is zero on every model. The self-trace identity gives $N_Z(U) = 0$, which is also the divisorial part of the constant potential. Both families are therefore the pullback classes of ω. The form ω is smooth and positive, so its affine class is nef and realizes the family on the base. The infinite extended mass is part of the construction of U, whereas the smooth measure ω^n has finite mass by compactness. □

Chapter 24
Envelope systems over Hermitian perturbations

Let X be a connected compact complex manifold of dimension $n \geq 1$, let θ be a smooth real $(1,1)$-form on X, and fix real numbers $a < c$.

An ideal envelope depends on its background. A compatible family over positive Hermitian perturbations of θ records representatives of one multiplier-ideal class. As the perturbation vanishes, these normalized representatives decrease to a potential in the original background. Projecting the limit back recovers the family exactly when its multiplier ideals survive; otherwise it gives the greatest realizable family below it.

We consider test curves in a real parameter τ, constant for $\tau \leq a$ with cutoff c. Recovery can fail even in a Kähler background: at zero volume, a ruled surface forces an additional divisorial pole. For big backgrounds under bounded mass, positive initial lower volume gives recovery by Theorem 29.5.3. Smooth changes of background commute with recovery, while the limiting mass can change when the background is nonclosed.

24.1 Curves constant on a left half-line

We restrict the envelope test curves of Definition 17.1.1 by requiring constancy for $\tau \leq a$ and multiplier-ideal fixedness of each slice. Constancy identifies the initial potential explicitly and bounds the growth of the associated ray between the slopes a and c.

Definition 24.1.1 Let $\mathscr{T}_{a,c}(\theta)$ be the set of families $\Gamma = (\Gamma_\tau)_{\tau<c}$ satisfying:

(1) $\Gamma_\tau \in \mathrm{PSH}(X,\theta)$ and $P_\theta[\Gamma_\tau]_{\mathcal{I}} = \Gamma_\tau$;
(2) the family is pointwise concave and decreasing in τ;
(3) $\Gamma_\tau = \Gamma_a$ for $\tau \le a$.

The cutoff is c, with the extension $\Gamma_c = \lim_{\tau\uparrow c}\Gamma_\tau$ and $\Gamma_\tau = -\infty$ for $\tau > c$. The order on these curves is pointwise order of their finite slices.

Every slice has supremum zero. The decreasing endpoint limit is genuine by Theorem 1.4.1 and Corollary 1.4.1, and it is also $P_\theta[\cdot]_{\mathcal{I}}$-fixed: order preservation gives

$$\Gamma_c \le P_\theta[\Gamma_c]_{\mathcal{I}} \le \Gamma_\tau \quad (\tau < c),$$

where the first inequality uses the normalized potential itself as a competitor. Taking the infimum proves equality.

Proposition 24.1.1 *A curve in $\mathscr{T}_{a,c}(\theta)$ is an envelope test curve with initial potential $\varphi = \Gamma_a$. Denote its Legendre transform by $U_t = (\sup_{\tau<c}(\Gamma_\tau + t\tau))^*$ for real $t > 0$. It is a Perron envelope ray and satisfies*

$$\varphi + at \le U_t \le \varphi + ct \qquad (t > 0).$$

In particular $U_t \sim \varphi$ at each finite time.

Proof A $P_\theta[\cdot]_{\mathcal{I}}$-fixed potential is P_θ-fixed by Proposition 15.2.1. The regularized supremum of the curve is exactly Γ_a, so all the hypotheses of Definition 17.1.1 hold. Apply Theorem 17.2.1. The term $\tau = a$ in the transform gives the lower bound, and $\Gamma_\tau \le \varphi$, $\tau < c$, give the upper bound. □

24.2 Transition maps and envelope systems

To organize curves over different backgrounds into one system, successive projections must agree with direct projection. Ideal preservation gives this identity: the final background has a unique fixed representative of each multiplier-ideal class. It also gives injectivity, because equality of projected slices forces equality of the original fixed representatives.

Proposition 24.2.1 *For a smooth real $(1,1)$-form $\beta \ge \theta$ on X, slicewise projection defines an injective map*

$$P_\beta[\cdot]_{\mathcal{I}} : \mathscr{T}_{a,c}(\theta) \longrightarrow \mathscr{T}_{a,c}(\beta).$$

For any smooth real $(1,1)$-form β' with $\beta \ge \beta' \ge \theta$, these maps satisfy

$$P_\beta\big[P_{\beta'}[\Gamma]_{\mathcal{I}}\big]_{\mathcal{I}} = P_\beta[\Gamma]_{\mathcal{I}}.$$

They preserve order.

Proof The input slices are β-plurisubharmonic, so their projections are genuine and $P_\beta[\cdot]_{\mathcal{I}}$-fixed. Decreasingness, constancy for $\tau \le a$, and order preservation follow from Proposition 15.2.2.

To check concavity, choose real $r, t < c$ and $0 < s < 1$, and put $\tau = (1-s)r + st$. That same proposition and the concavity of the original curve give

$$\begin{aligned}(1-s)P_\beta[\Gamma_r]_{\mathcal{I}} + sP_\beta[\Gamma_t]_{\mathcal{I}} &\le P_\beta[(1-s)\Gamma_r + s\Gamma_t]_{\mathcal{I}} \\ &\le P_\beta[\Gamma_\tau]_{\mathcal{I}}.\end{aligned}$$

Here the first inequality follows by combining the scaling and addition formulas with backgrounds $(1-s)\beta$ and $s\beta$.

Each projection preserves the multiplier ideals at every weight. Thus the two sides of the transition identity are $P_\beta[\cdot]_{\mathcal{I}}$-fixed representatives of the same multiplier-ideal class, and are equal. If two projected curves agree, their original slices have the same multiplier ideals; since those original slices are $P_\theta[\cdot]_{\mathcal{I}}$-fixed, they agree as well. This proves injectivity. □

Let $\mathscr{H}(X)$ denote the set of smooth positive Hermitian forms on X.

Definition 24.2.1 An *envelope system over* θ, with parameters $a < c$, is a family

$$\mathbf{\Gamma} = (\Gamma^\eta)_{\eta \in \mathscr{H}(X)}, \qquad \Gamma^\eta \in \mathscr{T}_{a,c}(\theta + \eta),$$

satisfying

$$\Gamma^\eta = P_{\theta+\eta}[\Gamma^{\eta'}]_{\mathcal{I}} \quad \text{whenever } \eta \ge \eta'. \tag{24.1}$$

Write $\mathcal{S}_{a,c}(\theta)$ for this set, ordered componentwise. A system is *realized in the background* θ if it has the form

$$\mathsf{E}(\Gamma)^\eta = P_{\theta+\eta}[\Gamma]_{\mathcal{I}} \quad \text{for some } \Gamma \in \mathscr{T}_{a,c}(\theta).$$

All components have the same multiplier ideals at each $\tau < c$: compare two of them with the component indexed by $\eta + \eta'$.

If $u \in \mathrm{PSH}(X, \theta)$, the constant family $\Gamma_\tau = P_\theta[u]_{\mathcal{I}}$ for $\tau < c$, with the specified cutoff, lies in $\mathscr{T}_{a,c}(\theta)$. Its projections form an envelope system. The transition proposition also proves that E is injective.

24.3 The limiting curve and recovery

Every positive Hermitian perturbation dominates all sufficiently small multiples of a fixed Hermitian form. A single decreasing sequence of components therefore determines the limit of the whole system. Projecting the resulting curve back gives the greatest realized system below the original one; a strict inequality means that recovery fails.

Theorem 24.3.1 *Fix any Hermitian form h on X. The index j below ranges over the positive integers. For every $\mathbf{\Gamma} \in \mathcal{S}_{a,c}(\theta)$ the limits*

$$(\mathsf{R}\mathbf{\Gamma})_\tau = \lim_{j\to\infty} \Gamma_\tau^{h/j} \qquad (\tau < c)$$

exist as decreasing genuine potentials and define an element of $\mathcal{T}_{a,c}(\theta)$. They are independent of h. The maps satisfy

$$\mathsf{RE} = \mathrm{id}, \qquad \mathsf{ER}(\mathbf{\Gamma}) \le \mathbf{\Gamma}.$$

Both maps preserve order. More precisely,

$$\mathsf{E}(\Gamma) \le \mathbf{\Gamma} \quad \Longleftrightarrow \quad \Gamma \le \mathsf{R}\mathbf{\Gamma}. \tag{24.2}$$

Thus $\mathsf{ER}(\mathbf{\Gamma})$ is the greatest system below $\mathbf{\Gamma}$ that is realized in the original background.

Proof Fix $\tau < c$ and put $q_j = \Gamma_\tau^{h/j}$. Compatibility and the fact that $q_{j+1} \le 0$ is a competitor for its larger-background envelope give $q_j \ge q_{j+1}$. The functions have supremum zero and a common smooth lower curvature bound. By Theorem 1.4.1 and Corollary 1.4.1, their decreasing limit q is genuine and has supremum zero. Passing the distributional inequalities

$$\theta + h/j + \mathrm{dd^c} q_j \ge 0$$

to the limit gives $q \in \mathrm{PSH}(X, \theta)$.

The defining envelope family grows when the background increases. Together with $q \le q_j$ and ideal-order preservation, this gives

$$q \le P_\theta[q]_{\mathcal{I}} \le P_{\theta+h/j}[q]_{\mathcal{I}} \le P_{\theta+h/j}[q_j]_{\mathcal{I}} = q_j.$$

Taking the infimum in j proves $P_\theta[q]_{\mathcal{I}} = q$. Concavity, decreasingness and constancy on the left half-line pass pointwise to these limits. They therefore form a curve in $\mathcal{T}_{a,c}(\theta)$.

For every $\eta > 0$ there is j with $h/j \le \eta$. Compatibility gives $\Gamma_\tau^\eta \ge \Gamma_\tau^{h/j}$. Consequently

$$(\mathsf{R}\mathbf{\Gamma})_\tau = \inf_{\eta \in \mathscr{H}(X)} \Gamma_\tau^\eta,$$

which proves independence of h. It also proves order preservation of R.

Write $q = (\mathsf{R}\boldsymbol{\Gamma})_\tau$. Since $q \le \Gamma^\eta_\tau$, order preservation gives $P_{\theta+\eta}[q]_{\mathcal{I}} \le \Gamma^\eta_\tau$. This proves $\mathsf{ER} \le \mathrm{id}$.

Now start with a base curve Γ. Its slice Γ_τ is a competitor in every larger background, so

$$\Gamma_\tau \le (\mathsf{RE}\Gamma)_\tau \le P_{\theta+\eta}[\Gamma_\tau]_{\mathcal{I}} \quad (\eta > 0).$$

The outside terms have the same multiplier ideals. By the two ideal inclusions supplied by these inequalities, the middle term has those ideals as well. Both it and Γ_τ are $P_\theta[\cdot]_{\mathcal{I}}$-fixed, so they are equal. This proves $\mathsf{RE} = \mathrm{id}$.

Order preservation of E follows from the transition proposition. Applying R proves the forward implication in (24.2). For the reverse implication apply E and use $\mathsf{ER} \le \mathrm{id}$. This equivalence proves the final maximality assertion. □

Proposition 24.3.1 *A system $\boldsymbol{\Gamma} \in \mathcal{S}_{a,c}(\theta)$ is realized in θ if and only if, for every $\tau < c$, its limiting slice $(\mathsf{R}\boldsymbol{\Gamma})_\tau$ is multiplier-ideal equivalent to one, and hence every, slice Γ^η_τ.*

Proof For a realized system this follows from $\mathsf{RE} = \mathrm{id}$ and ideal preservation by each projection. Conversely the assumed equivalence and fixedness in the larger background give

$$P_{\theta+\eta}[(\mathsf{R}\boldsymbol{\Gamma})_\tau]_{\mathcal{I}} = \Gamma^\eta_\tau.$$

Thus the system equals ER of itself. □

By Theorem 1.16.1, recovery is equivalent to preservation of every Lelong number on every local modification. These numbers are independent of the component, whereas the limiting slice can have larger values. The ruled-surface example in Section 24.5 increases one of them from zero to one. Under bounded mass and bigness, Theorem 29.5.3 prevents such an increase when the initial slices have a positive limiting lower volume.

When X is Kähler, the closed positive perturbations are cofinal: a sufficiently small positive multiple of one fixed Kähler form lies below any given Hermitian form. The compatibility formula therefore determines a system from its closed-positive components. Conversely, a compatible family on that collection extends uniquely. For $\eta > 0$, choose a closed positive $\kappa \le \eta$ and set $\Gamma^\eta = P_{\theta+\eta}[\Gamma^\kappa]_{\mathcal{I}}$. Two choices give the same result by passing to a common smaller closed positive form and using the transition identity.

24.4 Changes of representative and limiting mass

Subtracting a smooth potential transports a curve to the new background, and projection restores its envelope normalization. The gauge estimate is uniform over the perturbations, so this transport also respects the limiting curve. Limiting Monge–Ampère mass behaves differently: in a nonclosed background it can change even for the system with zero slices.

Proposition 24.4.1 *If $\theta' = \theta + \mathrm{dd^c} f$ with $f \in C^\infty(X, \mathbb{R})$, take $\Gamma \in \mathcal{T}_{a,c}(\theta)$ and $\mathbf{\Gamma} \in \mathcal{S}_{a,c}(\theta)$. The formulas*

$$(\mathsf{G}_f \Gamma)_\tau = P_{\theta'}[\Gamma_\tau - f]_{\mathcal{I}}, \qquad (\mathsf{G}_f \mathbf{\Gamma})^\eta_\tau = P_{\theta'+\eta}[\Gamma^\eta_\tau - f]_{\mathcal{I}}$$

define bijections on the corresponding curve and system spaces. They satisfy the cocycle law and commute with E *and* R.

Proof Subtracting f changes the background as indicated and preserves concavity, decreasingness and the constant left part. Slicewise projection preserves these properties by the convex-combination argument in Proposition 24.2.1. Every new slice has the multiplier ideals of the original one because f is smooth. Applying the construction with $-f$ gives a $P_\theta[\cdot]_{\mathcal{I}}$-fixed slice with those same ideals, hence the original slice. This proves the inverse statement; the same uniqueness proves the cocycle law.

Compatibility of the system and commutation with E follow by comparing multiplier ideals in the common final background. For commutation with R, the gauge estimate of Proposition 15.3.1, applied in each background $\theta + \eta$, gives

$$\Gamma^\eta_\tau + \min_X f \le (\mathsf{G}_f \mathbf{\Gamma})^\eta_\tau + f \le \Gamma^\eta_\tau + \max_X f.$$

The constants are independent of η. Pass to the cofinal decreasing limits. The new limit is ordinarily equivalent to the old limit minus f, hence has the same multiplier ideals. It is $P_{\theta'}[\cdot]_{\mathcal{I}}$-fixed by Theorem 24.3.1. It must therefore equal $P_{\theta'}[(\mathsf{R}\mathbf{\Gamma})_\tau - f]_{\mathcal{I}}$, proving commutation with R. □

For closed backgrounds on a compact Kähler manifold, [Xia, chapter "Non-Archimedean pluripotential theory", section "The definition of non-Archimedean metrics"] uses the closed-positive indexing above and defines volume by a limit of initial masses. Its components are general $\mathcal{I}$-model test curves with positive-mass initial potentials. Here the curves are constant on the prescribed left half-line, and the system space has no mass condition. For nonclosed positive backgrounds, the corresponding limit of masses need not be invariant under the gauge bijection.

Proposition 24.4.2 *There are Hermitian forms ω, h on a compact complex surface X and a smooth real function f, with $\omega' = \omega + \mathrm{dd^c} f > 0$, for which the following holds. For every $a < c$ there is a system $\mathbf{\Gamma} \in \mathcal{S}_{a,c}(\omega)$ such that both limits*

$$\lim_{\epsilon \downarrow 0} \int_X \mathrm{MA}^{\mathrm{ext}}_{\omega+\epsilon h}(\Gamma^{\epsilon h}_a), \qquad \lim_{\epsilon \downarrow 0} \int_X \mathrm{MA}^{\mathrm{ext}}_{\omega'+\epsilon h}((\mathsf{G}_f \mathbf{\Gamma})^{\epsilon h}_a)$$

exist and are unequal.

Proof Take the torus and the smooth perturbation from Proposition 15.3.2, so that $\omega' = \omega + \mathrm{dd^c} f$ is Hermitian and $\int_X (\omega')^2 \neq \int_X \omega^2$. Fix any Hermitian form h. In every background $\omega + \eta$, let every slice with $\tau < c$ be zero. This is an envelope system for any $a < c$: zero is the multiplier-ideal envelope of the unit-ideal class in a positive background. Its first limit is

$$\lim_{\epsilon \downarrow 0} \int_X (\omega + \epsilon h)^2 = \int_X \omega^2.$$

The gauge bijection sends every slice to $P_{\omega'+\eta}[-f]_{\mathcal{I}} = 0$, since f is smooth and the new background is positive. The second limit is therefore $\int_X (\omega')^2$, which is different. □

24.5 Failure of recovery in a positive background

For this section, take $E = \mathbb{C}/(\mathbb{Z} + \mathrm{i}\mathbb{Z})$ and let X be the compact ruled surface

$$X = (\mathbb{C}_z \times \mathbb{P}^1_w)/\sim, \qquad (z, w) \sim (z + 1, w), \qquad (z, w) \sim (z + \mathrm{i}, w + 1).$$

These commuting transformations define a holomorphic $\mathbb{P}^1$-bundle $\pi : X \to E$. Let D be its section $w = \infty$. The function $v = w - \operatorname{Im} z$ is invariant under both transformations and descends to a smooth function on $X \setminus D$.

The obstruction to recovery will be the class of D. It is nef, but its only positive current is $[D]$. Prescribing a pole along one fiber in a Kähler background will leave this residual class. Positive perturbations admit smooth residual representatives, so the perturbed envelopes have only the fiber pole. At the limit they must acquire D as an additional pole, and their residual volume tends to zero.

Lemma 24.5.1 *Let $\sigma_0 = (\mathrm{i}/2)\,\mathrm{d}z \wedge \mathrm{d}\bar{z}$ be the flat probability form on E. There is a smooth closed real $(1,1)$-form $\theta \in c_1(\mathcal{O}_X(D))$ whose class is nef, and a Kähler form*

$$\kappa = \theta + \pi^*\sigma_0.$$

The only positive current of the form $\theta + \mathrm{dd^c}u$ is $[D]$.

Proof Let s_D be the canonical section of $\mathcal{O}_X(D)$ with zero divisor D. For a real parameter $t > 0$, define a smooth Hermitian metric on this line bundle by prescribing its norm:

$$|s_D|_t^2 = (1 + t|v|^2)^{-1} \quad \text{on } X \setminus D.$$

To check smooth extension, use $\xi = 1/w$ near D. The local squared norm of a nonvanishing frame is

$$\left(|\xi|^2 + t|1 - (\operatorname{Im} z)\xi|^2\right)^{-1},$$

which is smooth and positive. The displayed norm is invariant on the complement, so these extensions define a global metric. Its curvature is

$$\theta_t = \mathrm{dd^c}\log(1 + t|v|^2) \quad \text{on } X \setminus D.$$

Set $\theta = \theta_1$ and $\sigma = \sigma_0/\pi$ on E, where the denominator is the number π.

Here is an explicit curvature estimate. Put $r = |v|^2$ and let H be the complex Hessian of $\log(1 + tr)$ in (z, w). Direct differentiation gives

$$\begin{aligned} H_{w\bar{w}} &= \frac{t}{(1+tr)^2}, \\ H_{z\bar{w}} &= \frac{\mathrm{i}t(1 - tv^2)}{2(1+tr)^2}, \\ H_{z\bar{z}} &= \frac{t}{2(1+tr)} - \frac{t^2(\operatorname{Re} v)^2}{(1+tr)^2}. \end{aligned}$$

Consequently

$$H_{z\bar{z}} - \frac{|H_{z\bar{w}}|^2}{H_{w\bar{w}}} = \frac{t(1-tr)}{4(1+tr)} \geq -\frac{t}{4}.$$

Completing the square proves $\theta_t \geq -(t/4)\pi^*\sigma$. This inequality extends smoothly across D. The restriction of θ_t to every fiber is a positive Fubini–Study form, including at infinity. Thus $\theta_t + A\pi^*\sigma$ is strictly positive off D whenever $A > t/4$. At $\xi = 0$, its coefficient matrix, with the common factor $\mathrm{i}/(2\pi)$ omitted, is

$$\begin{pmatrix} A & \mathrm{i}/2 \\ -\mathrm{i}/2 & 1/t \end{pmatrix}.$$

Its vertical coefficient is positive and its Schur complement is $A - t/4 > 0$. This checks strict positivity also at infinity. In particular $\kappa = \theta + \pi^*\sigma_0$ is Kähler.

The function

$$f_t = \log \frac{1 + t|v|^2}{1 + |v|^2}$$

extends smoothly across D and satisfies $\theta_t = \theta + \mathrm{dd^c} f_t$. The curvature lower bound, with $t \downarrow 0$, proves nefness in the smooth metric definition.

Write $\psi_D = \log |s_D|_1^2$. Poincaré–Lelong gives $\theta + \mathrm{dd^c}\psi_D = [D]$. If $u \in \mathrm{PSH}(X, \theta)$, then $u - \psi_D$ is plurisubharmonic on $X \setminus D$. Its lift $H(z, w)$ to $\mathbb{C}^2$ is invariant under the two deck transformations and, since u is bounded above on compact X, satisfies the following inequality for some real constant C:

$$H(z, w) \le C + \log\bigl(1 + |w - \operatorname{Im} z|^2\bigr).$$

For each fixed w, the restriction in z is either identically $-\infty$ or a genuine subharmonic function. It is 1-periodic and has logarithmic upper growth, so Lemma 1.3.6 makes it constant in z. Thus $H(z, w) = h(w)$, with h a genuine subharmonic function of w. The second deck transformation now gives $h(w + 1) = h(w)$, and the bound at $z = 0$ gives logarithmic growth. The same lemma makes h constant. Therefore $u = \psi_D + C$ off D, and Lemma 1.3.3 extends the equality across D. This proves the assertion. □

Fix a point $p \in E$ and let $F = \pi^{-1}(p)$. The Green-kernel construction in the proof of Theorem 18.3.1 gives a quasi-plurisubharmonic g with $\sigma_0 + \mathrm{dd^c} g = \delta_p$. Its pullback $\psi_F = g \circ \pi$ has a logarithmic pole of coefficient one along F, and

$$\pi^*\sigma_0 + \mathrm{dd^c}\psi_F = [F].$$

Theorem 24.5.1 *For the Kähler background κ above and every $a < c$, there is a system $\mathbf{\Gamma} \in \mathcal{S}_{a,c}(\kappa)$ for which*

$$\mathsf{ER}(\mathbf{\Gamma}) \ne \mathbf{\Gamma}.$$

Every finite slice of every component has multiplier ideals

$$\mathcal{I}(t\Gamma_\tau^\eta) = \mathcal{O}_X(-\lfloor t \rfloor F) \qquad (t > 0),$$

whereas its limiting slice has ideals

$$\mathcal{I}\bigl(t(\mathsf{R}\mathbf{\Gamma})_\tau\bigr) = \mathcal{O}_X(-\lfloor t \rfloor (F + D)).$$

In particular E *is not surjective and* R *is not injective on these unrestricted system spaces.*

Proof For each Hermitian form $\eta > 0$, the explicit nef approximation supplies a smooth f_η such that $\theta + \eta + \mathrm{dd^c} f_\eta \ge 0$. Then

$$u_\eta = \psi_F + f_\eta \in \mathrm{PSH}(X, \kappa + \eta).$$

Define every finite slice of Γ^η to be $q_\eta = P_{\kappa+\eta}[u_\eta]_{\mathcal{I}}$, with cutoff c. Its multiplier ideals are those of ψ_F, independent of the smooth choice f_η. Hence the transition

identity follows by comparing these ideals in the larger background. The constant curves form a system with the required parameters.

Let q be its limiting slice under R, and let $\nu(q, F)$ denote its generic Lelong number along F. Each q_η has generic Lelong number one along F, and Lemma 1.17.5 applied to the decreasing cofinal sequence gives $\nu(q, F) \geq 1$. By Lemma 1.17.1,

$$\kappa + \mathrm{dd^c} q - [F] \geq 0.$$

By Poincaré–Lelong this current is $\theta + \mathrm{dd^c}(q - \psi_F)$, where the difference is first taken as an integrable distribution. Positivity supplies its canonical θ-plurisubharmonic representative. By Lemma 24.5.1, its potential differs from ψ_D by a constant. Hence

$$q = \psi_F + \psi_D + C.$$

The two divisors are smooth and meet transversely, since one is a fiber and the other a section. Direct product integrability in local coordinates gives the stated ideals for every real $t > 0$. At a generic point of D, the Lelong number has thus increased from zero to one. The realization criterion proves the strict failure of recovery.

Finally, $\mathsf{RE} = \mathrm{id}$ implies that $\mathbf{\Gamma}$ and the distinct realized system $\mathsf{ER}\mathbf{\Gamma}$ have the same image under R. This proves its failure of injectivity. □

Proposition 24.5.1 *Let q_η be the common finite slice constructed in the preceding proof. Along the closed perturbations $\eta = \epsilon\kappa$, with real $\epsilon > 0$, the component slices of this example satisfy*

$$\int_X \mathrm{MA}^{\mathrm{ext}}_{\kappa+\epsilon\kappa}(q_{\epsilon\kappa}) = 2\epsilon \int_X \theta \wedge \kappa + \epsilon^2 \int_X \kappa^2 \longrightarrow 0.$$

The limiting slice has zero non-pluripolar Monge–Ampère measure.

Proof The potential $u_\eta = \psi_F + f_\eta$ has analytic singularities. Hence Proposition 15.2.3 gives $q_\eta \sim u_\eta \sim \psi_F$. Removing the divisorial component $[F]$ gives a positive current in background $\theta + \eta$, whose global potential $w_\eta = q_\eta - \psi_F$ is bounded by this ordinary equivalence.

Off F, plurifine locality identifies the Monge–Ampère measure with $(\theta + \eta + \mathrm{dd^c} w_\eta)^2$. This bounded product gives no mass to F by Proposition 1.7.1. For closed $\eta = \epsilon\kappa$, Stokes gives its total mass as $\int_X (\theta + \epsilon\kappa)^2$.

The normal bundle of D is trivial. Indeed the transition $w \mapsto w + 1$ gives $\xi \mapsto \xi/(1 + \xi)$, whose derivative at $\xi = 0$ is one; the other transition is the identity. Thus $\int_X \theta^2 = D^2 = 0$. Expanding the square gives the formula. The limit potential has current $[F] + [D]$, which vanishes away from $F \cup D$. Its non-pluripolar measure is consequently zero. □

In the closed setting of [Xia, section "The definition of non-Archimedean metrics"], positive volume characterizes systems arising from positive-mass base curves. The example above lies at the zero-volume boundary. For smooth big backgrounds under bounded mass, Theorem 29.5.3 gives recovery when the initial lower-volume limit is positive.

Part VI
Gauduchon duality and finite volumes

Chapter 25
A Gauduchon variational volume

Let X be a connected compact complex manifold of dimension $n \geq 1$, and fix a smooth strictly positive probability form μ on X.

For a closed Kähler background, Stokes' theorem determines the scalar in the normalized Monge–Ampère equation. For a nonclosed background, the total mass can vary with the potential, but the smooth equation still selects one scalar. Its nth root is the infimum of a Gauduchon quotient whose numerator is linear in the background. This formula extends the scalar to pseudoeffective backgrounds and makes its concavity visible. Where a smooth positive representative exists, the derivative recovers its solution form up to scale, and the logarithmic Hessian defines a metric on background classes.

For integrable densities, the corresponding dual problem computes the subsolution threshold. Allowing current tests makes that problem compact, while weighted tail estimates pass its determinant integral to a weak limit. A smooth strictly positive maximizing test gives more: equality in the trace inequality determines the solution current and forces an exact equation. The distinction between current attainment and this rigidity is essential when the test degenerates.

Varying the probability measure recovers the upper and lower Monge–Ampère masses. A threefold with zero lower mass shows that positivity for each smooth density need not be uniform over densities. Nef degeneration gives a second contrast with closed Kähler classes: intersection polynomials are replaced by fractional, logarithmic or oscillating decay, and a single blowup can change its order.

25.1 The quotient and the smooth equation

Gauduchon Stokes makes the numerator of the quotient invariant under a smooth $\mathrm{dd^c}$ change of background. Its denominator is chosen so that the quotient has the same homogeneity as the root of the equation scalar. For a Gauduchon metric γ, put

$$D_\mu(\gamma) = \int_X \left(\frac{\mu}{\gamma^n}\right)^{1/n} \gamma^n.$$

This number is positive. It scales by r^{n-1} when γ is multiplied by a positive constant r.

Definition 25.1.1 For a pseudoeffective smooth real $(1,1)$-form θ on X, define

$$\lambda_\mu(\theta) = \inf_{\gamma \text{ Gauduchon}} \frac{\displaystyle\int_X \theta \wedge \gamma^{n-1}}{D_\mu(\gamma)}, \qquad \mathcal{V}_\mu(\theta) = \lambda_\mu(\theta)^n.$$

In dimension one every Hermitian metric is permitted in the infimum; its quotient is simply $\int_X \theta$.

Proposition 25.1.1 *The number $\lambda_\mu(\theta)$ is finite and nonnegative. It is unchanged by a smooth $\mathrm{dd^c}$ change of θ, positively homogeneous of degree one, and concave on the pseudoeffective cone. If β is a smooth real $(1,1)$-form and $\beta - \theta$ is pseudoeffective, then*

$$\lambda_\mu(\beta) \geq \lambda_\mu(\theta).$$

The function is upper semicontinuous under smooth convergence within that cone.

Proof Choose $u \in \mathrm{PSH}(X,\theta)$. Distributional Stokes and the Gauduchon condition give

$$\int_X \theta \wedge \gamma^{n-1} = \int_X (\theta + \mathrm{dd^c} u) \wedge \gamma^{n-1} \geq 0.$$

Thus every quotient is nonnegative. Testing one fixed Gauduchon metric gives a finite upper bound. The same identity proves gauge invariance and monotonicity for a pseudoeffective difference.

Each quotient is a linear function of the background. The infimum of these linear functions is concave and positively homogeneous. Each individual quotient is continuous in the smooth topology. Given a smoothly converging sequence, bound the infimum by any one quotient, take the upper limit, and finally take the infimum over the test metric. This proves upper semicontinuity. □

The smooth equation identifies both the value of the quotient and every minimizing metric. Let $\mathscr{P}$ consist of backgrounds θ admitting a smooth real u with $\theta + \mathrm{dd^c} u > 0$. For $\theta \in \mathscr{P}$, the smooth equation theorem gives a unique positive form

$$\Omega_\theta = \theta + \mathrm{dd^c} u_\theta, \qquad \Omega_\theta^n = c_\theta(\mu)\mu, \qquad c_\theta(\mu) > 0.$$

The potential is unique after any fixed additive normalization. To apply Theorem 9.1.1, first choose a Hermitian representative of the background and express μ as a smooth positive density relative to its volume form.

Theorem 25.1.1 *For every $\theta \in \mathscr{P}$,*

$$\lambda_\mu(\theta) = c_\theta(\mu)^{1/n}, \qquad \mathcal{V}_\mu(\theta) = c_\theta(\mu).$$

The infimum in Definition 25.1.1 is attained. When $n \geq 2$, its minimizing Gauduchon metrics are precisely the constant multiples of the Gauduchon metric conformal to Ω_θ.

Proof For any Hermitian metric γ, the arithmetic–geometric mean inequality for the eigenvalues of Ω_θ relative to γ gives the pointwise inequality

$$\Omega_\theta \wedge \gamma^{n-1} \geq \left(\frac{\Omega_\theta^n}{\gamma^n}\right)^{1/n} \gamma^n = c_\theta(\mu)^{1/n} \left(\frac{\mu}{\gamma^n}\right)^{1/n} \gamma^n.$$

If γ is Gauduchon, integration replaces Ω_θ in the numerator by θ. Every test quotient is therefore at least $c_\theta(\mu)^{1/n}$.

For $n \geq 2$, take the Gauduchon representative $\gamma = \mathrm{e}^g \Omega_\theta$ given by Theorem 4.1.1. The eigenvalues are equal at every point, so equality holds pointwise and then in the quotient. This proves the formula and attainment. Conversely, if a metric attains the infimum, the nonnegative smooth difference in the displayed inequality has zero integral. It vanishes everywhere, and equality in the arithmetic–geometric mean inequality makes γ conformal to Ω_θ. Uniqueness of the Gauduchon metric in a conformal class, up to constant scale, proves the last assertion.

For $n = 1$, integrate the equation and use $\int_X \mathrm{dd^c} u_\theta = 0$. It gives $c_\theta(\mu) = \int_X \theta$, equal to every quotient in the definition. □

For nonzero nonnegative L^1 densities in a Hermitian background, the analogous quotient computes the subsolution threshold by Theorem 13.25.1. Attainment and regularity for those data will be studied below.

Corollary 25.1.1 *If θ is big, then $\lambda_\mu(\theta) > 0$.*

Proof Choose a Hermitian form h and $\delta > 0$ such that $\theta - \delta h$ is pseudoeffective. Monotonicity, homogeneity and the theorem give $\lambda_\mu(\theta) \geq \delta \lambda_\mu(h) > 0$. □

Whether positive $\mathcal{V}_\mu(\theta)$ forces bigness remains open here, even for a general nef background.

25.2 Concavity, equality and variation

A minimizing Gauduchon metric supports the concave function λ_μ. Equality in concavity forces the same metric to minimize at both endpoints; differentiating the smooth equation identifies that supporting functional with the derivative. For $\theta \in \mathscr{P}$, choose a minimizing metric γ_θ. The variation directions are smooth real $(1,1)$-forms.

Theorem 25.2.1 *Let $\theta_0, \theta_1 \in \mathscr{P}$ and $0 < s < 1$. Then*

$$\lambda_\mu((1-s)\theta_0 + s\theta_1) \geq (1-s)\lambda_\mu(\theta_0) + s\lambda_\mu(\theta_1).$$

Equality holds if and only if, for some $r > 0$ and some smooth real f,

$$\theta_1 = r\theta_0 + \mathrm{dd^c} f.$$

Proof Concavity was proved in Proposition 25.1.1. Suppose first that $n \geq 2$ and equality holds. Take a minimizing Gauduchon metric for the middle background. Its quotient is the indicated convex combination of its two endpoint quotients. Each endpoint quotient is at least the corresponding infimum. Equality therefore forces this same metric to minimize both endpoint quotients.

By Theorem 25.1.1, the solution forms $\Omega_i = \Omega_{\theta_i}$, with scalars $c_i = c_{\theta_i}(\mu)$ for $i = 0, 1$, are conformal to this metric. Their volume forms are $c_0\mu$ and $c_1\mu$, so their pointwise proportionality factor is the constant $r = (c_1/c_0)^{1/n}$. Hence $\Omega_1 = r\Omega_0$, which gives the stated smooth $\mathrm{dd^c}$ relation. Conversely that relation, gauge invariance and homogeneity prove equality.

When $n = 1$, the function is $\int_X \theta$. Take $r = \int_X \theta_1 / \int_X \theta_0$. The difference $\theta_1 - r\theta_0$ has integral zero, so the Green operator on the compact curve writes it as $\mathrm{dd^c} f$ with f smooth. Thus the same equivalence holds. □

Proposition 25.2.1 *The function λ_μ is smooth along smooth finite-dimensional families in $\mathscr{P}$. For a smooth real $(1,1)$-form χ and $n \geq 2$, its derivative at θ is*

$$(\mathrm{d}\lambda_\mu)_\theta(\chi) = \frac{\displaystyle\int_X \chi \wedge \gamma_\theta^{n-1}}{D_\mu(\gamma_\theta)},$$

where γ_θ is any minimizing Gauduchon metric. In dimension one the derivative is $\int_X \chi$.

Proof Choose a fixed smooth potential u_* making $\theta + \mathrm{dd^c} u_* > 0$. Nearby backgrounds have the positive representatives $\theta_t + \mathrm{dd^c} u_*$. The implicit-function argument in the proof of Proposition 14.8.1 applies to this family and gives smooth dependence of the solution and its scalar. Adding u_* to these solution potentials and fixing the mean gives the normalized solutions for θ_t.

Write $\lambda = \lambda_\mu(\theta)$, $\Omega = \Omega_\theta$, and $\gamma_\theta = \mathrm{e}^g\Omega$. Along $\theta_t = \theta + t\chi$ for small real t, put $\Omega_t = \theta_t + \mathrm{dd^c}u_t$ and $\lambda_t = \lambda_\mu(\theta_t)$. Dots denote derivatives at $t = 0$. Differentiating $\Omega_t^n = \lambda_t^n\mu$ gives

$$(\chi + \mathrm{dd^c}\dot{u}) \wedge \Omega^{n-1} = \lambda^{n-1}\dot{\lambda}\,\mu.$$

Multiply by $\mathrm{e}^{(n-1)g}$ and integrate. The term with $\mathrm{dd^c}\dot{u}$ vanishes by the Gauduchon condition. Moreover

$$D_\mu(\gamma_\theta) = \lambda^{n-1}\int_X \mathrm{e}^{(n-1)g}\mu.$$

Substitution proves the derivative formula. The dimension-one formula follows directly from the definition. □

25.3 Second variation on background classes

The negative logarithm of the root volume has a positive Hessian after quotienting out smooth $\mathrm{dd^c}$ directions. The smooth equation expresses this Hessian as the squared norm of its linearized solution, integrated against the adjoint probability. On the closed Kähler cone it reduces to $1/n$ times the Hessian of $-\log\int_X \theta^n$.

Keep $\Omega = \Omega_\theta$, $u = u_\theta$ and $\lambda = \lambda_\mu(\theta)$, and let ν_Ω be the adjoint probability of Proposition 4.5.1; it annihilates $\mathrm{tr}_\Omega\,\mathrm{dd^c}$. In dimension one this probability is $\Omega/\int_X\Omega = \mu$.

Let $\theta_t = \theta + t\chi$, where t is real and χ is a smooth real $(1,1)$-form on X. For sufficiently small t, the background stays in $\mathscr{P}$. Choose the solution potentials with one fixed mean normalization. Dots denote derivatives of the normalized solution and its root volume at $t = 0$. Define the Hermitian endomorphism

$$A_\chi = \Omega^{-1}(\chi + \mathrm{dd^c}\dot{u}).$$

Here Ω^{-1} means raising an index with the positive metric Ω; traces of its powers are independent of coordinates. We write I for the identity endomorphism of $T^{1,0}X$ and $\|\cdot\|_\Omega$ for its Hilbert–Schmidt norm induced by Ω.

Proposition 25.3.1 *Along this affine variation,*

$$\frac{\ddot{\lambda}}{\lambda} = -\frac{1}{n}\int_X \left\| A_\chi - \frac{\dot{\lambda}}{\lambda} I \right\|_\Omega^2 \, \mathrm{d}\nu_\Omega, \tag{25.1}$$

$$-\frac{\mathrm{d}^2}{\mathrm{d}t^2} \log \lambda_\mu(\theta_t) = \frac{1}{n}\int_X \mathrm{tr}(A_\chi^2)\, \mathrm{d}\nu_\Omega. \tag{25.2}$$

The first expression vanishes exactly when $\chi = r\theta + \mathrm{dd^c} f$ for a real constant r and a smooth real f. The second vanishes exactly when $\chi = \mathrm{dd^c} f$ for a smooth real f.

Proof Smooth dependence was proved in Proposition 25.2.1. Differentiate the logarithmic equation. Since μ is fixed,

$$\mathrm{tr}\, A_\chi = n\frac{\dot{\lambda}}{\lambda}.$$

In particular this trace is constant on X. Differentiate once more along the affine background path:

$$\mathrm{tr}_\Omega(\mathrm{dd^c}\ddot{u}) - \mathrm{tr}(A_\chi^2) = n\left(\frac{\ddot{\lambda}}{\lambda} - \frac{\dot{\lambda}^2}{\lambda^2}\right).$$

Integration against the fixed probability ν_Ω at $t = 0$ removes the first term. This proves (25.2). No derivative of the adjoint measure is taken in this calculation.

Put $r = \dot{\lambda}/\lambda$. Since A_χ is Hermitian and $\mathrm{tr}\, A_\chi = nr$,

$$\|A_\chi - rI\|_\Omega^2 = \mathrm{tr}(A_\chi^2) - nr^2.$$

Substitution gives (25.1). The adjoint measure has a strictly positive smooth density. Thus equality in that formula is equivalent to $A_\chi = rI$ everywhere, or

$$\chi + \mathrm{dd^c}\dot{u} = r(\theta + \mathrm{dd^c} u).$$

This is the stated relation with $f = ru - \dot{u}$. Conversely, if $\chi = r\theta + \mathrm{dd^c} f$, then θ_t is smoothly $\mathrm{dd^c}$-equivalent to $(1 + rt)\theta$ for small t. Homogeneity makes $\lambda_\mu(\theta_t) = (1 + rt)\lambda$, so its second derivative is zero.

The integrand in (25.2) is the squared norm of a Hermitian endomorphism. Its integral vanishes exactly when $A_\chi = 0$, which gives $\chi = -\mathrm{dd^c}\dot{u}$. Conversely a smooth $\mathrm{dd^c}$ change leaves λ constant. This proves both equality statements. □

The linearized solution and scalar depend linearly on χ. For smooth real $(1,1)$-forms χ_1, χ_2 on X, polarization therefore gives a bilinear form

$$\mathfrak{g}_\theta(\chi_1, \chi_2) = -D^2 \log \lambda_\mu|_\theta(\chi_1, \chi_2) = \frac{1}{n}\int_X \mathrm{tr}(A_{\chi_1} A_{\chi_2})\, \mathrm{d}\nu_\Omega. \tag{25.3}$$

It descends to smooth real $(1,1)$-forms modulo $\mathrm{dd^c} C^\infty(X, \mathbb{R})$ and is positive definite there. A change of representative of θ leaves its solution form Ω unchanged; a change of representative of a direction is absorbed by the derivative of the solution potential.

Thus the form is intrinsic on these background classes. On every finite-dimensional affine family whose directions are linearly independent modulo smooth $\mathrm{dd^c}$ terms, it is a smooth Riemannian metric.

These variations change the background class. Directions obtained by varying a potential inside one fixed class are exactly the directions killed by (25.3). Thus this construction concerns a different space from the potential-space geometry in Chapter 3.

25.4 Recovering a positive class from its Gauduchon derivative

The derivative determines a normalized Gauduchon form. Combining its conformal direction with the value of the volume reconstructs the solution form itself. Fix $\theta \in \mathscr{P}$ and a minimizing metric γ_θ.

For $n \geq 2$ define the smooth positive $(n-1, n-1)$-form

$$\Xi_\theta = \frac{\gamma_\theta^{n-1}}{D_\mu(\gamma_\theta)}.$$

The quotient is independent of its constant scale, is $\mathrm{dd^c}$-closed, and represents the derivative in every smooth real $(1,1)$-direction χ:

$$(\mathrm{d}\lambda_\mu)_\theta(\chi) = \int_X \chi \wedge \Xi_\theta.$$

Proposition 25.4.1 *For $n \geq 2$, the pair consisting of $\lambda_\mu(\theta)$ and its derivative on all smooth real $(1,1)$-forms determines the smooth $\mathrm{dd^c}$-class of $\theta \in \mathscr{P}$. The derivative alone determines the class up to multiplication by a positive constant. More precisely, if $\widehat{\gamma}$ is the positive root of Ξ_θ, then*

$$D_\mu(\widehat{\gamma}) = 1, \qquad \Omega_\theta = \lambda_\mu(\theta)\left(\frac{\mu}{\widehat{\gamma}^n}\right)^{1/n}\widehat{\gamma}.$$

The classes in $\mathscr{P}$ normalized by $\lambda_\mu = 1$ are in bijection with the rays of Gauduchon metrics.

Proof Pairing against every smooth $(1,1)$-form determines the smooth form Ξ_θ. Its positive root exists and is unique by the positive-root construction in the proof of Theorem 4.3.1. If $D = D_\mu(\gamma_\theta)$, that root is $\widehat{\gamma} = D^{-1/(n-1)}\gamma_\theta$. The homogeneity of D_μ gives $D_\mu(\widehat{\gamma}) = 1$.

The solution form Ω_θ is conformal to its minimizing metric. Its volume is $\lambda_\mu(\theta)^n\mu$. These two requirements determine the conformal factor pointwise and give the displayed formula. Thus the value and derivative recover Ω_θ, a representative of the desired class. Without the value, the same formula recovers its positive ray. Conversely positive scaling of the class scales Ω_θ and leaves its minimizing Gauduchon ray, and hence Ξ_θ, unchanged.

For surjectivity of the last assertion, take any Gauduchon metric γ and set

$$\Omega_\gamma = \left(\frac{\mu}{\gamma^n}\right)^{1/n} \gamma.$$

This is unchanged by constant rescaling of γ, is Hermitian, and satisfies $\Omega_\gamma^n = \mu$. Its zero potential solves the smooth normalized equation, so its class has $\lambda_\mu = 1$. Equality in the pointwise arithmetic–geometric mean inequality shows that γ is its minimizing Gauduchon metric. Injectivity follows from the recovery formula: two normalized classes with the same Gauduchon ray have the same solution form. This proves the bijection. □

In dimension one, the smooth $\mathrm{dd^c}$-class is already determined by its integral, by the scalar Green operator. The derivative is the fixed functional $\chi \mapsto \int_X \chi$, and the value determines the class.

25.5 The Hessian metric and its completion

Homogeneity separates the scale of a positive class from its position on a level set of λ_μ. For the Hessian metric, the logarithm of this scale is a Euclidean coordinate. Completeness therefore reduces to the unit-volume hypersurface. We work on a finite-dimensional cone, where the induced path distance is Riemannian.

Let V be a finite-dimensional real vector space of smooth real $(1,1)$-forms such that

$$V \cap \mathrm{dd^c} C^\infty(X, \mathbb{R}) = \{0\}, \qquad \mathscr{C} = V \cap \mathscr{P} \neq \varnothing.$$

Then $\mathscr{C}$ is an open convex cone in V. Write $\lambda = \lambda_\mu|_{\mathscr{C}}$ and let $g = -D^2 \log \lambda$ be the metric in (25.3). Put

$$\mathscr{S} = \{\theta \in \mathscr{C} : \lambda(\theta) = 1\},$$

and equip this smooth hypersurface with the induced metric $g_{\mathscr{S}}$.

Proposition 25.5.1 *The map*

$$\Phi : \mathbb{R} \times \mathscr{S} \longrightarrow \mathscr{C}, \qquad (r, q) \longmapsto \mathrm{e}^r q$$

is an isometry from the product metric $\mathrm{d}r^2 + g_{\mathscr{S}}$ *onto* g*. In particular, if* d_g *and* $d_{\mathscr{S}}$ *are the intrinsic distances, then*

$$d_g(\mathrm{e}^{r_0} q_0, \mathrm{e}^{r_1} q_1)^2 = |r_1 - r_0|^2 + d_{\mathscr{S}}(q_0, q_1)^2.$$

The metric on $\mathscr{C}$ *is complete if and only if the induced metric on* $\mathscr{S}$ *is complete. Every path for which* λ *tends to zero or to infinity has infinite length.*

Proof Homogeneity gives $\lambda(s\theta) = s\lambda(\theta)$ for $s > 0$. Hence $r = \log \lambda(\theta)$ and $q = \theta/\lambda(\theta)$ give the smooth inverse of Φ. The level set is smooth because $D\lambda_\theta(\theta) = \lambda(\theta) > 0$.

Differentiating the Euler identity $D \log \lambda_\theta(\theta) = 1$ in a direction $\chi \in V$ gives

$$g_\theta(\theta, \chi) = D \log \lambda_\theta(\chi), \qquad g_\theta(\theta, \theta) = 1.$$

For directions $\chi, \eta \in V$, differentiating homogeneity twice also gives

$$g_{s\theta}(s\chi, s\eta) = g_\theta(\chi, \eta).$$

For a point $q \in \mathcal{S}$, tangent vectors $\xi, \eta \in T_q\mathcal{S}$, and real numbers a, b, r the first displayed identity makes both orthogonal to the radial vector q. Thus

$$g_{e^r q}\big(e^r(aq + \xi), e^r(bq + \eta)\big) = ab + g_q(\xi, \eta),$$

which proves the metric identity.

For completeness, a product path $(r(t), q(t))$ has length $\int \sqrt{|r'|^2 + |q'|^2_{g_\mathcal{S}}}\, dt$. The triangle inequality in $\mathbb{R}^2$ bounds this below by the square root of the sum of the squared endpoint difference in r and the squared length of the projected path in $\mathcal{S}$. For the opposite bound choose a path in $\mathcal{S}$ of length within ϵ of its distance, parametrize it with constant speed, and let r vary linearly. Let ϵ decrease to zero. This proves the distance formula. It identifies Cauchy sequences in $\mathscr{C}$ with pairs of Cauchy sequences in the two factors; since $\mathbb{R}$ is complete, it proves the completeness assertion. Finally the length of any path is at least the total change in $\log \lambda$. Divergence of that logarithm proves the last assertion. □

25.5.1 Product and blowup examples

The two examples below give different completions of the unit-volume hypersurface. The first also shows why the derivative is needed in Proposition 25.4.1: knowing the volume for every smooth probability measure need not identify the class.

For this example take $X = \mathbb{P}^1 \times \mathbb{P}^1$, and let α, β be the pulled-back Fubini–Study forms of integral one on their respective factors. They satisfy

$$\alpha^2 = \beta^2 = 0, \qquad \int_X \alpha \wedge \beta = 1.$$

For $a, b > 0$ the form $\theta_{a,b} = a\alpha + b\beta$ is Kähler. Closed Stokes in the smooth equation gives, for every smooth positive probability measure μ,

$$\mathcal{V}_\mu(\theta_{a,b}) = 2ab, \qquad \lambda_\mu(\theta_{a,b}) = \sqrt{2ab}.$$

Consequently, for real t, all $\theta_{e^t, e^{-t}}$ have exactly the same volume profile $\mu \mapsto 2$. Their smooth dd^c-classes are distinct: for any fixed point p of the second factor, integration over $\mathbb{P}^1 \times \{p\}$ gives e^t, which is invariant under addition of $dd^c f$.

By direct differentiation, the restriction of (25.3) to this two-dimensional cone is

$$\mathfrak{g} = \frac{1}{2}\left(\frac{\mathrm{d}a^2}{a^2} + \frac{\mathrm{d}b^2}{b^2}\right).$$

The coordinates $(\log a/\sqrt{2}, \log b/\sqrt{2})$ identify this metric with the Euclidean metric on $\mathbb{R}^2$, so the induced metric on this family is complete. The curve $a = \mathrm{e}^t, b = \mathrm{e}^{-t}$ has unit speed. All distances here are intrinsic to this two-dimensional family.

The blowup of $\mathbb{P}^2$ gives an incomplete cone whose added boundary has positive volume.

Example 25.5.1 Let $\pi : X \to \mathbb{P}^2$ be the blowup at one point. Write H for the pullback of the hyperplane class and E for the exceptional divisor class. In the two-dimensional space they span, the cone of classes with a positive smooth representative is

$$\mathscr{C} = \{aH - bE : a > b > 0\}.$$

These are closed classes, and we may represent them by fixed smooth forms to apply the proposition. Their intersections are $H^2 = 1$, $H \cdot E = 0$, $E^2 = -1$. Consequently, for every smooth positive probability measure μ,

$$\lambda_\mu(aH - bE) = \sqrt{a^2 - b^2}.$$

In the coordinates

$$a = \mathrm{e}^r \cosh s, \qquad b = \mathrm{e}^r \sinh s, \qquad r \in \mathbb{R}, \quad s > 0,$$

the Hessian metric is $\mathrm{d}r^2 + \mathrm{d}s^2$. Its completion is therefore the Euclidean closed half-plane. The added boundary $s = 0$ consists of the nef classes $\mathrm{e}^r H$, each of positive volume e^{2r}.

Proof The projection from the blown-up point extends to a morphism $f : X \to \mathbb{P}^1$ with class $L = H - E$. The incidence map $(\pi, f) : X \to \mathbb{P}^2 \times \mathbb{P}^1$ is an embedding. If h and ℓ are pullbacks of Fubini–Study forms of integral one, then

$$(a - b)h + b\ell$$

is positive for $a > b > 0$, by the product embedding. Conversely, a positive representative has positive integrals over the exceptional curve and the strict transform of a line through the blown-up point. These integrals are respectively b and $a - b$, proving the cone description. The same curves show that no nonzero linear combination in this two-dimensional space is smoothly $\mathrm{dd^c}$-exact.

The intersection numbers follow from the hyperplane pullback and the normal bundle $\mathcal{O}_{\mathbb{P}^1}(-1)$ of the exceptional curve; equivalently $H^2 = H \cdot L = 1$ and $L^2 = 0$. Closed Stokes in the smooth equation gives $\mathcal{V}_\mu = a^2 - b^2$, independently of μ. Direct differentiation gives

$$g = \frac{(a^2 + b^2)(\mathrm{d}a^2 + \mathrm{d}b^2) - 4ab\,\mathrm{d}a\,\mathrm{d}b}{(a^2 - b^2)^2}.$$

Substituting the stated coordinates gives $\mathrm{d}r^2 + \mathrm{d}s^2$. The coordinate domain is exactly $\mathbb{R} \times (0, \infty)$, so the intrinsic distance and completion are those of that Euclidean half-plane. Finally H has a semipositive pullback representative and is a limit of positive classes, hence is nef. It has square one but zero integral over E, so it has no positive representative. This proves all assertions about the added boundary. □

25.6 A compact dual problem over Gauduchon currents

The dual problem becomes compact when its tests are allowed to be positive $(n-1, n-1)$-currents. To extend the denominator, take the determinant of the matrix density relative to the trace measure. Its integral is upper semicontinuous under weak convergence and gives an attained maximum. The equation will identify that maximum with the supremum over smooth tests.

From here until Section 25.10, let X again be a connected compact complex manifold of dimension $n \geq 2$, and fix a Hermitian form ω and a smooth positive probability volume $\mathrm{d}V$ on X. Integrability and convergence in measure use $\mathrm{d}V$. Let $\mathcal{G}_\omega$ be the compact convex set of normalized Gauduchon currents from Section 4.4, and let $F \in C(X, \mathbb{R})$.

For $T \in \mathcal{G}_\omega$, let $\sigma_T = \omega \wedge T$, a probability measure. In a unitary frame for ω, let B be the matrix of a smooth real $(1,1)$-form β relative to ω. The coefficient density of T is represented by a positive semidefinite Hermitian matrix P_T characterized by

$$\beta \wedge T = \frac{1}{n}\operatorname{tr}(BP_T)\,\mathrm{d}\sigma_T, \qquad \operatorname{tr} P_T = n \quad \sigma_T\text{-almost everywhere.}$$

The matrix transforms by unitary conjugation when the frame changes. Its determinant is therefore well-defined. Define

$$f_F = \left(\frac{\mathrm{e}^F \mathrm{d}V}{\omega^n}\right)^{1/n}, \qquad \Gamma_F(T) = f_F(\det P_T)^{1/n}\sigma_T, \qquad \mathfrak{D}_F(T) = \Gamma_F(T)(X). \tag{25.4}$$

These are finite: the arithmetic–geometric mean inequality gives $(\det P_T)^{1/n} \leq 1$. The measure $\Gamma_F(T)$ may vanish for a particular current.

For a smooth positive $T = \gamma^{n-1}$, the coefficient matrix relative to ω^n is the cofactor matrix of the matrix of γ. Its determinant is $(\gamma^n/\omega^n)^{n-1}$. The definition consequently

gives

$$\mathfrak{D}_F(\gamma^{n-1}) = \int_X \left(\frac{\mathrm{e}^F \mathrm{d}V}{\gamma^n}\right)^{1/n} \gamma^n.$$

The same expression scales linearly when T is multiplied by a positive constant. Thus it extends the denominator of the smooth Gauduchon formula, with a possibly unnormalized density $\mathrm{e}^F \mathrm{d}V$.

Lemma 25.6.1 *For every positive $(n-1, n-1)$-current T of finite mass on X,*

$$\mathfrak{D}_F(T) = \inf_{\substack{\beta>0 \text{ continuous} \\ \beta^n = \mathrm{e}^F \mathrm{d}V}} \int_X \beta \wedge T.$$

In particular $\mathfrak{D}_F$ is concave, positively homogeneous and weakly upper semicontinuous on $\mathcal{G}_\omega$. Its definition is independent of the auxiliary frame and of the metric used to express the matrix density.

Proof Let I be the identity matrix of size n. For any positive semidefinite Hermitian matrix P of that size,

$$(\det P)^{1/n} = \inf_{Q>0,\ \det Q=1} \frac{1}{n} \operatorname{tr}(QP).$$

The lower bound is the arithmetic–geometric mean inequality. For the reverse bound, for real $\varepsilon > 0$ use

$$Q_\varepsilon = \det(P + \varepsilon I)^{1/n} (P + \varepsilon I)^{-1}.$$

Its determinant is one. If P is positive definite the trace expression converges to $(\det P)^{1/n}$. If P is singular it converges to zero: each ratio of an eigenvalue to that eigenvalue plus ε is bounded, while the prefactor tends to zero. When $\operatorname{tr} P = n$, the trace expression is at most $1 + \varepsilon$.

Apply this formula to P_T and multiply by f_F. Every continuous positive β in the asserted infimum has matrix $f_F Q$ with $\det Q = 1$, so it gives an upper bound for $\mathfrak{D}_F(T)$. Conversely, for fixed $\varepsilon > 0$ the measurable field Q_ε is uniformly bounded above and below by positive constants. It can be approximated in $L^1(\sigma_T)$ by continuous positive definite Hermitian fields with the same type of bounds. Here is one standard measure-theoretic construction: apply Lusin's theorem in finitely many bundle charts, extend the continuous fields from the resulting compact sets, and use a partition of unity; spectral truncation to a fixed positive interval preserves the bounds. Finally divide each approximating field by the nth root of its determinant. Determinant normalization is uniformly continuous on the bounded positive interval, so the normalized fields still converge in $L^1(\sigma_T)$ and have determinant one. Their trace integrals converge to that of Q_ε.

Let $\varepsilon \downarrow 0$. The bound by $1 + \varepsilon$ permits dominated convergence. This proves equality of the infimum and the integral. For a current of arbitrary positive finite trace mass, use homogeneity; the zero current is immediate. The infimum consists of continuous linear functionals of T, which proves concavity and upper semicontinuity.

It also expresses the quantity without a matrix frame or auxiliary metric, proving the final assertion. □

Let $B_0(F)$ denote the scalar functional of Chapter 14 for the background ω and reference zero. Thus a bounded solution satisfies $\mathrm{MA}_\omega(u) = \mathrm{e}^{B_0(F)+F}\mathrm{d}V$.

Theorem 25.6.1 *For every $F \in C(X, \mathbb{R})$,*

$$M(F) := \max_{T \in \mathscr{G}_\omega} \mathfrak{D}_F(T) = \mathrm{e}^{-B_0(F)/n} > 0. \tag{25.5}$$

The maximum agrees with the supremum over smooth strictly positive normalized Gauduchon forms. Its maximizer set $\mathscr{M}(F)$ is nonempty, compact and convex.

If F is smooth, there is exactly one maximizing current. It is the normalized $(n-1)$st power of the Gauduchon metric conformal to the smooth solution form.

Proof Upper semicontinuity on the compact set $\mathscr{G}_\omega$ gives attainment; testing a smooth positive member gives a positive value. Concavity makes the maximizer set convex, and upper semicontinuity makes it closed.

Suppose first that F is smooth. Let $\Omega = \omega + \mathrm{dd^c}u > 0$ solve $\Omega^n = \mathrm{e}^{B_0(F)+F}\mathrm{d}V$. The form $\beta = \mathrm{e}^{-B_0(F)/n}\Omega$ is an admissible test in the preceding lemma. Distributional Stokes gives, for every $T \in \mathscr{G}_\omega$,

$$\mathfrak{D}_F(T) \le \int_X \beta \wedge T = \mathrm{e}^{-B_0(F)/n} \int_X \omega \wedge T = \mathrm{e}^{-B_0(F)/n}.$$

If γ is Gauduchon and conformal to Ω, equality holds in the pointwise matrix inequality for T proportional to γ^{n-1}. Normalize its trace integral to one. This proves the formula and smooth attainment.

For uniqueness, equality of the integrated inequalities forces equality in the matrix arithmetic–geometric mean inequality σ_T-almost everywhere. Since the test matrix of β is positive definite and $\operatorname{tr} P_T = n$, this means that the coefficient measure of T is a nonnegative scalar measure times the smooth matrix direction of γ^{n-1}. In distribution notation write $T = a\gamma^{n-1}$, where a is a scalar distribution represented by a positive measure relative to a fixed smooth volume. The equation $\mathrm{dd^c}(a\gamma^{n-1}) = 0$ is a scalar elliptic adjoint equation, so elliptic regularity makes a smooth. For a Gauduchon metric, Proposition 4.5.1 identifies this adjoint kernel with the constants: the original trace operator has constant kernel, index zero, and integration against γ^n annihilates its range. Thus a is constant. The trace normalization fixes that constant, proving uniqueness.

For continuous F, G, the pointwise definition gives

$$\mathrm{e}^{-\|F-G\|_\infty/n}\mathfrak{D}_G(T) \le \mathfrak{D}_F(T) \le \mathrm{e}^{\|F-G\|_\infty/n}\mathfrak{D}_G(T).$$

The same bounds hold for maxima, and for suprema over smooth positive members. Approximate F uniformly by smooth functions. The uniform Lipschitz continuity of B_0 and the formula just proved for smooth data give (25.5) in the limit, including

equality with the smooth supremum. The approximation concerns F. Some normalized Gauduchon currents have no smooth positive Gauduchon approximation, as Example 27.8.1 shows. □

25.7 Supporting probabilities represented by maximizing currents

The maximizing currents represent exactly the supporting probabilities of B_0. The point that needs proof is continuity of their determinant measures: upper semicontinuity of the total determinant integral becomes equality on the maximizing set. Let $\mathcal{A}_0(F)$ be the supporting probabilities from Theorem 14.2.1.

For $T \in \mathscr{M}(F)$ put

$$\nu_{T,F} = \Gamma_F(T)/M(F).$$

This is a probability measure.

Lemma 25.7.1 *Suppose $T_j \to T$ weakly in $\mathcal{G}_\omega$ and $\mathfrak{D}_F(T_j) \to M(F)$. Then $T \in \mathscr{M}(F)$ and $\Gamma_F(T_j) \rightharpoonup \Gamma_F(T)$. On $\mathscr{M}(F)$ the map $T \mapsto \Gamma_F(T)$ is affine and continuous.*

Proof Upper semicontinuity gives $\mathfrak{D}_F(T) \geq M(F)$, hence equality. The measures $\Gamma_F(T_j)$ have bounded mass. Take a weak cluster point Γ'; its mass is $M(F)$. For a continuous real h, choose $\varepsilon > 0$ so that $1 \pm \varepsilon h > 0$. Applying upper semicontinuity to the two logarithmic densities $F + n\log(1 \pm \varepsilon h)$ gives

$$\int_X (1 \pm \varepsilon h)\,\mathrm{d}\Gamma' \leq \int_X (1 \pm \varepsilon h)\,\mathrm{d}\Gamma_F(T).$$

The equal masses and the two signs imply equality of the integrals of h. Thus every cluster point equals $\Gamma_F(T)$, which proves convergence.

For affineness, take $T_1, T_2 \in \mathcal{G}_\omega$ and $t \in [0,1]$, and dominate their coefficient measures by the sum of their trace measures. Concavity and homogeneity of the determinant root, applied to their matrix densities with respect to this common measure, give

$$\Gamma_F(tT_1 + (1-t)T_2) \geq t\Gamma_F(T_1) + (1-t)\Gamma_F(T_2), \qquad 0 \leq t \leq 1.$$

If T_1, T_2 maximize, the current on the left also maximizes. Both measures in the inequality then have the same mass $M(F)$, so equality holds. This proves affineness. Continuity is the special case of the first assertion in which every current is a maximizer. □

Theorem 25.7.1 *For the Hermitian background ω and zero reference,*

$$\mathcal{A}_0(F) = \{\nu_{T,F} : T \in \mathscr{M}(F)\}.$$

Consequently all the determinant-weighted measures of maximizing currents are absolutely continuous with respect to $\mathrm{d}V$. *This assertion does not say that the currents themselves are absolutely continuous.*

Proof For a maximizing T and any $G \in C(X)$,

$$M(G) \geq \mathfrak{D}_G(T) = M(F) \int_X \mathrm{e}^{(G-F)/n} \, \mathrm{d}\nu_{T,F}.$$

Taking $-n$ times the logarithm and applying Jensen's inequality gives

$$B_0(G) \leq B_0(F) - n \log \int_X \mathrm{e}^{(G-F)/n} \, \mathrm{d}\nu_{T,F} \leq B_0(F) - \int_X (G-F) \, \mathrm{d}\nu_{T,F}.$$

Hence every displayed probability belongs to $\mathcal{A}_0(F)$.

Denote their set by $\mathcal{N}(F)$. The preceding lemma shows that it is compact and convex. We compute directional derivatives to prove the reverse inclusion. Uniform Taylor expansion, for $h \in C(X)$ and $t \downarrow 0$, gives on the whole compact current set

$$\mathfrak{D}_{F+th}(T) = \mathfrak{D}_F(T) + \frac{t}{n} \int_X h \, \mathrm{d}\Gamma_F(T) + O(t^2),$$

with the error bounded uniformly in T. Testing maximizers at F gives

$$\liminf_{t \downarrow 0} \frac{M(F+th) - M(F)}{t} \geq \frac{M(F)}{n} \max_{\nu \in \mathcal{N}(F)} \int_X h \, \mathrm{d}\nu.$$

For the reverse inequality choose a maximizer T_t at $F + th$. The multiplicative bounds in the previous proof imply $\mathfrak{D}_F(T_t) \to M(F)$. Every sequence $t \downarrow 0$ has a subsequence for which T_t converges; the preceding lemma identifies its limiting weighted measure with that of a maximizer at F. Since $\mathfrak{D}_F(T_t) - M(F) \leq 0$, the Taylor expansion gives the matching upper bound. Therefore

$$(B_0)'_+(F; h) = -\max_{\nu \in \mathcal{N}(F)} \int_X h \, \mathrm{d}\nu.$$

If $\nu \in \mathcal{A}_0(F)$, its supporting inequality in direction th implies

$$\int_X h \, \mathrm{d}\nu \leq \max_{\widetilde{\nu} \in \mathcal{N}(F)} \int_X h \, \mathrm{d}\widetilde{\nu} \quad \text{for every } h \in C(X).$$

Separation of a point from a compact convex subset of the dual of $C(X)$ now gives $\nu \in \mathcal{N}(F)$. This proves equality of the two sets. Their absolute continuity was proved in Theorem 14.2.1. □

For continuous F, uniqueness of the supporting probability is equivalent to all maximizing currents having the same normalized determinant-weighted measure. This is weaker than uniqueness of the current. Representing such a current as a conformal multiple of a weak power ω_u^{n-1} is a further question, discussed in Section 33.13; the construction here uses its coefficient measures directly.

25.8 Integrability of Gauduchon determinant measures

The lower bound for equation normalizations gives a uniform L^q estimate for the determinant measures of all normalized Gauduchon currents. Every exponent below $n/(n-1)$ is available. At the endpoint, the optimal constant is exactly $v_-(\omega)^{-1/n}$, linking integrability of the dual tests to the lower mass of the background.

Write $\Gamma_0(T)$ for the determinant measure in (25.4) with $F = 0$.

Theorem 25.8.1 *For every $1 < q < n/(n-1)$ there is $C_q > 0$ such that every $T \in \mathscr{G}_\omega$ satisfies*

$$\Gamma_0(T) = k_T \, \mathrm{d}V, \qquad k_T \geq 0, \qquad \|k_T\|_{L^q(\mathrm{d}V)} \leq C_q.$$

Proof Fix $p > 1$ and take any smooth strictly positive function h. The smooth equation gives $\Omega_h = \omega + \mathrm{dd^c} u_h > 0$ and $c_h > 0$ such that

$$\Omega_h^n = c_h h^n \, \mathrm{d}V.$$

By Theorem 10.3.1, the density of this bounded solution satisfies

$$c_h \|h^n\|_{L^p(\mathrm{d}V)} \geq c_p > 0.$$

Pointwise matrix arithmetic–geometric mean, applied to Ω_h and the coefficient density of T, gives the measure inequality

$$c_h^{1/n} h \, \Gamma_0(T) \leq \Omega_h \wedge T.$$

Integrating and using $\mathrm{dd^c} T = 0$ yields

$$\int_X h \, \mathrm{d}\Gamma_0(T) \leq c_h^{-1/n} \int_X \Omega_h \wedge T = c_h^{-1/n} \leq c_p^{-1/n} \|h\|_{L^{np}(\mathrm{d}V)}.$$

Approximation by smooth strictly positive functions extends this bound to every continuous nonnegative h. Applied to $|h|$, it bounds the positive measure functional on all continuous functions by the L^{np} norm. Density of continuous functions in L^{np} and the usual duality of these spaces show that the measure has a density in $L^{np/(np-1)}$, with norm at most $c_p^{-1/n}$. As p ranges over $(1, +\infty)$, the exponent $np/(np-1)$ ranges over $(1, n/(n-1))$. This proves the result. □

In particular, the determinant of the matrix density vanishes on the singular part of the trace measure: if $\sigma_T = \sigma_{T,\mathrm{ac}} + \sigma_{T,\mathrm{s}}$ is its Lebesgue decomposition, then

$$\det P_T = 0 \quad \sigma_{T,\mathrm{s}}\text{-almost everywhere.}$$

Indeed $f_0 = (\mathrm{d}V/\omega^n)^{1/n}$ is everywhere positive and the determinant-weighted measure is absolutely continuous. This assertion concerns the determinant; it does not remove a possible singular part of the current itself.

Theorem 25.8.2 *With the extended value $+\infty$ allowed, one has*

$$\sup_{T\in\mathscr{G}_\omega} \|k_T\|_{L^{n/(n-1)}(\mathrm{d}V)} = \nu_-(\omega)^{-1/n},$$

where the right side is interpreted as $+\infty$ when $\nu_-(\omega) = 0$. The same supremum is obtained by restricting to smooth strictly positive normalized Gauduchon forms. Thus a uniform estimate at the endpoint exponent is equivalent to $\nu_-(\omega) > 0$.

Proof Suppose first that $\nu_-(\omega) > 0$. In the smooth equation of the preceding proof, the definition of ν_- gives

$$c_h \int_X h^n \,\mathrm{d}V = \int_X \Omega_h^n \geq \nu_-(\omega).$$

The same pointwise inequality and Stokes therefore give

$$\int_X h \,\mathrm{d}\Gamma_0(T) \leq \nu_-(\omega)^{-1/n} \|h\|_{L^n(\mathrm{d}V)}.$$

Duality proves the asserted upper bound for every current in $\mathscr{G}_\omega$.

For the reverse bound, let g be any smooth positive probability density relative to $\mathrm{d}V$, and solve

$$\Omega_g^n = c(g) g \,\mathrm{d}V.$$

Let T_g be the trace-normalized $(n-1)$st power of the Gauduchon metric conformal to Ω_g. Equality in the pointwise matrix inequality gives

$$\int_X g^{1/n} \,\mathrm{d}\Gamma_0(T_g) = c(g)^{-1/n}.$$

Since $\|g^{1/n}\|_{L^n} = 1$, its endpoint dual norm is at least $c(g)^{-1/n}$. By Theorem 25.12.1,

$$\inf_{g>0 \text{ smooth}, \int_X g\,\mathrm{d}V=1} c(g) = \nu_-(\omega).$$

Taking the supremum over these smooth T_g proves the matching lower bound. If $\nu_-(\omega) = 0$, the same argument makes this supremum infinite. This also proves that allowing nonsmooth currents does not change it. □

The endpoint norm is independent of the smooth probability measure used to express it, as the formula shows. Subcritical constants may depend on that choice.

Corollary 25.8.1 *Let $\mathcal{A}_0(F)$ be the supporting probabilities in Theorem 25.7.1. For every $R > 0$ and $1 < q < n/(n-1)$ there is $C_{R,q}$ with the following property. If $F \in C(X)$ and $\|F\|_\infty \le R$, every $\nu \in \mathcal{A}_0(F)$ has a density a satisfying*

$$\nu = a\,\mathrm{d}V, \qquad \|a\|_{L^q(\mathrm{d}V)} \le C_{R,q}.$$

If $\nu_-(\omega) > 0$, this holds also for $q = n/(n-1)$.

Proof Write $M(F) = \exp(-B_0(F)/n)$ and represent ν by a maximizing current $T \in \mathscr{M}(F)$, using Theorem 25.7.1. Then

$$\nu = \frac{\mathrm{e}^{F/n} k_T}{M(F)}\,\mathrm{d}V, \qquad M(F) \ge \mathrm{e}^{-R/n} M(0) > 0.$$

The preceding theorems give the subcritical bound with constant $\mathrm{e}^{2R/n} M(0)^{-1} C_q$. Use the endpoint constant when it is finite. □

The endpoint fails for the full normalized Gauduchon family on the threefold of Theorem 25.15.1, while every subcritical estimate above remains valid. For a fixed smooth right-hand side, Theorem 25.6.1 gives a unique smooth maximizing current, whose determinant density is bounded. This does not give uniform higher moments along sequences approaching its value; Proposition 25.14.1 provides a counterexample.

The endpoint can be available on a manifold with infinite upper mass. On the Hopf threefold of Chapter 8, take its Gauduchon metric ω, for which $\nu_-(\omega) = 6$ and $\nu_+(\omega) = +\infty$, and set $\mathrm{d}V = \omega^3/6$. The normalized Gauduchon form $T_0 = \omega^2/6$ has $\sigma_{T_0} = \mathrm{d}V$ and $P_{T_0} = I$. Hence $k_{T_0} = 6^{-1/3}$, and

$$\sup_{T \in \mathscr{G}_\omega} \|k_T\|_{L^{3/2}(\mathrm{d}V)} = 6^{-1/3}.$$

Thus an attained endpoint determinant estimate can coexist with infinite upper Monge–Ampère mass.

25.9 Integrable densities in the compact Gauduchon problem

Return to the fixed Hermitian background ω on X. For an unbounded density, weak compactness of the test currents alone does not pass the determinant integral to the limit. Uniform control of the weighted tails supplies the missing upper semicontinuity. Use the root norm ρ_ω, its finite domain $\mathcal{D}_\omega$ and its vanishing-tail subspace $\mathcal{D}^0_\omega$ from Proposition 13.26.4, and the determinant densities k_T from Theorem 25.8.1.

If $p \geq 0$ is measurable, define

$$D_p(T) = \int_X p k_T \, \mathrm{d}V,$$

allowing $+\infty$. For a smooth positive normalized Gauduchon form this is the test $L_{\omega,\gamma}(p)$. For nonzero $p \in L^n$ with $p \geq 0$, the threshold formula is $b_\omega(p^n) = \rho_\omega(p)^{-n}$.

Theorem 25.9.1 *For every nonnegative $p \in L^n$,*

$$\sup_{T \in \mathcal{G}_\omega} D_p(T) = \rho_\omega(p), \tag{25.6}$$

with $+\infty$ allowed. If $p \in \mathcal{D}^0_\omega$, the functional D_p is finite, concave and weakly upper semicontinuous on $\mathcal{G}_\omega$. Its maximum is attained, and the measures $p k_T \, \mathrm{d}V$, as T ranges over this set, have uniformly integrable densities.

Proof For a smooth strictly positive p, Theorem 25.6.1 applied to $F = n \log p$ proves (25.6). By Theorem 25.8.1, for every real $r > n$ and every $q \in L^r$, there is a constant C_r such that

$$\sup_{T \in \mathcal{G}_\omega} \int_X |q| k_T \, \mathrm{d}V \leq C_r \|q\|_{L^r}. \tag{25.7}$$

Approximate a bounded nonnegative p in L^{2n} by smooth strictly positive functions. This estimate and (13.77) give uniform convergence of the current functionals and convergence of the root norms. Thus the supremum formula holds for bounded p. Their current functionals are also upper semicontinuous and concave, by Lemma 25.6.1 and uniform convergence. For general $p \in L^n$, use positive integer truncation levels K. Monotone truncation gives

$$\sup_T D_p(T) = \sup_T \sup_K D_{\min(p,K)}(T) = \sup_K \rho_\omega(\min(p,K)) = \rho_\omega(p).$$

Only two suprema have been interchanged; this step alone gives no attainment.

Now suppose $p \in \mathcal{D}^0_\omega$. By the supremum formula applied to the tail,

$$0 \leq D_p(T) - D_{\min(p,K)}(T) \leq \rho_\omega(p \mathbf{1}_{\{p>K\}}) \longrightarrow 0$$

uniformly in T. Hence D_p is finite and a uniform limit of upper semicontinuous concave functionals. Compactness of $\mathcal{G}_\omega$ gives a maximizer. Finally, fix $1 < q <$

$n/(n-1)$. For every measurable E,

$$\sup_T \int_E p k_T \, \mathrm{d}V \le \rho_\omega(p \mathbf{1}_{\{p>K\}}) + K C_q \, \mathrm{d}V(E)^{1-1/q}.$$

Choosing first K and then the volume of E proves uniform absolute continuity. The total masses are bounded by $\rho_\omega(p)$, so the densities are uniformly integrable. □

For a nonzero density, current attainment gives a supporting probability for the threshold. Its relation to an exact equation depends on the regularity of the maximizing current, which will enter through the trace inequalities below.

25.9.1 The tail condition and perturbations of the density

Vanishing tails are equivalent to continuity of the threshold under uniformly bounded multiplicative perturbations that converge in measure. This characterizes the precise role of the tail condition: if it fails, a perturbation supported on a shrinking set can lower the threshold by a fixed factor.

Proposition 25.9.1 *Let $0 \le p \in \mathcal{D}_\omega$ be nonzero. The following are equivalent:*

(1) $p \in \mathcal{D}_\omega^0$;
(2) *the densities $p k_T$, for $T \in \mathcal{G}_\omega$, are uniformly integrable;*
(3) *whenever F_j are smooth real functions, uniformly bounded and tending to zero in measure, one has*
$$b_\omega(\mathrm{e}^{F_j} p^n) \longrightarrow b_\omega(p^n).$$

If the first condition holds, the convergence in the third also holds for bounded measurable F_j with the same uniform bound and convergence in measure. If it fails, there are smooth $F_j \ge 0$, uniformly bounded and tending to zero in measure, such that

$$\mathrm{e}^{F_j} p^n \longrightarrow p^n \quad \text{in } L^1, \qquad b_\omega(\mathrm{e}^{F_j} p^n) \le 2^{-n} b_\omega(p^n).$$

Proof The first condition implies the second by Theorem 25.9.1. Conversely, $\mathrm{d}V(\{p > K\}) \to 0$, so uniform integrability and (25.6) give

$$\rho_\omega(p \mathbf{1}_{\{p>K\}}) = \sup_{T \in \mathcal{G}_\omega} \int_{\{p>K\}} p k_T \, \mathrm{d}V \longrightarrow 0.$$

This is the first condition by Proposition 13.26.4.

If $p \in \mathcal{D}_\omega^0$ and $|F_j| \le A$, then $p(\mathrm{e}^{F_j/n} - 1)$ tends to zero in measure and its absolute value is at most $(\mathrm{e}^{A/n} + 1)p$. The dominated-convergence criterion of Proposition 13.26.4 makes its root norm tend to zero. The triangle inequality gives $\rho_\omega(p\mathrm{e}^{F_j/n}) \to \rho_\omega(p) > 0$. Taking negative nth powers proves the third condition, including measurable perturbations.

Suppose now that the tails do not tend to zero, and put $R = \rho_\omega(p) > 0$. The smooth Gauduchon formula yields $\delta > 0$, numbers $K_j \to \infty$ and smooth strictly positive $T_j \in \mathcal{G}_\omega$ such that

$$\int_{\{p>K_j\}} pk_{T_j}\,\mathrm{d}V > \delta.$$

The finite measure $pk_{T_j}\,\mathrm{d}V$ is regular. Choose a compact set $C_j \subset \{p > K_j\}$ whose mass exceeds $\delta/2$, and an open neighborhood O_j with $\mathrm{d}V(O_j) \le \mathrm{d}V(\{p > K_j\}) + 1/j$. Take a smooth cutoff $0 \le \chi_j \le 1$ that equals one on C_j and is supported in O_j. Choose a fixed $A > 0$ with $\mathrm{e}^{A/n}\delta/2 > 2R$, and set $F_j = A\chi_j$. Then $F_j \to 0$ in measure, while

$$\rho_\omega(p\mathrm{e}^{F_j/n}) \ge \int_X p\mathrm{e}^{F_j/n} k_{T_j}\,\mathrm{d}V > 2R.$$

The threshold formula gives the asserted uniform gap. Finally, $|\mathrm{e}^{F_j} - 1|p^n \le (\mathrm{e}^A - 1)p^n\mathbf{1}_{O_j}$, whose integral tends to zero by absolute continuity of the integral. This proves the last assertion and the converse to the third condition. □

For $f = p^n$, these perturbations satisfy $\mathrm{e}^{-A}f \le \mathrm{e}^{F_j}f \le \mathrm{e}^A f$ with a fixed A. The continuity statement is restricted to such an order interval. The unrestricted L^1 instability in Section 13.26 allows approximations without these bounds.

For attainment at one density, uniform integrability of a single maximizing sequence is enough. This weaker condition also identifies the limit of its weighted measures.

Proposition 25.9.2 *Let $0 \le p \in \mathcal{D}_\omega$ be nonzero and put $R = \rho_\omega(p)$. Suppose $T_j \in \mathcal{G}_\omega$ converge weakly to T and satisfy $D_p(T_j) \to R$. If the densities pk_{T_j} are uniformly integrable, then*

$$D_p(T) = R, \qquad pk_{T_j}\,\mathrm{d}V \rightharpoonup pk_T\,\mathrm{d}V.$$

Consequently, a uniformly integrable maximizing sequence has a maximizing current among its weak cluster limits.

Proof For positive integers K, put $p_K = \min(p, K)$. Since p is finite almost everywhere, uniform integrability gives

$$\varepsilon_K := \sup_j \int_{\{p>K\}} pk_{T_j}\,\mathrm{d}V \longrightarrow 0.$$

For every nonnegative continuous h, the weight hp_K is bounded and measurable. The bounded-weight upper semicontinuity proved in Theorem 25.9.1 therefore gives

$$\limsup_j \int_X hpk_{T_j}\,\mathrm{d}V \le \int_X hp_K k_T\,\mathrm{d}V + \|h\|_\infty \varepsilon_K.$$

Let $K \to \infty$. For $h = 1$ this yields $R \le D_p(T)$; the supremum formula (25.6) gives the reverse inequality. If μ is a weak cluster limit of $pk_{T_j}\,\mathrm{d}V$, the same estimate for

all nonnegative continuous h gives $\mu \le p k_T \, \mathrm{d}V$. Both measures have mass R, so they agree. Their bounded masses give weak compactness, and every cluster limit has now been identified. The final assertion follows from compactness of $\mathscr{G}_\omega$. □

25.9.2 Supporting probabilities for an integrable density

The vanishing-tail condition also permits continuous multiplicative perturbations of the density. Their logarithmic thresholds have supporting probabilities given by the maximizing currents, just as for the positive continuous densities above.

Let $f \in L^1$ be nonzero and nonnegative, and assume $p = f^{1/n} \in \mathcal{D}^0_\omega$. For a continuous real F, set

$$B_f(F) = \log b_\omega(\mathrm{e}^F f) = -n \log \rho_\omega(p\mathrm{e}^{F/n}), \qquad M_f(F) = \rho_\omega(p\mathrm{e}^{F/n}).$$

Both quantities are finite and $M_f(F) > 0$, by the order-ideal property. Let $\mathscr{M}_f(F)$ be the currents maximizing $D_{p\mathrm{e}^{F/n}}$, and put

$$\nu_{T,F} = \frac{p\mathrm{e}^{F/n} k_T}{M_f(F)} \, \mathrm{d}V \quad (T \in \mathscr{M}_f(F)).$$

These are probability measures. Define $\mathscr{A}_f(F)$ to be all Borel probability measures satisfying

$$B_f(G) \le B_f(F) - \int_X (G - F) \, \mathrm{d}\nu \qquad (G \in C(X, \mathbb{R})). \tag{25.8}$$

Proposition 25.9.3 *The functional B_f is concave, decreasing and* 1*-Lipschitz for the uniform norm, and $B_f(F + a) = B_f(F) - a$ for constants a. Moreover,*

$$\mathcal{A}_f(F) = \{\nu_{T,F} : T \in \mathcal{M}_f(F)\}.$$

This set is nonempty, convex and weakly compact. Its measures are absolutely continuous. For each $R > 0$, their densities, with $\|F\|_\infty \le R$, are uniformly integrable. For $H \in C(X, \mathbb{R})$, its one-sided derivatives satisfy

$$(B_f)'_+(F; H) = -\max_{\nu \in \mathcal{A}_f(F)} \int_X H \, \mathrm{d}\nu, \qquad (B_f)'_-(F; H) = -\min_{\nu \in \mathcal{A}_f(F)} \int_X H \, \mathrm{d}\nu.$$

Proof For each T, Hölder's inequality makes $F \mapsto \int_X p\mathrm{e}^{F/n} k_T \, \mathrm{d}V$ logarithmically convex. Taking suprema preserves the inequality defining logarithmic convexity. Thus $B_f = -n \log M_f$ is concave. Order and homogeneity give the translation rule and, by comparison with constant translates, the Lipschitz estimate.

For fixed F, the weight $p\mathrm{e}^{F/n}$ belongs to $\mathcal{D}^0_\omega$, so its determinant-weighted densities are uniformly integrable. If $T_j \to T$ and $D_{p\mathrm{e}^{F/n}}(T_j) \to M_f(F)$, Proposition 25.9.2 makes T a maximizer and gives convergence of the weighted measures. For $T_1, T_2 \in \mathcal{G}_\omega$ and real $0 \le t \le 1$, matrix determinant concavity gives

$$p\mathrm{e}^{F/n} k_{tT_1 + (1-t)T_2} \, \mathrm{d}V \ge tp\mathrm{e}^{F/n} k_{T_1} \, \mathrm{d}V + (1-t) p\mathrm{e}^{F/n} k_{T_2} \, \mathrm{d}V.$$

For two maximizing currents, equal total masses turn this into equality. The set $\mathcal{N}_f(F)$ of displayed probabilities is therefore nonempty, compact and convex.

A maximizing T gives

$$M_f(G) \ge M_f(F) \int_X \mathrm{e}^{(G-F)/n} \, \mathrm{d}\nu_{T,F}.$$

Jensen's inequality proves (25.8), so $\mathcal{N}_f(F) \subset \mathcal{A}_f(F)$. For the reverse inclusion, uniform Taylor expansion gives

$$D_{p\mathrm{e}^{(F+tH)/n}}(T) = D_{p\mathrm{e}^{F/n}}(T) + \frac{t}{n} \int_X Hp\mathrm{e}^{F/n} k_T \, \mathrm{d}V + O(t^2),$$

where the error is uniform in T since $D_{p\mathrm{e}^{F/n}}(T) \le M_f(F)$. Maximizers at $F + tH$ are near-maximizers at F, by the uniform multiplicative bounds. The convergence of their weighted measures proved above and this uniform expansion are exactly the two inputs to the one-sided differentiation argument in Theorem 25.7.1. It gives

$$(B_f)'_+(F; H) = -\max_{\nu \in \mathcal{N}_f(F)} \int_X H \, \mathrm{d}\nu.$$

The same compact-convex separation argument in that theorem identifies all supporting probabilities with $\mathcal{N}_f(F)$. Replacing H by $-H$ gives the left derivative.

Finally $M_f(F) \geq \mathrm{e}^{-R/n}\rho_\omega(p)$ when $\|F\|_\infty \leq R$. Thus the displayed probability densities are at most $\mathrm{e}^{2R/n}\rho_\omega(p)^{-1}pk_T$. The uniform integrability in the preceding theorem proves all the stated absolute-continuity assertions. □

When $f = 1$, this is the zero-reference construction of Chapter 14. For general f, the functional is defined by the maximal subsolution coefficient; it does not presume exact equation existence. Under positive lower mass, every nonnegative L^1 density satisfies the root-space hypothesis; for nonzero densities, Corollary 30.7.1 also identifies $b_\omega(f)$ as the unique envelope-class equation scalar. Without that property, the uniform tail condition gives an explicit sufficient hypothesis for this compact dual description.

25.9.3 Trace inequalities at the subsolution threshold

A smooth Gauduchon test pairs with the full current $\omega + \mathrm{dd^c}u$. For a threshold subsolution, Lemma 6.4.2 bounds that trace below by the determinant-weighted density. The nonnegative difference has a mass depending only on the test. Along a maximizing sequence this mass tends to zero, uniformly over all threshold subsolutions. Write $\|\cdot\|_{\mathrm{TV}}$ for total variation of a finite signed measure.

Proposition 25.9.4 *Let $0 \leq f \in L^1$ be nonzero and $b = b_\omega(f) > 0$, and write $p = f^{1/n}$. Suppose $u \in \mathrm{PSH}(X,\omega)$ satisfies $\mathrm{MA}^{\mathrm{ext}}_\omega(u) \geq bf\,\mathrm{d}V$. For every smooth strictly positive $T \in \mathcal{G}_\omega$, the measure*

$$R_{u,T} = (\omega + \mathrm{dd^c}u) \wedge T - b^{1/n}pk_T\,\mathrm{d}V$$

is positive and has mass

$$R_{u,T}(X) = 1 - b^{1/n}D_p(T).$$

In particular, if (T_j) is a smooth maximizing sequence for D_p, then $\|R_{u,T_j}\|_{\mathrm{TV}} \to 0$, uniformly over all such threshold subsolutions u.

If also $p \in \mathcal{D}^0_\omega$ and $T_j \to T$ as currents, then T maximizes D_p and

$$(\omega + \mathrm{dd^c}u) \wedge T_j \rightharpoonup b^{1/n}pk_T\,\mathrm{d}V.$$

For a fixed current limit T, this limiting probability is independent of the threshold subsolution u along the same sequence.

Proof A smooth strictly positive T is γ^{n-1} for a Gauduchon metric γ. The normalization is $\int_X \omega \wedge T = 1$. Lemma 6.4.2 proves positivity of $R_{u,T}$; distributional Stokes gives $((\omega + \mathrm{dd^c}u) \wedge T)(X) = 1$. This proves the mass formula. Since $\rho_\omega(p) = b^{-1/n}$, a maximizing sequence makes this nonnegative mass tend to zero. It equals the total variation norm and is independent of u.

Under the tail hypothesis, Theorem 25.9.1 gives uniform integrability of pk_{T_j}. Thus Proposition 25.9.2 makes T a maximizer and gives $pk_{T_j}\,\mathrm{d}V \rightharpoonup pk_T\,\mathrm{d}V$. Adding the error measures, which tend to zero in total variation, proves the limit. □

The limiting formula is stated along smooth maximizing sequences. It does not define the product $(\omega + \mathrm{dd^c}u) \wedge T$ for an arbitrary singular current T, or assert independence of the chosen maximizing current.

Smooth strict positivity of a maximizing test has a stronger consequence: in dimension at least two, it forces an exact smooth equation. The local regularity is supplied by Lemma 1.1.1.

Corollary 25.9.1 *Let $0 \le f \in L^1(\mathrm{d}V)$ be nonzero and $b = b_\omega(f) > 0$. If $D_{f^{1/n}}$ has a smooth strictly positive maximizer $T \in \mathscr{G}_\omega$, then $f^{1/n}$ agrees almost everywhere with a smooth nonnegative function. Every threshold subsolution is smooth and satisfies*

$$\mathrm{MA}_\omega(u) = bf\,\mathrm{d}V.$$

Such a solution exists, and all threshold subsolutions differ by constants.

Proof Theorem 13.24.1 supplies a threshold subsolution u. The following argument applies to every such u. Write $T = \gamma^{n-1}$ for its smooth positive Gauduchon root and set

$$a = \left(\frac{bf\,\mathrm{d}V}{\gamma^n}\right)^{1/n}.$$

The normalization of T is $\int_X \omega \wedge T = 1$. Put $A = \omega + \mathrm{dd^c}u \ge 0$. Since T maximizes, Proposition 25.9.4 gives the equality

$$A \wedge \gamma^{n-1} = a\gamma^n.$$

The positive coefficient measures of A are controlled by this trace, which is absolutely continuous. Thus A has an almost-everywhere positive semidefinite matrix density, with coefficients in L^n because $a \in L^n$.

For every smooth Hermitian form β, Lemma 6.4.2 also gives

$$A \wedge \beta^{n-1} \ge \left(\frac{bf\,\mathrm{d}V}{\beta^n}\right)^{1/n} \beta^n.$$

These pointwise matrix inequalities hold simultaneously almost everywhere for all positive definite matrices: on each coordinate chart first use a countable dense family of constant positive matrices, extended to global smooth Hermitian forms on smaller charts, and then use continuity in the matrix. The matrix arithmetic–geometric mean duality therefore yields

$$\left(\frac{A^n}{\gamma^n}\right)^{1/n} \ge a \quad \text{almost everywhere.}$$

Here A^n means only the determinant of its measurable coefficient matrix; no equality with a nonlinear current product has yet been used. The trace equality gives the reverse inequality by arithmetic–geometric mean. Equality forces all eigenvalues relative to γ to equal a, including where $a = 0$. Consequently $A = a\gamma$ as currents.

Apply Lemma 1.1.1 with $\theta = \omega$. Both a and u are smooth, and $f^{1/n} = b^{-1/n} a(\gamma^n/\mathrm{d}V)^{1/n}$ has a smooth nonnegative representative. Only now take the smooth top power, with $\omega_u = \omega + \mathrm{dd^c}u$: $\omega_u^n = a^n\gamma^n = bf\,\mathrm{d}V$. For two threshold subsolutions, the identity $\omega + \mathrm{dd^c}u = a\gamma$ has the same right side. Their smooth difference is therefore pluriharmonic and hence constant on connected compact X.□

If f is smooth and strictly positive, Theorem 25.1.1 supplies a smooth strictly positive maximizing Gauduchon form from the smooth equation. Thus the corollary recovers that case, including constant-difference uniqueness among all genuine threshold subsolutions.

Corollary 25.9.2 *Under the hypotheses of Corollary 25.9.1, let u be the threshold solution with $\sup_X u = 0$. If u_λ is the unique bounded solution of*

$$\mathrm{MA}_\omega(u_\lambda) = \mathrm{e}^{\lambda u_\lambda} f\,\mathrm{d}V, \qquad \lambda > 0,$$

then $u_\lambda - \sup_X u_\lambda \to u$ uniformly as $\lambda \downarrow 0$.

Proof The preceding regularity corollary makes f equal almost everywhere to a smooth function, so it belongs to L^p for every finite p. The scalar of the bounded equation is $b_\omega(f)$ by Proposition 13.24.2. Thus Proposition 13.18.1, with the trivial symmetry group, gives uniform precompactness and identifies every cluster limit as a normalized solution at this scalar. Each cluster limit is a threshold subsolution and hence equals u by Corollary 25.9.1. If the whole family failed to converge uniformly, a sequence staying a fixed positive uniform distance from u would have a convergent subsequence, a contradiction. □

Corollary 25.9.3 *Suppose ω is Kähler. For a nonzero nonnegative $f \in L^1(\mathrm{d}V)$, the functional $D_{f^{1/n}}$ has a smooth strictly positive maximizer in $\mathscr{G}_\omega$ if and only if f agrees almost everywhere with a smooth strictly positive function.*

Proof The Kähler formula in Proposition 13.24.2 makes $b_\omega(f) > 0$. If a smooth strictly positive maximizer exists, the proof of Corollary 25.9.1 gives

$$\omega + \mathrm{dd^c}u = a\gamma, \qquad a \geq 0,$$

with a, u smooth and $a \not\equiv 0$. Since $\mathrm{d}\omega = 0$, the first-order equation in Lemma 1.1.1 becomes $\mathrm{d}a = -aC$ for a smooth one-form C. Along any smooth path σ, this is the scalar equation

$$(a \circ \sigma)' = -(a \circ \sigma)C(\dot\sigma).$$

Its solution is the initial value multiplied by a positive exponential. Connectedness therefore gives $a > 0$ everywhere, and $f = b_\omega(f)^{-1} a^n\gamma^n/\mathrm{d}V$ is smooth and strictly positive. Conversely, the smooth equation and Theorem 25.1.1 supply the maximizing Gauduchon form. □

The conclusion about strict positivity changes for a nonclosed background, even on a Kähler manifold.

Example 25.9.1 On every connected compact complex manifold X of dimension $n \geq 2$, with a fixed smooth strictly positive probability volume $\mathrm{d}V$, there is a nonclosed Hermitian form ω and a nonzero smooth nonnegative density f vanishing on an open set such that $b_\omega(f) = 1$ and $D_{f^{1/n}}$ has a smooth strictly positive maximizer.

Proof Fix a Gauduchon metric γ and a point x. Choose a smooth real function ψ such that $\mathrm{dd^c}\psi \leq -\gamma$ on a coordinate neighborhood U of x. In holomorphic coordinates $z = (z_1, \ldots, z_n)$ centered at x, one obtains it by multiplying $-M|z|^2$ by a cutoff that equals one near x, with M large enough, and then shrinking U.

Choose a smooth function $\chi \geq 0$ which vanishes on a smaller neighborhood of x, equals one outside U, and has values at most one. Since X is compact, for sufficiently large $L > 0$ the form

$$a = L\chi, \qquad \omega = a\gamma - \mathrm{dd^c}\psi$$

is strictly positive: on U the second term is at least γ, and outside U the term $L\gamma$ dominates $\mathrm{dd^c}\psi$. Set $f = a^n\gamma^n/\mathrm{d}V$. The smooth potential ψ satisfies $\omega + \mathrm{dd^c}\psi = a\gamma \geq 0$ and $\mathrm{MA}_\omega(\psi) = f\,\mathrm{d}V$, so $b_\omega(f) \geq 1$.

Stokes and $\mathrm{dd^c}\gamma^{n-1} = 0$ give

$$\int_X \omega \wedge \gamma^{n-1} = \int_X a\gamma^n = \int_X (f\,\mathrm{d}V/\gamma^n)^{1/n}\gamma^n.$$

The Gauduchon formula Theorem 13.25.1 gives the opposite inequality $b_\omega(f) \leq 1$. The displayed test thus attains equality, and its normalized current is a smooth strictly positive maximizer. If ω were closed, the preceding corollary would force f to be strictly positive, contrary to its vanishing on an open set. Thus ω is nonclosed. □

In Example 25.9.1, every genuine threshold subsolution is consequently a constant translate of ψ, and the normalized positive-exponent solutions converge uniformly to $\psi - \sup_X \psi$. Thus vanishing on an open set does not by itself force nonuniqueness or failure of this approximation.

The restriction $n \geq 2$ in Corollary 25.9.1 is also necessary. On a compact connected Riemann surface X with Hermitian form ω, a normalized Gauduchon current has degree zero and is the constant $(\int_X \omega)^{-1}$: the equation $\mathrm{dd^c}T = 0$ makes it smooth and harmonic. This single smooth strictly positive current maximizes for every nonzero nonnegative L^1 density, which need not have a smooth representative.

25.9.4 A smooth maximizing form with an analytic degeneracy set

Return to the fixed Hermitian background ω on X with $n \geq 2$. A smooth maximizing form determines the equation wherever it is positive. If its degeneracy is confined to a proper analytic set, the measures give that set no mass, so the local equation extends globally. Uniqueness follows by taking a maximum and applying Lemma 1.17.2.

Theorem 25.9.2 *Let $0 \le f \in L^1(\mathrm{d}V)$ be nonzero, and assume $b = b_\omega(f) > 0$. Suppose $D_{f^{1/n}}$ has a smooth semipositive maximizer $T \in \mathscr{G}_\omega$ which is strictly positive on $X \setminus Z$, where Z is a proper analytic subset. Then every threshold subsolution is smooth on $X \setminus Z$ and satisfies*

$$\mathrm{MA}^{\mathrm{ext}}_\omega(u) = bf\,\mathrm{d}V$$

on X. Such a solution exists and has finite mass. The function $f^{1/n}$ has a smooth nonnegative representative on $X \setminus Z$.

All threshold subsolutions differ by constants.

Proof Put $p = f^{1/n}$ and $A = \omega + \mathrm{dd^c}u$, where u is any threshold subsolution. Existence of at least one such u is Theorem 13.24.1. Fix a smooth strictly positive $T_0 \in \mathscr{G}_\omega$ and for real $\varepsilon > 0$, set $T_\varepsilon = (T + \varepsilon T_0)/(1+\varepsilon)$. Each T_ε belongs to $\mathscr{G}_\omega$ and has a smooth Hermitian root, so Proposition 25.9.4 gives

$$A \wedge T_\varepsilon \ge b^{1/n} p k_{T_\varepsilon}\,\mathrm{d}V.$$

For smooth semipositive forms, k_T is the determinant root of the coefficient matrix, in the normalization of Section 25.6. This is a continuous function of that matrix. Thus $k_{T_\varepsilon} \to k_T$ uniformly on compact X. Since $p \in L^1$, passage to the limit gives

$$A \wedge T \ge b^{1/n} p k_T\,\mathrm{d}V.$$

The left side has mass one by Stokes and $\mathrm{dd^c}T = 0$. The right side has mass $b^{1/n} D_p(T) = 1$, because T maximizes. Their difference is positive and has zero mass, hence

$$A \wedge T = b^{1/n} p k_T\,\mathrm{d}V. \tag{25.9}$$

On $U = X \setminus Z$ write $T = \gamma^{n-1}$, where γ is smooth and strictly positive, and set $a = (bf\,\mathrm{d}V/\gamma^n)^{1/n}$. The trace equality becomes $A \wedge \gamma^{n-1} = a\gamma^n$. On relatively compact subsets of U this controls every coefficient measure of A by an L^n density. The all-Hermitian trace inequalities and matrix arithmetic–geometric mean, exactly as in the proof of Corollary 25.9.1, force $A = a\gamma$ there. This step uses only the local positivity of γ; the other Hermitian tests extend smoothly from smaller coordinate charts to X.

The local regularity lemma Lemma 1.1.1 makes u, a smooth on U. Taking the smooth top power proves the equation there, and $f^{1/n} = b^{-1/n} a(\gamma^n/\mathrm{d}V)^{1/n}$ is smooth and nonnegative there. Both $\mathrm{MA}^{\mathrm{ext}}_\omega(u)$ and $f\,\mathrm{d}V$ vanish on the pluripolar set Z. Hence the equality holds on X, and its total mass is $b\int_X f\,\mathrm{d}V < +\infty$.

It remains to prove uniqueness. Let v be another threshold subsolution and put $w = \max(u, v)$. By Proposition 6.2.1,

$$\mathrm{MA}^{\mathrm{ext}}_\omega(w) \ge \mathbf{1}_{\{u \ge v\}}\mathrm{MA}^{\mathrm{ext}}_\omega(u) + \mathbf{1}_{\{u<v\}}\mathrm{MA}^{\mathrm{ext}}_\omega(v) \ge bf\,\mathrm{d}V.$$

The common pole set has zero mass for all the measures involved. Thus w is also a threshold subsolution, and the preceding argument gives the same smooth current $a\gamma$ for u, v, w on U.

Apply Lemma 1.17.2 to $u \le w$ and $v \le w$. Their Hessians agree on $X \setminus Z$, so both $w - u$ and $w - v$ are constant. Hence $u - v$ is constant. □

25.9.5 The limit of the exponential equations

The uniqueness just proved identifies the limit of the exponential equations. To control their normalization, the maximizing form supplies a probability against which normalized potentials have bounded moments. Jensen's inequality converts this bound into an estimate for the scalar and a common lower barrier for the potentials.

Proposition 25.9.5 *Assume the hypotheses of Theorem 25.9.2, and let u_0 be its unique threshold solution with $\sup_X u_0 = 0$. For real $\lambda > 0$, let u_λ be the greatest exponential solution of Corollary 13.28.1, and put*

$$v_\lambda = u_\lambda - \sup_X u_\lambda, \qquad c_\lambda = \mathrm{e}^{\lambda \sup_X u_\lambda}.$$

There is a constant C independent of $\lambda > 0$ such that

$$b \le c_\lambda \le b\,\mathrm{e}^{C\lambda}, \qquad u_0 - C \le v_\lambda \le 0.$$

Moreover $v_\lambda \to u_0$ in $L^1(X)$ as $\lambda \downarrow 0$.

Proof Write $p = f^{1/n}$ and introduce the probability measure

$$\mathrm{d}\nu = b^{1/n} p k_T \,\mathrm{d}V.$$

It has mass one because T maximizes. Its density belongs to L^n, since $p \in L^n$ and the smooth semipositive form T has bounded determinant density k_T. By Lemma 1.5.1, normalized ω-plurisubharmonic functions have uniformly bounded moments of every finite order. Hölder's inequality with exponents n and $n/(n-1)$ therefore gives

$$\int_X (-v_\lambda)\,\mathrm{d}\nu \le C.$$

Here $n \ge 2$ is used.

Apply the trace inequality to u_λ, or equivalently to v_λ, and pass from positive regularizations of T to T as in the preceding proof. After integration,

$$1 \ge (c_\lambda/b)^{1/n} \int_X \mathrm{e}^{\lambda v_\lambda/n}\,\mathrm{d}\nu \ge (c_\lambda/b)^{1/n} \mathrm{e}^{-C\lambda/n},$$

where the last step is Jensen's inequality. This proves the upper scalar bound. The lower bound and the comparison

$$v_\lambda \geq u_0 - \lambda^{-1} \log(c_\lambda / b)$$

are Corollary 13.28.1, giving the asserted common lower barrier.

Take any sequence $\lambda_j \downarrow 0$. By Theorem 1.4.1, a subsequence of v_{λ_j} converges in L^1 to a potential v with supremum zero. Proposition 13.24.1 makes v a threshold subsolution. Uniqueness in Theorem 25.9.2 therefore gives $v = u_0$.

Every sequence tending to zero has a subsequence with this same L^1 limit. Normalized compactness therefore proves convergence of the whole family. □

The lower barrier gives capacity convergence under the additional envelope condition below. If the maximizing form is strictly positive everywhere, Corollary 25.9.2 already gives uniform convergence.

Corollary 25.9.4 *Under the hypotheses of Proposition 25.9.5, suppose in addition that the obstacle envelope $P_{\delta\omega}(u_0)$ of Section 1.19 is genuine for some real $0 < \delta < 1$. Then $v_\lambda \to u_0$ in capacity as $\lambda \downarrow 0$. In particular this holds if the threshold solution u_0 is bounded.*

Proof Fix a sequence $\lambda_j \downarrow 0$. Proposition 25.9.5 gives $u_0 - C \leq v_{\lambda_j} \leq 0$ and $v_{\lambda_j} \to u_0$ in L^1. For all large j,

$$\mathrm{MA}_\omega^{\mathrm{ext}}(v_{\lambda_j}) = c_{\lambda_j} \mathrm{e}^{\lambda_j v_{\lambda_j}} f \, \mathrm{d}V \leq b \mathrm{e}^C f \, \mathrm{d}V.$$

The bounded functions $|\mathrm{e}^{v_{\lambda_j}} - \mathrm{e}^{u_0}|$ tend to zero in smooth-volume measure. Since $f \in L^1$, their integrals against $f \, \mathrm{d}V$ tend to zero. Thus the self-measure error in (13.7) vanishes. Apply Proposition 13.8.1 with $\theta = \omega$ and $\varphi = u_0$; its additional limit-type hypothesis is automatic because the limit is u_0 itself. This proves capacity convergence along every sequence and hence for the family. If u_0 is bounded, a constant below it is a competitor for $P_{\delta\omega}(u_0)$, so that envelope is genuine. □

The smaller-background condition is an additional hypothesis. Its validity for an arbitrary threshold solution in this semipositive dual setting is not established here.

Corollary 25.9.5 *Under the hypotheses of Theorem 25.9.2, suppose in addition that $f \in L^p(\mathrm{d}V)$ for some $p > 1$. Then the normalized threshold solution u_0 is continuous, and the normalized greatest exponential solutions converge uniformly to u_0 as the exponent tends to zero.*

Proof The scalar of the continuous equation is $b_\omega(f)$ by Proposition 13.24.2. Its normalized solution is therefore u_0, by Theorem 25.9.2. For L^p data the greatest exponential solution agrees with the bounded solution, as explained after Theorem 13.28.1. Apply Proposition 13.18.1 with the trivial group. The normalized family is uniformly precompact, and every cluster limit solves the threshold equation. Uniqueness identifies every such limit with u_0, proving convergence of the whole family. □

25.9.6 Quantitative convergence of the currents

The scalar estimate also bounds the trace error by the exponent. Away from the degeneracy set, the arithmetic–geometric mean deficit converts this estimate into convergence of all current coefficients. Use $u_0, v_\lambda, c_\lambda$ from Proposition 25.9.5. Total variation of current coefficients is measured in a fixed smooth Hermitian metric; on a relatively compact chart it is equivalent to the sum of their coefficient variations.

Proposition 25.9.6 *Under the hypotheses and notation of Proposition 25.9.5, put*

$$A_\lambda = \omega + \mathrm{dd^c} v_\lambda, \qquad A_0 = \omega + \mathrm{dd^c} u_0.$$

There is a constant C such that

$$\|A_\lambda \wedge T - A_0 \wedge T\|_{\mathrm{TV}} \le C\lambda \qquad (\lambda > 0).$$

For each compact set $K \Subset X \setminus Z$, there is a constant C_K such that

$$\|A_\lambda - A_0\|_{\mathrm{TV}(K)} \le C_K \sqrt{\lambda} \qquad (0 < \lambda \le 1).$$

Proof Put $p = f^{1/n}$ and use the probability $\nu = A_0 \wedge T = b^{1/n} p k_T \,\mathrm{d}V$ from the preceding limit proof, and set

$$q_\lambda = (c_\lambda/b)^{1/n} \mathrm{e}^{\lambda v_\lambda/n}, \qquad R_\lambda = A_\lambda \wedge T - q_\lambda \nu \ge 0.$$

The same proof gives $c_\lambda \ge b$ and $\int_X (-v_\lambda)\,\mathrm{d}\nu \le C_0$. Write $s_+ = \max(s, 0)$ for the positive part of a real number s. Therefore

$$(1 - q_\lambda)_+ \le 1 - \mathrm{e}^{\lambda v_\lambda/n}, \qquad \int_X (1 - q_\lambda)_+ \,\mathrm{d}\nu \le C_0 \lambda/n.$$

Since $A_\lambda \wedge T$ and ν both have mass one,

$$\|A_\lambda \wedge T - \nu\|_{\mathrm{TV}} \le R_\lambda(X) + \int_X |q_\lambda - 1| \, \mathrm{d}\nu = 2 \int_X (1 - q_\lambda)_+ \, \mathrm{d}\nu.$$

This proves the first estimate and also gives $R_\lambda(X) \le C_0\lambda/n$ and $\int_X |q_\lambda - 1| \, \mathrm{d}\nu \le 2C_0\lambda/n$.

On $U = X \setminus Z$, write $T = \gamma^{n-1}$ and $A_0 = a\gamma$ as in the exactness theorem. Put $\alpha_\lambda = q_\lambda a$. Decompose the positive current A_λ into its absolutely continuous matrix density B_λ relative to γ and its positive singular part. The all-Hermitian trace inequalities imply

$$(\det B_\lambda)^{1/n} \ge \alpha_\lambda \quad \text{almost everywhere on } U.$$

This is a matrix determinant statement; no nonlinear product of coefficient measures is involved. Put $s_\lambda = \operatorname{tr} B_\lambda/n$ and $\delta_\lambda = s_\lambda - \alpha_\lambda \ge 0$. The trace error has absolutely continuous part $\delta_\lambda \gamma^n$, and its singular part is the trace of the singular part of A_λ. Thus

$$\int_U s_\lambda \gamma^n \le 1, \qquad \int_U \delta_\lambda \gamma^n \le C_0\lambda/n,$$

and the singular trace mass on U has the same upper bound.

For nonnegative eigenvalues $\ell_1, \dots, \ell_n$ with geometric mean at least $\alpha \ge 0$ and arithmetic mean s, arithmetic–geometric mean and Cauchy–Schwarz give

$$\sum_i (\sqrt{\ell_i} - \sqrt{\alpha})^2 \le n(s - \alpha), \qquad \sum_i |\ell_i - \alpha| \le 2n\sqrt{s(s - \alpha)}.$$

Indeed $\sum_i \sqrt{\ell_i} \ge n\sqrt{\alpha}$, while $\sum_i (\sqrt{\ell_i} + \sqrt{\alpha})^2 \le 4ns$. Apply this to B_λ and integrate, using Cauchy–Schwarz once more. The absolutely continuous part of $A_\lambda - \alpha_\lambda\gamma$ has integrated matrix trace norm at most $2n\sqrt{C_0\lambda/n}$. Its singular part has norm at most $C_0\lambda$ in the same metric. Finally

$$\int_U |\alpha_\lambda - a|\gamma^n = \int_U |q_\lambda - 1| \, \mathrm{d}\nu \le 2C_0\lambda/n.$$

These estimates give the second conclusion after comparing γ with the fixed metric on K and using $\lambda \le \sqrt{\lambda}$. □

The coefficient estimate is local on $X \setminus Z$. Capacity convergence across Z uses the additional envelope condition in Corollary 25.9.4.

25.9.7 The classes of the divisorial components

Uniqueness constrains the divisor classes that can occur in the threshold solution. A linear relation between them would give a global potential that changes their coefficients while preserving the equation off the degeneracy set. Write $\{D\}$ for the real Bott–Chern class of a prime divisor D.

Corollary 25.9.6 *Under the hypotheses of Theorem 25.9.2, let u be its threshold solution and let $D_1, \dots, D_k$ be the prime divisors with $\nu(u, D_j) > 0$. Then the classes $\{D_j\}$ are linearly independent in $H^{1,1}_{\mathrm{BC}}(X, \mathbb{R})$. In particular their number is at most $\dim_{\mathbb{R}} H^{1,1}_{\mathrm{BC}}(X, \mathbb{R})$.*

Proof Smoothness off Z shows that these divisors are among its finitely many divisorial components. Put $a_j = \nu(u, D_j) > 0$. By Lemma 1.17.1,

$$R = \omega + \mathrm{dd^c} u - \sum_j a_j [D_j] \geq 0.$$

Suppose real numbers $c_1, \dots, c_k$ satisfy $\sum_j c_j \{D_j\} = 0$. Fix a Gauduchon metric η and apply Lemma 22.7.1 to the integration currents. Let $v_\eta([D_j])$ be the canonical integrable potential in that lemma. The linearity of its smooth representative on Bott–Chern classes gives

$$h = \sum_j c_j v_\eta([D_j]) \in L^1(X), \qquad \mathrm{dd^c} h = \sum_j c_j [D_j].$$

Choose $t \neq 0$ sufficiently small that $a_j + t c_j \geq 0$ for every j. The distribution $u_t = u + th$ satisfies

$$\omega + \mathrm{dd^c} u_t = R + \sum_j (a_j + t c_j)[D_j] \geq 0,$$

so it has a genuine ω-plurisubharmonic representative. On $X \setminus Z$, the function h is smooth and pluriharmonic. Thus u_t has the same Monge–Ampère measure as u there. Both extended measures ignore Z, so u_t is another threshold solution. Uniqueness makes $th = u_t - u$ constant. Hence $\sum_j c_j [D_j] = 0$, and testing at a generic point of each distinct divisor gives $c_j = 0$. □

The conclusion concerns divisors on which the solution has positive generic Lelong number. It imposes no independence condition on the other divisorial components of the degeneracy set.

25.9.8 A branched projective example

A branched endomorphism of projective space gives a smooth threshold solution whose maximizing form degenerates. Its density is smooth, but its nth root need not be smooth across the branch divisor.

For this example take $X = \mathbb{P}^n$, $n \geq 2$, with homogeneous coordinates $[Z_0 : \cdots : Z_n]$, and let ω be the Fubini–Study form normalized by $\int_X \omega^n = 1$. Set $\mathrm{d}V = \omega^n$.

Example 25.9.2 For every $n \geq 2$, there is a smooth nonnegative density f on $\mathbb{P}^n$, vanishing precisely on the coordinate hyperplanes, such that $b_\omega(f) = 1$ and $D_{f^{1/n}}$ has a smooth semipositive maximizer strictly positive off those hyperplanes. The threshold solution is smooth, but $f^{1/n}$ is not differentiable at a general point of any coordinate hyperplane. There is no smooth strictly positive maximizing form. The normalized greatest exponential solutions converge uniformly to the normalized threshold solution.

Proof Define the holomorphic map $F : \mathbb{P}^n \to \mathbb{P}^n$ and the semipositive form A by

$$F[Z_0 : \cdots : Z_n] = [Z_0^2 : \cdots : Z_n^2], \qquad A = \tfrac{1}{2} F^* \omega.$$

The homogeneous formula

$$u = \tfrac{1}{2} \log\left(\sum_{i=0}^{n} |Z_i|^4\right) - \log\left(\sum_{i=0}^{n} |Z_i|^2\right)$$

defines a smooth function on projective space, and $A = \omega + \mathrm{dd^c} u$. This form is semipositive and is strictly positive exactly where every homogeneous coordinate is nonzero. Define

$$f = A^n/\omega^n, \qquad T = A^{n-1}.$$

Closed Stokes gives $\int f\omega^n = \int \omega \wedge T = 1$. Thus $T \in \mathscr{G}_\omega$. Every genuine ω-plurisubharmonic potential has extended mass at most one: each bounded truncation has mass one, and restriction away from its truncation level can only decrease that mass. Consequently an inequality $\mathrm{MA}_\omega^{\mathrm{ext}}(v) \geq c f \omega^n$ forces $c \leq 1$. The smooth solution u gives the reverse bound, so $b_\omega(f) = 1$.

The determinant formula for smooth semipositive forms gives

$$k_T = f^{(n-1)/n}, \qquad D_{f^{1/n}}(T) = \int_{\mathbb{P}^n} f\omega^n = 1.$$

By (25.6) and $\rho_\omega(f^{1/n}) = b_\omega(f)^{-1/n} = 1$, this is a maximizer. Its strict positivity away from the coordinate hyperplanes follows from that of A; on a hyperplane A has deficient rank, and so does its cofactor matrix.

For the density root, use the chart $Z_0 = 1$ with coordinates $z_1, \ldots, z_n$. Pulling back the Fubini–Study volume by $(z_1, \ldots, z_n) \mapsto (z_1^2, \ldots, z_n^2)$ and including the factor

2^{-n} gives

$$f(z) = 2^n \left(\prod_{i=1}^{n} |z_i|^2 \right) \frac{(1 + \sum_i |z_i|^2)^{n+1}}{(1 + \sum_i |z_i|^4)^{n+1}}.$$

Near a point of $\{z_1 = 0\}$ with all the other coordinates nonzero, its nth root is $|z_1|^{2/n}$ times a smooth strictly positive function. Restriction to a real line transverse to that hyperplane proves nondifferentiability, including the absolute-value case $n = 2$. Coordinate symmetry gives the same conclusion on every component. The absence of a smooth strictly positive maximizer follows from Corollary 25.9.3, since f has zeros. Finally f is bounded, so Corollary 25.9.5 gives the uniform exponential limit.□

25.9.9 A moment condition on maximizing tests

On the general Hermitian manifold, a bound for moments of a smooth maximizing sequence can replace smoothness of its limiting current. Combined with a smaller-background envelope, it makes the exponential cluster limit an exact threshold solution.

For nonzero $f \geq 0$ with positive threshold, let $\mathcal{S}_f$ and Φ_f be the normalized subsolution class and its greatest member from Proposition 13.24.1. Write u_λ for the greatest exponential solution, $v_\lambda = u_\lambda - \sup_X u_\lambda$ and $c_\lambda = \mathrm{e}^{\lambda \sup_X u_\lambda}$.

Theorem 25.9.3 *Let $0 \leq f \in L^1(\mathrm{d}V)$ be nonzero, and $b = b_\omega(f) > 0$. Put $p = f^{1/n}$ and $R = \rho_\omega(p) = b^{-1/n}$. Suppose there are smooth strictly positive $T_j \in \mathcal{G}_\omega$ such that*

$$d_j = D_p(T_j) \longrightarrow R, \qquad \mathrm{d}\nu_j = d_j^{-1} p k_{T_j} \, \mathrm{d}V,$$

and a constant C satisfying

$$\int_X (-v) \, \mathrm{d}\nu_j \leq C \quad \textit{for all } j \textit{ and all } v \in \mathrm{PSH}(X, \omega) \textit{ with } \sup_X v = 0. \tag{25.10}$$

For the greatest exponential solutions and their normalizations, one then has

$$b \leq c_\lambda \leq b\mathrm{e}^{C\lambda}, \qquad \Phi_f - C \leq v_\lambda \leq 0.$$

Suppose in addition that $P_{\delta\omega}(\Phi_f)$ is genuine for some $0 < \delta < 1$. Whenever $\lambda_k \downarrow 0$ and $v_{\lambda_k} \to v$ in $L^1(X)$, the convergence holds in capacity and

$$\mathrm{MA}_\omega^{\mathrm{ext}}(v) = b f \, \mathrm{d}V, \qquad \Phi_f - C \leq v \leq \Phi_f.$$

In particular an exact threshold solution exists.

Proof The trace inequality Lemma 6.4.2 and Stokes give, for each j and $\lambda > 0$,

$$1 \geq c_\lambda^{1/n} d_j \int_X \mathrm{e}^{\lambda v_\lambda / n} \, \mathrm{d}\nu_j \geq c_\lambda^{1/n} d_j \mathrm{e}^{-C\lambda/n}.$$

The second inequality is Jensen's inequality and (25.10). Let $j \to \infty$. Since $R = b^{-1/n}$, this proves the upper scalar bound. The lower scalar bound and comparison with every normalized threshold subsolution were proved in Corollary 13.28.1. Apply that comparison to Φ_f to obtain

$$v_\lambda \geq \Phi_f - \lambda^{-1} \log(c_\lambda/b) \geq \Phi_f - C.$$

Now let $v_{\lambda_k} \to v$ in L^1. By Proposition 13.24.1, its limit lies in $\mathcal{S}_f$ and hence below Φ_f. The lower bound passes to the limit, so $v \simeq \Phi_f$. For all sufficiently large k,

$$\mathrm{MA}_\omega^{\mathrm{ext}}(v_{\lambda_k}) = c_{\lambda_k} \mathrm{e}^{\lambda_k v_{\lambda_k}} f \, \mathrm{d}V \leq b\mathrm{e}^C f \, \mathrm{d}V.$$

The functions $|\mathrm{e}^{v_{\lambda_k}} - \mathrm{e}^v|$ are uniformly bounded and tend to zero in smooth-volume measure. Their integrals against the fixed L^1 density f therefore tend to zero. This gives the self-measure error required by Proposition 13.8.1, with reference Φ_f. The additional limit-type condition has just been checked, and the smaller-background condition is an assumption. That proposition gives capacity convergence.

Damped continuity now identifies the equation without a mass-conservation argument. Write $g_k = c_{\lambda_k} \mathrm{e}^{\lambda_k v_{\lambda_k}} f$. We know $g_k \to bf$ in L^1 and $0 \leq g_k \leq b\mathrm{e}^C f$ for large k. Hence

$$\mathrm{e}^{v_{\lambda_k}} \mathrm{MA}_\omega^{\mathrm{ext}}(v_{\lambda_k}) = \mathrm{e}^{v_{\lambda_k}} g_k \, \mathrm{d}V \longrightarrow \mathrm{e}^v bf \, \mathrm{d}V$$

in total variation, while damped continuity gives the weak limit $\mathrm{e}^v \mathrm{MA}_\omega^{\mathrm{ext}}(v)$. Divide off the pole set to obtain the equation and its finite mass. Finally normalized compactness supplies an L^1 cluster limit along any sequence $\lambda_k \downarrow 0$, proving existence. □

The moment condition is automatic if the densities of v_j are uniformly bounded in $L^q(\mathrm{d}V)$ for some $q > 1$, by Hölder and the uniform finite moments of normalized quasi-plurisubharmonic functions. More generally, a uniform entropy bound suffices. Write $a_j = \mathrm{d}v_j/\mathrm{d}V$ and suppose $\int_X a_j \log a_j \, \mathrm{d}V \leq H$. By Proposition 14.6.1, a uniform entropy bound is equivalent to a uniform bound for the zero-reference dual entropy $H_0(v_j)$ defined there. Choose $\alpha > 0$ and A from Lemma 1.5.1, so that $\int_X \mathrm{e}^{-\alpha v} \, \mathrm{d}V \leq A$ for every normalized v. The elementary entropy inequality, first for truncation levels $T > 0$ and $v_T = \max(v, -T)$, gives

$$\alpha \int_X (-v_T) a_j \, \mathrm{d}V \leq \int_X a_j \log a_j \, \mathrm{d}V + \log \int_X \mathrm{e}^{-\alpha v_T} \, \mathrm{d}V \leq H + \log A.$$

For completeness, apply $t \log t - t + 1 \geq 0$ to $t = a_j/q$, integrate against the probability density $q = \mathrm{e}^{-\alpha v_T} / \int_X \mathrm{e}^{-\alpha v_T} \, \mathrm{d}V$, and rearrange. Monotone convergence as $T \to \infty$ proves (25.10) with $C = (H + \log A)/\alpha$.

The same entropy bound also gives dual attainment along this sequence. Indeed $t \log t \geq -\mathrm{e}^{-1}$ and, for $K > 1$,

$$\int_{\{a_j > K\}} a_j \, \mathrm{d}V \leq \frac{H + \mathrm{e}^{-1}}{\log K}.$$

This proves uniform integrability of a_j. Since $d_j \to R > 0$, it also proves uniform integrability of $pk_{T_j} = d_j a_j$. By Proposition 25.9.2, every weak current cluster limit maximizes D_p, and along a convergent current subsequence the probabilities converge to $R^{-1} pk_T \, \mathrm{d}V$. For each normalized v, the function $-v$ is nonnegative and lower semicontinuous. The lower bound for integrals under weak convergence of measures therefore passes the moment bound to this limiting probability. No regularity of T follows from these conclusions.

The envelope condition is automatic if Φ_f is bounded. In general it and the moment bound are additional hypotheses; root tails alone do not supply them in this argument. The conclusion identifies the equation of each cluster limit. Uniqueness, which gave convergence of the whole family in Proposition 25.9.5, remains a separate requirement here.

25.10 Nef limits and nonattainment

We return to arbitrary dimension $n \geq 1$. Along a nef perturbation, the infimum over Gauduchon metrics commutes with the decreasing limit of the background. Thus the boundary value is determined by smooth equations even when no metric attains it. For a nef θ and any Hermitian h, the backgrounds $\theta + \epsilon h$ lie in $\mathscr{P}$ for every $\epsilon > 0$, by Section 23.1.

Theorem 25.10.1 *For every nef background θ and every Hermitian form h,*

$$\lambda_\mu(\theta) = \lim_{\epsilon \downarrow 0} \lambda_\mu(\theta + \epsilon h), \qquad \mathcal{V}_\mu(\theta) = \lim_{\epsilon \downarrow 0} c_{\theta+\epsilon h}(\mu).$$

These limits are decreasing and independent of h.

Proof The quantities decrease by monotonicity. The infima over $\epsilon > 0$ and over Gauduchon metrics commute. For a fixed metric the quotient is linear in ϵ, so

$$\begin{aligned} \inf_{\epsilon>0} \lambda_\mu(\theta + \epsilon h) &= \inf_{\gamma} \inf_{\epsilon>0} \frac{\int_X \theta \wedge \gamma^{n-1} + \epsilon \int_X h \wedge \gamma^{n-1}}{D_\mu(\gamma)} \\ &= \lambda_\mu(\theta). \end{aligned}$$

The coefficient of ϵ is positive, so the inner infimum is its limit at zero. Apply the exact formula on $\mathscr{P}$ and raise to the nth power. The resulting expression is the original definition at θ, proving independence of h. □

If X is Kähler and θ is closed and nef, choose h Kähler. Closed Stokes in every perturbed equation gives

$$c_{\theta+\epsilon h}(\mu) = \int_X (\theta + \epsilon h)^n.$$

It follows that $\mathcal{V}_\mu(\theta) = \int_X \theta^n$ in this regime.

Proposition 25.10.1 *Suppose θ is pseudoeffective and $\lambda_\mu(\theta) = 0$. The infimum is attained if and only if $\theta = \mathrm{dd^c} f$ for a smooth real function f.*

Proof The denominator $D_\mu(\gamma)$ is strictly positive. Thus a minimizing metric at value zero has $\int_X \theta \wedge \gamma^{n-1} = 0$. Apply Lemma 4.3.1, including its dimension-one case. Conversely, Stokes' theorem makes every quotient zero when θ is smoothly $\mathrm{dd^c}$-exact. □

For the ruled surface X and section D of Lemma 24.5.1, the nef class $c_1(\mathcal{O}_X(D))$ has zero square and a nonzero positive representative $[D]$. Its root volume is zero, but no smooth Gauduchon metric minimizes the defining quotient.

25.11 Subtraction with a closed nef core

A closed nef class of positive top intersection can keep the root volume positive when a path reaches the boundary of the positive cone. Adding a semipositive form preserves this lower bound, even when the added form is nonclosed. Two tests chosen at the initial point then bound the total subtraction cost. The dependence on the density and on the derivatives of the added form remains explicit.

Assume in this section that X is Kähler of dimension $n \geq 2$, and fix a Kähler form Ω. Let α be smooth, closed and nef, let $B \geq 0$ be a smooth real $(1,1)$-form, and let $t > 0$. For real s, put

$$a_s = \alpha + (t - s)\Omega, \qquad \theta_s = a_s + B, \qquad V_s = \int_X a_s^n.$$

Choose a Kähler representative κ_0 of $[a_0]$ and solve $\gamma_0 = \kappa_0 + \mathrm{dd^c} u_0 > 0$, $\gamma_0^n = V_0 \mu$. Define the fixed quantities

$$\begin{gathered} I_B = \int_X B \wedge \kappa_0^{n-1}, \qquad U_\mu = \frac{V_0 + I_B}{D_\mu(\kappa_0)}, \\ \Delta_\mu(B) = V_0^{-(n-1)/n} \int_X B \wedge \gamma_0^{n-1}. \end{gathered} \tag{25.11}$$

Theorem 25.11.1 *Suppose $V := V_t > 0$ and there is a smooth compact complex curve $C \subset X$ with $\int_C \alpha = \int_C B = 0$. Then θ_s has a smooth positive representative for $s < t$, θ_t is nef but has no smooth positive representative, and θ_s is not nef for $s > t$. Moreover,*

$$\lambda_\mu(a_s) = V_s^{1/n} \quad (0 \leq s \leq t), \qquad \lambda_\mu(\theta_t) \geq V^{1/n} > 0.$$

For every choice of minimizing Gauduchon metrics g_s for θ_s, $0 \le s < t$, their subtraction ratios are continuous in s and

$$\begin{aligned} 0 \le C := \int_0^t \frac{\int_X \Omega \wedge g_s^{n-1}}{D_\mu(g_s)} \, \mathrm{d}s &= \lambda_\mu(\theta_0) - \lambda_\mu(\theta_t) \\ &\le \min\{U_\mu, V_0^{1/n} + \Delta_\mu(B)\} - V^{1/n} < \infty. \end{aligned} \tag{25.12}$$

The integral is improper at t. If $B \le M\kappa_0$, then

$$\Delta_\mu(B) \le M V_0^{1/n}, \qquad C \le (1+M)V_0^{1/n} - V^{1/n}.$$

There are a quasi-plurisubharmonic ρ and $\varepsilon > 0$ with $\alpha + \mathrm{dd^c}\rho \ge \varepsilon\Omega$ and $\theta_t + \mathrm{dd^c}\rho \ge \varepsilon\Omega$.

Proof For $s < t$, take a nef approximation $\alpha + \mathrm{dd^c} v_\delta \ge -\delta\Omega$ with $\delta < t - s$. It gives a Kähler representative of a_s, and adding B gives a positive representative of θ_s. The same approximations show nefness at t. On the curve, smooth Stokes gives

$$\int_C (\theta_s + \mathrm{dd^c} v) = (t - s) \int_C \Omega$$

for every smooth v. This excludes strict positivity at t. For $s > t$, nef approximation with $\delta < s - t$ would contradict this identity.

The closed nef formula following Theorem 25.10.1 gives $\lambda_\mu(a_s) = V_s^{1/n}$, including at t. Monotonicity gives the lower bound for θ_t; a fixed Gauduchon test bounds its value above. The same nef-limit theorem identifies this value with the limit as $s \uparrow t$.

Write $f(s) = \lambda_\mu(\theta_s)$ for $s < t$. By Proposition 25.2.1, this is a smooth function and

$$H(s) := \frac{\int_X \Omega \wedge g_s^{n-1}}{D_\mu(g_s)} = -f'(s).$$

Thus the ratio is continuous and independent of the choice of minimizing metric. For $r < t$, the fundamental theorem of calculus gives

$$\int_0^r H(s) \, \mathrm{d}s = f(0) - f(r).$$

Use $H \ge 0$ and the endpoint limit to obtain the identity in (25.12).

The Kähler test κ_0 gives $\lambda_\mu(\theta_0) \le U_\mu$. Since $\gamma_0^n = V_0\mu$ and γ_0 is cohomologous to a_0,

$$D_\mu(\gamma_0) = V_0^{(n-1)/n}, \qquad \int_X a_0 \wedge \gamma_0^{n-1} = V_0.$$

This test gives the second initial upper bound. If $B \le M\kappa_0$, closed Stokes yields $\int_X B \wedge \gamma_0^{n-1} \le M \int_X \kappa_0 \wedge \gamma_0^{n-1} = MV_0$. These inequalities prove the cost bounds.

Finally, the closed nef-and-big criterion on a compact Kähler manifold, recalled in [Xia], gives a Kähler current $\alpha + \mathrm{dd^c}\rho \ge \varepsilon\Omega$ because $V > 0$. Adding B proves the last assertion. The exterior derivative of this second current is $\mathrm{d}B$. □

A birational holomorphic contraction $p : X \to Y$ between compact Kähler manifolds supplies such data when it contracts a smooth curve C. Take $\alpha = p^*\omega_Y$, with ω_Y Kähler, and $B \geq 0$ vanishing on TC. The degree-one map gives $\int_X \alpha^n = \int_Y \omega_Y^n > 0$, while both curve integrals vanish.

The second fixed test separates the closed intersection term from the differential correction of B.

Proposition 25.11.1 *Set $S = \sum_{j=0}^{n-2} \gamma_0^j \wedge \kappa_0^{n-2-j}$. Then*

$$\Delta_\mu(B) = V_0^{-(n-1)/n} \left(I_B + \int_X u_0 \, \mathrm{dd^c} B \wedge S \right). \tag{25.13}$$

If $-C_B\kappa_0^2 \leq \mathrm{dd^c} B \leq C_B\kappa_0^2$ in the weak-positivity order, then

$$\Delta_\mu(B) \leq V_0^{-(n-1)/n} I_B + (n-1)C_B \operatorname{osc}_X(u_0) V_0^{1/n}.$$

Such a finite C_B exists for every fixed smooth B.

Proof The factors of S are closed and $\gamma_0^{n-1} - \kappa_0^{n-1} = \mathrm{dd^c} u_0 \wedge S$. Integration by parts gives the identity. Also $\int_X \mathrm{dd^c} B \wedge S = 0$, so normalize $\sup_X u_0 = 0$. The assumed inequalities and positivity of the summands of S bound the correction above by

$$C_B \operatorname{osc}_X(u_0) \int_X \kappa_0^2 \wedge S = (n-1)C_B \operatorname{osc}_X(u_0) V_0.$$

The equality uses the mixed intersections of the cohomologous Kähler forms. Finally, evaluate $\mathrm{dd^c} B$ on κ_0-unit decomposable complex two-vectors. Compactness bounds these evaluations, giving C_B. □

The potential u_0 solves one fixed equation with density $V_0\mu/\kappa_0^n$. Thus its oscillation retains the dependence on μ; the tensors $\mathrm{d}\theta_s = \mathrm{d}B$ and $\mathrm{dd^c}\theta_s = \mathrm{dd^c} B$ are fixed along the path. For the closed core, put $P_s = \int_X \Omega \wedge a_s^{n-1}$. Then

$$-\frac{\mathrm{d}}{\mathrm{d}s}\lambda_\mu(a_s) = \frac{P_s}{V_s^{(n-1)/n}}, \qquad \int_0^t \frac{P_s}{V_s^{(n-1)/n}} \, \mathrm{d}s = V_0^{1/n} - V^{1/n}.$$

Moreover $P_s' = -(n-1)\int_X \Omega^2 \wedge a_s^{n-2} \leq 0$, since all classes on the segment are nef. Integrating $V_s' = -nP_s$ gives

$$V \geq V_0 - nt \int_X \Omega \wedge a_0^{n-1}. \tag{25.14}$$

This calculation remains valid when $V = 0$, so a positive right side supplies the positive-intersection hypothesis. It is a numerical inequality on a segment with nef endpoint.

Example 25.11.1 There is a genuinely nonclosed path as above on the blowup of $\mathbb{P}^2$ at a point, with

$$\lambda_\mu(\theta_1) \geq \frac{3}{2}, \qquad 0 \leq C \leq \frac{11}{\sqrt{8}} - \frac{3}{2},$$

and an explicit strictly positive current at the nef endpoint.

Proof Realize the blowup as $X = \{(x, [u : v]) \in \mathbb{P}^2 \times \mathbb{P}^1 : x_1 v = x_2 u\}$, with projections p, q and exceptional curve $E = p^{-1}([1 : 0 : 0])$. Let $L = p^*\omega_{\mathrm{FS},\mathbb{P}^2}$, $M = q^*\omega_{\mathrm{FS},\mathbb{P}^1}$ and $\Omega = L + M$, with the Fubini–Study forms normalized on projective lines. The incidence embedding makes Ω Kähler. The degree-one blowdown, a fiber of q, and $q|_E : E \to \mathbb{P}^1$ give

$$\int_X L^2 = \int_X L \wedge M = 1, \quad M^2 = 0, \quad L|_{TE} = 0, \quad \int_E M = 1.$$

Take $t = 1$, $\alpha = L$, $\kappa_0 = 2L + M$ and $\mu = \kappa_0^2/8$. With $r = 1 - s$, the core volumes are $V_s = 1 + 4r + 3r^2$. Choose a nonconstant smooth bump $0 \leq \chi \leq 1$ supported away from E, and put

$$b = \frac{1}{3}\int_X \chi L \wedge \kappa_0, \qquad f = \chi - b, \qquad B = (1 + f/2)L.$$

Since $\int_X L \wedge \kappa_0 = 3$, we have $|f| \leq 1$ and $\int_X fL \wedge \kappa_0 = 0$. Thus $B \geq L/2$ and $B|_{TE} = 0$. Where $\mathrm{d}\chi \neq 0$, L is symplectic, and wedging with L is injective on real one-forms in dimension four. Hence $\mathrm{d}B = \frac{1}{2}\mathrm{d}f \wedge L \neq 0$.

At the endpoint $\theta_1 = (2 + f/2)L \geq 3L/2$, giving the stated lower bound. The initial solution has $\gamma_0 = \kappa_0$ and $u_0 = 0$, so $D_\mu(\kappa_0) = \sqrt{8}$, $I_B = 3$, $\Delta_\mu(B) = 3/\sqrt{8}$ and $U_\mu = 11/\sqrt{8}$. The cost identity gives the claimed estimate.

The pullback of $\{x_1 = 0\}$ is E plus a fiber of q, so $\mathcal{O}(E) = p^*\mathcal{O}_{\mathbb{P}^2}(1) \otimes q^*\mathcal{O}_{\mathbb{P}^1}(-1)$. Give this bundle its product metric h_E, of curvature $L - M$, and let s_E be the canonical section. Poincaré–Lelong gives, for $\rho = \frac{1}{2}\log|s_E|^2_{h_E}$,

$$\theta_1 + \mathrm{dd^c}\rho = B + \tfrac{1}{2}\Omega + \tfrac{1}{2}[E] \geq \tfrac{1}{2}\Omega.$$

Finally, $\int_X \Omega \wedge a_0 = 5$, so the right side of (25.14) is $8 - 10 = -2$, although $V_1 = 1$. That sufficient numerical condition is therefore not necessary. □

25.12 Varying the probability measure

We return to a general connected compact complex manifold X of dimension $n \geq 1$. Changing the smooth positive probability measure changes each Gauduchon denominator by a uniformly bounded factor. The value of the volume may change, but its vanishing and positivity do not. On $\mathscr{P}$, independence of the measure is equivalent to invariance of total mass under changes of potential.

Proposition 25.12.1 *For smooth positive probability measures μ, ν, choose constants $m, M > 0$ such that $m\mu \leq \nu \leq M\mu$. Then for every pseudoeffective θ,*

$$M^{-1/n}\lambda_\mu(\theta) \leq \lambda_\nu(\theta) \leq m^{-1/n}\lambda_\mu(\theta).$$

In particular the zero set and the positivity set are independent of the chosen smooth probability measure.

Proof The density inequalities give $m^{1/n}D_\mu(\gamma) \leq D_\nu(\gamma) \leq M^{1/n}D_\mu(\gamma)$. The numerators are nonnegative. Divide and take infima. □

Proposition 25.12.2 *Fix a Hermitian representative ω. The value $\mathcal{V}_\mu(\omega)$ is the same for all smooth positive probability measures μ if and only if*

$$\int_X (\omega + \mathrm{dd^c} v)^n$$

is independent of the smooth potential v with $\omega + \mathrm{dd^c} v > 0$.

Proof For such a potential, put

$$M_v = \int_X (\omega + \mathrm{dd^c} v)^n, \qquad \mu_v = \frac{(\omega + \mathrm{dd^c} v)^n}{M_v}.$$

It is a smooth positive probability measure, and v already solves its normalized equation with scalar M_v. Scalar uniqueness and Theorem 25.1.1 give $\mathcal{V}_{\mu_v}(\omega) = M_v$. Thus measure independence implies mass invariance. Conversely, if all these masses equal one constant, every smooth normalized equation has that constant as its scalar after integration. The volume is then independent of its smooth probability measure.□

The non-closed torus example of Example 2.3.1 therefore produces different values for different smooth probability measures. On a Gauduchon surface, smooth mass is conserved, so the value equals $\int_X \omega^2$ for every such measure. On the Hopf threefold of Chapter 8, the value is finite and positive for each fixed μ, even though $v_+(\omega) = +\infty$.

25.12.1 Upper and lower masses

As the probability measure varies, the equation realizes every smooth strictly positive potential mass. Exact-background regularization gives the same extrema over bounded potentials. The infimum has a further simplification: maximizing the denominator at each Gauduchon metric removes the density entirely.

Fix a Hermitian form ω, and let $\mathscr{M}$ be the set of smooth strictly positive probability measures on X.

Theorem 25.12.1 *One has*

$$\inf_{\mu\in\mathscr{M}} \mathcal{V}_\mu(\omega) = v_-(\omega), \qquad \sup_{\mu\in\mathscr{M}} \mathcal{V}_\mu(\omega) = v_+(\omega).$$

The supremum is allowed to be infinite. Moreover

$$v_-(\omega)^{1/n} = \inf_{\gamma \text{ Gauduchon}} \frac{\displaystyle\int_X \omega\wedge\gamma^{n-1}}{\left(\displaystyle\int_X \gamma^n\right)^{(n-1)/n}}. \tag{25.15}$$

Proof The smooth normalized equation realizes every value $\mathcal{V}_\mu(\omega)$ as the mass of a smooth strictly positive potential. Conversely every such potential v realizes its mass by the probability measure μ_v in the proof of Proposition 25.12.2. Thus the two ranges, as subsets of $(0,+\infty)$, are equal.

We justify replacing bounded potentials by smooth strictly positive ones in the extrema defining $v_\pm$. If $u \in \mathrm{PSH}(X,\omega)\cap L^\infty$, Theorem 1.11.1 gives uniformly bounded smooth strictly ω-plurisubharmonic u_j decreasing to u. Their Monge–Ampère measures converge by bounded monotone continuity, including their total masses on compact X. Every bounded mass is therefore a limit of strictly positive smooth masses. The reverse inclusion is immediate. This proves the first two identities, without a uniform bound on those masses.

For the last formula, take the nth root and commute the infima over μ and γ in the dual formula. At fixed γ, Hölder's inequality and $\int_X \mu = 1$ give

$$D_\mu(\gamma) \le \left(\int_X \gamma^n\right)^{(n-1)/n}.$$

Equality holds for the smooth probability measure $\mu = \gamma^n/\int_X \gamma^n$. The numerator is positive and does not depend on μ, so this maximization of the denominator computes the infimum of the quotient. Substitution gives (25.15). For $n = 1$ both denominators are one, and the same argument applies. □

Thus $v_-(\omega) > 0$ is equivalent to a positive lower bound for $\int_X \omega\wedge\gamma^{n-1}$ among Gauduchon metrics of volume one. This lower bound fails on the threefold constructed in Theorem 25.15.1. By Corollary 13.23.1, it is also equivalent to solvability of

the normalized equation for every finite non-pluripolar measure in a Hermitian background. Likewise every fixed smooth density gives a finite normalization, while their supremum can be infinite.

Proposition 25.12.3 *For a Hermitian ω, the infimum in (25.15) is attained if and only if the infimum of smooth strictly positive Monge–Ampère masses is attained. Every such minimizing potential has a Gauduchon metric $\omega + \mathrm{dd^c} v$.*

Proof Suppose a test metric γ attains the first infimum, and put $m = v_-(\omega)^{1/n}$. Its numerator is strictly positive, so $m > 0$. For $\mu_\gamma = \gamma^n / \int_X \gamma^n$, the same test metric gives $\lambda_{\mu_\gamma}(\omega) \le m$. The mass-extrema theorem gives the opposite inequality. Hence the smooth equation for μ_γ has mass $m^n = v_-(\omega)$, giving a minimizing smooth potential. For $n \ge 2$, equality in the dual formula makes its solution conformal to γ. The fixed proportional volume forms make the conformal factor constant, so the solution metric is Gauduchon.

Conversely, at a smooth strictly positive mass minimizer v, both signs of every smooth variation are admissible for sufficiently small parameters. The first variation of the raw mass therefore gives

$$\int_X f \, \mathrm{dd^c}\big((\omega + \mathrm{dd^c} v)^{n-1}\big) = 0 \quad \text{for every smooth real } f.$$

Thus $\Omega = \omega + \mathrm{dd^c} v$ is Gauduchon. Using $\gamma = \Omega$ in the quotient and Gauduchon Stokes gives

$$\frac{\int_X \omega \wedge \Omega^{n-1}}{(\int_X \Omega^n)^{(n-1)/n}} = (\int_X \Omega^n)^{1/n} = v_-(\omega)^{1/n},$$

so the test infimum is attained. In dimension one all metrics are Gauduchon, and all masses equal $\int_X \omega$. □

The density infimum also has a definition beyond the positive cone. For a pseudoeffective smooth real $(1,1)$-background θ on X, put

$$\lambda_*(\theta) = \inf_{\gamma \text{ Gauduchon}} \frac{\int_X \theta \wedge \gamma^{n-1}}{(\int_X \gamma^n)^{(n-1)/n}}.$$

The same Hölder argument proves

$$\lambda_*(\theta) = \inf_{\mu \in \mathscr{M}} \lambda_\mu(\theta).$$

It is finite, nonnegative, concave, homogeneous, and invariant under smooth $\mathrm{dd^c}$ changes. If θ is nef, fix a Hermitian form h and, for real $\epsilon > 0$, choose a smooth Hermitian representative ω_ϵ of $\theta + \epsilon h$. Commuting the infima over the test metric and $\epsilon > 0$ also gives

$$\lambda_*(\theta) = \lim_{\epsilon \downarrow 0} \lambda_*(\theta + \epsilon h) = \lim_{\epsilon \downarrow 0} v_-(\omega_\epsilon)^{1/n}.$$

Gauge invariance makes the last expression independent of that representative. This extends the lower-mass formula to the nef boundary. It does not identify the resulting value with every lower-volume definition for an arbitrary singular class.

25.13 Curvature conditions for mass extrema

At a smooth local extremum of the mass, the first two variations give a Gauduchon condition and a sign for one lower background power. On a threefold these necessary conditions are sufficient because the mass polynomial stops at its quadratic term. Fix a Hermitian form ω and use $\mathcal{H}_\omega$ and M_ω from Chapter 2.

For $n \geq 2$, Theorem 2.2.2 and Proposition 25.12.2 identify the differential conditions for $\mathcal{V}_\mu(\omega)$ to be independent of the smooth positive probability measure μ. By Proposition 14.6.2, these are also the conditions for $H_0 - \mathrm{Ent}_{\mathrm{d}V}$ to be constant on finite-entropy probabilities, for a fixed smooth positive probability volume $\mathrm{d}V$; testing smooth positive probabilities suffices. In dimension one the mass is always $\int_X \omega$, so both conclusions hold automatically.

For the next argument, a smooth real $(n-1, n-1)$-form T is nonnegative when $\mathrm{i}\alpha \wedge \bar{\alpha} \wedge T \geq 0$ for every $(1,0)$-covector α. This is the usual positivity in this bidegree. We need an elementary local test for its sign.

Lemma 25.13.1 *Suppose a smooth real $(n-1, n-1)$-form T satisfies*

$$\int_X \mathrm{d}f \wedge \mathrm{d}^{\mathrm{c}} f \wedge T \geq 0 \quad \textit{for every smooth real } f.$$

Then $T \geq 0$.

Proof If the conclusion fails at a point, choose a real linear function ℓ in local real coordinates, whose differential at that point has the offending complex direction. Shrinking the coordinate ball makes $\mathrm{d}\ell \wedge \mathrm{d}^{\mathrm{c}}\ell \wedge T$ strictly negative there. Choose a nonzero smooth cutoff χ supported in this ball and, for positive integers k, set

$$f_k = k^{-1}\chi \sin(k\ell),$$

extended by zero to X. Since $\mathrm{d}\ell$ is a nonzero constant covector in these coordinates,

$$\mathrm{d}f_k \wedge \mathrm{d}^{\mathrm{c}} f_k = \chi^2 \cos^2(k\ell)\, \mathrm{d}\ell \wedge \mathrm{d}^{\mathrm{c}}\ell + O(k^{-1}),$$

where the remainder has uniformly bounded coefficients after multiplication by k. The oscillatory part of $\cos^2(k\ell) = (1 + \cos(2k\ell))/2$ has integral tending to zero, by integration by parts in a coordinate direction where ℓ has nonzero derivative. Hence

$$\lim_{k\to\infty} \int_X \mathrm{d}f_k \wedge \mathrm{d}^{\mathrm{c}} f_k \wedge T = \frac{1}{2}\int_X \chi^2 \mathrm{d}\ell \wedge \mathrm{d}^{\mathrm{c}}\ell \wedge T < 0,$$

a contradiction. Every complex covector is the $(1,0)$ part of a real covector, so this tests all required directions. □

Proposition 25.13.1 *Let $n \geq 3$, and let $v \in \mathcal{H}_\omega$ be a local minimum of M_ω in the C^2 topology. Then $\Omega = \omega + \mathrm{dd^c} v$ satisfies*

$$\mathrm{dd^c}(\Omega^{n-1}) = 0, \qquad \mathrm{dd^c}(\Omega^{n-2}) \leq 0.$$

At a local maximum the first condition is unchanged and the second inequality is reversed.

Proof Every smooth direction is admissible for a sufficiently short two-sided interval. The first variation gives the Gauduchon condition. Differentiating a second time gives

$$\begin{aligned}\frac{\mathrm{d}^2}{\mathrm{d}t^2}\bigg|_{t=0} \mathrm{M}_\omega(v + tf) &= n(n-1)\int_X (\mathrm{dd^c} f)^2 \wedge \Omega^{n-2} \\ &= -n(n-1)\int_X \mathrm{d}f \wedge \mathrm{d^c} f \wedge \mathrm{dd^c}(\Omega^{n-2}).\end{aligned}$$

For the last equality, integrate once against $f\,\mathrm{dd^c} f$, then use that $\mathrm{dd^c}(\Omega^{n-2})$ is d- and $\mathrm{d^c}$-closed. At a minimum this second variation is nonnegative for every f. Apply Lemma 25.13.1 to $-\mathrm{dd^c}(\Omega^{n-2})$. The maximum case is identical with the opposite sign. □

In dimension three these necessary conditions are also sufficient, and a local extremum is automatically global.

Theorem 25.13.1 *Assume $n = 3$ and $v \in \mathcal{H}_\omega$. The following are equivalent:*

(1) *v is a local minimum of M_ω;*
(2) *v is a global minimum of M_ω on $\mathcal{H}_\omega$;*
(3) *$\Omega = \omega + \mathrm{dd^c} v$ is Gauduchon and $\mathrm{dd^c}\omega \leq 0$.*

In this case $v_-(\omega) = \int_X \Omega^3$, and the Gauduchon infimum in (25.15) is attained. The corresponding statements for maxima hold with $\mathrm{dd^c}\omega \geq 0$, and then $v_+(\omega) = \int_X \Omega^3 < +\infty$.

Proof The preceding proposition gives the necessary conditions, since $\mathrm{dd^c}\Omega = \mathrm{dd^c}\omega$. Conversely, if Ω is Gauduchon, the dimension-three mass polynomial becomes, for every smooth real f,

$$\int_X (\Omega + \mathrm{dd^c} f)^3 - \int_X \Omega^3 = -3\int_X \mathrm{d}f \wedge \mathrm{d^c} f \wedge \mathrm{dd^c}\omega. \tag{25.16}$$

The form $\mathrm{d}f \wedge \mathrm{d^c} f$ is nonnegative. The displayed difference is therefore nonnegative if $\mathrm{dd^c}\omega \leq 0$, proving a global minimum. Global minimality implies local minimality. Exact-background regularization identifies the infimum over smooth strictly positive potentials with $v_-(\omega)$, and Proposition 25.12.3 proves attainment in the Gauduchon

formula. Reversing the sign in the same identity proves the maximum statements, including the equality with the supremum over bounded potentials. □

The local test supplies the necessity for smooth attainment in (25.16). Neither that test nor the sufficient lower-bound argument produces such a representative in an arbitrary background.

For the Hopf metric ω and the pulled-back Fubini–Study form α of Chapter 8, $\mathrm{dd^c}\omega = -\alpha^2 \le 0$ and ω is Gauduchon. Thus the minimum is attained by the constant potential and equals 6, consistently with its infinite upper mass. On a surface, by contrast, the mass is affine in the potential; a local extremum exists precisely when ω is Gauduchon, in which case all masses are equal. In higher dimensions the necessary sign condition of Proposition 25.13.1 does not control all higher terms of the mass polynomial, and sufficiency is not claimed.

25.14 Geometric bounds and determinant concentration

The mass-invariance criterion gives geometric conditions for the critical determinant bound. That bound cannot in general be improved to a higher exponent, even inside one Kähler class. A concentrating family also separates the smooth maximizing current for one density from sequences approaching its value.

Assume $n \ge 2$ and use the determinant densities k_T of Section 25.8, relative to a smooth positive probability volume $\mathrm{d}V$. Put $q = n/(n-1)$.

Corollary 25.14.1 *Suppose g is a Hermitian form with $\mathrm{dd^c}g = \mathrm{dd^c}(g^2) = 0$, and put $V_g = \int_X g^n$. For every Hermitian form $h \ge cg$, $c > 0$,*

$$v_-(h) \ge c^n V_g, \qquad \sup_{T \in \mathscr{G}_h} \|k_T\|_{L^q(\mathrm{d}V)} \le c^{-1} V_g^{-1/n}.$$

Equality holds in the determinant bound when $h = g$ and $c = 1$. In particular, for each fixed Hermitian background on a compact Kähler manifold or a compact complex surface, the critical norm is uniformly bounded over its normalized Gauduchon currents.

Proof The identity $\mathrm{dd^c}(g^2) = 2g \wedge \mathrm{dd^c}g + 2\mathrm{d}g \wedge \mathrm{d^c}g$ and Theorem 2.2.2 give $v_-(g) = V_g$. The lower-mass quotient (25.15) is monotone and homogeneous in its numerator, so $h \ge cg$ implies $v_-(h)^{1/n} \ge cV_g^{1/n}$. Apply Theorem 25.8.2.

A Kähler form satisfies both differential identities. On a surface, Theorem 4.1.1 supplies a metric g with $\mathrm{dd^c}g = 0$, and the second identity holds by degree. Compactness gives $h \ge cg$ for each fixed h. □

For a fixed smooth F, Theorem 25.6.1 already gives a unique smooth strictly positive maximizing current for $\mathfrak{D}_F$. Its determinant density is bounded. The following construction shows that approaching this maximum imposes no uniform higher determinant moments.

Proposition 25.14.1 *Let $X = E^n$, where $E = \mathbb{C}/(\mathbb{Z} + \mathrm{i}\mathbb{Z})$ has its area-one form, and write ω_i for its pullbacks. Set*

$$h = \sum_{i=1}^{n} \omega_i, \qquad \mathrm{d}V = \omega_1 \wedge \cdots \wedge \omega_n, \qquad V = n!, \qquad \alpha = \frac{n-1}{n}.$$

For $F = 0$, the unique maximizing current is $T_* = h^{n-1}/V$, with $k_{T_*} = V^{-1/n}$ and $\mathfrak{D}_0(T_*) = V^{-1/n}$. There are smooth strictly positive closed currents $S_\varepsilon \in \mathscr{G}_h$, $0 < \varepsilon < \min\{1/4, \mathrm{e}^{-1}\}$, such that

$$\|k_{S_\varepsilon}\|_{L^q} = V^{-1/n}, \qquad \mathfrak{D}_0(S_\varepsilon) \longrightarrow V^{-1/n},$$

while for every fixed finite $p > q$,

$$\int_X k_{S_\varepsilon}^p \, \mathrm{d}V \geq C_{n,p} [\log(1/\varepsilon)]^{-\alpha p} \varepsilon^{2-2\alpha p} \longrightarrow +\infty. \tag{25.17}$$

Their L^∞ norms also diverge. Each current is an $(n-1)$st power of a Kähler form in the fixed class $V^{-1/(n-1)}[h]$, and has the Bott–Chern, de Rham and Aeppli classes of T_*. No S_ε attains the fixed-density maximum.

Proof Put $H_i = \bigwedge_{j \neq i} \omega_j$. Wedge expansion gives $h^{n-1} = (n-1)! \sum_i H_i$ and $h \wedge H_i = \mathrm{d}V$. Thus $T_* = (1/n) \sum_i H_i$ is normalized, and the determinant formula gives $k_{T_*} = V^{-1/n}$. The smooth equation for $F = 0$ is solved by $h^n = V \, \mathrm{d}V$; the compact duality theorem proves the value and uniqueness assertion.

Choose a nonnegative smooth bump β on the unit disk in $\mathbb{C}$, of integral one, with $\beta \geq \beta_0 > 0$ on a disk of radius $r_0 \in (0, 1)$. On an embedded disk in the first elliptic factor put $b_\varepsilon(z) = \varepsilon^{-2} \beta(z/\varepsilon)$ and extend by zero. It has integral one. For $0 < t < 1$, set

$$f_{\varepsilon,t} = 1 - t + t b_\varepsilon, \qquad a = V^{-1/(n-1)},$$
$$\gamma_{\varepsilon,t} = a(f_{\varepsilon,t}\omega_1 + \omega_2 + \cdots + \omega_n), \qquad T_{\varepsilon,t} = \gamma_{\varepsilon,t}^{n-1}.$$

The form $\gamma_{\varepsilon,t}$ is strictly positive and closed: $f_{\varepsilon,t}$ depends only on the first, two-real-dimensional factor. Direct expansion gives

$$T_{\varepsilon,t} = \frac{1}{n}\left(H_1 + f_{\varepsilon,t} \sum_{i=2}^{n} H_i\right), \qquad k_{T_{\varepsilon,t}} = V^{-1/n} f_{\varepsilon,t}^{\alpha}. \tag{25.18}$$

Since $\int_X f_{\varepsilon,t} \, \mathrm{d}V = 1$, the trace integral is one and $\|k_{T_{\varepsilon,t}}\|_{L^q} = V^{-1/n}$.

The mean-zero function $f_{\varepsilon,t} - 1$ has a smooth real Poisson potential ψ on E with $\mathrm{dd^c}\psi = (f_{\varepsilon,t} - 1)\omega_1$. For example its mean-zero Fourier series solves $\Delta\psi = 4\pi(f_{\varepsilon,t} - 1)$ in the area-one coordinates. Consequently $\gamma_{\varepsilon,t} - ah = \mathrm{dd^c}(a\psi)$. The closed form $P = \sum_{j=0}^{n-2} \gamma_{\varepsilon,t}^{n-2-j} \wedge (ah)^j$ satisfies

$$T_{\varepsilon,t} - T_* = \mathrm{dd^c}(a\psi P).$$

This proves all the class assertions, including $n = 2$, when $P = 1$.

Jensen's inequality and $f_{\varepsilon,t} \geq 1 - t$ give

$$V^{-1/n}(1-t)^{\alpha} \leq \mathfrak{D}_0(T_{\varepsilon,t}) = V^{-1/n} \int_X f_{\varepsilon,t}^{\alpha} \, \mathrm{d}V < V^{-1/n}.$$

The strict inequality follows because $f_{\varepsilon,t}$ is nonconstant and $0 < \alpha < 1$. On the disk of radius $r_0\varepsilon$ times the other factors, $f_{\varepsilon,t} \geq t\beta_0\varepsilon^{-2}$; its volume is $\pi r_0^2\varepsilon^2$. Hence

$$\int_X k_{T_{\varepsilon,t}}^p \, \mathrm{d}V \geq V^{-p/n}\pi r_0^2 \beta_0^{\alpha p} t^{\alpha p} \varepsilon^{2-2\alpha p},$$
$$\|k_{T_{\varepsilon,t}}\|_\infty \geq V^{-1/n}\beta_0^{\alpha} t^{\alpha} \varepsilon^{-2\alpha}.$$

Take $t = 1/\log(1/\varepsilon)$ and $S_\varepsilon = T_{\varepsilon,t}$. The objective converges to its maximum, while both lower bounds diverge for $p > q$ because $\alpha p > 1$. All constants are independent of ε. □

Keeping $t \in (0, 1)$ fixed in the same construction gives $\int_X k_{T_{\varepsilon,t}}^p \, \mathrm{d}V \geq C_{n,p,t}\varepsilon^{2-2\alpha p}$ and $\|k_{T_{\varepsilon,t}}\|_\infty \geq C_{n,t}\varepsilon^{-2\alpha}$. Thus failure of higher bounds occurs both on the full normalized cone and along a maximizing sequence for one smooth density. The critical norm is attained by every member of these families; the fixed-density objective is attained only by T_*.

25.15 Vanishing lower mass

The lower-mass quotient can tend to zero on a compact threefold. The construction first concentrates the trace of a normalized Gauduchon form, then adds an average of compact curves in complementary directions. Its degree stays bounded while the volume of its positive root tends to infinity. The smooth equation converts these forms into potentials whose masses tend to zero.

Theorem 25.15.1 *There is a connected compact complex threefold X such that, for every Hermitian form h on X, there are smooth real functions u_j satisfying*

$$h + \mathrm{dd^c} u_j > 0, \qquad \sup_X u_j = 0, \qquad \int_X (h + \mathrm{dd^c} u_j)^3 \longrightarrow 0.$$

In particular, $v_-(h) = 0$ for every Hermitian form on this manifold.

We first give the manifold and the concentration argument. The last step of the proof will use the smooth Monge–Ampère equation.

25.15.1 The elliptic quotient

Write points of $\mathbf{P}^3$ as $[z : w]$, with $z, w \in \mathbb{C}^2$, and set

$$\Omega = \{[z : w] : z \neq 0,\ w \neq 0\}, \qquad q = \mathrm{e}^{2\pi}, \qquad g[z : w] = [z : qw].$$

Here $g : \Omega \to \Omega$ is the indicated automorphism, and $\langle g \rangle$ is the cyclic group generated by it. Its action is free. Indeed, equality of two projective points in one orbit would require the same scalar to be 1 on the nonzero z block and q^k on the nonzero w block. Thus g^k has a fixed point only for $k = 0$.

The function

$$t = \log \frac{\|w\|^2}{\|z\|^2}$$

is well defined on Ω and satisfies $t \circ g = t + 4\pi$. It follows that the action is properly discontinuous: two compact subsets have bounded t ranges, so only finitely many translates can meet. Moreover, the strip $0 \leq t \leq 4\pi$ is compact. In homogeneous coordinates of total norm one its defining inequalities bound both blocks away from zero, making it a closed subset of $\mathbf{P}^3$ contained in Ω. The strip meets every orbit. Consequently

$$X = \Omega/\langle g \rangle$$

is a connected compact complex threefold.

The map $[z : w] \mapsto ([z], [w])$ induces a holomorphic submersion $X \to \mathbf{P}^1 \times \mathbf{P}^1$. Its fibers are $E = \mathbb{C}/(2\pi\mathbb{Z} + 2\pi\mathrm{i}\mathbb{Z})$, using a logarithm of the relative scalar. Let α, β be the pullbacks of the Fubini–Study forms on the two factors, each of integral one in our normalization. Although t does not descend, its differentials do. The form

$$\omega = \alpha + \beta + \mathrm{d}t \wedge \mathrm{d}^{\mathrm{c}}t \tag{25.19}$$

is Hermitian. Its first two terms are positive in the two base directions, and the last is semipositive and strictly positive on the vertical complex line. Their kernels have zero intersection.

On the open set $z_0 w_0 \neq 0$, use coordinates

$$x = \frac{z_1}{z_0}, \qquad y = \frac{w_1}{w_0}, \qquad \zeta = \log \frac{w_0}{z_0} \in E.$$

Near $0 \in E$, the last coordinate is represented by an ordinary complex number. For any one of these coordinates, denoted by v, put $\sigma_v = \frac{\mathrm{i}}{2\pi}\mathrm{d}v \wedge \mathrm{d}\bar{v}$. Choose a smooth probability density p relative to σ_v, positive on $|v| < 1$ and zero on $|v| \geq 1$. Choose a smooth function χ_0 supported in $|v| < 1$, equal to one on $|v| \leq 1/2$, and with values in $[0, 1]$. Write $\chi(x, y) = \chi_0(x)\chi_0(y)$. For $0 < \delta < 1/100$ define, with extension by zero,

$$B_\delta = \chi(x, y)\delta^{-2} p(\zeta/\delta)\sigma_\zeta.$$

All extensions are smooth because the chosen functions are smooth with compact support in their coordinate domains.

Lemma 25.15.1 *For smooth real* $(2,2)$*-forms* Θ*, the finite numbers*

$$C_\delta = \sup\left\{\int_X B_\delta \wedge \Theta : \Theta > 0,\ \mathrm{dd^c}\Theta = 0,\ \int_X \omega \wedge \Theta = 1\right\},$$

tend to infinity as $\delta \downarrow 0$*.*

Proof For fixed δ, choose a finite constant $M_\delta > 0$ with $B_\delta \le M_\delta \omega$. This inequality proves finiteness. Suppose instead that $C_{\delta_l} \le C$ along a sequence tending to zero. Scaling the test forms and applying Theorem 4.3.1 to $C\omega - B_{\delta_l}$ gives normalized quasi-plurisubharmonic functions v_l such that

$$C\omega + \mathrm{dd^c} v_l \ge B_{\delta_l}, \qquad \sup_X v_l = 0.$$

By Theorem 1.4.1, a subsequence converges in L^1 to a nontrivial $C\omega$-plurisubharmonic function v. The approximate identity in the ζ variable gives

$$B_{\delta_l} \rightharpoonup \chi[\zeta = 0], \qquad C\omega + \mathrm{dd^c} v \ge \chi[\zeta = 0].$$

On a coordinate ball where $\chi = 1$, add a smooth strictly plurisubharmonic function whose Hessian dominates $C\omega$. The resulting positive closed current dominates $[\zeta = 0]$. Hence v has Lelong number at least one on an open subset of this hypersurface. Smooth correction does not change Lelong numbers. Siu's analyticity theorem from Section 1.13.1 implies that $Z = \{\nu(v, \cdot) \ge 1\}$ is an analytic subset of X. It is proper because v is not identically $-\infty$.

Its inverse image $\widetilde{Z}$ in Ω contains an open subset of $H_0 \cap \Omega$, where $H_0 = \{w_0 = z_0\}$. This surface is a projective plane with two points removed, so it is connected and irreducible. The identity principle gives $H_0 \cap \Omega \subset \widetilde{Z}$. Deck invariance then gives

$$H_k \cap \Omega \subset \widetilde{Z}, \qquad H_k = \{w_0 = q^k z_0\}, \qquad k \in \mathbb{Z}.$$

Use the chart $z_1 = 1$ near $[0 : 1 : 1 : 0]$. There H_k is the graph $z_0 = q^{-k} w_0$. For each fixed (w_0, w_1) in a small neighborhood of $(1, 0)$, every local defining holomorphic function of $\widetilde{Z}$ vanishes at these distinct values of z_0, which tend to zero. The one-variable identity theorem makes each defining function identically zero on the polydisk. Thus $\widetilde{Z}$ contains an open subset of the connected manifold Ω, contradicting properness. This excludes every bounded subsequence of C_δ as $\delta \downarrow 0$.□

Choose $\delta_j \downarrow 0$ and smooth positive $\mathrm{dd^c}$-closed forms Θ_j with

$$\int_X \omega \wedge \Theta_j = 1, \qquad J_j = \int_X B_{\delta_j} \wedge \Theta_j > j^4. \tag{25.20}$$

The lemma permits each δ_j to be smaller than half its predecessor, and the definition of the supremum supplies Θ_j. No compactness of this family is assumed.

25.15.2 Averaging the trace and compact curves

Averaging by local translations spreads the large trace over a small product region. A second average, this time of compact rational curves, supplies the two complementary matrix directions needed for a determinant lower bound. Fix j and write $\delta = \delta_j$.

For complex parameters a, b, τ, the automorphisms

$$T_{a,b,\tau}[z_0 : z_1 : w_0 : w_1] = [z_0 : z_1 + az_0 : \mathrm{e}^{\tau} w_0 : \mathrm{e}^{\tau}(w_1 + bw_0)]$$

commute with g and act in the product chart by $(x, y, \zeta) \mapsto (x + a, y + b, \zeta + \tau)$.

Average $T^*_{a,b,\tau}\Theta_j$ against the probability density

$$3^{-4} p(a/3)p(b/3)(3\delta)^{-2} p(\tau/(3\delta))\sigma_a \wedge \sigma_b \wedge \sigma_\tau$$

and denote the result by $\widehat{\Theta}_j$. It is smooth, positive and $\mathrm{dd^c}$-closed. The parameters range in a fixed compact set of automorphisms. Thus $(T^{-1}_{a,b,\tau})^*\omega \le C\omega$ with one constant, and change of variables gives

$$\int_X \omega \wedge \widehat{\Theta}_j \le C. \tag{25.21}$$

Let F_j be the coefficient of $\sigma_x \wedge \sigma_y$ in the restriction of Θ_j to a ζ slice, and let $\widehat{F}_j$ be the corresponding coefficient for $\widehat{\Theta}_j$. Positivity gives $F_j \ge 0$. The translations express $\widehat{F}_j$ as an average of F_j. At a point of

$$V_j = \{|x|, |y| < 1/4,\ |\zeta| < \delta/4\},$$

every translation to supp B_δ has $|a|, |b| \le 5/4$ and $|\tau| \le 5\delta/4$. The first two averaging factors are bounded below by a positive constant on this range, and the third is bounded below by a positive constant times δ^{-2}. Since $\chi \le 1$ and p is bounded, comparison of these integrands in (25.20) gives

$$\widehat{F}_j \ge cJ_j \quad \text{on } V_j, \tag{25.22}$$

with $c > 0$ independent of j.

The second average will occupy the other two cofactor directions. Put

$$f(a, b) = p(4(a - 1))p(b/2), \qquad w_j(\tau) = 4\delta^{-2} p(2\tau/\delta).$$

For parameters in these supports and $|u| \le 1/16$, consider holomorphic maps $\iota_{\tau,u,a,b} : \mathbb{P}^1 \to X$ defined by

$$\iota_{\tau,u,a,b}[z_0 : z_1] = [z_0 : z_1 : \mathrm{e}^{\tau}(z_0 + uz_1) : \mathrm{e}^{\tau}(az_1 + bz_0)] \in X. \tag{25.23}$$

For $\varepsilon > 0$, average their integration currents against

$$w_j(\tau) f(a, b)\varepsilon^{-2} p(u/\varepsilon)\sigma_\tau \wedge \sigma_a \wedge \sigma_b \wedge \sigma_u, \qquad 0 < \varepsilon \le 1/16,$$

and call the result $T_{j,\varepsilon}$. Define $T_{j,0}$ by setting $u = 0$ and omitting the last probability integral.

Lemma 25.15.2 *For $\varepsilon > 0$, the form $T_{j,\varepsilon}$ is globally smooth, semipositive and closed, and*

$$\int_X \omega \wedge T_{j,\varepsilon} \le C'$$

with C' independent of j, ε. For each fixed j, it converges smoothly to $T_{j,0}$ on compact subsets of the product chart as $\varepsilon \downarrow 0$. There

$$T_{j,0} = w_j(\zeta)\sigma_\zeta \wedge K_f, \qquad K_f = \int_{\mathbb{C}} f(a, y - ax)\frac{\mathrm{i}}{2\pi}(\mathrm{d}y - a\,\mathrm{d}x) \wedge (\mathrm{d}\bar{y} - \bar{a}\,\mathrm{d}\bar{x})\,\sigma_a. \tag{25.24}$$

The form K_f has a uniform positive lower bound on $|x|, |y| \le 1/4$.

Proof On the parameter supports, $|a - bu| \ge 3/4 - 2/16 = 5/8$. Hence the matrix $\left(\begin{smallmatrix} 1 & u \\ b & a \end{smallmatrix}\right)$ is uniformly invertible. The projective line in (25.23) lies in Ω. Distinct deck translates of this line are disjoint: equality at a point fixes $[z_0 : z_1]$, and invertibility of the matrix then forces the deck scalar to be one. Thus the map is an embedded compact rational curve in X.

The evaluation map from $\mathbf{P}^1$ times the open parameter domain to X is a holomorphic submersion. At $z_1 \ne 0$, variations of u and a span the two w coordinates after fixing a projective gauge; at $z_1 = 0$, variations of τ and b span them. The $\mathbf{P}^1$ coordinate supplies the remaining direction. For $\varepsilon > 0$, integration along the fibers of this submersion, with the smooth compactly supported parameter density, proves global smoothness. Averaging closed positive integration currents proves closedness and semipositivity.

All curve parameters belong to one compact subset of the invertible-matrix domain. Their ω areas consequently have a common bound by smooth dependence of the maps. The τ and u weights have mass one, and the a, b weight has fixed finite mass. This proves the degree bound.

When $u = 0$, the local equations are $\zeta = \tau$ and $y = ax + b$. Integrating in b and τ yields (25.24); the displayed $(1, 1)$-factor is the integration-current formula for the graph of $y = ax + b$. On $|x|, |y| \le 1/4$, the density $f(a, y - ax)$ is bounded below by a positive constant for a in a fixed sufficiently small disk about 1. The covectors $\mathrm{d}y - a\,\mathrm{d}x$, as a ranges in that disk, have no common nonzero annihilator. Integrating their squared values and using compactness of the horizontal unit sphere gives the asserted lower bound.

For general u, the local equations are

$$\zeta = \tau + \log(1 + ux), \qquad y = \frac{ax + b}{1 + ux}.$$

On any fixed compact subset, $1 + ux$ stays nonzero for all sufficiently small u. Solving for τ and b gives $\tau = \zeta - \log(1 + ux)$ and $b = (1 + ux)y - ax$. Regard w_j as a smooth function on E, so this change of variables also applies near the boundary

of its support. The Jacobians, density coefficients and every fixed number of their derivatives depend smoothly on u. Averaging against the approximate identity in u proves local C^∞ convergence. This convergence is needed for fixed j only; it is not asserted uniformly in j. □

25.15.3 The volume of the positive root

The two averages contribute complementary blocks of the cofactor matrix. A Schur-complement estimate turns the large vertical trace and the positive horizontal block into a lower bound for the root volume. In the coframe (x, y, ζ) put $\mathrm{d}V_0 = \sigma_x \wedge \sigma_y \wedge \sigma_\zeta$. For a real $(1,1)$-form ξ, let H_ξ be its coefficient matrix in this normalization. A real $(2,2)$-form Θ corresponds to a Hermitian matrix Q_Θ by

$$\xi \wedge \Theta = \operatorname{tr}(H_\xi Q_\Theta)\,\mathrm{d}V_0.$$

Strict positivity is equivalent to $Q_\Theta > 0$.

If $\eta^2 = \Theta$ and $H = H_\eta$, the wedge expansion gives

$$Q_\Theta = 2(\det H)H^{-1}, \qquad \eta^3 = \frac{3}{\sqrt{2}}(\det Q_\Theta)^{1/2}\,\mathrm{d}V_0.$$

Conversely, these formulas uniquely recover a smooth positive H from $Q_\Theta > 0$. This is the dimension-three root construction already used in Theorem 4.3.1. Write $\Theta^{3/2} = \eta^3$.

Lemma 25.15.3 *There are smooth strictly positive* $\mathrm{dd^c}$*-closed* $(2,2)$*-forms* Γ_j *such that*

$$\int_X \omega \wedge \Gamma_j \le C_2, \qquad R_j = \int_X \Gamma_j^{3/2} \longrightarrow +\infty.$$

Proof Write the cofactor matrix of $\widehat{\Theta}_j$ in block form, with a 2×2 Hermitian block A_j and a two-entry column p_j:

$$Q_j = \begin{pmatrix} A_j & p_j \\ p_j^* & \widehat{F}_j \end{pmatrix}.$$

The cofactor matrix of $T_{j,0}$ is $\operatorname{diag}(B_j, 0)$. Let I_2 denote the 2×2 identity matrix. The lower bound for K_f and the definition of w_j give a constant $c_0 > 0$ such that $B_j \ge c_0 \delta_j^{-2} I_2$ on V_j. The Schur complement gives

$$\det\bigl(Q_j + \operatorname{diag}(B_j, 0)\bigr) = \widehat{F}_j \det\left(B_j + A_j - \frac{p_j p_j^*}{\widehat{F}_j}\right) \ge \widehat{F}_j \det B_j.$$

Here $A_j - p_j p_j^*/\widehat{F}_j$ is positive definite because $Q_j > 0$; monotonicity of determinant on positive Hermitian matrices proves the inequality. Combining the root formula

with (25.22) shows that

$$(\widehat{\Theta}_j + T_{j,0})^{3/2} \geq c\delta_j^{-2}\sqrt{J_j}\,\mathrm{d}V_0 \quad \text{on } V_j.$$

The smaller region $W_j = \{|x|, |y| < 1/8,\ |\zeta| < \delta_j/8\}$ has $\mathrm{d}V_0$ volume equal to a fixed positive constant times δ_j^2. Consequently

$$I_j = \int_{W_j} (\widehat{\Theta}_j + T_{j,0})^{3/2} \geq c_1\sqrt{J_j} > c_1 j^2.$$

For this fixed j, the preceding lemma gives smooth convergence on the closure of W_j. The matrix of $\widehat{\Theta}_j + T_{j,0}$ is strictly positive there, so the positive-root volume depends continuously on these smooth coefficients. Choose $0 < \varepsilon_j \leq 1/16$ sufficiently small that

$$\int_{W_j} (\widehat{\Theta}_j + T_{j,\varepsilon_j})^{3/2} \geq I_j/2.$$

Set $\Gamma_j = \widehat{\Theta}_j + T_{j,\varepsilon_j}$. It is smooth, strictly positive and $\mathrm{dd^c}$-closed. Its degree is bounded by (25.21) and Lemma 25.15.2, whereas its root volume is at least $c_1 j^2/2$. No uniform choice of ε_j was used. □

Proof (of Theorem 25.15.1) Fix any Hermitian form h on X, and choose $A > 0$ with $h \leq A\omega$. Let $\eta_j^2 = \Gamma_j$. These metrics are Gauduchon, and

$$P_j = \int_X h \wedge \eta_j^2 \leq AC_2, \qquad R_j = \int_X \eta_j^3 \longrightarrow +\infty.$$

Set $\mu_j = \eta_j^3/R_j$ and solve the smooth equation in background h:

$$h_j = h + \mathrm{dd^c}u_j > 0, \qquad h_j^3 = c_h(\mu_j)\mu_j, \qquad \sup_X u_j = 0.$$

Since $D_{\mu_j}(\eta_j) = R_j^{2/3}$, the test η_j in Theorem 25.1.1 gives

$$0 < \int_X h_j^3 = c_h(\mu_j) = \mathcal{V}_{\mu_j}(h) \leq \frac{P_j^3}{R_j^2} \longrightarrow 0.$$

The potentials u_j are smooth and hence bounded, proving $v_-(h) = 0$. □

25.15.4 Consequences for dimension and equations

The example occurs in the smallest possible dimension. On a surface, comparison with a Gauduchon metric gives positive lower mass. Products with projective spaces

carry the threefold degeneration to every higher dimension; a product with the Hopf threefold also permits unbounded upper mass.

Corollary 25.15.1 *Every Hermitian form on a compact complex manifold of dimension at most two has positive lower mass. In every dimension $n \geq 3$ there is a connected compact complex manifold on which every Hermitian form has zero lower mass.*

Proof In dimension one the mass is $\int_X h > 0$; the surface case is Corollary 25.14.1. We use the monotonicity and degree-n homogeneity of v_-, which follow directly from (25.15), to pass from a split metric to every Hermitian metric in higher dimension.

For $n > 3$, take the product of the preceding threefold with $\mathbf{P}^{n-3}$ and let κ be a Fubini–Study form on that projective factor. Use the split metric $h_0 = \omega + \kappa$, with pullbacks understood. For the potentials supplied by the theorem,

$$\int_{X\times\mathbf{P}^{n-3}} (h_0 + \mathrm{dd^c} u_j)^n = \binom{n}{3}\left(\int_X (\omega + \mathrm{dd^c} u_j)^3\right)\left(\int_{\mathbf{P}^{n-3}} \kappa^{n-3}\right) \longrightarrow 0.$$

For another Hermitian h, choose A with $h \leq Ah_0$. Monotonicity and homogeneity give $v_-(h) \leq A^n v_-(h_0) = 0$. □

Corollary 25.15.2 *In every dimension $n \geq 6$ there is a connected compact complex manifold with*

$$v_-(h) = 0, \qquad v_+(h) = +\infty$$

for every Hermitian form h.

Proof Take the product of the threefold above with the Hopf threefold of Chapter 8. For a split metric, potentials pulled back from the first factor make the masses tend to zero, while smooth potentials pulled back from the second make them tend to infinity. In each case the product integral is $\binom{6}{3}$ times the product of the two factor masses. The preceding metric comparison handles the lower mass, and metric independence in Proposition 6.5.1 handles the upper mass. Products with projective spaces give every higher dimension by the same wedge expansion. □

Finally fix a smooth positive probability measure μ on the threefold. The smooth equation still gives $\mathcal{V}_\mu(h) > 0$ for each Hermitian h. With $\mathscr{M}$ and λ_* as in Section 25.12.1, the mass-extrema theorem says

$$\inf_{\nu\in\mathscr{M}} \mathcal{V}_\nu(h) = 0, \qquad \lambda_*(h) = 0.$$

Thus even a smooth strictly positive background can have vanishing density infimum. For the metrics η_j and numbers R_j, P_j in the proof of Theorem 25.15.1, put $\mu_j = \eta_j^3/R_j$. Its equation constants satisfy $\mathcal{V}_{\mu_j}(h) \leq P_j^3/R_j^2 \to 0$. The densities μ_j/μ cannot have a uniform upper bound: such a bound $\mu_j \leq M\mu$ would give $\mathcal{V}_{\mu_j}(h) \geq M^{-1}\mathcal{V}_\mu(h) > 0$ by the denominator comparison in Proposition 25.12.1.

The example also makes the endpoint in Theorem 25.8.2 fail for the full family of normalized Gauduchon currents: its supremum of $L^{3/2}$ determinant-density norms is infinite, although every subcritical L^q norm, $1 < q < 3/2$, remains uniformly bounded by Theorem 25.8.1. An endpoint estimate restricted to the maximizing currents that represent adjoint measures remains a separate question.

25.15.5 A balanced metric and a finite upper mass

The quotient threefold has a balanced metric with finite upper mass, despite the vanishing of its lower mass. A direct calculation expresses the loss of mass as vertical Dirichlet energy along the elliptic fibers. It also shows why potentials separated into a periodic radial part and a base part cannot realize the degeneration. Put $D = \mathrm{d}t \wedge \mathrm{d}^{\mathrm{c}}t$, so that the metric already constructed is $\omega = \alpha + \beta + D$.

Proposition 25.15.1 *On the threefold of Theorem 25.15.1, the Hermitian form $\omega = \alpha + \beta + D$ satisfies*

$$\mathrm{d}(\omega^2) = 0, \qquad \mathrm{dd}^{\mathrm{c}}\omega = 2\alpha \wedge \beta \geq 0, \qquad \int_X \omega^3 = 24\pi.$$

For every smooth real function u,

$$\int_X (\omega + \mathrm{dd}^{\mathrm{c}}u)^3 = 24\pi - 6\int_X \mathrm{d}u \wedge \mathrm{d}^{\mathrm{c}}u \wedge \alpha \wedge \beta. \tag{25.25}$$

Consequently $v_+(\omega) = 24\pi$. In particular X has the bounded mass property, while $v_-(h) = 0$ for every Hermitian form h on X.

Proof Although t is defined on the covering space, both $\mathrm{d}t$ and $\mathrm{d}^{\mathrm{c}}t$ descend. The Fubini–Study potentials give

$$B := \mathrm{dd}^{\mathrm{c}}t = \beta - \alpha.$$

The forms α, β are closed, and $\alpha^2 = \beta^2 = 0$, since each is pulled back from a curve. As $D^2 = 0$, we obtain

$$\mathrm{d}D = -\mathrm{d}t \wedge B, \qquad \mathrm{dd}^{\mathrm{c}}D = -B^2 = 2\alpha \wedge \beta,$$

$$\mathrm{d}(\omega^2) = -2(\alpha + \beta + D) \wedge \mathrm{d}t \wedge (\beta - \alpha) = 0.$$

In the last identity, the term involving D has a repeated $\mathrm{d}t$, and $(\alpha + \beta) \wedge (\beta - \alpha) = 0$. Thus ω is balanced, hence Gauduchon, and the formula for $\mathrm{dd}^{\mathrm{c}}\omega$ follows.

On a fiber use $\zeta = a + \mathrm{i}b$ with both real periods 2π. There $t = 2a+$ a constant, so

$$D|_E = \frac{1}{\pi}\,\mathrm{d}a \wedge \mathrm{d}b, \qquad \int_E D = 4\pi.$$

The equality $\omega^3 = 6\alpha \wedge \beta \wedge D$, fiber integration and the integral-one normalization of each Fubini–Study form give $\int_X \omega^3 = 24\pi$.

Apply the Gauduchon threefold identity (25.16) with $\mathrm{dd^c}\omega = 2\alpha \wedge \beta$ to obtain (25.25). Its last integral is nonnegative. Thus every smooth strictly ω-plurisubharmonic mass is at most 24π, and the constant potential attains this value. Bounded exact-background regularization, as in Theorem 25.12.1, identifies the same supremum over bounded potentials. Metric independence in Proposition 6.5.1 proves bounded mass on X. The assertion about the lower mass is Theorem 25.15.1. □

For an admissible smooth u, equality in the upper-mass bound holds precisely when u is constant along every elliptic fiber. Indeed, $\alpha \wedge \beta$ is a positive volume form in the base directions, so the nonnegative integral in (25.25) vanishes exactly when the fiber derivative of u vanishes. The fibers are connected, and smooth local sections then show that u is the pullback of a smooth function on $\mathbf{P}^1 \times \mathbf{P}^1$. For the sequence with masses tending to zero, that same integral instead tends to 4π.

Even after adding an arbitrary base potential, a periodic function of t has a positive mass gap. The sharp gap follows from the bounds on its first two derivatives imposed by positivity.

Proposition 25.15.2 *Let $\pi : X \to B = \mathbf{P}^1 \times \mathbf{P}^1$ be the elliptic fibration above, and put $L = 4\pi$. Let $f : \mathbb{R} \to \mathbb{R}$ be smooth and L-periodic. Among smooth real potentials of the form*

$$u = f(t) + \pi^* v, \qquad f(s + L) = f(s), \qquad v \in C^\infty(B, \mathbb{R}), \qquad \omega + \mathrm{dd^c} u > 0,$$

the infimum of $\int_X (\omega + \mathrm{dd^c} u)^3$ is 8. Every such mass is strictly greater than 8, and this infimum is approached already with $v = 0$.

Proof Write $p = f'$. It is L-periodic and has mean zero. The chain rule gives

$$\omega + \mathrm{dd^c} u = (1 - p)\alpha + (1 + p)\beta + \pi^* \mathrm{dd^c} v + (1 + p')D.$$

Positivity on the vertical line implies $p' > -1$. On the complex plane $\ker \partial t$, the projection to the base is an isomorphism and D vanishes. For each fixed value of t modulo L, every base point has a point above it with that value. Thus, writing α_B, β_B for the two Fubini–Study forms on B, the form $(1 - p)\alpha_B + (1 + p)\beta_B + \mathrm{dd^c} v$ is positive on B for each value of p. Integration over the two rulings gives $1 - p > 0$ and $1 + p > 0$. Thus $|p| < 1$ and $p' > -1$; when $v = 0$, these inequalities also suffice for strict positivity.

In a logarithmic fiber coordinate $x + \mathrm{i}y$, both real periods are 2π, while $t = 2x +$ a constant and $D|_E = \pi^{-1} \mathrm{d}x \wedge \mathrm{d}y$. Hence $\int_E F(t)D = \int_0^L F(s)\,\mathrm{d}s$ for every L-periodic F. Only the vertical derivative of u contributes to (25.25). Fiber integration therefore gives

$$\int_X (\omega + \mathrm{dd^c} u)^3 = 6\left(L - \int_0^L p(s)^2\,\mathrm{d}s\right). \tag{25.26}$$

Suppose first that $p \not\equiv 0$, and put $A = \max p > 0$, $-B_0 = \min p < 0$ and $r = A + B_0 < 2$. The function $G(s) = (A - s)(s + B_0)$ is nonnegative on $[-B_0, A]$.

Choose a forward interval I of length at most L from a maximum of p to a subsequent minimum. Since $p' \geq -1$, we have $G(p) \geq -G(p)p'$. Using the zero mean of p, we obtain

$$\begin{aligned} AB_0L - \int_0^L p^2 \,\mathrm{d}s = \int_0^L G(p)\,\mathrm{d}s &\geq -\int_I G(p)p' \,\mathrm{d}s \\ &= \int_{-B_0}^A G(s)\,\mathrm{d}s = \frac{r^3}{6}. \end{aligned}$$

As $AB_0 \leq r^2/4$, it follows that

$$\int_0^L p^2 \,\mathrm{d}s \leq \frac{Lr^2}{4} - \frac{r^3}{6} < L - \frac{4}{3}.$$

The strict inequality follows because the middle expression is strictly increasing for $0 < r \leq 2$, as $L > 2$. It also holds when $p = 0$. Equation (25.26) now gives the strict lower bound 8.

To prove sharpness, put $a = (L-2)/2$ and define an L-periodic function p_0 by

$$p_0(s) = \begin{cases} 1, & 0 \leq s < a, \\ 1 - (s-a), & a \leq s < a+2, \\ -1, & a+2 \leq s < L. \end{cases}$$

It has mean zero, $|p_0| \leq 1$ and $\int_0^L p_0^2 \,\mathrm{d}s = L - 4/3$. Its distributional derivative is -1 on the descending interval, zero on the two constant intervals, and has a positive atom at the period join. Thus $p_0' \geq -1$ as distributions on the circle. Let ρ_ε be a nonnegative smooth periodic approximate identity of integral one, and set $p_\varepsilon = (1-\varepsilon)(p_0 * \rho_\varepsilon)$ for $0 < \varepsilon < 1$. Then p_ε is smooth, has mean zero, and

$$|p_\varepsilon| \leq 1 - \varepsilon, \qquad p_\varepsilon' \geq -1 + \varepsilon, \qquad p_\varepsilon \longrightarrow p_0 \quad \text{in } L^2.$$

Its primitive f_ε is smooth and periodic, so $u_\varepsilon = f_\varepsilon(t)$ is admissible. Substitution in (25.26) proves that its mass tends to 8. □

Proposition 25.15.3 *The threefold X admits neither a pluriclosed Hermitian metric nor a closed Kähler current.*

Proof The nonzero smooth semipositive $(2,2)$-form $\alpha \wedge \beta$ is $\mathrm{dd^c}$-exact by the preceding calculation. If a Hermitian metric η satisfied $\mathrm{dd^c}\eta = 0$, Stokes would give

$$0 < \int_X \eta \wedge \alpha \wedge \beta = \frac{1}{2} \int_X \eta \wedge \mathrm{dd^c}\omega = \frac{1}{2} \int_X \omega \wedge \mathrm{dd^c}\eta = 0,$$

a contradiction. The strict inequality holds because η is positive in the vertical direction at every point.

If a closed positive current T dominated $\varepsilon\omega$ for some $\varepsilon > 0$, positivity and current Stokes would likewise give

$$0 < \varepsilon \int_X \omega \wedge \alpha \wedge \beta \leq \int_X T \wedge \alpha \wedge \beta = \frac{1}{2} \int_X \omega \wedge \mathrm{dd^c}T = 0.$$

This excludes a closed Kähler current. □

The example is therefore outside the Fujiki class, by the characterization recalled in Section 32.3. It shows that a balanced metric does not guarantee positive lower mass, even together with bounded upper mass. In contrast, the Hopf threefold has positive lower mass and infinite upper mass. The two properties are logically independent. Their simultaneous failure on the products in Corollary 25.15.2 is a third regime. In particular, bounded mass alone cannot replace the positive lower-mass hypothesis in a theorem asserting solvability for every non-pluripolar measure.

25.16 Orders of degeneration at the nef boundary

A closed nef class on a Kähler manifold degenerates with an integral order determined by its intersection polynomial. For a nonclosed nef background, the logarithmic rate can be fractional or fail to converge. Lower and upper orders retain this information while remaining independent of the probability measure and perturbing metric: changing either only introduces fixed comparison factors.

Return to a connected compact complex manifold X of dimension $n \geq 1$, with a smooth strictly positive probability form μ. Let θ be nef and let h be Hermitian. For $0 < \epsilon < 1$, put

$$a_{\mu,h}(\theta;\epsilon) = \frac{\log \mathcal{V}_\mu(\theta + \epsilon h)}{\log \epsilon}.$$

The logarithm is finite because $\theta + \epsilon h$ belongs to $\mathscr{P}$.

Proposition 25.16.1 *The limits*

$$\underline{a}(\theta) = \liminf_{\epsilon \downarrow 0} a_{\mu,h}(\theta;\epsilon), \qquad \overline{a}(\theta) = \limsup_{\epsilon \downarrow 0} a_{\mu,h}(\theta;\epsilon)$$

are independent of μ and h, and satisfy

$$0 \le \underline{a}(\theta) \le \overline{a}(\theta) \le n.$$

They are unchanged by replacing θ with $\theta + \mathrm{dd^c} f$, for f smooth, or with $r\theta$, for $r > 0$. If $\mathcal{V}_\mu(\theta) > 0$, both orders are zero.

Proof Pseudoeffectivity of θ and monotonicity give

$$\epsilon\lambda_\mu(h) \le \lambda_\mu(\theta + \epsilon h) \le \lambda_\mu(\theta + h), \qquad 0 < \epsilon \le 1.$$

Thus, for constants $c, C > 0$,

$$c\epsilon^n \le \mathcal{V}_\mu(\theta + \epsilon h) \le C.$$

Taking logarithms and dividing by the negative number $\log \epsilon$ proves the asserted bounds for the lower and upper limits. It also shows that the quotients stay bounded as $\epsilon \downarrow 0$.

Changing the smooth probability measure multiplies the volume by factors between two fixed positive constants, by Proposition 25.12.1. The resulting logarithmic error, divided by $\log \epsilon$, tends to zero.

For another Hermitian metric h', choose $b \ge a > 0$ with $ah \le h' \le bh$. Then

$$\mathcal{V}_\mu(\theta + a\epsilon h) \le \mathcal{V}_\mu(\theta + \epsilon h') \le \mathcal{V}_\mu(\theta + b\epsilon h).$$

For any fixed $t > 0$,

$$\frac{\log \mathcal{V}_\mu(\theta + t\epsilon h)}{\log \epsilon} = a_{\mu,h}(\theta; t\epsilon)\frac{\log(t\epsilon)}{\log \epsilon}.$$

The last factor tends to one and the first factor is bounded. Its lower and upper limits are unchanged by the rescaling $\epsilon \mapsto t\epsilon$. Taking lower and upper limits in the preceding inequalities, after reversing them upon division by $\log \epsilon < 0$, proves metric independence.

Smooth gauge invariance follows from that of $\mathcal{V}_\mu$. Homogeneity gives

$$\mathcal{V}_\mu(r\theta + \epsilon h) = r^n \mathcal{V}_\mu(\theta + (\epsilon/r)h).$$

The constant logarithmic error and the same rescaling argument prove scale invariance. Finally, if $\mathcal{V}_\mu(\theta) > 0$, nef continuity bounds the perturbed volumes above and below by positive constants for small ϵ. Both quotients then tend to zero. □

The two orders can differ, and a common order can be irrational. Even when both are zero, the volume can tend to zero logarithmically. The elliptic examples in

Section 25.16.1 will exhibit each behavior. Closed nef classes on Kähler manifolds have a more rigid polynomial degeneration.

Proposition 25.16.2 *Suppose X is Kähler, θ is closed and nef, and h is Kähler. Set*

$$r = \max\left\{0 \le k \le n : \int_X \theta^k \wedge h^{n-k} > 0\right\}.$$

Then

$$\underline{a}(\theta) = \overline{a}(\theta) = n - r.$$

Writing a for their common value, $n - a$ recovers the usual numerical dimension of a closed nef class in this setting.

Proof All the displayed intersection numbers are nonnegative. Indeed, choose smooth positive representatives of $\theta + \delta h$. Their mixed integrals against h are nonnegative, agree with the cohomological integrals by closed Stokes, and converge to the stated numbers as $\delta \downarrow 0$. The coefficient for $k = 0$ is positive, so r exists.

By the closed nef formula already proved,

$$\mathcal{V}_\mu(\theta + \epsilon h) = \int_X (\theta + \epsilon h)^n = \sum_{k=0}^{n} \binom{n}{k} \epsilon^{n-k} \int_X \theta^k \wedge h^{n-k}.$$

The leading nonzero term as $\epsilon \downarrow 0$ is a positive constant times ϵ^{n-r}. Taking the logarithmic quotient proves the assertion. Independence from the Kähler test metric also follows from Proposition 25.16.1. □

The construction can also be evaluated on the non-Kähler Hopf threefold from Chapter 8. Keep its forms $\omega = \alpha + \beta$, where $\alpha \ge 0$ is closed of rank two, β has rank one, $\alpha^3 = \beta^2 = 0$, and $\int_X \omega^3 = 6$. Take $\theta = \alpha$, $h = \omega$, and $\mu_0 = \omega^3/6$. The metric

$$\Omega_\epsilon = \alpha + \epsilon\omega = (1 + \epsilon)\alpha + \epsilon\beta$$

is strictly positive. Its volume is

$$\Omega_\epsilon^3 = \epsilon(1 + \epsilon)^2\omega^3 = 6\epsilon(1 + \epsilon)^2\mu_0.$$

The zero potential therefore already solves the smooth normalized equation, giving

$$\mathcal{V}_{\mu_0}(\alpha + \epsilon\omega) = 6\epsilon(1 + \epsilon)^2, \qquad \underline{a}(\alpha) = \overline{a}(\alpha) = 1.$$

The same orders hold for every smooth positive probability measure and every Hermitian perturbation direction.

Each perturbed metric nevertheless has infinite upper mass. If V_j are the smooth potentials of the Hopf counterexample, then

$$\Omega_\epsilon + \mathrm{dd^c}(\epsilon V_j) = \alpha + \epsilon(\omega + \mathrm{dd^c} V_j) > 0,$$

and monotonicity of the determinant of positive matrices gives

$$\int_X \big(\Omega_\epsilon + \mathrm{dd^c}(\epsilon V_j)\big)^3 \geq \epsilon^3 \int_X (\omega + \mathrm{dd^c} V_j)^3 \longrightarrow +\infty.$$

Thus the finite boundary order coexists with failure of the bounded mass property at every positive perturbation.

25.16.1 Elliptic surface volumes and order spectra

A coefficient that vanishes along one elliptic factor determines the harmonic mean in the volume formula. Finite-order zeros give fractional or logarithmic decay; flat coefficients allow the logarithmic slope to oscillate.

Let $E = \mathbb{C}/(\mathbb{Z} + \mathrm{i}\mathbb{Z})$, write $X = E_z \times E_w$, and fix

$$\omega_1 = \frac{\mathrm{i}}{2}\mathrm{d}z \wedge \mathrm{d}\bar{z}, \qquad \omega_2 = \frac{\mathrm{i}}{2}\mathrm{d}w \wedge \mathrm{d}\bar{w}, \qquad \mu = \omega_1 \wedge \omega_2, \qquad h = \omega_1 + \omega_2.$$

Both area forms have integral one. Let a, q, r be smooth functions of z alone, with a, q real and nonnegative, r complex-valued, and $|r|^2 \leq aq$. Set

$$\theta = \frac{\mathrm{i}}{2}\big(a\,\mathrm{d}z \wedge \mathrm{d}\bar{z} + r\,\mathrm{d}z \wedge \mathrm{d}\bar{w} + \bar{r}\,\mathrm{d}w \wedge \mathrm{d}\bar{z} + q\,\mathrm{d}w \wedge \mathrm{d}\bar{w}\big).$$

This is a smooth semipositive, hence nef, form.

Proposition 25.16.3 *For $\epsilon > 0$, put $g = q + \epsilon$ and $I = \int_E g^{-1}\omega_1$. Then*

$$\mathcal{V}_\mu(\theta + \epsilon h) = \frac{2}{I}\left(\int_E (a + \epsilon)\omega_1 - \int_E \frac{|r|^2}{g}\omega_1\right). \tag{25.27}$$

The infimum defining this volume is attained by a smooth positive Gauduchon metric. In particular, if $a \equiv b$ for a constant $b > 0$ and $r = 0$, then $\mathcal{V}_\mu(\theta + \epsilon h) = 2(b + \epsilon)/I$. If $|r|^2 = aq$, then

$$\mathcal{V}_\mu(\theta + \epsilon h) = \frac{2\epsilon(1 + K)}{I}, \qquad K = \int_E \frac{a}{q + \epsilon}\omega_1. \tag{25.28}$$

Proof Define

$$c = I^{-1}\left(\int_E (a + \epsilon)\omega_1 - \int_E \frac{|r|^2}{g}\omega_1\right), \qquad A = \frac{c + |r|^2}{g}.$$

The inequality $(a + \epsilon)g - |r|^2 > 0$ gives $c > 0$, while the definition of c makes $A - a - \epsilon$ have zero ω_1-mean. Division of its nonconstant Fourier coefficients by the nonzero flat Laplace eigenvalues produces a smooth real function u on E with

$\mathrm{dd^c}u = (A - a - \epsilon)\omega_1$. Pull it back to X. The form $\Omega = \theta + \epsilon h + \mathrm{dd^c}u$ has coefficient matrix $\left(\begin{smallmatrix} A & r \\ \bar{r} & g \end{smallmatrix}\right)$, determinant c, and $\Omega^2 = 2c\mu$.

The metric $\gamma_* = \Omega/g$ has constant fiber coefficient 1; its remaining coefficients depend only on z. Applying $\mathrm{dd^c}$ to its base or mixed terms repeats $\mathrm{d}z$ or $\mathrm{d}\bar{z}$, so $\mathrm{dd^c}\gamma_* = 0$. It is smooth, positive, and pointwise conformal to Ω. The arithmetic–geometric mean argument in Theorem 25.1.1 and equality for γ_* prove (25.27). The two specializations follow by direct substitution. □

Proposition 25.16.4 *There are smooth semipositive nonclosed forms on the fixed surface X with common order $1 - 1/m$ for each integer $m > 1$. There is also one with*

$$\mathcal{V}_\mu(\theta + \epsilon h) \sim \frac{2b}{\pi \log(1/\epsilon)} \quad (\epsilon \downarrow 0),$$

for any chosen $b > 0$; it has boundary volume zero and both orders zero.

Proof In Proposition 25.16.3 take $a = b > 0$ and $r = 0$. Choose q smooth, positive outside $z_0 \in E$, and equal to $|\zeta|^{2m}$ in a small flat coordinate disk about z_0. A positive smooth extension exists away from the disk. For $m = 1$, polar coordinates give

$$I = \int_E \frac{\omega_1}{q + \epsilon} = \pi \log(1/\epsilon) + O(1).$$

Equation (25.27) gives the displayed asymptotic. If $m > 1$, the change $s = |\zeta|^2 = \epsilon^{1/m} t$ gives

$$I \sim C_m \epsilon^{1/m - 1}, \qquad C_m = \pi \int_0^\infty \frac{\mathrm{d}t}{1 + t^m} > 0,$$

and the common order is $1 - 1/m$. In the logarithmic case $\lambda_\mu(\theta) \geq 0$ by semipositivity, while $\lambda_\mu(\theta) \leq \lambda_\mu(\theta + \epsilon h) \to 0$ by monotonicity. Hence $\mathcal{V}_\mu(\theta) = 0$. The quotient of the displayed volume by ϵ^δ diverges for every $\delta > 0$. Since $\mathrm{d}\theta = \mathrm{d}q \wedge \omega_2 \neq 0$, these backgrounds are nonclosed. □

Proposition 25.16.5 *For every $0 < \alpha < \beta < 1$ there is a smooth semipositive nonclosed background on the same (X, μ, h) with*

$$\big(\underline{a}(\theta), \overline{a}(\theta)\big) = (1 + \alpha, 1 + \beta).$$

For every $0 < \sigma < 1$ there is such a background with common order $1 + \sigma$.

Proof For all sufficiently large t, define

$$p(t) = \frac{\alpha + \beta}{2} + \frac{\beta - \alpha}{2} \sin(\log \log t), \qquad P(t) = t p(t).$$

Choose T large enough that $0 < c_0 \leq P'(t) \leq d_0 < 1$ for $t \geq T$; this follows from

$$P'(t) = p(t) + \frac{\beta - \alpha}{2 \log t} \cos(\log \log t).$$

On $E \setminus \{z_0\}$ choose a smooth $t(z) \geq T$ equal to $|\zeta|^{-2}$ near z_0 and bounded outside every neighborhood of z_0. Put

$$q = \mathrm{e}^{-t}, \qquad a = \mathrm{e}^{-P(t)}, \qquad r = \mathrm{e}^{-(P(t)+t)/2},$$

and set all three equal to zero at z_0. These functions are smooth and flat there: $P(t) \geq \alpha t$ and every fixed derivative of P grows at most polynomially in t, so the chain rule bounds each coordinate derivative by a polynomial in $|\zeta|^{-1}$ times $\mathrm{e}^{-c/|\zeta|^2}$. We have $|r|^2 = aq$, so the resulting θ is semipositive. It is nonclosed: the component $\mathrm{d}q \wedge \omega_2$ of $\mathrm{d}\theta$ has two fiber differentials, which derivatives of the remaining terms cannot cancel.

Write $\epsilon = \mathrm{e}^{-L}$. Choose a fixed integration endpoint $T_0 \geq T$ inside the coordinate disk, with the transition to the bounded global $t(z)$ outside a smaller disk. The complementary contributions to I and K in (25.28) are $O(1)$, while polar coordinates on the disk give

$$I_{\mathrm{loc}} = \pi \int_{T_0}^{\infty} \frac{t^{-2}\,\mathrm{d}t}{\mathrm{e}^{-t} + \mathrm{e}^{-L}}, \qquad K_{\mathrm{loc}} = \pi \int_{T_0}^{\infty} \frac{\mathrm{e}^{-P(t)} t^{-2}\,\mathrm{d}t}{\mathrm{e}^{-t} + \mathrm{e}^{-L}}.$$

Integrating I over $t \geq L$ gives a lower bound by a fixed multiple of e^L/L. For the upper bound, use $(\mathrm{e}^{-t} + \mathrm{e}^{-L})^{-1} \leq \mathrm{e}^t$ for $t \leq L$ and at most e^L for $t \geq L$, then split $[T_0, L]$ at $L/2$. Thus

$$\log I = L - \log L + O(1). \tag{25.29}$$

The derivatives of P and $t - P(t)$ are positive and bounded away from zero. Splitting the K-integral at L therefore gives $K \leq C\mathrm{e}^{L-P(L)}$. On $[L, L+1]$ its denominator is at most $2\mathrm{e}^{-L}$ and $P(t) \leq P(L) + 1$, so

$$c_1 L^{-2} \mathrm{e}^{L-P(L)} \leq K \leq c_2 \mathrm{e}^{L-P(L)}.$$

The complementary $O(1)$ term is absorbed because $P(L) \leq \beta L$. Consequently $\log K = L - P(L) + O(\log L)$ and $K \to \infty$. Combining this with (25.28) and (25.29) yields

$$\log \mathcal{V}_\mu(\theta + \mathrm{e}^{-L} h) = -L - P(L) + O(\log L), \qquad \frac{\log \mathcal{V}_\mu(\theta + \epsilon h)}{\log \epsilon} = 1 + p(L) + o(1). \tag{25.30}$$

The sine in $p(L)$ has limits -1 and 1 along sequences, proving the first assertion. If instead $P(t) = \sigma t$, the same smoothness and integral estimates give common order $1 + \sigma$. In particular $\sigma = \sqrt{2} - 1$ gives an irrational common order. □

25.16.2 Comparison under an arbitrary modification

A pulled-back Gauduchon test becomes degenerate along the exceptional locus. Adding a small positive test repairs this degeneracy. To compare the perturbed

volumes uniformly, we first bound the upstairs metric pairing by the downstairs one over the entire test cone.

Here $\pi : Y \to X$ is a modification of connected compact complex n-folds, with $n \geq 2$. We use Hironaka's analytic elimination of indeterminacies in the following form: the map π is dominated by a composition of smooth-center blowups $p : Z \to X$, with a modification $F : Z \to Y$ satisfying $p = \pi \circ F$. It follows by resolving the inverse meromorphic map; the maps agree on the isomorphism locus and hence everywhere. This classical input is discussed in Section 33.13.

Lemma 25.16.1 *Let $p : Z \to X$ be a finite composition of blowups along smooth centers of codimension at least two, and let h_X be a Hermitian form. There are $C_0 > 0$, an effective p-exceptional real divisor D, and a smooth curvature representative η_D such that*

$$h_Z = C_0 p^* h_X - \eta_D > 0. \tag{25.31}$$

Every component of D maps into a set of complex codimension at least two in X.

Proof First let $q : M' \to M$ be one blowup, with exceptional divisor E. On E, the restriction of $\mathcal{O}(-E)$ is relatively positive on the projective fibers. A metric on the normal bundle of the center induces a Hermitian metric on this restriction whose curvature is positive on their tangent spaces. Extend the metric to $\mathcal{O}(-E)$ over M' by extending the smooth logarithm of its ratio to a fixed reference metric from the closed submanifold E. Let β be its curvature. The semipositive form $q^* h_M$ has, on E, kernel precisely the tangent space to the fiber, and β is positive on that kernel; off E, $q^* h_M$ is positive. Compactness now gives $C q^* h_M + \beta > 0$ for a sufficiently large C. Indeed a contrary sequence of unit tangent vectors, with $C \to \infty$, would converge to a vector in the kernel of $q^* h_M$ on which β is nonpositive. With $\eta_E = -\beta$, this is $C q^* h_M - \eta_E > 0$. Iterating the construction, pulling back preceding exceptional divisors with positive coefficients and adding the new exceptional divisor, proves (25.31) while retaining effectivity and exceptionality. □

Lemma 25.16.2 *Fix Hermitian forms h_X, h_Y for a modification $\pi : Y \to X$. There is $K > 0$ such that*

$$\int_Y h_Y \wedge \pi^* \Psi \leq K \int_X h_X \wedge \Psi \tag{25.32}$$

for every smooth strictly positive $(n-1, n-1)$-form Ψ on X with $\mathrm{dd^c}\Psi = 0$. The constant is independent of Ψ.

Proof Use the analytic domination $p = \pi \circ F$ above and Lemma 25.16.1. Compactness gives $a > 0$ with $h_Z \geq a F^* h_Y$. Pulling back the semipositive form Ψ and integrating yields

$$a \int_Y h_Y \wedge \pi^* \Psi \leq \int_Z h_Z \wedge p^* \Psi = C_0 \int_X h_X \wedge \Psi - \int_D p^* \Psi.$$

The change-of-variables equalities use the degree-one maps F, p, and Poincaré–Lelong converts the curvature pairing to the divisor pairing under $\mathrm{dd^c} p^* \Psi = 0$. The last integral vanishes: on the regular locus of each component of D, the restriction of

p has rank at most $n-2$, so the restriction of the $(n-1,n-1)$-form $p^*\Psi$ is zero. Taking $K = C_0/a$ proves the assertion. □

Proposition 25.16.6 *Let $\pi : Y \to X$ be a modification of connected compact complex manifolds of dimension $n \geq 2$, and let θ be a smooth nef background on X. Its pullback is nef. For fixed Hermitian forms h_X, h_Y and smooth positive probability forms μ_X, μ_Y, there are $C, K > 0$, independent of ϵ, such that*

$$\mathcal{V}_{\mu_Y}(\pi^*\theta + \epsilon h_Y) \leq C\mathcal{V}_{\mu_X}(\theta + K\epsilon h_X) \qquad (\epsilon > 0). \tag{25.33}$$

Consequently

$$\underline{a}(\pi^*\theta) \geq \underline{a}(\theta), \qquad \overline{a}(\pi^*\theta) \geq \overline{a}(\theta).$$

In dimension one a modification is an isomorphism and preserves both orders.

Proof Pullback preserves nefness: pull back a smooth approximation $\theta + \mathrm{dd^c}u_t \geq -th_X$, compare $\pi^*h_X \leq C'h_Y$, and choose $t < \epsilon/C'$ to obtain a positive representative of $\pi^*\theta + \epsilon h_Y$. Thus both volumes in (25.33) are positive.

Choose $c > 0$ with $\mu_Y \geq c\pi^*\mu_X$: the ratio $\pi^*\mu_X/\mu_Y$ is smooth, nonnegative, and bounded on compact Y. Fix a smooth positive Gauduchon metric γ_0 upstairs by Theorem 4.1.1. For an arbitrary downstairs Gauduchon metric γ, write $\Psi = \gamma^{n-1}$ and define Γ_δ, $\delta > 0$, by

$$\Gamma_\delta^{n-1} = \pi^*\Psi + \delta\gamma_0^{n-1}.$$

A strictly positive $(n-1,n-1)$-form has a unique smooth positive $(1,1)$-root of this form: solve for its Hermitian matrix from the positive cofactor matrix. Hence Γ_δ is a smooth positive Gauduchon metric.

For θ nef and $\epsilon > 0$, Lemma 25.16.2 gives the numerator limit

$$\lim_{\delta\downarrow 0}\int_Y (\pi^*\theta+\epsilon h_Y)\wedge\Gamma_\delta^{n-1} = \int_X \theta\wedge\Psi+\epsilon\int_Y h_Y\wedge\pi^*\Psi \leq \int_X(\theta+K\epsilon h_X)\wedge\Psi. \tag{25.34}$$

The first term is nonnegative by nefness and the second is strictly positive on the isomorphism locus, so the limit is positive.

The denominator satisfies

$$\lim_{\delta\downarrow 0} D_{\mu_Y}(\Gamma_\delta) = \int_Y (\pi^*\gamma^n)^{(n-1)/n}\mu_Y^{1/n} \geq c^{1/n}D_{\mu_X}(\gamma) > 0. \tag{25.35}$$

To justify the limit on the exceptional locus, represent Γ_δ^{n-1} by its positive cofactor matrix. The density $(\Gamma_\delta^n)^{(n-1)/n}$ is a fixed dimensional constant times the $1/n$ power of that matrix's determinant. It extends continuously to positive semidefinite matrices. Compactness gives dominated convergence. The inequality follows from $\mu_Y \geq c\pi^*\mu_X$, and the final equality is change of variables for the degree-one map π.

Test upstairs with Γ_δ, use (25.34)–(25.35), and let $\delta \downarrow 0$. For each downstairs γ this gives

$$\lambda_{\mu_Y}(\pi^*\theta + \epsilon h_Y) \leq c^{-1/n}\frac{\int_X(\theta + K\epsilon h_X)\wedge\gamma^{n-1}}{D_{\mu_X}(\gamma)}.$$

Taking the infimum over the full downstairs test family and then the nth power proves (25.33) with $C = c^{-1}$.

Divide its logarithm by $\log \epsilon < 0$ and use the constant-factor and direction comparisons from Proposition 25.16.1; the inequality reverses and gives $\underline{a}(\pi^*\theta) \geq \underline{a}(\theta)$ and $\overline{a}(\pi^*\theta) \geq \overline{a}(\theta)$. A proper degree-one map of smooth compact curves is an isomorphism, settling $n = 1$. □

25.16.3 Strict changes under one blowup

A blowup changes the amount of smooth volume near the vanishing set of a coefficient. A product equation turns that change into a strict increase of the degeneration order. Let $B = \mathbb{P}^m$, $m \geq 2$, and normalize the Fubini–Study form H by $\int_B H^m = 1$. Let $\pi : \widetilde{B} \to B$ be the blowup of the affine origin. The incidence projection $\rho : \widetilde{B} \to \mathbb{P}^{m-1}$ gives $L = \rho^*\omega_{\mathrm{FS}}$; we also write H for π^*H. The incidence embedding in $\mathbb{P}^m \times \mathbb{P}^{m-1}$ makes $\kappa = H + L$ Kähler.

Proposition 25.16.7 *For every choice of integers $m \geq 2$ and $p, k \geq 1$, there is a single smooth-center blowup of a projective $(m + p)$-fold and a smooth semipositive nonclosed background whose common orders downstairs and after pullback are*

$$a_{\mathrm{down}} = \max\{p - m/k, 0\}, \qquad a_{\mathrm{up}} = p - 1/k.$$

The orders exist on both sides. The increase can be any positive integer in varying dimension. The same family can lose positive limiting volume while both orders remain zero.

Proof Put $W_0 = \mathbb{P}^p$, with normalized Fubini–Study form ω satisfying $\int_{W_0} \omega^p = 1$. Write $s = |z|^2$ on the affine chart of B. Choose a smooth $Q(s) \geq 0$ equal to s^k near zero, positive for $s > 0$, and constant and positive for large s; it defines a smooth function q on B. Put $n = m + p$, choose $b > 0$, and define

$$\theta = bH + q\omega \quad \text{on } B \times W_0, \qquad \widetilde{\theta} = bH + (q \circ \pi)\omega \quad \text{on } \widetilde{B} \times W_0.$$

The map $\Pi = \pi \times \mathrm{id}$ is the single blowup of the smooth center $\{0\} \times W_0$, and $\widetilde{\theta} = \Pi^*\theta$. Fix the Kähler directions $h = H + \omega$, $\widetilde{h} = \kappa + \omega$ and smooth positive probability densities $\mu = H^m \wedge \omega^p$, $\widetilde{\mu} = \kappa^m \wedge \omega^p/(2^m - 1)$. Both backgrounds are smooth semipositive and nonclosed.

Write $\nu_{\mathrm{down}} = H^m$ and $\nu_{\mathrm{up}} = \kappa^m/(2^m - 1)$ for the base probability forms, whose normalization is checked below. For either product write its perturbation as $\xi = A + g\omega$. Downstairs $A = (b + \epsilon)H$; upstairs $A = (b + \epsilon)H + \epsilon L$; in both cases $g = q + \epsilon$, with pullback understood upstairs. The form A is Kähler on the relevant base. Use ν for the corresponding base form, and put $M = \int A^m$, $J = \int g^{-p}\nu$. The smooth equation Theorem 9.1.1 and closed Stokes give u with

$$\alpha = A + \mathrm{dd^c}u > 0, \qquad \alpha^m = (M/J)g^{-p}\nu.$$

The radial cumulative masses of the base forms are

$$\begin{aligned} \Xi_{\mathrm{down}}(s) &= \left(\frac{s}{1+s}\right)^m, \\ \Xi_{\mathrm{up}}(s) &= \frac{\left(1+\frac{s}{1+s}\right)^m - 1}{2^m - 1}. \end{aligned} \tag{25.36}$$

To check these formulas and the constants M, write $t = \log|z|^2$ on the punctured affine chart. For a radial potential $F(t)$, direct polar integration of its Hessian gives

$$\int_{s_1<|z|^2<s_2} (\mathrm{dd^c} F(\log|z|^2))^m = F'(\log s_2)^m - F'(\log s_1)^m.$$

The potentials of H and L are $\log(1+\mathrm{e}^t)$ and t. Thus $xH + yL$ has endpoint slopes y and $x+y$, and its total mass is $(x+y)^m - y^m$. The cases $(x,y) = (1,0), (1,1)$ give (25.36); the exceptional divisor and hyperplane at infinity have zero smooth volume. The form

$$\Omega = \alpha + g\omega = \xi + \mathrm{dd^c} u > 0, \qquad \Omega^n = \binom{n}{m}\frac{M}{J}\nu \wedge \omega^p$$

has constant volume density. The lower bound for all tests in Theorem 25.1.1 is attained by

$$\gamma_* = g^{-p/(n-1)}\Omega.$$

Indeed its $(n-1)$st power is

$$\gamma_*^{n-1} = \binom{n-1}{m} g^{-1}\alpha^m \wedge \omega^{p-1} + \binom{n-1}{m-1}\alpha^{m-1} \wedge \omega^p. \tag{25.37}$$

The first coefficient multiplies a top-degree form on the base, and every factor in the second term is closed; hence both terms are closed. Thus γ_* is balanced and Gauduchon. Theorem 25.1.1 gives the exact identities

$$\mathcal{V}_\mu(\theta + \epsilon h) = \binom{n}{m}\frac{(b+\epsilon)^m}{J_{\mathrm{down}}}, \qquad \mathcal{V}_{\widetilde{\mu}}(\widetilde{\theta} + \epsilon\widetilde{h}) = \binom{n}{m}\frac{(b+2\epsilon)^m - \epsilon^m}{J_{\mathrm{up}}}, \tag{25.38}$$

where $J_{\mathrm{down}} = \int_B (q+\epsilon)^{-p}\nu_{\mathrm{down}}$ and $J_{\mathrm{up}} = \int_{\widetilde{B}} (q\circ\pi+\epsilon)^{-p}\nu_{\mathrm{up}}$.

Near zero, differentiation of (25.36) gives

$$\mathrm{d}\Xi_{\mathrm{down}} = (ms^{m-1} + O(s^m))\,\mathrm{d}s, \qquad \mathrm{d}\Xi_{\mathrm{up}} = \left(\frac{m}{2^m-1} + O(s)\right)\mathrm{d}s. \tag{25.39}$$

For a density $(as^{d-1}+O(s^d))\,\mathrm{d}s$ and $q = s^k$, the singular integral has the following leading behavior as $\epsilon \downarrow 0$:

$$\begin{cases} \dfrac{a}{k}B(d/k, p-d/k)\epsilon^{d/k-p} & kp > d, \\ \dfrac{a}{k}\log(1/\epsilon) & kp = d, \\ \text{a finite positive limit} & kp < d. \end{cases} \tag{25.40}$$

Here $B(u,v) = \int_0^\infty t^{u-1}(1+t)^{-u-v}\,\mathrm{d}t$. In the power case substitute $s = \epsilon^{1/k}v$, then $t = v^k$. At equality the integrand on $\epsilon^{1/k} \ll s \ll 1$ is a/s. The $O(s^d)$ error is negligible: on a sufficiently small fixed interval its relative size is arbitrarily small, while the complementary integral is bounded. For $kp < d$, monotone convergence applies to the integrable limiting density. Since the numerators of (25.38) tend to $\binom{n}{m}b^m > 0$, formulas (25.39)–(25.40), with $d = m$ downstairs and $d = 1$ upstairs, give the stated orders.

For $m = k = 2$, $p = 1$, the denominators satisfy $J_{\text{down}} \sim \log(1/\epsilon)$ and $J_{\text{up}} \sim (\pi/3)\epsilon^{-1/2}$; the volumes are respectively asymptotic to $3b^2/\log(1/\epsilon)$ and $(9b^2/\pi)\epsilon^{1/2}$. The orders change from 0 to 1/2. For $m = 2$, $k = p = 1$, the downstairs denominator has a finite positive limit, while $J_{\text{up}} \sim (2/3)\log(1/\epsilon)$, so positive limiting volume downstairs becomes $9b^2/[2\log(1/\epsilon)]$ upstairs, with order zero on each side. For an arbitrary integer $j \geq 1$, choose $m = j+2$, $p = j+1$, $k = 1$. Then $p < m$ gives a positive downstairs limit, while the orders are 0 and j. □

25.16.4 Positive-current subtraction and divisorial residuals

A positive current difference orders every Gauduchon pairing. If the residual remains nef, the perturbed volumes stay positive and their logarithmic orders can be compared.

Proposition 25.16.8 *Let θ, β be smooth nef real $(1,1)$-forms on a compact complex n-fold, and suppose*

$$S = (\theta - \beta) + \mathrm{dd^c}\varphi \geq 0$$

is a positive $(1,1)$-current. For a Hermitian form h, a smooth strictly positive probability form μ and every $\epsilon > 0$,

$$\mathcal{V}_\mu(\beta + \epsilon h) \leq \mathcal{V}_\mu(\theta + \epsilon h). \tag{25.41}$$

Consequently $\underline{a}(\beta) \geq \underline{a}(\theta)$ and $\overline{a}(\beta) \geq \overline{a}(\theta)$.

Proof For every admissible Gauduchon test γ, distributional integration by parts gives

$$\int_X (\theta - \beta) \wedge \gamma^{n-1} = \langle S, \gamma^{n-1} \rangle \geq 0.$$

The quotient for $\theta + \epsilon h$ is therefore at least that for $\beta + \epsilon h$ for the same test. Taking infima over the identical full test family proves the corresponding inequality for λ_μ. Both sides are positive by Proposition 25.16.1, so raising to the nth power gives (25.41). Division by the negative number $\log \epsilon$ reverses the logarithmic inequality and proves both order comparisons. □

For an effective real divisor D, Poincaré–Lelong gives $[D] = \eta_D + \mathrm{dd^c}\varphi$, where η_D is a smooth curvature representative. Thus Proposition 25.16.8 applies to $\beta = \theta - \eta_D$ whenever β is smooth nef. The same reasoning applies to a convergent positive divisorial sum only when its curvature representative is smooth and a global potential realizes the current difference. If $T = \theta + \mathrm{dd^c}u$ is a chosen positive current and S is the part removed, then $T - S = \beta + \mathrm{dd^c}(u - \varphi)$. This identifies the residual to which the comparison applies.

Proposition 25.16.9 *Within the smooth-nef domain, specified effective divisorial removal can preserve an order, increase it by an arbitrary integer up to the dimension or by a fraction, and destroy equality of lower and upper orders. Removing an effective divisor can also leave a non-nef residual.*

Proof On E^n, let ω_i be the flat area forms with integral one, set $h = \sum_i \omega_i$ and $\mu = \omega_1 \wedge \cdots \wedge \omega_n$. A coordinate divisor F_i has smooth curvature representative ω_i: on the corresponding elliptic curve, solve a zero-mean Poisson equation to adjust any curvature representative to ω_i, then pull it back. For $\theta = h$ and $D = \sum_{i=1}^k F_i$, the residual $\beta = \sum_{i=k+1}^n \omega_i$ is smooth nef. Its perturbations have constant density, and the positive closed perturbation itself attains the all-test equality in Theorem 25.1.1. Therefore

$$\mathcal{V}_\mu(\theta + \epsilon h) = n!(1+\epsilon)^n, \qquad \mathcal{V}_\mu(\beta + \epsilon h) = n!\epsilon^k(1+\epsilon)^{n-k}. \tag{25.42}$$

For $n = 1$ the same formula follows from $\lambda_\mu(\xi) = \int_X \xi$. These examples realize each integer jump $1, \ldots, n$. A nonzero divisor with jump zero is given by $\theta = 2h$ and $D = F_1$, since the residual remains strictly positive.

For a fractional jump, use the diagonal background $\beta = b\omega_1 + q\omega_2$ of Proposition 25.16.4 with $q = |\zeta|^{2m}$ near its unique zero, $m > 1$. Choose $c > 0$ and put $\theta = \beta + c\omega_2 > 0$ and $D = c(E \times \{w_0\})$, whose smooth curvature representative is $c\omega_2$. The original background has both orders zero; the residual has common order $1 - 1/m$.

To lose a common order, let β_{res} be the oscillatory rank-one background of Proposition 25.16.5 and put

$$\theta = \beta_{\mathrm{res}} + ch > 0, \qquad D = c(\{z_0\} \times E) + c(E \times \{w_0\}), \qquad \eta_D = ch.$$

The original θ has both orders zero: $\theta \geq ch$ yields a uniform positive lower bound for every test quotient, while one fixed test gives a finite upper bound. Its smooth nef residual is exactly β_{res} and has unequal orders. In each of these examples, the smooth residual plus $[D]$ is a positive current in the smooth $\mathrm{dd^c}$-class of θ, with divisorial

part exactly $[D]$ because the other summand is smooth. Its minimality is not needed for the comparison; the minimal-current case is treated below.

Finally blow up $\mathbb{P}^2$ at two distinct points. Let H be the pullback of the normalized Fubini–Study class, E_1, E_2 the exceptional curves, and L the strict transform of the line through the points. The smooth semipositive pullback θ is nef. Subtract curvature representatives of $D = E_1 + E_2$. The residual β has class $H - E_1 - E_2$ and contains the positive current $[L]$, by Poincaré–Lelong for the total transform of the line. The intersection numbers $H^2 = 1$, $H \cdot E_i = 0$, $E_i^2 = -1$, and $E_1 \cdot E_2 = 0$ give

$$\int_L \beta = (H - E_1 - E_2)^2 = -1.$$

If β were nef, smooth representatives $\beta + \epsilon h + \mathrm{dd^c} u_\epsilon > 0$ would imply $-1 \geq -\epsilon \int_L h$ for all $\epsilon > 0$, which is impossible. □

For a nef background, perturbative minimal divisorial multiplicities vanish. Indeed, every $\theta + \delta h$ has a smooth positive representative, so the infimum of its generic divisorial Lelong numbers is zero. Passing to $\delta \downarrow 0$ still gives zero. Thus removing the divisor defined by these multiplicities leaves the background unchanged. Removing the divisorial part of an actual boundary current is different, even when that current has minimal singularities.

Corollary 25.16.1 *On the ruled surface of Lemma 24.5.1, retain its nef curvature form θ, infinity section D and Kähler form $\kappa = \theta + \pi^*\sigma_0$. For every smooth positive probability form μ,*

$$\mathcal{V}_\mu(\theta + \epsilon\kappa) = 2\epsilon(1 + \epsilon), \qquad \mathcal{V}_\mu(\epsilon\kappa) = 2\epsilon^2.$$

The unique positive current in the boundary class is $[D]$. It has minimal singularities, and removing its full divisor leaves the zero background and changes the common order from one to two.

Proof The earlier lemma proves uniqueness of $[D]$ and nefness. The calculation in Proposition 24.5.1 gives $\int_X \theta^2 = 0$. Put $B = \pi^*\sigma_0$. We have $B^2 = 0$ and

$$\int_X \theta \wedge B = \int_D B = 1,$$

by Poincaré–Lelong, Stokes and the fact that the section maps isomorphically to the elliptic base. The closed nef formula following Theorem 25.10.1 now computes both volumes by expanding their squares. It holds for every μ. The unique boundary current has minimal singularities because any two of its global potentials differ by a pluriharmonic function, hence by a constant. Choosing θ as the curvature of its entire divisor leaves zero. The two volume polynomials give the claimed orders. □

Problem 25.16.1 Which geometric conditions on a nonclosed nef background force a common degeneration order, or an integral one? When is equality attained in the modification and residual order comparisons?

25.17 Current tests for smooth positivity

Let X again be a connected compact complex manifold of dimension $n \geq 1$. Testing against all positive $\mathrm{dd^c}$-closed currents detects smooth positivity of a background. Hahn–Banach separation gives the exact threshold, with zero as the nef boundary. The full current set is needed here: the smooth tests in Lamari duality detect pseudoeffectivity.

Fix a Hermitian metric h and use $\mathcal{G}_h$ from Section 4.4. That set is compact for $n \geq 2$. For $n = 1$, elliptic regularity and the maximum principle give $\mathcal{G}_h = \{1/\int_X h\}$, so compactness holds in this case as well.

For any smooth real $(1,1)$-form θ, define

$$m_h(\theta) = \min_{T \in \mathcal{G}_h} \int_X \theta \wedge T.$$

The minimum exists by weak compactness, and it is finite since $-Ch \leq \theta \leq Ch$ for some C.

Theorem 25.17.1 *The background θ has a smooth positive representative if and only if $m_h(\theta) > 0$. Moreover θ is nef if and only if $m_h(\theta) \geq 0$, and*

$$m_h(\theta) = \sup\{a \in \mathbb{R} : \ \exists u \in C^\infty(X, \mathbb{R}), \ \theta + \mathrm{dd^c} u > ah\}.$$

Proof If $\theta + \mathrm{dd^c} u \geq ah$ for $a > 0$, then distributional integration by parts gives $\int_X \theta \wedge T \geq a$ for every normalized test current. This proves one implication.

For the converse, work in the real Fréchet space of smooth $(1,1)$-forms. Let $\mathcal{P}$ be its open convex cone of strictly positive forms, and let $L = \mathrm{dd^c} C^\infty(X, \mathbb{R})$. If the affine space $\theta + L$ is disjoint from $\mathcal{P}$, the Hahn–Banach separation theorem for an open convex set and a disjoint affine space gives a nonzero continuous real linear functional T with

$$T(\theta + \ell) \leq T(\eta) \qquad (\ell \in L, \ \eta \in \mathcal{P}).$$

Varying ℓ by arbitrary real multiples shows $T(L) = 0$. Scaling any $\eta \in \mathcal{P}$ to zero shows $T(\theta) \leq 0$, while scaling it to infinity shows $T(\eta) \geq 0$. Thus T is a positive current: every smooth semipositive test form is a limit of strictly positive forms. Positivity also makes it a current of order zero. The equality $T(L) = 0$ is precisely $\mathrm{dd^c} T = 0$. Since $T \neq 0$ and $h > 0$, its trace $T(h)$ is positive. Normalizing produces an element of $\mathcal{G}_h$ with nonpositive pairing against θ. This contradicts $m_h(\theta) > 0$ and proves the converse.

The identity $m_h(\theta - ah) = m_h(\theta) - a$ and the equivalence just proved show that the set inside the supremum is exactly $(-\infty, m_h(\theta))$. Finally, nefness means that $\theta + \varepsilon h$ has a smooth positive representative for every $\varepsilon > 0$. This is equivalent to $m_h(\theta) + \varepsilon > 0$ for all such ε, or $m_h(\theta) \geq 0$. □

For $n \geq 2$, let $\mathscr{C}_h$ be the weak closure of the smooth positive normalized tests and let $\kappa_h(\theta)$ be their linear positivity threshold, as in Section 4.4. The inclusion $\mathscr{C}_h \subseteq \mathscr{G}_h$ gives $m_h(\theta) \leq \kappa_h(\theta)$. The larger test set detects smooth positivity and nefness; the weak closure of the smooth tests detects positivity and bigness of current representatives. The inequality can be strict, as Example 27.8.1 shows.

Proposition 25.17.1 *Let θ be nef and suppose $\lambda_\mu(\theta) > 0$. If $T \in \mathscr{G}_h$ satisfies $\int_X \theta \wedge T = 0$, then its determinant-weighted measure vanishes.*

Proof In dimension one the hypotheses cannot hold, since $\int_X \theta \wedge T = \lambda_\mu(\theta)/\int_X h > 0$. Suppose $n \geq 2$, and write $D_\mu(T)$ for the determinant functional of Section 25.6, with h as reference metric and volume form μ. If $\eta \in \mathscr{P}$, its solution form satisfies $\Omega^n = \lambda_\mu(\eta)^n \mu$. Test the infimum in Lemma 25.6.1 with $\Omega/\lambda_\mu(\eta)$. Stokes gives

$$\int_X \eta \wedge T = \int_X \Omega \wedge T \geq \lambda_\mu(\eta) D_\mu(T).$$

Apply this to $\eta = \theta + \varepsilon h$. The left side is ε, while Theorem 25.10.1 gives $\lambda_\mu(\eta) \to \lambda_\mu(\theta) > 0$. Thus $D_\mu(T) = 0$. The determinant-weighted measure is positive and has this total mass, so it vanishes. □

A nef boundary with positive variational volume therefore has annihilating currents whose coefficient matrices are rank deficient almost everywhere. Their trace measures need not be supported on analytic subsets merely because these determinants vanish.

25.17.1 Maximal time of the Chern–Ricci flow

A smooth Chern–Ricci flow gives positive representatives along an affine ray. Conversely, a positive representative at a later time supplies a uniformly positive reference path and prevents an earlier singularity. The current criterion then computes the lifetime. Let ω_0 be the initial Hermitian metric. For a Hermitian form ω on X, use the conventional Chern–Ricci form

$$\mathrm{Ric}(\omega) = -\mathrm{i}\partial\bar{\partial} \log \omega^n = -2\pi \mathrm{dd^c} \log \omega^n,$$

interpreted in local coordinates; coordinate changes do not affect the form. This fixes the time normalization of the flow $\partial_t \omega = -\mathrm{Ric}(\omega)$. Put $\chi = \mathrm{Ric}(\omega_0)$ and $\theta_t = \omega_0 - t\chi$.

Theorem 25.17.2 *The Chern–Ricci flow starting from ω_0 has a unique maximal smooth solution on $[0, \tau)$, where*

$$\tau = \sup\{t \geq 0 : \ \theta_t \textit{ has a smooth positive representative}\}.$$

Proof The admissible times form an interval containing zero: positive representatives at two times give positive representatives at the intervening times by convex

combination. Positivity is also open in time. Fix an admissible $T > 0$ and choose a smooth real function a such that $\theta_T + \mathrm{dd^c} a > 0$. The reference forms

$$\gamma_t = \theta_t + \frac{t}{T}\mathrm{dd^c} a = \left(1 - \frac{t}{T}\right)\omega_0 + \frac{t}{T}(\theta_T + \mathrm{dd^c} a), \qquad 0 \le t \le T,$$

are uniformly positive, with all their derivatives uniformly bounded. Put

$$\Omega = \mathrm{e}^{a/(2\pi T)}\omega_0^n.$$

Then $\dot\gamma_t = -\operatorname{Ric}(\Omega)$, with the Ricci form of a volume form defined by the same local expression. The scalar equation is

$$\dot u = 2\pi \log \frac{(\gamma_t + \mathrm{dd^c} u)^n}{\Omega}, \qquad \gamma_t + \mathrm{dd^c} u > 0, \qquad u(0) = 0. \tag{25.43}$$

Applying $\mathrm{dd^c}$ shows that $\omega(t) = \gamma_t + \mathrm{dd^c} u(t)$ solves the metric flow. Conversely, integrating $\dot u = 2\pi \log(\omega(t)^n/\Omega)$ from zero recovers this identity from any metric solution. The linearization of the scalar equation is $\partial_t - 2\pi \operatorname{tr}_{\omega(t)} \mathrm{dd^c}$, a strictly parabolic operator. Classical short-time existence for smooth strictly parabolic equations therefore supplies a solution. Two scalar solutions have a difference satisfying a linear parabolic equation, whose positive principal coefficients are the averages of the inverse metrics along the segment between them. The maximum principle proves uniqueness.

Suppose this solution has maximal time $T_* < T$. We establish estimates independent of $t < T_*$ to continue it. For a large fixed C, the functions Ct and $-Ct$ are respectively a supersolution and a subsolution of (25.43): at a spatial extremum of u the complex Hessian has the required sign, and γ_t^n/Ω is bounded above and away from zero. The maximum principle gives $|u| \le CT$.

Set $v = \dot u$ and $\mathcal{L} = 2\pi \operatorname{tr}_{\omega(t)} \mathrm{dd^c}$. Differentiation gives

$$(\partial_t - \mathcal{L})v = 2\pi \operatorname{tr}_{\omega(t)} \dot\gamma_t, \qquad (\partial_t - \mathcal{L})u = v - 2\pi n + 2\pi \operatorname{tr}_{\omega(t)} \gamma_t.$$

Since γ_t is affine in time, these imply

$$\begin{aligned}(\partial_t - \mathcal{L})(tv - u - 2\pi n t) &= -2\pi \operatorname{tr}_{\omega(t)} \omega_0 < 0,\\ (\partial_t - \mathcal{L})((T-t)v + u + 2\pi n t) &= 2\pi \operatorname{tr}_{\omega(t)} \gamma_T > 0.\end{aligned}$$

The first expression is bounded above by its initial value, zero; the second is bounded below by $\min_X Tv(0)$. The bound for u and $T - t \ge T - T_* > 0$ give a lower bound for v. The first inequality gives its upper bound on $[T_*/2, T_*)$, and the initial smooth solution bounds it on $[0, T_*/2]$. In particular, $\omega(t)^n/\Omega = \mathrm{e}^{v/(2\pi)}$ is bounded above and away from zero.

We next bound the trace, retaining the time derivative of the reference metric. In the coordinates of Lemma 9.4.1, write $g(t)$ for the matrix of γ_t and $h = g + (u_{i\bar{j}})$ for that of $\omega(t)$. Put

$$S = \operatorname{tr}_g h, \qquad B = \operatorname{tr}_h g, \qquad F = \log \frac{\Omega}{\gamma_t^n}.$$

Thus $\log(\det h/\det g) = v/(2\pi) + F$. The family $g(t)$ has uniformly bounded Chern torsion and curvature. At each fixed time Lemma 9.4.1 applies with these uniform constants. Direct differentiation of S gives

$$\partial_t \log S = \frac{\Delta_g v + \operatorname{tr}_g \dot{g}}{S} - \frac{\operatorname{tr}(g^{-1}\dot{g}\, g^{-1}h)}{S}.$$

The $\Delta_g v/S$ term cancels the corresponding term in $\mathcal{L}\log S$. If $\tau_k = \sum_i T^i_{ik}$ is the torsion trace of $g(t)$, the resulting inequality on $\{S \geq 1\}$ is

$$(\partial_t - \mathcal{L})\log S \leq -\frac{4\pi}{S^2}\operatorname{Re}(h^{k\bar{q}}\tau_k S_{\bar{q}}) + CB + C. \tag{25.44}$$

Indeed $\Delta_g F$ and $\operatorname{tr}_g \dot{g}$ are uniformly bounded, $S^{-1} \leq B/n^2$, and $-Cg \leq \dot{g} \leq Cg$ bounds the last displayed quotient. These bounds account for every added time-dependent term.

Choose a constant C_0 so that $1 \leq s = u + C_0 \leq D$ on the whole time interval, and maximize

$$Q = \log S - Au + s^{-1}$$

on $X \times [0, t_0]$, where $t_0 < T_*$ and $A \geq 1$ will be fixed independently of t_0. A maximum at time zero gives the desired trace bound from the initial data. At a later maximum, $\partial \log S = (A + s^{-2})\partial u$. Since $|\tau|^2_h \leq CB$, Young's inequality bounds the absolute value of the first term on the right of (25.44) by

$$2\pi s^{-3}|\partial u|^2_h + \frac{C(A+1)^2 s^3}{S^2}B.$$

If $S^2 \leq (A+1)^2 D^3$, the trace is already bounded at the maximum. In the remaining case the last coefficient is at most C, and

$$\begin{aligned} 0 \leq (\partial_t - \mathcal{L})Q &\leq CB + C - (A + s^{-2})(v - 2\pi n + 2\pi B) - 2\pi s^{-3}|\partial u|^2_h \\ &\leq (C - 2\pi A)B + C(A+1). \end{aligned}$$

Choose A so that $2\pi A > C + 1$. This bounds B at the maximum. The eigenvalues give $S \leq n(\det h/\det g)B^{n-1}$ for $n \geq 2$, and $S = \det h/\det g$ for $n = 1$. The determinant bound therefore bounds S at the maximum in every case. The bound for u transfers this estimate from Q to S everywhere, uniformly in t_0. The determinant lower bound then bounds the smallest eigenvalue away from zero. Thus $\omega(t)$ and γ_t are uniformly equivalent for $t < T_*$.

Uniform ellipticity now gives the higher estimates by classical local parabolic regularity. In a coordinate ball, write the logarithmic determinant as a concave function of the complex part of the real Hessian. On the fixed compact set of positive matrices just obtained, its supporting affine functions have uniformly positive, bounded slopes. Their infimum extends the operator to a concave uniformly elliptic function of every real symmetric matrix. The spatial and time dependence enters through the smooth family $g(t)$ and the fixed density Ω. The parabolic Evans–Krylov estimate gives uniform $C^{2+\alpha,1+\alpha/2}$ bounds away from the initial time. The original smooth solution covers a fixed initial interval. Differentiating the equation and applying parabolic Schauder estimates then gives uniform bounds for all spatial and time derivatives. This is the parabolic counterpart of the local extension used in Corollary 9.4.1.

The bounds for the time derivatives make $u(t)$ Cauchy in every spatial C^k norm as $t \uparrow T_*$. Its smooth limit still has a strictly positive solution form by uniform ellipticity. Short-time existence with this limit as initial datum extends the solution beyond T_*, a contradiction. Hence the flow exists up to every admissible time T, and uniqueness patches these solutions on $[0, \tau)$.

Finally, integrating the metric flow gives

$$\omega(t) = \theta_t + \mathrm{dd^c}\left(2\pi \int_0^t \log \frac{\omega(s)^n}{\omega_0^n}\, \mathrm{d}s\right).$$

Every time at which a smooth positive solution exists is therefore admissible. This proves maximality. □

Combining the continuation theorem with the current criterion gives a numerical description of the lifetime.

Corollary 25.17.1 *Let*

$$b = \max_{T \in \mathcal{G}_{\omega_0}} \int_X \chi \wedge T.$$

Then the maximal smooth existence time is

$$\tau = \begin{cases} 1/b, & b > 0, \\ +\infty, & b \le 0. \end{cases}$$

If $\tau < \infty$, the limiting background θ_τ is nef and there is a normalized current T_* with $\int_X \theta_\tau \wedge T_* = 0$. If also $\lambda_\mu(\theta_\tau) > 0$, every such annihilating current has zero determinant-weighted measure.

Proof For $t \ge 0$, normalization of the tests gives the exact identity

$$m_{\omega_0}(\theta_t) = 1 - tb.$$

The smooth-positivity theorem and Theorem 25.17.2 prove the formula for τ. When $b > 0$, the maximum defining b is attained by weak compactness. Its maximizer T_* annihilates θ_τ. The same identity gives $m_{\omega_0}(\theta_\tau) = 0$, hence nefness. The last assertion is Proposition 25.17.1. □

Along the flow, $\omega(t) = \theta_t + \mathrm{dd^c}\varphi_t$ with a smooth potential. For every fixed $T \in \mathscr{G}_{\omega_0}$ one consequently has

$$\int_X \omega(t) \wedge T = 1 - t \int_X \chi \wedge T.$$

These finite traces describe a boundary obstruction without any total Monge–Ampère mass bound. They do not locate the singularities of the flow or prove smooth convergence away from an analytic set. Those geometric questions require local estimates in addition to the scalar pairings.

25.18 Divisorial constraints and residual volume

Prescribing divisorial coefficients subtracts their Gauduchon degrees from the numerator of the volume quotient. Lamari duality then decides feasibility, while the ordinary-type correspondence of Theorem 14.9.1 reduces the equation to the residual background. If that background has a smooth positive representative, a maximizing Gauduchon metric forces every threshold subsolution with the prescribed coefficients to have a smooth remainder.

Fix a smooth real $(1,1)$-form θ on X. Let $D = \sum_{j=1}^N a_j D_j$ be a finite effective real divisor, with distinct prime divisors D_j and $a_j > 0$. Choose a smooth Hermitian metric on each $\mathcal{O}_X(D_j)$, write s_j for its canonical section, and let η_j denote its curvature in our normalization. Set

$$\sigma_D = \sum_j a_j \log |s_j|^2, \qquad \eta_D = \sum_j a_j \eta_j, \qquad \beta_D = \theta - \eta_D.$$

The Poincaré–Lelong formula is

$$\mathrm{dd^c}\sigma_D = [D] - \eta_D.$$

For a Gauduchon metric γ on X, define

$$\deg_\gamma D = \int_X [D] \wedge \gamma^{n-1} = \sum_j a_j \int_{D_j} \gamma^{n-1}.$$

Integration over a singular divisor means integration over its regular locus, or equivalently its integration current. This degree is finite, positive for $D \neq 0$, and equals $\int_X \eta_D \wedge \gamma^{n-1}$ by distributional Stokes.

Proposition 25.18.1 *The following are equivalent:*

(1) *There is $u \in \mathrm{PSH}(X, \theta)$ whose generic Lelong coefficient along every D_j is at least a_j.*
(2) *The residual background β_D is pseudoeffective.*
(3) *For every Gauduchon metric γ,*

$$\int_X \theta \wedge \gamma^{n-1} \geq \deg_\gamma D.$$

These conditions do not assert that $u - \sigma_D$ is bounded, or that the coefficients of u are exactly the prescribed ones.

Proof Suppose first that u has the indicated coefficients. Lemma 1.17.1 permits subtraction of these coefficients: $\theta + \mathrm{dd^c} u - [D] \geq 0$. Consequently

$$\beta_D + \mathrm{dd^c}(u - \sigma_D) = \theta + \mathrm{dd^c} u - [D] \geq 0.$$

The difference is understood in L^1 and then with its quasi-plurisubharmonic representative. This proves the second condition. Conversely, if $v \in \mathrm{PSH}(X, \beta_D)$, then $u = \sigma_D + v$ satisfies

$$\theta + \mathrm{dd^c} u = [D] + \beta_D + \mathrm{dd^c} v \geq 0,$$

so its divisorial coefficients dominate those of D. For $n \geq 2$, Lamari duality in Theorem 4.3.1 says that a smooth background is pseudoeffective exactly when all its Gauduchon integrals are nonnegative. Apply it to β_D and use the degree identity. For $n = 1$ the same criterion is $\int_X \beta_D \geq 0$: necessity is Stokes, and sufficiency follows by solving for a smooth real function f in $\mathrm{dd^c} f = (\int_X \beta_D)\mu - \beta_D$ with the scalar Green operator. This also proves the remaining implication on a curve. □

For a nonzero effective real divisor D and a pseudoeffective θ, the exact threshold is therefore

$$\tau_\theta(D) = \inf_{\gamma \text{ Gauduchon}} \frac{\int_X \theta \wedge \gamma^{n-1}}{\deg_\gamma D}. \tag{25.45}$$

It is finite and nonnegative. A potential with divisorial coefficients at least tD exists if and only if $0 \leq t \leq \tau_\theta(D)$. In particular the endpoint is included: all the non-strict Gauduchon inequalities still hold there, so duality supplies a positive current. This is a threshold for existence of a positive current, not for a bounded residual potential.

Whenever β_D is pseudoeffective, define its residual volume by

$$\mathcal{V}_\mu(\theta; D) = \mathcal{V}_\mu(\beta_D).$$

The definition and the degree identity give the exact formula

$$\mathcal{V}_\mu(\theta; D)^{1/n} = \inf_{\gamma \text{ Gauduchon}} \frac{\int_X \theta \wedge \gamma^{n-1} - \deg_\gamma D}{D_\mu(\gamma)}. \tag{25.46}$$

It is finite and nonnegative. Changing the auxiliary line-bundle metrics changes η_D by a smooth $\mathrm{dd^c}$ term. Thus this volume is independent of those metrics and of a smooth $\mathrm{dd^c}$ change of θ. It depends on the specified divisorial data and on the smooth probability measure μ.

Theorem 25.18.1 *Suppose β_D admits a smooth positive representative, and put $c = \mathcal{V}_\mu(\theta; D)$. There is a θ-plurisubharmonic potential $u_0 = \sigma_D + v$, with v smooth, satisfying*

$$\mathrm{MA}_\theta^{\mathrm{ext}}(u_0) = c\mu.$$

Every $u \in \mathrm{PSH}(X, \theta)$ such that

$$v(u, D_j) \geq a_j \quad (1 \leq j \leq N), \qquad \mathrm{MA}_\theta^{\mathrm{ext}}(u) \geq c\mu$$

is a constant translate of u_0. In particular, a solution in the ordinary singularity type of σ_D has scalar c, has a smooth remainder, and is unique up to constants.

Proof The smooth equation and Theorem 25.1.1 give

$$\Omega = \beta_D + \mathrm{dd^c} v > 0, \qquad \Omega^n = c\mu.$$

Then $\theta + \mathrm{dd^c}(\sigma_D + v) = [D] + \Omega \geq 0$. Off D its extended measure is Ω^n, and both measures give D zero mass. This proves existence.

For a potential u satisfying the displayed inequalities, divisorial removal as in Proposition 25.18.1 makes

$$w = u - \sigma_D - v \in \mathrm{PSH}(X, \Omega), \qquad \Omega + \mathrm{dd^c} w = \theta + \mathrm{dd^c} u - [D] \geq 0.$$

The difference is first taken in L^1 and then with its canonical representative. On $X \setminus D$ the two currents have smooth changes of local potential decomposition, so their extended measures agree there. Both measures vanish on D. Hence

$$\mathrm{MA}_\Omega^{\mathrm{ext}}(w) = \mathrm{MA}_\theta^{\mathrm{ext}}(u) \geq c\mu.$$

Use $\mathrm{d}V = \mu$. The zero potential solves the smooth equation in background Ω, so $b_\Omega(1) = c$ by Proposition 13.24.2. For $n \geq 2$, its Gauduchon representative gives a smooth strictly positive dual maximizer. By Corollary 25.9.1, all threshold subsolutions differ by constants; comparing w with zero shows that w is constant. For $n = 1$, the positive current $\Omega + \mathrm{dd^c} w$ dominates its extended measure, hence dominates $c\mu = \Omega$. Its total mass is $\int_X \Omega = c$ by Stokes. Thus $\mathrm{dd^c} w = 0$, and w is again constant.

Finally, if u lies in the ordinary type of σ_D and solves an equation with scalar $c' \geq 0$, the same reduction gives a bounded w with $\mathrm{MA}_\Omega(w) = c'\mu$. If $c' \neq c$, the smaller of these two measures is at most a constant strictly less than one times the larger. Applying Theorem 10.2.2 after every constant translation of the larger-measure potential is impossible. Therefore $c' = c$, and the rigidity just proved gives $u = u_0 + \text{constant}$. □

When θ is Hermitian, write $\mu = f\,\mathrm{d}V$ relative to a smooth strictly positive probability volume $\mathrm{d}V$. Proposition 27.8.1 bounds this residual coefficient in terms of the global threshold $b_\theta(f)$ and the linear Gauduchon threshold of the divisor class. The residual problem specifies the divisor; the global threshold allows all genuine potentials.

Along a real family $D_t = tD$ in the feasible interval, the root of (25.46) is concave and nonincreasing. Every test quotient is affine with nonpositive slope, so both assertions follow by taking infima. On a subinterval where $\theta - t\eta_D$ has a positive representative, choose a minimizing Gauduchon metric γ_t for that background. Then Proposition 25.2.1 gives the sharper formula

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{V}_\mu(\theta; tD)^{1/n} = -\frac{\deg_{\gamma_t} D}{D_\mu(\gamma_t)} < 0. \tag{25.47}$$

In dimension one the derivative is $-\deg D$ and the denominator is one. If a residual background at an endpoint is nef, Theorem 25.10.1 also expresses its value as the limit of the normalized smooth equations after adding ϵh, for a fixed Hermitian form h and real $\epsilon > 0$.

25.18.1 Varying divisorial coefficients

With fixed divisor support, varying the coefficients changes the residual equation smoothly as long as the residual background stays positive. The smooth remainders then control capacity convergence and all divisorial coefficients of the limit. Keep D_j, s_j, η_j as above, and put $\sigma_j = \log |s_j|^2$.

For $a \in \mathbb{R}^N_{\geq 0}$ put

$$D_a = \sum_j a_j D_j, \qquad \sigma_a = \sum_{a_j > 0} a_j \sigma_j, \qquad \beta_a = \theta - \sum_j a_j \eta_j.$$

Write $\mathcal{A}$ for the set of such a for which β_a has a smooth positive representative. Zero coefficients are omitted from the logarithmic sum.

Proposition 25.18.2 *There are uniquely normalized smooth functions v_a, $a \in \mathcal{A}$, such that*

$$\int_X v_a \, \mathrm{d}\mu = 0, \qquad \Omega_a = \beta_a + \mathrm{dd^c} v_a > 0, \qquad \Omega_a^n = c(a)\mu, \quad c(a) = \mathcal{V}_\mu(\beta_a) > 0.$$

The maps $a \mapsto v_a$ and $a \mapsto c(a)$ are smooth, with bounds for every smooth norm of v_a on compact subsets of $\mathcal{A}$. Here smoothness at a zero coefficient means restriction of a smooth family from an open neighborhood in $\mathbb{R}^N$.

Set $u_a = \sigma_a + v_a$. If $a^{(k)} \in \mathcal{A}$ and $a^{(k)} \to a \in \mathcal{A}$, then $u_{a^{(k)}} \to u_a$ in L^1 and in capacity, and smoothly on compact subsets of $X \setminus \bigcup_j D_j$. Moreover,

$$\left\| \mathrm{MA}_\theta^{\mathrm{ext}}(u_{a^{(k)}}) - \mathrm{MA}_\theta^{\mathrm{ext}}(u_a) \right\|_{\mathrm{TV}} = |c(a^{(k)}) - c(a)| \longrightarrow 0.$$

For every modification $\pi : Y \to X$ with smooth source and every prime divisor $E \subset Y$,

$$\nu(u_a \circ \pi, E) = \sum_j a_j \, \mathrm{ord}_E(\pi^* D_j).$$

In particular the limit has exactly the prescribed divisorial ordinary type; no additional singularity occurs in this finite family.

Proof The set of parameters in $\mathbb{R}^N$ for which β_a has a smooth positive representative is open. The smooth equation gives v_a and $c(a)$ there. The proof of Proposition 25.2.1 gives smooth dependence of both quantities with the fixed mean normalization. Uniqueness makes the local solution families agree. A finite covering of any compact parameter set then gives all the asserted smooth bounds.

The Poincaré–Lelong identity gives

$$\theta + \mathrm{dd^c} u_a = [D_a] + \Omega_a.$$

Thus $u_a \in \mathrm{PSH}(X, \theta)$. Its extended measure equals $\Omega_a^n = c(a)\mu$ off the divisor union, and both measures vanish on that union. This also treats $D_a = 0$. Each fixed σ_j is locally integrable, so coefficient convergence and smooth convergence of v_a give the L^1 limit. Off the fixed divisor union every σ_j is smooth, which gives the asserted smooth convergence. For capacity convergence, remove an open neighborhood of that analytic set of arbitrarily small Hermitian capacity. On its compact complement the convergence is uniform. This proves capacity convergence, since the divisor union has capacity zero. The equality of total-variation distances follows from the equation and $\mu(X) = 1$.

Finally, after pullback the logarithmic potential $\sigma_j \circ \pi$ has generic coefficient $\mathrm{ord}_E(\pi^* D_j)$, whereas $v_a \circ \pi$ is smooth. Additivity for the finite logarithmic sum proves the displayed formula on every model. Since v_a is smooth also at the limiting parameter, u_a has the ordinary type of σ_a. □

The positive-cone hypothesis at the limit supplies the smooth bounds used here. For moving sections and weights, Theorem 14.9.2 gives L^1 convergence of supremum-normalized full potentials; it also allows uniformly bounded measurable

tests converging in L^1 when the residual background is fixed. Across a modification, the pulled-back probability density can vanish on the exceptional divisor, so the smooth positive equation no longer applies directly. In the minimal exceptional-pole setting, Proposition 27.11.3 gives a necessary vanishing order for a smooth residual; that order alone does not ensure smoothness.

Multiplier ideals need not be eventually constant in this family. At a smooth point of D_j outside the other divisors, take a coordinate z_1 defining D_j. There $u_a = a_j \log |z_1|^2 + O(1)$, so

$$\mathcal{I}(u_a) = (z_1^{\lfloor a_j \rfloor})$$

by the one-variable integral in Example 1.12.1 and Fubini's theorem. Thus, if the parameter stays in $\mathcal{A}$ while a_j increases to an integer from below, this local ideal jumps at the limit. Exactness of the limiting logarithmic coefficients does not mean eventual equality of the multiplier ideals.

25.19 A divisorial threshold on the Hopf threefold

On the Hopf threefold, a hyperplane pullback has an explicit divisorial threshold. The residual metric solves its normalized equation with zero remainder, so its volume can be computed directly, including at the nef endpoint.

For this section take X and $\pi : X \to \mathbb{P}^2$ from Chapter 8, and write its vertical form as ζ, so that $\omega = \alpha + \zeta$. The Hopf forms satisfy

$$\alpha^3 = 0, \quad \zeta^2 = 0, \quad \mathrm{dd^c}\zeta = -\alpha^2, \quad \int_X \alpha^2 \wedge \zeta = 2.$$

Let H be the inverse image of a projective hyperplane under $\pi : X \to \mathbb{P}^2$. The pulled-back Fubini–Study metric gives $\eta_H = \alpha$ and a logarithmic potential σ_H with $\mathrm{dd^c}\sigma_H = [H] - \alpha$.

Proposition 25.19.1 *The exact divisorial threshold is $\tau_\omega(H) = 1$. For $\mu_0 = \omega^3/6$ and $0 \le a \le 1$,*

$$\mathcal{V}_{\mu_0}(\omega; aH) = 6(1-a)^2.$$

For $a < 1$ this is the positive scalar in the smooth-remainder singular equation. At $a = 1$ the logarithmic potential has zero non-pluripolar measure, and its residual background is nef. Every residual background with $a < 1$ has infinite upper mass.

Proof The residual background is

$$R_a = (1-a)\alpha + \zeta.$$

It is positive for $0 \le a < 1$ and semipositive for $a = 1$. Thus every coefficient up to one is feasible. For the reverse inequality, the metrics $\gamma_t = \alpha + t\zeta$, $t > 0$, are Gauduchon, since

$$\mathrm{dd^c}(\gamma_t^2) = -2t\alpha^3 = 0.$$

Using $\zeta^2 = 0$ gives

$$\frac{\int_X \omega \wedge \gamma_t^2}{\deg_{\gamma_t} H} = \frac{1+2t}{2t} \longrightarrow 1.$$

The threshold formula proves $\tau_\omega(H) = 1$.

For $a < 1$,

$$R_a^3 = 3(1-a)^2\alpha^2 \wedge \zeta = (1-a)^2\omega^3 = 6(1-a)^2\mu_0.$$

The zero remainder already solves the residual smooth equation, so scalar uniqueness gives its volume. At $a = 1$, $R_1 = \zeta$ is nef because $\zeta + \epsilon\omega > 0$. The same computation gives

$$\mathcal{V}_{\mu_0}(\zeta + \epsilon\omega) = 6\epsilon^2(1+\epsilon) \longrightarrow 0.$$

Nef continuity gives its zero residual volume. Also $\omega + \mathrm{dd^c}\sigma_H = [H] + \zeta$, so the non-pluripolar measure of σ_H is $\zeta^3 = 0$.

Finally fix $a < 1$ and choose $0 < \delta \leq \min\{1-a, 1\}$. Then $R_a - \delta\omega \geq 0$. For the smooth functions V_j in Theorem 8.0.1,

$$R_a + \mathrm{dd^c}(\delta V_j) = (R_a - \delta\omega) + \delta(\omega + \mathrm{dd^c}V_j) > 0.$$

Monotonicity of the determinant on positive matrices bounds the mass below by $\delta^3 \int_X (\omega + \mathrm{dd^c}V_j)^3$, which tends to infinity. Thus finite residual volume and its explicit divisorial threshold coexist with unbounded upper mass. □

Part VII
Partial kernels and equilibrium

Chapter 26
Partial kernels and quotient equilibrium

Let X be a connected compact complex manifold of dimension $n \geq 1$, and let (L, h) be a holomorphic line bundle on X with a smooth Hermitian metric.

Partial Bergman kernels use a singular weight to select global holomorphic sections and a continuous weight in their norm. When the selecting potential is plurisubharmonic relative to the curvature of (L, h), the Bernstein–Markov estimate preserves the obstacle, while Lelong-number estimates on modifications preserve the multiplier-ideal constraint. Together they give the partial envelope as an upper bound. A matching lower bound requires global section approximation. With a Kähler modification and positive non-pluripolar mass, the closed theory also gives measure convergence; a Jacobian isometry transfers it to X.

Positivity against an unrelated Hermitian form permits different behavior. On a fibration, every section of a pulled-back line bundle descends even when the selecting potential does not. Moving a pole along a fiber can then preserve every kernel while changing the multiplier-ideal type. Point and divisor models exhibit this loss through explicit binomial tails. Section growth determines the normalization, and changing the integration volume separates the base equilibrium measure from the conditional fiber probabilities retained by its lift.

When vanishing orders grow sublinearly in the tensor degree, a weighted entire-function model gives a Gaussian transition on a shrinking scale. A uniform local norm comparison and a global L^2 correction show how far the same transition persists on a compact curve.

26.1 The selected space and the continuous weight

Fix a quasi-plurisubharmonic function φ and a smooth positive probability measure ν on X. For each integer $m \geq 1$ set

$$V_m = H^0\big(X, L^m \otimes \mathcal{I}(m\varphi)\big), \qquad N_m = \dim V_m.$$

For a real $v \in C(X)$ define the norm

$$\|s\|_{m,v,\nu}^2 = \int_X |s|_{h^m}^2 \mathrm{e}^{-mv}\,\mathrm{d}\nu \quad (s \in V_m).$$

The continuous weight is bounded, so the norm is finite on every global section. Its nondegeneracy follows from the strictly positive density and the identity theorem. This separation of the selecting weight from the norm follows [Xia, Section 14.3].

Fix an integer $r \geq 1$ for the normalization. Let $s_1, \dots, s_{N_m}$ be an orthonormal basis of V_m for this norm, and for $x \in X$ put

$$B_{m,v,\varphi,\nu}(x) = \mathrm{e}^{-mv(x)} \sum_j |s_j(x)|_{h^m}^2, \qquad \beta_m^{(r)} = \frac{r!}{m^r} B_{m,v,\varphi,\nu}\,\nu.$$

Empty spaces give zero kernels and zero measures. Unitary invariance and the evaluation argument of Proposition 19.1.2 give

$$B_{m,v,\varphi,\nu}(x) = \sup_{\substack{s \in V_m \\ \|s\|_{m,v,\nu} \leq 1}} |s(x)|_{h^m}^2 \mathrm{e}^{-mv(x)}, \qquad \int_X \beta_m^{(r)} = \frac{r! N_m}{m^r}.$$

The singular weight occurs only in V_m. In particular, ordering its multiplier ideals orders the kernels when h, v, ν are fixed, because it orders the spaces in the displayed supremum.

Replacing h by $h\mathrm{e}^{-a}$ and v by $v - a$, for smooth real a, leaves both the norm and the kernel measure unchanged.

26.2 A local Bernstein–Markov estimate

A Bernstein–Markov estimate compares the supremum norm with the integral norm up to an arbitrarily small exponential loss in the degree. The local submean inequality supplies it for a smooth positive volume and a continuous weight on the whole manifold.

For a compact set $K \subseteq X$ and $v \in C(K)$, write

$$\|s\|_{m,v,K} = \sup_K |s|_{h^m} \mathrm{e}^{-mv/2}.$$

A probability measure μ on K is Bernstein–Markov for (K, v) if, for every $\epsilon > 0$, there is C_ϵ such that

$$\|s\|_{m,v,K} \leq C_\epsilon \mathrm{e}^{\epsilon m} \left(\int_K |s|_{h^m}^2 \mathrm{e}^{-mv}\,\mathrm{d}\mu \right)^{1/2}$$

for all $m \geq 1$ and all $s \in H^0(X, L^m)$. If K is non-pluripolar, the supremum norm is nondegenerate: the zero set of a nonzero section is an analytic divisor, hence pluripolar. The Bernstein–Markov inequality then makes the integral norm nondegenerate as well. The same inequality holds on any selected subspace V_m.

Proposition 26.2.1 *Every smooth positive probability measure is Bernstein–Markov for (X, v) when $v \in C(X)$, on an arbitrary compact complex manifold.*

Proof Let ν be the smooth strictly positive probability measure under consideration. Choose finitely many nested coordinate charts with holomorphic frames for L. In a holomorphic frame e, write $|e|_h^2 = \mathrm{e}^{-\chi}$ and $s = fe^{\otimes m}$. Uniform continuity permits, for any $\epsilon > 0$, a uniform small radius r such that every point lies at the center of a coordinate ball of radius r contained in one of the larger charts, and the oscillation of $\chi + v$ on that ball is at most 2ϵ. The radius and the volume comparison constants can be chosen uniformly using the finite covering.

The submean inequality for the holomorphic function f and the lower bound for the density of ν give

$$\begin{aligned} |s(x)|_{h^m}^2 \mathrm{e}^{-mv(x)} &= |f(x)|^2 \mathrm{e}^{-m(\chi+v)(x)} \\ &\leq C_r \mathrm{e}^{2\epsilon m} \int_{B(x,r)} |f|^2 \mathrm{e}^{-m(\chi+v)} \, \mathrm{d}\nu \\ &\leq C_r \mathrm{e}^{2\epsilon m} \|s\|_{m,v,\nu}^2. \end{aligned}$$

Take the supremum over x and then the square root. □

For a general weighted compact subset, the Bernstein–Markov property will be an explicit hypothesis.

26.3 Partial envelopes on a weighted compact set

The norm weight imposes an obstacle on the set where sections are measured. The selecting weight imposes an ideal constraint on the whole manifold. Their joint envelope is bounded above by a translate of the normalized ideal envelope, because a non-pluripolar compact obstacle gives a global upper bound. The reverse inequality follows by translating that normalized envelope below the obstacle.

Let θ be a smooth real $(1,1)$-form on X, let $\varphi \in \mathrm{PSH}(X, \theta)$, let $K \subseteq X$ be compact and non-pluripolar, and let $w \in C(K, \mathbb{R})$. Use the multiplier-ideal order characterized in Theorem 1.16.1 and the quasi-everywhere convention of Section 1.19. Define

$$\begin{aligned} E_{\theta,K}[\varphi]_{\mathcal{I}}(w) = \Big(\sup\big\{a \in \mathrm{PSH}(X,\theta) : a \preceq_{\mathcal{I}} \varphi, \\ a \leq w \text{ quasi-everywhere on } K\big\}\Big)^*. \end{aligned} \tag{26.1}$$

Theorem 26.3.1 *The function $E = E_{\theta,K}[\varphi]_{\mathcal{I}}(w)$ is a genuine θ-plurisubharmonic function and the largest member of its defining family. Let $q = P_\theta[\varphi]_{\mathcal{I}}$, and let C be any upper bound from Lemma 1.19.1. Then*

$$q + \min_K w \le E \le q + C.$$

In particular, E has the ordinary singularity type of q, and

$$E \sim_{\mathcal{I}} \varphi.$$

Proof Theorem 15.2.1 gives $q \le 0$ and $q \sim_{\mathcal{I}} \varphi$. Thus $q + \min_K w$ belongs to the defining family. Lemma 1.19.1 bounds the entire family above by C, so its regularized supremum is genuine and gives the lower inequality.

For each competitor a, the upper bound $\sup_X a \le C$ and Propositions 15.2.1 and 15.2.2 give

$$a - C \le a - \sup_X a \le P_\theta[a]_{\mathcal{I}} \le P_\theta[\varphi]_{\mathcal{I}} = q.$$

Taking regularized suprema proves $E \le q + C$.

The two bounds show that E and q differ by a bounded amount. They therefore have the same multiplier ideals, and these equal those of φ.

It remains to check the obstacle after regularization. Choose a countable subfamily by Choquet's lemma, and replace it by successive finite maxima. Finite maxima remain admissible, by Theorem 1.16.1 and by taking the finite union of the obstacle's exceptional sets. Let a_j be the resulting increasing sequence. Outside a countable union of pluripolar subsets of K, every a_j is at most w. The unregularized limit is therefore at most w there. The regularized limit equals E and changes this limit only on a further pluripolar set, by the local supremum theorem in Section 1.19. Thus $E \le w$ quasi-everywhere on K.

We have proved that E is a member of its defining family. It dominates each competitor by construction, so it is the largest member; in particular the unregularized supremum in (26.1) already equals E at every point. □

Proposition 26.3.1 *The envelope depends only on the multiplier-ideal class of φ. It is increasing in the obstacle and in the multiplier-ideal order of φ. For $w, w' \in C(K)$ and $c \in \mathbb{R}$,*

$$E_{\theta,K}[\varphi]_{\mathcal{I}}(w+c) = E_{\theta,K}[\varphi]_{\mathcal{I}}(w) + c,$$
$$\left|E_{\theta,K}[\varphi]_{\mathcal{I}}(w) - E_{\theta,K}[\varphi]_{\mathcal{I}}(w')\right| \le \|w - w'\|_{C(K)}.$$

The second line means the two corresponding pointwise inequalities, including at poles. For $0 < t < 1$, smooth real $(1,1)$-backgrounds θ_i on X, for $i = 0, 1$, potentials $\varphi_i \in \mathrm{PSH}(X, \theta_i)$ and obstacles $w_i \in C(K)$, put $\theta_t = (1-t)\theta_0 + t\theta_1$, and similarly define φ_t, w_t. Then

$$(1-t)E_{\theta_0,K}[\varphi_0]_{\mathcal{I}}(w_0) + tE_{\theta_1,K}[\varphi_1]_{\mathcal{I}}(w_1) \le E_{\theta_t,K}[\varphi_t]_{\mathcal{I}}(w_t).$$

For a smooth real f, changing the background and obstacle gives the exact identity

$$E_{\theta+\mathrm{dd^c}f,K}[\varphi - f]_{\mathcal{I}}(w - f|_K) = E_{\theta,K}[\varphi]_{\mathcal{I}}(w) - f.$$

Proof Equality or inclusion of the defining families proves the first two assertions. Adding a constant gives a bijection of competitor families, proving translation. Apply monotonicity to $w' \pm \|w - w'\|_{C(K)}$ to obtain the two inequalities in the second line.

For concavity, the left side is θ_t-plurisubharmonic and obeys the obstacle w_t outside the union of two pluripolar exceptional sets. Each of its summands has the multiplier-ideal class of its prescribed singularity by the theorem. Addition and positive scaling in Lemma 1.16.3 give the multiplier-ideal class of φ_t. Thus the left side is an admissible competitor on the right. Finally, $a \mapsto a - f$ is a bijection of the two families in the gauge identity, and regularization commutes with subtraction of a smooth function. □

This envelope provides the upper bound for partial Fubini–Study limits. To apply it, we must retain both the ideal constraint and the compact obstacle under L^1 convergence.

26.4 An upper bound for partial Fubini–Study limits

The Bernstein–Markov estimate bounds the partial Fubini–Study potentials from above. Their ideal constraints survive in the limit because the loss in each Lelong-number estimate is of order m^{-1} on a fixed modification. The compact-set envelope retains the obstacle even when the set has zero smooth volume.

Now take $\theta = c_1(L, h)$, assume $\varphi \in \mathrm{PSH}(X, \theta)$, and choose a Bernstein–Markov probability μ for (K, w). Use the selected spaces V_m and dimensions N_m from Section 26.1. Choose an orthonormal basis $s_1, \dots, s_{N_m}$ for the continuous weighted norm of Section 26.2, and define the global partial Fubini–Study potential

$$U_m = \frac{1}{m} \log \sum_{j=1}^{N_m} |s_j|^2_{h^m} \quad (N_m > 0).$$

Set $U_m = -\infty$ if $N_m = 0$. For a nonempty space the local expression in a frame shows that $U_m \in \mathrm{PSH}(X, \theta)$ and has analytic singularity type. On K, it satisfies $\mathrm{e}^{m(U_m - w)} = B_{m,w,\varphi,\mu}$.

For a local smooth modification $\pi : Y \to U_{\mathrm{loc}}$ and $y \in Y$, write J_π for its local holomorphic Jacobian determinant and apply Lemma 1.17.8 to every basis coefficient. The Lelong number of the logarithm of a sum of squared holomorphic functions is the minimum of their orders. The smooth metric contributes zero, so

$$\nu(U_m \circ \pi, y) \geq \nu(\varphi \circ \pi, y) - \frac{n + \mathrm{ord}_y J_\pi}{m} \qquad (N_m > 0).$$

Theorem 26.4.1 *Along the indices for which $N_m > 0$, the potentials U_m form a relatively compact family in $L^1(X)$. Every cluster limit U along indices tending to infinity is genuine, has multiplier-ideal order*

$$U \preceq_{\mathcal{I}} \varphi,$$

and satisfies $U \leq w$ quasi-everywhere on K. Consequently

$$\left(\limsup_{m \to \infty} U_m\right)^* \leq E_{\theta,K}[\varphi]_{\mathcal{I}}(w).$$

If only finitely many spaces are nonempty, the last left side is $-\infty$.

Proof The extremal formula for the kernel and the Bernstein–Markov inequality give, with $C_\epsilon \geq 1$,

$$U_m \leq w + \delta_{m,\epsilon} \quad \text{on } K, \qquad \delta_{m,\epsilon} = 2\epsilon + \frac{2 \log C_\epsilon}{m}.$$

The extremal formula takes the supremum over a single unit section, so no dimension factor occurs in this estimate. Fixing one $\epsilon > 0$, Lemma 1.19.1 gives a uniform global upper bound for all nonempty indices. On the other hand, the trace identity gives

$$\int_K \mathrm{e}^{m(U_m - w)} \, \mathrm{d}\mu = N_m \geq 1.$$

Since μ is a probability measure, this implies $\sup_X U_m \geq \min_K w$. The normalized compactness theorem now gives relative L^1 compactness with genuine limits.

Let $U_{m_j} \to U$ in $L^1(X)$, with $m_j \to \infty$. Fix any local smooth modification $\pi : Y \to U_{\mathrm{loc}}$ over an open subset $U_{\mathrm{loc}} \subseteq X$, and any point $y \in Y$. Lemma 1.22.2 supplies L^1_{loc} convergence upstairs. Upper semicontinuity of Lelong numbers in Lemma 1.17.5 and the preceding order bound yield

$$\nu(U \circ \pi, y) \geq \limsup_j \nu(U_{m_j} \circ \pi, y) \geq \nu(\varphi \circ \pi, y).$$

The local valuative characterization in Theorem 1.16.1 therefore gives $U \preceq_{\mathcal{I}} \varphi$ at every positive weight. This argument fixes a modification and a point before passing to the limit; the already convergent subsequence works for every such choice.

To retain the obstacle on a possibly thin compact set, let V_θ be the normalized potential of minimal singularities and set

$$G = E_{\theta,K}[V_\theta]_{\mathcal{I}}(w).$$

Every $a \in \mathrm{PSH}(X, \theta)$ satisfies $a \leq V_\theta + \sup_X a$, and hence $a \preceq_{\mathcal{I}} V_\theta$. Thus the ideal constraint in the definition of G imposes no further restriction: G is the largest θ-plurisubharmonic function below w quasi-everywhere on K. In particular, the Bernstein–Markov estimate gives globally

$$U_m \leq G + \delta_{m,\epsilon}.$$

Passing to the L^1 limit gives $U \leq G + 2\epsilon$ almost everywhere, and then everywhere by Lemma 1.3.3. Let $\epsilon \downarrow 0$. Since $G \leq w$ quasi-everywhere on K by Theorem 26.3.1, the same holds for U. Both defining conditions for the partial envelope have now been checked, so $U \leq E_{\theta,K}[\varphi]_{\mathcal{I}}(w)$.

Finally, the local Hartogs lemma says that if $U_{m_j} \to U$ in L^1, then for any $x_j \to x$ one has $\limsup_j U_{m_j}(x_j) \leq U(x)$. Apply this after extracting an L^1 convergent subsequence from any sequence of nonempty indices and points approaching x. Every resulting limit is at most the partial envelope. This proves the regularized upper-limit inequality, including at its poles. Empty indices contribute only $-\infty$. □

A matching lower bound requires sections approximating the envelope. The projective models later in the chapter construct them explicitly. To study measures, we first integrate the kernels as derivatives of a determinant functional.

26.5 The determinant functional

The logarithm of the Gram determinant integrates the partial kernel measure as the continuous weight varies. Orthogonal projection expresses its second derivative as a negative squared norm, giving concavity and a uniform bound in terms of the section dimension.

Return to the full-manifold norm $\|\cdot\|_{m,v,\nu}$ of Section 26.1, with a quasi-plurisubharmonic selecting weight φ. Fix m with $N_m > 0$ and a basis of V_m. Let $G_m(v)$ be its positive Hermitian Gram matrix in this norm. Define

$$\mathcal{F}_m^{(r)}(v) = -\frac{r!}{m^{r+1}} \log \frac{\det G_m(v)}{\det G_m(0)}.$$

The quotient makes the definition independent of the chosen basis. Put $\mathcal{F}_m^{(r)} = 0$ if $N_m = 0$.

Proposition 26.5.1 *For $v, f \in C(X, \mathbb{R})$,*

$$\frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} \mathcal{F}_m^{(r)}(v + tf) = \int_X f \,\mathrm{d}\beta_m^{(r)}.$$

For constants $c \in \mathbb{R}$ and weights $w \in C(X, \mathbb{R})$, the functional is concave and satisfies

$$\mathcal{F}_m^{(r)}(v + c) = \mathcal{F}_m^{(r)}(v) + \frac{r!N_m}{m^r}c, \qquad |\mathcal{F}_m^{(r)}(v) - \mathcal{F}_m^{(r)}(w)| \le \frac{r!N_m}{m^r}\|v - w\|_\infty.$$

If $N_m > 0$, its second derivative along $v + tf$ is strictly negative unless f is constant.

Proof Use a basis $s_1, \dots, s_{N_m}$ orthonormal at v, so $G_m(v) = I$, the identity matrix of size N_m. Matrix indices i, j below range from 1 to N_m. Boundedness of the weights justifies differentiation under the integral. With pointwise inner products taken in h^m, put

$$A_{ij} = \int_X f\langle s_i, s_j\rangle_{h^m} \mathrm{e}^{-m\nu} \,\mathrm{d}\nu, \qquad C_{ij} = \int_X f^2\langle s_i, s_j\rangle_{h^m} \mathrm{e}^{-m\nu} \,\mathrm{d}\nu.$$

Then $G_m' = -mA$ and $G_m'' = m^2C$. The identity $(\log\det G)' = \mathrm{tr}(G^{-1}G')$ gives the first derivative. Differentiating it once more yields

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}\bigg|_{t=0} \mathcal{F}_m^{(r)}(v + tf) = -\frac{r!}{m^{r-1}}\left(\int_X f^2 B_{m,v,\varphi,\nu} \,\mathrm{d}\nu - \sum_{i,j} |A_{ij}|^2\right).$$

Let P_v be the orthogonal projection of the full weighted L^2 space of measurable sections onto V_m. The quantity in parentheses equals

$$\sum_j \|(I - P_v)(fs_j)\|_{m,v,\nu}^2 \ge 0.$$

This proves concavity. Multiplication of every Gram entry by e^{-mc} proves the translation rule. The first derivative and the total mass identity bound the derivative along the segment from v to w, giving the Lipschitz estimate.

If the second derivative vanishes, choose $0 \ne s \in V_m$ with $fs = \widetilde{s} \in V_m$ in L^2. Continuity and the positive density give equality everywhere. The real holomorphic quotient $\widetilde{s}/s = f$ is constant, say c, on a ball where $s \ne 0$. The identity theorem gives

$\widetilde{s} = cs$ on connected X, and continuity gives $f = c$ everywhere. The translation rule proves the converse. □

The derivative here is a partial kernel measure. The obstruction to a primitive for the raw Hermitian Monge–Ampère operator remains the one proved in Chapter 3.

Corollary 26.5.1 *If $N_m = o(m^n)$, then $\beta_m^{(n)} \to 0$ in total variation. Also $\mathcal{F}_m^{(n)}(v) \to 0$ uniformly when v ranges over a bounded subset of $C(X)$.*

Proof The measures are positive and their total masses are $n!N_m/m^n$. Use the Lipschitz estimate and $\mathcal{F}_m^{(n)}(0) = 0$ for the second assertion. □

For example, let X be any connected compact complex manifold that is not Moishezon. By Corollary 20.5.1, either every positive-degree section space of L is zero, or $h^0(X, L^m) = O(m^d)$ for some $d < n$. Every selected subspace therefore satisfies Corollary 26.5.1. This applies in particular whenever X fails the bounded mass property.

26.6 Partial kernels under a modification

A modification preserves holomorphic sections, but it changes the integrability condition defining a multiplier ideal. The correction is the relative canonical bundle. With its natural metric, the same correction gives an exact identity for partial kernel measures. This permits the Kähler convergence theorem to be transferred to a manifold admitting a Kähler modification.

Let $\pi : Y \to X$ be a modification of connected compact complex manifolds. Use the exceptional set $S \subset X$, its inverse image $D = \pi^{-1}(S) \subset Y$, the relative canonical bundle R_π and its Jacobian section J_π from Section 1.22.4. The ideal transformation and order invariance are Lemmas 1.22.4 and 1.22.5.

Proposition 26.6.1 *Let θ be a smooth closed real $(1,1)$-form on X and let $\varphi \in \mathrm{PSH}(X, \theta)$. For $v \in C(X)$, put $E_X(v) = E_{\theta,X}[\varphi]_{\mathcal{I}}(v)$. Then*

$$E_{\pi^*\theta,Y}[\varphi \circ \pi]_{\mathcal{I}}(v \circ \pi) = E_X(v) \circ \pi.$$

For any $u \in \mathrm{PSH}(X, \theta)$ the non-pluripolar measures also satisfy

$$\pi_* \mathrm{MA}_{\pi^*\theta}(u \circ \pi) = \mathrm{MA}_\theta(u),$$

initially as extended positive measures if finiteness has not yet been established.

Proof The potential bijection is Proposition 1.22.2.

For a continuous obstacle on the whole manifold, the quasi-everywhere and pointwise competitor conditions agree by Section 1.19. Surjectivity of π and Lemma 1.22.5

now identify the two competitor families. Each partial envelope is the largest member of its family by Theorem 26.3.1. This proves the envelope identity without interchanging pullback and an unverified regularized supremum.

Finally, the measures agree under the biholomorphism away from D and S by locality of the non-pluripolar construction. They give these analytic sets zero mass. Testing nonnegative Borel functions, or increasing compact subsets of the complements, proves the pushforward identity, including the possibility of infinite total mass. □

Choose smooth positive probability volumes ν_X, ν_Y. They determine a smooth metric h_R on R_π such that

$$|J_\pi|^2_{h_R}\, \nu_Y = \pi^* \nu_X. \tag{26.2}$$

For example, write the coordinate volume densities as f_X, f_Y. The local frame of R_π given by the quotient of coordinate canonical frames has squared norm $(f_X \circ \pi)/f_Y$. This is smooth and strictly positive even where J_π vanishes, and the coordinate transformation laws make it a metric. Equation (26.2) concerns smooth top forms; it does not define the pullback of an arbitrary measure.

Proposition 26.6.2 *Let the holomorphic line bundle $L \to X$, smooth Hermitian metric h, selecting weight φ and continuous weight v be as in Section 26.1, with $v \in C(X)$ and $\varphi \in \mathrm{QPSH}(X)$. Use ν_X downstairs. Upstairs use $\pi^* L$, the fixed auxiliary bundle (R_π, h_R), the singular weight $\varphi \circ \pi$, the continuous weight $v \circ \pi$, and ν_Y. Then the identification*

$$H^0(X, L^m \otimes \mathcal{I}(m\varphi)) \simeq H^0(Y, \pi^* L^m \otimes R_\pi \otimes \mathcal{I}(m\varphi \circ \pi))$$

is an isometry for the continuous weighted norms. Their kernels and measures obey

$$B_{Y,m} = |J_\pi|^2_{h_R}(B_{X,m} \circ \pi), \qquad \pi_* \beta^{(r)}_{Y,m} = \beta^{(r)}_{X,m}$$

for the same normalization integer r. The formulas include empty section spaces.

Proof The section identification is Lemma 1.22.4 with $M = L^m$ and $a = m\varphi$. By (26.2) and change of variables,

$$\int_Y |J_\pi \pi^* s|^2_{\pi^* h^m \otimes h_R} \mathrm{e}^{-mv\circ\pi}\, \mathrm{d}\nu_Y = \int_X |s|^2_{h^m} \mathrm{e}^{-mv}\, \mathrm{d}\nu_X.$$

The singular weight has already selected the section space; it does not enter this norm. Apply the isometry to an orthonormal basis and sum its squared evaluations to obtain the kernel formula, including at zeros of J_π. Multiplying by $r!/m^r$ and testing a continuous function pulled back from X proves the measure formula. □

26.6.1 Convergence with a Kähler modification

The Jacobian isometry allows the Kähler equilibrium theorem to be pushed down from a modification. The resulting limit is intrinsic because both the partial envelope and its non-pluripolar measure descend.

Theorem 26.6.1 *Suppose X admits a modification from a compact Kähler manifold. Let $L \to X$ be a holomorphic line bundle with smooth Hermitian metric h and curvature $\theta = c_1(L, h)$, and let $\varphi \in \mathrm{PSH}(X, \theta)$ satisfy*

$$\mathrm{MA}_\theta(\varphi)(X) > 0.$$

For every $v \in C(X)$ and smooth positive probability volume ν_X, the partial kernel measures on the whole manifold satisfy

$$\frac{n!}{m^n} B_{m,v,\varphi,\nu_X}\, \nu_X \rightharpoonup \mathrm{MA}_\theta(E_X(v)).$$

The measure on the right is finite and nonzero. Its total mass equals

$$M = \lim_{m\to\infty} \frac{n!}{m^n} h^0(X, L^m \otimes \mathcal{I}(m\varphi)),$$

which is independent of v and ν_X. The limit measure is independent of the chosen Kähler modification.

Proof Choose a Kähler modification $\pi : Y \to X$ and a smooth positive probability volume ν_Y. The pulled-back potential belongs to $\mathrm{PSH}(Y, \pi^*\theta)$, and its non-pluripolar mass is the given positive mass by Proposition 26.6.1. In particular that mass is finite, by the closed Kähler theory.

Apply [Xia, Theorem 14.3.1] on Y, with line bundle π^*L, auxiliary twisting line bundle R_π, continuous obstacle $v \circ \pi$, and smooth probability ν_Y. Its Bernstein–Markov hypothesis follows from the proof of Proposition 26.2.1; a fixed smooth auxiliary bundle only changes the local constants. The ideal convention and continuous weighted norm agree exactly with those in the preceding proposition. The source theorem gives

$$\beta^{(n)}_{Y,m} \rightharpoonup \mathrm{MA}_{\pi^*\theta}\big(E_{\pi^*\theta,Y}[\varphi \circ \pi]_{\mathcal{I}}(v \circ \pi)\big).$$

The equilibrium measure has finite positive mass under the stated hypothesis.

Push forward the convergence, testing continuous functions on X. The kernel identity, envelope identity and non-pluripolar pushforward identity give the displayed downstairs limit. Testing the constant function one proves the dimension formula. The dimensions do not involve v or either volume. Finally the limiting expression is defined entirely on X, proving independence of the modification. □

The positive dimension limit forces X to be Moishezon by Corollary 20.5.1. The Kähler modification supplies the asymptotic theorem and also implies BMP by Lemma 20.4.1; the Jacobian identities themselves hold for every smooth compact

modification. When ambient section growth is smaller, the trace identity instead points to normalization in that lower growth degree.

Corollary 26.6.1 *Under the hypotheses of Theorem 26.6.1, write M for the limiting normalized dimension in that theorem. The determinant functionals with $r = n$ converge pointwise on $C(X)$ to*

$$\mathcal{F}(v) = \int_0^1 \left(\int_X v \, \mathrm{dMA}_\theta(E_X(tv)) \right) \mathrm{d}t.$$

This functional is concave and vanishes at zero. For $v, w \in C(X, \mathbb{R})$ and $c \in \mathbb{R}$,

$$\mathcal{F}(v + c) = \mathcal{F}(v) + Mc, \qquad |\mathcal{F}(v) - \mathcal{F}(w)| \le M\|v - w\|_\infty.$$

Proof Put $N_m = \dim H^0(X, L^m \otimes \mathcal{I}(m\varphi))$. Integrate the finite-dimensional first-variation identity in Proposition 26.5.1 along tv. For each fixed t the integrand converges by the theorem. Its absolute value is bounded by

$$\|v\|_\infty \frac{n! N_m}{m^n},$$

and this sequence of constants is bounded because it converges to M. Dominated convergence on $[0, 1]$ gives the formula. The limiting integrand is measurable as the pointwise limit of continuous finite-dimensional derivatives. Concavity, the translation identity and the Lipschitz bound pass from the finite-dimensional functionals to their pointwise limit. □

This argument integrates weak convergence along a weight segment. A lower bound for the partial Fubini–Study potentials in Theorem 26.4.1 still requires a generation argument.

26.7 Kernels on a fibration

Compact connected fibers make every section of a pulled-back line bundle descend. Choosing probability measures on the fibers makes this descent an isometry; the kernel measures then lift by fiber integration.

Let $\pi : X \to B$ be a surjective holomorphic submersion of connected compact complex manifolds, with connected fibers. Write $n = \dim X$ and $q = \dim B \ge 1$. Let A be a holomorphic line bundle on B with smooth metric h_A, and choose a quasi-plurisubharmonic function ψ on B. Set

$$(L, h) = (\pi^* A, \pi^* h_A), \qquad \varphi = \pi^* \psi.$$

For $y \in B$, write $X_y = \pi^{-1}(y)$. Choose a smoothly varying family of smooth strictly positive probability measures ν_y on these fibers. Such a family is obtained by restricting any smooth positive vertical volume density and dividing by its fiber integral. For $F \in C(X)$ set

$$\mathcal{A}F(y) = \int_{X_y} F \, \mathrm{d}\nu_y.$$

The function $\mathcal{A}F$ is continuous. Indeed, a proper smooth submersion is locally a smooth fiber bundle; in a trivialization the smooth densities vary smoothly, and uniform continuity of F on compact pieces gives continuity of the integrals. For a finite Borel measure μ on B, define its lift $\mathcal{L}\mu$ by

$$\int_X F \, \mathrm{d}\mathcal{L}\mu = \int_B \mathcal{A}F \, \mathrm{d}\mu.$$

Positivity and the Riesz representation theorem give a finite measure with the same total mass as μ. In particular, weak convergence of finite measures on the base implies weak convergence of their lifts.

Proposition 26.7.1 *Pullback identifies*

$$H^0\big(B, A^m \otimes \mathcal{I}(m\psi)\big) \simeq H^0\big(X, L^m \otimes \mathcal{I}(m\varphi)\big).$$

Let $K \subseteq B$ be closed and non-pluripolar, let $w \in C(K)$, and let μ be a Bernstein–Markov probability measure for (K, w) on B. Set $K_X = \pi^{-1}(K)$, $w_X = \pi^* w$ and $\mu_X = \mathcal{L}\mu$. The identified spaces have identical norms for these weighted sets and measures. Their partial kernel measures, normalized with the same integer r, satisfy

$$\beta^{(r)}_{X,m} = \mathcal{L}\big(\beta^{(r)}_{B,m}\big).$$

The Bernstein–Markov inequality itself also transfers to the unweighted section spaces on X.

Proof Every section of $\pi^* A^m$ descends. On the inverse image of a trivializing base chart, its coefficient is a holomorphic function and is constant on every compact connected fiber. Local holomorphic sections of the submersion show that the descended coefficient is holomorphic. These coefficients glue by the transition functions of A^m, giving the unweighted identification.

By Lemma 1.16.4, the upstairs multiplier ideal is the extension of $\mathcal{I}(m\psi)$. A descended coefficient belongs to this extended ideal exactly when its base coefficient belongs to $\mathcal{I}(m\psi)$: pullback gives one implication, and restriction to a local holomorphic section gives the other. This proves the selected-space identification.

Pointwise norms of corresponding sections agree and are constant along fibers. Since ν_y has mass one,

$$\int_{K_X} |\pi^* s|^2_{h^m} \mathrm{e}^{-mw_X} \, \mathrm{d}\mu_X = \int_K |s|^2_{h_A^m} \mathrm{e}^{-mw} \, \mathrm{d}\mu.$$

Their supremum norms also agree by surjectivity. Thus the Bernstein–Markov inequality transfers, and the integral norms are nondegenerate. Pulling back an orthonormal basis proves $B_{X,m} = \pi^* B_{B,m}$ on K_X. Integrating a continuous test function first along the fibers proves the measure identity. Kernels may be set to zero off their respective weighted sets. □

26.7.1 The lifted equilibrium measure

On a Kähler base, the partial kernel measures converge to the Monge–Ampère measure of the partial envelope. Continuity of fiber integration then gives a limit upstairs with the base-dimensional normalization.

Keep the submersion and weighted compact set (K, w, μ) above. Assume that B is Kähler, put $\theta = c_1(A, h_A)$, and take $\psi \in \mathrm{PSH}(B, \theta)$ with positive non-pluripolar Monge–Ampère mass. Write $B_{B,m}, B_{X,m}$ for the corresponding kernels, $\mu_X = \mathcal{L}\mu$, and

$$E = E_{\theta,K}[\psi]_{\mathcal{I}}(w).$$

Theorem 26.7.1 *Under these hypotheses,*

$$\frac{q!}{m^q} B_{X,m}\, \mu_X \rightharpoonup \mathcal{L}\big(\mathrm{MA}_\theta(E)\big).$$

If $n > q$, the measures with normalization $n!/m^n$ tend to zero in total variation.

Proof Apply [Xia, Theorem 14.3.1] on the compact Kähler base, with its auxiliary twisting line bundle trivial. Its hypotheses are precisely the positive-mass potential ψ, the closed non-pluripolar set K, the continuous weight w, and the Bernstein–Markov probability μ. It gives

$$\frac{q!}{m^q} B_{B,m}\, \mu \rightharpoonup \mathrm{MA}_\theta(E).$$

Proposition 26.7.1 and continuity of the lifting operation give the asserted upstairs limit. In particular the q-normalized total masses are bounded. Multiplying them by $(n!/q!)m^{q-n}$ proves the last statement when $n > q$. □

26.7.2 The Hopf equilibrium measure

The vertical part of the Hopf metric converts the lifted base equilibrium measure into a Hermitian Monge–Ampère measure. Its coefficient records both the fiber probability and the top-product expansion.

Use X_n, $n \geq 2$, its projection $\pi : X_n \to \mathbb{P}^{n-1}$ and the forms $\sigma, \alpha, \zeta, \omega$ of Chapter 21. Take $A = \mathcal{O}(1)$ with curvature $\theta = \sigma$ and retain the selecting potential ψ and weighted data (K, w, μ) above. On each elliptic fiber set $\nu_y = \zeta|_{X_y}/2$. For any base measure τ its lift is

$$\mathcal{L}\tau = \tfrac{1}{2}\pi^*\tau \wedge \zeta$$

in the precise fiber-integration notation of that chapter. Since $E \in \mathrm{PSH}(B, \sigma)$, Theorem 21.2.1 identifies the limit as

$$\frac{(n-1)!}{m^{n-1}} B_{X_n,m}\, \mu_X \rightharpoonup \frac{1}{2n}\mathrm{MA}^{\mathrm{ext}}_{\omega}(\pi^* E). \tag{26.3}$$

The factor $1/(2n)$ combines the fiber probability with the top-product coefficient in $\omega = \alpha + \zeta$. The closed curvature α has zero top product on these descended potentials; the vertical form supplies the nonzero Hermitian measure. For $n = 3$ this gives a nonzero limit on a manifold failing BMP, with normalization $2/m^2$.

Every potential positive against the closed line-bundle curvature $k\alpha$, for a positive integer k, descends by Theorem 21.1.1. A potential positive only against ω can instead have an isolated pole. Its sections still descend, so the selected space records the pole through a condition on the base. We now determine that condition.

26.8 Isolated logarithmic poles on a fibration

In sufficiently large tensor powers, an isolated pole on a fiber forces every selected section to vanish at its image on the base. Integrability uses the ambient dimension, so its threshold differs from the base point-pole threshold by at most the fiber dimension. The intervening jets have dimension $O(m^{q-1})$ on a q-dimensional base; after normalization by m^q, they leave the equilibrium limit unchanged.

For nested section spaces, the normalized codimension is exactly the total variation of their kernel-measure difference.

Lemma 26.8.1 *Let B be a connected compact complex manifold of dimension $q \geq 1$. Let $A \to B$ be a holomorphic line bundle with a smooth Hermitian metric. For every integer $m \geq 1$, let $W_m \subseteq V_m \subseteq H^0(B, A^m)$ have the continuous weighted norms determined by the same smooth metric, weight $v \in C(B)$ and smooth positive probability volume ν. Write $d_m = \dim V_m - \dim W_m$. Use the kernels $B_{V_m,v}, B_{W_m,v}$ and measures $\beta^{(q)}_{V_m,v}, \beta^{(q)}_{W_m,v}$ of Section 26.1, with the selected space indicated by the subscript. Let $\|\cdot\|_{\mathrm{TV}}$ denote total variation. Then*

$$B_{V_m,v} - B_{W_m,v} \geq 0,$$
$$\|\beta^{(q)}_{V_m,v} - \beta^{(q)}_{W_m,v}\|_{\mathrm{TV}} = \frac{q!d_m}{m^q}.$$

Their determinant functionals, each normalized to vanish at the zero weight, satisfy

$$|\mathcal{F}^{(q)}_{V_m}(v) - \mathcal{F}^{(q)}_{W_m}(v)| \leq \frac{q!d_m}{m^q}\|v\|_\infty.$$

These statements include zero-dimensional spaces.

Proof Extend an orthonormal basis of W_m to one of V_m for the norm at v. The kernel difference is the sum of the squared weighted evaluations of the added basis vectors. It is nonnegative, and its integral is their number d_m. This gives the total-variation formula. Apply the first-variation proof of Proposition 26.5.1 to each of the two fixed subspaces and integrate their difference along tv, $0 \leq t \leq 1$. At each t, the just-proved positive kernel difference has total mass $q!d_m/m^q$. Its integral against v is therefore bounded in absolute value by that mass times $\|v\|_\infty$. The normalization at zero proves the last assertion. □

Return to a submersion $\pi : X \to B$ and a pullback line bundle $L = \pi^*A$ as in Section 26.7, and put $f = n - q$. Fix distinct points $p_1, \ldots, p_N$ on X and positive rational numbers $a_1, \ldots, a_N$. Suppose $\varphi \in \mathrm{QPSH}(X)$ is locally bounded away from these points and, in local coordinates x centered at p_j, has the form

$$\varphi(x) = a_j \log|x|^2 + O(1). \tag{26.4}$$

Write

$$Z = \{\pi(p_1), \ldots, \pi(p_N)\}, \qquad a_b = \max_{\pi(p_j)=b} a_j \quad (b \in Z).$$

For an integer $m \geq 1$, set

$$r_{m,b} = \max\{\lfloor ma_b \rfloor - n + 1, 0\}, \qquad s_{m,b} = \max\{\lfloor ma_b \rfloor - q + 1, 0\}.$$

Here $\mathfrak{m}_b$ denotes the ideal sheaf of the point b, with zeroth power $\mathcal{O}_B$.

Proposition 26.8.1 *Pullback identifies the selected space on X with*

$$V_m = H^0\left(B, A^m \otimes \bigcap_{b\in Z} \mathfrak{m}_b^{r_{m,b}}\right).$$

If

$$W_m = H^0\left(B, A^m \otimes \bigcap_{b\in Z} \mathfrak{m}_b^{s_{m,b}}\right),$$

then $W_m \subseteq V_m$ and

$$0 \le \dim V_m - \dim W_m \le \sum_{b\in Z} \sum_{j=r_{m,b}}^{s_{m,b}-1} \binom{j+q-1}{q-1} = O(m^{q-1}).$$

An empty inner sum is zero. The constants depend on the fixed point data, not on m.

Proof Lemma 1.16.1, with a point as center, gives $\mathcal{I}(m\varphi)_{p_j} = \mathfrak{m}_{p_j}^{\max\{\lfloor ma_j\rfloor - n+1,0\}}$. Away from the specified points the ideal is the full structure sheaf.

Every section of L^m descends to B, by the compact-fiber argument in Proposition 26.7.1. Near p_j, choose submersion coordinates (z,τ) with $\pi(z,\tau) = z$. A pulled-back section has a local coefficient $g(z)$ independent of τ. Its order at $(0,0)$ equals its order at $z = 0$. The conditions from points in the same fiber are thus the same kind of point-vanishing condition, and their maximum is $r_{m,b}$. This proves the exact space identification.

For the dimension bound, send a section of V_m to its germs modulo $\mathfrak{m}_b^{s_{m,b}}$ at the finitely many points b. The kernel is W_m. Local trivializations of A^m therefore give an injection

$$V_m/W_m \hookrightarrow \bigoplus_{b\in Z} \mathfrak{m}_b^{r_{m,b}}/\mathfrak{m}_b^{s_{m,b}}.$$

No surjectivity of a jet-evaluation map is required. In dimension q, the degree-j homogeneous monomials number $\binom{j+q-1}{q-1}$. They compute the displayed quotient dimensions. Finally $0 \le s_{m,b} - r_{m,b} \le f$ and $s_{m,b} = O(m)$, proving the bound, also when $q = 1$ or $f = 0$. □

Equip A with a smooth metric h_A and put $\theta = c_1(A, h_A)$. Choose a smooth positive probability volume ν_B, smooth fiber probabilities as in the preceding sections, and their smooth positive lift $\nu_X = \mathcal{L}\nu_B$. Use the pullback metric on L. For $v \in C(B)$, use $v \circ \pi$ in the continuous norm on X.

Theorem 26.8.1 *Assume B is Kähler and there is $\psi \in \mathrm{PSH}(B,\theta)$ of positive non-pluripolar mass, locally bounded off Z, with*

$$\psi(z) = a_b \log|z-b|^2 + O(1) \quad \textit{near each } b \in Z.$$

Let $E(v) = E_{\theta,B}[\psi]_{\mathcal{I}}(v)$. For the selecting weight φ in (26.4),

$$\beta^{(q)}_{X,m,v\circ\pi,\varphi} \rightharpoonup \mathcal{L}\big(\mathrm{MA}_\theta(E(v))\big).$$

More precisely, its total-variation distance from the lift of the base partial kernel measure selected by ψ is $O(m^{-1})$. Its determinant functionals satisfy

$$\mathcal{F}^{(q)}_{X,m}(v\circ\pi) \longrightarrow \int_0^1 \left(\int_B v\,\mathrm{dMA}_\theta(E(tv))\right)\mathrm{d}t.$$

The normalized dimensions converge to the finite positive equilibrium mass. If $n > q$, normalization by $n!/m^n$ gives zero kernel measures in total variation.

Proof Lemma 1.16.1 in dimension q identifies the base multiplier-ideal-selected space with W_m in the proposition. Corresponding sections in V_m and on X have equal continuous norms, since the pointwise norm and continuous weight descend and every fiber probability has mass one. The same orthonormal-basis argument as in Proposition 26.7.1 gives the exact lift of the kernel associated with V_m on B.

By Lemma 26.8.1 and Proposition 26.8.1, the normalized kernel difference between V_m and W_m is positive with total mass $O(m^{-1})$. Its lift remains positive with the same mass. Apply Theorem 26.6.1 to B, using the identity modification: all its closed Kähler and positive-mass hypotheses are among the assumptions here. It gives the base W_m convergence, and the vanishing kernel difference proves the asserted limit on X.

The determinant comparison bounds the difference of the two functionals by $O(m^{-1})\|v\|_\infty$. The exact norm identification and Corollary 26.6.1 on B give the displayed limit. Testing one proves the dimension assertion. The q-normalized masses are bounded, so multiplication by $(n!/q!)m^{q-n}$ proves the last assertion. □

For a positive curvature form θ, Lemma 1.16.2 constructs the required ψ when the coefficients are sufficiently small. Its curvature dominates a positive multiple of θ off the poles, so its non-pluripolar mass is positive; closed Kähler theory makes it finite. Applying the same lemma on X with a Hermitian metric constructs φ. If the fibers have positive dimension and the pole set is nonempty and finite, φ cannot descend, since it is finite at other points of a fiber containing a pole.

The exact section formula also explains a failure of singularity recovery. Moving an isolated pole within its fiber does not change any selected section space. Adding finitely many poles in that same fiber does not change the spaces when the maximal coefficient there is unchanged. For the fixed pullback metric $h = \pi^* h_A$, continuous weight v and lifted measure ν_X, all kernels and determinant functionals are then exactly the same in every degree. The multiplier-ideal types on X are nevertheless different: at sufficiently large weight one ideal is proper at a point where the other is the

full local ring. Under the theorem's hypotheses this loss of information coexists with a positive q-dimensional growth limit. Closed-curvature positivity of the selecting potential, present in the source theory, cannot be replaced merely by positivity against a Hermitian background.

26.9 A point pole on the Hopf manifold

On the Hopf manifold, a point pole selects homogeneous polynomials with a lower bound on their affine degree. Summing the squared pointwise norms of the normalized monomials gives a binomial tail. Its exponential rate determines the potential limit, and its concentration determines the measure. The limiting pole fills the entire fiber, making the loss of the original point singularity explicit.

Put $q = n - 1$ in the Hopf setting above, and use $L = L_1 = \pi^* \mathcal{O}(1)$ with its pulled-back Fubini–Study metric and the probability $\nu_X = \omega^n/(2n)$. Let $p \in X_n$ lie over $b = [1 : 0 : \cdots : 0]$. Suppose φ has a single logarithmic pole of rational coefficient $0 < a < 1$ at p and is locally bounded elsewhere, as in (26.4). The continuous weight is zero. In homogeneous coordinates $[Z_0 : \cdots : Z_q]$ on $\mathbb{P}^q$, set

$$t([Z_0 : \cdots : Z_q]) = \frac{|Z_1|^2 + \cdots + |Z_q|^2}{|Z_0|^2 + \cdots + |Z_q|^2},$$

$$r_m = \max\{\lfloor ma \rfloor - q, 0\}, \qquad D_m = \binom{m+q}{q}.$$

Thus $t = 0$ exactly at b. Define the function e_a on $[0, 1]$ by

$$e_a(t) = \begin{cases} a \log(t/a) + (1-a) \log((1-t)/(1-a)), & 0 < t < a, \\ 0, & a \le t \le 1, \\ -\infty, & t = 0, \end{cases} \tag{26.5}$$

and put $E_a = e_a \circ t$ on $\mathbb{P}^q$. When $q = 1$, this is the same function as Γ_a in (17.2); here it arises from a point pole on the total space of the fibration.

Proposition 26.9.1 *The selected kernel and dimension are*

$$B_{X_n,m} = D_m \left(\sum_{j=r_m}^{m} \binom{m}{j} t^j (1-t)^{m-j} \right) \circ \pi, \qquad N_m = D_m - \binom{r_m + q - 1}{q}.$$

The last binomial coefficient is zero when $r_m = 0$. In particular all these spaces are nonempty, and

$$\lim_{m\to\infty} \frac{q! N_m}{m^q} = 1 - a^q > 0.$$

Proof By Proposition 26.8.1, the space on the base consists of homogeneous polynomials of degree m vanishing to order at least r_m at b. In the affine chart $Z_0 = 1$, write $\ell = (\ell_0, \dots, \ell_q) \in \mathbb{Z}_{\geq 0}^{q+1}$, $|\ell| = \sum_i \ell_i$ and $Z^\ell = \prod_i Z_i^{\ell_i}$. The selected monomials are exactly those Z^ℓ with $\ell_1 + \cdots + \ell_q \geq r_m$.

Coordinate circle invariance makes distinct monomials orthogonal. Their squared norms for σ^q are

$$\int_{\mathbb{P}^q} \frac{|Z^\ell|^2}{|Z|^{2m}} \sigma^q = \frac{q!\, \ell_0! \cdots \ell_q!}{(m+q)!}, \qquad |\ell| = m. \tag{26.6}$$

For completeness, integrate $|z^\ell|^2 e^{-|z|^2}$ over $\mathbb{C}^{q+1}$ in two ways. Product integration gives $\pi^{q+1} \prod_i \ell_i!$; radial integration separates a sphere average and the factor $(m+q)!$. Dividing by the same computation for $\ell = 0$ gives the displayed normalized sphere average. Its projective pushforward is σ^q, since both are the invariant probability under the unitary group. This proves the norm formula.

The reciprocal squared norm is $D_m m!/(\ell_0! \cdots \ell_q!)$. Sum first over the monomials with $\ell_1 + \cdots + \ell_q = j$. The multinomial formula gives

$$D_m \binom{m}{j} t^j (1-t)^{m-j}.$$

The norms and kernels lift exactly to X_n, proving the first formula. There are $\binom{j+q-1}{q-1}$ monomials of affine degree j. Summing the excluded degrees $0 \leq j < r_m$ gives $\binom{r_m+q-1}{q}$. Since $r_m/m \to a$ and $0 \leq r_m \leq m$, this proves the dimension formula, nonemptiness and the limit. □

Theorem 26.9.1 *Let $U_m = m^{-1} \log B_{X_n,m}$ be the global partial Fubini–Study potential for the zero continuous weight. Then*

$$U_m \longrightarrow E_a \circ \pi$$

in $L^1(X_n)$, pointwise with extended values, and locally uniformly off the fiber $\pi^{-1}(b)$. The normalized measures converge in total variation:

$$\frac{q!}{m^q} B_{X_n,m}\, \nu_X \longrightarrow \mathbf{1}_{\{t\circ\pi > a\}} \nu_X.$$

Proof Write the sum in the proposition as $P_m(t)$. For fixed $0 < t < 1$, it is the probability that a binomial random variable of parameters (m, t) is at least r_m. Its variance is $mt(1-t)$. Chebyshev's inequality and $r_m/m \to a$ imply

$$P_m(t) \longrightarrow \begin{cases} 0, & t < a, \\ 1, & t > a. \end{cases}$$

The pushforward of σ^q by t is $qt^{q-1}\,\mathrm{d}t$. One can see this directly in the affine chart: the Fubini–Study radial density is $2qr^{2q-1}(1+r^2)^{-q-1}\,\mathrm{d}r$, and $t = r^2/(1+r^2)$. In particular the level $t = a$ has measure zero. Now $0 \le P_m \le 1$ and $q!D_m/m^q \to 1$. Dominated convergence gives convergence of the densities in $L^1(\sigma^q)$, hence also in $L^1(\nu_X)$ after lifting. This is exactly the asserted total-variation convergence. Its limiting mass is $\int_a^1 qt^{q-1}\,\mathrm{d}t = 1 - a^q$.

To compute the potential, put

$$H(s) = -s\log s - (1-s)\log(1-s), \qquad 0 \le s \le 1,$$

with $0\log 0 = 0$. The elementary binomial bounds are

$$\frac{\mathrm{e}^{mH(j/m)}}{m+1} \le \binom{m}{j} \le \mathrm{e}^{mH(j/m)} \quad (0 \le j \le m).$$

Indeed the jth probability for parameters $(m, j/m)$ is at most one and at least $1/(m+1)$. For the latter assertion, the ratio of neighboring probabilities shows that the jth term is a largest term; the $m+1$ terms sum to one. The endpoint cases have probability one.

The logarithm of a sum of at most $m+1$ nonnegative terms differs from the largest logarithm by at most $\log(m+1)$. It follows that $m^{-1}\log P_m(t)$ differs by at most $\log(m+1)/m$ from

$$\max_{r_m \le j \le m} \{H(j/m) + (j/m)\log t + (1 - j/m)\log(1-t)\}.$$

On compact subintervals of $(0, 1)$, uniform continuity and $r_m/m \to a$ make these maxima converge uniformly to the maximum over $a \le s \le 1$. Differentiation in s shows that the maximum is zero when $t \ge a$ and equals the first expression in (26.5) when $t < a$. Thus the limit is $e_a(t)$, including at $t = a$.

There is also uniform convergence near $t = 1$. Choose c with $a < c < 1$. The same variance estimate gives $P_m(t) \ge 1 - C/m$ uniformly for $c \le t \le 1$ and large m; hence $m^{-1}\log P_m \to 0$ uniformly there. Together with the preceding compact-interval argument and $m^{-1}\log D_m \to 0$, this proves local uniform convergence off $t = 0$. At $t = 0$, eventually $r_m > 0$, so $P_m(0) = 0$ and the extended pointwise limit is $-\infty$.

Finally, $t^m \le P_m(t) \le 1$ because the term $j = m$ occurs in every binomial sum. Since $m^{-1}\log D_m$ is nonnegative and uniformly bounded, this gives $\log(t\circ\pi) \le U_m \le C$ off the fiber over b. In local submersion coordinates there, $\log(t\circ\pi) = \log|z|^2 + O(1)$,

which is integrable against a smooth volume. The pointwise limit and these bounds prove L^1 convergence by dominated convergence. □

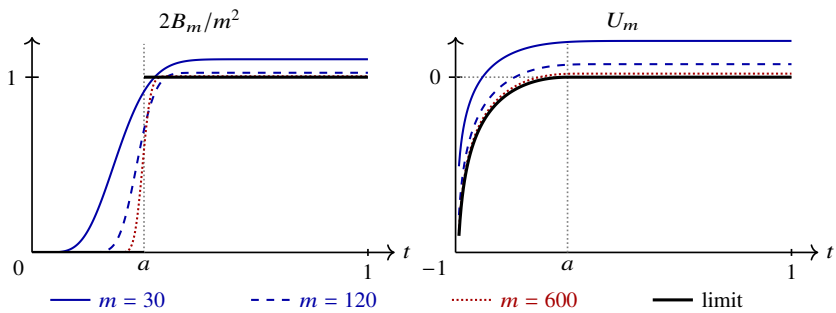

Fig. 26.1 The one-point model with $q = 2$ and $a = 1/3$. The normalized kernel density (left) and the Fubini–Study potential (right) are plotted against t. The black curves are their limits. The potential panel omits $t < 0.01$, where the limit tends to $-\infty$.

The formula for E_a has an intrinsic envelope interpretation. Set $\psi_a = a \log t$ on $\mathbb{P}^q$. This is σ-plurisubharmonic for $0 < a < 1$: away from b, its curvature is the convex combination of σ and the semipositive form induced by the projection $[Z_0 : Z'] \mapsto [Z']$, where $Z' = (Z_1, \dots, Z_q)$. For $q = 1$, the second form is zero. The local logarithmic expression gives the positive extension at b.

Proposition 26.9.2 *The function E_a belongs to* $\mathrm{PSH}(\mathbb{P}^q, \sigma)$ *and*

$$E_a = E_{\sigma,\mathbb{P}^q}[\psi_a]_{\mathcal{I}}(0), \qquad \mathrm{MA}_\sigma(E_a) = \mathbf{1}_{\{t>a\}}\sigma^q.$$

Moreover, $E_a \circ \pi$ is the largest $u \in \mathrm{PSH}(X_n, \alpha)$ satisfying $u \le 0$ and $u \preceq_{\mathcal{I}} \varphi$. Thus

$$\mathbf{1}_{\{t\circ\pi>a\}}\nu_X = \frac{1}{2n}\mathrm{MA}^{\mathrm{ext}}_\omega(E_a \circ \pi).$$

Proof In the chart $Z_0 = 1$, put $Z' = (Z_1, \dots, Z_q)$, $z = Z'/Z_0$ and $R^2 = a/(1-a)$. For $|z| < R$, the local potential of $\sigma + \mathrm{dd^c} E_a$ is

$$\log(1 + |z|^2) + E_a = a \log |z|^2 - a \log a - (1-a)\log(1-a).$$

Its complex Hessian is positive and has rank $q - 1$ away from zero. For $|z| > R$, the form is σ. The two expressions for E_a have equal values and equal first derivatives at $|z| = R$. They therefore glue as a $C^{1,1}$ function near the interface, with no hypersurface term in its distributional Hessian. Near b it has the logarithmic pole of coefficient a. This proves global σ-plurisubharmonicity and $E_a \simeq \psi_a$.

To verify the envelope bound, let $w \in \mathrm{PSH}(\mathbb{P}^q, \sigma)$ satisfy $w \le 0$ and $w \preceq_{\mathcal{I}} \psi_a$. By Lemma 1.17.9, $w \le \psi_a + C$. Fix a complex line through zero in the affine chart. If the restriction of $w + \log(1 + |z|^2)$ is identically $-\infty$, there is nothing to prove on

that line. Otherwise it is subharmonic. On its punctured disk of radius R, subtract $a \log |z|^2$. The result is subharmonic and bounded above near zero by the ordinary singularity inequality, so it extends subharmonically across zero. On the boundary its value is at most $\log(1 + R^2) - a \log R^2$. The maximum principle gives exactly $w \le E_a$ in the disk. Every point with $|z| < R$ lies on such a line; outside this ball $w \le 0 = E_a$. Since E_a itself is an admissible competitor, it is the stated envelope. Continuous zero obstacles make the quasi-everywhere and everywhere inequalities equivalent, by Section 1.19.

On the complement of b, the local potentials are $C^{1,1}$. By Proposition 1.7.2, their Bedford–Taylor top products are the almost-everywhere determinants of their complex Hessians. The rank computation makes this determinant zero for $t < a$ and equal to σ^q for $t > a$. The interface has smooth volume zero, and a $C^{1,1}$ local potential adds no measure there. The non-pluripolar measure gives b zero mass. This proves the measure formula.

For the assertion on X_n, every α-plurisubharmonic function is $u = w \circ \pi$ by Proposition 1.22.1. Since φ has analytic singularities, $u \preceq_{\mathcal{I}} \varphi$ implies $u \preceq \varphi$ by Lemma 1.17.9. In product coordinates (z, τ) at p, restrict this inequality to $\tau = 0$. It yields $w(z) \le a \log |z|^2 + C$, and hence $w \preceq \psi_a$ globally. Conversely, $w \preceq \psi_a$ gives

$$w(z) \le a \log |z|^2 + C \le a \log(|z|^2 + |\tau|^2) + C$$

near p. Away from p, local boundedness of φ and compactness make the upper inequality global. Thus $w \circ \pi \preceq \varphi$, which implies the ideal order. Applying analytic ideal domination on the base as well shows that the two ideal-constrained competitor families are exactly those already considered. Their largest members therefore agree under pullback. Finally Theorem 21.2.1 and the base measure formula give the last identity. □

For sufficiently small rational a, Lemma 1.16.2 gives $\varphi \in \mathrm{PSH}(X_n, \omega)$ with its only pole at p. The theorem consequently applies on the Hopf threefold where BMP fails, and gives a nonzero limit with the intrinsic normalization $2/m^2$. The limiting potential has a pole on the whole elliptic fiber over b. At other points of that fiber the input φ is locally bounded, whereas the limiting potential has a nontrivial multiplier ideal at sufficiently large weight. Thus even this positive-growth limit does not recover the multiplier-ideal type of an arbitrary Hermitian-positive selecting weight.

The measure limit is also different from the raw Hermitian Monge–Ampère measure of φ. Choose its cutoff support in a small neighborhood of p. There is a nonempty open set outside that support with $t \circ \pi < a$: take a neighborhood of another point of the same elliptic fiber. On this open set $\varphi = 0$, so locality gives $\mathrm{MA}^{\mathrm{ext}}_{\omega}(\varphi) = \omega^n$, while the limiting kernel measure is zero. The equality above concerns the projected equilibrium potential $E_a \circ \pi$, whose construction and singularity type have been identified, not the original selecting weight.

26.10 Singularities along a section over a divisor

A singularity along a section over a base divisor translates into a divisorial vanishing condition on the descended sections. As for point poles, the fiber normal variables shift the integrability threshold by a bounded amount. Divisor restriction sequences control the resulting loss of sections and reduce kernel convergence to the base equilibrium theorem.

To bound the lost sections, we need a dimension estimate uniform over a linear range of twists on the divisor.

Lemma 26.10.1 *Let M be a compact complex manifold of dimension $d \geq 0$, and let E, F be holomorphic line bundles on M. For each fixed $C_0 > 0$ there is C_1 such that*

$$h^0(M, E^m \otimes F^j) \leq C_1 m^d \qquad (m \geq 1,\ j \in \mathbb{Z},\ |j| \leq C_0 m).$$

Proof For $d = 0$, the manifold is a finite set and the dimension equals its number of points. Suppose $d \geq 1$. Fix smooth Hermitian metrics on E, F, a smooth positive volume $\mathrm{d}V$, and finitely many coordinate charts with frames and smaller subcharts covering M. The local metric weight of $E^m \otimes F^j$ has second derivatives bounded by Cm, uniformly for $|j| \leq C_0 m$. At a point p in a smaller subchart, multiply the frame by the exponential of a holomorphic affine function to remove the constant and real-linear terms of this weight. The new weight is zero at p and has absolute value at most $Cm|z - p|^2$ on a fixed neighborhood. A ball of radius $\varepsilon m^{-1/2}$ lies in the chart, with $\varepsilon > 0$ independent of p, m, j. The submean inequality for a section's holomorphic coefficient gives

$$|s(p)|^2 \leq Cm^d \int_M |s|^2 \,\mathrm{d}V.$$

All constants depend only on the fixed charts, metrics, volume and C_0. Evaluation at p has one-dimensional target, so for an L^2-orthonormal basis $s_1, \ldots, s_N$ its squared operator norm is $\sum_{\ell=1}^N |s_\ell(p)|^2$. The displayed bound and integration give $N \leq C \operatorname{Vol}(M) m^d$. □

Let $\pi : X \to B$ be a surjective holomorphic submersion of connected compact complex manifolds, with connected fibers of dimension $r \geq 1$. Suppose B is Kähler of dimension $q \geq 1$. Fix a holomorphic line bundle A on B with a smooth Hermitian metric h_A and positive curvature $\theta = c_1(A, h_A)$. Let $Z \subset B$ be a nonempty smooth divisor, and suppose there is a holomorphic section $\sigma : Z \to X$ of π over Z. Put $Y = \sigma(Z)$. Fix $c > 0$ and a smooth metric h_Z on $\mathcal{O}_B(Z)$, with curvature Θ_Z, such that $\alpha := \theta - c\Theta_Z > 0$. Let s_Z be the canonical section of $\mathcal{O}_B(Z)$, and set $\psi = c \log |s_Z|^2_{h_Z}$.

Fix a Hermitian form ω on X independently of h_A. Suppose $\varphi \in \mathrm{PSH}(X, \omega)$ is locally bounded on $X \setminus Y$ and, in submersion coordinates $(x, z', w_1, \dots, w_r)$ near each point of Y, satisfies

$$\begin{gathered} \varphi = c \log\left(|x|^2 + \sum_{j=1}^{r} |w_j|^2 \right) + O(1), \\ Z = \{x = 0\}, \qquad Y = \{x = w_1 = \cdots = w_r = 0\}. \end{gathered} \tag{26.7}$$

The remainder is locally bounded. The vector z' consists of the $q-1$ tangential base coordinates; it is absent when $q = 1$.

Theorem 26.10.1 *The weight ψ is θ-plurisubharmonic and has positive non-pluripolar mass*

$$\int_B \mathrm{MA}_\theta(\psi) = \int_B \alpha^q > 0.$$

For every integer $m \geq 1$, put $\ell_m = \lfloor mc \rfloor$ and $k_m = \max\{\ell_m - r, 0\}$. Under holomorphic descent, the selected spaces are

$$\begin{aligned} V_m &:= H^0\big(X, \pi^* A^m \otimes \mathcal{I}(m\varphi)\big) = H^0\big(B, A^m(-k_m Z)\big), \\ W_m &:= H^0\big(B, A^m \otimes \mathcal{I}(m\psi)\big) = H^0\big(B, A^m(-\ell_m Z)\big) \subseteq V_m. \end{aligned}$$

There is a constant C, depending on the fixed data and independent of m, such that

$$0 \leq \dim V_m - \dim W_m \leq C m^{q-1}.$$

For $q = 1$ the difference is at most $r\,\#Z$ for every m.

Proof Poincaré–Lelong gives $\theta + \mathrm{dd^c}\psi = \alpha + c[Z] \geq 0$. Apply the divisorial measure identity (14.13) with residual background $\alpha > 0$ and bounded remainder zero. It gives $\mathrm{MA}_\theta(\psi) = \alpha^q$, proving the mass formula.

Every section of $\pi^* A^m$ descends by Proposition 26.7.1. In the $r+1$ normal variables (x, w), Lemma 1.16.1 gives

$$\mathcal{I}(m\varphi) = (x, w_1, \dots, w_r)^{k_m}.$$

A pulled-back local coefficient $g(x, z')$ is independent of w. Its Taylor expansion lies in this ideal exactly when every power of x below k_m has zero coefficient, or equivalently when x^{k_m} divides g. This is an identity of holomorphic germs in all tangential variables, so it applies at every point of Z. Outside Y the upstairs weight is locally bounded. Applying the same lemma in one normal variable on the base gives $\mathcal{I}(m\psi) = \mathcal{O}_B(-\ell_m Z)$. These facts prove the two space identities and their inclusion.

For $k_m \leq j < \ell_m$, the divisor restriction sequence gives

$$\begin{aligned} 0 \longrightarrow A^m(-(j+1)Z) &\longrightarrow A^m(-jZ) \\ &\longrightarrow \big(A^m(-jZ)\big)|_Z \longrightarrow 0. \end{aligned}$$

Consequently the loss in section dimension at this step is at most $h^0(Z, (A^m(-jZ))|_Z)$. Apply Lemma 26.10.1 on Z with $E = A|_Z$ and $F = \mathcal{O}_B(-Z)|_Z$: the integers satisfy $0 \le j \le cm$, so the bound is Cm^{q-1} uniformly. There are at most r steps. Summing proves the estimate. When $q = 1$, each restriction space has dimension $\#Z$, giving the last assertion without an asymptotic argument. □

Example 26.10.1 For every $q, r \ge 1$, integer $k \ge 1$, and real $0 < c < k$, the hypotheses above occur on $X = \mathbb{P}^q \times \mathbb{P}^r$ with $B = \mathbb{P}^q$ and $A = \mathcal{O}_{\mathbb{P}^q}(k)$. They give, for $q \ge 2$,

$$\dim V_m - \dim W_m = \frac{r(k-c)^{q-1}}{(q-1)!} m^{q-1} + O(m^{q-2}).$$

For $q = 1$ the difference is $\min\{r, \lfloor mc \rfloor\}$ for every m.

Proof Use homogeneous coordinates $[Z_0 : \cdots : Z_q]$ on the base and $[W_0 : \cdots : W_r]$ on the fiber. Set $Z = \{Z_0 = 0\}$ and $Y = Z \times \{[1 : 0 : \cdots : 0]\}$. Write ω_B, ω_F for the Fubini–Study forms, normalized to have top integrals one. Define

$$R_B = \frac{|Z_0|^2}{\sum_{i=0}^q |Z_i|^2}, \qquad R_F = \frac{\sum_{j=1}^r |W_j|^2}{\sum_{j=0}^r |W_j|^2}, \qquad \varphi = c \log(R_B + R_F), \quad \psi = c \log R_B.$$

The function φ is smooth off Y and satisfies (26.7). After taking a common denominator, the numerator of $R_B + R_F$ is

$$\sum_{j=0}^r |Z_0 W_j|^2 + \sum_{i=0}^q \sum_{j=1}^r |Z_i W_j|^2.$$

Its logarithm in each affine frame is plurisubharmonic. Hence $\mathrm{dd^c}\varphi \ge -c(\omega_B + \omega_F)$. Choose an integer $b \ge c$ and set $\omega = k\omega_B + b\omega_F$. This is a Hermitian form, and $\omega + \mathrm{dd^c}\varphi \ge (k-c)\omega_B + (b-c)\omega_F \ge 0$. On the base, $\theta = k\omega_B$, $\Theta_Z = \omega_B$ and $\alpha = (k-c)\omega_B > 0$.

Put $\ell_m = \lfloor mc \rfloor$. Divisibility of homogeneous polynomials by powers of Z_0 gives the exact counts

$$\dim W_m = \binom{km - \ell_m + q}{q}, \qquad \dim V_m = \binom{km - \max\{\ell_m - r, 0\} + q}{q}.$$

For large m, the two residual degrees differ by r and equal $(k-c)m + O(1)$. Subtracting the degree-q binomial polynomials gives the asserted leading coefficient and remainder. For $q = 1$, subtraction gives $\ell_m - \max\{\ell_m - r, 0\} = \min\{r, \ell_m\}$ for every m. □

The bounded shift of divisorial order has negligible normalized codimension. To apply the kernel comparison under the hypotheses of Theorem 26.10.1, fix a smooth positive probability volume ν_B, smooth fiber probabilities, and the lifting map $\mathcal{L}$ of Section 26.7; put $\nu_X = \mathcal{L}\nu_B$. Use the pullback metric on π^*A and, for a real

$v \in C(B)$, its pullback $v \circ \pi$ as continuous norm weight, and write

$$E(v) = E_{\theta,B}[\psi]_{\mathcal{I}}(v).$$

Corollary 26.10.1 *The q-normalized kernel measure selected by φ on X has total-variation distance $O(m^{-1})$ from the lift of the base kernel measure selected by ψ. In particular,*

$$\beta^{(q)}_{X,m,v\circ\pi,\varphi} \rightharpoonup \mathcal{L}\big(\mathrm{MA}_\theta(E(v))\big).$$

The determinant functionals, normalized to vanish at zero weight, satisfy

$$\mathcal{F}^{(q)}_{X,m}(v \circ \pi) \longrightarrow \int_0^1 \left(\int_B v \, \mathrm{dMA}_\theta(E(tv)) \right) \mathrm{d}t.$$

Normalization by m^{q+r} instead gives zero kernel measures in total variation.

Proof Fiber integration identifies the upstairs continuous norm with the base norm on V_m, just as in Proposition 26.7.1. The kernel measure upstairs is therefore the lift of the kernel measure for V_m. By Lemma 26.8.1 and Theorem 26.10.1, its difference from the lifted W_m measure is positive and has total mass $q!(\dim V_m - \dim W_m)/m^q = O(m^{-1})$. The base is Kähler and ψ has positive non-pluripolar mass, so the equilibrium theorem used in Theorem 26.7.1 applies to W_m. It gives the weak limit. The determinant comparison in Lemma 26.8.1 is $O(m^{-1})\|v\|_\infty$; applying Corollary 26.6.1 to the identity modification of B gives the displayed limit. Its dominated-convergence bound is the uniformly bounded q-normalized section dimension times $\|v\|_\infty$. Finally the q-normalized masses are bounded, so multiplication by m^{-r}, with the fixed factorial adjustment, proves the last assertion. □

This proves the approximation in Problem 32.9.2 for the specified singular center and curvature margin $\theta - c\Theta_Z > 0$.

26.11 Horizontal poles over projective space

For a hyperplane in projective space, the descended vanishing conditions give binomial kernels. Their large-deviation rate is the obstacle envelope of the original upstairs singularity. Concentration of evaluation sections then identifies the limiting measure for every smooth integration volume, improving the preceding weak limit to total variation.

Specialize Section 26.10 to $B = \mathbb{P}^q$ with homogeneous coordinates $[Z_0 : \cdots : Z_q]$, $A = \mathcal{O}_B(k)$ and $Z = \{Z_0 = 0\}$, where q, r, k are positive integers and $0 < c < k$. Use the Fubini–Study metric on A, with curvature $\theta = k\omega_B$ and $\int_B \omega_B^q = 1$. Retain the submersion $\pi : X \to B$, its connected r-dimensional fibers, and the selecting

singularity (26.7) along a section Y over Z. Write

$$R = \frac{|Z_0|^2}{\sum_{i=0}^q |Z_i|^2}, \qquad a = \frac{c}{k}, \qquad D_c = \{R \geq a\}, \qquad C_c = \frac{(k-c)^q}{q!}.$$

For any smooth positive probability volume ν_X, fiber integration gives $\nu_B = \pi_* \nu_X = w\omega_B^q$ with $w > 0$ smooth. Let B_m^B and $B_m^X = B_m^B \circ \pi$ be the kernels in the unweighted norms on the descended spaces V_m. Put $M = \max_B w$.

The selecting potential need not be $\pi^*\theta$-plurisubharmonic. Its domination envelope in this curvature background is

$$\mathcal{P}(\varphi) = \left(\sup\{u \in \mathrm{PSH}(X, \pi^*\theta) : u \leq 0,\ u \preceq \varphi\}\right)^*.$$

Define

$$u_c(R) = \begin{cases} -k\left(a \log \dfrac{a}{R} + (1-a)\log \dfrac{1-a}{1-R}\right), & 0 < R < a, \\ 0, & a \leq R \leq 1, \\ -\infty, & R = 0. \end{cases} \tag{26.8}$$

Theorem 26.11.1 *The envelope satisfies $\mathcal{P}(\varphi) = u_c \circ \pi$. With $\ell_m = \lfloor mc \rfloor$ and $t_m = \max\{\ell_m - r, 0\}$,*

$$N_m = \dim V_m = \binom{km - t_m + q}{q} = C_c m^q + O(m^{q-1}).$$

The intrinsic growth degree is $q < \dim X$. For every $m \geq 1$, in the extended-value sense,

$$\frac{1}{m} \log B_m^X \geq \mathcal{P}(\varphi) - \frac{\log M}{m}. \tag{26.9}$$

These potentials converge to $\mathcal{P}(\varphi)$ globally in L^1 for every smooth positive volume and locally uniformly off $\pi^{-1}(Z)$. In the normalization of Section 26.1, the kernel measures converge in total variation:

$$\begin{aligned} \frac{q!}{m^q} B_m^B \nu_B &\longrightarrow \mu_c := \mathbf{1}_{D_c} \theta^q, \\ \frac{q!}{m^q} B_m^X \nu_X &\longrightarrow \Lambda_{\nu_X} := \frac{k^q \mathbf{1}_{D_c} \circ \pi}{w \circ \pi} \nu_X. \end{aligned} \tag{26.10}$$

The limiting masses are $(k-c)^q$, and $\pi_* \Lambda_{\nu_X} = \mu_c$. Dividing the unscaled kernel measures by N_m gives the probability limits $\mu_c/(k-c)^q$ and $\Lambda_{\nu_X}/(k-c)^q$. Normalization by m^{q+r} gives zero in total variation.

For every sequence of intermediate spaces $W_m \subseteq U_m \subseteq V_m$, with the same restricted norm, the lower bound and all these convergences remain valid. Their dimensions are $C_c m^q + O(m^{q-1})$; dimension normalization uses $\dim U_m$.

26.11.1 The domination envelope

The base envelope can be computed on the affine chart $Z_0 \neq 0$. Set $z_i = Z_i/Z_0$, $s = |z|^2$, $l = k - c$, $s_0 = l/c$, and $C = k \log k - c \log c - l \log l$. The reference weight is $\phi_A = k \log(1 + s)$, and put

$$F_0(z) = \begin{cases} k \log(1 + s), & s \leq s_0, \\ l \log s + C, & s \geq s_0. \end{cases}$$

As a function of $\log s$, it is increasing and convex: its derivative increases to l and then stays constant. Thus it is plurisubharmonic. The branches match to first order, and the tangent-line inequality gives $F_0 \leq \phi_A$. On a chart $Z_j \neq 0$, $j \geq 1$, with $x = Z_0/Z_j$ and the other ratios denoted by ζ, its transformed weight near Z is

$$F_j = c \log |x|^2 + l \log(1 + |\zeta|^2) + C.$$

It therefore defines a global plurisubharmonic metric weight with a divisor pole of coefficient c and a smooth remainder.

For any competing plurisubharmonic weight $v \leq \phi_A$ with $v \leq c \log |x|^2 + O(1)$ near Z, the obstacle gives $v_0 \leq F_0$ on $s \leq s_0$. On each complex line $z = t\xi$, $|\xi| = 1$, either $v_0(t\xi) \equiv -\infty$ or $v_0(t\xi) - l \log |t|^2$ is subharmonic on $|t| > \sqrt{s_0}$ and bounded above near infinity. In the coordinate $1/t$, the puncture is removable. The maximum principle bounds this difference by C. Varying ξ proves $v \leq F$ globally. Consequently $F = \phi_A + u_c$ is the divisor obstacle envelope, and its contact set is D_c.

By Proposition 1.22.1, every $\pi^*\theta$-plurisubharmonic potential is $v \circ \pi$ for a θ-plurisubharmonic v. Restriction to the local section $w_1 = \cdots = w_r = 0$ shows that $v \circ \pi \preceq \varphi$ implies $v \leq c \log |x|^2 + O(1)$. Conversely this divisor bound and $|x|^2 \leq |x|^2 + \sum_j |w_j|^2$ give the domination near Y. A finite cover of Y gives one constant there; on the compact complement of a smaller neighborhood, φ is bounded below and $v \leq 0$. Thus one constant works on X. The competitor families correspond, proving $\mathcal{P}(\varphi) = u_c \circ \pi$. Its pole fills $\pi^{-1}(Z)$, although φ is finite on $\pi^{-1}(Z) \setminus Y$.

26.11.2 Binomial kernels

For a fixed integer $j \geq 0$, set $d = km$, $t_{m,j} = \max\{\ell_m - j, 0\}$, and $\mathcal{H}_m^{(j)} = H^0(B, \mathcal{O}(d)(-t_{m,j}Z))$. The choices $j = 0, r$ give W_m, V_m. Under the reference norm for $v_0 = \omega_B^q$, their kernels are

$$B^0_{m,j}(R) = A_d Q_{m,j}(R), \qquad A_d = \binom{d+q}{q}, \qquad Q_{m,j}(R) = \mathbb{P}\{\mathrm{Bin}(d, R) \geq t_{m,j}\}. \tag{26.11}$$

Apply the monomial norm formula (26.6) in degree d. Coordinate circle invariance makes the monomials orthogonal, and selection retains those with exponent of Z_0 at

least $t_{m,j}$. The multinomial identity, summed over the other exponents, gives (26.11). Counting these monomials gives $\dim \mathcal{H}_m^{(j)} = \binom{d-t_{m,j}+q}{q} = C_c m^q + O(m^{q-1})$.

Write $D(v \mid R) = v \log(v/R) + (1-v)\log((1-v)/(1-R))$ and $b = \lfloor da \rfloor$. If $R \geq b/d$, a binomial mode $\min\{d, \lfloor (d+1)R \rfloor\}$ is at least b, so the tail above $t_{m,j} \leq b$ is at least $1/(d+1)$. If $0 < R < b/d$, the mass at b for parameter $v = b/d$ is a largest mass, hence at least $1/(d+1)$. For parameter R, that mass is at least $(d+1)^{-1}\exp(-dD(v \mid R))$. Since

$$\frac{\partial D(v \mid R)}{\partial v} = \log \frac{v(1-R)}{R(1-v)} \geq 0 \quad (v \geq R),$$

it is at least $(d+1)^{-1}\exp(-dD(a \mid R))$. Together with $u_c \leq 0$, these cases give $Q_{m,j} \geq (d+1)^{-1}\exp(mu_c)$, including $b = 0$ and $R = 1$. Since $A_d \geq d+1$, the reference kernels satisfy $m^{-1}\log B^0_{m,j} \geq u_c$; at $R = 0$ this holds in the extended sense.

For fixed $0 < R < a$, Stirling's formula at the first retained index $t_{m,j}$ gives logarithmic rate $-kD(a \mid R)$. Ratios of successive binomial masses above that index are eventually bounded by a number less than one, because $t_{m,j}/d \to a > R$. The whole tail has the same logarithmic rate. For $R > a$, the binomial variance estimate gives $Q_{m,j} \to 1$. At $R = a$, the lower mode bound and $Q_{m,j} \leq 1$ give logarithmic limit zero. The functions $m^{-1}\log Q_{m,j}(R)$ increase with R, and their limit u_c is continuous on each $[\delta, 1]$. A finite partition on which the oscillation of u_c is small upgrades pointwise convergence to uniform convergence there.

The event that the first $t_{m,j}$ trials succeed gives

$$c \log R \leq \frac{t_{m,j}}{m} \log R \leq \frac{1}{m} \log Q_{m,j}(R) \leq 0.$$

Since $\log R$ is integrable against every smooth volume, dominated convergence proves global L^1 convergence of the relative weights. The added term $m^{-1}\log A_d$ tends to zero. For the measures, the variance estimate gives $Q_{m,j} \to \mathbf{1}_{D_c}$ away from $R = a$. The marginal of R is $q(1-R)^{q-1}\,\mathrm{d}R$: apply the radial computation in the proof of Theorem 26.9.1 to $1 - R$. Thus the boundary has zero volume. Since $q!A_d/m^q \to k^q$ and these constants are bounded, dominated convergence proves total-variation convergence to μ_c and

$$\mu_c(B) = k^q \int_a^1 q(1-R)^{q-1}\,\mathrm{d}R = (k-c)^q.$$

26.11.3 Changing the integration volume

Put $\delta = \min_B w > 0$, and write $B^w_{m,j}$ for the kernel with volume $w\nu_0$. Comparing norms gives

$$M^{-1}B^0_{m,j} \le B^w_{m,j} \le \delta^{-1}B^0_{m,j}.$$

This proves the weight lower bound and both weight convergences. To identify the measure limit, more than comparability is needed.

Let Π_m be the reference L^2 orthogonal projection onto $\mathcal{H}^{(j)}_m$. In a unit frame at x, write $F_{m,x}$ for the evaluation vector of $H^0(B, \mathcal{O}(d))$ and $K_{m,x} = \Pi_m F_{m,x}$. Formula (26.11) gives

$$\|F_{m,x}\|_0^2 = A_d, \qquad \|K_{m,x}\|_0^2 = A_d Q_{m,j}(R(x)),$$
$$\|F_{m,x} - K_{m,x}\|_0^2 = A_d(1 - Q_{m,j}(R(x))).$$

The full coherent section satisfies

$$|F_{m,x}(y)|^2_{h^m_A} = A_d^2 \tau(x,y)^d, \qquad \tau([Z],[W]) = \frac{|\langle Z, W\rangle|^2}{|Z|^2|W|^2}.$$

The measure $A_d^{-1}|F_{m,x}|^2\nu_0$ is a probability measure concentrating at x: outside any neighborhood of x, $\tau \le \rho < 1$ and its mass is at most $A_d\rho^d \to 0$. Thus continuity of w gives $\|(w - w(x))F_{m,x}\|_0 = o(\sqrt{A_d})$ for each fixed x.

On $\mathcal{H}^{(j)}_m$, let $A_m s = \Pi_m(ws)$. Its quadratic form is the weighted norm, so $\delta I \le A_m \le MI$, and

$$B^w_{m,j}(x) = \langle A_m^{-1}K_{m,x}, K_{m,x}\rangle_0.$$

For $R(x) > a$, the preceding norm identities give $\|F_{m,x} - K_{m,x}\|_0 = o(\sqrt{A_d})$. It follows that $\|(A_m - w(x)I)K_{m,x}\|_0 = o(\sqrt{A_d})$. Using $\|A_m^{-1}\| \le \delta^{-1}$ yields

$$B^w_{m,j}(x) = w(x)^{-1}B^0_{m,j}(x) + o(A_d) \qquad (R(x) > a).$$

For $R(x) < a$, the upper kernel comparison gives $m^{-q}B^w_{m,j}(x) \to 0$. Finally $0 \le q!m^{-q}wB^w_{m,j} \le (M/\delta)q!A_d/m^q$ is uniformly bounded. Dominated convergence away from the null set $R = a$ proves $q!m^{-q}B^w_{m,j}w\nu_0 \to \mu_c$ in total variation.

Proof (Proof of Theorem 26.11.1) The exact space identity is Theorem 26.10.1; its projective dimension follows from the monomial count. The envelope computation and the two endpoint kernel calculations prove the downstairs assertions. Isometric descent gives $B^X_m = B^B_m \circ \pi$. For $f_c = k^q\mathbf{1}_{D_c}/w$, fiber integration gives

$$\int_X \left|\frac{q!}{m^q}B^X_m - f_c \circ \pi\right| \nu_X = \int_B \left|\frac{q!}{m^q}B^B_m - f_c\right| \nu_B.$$

This proves the upstairs total-variation limit and its pushforward. The same integration transfers L^1 convergence; any other smooth positive volume on compact X is comparable to ν_X. Local uniform convergence pulls back off $\pi^{-1}(Z)$. The dimension

asymptotic gives the probability limits, and $N_m/m^{q+r} = O(m^{-r})$ proves the zero limit in ambient degree.

For $W_m \subseteq U_m \subseteq V_m$, monotonicity gives $B_{W,m} \le B_{U,m} \le B_{V,m}$. By Lemma 26.8.1, the positive difference of the q-normalized U_m and W_m measures has total mass at most $q!(\dim V_m - \dim W_m)/m^q = O(m^{-1})$. The two endpoint weights give the lower bound and local uniform convergence by sandwiching. Their L^1 errors bound the intermediate error, proving global convergence as well. □

The base limit is independent of ν_X. If $\nu_X(\mathrm{d}b, \mathrm{d}x) = \nu_B(\mathrm{d}b)\lambda_b(\mathrm{d}x)$ is its disintegration, then $\Lambda_{\nu_X}(\mathrm{d}b, \mathrm{d}x) = \mu_c(\mathrm{d}b)\lambda_b(\mathrm{d}x)$. Thus the lifted limit retains exactly the conditional fiber probabilities on D_c. Equality of two lifts implies equality of those probabilities for μ_c-almost every b: test against $(g \circ \pi)f$, with $g \in C(B)$ and f in a countable dense subset of $C(X)$, and then take one common exceptional null set. The converse follows directly by integration.

This dependence occurs in the product example of Example 26.10.1. Keep its weight $\varphi = c\log(R_B + R_F)$ and let $\eta_f = k\omega_B + bf\omega_F$, where $f > 0$ is smooth on $\mathbb{P}^r$ and $b \min f > c$. Then $\eta_f + \mathrm{dd^c}\varphi \ge (k-c)\omega_B + (bf-c)\omega_F > 0$, and its normalized volume and limit are

$$\nu_{X,f} = \omega_B^q \otimes \frac{f^r\omega_F^r}{\int_{\mathbb{P}^r} f^r\omega_F^r}, \qquad \Lambda_{\nu_{X,f}} = \mu_c \otimes \frac{f^r\omega_F^r}{\int_{\mathbb{P}^r} f^r\omega_F^r}.$$

A nonconstant f changes the lift. For $r \ge 2$, it also makes η_f nonclosed: $\mathrm{d}\eta_f = b\,\mathrm{d}f \wedge \omega_F$, and wedging by ω_F^{r-2} reduces its vanishing to the injectivity of $\xi \mapsto \xi \wedge \omega_F^{r-1}$ on one-forms. For $r = 1$ this derivative vanishes by degree.

The spaces, envelope and kernels are unchanged by varying an admissible Hermitian background with $\pi, A, h_A, \varphi, \nu_X$ fixed. Moving p in $Y = Z \times \{p\}$ also leaves them unchanged. For $mc \ge r+1$, however, the constant germ fails integrability at the old center and becomes integrable there after the move. Thus all kernels together need not determine the original multiplier-ideal type.

26.12 Degenerate curvature and growing vanishing orders

The compact models above use vanishing orders proportional to the tensor degree. To see the transition when the order grows more slowly, we pass to an entire-function model. The local metric fixes the scale of the full kernel; the selecting ideal creates a second, larger radius. Grouping the monomials by degree expresses the transition at that radius through Poisson tails.

Fix an integer $r \ge 1$. On $\mathbb{C}$, with coordinate $z = x + \mathrm{i}y$ and area measure $\mathrm{d}A = \mathrm{d}x\,\mathrm{d}y$, put $\phi(z) = |z|^{2r}$. For a real $m > 0$, let $H_{m,r}$ be the Hilbert space of entire functions with norm

$$\|f\|_{m,r}^2 = \int_{\mathbb{C}} |f|^2 \mathrm{e}^{-m\phi}\,\mathrm{d}A.$$

For an integer $q \geq 0$, let $H^{(q)}_{m,r}$ consist of the functions vanishing to order at least q at zero. Point and jet evaluations are bounded by the local submean and Cauchy inequalities, since the weight is bounded below on each fixed disc. Thus this subspace is closed. Write $K^{(q)}_{m,r}(z, w)$ for its reproducing kernel, holomorphic in z and antiholomorphic in w, and put

$$B^{(q)}_{m,r}(z) = \mathrm{e}^{-m\phi(z)} K^{(q)}_{m,r}(z, z).$$

Here the selecting condition is the ideal (z^q), while the smooth weight determines the norm. We use $\Gamma(a) = \int_0^\infty t^{a-1}\mathrm{e}^{-t}\,\mathrm{d}t$ for $a > 0$.

Proposition 26.12.1 *For every $m > 0$ and integer $q \geq 0$,*

$$K^{(q)}_{m,r}(z, w) = \frac{r}{\pi} \sum_{k=q}^{\infty} \frac{m^{(k+1)/r}(z\bar{w})^k}{\Gamma((k+1)/r)}. \tag{26.12}$$

The series and all its derivatives converge uniformly on compact sets. For $\zeta \in \mathbb{C}$,

$$m^{-1/r} B^{(q)}_{m,r}(m^{-1/(2r)}\zeta) = \frac{r}{\pi}\mathrm{e}^{-|\zeta|^{2r}} \sum_{k=q}^{\infty} \frac{|\zeta|^{2k}}{\Gamma((k+1)/r)}. \tag{26.13}$$

Consequently, if m tends to infinity through positive integers and $q = q_m \to \infty$, the left side tends to zero uniformly on compact ζ-sets.

Proof Rotational invariance makes the monomials orthogonal. For every integer $k \geq 0$, polar integration with radial variable ρ and the substitution $t = m\rho^{2r}$ give

$$\|z^k\|^2_{m,r} = 2\pi \int_0^\infty \rho^{2k+1}\mathrm{e}^{-m\rho^{2r}}\,\mathrm{d}\rho = \frac{\pi}{r} m^{-(k+1)/r}\Gamma((k+1)/r).$$

For an entire function $f(z) = \sum_{k\geq 0} a_k z^k$, angular Parseval on circles and Tonelli's theorem yield $\|f\|^2_{m,r} = \sum_{k\geq 0} |a_k|^2 \|z^k\|^2_{m,r}$. Thus the normalized monomials form an orthonormal basis, and the selected subspace has the basis with $k \geq q$. This proves (26.12). Stirling's formula shows that the reciprocal Gamma coefficients decay faster than every fixed geometric progression, even after multiplication by any fixed power of k. This proves compact convergence with all derivatives. Substitution gives (26.13). On $|\zeta| \leq R$ for a fixed real $R > 0$, its tail is bounded by the tail of the convergent series with $|\zeta|$ replaced by R, proving the last assertion. □

For $r = 1$ and $q = 0$, the density is m/π. For general r the value at zero is $rm^{1/r}/(\pi\Gamma(1/r))$ when $q = 0$, and zero when $q > 0$. The scale $m^{-1/(2r)}$ therefore describes fixed vanishing orders, but a growing order requires a further rescaling.

26.12.1 A uniform tail estimate

Grouping monomial degrees by their residue modulo r expresses the kernel through weighted Poisson tails. The Gamma-ratio estimate below removes those weights with an error independent of the vanishing threshold.

For $T > 0$, let N_T be a Poisson random variable, defined by $\mathbb{P}(N_T = \ell) = \mathrm{e}^{-T}T^{\ell}/\ell!$ for integers $\ell \geq 0$; $\mathbb{P}$ and $\mathbb{E}$ denote probability and expectation. For a real $0 < \alpha \leq 1$, define

$$w_{\alpha,T}(\ell) = T^{\alpha-1}\frac{\Gamma(\ell+1)}{\Gamma(\ell+\alpha)}.$$

All constants in the next lemma may depend on α, but not on T.

Lemma 26.12.1 *There is a constant $C_\alpha > 0$ such that, for $T \geq 1$,*

$$\mathbb{E}\big|w_{\alpha,T}(N_T) - 1\big| \leq C_\alpha T^{-1/2}.$$

Proof The Poisson series gives $\mathbb{E}N_T = T$ and $\mathrm{Var}(N_T) = T$. Stirling's formula, applied to the ratio of Gamma functions, gives

$$\begin{aligned}
\frac{\Gamma(\ell+1)}{\Gamma(\ell+\alpha)} &= \ell^{1-\alpha}(1 + O_\alpha(\ell^{-1})) && (\ell \to \infty),\\
\frac{\Gamma(\ell+1)}{\Gamma(\ell+\alpha)} &\leq C_\alpha(\ell+1)^{1-\alpha} && (\ell \geq 0).
\end{aligned}$$

The second bound follows from the first by increasing the constant for finitely many small integers. Since $0 \leq 1 - \alpha < 1$, it implies

$$w_{\alpha,T}(\ell) \leq C_\alpha\left(1 + \frac{\ell+1}{T}\right), \qquad \mathbb{E}(w_{\alpha,T}(N_T) + 1)^2 \leq C_\alpha \quad (T \geq 1).$$

For $T \geq 2$, put $E_T = \{|N_T - T| \leq T/2\}$. On this event the ratio expansion and the bounded derivative of $s^{1-\alpha}$ on $[1/2, 3/2]$ give

$$|w_{\alpha,T}(N_T) - 1| \leq C_\alpha\left(\frac{|N_T - T|}{T} + \frac{1}{T}\right).$$

Its expectation is at most $C_\alpha T^{-1/2}$ by Cauchy–Schwarz. On the complement, Chebyshev's inequality gives $\mathbb{P}(E_T^c) \leq 4/T$. The preceding second-moment bound and Cauchy–Schwarz give the same estimate there. Enlarging the constant handles $1 \leq T \leq 2$. □

For integers $q \geq 0$ and $0 \leq s < r$, put

$$L_s(q) = \left\lceil \frac{q-s}{r} \right\rceil \geq 0.$$

These are the thresholds obtained by writing a monomial degree as $k = r\ell + s$.

Proposition 26.12.2 *There is a constant $C_r > 0$, depending only on r, such that for $m > 0$, $z \neq 0$ and $T = m|z|^{2r} \geq 1$,*

$$\left| \frac{\pi B_{m,r}^{(q)}(z)}{mr^2|z|^{2r-2}} - \frac{1}{r}\sum_{s=0}^{r-1} \mathbb{P}\{N_T \geq L_s(q)\} \right| \leq C_r T^{-1/2} \quad (q \geq 0). \tag{26.14}$$

In particular, the error is uniform in the vanishing order q.

Proof Let $\rho = |z| > 0$ and put $\alpha_s = (s+1)/r$ for integers $0 \leq s < r$. Group the positive series (26.12) by these residues. Since $T = m\rho^{2r}$, it gives the exact identity

$$\begin{aligned} B_{m,r}^{(q)}(z) &= \frac{r}{\pi\rho^2} \sum_{s=0}^{r-1} \mathrm{e}^{-T} \sum_{\ell \geq L_s(q)} \frac{T^{\ell+\alpha_s}}{\Gamma(\ell+\alpha_s)} \\ &= \frac{rT}{\pi\rho^2} \sum_{s=0}^{r-1} \mathbb{E}\big[w_{\alpha_s,T}(N_T) \mathbf{1}_{\{N_T \geq L_s(q)\}} \big]. \end{aligned}$$

Divide by $mr^2\rho^{2r-2}/\pi = r^2T/(\pi\rho^2)$ and apply Lemma 26.12.1 to each of the finitely many residues. The indicator is bounded by one, independently of its threshold. □

26.12.2 The limiting envelope and the transition scale

A vanishing order proportional to m removes the kernel density inside a fixed circle. Inside that circle the potential limit is logarithmic; outside it agrees with the smooth weight. Let m tend to infinity through positive integers. For $\lambda > 0$, set $q_m = \lfloor m\lambda \rfloor$ and $R_\lambda = (\lambda/r)^{1/(2r)}$. Define

$$V_\lambda(z) = \begin{cases} |z|^{2r}, & |z| \geq R_\lambda, \\ \lambda \log(|z|^2/R_\lambda^2) + \lambda/r, & 0 < |z| < R_\lambda, \\ -\infty, & z = 0. \end{cases}$$

Here δ_0 denotes the unit point mass at zero. In dimension one, the non-pluripolar measure discards the mass carried by the pole set.

Theorem 26.12.1 *One has*

$$\frac{1}{m}\log K_{m,r}^{(q_m)}(z,z) \longrightarrow V_\lambda(z)$$

pointwise with extended values and in $L^1_{\mathrm{loc}}(\mathbb{C})$. *The kernel measures converge locally in total variation:*

$$\frac{1}{m}B_{m,r}^{(q_m)}\,\mathrm{d}A \longrightarrow \mathbf{1}_{\{|z|>R_\lambda\}}\mathrm{dd^c}\phi. \tag{26.15}$$

The function V_λ *is the largest subharmonic function* v *on* $\mathbb{C}$ *such that* $v \le \phi$ *and* $v(z) \le \lambda\log|z|^2 + C$ *near zero for some constant* C. *Its distributional Laplacian is*

$$\mathrm{dd^c}V_\lambda = \lambda\delta_0 + \mathbf{1}_{\{|z|>R_\lambda\}}\mathrm{dd^c}\phi. \tag{26.16}$$

Thus the limit in (26.15) *is the non-pluripolar measure of this envelope, rather than its full distributional Laplacian.*

Proof For fixed $z \neq 0$, put $\rho = |z|$ and $T = m\rho^{2r}$. If $\rho \neq R_\lambda$, then $L_s(q_m)/T \to \lambda/(r\rho^{2r})$. Chebyshev's inequality and (26.14) give

$$\frac{1}{m}B_{m,r}^{(q_m)}(z) \longrightarrow \begin{cases} 0, & 0<\rho<R_\lambda, \\ r^2\rho^{2r-2}/\pi, & \rho>R_\lambda. \end{cases} \tag{26.17}$$

The same comparison with $q = 0$ bounds the selected density by $C_r m\rho^{2r-2}$ when $T \ge 1$. When $T \le 1$, (26.13) with $q = 0$ bounds it by $C_r m^{1/r} \le C_r m$. Thus $m^{-1}B_{m,r}^{(q_m)}$ is uniformly bounded on each compact set. The circle $\rho = R_\lambda$ and the origin have zero area. Dominated convergence and $\mathrm{dd^c}\phi = (r^2/\pi)|z|^{2r-2}\mathrm{d}A$ prove (26.15).

For the potential limit inside the circle, fix $0 < \rho < R_\lambda$ and write

$$d_{m,k} = \frac{m^{(k+1)/r}\rho^{2k}}{\Gamma((k+1)/r)} \quad (k \ge q_m).$$

The Gamma ratio estimate from Stirling's formula gives

$$\frac{d_{m,k+1}}{d_{m,k}} \le \left(\frac{rm\rho^{2r}}{q_m+1}\right)^{1/r}\left(1+\frac{C_r}{q_m+1}\right) \quad (k \ge q_m)$$

for all sufficiently large m. Its right side tends to a number less than one. Hence the sum of these terms lies between d_{m,q_m} and a fixed multiple of that term. Stirling's formula then yields

$$\frac{1}{m}\log K_{m,r}^{(q_m)}(z,z) \longrightarrow \lambda\log\rho^2 + \frac{\lambda}{r}\big(1-\log(\lambda/r)\big) = V_\lambda(z).$$

For $\rho > R_\lambda$, the positive density limit in (26.17) instead gives $m^{-1}\log B_{m,r}^{(q_m)}(z) \to 0$, so the unweighted kernel potential tends to $\phi(z)$.

At $\rho = R_\lambda$, put $k_m = \max\{q_m, \lfloor rm\rho^{2r} \rfloor\}$. Then $k_m/m \to r\rho^{2r} = \lambda$. The single monomial with this degree and Stirling's formula give $\liminf m^{-1} \log K_{m,r}^{(q_m)}(z,z) \geq \phi(z)$. The full-kernel bound already proved gives the reverse upper limit. At zero the selected kernel vanishes for all sufficiently large m. This proves pointwise convergence everywhere.

For local L^1 convergence, the full-kernel bounds give a uniform upper bound for $m^{-1} \log K_{m,r}^{(q_m)}(z,z)$ on each compact set. The single selected monomial of degree q_m gives the lower bound

$$\frac{1}{m} \log K_{m,r}^{(q_m)}(z,z) \geq c_m + \frac{q_m}{m} \log |z|^2,$$

where c_m is bounded as $m \to \infty$, again by Stirling's formula. Since $0 \leq q_m/m \leq \lambda$, the negative part on a fixed compact set is bounded by a constant plus $2\lambda|\log|z||$ near zero. This is locally area-integrable. Dominated convergence proves the L^1_{loc} assertion.

The two radial expressions for V_λ and their first radial derivatives agree at R_λ, because $rR_\lambda^{2r} = \lambda$. Away from zero and that circle, direct differentiation proves (26.16). Matching first derivatives gives no circle measure, and $\mathrm{dd^c} \log |z|^2 = \delta_0$ supplies the atom at zero. The right side is positive, so V_λ is subharmonic. For $t = |z|^{2r}/(\lambda/r) < 1$, the inequality $\log t + 1 \leq t$ proves $V_\lambda \leq \phi$ inside the circle; outside it there is equality.

Finally let v satisfy the stated subharmonic and pole conditions. On the punctured disc $|z| < R_\lambda$, the function $v - \lambda \log |z|^2$ is subharmonic and bounded above near zero. Its subharmonic extension across zero is at most $\lambda/r - \lambda \log R_\lambda^2$ on the boundary. The maximum principle gives $v \leq V_\lambda$ inside the disc. Outside, this follows from $v \leq \phi$. Since V_λ itself satisfies the conditions, it is the largest admissible function. Removing its pole atom gives the last assertion. □

The measures in this theorem have infinite total mass on $\mathbb{C}$; their convergence is asserted on compact sets. The weighted potentials $m^{-1} \log B_{m,r}^{(q_m)}$ converge to $V_\lambda - \phi$. In particular, the radius R_λ is created by the selecting ideal even when $r = 1$, while $r > 1$ also records a curvature zero at the origin.

For any integer sequence $q_m \to \infty$, the boundary transition is governed by the central limit of the same Poisson tails. Define the Gaussian distribution function

$$\Phi(\tau) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\tau} \mathrm{e}^{-s^2/2}\, \mathrm{d}s \quad (\tau \in \mathbb{R}).$$

Proposition 26.12.3 *Fix a real τ. Choose $z_m \neq 0$ with*

$$m|z_m|^{2r} = \frac{q_m}{r} + \tau\sqrt{\frac{q_m}{r}},$$

which is positive for all sufficiently large m. Then

$$\frac{\pi B_{m,r}^{(q_m)}(z_m)}{mr^2|z_m|^{2r-2}} \longrightarrow \Phi(\tau).$$

In particular, this applies both when $q_m/m \to \lambda > 0$ and when $q_m/m \to 0$.

Proof We first record the Poisson limit used here. For a fixed real $M > 0$, Stirling's formula for $\ell!$ gives, uniformly over integers ℓ with $|\ell - T| \leq M\sqrt{T}$,

$$\mathbb{P}(N_T = \ell) = \frac{1}{\sqrt{2\pi T}} \exp\left(-\frac{(\ell - T)^2}{2T}\right)(1 + O_M(T^{-1/2})) \quad (T \to \infty).$$

Indeed insert the factorial expansion in $\mathrm{e}^{-T}T^\ell/\ell!$ and expand the logarithm in $(\ell - T)/T$; the cubic remainder is $O_M(T^{-1/2})$. Riemann sums on bounded intervals therefore converge to Gaussian integrals. Chebyshev's bound $\mathbb{P}(|N_T - T| > M\sqrt{T}) \leq M^{-2}$ then permits $M \to \infty$. Thus if real thresholds a_T satisfy $(a_T - T)/\sqrt{T} \to a \in \mathbb{R}$, one has $\mathbb{P}(N_T \geq a_T) \to 1 - \Phi(a)$; rounding changes the normalized threshold by at most $T^{-1/2}$, and the Gaussian distribution is continuous.

Set $T_m = m|z_m|^{2r}$. Then $T_m \to \infty$ and, for every integer $0 \leq s < r$,

$$\frac{L_s(q_m) - T_m}{\sqrt{T_m}} \longrightarrow -\tau.$$

Apply the preceding Poisson limit in (26.14). Its uniform error tends to zero, and each of the finitely many probabilities tends to $1 - \Phi(-\tau) = \Phi(\tau)$. □

For a sequence $q_m \to \infty$, put $R_m = (q_m/(rm))^{1/(2r)}$. At fixed τ, the radii in Proposition 26.12.3 satisfy

$$|z_m| = R_m\left(1 + \frac{\tau}{2\sqrt{rq_m}} + O_{r,\tau}(q_m^{-1})\right).$$

If $q_m = o(m)$, these radii tend to zero, while their ratio to the full-kernel scale $m^{-1/(2r)}$ tends to infinity. The limiting point is a curvature zero when $r > 1$. To transfer this transition to a compact manifold, the local norm comparison and the global extension error must be uniform in the growing vanishing order. The next section establishes both estimates on a curve.

26.13 Localization on curves

The entire-function transition persists on a compact curve as long as the metric remainder is negligible on the shrinking disc carrying the selected sections. Restriction gives the upper bound. Positivity away from the distinguished point permits a global L^2 correction that preserves the growing vanishing order and gives the lower bound.

Let C be a connected compact Riemann surface, let $x \in C$, and let (L, h) be a smooth Hermitian holomorphic line bundle. Fix an integer $r \geq 1$. Suppose that in a holomorphic coordinate z centered at x and a holomorphic frame e,

$$|e|_h^2 = \mathrm{e}^{-\phi}, \qquad \phi(z) = |z|^{2r} + O(|z|^{2r+1}).$$

Let ν be a smooth positive volume form, locally $a(z)\,\mathrm{d}A$, where $a(0) = 1$ and $\mathrm{d}A$ is Euclidean area. For positive integers m, choose nonnegative integers q_m such that

$$q_m \longrightarrow \infty, \qquad q_m = o(m^{1/(2r+1)}).$$

Write $\mathfrak{m}_x$ for the ideal sheaf of x and give

$$V_m = H^0(C, L^m \otimes \mathfrak{m}_x^{q_m}) \quad \text{the norm} \quad \|s\|_m^2 = \int_C |s|_{h^m}^2 \,\mathrm{d}\nu.$$

Its density is $B_m = \sum_j |s_{m,j}|_{h^m}^2$ for any orthonormal basis $(s_{m,j})$, with value zero if $V_m = 0$.

Theorem 26.13.1 *For a fixed real τ, let x_m lie in the coordinate neighborhood and satisfy*

$$T_m := m|z(x_m)|^{2r} = q_m/r + \tau\sqrt{q_m/r}.$$

Then

$$\limsup_{m\to\infty} \frac{\pi B_m(x_m)}{mr^2|z(x_m)|^{2r-2}} \leq \Phi(\tau). \tag{26.18}$$

If the curvature of (L, h) is semipositive on C and strictly positive on $C \setminus \{x\}$, this ratio converges to $\Phi(\tau)$.

Proof Put $z_m = z(x_m)$, $\rho_m = |z_m|$, $R_m = (q_m/(rm))^{1/(2r)}$ and

$$S_m = \frac{mr^2\rho_m^{2r-2}}{\pi}.$$

For all sufficiently large m, $T_m > 0$, $R_m \to 0$, and $\rho_m/R_m \to 1$. Positive constants denoted by M, with subscripts when needed, may change between estimates. They are independent of m and of the argument of z_m; they may depend on the fixed geometric data, r, τ, and fixed auxiliary cutoffs or radii. We write $q = q_m$ when no confusion is possible. The model densities $B_{m,r}^{(q)}$ are those of Section 26.12.

The kernel on a shrinking disc. Fix a real $A > 1$ and set $D_m = \{|z| < AR_m\}$. On D_m,

$$m|\phi(z) - |z|^{2r}| \le M_A m R_m^{2r+1} = M_A \frac{q_m R_m}{r} = o(1), \qquad \sup_{D_m} |a - 1| = o(1).$$

The first limit follows from $q_m R_m = r^{-1/(2r)}(q_m^{2r+1}/m)^{1/(2r)} \to 0$. Hence the actual local norm and the radial norm with weight $\mathrm{e}^{-m|z|^{2r}}\,\mathrm{d}A$ differ by a factor $1 + o(1)$, uniformly over all holomorphic functions on D_m. The same comparison holds for their pointwise weights at z_m.

For integers $k \ge 0$, let $I_{m,k}$ and $J_{m,k}$ be the squared norms of z^k on D_m and on the whole plane, respectively, with this radial weight. Polar integration gives

$$I_{m,k} = \frac{\pi}{r} m^{-(k+1)/r} \gamma((k+1)/r, U_m), \qquad U_m = m(AR_m)^{2r},$$
$$J_{m,k} = \frac{\pi}{r} m^{-(k+1)/r} \Gamma((k+1)/r), \qquad \gamma(b, U) = \int_0^U t^{b-1} \mathrm{e}^{-t}\,\mathrm{d}t.$$

Here $b, U > 0$. Angular Parseval and Tonelli show that the normalized monomials form a complete orthonormal system in the weighted holomorphic disc space: the squared norm of a Taylor series is the sum of the squared monomial norms, so its partial sums converge in that norm. The density of the subspace vanishing to order at least q is therefore

$$B_m^D(z_m) = \mathrm{e}^{-T_m} \sum_{k \ge q} \frac{\rho_m^{2k}}{I_{m,k}}.$$

Point evaluation is continuous by the local submean inequality; thus this sum also gives its extremal norm.

Choose fixed real numbers $1 < c < d < A^{2r}$ and put $k_m = \lfloor r c T_m \rfloor$. For large m, $k_m \ge q$ and $U_m \ge d T_m$. Uniformly for $0 \le k \le k_m + 1$,

$$\frac{I_{m,k}}{J_{m,k}} = 1 + O(\mathrm{e}^{-\eta T_m}) \tag{26.19}$$

for some fixed $\eta > 0$. Indeed, let G_b have Gamma density $t^{b-1}\mathrm{e}^{-t}/\Gamma(b)$ on $(0, \infty)$. A change of variable gives $\mathbb{E}\mathrm{e}^{sG_b} = (1 - s)^{-b}$ for $0 < s < 1$. With $b = (k+1)/r \le c T_m + O_r(1)$, Markov's inequality yields

$$1 - \frac{\gamma(b, U_m)}{\Gamma(b)} \le \mathrm{e}^{-sU_m}(1 - s)^{-b}.$$

Since $d > c$, choose $s > 0$ small enough that $sd + c\log(1 - s) > 0$. The right side is at most $M\mathrm{e}^{-\eta T_m}$, as required.

Cauchy–Schwarz for the positive radial measure gives $I_{m,k+1}^2 \le I_{m,k} I_{m,k+2}$. Hence $I_{m,k+1}/I_{m,k}$ increases with k. By (26.19) and the Gamma-ratio asymptotic,

$$\rho_m^2 \frac{I_{m,k_m}}{I_{m,k_m+1}} = (1+o(1)) T_m^{1/r} \frac{\Gamma((k_m+1)/r)}{\Gamma((k_m+2)/r)} \longrightarrow c^{-1/r} < 1.$$

Choose $\kappa \in (c^{-1/r}, 1)$. For all large m, the ratio of consecutive terms $\rho_m^{2k}/I_{m,k}$ is at most κ for every $k \ge k_m$. Consequently,

$$\mathrm{e}^{-T_m} \sum_{k>k_m} \frac{\rho_m^{2k}}{I_{m,k}} \le \frac{\kappa}{1-\kappa} \mathrm{e}^{-T_m} \frac{\rho_m^{2k_m}}{I_{m,k_m}} = o(S_m).$$

For the last equality, replace I_{m,k_m} by J_{m,k_m} using (26.19). The resulting monomial is bounded by $B_{m,r}^{(k_m)}(z_m)$. In (26.14), each residue threshold divided by T_m tends to $c > 1$, so Chebyshev's inequality makes all those Poisson tails tend to zero. The comparison error is $O(T_m^{-1/2})$.

For $q \le k \le k_m$, use (26.19). Since $I_{m,k} \le J_{m,k}$ for every k, we obtain

$$B_{m,r}^{(q)}(z_m) \le B_m^D(z_m) \le (1+o(1)) B_{m,r}^{(q)}(z_m) + o(S_m).$$

By Proposition 26.12.3,

$$B_m^D(z_m)/S_m \longrightarrow \Phi(\tau). \tag{26.20}$$

The global norm of a selected section dominates its actual norm on D_m. Restriction and the preceding norm comparisons therefore give $B_m(x_m) \le (1+o(1)) B_m^D(z_m)$, including when $V_m = 0$. This proves (26.18).

Polynomials with almost all their norm near the point. Assume from now on the curvature hypotheses in the last assertion. Fix a real $c > 1$ and put $K_m = \lfloor c q_m \rfloor$. Define

$$C_m = \sum_{k=q_m}^{K_m} \frac{\rho_m^{2k}}{J_{m,k}}, \qquad p_m(z) = C_m^{-1/2} \sum_{k=q_m}^{K_m} \frac{\bar{z}_m^k z^k}{J_{m,k}}.$$

For large m these are well defined, $\|p_m\|_{m,r} = 1$, and

$$\mathrm{e}^{-T_m} |p_m(z_m)|^2 = B_{m,r}^{(q_m)}(z_m) - B_{m,r}^{(K_m+1)}(z_m) = (\Phi(\tau) + o(1)) S_m. \tag{26.21}$$

The omitted tail is $o(S_m)$ by (26.14), since its Poisson thresholds divided by T_m tend to $c > 1$.

Choose $\delta \in (0, 1/2)$ so that the coordinate disc $|z| < \delta$ satisfies $\phi(z) \ge \beta |z|^{2r}$ for a fixed $\beta \in (1/2, 1)$, and a is bounded above there. For a sufficiently large fixed $A > 1$,

$$\int_{AR_m < |z| < \delta} |p_m|^2 \mathrm{e}^{-m\phi} a \, \mathrm{d}A = o(1). \tag{26.22}$$

To see this, write $p_m(z) = \sum_{k=q_m}^{K_m} b_{m,k} z^k$, so that $\sum_k |b_{m,k}|^2 J_{m,k} = 1$. Radial domination, angular orthogonality and polar integration bound the left side by a fixed

constant times

$$\sum_{k=q_m}^{K_m} |b_{m,k}|^2 J_{m,k}\, \beta^{-(k+1)/r} \mathbb{P}\{G_{(k+1)/r} \geq \beta A^{2r} q_m/r\}.$$

The Gamma moment at $s = 1/2$ gives, for $b, U > 0$,

$$\beta^{-b}\mathbb{P}\{G_b \geq U\} \leq \exp\bigl(-U/2 + b\log(2/\beta)\bigr).$$

Choose A so large that $\beta A^{2r}/2 > c\log(2/\beta) + 1$. Since $(k+1)/r \leq (cq_m + 1)/r$, the last factor in the sum is $O(\mathrm{e}^{-q_m/r})$, uniformly in k. This proves (26.22). On $|z| < AR_m$ the actual and radial model norms differ by a factor $1 + o(1)$, as proved above. The model norm on this disc is at most 1.

Fix a smooth cutoff χ, with $0 \leq \chi \leq 1$, compactly supported in $|z| < \delta$ and equal to one near zero. Extend $\chi p_m e^{\otimes m}$ by zero to C. The preceding estimates give

$$\|\chi p_m e^{\otimes m}\|_m^2 \leq 1 + o(1). \tag{26.23}$$

All these choices precede the limit in m.

A global correction with the same vanishing condition. Let $f_m = \bar\partial(\chi p_m e^{\otimes m})$. Its support is contained in a fixed annulus where ϕ has a positive lower bound. Cauchy–Schwarz and the normalized coefficient identity give, for $|z| < \delta < 1$,

$$|p_m(z)|^2 \leq \sum_{k=q_m}^{K_m} \frac{|z|^{2k}}{J_{m,k}} \leq M(K_m + 1)m^{(K_m+1)/r}.$$

Here $\Gamma(b)$ has a positive lower bound for $b \geq 1/r$. Since $q_m \log m = o(m)$, the factor $\mathrm{e}^{-m\phi}$ on the annulus dominates this coefficient growth.

Use the Kähler form ν on the curve to define form norms and the induced metric $h_{K_C^{-1}}$ on the inverse canonical line bundle K_C^{-1}. Choose a smooth cutoff η supported in the coordinate chart, equal to one near x, and satisfying $0 \leq \eta \leq 1$. Shrink its support if necessary. For a sufficiently small fixed $t_0 > 0$, the global smooth functions

$$\ell_t = \eta \log(|z|^2 + t^2), \qquad 0 < t \leq t_0,$$

extended by zero outside the chart, are nonpositive. Their complex Hessians are nonnegative where $\eta = 1$. On the fixed transition annulus they are uniformly bounded below, independently of t.

Choose a fixed smooth real function ψ, equal to a sufficiently large positive multiple of $|z|^2$ near x and cut off in the chart, such that

$$\mathrm{i}\Theta(K_C^{-1}) + \mathrm{i}\partial\bar\partial\psi \geq 2\nu$$

on a neighborhood of x where $\eta = 1$. Here $\mathrm{i}\Theta$ denotes the Chern curvature, so the weight $\mathrm{e}^{-\psi}$ contributes $\mathrm{i}\partial\bar\partial\psi$. For all large m and all $0 < t \leq t_0$, equip $E_m = L^m \otimes K_C^{-1}$

with the smooth metric

$$H_{m,t} = (h^m \otimes h_{K_C^{-1}})\, \mathrm{e}^{-\psi - q_m \ell_t}.$$

It satisfies

$$\mathrm{i}\Theta(E_m, H_{m,t}) \geq \nu. \tag{26.24}$$

Near x, the two terms involving h and ℓ_t are nonnegative and the other two dominate 2ν. On the complement of a fixed such neighborhood, compactness and strict positivity give a lower bound $c_0 m\nu$ for $m\mathrm{i}\Theta(L, h)$, with $c_0 > 0$. It absorbs all remaining negative terms, bounded by $M(q_m + 1)\nu$, because $q_m = o(m)$.

On the support of f_m, the functions ℓ_t are bounded below uniformly in t. The polynomial estimate, the bounded derivatives of χ, and $q_m \log m = o(m)$ therefore give a constant $a_0 > 0$ such that, for all large m,

$$\int_C |f_m|^2_{h^m,\nu} \mathrm{e}^{-\psi - q_m \ell_t}\, \mathrm{d}\nu \leq \mathrm{e}^{-a_0 m} \qquad (0 < t \leq t_0).$$

Identify an L^m-valued $(0,1)$-form with an E_m-valued $(1,1)$-form using the canonical tensor identity; the induced metrics preserve its norm. In bidegree $(1,1)$ on a curve, (26.24) makes the curvature commutator at least the identity. The smooth complete-Kähler L^2 theorem [Dem11, Theorem 5.1] applies: the curve is compact and the data are automatically $\bar\partial$-closed. It gives an L^m-valued weak solution $u_{m,t}$ with

$$\bar\partial u_{m,t} = f_m, \qquad \int_C |u_{m,t}|^2_{h^m} \mathrm{e}^{-\psi - q_m \ell_t}\, \mathrm{d}\nu \leq \mathrm{e}^{-a_0 m}.$$

The canonical-bundle term in (26.24) accounts for this change of bidegree.

For each fixed m, let $t \downarrow 0$. Since $\ell_t \leq 0$ and ψ is bounded, extract a weakly convergent subsequence in the original L^2 norm, with limit u_m. Distributional continuity gives $\bar\partial u_m = f_m$. For fixed $t' > 0$ and $t < t'$, one has $\ell_t \leq \ell_{t'}$, so the same bound holds in the fixed smooth weighted space with weight $\mathrm{e}^{-\psi - q_m \ell_{t'}}$. Its norm is equivalent to the original norm for this fixed m, t'. Weak lower semicontinuity followed by monotone convergence as $t' \downarrow 0$ gives

$$\int_C |u_m|^2_{h^m} \mathrm{e}^{-\psi - q_m \ell}\, \mathrm{d}\nu \leq \mathrm{e}^{-a_0 m}, \qquad \ell = \eta \log |z|^2.$$

In particular, $\|u_m\|^2_m \leq M\mathrm{e}^{-a_0 m}$. Near x the right side f_m vanishes, so the distributional Cauchy–Riemann equation makes u_m holomorphic in the frame $e^{\otimes m}$. By Lemma 1.16.1 in dimension one, integrability against $|z|^{-2q_m}$ forces vanishing order at least q_m. The smooth factors are bounded above and below near x for each fixed m.

It follows that $s_m = \chi p_m e^{\otimes m} - u_m$ is a global holomorphic section in V_m. By (26.23), $\|s_m\|_m \leq 1 + o(1)$.

Evaluation at the moving point. For large m, the disc D_m lies where $\chi = 1$ and $f_m = 0$. Apply the local norm comparison and the disc extremal estimate to the

holomorphic representative of u_m, which vanishes to order at least q_m. By (26.20),

$$|u_m(x_m)|^2_{h^m} \le (1 + o(1))B^D_m(z_m)\|u_m\|^2_m \le MS_m e^{-a_0 m}.$$

The actual pointwise metric factor differs from e^{-T_m} by $1 + o(1)$. Since $\Phi(\tau) > 0$, (26.21) and the triangle inequality give

$$|s_m(x_m)|^2_{h^m} \ge (\Phi(\tau) + o(1))S_m.$$

These sections are nonzero for large m. The extremal characterization of B_m and $\|s_m\|_m \le 1 + o(1)$ yield the matching lower bound. This proves the limit. □

The condition $q_m = o(m^{1/(2r+1)})$ enters through $mR_m^{2r+1} = o(1)$: it removes the first uncontrolled term of the metric weight throughout the disc carrying the norm. At larger orders this term can affect the transition, so the present comparison does not determine the limit. On a fibration over a curve, the kernel identities of Section 26.7 transfer the proved transition whenever the descended spaces have these vanishing conditions.

Part VIII
Closed classes and locally conformally Kähler geometry

Chapter 27
Bott–Chern classes and singular loci

Let X be a connected compact complex manifold of dimension $n \geq 1$, and fix a Hermitian metric ω. A closed class may admit positive currents while forcing singularities on every positive representative. Gauduchon traces bound its almost positive representatives; regularization then identifies the non-nef and non-Kähler loci, separating singularities forced by the class from those of a chosen current. A Kähler resolution also gives finite top masses: the supremum over the restricted class detects bigness, while the mass of a specified divisorial trace is the top intersection of its residual class. Positive mass can produce a strict current without preserving the original singularity type.

This distinction matters for equations in a Hermitian background. A divisor consumes Gauduchon trace but contributes no density to the non-pluripolar measure. It therefore constrains the subsolution coefficient; at the maximal coefficient, weak limits of maximizing tests annihilate its class. On the elliptic threefold, closedness forces every positive current to descend to a surface. The resulting absence of big closed classes coexists with positive Hermitian volume, whose degeneration can be computed explicitly.

The singularity type also leaves a metric question open. A smooth equation in a big class on a Kähler manifold defines a metric on the ample locus, but smoothness there does not determine the distance to its complement. Density zeros can come from ramification, collapsed exceptional fibers or a singular Hessian, with different normal lengths even at the same determinant order. Radial equations express boundedness and finite distance by two distinct integrals: a continuous minimal remainder can have a complete end, whereas a Hölder remainder forces finite distance. For exponential equations, dividing the determinant by its tangential part gives a normal-length estimate and a sharp logarithmic threshold.

27.1 Closed classes with Hermitian lower bounds

Testing a current against a Gauduchon form eliminates its potential. The resulting fixed trace bounds all almost positive representatives of a class.

Use the Bott–Chern cohomology and the pseudoeffective and nef classes of Chapter 22, and bigness as in Section 1.19. Fix a class $\alpha \in H^{1,1}_{\mathrm{BC}}(X, \mathbb{R})$ and a smooth closed representative θ. Current representatives have the form $T = \theta + \mathrm{dd^c}u$ by Lemma 22.4.1. A strictly positive closed current is called a *Kähler current*; its class is big. These class properties are unchanged by replacing θ by $\theta + \mathrm{dd^c} f$ for a smooth real f.

For a pseudoeffective α, write

$$\alpha[-\varepsilon\omega] = \{T \in \alpha : T \geq -\varepsilon\omega\}.$$

Proposition 27.1.1 *For each fixed $\varepsilon > 0$, the currents in $\alpha[-\varepsilon\omega]$ have uniformly bounded distributional mass.*

Proof Choose a Gauduchon metric γ. For $T = \theta + \mathrm{dd^c}u$ in this family, the current $T + \varepsilon\omega$ is positive and

$$\int_X (T + \varepsilon\omega) \wedge \gamma^{n-1} = \int_X (\theta + \varepsilon\omega) \wedge \gamma^{n-1}.$$

The right side is fixed. Comparison of smooth Hermitian metrics converts this trace into a bound for all coefficients of the positive current $T + \varepsilon\omega$. Subtracting the fixed smooth form gives the asserted bound for T. In dimension one the same argument uses Stokes directly. □

Proposition 27.1.2 *The pseudoeffective cone in $H^{1,1}_{\mathrm{BC}}(X, \mathbb{R})$ is closed and convex and contains no nonzero line. For a fixed Gauduchon metric γ, the trace functional ℓ_γ of Section 22.4 is strictly positive on its nonzero elements, and the slice of pseudoeffective classes with $\ell_\gamma = 1$ is compact. This slice may be empty. Pullback and pushforward by a compact modification are continuous linear maps on Bott–Chern cohomology and preserve pseudoeffectivity.*

Proof Use the finite-dimensional smooth representative space of Lemma 22.7.1 to give Bott–Chern cohomology its usual topology. The trace is continuous there. Its positivity and the absence of lines were proved in Section 22.4; convexity follows by adding positive currents.

Let pseudoeffective classes α_j converge to α, and choose positive closed representatives T_j. Their masses $\ell_\gamma(\alpha_j)$ are bounded. Weak compactness gives a subsequence converging to a positive closed current T. Writing B_γ for the continuous smooth representative map of Lemma 22.7.1, continuity of $B_\gamma(T_j)$ shows that the class of T is α. Thus the cone is closed. For its mass-one slice, (22.3) bounds $B_\gamma(T)$ in the finite-dimensional representative space. The slice is closed and bounded there, hence compact.

The class maps of a modification are linear and well-defined by the smooth-to-current description of cohomology, so they are continuous between finite-dimensional spaces. Pullback preserves positive closed currents by local psh potentials, as in Proposition 22.7.1; proper pushforward preserves positivity and closedness. These representatives prove the assertions about the cones. □

Compactness of this linear slice does not give a bound for Monge–Ampère masses of potentials. The functional tests one closed current against a fixed smooth form; it is not a top-degree volume of the class.

In particular, a nef class is pseudoeffective. Its smooth representatives with errors decreasing to zero have uniformly bounded trace. A weak cluster current is positive and closed, and continuity of the smooth representative map in Lemma 22.7.1 preserves its class.

27.2 Minimal multiplicities and the non-nef locus

Assume α is pseudoeffective. Minimal potentials in the families $\alpha[-\varepsilon\omega]$ measure the singularities that persist as the allowed negative error tends to zero. Their Lelong numbers define the non-nef locus.

For $\varepsilon > 0$ put

$$V_\varepsilon = V_{\theta+\varepsilon\omega}, \qquad T_{\min,\varepsilon} = \theta + \mathrm{dd^c} V_\varepsilon,$$

using the normalized envelope of Section 1.19. Pseudoeffectivity makes its competitor family nonempty. V_ε has minimal singularities in this family: for any $u \in \mathrm{PSH}(X, \theta + \varepsilon\omega)$, $u - \sup_X u \le V_\varepsilon$. Thus $T_{\min,\varepsilon} \in \alpha[-\varepsilon\omega]$.

Definition 27.2.1 For $x \in X$, the *minimal multiplicity* of α at x is

$$\nu(\alpha, x) = \sup_{\varepsilon>0} \nu(T_{\min,\varepsilon}, x).$$

The *non-nef locus* is

$$E_{\mathrm{nn}}(\alpha) = \{x \in X : \nu(\alpha, x) > 0\}.$$

If ε decreases, the admissible family shrinks, so its minimal multiplicities increase. Thus the supremum is the limit as $\varepsilon \downarrow 0$. Comparison of two auxiliary metrics proves independence of ω. Changing the representative changes minimal potentials only by a smooth function and bounded normalization errors, so it does not affect these numbers either.

The smooth regularization in Section 1.15.1 turns vanishing Lelong numbers into an arbitrarily small curvature loss. This gives the converse in the following characterization.

Proposition 27.2.1 *For a pseudoeffective class α, the set $E_{\mathrm{nn}}(\alpha)$ is a countable union of analytic subsets. Moreover,*

$$\alpha \text{ is nef} \quad \Longleftrightarrow \quad E_{\mathrm{nn}}(\alpha) = \varnothing.$$

Proof By Siu's analyticity theorem from Section 1.13.1 and monotonicity of the minimal multiplicities,

$$E_{\mathrm{nn}}(\alpha) = \bigcup_{k,\ell \geq 1} \{x : \nu(T_{\min,1/\ell}, x) \geq 1/k\}.$$

If α is nef, a smooth representative is available in every $\alpha[-\varepsilon\omega]$. A minimal potential is bounded below by its smooth potential up to a constant, so all its Lelong numbers vanish. Conversely, if the non-nef locus is empty, all $T_{\min,\varepsilon}$ have zero Lelong numbers. Apply the regularization consequence just stated to obtain a smooth representative bounded below by $-2\varepsilon\omega$. Letting ε vary proves nefness. □

The parameter ε is essential. Even for a nef class, a positive current with minimal singularities in α itself may have nonzero Lelong numbers; see the example discussed in Section 33.14. Definition 27.2.1 does not replace the almost positive approximating families by the single family of positive currents.

27.3 The non-Kähler locus

For a big class, strict currents determine where the class admits a smooth positive representative. Analytic regularization and a finite-intersection argument produce one strict current whose polar set is exactly the obstruction.

For an almost positive closed $(1,1)$-current T on X, write $E_+(T) = \{x : \nu(T,x) > 0\}$.

Definition 27.3.1 If α is big, let T range over its Kähler-current representatives. Its *non-Kähler locus* is

$$E_{\mathrm{nK}}(\alpha) = \bigcap_T E_+(T).$$

For a pseudoeffective class that is not big, we use the convention $E_{\mathrm{nK}}(\alpha) = X$.

Theorem 27.3.1 *Let α be big. There is a Kähler current with analytic singularities, smooth off its polar set, whose positive Lelong set is exactly $E_{\mathrm{nK}}(\alpha)$. In particular this locus is a proper analytic subset, and*

$$E_{\mathrm{nn}}(\alpha) \subset E_{\mathrm{nK}}(\alpha), \qquad E_{\mathrm{nK}}(\alpha) = \varnothing \quad \Longleftrightarrow \quad \alpha \text{ is Kähler.}$$

Proof Analytic regularization reduces the defining intersection to Kähler currents with analytic singularities and smooth remainder away from their poles. Indeed, regularization of any strict current remains strict after a sufficiently small loss, and its positive Lelong set is contained in that of the original current.

Given two strict currents with potentials u_1, u_2, their maximum is a potential for a current dominating $\delta\omega$ for a common $\delta > 0$, by Lemma 1.3.1. Its Lelong numbers are the minima of the two Lelong numbers. Regularizing this current gives a strict current with analytic singularities whose positive Lelong set is contained in the intersection of the previous two sets.

The descending chain condition for analytic subsets of a compact complex manifold now gives a smallest finite intersection of the analytic sets under consideration. This intersection equals the full intersection. The maximum-and-regularization construction realizes a subset of that finite intersection as the polar set of a single strict analytic current. Since the full intersection is contained in the polar set of every such current, equality follows.

For any positive current $T \in \alpha$ and any $\varepsilon > 0$, minimality gives $\nu(T_{\min,\varepsilon}, x) \leq \nu(T, x)$. Take the supremum in ε, then the intersection over strict T, to obtain the inclusion of loci. If the non-Kähler locus is empty, the realizing analytic current has no poles and is a smooth Kähler representative. The reverse implication follows by using a smooth Kähler form in the defining intersection. □

Proposition 27.3.1 *The non-Kähler locus of a big class has no isolated points.*

Proof Let $\theta + \mathrm{dd^c}\Phi \geq \delta\omega$ be a current as in Theorem 27.3.1. Suppose x is an isolated point of its polar set. Choose holomorphic coordinates $z = (z_1, \ldots, z_n)$ on a ball centered at x, with no other pole. For a sufficiently large positive constant A and a real constant C to be chosen, the local function $q(z) = A|z|^2 - C$ satisfies $\theta + \mathrm{dd^c}q \geq \delta'\omega$ on a smaller ball, for some $\delta' > 0$.

Choose C so that $q < \Phi$ on a closed annular collar of the boundary. This is possible because Φ is smooth on that collar. Since $\Phi \to -\infty$ at x, the maximum $\max(\Phi, q)$ agrees with q near x and with Φ on the collar. By Lemma 1.3.2, it glues to Φ outside the ball and gives a global Kähler current smooth near x. Its Lelong number at x is zero, contradicting the definition of $E_{\mathrm{nK}}(\alpha)$. □

Closed bigness has a strong global consequence: the existence of a Kähler current is equivalent to membership in Fujiki's class $\mathcal{C}$, by the standard modification criterion recorded in Section 33.14. Thus a compact complex manifold outside $\mathcal{C}$ has no big ordinary Bott–Chern class. The strictly positive nonclosed backgrounds of this book still exist on every compact complex manifold.

27.4 Null loci and numerical restricted volumes

Assume in this section that X is Kähler, and choose ω Kähler. Closedness makes intersection numbers independent of representatives. For a nef class α, their vanishing defines the classical null locus

$$\mathrm{Null}(\alpha) = \bigcup_{\substack{V \subset X \text{ irreducible, } \dim V > 0 \\ \int_V \alpha^{\dim V} = 0}} V.$$

The intersection number is computed on a resolution of V by pulling back a smooth closed representative; Stokes' theorem gives its independence of that representative. For a nef and big class, the equality

$$E_{\mathrm{nK}}(\alpha) = \mathrm{Null}(\alpha)$$

is the classical nef-and-big locus theorem recorded in Section 33.14. It also holds on compact manifolds in class C.

For a general pseudoeffective class on a compact Kähler manifold, the appropriate replacement is the numerical restricted volume in (27.1). For an irreducible compact analytic subvariety $V \subset X$ of dimension $k > 0$, let V_{reg} be its regular locus. The numerical restricted volume is zero when $V \subset E_{\mathrm{nn}}(\alpha)$. Otherwise, for real $\varepsilon > 0$, let T range over closed representatives of α with $T \geq -\varepsilon\omega$, analytic singularities with smooth remainder, and polar set not containing V. The subscript ac denotes the absolutely continuous part on V_{reg}. The volume is

$$\langle \alpha^k \rangle_{X|V} = \lim_{\varepsilon \downarrow 0} \sup_T \int_{V_{\mathrm{reg}}} \big((T + \varepsilon\omega)|_{V_{\mathrm{reg}}}\big)^k_{\mathrm{ac}}. \tag{27.1}$$

The smooth remainder makes this measure the extension by zero of a smooth measure away from an analytic subset.

The numerical null locus is the union of the positive-dimensional irreducible subvarieties on which (27.1) vanishes. The trace theory of [Xia, Chapter 8] relates these restricted volumes to masses on V, which explains why singularity types and restriction of multiplier ideals are relevant to identifying this locus. Such a trace need not equal an ordinary pointwise restriction of a singular potential; see Chapter 18.

27.5 Restricted volumes and Hermitian perturbations

Return to an arbitrary compact complex X and Hermitian metric ω. Replacing the closed perturbation in (27.1) by ω can give infinite volume even for the zero Bott–Chern class $\{0\}$. The unbounded smooth masses on the Hopf threefold already imply this failure in the ambient case $V = X$.

Proposition 27.5.1 *Suppose there are smooth real functions v_j on X, for positive integers j, such that*

$$\omega + \mathrm{dd^c} v_j > 0, \qquad \int_X (\omega + \mathrm{dd^c} v_j)^n \longrightarrow +\infty.$$

For every $\varepsilon > 0$,

$$\sup_{\substack{T \in \{0\} \text{ smooth, closed} \\ T \geq -\varepsilon\omega}} \int_X (T + \varepsilon\omega)^n = +\infty. \tag{27.2}$$

This occurs for the Gauduchon metric on the Hopf threefold of Chapter 8.

Proof Fix $\varepsilon > 0$ and take $T_j = \varepsilon \mathrm{dd^c} v_j$. Each T_j is smooth, closed and represents the zero Bott–Chern class. Moreover,

$$T_j + \varepsilon\omega = \varepsilon(\omega + \mathrm{dd^c} v_j) > 0,$$

and its top mass is $\varepsilon^n \int_X (\omega + \mathrm{dd^c} v_j)^n \to +\infty$. The Hopf specialization follows from Theorem 8.0.1. □

All currents in this proof have empty polar set, so imposing analytic singularities with smooth remainder does not eliminate the example. The supremum is already infinite for each fixed error size; letting ε tend to zero afterwards does not repair it. In contrast, if ω is Kähler and $T = \mathrm{dd^c} v$ is smooth, Stokes gives

$$\int_X (T + \varepsilon\omega)^n = \varepsilon^n \int_X \omega^n \longrightarrow 0.$$

The zero Bott–Chern class is nef, has empty non-nef locus, and contains only the zero positive current: a positive representative would be $\mathrm{dd^c} u \geq 0$, so u would be a global plurisubharmonic function on a compact manifold and hence constant. Thus (27.2) measures the nonlinear mass of the Hermitian error family, not positivity of the closed class.

A numerical restricted-volume theory in a Hermitian background must therefore control the approximating family or use another invariant. The qualitative non-Hermitian locus, defined through strict currents with a prescribed exterior derivative, does not use this supremum. Dang's qualitative null locus instead tests failure of bigness after restriction; the exact singularity condition and its equality with the non-Hermitian locus for nef and big backgrounds are recalled in Section 32.3.

27.6 Finite restricted masses of closed classes

The nonclosed perturbation in the preceding example destroys the cohomological mass bound. Restricting a closed class to a Kähler resolution restores a finite comparison mass. Taking the supremum over its positive currents gives a measure of positivity on each subvariety, including subvarieties contained in the exceptional image of a modification of X.

Assume in this section and the next that X belongs to Fujiki's class $\mathcal{C}$. Every irreducible reduced subvariety $Y \subset X$ has a resolution $\pi : Z \to Y$ with Z compact Kähler. Indeed, closed subspaces of a class-$\mathcal{C}$ space remain in that class, and the Kähler modification theorem applies to Y. These classical geometric inputs are recorded in Section 33.14. In particular, taking the proper transform under one fixed modification of X is not needed to obtain a model of Y.

Two such resolutions admit a common Kähler refinement: resolve the closure of their common graph in $Z_1 \times Z_2$ by embedded resolution. The resulting smooth strict transform is Kähler, since the ambient resolution is a sequence of blowups of the compact Kähler product along smooth centers. The Kähler property of a blowup follows by adding a small positive multiple of a curvature form of $\mathcal{O}(-E)$ to the pulled-back metric; the curvature is positive in the vertical directions along the exceptional divisor E.

For a closed real $(1,1)$-class b on a compact Kähler manifold Z of dimension $k > 0$, set

$$\mathcal{R}_Z(b) = \sup_{\substack{S \in b \\ S \geq 0}} \int_Z \langle S^k \rangle, \qquad \sup \varnothing = 0. \tag{27.3}$$

For a class $a \in H^{1,1}_{\mathrm{BC}}(X, \mathbb{R})$ and an irreducible reduced k-dimensional $Y \subset X$, write $f = j \circ \pi$, where $j : Y \hookrightarrow X$, and define

$$\mathcal{R}(a; Y) = \mathcal{R}_Z(f^* a). \tag{27.4}$$

Proposition 27.6.1 *The quantities in* (27.3) *and* (27.4) *are finite. The latter is independent of the Kähler resolution. If $\eta \in b$ is smooth, κ is Kähler and $C > 0$ satisfies $C\kappa - \eta > 0$, then*

$$\mathcal{R}_Z(b + t\{\kappa\}) \leq (C + t)^k \int_Z \kappa^k \qquad (t \geq 0). \tag{27.5}$$

For a modification $\mu : X' \to X$ and the proper transform Y' of a subvariety meeting the isomorphism locus,

$$\mathcal{R}(\mu^* a; Y') = \mathcal{R}(a; Y).$$

Proof First consider a Kähler form λ on Z. For $q \in \mathrm{PSH}(Z, \lambda)$ normalized by $q \leq 0$, the truncations $q_j = \max(q, -j)$ have mass $\int_Z \lambda^k$ by Theorem 2.2.2, or Stokes' theorem when $k = 1$. Restricting their measures to $\{q > -j\}$ and passing to the

increasing limit gives

$$\int_Z \langle (\lambda + \mathrm{dd^c} q)^k \rangle \le \int_Z \lambda^k.$$

The smooth current λ attains equality.

Put $A = C\kappa - \eta$. For a positive current S in $b + t\{\kappa\}$, the current $S + A$ is positive and belongs to $(C + t)\{\kappa\}$. Positivity and multilinearity of local non-pluripolar products give $\langle S^k \rangle \le \langle (S + A)^k \rangle$. The Kähler bound just proved yields (27.5). The definition uses the class itself, so changing its smooth representative or the comparison form does not change the supremum.

Let $\rho : W \to Z$ be a modification between compact Kähler manifolds. By Proposition 1.22.2, every positive current in $\rho^* b$ is the pullback of a unique positive current in b. Their non-pluripolar measures correspond on the isomorphism locus, and both give zero mass to the exceptional analytic sets. Hence their total masses agree and

$$\mathcal{R}_W(\rho^* b) = \mathcal{R}_Z(b).$$

Apply this identity on a common Kähler refinement to obtain independence of resolution. The same argument compares resolutions of Y and Y', since their maps to X agree on a dense open set and therefore everywhere on the common refinement.□

Positivity of this mass has a classical interpretation on a Kähler model. We use two facts from Kähler pluripotential theory: a class containing a positive current with positive non-pluripolar top mass is big, and a current with minimal singularities in a big class attains its positive-product volume. For a nef class this volume is its top intersection. Their precise sources are recorded in Section 33.14. The strict current needed for a nef class can also be obtained by the following normalized diagonal construction.

Proposition 27.6.2 *For a closed class b on a compact Kähler k-fold Z,*

$$\mathcal{R}_Z(b) > 0 \quad \Longleftrightarrow \quad b \text{ is big}.$$

If b is nef, then

$$\mathcal{R}_Z(b) = \int_Z b^k. \tag{27.6}$$

For nef b with $I = \int_Z b^k > 0$, fix a Kähler form κ, let $p_1, p_2 : Z \times Z \to Z$ be the projections and $\Delta \subset Z \times Z$ the diagonal, and put

$$M = k \int_Z b^{k-1} \wedge \{\kappa\}.$$

There is a closed positive (k,k)-current Θ in $(p_1^* b + p_2^* b)^k$ with $\Theta \ge d[\Delta]$, $d > 0$, such that

$$K_b = M^{-1}(p_1)_*(\Theta \wedge p_2^* \kappa) \in b, \qquad K_b \ge (d/M)\kappa. \tag{27.7}$$

Proof If $\mathcal{R}_Z(b) > 0$, some positive representative has positive top mass, so the positive-mass criterion makes b big. Conversely, if $S \in b$ satisfies $S \geq \delta\kappa$, then $S - \delta\kappa$ is positive and closed. Expansion of the non-pluripolar product gives $\langle S^k \rangle \geq \delta^k \kappa^k$.

Suppose b is nef. For $t > 0$, the class $b + t\{\kappa\}$ is Kähler. Adding $t\kappa$ to an arbitrary positive representative of b and using the Kähler mass bound yields

$$\mathcal{R}_Z(b) \leq \int_Z (b + t\{\kappa\})^k.$$

Letting t decrease to zero gives the upper bound in (27.6). In particular, equality holds if $I = 0$.

For $I > 0$, choose L so that $L\{\kappa\} - b$ is Kähler. Nonnegativity of intersections of nef classes gives

$$0 \leq \int_Z b^{k-1} \wedge (L\{\kappa\} - b) = LM/k - I,$$

so $M > 0$. The class $B = p_1^* b + p_2^* b$ is nef and satisfies

$$\int_{Z \times Z} B^{2k} = \binom{2k}{k} I^2 > 0.$$

The classical mass-concentration theorem along the diagonal gives $\Theta \in B^k$ with $\Theta \geq d[\Delta]$ for some $d > 0$. In pushing forward $B^k \wedge p_2^*\{\kappa\}$, only the term with one factor from the first copy of Z survives. Thus

$$(p_1)_*(B^k \wedge p_2^*\{\kappa\}) = Mb, \qquad (p_1)_*([\Delta] \wedge p_2^*\kappa) = \kappa.$$

This proves (27.7). The Kähler $\mathrm{dd^c}$-lemma for currents identifies the computed cohomology class with the Bott–Chern class b. Therefore b is big, and a current with minimal singularities has mass $\int_Z b^k$. This gives the remaining lower bound. □

The criterion descends to a smooth subvariety Y. Write $a_Y = j^* a$. If $\pi : Z \to Y$ is a Kähler modification and $K \in \pi^* a_Y$ satisfies $K \geq \delta\kappa$, choose a Hermitian metric h on Y and $C > 0$ with $\pi^* h \leq C\kappa$. Potential descent gives a positive representative $\pi_* K$ of a_Y, and

$$\pi_* K \geq (\delta/C)\pi_*\pi^* h = (\delta/C)h. \tag{27.8}$$

Conversely, a strict positive representative of a_Y pulls back to a positive current whose non-pluripolar mass is positive on a coordinate ball in the isomorphism locus. On that ball it dominates a positive multiple of the closed form κ, so the local product expansion applies. Consequently, $\mathcal{R}(a; Y) > 0$ is equivalent to bigness of the restricted class a_Y. This argument uses the Hermitian metric only for the current inequalities; the product comparison takes place with a closed local form.

Proposition 27.6.3 *Let W be a connected compact Kähler manifold of dimension $k > 0$, let $g : W \to X$ be holomorphic, and suppose* $\dim g(W) < k$. *Then for every closed class a on X,*

$$\mathcal{R}_W(g^*a) = 0.$$

Proof If the image is a point, $g^*a = 0$ and (27.6) gives the assertion. Otherwise, choose a Kähler resolution $h : Q \to g(W)$, and resolve the graph of the meromorphic lift of g. This gives a Kähler modification $\rho : \widehat{W} \to W$ and a holomorphic map $F : \widehat{W} \to Q$ with $g \circ \rho = h \circ F$, where h is also viewed as a map into X.

Choose a smooth representative $\eta \in a$, a Kähler form κ_Q on Q, and $C > 0$ such that $C\kappa_Q - h^*\eta \geq 0$. The forms

$$P = CF^*\kappa_Q, \qquad A = P - \rho^* g^* \eta$$

are closed and semipositive. Since $\dim Q < k$, $P^k = 0$. For any positive current $S \in g^*a$, the current $\rho^*S + A$ is positive in the nef class $\{P\}$. Hence

$$\int_{\widehat{W}} \langle (\rho^*S)^k \rangle \leq \int_{\widehat{W}} \langle (\rho^*S + A)^k \rangle \leq \mathcal{R}_{\widehat{W}}(\{P\}) = \int_{\widehat{W}} P^k = 0.$$

Modification invariance proves the result. If the class has no positive representative, its invariant is zero by definition. □

Corollary 27.6.1 *Let a be nef on X. For every positive-dimensional irreducible reduced $Y \subset X$,*

$$\mathcal{R}(a; Y) = \int_Y a^{\dim Y}.$$

If $\int_X a^n > 0$, then

$$\bigcup_{\substack{Y \subset X \text{ irreducible, } \dim Y > 0 \\ \mathcal{R}(a;Y)=0}} Y = \mathrm{Null}(a) = E_{\mathrm{nK}}(a). \tag{27.9}$$

Proof Pullback preserves nefness: pull back smooth representatives with arbitrarily small negative error and bound the pulled-back Hermitian metric by a Kähler metric on the resolution. Now apply (27.6). The intersection number on Y is defined by this same pullback. The first equality in (27.9) follows from the definition of the null locus; the second is the classical nef-and-big locus theorem recalled earlier in this chapter. □

When X is Kähler and a is nef, (27.1) is also $\int_Y a^{\dim Y}$, so the two restricted invariants agree. For a general class, (27.4) takes the supremum over all positive currents on a resolution of Y; it does not require those currents to extend to X. The nef comparison therefore does not identify the two constructions outside that regime.

27.7 Divisorial traces with bounded remainder

A class invariant takes a supremum over currents; an actual trace retains a specified singularity. For a divisorial trace with bounded remainder, the non-pluripolar product discards the divisor and retains the residual current with bounded potentials. Its class is nef, and its top intersection computes the mass.

Let $Y \subset X$ be irreducible and reduced, with $k = \dim Y > 0$, let α_0 be smooth and closed on X, and let $u \in \mathrm{PSH}(X, \alpha_0)$ have a genuine ordinary restriction $v = u|_Y$. On a Kähler resolution $f = j \circ \pi : Z \to X$, put $\alpha_Z = f^*\alpha_0$ and $v_\pi = u \circ f$. For a smooth Y, one can instead start directly with $v \in \mathrm{PSH}(Y, \alpha_Y)$ for a smooth closed form α_Y and put $\alpha_Z = \pi^*\alpha_Y$, $v_\pi = v \circ \pi$. These are actual potentials; the intrinsic multiplier-ideal trace of Chapter 18 is a different construction.

Suppose on one such resolution that

$$v_\pi = \ell + w, \qquad \ell = \sum_{j=1}^{N} c_j \log |s_j|^2_{h_j}, \qquad w \text{ bounded}, \tag{27.10}$$

where $c_j > 0$, the D_j are prime divisors, s_j is the canonical section of $\mathcal{O}(D_j)$, and h_j is a smooth Hermitian metric. The empty sum is allowed. Write $\Gamma = \sum_j c_j D_j$ and $\gamma_j = c_1(\mathcal{O}(D_j), h_j)$, and set

$$\eta_v = \alpha_Z - \sum_j c_j \gamma_j, \qquad b_v = \{\eta_v\} = \{\alpha_Z\} - \{[\Gamma]\}. \tag{27.11}$$

For singular Y, all assertions below are made on its resolutions.

Lemma 27.7.1 *The current $R_v = \eta_v + \mathrm{dd^c} w$ is positive and has bounded local potentials, and its class b_v is nef. Moreover,*

$$\int_Z \langle (\alpha_Z + \mathrm{dd^c} v_\pi)^k \rangle = \int_Z R_v^k = \int_Z b_v^k = \mathcal{R}_Z(b_v). \tag{27.12}$$

If $v_1, \dots, v_k$ satisfy the same conditions on Z, for possibly different closed backgrounds, then

$$\int_Z R_{v_1} \wedge \cdots \wedge R_{v_k} = \int_Z b_{v_1} \wedge \cdots \wedge b_{v_k}. \tag{27.13}$$

Proof Off $|\Gamma|$, the current $\alpha_Z + \mathrm{dd^c} v_\pi$ equals $\eta_v + \mathrm{dd^c} w$ and is positive. After adding a smooth local potential for η_v, the bounded function w is plurisubharmonic there. By Lemma 1.3.7, it extends across the divisor as a bounded plurisubharmonic function, agreeing almost everywhere with the original one. We use this extension to represent w canonically. Thus $R_v \geq 0$ globally and

$$\alpha_Z + \mathrm{dd^c} v_\pi = [\Gamma] + R_v.$$

Nefness of b_v now follows from Lemma 22.6.1. The non-pluripolar measure on the left of (27.12) agrees with R_v^k off the divisor, and neither charges the divisor. This gives the first equality.

Fix a Kähler form κ on Z. For all sufficiently large real t, the smooth closed form $\eta_v + t\kappa$ is Kähler. Its bounded potential w therefore satisfies

$$\int_Z (R_v + t\kappa)^k = \int_Z (\eta_v + t\kappa)^k$$

by Theorem 2.2.2, with the one-dimensional case given by Stokes. Both sides are polynomials in t. Their constant coefficients agree, giving $\int_Z R_v^k = \int_Z b_v^k$. The last equality in (27.12) is (27.6). Apply the same polynomial argument to positive linear combinations of the R_{v_i} and compare coefficients to obtain (27.13). □

Denote the common value in (27.12) by $\mathfrak{r}(v;Y)$. When Y is smooth, it is also the mass of the original trace current:

$$\mathfrak{r}(v;Y) = \int_Y \langle (\alpha_Y + \mathrm{dd^c} v)^k \rangle,$$

where $\alpha_Y = \alpha_0|_Y$ in the restriction setting. Indeed, the currents upstairs and downstairs correspond on the isomorphism locus of π, and their non-pluripolar measures give no mass to its analytic complement. Change of variables therefore identifies the total masses.

Theorem 27.7.1 *The number $\mathfrak{r}(v;Y)$ is independent of the chosen divisor metrics and of the Kähler resolution carrying* (27.10)*. Two potentials of the same ordinary singularity type, in the same closed background, have the same number whenever each has a decomposition of the form* (27.10)*. Their residual classes agree on a common refinement. Residual classes add when the traces and their backgrounds are added, and their mixed masses are given by* (27.13)*.*

Proof Changing a divisor metric changes ℓ by a smooth function, changes w by its negative, and changes η_v by the corresponding $\mathrm{dd^c}$-term. Thus R_v is unchanged. The same cancellation holds when a smooth change of background is accompanied by the opposite change of potential.

For a modification $\rho : W \to Z$ with W Kähler, pull back the total divisor Γ, including all exceptional coefficients. The pulled-back decomposition has bounded remainder $w \circ \rho$, residual current $\rho^* R_v$ and nef residual class $\rho^* b_v$. Its mass is

$$\int_W (\rho^* b_v)^k = \int_Z b_v^k.$$

For two decompositions on different resolutions, pass to a common Kähler refinement. At a generic smooth point of a prime component of the union of their divisors, the difference of the logarithmic terms is $(a_1 - a_2)\log|t|^2 + O(1)$ in a local defining coordinate t. The two bounded-difference inequalities force $a_1 = a_2$. Hence the total divisors, and therefore the residual classes in the common background, coincide.

Finally, adding the decompositions adds both the divisors and the bounded remainders. The residual class is the sum of the nef residual classes, and (27.13) computes its mixed terms. □

Proposition 27.7.1 *If $\mathfrak{r}(v;Y) > 0$, there is a strict positive current in $\{\alpha_Z\}$ with a potential p satisfying $p \le v_\pi + A$ for some constant A. If b_v has a smooth positive representative on one resolution carrying (27.10), that resolution admits such a strict current with $p \sim v_\pi$. When Y is smooth, the strict current and the respective potential inequalities descend to Y.*

Proof Positive residual mass gives $\int_Z b_v^k > 0$. Apply (27.7) to obtain $K = \eta_v + \mathrm{dd^c} a \ge \delta\kappa$. The current

$$[\Gamma] + K = \alpha_Z + \mathrm{dd^c}(\ell + a)$$

is strict. Since a is bounded above on Z and w is bounded below, its potential $p = \ell + a$ satisfies $p \le \ell + w + A = v_\pi + A$. If b_v has a smooth positive representative, choose a smooth, giving the bounded difference in both directions.

For smooth Y, take $\alpha_Y = \alpha_0|_Y$ in the restriction setting. Then Proposition 1.22.2 gives $p = q \circ \pi$ for $q \in \mathrm{PSH}(Y, \alpha_Y)$. Strict positivity follows from (27.8); surjectivity of π descends each potential inequality. The smooth-positive hypothesis concerns the specified model: pulling a Kähler class through a further blowup need not give a Kähler class. □

Example 27.7.1 Let $Y = \mathbb{P}^k$ be a hyperplane in $X = \mathbb{P}^{k+1}$, and let another hyperplane H_X cut out $H \subset Y$. Normalize the Fubini–Study form by $\int_Y \omega^k = 1$. For $c > 0$ and $d \ge 0$, take $\alpha_0 = (c + d)\omega$ on X and $u = c \log |s_{H_X}|^2$ with the Fubini–Study metric. Its genuine restriction satisfies

$$\alpha_0|_Y + \mathrm{dd^c} v = c[H] + d\omega|_Y, \qquad \mathfrak{r}(v;Y) = d^k, \qquad \mathcal{R}(\{\alpha_0\};Y) = (c + d)^k.$$

Thus $d = 0$ gives zero trace mass in a class with positive restricted invariant. The divisor remains a nonzero part of the current even though the non-pluripolar top product discards it.

Example 27.7.2 Let $\mu : Y \to \mathbb{P}^2$ be the blowup of a point, with exceptional curve E, and put $\alpha_Y = \mu^*\omega_{\mathrm{FS}}$, $v = 0$. The residual class is nef and big, and

$$\mathfrak{r}(0;Y) = \mathcal{R}(\{\alpha_Y\};Y) = 1.$$

There is nevertheless no bounded $q \in \mathrm{PSH}(Y, \alpha_Y)$ for which $\alpha_Y + \mathrm{dd^c} q \ge \delta\kappa$ with $\delta > 0$ and κ Kähler. Indeed, boundedness permits restriction to E, and the disk criterion in Lemma 1.22.1 gives

$$\mathrm{dd^c}(q|_E) \ge \delta\kappa|_E,$$

because $\alpha_Y|_E = 0$. Integration over the compact curve contradicts Stokes' theorem. Positive mass therefore supplies the one-sided comparison in Proposition 27.7.1, but may require additional singularities.

Example 27.7.3 Let $\mu : X \to \mathbb{P}^3$ be the blowup of a line L, and set $a = \mu^*\{\omega_{\mathrm{FS}}\}$. Its exceptional divisor is $E = \mathbb{P}(N_{L/\mathbb{P}^3}) \simeq \mathbb{P}^1 \times \mathbb{P}^1$, since $N_{L/\mathbb{P}^3} \simeq \mathcal{O}(1) \oplus \mathcal{O}(1)$. On E, the map μ is the projection to L. Hence $\int_E a^2 = 0$, whereas a section Y of $E \to L$ satisfies $\int_Y a = 1$. Every irreducible subvariety not contained in E has positive intersection, because $\mu^*\omega_{\mathrm{FS}}$ is positive on its nonempty open part outside E. By Corollary 27.6.1,

$$E_{\mathrm{nK}}(a) = E, \qquad \mathcal{R}(a; E) = 0, \qquad \mathcal{R}(a; Y) = 1.$$

The zero-invariant union detects the whole null locus. Positivity on one subvariety does not assert that it lies outside that locus.

27.8 Divisorial components at the threshold

A divisorial component consumes part of the fixed Gauduchon trace while contributing no density to the non-pluripolar measure. This bounds the coefficient of a subsolution. At the maximal coefficient, every weak limit of a maximizing sequence annihilates the divisor class. Uniform integrability along the sequence ensures that the limit also attains the maximum, by Proposition 25.9.2.

Return to an arbitrary compact complex X of dimension $n \geq 2$ with Hermitian metric ω. Fix a smooth strictly positive probability volume $\mathrm{d}V$ and a nonzero $0 \leq f \in L^1(\mathrm{d}V)$. Write $p = f^{1/n}$ and $b = b_\omega(f)$ for its root density and subsolution threshold. Use the current family $\mathcal{G}_\omega$, determinant density k_T, functional D_p, its supremum $\rho_\omega(p)$ and the root space $\mathcal{D}^0_\omega$ from Section 25.9. The linear positivity threshold κ_ω is defined in Corollary 4.3.1.

Proposition 27.8.1 *Let $c > 0$ be real and let E be an effective real divisor on X. Suppose $u \in \mathrm{PSH}(X, \omega)$ satisfies*

$$\mathrm{MA}^{\mathrm{ext}}_\omega(u) \geq c f \, \mathrm{d}V, \qquad \omega + \mathrm{dd^c} u \geq [E].$$

Let β_E be a smooth closed representative of its Bott–Chern class, and put $\kappa = \kappa_\omega(\beta_E)$. Then $b > 0$, $0 \leq \kappa < 1$ and

$$c \leq b(1 - \kappa)^n. \tag{27.14}$$

If $c = b$, every smooth maximizing sequence (T_j) for D_p satisfies

$$\int_X \beta_E \wedge T_j = \int_X [E] \wedge T_j \longrightarrow 0.$$

Every weak cluster current T therefore satisfies $\int_X \beta_E \wedge T = 0$, and the class of E is not big. If the densities $p k_{T_j}$ of the chosen sequence are uniformly integrable, every such cluster current maximizes D_p. In particular, this holds for every smooth maximizing sequence when $p \in \mathcal{D}^0_\omega$.

Proof The subsolution characterization gives $b \geq c > 0$. For a smooth strictly positive normalized Gauduchon form T, Lemma 6.4.2 shows that

$$R = (\omega + \mathrm{dd^c} u) \wedge T - c^{1/n} p k_T \, \mathrm{d}V$$

is positive, with mass $1 - c^{1/n} D_p(T)$. The density term gives no mass to the support of E, whereas $[E] \wedge T$ is supported there. Restricting the positive measure R to that support and using $\omega + \mathrm{dd^c} u \geq [E]$ gives

$$\int_X [E] \wedge T \leq R(X) = 1 - c^{1/n} D_p(T).$$

Let ℓ_E be a global logarithmic potential, so that $[E] = \beta_E + \mathrm{dd^c} \ell_E$ by the Poincaré–Lelong formula. Since ℓ_E is locally integrable and T is smooth and $\mathrm{dd^c}$-closed, distributional Stokes identifies the left side with $\int_X \beta_E \wedge T$. It is nonnegative for every such test, so $\kappa \geq 0$. Also $D_p(T) > 0$, which gives $\kappa < 1$. Taking the supremum of $D_p(T)$ and using $\rho_\omega(p) = b^{-1/n}$ proves (27.14).

If $c = b$ and (T_j) is maximizing, the right side of the preceding measure bound tends to zero. Pairing with the fixed smooth form β_E is continuous under current convergence, proving the cluster assertion. Equation (27.14) forces $\kappa = 0$, so Corollary 4.3.1 excludes bigness. Uniform integrability along the chosen sequence gives the maximizing assertion by Proposition 25.9.2. When $p \in \mathcal{D}^0_\omega$, Theorem 25.9.1 supplies this condition for every sequence. □

At the threshold $c = b$, if $E \neq 0$, the vanishing pairing defines a nonempty proper exposed face of the weak closure of smooth strictly positive normalized Gauduchon forms. Indeed, this closure is compact and convex; the pairing with β_E is continuous and nonnegative there, and it is strictly positive on every smooth strictly positive member. The proposition places every cluster of a smooth maximizing sequence on this face. This assertion concerns that closure; it does not assert that every normalized Gauduchon current is a limit of smooth ones. Nor does the smooth pairing $\int_X \beta_E \wedge T$ define $[E] \wedge T$ for a singular T.

Example 27.8.1 The weak closure of smooth strictly positive normalized Gauduchon forms can be smaller than $\mathcal{G}_\omega$, even on a Kähler surface. Let X be the blowup of $\mathbb{P}^2$ at one point and C its exceptional curve. Fix a Kähler form ω and a smooth closed representative β_C of $\{C\}$. Then

$$T_C = \frac{[C]}{\int_C \omega} \in \mathcal{G}_\omega, \qquad \int_X \beta_C \wedge T_C = \frac{-1}{\int_C \omega} < 0.$$

Every smooth strictly positive $T \in \mathcal{G}_\omega$ instead satisfies $\int_X \beta_C \wedge T = \int_C T > 0$.

Proof The integration current $[C]$ is positive and closed, so its normalization belongs to $\mathcal{G}_\omega$. Its pairing with the smooth form β_C is the self-intersection $C^2 = -1$, computed in Example 25.5.1. For smooth T, the Poincaré–Lelong formula and $\mathrm{dd^c} T = 0$ give the second identity. The pairing with the fixed smooth β_C is continuous under current

convergence. It is therefore nonnegative on the weak closure of these smooth tests, which excludes T_C. □

Let $\mathscr{C}_\omega$ denote the weak closure of smooth normalized Gauduchon tests from Section 4.4, and let m_ω be the minimum over all normalized currents from Section 25.17. In particular $T_C \notin \mathscr{C}_\omega$. Since $[C]$ is a positive representative of β_C, the two thresholds satisfy

$$m_\omega(\beta_C) \le -\frac{1}{\int_C \omega} < 0 \le \kappa_\omega(\beta_C).$$

Thus extending the minimum in (4.4) to all normalized Gauduchon currents changes its value in this example.

Return to the original (X, ω, f). The next two corollaries use uniform integrability along one smooth maximizing sequence. The root-tail condition $p \in \mathcal{D}^0_\omega$ is sufficient, since it gives $b > 0$ and uniform integrability for the entire test family.

Corollary 27.8.1 *Assume $b > 0$ and that D_p has a smooth maximizing sequence (T_j) with uniformly integrable densities pk_{T_j}. Let E be an effective real divisor whose class has a smooth semipositive representative β_E. If*

$$\int_{\{\beta_E=0\}} f \, \mathrm{d}V = 0,$$

no subsolution at the threshold $b_\omega(f)$ satisfies $\omega + \mathrm{dd^c} u \ge [E]$.

Proof If such a subsolution existed, Proposition 27.8.1 would supply a maximizing current $T \in \mathscr{G}_\omega$ with $\int_X \beta_E \wedge T = 0$. In the matrix-density notation of Section 25.6, write $\sigma_T = \omega \wedge T$ for the trace measure, P_T for the matrix density of T and B for the matrix of β_E. Positivity makes $\mathrm{tr}(BP_T) = 0$ for σ_T-almost every point. Wherever $B \neq 0$, this forces P_T to be singular: a positive definite P_T would have strictly positive trace against the nonzero positive semidefinite B. Thus $\Gamma_0(T)$ is supported on $\{\beta_E = 0\}$. It equals $k_T \, \mathrm{d}V$ by Theorem 25.8.1. The hypothesis gives $pk_T = 0$ almost everywhere, contradicting $D_p(T) = \rho_\omega(p) > 0$. □

Corollary 27.8.2 *Assume $b > 0$ and that D_p has a smooth maximizing sequence (T_j) with uniformly integrable densities pk_{T_j}. Let $u \in \mathrm{PSH}(X, \omega)$ be a subsolution at b. Suppose it satisfies $\omega + \mathrm{dd^c}u \geq [E]$ for an effective real divisor E whose Bott–Chern class is nef. Let β_E be any smooth closed representative of that class. For every smooth strictly positive probability measure μ, the root volume of Definition 25.1.1 satisfies*

$$\lambda_\mu(\beta_E) = 0.$$

Proof By Proposition 27.8.1, there is a maximizing current $T \in \mathscr{G}_\omega$ with $\int_X \beta_E \wedge T = 0$. If $\lambda_\mu(\beta_E) > 0$, Proposition 25.17.1 gives $\Gamma_0(T) = 0$. It follows that $D_{f^{1/n}}(T) = 0$, contradicting its positive maximal value. Since nefness gives $\lambda_\mu(\beta_E) \geq 0$, the conclusion follows for each μ. □

The semipositive criterion and the nef-volume criterion detect different obstructions. The Hopf hyperplane class has zero variational volume, but its nowhere vanishing semipositive representative excludes a component at the threshold.

For example, the hyperplane divisor H on the Hopf threefold has a smooth semipositive representative of rank two everywhere. By Corollary 8.6.1 and Proposition 13.26.4, one has $\mathcal{D}^0_\omega = L^3$ on this example. Hence, for every nonzero nonnegative L^1 density, a threshold subsolution has zero generic divisorial coefficient along H. Otherwise divisorial removal, as in Proposition 25.18.1, would give $\omega + \mathrm{dd^c}u \geq a[H]$ for some $a > 0$, contradicting Corollary 27.8.1 applied to aH. This exclusion uses no bounded mass property. It is stronger here than nonbigness alone: the residual formulas of Proposition 25.19.1 allow a positive coefficient along H at smaller equation constants.

These conditions concern the maximal global coefficient. Realizing divisorial components from the resulting annihilating currents remains open; a threshold within a prescribed singularity type is a different problem.

27.9 Closed currents on the elliptic threefold

The elliptic threefold has no strict closed current: an exact smooth form forces every positive closed current to have zero vertical component. Matrix positivity removes the mixed components, and closedness makes the remaining coefficients descend to the base. The same descent then identifies the cones, envelopes and selected section spaces.

Let X be the threefold of Section 25.15, and let

$$\pi : X \longrightarrow B = \mathbf{P}^1 \times \mathbf{P}^1$$

be its elliptic fibration. Write σ_1, σ_2 for the integral-one Fubini–Study forms from the two factors, so $\alpha = \pi^*\sigma_1$ and $\beta = \pi^*\sigma_2$. The product coordinates used in that construction show that π is locally a holomorphic product with the connected elliptic curve E. Let t denote the logarithmic ratio on the covering space in that construction;

its differentials descend to X. We retain the balanced metric $\omega = \alpha + \beta + D$, where $D = \mathrm{d}t \wedge \mathrm{d}^{\mathrm{c}}t$. It satisfies

$$\alpha \wedge \beta = \tfrac{1}{2}\mathrm{dd}^{\mathrm{c}}\omega.$$

The argument below uses this identity, not the value of the lower Monge–Ampère mass.

Lemma 27.9.1 *Every positive closed real* $(1,1)$*-current* T *on* X *has the form*

$$T = \pi^* S$$

for a unique positive closed $(1,1)$*-current* S *on* B*. Pullback here is the ordinary pullback by a smooth holomorphic submersion.*

Proof The measure $T \wedge \alpha \wedge \beta$ is positive. Current Stokes gives

$$\int_X T \wedge \alpha \wedge \beta = \tfrac{1}{2}\int_X \omega \wedge \mathrm{dd}^{\mathrm{c}}T = 0.$$

Thus this measure is zero. All products in this identity pair a current with smooth forms.

In local product coordinates (z_1, z_2, ζ), write the coefficient measures of T as densities $p_{j\bar{k}}$ of a positive Hermitian matrix relative to their trace measure. The smooth form $\alpha \wedge \beta$ is a strictly positive volume form in the two base directions. Its wedge with T is a positive smooth factor times the vertical diagonal coefficient. That coefficient therefore vanishes. Positivity of the matrix and $|p_{j\bar{\zeta}}|^2 \le p_{j\bar{j}}p_{\zeta\bar{\zeta}}$ then make its entire vertical row and column zero. Consequently T has only terms of the form

$$\frac{\mathrm{i}}{2\pi}\, t_{j\bar{k}}(z,\zeta)\mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k, \qquad 1 \le j, k \le 2,$$

where the coefficients are distributions of order zero.

The equations $\partial T = \bar{\partial} T = 0$ imply

$$\partial_\zeta t_{j\bar{k}} = \partial_{\bar{\zeta}} t_{j\bar{k}} = 0$$

in the sense of distributions: these derivatives are exactly the coefficients of the terms containing one vertical differential. A distribution on a product whose derivatives in both real fiber directions vanish is pulled back from the base. For completeness, convolve it on a smaller product polydisk. The convolutions have both fiber derivatives zero and are independent of the fiber variable. Testing them against a fixed fiber test function of integral one gives distributions on the base; passage to the limit identifies the original distribution with their pullback. Independence on overlapping fiber disks and connectedness of the fiber extend this conclusion across the whole local product $U \times E$.

We therefore have $T|_{\pi^{-1}(U)} = \pi^* S_U$. Positivity and closedness of S_U follow by testing the coefficient distributions against a nonnegative fiber test function of integral one and using the horizontal equations. Pullback on distributions is injective in a

product, by the same test. Hence the S_U agree on overlaps and define the required positive closed current S. Injectivity also proves uniqueness. □

The Gauduchon forms in Lemma 25.15.3 have bidegree $(2,2)$ and satisfy $\mathrm{dd^c}\Gamma_j = 0$. Their bounded degree and divergent positive-root volume are compatible with this description of the closed $(1,1)$-currents. The two statements concern different cones of currents.

Proposition 27.9.1 *Pullback*

$$\pi^* : H^{1,1}_{\mathrm{BC}}(B,\mathbb{R}) \longrightarrow H^{1,1}_{\mathrm{BC}}(X,\mathbb{R})$$

is injective. The pseudoeffective and nef cones on X coincide and are

$$\mathrm{Psef}(X) = \mathrm{Nef}(X) = \{a\{\alpha\}_{\mathrm{BC}} + b\{\beta\}_{\mathrm{BC}} : a, b \geq 0\}.$$

Its trace-one base for the Gauduchon metric ω is the line segment $a + b = (8\pi)^{-1}$. Every class in this cone has a smooth semipositive representative, and none is big. The real form $\beta - \alpha$ is de Rham exact but represents a nonzero Bott–Chern class.

Proof Suppose a smooth closed real form τ on B satisfies $\pi^*\tau = \mathrm{dd^c}u$ with u smooth on X. Restriction to each compact connected elliptic fiber makes u harmonic, hence constant there. Local holomorphic sections show that $u = v \circ \pi$ for a smooth function v on B. Thus $\pi^*(\tau - \mathrm{dd^c}v) = 0$, which implies $\tau = \mathrm{dd^c}v$ because π is a submersion. This proves injectivity.

The standard cohomology calculation for $\mathbf{P}^1 \times \mathbf{P}^1$, together with the Kähler $\partial\bar{\partial}$ lemma, gives

$$H^{1,1}_{\mathrm{BC}}(B,\mathbb{R}) = \mathbb{R}\{\sigma_1\} \oplus \mathbb{R}\{\sigma_2\}, \qquad \int_B \sigma_1 \wedge \sigma_2 = 1, \qquad \sigma_1^2 = \sigma_2^2 = 0.$$

These are the usual projective-space and product computations recalled in the closed theory. If S is positive and closed, its two class coefficients are

$$a = \int_B S \wedge \sigma_2 \geq 0, \qquad b = \int_B S \wedge \sigma_1 \geq 0.$$

The current description of Bott–Chern cohomology and the preceding lemma show that every pseudoeffective class on X is therefore in the displayed cone. Conversely every such class has the smooth semipositive representative $a\alpha + b\beta$, and is nef. A nef class is pseudoeffective by the compactness argument following Proposition 27.1.2. This proves the equality of cones. It does not require a calculation of the whole Bott–Chern group of X.

Since $\omega^2 = 2\alpha \wedge \beta + 2(\alpha + \beta) \wedge D$ and $\int_E D = 4\pi$, fiber integration gives

$$\int_X (a\alpha + b\beta) \wedge \omega^2 = 8\pi(a + b).$$

This proves the assertion about the trace-one base. No closed Kähler current exists on X, by Proposition 25.15.3, so none of these classes is big. Finally, the globally defined one-form $\mathrm{d}^{\mathrm{c}}t$ satisfies

$$\mathrm{d}(\mathrm{d}^{\mathrm{c}}t) = \beta - \alpha.$$

The class of $\sigma_2 - \sigma_1$ on B is nonzero, and injectivity makes its pullback nonzero in Bott–Chern cohomology. Thus de Rham exactness does not imply dd^{c}-exactness on this example. □

Every pseudoeffective class on this threefold therefore has empty non-nef locus and non-Kähler locus equal to X. All its closed positive representatives come from a two-dimensional base, so $(a\alpha + b\beta)^3 = 0$.

27.9.1 Potentials and envelopes from the base

Potential descent preserves both ordinary and ideal constraints, so the singularity envelopes are computed on the base. Their three-dimensional Monge–Ampère measures nevertheless vanish because they have only two horizontal directions.

Proposition 27.9.2 *Let $a, b \geq 0$, put $\theta_B = a\sigma_1 + b\sigma_2$ and $\theta = \pi^*\theta_B$. Pullback is a bijection*

$$\mathrm{PSH}(B, \theta_B) \longrightarrow \mathrm{PSH}(X, \theta), \qquad v \longmapsto v \circ \pi.$$

It preserves ordinary singularity order and equivalence, and ideal singularity order and equivalence. For every $v \in \mathrm{PSH}(B, \theta_B)$,

$$P_\theta[v \circ \pi] = P_{\theta_B}[v] \circ \pi, \qquad P_\theta[v \circ \pi]_{\mathcal{I}} = P_{\theta_B}[v]_{\mathcal{I}} \circ \pi.$$

Nevertheless, every $u \in \mathrm{PSH}(X, \theta)$ satisfies

$$\mathrm{MA}_\theta^{\mathrm{ext}}(u) = 0.$$

Proof The form θ_B is closed, and π is a proper holomorphic submersion with connected fibers. Thus Proposition 1.22.1 gives the asserted bijection. Surjectivity of π also shows that an ordinary singularity inequality holds upstairs exactly when it holds downstairs, with the same constant.

By Lemma 1.16.4, multiplier ideals extend under π at every positive weight. Restriction to local holomorphic sections reflects ideal inclusion, so the ideal orders and equivalences correspond as well.

The competitors defining both envelopes correspond under this bijection, including the bound by zero and the required ordinary or ideal condition. Their pointwise suprema commute with pullback. Upper semicontinuous regularization also commutes with pullback in a product chart, since a base neighborhood is obtained from every sufficiently small product neighborhood and local sections give the reverse comparison. This proves the two envelope identities.

Finally, write $u = v \circ \pi$ and, for positive integers j, set $u_j = \max\{u, -j\}$. It descends to $v_j = \max\{v, -j\}$. The semipositivity of θ_B makes these admissible bounded potentials. By Lemma 1.7.2, $(\theta + \mathrm{dd^c} u_j)^3 = 0$ because the base has dimension two. The increasing truncation definition of the extended non-pluripolar measure now proves the last assertion. □

Thus equality of the two envelopes in these backgrounds is exactly the corresponding equality on the base, while every three-dimensional Monge–Ampère measure is zero. A positive-volume criterion from the big closed theory cannot be applied to these non-big classes. The strictly Hermitian background ω belongs to a different theory: its smooth positive-density equations are solvable, even though its infimum of masses is zero.

27.9.2 Selected sections and their growth

The same descent identifies multiplier-ideal-selected sections and their continuous norms. The resulting quadratic growth agrees with the dimension of the base. Write $\mathcal{O}_B(a, b)$ for the tensor product of the degree-a and degree-b hyperplane bundles from its two factors.

Proposition 27.9.3 *Let a, b be positive integers, let $L_B = \mathcal{O}_B(a, b)$ and $L = \pi^* L_B$. For every quasi-plurisubharmonic v on B and positive integer m, pullback gives an isomorphism*

$$H^0\big(B, L_B^m \otimes \mathcal{I}(mv)\big) \longrightarrow H^0\big(X, L^m \otimes \mathcal{I}(mv \circ \pi)\big).$$

For $v = 0$ these spaces have dimension $(am + 1)(bm + 1)$.

Equip L_B with a smooth Hermitian metric, choose a continuous real function φ on B, and put $\nu = \pi_* \mathrm{d}V_X$ for a smooth positive volume form on X. Use the norms

$$\|s\|_B^2 = \int_B |s|^2 \mathrm{e}^{-m\varphi}\, \mathrm{d}\nu, \qquad \|\pi^* s\|_X^2 = \int_X |\pi^* s|^2 \mathrm{e}^{-m\varphi\circ\pi}\, \mathrm{d}V_X.$$

The isomorphism is isometric. If $K_{m,B}$ and $K_{m,X}$ are the corresponding pointwise squared-norm sums over orthonormal bases, including the displayed continuous weight, then

$$K_{m,X} = K_{m,B} \circ \pi, \qquad \pi_*(K_{m,X}\, \mathrm{d}V_X) = K_{m,B}\, \mathrm{d}\nu.$$

Proof The selected-space identification is Proposition 26.7.1, applied to L_B and v on the base. For $v = 0$, the usual bihomogeneous polynomial basis on $\mathbf{P}^1 \times \mathbf{P}^1$ has $(am + 1)(bm + 1)$ elements.

Integration along the fibers of the proper submersion shows that ν is a smooth positive volume form. Its defining integration identity proves equality of the two norms. Pull back an orthonormal basis and compare its pointwise norms to obtain the

first kernel identity; applying the same integration identity proves the second. The assertions include a zero-dimensional selected space, for which both kernels vanish.□

Even without a singular selecting weight, the growth is quadratic:

$$\lim_{m\to\infty} m^{-2}h^0(X, L^m) = ab, \qquad \lim_{m\to\infty} m^{-3}h^0(X, L^m) = 0.$$

The variational volume of a small Hermitian perturbation measures the missing growth dimension in the following calculation.

27.9.3 The boundary volume and section growth

For the fixed balanced volume on this threefold, the zero potential solves the equation for every positive linear combination of α, β, D. The variational volume is therefore an explicit cubic polynomial. Its order at the nef boundary records which of the two horizontal coefficients vanish.

Proposition 27.9.4 *On the elliptic threefold, put $\theta = a\alpha + b\beta$ with $a, b \geq 0$, and let $\mu_0 = \omega^3/(24\pi)$. Then for every $\epsilon > 0$,*

$$\mathcal{V}_{\mu_0}(\theta + \epsilon\omega) = 24\pi\epsilon(a + \epsilon)(b + \epsilon). \tag{27.15}$$

In particular $\mathcal{V}_\mu(\theta) = 0$ for every smooth positive probability measure μ. If q is the number of positive entries among a, b, then the orders of Proposition 25.16.1 satisfy

$$\underline{a}(\theta) = \overline{a}(\theta) = 3 - q.$$

These orders hold for every smooth positive probability measure and every Hermitian perturbation direction.

Proof The forms α, β, D are semipositive of rank one, their squares vanish, and $\omega = \alpha + \beta + D$ is strictly positive. Thus

$$\Omega_\epsilon = \theta + \epsilon\omega = (a + \epsilon)\alpha + (b + \epsilon)\beta + \epsilon D > 0.$$

The wedge expansion gives

$$\Omega_\epsilon^3 = 6\epsilon(a + \epsilon)(b + \epsilon)\alpha \wedge \beta \wedge D = 24\pi\epsilon(a + \epsilon)(b + \epsilon)\mu_0.$$

The zero potential therefore solves its smooth normalized equation. By Theorem 25.1.1, the equation's scalar is precisely $\mathcal{V}_{\mu_0}(\Omega_\epsilon)$, proving (27.15). Nef continuity in Theorem 25.10.1 and smooth-density comparison give $\mathcal{V}_\mu(\theta) = 0$.

If both a, b are positive, the leading term is $24\pi ab\epsilon$. If exactly one is positive, it is $24\pi(a + b)\epsilon^2$. If both are zero, the formula is $24\pi\epsilon^3$. The logarithmic quotients have limits $1, 2, 3$, respectively. Their independence of the density and perturbing metric is Proposition 25.16.1. □

For nonnegative integers a, b, the bundle $L = \pi^* \mathcal{O}_B(a, b)$ has

$$h^0(X, L^m) = (am + 1)(bm + 1).$$

Indeed the compact-fiber descent and bihomogeneous polynomial argument of Proposition 27.9.3 also apply when one or both integers vanish. Its growth degree is exactly q, while the boundary order of its smooth curvature representative is $3 - q$. Thus the boundary order records the loss of algebraic growth dimension in this explicit family, even though the unperturbed three-dimensional Monge–Ampère measures vanish.

Every positive perturbation Ω_ϵ still has zero lower mass, by Theorem 25.15.1, and finite upper mass, since X has BMP. Formula (27.15) concerns one fixed smooth density, and does not assert conservation of the raw mass over all potentials. Whether comparable identifications hold beyond such explicit families is part of Problem 25.16.1.

27.9.4 Limits of the minimizing Gauduchon forms

The minimizing tests for the preceding boundary formula are Gauduchon powers of the equation metrics themselves. Their normalized $(2, 2)$-forms converge smoothly, although their metric roots may diverge. The root volume λ_{μ_0} and denominator D_{μ_0} are those of Definition 25.1.1; $\asymp$ denotes two-sided bounds by fixed positive multiples as $\epsilon \downarrow 0$.

Proposition 27.9.5 *Let $a, b \geq 0$, $\theta = a\alpha + b\beta$ and $\Omega_\epsilon = \theta + \epsilon\omega$. For every $\epsilon > 0$, the metric Ω_ϵ is Gauduchon. It minimizes the quotient defining $\lambda_{\mu_0}(\Omega_\epsilon)$. Put*

$$I_\epsilon = 8\pi\big(ab + 2(a + b)\epsilon + 3\epsilon^2\big), \qquad T_\epsilon = \frac{\Omega_\epsilon^2}{I_\epsilon}, \qquad \gamma_\epsilon = \frac{\Omega_\epsilon}{\sqrt{I_\epsilon}}.$$

Then $\int_X \omega \wedge T_\epsilon = 1$, $\mathrm{dd^c} T_\epsilon = 0$ and $T_\epsilon = \gamma_\epsilon^2$. As $\epsilon \downarrow 0$, the forms T_ϵ converge smoothly to

$$T_0 = \begin{cases} (4\pi)^{-1}\alpha \wedge \beta, & a, b > 0, \\ (8\pi)^{-1}\alpha \wedge (\beta + D), & a > 0,\ b = 0, \\ (8\pi)^{-1}\beta \wedge (\alpha + D), & a = 0,\ b > 0, \\ (24\pi)^{-1}\omega^2, & a = b = 0. \end{cases}$$

In every case $\theta \wedge T_0 = 0$. If q is the number of positive coefficients among a, b, the denominator of the normalized minimizing test satisfies

$$D_{\mu_0}(\gamma_\epsilon) = \frac{\big(24\pi\epsilon(a + \epsilon)(b + \epsilon)\big)^{2/3}}{I_\epsilon} \asymp \epsilon^{q/3},$$

where the two comparison constants may depend on the fixed pair (a, b).

Proof Write $x = a + \epsilon$, $y = b + \epsilon$ and $z = \epsilon$. Since $\alpha^2 = \beta^2 = D^2 = 0$,

$$\Omega_\epsilon^2 = 2xy\alpha \wedge \beta + 2xz\alpha \wedge D + 2yz\beta \wedge D.$$

The forms α, β are closed and $\mathrm{dd^c} D = 2\alpha \wedge \beta$. Applying $\mathrm{dd^c}$ to each term makes its last two terms zero because they contain α^2 or β^2. Thus $\mathrm{dd^c}(\Omega_\epsilon^2) = 0$. The zero potential solves the normalized equation for Ω_ϵ and μ_0, so the characterization of minimizing metrics in Theorem 25.1.1 gives the minimizing assertion.

Fiber integration gives $\int_X \alpha \wedge \beta \wedge D = 4\pi$. Hence

$$\int_X \omega \wedge \Omega_\epsilon^2 = 8\pi(xy + xz + yz) = I_\epsilon.$$

This proves the trace normalization. Dividing the three coefficients in the preceding wedge expansion by I_ϵ gives each displayed limit, with convergence in every smooth norm since the forms are fixed. The identities $\alpha^2 = \beta^2 = 0$ then give $\theta \wedge T_0 = 0$ in all four cases.

Set $c_\epsilon = 24\pi\epsilon(a + \epsilon)(b + \epsilon)$. The equation gives

$$\gamma_\epsilon^3 = \frac{c_\epsilon}{I_\epsilon^{3/2}} \mu_0.$$

Substitution in the definition of D_{μ_0} and $\int_X \mu_0 = 1$ yield $D_{\mu_0}(\gamma_\epsilon) = c_\epsilon^{2/3}/I_\epsilon$. For fixed (a, b) one has $c_\epsilon \asymp \epsilon^{3-q}$ and $I_\epsilon \asymp \epsilon^{2-q}$. The quotient therefore has order $q/3$, as asserted. □

The limiting positive $(2, 2)$-forms have coefficient-matrix ranks $1, 2, 3$ when $q = 2, 1, 0$, respectively. This follows in a coframe adapted to the three rank-one forms α, β, D from the three complementary products in the proof. If $a, b > 0$, the metric roots themselves converge smoothly to

$$\frac{a\alpha + b\beta}{\sqrt{8\pi ab}}.$$

If $a > 0$ and $b = 0$, however, the coefficient of α in γ_ϵ grows as a positive constant times $\epsilon^{-1/2}$, while its other two coefficients tend to zero. Thus the roots are unbounded relative to the fixed metric ω, although their squares converge smoothly. The same statement holds with α and β interchanged. A rank-two positive $(2, 2)$-form cannot be the square of a finite semipositive $(1, 1)$-form in dimension three: the latter's eigenvalues give complementary pairwise products, whose possible nonzero ranks are three or one. At the vertex $a = b = 0$, the normalized root instead stays equal to $\omega/\sqrt{24\pi}$.

For a nonzero boundary class, the smooth limiting test annihilates θ and its determinant denominator vanishes. No concentration of its coefficient measures occurs in this family. This degeneration of fixed-density minimizing tests is different from the sequence of Gauduchon forms used to prove zero lower mass in Lemma 25.15.3.

At the vertex, the denominator remains positive; the order three comes entirely from scaling the background.

27.10 Smooth equations on the ample locus

Return to a compact Kähler manifold X of dimension $n \geq 1$. Let α be a big closed class, with smooth representative θ. Write $\mathrm{Amp}(\alpha) = X \setminus E_{\mathrm{nK}}(\alpha)$. The full-mass solution for a smooth positive volume form has minimal singularities. A strict analytic current supplies a barrier on the ample locus, where the interior estimate of Corollary 13.17.1 gives smoothness.

Let $\mathrm{vol}(\alpha)$ be the classical volume and $\mathcal{E}(X, \theta)$ its full-mass class.

Theorem 27.10.1 *Let μ be a smooth positive volume form with $\mu(X) = \mathrm{vol}(\alpha)$. The normalized full-mass solution $\varphi \in \mathcal{E}(X, \theta)$ of*

$$\left\langle (\theta + \mathrm{dd^c}\varphi)^n \right\rangle = \mu, \qquad \sup_X \varphi = 0, \tag{27.16}$$

is smooth on $\mathrm{Amp}(\alpha)$. *The restriction of $\theta + \mathrm{dd^c}\varphi$ to this locus is a Kähler form.*

Existence and uniqueness are the classical closed-big equation theorem recorded in Section 33.17. We prove the regularity assertion.

Proof Fix a Kähler form ω and put $f = \mu/\omega^n$. Compactness and smoothness give $f \in L^p(X, \omega^n)$ for every finite $p > 1$. The classical minimal-singularity estimate for this closed-big equation, recorded in Section 33.17, gives a constant $C > 0$ such that

$$V_\theta - C \leq \varphi \leq V_\theta. \tag{27.17}$$

The upper bound follows from the normalization.

Choose a strict analytic current T of Theorem 27.3.1. Since its Bott–Chern class is α, it has a potential ρ with analytic singularities and smooth remainder. For some constant $\varepsilon > 0$ it satisfies

$$T = \theta + \mathrm{dd^c}\rho \geq \varepsilon\omega, \qquad \{\rho = -\infty\} = E_{\mathrm{nK}}(\alpha).$$

Thus ρ is smooth on $\Omega = \mathrm{Amp}(\alpha)$. Subtract a constant so that $\rho \leq 0$, hence $\rho \leq V_\theta$. By (27.17),

$$\rho - C \leq \varphi \leq 0.$$

Thus φ is locally bounded on Ω, and locality identifies its non-pluripolar equation there with the bounded-potential equation. The density f is globally in L^p and is smooth and strictly positive on Ω. All hypotheses of Corollary 13.17.1 hold for the strict barrier ρ, so φ is smooth and $\theta + \mathrm{dd^c}\varphi > 0$ on Ω. Closedness of θ makes this form Kähler. □

Here the density is smooth and positive on the entire regular locus of the strict barrier; its global integrability follows from smoothness on compact X. The resulting metric can still approach the omitted set in finite length. To determine that length, one must compare its normal and tangential directions. The following examples begin with density zeros; singular radial densities will then separate boundedness of the potential from completeness of the metric.

27.11 Density zeros and metric degeneration

In complex dimension $n \geq 2$, a zero of the determinant can arise from a holomorphic map, from an exceptional fiber, or from a singular Hessian. These mechanisms give different relations between the regularity of the potential and the distance to the zero set. Bounded Hölder solutions on a positive background are supplied by Theorems 10.1.1 and 13.4.1; their existence does not give a pointwise estimate for the metric at such a zero.

Proposition 27.11.1 *Let $F : X \to Y$ be finite surjective between compact Kähler n-folds, let ω_Y be Kähler, and suppose $F^*[\omega_Y]$ has a Kähler representative θ. The full-mass solution for $(F^*\omega_Y)^n$ has a smooth bounded minimal potential and smooth semipositive current $F^*\omega_Y$. Fix a smooth positive local volume form $\mathrm{d}V$. At a generic branch chart, disjoint from other branch components, where $F(z, w) = (z^k, w)$, $k \geq 2$,*

$$(F^*\omega_Y)^n = |z|^{2(k-1)} G \, \mathrm{d}V, \qquad F^*\omega_Y \asymp |z|^{2(k-1)} |\mathrm{d}z|^2 + |\mathrm{d}w|^2, \tag{27.18}$$

with G smooth positive. The distance to the branch divisor is comparable to $|z|^k$. The deleted regular metric is incomplete, and the smooth current has no divisorial component. There is a strict logarithmic barrier with pole along the global ramification divisor.

Proof The Kähler $\partial\bar{\partial}$ lemma identifies the chosen class with its Bott–Chern class, as in Section 22.4. Thus $F^*\omega_Y = \theta + \mathrm{dd^c}u$ for a global smooth u. Its class mass is the full background mass; closed full-mass uniqueness applies as recalled in Section 33.17. The Jacobian of the displayed branch form is kz^{k-1}, and uniform comparison of ω_Y to the Euclidean form gives (27.18). Every path to $z = 0$ has length bounded below by the variation of a fixed multiple of $|z|^k$, and a fixed-w radial path attains this order. The determinant section of $\mathrm{d}F$ defines the global ramification divisor R. With any smooth metric on $\mathcal{O}(R)$, $\theta - \epsilon\Theta_R > 0$ for sufficiently small $\epsilon > 0$, so $\epsilon \log |s_R|^2 - C$ has a strict positive θ-current. This uses the global divisor; no local branch coordinate is treated as a global line bundle. □

Example 27.11.1 For $n \geq 2$, $q > 1$ and integer $k \geq 2$, let $H_q = (\mathbb{C}^n \setminus \{0\})/(Z \sim qZ)$. On the cover put $R = \sum_j |Z_j|^2$, $S = \sum_j |Z_j|^{2k}$,

$$h_0 = \frac{\mathrm{i}}{2\pi} \frac{\sum_j \mathrm{d}Z_j \wedge \mathrm{d}\bar{Z}_j}{R}, \quad u_k = \log S - k \log R,$$

$$\theta_k = k\mathrm{dd}^{\mathrm{c}} \log R + \frac{\mathrm{i}}{2\pi} \partial \log S \wedge \bar{\partial} \log S.$$

These descend under $Z \mapsto qZ$. The coordinate-power map $F_k([Z]) = [(Z_1^k, \ldots, Z_n^k)]$ on the Hopf quotient is finite surjective: $F_k(qZ) = q^k F_k(Z)$, and root choices are finite modulo source scaling. The form $\mathrm{dd}^{\mathrm{c}} \log R$ is positive transverse to the complex Euler direction, while $\partial \log S$ is nonzero on that direction; hence $\theta_k > 0$. Direct differentiation gives

$$\theta_k + \mathrm{dd}^{\mathrm{c}} u_k = F_k^* h_0 = \frac{\mathrm{i}k^2}{2\pi S} \sum_j |Z_j|^{2k-2} \mathrm{d}Z_j \wedge \mathrm{d}\bar{Z}_j. \tag{27.19}$$

The background is genuinely nonclosed, since $\mathrm{d}\theta_k = F_k^*(\mathrm{d}h_0)$ is not identically zero. At a generic coordinate divisor the determinant has order $2(k-1)$, the regular metric is comparable to $|z|^{2(k-1)}|\mathrm{d}z|^2 + |\mathrm{d}w|^2$, and its distance has order $|z|^k$. For a coordinate divisor D, a small logarithmic potential along D is a strict barrier. For small $c > 0$, set $h_c = \theta_k + c\Theta_D$ and $U_c = c \log |s_D|^2 + u_k$. Then $h_c > 0$ and

$$h_c + \mathrm{dd}^{\mathrm{c}} U_c = c[D] + F_k^* h_0.$$

The solution has exact nonminimal ordinary type, the same smooth regular metric, and original singular measure $(F_k^* h_0)^n$ by Theorem 6.1.1. Finally $H_q \cong S^{2n-1} \times S^1$ has $H^2(H_q; \mathbb{R}) = 0$. A Kähler form would be exact, so Stokes would make its positive top-volume integral zero; thus the manifold is genuinely non-Kähler.

The residual currents in these constructions have bounded potentials. More generally, Theorem 1.18.1 bounds the path distance by the square root of a Hölder modulus wherever the current defines a smooth positive metric. Mere continuity does not have the same consequence, as the radial examples below will show.

27.11.1 Determinant and normal distance

Use the local length distance of Theorem 1.18.1. A trace bound limits all eigenvalues from above; combined with the determinant, it gives a lower bound in the normal direction.

Proposition 27.11.2 *Let $U \subset \mathbb{C} \times \mathbb{C}^{n-1}$ be a polydisk with coordinates (z, w). Let η be smooth semipositive on U, positive off $D = \{z = 0\}$, with uniformly bounded Euclidean trace and determinant comparable to $|z|^{2\ell}$, $\ell \in \mathbb{N}$ and $\ell \geq 1$. On a nested smaller chart,*

$$c|z|^{\ell+1} \leq d_\eta^U((z, w), D) \leq C|z|. \tag{27.20}$$

Both endpoints occur for smooth closed forms with the same determinant order. If the tangential block is uniformly positive, the distance has exact order $|z|^{\ell+1}$.

Proof The least eigenvalue obeys $\lambda_{\min} \geq \det \eta/(\operatorname{tr} \eta)^{n-1} \geq c|z|^{2\ell}$. Every path to D therefore costs at least the normal integral of $|z|^\ell |\mathrm{d}z|$, while a radial path and the trace upper bound give the other inequality. In dimension two, for an integer $0 \leq a \leq \ell$, take $\eta_a = \mathrm{dd^c}|G_a|^2$ with $G_a(z, w) = (z^{a+1}/(a + 1), z^{\ell-a}w)$ and append Euclidean coordinates for $n > 2$. Its Jacobian is z^ℓ, but at $(z, 0)$ the first target coordinate forces distance of order $|z|^{a+1}$, attained radially. Write the Hermitian coefficient matrix in the normal and tangential directions as $\left(\begin{smallmatrix} A & b^* \\ b & C \end{smallmatrix}\right)$. When C is uniformly positive, its Schur complement $A - b^*C^{-1}b = \det \eta/\det C$ is comparable to $|z|^{2\ell}$. Completing the square gives the lower bound, and solving $\mathrm{d}w/\mathrm{d}s = -C^{-1}b\, \mathrm{d}z/\mathrm{d}s$ along a radial normal curve on nested charts gives the matching upper bound. □

27.11.2 Smoothness of exceptional residuals

Return to the blowup $\pi : X = \mathrm{Bl}_Z Y \to Y$ of Section 13.21, where (Y, h) is a compact Hermitian n-fold and the smooth center Z has codimension $r \geq 2$. Write E for the exceptional divisor, and retain the background $\theta = \pi^*h + c\Theta_E$, $c > 0$, with divisor logarithm $L = \log |s_E|^2$. Fix a smooth positive volume form $\mathrm{d}V$ on X, and use adapted coordinates $\pi(y, z, w) = (y, z, zw)$, so $E = \{z = 0\}$.

Proposition 27.11.3 *Suppose a minimal-type solution has residual $S = (\theta + \mathrm{dd^c}u) - c[E]$ extending smoothly and semipositively across E. On smaller adapted charts, $0 \leq S^n/\mathrm{d}V \leq C|z|^{2(r-1)}$. Therefore a prescribed density $|z|^{2\ell}g$, with g smooth strictly positive, requires $\ell \geq r - 1$ for a smooth residual. The order condition is necessary but not sufficient; in particular a smooth strictly positive density is incompatible with a smooth minimal residual in this exceptional-pole regime.*

Proof Write $u = cL + v$. Smoothness of $S = \pi^*h + \mathrm{dd^c}v$ makes the distributional Laplacian of v smooth, so local elliptic regularity makes v smooth. On a compact exceptional fiber, π^*h restricts to zero and $\mathrm{dd^c}(v|_E) = S|_E \geq 0$; the maximum

principle makes v constant there. Every vertical diagonal entry of S is a smooth nonnegative function vanishing at $z = 0$, hence is $O(|z|^2)$. The other diagonal entries are bounded. Hadamard's determinant inequality gives the stated order. The pullback $\pi^* h$ realizes equality, while the families below show nonsufficiency. This argument does not apply to the prescribed nonminimal regime where the residual is positive on exceptional fibers. □

Example 27.11.2 On a polydisk with coordinates $(z, w) \in \mathbb{C} \times \mathbb{C}^{n-1}$, set $P(z, w) = |z|^{2/n}(1 + |w|^2)$. Off $z = 0$, its tangential Hessian block is $|z|^{2/n} I$ and its normal Schur complement is $n^{-2}|z|^{2/n-2}$, so its determinant is the positive constant n^{-2}. Bounded plurisubharmonic removal extends it across $z = 0$ without an extra Monge–Ampère atom. Yet P is not C^1 along that divisor.

This local example extends to a compact Kähler model on $\mathbb{P}^n$. In an affine ball put $y = (z, w)$ and write $H = \lambda \log(1 + |y|^2)$ and $Q = H - B$. Choose nested radii and parameters so $Q > P + 2\epsilon$ on the outer collar and $Q < P - 2\epsilon$ on the inner ball. A smooth convex regularized maximum $U = M_\epsilon(P, Q)$ equals P inside and Q outside. Put $v = U - H$ on the ball and $v = -B$ outside. Its current is positive and smooth off $\{z = 0\}$; on the gluing annulus its density is at most $C|z|^{2/n-2}$, hence globally L^p for every $1 < p < n/(n-1)$. On the inner chart the density is smooth strictly positive but the potential is not C^1. There the metric is comparable to $|z|^{2/n-2}|\mathrm{d}z|^2 + |z|^{2/n}|\mathrm{d}w|^2$ and distance to the divisor has order $|z|^{1/n}$. The construction also works on a Hopf manifold. Choose a global Hermitian form equal to $\mathrm{dd^c}(\lambda|y|^2)$ on a coordinate ball and equal to a fixed nonclosed Hermitian form outside a slightly larger ball, using a convex interpolation of positive forms. Carry out the same regularized maximum with $H = \lambda|y|^2$ on the inner ball and set $v = -B$ outside it. The background is still nonclosed on the outer region and the local potential and density are unchanged. Global smoothness of the density is not claimed.

In Example 27.11.2, the density loses smoothness along the divisor in the gluing region. An analytic polar set containing this part of the divisor also contains its continuation into the inner chart. Thus Corollary 13.17.1 cannot give smoothness across that divisor: its density hypothesis concerns the entire complementary regular locus of the strict barrier. Local smoothness of the density alone cannot replace that hypothesis.

Example 27.11.3 Let $\pi : X = \mathrm{Bl}_0 \mathbb{P}^n \to \mathbb{P}^n$ be the blowup at the origin of the affine chart with coordinates y, and let E be its exceptional divisor. For every $\gamma, c > 0$, one can choose $\lambda > 0$ so that the background $\theta = \pi^*(\lambda\omega_{\mathrm{FS}}) + c\Theta_E$ admits a potential with bounded Hölder minimal remainder and total residual potential $A|y|^{2\gamma} + B$ near the target point, for constants $A > 0$ and $B \in \mathbb{R}$. To obtain it globally, choose $m(s) = sU'(s) > 0$ with $m'(s) > 0$, equal to $A\gamma s^\gamma$ near $s = 0$ and $\lambda s/(1 + s)$ outside a slightly larger ball. Small A orders the endpoint values. Match the positive derivative germs on narrow collars, fill the positive integral deficit with a positive interior derivative and bump, and integrate $U' = m/s$ to match $\lambda \log(1 + s)$ outside. Set $q(y) = U(|y|^2) - \lambda \log(1 + |y|^2)$ on the affine chart and extend it by zero outside. Positivity and matching hold throughout the compact interpolation.

On the blowup $u = cL + q \circ \pi$ has, in coordinates $y = (z, zw)$, total residual potential $A|z|^{2\gamma}(1 + |w|^2)^\gamma + B$. Its density is $|z|^{2\ell}G$ with G smooth positive and $\ell = n\gamma - 1 > -1$, since the target determinant is $A^n\gamma^{n+1}s^{n(\gamma-1)}$ and the squared blowup Jacobian is $|z|^{2(n-1)}$. Its regular metric is comparable to

$$|z|^{2\gamma-2}|\mathrm{d}z|^2 + |z|^{2\gamma}|\mathrm{d}w|^2, \tag{27.21}$$

distance to E is $|z|^\gamma$, and the completion collapses E. Nonnegative integral ℓ makes the density smooth. At $\ell = 0$, the density is smooth strictly positive, but the remainder has exact normal Hölder exponent $2/n$ for $n > 2$, is Lipschitz but not C^1 for $n = 2$, and has an unbounded normal metric coefficient. For $-1 < \ell < 0$ its L^p condition is $p\ell > -1$. Taking $\ell = n$ gives a smooth density zero while the residual Hessian remains nonsmooth, proving the nonsufficiency in Proposition 27.11.3. A small nonclosed perturbation supported away from the singular point preserves the local calculations and yields a genuinely nonclosed background with scalar one for its specified measure.

27.12 Radial densities and metric completeness

For a radial equation, the cumulative density determines the radial derivative of the potential and both Hessian eigenvalues. The potential integral and the radial length integral then give separate tests for boundedness and finite metric distance. On $\mathbb{P}^n$, $n \geq 2$, normalize $\int_{\mathbb{P}^n} \omega_{\mathrm{FS}}^n = 1$. On the standard affine chart with coordinates $y \in \mathbb{C}^n$, put $\omega_{\mathrm{E}} = \mathrm{dd^c}|y|^2$; then $\int_{\{|y|^2<S\}} \omega_{\mathrm{E}}^n = S^n$ for $S > 0$.

Theorem 27.12.1 *Fix $\lambda > 0$ and a smooth positive $D : (0, \infty) \to (0, \infty)$ such that $D(s) = \lambda^n/(1 + s)^{n+1}$ for all sufficiently large s, and*

$$n \int_0^\infty s^{n-1} D(s)\, \mathrm{d}s = \lambda^n. \tag{27.22}$$

Continue $D(|y|^2)\omega_{\mathrm{E}}^n$ by the specified Fubini–Study density outside the affine construction. It has a unique normalized full-mass closed non-pluripolar solution in $[\lambda\omega_{\mathrm{FS}}]$. The solution is smooth off $y = 0$ and has total radial local potential $U(s)$ satisfying

$$\begin{aligned} m(s)^n &= n \int_0^s t^{n-1} D(t)\, \mathrm{d}t, \qquad m(s) = sU'(s), \\ m'(s) &= \frac{s^{n-1}D(s)}{m(s)^{n-1}} > 0. \end{aligned} \tag{27.23}$$

Its tangential and radial Hessian eigenvalues are m/s and m'. For any sufficiently small $s_0 > 0$, it has bounded minimal ordinary type precisely when

$$\int_0^{s_0} \frac{m(s)}{s}\,\mathrm{d}s < \infty. \tag{27.24}$$

The distance of the punctured regular metric to the origin is finite precisely when

$$\int_0^{s_0} \frac{\sqrt{m'(s)}}{2\sqrt{s}}\,\mathrm{d}s < \infty. \tag{27.25}$$

If the integral diverges, the entire punctured compact regular metric is complete.

Write q for the global solution potential, and let $\pi : X = \mathrm{Bl}_0\,\mathbb{P}^n \to \mathbb{P}^n$ have exceptional divisor E. Retain $L = \log|s_E|^2$ and the divisor curvature Θ_E of Section 13.21. For $\theta = \pi^*(\lambda\omega_{\mathrm{FS}}) + c\Theta_E$ with $c > 0$, the corresponding full-mass solution has potential $cL + q \circ \pi + C$, exact divisor coefficient c, mass λ^n, and, in a blowup chart $\pi(z, w) = (z, zw)$, density

$$|z|^{2(n-1)} D\big(|z|^2(1 + |w|^2)\big) \tag{27.26}$$

relative to the fixed local Euclidean volume convention. Its ordinary type is minimal precisely under (27.24). In unbounded cases the residual is sublogarithmic with zero generic Lelong number, but the type is nonminimal. The regular metric has the same completeness criterion (27.25).

Proof Define m by the first equation in (27.23). It is positive for $s > 0$ and tends to zero at the origin. Its total mass and the prescribed outer density make $m(s) = \lambda s/(1 + s)$ on the outer region: both m^n expressions have equal limits at infinity and equal tails there. Integrate $U' = m/s$ and choose the additive constant so that $U(s) = \lambda \log(1+s)$ on that region. The radial Hessian eigenvalues are m/s and m', and the determinant is $(m/s)^{n-1}m' = D$. Thus $q = U(|y|^2) - \lambda \log(1 + |y|^2)$ extends by zero outside the affine construction and gives a positive current smooth off the point. Subtracting its supremum supplies the stated normalization.

The increasing U is bounded above near zero, so analytic-set removal gives a plurisubharmonic extension, possibly taking the value $-\infty$ at the origin by Lemma 1.3.7. If unbounded, $m(s) \to 0$ implies $U(s) = o(|\log s|)$: for each $\epsilon > 0$, bound m by ϵ on a small interval and integrate m/s. The closed non-pluripolar product equals the prescribed density off the point and ignores the point. The radial density has no atom there and has mass λ^n by (27.22). Closed full-mass uniqueness proves the assertion as recalled in Section 33.17. Integrating $U' = m/s$ gives precisely (27.24).

For a path approaching the origin, the metric norm of its velocity is at least a fixed multiple of $\sqrt{m'(R^2)}|\dot{R}|$, where $R = |y|$. Radial paths attain the resulting bound up to the fixed convention. Since $\mathrm{d}R = \mathrm{d}s/(2\sqrt{s})$, this is (27.25). The metric is smooth positive on compact subsets away from the origin, so divergence is equivalent to completeness of the entire punctured compact metric.

On the blowup use Proposition 13.21.1 and the squared Jacobian $|z|^{2(n-1)}$. The closed product is the zero extension of the pullback density, has mass $\lambda^n = \mathrm{vol}([\theta])$,

and is unique in its full-mass class. The original operator agrees by Theorem 6.1.1 on the continuous-off-E class, even when the residual tends to $-\infty$ there. Sublogarithmic growth gives zero residual Lelong number, fixing the coefficient c without giving bounded difference. The regular metric is isometric to the punctured target metric, proving the same distance criterion.

The permitted inner asymptotics below can be glued to the prescribed outer density while keeping (27.22): take a small inner amplitude or radius, join through a smooth positive annulus, and add a nonnegative smooth annular bump to fill the remaining mass. Thus the family is a genuine prescribed-density family. The assertions concern the total local potential U; subtracting a smooth background potential can alter a leading smooth term. □

Theorem 27.12.2 *For the normalized family in Theorem 27.12.1, put $L = \log(1/s)$.*

(1) *If $D(s) \sim Cs^{\ell-n+1}L^{-b}$ with $C > 0$, $\ell > -1$ and $b \in \mathbb{R}$, put $\gamma = (\ell+1)/n$ and $K = (C/\gamma)^{1/n}$. Then*

$$m \sim Ks^{\gamma}L^{-b/n}, \qquad U(s) - U(0) \sim \frac{K}{\gamma}s^{\gamma}L^{-b/n}, \qquad m' \sim \gamma K s^{\gamma-1}L^{-b/n}. \tag{27.27}$$

The residual is bounded Hölder. In bounded blowup charts its metric is comparable to $L^{-b/n}(|z|^{2\gamma-2}|\mathrm{d}z|^2 + |z|^{2\gamma}|\mathrm{d}w|^2)$, and its distance to E has order $|z|^{\gamma}(\log(1/|z|))^{-b/(2n)}$. The end is incomplete.

(2) *If $D(s) \sim Cs^{-n}L^{-b}$, local mass is finite precisely for $b > 1$. In that range put $a = (b-1)/n$ and $K = (nC/(b-1))^{1/n}$. Then*

$$m \sim KL^{-a}, \qquad m' \sim Kas^{-1}L^{-a-1}. \tag{27.28}$$

The potential has a finite limit, minimal ordinary type, and finite distance precisely when $b > n+1$; its bounded tail is asymptotic to $KL^{1-a}/(a-1)$. At $b = n+1$ it diverges as $-K\log L$, and for $1 < b < n+1$ as $-KL^{1-a}/(1-a)$. The unbounded cases are full mass but nonminimal and have complete regular ends. Their upstairs densities lie in no L^p, $p > 1$.

(3) *If $D(s) \sim Cs^{-n}L^{-(n+1)}(\log L)^{-d}$, $d \in \mathbb{R}$, put $C_* = C^{1/n}$. Then*

$$m \sim C_*L^{-1}(\log L)^{-d/n}, \qquad m' \sim C_*s^{-1}L^{-2}(\log L)^{-d/n}. \tag{27.29}$$

The potential is bounded precisely when $d > n$, with tail of order $(\log L)^{1-d/n}$. At $d = n$ it diverges as $-C_*\log\log L$, and below n as $-C_*(\log L)^{1-d/n}/(1-d/n)$. Distance is finite precisely when $d > 2n$, with tail of order $(\log L)^{1-d/(2n)}$. Thus $n < d \le 2n$ gives bounded continuous minimal, non-Hölder residuals with complete regular ends. All these critical densities have finite mass and lie in no L^p, $p > 1$.

The logarithmic solution coefficient is c. A strict barrier can be chosen with coefficient $c + \delta$, for some $\delta > 0$ independent of ℓ, b, d, by Proposition 13.21.1.

Proof Use the exact expressions (27.23). Each asymptotic follows by bounding D between $(1 \pm \epsilon)$ times its displayed model and integrating the model; no derivative of an unspecified asymptotic error is taken. In the first case $m^n \sim nCs^{\ell+1}L^{-b}/(\ell+1)$, yielding (27.27). Both Hessian eigenvalues are comparable to $s^{\gamma-1}L^{-b/n}$, which gives the tensor and distance assertions after the blowup coordinate substitution. The positive power makes the potential Hölder, after absorbing logarithms by a slightly smaller exponent.

In the second case $n\int_0^s t^{-1}C(\log(1/t))^{-b}\,\mathrm{d}t$ is finite precisely for $b > 1$, and is asymptotic to $nCL^{1-b}/(b-1)$. This gives (27.28). Since $\mathrm{d}U = -m\,\mathrm{d}L$, its boundedness is equivalent to convergence of $\int^\infty L^{-a}\,\mathrm{d}L$, namely $a > 1$. The distance is a constant multiple of $\int^\infty L^{-(a+1)/2}\,\mathrm{d}L$ and has the same strict threshold. Direct primitives give the stated finite and divergent branches, including the $a = 1$ logarithm. Upstairs the density is comparable to $|z|^{-2}(\log(1/|z|))^{-b}$; every $p > 1$ introduces an exponentially growing factor in logarithmic coordinates, so no such L^p integrability holds.

At the critical refinement, $m^n \sim CL^{-n}(\log L)^{-d}$. The potential and distance criteria become convergence of, respectively,

$$\int^\infty L^{-1}(\log L)^{-d/n}\,\mathrm{d}L \quad \text{and} \quad \int^\infty L^{-1}(\log L)^{-d/(2n)}\,\mathrm{d}L.$$

Their thresholds are $d > n$ and $d > 2n$. Direct integration gives the stated tails and negative divergent branches. A bounded tail in a positive power of $\log L$ decays more slowly than every positive power of $|z|$, so it is not Hölder. The mass integral is finite for every real d because the principal factor is $L^{-(n+1)}$ with $n \geq 2$; the same exponential argument excludes L^p, $p > 1$. These calculations test all equality endpoints in the actual prescribed family. □

27.13 Equations on products with a curve

On a product with a curve, the determinant equation reduces to a scalar Poisson equation in the normal variable, even when the other factor is nonclosed. Its logarithmic primitives give a comparison with the radial families: for the single logarithmic pole below, boundedness and finite distance have the same threshold, while the iterated logarithmic family of Theorem 27.12.2 separates them.

Theorem 27.13.1 *Let C be a compact Riemann surface with positive form ω_C, let (N, ω_N) be a compact Hermitian $(n-1)$-fold, and put $h = \omega_C + \omega_N$ on $C \times N$. Let $p_0 \in C$ and let $F \geq 0$ be integrable, smooth positive away from p_0, with $\int_C F\omega_C = \int_C \omega_C$. There is a separable solution for the original measure $(F \circ \mathrm{pr}_C)h^n$ whose current has no divisor component on $\{p_0\} \times N$.*

In a coordinate z at p_0, write $r = |z|$ and $\omega_C = (\mathrm{i}/2\pi)g(z)\,\mathrm{d}z \wedge \mathrm{d}\bar{z}$.

(1) *If $Fg = C_0 r^{2\tau}$, $C_0 > 0$ and $\tau > -1$, the total normal potential, modulo a smooth harmonic term, is $C_0 r^{2(\tau+1)}/(\tau+1)^2$. It is bounded Hölder and minimal, while*

normal distance has order $r^{\tau+1}$. If $\tau = -a$ with $0 < a < 1$, its density is locally L^p exactly when $pa < 1$. Nonnegative integral vanishing orders give smooth potentials.

(2) *If $Fg = C_0 r^{-2} t^{-b}$ with $t = \log(1/r)$, finite mass holds exactly for $b > 1$. The atomless total normal potential is*

$$P_b(t) = \begin{cases} \dfrac{4C_0 t^{2-b}}{(b-1)(b-2)}, & b \neq 2, \\ -4C_0 \log t, & b = 2, \end{cases} \tag{27.30}$$

modulo constants and smooth harmonic terms. It is bounded continuous and minimal precisely when $b > 2$, and then is not Hölder. The normal end has finite distance precisely for $b > 2$. For $1 < b \leq 2$, it is complete; the potential is unbounded and nonminimal but has zero Lelong number and trivial multiplier ideals. Every density in this logarithmic family lies outside L^p for $p > 1$.

In the Kähler product case the unbounded solutions have full background mass, and closed uniqueness excludes a different bounded solution for the same measure. A small strict logarithmic barrier along $\{p_0\} \times N$ can coexist with the bounded minimal solutions.

Proof On the curve, the real current $F\omega_C - \omega_C$ has integral zero and hence is $\mathrm{dd^c} v$ for a global distribution v, unique up to a constant. Its quasi-subharmonic representative is smooth off p_0 by the local Poisson equation. Since $F\omega_C$ is atomless at p_0, its current has no atom. Pull v back to $C \times N$. Positivity and the vanishing of any wedge with two curve factors give on the regular locus

$$(h + \mathrm{dd^c} v)^n = n(F\omega_C) \wedge \omega_N^{n-1} = (F \circ \mathrm{pr}_C) h^n. \tag{27.31}$$

By Theorem 6.1.1, the extended product agrees with this smooth product off $\{p_0\} \times N$ and gives that analytic set zero mass. The right side has the same property and finite mass. This proves the global equation, including when the curve potential has a finite discontinuous value at p_0.

For the power input put $P_\tau(r) = C_0 r^{2(\tau+1)}/(\tau+1)^2$. Direct differentiation gives $\partial_z \bar{\partial}_z P_\tau = C_0 r^{2\tau}$. Its radial boundary flux vanishes as $r \downarrow 0$ when $\tau > -1$, so no Dirac term appears. The difference between the total potential and P_τ is harmonic on the full disk and smooth. Cut off the local potential and solve the remaining smooth mean-zero curve equation to realize the model globally. Its normal line element is comparable to $r^\tau |\mathrm{d}r|$, giving distance order $r^{\tau+1}$ by an all-path lower bound and a radial upper path. The L^p assertion is the integral $\int_0^\epsilon r^{1-2pa}\,\mathrm{d}r$.

For the logarithmic input, $P_b''(t) = 4C_0 t^{-b}$ and $\partial_z \bar{\partial}_z P_b = P_b''/(4r^2) = C_0 r^{-2} t^{-b}$. Its flux tends to zero for $b > 1$, excluding a logarithmic Dirac mass. The same cutoff and smooth correction make a compact model. Density mass is $\int^\infty t^{-b}\,\mathrm{d}t$, finite exactly for $b > 1$; its pth power for $p > 1$ diverges after the logarithmic change of variables. The primitive has a finite limit exactly for $b > 2$, with tail t^{2-b}, slower than every r^α. Normal length is a constant times $\int^\infty t^{-b/2}\,\mathrm{d}t$, giving the same threshold.

The projection to the curve bounds every path from below by that normal integral, while a fixed-N radial path attains it.

For $1 < b \le 2$, $|P_b(t)|/t \to 0$, giving zero Lelong number. Every multiplier weight is bounded by an integrand of the form $\exp(-2t + o(t))$, which is integrable, so its multiplier ideals are trivial. These properties do not imply bounded ordinary type. In a Kähler product, the normalized curve integral gives full non-pluripolar background mass, and closed full-mass uniqueness excludes another bounded solution, as recalled in Section 33.17. A bounded competitor for the original operator has deep truncations equal to itself globally, so the same exclusion applies there. Finally, for $D = \{p_0\} \times N$ and sufficiently small $\delta > 0$, $h - \delta\Theta_D > 0$; the potential $\delta \log |s_D|^2 - C$ is a strict barrier. □

27.14 An incomplete metric in a big class

A minimal divisorial singularity can cancel the pole of an exponential equation and leave the deleted divisor at finite metric distance. On the blowup of a projective plane, the resulting metric is the pullback of the Fubini–Study metric away from the exceptional divisor. The example has a big non-nef class and a unique finite-energy solution.

Finite energy is understood relative to a potential with minimal singularities in the classical closed theory of [Xia].

Theorem 27.14.1 *There exist a smooth projective surface Y, a smooth irreducible divisor E, a smooth closed real $(1,1)$-form θ with big class α, a Hermitian metric on $\mathcal{O}_Y(E)$, a defining section σ, and a smooth positive volume form $\mathrm{d}V_Y$ with the following properties. The equation*

$$\left\langle (\theta + \mathrm{dd^c}\rho)^2 \right\rangle = \mathrm{e}^{\rho} |\sigma|^{-2}\, \mathrm{d}V_Y \tag{27.32}$$

has a unique solution $\rho \in \mathcal{E}^1(Y, \theta)$, and this solution has minimal singularities. Moreover,

$$\mathrm{Amp}(\alpha) = Y \setminus E$$

is precisely the open locus on which $\theta + \mathrm{dd^c}\rho$ is smooth and strictly positive. Its restriction to every nonempty connected open subset of $Y \setminus E$ is incomplete. In particular, no dense Zariski open subset carries a complete induced metric.

Let $p = [1 : 0 : 0] \in \mathbb{P}^2$, and let ω_2 be the Fubini–Study form on $\mathbb{P}^2$, normalized by $\int_{\mathbb{P}^2} \omega_2^2 = 1$.

Proof (Proof of Theorem 27.14.1) *The surface and the divisor.* Use the incidence model

$$Y = \{([Z_0 : Z_1 : Z_2], [W_1 : W_2]) \in \mathbb{P}^2 \times \mathbb{P}^1 : Z_1 W_2 = Z_2 W_1\}.$$

The projections $\pi : Y \to \mathbb{P}^2$ and $q : Y \to \mathbb{P}^1$ identify Y with the blow-up at p, with exceptional divisor $E = \pi^{-1}(p)$. Let ω_1 be the Fubini–Study form on $\mathbb{P}^1$ with the same normalization as ω_2. Write

$$\beta = \pi^*\omega_2, \qquad \gamma = q^*\omega_1, \qquad \Omega = \beta + \gamma.$$

The form Ω is Kähler, being the restriction of the product metric.

The projection $(Z_0, Z_1, Z_2) \mapsto (Z_1, Z_2)$ gives a section σ of

$$\pi^*\mathcal{O}_{\mathbb{P}^2}(1) \otimes q^*\mathcal{O}_{\mathbb{P}^1}(-1).$$

Indeed, its image belongs to the tautological line represented by $[W_1 : W_2]$. The section vanishes simply along E, identifying this line bundle with $\mathcal{O}_Y(E)$. For the product of the Fubini–Study and induced tautological metrics, its squared norm is

$$s = |\sigma|^2 = \frac{|Z_1|^2 + |Z_2|^2}{|Z_0|^2 + |Z_1|^2 + |Z_2|^2}.$$

In a blow-up chart $\pi(t, w) = (t, tw)$ and $q(t, w) = [1 : w]$, one has

$$s = \frac{|t|^2(1 + |w|^2)}{1 + |t|^2(1 + |w|^2)}, \qquad \mathrm{dd^c} \log s = [E] + \gamma - \beta. \tag{27.33}$$

The current identity follows either from this formula and the second blow-up chart or from Poincaré–Lelong.

The big class and its equation. Set

$$\theta = 3\beta - 2\gamma, \qquad \rho = 2 \log s, \qquad T = \theta + \mathrm{dd^c}\rho = \beta + 2[E]. \tag{27.34}$$

Thus $\rho \in \mathrm{PSH}(Y, \theta)$. The class $\alpha = \{\theta\}$ is big: the potential $\psi = (5/2) \log s$ satisfies

$$\theta + \mathrm{dd^c}\psi = \tfrac{1}{2}(\beta + \gamma) + \tfrac{5}{2}[E] \geq \tfrac{1}{2}\Omega. \tag{27.35}$$

It is not nef, since $\int_E \theta = -2$: the restriction of β to E vanishes and $q|_E$ is an isomorphism to $\mathbb{P}^1$. A nef class has nonnegative degree on every curve by its defining smooth approximations.

On $Y \setminus E$ define $\mathrm{d}V_Y = \beta^2/s$. To check extension across E, put

$$\mathrm{d}\lambda_{t,w} = \tfrac{\mathrm{i}}{2}\mathrm{d}t \wedge \mathrm{d}\bar{t} \wedge \tfrac{\mathrm{i}}{2}\mathrm{d}w \wedge \mathrm{d}\bar{w}.$$

Use the analogous Euclidean volume $\mathrm{d}\lambda_{z_1,z_2}$ in the affine coordinates of $\mathbb{P}^2$. The normalized Fubini–Study volume there is

$$\omega_2^2 = \frac{2}{\pi^2} \frac{\mathrm{d}\lambda_{z_1,z_2}}{(1 + |z_1|^2 + |z_2|^2)^3}.$$

Since the complex Jacobian of $(t, w) \mapsto (t, tw)$ is t, we obtain

$$\beta^2 = \frac{2}{\pi^2} \frac{|t|^2 \, \mathrm{d}\lambda_{t,w}}{[1 + |t|^2(1 + |w|^2)]^3},$$
$$\mathrm{d}V_Y = \frac{2}{\pi^2} \frac{\mathrm{d}\lambda_{t,w}}{(1 + |w|^2)[1 + |t|^2(1 + |w|^2)]^2}.$$

The second expression is smooth and strictly positive at $t = 0$. Interchanging the two homogeneous coordinates gives the extension on the other chart. The extensions agree off E, hence everywhere on their overlap.

The non-pluripolar product of T agrees with β^2 off E and gives no mass to E. The smooth measure β^2 also gives no mass to E, so

$$\langle T^2 \rangle = \beta^2, \qquad \mathrm{e}^{\rho} |\sigma|^{-2} \, \mathrm{d}V_Y = s^2 s^{-1} \frac{\beta^2}{s} = \beta^2.$$

The right side of (27.32) is defined on $Y \setminus E$ and extended by zero on E; no product of zero and infinity is taken there. This proves the equation and gives $\int_Y \langle T^2 \rangle = \int_{\mathbb{P}^2} \omega_2^2 = 1$.

Minimal singularities and energy. For $u \in \mathrm{PSH}(Y, \theta)$, the function $u - \rho$ on $Y \setminus E$ descends to an ω_2-plurisubharmonic function on $\mathbb{P}^2 \setminus \{p\}$, because

$$\beta + \mathrm{dd^c}(u - \rho) = \theta + \mathrm{dd^c} u \geq 0 \quad \text{on } Y \setminus E.$$

By Lemma 1.3.8, $u \leq \rho + C_u$ there. This inequality holds almost everywhere on Y, so Lemma 1.3.3 makes it hold everywhere. In particular $u = -\infty$ on E, and ρ has minimal singularities.

Let V_θ be the normalized minimal envelope. Since $s \leq 1$, minimality gives

$$\rho \leq V_\theta \leq \rho + C.$$

Thus $\rho - V_\theta$ is bounded off E. The defining relative energy is

$$E_\theta(\rho) = \frac{1}{3} \sum_{j=0}^{2} \int_Y (\rho - V_\theta) \left\langle (\theta + \mathrm{dd^c}\rho)^j \wedge (\theta + \mathrm{dd^c} V_\theta)^{2-j} \right\rangle.$$

The mixed measures have finite mass in the closed Kähler theory and do not charge E. Hence $E_\theta(\rho) > -\infty$, so $\rho \in \mathcal{E}^1(Y, \theta)$. A potential with minimal singularities has full mass; in this example that mass is 1.

Uniqueness. Suppose $u \in \mathcal{E}^1(Y, \theta)$ is another solution, and let $\mathrm{d}\mu = s^{-1} \mathrm{d}V_Y$ on $Y \setminus E$, extended by zero on E. This measure is locally smooth and positive off E, but has infinite total mass. The classical closed full-mass comparison principle gives

$$\int_{\{u<\rho\}} \mathrm{e}^{\rho} \, \mathrm{d}\mu \leq \int_{\{u<\rho\}} \mathrm{e}^{u} \, \mathrm{d}\mu.$$

Both integrals are finite, since the two Monge–Ampère measures have mass 1. The strictly larger integrand on the left forces $\mu(\{u < \rho\}) = 0$. Interchanging u and ρ gives $\mu(\{\rho < u\}) = 0$ as well. They therefore agree almost everywhere for smooth volume on $Y \setminus E$, and hence on Y, since E has zero smooth volume. By Lemma 1.3.3, $u = \rho$ everywhere.

The ample locus and incompleteness. The current in (27.35) is smooth off E, while every global θ-plurisubharmonic potential is $-\infty$ on E. Thus no Kähler current in α can be smooth near a point of E. The characterization in Theorem 27.3.1 gives

$$\mathrm{Amp}(\alpha) = Y \setminus E_{\mathrm{nK}}(\alpha) = Y \setminus E.$$

The current T equals β there and is strictly positive because π is a biholomorphism away from E. Its nonzero divisorial part $2[E]$ prevents smoothness at every point of E.

Let $U \subset Y \setminus E$ be a nonempty connected open set, and set $W = \pi(U)$. The map $\pi : (U, T|_U) \to (W, \omega_2|_W)$ is an isometry. Choose $x \in W$ and a smooth path $c : [0,1] \to \mathbb{P}^2$ from x to p. Its first exit time

$$t_0 = \min\{t \in [0,1] : c(t) \notin W\}$$

is positive, and $c([0,t_0)) \subset W$. This path has finite Fubini–Study length. For $t_k \nearrow t_0$, its tail lengths show that $c(t_k)$ is Cauchy for the intrinsic metric of W. It has no limit there: intrinsic convergence implies ambient convergence, whose limit is $c(t_0) \notin W$. Thus U is incomplete. Applying this argument to each connected component proves the assertion for a possibly disconnected dense Zariski open set. □

The cancellation $e^{\rho}|\sigma|^{-2} = s$ leaves the deleted point at finite distance. Thus the exponential equation in a big class can have an incomplete metric even when its finite-energy solution is unique and has minimal singularities. Here the class is non-nef. Completeness under nefness or the additional compatibility conditions of a geometric flow remains a separate question; see Section 33.15.

27.15 Normal length near a divisor

A lower bound for the determinant alone does not control normal length: the tangential eigenvalues may absorb the volume.

Let X be a connected compact complex manifold of dimension $n \geq 1$, let D be a nonempty smooth effective divisor, and put $M = X \setminus D$. Fix a Hermitian form ω on X. Let s_D be the canonical section of $\mathcal{O}_X(D)$, equipped with a smooth Hermitian norm, and set

$$R = |s_D|, \qquad \tau = -\log R^2.$$

Rescale the norm so that $\tau \geq 1$ on M. Write J for the complex structure. For a smooth positive real $(1,1)$-form T on M, use the Riemannian metric $g_T(v,v) = \pi T(v, Jv)$

for real tangent vectors v. Thus, if $T = (\mathrm{i}/2\pi)\sum H_{j\bar{k}}\,\mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k$, then $g_T(v,v) = \sum H_{j\bar{k}}\,\mathrm{d}z_j(v)\overline{\mathrm{d}z_k(v)}$. Hermitian norms below use this coefficient normalization.

Near D, define

$$E_\tau = \ker \partial\tau, \qquad B_\tau = T|_{E_\tau}, \qquad \det_\omega T = \frac{T^n}{\omega^n}.$$

The notation $\det_\omega B_\tau$ means the determinant relative to the restriction of ω to this complex hyperplane; when $n = 1$, it means 1.

Dividing by $\det_\omega B_\tau$ gives the normal Schur complement. A lower bound of order $R^{-2}\tau^{-p}$ forces completeness when $p \le 2$; the exponential equation will supply this bound from its density, potential and tangential determinant.

Proposition 27.15.1 *Suppose that there are constants $c > 0$, $\tau_0 \ge 1$ and $p \le 2$ such that, on $\{\tau > \tau_0\}$,*

$$\frac{\det_\omega T}{\det_\omega B_\tau} \ge cR^{-2}\tau^{-p}. \tag{27.36}$$

Then (M, g_T) is complete.

Proof In a coordinate neighborhood meeting D, write $s_D = z_1 e$, where e is a local holomorphic frame with $|e|^2 = \mathrm{e}^{-\psi}$ and ψ is smooth. Then

$$\tau = -\log|z_1|^2 + \psi, \qquad \partial\tau = -\frac{\mathrm{d}z_1}{z_1} + \partial\psi.$$

A finite cover of the compact divisor and uniform comparison with the smooth background give, after increasing τ_0,

$$c_1 R^{-2} \le |\partial\tau|^2_\omega \le C_1 R^{-2} \quad \text{on } \{\tau > \tau_0\} \tag{27.37}$$

for positive constants c_1, C_1. In particular, E_τ is a hyperplane there.

At a point of this set, choose an ω-unitary frame whose last $n-1$ vectors span E_τ. Let H be the Hermitian matrix representing g_T as the quadratic form w^*Hw in complex components w, and write

$$H = \begin{pmatrix} h & c^* \\ c & B \end{pmatrix}, \qquad S = h - c^*B^{-1}c = \frac{\det H}{\det B}.$$

Here c is a column vector and c^* is its conjugate transpose. Completing the square gives, for a vector with components (ξ, η),

$$\begin{pmatrix} \bar{\xi} & \eta^* \end{pmatrix} H \begin{pmatrix} \xi \\ \eta \end{pmatrix} = S|\xi|^2 + (\eta + B^{-1}c\xi)^* B(\eta + B^{-1}c\xi) \ge S|\xi|^2.$$

For $n = 1$ this is the same identity without the second term. Since $|\xi| = |\partial\tau(v)|/|\partial\tau|_\omega$ for the complex components of a real vector v, (27.36) and (27.37) imply

$$g_T(v,v) \ge c_2\tau^{-p}|\partial\tau(v)|^2 \ge \frac{c_2}{4}\tau^{-p}|\mathrm{d}\tau(v)|^2, \tag{27.38}$$

where $c_2 > 0$ is uniform. The last inequality uses $\mathrm{d}\tau(v) = 2\,\mathrm{Re}\,\partial\tau(v)$.

Define a continuous function Φ_p on $[1, \infty)$ by

$$\Phi_p(t) = \begin{cases} 0, & t \le \tau_0, \\ \displaystyle\int_{\tau_0}^{t} x^{-p/2}\,\mathrm{d}x, & t > \tau_0. \end{cases}$$

Integrating (27.38) along any piecewise smooth curve shows that $\Phi_p \circ \tau$ is Lipschitz for the length distance of g_T. This holds also for curves crossing $\{\tau = \tau_0\}$, since Φ_p is locally Lipschitz and is constant below that level. Since $p \le 2$, the integral tends to infinity with t. Its composition with τ is therefore proper on M.

A Cauchy sequence for the length distance has bounded values of this Lipschitz function, hence lies in a compact subset of M. It has a subsequence converging in the manifold topology. On a coordinate ball about its limit, smooth positivity of T compares its metric with the Euclidean metric, so this subsequence also converges in length distance. The Cauchy property gives convergence of the full sequence. This proves completeness. □

Corollary 27.15.1 *In the preceding setting, let θ be a smooth real $(1,1)$-form on X, and let $u \in \mathrm{PSH}(X, \theta)$ be smooth on M, with $T = \theta + \mathrm{dd^c}u > 0$ there. Suppose, for some $\lambda > 0$ and positive $f \in C^\infty(M) \cap L^1(X, \omega^n)$, that*

$$T^n = \mathrm{e}^{\lambda u} f \omega^n \quad \text{on } M.$$

Assume that, near D, there are constants $c, C > 0$, $q > 1$, $a \ge 0$ and $b \in \mathbb{R}$ such that

$$f \ge c R^{-2} \tau^{-q}, \qquad u \ge -\frac{a}{\lambda} \log \tau - C, \qquad \det_\omega B_\tau \le C \tau^b. \tag{27.39}$$

If $q + a + b \le 2$, then (M, g_T) is complete.

Proof The equation and the first two inequalities give $\det_\omega T \ge c\mathrm{e}^{-\lambda C} R^{-2} \tau^{-(q+a)}$. Dividing by the third bound gives (27.36) with $p = q + a + b$. Apply Proposition 27.15.1. □

Since a global quasi-plurisubharmonic function is bounded above, the assumed integrability of f also makes the displayed measure finite. The determinant of the intrinsic hyperplane E_τ, rather than an arbitrary coordinate tangential block, is the quantity controlling normal length.

27.15.1 The critical exponent on a compact torus

A product torus gives both a complete metric at the critical exponent and incomplete metrics at every larger exponent. The two logarithmic profiles have full mass; the incomplete one is bounded.

Let $E = \mathbb{C}/(\mathbb{Z}+\mathrm{i}\mathbb{Z})$, let $\omega_E = (\mathrm{i}/2\pi)\,\mathrm{d}z\wedge\mathrm{d}\bar{z}$, and put $X = E^n$, $D = \{0\}\times E^{n-1}$ and $M = X \setminus D$. Write $\pi_j : X \to E$ for the coordinate projections and $\omega_0 = \sum_{j=1}^n \pi_j^*\omega_E$. Choose radii $0 < r_1 < r_2 < \mathrm{e}^{-1}$ and a smooth radial cutoff χ on the coordinate disc $\{|z| < r_2\} \subset E$, equal to 1 on $\{|z| \le r_1\}$ and to 0 near its outer boundary, with $0 \le \chi \le 1$. For fixed numbers $A, \eta, \beta > 0$, set $L = -\log|z|^2$ and define on E

$$v_{\mathrm{c}} = -A\chi \log L, \qquad v_\beta = \eta\chi L^{-\beta}. \tag{27.40}$$

Both functions are extended by zero outside the disc; at its center set $v_{\mathrm{c}}(0) = -\infty$ and $v_\beta(0) = 0$. The bounded profile also occurs in Example 13.19.1. We verify their global mass and metric properties directly.

Lemma 27.15.1 *For $C > 0$ sufficiently large, put $\theta = C\omega_0$. For either profile v in (27.40), the function $u = \pi_1^* v$ belongs to $\mathrm{PSH}(X, \theta)$, the form $T = \theta + \mathrm{dd^c}u$ is smooth and positive on $X \setminus D$, and*

$$\int_X \mathrm{MA}_\theta(u) = \int_X \theta^n.$$

The same C may be used for the two fixed profiles. The bounded profile has minimal ordinary singularities; the profile v_{c} does not.

Proof For a twice differentiable function G of $L = -\log|z|^2$, differentiation away from zero gives

$$\frac{\partial^2 G(L)}{\partial z \partial \bar{z}} = \frac{G''(L)}{|z|^2}.$$

On the inner disc this yields

$$(v_{\mathrm{c}})_{z\bar{z}} = \frac{A}{|z|^2 L^2}, \qquad (v_\beta)_{z\bar{z}} = \frac{\eta\beta(\beta+1)}{|z|^2 L^{\beta+2}}. \tag{27.41}$$

Both densities are integrable at zero. Both profiles are locally integrable, and their distributional Hessians have no additional atom there: integrate twice by parts outside $|z| = \varepsilon$. The boundary terms against a smooth test function are bounded by constants times $\varepsilon|G(-\log\varepsilon^2)|$ and $|G'(-\log\varepsilon^2)|$, respectively, and both tend to zero for the two specified G. Thus (27.41) holds as a distribution across the center. On the cutoff annulus all derivatives are bounded. A sufficiently large common C therefore makes

$$\alpha_v = C\omega_E + \mathrm{dd^c}v$$

a positive measure, smooth and strictly positive away from zero, with no atom at zero. Pairing $\mathrm{dd^c}v$ with the constant test function gives $\int_E \alpha_v = C\int_E \omega_E$.

Let $Y = E^{n-1}$, let $\pi_Y : X \to Y$ be the remaining projection, and let θ_Y be the sum of its $n-1$ flat factors, each multiplied by C. On $X \setminus D$ the ordinary smooth wedge product equals

$$T^n = n\,\pi_1^*\alpha_v \wedge \pi_Y^*\theta_Y^{n-1}.$$

For $n = 1$ this means just α_ν, with Y a point. The right side defines a finite product measure on X and gives D zero mass. By Theorem 6.1.1, the non-pluripolar measure agrees with this product off D and gives the pluripolar set D zero mass. The two global measures are consequently equal. Integrating the product proves the full-mass identity. Since $\theta > 0$, its minimal envelope is $V_\theta = 0$ by Section 1.19. The bounded profile therefore has minimal ordinary singularities, whereas v_{c} cannot dominate 0 up to a constant. □

Choose a smooth norm on $\mathcal{O}_E(0)$ for which its canonical section has squared norm $|z|^2$ near zero. Such a norm is obtained from any smooth norm by multiplying by a smooth positive function, prescribed on a smaller disc. Pull it back to X and rescale by a constant. For some fixed $K > 0$, we may then arrange

$$R^2 = \mathrm{e}^{-K}|z_1|^2, \qquad \tau = L + K$$

near D, as well as $\tau \geq 1$ globally on M. For both profiles, E_τ is exactly the tangent space to the remaining torus factors near D, so $\det_\theta B_\tau = 1$.

Example 27.15.1 Fix $\lambda > 0$ and $0 < A < 1/\lambda$, and take $u_{\mathrm{c}} = \pi_1^* v_{\mathrm{c}}$ with the background of Lemma 27.15.1. This gives a complete metric on M solving

$$\mathrm{MA}_\theta(u_{\mathrm{c}}) = \mathrm{e}^{\lambda u_{\mathrm{c}}} f_{\mathrm{c}} \theta^n, \qquad f_{\mathrm{c}} \in C^\infty(M) \cap L^1(X, \theta^n), \quad f_{\mathrm{c}} > 0,$$

where the right side is extended by zero on D. The three exponents in (27.39) can be taken to be

$$q = 2 - \lambda A > 1, \qquad a = \lambda A, \qquad b = 0.$$

Thus their sum equals the critical exponent 2.

Indeed, define $f_{\mathrm{c}} = \mathrm{e}^{-\lambda u_{\mathrm{c}}} \det_\theta(\theta + \mathrm{dd^c} u_{\mathrm{c}})$ on M. On the inner disc the product structure gives

$$f_{\mathrm{c}} = L^{\lambda A} + \frac{A}{C|z_1|^2 L^{2-\lambda A}}.$$

The second term is integrable precisely when $\lambda A < 1$, by the substitution $L = -\log r^2$ in the radial integral. The first term is integrable as well, and the density is smooth elsewhere. The measure identity across D follows from the preceding lemma. Since $\tau = L + K$, all three bounds hold with the displayed exponents. Completeness follows from Corollary 27.15.1. Along a radial normal segment, its critical length divergence is also visible directly:

$$\int_0^{r_1} \frac{\sqrt{A}}{r(-\log r^2)}\, \mathrm{d}r = \infty.$$

The corollary, rather than this single family of radial paths, proves completeness for arbitrary paths.

Example 27.15.2 Fix $\lambda, \eta, \beta > 0$ and take $u_\beta = \pi_1^* v_\beta$ with the background of Lemma 27.15.1. This is a continuous bounded solution of full mass and minimal ordinary singularities. Its metric on M is incomplete, although

$$\mathrm{MA}_\theta(u_\beta) = \mathrm{e}^{\lambda u_\beta} f_\beta \theta^n, \qquad f_\beta \in C^\infty(M) \cap L^1(X, \theta^n), \quad f_\beta > 0.$$

To see this, put $K_\beta = \eta\beta(\beta+1)$ and define the density by the equation on M. Near D it is

$$f_\beta = \mathrm{e}^{-\lambda\eta L^{-\beta}} \left(1 + \frac{K_\beta}{C|z_1|^2 L^{\beta+2}}\right).$$

It is integrable, and the global identity follows as above. Fix the other coordinates and let $z_1 = r \downarrow 0$ on a radial segment starting at $0 < r_0 < r_1$. Its length is at most

$$\sqrt{C}\, r_0 + \sqrt{K_\beta} \int_0^{r_0} \frac{\mathrm{d}r}{r(-\log r^2)^{1+\beta/2}} = \sqrt{C}\, r_0 + \frac{\sqrt{K_\beta}}{\beta}(-\log r_0^2)^{-\beta/2} < \infty.$$

Points along this segment approaching D form a Cauchy sequence with no limit in M.

Here the smallest admissible exponents in (27.39) are $q = \beta + 2$, $a = 0$ and $b = 0$: the density is asymptotic to a positive constant times $R^{-2}\tau^{-(\beta+2)}$, the potential is bounded, and the tangential determinant is 1. Their sum is $2 + \beta$. Since $\beta > 0$ is arbitrary, the threshold 2 in the criterion cannot be replaced by any larger universal number, even in a Kähler class with a bounded full-mass solution.

In these examples one may write $f = F/R^2$ with $F = R^2 f$ smooth and positive on M. This does not assert that F extends smoothly and positively across D: the logarithmic factors make it tend to zero there. The examples concern integrable, logarithmically weakened divisorial poles. They do not settle completeness for a prescribed nonintegrable pole with a smooth positive coefficient across the divisor, or the flow compatibility question discussed in Section 33.15.

Chapter 28
Locally conformally Kähler geometry

Let X be a connected compact complex manifold of dimension $n \geq 2$. An LCK metric becomes Kähler after local conformal changes. The Lee form measures the obstruction to making this change globally. On a covering where the Lee form is exact, the Kähler metric and its admissible potentials change by the same character under deck transformations. Their descent gives a twisted Hessian with a zeroth-order term, so even the additive normalization of a potential changes. The Hopf Vaisman metric illustrates the distinction: its local Kähler structure coexists with unbounded masses under ordinary potential perturbations.

Prescribing volume while retaining the Lee class and Gauduchon gauge couples the determinant with a conformal equation. For a volume invariant under the Lee and anti-Lee flows of a Vaisman metric, uniqueness in the ordinary smooth equation gives a basic potential. Transverse Stokes integration fixes its scalar, and changing the Lee form reconstructs the required Vaisman metric. A different compatibility condition gives rigidity: when one LCK metric is both Gauduchon and Chern–Ricci flat, the Bianchi and Bochner identities force its harmonic Lee form to be parallel.

The Lee form also produces a closed curvature class that can be nonzero in Bott–Chern cohomology while vanishing in de Rham cohomology. Pairing such classes with all Gauduchon metrics gives a cone that detects their positivity and survives bimeromorphic changes. Strongly Gauduchon metrics exclude all nonzero positive de Rham-exact currents and bound the linear mass of closed positive currents by their de Rham classes. The Calabi–Eckmann example separates this condition from the presence of zero in the Lee–Gauduchon cone; the balanced elliptic example shows that it still permits zero lower Monge–Ampère mass.

28.1 The Lee form and the Kähler covering

For a Hermitian form, local conformal Kähler geometry is encoded by a differential equation with a closed one-form.

Definition 28.1.1 A Hermitian form ω on X is *locally conformally Kähler*, or LCK, if there is a smooth real one-form λ on X such that

$$\mathrm{d}\omega = \lambda \wedge \omega, \qquad \mathrm{d}\lambda = 0.$$

The form λ is called its *Lee form*. It is *strictly LCK* if λ is not exact. A strictly LCK metric is *Vaisman* if its Lee form is parallel for its Levi–Civita connection.

Fix an LCK metric ω with Lee form λ. On a compact manifold a nonzero parallel one-form cannot be exact: the differential of a global potential vanishes at a maximum, whereas a parallel form has constant length. Thus a nonzero parallel Lee form is automatically nowhere zero and nonexact.

The Lee form is determined by ω. Indeed, the map $a \mapsto a \wedge \omega$ is injective on one-forms: wedging with ω^{n-2} reduces this assertion to the symplectic isomorphism $a \mapsto a \wedge \omega^{n-1}$.

Locally write $\lambda = \mathrm{d}f$. Then

$$\Omega = \mathrm{e}^{-f}\omega > 0, \qquad \mathrm{d}\Omega = 0.$$

The same construction on a connected covering $\pi : \widetilde{X} \to X$ on which $\pi^*\lambda = \mathrm{d}F$ gives a Kähler form

$$\widetilde{\omega} = \mathrm{e}^{-F}\pi^*\omega. \tag{28.1}$$

For a deck transformation g, the difference $g^*F - F = c_g$ is constant, and

$$g^*\widetilde{\omega} = \chi(g)\widetilde{\omega}, \qquad \chi(g) = \mathrm{e}^{-c_g}.$$

This positive character describes the change of scale. Forms transforming by χ may equivalently be viewed as forms with values in the flat weight line bundle. This is the sense in which an LCK form has a closed interpretation on the cover [OV24b, Section 3.4].

Let g be the Riemannian metric of ω and J the complex structure. On the local conformal chart, $\widetilde{g} = \mathrm{e}^{-f}g$ is Kähler. For real vector fields U, V, the Levi–Civita connections ∇ and $\widetilde{\nabla}$ are related by

$$\widetilde{\nabla}_U V = \nabla_U V - \tfrac{1}{2}\big(\lambda(U)V + \lambda(V)U - g(U,V)\lambda^\sharp\big).$$

Substituting JV for V and using $\widetilde{\nabla}J = 0$ gives

$$(\nabla_U J)V = \tfrac{1}{2}\big(\lambda(JV)U - \lambda(V)JU + g(U,V)J\lambda^\sharp + g(JU,V)\lambda^\sharp\big). \tag{28.2}$$

Both sides are global tensors, so the identity holds on X.

For $a \in C^\infty(X, \mathbb{R})$, the conformal change $\omega' = e^a \omega$ changes the Lee form to $\lambda + da$. In particular, the de Rham class $[\lambda]$ depends only on the conformal class. If λ is exact, the construction above gives a Kähler form on X itself.

28.2 Gauduchon representatives and the Lee class

Passing to the conformal Gauduchon representative makes the Lee form harmonic. For a strictly LCK structure, its curvature has a positive Gauduchon pairing, even though it is de Rham exact. This produces a nonzero ordinary Bott–Chern class and an obstruction to Kähler modifications.

Set

$$\lambda^c = \frac{i}{4\pi}(\lambda^{0,1} - \lambda^{1,0}).$$

Thus $\lambda^c = d^c f$ wherever $\lambda = df$. The closed real $(1,1)$-form

$$\kappa_\lambda := d^c\lambda = -d\lambda^c \tag{28.3}$$

will connect the Lee class with ordinary Bott–Chern cohomology. It is de Rham exact, but need not be dd^c-exact globally.

Proposition 28.2.1 *An LCK metric is Gauduchon if and only if its Lee form is co-closed for that metric. Its conformal Gauduchon representative therefore has harmonic Lee form. Moreover, for any LCK metric,*

$$\int_X \kappa_\lambda \wedge \omega^{n-1} = (n-1)\int_X \lambda \wedge \lambda^c \wedge \omega^{n-1}. \tag{28.4}$$

If $\lambda \not\equiv 0$, the right side is strictly positive.

Proof On a local conformal Kähler chart,

$$d^c(\omega^{n-1}) = (n-1)\lambda^c \wedge \omega^{n-1}.$$

Let $*$ be the real Hodge star of the Riemannian metric associated with ω, and let d^* be the adjoint of the exterior derivative. The pointwise Hermitian Hodge-star identity identifies $\lambda^c \wedge \omega^{n-1}$ with a fixed nonzero constant times $*\lambda$. Consequently $dd^c(\omega^{n-1}) = 0$ is equivalent to $d * \lambda = 0$, or $d^*\lambda = 0$. Since $d\lambda = 0$, co-closedness is equivalent to harmonicity. Existence of the conformal representative follows from Theorem 4.1.1; conformal changes preserve the LCK condition.

Stokes' theorem and $d(\omega^{n-1}) = (n-1)\lambda \wedge \omega^{n-1}$ give

$$\begin{aligned} -\int_X d\lambda^c \wedge \omega^{n-1} &= -\int_X \lambda^c \wedge d(\omega^{n-1}) \\ &= (n-1)\int_X \lambda \wedge \lambda^c \wedge \omega^{n-1}. \end{aligned}$$

For a real one-form, $\lambda \wedge \lambda^{\mathrm{c}} = \mathrm{i}\lambda^{1,0} \wedge \lambda^{0,1}/(2\pi)$ is semipositive. Its trace against ω is positive wherever $\lambda \neq 0$, proving the last assertion. □

For any closed real one-form, the same curvature construction defines a linear map

$$j_X : H^1_{\mathrm{dR}}(X, \mathbb{R}) \longrightarrow H^{1,1}_{\mathrm{BC}}(X, \mathbb{R}), \qquad [\lambda] \longmapsto \{\kappa_\lambda\}. \tag{28.5}$$

Indeed, replacing λ by $\lambda + \mathrm{d}a$ replaces κ_λ by $\kappa_\lambda - \mathrm{dd^c}a$. Pairing this class with any Gauduchon form γ^{n-1} is therefore well defined. For a strictly LCK structure, its own conformal Gauduchon representative gives a strictly positive pairing by (28.4). In particular $\{\kappa_\lambda\} \neq 0$, although its de Rham image is zero.

Corollary 28.2.1 *A compact strictly LCK manifold admits no big ordinary Bott–Chern class. In particular it lies outside Fujiki's class C.*

Proof If there were a big closed class, the modification criterion recalled in Chapter 27 would give a Kähler modification $\pi : Y \to X$. The form $\pi^*\kappa_\lambda$ is closed and de Rham exact. The Kähler $\partial\bar{\partial}$-lemma gives a smooth real function h on Y with $\pi^*\kappa_\lambda = \mathrm{dd^c}h$. Pushing forward gives $\kappa_\lambda = \mathrm{dd^c}(\pi_* h)$, since π has degree one. Pairing this current identity with a Gauduchon form gives zero by Stokes, contradicting (28.4). □

For the Hopf example below, $\kappa_\lambda = \alpha$ is smooth semipositive and therefore nef. It has empty non-nef locus, whereas its non-Kähler locus is all of X, by the non-big convention in Chapter 27. These closed-class statements coexist with the strictly positive nonclosed background ω used for its Monge–Ampère equations.

Proposition 28.2.2 *If X has a strictly LCK structure and $\{\kappa_\lambda\}$ is pseudoeffective, then X admits no balanced metric.*

Proof Let $T = \kappa_\lambda + \mathrm{dd^c}u \geq 0$ represent the class. It is a nonzero closed current, since the class is nonzero. It is also de Rham exact. If a Hermitian metric b were balanced, then $\mathrm{d}(b^{n-1}) = 0$, so distributional Stokes would give

$$\int_X T \wedge b^{n-1} = 0.$$

The mass of a nonzero positive $(1,1)$-current against a smooth strictly positive $(n-1, n-1)$-form is strictly positive, a contradiction. □

The Gauduchon duality framework leads to the following conjecture.

Conjecture 28.2.1 For every compact strictly LCK manifold, the ordinary Bott–Chern class $\{\kappa_\lambda\}$ is pseudoeffective.

The positivity in (28.4) tests the class against the LCK metric in its Gauduchon gauge. Pseudoeffectivity requires nonnegative pairing with *every* Gauduchon metric, by Theorem 4.3.1.

28.3 Ordinary and twisted potentials

The Kähler covering supplies a potential calculus compatible with the Lee equation. Conjugating the ordinary differential operators by the local conformal factor produces a twisted Hessian on X.

For the fixed closed one-form λ, define

$$\mathrm{d}_\lambda = \mathrm{d} - \lambda\wedge, \qquad \mathrm{d}^{\mathrm{c}}_\lambda = \mathrm{d}^{\mathrm{c}} - \lambda^{\mathrm{c}}\wedge, \qquad D_\lambda = \mathrm{d}_\lambda \mathrm{d}^{\mathrm{c}}_\lambda.$$

On a chart where $\lambda = \mathrm{d}f$, these operators satisfy

$$\mathrm{d}_\lambda = \mathrm{e}^f \mathrm{d}\mathrm{e}^{-f}, \qquad \mathrm{d}^{\mathrm{c}}_\lambda = \mathrm{e}^f \mathrm{d}^{\mathrm{c}}\mathrm{e}^{-f}, \qquad D_\lambda = \mathrm{e}^f \mathrm{d}\mathrm{d}^{\mathrm{c}}\mathrm{e}^{-f}. \tag{28.6}$$

For a smooth real function u, the identities show that $D_\lambda u$ is a real $(1,1)$-form and that $\mathrm{d}_\lambda D_\lambda u = 0$.

Proposition 28.3.1 *Let u be smooth and real. Then*

$$\mathrm{d}(\omega + \mathrm{d}\mathrm{d}^{\mathrm{c}}u) - \lambda \wedge (\omega + \mathrm{d}\mathrm{d}^{\mathrm{c}}u) = -\lambda \wedge \mathrm{d}\mathrm{d}^{\mathrm{c}}u,$$

whereas

$$\mathrm{d}(\omega + D_\lambda u) = \lambda \wedge (\omega + D_\lambda u).$$

Consequently, an ordinary potential preserves the LCK equation with the same Lee form precisely when $\lambda \wedge \mathrm{d}\mathrm{d}^{\mathrm{c}}u = 0$. A twisted potential always preserves that equation, and gives an LCK metric whenever $\omega + D_\lambda u > 0$.

Proof The first identity follows from $\mathrm{d}\mathrm{d}\mathrm{d}^{\mathrm{c}}u = 0$ and $\mathrm{d}\omega = \lambda \wedge \omega$. The second follows from $\mathrm{d}_\lambda \omega = 0$ and (28.6). □

On the Kähler covering, a twisted potential becomes the automorphic function $\widetilde{u} = \mathrm{e}^{-F}\pi^* u$, and

$$\mathrm{e}^{-F}\pi^*(\omega + D_\lambda u) = \widetilde{\omega} + \mathrm{d}\mathrm{d}^{\mathrm{c}}\widetilde{u}. \tag{28.7}$$

The numerator in the corresponding Monge–Ampère measure transforms by χ^n. Its ratio to $\widetilde{\omega}^n$ is invariant and descends to X. A scalar measure on X is obtained after fixing the metric gauge, rather than by integrating the automorphic top form over the noncompact cover.

The distinction also affects normalization. Expanding the twisted Hessian gives

$$D_\lambda u = \mathrm{d}\mathrm{d}^{\mathrm{c}}u - \mathrm{d}u \wedge \lambda^{\mathrm{c}} - \lambda \wedge \mathrm{d}^{\mathrm{c}}u + u\big(\kappa_\lambda + \lambda \wedge \lambda^{\mathrm{c}}\big). \tag{28.8}$$

In particular, $D_\lambda 1 = \kappa_\lambda + \lambda \wedge \lambda^{\mathrm{c}}$ need not vanish. Adding a constant can change the metric. The ordinary convention $\sup_X u = 0$ therefore cannot be imposed as a harmless normalization for a twisted equation.

In a chosen gauge the real twisted Bott–Chern group in bidegree $(1,1)$ is

$$H^{1,1}_{\mathrm{BC},\lambda}(X,\mathbb{R}) = \frac{\{\eta \in C^\infty(X, \Lambda^{1,1}_{\mathbb{R}}) : \mathrm{d}_\lambda \eta = 0\}}{D_\lambda C^\infty(X,\mathbb{R})}.$$

This is cohomology with values in the weight line bundle [OV24b, Section 18.5]. It is different both from ordinary Bott–Chern cohomology and from the quotient of all backgrounds by ordinary $\mathrm{dd^c}$-terms in Chapter 23.

28.3.1 The Vaisman Hopf metric

Return to the Hopf threefold and the forms $\omega = \alpha + \beta$ of Chapter 8. Write $s = \log |z|^2$ on the covering. The radial description makes the Lee form explicit and shows that this Gauduchon metric is also Vaisman.

Proposition 28.3.2 *The Gauduchon form $\omega = \alpha + \beta$ in Chapter 8 is Vaisman, with Lee form $\lambda = -\mathrm{d}s$. It satisfies*

$$\kappa_\lambda = \alpha, \qquad D_\lambda 1 = \omega.$$

In particular, the bounded mass property fails for an ordinary potential theory whose background is simultaneously Gauduchon and Vaisman.

Proof Using $\alpha = \mathrm{dd^c} s$ and $\beta = \mathrm{d}s \wedge \mathrm{d^c} s$, direct differentiation gives

$$\mathrm{d}\omega = -\mathrm{d}s \wedge \alpha = -\mathrm{d}s \wedge \omega, \qquad \mathrm{e}^s \omega = \mathrm{dd^c}(\mathrm{e}^s) = \mathrm{dd^c}|z|^2.$$

Although s is not a function on the quotient, $\mathrm{d}s$ descends. Its integral around the loop induced by $z \mapsto \mathrm{e}^{-1} z$ is -2, so the Lee form is not exact.

Let g_{S^5} be the round metric on the unit sphere. Up to a constant factor, the associated Riemannian metric on the cover is

$$\frac{|\mathrm{d}z|^2}{|z|^2} = \mathrm{d}t^2 + g_{S^5}, \qquad t = \log |z|.$$

The one-form $\mathrm{d}t$ is parallel for this product metric, proving the Vaisman condition. Finally, $\kappa_\lambda = \mathrm{dd^c} s = \alpha$ and $\lambda \wedge \lambda^{\mathrm{c}} = \beta$. The unbounded masses follow from Theorem 8.0.1. □

The unbounded masses arise in the ordinary family $\omega + \mathrm{dd^c} u$, whose members need not preserve the Lee equation. The volume problem within the Vaisman class imposes an additional geometric constraint.

28.4 Volume equations in Gauduchon gauge

Return to a compact LCK manifold X of arbitrary dimension $n \geq 2$. Prescribing its volume in Gauduchon gauge couples two equations: one for the determinant and one for the conformal factor. To compare metrics with a fixed Lee class, first identify the twisted groups for cohomologous Lee forms. For a closed real one-form λ, smooth real functions a, u and a smooth differential form η, multiplication by e^{-a} gives

$$\mathrm{d}_\lambda(\mathrm{e}^{-a}\eta) = \mathrm{e}^{-a}\mathrm{d}_{\lambda+\mathrm{d}a}\eta, \qquad D_\lambda(\mathrm{e}^{-a}u) = \mathrm{e}^{-a}D_{\lambda+\mathrm{d}a}u.$$

Fix this identification when comparing classes; the additive constant in a corresponds to a constant rescaling of the weight line bundle.

Let ω_1 be an LCK Gauduchon metric with Lee form λ, and let ω_2 have Lee form $\lambda + \mathrm{d}a$. If their twisted classes agree under the chosen identification, then

$$\mathrm{e}^{-a}\omega_2 = \omega_1 + D_\lambda u =: A > 0$$

for a smooth real u. Substitution of $\omega_2 = \mathrm{e}^a A$ shows that equal volume and the Gauduchon condition are exactly the coupled equations

$$\begin{cases} A^n = \mathrm{e}^{-na}\omega_1^n, \\ \mathrm{dd^c}\left(\mathrm{e}^{(n-1)a}A^{n-1}\right) = 0. \end{cases} \tag{28.9}$$

The scalar Hermitian equation of Chapter 9 does not impose the second equation and uses $\mathrm{dd^c}u$ in place of $D_\lambda u$. The prescribed-volume Gauduchon theorem, Theorem 9.9.1, also does not assert that its solution has an LCK structure or the specified twisted class.

Problem 28.4.1 Does a compact LCK manifold have at most one Gauduchon LCK metric with a specified Lee class, specified twisted Bott–Chern class and specified smooth positive volume form?

This is the uniqueness problem asked in [OV24b, Question 47.56(3)], with the class identification made explicit. The question concerns metrics: uniqueness of a scalar representative u would additionally require accounting for the kernel of D_λ. The degenerate-density counterexamples in Chapter 11 concern ordinary potentials and do not decide this question.

28.4.1 Invariant volumes on Vaisman manifolds

The Lee and anti-Lee flows preserve a Vaisman metric. For an invariant prescribed volume, uniqueness in the smooth Hermitian equation forces its potential to be constant along these flows. Transverse Stokes integration then fixes the normalization scalar. Changing the Lee form by the differential of the potential recovers a Vaisman metric with the prescribed volume.

Theorem 28.4.1 *Let (X, ω, λ) be a compact Vaisman manifold, and let μ be a smooth positive volume form invariant under both the Lee field $\lambda^\sharp$ and the anti-Lee field $J\lambda^\sharp$. Assume*

$$\int_X \mu = \int_X \omega^n.$$

There is a unique Vaisman metric ω' with Lee class $[\lambda]$ and $(\omega')^n = \mu$.

We first describe the transverse forms and their integration. Let g be the metric of ω. The Lee form has constant squared norm $s = |\lambda|_g^2 > 0$. Write

$$A = s^{-1}\lambda^\sharp, \qquad B = JA, \qquad \tau = \frac{s}{4\pi}, \qquad \eta = \kappa_\lambda.$$

Thus $\lambda(A) = 1$ and $\lambda(B) = 0$. Since $\nabla\lambda = 0$, the identity (28.2) gives

$$\begin{aligned} \tau\omega &= \eta + \lambda \wedge \lambda^{\mathrm{c}}, \\ \eta(U, JV) &= \frac{1}{4\pi}\big(sg(U, V) - \lambda(U)\lambda(V) - \lambda(JU)\lambda(JV)\big). \end{aligned} \tag{28.10}$$

In particular, η is a closed semipositive $(1, 1)$-form with kernel $\mathcal{F} = \langle A, B\rangle$. To obtain the displayed identity, use $\lambda^{\mathrm{c}} = -\lambda \circ J/(4\pi)$ and antisymmetrize $\nabla\lambda^{\mathrm{c}}$; the terms containing $\nabla\lambda$ vanish.

The field A is parallel, and the same connection formula yields

$$\nabla_U B = \tfrac{1}{2}\big(-JU + \lambda(U)B + \lambda(JU)A\big), \qquad \nabla_A J = 0.$$

The first identity makes B Killing and gives $[A, B] = 0$. The second makes A holomorphic; integrability of J then makes $B = JA$ holomorphic. Their flows preserve g, J, λ and η. Consequently $\mathcal{F}$ is a holomorphic foliation and η descends to a Kähler form on each local transverse space.

A smooth function v is *basic* for $\mathcal{F}$ if $Av = Bv = 0$. For such a function, $\mathrm{d}v$, $\mathrm{d}^{\mathrm{c}}v$ and $\mathrm{dd}^{\mathrm{c}}v$ are horizontal and invariant under both flows; Cartan's formula gives the assertion for $\mathrm{dd}^{\mathrm{c}}v$. Put $m = n - 1$ and $\eta_v = \eta + \mathrm{dd}^{\mathrm{c}}v$. Then

$$\int_X \eta_v^m \wedge \lambda \wedge \lambda^{\mathrm{c}} = \int_X \eta^m \wedge \lambda \wedge \lambda^{\mathrm{c}}. \tag{28.11}$$

Indeed, $\eta_v^m - \eta^m = \mathrm{dd}^{\mathrm{c}}v \wedge T$, where $T = \sum_{j=0}^{m-1} \eta_v^j \wedge \eta^{m-1-j}$ is closed and horizontal. Since $\mathrm{d}(\lambda \wedge \lambda^{\mathrm{c}}) = \lambda \wedge \eta$, the extra term in $\mathrm{d}(\mathrm{d}^{\mathrm{c}}v \wedge T \wedge \lambda \wedge \lambda^{\mathrm{c}})$ vanishes: it has transverse degree $2m + 1$. Stokes' theorem proves (28.11).

The following transverse maximum principle will identify the metric after its scale has been fixed.

Proposition 28.4.1 *Let $\mathcal{F}$ be a foliation on a connected compact manifold M with transverse complex dimension $m \geq 1$. Let η_0, η_1 be transverse Kähler forms and let v be a smooth real basic function. Compute $\mathrm{dd^c} v$ in transverse complex coordinates. Suppose*

$$\eta_1 = \eta_0 + \mathrm{dd^c} v, \qquad \eta_1^m = \eta_0^m.$$

Then v is constant and $\eta_1 = \eta_0$.

Proof The function v is constant on each local plaque. On every transverse coordinate chart it satisfies

$$\mathrm{dd^c} v \wedge \sum_{j=0}^{m-1} \eta_1^j \wedge \eta_0^{m-1-j} = 0.$$

This is a linear elliptic equation with no zeroth-order term. Let $c = \max_M v$. At a point where $v = c$, the strong maximum principle on a connected transverse chart makes v equal to c throughout that chart. Basicness makes it equal to c on the corresponding foliation chart. Hence $\{v = c\}$ is nonempty, open and closed. Connectedness proves the assertion. □

Proof (of Theorem 28.4.1) Multiplying ω by τ and μ by τ^n preserves the assertion, so we may assume $\tau = 1$. Thus $\omega = \eta + \lambda \wedge \lambda^{\mathrm{c}}$. Write $\mu = f\omega^n$; the positive smooth function f is basic.

Existence. By Theorem 9.1.1, there are a unique normalized smooth function u and a scalar b such that

$$(\omega + \mathrm{dd^c} u)^n = \mathrm{e}^b f \omega^n, \qquad \omega + \mathrm{dd^c} u > 0, \qquad \sup_X u = 0.$$

Pullback by either distinguished flow preserves the equation and the normalization. Uniqueness makes u basic. Hence $\eta_u = \eta + \mathrm{dd^c} u$ is a transverse Kähler form, and

$$(\omega + \mathrm{dd^c} u)^n = n\eta_u^m \wedge \lambda \wedge \lambda^{\mathrm{c}}.$$

Integrating and using (28.11) and $\int_X \mu = \int_X \omega^n$ gives $b = 0$.

Set $\lambda_u = \lambda - \mathrm{d}u$ and

$$\omega_u' = \eta_u + \lambda_u \wedge \lambda_u^{\mathrm{c}}. \tag{28.12}$$

Here $\eta_u = \kappa_{\lambda_u}$ has kernel $\mathcal{F}$, while $\lambda_u(A) = 1$ and $\lambda_u(B) = 0$. These facts give $\omega_u' > 0$. Direct differentiation gives $\mathrm{d}\omega_u' = \lambda_u \wedge \omega_u'$. Basicness of u makes A a holomorphic Killing field for the new metric g_u', and (28.12) gives

$$g_u'(A, \cdot) = \frac{1}{4\pi} \lambda_u.$$

The closedness of λ_u makes $\nabla^{g_u'} A$ symmetric, whereas the Killing equation makes it skew-symmetric. Thus A and λ_u are parallel, so ω_u' is Vaisman with Lee class $[\lambda]$.

Finally, η_u^m has top transverse degree. Every term in $\lambda_u \wedge \lambda_u^c - \lambda \wedge \lambda^c$ contains a horizontal one-form. Therefore

$$(\omega_u')^n = n\eta_u^m \wedge \lambda \wedge \lambda^c = (\omega + \mathrm{dd^c}u)^n = \mu.$$

Uniqueness. Let $\widehat{\omega}$ be another Vaisman solution, with Lee form $\widehat{\lambda} = \lambda + \mathrm{d}v$. Use hats for its s, τ, A, B, η. The form $\eta + \widehat{\eta}$ is exact and semipositive. Its top power has zero integral and is nonnegative, so it vanishes everywhere. The kernel of a sum of two semipositive Hermitian forms is the intersection of their kernels. Since both kernels are complex lines, they coincide. Thus both metrics have the same foliation $\mathcal{F}$.

The nowhere vanishing holomorphic field $A - \mathrm{i}B$ trivializes the holomorphic tangent line of $\mathcal{F}$. The analogous field $\widehat{A} - \mathrm{i}\widehat{B}$ is a holomorphic multiple of it; compactness makes that multiple constant. Write $\widehat{A} = aA + bB$, with $a, b \in \mathbb{R}$. Since $\widehat{\lambda}(\widehat{A}) = 1$ and $\widehat{\lambda}(\widehat{B}) = 0$,

$$\widehat{A}v = 1 - a, \qquad \widehat{B}v = b.$$

Both fields are complete and v is bounded. These derivatives can be constant only if $a = 1$ and $b = 0$. Hence $\widehat{A} = A$, $\widehat{B} = B$, and v is basic.

The contraction formula following from (28.10) is

$$\iota_B \iota_A \widehat{\omega}^n = \frac{n}{4\pi\widehat{\tau}^n}\widehat{\eta}^m, \qquad \iota_B \iota_A (\omega_u')^n = \frac{n}{4\pi}\eta_u^m. \tag{28.13}$$

Equality of their volume forms gives $\widehat{\eta}^m = \widehat{\tau}^n \eta_u^m$. Since $\widehat{\eta} = \eta - \mathrm{dd^c}v$ and v is basic, integrating against $\lambda \wedge \lambda^c$ and applying (28.11) twice gives $\widehat{\tau}^n = 1$. Thus $\widehat{\tau} = 1$ and $\widehat{\eta}^m = \eta_u^m$. Their difference is $-\mathrm{dd^c}(v + u)$, so Proposition 28.4.1 gives $v + u$ constant. It follows that $\widehat{\lambda} = \lambda_u$, $\widehat{\eta} = \eta_u$ and $\widehat{\omega} = \omega_u'$. □

28.5 Chern–Ricci flatness and Vaisman rigidity

Return to an LCK metric ω with Lee form λ. In Gauduchon gauge, λ is harmonic. Chern–Ricci flatness then expresses the Riemannian Ricci tensor in terms of that form and its covariant derivative. Combining the contracted Bianchi identity with the Bochner formula forces the derivative to vanish.

Use $\mathrm{Ric}^{\mathrm{Ch}} = \mathrm{Ric}/(2\pi)$ for the normalized Chern–Ricci form representing $c_1^{\mathrm{BC}}(X)$, with Ric as in Section 25.17.1.

Theorem 28.5.1 *A compact strictly LCK metric that is both Gauduchon and Chern–Ricci flat is Vaisman.*

Proof Let g be the Riemannian metric of ω and ∇ its Levi–Civita connection. Norms, traces and contractions use g. For real tangent vectors U, V, put

$$A = \lambda^\sharp, \qquad B = JA, \qquad S = \nabla\lambda, \qquad S^+(U,V) = \tfrac{1}{2}\big(S(U,V) + S(JU,JV)\big).$$

The tensor S is symmetric because $\mathrm{d}\lambda = 0$. By Proposition 28.2.1, λ is harmonic, so $\mathrm{tr}_g\, S = 0$ and $\mathrm{div}\, A = 0$. Write $J\lambda = -\lambda \circ J$, without the constant factor used in λ^{c} elsewhere in the chapter.

We first compute the Riemannian Ricci tensor:

$$\mathrm{Ric}_g = nS^+ - (n-1)S - \tfrac{1}{2}|\lambda|^2 g + \tfrac{1}{2}\lambda\otimes\lambda + \tfrac{n}{2}J\lambda\otimes J\lambda. \tag{28.14}$$

To obtain this identity, take a local function f with $\lambda = \mathrm{d}f$ and put $\widetilde{g} = \mathrm{e}^{-f}g$. The conformal Ricci formula in real dimension $2n$ gives

$$\mathrm{Ric}_{\widetilde{g}} = \mathrm{Ric}_g + (n-1)S + \tfrac{n-1}{2}\lambda\otimes\lambda - \tfrac{n-1}{2}|\lambda|^2 g,$$

where the trace-Hessian term vanishes because $\mathrm{tr}_g\, S = 0$. The local metric $\widetilde{g}$ is Kähler. Since the Chern–Ricci form of g vanishes, its Kähler Ricci form, written using the unnormalized operator $\mathrm{i}\partial\bar\partial$, is $ni\partial\bar\partial f = \frac{n}{2}\mathrm{d}(J\lambda)$.

Applying (28.2) to $(\nabla_U J\lambda)(V) = -S(U,JV) - \lambda((\nabla_U J)V)$ and antisymmetrizing yields

$$\mathrm{d}(J\lambda)(U,JV) = 2S^+(U,V) - |\lambda|^2 g(U,V) + \lambda(U)\lambda(V) + J\lambda(U)J\lambda(V).$$

For a Kähler metric its Ricci tensor is obtained from its Ricci form by evaluation on (U,JV). Comparing the last two displays with the conformal Ricci formula proves (28.14). The calculation is local, so the resulting tensor identity holds globally.

All integrals below use the Riemannian volume of g. Since $S(A,A) = \frac{1}{2}A(|\lambda|^2)$ and $\mathrm{div}\, A = 0$,

$$\int_X S(A,A)\,\mathrm{d}\,\mathrm{vol}_g = 0. \tag{28.15}$$

The contracted Bianchi identity and $\mathrm{d}^*\lambda = 0$ imply $\int_X \langle \mathrm{Ric}_g, S\rangle\,\mathrm{d}\,\mathrm{vol}_g = 0$: integrate the divergence of the Ricci tensor against λ, then integrate the resulting scalar-curvature derivative against this co-closed form. Since S^+ is the orthogonal projection

of S onto the J-invariant symmetric tensors, (28.14), the vanishing trace of S and (28.15) give

$$n \int_X |S^+|^2 \, \mathrm{d}\,\mathrm{vol}_g - (n-1) \int_X |S|^2 \, \mathrm{d}\,\mathrm{vol}_g + \tfrac{n}{2} \int_X S(B,B) \, \mathrm{d}\,\mathrm{vol}_g = 0. \tag{28.16}$$

The integrated Bochner identity for the harmonic one-form λ is

$$\int_X |S|^2 \, \mathrm{d}\,\mathrm{vol}_g + \int_X \mathrm{Ric}_g(A,A) \, \mathrm{d}\,\mathrm{vol}_g = 0.$$

Evaluating (28.14) on (A,A) cancels its two quartic terms, and the $J\lambda \otimes J\lambda$ term vanishes. Thus

$$\mathrm{Ric}_g(A,A) = (1 - \tfrac{n}{2}) S(A,A) + \tfrac{n}{2} S(B,B).$$

Together with (28.15), this gives

$$\int_X |S|^2 \, \mathrm{d}\,\mathrm{vol}_g + \tfrac{n}{2} \int_X S(B,B) \, \mathrm{d}\,\mathrm{vol}_g = 0. \tag{28.17}$$

Subtracting (28.17) from (28.16) and using the orthogonal projection identity gives

$$\int_X |S - S^+|^2 \, \mathrm{d}\,\mathrm{vol}_g = 0.$$

Hence $S = S^+$ everywhere. Consequently $S(B,B) = S(A,A)$ pointwise. Its integral vanishes by (28.15); (28.17) then gives $\int_X |S|^2 = 0$. Hence $\nabla\lambda = 0$, which is the Vaisman condition. □

The hypotheses concern the same metric. Vanishing of $c_1^{\mathrm{BC}}(X)$ gives a Chern–Ricci-flat representative in each Hermitian conformal class: if a smooth real function h satisfies $\mathrm{Ric}^{\mathrm{Ch}}(\omega) = \mathrm{dd}^{\mathrm{c}} h$ in this normalization, then

$$\mathrm{Ric}^{\mathrm{Ch}}(\mathrm{e}^{h/n}\omega) = \mathrm{Ric}^{\mathrm{Ch}}(\omega) - \mathrm{dd}^{\mathrm{c}} h = 0.$$

Gauduchon's theorem gives a Gauduchon representative in the same conformal class. The rigidity theorem applies when these representatives coincide. Arranging this coincidence is an approach to the following conjecture.

Conjecture 28.5.1 A compact complex manifold with $c_1^{\mathrm{BC}}(X) = 0$ that admits a strictly LCK metric admits a Vaisman metric.

The equation problem is to arrange Chern–Ricci flatness and the Gauduchon condition within an LCK family. The conjecture asks for one Vaisman metric; it does not assert parallelness of every Lee form on the manifold.

28.6 The Lee–Gauduchon cone

For the rest of the chapter, let X be an arbitrary compact complex manifold of dimension $n \geq 2$. To detect pseudoeffectivity in the image of the Lee map j_X of (28.5), one must pair against every Gauduchon metric. The closed forms $\mathrm{d}^{\mathrm{c}}(\gamma^{n-1})$ record these tests in de Rham cohomology. Their natural ambient space is the annihilator of $\ker j_X$, where variation of the metric gives an open convex cone and strict duality detects every nonzero positive class in the image.

Set $V_X = H^1_{\mathrm{dR}}(X, \mathbb{R})$ and

$$K_X = \ker j_X, \qquad W_X = K_X^{\perp} \subset H^{2n-1}_{\mathrm{dR}}(X, \mathbb{R}),$$

where the annihilator uses the Poincaré pairing with a class $a \in H^{2n-1}_{\mathrm{dR}}(X, \mathbb{R})$

$$\langle [\lambda], a \rangle = \int_X \lambda \wedge a.$$

Finally, let

$$C_X = \{[\lambda] \in V_X : j_X([\lambda]) \text{ is pseudoeffective}\}.$$

The image of j_X can be strictly smaller than the kernel of the Bott–Chern-to-de-Rham map, as Example 28.7.1 will show.

Lemma 28.6.1 *The space K_X consists of the de Rham classes of real parts of closed holomorphic one-forms. Moreover,*

$$C_X \cap (-C_X) = K_X.$$

Proof If $j_X([\lambda]) = 0$, write $\mathrm{d}^{\mathrm{c}}\lambda = \mathrm{dd}^{\mathrm{c}} f$ for a smooth real function f. Since $\mathrm{d}^{\mathrm{c}}\mathrm{d} = -\mathrm{dd}^{\mathrm{c}}$, the form $\lambda + \mathrm{d}f$ is both d-closed and d^{c}-closed. Its $(1,0)$-part is therefore holomorphic and closed: the two equations separate the ∂ and $\bar{\partial}$ derivatives. Thus $\lambda + \mathrm{d}f$ is the real part of the closed holomorphic form $2(\lambda + \mathrm{d}f)^{1,0}$. Conversely, the real part of a closed holomorphic form is closed for both operators, and its image under j_X is zero.

By Proposition 27.1.2, the pseudoeffective Bott–Chern cone contains no nonzero line. Thus $j_X(\xi)$ and $-j_X(\xi)$ are both pseudoeffective exactly when $j_X(\xi) = 0$. This is the asserted identity $C_X \cap (-C_X) = K_X$. □

Definition 28.6.1 The *Lee–Gauduchon cone* is

$$\mathrm{LG}(X) = \left\{[\mathrm{d}^{\mathrm{c}}(\gamma^{n-1})] : \gamma \text{ is a Gauduchon metric on } X\right\}.$$

For a closed one-form λ and any smooth real $(n-1, n-1)$-form Γ with $\mathrm{dd}^{\mathrm{c}}\Gamma = 0$, integration by parts gives

$$\int_X \lambda \wedge \mathrm{d}^{\mathrm{c}}\Gamma = \int_X \kappa_\lambda \wedge \Gamma. \tag{28.18}$$

The form $d^c\Gamma$ is closed, and both sides depend only on the relevant cohomology classes. In particular the class on the left annihilates K_X.

Proposition 28.6.1 *The cone* $\mathrm{LG}(X)$ *is nonempty, convex and open in* W_X*. Its linear span is* W_X*.*

Proof Let $\mathcal{Z}_X$ be the real vector space of smooth $(n-1, n-1)$-forms Γ satisfying $dd^c\Gamma = 0$, and set

$$q_X(\Gamma) = [d^c\Gamma].$$

First we show $q_X(\mathcal{Z}_X) = W_X$. The inclusion into W_X follows from (28.18). If $[\lambda]$ annihilates the image of q_X, its pairing with every Gauduchon form is zero. The generalized Lamari criterion, Theorem 4.3.1, implies that both κ_λ and $-\kappa_\lambda$ have pseudoeffective Bott–Chern classes. By Lemma 28.6.1, $[\lambda] \in K_X$. Thus the image and W_X have the same annihilator in V_X. The finite-dimensional nondegenerate Poincaré pairing proves their equality.

Let $\mathcal{Z}_X^+$ denote the strictly positive forms in $\mathcal{Z}_X$. Every such form has a unique positive $(n-1)$st root, by the matrix construction used in the proof of Theorem 4.3.1. This root is Gauduchon. Hence

$$\mathrm{LG}(X) = q_X(\mathcal{Z}_X^+).$$

Gauduchon's theorem makes $\mathcal{Z}_X^+$ nonempty. It is convex and stable under positive scaling.

To see openness directly, choose forms $\Gamma_1, \dots, \Gamma_r \in \mathcal{Z}_X$ whose images form a basis of W_X. If $\Gamma \in \mathcal{Z}_X^+$, then $\Gamma + \sum_i t_i\Gamma_i$ remains strictly positive for all sufficiently small real t_i. Its image describes a neighborhood of $q_X(\Gamma)$ in W_X. This also covers $W_X = \{0\}$, with the empty basis. Since a nonempty open subset spans its ambient vector space, the span assertion follows. □

Theorem 28.6.1 *The cone* C_X *is closed, and*

$$C_X = \{\xi \in V_X : \langle \xi, a\rangle \geq 0 \textit{ for every } a \in \mathrm{LG}(X)\}.$$

Conversely,

$$\mathrm{LG}(X) = \{a \in W_X : \langle \xi, a\rangle > 0 \textit{ for every } \xi \in C_X \setminus K_X\}.$$

Proof The first equality is exactly Theorem 4.3.1 applied to the smooth closed background κ_λ, using (28.18). It expresses C_X as an intersection of closed half-spaces, so this cone is closed.

If $a = [d^c(\gamma^{n-1})] \in \mathrm{LG}(X)$ and $\xi = [\lambda] \in C_X \setminus K_X$, choose $T = \kappa_\lambda + dd^c u \geq 0$. This current is nonzero because it represents the nonzero class $j_X(\xi)$. Therefore

$$\langle \xi, a\rangle = \int_X T \wedge \gamma^{n-1} > 0.$$

For the converse, suppose $a \in W_X \setminus \mathrm{LG}(X)$. Finite-dimensional separation of a point from a nonempty open convex cone gives a nonzero functional $\ell \in W_X^*$ such that

$$\ell(a) \le 0, \qquad \ell(b) \ge 0 \quad (b \in \mathrm{LG}(X)).$$

Indeed, in the usual separating inequalities the constant can be taken to be zero by scaling elements of the cone. By Poincaré duality, ℓ is represented by a class $\xi \in V_X$, modulo K_X. Its nonzero restriction means $\xi \notin K_X$. The first equality in the theorem implies $\xi \in C_X$, contradicting the required strict inequalities for a. This proves the second equality. If $\mathrm{LG}(X) = W_X$, there is no point to separate and the same conclusion holds. □

28.6.1 Bimeromorphic invariance

A modification preserves degree-one cohomology and detects pseudoeffectivity of classes pulled back from the base. These two facts identify the cones C_X; duality then identifies the Lee–Gauduchon cones.

Lemma 28.6.2 *Let $\pi : Y \to X$ be a modification between connected compact complex manifolds, and let θ be a smooth closed real $(1,1)$-form on X. Then*

$$\{\theta\} \text{ is pseudoeffective} \iff \{\pi^*\theta\} \text{ is pseudoeffective}.$$

The pullback on real Bott–Chern cohomology in bidegree $(1,1)$ is injective.

Proof By Proposition 27.1.2, both pullback and pushforward preserve pseudoeffectivity. The underlying potential pullback is the one used in Lemma 1.22.2. Since π has degree one,

$$\pi_*\pi^*\{\theta\} = \{\theta\}.$$

Indeed, the current identity $\pi_*\pi^*\theta = \theta$ follows by integration on the isomorphism locus, whose complement has smooth volume zero. This left inverse proves injectivity and the reverse implication for pseudoeffectivity. □

The degree-one comparison follows by moving cycles off the exceptional image.

Lemma 28.6.3 *If $\pi : Y \to X$ is a modification between connected compact complex manifolds, then*

$$\pi^* : H^1_{\mathrm{dR}}(X,\mathbb{R}) \longrightarrow H^1_{\mathrm{dR}}(Y,\mathbb{R})$$

is an isomorphism. Its Poincaré dual is the isomorphism $\pi_* : H^{2n-1}_{\mathrm{dR}}(Y,\mathbb{R}) \to H^{2n-1}_{\mathrm{dR}}(X,\mathbb{R})$.

Proof For a proper analytic subset A of a connected complex manifold M, restriction $H^1(M,\mathbb{R}) \to H^1(M \setminus A,\mathbb{R})$ is injective, and is an isomorphism if A has complex codimension at least two. Indeed, choose a Whitney stratification of A. General position moves every one-cycle off its strata, which have real codimension at least two.

If those codimensions are at least four, it also moves every bounding two-chain off the strata, relative to its boundary. Thus the map on first homology is surjective in the first case and bijective in the second. Dualizing over $\mathbb{R}$ gives the asserted restriction statements.

Since X is smooth, there is an analytic subset $A \subset X$ of complex codimension at least two such that π is an isomorphism over $U = X \setminus A$. Restriction from X to U is therefore an isomorphism, whereas restriction from Y to $\pi^{-1}U$ is injective. Every class on $\pi^{-1}U \simeq U$ extends to X and hence pulls back to a class on Y. The commutative restriction diagram proves that π^* is an isomorphism. Poincaré duality and the projection formula give the assertion for π_*. □

Theorem 28.6.2 *Let X and Y be bimeromorphic connected compact complex manifolds. Under the natural identification of H^1_{dR} and its Poincaré-dual identification of H^{2n-1}_{dR}, their spaces K_X, W_X and cones $C_X, \mathrm{LG}(X)$ agree.*

Proof First let $\pi : Y \to X$ be a modification. Holomorphic pullback commutes with d^{c}, so

$$j_Y(\pi^*\xi) = \pi^* j_X(\xi).$$

By Lemma 28.6.2, membership in C_X is equivalent to membership of the pullback in C_Y, and membership in K_X is likewise preserved in both directions. By Lemma 28.6.3, every degree-one class upstairs occurs in this way. Thus the degree-one isomorphism identifies the two cones and kernels. Its dual identifies W_Y with W_X, and Theorem 28.6.1 gives

$$\pi_* \mathrm{LG}(Y) = \mathrm{LG}(X).$$

For a bimeromorphic map between X and Y, resolve its graph to obtain modifications $p : Z \to X$ and $q : Z \to Y$ from one smooth compact manifold. The preceding argument applies to both maps and identifies the cones and kernels via $(q^*)^{-1}p^*$ and its Poincaré dual. These identifications are independent of the resolution: they agree on a common open set where the original map is an isomorphism, and the injectivity of degree-one restriction proved above determines each class uniquely. The dual identifications are consequently canonical as well. □

28.6.2 Balanced metrics and the Hopf cone

The presence of zero in the Lee–Gauduchon cone rules out nonzero pseudoeffective classes in the image of j_X. Balanced metrics satisfy this condition. On the Hopf threefold, by contrast, the Lee curvature gives a positive ray and the dual cone is an open half-line.

Corollary 28.6.1 *The following conditions are equivalent:*

$$0 \in \mathrm{LG}(X), \qquad \mathrm{LG}(X) = W_X, \qquad C_X = K_X.$$

A balanced metric implies these conditions. On a manifold in Fujiki's class C, one has $W_X = \{0\}$ and $\mathrm{LG}(X) = \{0\}$.

Proof An open cone in a finite-dimensional vector space that contains zero is the entire space: scale a neighborhood of zero to reach any point. The two dual descriptions in Theorem 28.6.1 now prove the equivalences, including the case $W_X = \{0\}$. If b is balanced, its form b^{n-1} is closed and of pure type, hence also d^{c}-closed. It is Gauduchon and represents zero in $\mathrm{LG}(X)$.

On a compact Kähler manifold the $\partial\bar{\partial}$-lemma gives $j_X = 0$: every κ_λ is closed and de Rham exact. Thus $K_X = V_X$ and $W_X = 0$. Nonemptiness gives $\mathrm{LG}(X) = \{0\}$. Bimeromorphic invariance extends this conclusion to Fujiki's class C. □

Consequently, a nonzero pseudoeffective class in the image of j_X obstructs balanced metrics throughout the bimeromorphic class. For LCK manifolds, Conjecture 28.2.1 would provide such a class.

For the Hopf threefold of Chapter 8, the covering description identifies the underlying manifold with $S^1 \times S^5$, so $V_X = \mathbb{R}[\lambda]$. The class $j_X([\lambda]) = \{\alpha\}$ is nonzero by (28.4), and is pseudoeffective because $\alpha \geq 0$. Negative multiples are not pseudoeffective, since their pairing against the fixed Gauduchon metric is negative. Thus

$$K_X = 0, \qquad C_X = \mathbb{R}_{\geq 0}[\lambda], \qquad \mathrm{LG}(X) = \{a \in H^5_{\mathrm{dR}}(X, \mathbb{R}) : \langle [\lambda], a \rangle > 0\}.$$

The same manifold has unbounded Monge–Ampère mass by Theorem 8.0.1. The positive ray detected here controls linear cohomological pairings, which do not bound those nonlinear masses.

28.7 Strongly Gauduchon metrics

Return to an arbitrary compact complex manifold X of dimension $n \geq 2$. The closed form underlying a strongly Gauduchon metric pairs positively with every nonzero closed positive $(1,1)$-current, and its pairing depends only on the de Rham class. It therefore bounds linear mass and excludes nonzero positive exact currents. The Lee–Gauduchon cone tests only the de Rham-exact Bott–Chern classes in the image of j_X. On the Calabi–Eckmann threefold this image vanishes while a positive exact current remains, so the two conditions differ.

Definition 28.7.1 A Hermitian metric γ is *strongly Gauduchon* if γ^{n-1} is the $(n-1, n-1)$-component of a smooth closed real $(2n-2)$-form.

Write the closed real form in the definition as

$$\Theta = \gamma^{n-1} + \beta + \overline{\beta}, \qquad \beta \text{ of type } (n, n-2).$$

Then $\partial(\gamma^{n-1}) = -\bar{\partial}\beta$. Conversely, this equation and its conjugate give $\mathrm{d}\Theta = 0$, because the remaining types exceed dimension n. Thus the definition is equivalent to $\partial(\gamma^{n-1})$ being $\bar{\partial}$-exact. Applying $\bar{\partial}$ also shows that γ is Gauduchon. Every balanced metric is strongly Gauduchon by taking $\Theta = \gamma^{n-1}$.

Proposition 28.7.1 *A strongly Gauduchon metric has zero Lee–Gauduchon class. A connected compact complex manifold of dimension $n \geq 2$ admits such a metric if and only if it has no nonzero positive de Rham-exact $(1,1)$-current.*

Proof Put $\Gamma = \gamma^{n-1}$ and use β as above. Then

$$\mathrm{d}^{\mathrm{c}}\Gamma = \frac{\mathrm{i}}{4\pi}(\bar{\partial}\beta - \partial\overline{\beta}) = \mathrm{d}\left(\frac{\mathrm{i}}{4\pi}(\beta - \overline{\beta})\right).$$

Thus $[\mathrm{d}^{\mathrm{c}}\Gamma] = 0$.

For the obstruction, let S be a real current of degree one and suppose $T = \mathrm{d}S \geq 0$ is nonzero and Θ is a closed real form with strictly positive $(n-1, n-1)$-part Γ. Type considerations and distributional Stokes give

$$\int_X T \wedge \Gamma = \int_X T \wedge \Theta = 0.$$

The first integral is strictly positive because $T \neq 0$ and $\Gamma > 0$, a contradiction.

For the converse, suppose that no strongly Gauduchon metric exists. In the real Fréchet space of smooth $(2n-2)$-forms, let $\mathcal{Z}$ be the closed subspace of closed forms and let $\mathcal{P}$ be the open convex cone of forms with strictly positive $(n-1, n-1)$-part. This cone is nonempty, since it contains h^{n-1} for any Hermitian metric h. The two sets are disjoint: the positive $(n-1)$st root of the pure-type part of a form in their intersection would be a strongly Gauduchon metric. Hahn–Banach separation therefore gives a nonzero real current T of degree two such that

$$T|_{\mathcal{Z}} = 0, \qquad T(\Theta) \geq 0 \quad (\Theta \in \mathcal{P}).$$

Fix $\Theta_0 \in \mathcal{P}$. If a real form Ψ has zero $(n-1, n-1)$-part, then $\Theta_0 + t\Psi \in \mathcal{P}$ for every $t \in \mathbb{R}$, so $T(\Psi) = 0$. Thus T is of type $(1,1)$. For any semipositive $(n-1, n-1)$-form Γ, $\Gamma + \varepsilon\Theta_0 \in \mathcal{P}$ when $\varepsilon > 0$. Letting $\varepsilon \downarrow 0$ proves $T(\Gamma) \geq 0$.

Since T annihilates every exact test form, it is closed. The de Rham theorem for currents and Poincaré duality identify its cohomology class with its pairings against closed complementary forms. All these pairings vanish, so $T = \mathrm{d}S$ for a real current S. This gives the required nonzero positive exact current and proves the converse. □

Equivalently, every nonzero pseudoeffective Bott–Chern class must have nonzero de Rham image. Indeed, a positive current representing the zero Bott–Chern class has zero pairing with a Gauduchon metric and must vanish. The Bott–Chern classes of

the nonzero positive exact currents are therefore exactly the nonzero pseudoeffective classes with zero de Rham image.

The criterion also characterizes when de Rham classes control the linear masses of closed positive currents.

Corollary 28.7.1 *Fix a Hermitian metric h and a norm on $H^2_{\mathrm{dR}}(X,\mathbb{R})$. Then X admits a strongly Gauduchon metric if and only if there is a constant $C > 0$ such that every closed positive $(1,1)$-current T satisfies*

$$\int_X T \wedge h^{n-1} \leq C\|[T]_{\mathrm{dR}}\|. \tag{28.19}$$

Under this condition, the family of closed positive $(1,1)$-currents whose de Rham classes belong to a fixed compact set is weakly compact. The image of the pseudoeffective Bott–Chern cone in $H^2_{\mathrm{dR}}(X,\mathbb{R})$ is a closed convex cone containing no nonzero line.

Proof Suppose γ is strongly Gauduchon, and choose a smooth closed real form Θ whose $(n-1,n-1)$-part is γ^{n-1}. Compactness gives $h^{n-1} \leq A\gamma^{n-1}$ for some $A > 0$. Positivity and type then give

$$\int_X T \wedge h^{n-1} \leq A \int_X T \wedge \gamma^{n-1} = A\langle [T]_{\mathrm{dR}}, [\Theta]_{\mathrm{dR}}\rangle.$$

The last pairing is a fixed linear functional on the finite-dimensional de Rham group, proving (28.19). Conversely, that bound makes every closed positive de Rham-exact current zero. Apply Proposition 28.7.1.

For the compactness assertion, the bound gives uniformly bounded trace mass. A weak limit of such currents is positive and closed. Its de Rham class is the limit of their classes: this follows by pairing against a finite smooth closed basis of complementary cohomology and using Poincaré duality. Thus the family is also weakly closed, and the usual weak compactness of positive currents proves the assertion.

Let c_j be a convergent sequence in the image cone, and choose closed positive representatives T_j with $[T_j]_{\mathrm{dR}} = c_j$. The compactness assertion gives a closed positive limit representing the limiting de Rham class. Its Bott–Chern class is pseudoeffective, so the image cone is closed. Convexity follows from addition and nonnegative scaling. Finally, if both c and $-c$ lie in the image, choose positive representatives T and S for them. Their sum is exact, so (28.19) makes $T + S = 0$. Positivity gives $T = S = 0$ and hence $c = 0$. □

Example 28.7.1 There is a connected compact complex threefold X with

$$\mathrm{LG}(X) = \{0\}, \qquad V_X = K_X = W_X = 0,$$

but with no strongly Gauduchon metric. Moreover, X has a nonzero pseudoeffective Bott–Chern class whose de Rham image is zero and which is not in the image of j_X.

Proof Consider the holomorphic action of the additive group $\mathbb{C}$ on

$$\widetilde{X} = (\mathbb{C}^2 \setminus \{0\}) \times (\mathbb{C}^2 \setminus \{0\}), \qquad t \cdot (z, w) = (\mathrm{e}^t z, \mathrm{e}^{\mathrm{i}t} w).$$

The action is free. Indeed, $\mathrm{e}^t = \mathrm{e}^{\mathrm{i}t} = 1$ forces $t = 0$. It is proper: if both (z, w) and $t \cdot (z, w)$ range in compact subsets of $\widetilde{X}$, the norms of all four vector components are bounded above and away from zero. Writing $t = a + \mathrm{i}b$, the ratios of the norms bound e^a and e^{-b} above and away from zero. Thus a and b stay bounded, which proves properness. The free proper holomorphic action has local holomorphic slices and a Hausdorff complex quotient X of dimension three.

Every orbit meets $S^3 \times S^3$ exactly once. The required group element is determined by

$$a = -\log |z|, \qquad b = \log |w|.$$

These expressions are smooth, so the slice identifies the quotient smoothly with $S^3 \times S^3$. In particular X is compact and connected, and

$$H^1_{\mathrm{dR}}(X, \mathbb{R}) = H^2_{\mathrm{dR}}(X, \mathbb{R}) = H^5_{\mathrm{dR}}(X, \mathbb{R}) = 0.$$

It follows directly that $V_X = K_X = W_X = 0$ and $\mathrm{LG}(X) = \{0\}$: the latter cone is nonempty by Gauduchon's theorem and lies in the zero group $H^5_{\mathrm{dR}}(X, \mathbb{R})$.

Projectivizing both vector components descends to a holomorphic submersion

$$p : X \longrightarrow \mathbf{P}^1 \times \mathbf{P}^1.$$

Its differential is surjective because the product projectivization on $\widetilde{X}$ is a submersion and the group orbits lie in its fibers. Let η be the sum of the Fubini–Study forms on the two factors. The smooth form

$$\alpha = p^* \eta$$

is closed, semipositive and of rank two everywhere. It is nonzero and de Rham exact, since $H^2_{\mathrm{dR}}(X, \mathbb{R}) = 0$. By Proposition 28.7.1, it obstructs a strongly Gauduchon metric.

Its Bott–Chern class is nonzero. Otherwise $\alpha = \mathrm{dd^c} f$ for a smooth real function f, and integration against any Gauduchon form would give zero, contradicting positivity and $\alpha \neq 0$. The class is pseudoeffective because it has the positive representative α. Finally j_X has zero domain, so this class is not in its image. □

Thus $0 \in \mathrm{LG}(X)$ does not imply existence of a strongly Gauduchon metric, and hence does not imply existence of a balanced metric.

On the balanced threefold of Proposition 25.15.1, let α, β be the two pulled-back Fubini–Study forms. The nonzero class $\{\beta - \alpha\}$ has zero de Rham image, and neither it nor its negative is pseudoeffective, by Proposition 27.9.1. This example also has zero lower Monge–Ampère mass. Thus the strongly Gauduchon condition guarantees neither injectivity of the Bott–Chern-to-de-Rham map nor the lower bound needed for solvability with every finite non-pluripolar measure.

Part IX
Theory under mass bounds

Chapter 29
Volumes under bounded mass

Let X be a connected compact complex manifold of dimension $n \geq 1$ with bounded mass, and let θ be a big smooth real $(1,1)$-form on X. We use the singularity volumes of Definition 12.3.1.

The non-pluripolar mass of a potential in a closed background depends only on its ordinary singularity type. For a nonclosed background, equivalent potentials can have different masses. Upper and lower singularity volumes retain the order information needed for comparison. Bounded mass makes the upper volume finite; a positive lower volume prevents auxiliary envelopes from collapsing. Together these bounds turn a small upper-volume deficit into a large admissible stretch, relating equality of volumes to relative full mass and ordinary envelopes.

Comparison through a common maximum measures the separation of normalized models. On a sector with a uniform positive lower-volume bound, stretching controls three-point comparisons, monotone limits and convergence in capacity. The comparison itself need not satisfy the triangle inequality, even on a projective surface. Finite chains give a complete metric once reciprocal lower volume is included to exclude escape to zero volume. Analytic regularization identifies the ideal-envelope models among the resulting limits, while strict barriers allow these comparisons to persist under small changes of background.

On a curve, the polar part of the curvature determines the ordinary model, and the volume comparison is exactly total variation of polar measures. This description extends to the zero-volume boundary. Taking the atomic part gives the nearest ideal model; diffuse polar curvature measures the failure of analytic approximation. Area rescaling and smooth gauges then describe changes of background, including the collapse of all normalized potentials when the backgrounds converge uniformly to a form of zero area.

29.1 Comparison of singularity volumes

Choose a Hermitian form $\Omega \geq \theta$. The estimates in Proposition 6.5.2 and Theorem 6.1.1 bound the mass of every θ-plurisubharmonic potential by $v_+(\Omega)$. Hence, for $u \in \mathrm{PSH}(X, \theta)$,

$$0 \leq \underline{\mathrm{vol}}_\theta(u) \leq \int_X \mathrm{MA}_\theta(u) \leq \overline{\mathrm{vol}}_\theta(u) < +\infty.$$

If θ is closed, both volumes equal $\int_X \mathrm{MA}_\theta(u)$, by invariance of non-pluripolar mass under equivalent singularities in the closed theory [Xia]. For a non-closed background this equality can fail even at $u = 0$, as Example 2.3.1 shows.

Volume increases when the background increases or the singularities decrease. The rooftop and maximum constructions preserve the required ordinary types; locality confines their errors to sets with vanishing mass.

Theorem 29.1.1 *Let θ_1, θ_2 be smooth real $(1,1)$-forms on X. Suppose θ_1 is big, $\theta_1 \leq \theta_2$, and $u_i \in \mathrm{PSH}(X, \theta_i)$ satisfy $u_1 \preceq u_2$. Then*

$$\underline{\mathrm{vol}}_{\theta_1}(u_1) \leq \underline{\mathrm{vol}}_{\theta_2}(u_2), \qquad \overline{\mathrm{vol}}_{\theta_1}(u_1) \leq \overline{\mathrm{vol}}_{\theta_2}(u_2).$$

Proof Shift u_1 so that $u_1 \leq u_2$. Choose $v_2 \sim u_2$ in $\mathrm{PSH}(X, \theta_2)$ and, for real $t > 0$, set

$$p_t = P_{\theta_1}(u_1 + t, v_2).$$

For $t > 0$, a sufficiently low constant translate of u_1 is a competitor; also $p_t \leq u_1 + t$. Thus $p_t \sim u_1$. The contact theorem, applied in each background on the corresponding contact part, and positivity of the mixed products give

$$\mathrm{MA}_{\theta_1}(p_t) \leq \mathbf{1}_{\{u_1 \leq v_2 - t\}} \mathrm{MA}_{\theta_1}(u_1) + \mathrm{MA}_{\theta_2}(v_2).$$

Indeed p_t is also θ_2-plurisubharmonic and is below v_2, so the contact inequality in that background applies. The first measure on the right has mass tending to zero: its indicators tend to zero off the pole sets, and the measure is finite and non-pluripolar. Integrate, let $t \to +\infty$, and then take the infimum over v_2.

For the upper volume, choose any $v_1 \sim u_1$ in $\mathrm{PSH}(X, \theta_1)$ and put $w_t = \max(v_1, u_2 - t)$. It is θ_2-plurisubharmonic and $w_t \sim u_2$. On $D_t = \{v_1 > u_2 - t\}$, locality and mixed-product positivity imply

$$\mathbf{1}_{D_t} \mathrm{MA}_{\theta_1}(v_1) \leq \mathbf{1}_{D_t} \mathrm{MA}_{\theta_2}(w_t).$$

The sets D_t exhaust the complement of a pluripolar set. Hence integration and passage to the limit give $\int_X \mathrm{MA}_{\theta_1}(v_1) \leq \overline{\mathrm{vol}}_{\theta_2}(u_2)$. Take the supremum over v_1. □

Adding a positive current also increases both volumes. For the lower volume, subtract its potential inside an envelope to recover a competitor in the original background.

Proposition 29.1.1 *Let $u \in \mathrm{PSH}(X, \theta)$ and $v \in \mathrm{PSH}(X, \eta)$ for a smooth real $(1,1)$-form η on X. Then, for either the lower or the upper volume,*

$$\mathrm{vol}_\theta(u) \le \mathrm{vol}_{\theta+\eta}(u+v).$$

Proof The form $\theta + \eta$ is big, by adding the positive current $\eta + \mathrm{dd^c} v$ to a strictly positive current in the class of θ. For the upper volume, add v to each competitor for u. Positivity of mixed products shows that the resulting mass is at least the original mass.

For the lower volume, choose $w \sim u + v$ in $\mathrm{PSH}(X, \theta + \eta)$, and let $p = P_\theta(w - v)$. Since $w - v = u + O(1)$ outside the pole sets, the envelope is genuine and $p \sim u$. The contact theorem says that its measure is carried by $D = \{p + v = w\}$.

Furthermore $p + v \le w$ everywhere by Lemma 1.3.3. Thus

$$\mathrm{MA}_\theta(p) \le \mathbf{1}_D \mathrm{MA}_{\theta+\eta}(p+v) \le \mathbf{1}_D \mathrm{MA}_{\theta+\eta}(w).$$

Integrating and taking the infimum over w proves the assertion. □

29.2 Volume deficits and relative full mass

The upper-volume deficit measures how far one potential can be stretched away from another. The positive lower volume of the less singular potential prevents the stretching envelopes from collapsing. We use Q_b and its competitor family from Definition 12.4.1.

Lemma 29.2.1 *Let $u, v \in \mathrm{PSH}(X, \theta)$ satisfy $u \preceq v$, and put*

$$\delta = \underline{\mathrm{vol}}_\theta(v) > 0, \qquad D = \overline{\mathrm{vol}}_\theta(v) - \overline{\mathrm{vol}}_\theta(u).$$

If $b > 1$ and $b^n D < \delta$, then $Q_b(u, v)$ is genuine.

Proof Adding constants to u and v only translates the stretching envelope. Normalize $v \le 0$, and write $V = \overline{\mathrm{vol}}_\theta(v)$. Since $b^n D < \delta$, choose $w \sim u$ in $\mathrm{PSH}(X, \theta)$ with

$$V - \int_X \mathrm{MA}_\theta(w) < b^{-n}\delta. \tag{29.1}$$

Translate w so that $w \le v$. For positive integers j, put

$$w_j = \max(w, v - j), \qquad h_j = Q_b(w_j, v).$$

The competitor $v - bj$ and the defining inequality give $v - bj \le h_j \le v$. Thus $w_j, h_j \sim v$, and h_j decreases. If its limit were $-\infty$, Theorem 1.4.1 would give $\sup_X h_j \to -\infty$.

By Theorem 13.7.2, the measure of h_j is carried by $C_j = \{b^{-1}h_j + (1-b^{-1})v = w_j\}$. Ordered contact and positivity of mixed products give

$$\mathrm{MA}_\theta(h_j) \le b^n \mathbf{1}_{C_j} \mathrm{MA}_\theta(w_j).$$

Fix a positive integer k. For large $j > k$, collapse would imply $h_j \le -bk$ everywhere. Since $v \le 0$, contact then implies $w_j \le -k$, hence $w \le -k$. Locality on $\{w > -k\} \subset \{w > v - j\}$ gives

$$\begin{aligned}\delta \le \int_X \mathrm{MA}_\theta(h_j) &\le b^n \int_{\{w \le -k\}} \mathrm{MA}_\theta(w_j) \\ &\le b^n \left(V - \int_{\{w > -k\}} \mathrm{MA}_\theta(w) \right).\end{aligned}$$

Here the first inequality uses $h_j \sim v$, and the last uses $w_j \sim v$. Letting $k \to \infty$ contradicts (29.1), since the finite non-pluripolar measure $\mathrm{MA}_\theta(w)$ gives zero mass to the pole set. The limit h is therefore genuine and satisfies $b^{-1}h + (1-b^{-1})v \le w$. If $w \le u + C$, then $h - bC \in \mathcal{A}_b(u, v)$, proving the assertion. □

Proposition 29.2.1 *Fix a smooth compact modification $\pi : Z \to X$. There is a finite constant C_π, depending only on π and θ, such that, for u, v, δ, D as in Lemma 29.2.1,*

$$0 \le \nu(u \circ \pi, z) - \nu(v \circ \pi, z) \le C_\pi \min\left\{1, \left(\frac{D}{\delta}\right)^{1/n}\right\} \qquad (z \in Z).$$

The same estimate holds for generic Lelong numbers along every prime divisor on Z.

Proof Apply Lemma 1.17.6 to the smooth background $\pi^*\theta$ on Z, and choose C_π bounding all its Lelong numbers. Order gives the lower bound and the upper bound C_π, which proves the assertion when $D \ge \delta$.

If $D < \delta$, every $b > 1$ with $b^n D < \delta$ admits $h \in \mathcal{A}_b(u, v)$ by the lemma. Pulling back the defining convex inequality and taking Lelong numbers gives

$$\nu(u \circ \pi, z) - \nu(v \circ \pi, z) \le b^{-1}\big(\nu(h \circ \pi, z) - \nu(v \circ \pi, z)\big) \le b^{-1} C_\pi.$$

For $D > 0$, let $b \uparrow (\delta/D)^{1/n}$; for $D = 0$, let $b \to \infty$. Taking generic values proves the divisorial assertion. □

At zero deficit, every finite stretch is possible. Fix a reference $\varphi \in \mathrm{PSH}(X, \theta)$.

Definition 29.2.1 Define

$$\mathcal{E}_{\mathrm{vol}}(X, \theta, \varphi) = \{u \in \mathrm{PSH}(X, \theta) : u \preceq \varphi,\ \overline{\mathrm{vol}}_\theta(u) = \overline{\mathrm{vol}}_\theta(\varphi)\}.$$

Convex comparison turns the envelope condition back into equality of volumes.

Theorem 29.2.1 *Suppose* $\underline{\mathrm{vol}}_\theta(\varphi) > 0$. *Then*

$$\mathcal{E}_{\mathrm{env}}(X, \theta, \varphi) = \mathcal{E}_{\mathrm{vol}}(X, \theta, \varphi).$$

Every member also satisfies $\underline{\mathrm{vol}}_\theta(u) = \underline{\mathrm{vol}}_\theta(\varphi)$.

Proof First assume the envelope condition. For each $a \in (0, 1)$ there is $h \in \mathrm{PSH}(X, \theta)$ such that $a\varphi + (1 - a)h \le u$. By Theorem 29.1.1 and Proposition 29.1.1 and homogeneity, for either volume,

$$a^n \,\mathrm{vol}_\theta(\varphi) \le \mathrm{vol}_\theta(a\varphi + (1 - a)h) \le \mathrm{vol}_\theta(u) \le \mathrm{vol}_\theta(\varphi).$$

Let $a \to 1$. This proves equality of both volumes in this direction, even without positive lower volume.

Conversely, equality of the upper volumes makes $D = 0$ in Lemma 29.2.1, with $v = \varphi$. Every $Q_b(u, \varphi)$ is genuine, which is the envelope condition. □

29.3 Domination and ordinary envelopes

Passing to an ordinary envelope gives one volume inequality by monotonicity. For the reverse inequality, contact support and domination identify the relative obstacle limit. Positive lower volume makes the stretching envelopes in the domination argument genuine. Write $\mu_w = \mathrm{MA}_\theta(w)$ for $w \in \mathrm{PSH}(X, \theta)$.

Theorem 29.3.1 *Let* $\varphi \in \mathrm{PSH}(X, \theta)$ *have positive lower volume, let* $u \in \mathcal{E}_{\mathrm{env}}(X, \theta, \varphi)$, $v \in \mathrm{PSH}(X, \theta)$ *with* $v \preceq \varphi$, *and* $0 \le c < 1$. *If*

$$\mathbf{1}_{\{u<v\}}\mu_u \le c\,\mathbf{1}_{\{u<v\}}\mu_v,$$

then $u \ge v$.

Proof Put $\ell = \underline{\mathrm{vol}}_\theta(\varphi) > 0$ and $\mathcal{E}_\varphi = \mathcal{E}_{\mathrm{env}}(X, \theta, \varphi)$. Replace v by $\max(u, v)$. Locality preserves the hypothesis and upward stability puts the new v in $\mathcal{E}_\varphi$. Translate u, v by the same constant so that $v \le \varphi$, and then translate all three potentials so that $\varphi \le 0$. Set

$$q_b = Q_b(u, v), \qquad D_b = \{b^{-1}q_b + (1 - b^{-1})v = u\}.$$

Lemma 12.4.1 gives $q_b \in \mathcal{E}_\varphi$ and $q_b \le u$. Contact support and positivity of the mixed products give

$$b^{-n}\mu_{q_b} + (1 - b^{-1})^n \mathbf{1}_{D_b}\mu_v \le \mathbf{1}_{D_b}\mu_u.$$

Here μ_{q_b} is supported on D_b. Choose b so large that $(1 - b^{-1})^n > c$. Subtracting finite measures on $D_b \cap \{u < v\}$ shows that μ_{q_b} vanishes there. Thus it is supported

on $D_b \cap \{u = v\} \subset \{q_b = u\}$. Since $q_b \le u$, ordered contact with coefficient one and Theorem 29.2.1 now give

$$\ell \le \mu_{q_b}(X) \le \mu_u(\{q_b = u\}). \tag{29.2}$$

If $u < v$ anywhere, there is $\delta > 0$ such that $\{u < v - \delta\}$ has positive smooth volume. Indeed, the contrary would give $u \ge v$ almost everywhere and then everywhere by Lemma 1.3.3. On this set $q_b \le u - (b-1)\delta$. The family q_b decreases in b and is bounded above by u. A genuine limit could not be $-\infty$ on a set of positive smooth volume. Hence its limit is identically $-\infty$, and Theorem 1.4.1 gives $\sup_X q_b \to -\infty$. The sets $\{q_b = u\}$ then have μ_u-mass tending to zero by dominated convergence off the pole set of u. This contradicts (29.2). □

Corollary 29.3.1 *For u, v as in Theorem 29.3.1 and $\lambda > 0$,*

$$\mathrm{e}^{-\lambda u}\mu_u \le \mathrm{e}^{-\lambda v}\mu_v \quad \Longrightarrow \quad u \ge v.$$

If $\mu_u \le c\mu_v$ on X for a finite $c \ge 0$, then $c \ge 1$.

Proof On $\{u < v - \delta\}$ the weighted inequality gives $\mu_u \le \mathrm{e}^{-\lambda\delta}\mu_v$. Apply strict domination to $v - \delta$ and let $\delta \downarrow 0$. If the second assertion failed, strict domination against every translate $v + A$ would contradict finiteness of u, v at a common point.□

Lemma 29.3.1 *Let $u, w \in \mathrm{PSH}(X, \theta)$ with $P_\theta(u, w)$ genuine, and put*

$$P_\theta[u](w) = \left(\lim_{C\to\infty} P_\theta(u + C, w)\right)^*.$$

Then

$$\mu_{P_\theta[u](w)} \le \mathbf{1}_{\{P_\theta[u](w)=w\}}\mu_w.$$

Proof Set $p_C = P_\theta(u + C, w)$ for $C \ge 0$ and $p = P_\theta[u](w)$. The envelopes increase, are bounded below by p_0 and above by w, and converge in capacity by Lemma 1.8.6. Assigning ties in the contact set to w gives

$$\mu_{p_C} \le \mathbf{1}_{\{p_C=u+C<w\}}\mu_u + \mathbf{1}_{\{p_C=w\}}\mu_w.$$

The first measure has total mass at most

$$\varepsilon_C = \mu_u(\{u \le \sup_X w - C\}) \longrightarrow 0,$$

because μ_u is finite and non-pluripolar. Testing against nonnegative continuous functions and applying Lemma 6.3.1 gives $\mu_p \le \mu_w$.

The bounded quasi-continuous weight $\chi = \min(1, (w - p)_+)$, defined arbitrarily on the polar indeterminacy set, vanishes on $\{p_C = w\}$, since $p_C \le p \le w$. Thus $\int_X \chi\, \mu_{p_C} \le \varepsilon_C$. Weighted lower semicontinuity gives $\int_X \chi\, \mu_p = 0$, so μ_p is carried by $\{p = w\}$. This proves the asserted restriction of μ_w. □

Theorem 29.3.2 *For $u \in \mathrm{PSH}(X,\theta)$ and $p = P_\theta[u]$,*

$$\underline{\mathrm{vol}}_\theta(u) = \underline{\mathrm{vol}}_\theta(p).$$

If this volume is positive, then also

$$\overline{\mathrm{vol}}_\theta(u) = \overline{\mathrm{vol}}_\theta(p), \qquad u \in \mathcal{E}_{\mathrm{env}}(X,\theta,p), \qquad P_\theta[p] = p.$$

Proof If $\underline{\mathrm{vol}}_\theta(p) = 0$, lower-volume equality follows from monotonicity. Suppose henceforth that it is positive. Normalize $u \le 0$ and choose any $w \sim p$ in $\mathrm{PSH}(X,\theta)$. Fix $K \ge 0$ with $p - K \le w \le p + K$. Since $u \le p$, the rooftop $p_C = P_\theta(u + C, w)$ satisfies $u - K \le p_C \le u + C$ for each $C \ge 0$. In particular it is genuine and equivalent to u. Its increasing canonical limit $r = P_\theta[u](w)$ satisfies

$$P_\theta(u + C + K, 0) - K \le p_C \le w, \qquad p - K \le r \le w \le p + K.$$

Indeed, the leftmost potential lies below $u + C$ and below $p - K \le w$, so is a rooftop competitor. Thus $r \sim p$, and Proposition 12.4.1 puts r in $\mathcal{E}_{\mathrm{env}}(X,\theta,p)$.

By Lemma 29.3.1, μ_r vanishes on $\{r < w\}$. Theorem 29.3.1, with coefficient zero and reference p, gives $r \ge w$; hence $r = w$. Capacity convergence and lower semicontinuity now yield

$$\int_X \mu_w \le \liminf_{C\to\infty} \int_X \mu_{p_C} \le \overline{\mathrm{vol}}_\theta(u).$$

Taking the supremum over $w \sim p$ proves the upper-volume inequality in the required direction; the reverse inequality is monotonicity. Theorem 29.2.1 then gives both relative-class membership and lower-volume equality. Finally Proposition 12.4.3 gives $P_\theta[p] = P_\theta[u] = p$. □

Corollary 29.3.2 *If $\varphi = P_\theta[\varphi]$ has positive lower volume, then*

$$\mathcal{E}_{\mathrm{env}}(X,\theta,\varphi) = \mathcal{E}_{\mathrm{vol}}(X,\theta,\varphi) = \{u \in \mathrm{PSH}(X,\theta) : P_\theta[u] = \varphi\}.$$

Proof The first equality is Theorem 29.2.1. Membership implies envelope equality by Proposition 12.4.3; the converse follows from Theorem 29.3.2. □

29.4 Strict barriers and background variation

Positive lower volume permits a small stretch away from any less singular potential, with a scale controlled by a common mass bound. Mixing the stretch with a strict potential gives a strict potential lying below the original one. The same mixed-mass bound controls the lower-volume error under a smooth background perturbation. Fix a Hermitian form ω.

Lemma 29.4.1 *Let $u, r \in \mathrm{PSH}(X, \theta)$, with $u \le r$, and assume $\underline{\mathrm{vol}}_\theta(u) > 0$. For a Hermitian form $\Omega \ge \theta$, set*

$$b_0 = 1 + \frac{\underline{\mathrm{vol}}_\theta(u)}{2^{n+1} n\, v_+(\Omega)}.$$

Then $Q_b(u, r)$ is genuine for $1 \le b \le b_0$, and

$$\lim_{b \downarrow 1} \overline{\mathrm{vol}}_\theta(Q_b(u, r)) = \overline{\mathrm{vol}}_\theta(u).$$

Proof Put $K = v_+(\Omega)$, so that $1 < b_0 \le 2$, and translate so that $u \le r \le 0$. Suppose $Q_b(u, r)$ is not genuine for some $1 < b \le b_0$. Set

$$u_j = \max(u, r - j), \qquad v_j = u + (b-1)u_j, \qquad q_j = P_\theta(v_j - (b-1)r).$$

Differences at common poles are interpreted through the corresponding convex inequality. Off the poles the obstacle lies between $u - (b-1)j$ and u. Hence q_j is genuine and $q_j \sim u$. As j increases it decreases. A genuine limit would be a competitor for $Q_b(u, r)$, so $\sup_X q_j \to -\infty$.

Let $C_j = \{q_j + (b-1)r = v_j\}$. The contact theorem puts μ_{q_j} on C_j. The positive current with potential $q_j + (b-1)r$ in background $b\theta$ contains the current with potential q_j in background θ. Ordered contact in background $b\theta$ therefore gives

$$\mu_{q_j} \le \mathbf{1}_{C_j} \big((\theta + \mathrm{dd^c} u) + (b-1)(\theta + \mathrm{dd^c} u_j)\big)^n.$$

On C_j we have $q_j = u + (b-1)u_j - (b-1)r \ge bu$. For any $k > 0$ and then all sufficiently large j, $q_j \le -bk$ everywhere, and consequently

$$\underline{\mathrm{vol}}_\theta(u) \le \mu_{q_j}(X) \le \int_{\{u \le -k\}} \big((\theta + \mathrm{dd^c} u) + (b-1)(\theta + \mathrm{dd^c} u_j)\big)^n. \tag{29.3}$$

For any $s, t \in \mathrm{PSH}(X, \theta)$ and $0 \le q \le n$, positivity of the non-pluripolar mixed products and Proposition 6.5.2 imply

$$\begin{aligned}
&\binom{n}{q} \int_X (\theta + \mathrm{dd^c} s)^{n-q} \wedge (\theta + \mathrm{dd^c} t)^q \\
&\quad \le \binom{n}{q} \int_X (\Omega + \mathrm{dd^c} s)^{n-q} \wedge (\Omega + \mathrm{dd^c} t)^q \\
&\quad \le 2^n \int_X \mathrm{MA}_\Omega((s+t)/2) \le 2^n K.
\end{aligned}$$

Here the estimate for the possibly unbounded average follows by constant truncation in the positive background Ω. All products in the display are positive. Expanding the last integrand in (29.3), and using $(b-1)^q \le b-1$ for $q \ge 1$, gives

$$\underline{\mathrm{vol}}_\theta(u) \le \mu_u(\{u \le -k\}) + n2^n K(b-1).$$

The measure μ_u is finite and ignores its pole set. Let $k \to \infty$. The remaining error is at most half of $\underline{\mathrm{vol}}_\theta(u)$ by the choice of b_0, a contradiction. Thus every stated stretch is genuine; at $b = 1$ the envelope is u.

Put $h_0 = Q_{b_0}(u, r)$ and, for $1 < b \le b_0$, $t = (b-1)/(b_0-1)$. The defining inequalities show that

$$(1-t)u + th_0 \le Q_b(u, r) \le u.$$

For $b < b_0$, addition, homogeneity and volume monotonicity give

$$(1-t)^n \overline{\mathrm{vol}}_\theta(u) \le \overline{\mathrm{vol}}_\theta(Q_b(u, r)) \le \overline{\mathrm{vol}}_\theta(u).$$

Letting $b \downarrow 1$ proves the limit. Undoing the common translation changes neither ordinary-type volume nor genuineness. □

Corollary 29.4.1 *If $u \in \mathrm{PSH}(X, \theta)$ has positive lower volume, then there are $w \in \mathrm{PSH}(X, \theta)$ and $\varepsilon > 0$ with*

$$w \le u, \qquad \theta + \mathrm{dd^c} w \ge \varepsilon\omega.$$

Proof Translate u so that $u \le V_\theta$. Choose a strict potential $\rho \le V_\theta$ with $\theta + \mathrm{dd^c}\rho \ge \delta\omega$, using bigness. For a genuine $h = Q_b(u, V_\theta)$ with $b > 1$, put

$$w = b^{-1}h + (1 - b^{-1})\rho.$$

Its defining convex inequality gives $w \le u$, and its current is at least $(1 - b^{-1})\delta\omega$. Restore the translation of u by the same translation of w. □

Proposition 29.4.1 *Assume also that $\theta \ge 0$. If $u \in \mathrm{PSH}(X, \theta)$ and $\underline{\mathrm{vol}}_\theta(u) > 0$, then $P_{\varepsilon\theta}(u)$ is genuine for some $\varepsilon \in (0, 1)$.*

Proof By Corollary 29.4.1, choose $w \le u$ and $\delta > 0$ with $\theta + \mathrm{dd^c} w \ge \delta\omega$. Choose $C > 0$ with $\theta \le C\omega$, and $0 < t < \min(1, \delta/C)$. Then

$$(1-t)\theta + \mathrm{dd^c} w \ge (\delta - tC)\omega \ge 0.$$

Thus w is a competitor for $P_{(1-t)\theta}(u)$. □

For model references, the positive-volume condition is

$$P_\theta[\varphi] = \varphi, \qquad \underline{\mathrm{vol}}_\theta(\varphi) > 0. \tag{29.4}$$

When $\theta \geq 0$, the proposition turns this into the smaller-background condition (12.1). If also $v_-(\theta) > 0$, the converse follows from Lemma 12.5.2: every potential equivalent to φ admits a smaller-background competitor with the same ε, and hence

$$\underline{\mathrm{vol}}_\theta(\varphi) \geq (1-\varepsilon)^n v_-(\theta) > 0.$$

Without this additional lower bound, the smaller-background argument still gives individual positivity, as proved in Section 12.5.

A smooth perturbation changes the lower volume by at most a linear error. A strict lower potential gives a multiplicative comparison for both volumes.

Proposition 29.4.2 *There is $C > 0$, depending only on θ and ω, such that, for every $u \in \mathrm{PSH}(X,\theta)$ and smooth real $(1,1)$-form η with $\theta \leq \eta \leq \theta + \varepsilon\omega$, $0 \leq \varepsilon \leq 1$,*

$$0 \leq \underline{\mathrm{vol}}_\eta(u) - \underline{\mathrm{vol}}_\theta(u) \leq C\varepsilon.$$

If u has positive lower volume, there is also $\delta > 0$ such that, for every $\varepsilon \geq 0$ and $\theta \leq \eta \leq \theta + \varepsilon\omega$, either volume satisfies

$$\mathrm{vol}_\theta(u) \leq \mathrm{vol}_\eta(u) \leq (1+\varepsilon/\delta)^n \,\mathrm{vol}_\theta(u).$$

Proof Choose a Hermitian form $\Omega \geq \theta, \omega$ and set $C = n2^n v_+(\Omega)$. For every $v \in \mathrm{PSH}(X,\theta)$ and $1 \leq k \leq n$, positivity and the mixed-mass estimate in the proof of Lemma 29.4.1, applied in background Ω to v and zero, give

$$\binom{n}{k}\int_X (\theta + \mathrm{dd^c} v)^{n-k} \wedge \omega^k \leq 2^n v_+(\Omega).$$

Expanding in the smooth positive perturbation and using $\varepsilon^k \leq \varepsilon$ yields

$$\int_X \mathrm{MA}_\eta(v) \leq \int_X \mathrm{MA}_{\theta+\varepsilon\omega}(v) \leq \int_X \mathrm{MA}_\theta(v) + C\varepsilon.$$

Take the infimum over $v \sim u$ in $\mathrm{PSH}(X,\theta)$. Each such v is also an admissible competitor in background η, proving the upper bound for the lower volume. The lower bound is background monotonicity.

Now assume u has positive lower volume. Choose $w \leq u$ with $\theta + \mathrm{dd^c} w \geq \delta\omega$ by Corollary 29.4.1. The case $\varepsilon = 0$ is immediate. For $t = \varepsilon/\delta > 0$, the potential tw belongs to $\mathrm{PSH}(X, t(\theta - \delta\omega))$. Addition and monotonicity give

$$\begin{aligned}\mathrm{vol}_\eta(u) &\leq \mathrm{vol}_{\eta + t(\theta-\delta\omega)}(u + tw)\\ &\leq \mathrm{vol}_{(1+t)\theta}((1+t)u) = (1+t)^n \,\mathrm{vol}_\theta(u).\end{aligned}$$

Here $\eta + t(\theta - \delta\omega) \leq (1+t)\theta$ and $u + tw \leq (1+t)u$. The lower bound is background monotonicity. □

29.5 Monotone limits

Passing to a monotone limit must preserve the ordinary-envelope fixed point property as well as the required volume bounds. For increasing sequences, contact support identifies the limiting envelope before its volume is known. For decreasing models, a uniform small stretch retains a positive lower-volume bound. Normalized compactness then preserves the convex inequalities supplied by the deficit lemma; strict barriers allow the background to decrease at the same time.

Theorem 29.5.1 *Suppose* $u_j \in \mathrm{PSH}(X,\theta)$ *increases and* $u = (\lim_j u_j)^* \in \mathrm{PSH}(X,\theta)$ *has positive lower volume. Then*

$$\Big(\lim_j P_\theta[u_j]\Big)^* = P_\theta[u], \qquad P_\theta[u_j] \longrightarrow P_\theta[u] \quad \textit{in capacity},$$

and

$$\underline{\mathrm{vol}}_\theta(u_j) \nearrow \underline{\mathrm{vol}}_\theta(u), \qquad \overline{\mathrm{vol}}_\theta(u_j) \nearrow \overline{\mathrm{vol}}_\theta(u).$$

For every fixed smooth compact modification $\pi : Z \to X$,

$$\sup_{z\in Z}\big|\nu(u_j\circ\pi, z) - \nu(u\circ\pi, z)\big| \longrightarrow 0.$$

Proof Translate so that $u \le 0$, and set $p_j = P_\theta[u_j]$, $p = P_\theta[u]$ and $s = (\lim_j p_j)^*$. Then $u \le s \le p \le V_\theta$. By Theorem 29.3.2, the lower volume of p is positive and $u \in \mathcal{E}_{\mathrm{env}}(X,\theta,p)$. Upward stability puts s in the same class.

The ordinary envelope can use V_θ as obstacle, by Definition 1.19.1. Hence Lemma 29.3.1 gives

$$\mu_{p_j} \le \mathbf{1}_{\{p_j = V_\theta\}}\mu_{V_\theta}.$$

The fixed bounded quasi-continuous weight $\chi = \min(1,(p-s)_+)$, defined arbitrarily on the polar indeterminacy set, vanishes on $\{p_j = V_\theta\}$. Since $p_j \to s$ in capacity, Lemma 6.3.1 gives $\int_X \chi\,\mu_s = 0$. Thus μ_s vanishes on $\{s < p\}$. Strict domination with reference p and coefficient zero yields $s = p$.

Choose any $w \sim p$ in $\mathrm{PSH}(X,\theta)$ and $K \ge 0$ with $p - K \le w \le p + K$. Define $r_j = P_\theta[u_j](w)$. The initial rooftop is genuine, since $u_j - K \le w$. For $C \ge 0$ the competitor comparison gives

$$P_\theta(u_j + C + K, 0) - K \le P_\theta(u_j + C, w) \le p_j + K.$$

The leftmost potential lies below $u_j + C$ and below $p_j - K \le p - K \le w$. For the upper bound, subtract K from the middle term: it lies below zero and is more singular than u_j. Taking the increasing limit in C gives

$$p_j - K \le r_j \le p_j + K, \qquad r_j \le w.$$

The r_j increase to $r = (\lim_j r_j)^*$ in capacity, and $p - K \le r \le w \le p + K$. Thus $r \sim p$. The contact lemma gives $\mu_{r_j} \le \mathbf{1}_{\{r_j = w\}}\mu_w$. Applying weighted lower semicontinuity

with $\min(1,(w-r)_+)$ shows that μ_r vanishes on $\{r<w\}$. Domination with reference p gives $r=w$. Therefore

$$\int_X \mu_w \le \liminf_j \int_X \mu_{r_j} \le \lim_j \overline{\mathrm{vol}}_\theta(p_j),$$

where $r_j \sim p_j$ justifies the last inequality. Taking the supremum over $w \sim p$ and using volume monotonicity proves

$$\overline{\mathrm{vol}}_\theta(p_j) \nearrow \overline{\mathrm{vol}}_\theta(p).$$

Put $\delta = \underline{\mathrm{vol}}_\theta(p) > 0$ and $D_j = \overline{\mathrm{vol}}_\theta(p) - \overline{\mathrm{vol}}_\theta(p_j)$. Since $D_j \to 0$, Lemma 29.2.1 supplies $a_j \to 1$, $0 < a_j < 1$, and $h_j \in \mathrm{PSH}(X,\theta)$ such that

$$a_j p + (1-a_j)h_j \le p_j$$

for all sufficiently large j. Addition, homogeneity and monotonicity of the lower volume, together with Theorem 29.3.2, give

$$a_j^n \delta \le \underline{\mathrm{vol}}_\theta(p_j) = \underline{\mathrm{vol}}_\theta(u_j) \le \delta.$$

This proves lower-volume convergence and positivity for all large j. The same envelope theorem then identifies the upper volumes of u_j and p_j, giving the asserted upper-volume limit. Finally Proposition 29.2.1, applied to $u_j \le u$, gives the uniform Lelong-number convergence on each fixed model. □

For the decreasing results, let θ_j be smooth real $(1,1)$-forms with $\theta \le \theta_{j+1} \le \theta_j$ and $\theta_j \to \theta$ uniformly. The constant sequence is allowed. Normalizing the auxiliary potentials first proves continuity in a fixed background; corrected tail envelopes then handle the varying forms.

Theorem 29.5.2 *Suppose $q_j \in \mathrm{PSH}(X,\theta_j)$ satisfies $q_j = P_{\theta_j}[q_j]$ and decreases to $q \in \mathrm{PSH}(X,\theta)$. Then $P_\theta[q] = q$ and*

$$\underline{\mathrm{vol}}_{\theta_j}(q_j) \searrow \underline{\mathrm{vol}}_\theta(q).$$

If $\underline{\mathrm{vol}}_\theta(q) > 0$, then also

$$\overline{\mathrm{vol}}_{\theta_j}(q_j) \searrow \overline{\mathrm{vol}}_\theta(q),$$

and, for every fixed smooth compact modification $\pi : Z \to X$,

$$\sup_{z\in Z}\left|\nu(q_j\circ\pi,z) - \nu(q\circ\pi,z)\right| \longrightarrow 0.$$

In this case all generic divisorial Lelong numbers converge.

Proof Since $q \le 0$, background and obstacle monotonicity give $q \le P_\theta[q] \le P_{\theta_j}[q] \le q_j$. Letting $j \to \infty$ proves the fixed-point assertion. Set $\ell = \lim_j \underline{\mathrm{vol}}_{\theta_j}(q_j)$. Monotonicity gives $\underline{\mathrm{vol}}_\theta(q) \le \ell$. If $\ell = 0$, the lower-volume assertion follows.

Suppose $\ell > 0$. Choose a Hermitian form $\Omega \geq \theta_1$, put $K = v_+(\Omega)$, and choose

$$b = 1 + \frac{\ell}{2^{n+1} n K}, \qquad a = 1 - b^{-1}.$$

The uniform scale in Lemma 29.4.1 makes $Q_b(q_j, V_{\theta_j})$ genuine for every j. Put $h_j = P_{\theta_j}[Q_b(q_j, V_{\theta_j})]$. For each normalized competitor s for this envelope, $aV_{\theta_j} + (1-a)s$ lies below zero and is more singular than q_j. Since $q_j = P_{\theta_j}[q_j]$, taking the envelope gives

$$aV_{\theta_j} + (1-a)h_j \leq q_j.$$

The functions h_j have supremum zero and a common curvature lower bound $-\Omega$. Normalized compactness and uniform background convergence give a subsequential L^1 limit $h \in \mathrm{PSH}(X, \theta)$. Also $V_{\theta_j} \searrow V_\theta$: their limit is θ-plurisubharmonic and lies between V_θ and zero. Passing to the limit in the convex inequality and using almost-everywhere order gives

$$aV_\theta + (1-a)h \leq q.$$

Moreover, $v_-(\Omega) \geq \underline{\mathrm{vol}}_{\theta_1}(q_1) \geq \ell > 0$ by volume monotonicity. Choose $\rho \leq V_\theta$ with $\theta + \mathrm{dd^c}\rho \geq \kappa\Omega$, $\kappa > 0$. Addition applied to $0 \in \mathrm{PSH}(X, \kappa\Omega)$ and $\rho \in \mathrm{PSH}(X, \theta - \kappa\Omega)$ gives $\underline{\mathrm{vol}}_\theta(\rho) \geq \kappa^n v_-(\Omega)$. Monotonicity and addition in the displayed convex combination now yield

$$\underline{\mathrm{vol}}_\theta(q) \geq a^n \underline{\mathrm{vol}}_\theta(V_\theta) \geq (a\kappa)^n v_-(\Omega) > 0.$$

Thus it remains to prove the volume limits when the limit has positive lower volume.

First suppose $\theta_j = \theta$ for every j. Put

$$\delta = \underline{\mathrm{vol}}_\theta(q) > 0, \qquad L = \lim_j \overline{\mathrm{vol}}_\theta(q_j), \qquad e_k = \overline{\mathrm{vol}}_\theta(q_k) - L.$$

The limit exists by Theorem 29.1.1 and bounded mass. After discarding finitely many indices, choose $b_k > 1$ tending to infinity with $b_k^n e_k < \delta$, and set $a_k = 1 - b_k^{-1}$.

For every $j \geq k$, the volume deficit between q_j and q_k is at most e_k, whereas the lower volume of q_k is at least δ. Thus Lemma 29.2.1 supplies $h_{k,j} \in \mathrm{PSH}(X, \theta)$ with

$$a_k q_k + (1 - a_k) h_{k,j} \leq q_j.$$

Combining normalized ordinary-envelope competitors, as in the proof of Proposition 12.4.3, gives

$$a_k q_k + (1 - a_k) P_\theta[h_{k,j}] \leq q_j.$$

The potentials $P_\theta[h_{k,j}]$ have supremum zero. For fixed k, normalized compactness gives a subsequential L^1 limit h_k as $j \to \infty$. Passing to the limit and applying Lemma 1.3.3 gives

$$a_k q_k + (1 - a_k) h_k \leq q.$$

For either volume, homogeneity, addition and monotonicity now yield

$$a_k^n \operatorname{vol}_\theta(q_k) \le \operatorname{vol}_\theta\big(a_k q_k + (1-a_k)h_k\big) \le \operatorname{vol}_\theta(q) \le \operatorname{vol}_\theta(q_k).$$

Letting $k \to \infty$ proves both volume limits in the fixed background.

For the varying backgrounds, choose $w \le q$ with $\theta + \mathrm{dd^c} w \ge d\omega$, $d > 0$, by Corollary 29.4.1. Choose $\varepsilon_j \downarrow 0$ with $\varepsilon_j > 0$ and $\theta_j \le \theta + \varepsilon_j \omega$, and put

$$c_j = \frac{d}{d+\varepsilon_j}, \qquad f_j = c_j q_j + (1-c_j)w,$$

$$F_j = \left(\sup_{k \ge j} f_k\right)^*, \qquad r_j = P_\theta[F_j].$$

The background of the displayed convex combination is bounded above by θ:

$$c_j \theta_j + (1-c_j)(\theta - d\omega) \le \theta.$$

Thus $f_j \in \mathrm{PSH}(X,\theta)$. Since $f_k \le q_k \le q_j$ for $k \ge j$, upper regularization gives $F_j \le q_j$. Also $f_k \to q$ in L^1, so a subsequence converges almost everywhere and $q \le F_j$ by Lemma 1.3.3.

Background monotonicity now gives

$$q \le r_j = P_\theta[F_j] \le P_{\theta_j}[F_j] \le P_{\theta_j}[q_j] = q_j.$$

The r_j decrease to q. Their lower volumes are positive, so Theorem 29.3.2 makes them ordinary model potentials. The fixed-background case just proved applies to r_j. Addition and background monotonicity also give, for either volume,

$$\operatorname{vol}_\theta(q) \le \operatorname{vol}_{\theta_j}(q_j) \le c_j^{-n} \operatorname{vol}_\theta(f_j) \le c_j^{-n} \operatorname{vol}_\theta(r_j).$$

The outer terms tend to the same limit. Monotonicity in both background and potential gives the stated decreasing convergence.

By Proposition 29.4.2, the volumes of the fixed potential q in background θ_j tend to its volumes in background θ. Hence the upper-volume deficit between q and q_j, measured in θ_j, tends to zero, while the lower volume of q_j is bounded below by $\underline{\operatorname{vol}}_\theta(q) > 0$. Apply Proposition 29.2.1 in each background. Its proof allows one common Lelong bound from $\pi^*\theta_1$, since $\theta_j \le \theta_1$. This proves uniform convergence on each fixed model. □

For an arbitrary decreasing sequence, envelope convergence is exactly the missing upper-volume continuity.

Proposition 29.5.1 *Suppose $u_j \in \mathrm{PSH}(X, \theta_j)$ decreases to $u \in \mathrm{PSH}(X, \theta)$ with positive lower volume. Then*

$$P_{\theta_j}[u_j] \searrow P_\theta[u] \quad \Longleftrightarrow \quad \overline{\mathrm{vol}}_{\theta_j}(u_j) \searrow \overline{\mathrm{vol}}_\theta(u).$$

When these conditions hold, the lower volumes converge to that of u in background θ, and the Lelong numbers converge uniformly on every fixed smooth compact modification.

Proof Normalize $u_j \le 0$ by a common translation. Set $p_j = P_{\theta_j}[u_j]$, $p = P_\theta[u]$ and $q = \lim_j p_j$. The envelope-volume theorem in background θ_j makes every p_j a fixed point of positive lower volume. The envelopes decrease, and the uniform convergence of the backgrounds gives $q \in \mathrm{PSH}(X, \theta)$. Since $u \le p \le q$, the preceding decreasing-model theorem applies and gives, for either volume,

$$\mathrm{vol}_\theta(q) = \lim_j \mathrm{vol}_{\theta_j}(p_j) = \lim_j \mathrm{vol}_{\theta_j}(u_j), \qquad P_\theta[q] = q.$$

If $q = p$, these equalities and Theorem 29.3.2 give both volume limits. Conversely, equality of the upper volumes of u and q puts u in $\mathcal{E}_{\mathrm{vol}}(X, \theta, q)$. By Corollary 29.3.2, $P_\theta[u] = q$. For the Lelong assertion, Proposition 29.4.2 makes the upper-volume deficit between u and u_j in background θ_j tend to zero. Apply Proposition 29.2.1 with the common lower bound $\underline{\mathrm{vol}}_\theta(u) > 0$ and the uniform Lelong bound from $\pi^*\theta_1$. □

Corollary 29.5.1 *For every $u \in \mathrm{PSH}(X, \theta)$,*

$$\underline{\mathrm{vol}}_{\theta_j}(u) \searrow \underline{\mathrm{vol}}_\theta(u).$$

If this limiting volume is positive, then also

$$\overline{\mathrm{vol}}_{\theta_j}(u) \searrow \overline{\mathrm{vol}}_\theta(u), \qquad P_{\theta_j}[u] \searrow P_\theta[u].$$

Proof The volume limits follow from the two estimates in Proposition 29.4.2. In the positive-lower-volume case, apply Proposition 29.5.1 to the constant sequence $u_j = u$. □

For the next corollary, let $Y \subset X$ be a connected compact complex submanifold of positive dimension, and let $u \in \mathrm{PSH}(X, \theta)$ satisfy $\nu(u, Y) = 0$. Use the trace data and restricted approximants of Chapter 18. The majorant in a smaller background gives the trace envelope positive lower volume, so Theorem 29.5.2 applies on Y.

Corollary 29.5.2 *Suppose Y has bounded mass. Choose a Hermitian form $\Omega_0 \geq \theta|_Y$ and a Hermitian form $\Omega \geq 2\Omega_0$ with $v_-(\Omega) > 0$. Then*

$$\nu\big(\mathcal{T}_\Omega(u; Y) \circ \pi, E\big) = \tau_{Y,u}(E)$$

for every prime divisor E on every smooth compact modification $\pi : Z \to Y$.

Proof Put $T = \mathcal{T}_\Omega(u; Y)$ and $r = \dim Y$. The normalized majorant v of Proposition 18.6.1 lies below T and belongs to $\mathrm{PSH}(Y, \Omega/2)$. For every $w \sim T$ in $\mathrm{PSH}(Y, \Omega)$, a constant translate of v lies below w. By Lemma 12.5.2,

$$\int_Y \mathrm{MA}_\Omega(w) \geq 2^{-r} v_-(\Omega), \qquad \underline{\mathrm{vol}}_\Omega(T) \geq 2^{-r} v_-(\Omega) > 0.$$

The restricted approximants belong to $\mathrm{PSH}(Y, \Omega)$ for all sufficiently large indices. By Theorem 18.6.1, their dyadic ordinary envelopes $q_j = P_\Omega[u_{2^j}|_Y]$ decrease to T. The decreasing-model theorem applies on Y and gives

$$\nu(T \circ \pi, E) = \lim_j \nu(q_j \circ \pi, E) = \lim_j 2^{-j} \operatorname{ord}_E \mathfrak{a}_{2^j}(u; Y) = \tau_{Y,u}(E),$$

as required. □

29.5.1 Recovery of envelope systems

Return to the background θ on X. For real $a < c$, use the envelope systems $\mathcal{S}_{a,c}(\theta)$, projection E and retraction R of Definition 24.2.1 and Theorem 24.3.1.

For an envelope system, compatibility fixes every local divisorial order before taking the perturbation limit. A positive limiting lower volume rules out an increase of these orders. Concavity propagates positivity from the constant left part to every finite parameter, so one initial-volume condition suffices for the whole system.

Theorem 29.5.3 *Let $\boldsymbol{\Gamma} \in \mathcal{S}_{a,c}(\theta)$. Fix a Hermitian form h and put $\Gamma = \mathsf{R}\boldsymbol{\Gamma}$. Then*

$$\lim_{j\to\infty} \underline{\mathrm{vol}}_{\theta+h/j}(\Gamma_a^{h/j}) = \underline{\mathrm{vol}}_\theta(\Gamma_a).$$

In particular this limit is independent of h. If it is positive, then $\boldsymbol{\Gamma} = \mathsf{E}(\Gamma)$. In this case, for every $\tau < c$, both singularity volumes satisfy

$$\mathrm{vol}_{\theta+h/j}(\Gamma_\tau^{h/j}) \searrow \mathrm{vol}_\theta(\Gamma_\tau), \qquad \underline{\mathrm{vol}}_\theta(\Gamma_\tau) > 0.$$

Proof Every ideal-envelope fixed point is an ordinary-envelope fixed point by Proposition 15.2.1. The retraction theorem gives $\Gamma_\tau^{h/j} \searrow \Gamma_\tau$ in the decreasing backgrounds $\theta + h/j$. Thus Theorem 29.5.2 proves the initial lower-volume identity, including when the limit is zero.

Suppose this limit is $\ell > 0$. For $\tau \le a$, the slice is Γ_a. For $a < \tau < c$, choose $r \in (\tau, c)$ and put $s = (\tau - a)/(r - a)$. Concavity gives

$$(1 - s)\Gamma_a + s\Gamma_r \le \Gamma_\tau.$$

Homogeneity, Proposition 29.1.1 and volume monotonicity imply

$$\underline{\mathrm{vol}}_\theta(\Gamma_\tau) \ge (1 - s)^n \ell > 0.$$

The decreasing-model theorem now gives both volume limits for every fixed $\tau < c$.

It remains to retain the multiplier ideals. Fix $\tau < c$ and write $q = \Gamma_\tau$, $q_j = \Gamma_\tau^{h/j}$ and $\theta_j = \theta + h/j$. By Proposition 29.4.2, the upper volumes of the fixed q in θ_j converge to its upper volume in θ. Consequently

$$D_j = \overline{\mathrm{vol}}_{\theta_j}(q_j) - \overline{\mathrm{vol}}_{\theta_j}(q) \longrightarrow 0, \qquad \underline{\mathrm{vol}}_{\theta_j}(q_j) \ge \underline{\mathrm{vol}}_\theta(q) > 0.$$

After discarding finitely many indices, choose $b_j \to +\infty$ with $b_j > 1$ and $b_j^n D_j < \underline{\mathrm{vol}}_{\theta_j}(q_j)$. The stretching lemma Lemma 29.2.1 supplies $v_j \in \mathrm{PSH}(X, \theta_j)$ such that

$$(1 - b_j^{-1})q_j + b_j^{-1} v_j \le q.$$

Choose a Hermitian form $\Omega \ge \theta + h$. By Lemma 1.5.1, there is $\alpha > 0$ such that $\mathcal{I}(\alpha v_j) = \mathcal{O}_X$ for all j; additive normalization does not change these ideals. For a fixed local divisor E over a coordinate neighborhood U, Lemma 1.17.7 gives $\nu_E(v_j) \le A_U(E)/\alpha$. Taking Lelong numbers in the convex inequality therefore yields

$$0 \le \nu_E(q) - \nu_E(q_j) \le \frac{A_U(E)}{\alpha b_j} \longrightarrow 0.$$

Compatibility makes $\nu_E(q_j)$ independent of j, so it equals $\nu_E(q)$. Blowing up a point on a local modification converts its pointwise Lelong number into such a divisorial number; on a curve the point itself is a divisor. Thus Theorem 1.16.1 gives $q \sim_{\mathcal{I}} q_j$. This holds at every finite parameter, and Proposition 24.3.1 proves recovery. □

The lower bound may tend to zero as $\tau \uparrow c$; the theorem imposes no uniform positive bound up to the cutoff. For $\theta' = \theta + \mathrm{dd^c} f$ with f smooth and real, the hypothesis is preserved by the gauge bijection: the estimate in Proposition 24.4.1 makes the transported initial slice ordinarily equivalent to $\Gamma_a - f$, and the map $u \mapsto u - f$ preserves the currents and the ordinary-type volume families. The ruled-surface system of Theorem 24.5.1 has zero initial limit and lies outside this positive-volume conclusion.

29.6 Cauchy convergence of model potentials

For two potentials, the maximum gives a common upper bound. The sum of the two upper-volume deficits measures their separation after ordinary projection. A uniform positive lower-volume bound controls how this comparison changes when a third potential is introduced. This replaces the triangle inequality in the Cauchy argument below. Write $U(u) = \overline{\mathrm{vol}}_\theta(u)$ and $M = U(V_\theta)$, and fix a smooth positive probability volume $\mathrm{d}V$ for L^1 norms.

Definition 29.6.1 For $u, v \in \mathrm{PSH}(X, \theta)$, define

$$d_\theta(u, v) = 2U(\max(u, v)) - U(u) - U(v).$$

For $\delta > 0$, put

$$\mathcal{M}_\delta(\theta) = \{u \in \mathrm{PSH}(X, \theta) : P_\theta[u] = u,\ \underline{\mathrm{vol}}_\theta(u) \geq \delta\}.$$

A sequence is d_θ-Cauchy if $d_\theta(u_j, u_k) \to 0$ as $j, k \to \infty$, and converges to u for this comparison if $d_\theta(u_j, u) \to 0$.

The function d_θ is finite, nonnegative and symmetric. It depends only on the ordinary singularity types, since replacing either input by an equivalent one changes the maximum by a bounded amount. Monotonicity also gives

$$|U(u) - U(v)| \leq d_\theta(u, v), \qquad u \leq v \implies d_\theta(u, v) = U(v) - U(u). \tag{29.5}$$

Proposition 29.6.1 *For potentials u, v of positive lower volume,*

$$d_\theta(u, v) = d_\theta(P_\theta[u], P_\theta[v]), \qquad d_\theta(u, v) = 0 \iff P_\theta[u] = P_\theta[v].$$

There is $C_\theta > 0$, depending on θ and $\mathrm{d}V$, such that, for $u, v \in \mathcal{M}_\delta(\theta)$,

$$\|u - v\|_{L^1} \leq 2C_\theta \min\left\{1, \left(\frac{2d_\theta(u, v)}{\delta}\right)^{1/n}\right\}.$$

Proof Translate the inputs below zero, and put $p = P_\theta[u]$, $q = P_\theta[v]$, $m = \max(u, v)$ and $r = P_\theta[m]$. Envelope order gives

$$m \leq \max(p, q) \leq r.$$

All these potentials have positive lower volume. The ordinary-envelope volume identity and monotonicity give $U(m) = U(\max(p, q)) = U(r)$, while $U(u) = U(p)$ and $U(v) = U(q)$. This proves projection invariance. If $d_\theta(u, v) = 0$, then $U(u) = U(v) = U(m)$. The relative full-mass equivalence and Proposition 12.4.3 give $P_\theta[u] = P_\theta[m] = P_\theta[v]$. Conversely, if $p = q$, projection invariance gives zero.

Now let $u, v \in \mathcal{M}_\delta(\theta)$ and set $s = d_\theta(u, v)$. Normalized compactness supplies C_θ bounding the L^1 norm of every θ-plurisubharmonic potential with supremum zero. If $s \geq \delta/2$, the asserted estimate follows from the two individual norm bounds. Suppose $0 < s < \delta/2$ and put $t = (2s/\delta)^{1/n}$. The deficit from u to m is at most s, and the lower volume of m is at least δ. Thus Lemma 29.2.1, with $b = t^{-1}$, gives $h \in \mathrm{PSH}(X, \theta)$ such that

$$(1 - t)m + th \leq u.$$

For each normalized competitor for $P_\theta[h]$, the corresponding convex combination with m lies below zero and is more singular than u. Since $u = P_\theta[u]$, taking the envelope gives

$$(1 - t)m + tP_\theta[h] \leq u.$$

Hence $\int_X (m - u)\, \mathrm{d}V \leq tC_\theta$, because $m \leq 0$ and $P_\theta[h]$ has supremum zero. Applying the same argument to v and adding gives the estimate. If $s = 0$, use any $t \in (0, 1)$ and let $t \downarrow 0$. □

The next bound supplies a uniform modulus for three-point comparisons. The triangle inequality itself fails, as Example 29.11.1 shows.

Proposition 29.6.2 *For $u, v, w \in \mathcal{M}_\delta(\theta)$,*

$$d_\theta(u, w) \leq d_\theta(u, v) + d_\theta(v, w) + 2nM \min\left\{1, \left(\frac{2d_\theta(v, w)}{\delta}\right)^{1/n}\right\}. \tag{29.6}$$

Consequently every d_θ-convergent sequence in $\mathcal{M}_\delta(\theta)$ is d_θ-Cauchy, and its limit is unique.

Proof Put $s = d_\theta(v, w)$. If $s \geq \delta/2$, use $d_\theta(u, w) \leq 2M$. If $0 < s < \delta/2$, set $t = (2s/\delta)^{1/n}$. The deficit lemma supplies $h \leq v$ with

$$(1 - t)\max(v, w) + th \leq v.$$

Set $r = \max(u, v)$ and $z = \max(u, v, w)$. On the branch $z = \max(v, w)$ the displayed inequality applies; on the branch $z = u$, both u and h lie below r. Thus

$$(1 - t)z + th \leq r.$$

Addition, homogeneity and volume monotonicity give

$$U(z) - U(r) \leq \big(1 - (1 - t)^n\big)U(z) \leq nMt.$$

Since $\max(u, w) \leq z$, it follows that

$$\begin{aligned} d_\theta(u, w) &\leq 2U(r) + 2nMt - U(u) - U(w) \\ &= d_\theta(u, v) + U(v) - U(w) + 2nMt \\ &\leq d_\theta(u, v) + s + 2nMt. \end{aligned}$$

For $s = 0$, the same proof works with every $t \in (0, 1)$; let $t \downarrow 0$. This proves (29.6). Applying it with the proposed limit as the middle point shows that a convergent sequence is Cauchy. Applying it with a sequence member as the middle point and then using separation proves uniqueness. □

Theorem 29.6.1 *Every d_θ-Cauchy sequence in $\mathcal{M}_\delta(\theta)$ converges for d_θ to a unique element of $\mathcal{M}_\delta(\theta)$. It also converges to that element in $L^1(X, \mathrm{d}V)$.*

Proof Choose a subsequence a_j of the given sequence such that

$$d_\theta(a_j, a_{j+1}) < \delta\, 2^{-(j+1)n} \qquad (j \geq 1).$$

For $k \geq 0$, set

$$v_j^k = \max(a_j, \dots, a_{j+k}), \qquad v_j = \left(\sup_{l \geq j} a_l\right)^*.$$

For every $l \geq j$, the deficit lemma with $b = 2^l$ gives $h_l \leq a_l$ satisfying

$$(1 - 2^{-l}) \max(a_l, a_{l+1}) + 2^{-l} h_l \leq a_l.$$

The maximum argument used above gives

$$(1 - 2^{-l}) v_j^{l+1-j} + 2^{-l} h_l \leq v_j^{l-j}.$$

Consequently

$$U(v_j^{l+1-j}) - U(v_j^{l-j}) \leq nM2^{-l}.$$

Summing from $l = j$ to $j + k - 1$ and then applying increasing-volume continuity as $k \to \infty$ yields

$$0 \leq U(v_j) - U(a_j) \leq nM2^{1-j}.$$

Here v_j is genuine and has lower volume at least δ, because $a_j \leq v_j \leq 0$.

Put $q_j = P_\theta[v_j]$. The ordinary-envelope volume theorem makes q_j a fixed point with the same two volumes as v_j. The sequence decreases, and each q_j has supremum zero. Normalized compactness therefore gives a genuine decreasing limit q. By Theorem 29.5.2,

$$q = P_\theta[q], \qquad \underline{\mathrm{vol}}_\theta(q) = \lim_j \underline{\mathrm{vol}}_\theta(q_j) \geq \delta, \qquad U(q_j) \longrightarrow U(q).$$

Both a_j and q lie below q_j. Hence

$$\begin{aligned} d_\theta(a_j, q) &\leq \big(U(q_j) - U(a_j)\big) + \big(U(q_j) - U(q)\big) \\ &\leq nM2^{1-j} + U(q_j) - U(q) \longrightarrow 0. \end{aligned}$$

To recover the whole sequence, choose a late subsequence member a_j as the middle point in (29.6). The Cauchy assumption makes the first term uniformly small for

all sufficiently late original members, while $d_\theta(a_j, q)$ makes the other terms small. Thus the original sequence converges to q. Uniqueness follows from the preceding proposition, and the L^1 estimate in Proposition 29.6.1 proves the last assertion. □

The closed lower-volume bound is essential for this completeness statement.

Example 29.6.1 On a compact Riemann surface with a smooth positive form θ of integral one, the ordinary envelope fixed points of mass strictly greater than δ, for $0 < \delta < 1$, do not form a complete space for d_θ.

Proof Fix a point p and use the nonpositive Green potential g_p from Theorem 18.3.1, so that $\mathrm{dd^c} g_p = \delta_p - \theta$. For $0 \le a < 1$, put $q_a = P_\theta[a g_p]$. The current of $a g_p$ is $a\delta_p + (1-a)\theta$, with non-pluripolar mass $1-a$. Since the background is closed, its two singularity volumes equal this mass. The ordinary-envelope volume theorem gives

$$q_a = P_\theta[q_a], \qquad \underline{\mathrm{vol}}_\theta(q_a) = U(q_a) = 1 - a.$$

The potentials q_a decrease as a increases. Therefore (29.5) gives $d_\theta(q_a, q_b) = |a-b|$. Choose $a_j \uparrow 1-\delta$ with $a_j < 1-\delta$. The resulting sequence is Cauchy and every term has mass greater than δ. Any d_θ limit has upper volume δ by (29.5). On a curve the two volumes coincide, so no such limit belongs to the strict-cutoff space. □

29.7 Rooftops and capacity convergence

Return to the fixed big background θ on X. A small upper-volume deficit also controls the common lower envelope of two potentials. For a finite family, the sum of the deficits must stay below the lower volume of a common upper bound. Applying this criterion after stretching gives increasing lower bounds for a suitable subsequence of models. Together with the decreasing upper bounds from the preceding section, they turn comparison convergence into capacity convergence.

Lemma 29.7.1 *Let $u_1, \dots, u_N, w \in \mathrm{PSH}(X, \theta)$, with $N \ge 1$ and $u_i \preceq w$. If*

$$\sum_{i=1}^{N} \big(U(w) - U(u_i)\big) < \underline{\mathrm{vol}}_\theta(w),$$

then $P_\theta(u_1, \dots, u_N)$ is genuine.

Proof Genuineness of a finite rooftop is unchanged by bounded changes of its obstacles. Choose $v_i \sim u_i$ whose masses satisfy

$$\sum_{i=1}^{N} \left(U(w) - \int_X \mu_{v_i} \right) < \underline{\mathrm{vol}}_\theta(w).$$

Translate so that $v_i \le w \le 0$, and put

$$v_{i,k} = \max(v_i, w-k), \qquad p_k = P_\theta(v_{1,k}, \dots, v_{N,k}).$$

The common competitor $w-k$ gives $w-k \le p_k \le w$, so $p_k \sim w$. If the desired rooftop were not genuine, p_k would decrease to $-\infty$, and normalized compactness would give $\sup_X p_k \to -\infty$.

Fix $s > 0$ and choose $k > s$ so large that $p_k \le -s$. Iterating Proposition 12.1.3 and using locality on $\{v_i > -s\}$ gives

$$\begin{aligned}\underline{\mathrm{vol}}_\theta(w) &\le \int_X \mu_{p_k} \le \sum_{i=1}^N \int_{\{v_{i,k} \le -s\}} \mu_{v_{i,k}} \\ &\le \sum_{i=1}^N \left(U(w) - \int_{\{v_i > -s\}} \mu_{v_i} \right).\end{aligned}$$

Here $v_{i,k} \sim w$, and $w-k < -s$ makes $\{v_{i,k} > -s\} = \{v_i > -s\}$. Let $s \to \infty$. The measures are finite and non-pluripolar, so the resulting inequality contradicts the choice of v_i. □

To control a rooftop's volume, first stretch each obstacle, then move both stretches a fixed fraction back toward their maximum. The softened stretches have a genuine common rooftop by the lemma.

Proposition 29.7.1 *Let $u_j, u \in \mathrm{PSH}(X,\theta)$, with $\underline{\mathrm{vol}}_\theta(u) > 0$, and suppose $d_\theta(u_j, u) \to 0$. For all sufficiently large j, the rooftop $p_j = P_\theta(u_j, u)$ is genuine, and*

$$d_\theta(p_j, u) \longrightarrow 0, \qquad \underline{\mathrm{vol}}_\theta(p_j) \longrightarrow \underline{\mathrm{vol}}_\theta(u).$$

Proof Put $\delta = \underline{\mathrm{vol}}_\theta(u)$, $m_j = \max(u_j, u)$ and $s_j = d_\theta(u_j, u)$. After discarding finitely many indices, choose $b_j \to \infty$ with $b_j > 1$ and $b_j^n s_j < \delta$. The deficits of both u_j and u from m_j are at most s_j, while m_j has lower volume at least δ. Thus Lemma 29.2.1 makes

$$G_j = Q_{b_j}(u_j, m_j), \qquad H_j = Q_{b_j}(u, m_j)$$

genuine. Choose $a \in (0,1)$ with $2nMa < \delta$, and set

$$g_j = (1-a)m_j + aG_j, \qquad h_j = (1-a)m_j + aH_j.$$

These potentials lie below m_j. Addition and monotonicity give

$$2U(m_j) - U(g_j) - U(h_j) \le 2\big(1 - (1-a)^n\big)U(m_j) \le 2nMa < \delta.$$

The preceding lemma therefore gives a genuine $r_j = P_\theta(g_j, h_j)$. For large j, put $t_j = (ab_j)^{-1} < 1$. The defining inequalities for G_j, H_j show that

$$(1-t_j)m_j + t_j r_j \le u_j, u.$$

Hence p_j is genuine. For either volume, addition and monotonicity give

$$(1-t_j)^n \operatorname{vol}_\theta(u) \le (1-t_j)^n \operatorname{vol}_\theta(m_j) \le \operatorname{vol}_\theta(p_j) \le \operatorname{vol}_\theta(u).$$

Since $t_j \to 0$ and $p_j \le u$, this proves both assertions. □

For ordinary models, the finite rooftops are again models. Their normalization prevents collapse when the number of obstacles tends to infinity.

Theorem 29.7.1 *Suppose $u_j, u \in \mathcal{M}_\delta(\theta)$ and $d_\theta(u_j, u) \to 0$. After passing to a subsequence and relabeling it, there are ordinary model potentials $v_j, w_j \in \mathcal{M}_{\delta/2}(\theta)$ such that*

$$w_j \le u_j \le v_j, \qquad w_j \le u \le v_j,$$

v_j decreases to u, w_j increases with $(\lim_j w_j)^* = u$, and

$$d_\theta(v_j, u) + d_\theta(w_j, u) \longrightarrow 0.$$

The original sequence converges to u in capacity.

Proof For large j, put $r_j = P_\theta(u_j, u)$, using Proposition 29.7.1. It is a model: envelope order gives $r_j \le P_\theta[r_j] \le u_j, u$, so $P_\theta[r_j]$ is a competitor for the rooftop and equals r_j. The sequence u_j is Cauchy by Proposition 29.6.2. Choose a subsequence so that

$$d_\theta(u_j, u_{j+1}) < \delta\, 2^{-(j+1)n}, \qquad U(u) - U(r_j) < \delta\, 4^{-(j+1)n}.$$

The tail-maximum construction in the proof of Theorem 29.6.1 gives

$$v_j = P_\theta\left[\left(\sup_{l \ge j} u_l\right)^*\right] \searrow u, \qquad d_\theta(v_j, u) \longrightarrow 0.$$

Indeed that construction gives the unique comparison limit, which here is u. Its upper models satisfy $v_j \ge u_j, u$ and have lower volume at least δ.

For the lower bounds, the deficit lemma gives genuine potentials

$$H_l = Q_{4^l}(r_l, u), \qquad K_l = Q_{2^l}(r_l, u).$$

Their defining inequalities give

$$(1-2^{-l})u + 2^{-l} H_l \le K_l \le u, \qquad U(u) - U(K_l) \le nM2^{-l}.$$

Choose j so large that $nM2^{1-j} < \delta$. For every $k \ge 0$, the sum of the deficits of $K_j, \dots, K_{j+k}$ is less than δ, so Lemma 29.7.1 makes their rooftop genuine. Since $r_l \le u$, stretching decreases with its parameter; for $l \ge j$,

$$Q_{2^j}(r_l, u) \ge K_l.$$

Thus $R_j^k = P_\theta(Q_{2^j}(r_j, u), \dots, Q_{2^j}(r_{j+k}, u))$ is genuine. The convex inequality for each obstacle gives

$$(1 - 2^{-j})u + 2^{-j} R_j^k \le w_j^k := P_\theta(r_j, \dots, r_{j+k}) \le u.$$

For either volume,

$$(1 - 2^{-j})^n \operatorname{vol}_\theta(u) \le \operatorname{vol}_\theta(w_j^k) \le \operatorname{vol}_\theta(u). \tag{29.7}$$

The same envelope-order argument used for r_j shows that w_j^k is a model. For fixed j it decreases in k, with supremum zero. Normalized compactness gives a genuine limit w_j. The lower-volume conclusion of Theorem 29.5.2 passes the lower bound in (29.7) to w_j. It is positive, so the theorem's upper-volume conclusion passes that bound as well.

The w_j increase and satisfy $w_j \le r_j \le u_j, u$. After discarding finitely many indices, they have lower volume at least $\delta/2$, and (29.7) gives $d_\theta(w_j, u) \le nM2^{-j}$. By Proposition 29.6.1, $w_j \to u$ in L^1, so $(\lim_j w_j)^* = u$. The two monotone sequences converge in capacity by Lemma 1.8.6. Their pointwise bounds squeeze the selected u_j to u in capacity. Applying the construction to every subsequence proves capacity convergence for the original sequence. □

29.8 Volumes of multiplier-ideal envelopes

Analytic regularizations recover the ideal envelope, whose volume may exceed that of the original potential. Decreasing-model continuity computes this excess. A strict analytic potential absorbs the curvature error, so the same comparison can be made entirely in the fixed background. For $u \in \mathrm{PSH}(X, \theta)$, set

$$\mathcal{V}^-_{\mathcal{I},\theta}(u) = \underline{\mathrm{vol}}_\theta(P_\theta[u]_{\mathcal{I}}), \qquad \mathcal{V}^+_{\mathcal{I},\theta}(u) = \overline{\mathrm{vol}}_\theta(P_\theta[u]_{\mathcal{I}}).$$

They depend only on the multiplier-ideal class of u in the fixed background. By Proposition 15.3.1 and gauge invariance of singularity volumes, they also depend only on the associated $\mathrm{dd^c}$-class and the multiplier-ideal data, rather than on its smooth decomposition. Moreover, Proposition 15.2.1 and Theorem 29.1.1 give

$$\underline{\mathrm{vol}}_\theta(u) \le \mathcal{V}^-_{\mathcal{I},\theta}(u), \qquad \overline{\mathrm{vol}}_\theta(u) \le \mathcal{V}^+_{\mathcal{I},\theta}(u).$$

The ideal envelope is an ordinary fixed point by Proposition 15.2.1. Its volumes are limits of the volumes of the dyadic analytic types.

Corollary 29.8.1 *Fix a Hermitian form h, and take the dyadic regularizations a_j of u in backgrounds $\theta_j = \theta + \epsilon_j h$ as in (15.3). Then*

$$\underline{\mathrm{vol}}_{\theta_j}(a_j) \searrow \mathcal{V}^-_{\mathcal{I},\theta}(u).$$

If $\mathcal{V}^-_{\mathcal{I},\theta}(u) > 0$, then also

$$\overline{\mathrm{vol}}_{\theta_j}(a_j) \searrow \mathcal{V}^+_{\mathcal{I},\theta}(u).$$

Proof By Theorem 15.5.1, the analytic ordinary models $q_j = P_{\theta_j}[a_j]$ decrease to $P_\theta[u]_{\mathcal{I}}$. Apply Theorem 29.5.2. Since $q_j \sim a_j$ by Proposition 15.2.3, their two volumes agree term by term. □

Positive lower volume of the ideal envelope lets relative full mass detect when this limit also computes the upper volume of u.

Proposition 29.8.1 *Let $u \in \mathrm{PSH}(X,\theta)$, and put $q = P_\theta[u]_{\mathcal{I}}$. Suppose $\underline{\mathrm{vol}}_\theta(q) > 0$. Then the following are equivalent:*

$$P_\theta[u] = P_\theta[u]_{\mathcal{I}}, \qquad \overline{\mathrm{vol}}_\theta(u) = \mathcal{V}^+_{\mathcal{I},\theta}(u).$$

They are also equivalent to $u \in \mathcal{E}_{\mathrm{env}}(X,\theta,q)$ and to

$$P_{\theta,F}[u] = P_{\theta,F}[u]_{\mathcal{I}} \quad \textit{for every smooth real } F.$$

If they hold, the corresponding equality for the lower volume also holds.

Proof If the envelopes agree, Theorem 29.3.2 gives equality of both volumes with those of q.

Conversely, upper-volume equality and $u \preceq q$ put u in $\mathcal{E}_{\mathrm{vol}}(X,\theta,q)$. Since $q = P_\theta[q]$, Corollary 29.3.2 gives $P_\theta[u] = q$.

The same corollary identifies zero-obstacle equality with membership in $\mathcal{E}_{\mathrm{env}}(X,\theta,q)$. By Proposition 15.6.6, this implies equality for every smooth obstacle; taking $F = 0$ gives the converse. □

Under the hypotheses of the proposition, Corollary 15.5.2 also identifies equality of the upper volumes with convergence of the dyadic ordinary envelopes to $P_\theta[u]$. The next construction removes their background error while retaining the upper-volume limit. The corrected analytic types need not be ordered.

Theorem 29.8.1 *Let $u \in \mathrm{PSH}(X,\theta)$, put $q = P_\theta[u]_{\mathcal{I}}$, and suppose $\underline{\mathrm{vol}}_\theta(q) > 0$. There are analytic $b_j \in \mathrm{PSH}(X,\theta)$ with $b_j \to u$ in L^1 such that the ordinary models $r_j = P_\theta[b_j]$ satisfy*

$$d_\theta(r_j,q) \longrightarrow 0, \qquad d_\theta(r_j,u) \longrightarrow \mathcal{V}^+_{\mathcal{I},\theta}(u) - \overline{\mathrm{vol}}_\theta(u). \tag{29.8}$$

Their lower volumes converge to $\underline{\mathrm{vol}}_\theta(q)$, and $r_j \to q$ in L^1 and capacity. In particular $d_\theta(r_j,u) \to 0$ if and only if $P_\theta[u] = P_\theta[u]_{\mathcal{I}}$.

Proof Fix a Hermitian form h. Bigness and Theorem 1.15.1 supply an analytic $\rho \le 0$ and a rational $\kappa > 0$ with $\theta + \mathrm{dd^c}\rho \ge \kappa h$. Choose the dyadic a_j in (15.3), enlarging their errors to positive rational numbers $\epsilon_j \downarrow 0$. Put

$$c_j = \frac{\kappa}{\kappa + \epsilon_j}, \qquad b_j = c_j a_j + (1 - c_j)\rho.$$

Then $\theta + \mathrm{dd^c} b_j \ge ((1 - c_j)\kappa - c_j\epsilon_j)h = 0$. The coefficients are rational, so Proposition 1.13.1 gives analytic singularities. Since $a_j \to u$ in L^1 and $c_j \to 1$, also $b_j \to u$ there.

Choose $A > 0$ with $\theta \le Ah$, and set

$$\widehat{\theta}_j = \theta + (1 + A/\kappa)\epsilon_j h, \qquad t_j = P_{\widehat{\theta}_j}[a_j].$$

These backgrounds decrease to θ and dominate $\theta + \epsilon_j h$. The dyadic envelope theorem therefore gives $t_j \searrow q$. Moreover

$$c_j\widehat{\theta}_j - \theta = (1 - c_j)(Ah - \theta) + c_j\epsilon_j h \ge 0.$$

The analytic-envelope identity gives $r_j \sim b_j$. Thus r_j/c_j is a normalized $\widehat{\theta}_j$-plurisubharmonic competitor more singular than a_j, since ρ is bounded above. It follows that $r_j \le c_j t_j$. Also $q \le t_j \le c_j t_j$.

For a lower bound, $q \sim_{\mathcal{I}} u \preceq_{\mathcal{I}} a_j$. Addition and scaling preserve this order by Theorem 1.16.1 and Lemma 1.16.3. Thus $c_j q + (1 - c_j)\rho \preceq_{\mathcal{I}} b_j$. This potential lies below zero, so envelope order and Proposition 15.2.3 give

$$c_j q + (1 - c_j)\rho \le P_\theta[b_j]_{\mathcal{I}} = r_j \le c_j t_j.$$

For either singularity volume, homogeneity, addition and monotonicity now yield

$$c_j^n \operatorname{vol}_\theta(q) \le \operatorname{vol}_\theta(r_j) \le \operatorname{vol}_\theta(\max(r_j, q)) \le c_j^n \operatorname{vol}_{\widehat{\theta}_j}(t_j).$$

The last comparison uses both $\theta \le c_j\widehat{\theta}_j$ and $\max(r_j, q) \le c_j t_j$. By Theorem 29.5.2, the outer terms tend to $\operatorname{vol}_\theta(q)$. This proves convergence of both volumes and the first limit in (29.8). In particular, r_j and q have a common positive lower-volume bound after discarding finitely many indices. Their L^1 and capacity convergence follow from Proposition 29.6.1 and Theorem 29.7.1.

To prove the second limit, shift u below q; the comparison is invariant under constants. Then

$$\overline{\operatorname{vol}}_\theta(r_j) \le \overline{\operatorname{vol}}_\theta(\max(r_j, u)) \le \overline{\operatorname{vol}}_\theta(\max(r_j, q)).$$

Both bounds tend to $\overline{\operatorname{vol}}_\theta(q)$. Substitute in the definition of d_θ. The last equivalence is Proposition 29.8.1. □

For a Hermitian background the simpler rescaling in Corollary 15.5.3 gives decreasing analytic models, with the same limits by Theorem 29.5.2. In a general

big background, the strict correction above may introduce additional poles. The upper-volume comparison still detects exactly the ideal-envelope excess.

Theorem 29.8.2 *Let $p = P_\theta[p]$ have positive lower volume. Then $p = P_\theta[p]_{\mathcal{I}}$ if and only if there are analytic ordinary models $p_j \in \mathrm{PSH}(X, \theta)$ with $d_\theta(p_j, p) \to 0$.*

More generally, if $p_j = P_\theta[p_j]_{\mathcal{I}}$ and $d_\theta(p_j, p) \to 0$, then $p = P_\theta[p]_{\mathcal{I}}$.

Proof If $p = P_\theta[p]_{\mathcal{I}}$, apply Theorem 29.8.1 with $u = p$. For the converse it suffices to prove the more general assertion, because analytic ordinary models are ideal envelope fixed points.

Put $q = P_\theta[p]_{\mathcal{I}}$ and $\delta = \underline{\mathrm{vol}}_\theta(p) > 0$. Choose $s_j \in (0, 1)$ tending to zero with $d_\theta(p_j, p) < \delta s_j^n$, after discarding finitely many indices. For $m_j = \max(p_j, p)$ one has

$$\underline{\mathrm{vol}}_\theta(m_j) \geq \delta, \qquad U(m_j) - U(p_j) \leq d_\theta(p_j, p).$$

By Lemma 29.2.1, there is $h_j \in \mathrm{PSH}(X, \theta)$ with $(1 - s_j)m_j + s_j h_j \leq p_j$. Set $H_j = P_\theta[h_j]_{\mathcal{I}}$. Monotonicity, addition and scaling in Proposition 15.2.2 give

$$(1 - s_j)q + s_j H_j \leq P_\theta[(1 - s_j)m_j + s_j h_j]_{\mathcal{I}} \leq P_\theta[p_j]_{\mathcal{I}} = p_j.$$

Consequently $\underline{\mathrm{vol}}_\theta(p_j) \geq (1 - s_j)^n \delta$ by addition and volume monotonicity. The models p_j, p eventually lie in $\mathcal{M}_{\delta/2}(\theta)$, so Proposition 29.6.1 gives $p_j \to p$ in L^1.

The H_j have supremum zero. Normalized compactness bounds $\int_X (-H_j)\,\mathrm{d}V$ by a constant C_θ. Since $q \leq 0$, the displayed convex inequality gives

$$\int_X (q - p_j)_+ \,\mathrm{d}V \leq s_j \int_X (-H_j)\,\mathrm{d}V \leq C_\theta s_j \longrightarrow 0.$$

Passing to the L^1 limit gives $q \leq p$ almost everywhere and hence everywhere. The opposite inequality holds by the envelope comparison, so $p = q$. □

Thus the positive-lower-volume ideal envelope models are precisely the limits of analytic ordinary models for d_θ. On each $\mathcal{M}_\delta(\theta)$ they form a closed set for this convergence and inherit Cauchy completeness from Theorem 29.6.1. The approximants constructed above have a common positive lower-volume bound, which need not equal the lower volume of the limiting model.

Problem 29.8.1 For a big nonclosed background on a manifold with bounded mass, find criteria for $\underline{\mathrm{vol}}_\theta(P_\theta[u]_{\mathcal{I}}) > 0$ in terms of restricted multiplier ideals or divisorial orders. When this lower volume vanishes, which implications between envelope equality and upper-volume equality in Proposition 29.8.1 remain valid?

29.9 Model comparison under background changes

A smooth change of representative preserves the current but changes the zero obstacle used to normalize its model. Taking the ordinary envelope again gives an exact correspondence for the volume comparison. An increase of the background changes the volumes; a common positive lower-volume bound controls this error uniformly and allows analytic approximations to have vanishing curvature loss.

Proposition 29.9.1 *Let $f \in C^\infty(X, \mathbb{R})$ and $\theta' = \theta + \mathrm{dd^c} f$. For every $\delta > 0$, the map*

$$G_f : \mathcal{M}_\delta(\theta) \longrightarrow \mathcal{M}_\delta(\theta'), \qquad G_f(p) = P_{\theta'}[p - f],$$

is a bijection with inverse $q \mapsto P_\theta[q + f]$. It preserves both singularity volumes, and

$$d_{\theta'}(G_f(p), G_f(q)) = d_\theta(p, q).$$

It takes analytic models to analytic models and ideal envelope models to ideal envelope models.

Proof The map $u \mapsto u - f$ identifies the competitor currents defining the two volumes. Also $\max(u - f, v - f) = \max(u, v) - f$, so

$$d_{\theta'}(u - f, v - f) = d_\theta(u, v).$$

By Proposition 15.3.1, $G_f(p) + f \sim p$. Thus $G_f(p)$ has the same volumes as p and, by Theorem 29.3.2, is an ordinary model. Dependence of d only on ordinary types proves the displayed identity. Taking the ordinary envelope of $G_f(p) + f \sim p$ gives $P_\theta[G_f(p) + f] = p$. The same argument with the two backgrounds interchanged proves bijectivity.

Smooth subtraction and bounded changes preserve analytic singularities. If $p = P_\theta[p]_{\mathcal{I}}$, Proposition 15.3.1 also gives

$$P_{\theta'}[p - f]_{\mathcal{I}} \sim p - f \sim G_f(p).$$

The ideal envelope is an ordinary model by Proposition 15.2.1. Ordinary models with the same ordinary type coincide, so it equals $G_f(p)$. The inverse map gives the converse. □

Fix a Hermitian form h. Choose a Hermitian $\Omega \geq \theta + 2h$ and an analytic $\rho \leq 0$ with $\theta + \mathrm{dd^c}\rho \geq \kappa h$, where $\kappa > 0$ is rational. Such a ρ exists by bigness and Theorem 1.15.1. Put $K = v_+(\Omega)$. For $\delta > 0$, set

$$b_\delta = 1 + \frac{\delta}{2^{n+1} n K}, \qquad \kappa_\delta = (1 - b_\delta^{-1})\kappa.$$

These constants will be fixed while the potentials and backgrounds vary.

Proposition 29.9.2 *Let β be a smooth real $(1,1)$-form with $\theta \le \beta \le \theta + h$. Suppose $u \in \mathrm{PSH}(X,\beta)$ and $\underline{\mathrm{vol}}_\beta(u) \ge \delta > 0$. For every smooth background γ with $\beta \le \gamma \le \beta + \varepsilon h$, $0 \le \varepsilon \le 1$, either singularity volume satisfies*

$$\mathrm{vol}_\beta(u) \le \mathrm{vol}_\gamma(u) \le (1 + \varepsilon/\kappa_\delta)^n \, \mathrm{vol}_\beta(u).$$

Proof Normalize $u \le 0$, so $u \le V_\beta$, and note that $\rho \le V_\beta$. Since $\Omega \ge \beta$ and $\underline{\mathrm{vol}}_\beta(u) \ge \delta$, Lemma 29.4.1 makes $H = Q_{b_\delta}(u, V_\beta)$ genuine. The potential

$$w = b_\delta^{-1} H + (1 - b_\delta^{-1})\rho$$

satisfies $w \le u$ and $\beta + \mathrm{dd^c} w \ge \kappa_\delta h$. The addition argument of Proposition 29.4.2, with $t = \varepsilon/\kappa_\delta$, therefore gives the upper bound. Its constants are independent of β and u. The lower bound is volume monotonicity. □

Corollary 29.9.1 *Let $u, v \in \mathrm{PSH}(X,\theta)$ have lower volume at least $\delta > 0$, and let η be a smooth background with $\theta \le \eta \le \theta + \varepsilon h$, where $0 \le \varepsilon \le 1$. Then*

$$\left| d_\eta(P_\eta[u], P_\eta[v]) - d_\theta(u,v) \right| \le 2K\left((1 + \varepsilon/\kappa_\delta)^n - 1 \right). \tag{29.9}$$

Proof The preceding proposition applies to u, v and $m = \max(u,v)$, and their upper volumes in background θ are at most K. Thus, for $x = u, v, m$, the errors

$$e_x = \overline{\mathrm{vol}}_\eta(x) - \overline{\mathrm{vol}}_\theta(x)$$

lie between zero and $K((1+\varepsilon/\kappa_\delta)^n - 1)$. The difference $d_\eta(u,v) - d_\theta(u,v)$ is $2e_m - e_u - e_v$, whose absolute value is at most twice this bound. Finally Proposition 29.6.1, in background η, identifies $d_\eta(u,v)$ with the projected comparison in (29.9). □

In particular, for $p_j, p \in \mathcal{M}_\delta(\theta)$ and $\theta \le \eta_j \le \theta + \varepsilon_j h$ with $\varepsilon_j \to 0$,

$$d_\theta(p_j, p) \longrightarrow 0 \quad \Longleftrightarrow \quad d_{\eta_j}(P_{\eta_j}[p_j], P_{\eta_j}[p]) \longrightarrow 0.$$

Monotonicity of the sequence of backgrounds is unnecessary for this estimate. The next result also allows the analytic approximants themselves to belong only to the larger backgrounds.

Theorem 29.9.1 *Let $p = P_\theta[p]$ have positive lower volume, and let η_j be smooth backgrounds with $\theta \le \eta_j \le \theta + \varepsilon_j h$, where $\varepsilon_j \ge 0$ tends to zero. Then $p = P_\theta[p]_{\mathcal{I}}$ if and only if there are analytic $a_j \in \mathrm{PSH}(X,\eta_j)$ with*

$$d_{\eta_j}(a_j, p) \longrightarrow 0.$$

The a_j may be required to be ordinary models in their respective backgrounds. Given any such analytic approximants, rational convex combinations with the fixed ρ produce analytic $b_j \in \mathrm{PSH}(X,\theta)$ whose models satisfy

$$d_\theta(P_\theta[b_j], p) \longrightarrow 0, \qquad P_\theta[b_j] \longrightarrow p \quad \text{in } L^1 \text{ and capacity.}$$

Proof Suppose first that $p = P_\theta[p]_{\mathcal{I}}$. By Theorem 29.8.1, there are analytic ordinary models r_j in background θ with $d_\theta(r_j, p) \to 0$ and a common positive lower-volume bound. Applying Corollary 29.9.1 gives $d_{\eta_j}(r_j, p) \to 0$. One can also use $a_j = P_{\eta_j}[r_j]$, since the analytic-envelope identity preserves their ordinary types in background η_j.

Conversely, normalize the given a_j below zero and put $m_j = \max(a_j, p)$ and $\delta = \underline{\mathrm{vol}}_\theta(p) > 0$. The basic comparison bound and the preceding uniform volume estimate give

$$\overline{\mathrm{vol}}_{\eta_j}(a_j), \quad \overline{\mathrm{vol}}_{\eta_j}(m_j) \longrightarrow \overline{\mathrm{vol}}_\theta(p).$$

Indeed the upper volume of p in background η_j tends to its upper volume in background θ, and both displayed differences from that volume are bounded in absolute value by $d_{\eta_j}(a_j, p)$.

Enlarge the errors to positive rational numbers tending to zero, and put

$$c_j = \frac{\kappa}{\kappa + \varepsilon_j}, \qquad b_j = c_j a_j + (1 - c_j)\rho, \qquad r_j = P_\theta[b_j].$$

As in Theorem 29.8.1, the current of b_j in background θ is positive and its singularities are analytic. Choose $A > 0$ with $\theta \le Ah$, and set

$$\widehat{\theta}_j = \theta + (1 + A/\kappa)\varepsilon_j h, \qquad M_j = P_{\widehat{\theta}_j}[m_j].$$

Then $\widehat{\theta}_j \ge \eta_j$ and $c_j\widehat{\theta}_j \ge \theta$. Since $r_j \sim b_j$, the potential r_j/c_j is a nonpositive $\widehat{\theta}_j$-plurisubharmonic competitor more singular than a_j, hence more singular than m_j. Therefore $r_j \le c_j M_j$. Also $p \le M_j \le c_j M_j$.

The lower volume of m_j in background η_j is at least δ. For large j, Proposition 29.9.2 applies with $\beta = \eta_j$, $\gamma = \widehat{\theta}_j$ and perturbation bound $(1 + A/\kappa)\varepsilon_j h$. Consequently

$$\overline{\mathrm{vol}}_{\widehat{\theta}_j}(m_j) \longrightarrow \overline{\mathrm{vol}}_\theta(p).$$

The ordinary-envelope theorem gives the same upper volume for M_j. On the other hand,

$$c_j\eta_j + (1 - c_j)(\theta - \kappa h) \le \theta.$$

Addition and background monotonicity therefore bound the upper volume of b_j below by $c_j^n\overline{\mathrm{vol}}_{\eta_j}(a_j)$. Since $r_j \sim b_j$, the two comparisons yield

$$\begin{aligned} c_j^n\overline{\mathrm{vol}}_{\eta_j}(a_j) &\le \overline{\mathrm{vol}}_\theta(r_j) \le \overline{\mathrm{vol}}_\theta(\max(r_j, p)) \\ &\le c_j^n\overline{\mathrm{vol}}_{\widehat{\theta}_j}(m_j). \end{aligned}$$

Both outer terms tend to $\overline{\mathrm{vol}}_\theta(p)$, proving $d_\theta(r_j, p) \to 0$. The analytic criterion Theorem 29.8.2 gives $p = P_\theta[p]_{\mathcal{I}}$.

For large j, Lemma 29.2.1 with $b = 2$ supplies a convex lower bound for r_j whose first term is $\frac{1}{2}\max(r_j, p)$. Volume addition gives $\underline{\mathrm{vol}}_\theta(r_j) \ge 2^{-n}\delta$. Thus

Proposition 29.6.1 and Theorem 29.7.1 give the asserted L^1 and capacity convergence of the models. □

The comparison is unchanged by bounded changes of the approximating potentials. The convergence of canonical models in the theorem therefore does not assert convergence of arbitrary analytic representatives. For two-sided background errors, the next section constructs admissible reference models in the varying backgrounds.

29.10 Two-sided background perturbations

A potential in the original background may cease to be admissible when the background decreases. Mixing it with a strict potential restores positivity, and ordinary projection gives a model in the new background. The volume comparison controls the error of this transport uniformly on a positive lower-volume sector. It also makes the limiting notion of approximation independent of the strict potential used in the correction.

To use the correction in a varying background, we first make the preceding estimate uniform for pairs of forms near θ.

Lemma 29.10.1 *For every $\delta > 0$ there are $C_\delta > 0$ and $\varepsilon_0 > 0$ such that, whenever smooth backgrounds satisfy*

$$\theta - \varepsilon h \le \beta \le \gamma \le \theta + \varepsilon h, \qquad 0 \le \varepsilon \le \varepsilon_0,$$

and $u \in \mathrm{PSH}(X, \beta)$ has lower volume at least δ, either volume satisfies

$$0 \le \mathrm{vol}_\gamma(u) - \mathrm{vol}_\beta(u) \le C_\delta \varepsilon.$$

For two such potentials u, v, one also has

$$\left| d_\gamma(P_\gamma[u], P_\gamma[v]) - d_\beta(u, v) \right| \le 2C_\delta \varepsilon.$$

Proof Put $\theta_0 = \theta - \kappa h/2$ and $h_0 = (1 + \kappa/2)h$. The fixed ρ is strict in θ_0, so this background is big. If $\varepsilon \le \min(\kappa/2, 1)$, then $\theta_0 \le \beta \le \theta_0 + h_0$ and $\gamma \le \beta + 2\varepsilon h_0$. Apply Proposition 29.9.2 with reference background θ_0 and comparison form h_0. Its strict margin and dominant Hermitian form can be fixed independently of β and γ. Their common upper mass bound turns the multiplicative estimate into the stated additive estimate for $\varepsilon \le 1/2$. Apply it to u, v and $\max(u, v)$; the same three-error calculation as in Corollary 29.9.1, followed by ordinary projection, gives the second assertion. □

Choose $A > 0$ with $\theta \le Ah$ and put $B = 2(1 + A/\kappa)$. For $p \in \mathrm{PSH}(X, \theta)$ with $p \le 0$ and a smooth background η with $\theta - \varepsilon h \le \eta \le \theta + \varepsilon h$, where $0 < \varepsilon < \min(\kappa/2, 1/B)$, set

$$c_\varepsilon = 1 - \varepsilon/\kappa, \qquad T^\rho_{\eta,\varepsilon}(p) = P_\eta[c_\varepsilon p + (1 - c_\varepsilon)\rho], \qquad \widehat{\theta}_\varepsilon = \theta + B\varepsilon h. \quad (29.10)$$

The mixed potential is η-plurisubharmonic, since its current is at least $\eta - \theta + \varepsilon h$.

Theorem 29.10.1 *For $p \in \mathcal{M}_\delta(\theta)$, the transport in (29.10) is an ordinary model in background η, of lower volume at least $c_\varepsilon^n \delta$. For either volume,*

$$c_\varepsilon^n \operatorname{vol}_\theta(p) \le \operatorname{vol}_\eta(T^\rho_{\eta,\varepsilon}(p)) \le c_\varepsilon^n \operatorname{vol}_{\widehat{\theta}_\varepsilon}(p). \tag{29.11}$$

For $p, q \in \mathcal{M}_\delta(\theta)$,

$$\begin{aligned} &\left| d_\eta(T^\rho_{\eta,\varepsilon}(p), T^\rho_{\eta,\varepsilon}(q)) - c_\varepsilon^n d_\theta(p,q) \right| \\ &\qquad \le 2c_\varepsilon^n K\left((1 + B\varepsilon/\kappa_\delta)^n - 1\right). \end{aligned} \tag{29.12}$$

For fixed p, these models tend to p in L^1 and capacity as $\varepsilon \to 0$. Two choices of strict analytic potential, rational margin and vanishing error bounds give d_η-comparison tending to zero between the transports of p in any common sequence of admissible backgrounds.

Proof Write $c = c_\varepsilon$, $s = 1 - c$, and $v = cp + s\rho$. Addition in background $\theta - s\kappa h \le \eta$ gives the lower bound in (29.11) for v. Its lower volume is positive, so Theorem 29.3.2 gives the same two volumes for $T(p) = T^\rho_{\eta,\varepsilon}(p)$.

The choices of B and $c \ge 1/2$ give

$$c\widehat{\theta}_\varepsilon - (\theta + \varepsilon h) \ge \varepsilon(cB - 1 - A/\kappa)h \ge 0.$$

Every nonpositive η-plurisubharmonic competitor $w \preceq v$ satisfies $w/c \in \mathrm{PSH}(X, \widehat{\theta}_\varepsilon)$ and $w/c \preceq p$, since $\rho \le 0$. Taking envelopes yields

$$cp + s\rho \le T(p) \le cP_{\widehat{\theta}_\varepsilon}[p]. \tag{29.13}$$

Volume monotonicity and the ordinary-envelope identity in the larger background prove (29.11).

Let $m = \max(p, q)$. The same pointwise bounds give

$$cm + s\rho \le \max(T(p), T(q)) \le cP_{\widehat{\theta}_\varepsilon}[m].$$

Consequently the upper volumes of $T(p), T(q)$ and their maximum lie between c^n times the respective upper volumes of p, q, m in backgrounds θ and $\widehat{\theta}_\varepsilon$. The lower volumes of all three original inputs are at least δ, and their upper volumes are at most K. By Proposition 29.9.2, each of the three errors is nonnegative and at most $c^n K((1 + B\varepsilon/\kappa_\delta)^n - 1)$. This proves (29.12).

For fixed p, Corollary 29.5.1 and Lemma 1.8.6 give $P_{\widehat{\theta}_\varepsilon}[p] \to p$ in L^1 and capacity as $\varepsilon \downarrow 0$. Decreasing tail suprema handle arbitrary error sequences. Quasi-continuity and integrability give $cp + s\rho \to p$ in both senses. The bounds (29.13) prove the asserted limits.

To compare two choices along a sequence η_j, denote their coefficients by $c_j^{(i)}$ and their upper backgrounds by $\theta + B_i\varepsilon_j^{(i)}h$, $i = 0, 1$. Put $c_j = \min_i c_j^{(i)}$ and $a_j = \max_i B_i\varepsilon_j^{(i)}$. Since the envelopes are nonpositive, both transported models lie

below $c_j P_{\theta+a_j h}[p]$. Moreover $\eta_j \le c_j(\theta + a_j h)$: use the previous form inequality for an index attaining the smaller coefficient. The upper volume of their maximum is therefore at most $c_j^n \overline{\mathrm{vol}}_{\theta+a_j h}(p)$, which tends to $\overline{\mathrm{vol}}_\theta(p)$. Each individual upper volume is at least $(c_j^{(i)})^n \overline{\mathrm{vol}}_\theta(p)$ and has the same limit. Their d_{η_j}-comparison tends to zero. □

For $p_j, p \in \mathcal{M}_\delta(\theta)$, (29.12) shows that $d_\theta(p_j, p) \to 0$ if and only if the transported comparisons tend to zero along backgrounds with errors tending to zero. The next criterion permits approximants chosen directly in those backgrounds.

Theorem 29.10.2 *Let $p = P_\theta[p]$ have positive lower volume. Let smooth backgrounds η_j satisfy $\theta - \varepsilon_j h \le \eta_j \le \theta + \varepsilon_j h$, where $0 < \varepsilon_j < \min(\kappa/2, 1/B)$ are rational and tend to zero. Put $p_j = T^\rho_{\eta_j,\varepsilon_j}(p)$. Then $p = P_\theta[p]_{\mathcal{I}}$ if and only if there are analytic $a_j \in \mathrm{PSH}(X, \eta_j)$ with*

$$d_{\eta_j}(a_j, p_j) \longrightarrow 0.$$

The approximants may be required to be ordinary models. This convergence condition is independent of the fixed strict analytic potential and admissible vanishing error bounds used to transport p. Every such analytic approximation admits rational corrections in the original background whose ordinary models converge to p for d_θ, in L^1 and in capacity.

Proof Suppose $p = P_\theta[p]_{\mathcal{I}}$. Choose analytic ordinary models r_j in background θ with $d_\theta(r_j, p) \to 0$ by Theorem 29.8.1. Their lower volumes have a common positive bound. The transports $a_j = T^\rho_{\eta_j,\varepsilon_j}(r_j)$ have analytic singularities: the convex coefficients are rational, and ordinary projection preserves the bounded-difference type of an analytic potential. The comparison estimate gives $d_{\eta_j}(a_j, p_j) \to 0$.

Conversely, assume these analytic approximants exist, and put $\delta = \underline{\mathrm{vol}}_\theta(p)$. For large j, the lower volume of p_j is at least $2^{-n}\delta$. The upper-volume deficit from a_j to $\max(a_j, p_j)$ tends to zero. Apply Lemma 29.2.1 with $b = 2$ in background η_j; addition gives

$$\underline{\mathrm{vol}}_{\eta_j}(a_j) \ge 2^{-2n}\delta$$

for all sufficiently large j.

Put $\beta_j = \theta + \varepsilon_j h$. Lemma 29.10.1 and projection invariance imply $d_{\beta_j}(a_j, p_j) \to 0$. Also $d_{\beta_j}(p_j, p) \to 0$. Indeed (29.13) bounds $\max(p_j, p)$ above by $c_{\varepsilon_j} P_{\widetilde{\theta}_{\varepsilon_j}}[p]$, and the form inequality in its proof also dominates β_j. The upper volume of this maximum in background β_j is squeezed to $\overline{\mathrm{vol}}_\theta(p)$, as are the upper volumes of p_j and p.

Project the three inputs to ordinary models in background β_j. Their lower volumes have a common positive bound, and a fixed dominant Hermitian form bounds their upper volumes. The three-point estimate (29.6) therefore gives $d_{\beta_j}(a_j, p) \to 0$, with constants independent of j. Apply Theorem 29.9.1 to conclude that p is an ideal envelope model and to obtain the stated corrections and convergence in the original background.

For another choice of transport, Theorem 29.10.1 gives comparison tending to zero between the two reference models in background η_j. The lower bound just proved and the same uniform three-point estimate transfer the convergence condition in both directions. This proves independence of the choice. □

29.11 A complete metric on positive-volume models

The three-point modulus controls convergence even though d_θ need not satisfy the triangle inequality. Intersecting divisors already show this on a projective surface. A complete metric can nevertheless be constructed from finite chains: lower-volume continuity controls the chain estimates locally, and the reciprocal lower volume prevents Cauchy sequences from escaping to the zero-volume boundary.

Example 29.11.1 On $\mathbf{P}^2$ with its Fubini–Study form θ normalized by $\int_{\mathbf{P}^2} \theta^2 = 1$, there are analytic ordinary models p, q, r of positive lower volume such that

$$d_\theta(p, q) > d_\theta(p, r) + d_\theta(r, q).$$

Proof Write $|Z|^2 = |Z_0|^2 + |Z_1|^2 + |Z_2|^2$ in homogeneous coordinates. Fix a rational $a \in (0, 1/2)$ and put

$$u = a \log \frac{|Z_0|^2}{|Z|^2}, \qquad v = a \log \frac{|Z_1|^2}{|Z|^2}, \qquad w = u + v.$$

The corresponding currents have residual forms $(1 - a)\theta$, $(1 - a)\theta$ and $(1 - 2a)\theta$, respectively. The background is closed, so each singularity volume equals the non-pluripolar mass. Discarding the divisors gives

$$U(u) = U(v) = (1 - a)^2, \qquad U(w) = (1 - 2a)^2.$$

The maximum $\max(u, v)$ has the ordinary type of

$$z = a \log \frac{|Z_0|^2 + |Z_1|^2}{|Z|^2}.$$

Blow up the common point $[0 : 0 : 1]$. Denote the blowup by $\pi : Y \to \mathbf{P}^2$, its exceptional divisor by E, and the pencil map $[Z_0 : Z_1]$ by $f : Y \to \mathbf{P}^1$. For the Fubini–Study form ω_1 of mass one on $\mathbf{P}^1$, the Poincaré–Lelong formula gives

$$\pi^*\theta + \mathrm{dd^c}(z \circ \pi) = a[E] + \beta, \qquad \beta = (1 - a)\pi^*\theta + a f^*\omega_1.$$

Here β is smooth and semipositive. Also

$$\int_Y (\pi^*\theta)^2 = 1, \qquad \int_Y \pi^*\theta \wedge f^*\omega_1 = 1, \qquad (f^*\omega_1)^2 = 0.$$

The mixed integral is one because every fiber of the pencil maps to a line. By Lemma 27.7.1 and change of variables off E,

$$U(\max(u, v)) = \int_Y \beta^2 = 1 - a^2.$$

Since $w \le u, v$, it follows that

$$d_\theta(u, v) = 4a - 4a^2,$$
$$d_\theta(u, w) + d_\theta(w, v) = 4a - 6a^2.$$

Set $p = P_\theta[u]$, $q = P_\theta[v]$ and $r = P_\theta[w]$. These models retain the analytic types by Proposition 15.2.3, and their lower volumes are at least $(1-2a)^2$. Projection invariance in Proposition 29.6.1 preserves the displayed comparisons, whose difference is $2a^2 > 0$. □

Return to the big background θ on the n-dimensional manifold X. Write $L(p) = \underline{\mathrm{vol}}_\theta(p)$ and let

$$\mathcal{M}_+(\theta) = \bigcup_{\delta>0} \mathcal{M}_\delta(\theta)$$

be the space of ordinary models of positive lower volume. We retain U and M from Section 29.6.

Proposition 29.11.1 *For $p, q \in \mathcal{M}_\delta(\theta)$,*

$$|L(p) - L(q)| \le nM \min\left\{1, \left(\frac{2d_\theta(p, q)}{\delta}\right)^{1/n}\right\}. \tag{29.14}$$

If $p_j, p \in \mathcal{M}_+(\theta)$ and $d_\theta(p_j, p) \to 0$, then $L(p_j) \to L(p) > 0$.

Proof Put $s = d_\theta(p, q)$ and $m = \max(p, q)$. If $0 < s < \delta/2$, take $t = (2s/\delta)^{1/n}$. The upper-volume deficit from either p or q to m is at most s, while $L(m) \ge \delta$. Thus Lemma 29.2.1 gives a potential $h \in \mathrm{PSH}(X, \theta)$ with

$$(1 - t)m + th \le p.$$

Volume addition and monotonicity imply $L(p) \ge (1 - t)^n L(m) \ge (1 - t)^n L(q)$. Interchanging p and q and using $L(p), L(q) \le M$ proves the bound. If $s \ge \delta/2$, use $|L(p) - L(q)| \le M$. If $s = 0$, separation gives $p = q$.

For the last assertion, put $s_j = d_\theta(p_j, p)$. Whenever $0 < s_j < L(p)/2$, the same stretching argument applied to $\max(p_j, p)$ gives

$$L(p_j) \ge (1 - t_j)^n L(p), \qquad t_j = \left(\frac{2s_j}{L(p)}\right)^{1/n}.$$

If $s_j = 0$, then $p_j = p$. Hence $L(p_j) \ge L(p)/2$ for all sufficiently large j. Apply (29.14) with this common positive bound. □

Theorem 29.11.1 *There is a complete metric $\mathfrak{d}_\theta$ on $\mathcal{M}_+(\theta)$ such that*

$$\mathfrak{d}_\theta(p_j, p) \longrightarrow 0 \quad \Longleftrightarrow \quad d_\theta(p_j, p) \longrightarrow 0. \tag{29.15}$$

A sequence is $\mathfrak{d}_\theta$-Cauchy if and only if it is d_θ-Cauchy and the sequence $1/L(p_j)$ is Cauchy in $\mathbf{R}$. On each $\mathcal{M}_\delta(\theta)$, the two comparisons are uniformly equivalent, and the restriction of $\mathfrak{d}_\theta$ is complete.

Proof Set $C = 1 + M$ and $m(p) = L(p)/C$, so $0 < m(p) < 1$. Put $a_1 = 1$, $a_{k+1} = na_k + 1$, and choose $0 < \varepsilon_1 < 1$. Recursively choose $\varepsilon_{k+1} > 0$ so small that, with $r_k = 4^{a_{k+1}}\varepsilon_{k+1}$,

$$3\varepsilon_{k+1} < \varepsilon_k, \qquad 3r_k + 4nM\left(\frac{2r_k}{C}\right)^{1/n} < \varepsilon_k.$$

Let $\mathcal{U}_0 = \mathcal{M}_+(\theta)^2$ and, for $k \geq 1$, set

$$\mathcal{U}_k = \left\{(p,q) : \left|\frac{1}{m(p)} - \frac{1}{m(q)}\right| < \varepsilon_k, \quad d_\theta(p,q) < \varepsilon_k \min\{m(p), m(q)\}^{a_k}\right\}.$$

These symmetric sets contain the diagonal, decrease with k, and have diagonal intersection. We claim that

$$\mathcal{U}_{k+1} \circ \mathcal{U}_{k+1} \circ \mathcal{U}_{k+1} \subset \mathcal{U}_k \qquad (k \geq 0), \tag{29.16}$$

where composition means joining successive pairs. For $k = 0$ this is immediate. For $k \geq 1$, suppose $(p_{i-1}, p_i) \in \mathcal{U}_{k+1}$ for $i = 1, 2, 3$, and put $m_* = \min_{0 \leq i \leq 3} m(p_i)$. The reciprocal differences along this chain have total less than $3\varepsilon_{k+1} < 1$. In particular $m(p_i) \leq 4m_*$ for every i. Each successive comparison is therefore less than

$$e = r_k m_*^{a_{k+1}}.$$

All four models have lower volume at least Cm_*. Applying (29.6) twice, with the last edge providing its error term, yields

$$\begin{aligned} d_\theta(p_0, p_3) &\leq 3e + 4nM\left(\frac{2e}{Cm_*}\right)^{1/n} \\ &\leq \left(3r_k + 4nM\left(\frac{2r_k}{C}\right)^{1/n}\right) m_*^{a_k} \\ &< \varepsilon_k \min\{m(p_0), m(p_3)\}^{a_k}. \end{aligned}$$

Here $(a_{k+1} - 1)/n = a_k$, $a_{k+1} \geq a_k$ and $m_* < 1$. The endpoint reciprocal difference is less than $3\varepsilon_{k+1} < \varepsilon_k$, proving (29.16).

Define

$$\mathfrak{d}_\theta(p,q) = \inf \sum_{i=1}^{N} 2^{-k_i}, \tag{29.17}$$

where the infimum runs over finite chains $p = p_0, p_1, \ldots, p_N = q$ and integers $k_i \geq 0$ with $(p_{i-1}, p_i) \in \mathcal{U}_{k_i}$. A single edge in $\mathcal{U}_0$ gives $\mathfrak{d}_\theta \leq 1$. Reversing and concatenating chains gives symmetry and the triangle inequality, while diagonal edges with arbitrarily large indices give $\mathfrak{d}_\theta(p, p) = 0$.

We verify the chain bound

$$\{\mathfrak{d}_\theta < 2^{-(k+1)}\} \subset \mathcal{U}_k \subset \{\mathfrak{d}_\theta \leq 2^{-k}\} \qquad (k \geq 1). \tag{29.18}$$

The second inclusion follows from a single edge. For the first, we prove by induction on the number of edges that every chain of cost less than 2^{-k} has endpoints in $\mathcal{U}_{k-1}$, simultaneously for all $k \geq 1$. The one-edge case follows from nesting. For a chain of cost $A < 2^{-k}$, choose $l \geq k$ with

$$2^{-(l+1)} \leq A < 2^{-l}.$$

Split the chain at the first edge at which the cumulative cost reaches $2^{-(l+1)}$. Both the preceding and the following subchains have cost less than $2^{-(l+1)}$. By induction their endpoint pairs lie in $\mathcal{U}_l$, with an empty subchain interpreted as a diagonal pair. Every edge cost is a power of two smaller than 2^{-l}, so the middle edge lies in $\mathcal{U}_{l+1} \subset \mathcal{U}_l$. The triple-composition bound places the full endpoint pair in $\mathcal{U}_{l-1} \subset \mathcal{U}_{k-1}$. This proves the induction and (29.18). Since the $\mathcal{U}_k$ have diagonal intersection, $\mathfrak{d}_\theta$ separates points.

The inclusions (29.18) identify convergence for $\mathfrak{d}_\theta$ with eventual membership in every $\mathcal{U}_k$. This implies d_θ convergence. Conversely, if $d_\theta(p_j, p) \to 0$, then Proposition 29.11.1 gives $L(p_j) \to L(p) > 0$. For each fixed k, both inequalities defining $\mathcal{U}_k$ hold eventually. This proves (29.15).

The same inclusions show that a $\mathfrak{d}_\theta$-Cauchy sequence is d_θ-Cauchy and has Cauchy reciprocal lower volumes. Those reciprocals are bounded, so its tail lies in some $\mathcal{M}_\delta(\theta)$. By Theorem 29.6.1, it converges there for d_θ, hence for $\mathfrak{d}_\theta$. This proves completeness. Conversely, a d_θ-Cauchy sequence with Cauchy reciprocal lower volumes has a common positive lower-volume bound on a tail. Both inequalities defining each $\mathcal{U}_k$ then hold uniformly on that tail, proving the stated Cauchy criterion.

On $\mathcal{M}_\delta(\theta)$, (29.14) controls the reciprocal differences uniformly in terms of d_θ, while $m(p) \geq \delta/C$ controls the second threshold in $\mathcal{U}_k$. Thus the comparisons are uniformly equivalent there. Lower-volume continuity makes $\mathcal{M}_\delta(\theta)$ closed for $\mathfrak{d}_\theta$, so its restriction is complete. □

Corollary 29.11.1 *The ideal envelope models form a closed complete subspace of* $(\mathcal{M}_+(\theta), \mathfrak{d}_\theta)$. *Positive-volume analytic ordinary models are dense in this subspace. Equivalently, this subspace is the metric completion of the positive-volume analytic ordinary models with the restricted metric.*

Proof By Theorem 29.8.2, a positive-lower-volume ordinary model limit of ideal models for d_θ is again an ideal model. The same theorem and Theorem 29.8.1 give analytic density. Apply (29.15) and completeness of the ambient metric space. □

29.12 Model potentials on curves

Let Y be a compact connected Riemann surface, let $\theta > 0$ be smooth, and put $A = \int_Y \theta$.

The curvature carried by the pole set determines the ordinary envelope on a curve. Its atomic part determines the ideal envelope. Comparing these two measures describes both the zero-volume boundary and the failure of analytic approximation for diffuse polar singularities. Total variation retains this singularity data while controlling convergence of the model potentials in capacity.

Write $T_u = \theta + \mathrm{dd^c} u$ and $\mu_u = \mathbf{1}_{\{u=-\infty\}} T_u$ for $u \in \mathrm{PSH}(Y, \theta)$. By Section 1.6.1, $\mu_u = T_u^{\mathrm{pol}}$. The two singularity volumes coincide with

$$U(u) = \int_Y \mathrm{MA}_\theta(u) = A - \mu_u(Y).$$

Use a fixed nonpositive Green kernel G for $\eta = \theta/A$, as in Section 18.3, and write $\|\cdot\|_{\mathrm{TV}}$ for total variation.

Lemma 29.12.1 *For a positive measure μ carried by a polar set with $\mu(Y) \le A$, put*

$$w_\mu(x) = \int_Y G(x, y)\, \mathrm{d}\mu(y), \qquad p_\mu = P_\theta[w_\mu].$$

Then $p_\mu \sim w_\mu$, $\mu_{p_\mu} = \mu$ and $U(p_\mu) = A - \mu(Y)$. Every $u \in \mathrm{PSH}(Y, \theta)$ satisfies

$$P_\theta[u] = p_{\mu_u}. \tag{29.19}$$

Consequently $\mu \mapsto p_\mu$ is a bijection from these polar measures to all normalized ordinary models on Y.

Proof The Green-kernel bounds give $w_\mu \in L^1(Y)$ and

$$T_{w_\mu} = \mu + (A - \mu(Y))\eta.$$

In particular w_μ is θ-plurisubharmonic and its polar curvature is μ. For any $v \in \mathrm{PSH}(Y, \theta)$,

$$v \preceq w_\mu \quad \Longleftrightarrow \quad T_v \ge \mu. \tag{29.20}$$

The forward implication follows from Corollary 1.6.1. Conversely, if $T_v \ge \mu$, the distribution $v - w_\mu$ has Hessian bounded below by $-(A - \mu(Y))\eta$. Its quasi-subharmonic representative is bounded above on Y. Almost-everywhere order between quasi-plurisubharmonic functions then gives $v \preceq w_\mu$.

Thus p_μ is the upper envelope of all $v \le 0$ with $T_v \ge \mu$. This family is closed under maxima by (1.8). Choquet selection and L^1 convergence of increasing maxima retain the current inequality $T_{p_\mu} \ge \mu$. Also $p_\mu \ge w_\mu - \sup_Y w_\mu$. Hence $b = p_\mu - w_\mu$ has a quasi-subharmonic representative bounded below, and compactness bounds it

above. Therefore $p_\mu \sim w_\mu$. The positive measure

$$T_{p_\mu} - \mu = (A - \mu(Y))\eta + \mathrm{dd^c} b$$

charges no polar set because b is bounded. This proves the assertions about μ_{p_μ} and $U(p_\mu)$, and ordinary equivalence gives $P_\theta[p_\mu] = p_\mu$.

For arbitrary u, take $\mu = \mu_u$. Then $T_u \geq \mu$, so (29.20) gives $u \preceq p_\mu$, and $U(u) = U(p_\mu)$. If this mass is positive, Corollary 29.3.2 gives (29.19). If it is zero, $\mu(Y) = A$ and $T_u = \mu = T_{w_\mu}$. Their difference is harmonic on Y and hence constant, giving the same conclusion. Distinct polar measures give distinct models by the already proved identity $\mu_{p_\mu} = \mu$. □

Theorem 29.12.1 *For every $u, v \in \mathrm{PSH}(Y, \theta)$,*

$$d_\theta(u, v) = \|\mu_u - \mu_v\|_{\mathrm{TV}}. \tag{29.21}$$

In particular d_θ is a metric on all normalized ordinary models on Y. For the ideal models q_D, q_E of Lemma 18.5.1,

$$d_\theta(q_D, q_E) = \|D - E\|_{\mathrm{TV}}. \tag{29.22}$$

Proof By (1.8), the polar curvature of $\max(u, v)$ is $\mu_u \wedge \mu_v$. The one-dimensional volume formula therefore gives

$$\begin{aligned} d_\theta(u, v) &= 2U(\max(u, v)) - U(u) - U(v) \\ &= \mu_u(Y) + \mu_v(Y) - 2(\mu_u \wedge \mu_v)(Y) \\ &= \|\mu_u - \mu_v\|_{\mathrm{TV}}. \end{aligned}$$

The classification in Lemma 29.12.1 proves separation on normalized ordinary models, and total variation supplies the triangle inequality. Ideal models have precisely the atomic polar measures, giving (29.22). □

Corollary 29.12.1 *All normalized ordinary models on Y form a complete geodesic metric space for d_θ. The ideal models form a closed geodesically convex subspace, which is the metric completion of the positive-mass analytic ordinary models.*

Proof The positive polar measures of mass at most A are closed in total variation. A total-variation limit retains positivity and the mass bound, and is carried by the countable union of polar carriers of the sequence. Completeness follows from that of finite signed measures. For polar measures μ, ν, the measures $\mu_t = (1 - t)\mu + t\nu$ remain in this set and satisfy

$$\|\mu_t - \mu_s\|_{\mathrm{TV}} = |t - s|\, \|\mu - \nu\|_{\mathrm{TV}}.$$

Their models give a constant-speed geodesic.

The atomic measures form a closed convex subset: a total-variation limit is carried by the countable union of the sets of atoms of the sequence. Finite restriction and

rational approximation of the coefficients from below, with an arbitrarily small further decrease if needed, approximate every such measure by finite atomic measures of mass strictly less than A. Their models have analytic singularities by Lemma 18.5.1. Apply (29.22). □

The positive-mass part itself is incomplete for d_θ. For a fixed $x \in Y$, the models of $tA\delta_x$, $t \uparrow 1$, give a Cauchy family with a zero-mass limit. In contrast, the complete metric of Theorem 29.11.1 makes this family non-Cauchy because its reciprocal lower volumes diverge. The two metrics induce the same topology on positive-mass models, but have different Cauchy sequences near the boundary.

The metric also controls convergence of the potentials at the zero-mass boundary. There the polar part exhausts the limiting curvature, so its total variation controls the full current.

Theorem 29.12.2 *Let p_j, p be normalized ordinary models on Y. If $d_\theta(p_j, p) \to 0$, then $p_j \to p$ in L^1 and in capacity. If $U(p) = 0$, then in addition*

$$\|T_{p_j} - T_p\|_{\mathrm{TV}} \le 2d_\theta(p_j, p) \longrightarrow 0. \tag{29.23}$$

Proof Write $\mu_j = \mu_{p_j}$ and $\mu = \mu_p$. By (29.21), $\mu_j \to \mu$ in total variation. Thus $U(p_j) = A - \mu_j(Y) \to U(p)$. If $U(p) > 0$, the models lie in a common $\mathcal{M}_\delta(\theta)$ for all large j. Apply Proposition 29.6.1 and Theorem 29.7.1.

Suppose $U(p) = 0$. Then $T_p = \mu$ and $R_j = T_{p_j} - \mu_j$ is positive of mass $A - \mu_j(Y)$. Consequently

$$\|T_{p_j} - T_p\|_{\mathrm{TV}} \le \|\mu_j - \mu\|_{\mathrm{TV}} + A - \mu_j(Y) \le 2\|\mu_j - \mu\|_{\mathrm{TV}},$$

which proves (29.23). By normalized compactness, each subsequence has an L^1-convergent further subsequence with a supremum-zero limit q. The current identity gives $\mathrm{dd^c}(q - p) = 0$. Harmonic uniqueness and normalization give $q = p$, proving L^1 convergence of the full sequence. Apply Lemma 1.6.2 on finitely many nested coordinate discs. Both terms on its right tend to zero, and the local-global capacity comparison in Section 1.8 gives the result. □

For normalized potentials, $T_{p_j} \rightharpoonup T_p$ is equivalent to $p_j \to p$ in L^1, by Theorem 5.1.1 and normalized compactness. It need not imply capacity convergence even when the sequence and limit are ideal models. Scaling the area-one construction in Theorem 15.4.1 gives zero-mass ideal models converging to zero in L^1. Capacity convergence would contradict Theorem 7.3.1, since their damped measures vanish and the limiting measure is θ.

Capacity convergence in turn need not retain total variation of the polar part. Moving a single pole gives an example within the zero-mass boundary itself.

Example 29.12.1 On $\mathbb{P}^1$ with the Fubini–Study form θ of area one, there are zero-mass ideal models p_j, p with $p_j \to p$ in capacity but $d_\theta(p_j, p) = 2$ for every j.

Proof In the affine coordinate z, for $a \in \mathbb{C}$ put

$$p_a(z) = \log \frac{|z-a|^2}{(1+|z|^2)(1+|a|^2)}.$$

This extends to $\mathbb{P}^1$, has supremum zero, and satisfies $T_{p_a} = \delta_a$. Hence it is the ideal model for δ_a by Lemma 18.5.1. Choose $a_j \neq 0$ tending to zero and set $p_j = p_{a_j}$, $p = p_0$. On the complement of any fixed disc about zero the convergence is uniform, including at infinity. The capacities of these discs tend to zero by (1.14) applied to p_0. Therefore $p_j \to p$ in capacity. Since $\|\delta_{a_j} - \delta_0\|_{\mathrm{TV}} = 2$, (29.21) gives the asserted distance. □

On the general curve Y, write $\mu_u = D_u + \sigma_u$, where $D_u = \sum_{x \in Y} \nu(u,x)\delta_x$ is the atomic part and σ_u has no atoms. Set $\tau(u) = \sigma_u(Y) = A - U(u) - D_u(Y)$, and let $\mathcal{A}_+(Y,\theta)$ denote the positive-mass analytic ordinary models.

Proposition 29.12.1 *For every $u \in \mathrm{PSH}(Y,\theta)$,*

$$\inf_{a \in \mathcal{A}_+(Y,\theta)} d_\theta(u,a) = d_\theta(u, P_\theta[u]_{\mathcal{I}}) = \tau(u). \tag{29.24}$$

On normalized ordinary models, $p \mapsto P_\theta[p]_{\mathcal{I}}$ is a 1-Lipschitz retraction onto the ideal models, and gives the unique nearest ideal model. There are $a_j \in \mathcal{A}_+(Y,\theta)$ with $d_\theta(u,a_j) \to 0$ if and only if $P_\theta[u] = P_\theta[u]_{\mathcal{I}}$.

Proof The ideal envelope is q_{D_u}. For any atomic measure E of mass at most A, the nonatomic measure σ_u and the signed atomic measure $D_u - E$ are mutually singular. Thus (29.21) gives

$$d_\theta(u, q_E) = \tau(u) + \|D_u - E\|_{\mathrm{TV}}. \tag{29.25}$$

The unique minimum among ideal models occurs at $E = D_u$. Density of the positive-mass analytic models in the ideal subspace gives (29.24). Taking the atomic part contracts total variation, proving the Lipschitz assertion. Finally, Lemma 29.12.1 identifies $P_\theta[u] = q_{D_u}$ with $\mu_u = D_u$, equivalently $\tau(u) = 0$. □

Example 29.12.2 There is an ordinary model p on Y with zero mass and trivial multiplier ideals at every positive weight. It satisfies

$$P_\theta[p]_{\mathcal{I}} = 0, \qquad \inf_{a \in \mathcal{A}_+(Y,\theta)} d_\theta(p,a) = A.$$

Proof Choose a compact real interval inside a coordinate disc. Construct a binary Cantor set K with 2^k pairwise disjoint level-k intervals of length $r_k = \exp(-4^{k+1})$, each containing its two level-$(k+1)$ children. Let σ be the probability measure assigning mass 2^{-k} to each level-k interval. It has no atoms. Put $u(x) = A \int_Y G(x,y)\,\mathrm{d}\sigma(y)$, so $T_u = A\sigma$. On the chosen compact coordinate disc, $G(x,y) \le \log|x-y|^2 + C$ for a fixed C. For $x \in K$, integration over its level-k interval, using $G \le 0$ on the complement, gives

$$u(x) \le A2^{-k}(2\log r_k + C) \longrightarrow -\infty.$$

Thus $u = -\infty$ on K, while it is smooth off K by its current equation. Hence K is polar, $\mu_u = A\sigma$ and $U(u) = 0$. By Lemma 29.12.1, $p = P_\theta[u]$ has the same polar curvature. Since this already has mass A, $T_p = T_u$ and $p = u - \sup_Y u$. Its Lelong numbers vanish, so all its multiplier ideals are trivial and its ideal envelope is zero. Finally $\tau(p) = A$, which gives the distance by (29.24). □

29.12.1 The ordinary-type topology on a curve

The topology of Section 15.6 also uses the ordinary envelope, but it asks for total variation only of the atomic curvature. On ideal models that is the whole polar measure, so it recovers the complete metric topology above. On arbitrary ordinary models, diffuse polar mass can concentrate without being seen by this variation coordinate.

Retain Y, θ, A and the decomposition $\mu_u = D_u + \sigma_u$ above. Use $\eta = \theta/A$ for L^1 norms. For ordinary classes put

$$d_\theta^{\mathrm{ot}}([u],[v]) = \|P_\theta[u] - P_\theta[v]\|_{L^1(\eta)} + \|D_u - D_v\|_{\mathrm{TV}}. \tag{29.26}$$

Theorem 29.12.3 *The finite pseudometric d_θ^{ot} induces $\tau_{\theta,0}$ on the whole ordinary-type space. For $0 \le s \le A$, the canonical ideal classes*

$$C_s = \{[q_D] : D \text{ positive atomic},\ D(Y) \le s\}$$

form a complete metric space both for d_θ^{ot} and for $\|D - E\|_{\mathrm{TV}}$. These metrics induce the same topology. The space C_s is not totally bounded if $s > 0$, and $C_0 = \{[0]\}$.

Proof A smooth compact modification of a smooth curve is an isomorphism. Its divisorial distance is exactly atomic total variation, since the divisor weight in dimension one is one. Thus the two summands of (29.26) are defining coordinates of $\tau_{\theta,0}$.

Conversely, suppose a net converges for d_θ^{ot}. The ideal envelope of u is q_{D_u} by Lemma 18.5.1. Atomic total variation implies L^1 convergence of these envelopes by Theorem 29.12.2. This theorem also gives continuity for nets: its domain and target are metric spaces, so its sequential assertion is equivalent to continuity. It remains to check the ideal relations.

At a point p with local coordinate z, the curve ideal formula is

$$\mathcal{I}(tu)_p = (z^{\lfloor tD_u(\{p\})\rfloor}).$$

This follows from Theorem 1.16.1 and Lemma 1.16.1: on a smooth curve the point Lelong number determines the ideal, and the logarithmic germ with that number has the displayed ideal. Write $D_j = D_{u_j}$, $D = D_u$ and fix $t > 0$, $a > 1$. The set $F = \{p : D(\{p\}) \ge 1/(2t)\}$ is finite. At its points, $aD_j(\{p\}) \ge D(\{p\})$ and

$aD(\{p\}) \geq D_j(\{p\})$ eventually. Outside F, the bound

$$\sup_p |D_j(\{p\}) - D(\{p\})| \leq \|D_j - D\|_{\mathrm{TV}} \longrightarrow 0$$

makes both $\mathcal{I}(tu)_p$ and $\mathcal{I}(tu_j)_p$ unit ideals for large j. The two desired inclusions therefore hold everywhere. This proves the asserted topology.

The map $D \mapsto q_D$ is injective. Positive atomic measures of mass at most s are closed in total variation: a limit is positive, has this mass bound, and is supported on the countable union of the supports of a converging sequence. Completeness of finite measures gives completeness for total variation. A d_θ^{ot}-Cauchy sequence is total-variation Cauchy; the resulting limit remains atomic, and Theorem 29.12.2 supplies convergence of the envelope coordinate. This proves its completeness as well as equality of the two topologies.

If $s > 0$, fix $0 < b \leq s$ and infinitely many distinct points p_j. The measures $b\delta_{p_j}$ have pairwise total-variation distance $2b$, proving failure of total boundedness for both metrics. At $s = 0$ only the zero measure remains. □

By Lemma 18.5.1, $C_{A-\delta}$ is exactly the canonical ideal locus of mass at least δ, for $0 \leq \delta \leq A$. The entire canonical locus C_A is complete, including zero mass. Its positive-mass part is incomplete: $D_j = A(1 - 1/j)\delta_p$ is Cauchy and converges only to the zero-mass boundary $A\delta_p$.

The same statement on actual ordinary classes is a pseudometric assertion. More precisely, put

$$\mathcal{G}_{\theta,\delta} = \left\{ [u] : P_\theta[u] = P_\theta[u]_{\mathcal{I}}, \quad \int_Y \mathrm{MA}_\theta(P_\theta[u]_{\mathcal{I}}) \geq \delta \right\}.$$

For $0 \leq \delta \leq A$ this is complete for d_θ^{ot}, since every point is at distance zero from $[q_{D_u}] \in C_{A-\delta}$. Limits need not be unique as ordinary classes. It is not totally bounded when $\delta < A$; at $\delta = A$ the pseudometric vanishes identically, even though Proposition 15.6.2 supplies distinct classes. Its strictly positive-mass part without a uniform lower bound is incomplete by the same one-atom sequence.

29.12.2 Concentration of diffuse polar curvature

Atomic variation is indispensable in Theorem 29.12.3. On $\mathbb{P}^1$ with its area-one Fubini–Study form, fix a rational $0 < \tau < 1$ and take the atomic probabilities ν_j and models r_j from Proposition 15.4.2, choosing the rational approximations so that the largest atom tends to zero. The ordinary and ideal envelopes both tend to zero in L^1, and for each $t > 0$, $a > 1$, all the controlled ideal relations with zero eventually hold. Their currents converge weakly to θ. Nevertheless,

$$d_\theta^{\mathrm{ot}}([r_j], [0]) \geq \|D_{r_j}\|_{\mathrm{TV}} = 1 - \tau > 0.$$

Thus these weaker conditions do not imply the ordinary-type topology even at a fixed positive canonical mass.

The converse difficulty concerns diffuse polar mass: the pseudometric controls its envelope in L^1, but not the total variation of the diffuse measure itself.

Proposition 29.12.2 *For the probability Fubini–Study form θ on $\mathbb{P}^1$, the whole space $(\mathfrak{S}(\mathbb{P}^1, \theta), d_\theta^{\mathrm{ot}})$ is incomplete. The failure occurs for ordinary models of one fixed positive non-pluripolar mass, all with trivial multiplier ideals.*

Proof Choose a nonatomic polar probability measure σ with compact support in an affine chart, using the Cantor construction in Example 29.12.2. Fix $0 < c < 1$ and a point p in that chart. Translate and shrink the support towards p, obtaining nonatomic polar probabilities σ_j supported in discs of radii tending to zero about p. Let $p_j = p_{c\sigma_j}$ be the ordinary models of Lemma 29.12.1. Their polar measures are $c\sigma_j$, their masses are $1 - c$, and $D_{p_j} = 0$. Their ideal envelopes are zero.

We claim $p_j \to q_{c\delta_p}$ in L^1. Write $q_y = q_{c\delta_y}$. Rotations preserving θ can be chosen continuously near p to send p to y. They transport q_p to q_y, and continuity of translations in L^1 gives $q_y \to q_p$ there as $y \to p$. Set

$$w_j = \int_{\mathbb{P}^1} q_y \, \mathrm{d}\sigma_j(y).$$

The common normalized L^1 bound makes this integral well-defined. It is nonpositive, its curvature dominates $c\sigma_j$, and $w_j \to q_p$ in L^1. The envelope characterization (29.20) therefore gives $w_j \le p_j$.

Normalized compactness supplies an L^1 cluster limit v along every subsequence. The inequalities $\theta + \mathrm{dd^c} p_j \ge c\sigma_j$ pass to $\theta + \mathrm{dd^c} v \ge c\delta_p$. Thus $v \le q_p$ by the ideal-envelope characterization. The lower bound from w_j gives $v \ge q_p$ almost everywhere, hence everywhere. This identifies every cluster limit and proves the claim.

Since $P_\theta[p_j] = p_j$ and $D_{p_j} = 0$, the classes $[p_j]$ are d_θ^{ot}-Cauchy. A limit $[v]$ would have $D_v = 0$ and $P_\theta[v] = q_p$. This is impossible: ordinary envelopes preserve all multiplier ideals and hence all atoms, whereas q_p has atom c at p. The sequence therefore has no limit in the ordinary-type space. □

The two topologies on ordinary models are also different. Choose the Cantor support of σ inside a positive real interval in the affine chart, away from zero. The rotations $R_j(z) = \mathrm{e}^{\mathrm{i}\varepsilon_j} z$, with $0 < \varepsilon_j \to 0$, move this support off itself. Put $p = p_{c\sigma}$ and $\widetilde{p}_j = p \circ R_j^{-1}$. Rotation invariance of the envelope gives ordinary models with polar measures $c(R_j)_*\sigma$ and no atoms. Continuity of rotations in L^1 yields $d_\theta^{\mathrm{ot}}([\widetilde{p}_j], [p]) \to 0$, whereas disjoint polar supports give $d_\theta(\widetilde{p}_j, p) = 2c$ by Theorem 29.12.1. Conversely, polar total variation controls both terms of (29.26), by Theorem 29.12.2. Thus $\tau_{\theta,0}$ is strictly weaker on ordinary models, although its restriction to ideal models agrees with their complete metric topology. The failure of fixed-zero-obstacle gauge continuity in Proposition 15.6.4 gives a further distinction from the polar metric, whose background invariance is proved in Section 29.13.

29.12.3 Convergence for all obstacles

Return to the compact curve Y and its positive form θ of area A. The topology $\tau_\theta^{\mathrm{all}}$ retains every smooth-obstacle envelope. On a curve these envelopes are still determined by the polar measure, whereas their ideal counterparts use only its atomic part. This gives a countable pseudometric description and lets us compare the strengthened topology with polar total variation.

Choose the countable dense family in Proposition 15.6.5 with $F_1 = 0$. Write $d_{O,F}$ and $d_{I,F}$ for the two envelope distances defining $\tau_{\theta,F}$, and put

$$d_\theta^{\mathrm{all}}([u],[v]) = \|D_u - D_v\|_{\mathrm{TV}} + \sum_{k=1}^{\infty} 2^{-k}\big(\min\{1, d_{O,F_k}([u],[v])\} + \min\{1, d_{I,F_k}([u],[v])\}\big). \tag{29.27}$$

Proposition 29.12.3 *The pseudometric d_θ^{all} induces the uniformity of $\tau_\theta^{\mathrm{all}}$. Its zero-distance classes are exactly the classes with the same polar curvature. The closure of analytic classes is $\{[u] : \mu_u = D_u\} = \mathcal{G}_\theta$. On each canonical ideal locus C_s, $0 \le s \le A$, it is a complete metric inducing the total-variation topology. On ordinary models, polar total variation implies convergence for d_θ^{all}.*

Proof For any smooth F, the competitor translation gives

$$P_{\theta,F}[u] - F = P_{\theta+\mathrm{dd^c}F}[u - F],$$

and likewise for the ideal envelope. The background on the right is a smooth real form of area A, though it may not be positive. We identify these envelopes with the models of μ_u and D_u, respectively, in that background. Set $\mu = \mu_u$ and use its Green potential w_μ. In the new background let $r = P_{\theta+\mathrm{dd^c}F}[w_\mu - F]$. The obstacle bounds give $r \sim w_\mu - F$, so r is an ordinary model. Moreover $u - F \preceq r$ and both have mass $A - \mu(Y)$. If this mass is positive, Corollary 29.3.2 identifies their envelopes. If it is zero, their curvature measures both equal μ, and harmonic uniqueness makes them differ by a constant. The ideal assertion follows from ideal projection and the atomic classification. It follows also that the envelope below F is the upper envelope of all $v \le F$ satisfying $\theta + \mathrm{dd^c}v \ge \mu$.

Polar total variation gives L^1 convergence of these models for every fixed F, by Proposition 29.13.1. The same assertion for atomic measures supplies continuity of every ideal-envelope coordinate in the atomic total-variation norm. The controlled ideal relations are uniformly implied by atomic total variation: for fixed $t > 0$, $a > 1$, choosing $\|D - E\|_{\mathrm{TV}} < (a-1)/(at)$ guarantees both inclusions in (15.5) by the local floor formula. Indeed, if $tE(\{p\}) \ge 1$, this bound gives $aD(\{p\}) > E(\{p\})$; otherwise the target ideal is the unit ideal. Interchange D, E for the other inclusion. Each divisorial coordinate is atomic total variation. The two envelope terms in the summable series, together with the atomic term, therefore generate exactly the all-obstacle uniformity: finitely many coordinates control the sum up to its uniformly small tail, and the sum controls each fixed coordinate.

The zero-obstacle classification shows that zero distance forces equal polar measures; the classification at every obstacle proves the converse. The same classification identifies equality of the ordinary and ideal envelopes at every F with $\mu_u = D_u$. Apply Theorem 15.6.2 to obtain the analytic closure.

For a d_θ^{all}-Cauchy sequence in C_s, its atomic measures have a total-variation limit of mass at most s, which is again atomic. The model convergence above gives convergence of every obstacle coordinate; the summable tail then gives convergence for d_θ^{all}. The converse implication to atomic total variation follows from the first summand. This proves completeness and equality of topologies on C_s. Polar total variation controls the atomic part and all the ordinary-envelope coordinates, proving the last assertion. □

Proposition 29.12.4 *For the probability Fubini–Study form on $\mathbb{P}^1$, d_θ^{all} is incomplete on the whole ordinary-type space, even on ordinary models of one fixed positive mass with trivial ideals. Its topology on ordinary models is strictly weaker than polar total variation and strictly stronger than $\tau_{\theta,0}$.*

Proof Use the diffuse polar probabilities σ_j concentrating at p and the models $p_j = p_{c\sigma_j}$ from Proposition 29.12.2, with $0 < c < 1$. Fix a smooth obstacle F and put $q_{y,F} = P_{\theta,F}[q_{c\delta_y}]$. Rotations preserving θ and sending p to y can be chosen continuously near p. They transport the obstacle to its composition with the rotation. The uniform obstacle bound and continuity of rotations in L^1 consequently give $q_{y,F} \to q_{p,F}$ in L^1 as $y \to p$.

The uniform L^1 bound near p makes the average $w_{j,F} = \int q_{y,F}\, \mathrm{d}\sigma_j(y)$ well-defined. It lies below F, its curvature dominates $c\sigma_j$, and it converges to $q_{p,F}$ in L^1. The obstacle characterization in the preceding proof gives $w_{j,F} \le P_{\theta,F}[p_j] \le F$. These bounds and normalized compactness give L^1 precompactness. Any cluster limit of the envelopes has curvature at least $c\delta_p$, so is at most $q_{p,F}$. The lower bound gives the reverse inequality. Hence $P_{\theta,F}[p_j] \to q_{p,F}$ for every F. Their ideal envelopes are all $P_\theta(F)$, and their atomic parts vanish. Thus $[p_j]$ is d_θ^{all}-Cauchy. A limit would also be a d_θ^{ot} limit, which the cited proposition excludes.

For strict comparison with polar variation, use the rotated diffuse model $\widetilde{p}_j = p \circ R_j^{-1}$ from the end of that proof. For every smooth F,

$$P_{\theta,F}[\widetilde{p}_j] = P_{\theta,F\circ R_j}[p] \circ R_j^{-1}.$$

The obstacle bound and L^1 continuity of rotations give convergence to $P_{\theta,F}[p]$. All atomic parts vanish, so $d_\theta^{\mathrm{all}}([\widetilde{p}_j],[p]) \to 0$, whereas their polar total-variation distance remains $2c$. The preceding proposition supplies continuity in the other direction.

Finally, the sequence in Proposition 15.6.4 converges for the zero obstacle and fails to converge for a smooth obstacle. Replacing its members by their ordinary models preserves every obstacle envelope by the curve classification, so the strict comparison already holds on ordinary models. □

29.13 Background changes on curves

Polar curvature separates the area of the background from the singularity data of a curve model. A smooth gauge reduces any positive-area background to a positive form. Keeping polar data fixed preserves distances on the common admissible range, while area rescaling identifies the entire model spaces. The gauge controls the normalized representatives. At zero area the whole space collapses to the unique normalized potential.

Keep the curve Y of Section 29.12 and fix a smooth positive form η of integral one. In this section allow any smooth real background σ with $A_\sigma = \int_Y \sigma \geq 0$, and let $\mathcal{M}(\sigma)$ be its normalized ordinary model space. Use η for L^1 norms and capacity. Put $U_\sigma(p) = \int_Y \mathrm{MA}_\sigma(p)$; at zero area the volume-comparison formula of Definition 29.6.1 is identically zero.

Proposition 29.13.1 *Polar curvature identifies $\mathcal{M}(\sigma)$ with the positive polar measures of mass at most A_σ. Write $p_\sigma(\mu)$ for the model of μ. Then*

$$U_\sigma(p_\sigma(\mu)) = A_\sigma - \mu(Y), \qquad d_\sigma(p_\sigma(\mu), p_\sigma(\nu)) = \|\mu - \nu\|_{\mathrm{TV}},$$

and ideal projection corresponds to taking the atomic part. The model space is complete and geodesic, and metric convergence implies L^1 and capacity convergence. If $A_\sigma = 0$, its only member is the smooth potential u_σ of Lemma 1.8.8.

Proof Suppose first that $A = A_\sigma > 0$. By Lemma 1.8.8, write $\sigma = A\eta + \mathrm{dd^c} h$. For $v \in \mathrm{PSH}(Y, A\eta)$, the ordinary-envelope inequalities of Proposition 15.3.1 give

$$P_{A\eta}[v] + \min_Y h \leq P_\sigma[v - h] + h \leq P_{A\eta}[v] + \max_Y h.$$

For a model p in background $A\eta$, the potential $q = P_\sigma[p - h]$ therefore satisfies $q + h \sim p$. Ordinary-type invariance of the envelope gives $P_\sigma[q] = q$. Conversely, for $q \in \mathcal{M}(\sigma)$, the same bounds show that $P_{A\eta}[q + h] \sim q + h$, and taking the envelope back recovers q. Thus this construction is a bijection of the model spaces.

Smooth gauges preserve curvature measures, and ordinary equivalence preserves their polar parts by Corollary 1.6.1. The bijection consequently preserves polar curvature. It also preserves singularity volumes; subtracting the same smooth function commutes with maxima. The classification and distance formula in Lemma 29.12.1 and Theorem 29.12.1 now give the first assertions. The identical gauge bounds for ideal envelopes make the corresponding ideal projections ordinarily equivalent after transport. They are ordinary models by Proposition 15.2.1, so uniqueness in each ordinary type makes them equal. Thus the bijection commutes with ideal projection, which still takes the atomic part. Completeness and geodesics follow by the same identification with polar measures as in Corollary 29.12.1.

For metric convergence to a model of positive mass, σ is big because $\sigma + \mathrm{dd^c}(-h) = A\eta > 0$, and the converging polar masses give a common positive lower-volume bound. Apply Proposition 29.6.1 and Theorem 29.7.1. At a zero-mass limit the current estimate (29.23) follows from the same polar decomposition. Normalized

compactness and harmonic uniqueness give L^1 convergence; Lemma 1.6.2 then gives capacity convergence on a finite cover.

If $A_\sigma = 0$, Lemma 1.8.8 leaves only u_σ after normalization. Its current vanishes, and both envelope operations fix it. All assertions reduce to the zero measure and the one-point metric space. □

Theorem 29.13.1 *Let θ_0, θ_1 have positive areas $A_i = A_{\theta_i}$ and put $\lambda = A_1/A_0$. The map*

$$F_{10} : \mathcal{M}(\theta_0) \longrightarrow \mathcal{M}(\theta_1), \qquad F_{10}(p_{\theta_0}(\mu)) = p_{\theta_1}(\lambda\mu),$$

is a bijection and satisfies

$$\begin{aligned} d_{\theta_1}(F_{10}(p), F_{10}(q)) &= \lambda d_{\theta_0}(p, q), \\ U_{\theta_1}(F_{10}(p)) &= \lambda U_{\theta_0}(p), \\ P_{\theta_1}[F_{10}(p)]_{\mathcal{I}} &= F_{10}(P_{\theta_0}[p]_{\mathcal{I}}). \end{aligned} \tag{29.28}$$

For a third smooth background θ_2 of positive area, the same construction gives $F_{21}F_{10} = F_{20}$.

Proof Multiplication by λ is a bijection between the positive polar measures of mass at most A_0 and those of mass at most A_1. By Proposition 29.13.1, it gives the asserted bijection, with distance

$$\|\lambda\mu - \lambda\nu\|_{\mathrm{TV}} = \lambda\|\mu - \nu\|_{\mathrm{TV}}.$$

The volume identity follows from $A_1 - \lambda\mu(Y) = \lambda(A_0 - \mu(Y))$. Taking the atomic part commutes with multiplication by λ, which gives the ideal-envelope identity. The area ratios multiply, proving the composition rule. □

For $A_\sigma > 0$, the distances d_σ/A_σ give isometric model spaces under these correspondences. The unscaled spaces have diameter $2A_\sigma$: this is an upper bound for the total variation of two positive measures of mass at most A_σ, and the models of $A_\sigma\delta_x$ and $A_\sigma\delta_y$, $x \neq y$, attain it. Consequently the full unscaled spaces can be isometric only when their areas agree.

Alternatively, $p_{\theta_0}(\mu) \mapsto p_{\theta_1}(\mu)$ is an isometry on the common range $\mu(Y) \leq \min(A_0, A_1)$. These two models have the same ordinary type. Indeed, their Green representatives have the same polar term, so their difference has smooth $\mathrm{dd^c}$ and is smooth by elliptic regularity; Lemma 29.12.1 and Proposition 29.13.1 identify the model with its Green representative, corrected by a smooth gauge, up to a bounded function. Under this correspondence the non-pluripolar volume changes by $A_1 - A_0$.

The area-rescaling map preserves analytic models when λ is rational, by the convention in Section 1.13: on a curve their polar measures have finite support and rational coefficients. If λ is irrational, an atom $a\delta_x$ with rational $0 < a < A_0$ acquires an irrational Lelong number, so its image is not analytic in this convention. Ideal models are preserved for every λ by (29.28).

Proposition 29.13.2 *Choose a smooth real h with $\theta_1 = \lambda\theta_0 + \mathrm{dd^c} h$. For $p \in \mathcal{M}(\theta_0)$ and $q = F_{10}(p)$,*

$$\begin{gathered} q = P_{\theta_1}[\lambda p - h], \\ \lambda p + \min_Y h \le q + h \le \lambda p + \max_Y h. \end{gathered} \tag{29.29}$$

If $U_{\theta_0}(p) = 0$, then

$$q = \lambda p - h - \sup_Y(\lambda p - h).$$

Proof The form $\theta_1 - \lambda\theta_0$ has integral zero, so the scalar Poisson equation supplies h, as in Lemma 1.8.8. Write $p = p_{\theta_0}(\mu)$. Scaling the competitors in the ordinary envelope shows that λp is a normalized ordinary model for $\lambda\theta_0$. The smooth-gauge construction in Proposition 29.13.1 therefore makes $P_{\theta_1}[\lambda p - h]$ a model. The current of $\lambda p - h$ is $\lambda(\theta_0 + \mathrm{dd^c} p)$, so its polar measure is $\lambda\mu$. The construction preserves this measure, giving the first identity in (29.29).

The function $\lambda p - h + \min_Y h$ is a competitor for q, giving the lower bound. If $v \le 0$ is any competitor, then $v + h - \max_Y h$ is $\lambda\theta_0$-plurisubharmonic, nonpositive and more singular than λp. It is therefore at most λp. Taking the upper envelope gives the upper bound. The inequalities hold at the poles as well and, off their common pole set, imply $|q - \lambda p| \le \operatorname{osc}_Y h$.

At zero mass, the polar measures exhaust the currents of both p and q. Hence $\mathrm{dd^c}(q - \lambda p + h) = 0$. Harmonic uniqueness and the supremum-zero normalization give the last formula. □

Corollary 29.13.1 *Suppose smooth real forms θ_j of positive area converge uniformly to a smooth form θ_0 of positive area. Put $\lambda_j = A_{\theta_j}/A_{\theta_0}$ and write F_{j0} for the preceding correspondence. For every $t > 0$,*

$$\begin{gathered} \sup_{p \in \mathcal{M}(\theta_0)} \|F_{j0}(p) - p\|_{L^1(Y)} \longrightarrow 0, \\ \sup_{p \in \mathcal{M}(\theta_0)} \operatorname{Cap}_\eta(\{|F_{j0}(p) - p| > t\}) \longrightarrow 0. \end{gathered} \tag{29.30}$$

More generally, let $q_j \in \mathcal{M}(\theta_j)$ have polar measures ν_j, and let $p = p_{\theta_0}(\mu)$. If $\lambda_j^{-1}\nu_j \to \mu$ in total variation, then $q_j \to p$ in L^1 and in capacity.

Proof The mean-zero inverse of the self-adjoint scalar Laplacian for η has a symmetric Green kernel G. Set

$$h_j(x) = \int_Y G(x, y)\,(\theta_j - \lambda_j\theta_0)(y).$$

The integrand form has integral zero, so $\mathrm{dd^c} h_j = \theta_j - \lambda_j\theta_0$, and elliptic regularity makes h_j smooth. Symmetry and the uniform L^1 bound from Section 18.3 give

$$\|h_j\|_{L^\infty(Y)} \le C \left\| \frac{\theta_j - \lambda_j\theta_0}{\eta} \right\|_{L^\infty(Y)} \longrightarrow 0.$$

Here C depends only on η. The symmetric kernel may be made nonpositive by subtracting a constant, which does not change h_j.

By (29.29), outside the common poles,

$$|F_{j0}(p) - p| \le \operatorname{osc}_Y h_j + |\lambda_j - 1|\,|p|.$$

Normalized compactness bounds $\|p\|_{L^1}$ uniformly over $\mathcal{M}(\theta_0)$, proving the first limit. Choose $C_0 > 0$ with $\theta_0 \le C_0\eta$. Then p/C_0 is normalized and η-plurisubharmonic, so the uniform sublevel estimate (1.14) proves the second limit. Pole sets have zero capacity.

For the last assertion put $p_j = F_{0j}(q_j)$. Its polar measure is $\lambda_j^{-1}\nu_j$, so Proposition 29.13.1 gives $p_j \to p$ in L^1 and capacity. The two uniform limits in (29.30), applied to p_j, give $q_j - p_j \to 0$ in the same senses. The triangle inequality for L^1 and subadditivity of capacity finish the proof. □

Finally suppose $\theta_j \to \theta_0$ uniformly, where $A_{\theta_j} \ge 0$ and $A_{\theta_0} = 0$. The limiting model space is $\{u_{\theta_0}\}$, and Theorem 1.8.1 gives uniform L^1 and capacity convergence to this point for all normalized potentials, hence for all models. The model spaces also collapse for their unscaled metrics: their diameters are $2A_{\theta_j} \to 0$. At positive area, dividing the metric by the area instead retains the same polar-measure space as $\mathcal{M}(\eta)$ under Theorem 29.13.1. Thus the normalized metric retains singularity data that disappear in the limiting potentials.

Chapter 30
Relative equations under positive lower mass

Let X be a connected compact complex manifold of dimension $n \geq 1$. Fix a smooth semipositive big real $(1,1)$-form θ and a model reference φ satisfying (12.1).

Approximating a finite non-pluripolar measure on X gives solutions of the ordinary type of φ, but their bounded-difference constants need not remain uniform. Relative envelopes retain a singularity condition at the limit: stretching away from φ by any finite factor must leave a genuine envelope.

Positive lower mass prevents these envelopes from collapsing. Disjoint contact sets let one finite measure control arbitrarily many rooftop obstacles. The resulting common lower envelope preserves the relative class and controls deep sublevel tails, so capacity convergence passes to the undamped measures. Exponential comparison supplies a common envelope when bounded carrier measures vary.

A positive exponent determines a unique potential. At exponent zero the geometry determines a unique scalar, while smooth projective examples admit several normalized potentials for the same measure in a nonclosed Hermitian background.

Unless explicitly stated otherwise, assume

$$v_-(\theta) > 0. \tag{30.1}$$

Fix $\varepsilon \in (0,1)$ with $P_{\varepsilon\theta}(\varphi)$ genuine, and write

$$\mathcal{E}_\varphi = \mathcal{E}_{\mathrm{env}}(X,\theta,\varphi), \qquad \mathcal{E}_0 = \mathcal{E}_{\mathrm{env}}(X,\theta,0), \qquad \mu_u = \mathrm{MA}_\theta^{\mathrm{ext}}(u).$$

Fix also a Hermitian form $\omega \geq \theta$ for capacity and a smooth positive volume form $\mathrm{d}V$ for L^1 convergence. The envelope constructions of Section 12.5 remain available when members of $\mathcal{E}_\varphi$ have infinite mass. The solutions below have finite mass because their equations have finite right-hand sides.

For $a \in \mathcal{E}_\varphi$, Proposition 12.4.3 gives

$$a - \sup_X a \leq P_\theta[a] = \varphi. \tag{30.2}$$

The inequality follows by using $a - \sup_X a$ as a normalized envelope competitor.

30.1 Equations for finite measures

For a fixed positive exponent, the right-hand side changes under translation of the potential. At exponent zero, translation leaves the equation unchanged, and a positive scalar is needed to normalize the prescribed measure.

Theorem 30.1.1 *Let μ be a nonzero finite non-pluripolar measure. Then:*

(1) *For every $\lambda > 0$ there is a unique $u \in \mathcal{E}_\varphi$ with $\mu_u = e^{\lambda u}\mu$.*
(2) *There are $u \in \mathcal{E}_\varphi$ and a unique $c > 0$ with $\mu_u = c\mu$.*

The proof first treats the minimal reference in Theorem 30.5.1; those solutions then serve as comparison potentials in Section 30.6. For general measure data, the conclusion is membership in $\mathcal{E}_\varphi$, rather than bounded difference from φ.

30.2 Domination

We first compare with one finite measure. Contact support bounds the measures of stretched envelopes by that fixed measure, and its small tails exclude collapse. The stretching envelopes Q_b are those of Definition 12.4.1; the uniform lower bound from Theorem 12.5.1 is

$$\mu_a(X) \geq \kappa := (1-\varepsilon)^n v_-(\theta) > 0 \qquad (a \in \mathcal{E}_\varphi).$$

Theorem 30.2.1 *Let $u \in \mathcal{E}_\varphi$ have finite μ_u, and let $v \in \mathrm{PSH}(X,\theta)$ satisfy $v \preceq \varphi$. If, for some $0 \leq c < 1$,*

$$\mu_u \leq c\mu_v \quad \text{on } \{u < v\},$$

then $u \geq v$.

Proof Replace v by $w = \max(u, v)$. Upward stability gives $w \in \mathcal{E}_\varphi$ and locality gives $\mu_u \leq c\mu_w$ on $\{u < w\}$. For $b > 1$ put $q_b = Q_b(u, w) \in \mathcal{E}_\varphi$. It satisfies $q_b \leq u$, and its measure is carried by

$$D_b = \{b^{-1}q_b + (1-b^{-1})w = u\}.$$

On D_b, positivity and ordered contact give

$$b^{-n}\mu_{q_b} + (1-b^{-1})^n\mu_w \leq \mu_u.$$

Choose b so large that $(1-b^{-1})^n > c$. On $D_b \cap \{u < w\}$ the preceding inequality forces both μ_w and μ_{q_b} to vanish. To justify this even when $\mu_w(X) = +\infty$, restrict first to common finite truncation levels of the potentials. On each such set the measures are finite and subtraction is legitimate. Exhausting the complements of the pole sets proves the assertion for the extended measures.

It follows that μ_{q_b} is carried by $D_b \cap \{u = w\} \subseteq \{q_b = u\}$. Since $q_b \le u$, another application of ordered contact gives the bound with coefficient one

$$\mu_{q_b} \le \mathbf{1}_{\{q_b=u\}}\mu_u. \tag{30.3}$$

This coefficient is independent of b.

Along integers $b \to \infty$, the potentials q_b decrease. If they collapsed identically to $-\infty$, Theorem 1.4.1 would give $M_b := \sup_X q_b \to -\infty$. But (30.3) would then imply

$$\kappa \le \int_X \mu_{q_b} \le \mu_u(\{u \le M_b\}) \longrightarrow 0.$$

Here finiteness of μ_u and its vanishing on the pole set are exactly what gives the limit. This is impossible. Hence the decreasing limit q is genuine. On $\{u < w - a\}$, for any $a > 0$, the defining inequality gives $q_b \le u - (b-1)a$. Outside the poles its limit is $-\infty$ there. Such a set has smooth volume zero because q is genuine. Thus $u \ge w$ almost everywhere, and then everywhere by Lemma 1.3.3. This proves the theorem.□

30.2.1 Domination for extended measures

To compare potentials whose measures may be infinite, first consider an ordered pair whose lower potential has no mass on the strict inequality set. A difference envelope then has no mass where it is negative. Its remaining measure is finite, so finite-measure domination applies and proves equality of the original pair.

Lemma 30.2.1 *Let $u, v \in \mathcal{E}_\varphi$ satisfy $u \le v$. If*

$$\mu_u(\{u < v\}) = 0,$$

then $u = v$.

Proof Translate both potentials by a common constant so that $u \le v \le \varphi \le 0$. Fix $\varepsilon < \delta < 1$, put $\beta = 1 - \delta$, and set

$$q = P_{\delta\theta}(v).$$

This envelope is genuine by Theorem 12.5.1. By Lemma 12.5.1,

$$w = P_\theta(u - \varphi) \in \mathcal{E}_0, \qquad w + \varphi \le u, \qquad w \le 0.$$

The stretching lemma with reference zero shows that $z = P_\theta(w/\beta)$ belongs to $\mathcal{E}_0$. Scaling the background and each defining competitor gives

$$\beta z \in \mathcal{E}_{\mathrm{env}}(X, \beta\theta, 0).$$

Indeed, if $a \in \mathrm{PSH}(X, \theta)$ lies below tz, then $\beta a \in \mathrm{PSH}(X, \beta\theta)$ lies below $t\beta z$, for every $t \ge 1$.

Since $q \le v \le \varphi$, we have

$$\beta z \le w \le u - \varphi \le u - q$$

off the pole sets. Thus

$$k = P_{\beta\theta}(u - q)$$

is genuine. Here the difference obstacle is taken off the pluripolar pole sets; it is quasi-continuous and finite almost everywhere. The general envelope theorem gives a finite upper bound for k and the canonical inequality $k + q \le u$. Upward stability therefore gives $k \in \mathcal{E}_{\mathrm{env}}(X, \beta\theta, 0)$.

Write $\nu_k = \mathrm{MA}^{\mathrm{ext}}_{\beta\theta}(k)$. Contact support carries ν_k by

$$D = \{k + q = u\}$$

outside the pole sets. Positivity of mixed products and ordered contact give

$$\nu_k \le \mathbf{1}_D \mu_u.$$

These inequalities are first restricted to common finite truncation levels and then exhausted. On $D \cap \{k < 0\}$ we have $u = k + q < q \le v$, so $\nu_k(\{k < 0\}) = 0$.

This vanishing proves that ν_k is finite. In fact, $\widehat{k} = \max(k, -1)$ is a single bounded $\beta\theta$-plurisubharmonic function, and locality gives

$$\nu_k(X) = \nu_k(\{k \ge 0\}) = \int_{\{k \ge 0\}} \mathrm{MA}_{\beta\theta}(\widehat{k}) \le \int_X \mathrm{MA}_{\beta\theta}(\widehat{k}) < \infty.$$

Apply Theorem 30.2.1 in the background $\beta\theta$, with reference zero and potentials $k, 0$, using coefficient zero. Its hypotheses hold: $\beta\theta$ is semipositive and big, zero satisfies the model condition, and

$$v_-(\beta\theta) = \beta^n v_-(\theta) > 0.$$

The scaling identity follows by the bijection $a \mapsto \beta a$ between bounded potentials in the two backgrounds. We have just proved finiteness of ν_k. The theorem yields $k \ge 0$, hence $u \ge q$.

To pass to $\delta \uparrow 1$, fix $\delta_0 \in (\varepsilon, 1)$ and $r = P_{\delta_0\theta}(v)$. For $\delta_0 < \delta < 1$, set $t = (\delta - \delta_0)/(1 - \delta_0)$. The potential $tv + (1 - t)r$ has background $\delta\theta$ and lies below v. Therefore

$$u \ge P_{\delta\theta}(v) \ge tv + (1 - t)r.$$

Let $t \uparrow 1$ at points where v, r are finite. This gives $u \ge v$ almost everywhere, and then everywhere by Lemma 1.3.3. Together with $u \le v$, it proves equality. □

Theorem 30.2.2 *Let $u \in \mathcal{E}_\varphi$ and $v \in \mathrm{PSH}(X, \theta)$ with $v \preceq \varphi$. If $0 \le c < 1$ and*

$$\mu_u \le c\mu_v \quad \text{on } \{u < v\},$$

then $u \ge v$.

Proof Put $w = \max(u, v) \in \mathcal{E}_\varphi$ and choose $b > 1$ with $(1-b^{-1})^n > c$. The stretching lemma gives $p = Q_b(u, w) \in \mathcal{E}_\varphi$ and $p \le u$. Its measure is carried by

$$D = \{b^{-1}p + (1 - b^{-1})w = u\}.$$

On D, positivity and ordered contact imply

$$b^{-n}\mu_p + (1 - b^{-1})^n \mu_w \le \mu_u.$$

On $\{u < w\}$, locality gives $\mu_u \le c\mu_w$. Restricting both inequalities to $D \cap \{u < w\}$ and to common finite truncation levels makes subtraction legitimate. Since $(1-b^{-1})^n > c$, it follows that μ_p vanishes there. Exhausting the finite levels proves that μ_p is carried by $D \cap \{u = w\} \subseteq \{p = u\}$. Thus $\mu_p(\{p < u\}) = 0$.

Apply Lemma 30.2.1 to $p \le u$, both in $\mathcal{E}_\varphi$. It yields $p = u$. Substitution into the defining convex inequality for p gives $w \le u$ off the poles, hence everywhere by Lemma 1.3.3. Thus $v \le w \le u$. □

Corollary 30.2.1 *Under the hypotheses on u, v in Theorem 30.2.2, for every $\lambda > 0$,*

$$\mathrm{e}^{-\lambda u}\mu_u \le \mathrm{e}^{-\lambda v}\mu_v \quad \Longrightarrow \quad u \ge v.$$

Also $\mu_u \le c\mu_v$ on X for a finite constant $c \ge 0$ implies $c \ge 1$.

Proof On $\{u < v - a\}$ the exponential inequality gives $\mu_u \le \mathrm{e}^{-\lambda a}\mu_v$. Apply the theorem to $v - a$ and let $a \downarrow 0$. All weights are taken off the pluripolar pole sets. For the second assertion, if $c < 1$, applying the theorem to $v + A$ for every real A contradicts finiteness of both potentials at a common point. □

Applying the exponential implication in both directions proves uniqueness for a fixed positive exponent. Applied to two equations $\mu_u = c_1\nu$ and $\mu_v = c_2\nu$, the scalar assertion gives $c_1 = c_2$. The final section shows why this does not determine the normalized potential at exponent zero.

30.3 Common envelopes and dominated continuity

Domination by one finite measure gives a common lower envelope after extraction. On disjoint contact sets, each finite rooftop inherits the same dominating measure. A summable integration error then prevents its stretched envelopes from collapsing, uniformly in the number of obstacles.

Lemma 30.3.1 *Suppose $u_j \in \mathcal{E}_\varphi$, $u_j \to u \in \mathrm{PSH}(X, \theta)$ in $L^1(X)$, and $\mu_{u_j} \le \nu$ for a fixed finite non-pluripolar measure ν. There is a subsequence, denoted again by u_j, such that*

$$h = P_\theta\left(\inf_j u_j\right) \in \mathcal{E}_\varphi.$$

In particular $u \in \mathcal{E}_\varphi$.

Proof By (30.2) and the boundedness of the suprema of an L^1-convergent family, there is a constant C with $u_j \le \varphi + C$ for all j. A common translation reduces to $u_j \le \varphi \le 0$, and then $u \le \varphi$. By Lemma 1.23.1,

$$I_j = \int_X |\mathrm{e}^{u_j} - \mathrm{e}^u| \, \mathrm{d}\mu_{u_j} \longrightarrow 0.$$

Extract so that $I_j \le 2^{-j}$ and $u_j \to u$ almost everywhere for smooth volume. This extraction is made once.

For $N \ge j$ put $p_{j,N} = P_\theta(u_j, \dots, u_N)$. These are in $\mathcal{E}_\varphi$ by Proposition 12.4.2. Partition their contact sets by the first index at which the minimum is attained. Ordered contact on these disjoint pieces gives both

$$\mu_{p_{j,N}} \le \nu, \qquad \int_X |\mathrm{e}^{p_{j,N}} - \mathrm{e}^u| \, \mathrm{d}\mu_{p_{j,N}} \le \sum_{\ell=j}^{N} I_\ell \le 2^{1-j}. \tag{30.4}$$

The first bound has no factor N: each disjoint piece is bounded by the restriction of the same measure ν to that piece.

Fix a real number $B \ge 1$ and set $r_{j,N} = Q_B(p_{j,N}, \varphi)$. It belongs to $\mathcal{E}_\varphi$, so $\mu_{r_{j,N}}(X) \ge \kappa$. Put $D_{j,N} = \{B^{-1} r_{j,N} + (1 - B^{-1})\varphi = p_{j,N}\}$. Contact support and positivity give

$$\mu_{r_{j,N}} \le B^n \mathbf{1}_{D_{j,N}} \mu_{p_{j,N}} \le B^n \nu.$$

On this contact set, since $\varphi \le 0$, $\mathrm{e}^{p_{j,N}} \le \mathrm{e}^{r_{j,N}/B}$. Consequently

$$\int_X \mathrm{e}^u \, \mathrm{d}\mu_{r_{j,N}} \le B^n 2^{1-j} + B^n \nu(X) \exp\Big(\frac{\sup_X r_{j,N}}{B}\Big). \tag{30.5}$$

On the other hand choose $K > 0$ with $B^n \nu(\{u \le -K\}) < \kappa/2$. This is possible since u is genuine and ν does not charge its pole set. The preceding measure domination and lower mass bound imply

$$\int_X \mathrm{e}^u \, \mathrm{d}\mu_{r_{j,N}} \ge \mathrm{e}^{-K} \kappa/2.$$

Choose j so large that $B^n 2^{1-j} < \mathrm{e}^{-K}\kappa/4$. Formula (30.5) then bounds $\sup_X r_{j,N}$ from below independently of N. The sequence decreases in N and has the common upper bound φ. Its limit R is therefore genuine. It satisfies, for every $\ell \ge j$,

$$B^{-1} R + (1 - B^{-1})\varphi \le u_\ell, \qquad R \le \varphi. \tag{30.6}$$

The second inequality follows already from $r_{j,N} \le p_{j,N} \le \varphi$.

It remains to include the finitely many earlier terms and to establish every defining envelope condition. Fix $b \ge 1$ and apply the preceding construction with $B = 2b$. Let j and R be as above. If $j > 1$, let $p_0 = P_\theta(u_1, \dots, u_{j-1})$ and $S = Q_{2b}(p_0, \varphi)$. Both are genuine by the finite-rooftop and stretching lemmas, and $S \le \varphi$. For $j = 1$ set $S = \varphi$. The potential $H = (R + S)/2$ is θ-plurisubharmonic. For a tail index use

(30.6) and $S \le \varphi$; for a prefix index use its counterpart for S and $R \le \varphi$. In either case,

$$b^{-1}H + (1 - b^{-1})\varphi \le u_\ell \quad \text{for every } \ell \ge 1.$$

For $b = 1$ this first proves that $h = P_\theta(\inf_\ell u_\ell)$ is genuine. For general b the left side is a competitor for h, so the same inequality with h in place of every u_ℓ supplies the defining b-condition. Also $h \le \varphi$. Thus $h \in \mathcal{E}_\varphi$.

Finally $h \le u_j$ for all j, hence $h \le u \le \varphi$ by the almost-everywhere limit and Lemma 1.3.3. Upward stability proves $u \in \mathcal{E}_\varphi$. Restoring the common translation proves the lemma as stated. □

Once a common lower envelope is available, a single integral criterion gives convergence in capacity. Its proof uses zero domination and allows infinite masses.

Proposition 30.3.1 *Suppose $u_j \in \mathcal{E}_\varphi$ and, for a fixed $h \in \mathcal{E}_\varphi$ and constant C,*

$$h \le u_j \le \varphi + C, \qquad u_j \longrightarrow u \in \mathrm{PSH}(X, \theta) \quad \text{in } L^1(X).$$

If

$$I_j := \int_X |\mathrm{e}^{u_j} - \mathrm{e}^u| \, \mu_{u_j} \longrightarrow 0,$$

then $u \in \mathcal{E}_\varphi$ and $u_j \to u$ in capacity.

Proof The bounds pass to u almost everywhere along an extraction and then everywhere by Lemma 1.3.3. Upward stability gives $u \in \mathcal{E}_\varphi$. Translate the entire family and its bounds by a common constant to make all potentials nonpositive.

It suffices to obtain capacity convergence along a further extraction of every subsequence. Extract so that $I_j \le 2^{-j}$ and $u_j \to u$ almost everywhere for smooth volume. Form

$$p_{j,N} = P_\theta(u_j, \dots, u_N), \qquad p_j = P_\theta\left(\inf_{\ell \ge j} u_\ell\right).$$

Every envelope lies above h and below $\varphi + C$, so is genuine and belongs to $\mathcal{E}_\varphi$. For the finite envelope, partition its contact set by the first index ℓ with $p_{j,N} = u_\ell$. Ordered contact on each part gives

$$\int_X |\mathrm{e}^{p_{j,N}} - \mathrm{e}^u| \, \mu_{p_{j,N}} \le \sum_{\ell=j}^{N} I_\ell \le 2^{1-j}.$$

Only positive integrals are used here, even if a rooftop has infinite mass.

As $N \to \infty$, $p_{j,N} \downarrow p_j$. As $j \to \infty$, the p_j increase to a genuine canonical limit p. Both convergences hold in capacity by Lemma 1.8.6. Also $h \le p \le u \le \varphi + C$; the upper inequality follows first from almost-everywhere convergence of the extracted u_j. Thus $p \in \mathcal{E}_\varphi$.

Apply Lemma 6.3.1 first as $N \to \infty$ and then as $j \to \infty$, with the bounded quasi-continuous weights displayed above. The weights converge in capacity because

the exponential is Lipschitz on a common upper half-line. We obtain

$$\int_X |e^p - e^u|\,\mu_p = 0, \qquad \mu_p(\{p < u\}) = 0.$$

The second equality follows off the poles, which the extended measures ignore. The relative zero-domination lemma gives $p = u$, without a finiteness hypothesis on μ_p.

Finally $U_j = (\sup_{\ell \geq j} u_\ell)^*$ decreases to u by Theorem 1.4.1 and the almost-everywhere extraction. Apply Lemma 1.8.6 to $p_j \uparrow u$ and $U_j \downarrow u$. The inequalities $p_j \leq u_j \leq U_j$ prove capacity convergence on this extraction, and hence on the original sequence. □

30.3.1 Convergence under a fixed measure

The extraction lemma supplies the lower envelope needed in the integral criterion. The same fixed measure then controls the deep sublevel tails, so convergence in capacity passes to the undamped Monge–Ampère measures.

Theorem 30.3.1 *Suppose $u_j \in \mathcal{E}_\varphi$, $u_j \to u \in \mathrm{PSH}(X, \theta)$ in $L^1(X)$, and $\mu_{u_j} \leq \nu$ for a fixed finite non-pluripolar measure ν. Then*

$$u \in \mathcal{E}_\varphi, \qquad u_j \longrightarrow u \text{ in capacity}, \qquad \mu_{u_j} \rightharpoonup \mu_u \leq \nu.$$

Proof Start with any subsequence. By Lemma 30.3.1, a further subsequence has a common lower envelope $h \in \mathcal{E}_\varphi$, and $u \in \mathcal{E}_\varphi$. Equation (30.2) and Hartogs' lemma give a common upper bound $u_j \leq \varphi + C$.

After a common translation, assume $u_j, u \leq 0$. Lemma 1.23.1 gives

$$\int_X |e^{u_j} - e^u|\,d\mu_{u_j} \leq \int_X |e^{u_j} - e^u|\,d\nu \longrightarrow 0.$$

Thus Proposition 30.3.1 proves capacity convergence along this extraction. For the raw measures, the common lower bound gives

$$\mu_{u_j}(\{u_j \leq -T\}) \leq \nu(\{h \leq -T\}) \longrightarrow 0 \quad \text{uniformly in } j.$$

The limit on the right uses finiteness of ν and its vanishing on the pole set of the genuine h. The raw-tail criterion for smooth backgrounds, Theorem 7.6.1, now gives weak convergence to μ_u. Weak limits preserve domination by the fixed finite ν. Every original subsequence has a further subsequence with these conclusions and the same limit u, so both convergences hold for the original sequence. □

30.4 Moving bounded carrier measures

When the bounded carrier measure varies, exponential domination supplies a common lower envelope. A smaller-background potential and a strict big potential give comparison functions with constants uniform over the family. If every approximant has the ordinary type of φ, smaller-background comparison gives the same construction even when $v_-(\theta) = 0$. Convergence of the carrier potentials in capacity will then identify the limiting equation.

For a bounded ω-plurisubharmonic potential ψ, write $\omega_\psi = \omega + \mathrm{dd^c}\psi$.

Proposition 30.4.1 *Fix $A, \lambda > 0$, $C \in \mathbb{R}$ and $L \geq 0$. Suppose $u_j \in \mathcal{E}_\varphi$, $u_j \leq \varphi + C$, and*

$$\mu_{u_j} \leq A\mathrm{e}^{\lambda u_j}\omega^n_{\psi_j}, \qquad \psi_j \in \mathrm{PSH}(X, \omega), \qquad \|\psi_j\|_\infty \leq L. \tag{30.7}$$

The index j ranges over the positive integers. Assume either $v_-(\theta) > 0$ or $u_j \sim \varphi$ for every j. Then

$$h = P_\theta\left(\inf_j u_j\right) \in \mathcal{E}_\varphi.$$

No convergence of ψ_j or u_j is required. In the ordinary-type alternative, the constants in $u_j \sim \varphi$ need not be uniform.

Proof Translate all u_j by one constant so that $u_j \leq \varphi \leq 0$; this changes A by a fixed factor. Translate each ψ_j to lie in $[-2L, 0]$, without changing its measure. Set $q = P_{\varepsilon\theta}(\varphi)$ and choose $\rho \in \mathrm{PSH}(X, \theta)$ and $a > 0$ such that

$$q \leq \varphi, \qquad \rho \leq 0, \qquad \theta + \mathrm{dd^c}\rho \geq a\omega.$$

The first potential is genuine by the model hypothesis, and the second by bigness. Fix $0 < \delta < (1 - \varepsilon)a$.

For a fixed real $b \geq 1$, put $r_{j,b} = Q_b(u_j, \varphi)$. The stretching lemma puts it in $\mathcal{E}_\varphi$, and $r_{j,b} \leq u_j \leq 0$. Contact support and positivity imply

$$\mu_{r_{j,b}} \leq b^n \mathbf{1}_{D_{j,b}}\mu_{u_j}, \qquad D_{j,b} = \{b^{-1}r_{j,b} + (1 - b^{-1})\varphi = u_j\}.$$

On this contact set, $u_j \leq r_{j,b}/b$. Thus

$$\mathrm{e}^{-\lambda r_{j,b}/b}\mu_{r_{j,b}} \leq b^n A\omega^n_{\psi_j}. \tag{30.8}$$

The weights are taken off the pluripolar pole sets.

Choose $C_b \geq 0$ so large that

$$\delta^n \exp(\lambda C_b/b) \geq b^n A,$$

and define

$$s_{j,b} = q + (1 - \varepsilon)\rho + \delta\psi_j - C_b.$$

This is a genuine θ-plurisubharmonic function, since

$$\begin{aligned}\theta + \mathrm{dd^c} s_{j,b} &= (\varepsilon\theta + \mathrm{dd^c} q) + (1-\varepsilon)(\theta + \mathrm{dd^c}\rho) + \delta \mathrm{dd^c}\psi_j \\ &\geq \delta\omega_{\psi_j}.\end{aligned}$$

Consequently $\mu_{s_{j,b}} \geq \delta^n \omega^n_{\psi_j}$. This inequality follows first away from the poles by the local positive product inequality, and then globally because the bounded reference measure does not charge pluripolar sets. Also $s_{j,b} \leq -C_b$ and $s_{j,b} \preceq \varphi$. We conclude

$$\mathrm{e}^{-\lambda s_{j,b}/b}\mu_{s_{j,b}} \geq b^n A\omega^n_{\psi_j} \geq \mathrm{e}^{-\lambda r_{j,b}/b}\mu_{r_{j,b}}.$$

If $v_-(\theta) > 0$, apply Corollary 30.2.1 with first potential $r_{j,b}$ and second potential $s_{j,b}$. It gives $r_{j,b} \geq s_{j,b}$.

In the ordinary-type alternative, $r_{j,b} \sim \varphi$: if $u_j \geq \varphi - K_j$, the potential $\varphi - bK_j$ is a competitor for $r_{j,b}$. The smaller-background envelope of $r_{j,b}$ is therefore genuine, and $s_{j,b} \preceq r_{j,b}$. The same conclusion follows instead from Corollary 13.1.1, without a lower-volume hypothesis.

The bound $\psi_j \geq -2L$ now supplies one genuine lower bound, independent of j:

$$H_b = q + (1-\varepsilon)\rho - 2\delta L - C_b \leq r_{j,b}.$$

The potential H_b is θ-plurisubharmonic. Therefore

$$b^{-1}H_b + (1-b^{-1})\varphi \leq u_j \quad \text{for every } j.$$

For $b = 1$ this proves that the infimum envelope h is genuine. For general b, the left side is a competitor for h, so it also lies below h. This is a competitor for every defining stretched-envelope condition of h. Since $h \leq \varphi$, we obtain $h \in \mathcal{E}_\varphi$. Restore the common translation to finish. □

Corollary 30.4.1 *Under the hypotheses of Proposition 30.4.1, if $u_j \to u \in \mathrm{PSH}(X,\theta)$ in $L^1(X)$, then $u \in \mathcal{E}_\varphi$.*

Proof The common envelope gives $h \leq u_j \leq \varphi + C$. Pass to an almost-everywhere convergent subsequence and use Lemma 1.3.3 to get $h \leq u \leq \varphi + C$. Upward stability gives the conclusion. □

30.4.1 Convergence with a common envelope

Capacity convergence of uniformly bounded carrier potentials controls the integration error in Proposition 30.3.1. A common relative lower envelope then controls the deep sublevel tails of the solution measures. The domination may also contain a fixed finite non-pluripolar measure.

Theorem 30.4.1 *Under the geometric and common-envelope hypotheses of Proposition 30.3.1, fix $A \geq 0$ and a finite non-pluripolar measure ν on X. Let $\psi_j \in \mathrm{PSH}(X, \omega)$ be uniformly bounded and converge in capacity to a bounded $\psi \in \mathrm{PSH}(X, \omega)$. Suppose*

$$\mu_{u_j} \leq A\omega^n_{\psi_j} + \nu.$$

Then $u \in \mathcal{E}_\varphi$, $u_j \to u$ in capacity, μ_u is finite, and

$$\mu_{u_j} \rightharpoonup \mu_u.$$

Proof Translate so that $u_j, u \leq 0$. The functions $\mathrm{e}^{u_j}, \mathrm{e}^u$ are uniformly bounded quasi-plurisubharmonic functions with a common lower curvature bound. They converge in smooth-volume L^1, since the exponential is Lipschitz on $(-\infty, 0]$. Lemmas 1.23.1 and 1.23.3 give

$$\int_X |\mathrm{e}^{u_j} - \mathrm{e}^u|\, \omega^n_{\psi_j} \longrightarrow 0, \qquad \int_X |\mathrm{e}^{u_j} - \mathrm{e}^u|\, \mathrm{d}\nu \longrightarrow 0.$$

The integration error tends to zero, so Proposition 30.3.1 gives capacity convergence and class membership.

The bounded reference measures have uniformly bounded mass and satisfy a common capacity estimate by Lemma 1.8.3. For a constant B independent of j, T,

$$\mu_{u_j}(\{u_j < -T\}) \leq B\,\mathrm{Cap}_\omega(\{h < -T\}) + \nu(\{h < -T\}) \longrightarrow 0$$

uniformly in j. The first term tends to zero because h is a genuine quasi-plurisubharmonic function, and the second because ν is finite and ignores its pole set. The total masses are uniformly bounded by the assumed domination. The smooth-background tail criterion, Theorem 7.6.1, gives finiteness of μ_u and the asserted weak convergence. □

30.5 General measures in the minimal envelope class

In the minimal envelope class, a bounded carrier measure reduces the equation to an integrable density. Truncating that density gives bounded solutions dominated by one fixed finite measure. Dominated continuity then passes to the full density; a vanishing exponent produces the normalized equation.

Theorem 30.5.1 *Let μ be a finite non-pluripolar measure with $\mu(X) > 0$. For every $\lambda > 0$ there is a unique $u \in \mathcal{E}_0$ satisfying*

$$\mathrm{MA}_\theta^{\mathrm{ext}}(u) = \mathrm{e}^{\lambda u}\mu.$$

There also exist $v \in \mathcal{E}_0$, normalized by $\sup_X v = 0$, and a unique positive scalar c such that

$$\mathrm{MA}_\theta^{\mathrm{ext}}(v) = c\mu.$$

Proof For reference zero, $P_{\epsilon\,\theta}(0) = 0$ for every $\epsilon > 0$. Letting $\epsilon \downarrow 0$ in Theorem 12.5.1 gives

$$\mathrm{MA}_\theta^{\mathrm{ext}}(w)(X) \geq v_-(\theta) \qquad (w \in \mathcal{E}_0). \tag{30.9}$$

Positive exponent. By Lemma 13.22.1, write $\mu = f\nu$ with $\nu = \mathrm{MA}_\theta(\psi)$, a bounded $\psi \in \mathrm{PSH}(X,\theta)$ and $0 \leq f \in L^1(\nu)$. Let $f_j = \min(f, j)$ for positive integers j. These densities are bounded and nonzero as measures. Theorem 13.9.1 gives bounded $u_j \in \mathrm{PSH}(X,\theta)$ with

$$\mathrm{MA}_\theta(u_j) = \mathrm{e}^{\lambda u_j} f_j \nu.$$

By Proposition 12.4.1, these bounded potentials belong to $\mathcal{E}_0$. Exponential order shows that u_j decreases, because f_j increases. In particular $u_j \leq u_1$. On the other hand

$$v_-(\theta) \leq \mathrm{MA}_\theta(u_j)(X) \leq \mathrm{e}^{\lambda \sup_X u_j}\mu(X).$$

Thus their suprema have a common lower bound. The decreasing limit u is a genuine θ-plurisubharmonic function, and convergence holds in $L^1(X)$. Moreover

$$\mathrm{MA}_\theta(u_j) \leq \mathrm{e}^{\lambda \sup_X u_1}\mu.$$

Apply Theorem 30.3.1 with reference zero. It gives $u \in \mathcal{E}_0$ and weak convergence of the Monge–Ampère measures to $\mathrm{MA}_\theta^{\mathrm{ext}}(u)$. The right sides converge in total variation to $\mathrm{e}^{\lambda u} f\nu$: the functions $\mathrm{e}^{\lambda u_j} f_j$ converge almost everywhere for ν and are bounded by $\mathrm{e}^{\lambda \sup_X u_1} f \in L^1(\nu)$. This proves the equation. Corollary 30.2.1 gives uniqueness.

Zero exponent. For each positive integer j, let $u_j \in \mathcal{E}_0$ be the just-constructed solution with exponent $1/j$. To bound the normalization, use the solution $s \in \mathcal{E}_0$ with exponent one, so that $\mathrm{MA}_\theta^{\mathrm{ext}}(s) = \mathrm{e}^s\mu$. Set $H = \max(0, \sup_X s)$ and $w_j = s + (j-1)H$. Then $w_j/j \geq s$, and

$$\mathrm{e}^{-w_j/j}\mathrm{MA}_\theta^{\mathrm{ext}}(w_j) \leq \mathrm{e}^{-s}\mathrm{MA}_\theta^{\mathrm{ext}}(s) = \mu = \mathrm{e}^{-u_j/j}\mathrm{MA}_\theta^{\mathrm{ext}}(u_j).$$

Exponential order, now with exponent $1/j$, gives $u_j \leq w_j$. Put

$$M_j = \sup_X u_j, \qquad v_j = u_j - M_j, \qquad c_j = \mathrm{e}^{M_j/j}.$$

Then $\sup_X v_j = 0$ and $0 < c_j \le e^H$. Equation (30.9) also gives

$$v_-(\theta) \le c_j \int_X e^{v_j/j}\, d\mu \le c_j \mu(X),$$

so $c_j \ge v_-(\theta)/\mu(X) > 0$. After extraction, Theorem 1.4.1 gives $v_j \to v$ in $L^1(X)$, with v genuine and normalized, and $c_j \to c > 0$.

By Lemma 1.23.2, $a_j = e^{v_j/j} \to 1$ in $L^1(\mu)$. Consequently

$$\mathrm{MA}_\theta^{\mathrm{ext}}(v_j) = c_j a_j \mu \longrightarrow c\mu \quad \text{in total variation,} \qquad \mathrm{MA}_\theta^{\mathrm{ext}}(v_j) \le e^H \mu.$$

Apply Theorem 30.3.1 again. It gives $v \in \mathcal{E}_0$ and identifies the limit as $\mathrm{MA}_\theta^{\mathrm{ext}}(v)$. The scalar assertion in Corollary 30.2.1, applied in both directions to two solutions, proves uniqueness of c. □

These minimal-class solutions also give upper bounds for the prescribed-reference approximants: exponential order with reference zero compares them with every θ-plurisubharmonic potential. This comparison will control the normalization in Section 30.6.

30.6 Equations with a model reference

To retain the reference φ, first smooth the bounded carrier potential. Exponential comparison supplies a common lower envelope, so moving-measure continuity preserves the equation. With the carrier fixed, approximate its integrable density by bounded ones. Minimal-class solutions give upper bounds, while positive lower mass prevents collapse.

Proof (of Theorem 30.1.1) Let μ be the given nonzero finite non-pluripolar measure. Apply Lemma 13.22.1 to the big background ω to write

$$\mu = f\nu, \qquad \nu = \omega_\psi^n, \qquad \psi \in \mathrm{PSH}(X,\omega) \cap L^\infty(X), \qquad 0 \le f \in L^1(\nu).$$

Positive continuous density. Fix $\lambda > 0$ and first suppose f is continuous and strictly positive. Choose smooth strictly ω-plurisubharmonic $\psi_j \downarrow \psi$ by Theorem 1.11.1. These potentials are uniformly bounded; write $\|\psi_j\|_\infty \le L$ and $\nu_j = \omega_{\psi_j}^n$. Bounded Bedford–Taylor continuity gives $\nu_j \rightharpoonup \nu$. The prescribed ordinary-type theorem Theorem 13.11.1 supplies $u_j \sim \varphi$ with

$$\mu_{u_j} = e^{\lambda u_j} f \nu_j.$$

Indeed $f\nu_j$ has a continuous positive density relative to smooth volume, hence belongs to every finite L^p space.

We first give a common upper bound. Choose K with $\mathrm{e}^{-\lambda(\psi_j+K)} \le \min_X f$ for all j. Since $\omega \ge \theta$, positivity of extended mixed products gives

$$\mathrm{e}^{-\lambda(\psi_j+K)}\omega^n_{\psi_j+K} \le f\nu_j = \mathrm{e}^{-\lambda u_j}\mu_{u_j} \le \mathrm{e}^{-\lambda u_j}\mathrm{MA}^{\mathrm{ext}}_\omega(u_j).$$

Apply Corollary 13.1.1 in background ω, with first potential $\psi_j + K$ and second potential u_j. The first is bounded and admits a smaller-background envelope, and $u_j \preceq \psi_j + K$. Thus $u_j \le \psi_j + K \le L + K =: C$.

By (30.2), $u_j \le \varphi + C$. The exponential common-envelope proposition gives a single $h \in \mathcal{E}_\varphi$ below all u_j, since

$$\mu_{u_j} \le \|f\|_\infty \mathrm{e}^{\lambda u_j}\omega^n_{\psi_j}.$$

The bounds $h \le u_j \le \varphi + C$ and Theorem 1.4.1 give a genuine L^1 limit u after extraction. The moving-measure theorem now applies: $\psi_j \to \psi$ in capacity and $\mu_{u_j} \le \mathrm{e}^{\lambda C}\|f\|_\infty \nu_j$. It gives $u \in \mathcal{E}_\varphi$, capacity convergence and $\mu_{u_j} \rightharpoonup \mu_u$.

The right-hand sides converge weakly to $\mathrm{e}^{\lambda u} f\nu$. To see this, $\mathrm{e}^{\lambda u_j} - \mathrm{e}^{\lambda u} \to 0$ in capacity with a common bound, so its integral against ν_j tends to zero by their uniform capacity estimate. The fixed function $\mathrm{e}^{\lambda u}$ is bounded and quasi-continuous. Approximate it by a continuous function outside an open set of arbitrarily small capacity; bounded product continuity and the same capacity bound allow this fixed weight in $\nu_j \rightharpoonup \nu$. Multiplication by the continuous f and a continuous test function preserves both estimates. Hence $\mu_u = \mathrm{e}^{\lambda u} f\nu$.

Bounded measurable density. Suppose next $0 \le f \in L^\infty(\nu)$ and $\int_X f\,\mathrm{d}\nu > 0$. Choose positive continuous f_j, uniformly bounded by a constant F, with $f_j \to f$ almost everywhere for ν. Such a sequence follows from density of continuous functions in $L^1(\nu)$, clipping to a common interval, adding positive constants tending to zero and extracting a summably convergent subsequence.

There is a bounded nonnegative g, nonzero as a measure, such that $g \le f_j$ for all j. Indeed, choose $a > 0$ with $\nu(\{f \ge 2a\}) > 0$. Egorov's theorem gives a measurable subset E of positive measure on which the convergence is uniform. For some N, $f_j \ge a$ on E for $j \ge N$. The finitely many remaining positive continuous functions have positive minima on X. Their minima and a have a positive lower bound b, so $g = b\mathbf{1}_E$ works.

By the first step, let $u_j \in \mathcal{E}_\varphi$ solve $\mu_{u_j} = \mathrm{e}^{\lambda u_j} f_j\nu$. The already proved minimal-reference theorem Theorem 30.5.1, applied to the finite nonzero measure $g\nu$, gives $s \in \mathcal{E}_0$ satisfying

$$\mathrm{MA}^{\mathrm{ext}}_\theta(s) = \mathrm{e}^{\lambda s} g\nu.$$

Apply Corollary 30.2.1 with reference zero, first potential $s \in \mathcal{E}_0$ and second potential $u_j \preceq 0$. The measure inequality is $g\nu \le f_j\nu$, so $u_j \le s$. With $C = \sup_X s$ we have

$$\mu_{u_j} \le \mathrm{e}^{\lambda C} F\nu, \qquad \kappa \le \mathrm{e}^{\lambda \sup_X u_j} F\nu(X).$$

The two bounds on the suprema give a genuine L^1 convergent subsequence. The fixed-measure continuity theorem puts its limit u in $\mathcal{E}_\varphi$ and identifies the weak limit as μ_u.

Lemma 1.23.1 gives $e^{\lambda u_j} \to e^{\lambda u}$ in $L^1(\nu)$. Also $f_j \to f$ in $L^1(\nu)$ by dominated convergence. Since both factors have uniform bounds,

$$e^{\lambda u_j} f_j \longrightarrow e^{\lambda u} f \quad \text{in } L^1(\nu).$$

Thus the right-hand sides converge in total variation, proving the equation for f.

Integrable density. For general $f \in L^1(\nu)$, apply the second step to $f_j = \min(f, j)$ and obtain $u_j \in \mathcal{E}_\varphi$ with $\mu_{u_j} = e^{\lambda u_j} f_j \nu$. Exponential order gives $u_{j+1} \le u_j$, because $f_j \le f_{j+1}$. Furthermore

$$\kappa \le e^{\lambda \sup_X u_j} \mu(X), \qquad \mu_{u_j} \le e^{\lambda \sup_X u_1} \mu.$$

The decreasing limit is genuine, and the fixed-measure continuity theorem gives $u \in \mathcal{E}_\varphi$ and $\mu_{u_j} \rightharpoonup \mu_u$. Dominated convergence gives $e^{\lambda u_j} f_j \to e^{\lambda u} f$ in $L^1(\nu)$, so $\mu_u = e^{\lambda u}\mu$. Corollary 30.2.1, applied in both directions, proves uniqueness.

Zero exponent. For each positive integer j, solve the equation just proved with exponent $1/j$, obtaining $u_j \in \mathcal{E}_\varphi$ with $\mu_{u_j} = e^{u_j/j}\mu$. By the minimal-reference theorem, choose $s \in \mathcal{E}_0$ solving

$$\mathrm{MA}_\theta^{\mathrm{ext}}(s) = e^s \mu.$$

Put $H = \max(0, \sup_X s)$ and $w_j = s + (j-1)H$. Then $w_j/j \ge s$, whence

$$e^{-w_j/j}\mathrm{MA}_\theta^{\mathrm{ext}}(w_j) \le \mu = e^{-u_j/j}\mu_{u_j}.$$

As in the bounded-density step, exponential order applies with reference zero, first potential $w_j \in \mathcal{E}_0$ and second potential $u_j \preceq 0$. It gives $u_j \le w_j \le jH$. Only an upper bound for s is needed; its mass is finite by its equation.

Set $M_j = \sup_X u_j$, $v_j = u_j - M_j$ and $c_j = e^{M_j/j}$. Then

$$\sup_X v_j = 0, \qquad \frac{\kappa}{\mu(X)} \le c_j \le e^H, \qquad \mu_{v_j} = c_j e^{v_j/j}\mu \le e^H \mu.$$

The lower bound follows from $\mu_{v_j}(X) \ge \kappa$ and $v_j \le 0$. Extract so that $v_j \to v$ in $L^1(X)$ and $c_j \to c > 0$. The limit v is genuine and normalized.

By Lemma 1.23.2, $a_j = e^{v_j/j} \to 1$ in $L^1(\mu)$. Consequently $\mu_{v_j} = c_j a_j \mu \to c\mu$ in total variation. The fixed-measure continuity theorem gives $v \in \mathcal{E}_\varphi$ and identifies $\mu_v = c\mu$. Finally, applying the scalar assertion of Corollary 30.2.1 in both directions proves that c is unique. □

30.7 Normalization at exponent zero

Now take a Hermitian background h on X with $v_-(h) > 0$ and reference zero. For nonzero $0 \le f \in L^1(\mathrm{d}V)$, the scalar of the minimal-class equation equals the subsolution threshold $b_h(f)$ from Section 13.24. Scalar comparison bounds every subsolution coefficient by this scalar. Dominated continuity identifies the equations of normalized exponential limits; uniqueness of these limits remains a question about the potentials.

Corollary 30.7.1 *Let h be Hermitian with $v_-(h) > 0$, and let $0 \le f \in L^1(\mathrm{d}V)$ be nonzero. There is $v \in \mathcal{E}_{\mathrm{env}}(X, h, 0)$ such that*

$$\mathrm{MA}_h^{\mathrm{ext}}(v) = b_h(f) f \,\mathrm{d}V.$$

The number $b_h(f) > 0$ is the unique positive scalar for which this equation has a solution in that envelope class.

Let u_λ be the greatest exponential solution from Theorem 13.28.1, and set $v_\lambda = u_\lambda - \sup_X u_\lambda$. Each u_λ belongs to $\mathcal{E}_{\mathrm{env}}(X, h, 0)$. If $\lambda_j \downarrow 0$ and $v_{\lambda_j} \to v$ in L^1, then the convergence holds in capacity, v belongs to the same envelope class and satisfies the displayed equation.

Proof By Theorem 30.5.1, there are $c > 0$ and $w \in \mathcal{E}_{\mathrm{env}}(X, h, 0)$ with $\mathrm{MA}_h^{\mathrm{ext}}(w) = c f \,\mathrm{d}V$. If $s \in \mathrm{PSH}(X, h)$ satisfies $\mathrm{MA}_h^{\mathrm{ext}}(s) \ge a f \,\mathrm{d}V$ with $a > 0$, then

$$\mathrm{MA}_h^{\mathrm{ext}}(w) \le \frac{c}{a} \mathrm{MA}_h^{\mathrm{ext}}(s).$$

Every h-plurisubharmonic s satisfies $s \preceq 0$. Thus the scalar assertion of Corollary 30.2.1, with reference zero, gives $c/a \ge 1$. Maximizing over subsolution coefficients yields $b_h(f) \le c$. The reverse inequality holds because w is itself a subsolution. The same argument identifies the scalar of every solution in this envelope class.

For $\lambda > 0$, Theorem 30.5.1 supplies an exponential solution s_λ in the envelope class. The greatest-solution property gives $s_\lambda \le u_\lambda$. Upward stability, Proposition 12.4.1, places u_λ and its constant translate v_λ in that class.

Write $c_\lambda = \exp(\lambda \sup_X u_\lambda)$. By Corollary 13.28.1, $c_\lambda \to b_h(f) > 0$ and

$$\mathrm{MA}_h^{\mathrm{ext}}(v_\lambda) = c_\lambda \mathrm{e}^{\lambda v_\lambda} f \,\mathrm{d}V \longrightarrow b_h(f) f \,\mathrm{d}V \quad \text{in total variation.}$$

Since $v_\lambda \le 0$ and c_λ stays bounded for small λ, these measures are dominated by one finite non-pluripolar measure $C f \,\mathrm{d}V$. Apply Theorem 30.3.1 to the stated L^1 convergent sequence. It gives class membership, capacity convergence and the displayed equation for its limit. □

Theorem 1.4.1 supplies such convergent subsequences. The corollary identifies their equations; the examples below show that the equation alone need not determine their normalized potential.

30.7.1 Nonuniqueness in the minimal envelope class

Allow the manifold and background to vary. The smooth examples from Theorem 11.9.1 have nonclosed Hermitian backgrounds on projective manifolds. Comparison with a Kähler form gives both positive lower mass and bounded upper mass, and their bounded potentials belong to the minimal envelope class. Thus even these mass bounds leave several normalized potentials with the same equation scalar.

Corollary 30.7.2 *In every complex dimension $n \geq 3$, there is a projective manifold X, a smooth nonclosed Hermitian form ω, and a nonzero finite non-pluripolar measure ν such that*

$$0 < \nu_-(\omega) \leq \nu_+(\omega) < +\infty,$$

but the equation

$$\mathrm{MA}_\omega^{\mathrm{ext}}(u) = c\nu, \qquad u \in \mathcal{E}_{\mathrm{env}}(X, \omega, 0), \qquad \sup_X u = 0,$$

has at least two smooth solutions. Its only possible positive constant is $c = 1$. Relative to a Kähler volume form on X, the measure ν has a real analytic density, positive off two connected smooth real hypersurfaces and vanishing quadratically along them.

Proof Use the form and the two potentials of Theorem 11.9.1. The manifold is projective, so Lemma 20.4.1 gives bounded mass. For the indicated Kähler form Ω, choose $\varepsilon > 0$ with $\varepsilon\Omega \leq \omega$. Lower-volume monotonicity in Theorem 29.1.1, applied to the two zero potentials, gives

$$\nu_-(\omega) \geq \nu_-(\varepsilon\Omega) = \varepsilon^n \int_X \Omega^n > 0.$$

The same-type competitors of zero are exactly the bounded potentials, so these lower volumes are the stated infima.

The two smooth potentials belong to $\mathcal{E}_{\mathrm{env}}(X, \omega, 0)$ by Proposition 12.4.1. Their extended measures equal the ordinary top powers. Their common measure ν is finite and ignores pluripolar sets. The construction gives $c = 1$, while Corollary 30.2.1 makes this the unique possible positive scalar. The density and its zero set were computed in Theorem 11.9.1. □

Chapter 31
Big backgrounds under bounded mass

Let X be a connected compact complex manifold of dimension $n \geq 1$ with bounded mass. Fix a smooth big real $(1,1)$-form θ and a model reference φ satisfying (29.4). Fix a Hermitian form ω for capacity and a smooth positive volume form $\mathrm{d}V$.

A smooth big background need not be semipositive. In this setting, bounded upper mass and positive lower reference volume supply the two bounds needed for approximation: first moments prevent collapse and preserve the relative class, while higher moments control convergence against a smoothed measure. These estimates first give solutions in the minimal class.

To prescribe a model reference, project a minimal-class solution to that reference. Projection retains only part of its exponentially weighted measure. Weighted addition fills any remaining deficit; maximizing the weighted mass therefore gives the required equation. The same contact estimates preserve the relative class when the models vary in the volume topology.

For general finite non-pluripolar data the solutions belong to the relative full-mass class. For L^p densities, a stronger moment estimate bounds their difference from the reference. Positive exponents determine the potential uniquely; at exponent zero, uniqueness concerns the scalar.

For $a \in \mathrm{PSH}(X,\theta)$, write

$$\mathcal{E}_\varphi = \mathcal{E}_{\mathrm{vol}}(X,\theta,\varphi) = \mathcal{E}_{\mathrm{env}}(X,\theta,\varphi), \qquad \theta_a = \theta + \mathrm{dd^c}a, \qquad \mu_a = \mathrm{MA}_\theta(a),$$

where the equality of classes is Theorem 29.2.1, and put

$$\ell = \underline{\mathrm{vol}}_\theta(\varphi) > 0, \qquad M = \overline{\mathrm{vol}}_\theta(\varphi) < \infty.$$

Every $u \in \mathcal{E}_\varphi$ satisfies

$$\ell \leq \mu_u(X) \leq M. \tag{31.1}$$

We use the stretching envelopes Q_b of Definition 12.4.1.

31.1 Equations for finite measures

A positive exponent determines a unique solution in the relative class. At exponent zero, the equation determines its positive scalar; normalized potentials may still differ, as in Corollary 30.7.2.

Theorem 31.1.1 *For every nonzero finite non-pluripolar measure μ:*

(1) *There are $u \in \mathcal{E}_\varphi$ and a unique $c > 0$ such that $\mathrm{MA}_\theta(u) = c\mu$.*
(2) *For each $\lambda > 0$ there is a unique $u \in \mathcal{E}_\varphi$ such that $\mathrm{MA}_\theta(u) = \mathrm{e}^{\lambda u}\mu$.*

If $\mu = f\,\mathrm{d}V$ with $0 \le f \in L^p(\mathrm{d}V)$ for some $p > 1$, the solutions in both parts may be taken with $u \sim \varphi$.

These assumptions also give $v_-(\omega) > 0$. Indeed, choose $C > 0$ with $\theta \le C\omega$. Since $\varphi \preceq 0$, Theorem 29.1.1 gives

$$0 < \underline{\mathrm{vol}}_\theta(\varphi) \le \underline{\mathrm{vol}}_{C\omega}(0) = C^n v_-(\omega).$$

31.2 Convergence under a fixed measure

A fixed dominating measure gives a common lower envelope after extraction and then convergence in capacity. The proof follows the semipositive-background construction of Theorem 30.3.1, with the interval (31.1) supplying the mass bounds. Weighted lower semicontinuity is available in the present background by Lemma 6.3.1.

Theorem 31.2.1 *Suppose $u_j \in \mathcal{E}_\varphi$, $u_j \to u \in \mathrm{PSH}(X,\theta)$ in $L^1(X)$, and $\mu_{u_j} \le \nu$ for one finite non-pluripolar measure ν. There is a subsequence with*

$$h = P_\theta\left(\inf_j u_j\right) \in \mathcal{E}_\varphi.$$

For the original sequence,

$$u \in \mathcal{E}_\varphi, \qquad u_j \longrightarrow u \text{ in capacity}, \qquad \mu_{u_j} \rightharpoonup \mu_u \le \nu.$$

Proof We apply the arguments of Lemma 30.3.1 and Theorem 30.3.1, checking their ingredients in the present background. The envelope identity and Hartogs' lemma give $u_j \le \varphi + C$. Finite rooftops and stretching preserve $\mathcal{E}_\varphi$ by Proposition 12.4.2 and Lemma 12.4.1; their masses are at least ℓ. These replace the corresponding semipositive-background facts.

After a common translation, assume $u_j, u \le \varphi \le 0$. Using Lemma 1.23.1, extract once so that smooth-volume almost-everywhere convergence holds and

$$I_j := \int_X |\mathrm{e}^{u_j} - \mathrm{e}^u|\,\mu_{u_j} \le 2^{-j}.$$

For $p_{j,N} = P_\theta(u_j, \dots, u_N)$, disjoint first-contact pieces give

$$\mu_{p_{j,N}} \le \nu, \qquad \int_X |\mathrm{e}^{p_{j,N}} - \mathrm{e}^u|\,\mu_{p_{j,N}} \le 2^{1-j}. \tag{31.2}$$

In the extraction lemma's proof, replace κ by ℓ. For each fixed $B \ge 1$, the stretched roofs $r_{j,N} = Q_B(p_{j,N}, \varphi)$ satisfy

$$\ell \le \mu_{r_{j,N}}(X), \qquad \mu_{r_{j,N}} \le B^n \nu.$$

Its exponential test therefore bounds $\sup_X r_{j,N}$ below, independently of N, once j is sufficiently large. The same prefix-and-tail construction with $B = 2b$ uses only convex combinations and genuine finite rooftops. It supplies every defining b-competitor for

$$h = P_\theta(\inf_j u_j) \in \mathcal{E}_\varphi.$$

Since $h \le u \le \varphi$, upward stability gives $u \in \mathcal{E}_\varphi$.

For continuity, put $p_j = P_\theta(\inf_{k \ge j} u_k)$ and $p = (\lim_j p_j)^*$. Both $p_{j,N} \downarrow p_j$ and $p_j \uparrow p$ have the genuine lower bound h, so converge in capacity by Lemma 1.8.6. All the limits lie in $\mathcal{E}_\varphi$ and $p \le u$. The general-background weighted lower-semicontinuity lemma, applied twice to (31.2), gives

$$\int_X |\mathrm{e}^p - \mathrm{e}^u|\,\mu_p = 0.$$

Now use Theorem 29.3.1 with coefficient zero to get $p = u$. The upper tail envelopes decrease to u by Hartogs' lemma. Squeezing between them and p_j proves capacity convergence.

Finally, the fixed finite dominator gives

$$\mu_{u_j}(\{u_j \le -T\}) \le \nu(\{h \le -T\}) \longrightarrow 0$$

uniformly in j. Theorem 7.6.1 proves weak convergence to $\mu_u \le \nu$. Starting with an arbitrary subsequence gives a further subsequence with both convergences and the same limit. Hence the conclusions hold for the original sequence. □

31.3 Relative moments

A relative first moment prevents a sequence from collapsing and keeps its limits in $\mathcal{E}_\varphi$. Choose a representative whose mass is close to the upper volume of its ordinary type. Comparing rooftop masses with this fixed measure then contradicts collapse. For a real number t, write $t_+ = \max(t, 0)$.

Proposition 31.3.1 *Fix $w \in \mathcal{E}_\varphi$. Suppose $u_j \in \mathcal{E}_\varphi$, $u_j \le w$, and*

$$K := \sup_j \int_X (w - u_j)\, \mu_{u_j} < +\infty.$$

Then $\sup_X u_j$ is bounded below. If $u_j \to u \in \mathrm{PSH}(X, \theta)$ in $L^1(X)$, then $u \in \mathcal{E}_\varphi$.

The same conclusions hold if $u_j \le w$ is replaced by a common upper bound $u_j \le \varphi + C$ and the integral is replaced by $\int_X (w - u_j)_+ \mu_{u_j}$.

Proof We first treat $u_j \le w$. Since $\overline{\mathrm{vol}}_\theta(w) = M$, choose $w' \sim w$ with $\mu_{w'}(X) > M - \ell/2$. Translate w' so that $w' \le w$, and put $p_j = P_\theta(u_j, w')$. These envelopes belong to $\mathcal{E}_\varphi$. Contact support and ordered contact imply

$$\int_X (w' - p_j)\, \mu_{p_j} \le \int_X (w - u_j)\, \mu_{u_j} \le K. \tag{31.3}$$

Indeed, the integrand vanishes on the contact part $\{p_j = w'\}$, and on the remaining contact part $p_j = u_j < w' \le w$.

We will use a consequence of the maximum principle. If $a \in \mathcal{E}_\varphi$, $a \le w'$, and $t > 0$, then

$$\mu_a(\{a > w' - t\}) \le \mu_{w'}(\{a > w' - t\}) + M - \mu_{w'}(X). \tag{31.4}$$

To see this, put $m = \max(a, w' - t)$. Its ordinary singularity type is that of w', so $\mu_m(X) \le M$. On $\{a > w' - t\}$ locality gives $\mu_m = \mu_a$. On the complementary set, ordered contact with $w' - t \le m$ gives $\mu_m \ge \mu_{w'}$. Integration proves the formula, including the equality set in the latter part.

If a subsequence had $\sup_X u_j \to -\infty$, then also $\sup_X p_j \to -\infty$. Splitting the mass of p_j at $w' - s$ and using (31.3) and (31.4), we obtain

$$\ell \le \mu_{p_j}(X) \le \frac{K}{s} + \mu_{w'}(\{p_j > w' - s\}) + M - \mu_{w'}(X).$$

For fixed s, the middle term tends to zero, since w' is finite off a pluripolar set and $\mu_{w'}$ is finite and non-pluripolar. Let $j \to \infty$ and then $s \to \infty$. This gives $\ell \le M - \mu_{w'}(X) < \ell/2$, a contradiction, and proves the first assertion.

Now suppose $u_j \to u$ in L^1. Then $u \le w \preceq \varphi$. Fix $b > 1$ and suppose that $Q_b(u, w)$ is not genuine. Since $w - w'$ is bounded, a constant translation of a competitor shows that $Q_b(u, w')$ is not genuine either. Keep the same p_j and set

$$h_j = Q_b(p_j, w') \in \mathcal{E}_\varphi.$$

We claim that $\sup_X h_j \to -\infty$. Otherwise, since $h_j \le p_j \le w'$, Theorem 1.4.1 would give a subsequence with a genuine L^1 limit h. Passing to the limit in

$$b^{-1}h_j + (1 - b^{-1})w' \le p_j \le u_j$$

first almost everywhere and then by Lemma 1.3.3 gives a competitor for $Q_b(u, w')$, a contradiction.

On the contact set of h_j one has $p_j = b^{-1}h_j + (1 - b^{-1})w'$. Thus $h_j \le w' - bs$ implies $p_j \le w' - s$ there. Contact and (31.3) imply

$$\mu_{h_j}(\{h_j \le w' - bs\}) \le b^n K/s.$$

Apply (31.4) with $a = h_j$ and $t = bs$ to the complementary set. We obtain

$$\ell \le b^n K/s + \mu_{w'}(\{h_j > w' - bs\}) + M - \mu_{w'}(X).$$

First let $j \to \infty$ and then $s \to \infty$. The chosen mass of w' again gives $\ell < \ell/2$, a contradiction. Therefore $Q_b(u, w)$ is genuine for every $b > 1$. The resulting convex inequalities and Theorem 29.1.1 and Proposition 29.1.1 give

$$(1 - b^{-1})^n M \le \overline{\mathrm{vol}}_\theta(u) \le M.$$

Let $b \to \infty$. The upper volumes are equal, hence $u \in \mathcal{E}_\varphi$.

For the last assertion put $q_j = P_\theta(u_j, w)$. Contact gives

$$\int_X (w - q_j)\, \mu_{q_j} \le \int_X (w - u_j)_+ \, \mu_{u_j} \le K.$$

The ordered case bounds $\sup_X q_j$ from below and therefore also $\sup_X u_j$. If $u_j \to u$ in L^1, extract a genuine L^1 limit q of the q_j, using $q_j \le w$. The ordered case gives $q \in \mathcal{E}_\varphi$. Since $q \le u \le \varphi + C$, upward stability gives $u \in \mathcal{E}_\varphi$ as well. □

When the first moment of the difference tends to zero, tail rooftops identify the limit in capacity.

Theorem 31.3.1 *Let $u_j, u \in \mathcal{E}_\varphi$, with $u_j \to u$ in $L^1(X)$. If*

$$\int_X |u_j - u|\, \mu_{u_j} \longrightarrow 0,$$

then $u_j \to u$ in capacity. After extraction, $P_\theta(\inf_{k\geq j} u_k)$ belongs to $\mathcal{E}_\varphi$ for every j.

Proof Start with any subsequence and extract again so that the displayed integrals are at most 2^{-j} and $u_j \to u$ almost everywhere for smooth volume. The model envelope identity and Hartogs' lemma give $u_j, u \leq \varphi + C$ uniformly. For $N \geq j$ let $p_{j,N} = P_\theta(u_j, \dots, u_N)$. Disjoint first-contact pieces give

$$\int_X |p_{j,N} - u|\, \mu_{p_{j,N}} \leq 2^{1-j}. \tag{31.5}$$

Apply the first-moment proposition with reference u and its one-sided version. It bounds $\sup_X p_{j,N}$ from below independently of N. Consequently the decreasing sequence has a genuine limit

$$p_j = P_\theta\left(\inf_{k\geq j} u_k\right),$$

and the same proposition puts p_j in $\mathcal{E}_\varphi$. Its increasing canonical limit p belongs to that class by upward stability, since $p_1 \leq p \leq u \preceq \varphi$. The inequality $p \leq u$ follows from the almost-everywhere convergence of the extracted sequence.

Both $p_{j,N} \downarrow p_j$ and $p_j \uparrow p$ converge in capacity by Lemma 1.8.6, with the genuine lower bound p_1. Apply Lemma 6.3.1 to (31.5), using the bounded weights $\min(1, |p_{j,N} - u|)$ and then $\min(1, |p_j - u|)$. Their capacity convergence follows by first discarding deep sublevels of p_1 and u and then using the Lipschitz function $\min(1, |x - y|)$ on finite values. The result is

$$\int_X \min(1, |p - u|)\, \mu_p = 0.$$

Zero domination gives $p = u$. The upper tail envelopes $(\sup_{k\geq j} u_k)^*$ decrease to u. Squeezing the extracted u_j between these envelopes and p_j proves capacity convergence. Applying this to every initial subsequence proves the conclusion for the original sequence. □

31.3.1 Higher moments in the minimal singularity type

Set $V = V_\theta$ and $\mathcal{E}_V = \mathcal{E}_{\mathrm{env}}(X, \theta, V)$, and write

$$\ell_V = \underline{\mathrm{vol}}_\theta(V) \geq \ell > 0, \qquad M_V = \overline{\mathrm{vol}}_\theta(V) < \infty.$$

The inequalities follow from volume monotonicity and bounded mass. A first moment against the moving Monge–Ampère measure controls a higher moment against each fixed probability ν with $A_\theta(\nu) < \infty$, as defined in Section 1.23.4. A concave obstacle converts the distribution of $V - u$ into this stronger integrability estimate.

Lemma 31.3.1 *Let $u \in \mathrm{PSH}(X, \theta)$ satisfy $u \sim V$ and $u \leq V$. Let $\chi : \mathbb{R}_{\leq 0} \to \mathbb{R}_{\leq 0}$ be increasing and concave, with $\chi(0) = 0$ and $\chi' \geq 1$. Assume χ is $C^{1,1}$ on compact intervals. Set*

$$b = V + \chi(u - V), \qquad p = P_\theta(b).$$

The obstacle is defined off the pole set of V; its value on that set may be taken to be $-\infty$. Then $p \sim V$ and

$$\mu_p \leq \mathbf{1}_{\{p=b\}} \chi'(u - V)^n \mu_u.$$

All measures and contact sets in this formula may be restricted to the complement of any fixed analytic pole barrier for θ.

Proof The difference $u - V$ is bounded, so $V - C \leq b \leq u \leq V$ for some C. Hence $p \sim V$. The obstacle is quasi-continuous: away from sets of arbitrarily small capacity, u and V are finite and continuous, and their difference has a fixed bounded range. The general contact theorem Theorem 13.7.2 therefore applies.

Choose a strict analytic pole barrier $\rho \leq V$ and put $\Omega = X \setminus \{\rho = -\infty\}$. On this open set u, V, p are locally bounded. Let $\tau = \chi^{-1}$; then τ is convex and $0 < \tau' \leq 1$. By Lemma 1.7.6, the function

$$w = V + \tau(p - V)$$

is θ-plurisubharmonic on Ω and satisfies

$$\mathrm{MA}_\theta(w) \geq \tau'(p - V)^n \mu_p.$$

Since $p \leq w \leq V$ on Ω, local removability in Lemma 1.3.7 extends w across the analytic complement, preserving its smooth lower curvature bound.

The inequality $p \leq b$ implies $w \leq u$, with equality on $D = \{p = b\}$. Ordered contact gives

$$\mathbf{1}_D \tau'(p - V)^n \mu_p \leq \mathbf{1}_D \mathrm{MA}_\theta(w) \leq \mathbf{1}_D \mu_u.$$

On D the derivative is $1/\chi'(u - V)$. Contact support for p proves the claim on Ω. Its analytic complement is ignored by all the non-pluripolar measures involved. □

Proposition 31.3.2 *Set $N = 2n + 1$ and $r = 1 + (4N)^{-1}$. For $u \in \mathrm{PSH}(X, \theta)$ with $u \sim V$ and $\sup_X u = 0$, put $B = \int_X (V - u)\,\mu_u$. There is a constant C, depending only on n, ℓ_V, M_V, such that, for every non-pluripolar probability measure ν with $A = A_\theta(\nu) < \infty$,*

$$\int_X (V - u)^r \,\mathrm{d}\nu \le C(1 + A)(1 + B)^{1+1/N}.$$

In particular a uniform bound for the relative first moments on the right gives uniform integrability of these normalized potentials with respect to ν.

Proof Normalization gives $u \le V$. Put $x = V - u \ge 0$ and

$$T = \sup\{t \ge 0 : \mu_u(\{x > t\}) > 0\},$$

with $T = 0$ if this set is empty. Since $u \sim V$, T is finite. Zero domination, applied to $V - t$ for $t > T$, shows that $u \ge V - T$. Thus $x \le T$ off the pole set. If $T \le 2$ the claim is immediate. We may suppose $T > 2$; in particular $B > 0$.

Fix a real number s with $0 < s < T - 2$. For $t \ge 0$, define

$$g_s(t) = \begin{cases} \big((1 + t)^{3/2}\mu_u(\{x > t\})\big)^{-1}, & 0 \le t \le T - s, \\ (1 + t)^{-3/2}, & t > T - s, \end{cases} \qquad F_s(t) = 1 + \int_0^t g_s(a)\,\mathrm{d}a,$$

and set

$$W_s(t) = \int_0^t F_s(a)^{1/N}\,\mathrm{d}a, \qquad \chi_s(-t) = -W_s(t).$$

The tail in the first denominator has a positive lower bound on $[0, T - s]$. Hence F_s is locally Lipschitz, and χ_s has precisely the regularity and concavity required in the lemma.

Tonelli's theorem gives

$$\int_X F_s(x)\,\mu_u = \mu_u(X) + \int_0^T g_s(t)\mu_u(\{x > t\})\,\mathrm{d}t \le 3\max(1, M_V). \tag{31.6}$$

Here $\int_0^\infty (1+t)^{-3/2}\,\mathrm{d}t = 2$; on the second interval the tail is at most M_V. Chebyshev's inequality gives $\mu_u(\{x > t\}) \le B/t$. It follows, by integration first for F_s and then for W_s, that

$$W_s(t) \ge c(1 + B)^{-1/N} t^\beta, \qquad 1 \le t \le T - s, \qquad \beta = 1 + \frac{1}{2N}. \tag{31.7}$$

Indeed $g_s(t) \ge cB^{-1}t^{-1/2}$ for $t \ge 1$ in this interval, so $F_s(t) \ge c(1 + B)^{-1}t^{1/2}$. For $t \ge 2$ integrate this estimate on $[t/2, t]$; for $1 \le t \le 2$ use $W_s(t) \ge t$.

Let $p_s = P_\theta(V - W_s(x))$. The preceding lemma and $W_s(t) \le tF_s(t)^{1/N}$ imply, by Cauchy–Schwarz and $N = 2n + 1$,

$$\begin{aligned}\int_X (V - p_s)^{1/2}\,\mu_{p_s} &\le \int_X x^{1/2} F_s(x)^{(n+1/2)/N}\,\mu_u \\ &\le B^{1/2}\left(\int_X F_s(x)\,\mu_u\right)^{1/2} \le CB^{1/2}.\end{aligned}$$

Write $a_s = \sup_X p_s \le 0$. The definition of V gives $p_s - a_s \le V$, so $V - p_s \ge -a_s$. Since $\mu_{p_s}(X) \ge \ell_V$, the last energy estimate yields $\ell_V\sqrt{-a_s} \le C\sqrt{B}$, and hence $-a_s \le CB$.

The definition of A now gives

$$\int_X W_s(x)\,\mathrm{d}\nu \le \int_X (V - p_s)\,\mathrm{d}\nu \le A - a_s \le A + CB.$$

Combining this with (31.7), and then taking $s \downarrow 0$ for each fixed $1 \le t < T$, proves

$$\nu(\{x > t\}) \le C(1 + B)^{1/N}(A + B)t^{-\beta}.$$

There is no tail for $t \ge T$. The layer-cake formula therefore gives

$$\int_X x^r\,\mathrm{d}\nu = r\int_0^\infty t^{r-1}\nu(\{x > t\})\,\mathrm{d}t \le 1 + C(1 + B)^{1/N}(A + B),$$

because $1 < r < \beta$. This implies the stated bound.

Finally $-u = -V + (V - u)$, where $-V \in L^1(\nu)$ because $A < \infty$. The fixed summand is integrable and the other summands have a uniform $L^r(\nu)$ bound. This proves uniform integrability, without claiming an L^r bound for V itself. □

31.4 General measures in the minimal class

Smooth the measure by the coordinate convolution of Section 1.23.4. Its integral estimates, combined with the higher-moment bound, give convergence against the original measure. The resulting self-measure error then yields capacity convergence, and a first-moment tail estimate identifies the Monge–Ampère limit.

Proposition 31.4.1 *Let ν be a non-pluripolar probability measure with $A_\theta(\nu) < \infty$, and let v_j be its smooth approximations from (1.29). Fix a constant $C > 0$ and let $v_j, v \in \mathrm{PSH}(X, \theta)$. Suppose*

$$v_j \sim V, \qquad \sup_X v_j = 0, \qquad \mu_{v_j} \le C\nu_j, \qquad v_j \longrightarrow v \text{ in } L^1(X).$$

Then $v \in \mathcal{E}_V$, convergence also holds in $L^1(\nu)$ and in capacity, and

$$\mu_{v_j} \rightharpoonup \mu_v.$$

Proof The limit is genuine and normalized, and $v_j, v \le V \le 0$. The smoothing lemma gives the uniform estimate

$$\int_X (-v_j)\, \mu_{v_j} \le C \int_X (-v_j)\, \mathrm{d}\nu_j \le C'. \tag{31.8}$$

In particular the relative first moments $\int_X (V - v_j)\mu_{v_j}$ are bounded. Proposition 31.3.1, with fixed reference V, puts v in $\mathcal{E}_V$.

Proposition 31.3.2, applied to the fixed measure ν, gives uniform integrability of v_j with respect to ν. For each fixed T, the bounded functions $\max(v_j, -T)$ have a common lower curvature bound in a Hermitian background $\Omega \ge \theta$ and converge in smooth-volume L^1 to $\max(v, -T)$. Lemma 1.23.1 gives their $L^1(\nu)$ convergence. Uniform integrability, followed by $T \to \infty$, gives $v_j \to v$ in $L^1(\nu)$. The limit is integrable because $A_\theta(\nu) < \infty$. The truncations need not be θ-plurisubharmonic; they are used only in the larger Hermitian background.

The smoothing lemma now yields

$$\int_X |v_j - v|\, \mu_{v_j} \le C \int_X |v_j - v|\, \mathrm{d}\nu_j \longrightarrow 0.$$

The self-measure convergence theorem Theorem 31.3.1 proves convergence in capacity. Finally (31.8) gives the raw sublevel estimate

$$\sup_j \mu_{v_j}(\{v_j \le -T\}) \le C'/T.$$

Their masses are bounded by M_V. The undamped continuity theorem Theorem 7.6.1 therefore gives the asserted weak convergence. □

31.4.1 Exponential and normalized equations

The smoothing argument solves the equations for measures with finite A_θ. A bounded carrier representation then reduces an arbitrary finite non-pluripolar measure to increasing bounded densities. Their solution measures have one fixed finite dominator, so Theorem 31.2.1 passes to the limit.

Theorem 31.4.1 *For every finite non-pluripolar measure μ with $\mu(X) > 0$, the following assertions hold.*

(1) *For every $\lambda > 0$ there is a unique $u \in \mathcal{E}_V$ such that*

$$\mathrm{MA}_\theta(u) = \mathrm{e}^{\lambda u}\mu.$$

(2) *There are $v \in \mathcal{E}_V$ with $\sup_X v = 0$ and a unique scalar $c > 0$ such that*

$$\mathrm{MA}_\theta(v) = c\mu.$$

Proof First suppose μ is a probability measure and $A_\theta(\mu) < \infty$. Let μ_j be its smooth probability approximations. The minimal ordinary-type L^p theorems Theorems 13.15.1 and 13.15.2 apply to their smooth, nonzero densities.

For the normalized equation they give $v_j \sim V$, $\sup_X v_j = 0$, with $\mu_{v_j} = c_j\mu_j$. The mass interval gives $\ell_V \le c_j \le M_V$. Extract $c_j \to c > 0$ and $v_j \to v$ in $L^1(X)$. The preceding proposition applies, so $v \in \mathcal{E}_V$ and $\mu_{v_j} \rightharpoonup \mu_v$. The right sides converge weakly to $c\mu$, proving the equation.

For the positive-exponent equation take $u_j \sim V$ satisfying $\mu_{u_j} = \mathrm{e}^{\lambda u_j}\mu_j$. Write $a_j = \sup_X u_j$ and $v_j = u_j - a_j$. The mass interval and Jensen's inequality give

$$\ell_V \le \mathrm{e}^{\lambda a_j}, \qquad M_V \ge \mathrm{e}^{\lambda a_j}\int_X \mathrm{e}^{\lambda v_j}\,\mathrm{d}\mu_j \ge \mathrm{e}^{\lambda a_j - \lambda A_\theta(\mu_j)}.$$

Thus a_j is bounded above and below. After extraction $a_j \to a$, $v_j \to v$ in $L^1(X)$, and $\mu_{v_j} \le C\mu_j$. The preceding proposition gives $v \in \mathcal{E}_V$, convergence in $L^1(\mu)$ and weak convergence of the Monge–Ampère measures. Set $u = v + a$. The smoothing lemma identifies the right sides as $\mathrm{e}^{\lambda u}\mu$. This proves the equation and membership in $\mathcal{E}_V$.

The same conclusions hold for any nonzero finite measure whose normalization has finite A_θ: scaling the measure only changes the positive scalar or translates the positive-exponent solution by a constant.

Now let μ be an arbitrary finite non-pluripolar measure of positive mass. The bounded-carrier decomposition Lemma 13.22.1, applied to ω, writes

$$\mu = f\nu, \qquad \nu = \mathrm{MA}_\omega(\psi), \qquad \psi \in \mathrm{PSH}(X,\omega) \cap L^\infty(X), \qquad 0 \le f \in L^1(\nu).$$

For $\mu_k = \min(f,k)\nu$, we have $\mu_k(X) > 0$. By Lemma 1.23.4, its normalization has finite A_θ for each fixed k.

For the normalized equation the preceding case gives $v_k \in \mathcal{E}_V$, $\sup_X v_k = 0$, and $\mu_{v_k} = c_k \mu_k$, with

$$\frac{\ell_V}{\mu(X)} \le c_k \le \frac{M_V}{\mu_1(X)}.$$

Extract $c_k \to c > 0$ and a genuine L^1 limit v of v_k. All their Monge–Ampère measures are bounded by the fixed finite non-pluripolar measure $M_V \mu/\mu_1(X)$. The fixed-dominator theorem Theorem 31.2.1 gives $v \in \mathcal{E}_V$ and weak convergence to μ_v. Since $c_k \mu_k \to c\mu$ in total variation, $\mu_v = c\mu$.

For the positive-exponent equation let $u_k \in \mathcal{E}_V$ solve $\mu_{u_k} = \mathrm{e}^{\lambda u_k} \mu_k$. Exponential order gives $u_{k+1} \le u_k$, because μ_k increases. Also

$$\ell_V \le \mu_{u_k}(X) \le \mathrm{e}^{\lambda \sup_X u_k} \mu(X), \qquad \mu_{u_k} \le \mathrm{e}^{\lambda \sup_X u_1} \mu.$$

The lower bound on the suprema makes the decreasing limit u genuine. The same fixed-dominator theorem puts it in $\mathcal{E}_V$ and gives weak convergence of the left sides. Dominated convergence with respect to v, using the integrable bound $\mathrm{e}^{\lambda \sup_X u_1} f$, gives total-variation convergence of the right sides to $\mathrm{e}^{\lambda u} \mu$.

Apply Corollary 29.3.1 with reference V in both directions to obtain uniqueness of the exponential solution and of the positive scalar. □

If positive volume is assumed for a Hermitian metric, the strict potential of any smooth big background gives positive lower volume of its minimal reference.

Corollary 31.4.1 *If X has both the bounded mass and positive volume properties, then the preceding theorem applies to every smooth big θ.*

Proof Choose $\rho \in \mathrm{PSH}(X, \theta)$ with $\theta_\rho \ge a\omega$ for some $a > 0$. Then $\rho \in \mathrm{PSH}(X, \theta - a\omega)$. Volume addition, followed by monotonicity of singularities, gives

$$0 < a^n v_-(\omega) = \underline{\mathrm{vol}}_{a\omega}(0) \le \underline{\mathrm{vol}}_\theta(\rho) \le \underline{\mathrm{vol}}_\theta(V_\theta).$$

For the addition step the big background is $a\omega$ and the added positive current is $(\theta - a\omega)_\rho$. Bounded mass supplies finiteness of the upper volume. □

31.5 Relative projection and supersolutions

Relative projection turns the minimal-class solutions just constructed into supersolutions in $\mathcal{E}_\varphi$. The small stretches of Lemma 29.4.1 first produce a genuine rooftop with the full upper volume of φ. The same construction works between ordered model references and will allow those references to vary. Contact support then bounds the projected measure by the original measure on the coincidence set.

Let $\eta \in \mathrm{PSH}(X, \theta)$ satisfy $P_\theta[\eta] = \eta \ge \varphi$, and write $\mathcal{E}_\eta = \mathcal{E}_{\mathrm{env}}(X, \theta, \eta)$.

Proposition 31.5.1 *For every $z \in \mathcal{E}_\eta$, the rooftop $P_\theta(\varphi, z)$ is genuine and belongs to $\mathcal{E}_\varphi$.*

Proof First suppose $z \le \eta$; also $\varphi \le \eta \le 0$. For $b > 1$ sufficiently close to one, Lemma 29.4.1 gives a genuine

$$h_b = Q_b(\varphi, \eta) \le \varphi.$$

Set $a = 1 - b^{-1}$. The same-class stretching lemma, now applied in $\mathcal{E}_\eta$, gives $z_{1/a} = Q_{1/a}(z, \eta) \in \mathcal{E}_\eta$. The potential

$$k_a = a z_{1/a} + (1 - a) h_b$$

is below both φ and z. Indeed $z_{1/a} \le \eta$ and $(1 - a)h_b + a\eta \le \varphi$ prove the first bound; $h_b \le \eta$ and $a z_{1/a} + (1 - a)\eta \le z$ prove the second. Thus $p = P_\theta(\varphi, z)$ is genuine. Addition and monotonicity give

$$(1 - a)^n \overline{\mathrm{vol}}_\theta(h_b) \le \overline{\mathrm{vol}}_\theta(k_a) \le \overline{\mathrm{vol}}_\theta(p) \le \overline{\mathrm{vol}}_\theta(\varphi).$$

As $b \downarrow 1$, $a \downarrow 0$, and the first term tends to the last by Lemma 29.4.1. The upper volumes are therefore equal. Since $p \le \varphi$, Theorem 29.2.1 gives $p \in \mathcal{E}_\varphi$.

For general z, the envelope identity gives $z - \sup_X z \le \eta$. Set $S = \max(\sup_X z, 0)$. The preceding case gives $P_\theta(\varphi, z - S) \in \mathcal{E}_\varphi$, and

$$P_\theta(\varphi, z - S) \le P_\theta(\varphi, z) \le \varphi.$$

Upward stability finishes the proof. □

Proposition 31.5.2 *Let $z \in \mathcal{E}_\eta$ and set $Q = P_\theta[\varphi](z)$. Then $Q \in \mathcal{E}_\varphi$, $Q \le z$, and*

$$\mu_Q \le \mathbf{1}_{\{Q=z\}} \mu_z.$$

Proof Put $p_C = P_\theta(\varphi + C, z)$ for $C \ge 0$. The preceding proposition gives $p_0 \in \mathcal{E}_\varphi$. Since $p_0 \le p_C \preceq \varphi$, upward stability gives $p_C \in \mathcal{E}_\varphi$. Its envelope identity, together with $p_C \le z$, yields the uniform bound

$$p_C \le \varphi + \sup_X p_C \le \varphi + \sup_X z.$$

The increasing canonical limit is consequently genuine, belongs to $\mathcal{E}_\varphi$ and is at most z. The monotone convergence is in capacity by Lemma 1.8.6.

Applying Lemma 29.3.1 with $u = \varphi$ and $w = z$ gives

$$\mu_Q \le \mathbf{1}_{\{Q=z\}} \mu_z. \tag{31.9}$$

Corollary 31.5.1 *For every nonzero finite non-pluripolar measure ν, there is $Q \in \mathcal{E}_\varphi$ such that*

$$0 \neq \mathrm{e}^{-Q}\mu_Q \leq \nu.$$

Proof The minimal-reference theorem supplies $z \in \mathcal{E}_V$ with $\mu_z = \mathrm{e}^z\nu$. Apply the proposition with $\eta = V$ to obtain $Q \in \mathcal{E}_\varphi$. On its contact set $Q = z$, so

$$\mu_Q \leq \mathbf{1}_{\{Q=z\}}\mathrm{e}^z\nu \leq \mathrm{e}^Q\nu.$$

The weights are finite off the pole sets and are assigned zero measure there. Thus $\mathrm{e}^{-Q}\mu_Q \leq \nu$. It is nonzero because $\mu_Q(X) \geq \ell > 0$. □

31.5.1 Varying model references

The monotone model bounds of Theorem 29.7.1 provide lower references whose upper volumes approach that of the limit. Projecting finite families to these references makes their rooftops comparable. To preserve the limiting relative class, domination is needed only where the potentials lie below one fixed member of that class. Domination on all of X also gives capacity and measure convergence. Use the comparison d_θ and the classes $\mathcal{M}_\delta(\theta)$ of Definition 29.6.1.

Theorem 31.5.1 *Let $\varphi_j, \varphi \in \mathcal{M}_\delta(\theta)$ for some $\delta > 0$, with $d_\theta(\varphi_j, \varphi) \to 0$. Suppose $u_j \in \mathcal{E}_{\varphi_j}$, $u_j \to u \in \mathrm{PSH}(X, \theta)$ in $L^1(X)$, and $\chi \in \mathcal{E}_\varphi$. If one finite non-pluripolar measure ν satisfies*

$$\mathbf{1}_{\{u_j \leq \chi\}}\mu_{u_j} \leq \nu,$$

then $u \in \mathcal{E}_\varphi$. After extraction and relabeling, the potentials

$$\psi_j = P_\theta\left(\inf_{l \geq j} \varphi_l\right), \qquad h_j = P_\theta\left(\inf_{l \geq j} u_l\right)$$

satisfy $\psi_j \in \mathcal{M}_{\delta/2}(\theta)$ and $h_j \in \mathcal{E}_{\psi_j}$. If $\mu_{u_j} \leq \nu$ on X, the original sequence converges to u in capacity and $\mu_{u_j} \rightharpoonup \mu_u$.

Proof Hartogs' lemma gives a common upper bound for u_j, u, χ; translate them to be nonpositive. The envelope identity gives $u_j \leq \varphi_j$. Since $\varphi_j \to \varphi$ in L^1 by Proposition 29.6.1, it follows that $u \leq \varphi$. Put $\Lambda = \nu + \mu_\chi$. By Lemma 1.23.1, extract so that $u_j \to u$ almost everywhere for smooth volume and

$$I_j := \int_X |\mathrm{e}^{u_j} - \mathrm{e}^u| \, \mathrm{d}\Lambda \leq 2^{-j}.$$

Apply Theorem 29.7.1 to extract further, retaining these estimates, and obtain increasing models $w_j \leq \varphi_j, \varphi$ with lower volume at least $\delta/2$ and $d_\theta(w_j, \varphi) \to 0$. For $l \geq j$, $w_j \leq w_l \leq \varphi_l$. Hence ψ_j is genuine and

$$w_j \leq \psi_j \leq \varphi.$$

The upper bound follows by passing to the L^1 limit in $\psi_j \le \varphi_l$. Envelope order gives $P_\theta[\psi_j] \le \varphi_l$ for all $l \ge j$, so ψ_j is a model. It increases with j, has lower volume at least $\delta/2$, and

$$\overline{\mathrm{vol}}_\theta(\psi_j) \longrightarrow \overline{\mathrm{vol}}_\theta(\varphi). \tag{31.10}$$

Set $g_j = P_\theta[\psi_j](\chi)$. The ordered-reference projection gives $g_j \in \mathcal{E}_{\psi_j}$, $g_j \le \chi$ and $\mu_{g_j} \le \mu_\chi$. Moreover g_j increases with j. For $N \ge j$, consider

$$q_{j,N} = P_\theta(g_j, u_j, \dots, u_N).$$

For each $l \ge j$, $P_\theta[\psi_j](u_l)$ belongs to $\mathcal{E}_{\psi_j}$ by the same projection, now with upper reference φ_l. The finite rooftop of these projections and g_j is a lower competitor for $q_{j,N}$. Since $q_{j,N} \le g_j$, finite-rooftop and upward stability give $q_{j,N} \in \mathcal{E}_{\psi_j}$, and its mass is at least $\delta/2$.

Assign the contact set to disjoint pieces, one for each obstacle. On a piece where $q_{j,N} = u_l$, we have $u_l \le g_j \le \chi$, so the restricted hypothesis controls its measure by ν. The remaining piece is controlled by $\mu_{g_j} \le \mu_\chi$. Consequently $\mu_{q_{j,N}} \le \Lambda$. Put $F_j = \min(\mathrm{e}^u, \mathrm{e}^{g_j})$. On the u_l piece,

$$|\mathrm{e}^{q_{j,N}} - F_j| \le |\mathrm{e}^{u_l} - \mathrm{e}^u|.$$

On the g_j piece, $g_j = q_{j,N} \le u_N$, so the error is $(\mathrm{e}^{g_j} - \mathrm{e}^u)_+ \le |\mathrm{e}^{u_N} - \mathrm{e}^u|$. The two measures in Λ account for both contributions at N. Thus

$$\int_X |\mathrm{e}^{q_{j,N}} - F_j|\, \mu_{q_{j,N}} \le \sum_{l=j}^{N} I_l \le 2^{1-j}. \tag{31.11}$$

Since $F_j \ge F_1 > 0$ outside a pluripolar set, choose $\epsilon > 0$ with $\Lambda(\{F_1 \le \epsilon\}) < \delta/4$. The mass lower bound and domination by Λ give

$$\epsilon\delta/4 \le \int_X F_j\, \mu_{q_{j,N}} \le 2^{1-j} + \Lambda(X)\mathrm{e}^{\sup_X q_{j,N}}.$$

For every sufficiently large j this bounds $\sup_X q_{j,N}$ below uniformly in N. The decreasing limit q_j is genuine. Applying Theorem 31.2.1 in the fixed class $\mathcal{E}_{\psi_j}$, with dominator Λ, gives $q_j \in \mathcal{E}_{\psi_j}$. Now

$$q_j \le h_j \le \psi_j,$$

so $h_j \in \mathcal{E}_{\psi_j}$ by upward stability. Discard the finitely many earlier indices. Since $h_j \le u \le \varphi$, monotonicity and (31.10) identify the upper volumes of u and φ. Thus Theorem 29.2.1 gives $u \in \mathcal{E}_\varphi$.

Suppose now $\mu_{u_j} \le \nu$ on X. The finite rooftops $p_{j,N} = P_\theta(u_j, \dots, u_N)$ are genuine since they lie above h_j. Their disjoint contact pieces give

$$\mu_{p_{j,N}} \le \nu, \qquad \int_X |\mathrm{e}^{p_{j,N}} - \mathrm{e}^u|\, \mu_{p_{j,N}} \le 2^{1-j}.$$

For fixed j, they decrease to h_j. These lower envelopes increase canonically to $h \le u \le \varphi$, and their upper volumes tend to that of φ; hence $h \in \mathcal{E}_\varphi$. Both monotone passages converge in capacity by Lemma 1.8.6. Applying Lemma 6.3.1 twice gives

$$\int_X |\mathrm{e}^h - \mathrm{e}^u|\, \mu_h = 0.$$

Theorem 29.3.1 with coefficient zero gives $h = u$. The upper tail envelopes decrease to u by Hartogs' lemma and the extracted almost-everywhere convergence. Squeezing proves convergence in capacity. Finally,

$$\mu_{u_j}(\{u_j \le -T\}) \le \nu(\{h_1 \le -T\}) \longrightarrow 0$$

uniformly in j. The general-background tail criterion of Theorem 7.6.1 gives weak convergence of the measures. The extraction can start from any subsequence, so both conclusions hold for the original sequence. □

31.6 Weighted measure addition

The sum of two exponentially weighted Monge–Ampère measures can have infinite mass. To realize it by a potential in $\mathcal{E}_\varphi$, we approximate the inputs while keeping a common lower bound in that class. This bound gives capacity convergence even across different ordinary types and localizes the limiting equation away from deep sublevels. Weighted measures are defined off the pole sets and assigned zero mass there.

Lemma 31.6.1 *Let $a_j \in \mathrm{PSH}(X,\theta)$ have a common upper bound and converge in $L^1(X)$ to $a \in \mathrm{PSH}(X,\theta)$ with $a \preceq \varphi$. Suppose there is $h \in \mathcal{E}_\varphi$ such that $h \le a_j$ for every j. If, for some $\lambda > 0$,*

$$\int_X |a_j - a|\mathrm{e}^{\lambda a}\, \mu_{a_j} \longrightarrow 0,$$

then $a \in \mathcal{E}_\varphi$ and $a_j \to a$ in capacity.

Proof By Lemma 1.3.3, the bounds pass to the limit, so $h \le a \preceq \varphi$. Upward stability gives $a \in \mathcal{E}_\varphi$. Start with any subsequence and extract again so that the displayed integrals are at most 2^{-j} and smooth-volume almost-everywhere convergence holds. The finite roofs

$$p_{j,N} = P_\theta(a_j, \dots, a_N)$$

are genuine because h is a competitor. Disjoint first-contact pieces give

$$\int_X \min(1, |p_{j,N} - a|)\mathrm{e}^{\lambda a}\, \mu_{p_{j,N}} \le \sum_{k=j}^{N} \int_X |a_k - a|\mathrm{e}^{\lambda a}\, \mu_{a_k} \le 2^{1-j}.$$

No relative-class assertion about these finite roofs is needed. Their decreasing limits $p_j = P_\theta(\inf_{k\ge j} a_k)$ are genuine and at least h. Almost-everywhere convergence gives $p_j \le a$. Hence p_j and their increasing canonical limit p lie in $\mathcal{E}_\varphi$.

The limits in N and then in j converge in capacity by Lemma 1.8.6, using the genuine lower bound h. Apply weighted lower semicontinuity Lemma 6.3.1 twice to the bounded weights $\min(1, |p_{j,N} - a|)e^{\lambda a}$ and their limits. They converge in capacity: discard deep sublevels of h and a, then use continuity on the remaining finite ranges. We obtain

$$\int_X \min(1, |p - a|)e^{\lambda a}\, \mu_p = 0.$$

The exponential is positive off a pluripolar set, which μ_p ignores. Zero domination therefore gives $p = a$. The upper tail envelopes $(\sup_{k\ge j} a_k)^*$ decrease to a. Squeezing proves capacity convergence along this extraction. The subsequence criterion gives it for the original sequence. □

The following limiting equation is written for exponent one. Rescaling the background and potentials gives the general exponent in Theorem 31.6.1.

Lemma 31.6.2 *Let $u, v, \sigma \in \mathrm{PSH}(X, \theta)$ be nonpositive, with $u, v \preceq \varphi$, and let $t_j \uparrow \infty$. Each of u_j, v_j is either the fixed corresponding potential or its truncation*

$$u_j = \max(u, \sigma - t_j), \qquad v_j = \max(v, \sigma - t_j).$$

Either or both inputs may be fixed. Suppose their Monge–Ampère masses are uniformly bounded. Fix $h \in \mathcal{E}_\varphi$ and let $a_j \in \mathrm{PSH}(X, \theta)$ satisfy

$$\mu_{a_j} = e^{a_j - u_j}\mu_{u_j} + e^{a_j - v_j}\mu_{v_j}, \qquad h \le a_j \le \min(u_j, v_j).$$

If $a_j \to a$ in $L^1(X)$, then $a \in \mathcal{E}_\varphi$, convergence holds in capacity, and

$$a \le \min(u, v), \qquad \mu_a = e^{a-u}\mu_u + e^{a-v}\mu_v.$$

The measure on the right is finite, even when $e^{-u}\mu_u + e^{-v}\mu_v$ has infinite total mass.

Proof The bounds give $h \le a \le \min(u, v) \preceq \varphi$, so $a \in \mathcal{E}_\varphi$. After discarding a finite prefix, we may assume $t_j \ge 0$, so all potentials a_j, a are nonpositive. Let E_j be the intersection of the strict sets $\{u > \sigma - t_j\}$ and $\{v > \sigma - t_j\}$ for those potentials that are actually truncated. A fixed potential imposes no condition. On E_j locality gives $\mu_{u_j} = \mu_u$ and $\mu_{v_j} = \mu_v$. Since $a, a_j \le u_j, v_j$, we have

$$e^a \mu_{a_j} \le e^{(a+a_j)/2}(\mu_{u_j} + \mu_{v_j}).$$

If $x \notin E_j$, one truncated input and its limit are at most $-t_j$, so both $a_j(x), a(x) \le -t_j$. Also $|a_j - a|e^{(a+a_j)/4}$ is bounded by an absolute constant, since $a_j, a \le 0$. Thus

$$\int_X |a_j - a|e^a\, \mu_{a_j} \le \int_X |a_j - a|e^{(a+a_j)/2}(\mu_u + \mu_v) + Ce^{-t_j/2}. \tag{31.12}$$

Here the assumed uniform mass bound is used only in the last term. The fixed measure $\mu_u + \mu_v$ is finite by bounded mass. Lemma 1.23.1, applied to e^{a_j}, e^a, gives convergence in measure of a_j to a for that measure, since a is finite almost everywhere. The integrand in the first term of (31.12) is uniformly bounded and tends to zero in measure. Its integral tends to zero. The preceding lemma now proves capacity convergence. Truncation levels t_j are retained when a subsequence is taken.

To identify the equation, for $T > 0$ choose a continuous nondecreasing function $g_T : [-\infty, 0] \to [0, 1]$, zero on $[-\infty, -2T]$ and one on $[-T, 0]$, with $g_T \uparrow 1$ at finite points. Put $\chi_T = g_T(h)$. We have $h \le u, v$ by passing to the limit in the upper bounds. On $\{\chi_T > 0\}$, both u and v exceed $-2T$. Therefore, once $t_j > 2T$, every truncated input agrees there with its limit, and

$$\chi_T \mu_{a_j} = \chi_T e^{a_j} (e^{-u} \mu_u + e^{-v} \mu_v).$$

Choose a Hermitian form $\Omega \ge \theta$. On the left replace a_j by $\max(a_j, -2T)$ and expand the bounded product as

$$(\theta + dd^c \max(a_j, -2T))^n = \sum_{k=0}^{n} \binom{n}{k} (\Omega + dd^c \max(a_j, -2T))^k \wedge (\theta - \Omega)^{n-k}.$$

The cutoff vanishes at the truncation level. Bounded capacity continuity with the fixed quasi-continuous weight χ_T therefore gives convergence to $\chi_T \mu_a$, after testing against a continuous function. The uncut product may be signed; its terms have uniformly capacity-controlled total variations.

On the right, $\chi_T e^{-u}$ and $\chi_T e^{-v}$ are bounded by e^{2T}. Fixed-measure exponential convergence gives convergence in total variation to $\chi_T e^a (e^{-u} \mu_u + e^{-v} \mu_v)$. Let $T \to \infty$. The limiting right side is at most $\mu_u + \mu_v$, because $a \le u, v$. Both sides ignore the polar set of h, so the cutoff identities prove the global equation and its finiteness. □

31.6.1 Ordinary types and bounded carriers

When one input has ordinary reference type or a bounded carrier measure, first moments control the rooftops of the approximating solutions. These rooftops provide the common lower bound required by the preceding limiting argument. The resulting solution lies below both inputs, so its unweighted measure is finite.

Theorem 31.6.1 *Let $u, v \in \mathcal{E}_\varphi$ and $\lambda > 0$. Suppose either $v \sim \varphi$, or there are a Hermitian metric ω, a bounded $\psi \in \mathrm{PSH}(X, \omega)$ and a finite constant $A > 0$ such that*

$$\mu_v \le A\,\mathrm{MA}_\omega(\psi).$$

There is a unique $a \in \mathcal{E}_\varphi$ satisfying

$$\mu_a = \mathrm{e}^{\lambda(a-u)}\mu_u + \mathrm{e}^{\lambda(a-v)}\mu_v.$$

It satisfies $a \le \min(u, v)$, and hence $\mu_a \le \mu_u + \mu_v$. If both $u, v \sim \varphi$, then also $a \sim \varphi$.

Proof It suffices to prove the result for $\lambda = 1$. Indeed replace (θ, φ, u, v) by $(\lambda\theta, \lambda\varphi, \lambda u, \lambda v)$. Envelopes and relative classes scale with the potentials, and their volumes and Monge–Ampère measures scale by λ^n. A bounded-carrier domination remains such a domination after changing its constant. Dividing the resulting solution by λ gives the desired equation. A common downward translation of u, v and the solution similarly allows us to assume $u, v \le \varphi \le 0$; the envelope identity gives this inequality after a sufficiently large common translation. We restore the translation at the end.

Ordinary minimal singularities. First take $\varphi = V$ and $u, v \sim V$. Truncate only the weights and define the finite nonzero measures

$$\tau_j = \mathrm{e}^{-\max(u,-j)}\mu_u + \mathrm{e}^{-\max(v,-j)}\mu_v.$$

The functions $\max(u, -j)$ need not be θ-plurisubharmonic; they occur only as weights in this step. The minimal-reference theorem supplies $a_j \in \mathcal{E}_V$ with $\mu_{a_j} = \mathrm{e}^{a_j}\tau_j$.

Choose $D \ge \frac{1}{2}\|u - v\|_\infty + n\log 2$, where the difference is taken off the common poles, and put $w = (u + v)/2 - D$. Positivity of mixed products gives

$$\mu_w \ge 2^{-n}(\mu_u + \mu_v) \ge \mathrm{e}^w\tau_j.$$

Exponential order in $\mathcal{E}_V$ gives $a_j \ge w$. As τ_j increases, the same order shows that a_j decreases. The envelope identity gives $a_1 \le V + \sup_X a_1$, so $a_1 \sim V$ by the lower bound w. Consequently

$$\mu_{a_j} \le \mathrm{e}^{a_1-u}\mu_u + \mathrm{e}^{a_1-v}\mu_v \le C(\mu_u + \mu_v).$$

The decreasing limit $a \ge w$ is genuine and has ordinary type V. Fixed-dominator continuity gives weak convergence of the left sides. The right sides converge in total variation by dominated convergence with respect to $\mu_u + \mu_v$. Therefore

$$\mu_a = \mathrm{e}^{a-u}\mu_u + \mathrm{e}^{a-v}\mu_v.$$

Exponential order, now comparing this equality with each input measure, gives $a \le u, v$. This order is asserted after passing to the equation; the truncated weights alone would not give it against u and v.

Two inputs of the same ordinary type. Suppose $u, v \sim \varphi$, and put $u_j = \max(u, V-j)$ and $v_j = \max(v, V-j)$. The preceding step gives solutions $a_j \sim V$ with

$$\mu_{a_j} = \mathrm{e}^{a_j - u_j}\mu_{u_j} + \mathrm{e}^{a_j - v_j}\mu_{v_j}, \qquad a_j \le \min(u_j, v_j).$$

The differences $u_j - v_j$ have the same uniform bound as $u - v$. With D as above, $w_j = (u_j + v_j)/2 - D$ satisfies the opposite equation inequality, and exponential order gives

$$(u+v)/2 - D \le w_j \le a_j \le \min(u_j, v_j).$$

The fixed lower bound $w = (u+v)/2 - D$ has ordinary type φ, hence lies in $\mathcal{E}_\varphi$. Extract a genuine L^1 limit a; it satisfies $w \le a \le \min(u, v)$, so $a \sim \varphi$. The input masses are at most M_V. Lemma 31.6.2, with truncation reference V and lower bound w, proves capacity convergence and the equation. Its use is legitimate although the finite-index solutions have type V and the limit has type φ.

One input of ordinary type. Now let $u \in \mathcal{E}_\varphi$ and $v \sim \varphi$. Set $u_j = \max(u, \varphi - j)$. The second step supplies $a_j \sim \varphi$ with

$$\mu_{a_j} = \mathrm{e}^{a_j - u_j}\mu_{u_j} + \mathrm{e}^{a_j - v}\mu_v, \qquad a_j \le \min(u_j, v).$$

We construct a common lower bound in $\mathcal{E}_\varphi$. Put $\delta = n \log 2 > 0$ and

$$s_j = Q_2(a_j, u_j) \in \mathcal{E}_\varphi, \qquad C_j = \{s_j + u_j = 2a_j\}.$$

Stretching and contact give $s_j \le a_j$ and

$$2^{-n}\mu_{s_j} + 2^{-n}\mathbf{1}_{C_j}\mu_{u_j} \le \mathbf{1}_{C_j}\mu_{a_j}. \tag{31.13}$$

On $D_j = \{s_j \le u - 2\delta\}$, contact implies $a_j \le u_j - \delta$. Since $\mathrm{e}^{-\delta} = 2^{-n}$, the μ_{u_j} term cancels in (31.13). Also $v \ge \varphi - C \ge u_j - C$ for a fixed C. Hence

$$\mathbf{1}_{D_j}\mu_{s_j} \le 2^n \mathrm{e}^C \mathbf{1}_{D_j \cap C_j} \mathrm{e}^{(s_j - u_j)/2}\mu_v. \tag{31.14}$$

All measures being subtracted are finite.

For $N \ge 1$ form the finite roofs

$$R_N = P_\theta(s_1, \dots, s_N), \qquad q_N = P_\theta(u, R_N).$$

They belong to $\mathcal{E}_\varphi$ by the same-class finite-rooftop theorem. On the set $\{R_N \le u - 2\delta\}$, partition contact according to the first index i with $R_N = s_i$. Since $u_i \ge u_N$, (31.14) gives

$$\mathbf{1}_{\{R_N \le u - 2\delta\}}\mu_{R_N} \le 2^n \mathrm{e}^C \mathrm{e}^{(R_N - u_N)/2}\mu_v.$$

The contact pieces are disjoint, so there is no factor N. On $\{q_N \le u - 2\delta\}$ only the contact part $q_N = R_N$ contributes. Since $u \le u_N$ and $t\mathrm{e}^{-t/2} \le 2/\mathrm{e}$ for $t \ge 0$, we obtain

$$\int_X (u - q_N)\,\mu_{q_N} \le 2\delta M + 2^{n+1}\mathrm{e}^{C-1}\mu_v(X).$$

The first term bounds the complementary set, including any contact with u. The first-moment criterion, with the fixed reference u, implies that the decreasing sequence q_N has a genuine limit $h \in \mathcal{E}_\varphi$. For each fixed j and every $N \ge j$, $q_N \le s_j \le a_j$, so $h \le a_j$ for every j.

The a_j now have a genuine L^1 subsequential limit, bounded above by u, v. Apply Lemma 31.6.2, with reference φ, fixed second input v, and this common lower bound h. The input masses are at most M. It proves the required equation in $\mathcal{E}_\varphi$.

A bounded reference measure for the second input. Finally let $u, v \in \mathcal{E}_\varphi$ and $\mu_v \le A\nu$, with $\nu = \mathrm{MA}_\omega(\psi)$ and ψ bounded. Again put $u_j = \max(u, \varphi - j)$. By the preceding step, applied with the inputs interchanged, there are $a_j \in \mathcal{E}_\varphi$ satisfying the same equations and bounds as in the third step.

Let $r = P_\theta(u, v) \in \mathcal{E}_\varphi$. On $\{a_j < r\}$ one has $r \le u_j, v$, so $t\mathrm{e}^{-t} \le 1/\mathrm{e}$ gives

$$\int_X (r - a_j)_+\,\mu_{a_j} \le \mathrm{e}^{-1}\big(\mu_{u_j}(X) + \mu_v(X)\big) \le 2M/\mathrm{e}.$$

The one-sided first-moment criterion gives a uniform lower bound for $\sup_X a_j$, since $a_j \le \varphi$. By Lemma 1.23.4,

$$\sup_{\substack{z \in \mathrm{PSH}(X,\theta) \\ \sup_X z = 0}} \int_X (-z)\,\mathrm{d}\nu < \infty.$$

Applying it to $a_j - \sup_X a_j$ and using the lower bound for the suprema, we obtain

$$\sup_j \int_X (u_j - a_j)\,\mu_v \le \sup_j \int_X (-a_j)\,\mu_v < \infty. \tag{31.15}$$

Keep $s_j = Q_2(a_j, u_j)$, C_j and D_j from the third step. The same cancellation in (31.13), now using only $a_j \le v$, gives

$$\mathbf{1}_{D_j}\mu_{s_j} \le 2^n \mathbf{1}_{D_j \cap C_j}\mathrm{e}^{a_j - v}\mu_v \le 2^n \mu_v.$$

Take one additional rooftop,

$$q_j = P_\theta(s_j, u - 2\delta) \in \mathcal{E}_\varphi.$$

On its contact with $u - 2\delta$ its measure is at most μ_u; on its other contact part $q_j = s_j \le u - 2\delta$, the preceding inequality applies. Thus

$$\mu_{q_j} \le \mu_u + 2^n \mu_v. \tag{31.16}$$

Moreover the energy vanishes on the first contact part. On the other part, the measure is supported on C_j, where $u_j - s_j = 2(u_j - a_j)$. Therefore

$$\int_X (u - 2\delta - q_j)\,\mu_{q_j} \le 2^{n+1} \int_X (u_j - a_j)\,\mu_v.$$

This is uniformly bounded by (31.15). The first-moment criterion with reference $u-2\delta$ prevents collapse of q_j. Extract genuine L^1 limits of a_j and q_j. The fixed-dominator theorem applies to (31.16) and, after a further extraction, gives

$$h = P_\theta\left(\inf_j q_j\right) \in \mathcal{E}_\varphi.$$

Since $q_j \le s_j \le a_j$, this is a common lower bound for the extracted solutions. Lemma 31.6.2 completes the passage to the equation. Its truncation levels are the original indices of the retained subsequence, not newly assigned levels.

In every stage the solution is at most u, v, so its measure is bounded by $\mu_u + \mu_v$. Two solutions have the same exponentially weighted measure, so Corollary 29.3.1 in both directions gives uniqueness, even when that weighted measure has infinite mass. Restoring the common translation and the exponent finishes the proof. □

31.7 General measures with a model reference

The projection produces a nonzero weighted measure below the prescribed datum. Maximize its mass among supersolutions with that same upper bound. If any part of the datum remains, weighted addition increases the mass and contradicts maximality. Density truncation then gives the equation for every finite non-pluripolar measure.

Theorem 31.7.1 *Let μ be a nonzero finite non-pluripolar measure and let $\lambda > 0$. There is a unique $u \in \mathcal{E}_\varphi$ such that*

$$\mathrm{MA}_\theta(u) = \mathrm{e}^{\lambda u}\mu.$$

Proof We first take $\lambda = 1$ and suppose that

$$\mu \le A\sigma, \qquad A > 0, \qquad \sigma = \mathrm{MA}_\omega(\psi), \quad \psi \in \mathrm{PSH}(X,\omega) \cap L^\infty(X).$$

The supersolution corollary gives $Q \in \mathcal{E}_\varphi$ with a finite nonzero measure

$$\nu_0 = \mathrm{e}^{-Q}\mu_Q \le \mu.$$

Consider

$$\mathcal{T} = \{w \in \mathcal{E}_\varphi : \nu_0 \le \mathrm{e}^{-w}\mu_w \le \mu\}.$$

The family contains Q. Exponential order gives $w \le Q$ for every $w \in \mathcal{T}$. Also

$$\ell \le \mu_w(X) \le e^{\sup_X w}\mu(X), \qquad \sup_X w \ge \log \frac{\ell}{\mu(X)}.$$

These bounds prevent collapse of any sequence in $\mathcal{T}$.

Let

$$\alpha = \sup_{w \in \mathcal{T}} \int_X e^{-w}\, \mu_w \le \mu(X),$$

and choose $w_j \in \mathcal{T}$ whose weighted masses tend to α. Extract a genuine L^1 limit w. Since

$$\mu_{w_j} \le e^{\sup_X Q}\mu,$$

the fixed-dominator theorem gives $w \in \mathcal{E}_\varphi$, convergence in capacity, and $\mu_{w_j} \rightharpoonup \mu_w$. Put $\nu_j = e^{-w_j}\mu_{w_j}$. After another extraction, these finite measures converge weakly to a measure ν, and

$$\nu_0 \le \nu \le \mu, \qquad \nu(X) = \alpha.$$

We claim $\mu_w = e^w \nu$. Lemma 1.23.1 gives

$$\int_X |e^{w_j} - e^w|\, d\mu \longrightarrow 0.$$

If f is continuous, replacing e^{w_j} by e^w in $\int_X f e^{w_j}\, d\nu_j$ therefore has error tending to zero. For the remaining integral, approximate the fixed bounded measurable function $f e^w$ in $L^1(\mu)$ by continuous functions. Since $\nu_j, \nu \le \mu$, the approximation error is uniform in j, while the integrals of each continuous approximation converge by weak convergence. Thus

$$\int_X f\, \mu_{w_j} = \int_X f e^{w_j}\, d\nu_j \longrightarrow \int_X f e^w\, d\nu.$$

This proves the claim. The measure ν ignores the poles of w, so $e^{-w}\mu_w = \nu$. Hence $w \in \mathcal{T}$ attains the maximum α.

If $\mu - \nu$ were nonzero, the supersolution corollary would supply $v \in \mathcal{E}_\varphi$ with

$$0 \ne \nu_v = e^{-v}\mu_v \le \mu - \nu.$$

In particular $\mu_v \le e^{\sup_X v}\mu \le A e^{\sup_X v}\sigma$, so the bounded-carrier hypothesis of the addition theorem is satisfied. That theorem gives $a \in \mathcal{E}_\varphi$ with

$$e^{-a}\mu_a = e^{-w}\mu_w + e^{-v}\mu_v = \nu + \nu_v.$$

The right side is between ν_0 and μ, but its mass exceeds α. This contradicts maximality. Therefore $\nu = \mu$, and w solves the equation.

Now let μ be any nonzero finite non-pluripolar measure. The bounded-carrier decomposition writes

$$\mu = f\sigma, \qquad \sigma = \mathrm{MA}_\omega(\psi), \quad \psi \in \mathrm{PSH}(X,\omega) \cap L^\infty(X), \quad 0 \le f \in L^1(\sigma).$$

Set $f_k = \min(f, k)$ and $\mu_k = f_k\sigma$. These measures are nonzero, and the first part gives $u_k \in \mathcal{E}_\varphi$ with $\mu_{u_k} = \mathrm{e}^{u_k}\mu_k$. Exponential order shows that u_k decreases. The uniform bound

$$\sup_X u_k \ge \log \frac{\ell}{\mu(X)}$$

gives a genuine limit u. Moreover $\mu_{u_k} \le \mathrm{e}^{\sup_X u_1}\mu$. Fixed-dominator continuity gives $u \in \mathcal{E}_\varphi$ and weak convergence of these measures. On the right, $\mathrm{e}^{u_k} f_k \to \mathrm{e}^u f$ in $L^1(\sigma)$ by dominated convergence, with dominator $\mathrm{e}^{\sup_X u_1} f$. Thus $\mu_u = \mathrm{e}^u\mu$.

For general $\lambda > 0$, apply the exponent-one result in background $\lambda\theta$, with model $\lambda\varphi$ and datum $\lambda^n\mu$, and divide its potential by λ. Scaling of the class and the Monge–Ampère measure gives the displayed equation. Uniqueness follows by applying Corollary 29.3.1 to the two solutions in both directions. □

Theorem 31.7.2 *For every nonzero finite non-pluripolar measure μ, there are $u \in \mathcal{E}_\varphi$ and $c > 0$ such that*

$$\sup_X u = 0, \qquad \mathrm{MA}_\theta(u) = c\mu.$$

The scalar c is unique and satisfies

$$\frac{\ell}{\mu(X)} \le c \le \frac{M}{\mu(X)}.$$

Proof For each integer $j \ge 1$, solve

$$\mu_{u_j} = \mathrm{e}^{u_j/j}\mu, \qquad u_j \in \mathcal{E}_\varphi,$$

using the preceding theorem. Put $S = \sup_X u_1$ and $w_j = u_1 + (j-1)S$. Then $w_j/j \ge u_1$, and hence

$$\mu_{w_j} = \mathrm{e}^{u_1}\mu \le \mathrm{e}^{w_j/j}\mu.$$

Exponential order at exponent $1/j$ gives $u_j \le w_j$. Define

$$v_j = u_j - \sup_X u_j, \qquad c_j = \mathrm{e}^{(\sup_X u_j)/j}.$$

The upper bound just proved and the positive lower mass imply

$$\frac{\ell}{\mu(X)} \le c_j \le \mathrm{e}^S, \qquad \mu_{v_j} = c_j \mathrm{e}^{v_j/j}\mu \le \mathrm{e}^S\mu.$$

The envelope identity gives $v_j \le \varphi$. Theorem 1.4.1 supplies a subsequence j_k for which $v_{j_k} \to u$ in L^1, with u genuine and $\sup_X u = 0$, and $c_{j_k} \to c > 0$. Fixed-dominator continuity gives $u \in \mathcal{E}_\varphi$ and $\mu_{v_{j_k}} \rightharpoonup \mu_u$.

Lemma 1.23.2 gives $e^{v_j/j} \to 1$ in $L^1(\mu)$, also for this non-semipositive background. Consequently $c_{j_k} e^{v_{j_k}/j_k}\mu \to c\mu$ in total variation. The retained reciprocal exponents are $1/j_k$, so the limit equation is $\mu_u = c\mu$. Integrating it gives the stated interval for c. The scalar comparison in Corollary 29.3.1 gives uniqueness of c. □

For L^p data, Section 31.8 will prove the stronger conclusion $u \sim \varphi$.

31.7.1 Stability under changes of reference

Convergence of the models in d_θ controls the relative class of a limit. For equations, convergence of the data supplies the fixed dominating measures needed after extraction. The positive exponent then identifies the potential; at exponent zero, scalar uniqueness identifies the limiting measure.

Theorem 31.7.3 *Let $\varphi_j, \varphi \in \mathcal{M}_\delta(\theta)$ for some $\delta > 0$, with $d_\theta(\varphi_j, \varphi) \to 0$. Let ν be a finite non-pluripolar measure and let f_j, f be nonnegative nonzero functions in $L^1(\nu)$ with $f_j \to f$ in $L^1(\nu)$. Put $\mu_j = f_j\nu$ and $\mu = f\nu$.*

(1) *For a fixed $\lambda > 0$, the solutions*

$$u_j \in \mathcal{E}_{\varphi_j}, \qquad \mathrm{MA}_\theta(u_j) = e^{\lambda u_j}\mu_j$$

converge in $L^1(X)$ and in capacity to the solution $u \in \mathcal{E}_\varphi$ with datum μ. Their Monge–Ampère measures converge to $\mathrm{MA}_\theta(u)$ in total variation.

(2) *Let $u_j \in \mathcal{E}_{\varphi_j}$ satisfy $\sup_X u_j = 0$ and $\mathrm{MA}_\theta(u_j) = c_j\mu_j$. Then $c_j \to c$, where c is the scalar for reference φ and datum μ, and*

$$\mathrm{MA}_\theta(u_j) \longrightarrow c\mu \quad \textit{in total variation.}$$

The potentials form an L^1 relatively compact family. Every L^1 cluster limit u lies in $\mathcal{E}_\varphi$, has supremum zero and solves $\mathrm{MA}_\theta(u) = c\mu$. Each subsequence converging in L^1 converges to that limit in capacity.

Proof Write $M_0 = \overline{\mathrm{vol}}_\theta(V_\theta)$. Every solution above has mass between δ and M_0. In the first case let $s_j = \sup_X u_j$. Then

$$s_j \geq \lambda^{-1} \log \frac{\delta}{\mu_j(X)},$$

which is a uniform lower bound. To bound s_j above, suppose after extraction that $s_j \to +\infty$. Normalized compactness gives $v_j = u_j - s_j \to v$ in L^1 with $\sup_X v = 0$. The fixed-measure lemma and $f_j \to f$ in $L^1(\nu)$ imply

$$\int_X e^{\lambda v_j} f_j \, d\nu \longrightarrow \int_X e^{\lambda v} f \, d\nu > 0.$$

The limit is positive since $f\nu$ is nonzero and ignores the poles of ν. But the left side equals $e^{-\lambda s_j}\mu_{u_j}(X) \le M_0 e^{-\lambda s_j}$, a contradiction. Thus s_j is bounded and the u_j form an L^1 relatively compact family of genuine potentials.

Start with any subsequence and extract $u_j \to u$ in L^1. Extract again so that $\sum_j \|f_j - f\|_{L^1(\nu)} < \infty$. The function

$$g = f + \sum_j |f_j - f| \in L^1(\nu)$$

dominates all retained f_j. If C is their common upper bound for u_j, then $\mu_{u_j} \le e^{\lambda C} g\nu$. By Theorem 31.5.1, $u \in \mathcal{E}_\varphi$, the potentials converge in capacity and the measures converge weakly to μ_u. On the other hand,

$$e^{\lambda u_j} f_j \longrightarrow e^{\lambda u} f \quad \text{in } L^1(\nu).$$

Indeed, the error from $f_j - f$ is at most $e^{\lambda C}\|f_j - f\|_1$, and Lemma 1.23.1 for the fixed measure $f\nu$ controls the other term. Hence $\mu_u = e^{\lambda u}\mu$, with convergence in total variation. Uniqueness in Theorem 31.7.1 identifies every cluster limit. The subsequence criterion gives all three convergences for the original sequence.

For the second case, integrating the equation gives

$$\frac{\delta}{\mu_j(X)} \le c_j \le \frac{M_0}{\mu_j(X)}.$$

Since $\mu_j(X) \to \mu(X) > 0$, these scalars stay in a compact subinterval of $(0, +\infty)$. Normalized compactness gives, after extraction, $u_j \to u$ in L^1, $\sup_X u = 0$, and $c_j \to c' > 0$. Extract a summable density error as above. Then $\mu_{u_j} \le Cg\nu$ for a fixed C, so Theorem 31.5.1 gives $u \in \mathcal{E}_\varphi$, capacity convergence and weak convergence to μ_u. Since $c_j f_j \to c' f$ in $L^1(\nu)$, the limit equation is $\mu_u = c'\mu$. Scalar uniqueness in Theorem 31.7.2 gives $c' = c$. Thus the whole scalar sequence converges, and so does $c_j\mu_j$ in total variation. Applying the same argument within any L^1 convergent subsequence proves its capacity convergence and the asserted cluster equation. □

31.7.2 Arbitrary weighted sums

General-measure existence now permits addition for arbitrary $u, v \in \mathcal{E}_\varphi$. Truncating the exponential weights gives finite data. A first-moment estimate prevents collapse of the solutions; a rooftop involving their limit localizes the equation before comparison with the inputs is available.

Theorem 31.7.4 *For every $u, v \in \mathcal{E}_\varphi$ and $\lambda > 0$, there is a unique $a \in \mathcal{E}_\varphi$ satisfying*

$$\mu_a = \mathrm{e}^{\lambda(a-u)}\mu_u + \mathrm{e}^{\lambda(a-v)}\mu_v.$$

Moreover $a \le \min(u, v)$ and $\mu_a \le \mu_u + \mu_v$.

Proof It suffices to take $\lambda = 1$, by the scaling already used in Theorem 31.6.1. For positive integers j, let

$$\tau_j = \mathrm{e}^{-\max(u,-j)}\mu_u + \mathrm{e}^{-\max(v,-j)}\mu_v.$$

These are nonzero finite non-pluripolar measures, and they increase with j. Only the weights are truncated; their truncations are not asserted to be θ-plurisubharmonic. Theorem 31.7.1 gives $a_j \in \mathcal{E}_\varphi$ with $\mu_{a_j} = \mathrm{e}^{a_j}\tau_j$. Exponential order implies that a_j decreases.

Set $r = P_\theta(u, v) \in \mathcal{E}_\varphi$. On $\{a_j < r\}$, one has $r \le u, v$ and

$$(r - a_j)\mathrm{e}^{a_j - \max(u,-j)} \le (r - a_j)\mathrm{e}^{a_j - r} \le \mathrm{e}^{-1},$$

with the same inequality for v. Consequently

$$\int_X (r - a_j)_+ \mu_{a_j} \le \mathrm{e}^{-1}\big(\mu_u(X) + \mu_v(X)\big).$$

Also $a_j \le a_1 \le \varphi + \sup_X a_1$. The one-sided first-moment criterion, with fixed reference r, bounds $\sup_X a_j$ below. Thus the decreasing sequence has a genuine L^1 limit a, and the same criterion puts a in $\mathcal{E}_\varphi$.

We identify the equation before comparing a with the inputs. The same-class rooftop

$$q = P_\theta(a, u, v) \in \mathcal{E}_\varphi$$

lies below a_j, a, u, v.

Let χ_T be a continuous nondecreasing cutoff of q, zero for $q \le -2T$, one for $q \ge -T$, and increasing to one as $T \to \infty$ off the pole set. On $\{\chi_T > 0\}$ both inputs exceed $-2T$. Hence, for $j > 2T$,

$$\chi_T \mu_{a_j} = \chi_T \mathrm{e}^{a_j}(\mathrm{e}^{-u}\mu_u + \mathrm{e}^{-v}\mu_v). \tag{31.17}$$

The factors $\chi_T \mathrm{e}^{-u}$ and $\chi_T \mathrm{e}^{-v}$ are bounded by e^{2T}. Since $a_j \downarrow a$ and $a_j \le \sup_X a_1$, dominated convergence against the fixed measure $\mu_u + \mu_v$ gives total-variation convergence of the right sides to the corresponding expression with a.

For the left side, $a_j \to a$ in capacity by Lemma 1.8.6. Choose a Hermitian $\Omega \ge \theta$ and replace a_j by $b_j = \max(a_j, -2T)$ before forming bounded products. The functions b_j are uniformly bounded Ω-plurisubharmonic functions and converge in capacity. The expansion

$$(\theta + \mathrm{dd^c} b_j)^n = \sum_{k=0}^{n} \binom{n}{k} (\Omega + \mathrm{dd^c} b_j)^k \wedge (\theta - \Omega)^{n-k}$$

and bounded capacity continuity apply against the fixed quasi-continuous weight χ_T. On its nonzero set no potential is truncated, so locality gives convergence to $\chi_T \mu_a$. This treats the possibly signed bounded product before inserting the cutoff.

Thus (31.17) holds in the limit. Let $T \to \infty$. Both sides ignore the poles of q, so monotone convergence gives

$$\mu_a = \mathrm{e}^{a-u}\mu_u + \mathrm{e}^{a-v}\mu_v.$$

This also proves finiteness of the weighted right side, since $a \in \mathcal{E}_\varphi$.

For $\varepsilon > 0$, the equation implies

$$\mathbf{1}_{\{u<a-\varepsilon\}}\mu_u \le \mathrm{e}^{-\varepsilon}\mathbf{1}_{\{u<a-\varepsilon\}}\mu_a.$$

Strict domination gives $u \ge a - \varepsilon$, and then $u \ge a$. The same argument gives $v \ge a$, so $\mu_a \le \mu_u + \mu_v$. For two solutions, equality of their exponentially weighted measures and Corollary 29.3.1 in both directions give uniqueness. □

Corollary 31.7.1 *For fixed $\lambda > 0$, write $u \star_\lambda v$ for the solution in the theorem. This operation on $\mathcal{E}_\varphi$ is commutative and associative, and for $c \in \mathbb{R}$,*

$$(u + c) \star_\lambda (v + c) = (u \star_\lambda v) + c, \qquad u \star_\lambda u = u - \lambda^{-1}\log 2.$$

Proof Assign zero measure on poles and put $F_\lambda(u) = \mathrm{e}^{-\lambda u}\mu_u$. The theorem says $F_\lambda(u \star_\lambda v) = F_\lambda(u) + F_\lambda(v)$. By Corollary 29.3.1, the map F_λ is injective. Commutativity and associativity therefore follow from addition of positive measures, including measures of infinite total mass. Finally, $F_\lambda(u + c) = \mathrm{e}^{-\lambda c}F_\lambda(u)$ gives both displayed identities and uniqueness identifies the potentials. □

31.8 Relative boundedness for L^p data

A uniform moment bound on the datum upgrades the relative-class solution to ordinary reference type. The estimate applies directly to $u \in \mathcal{E}_\varphi$: truncating u against $\varphi - j$ would introduce a contact measure that need not be controlled by the datum. Concave obstacles instead retain this domination and give a differential inequality for the depth of $\varphi - u$.

Lemma 31.8.1 *Let $u \in \mathcal{E}_\varphi$, $u \le \varphi$, and let $\chi : \mathbb{R}_{\le 0} \to \mathbb{R}_{\le 0}$ be increasing and concave, with $\chi(0) = 0$. Fix a finite constant $L \ge 1$. Suppose χ is $C^{1,1}$ on compact intervals and*

$$1 \le \chi' \le L < \infty.$$

Then the envelope

$$p = P_\theta(\varphi + \chi(u - \varphi))$$

belongs to $\mathcal{E}_\varphi$, and

$$\mu_p \le \mathbf{1}_{\{p=\varphi+\chi(u-\varphi)\}}\chi'(u - \varphi)^n \mu_u.$$

The obstacle and the contact set are defined off the polar indeterminacy sets, which all these measures ignore.

Proof For $t \le 0$ the derivative bounds give $Lt \le \chi(t) \le t$. Thus

$$Q_L(u, \varphi) \le p \le u \le \varphi.$$

The lower competitor belongs to $\mathcal{E}_\varphi$ by stretching, so p is genuine and belongs to that class by upward stability. The obstacle is finite and quasi-continuous off poles, and the contact theorem applies.

Let $\tau = \chi^{-1}$. It is convex, with $L^{-1} \le \tau' \le 1$. We justify the transformation

$$w = \varphi + \tau(p - \varphi)$$

without assuming any open set on which φ is locally bounded. On a coordinate ball choose a smooth q with $\theta + \mathrm{dd^c} q \ge 0$. The functions

$$p_m = \max(p, q - m), \qquad \varphi_m = \max(\varphi, q - m)$$

are locally bounded θ-plurisubharmonic functions, and $p_m \le \varphi_m$. Lemma 1.7.6 applies to $w_m = \varphi_m + \tau(p_m - \varphi_m)$ and gives

$$\theta_{w_m} \ge 0, \qquad \mu_{w_m} \ge \tau'(p_m - \varphi_m)^n \mu_{p_m}.$$

The scalar function $(s, t) \mapsto t + \tau(s - t)$ on $s \le t$ is nondecreasing in each variable. Consequently w_m decreases as m increases. Since $p \le p_m \le w_m \le \varphi_m$, it has a genuine local limit. Off the poles this limit is exactly w as displayed above. The local limits therefore agree canonically on overlaps and define a global θ-plurisubharmonic function.

On $D_m = \{p > q - m\}$ neither p nor φ is truncated. Thus $w_m = w$, and plurifine locality transfers the product inequality to D_m. These sets exhaust the complement of the poles of p, locally on the ball. All products ignore the discarded polar set, so

$$\mu_w \ge \tau'(p - \varphi)^n \mu_p.$$

The obstacle bound gives $w \le u$, with equality on the contact set of p. Ordered contact, the identity $\tau'(p-\varphi) = 1/\chi'(u-\varphi)$ there, and contact support of μ_p prove the assertion. □

Proposition 31.8.1 *Let μ be a finite non-pluripolar measure. Suppose, for some real $m > 2n+1$,*

$$A_m := \sup_{\substack{v \in \mathrm{PSH}(X,\theta) \\ \sup_X v = 0}} \int_X (-v)^m \, \mathrm{d}\mu < \infty.$$

There is a finite constant C, depending only on $n, \ell, M, m, \mu(X)$ and A_m, such that every $u \in \mathcal{E}_\varphi$ with $\sup_X u = 0$ and $\mu_u \le \mu$ satisfies

$$\varphi - C \le u \le \varphi.$$

Proof The envelope identity gives $u \le \varphi \le 0$. Put $x = \varphi - u$, defined off poles, and $N = 2n+1$. Hölder's inequality gives the uniform first-moment bound

$$B := \int_X x \, \mu_u \le \int_X (-u) \, \mathrm{d}\mu \le \mu(X)^{1-1/m} A_m^{1/m} =: A_1.$$

Let $H(t) = \mu_u(\{x > t\})$ and $T = \sup\{t \ge 0 : H(t) > 0\}$, with $T = 0$ if the set is empty. At this point T may be infinite. If $T < \infty$, zero domination applied to $\varphi - t$ for $t > T$ gives $u \ge \varphi - T$. It is therefore enough to bound T. We may assume $T > 1$ and choose any S with $1 < S < T$.

Define, for $t \ge 0$,

$$g(t) = \begin{cases} ((1+t)^{3/2} H(t))^{-1}, & 0 \le t \le S, \\ (1+t)^{-3/2}, & t > S, \end{cases} \qquad F(t) = 1 + \int_0^t g(s) \, \mathrm{d}s,$$

and set $W(t) = \int_0^t F(s)^{1/N} \, \mathrm{d}s$ and $\chi(-t) = -W(t)$. Since $H(S) > 0$, F is bounded on the entire half-line, although its bound may depend on S and u. It is locally Lipschitz and nondecreasing. Thus χ has bounded derivative, is increasing and concave, and meets the preceding lemma's hypotheses. Also $W(t) \ge t$ and $W(t) \le tF(t)^{1/N}$.

Tonelli's theorem and $H(t) \le M$ give a bound independent of S and u:

$$\int_X F(x) \, \mu_u = \mu_u(X) + \int_0^\infty g(t) H(t) \, \mathrm{d}t \le 3 \max(1, M) =: K.$$

Let $p = P_\theta(\varphi - W(x)) \in \mathcal{E}_\varphi$. On its contact set the preceding lemma bounds its measure by $F(x)^{n/N}\mu_u$. Hence, using $N = 2n + 1$,

$$\begin{aligned}\int_X (\varphi - p)^{1/2}\,\mu_p &\le \int_X x^{1/2} F(x)^{(n+1/2)/N}\,\mu_u \\ &\le \left(\int_X x\,\mu_u\right)^{1/2}\left(\int_X F(x)\,\mu_u\right)^{1/2} \le \sqrt{KB}.\end{aligned}$$

The identity $P_\theta[p] = \varphi$ implies $p - \sup_X p \le \varphi$. Since $p \le \varphi \le 0$, the last integral is at least $\ell\sqrt{-\sup_X p}$. Consequently

$$0 \le -\sup_X p \le KA_1/\ell^2 =: D.$$

Using $W(x) \le \varphi - p \le -p$, and applying the assumed moment bound to $p - \sup_X p$, we obtain

$$\int_X W(x)^m\,\mathrm{d}\mu \le 2^{m-1}\big(A_m + D^m\mu(X)\big) < C_0, \tag{31.18}$$

where we choose $C_0 = 1 + 2^{m-1}(A_m + D^m\mu(X))$. This constant is independent of S and u.

For $0 < t \le S$, Chebyshev's inequality and $\mu_u \le \mu$ give

$$H(t) \le C_0 W(t)^{-m}, \qquad F'(t) \ge \frac{W(t)^m}{C_0(1+t)^{3/2}} \quad \text{for almost every } t.$$

Since $W' = F^{1/N}$, the derivative of $(W')^{N+1}$ is at least

$$\frac{N+1}{NC_0}\,\frac{W^m W'}{(1+t)^{3/2}}.$$

All functions being differentiated are absolutely continuous on compact intervals. Integrating from zero to t, using $W(0) = 0$, gives

$$(W'(t))^{N+1} \ge \frac{N+1}{NC_0(m+1)}\,\frac{W(t)^{m+1}}{(1+t)^{3/2}}.$$

In particular

$$W'(t) \ge c_0(1+t)^{-\gamma}W(t)^\kappa, \quad \gamma = \frac{3}{2(N+1)} < 1, \quad \kappa = \frac{m+1}{N+1} > 1,$$

with $c_0 > 0$ independent of S, u. Integrating the derivative of $W^{1-\kappa}$ on $[1, S]$ yields

$$(\kappa - 1)c_0\int_1^S (1+t)^{-\gamma}\,\mathrm{d}t \le W(1)^{1-\kappa} - W(S)^{1-\kappa} \le 1,$$

because $W(1) \geq 1$. The integral diverges as $S \to \infty$. It follows that every $S < T$ has a common finite upper bound, and hence so does T. The zero-domination argument at the start now gives $u \geq \varphi - C$. The cases $T \leq 1$, including $B = 0$, were already covered. □

Corollary 31.8.1 *Let $0 \leq f \in L^p(X, \mathrm{d}V)$, with $p > 1$ and $\int_X f \,\mathrm{d}V > 0$. For $\lambda > 0$, the solution in Theorem 31.7.1 to*

$$\mathrm{MA}_\theta(u) = \mathrm{e}^{\lambda u} f \,\mathrm{d}V$$

satisfies $u \sim \varphi$. Every normalized solution in $\mathcal{E}_\varphi$ of

$$\mathrm{MA}_\theta(u) = c f \,\mathrm{d}V, \qquad c > 0,$$

also satisfies $u \sim \varphi$. In particular both existence statements can be made in the ordinary singularity type of φ.

Proof Lemma 1.5.1, applied in a Hermitian background dominating θ, gives uniform moments of every finite order against $\mathrm{d}V$. Hölder's inequality therefore implies, for every finite m,

$$\sup_{\substack{v \in \mathrm{PSH}(X,\theta) \\ \sup_X v = 0}} \int_X (-v)^m f \,\mathrm{d}V < \infty.$$

Choose $m > 2n + 1$. For the positive-exponent solution, set $S = \sup_X u$ and $v = u - S$. Then $v \in \mathcal{E}_\varphi$, $\sup_X v = 0$, and

$$\mu_v = \mathrm{e}^{\lambda u} f \,\mathrm{d}V \leq \mathrm{e}^{\lambda S} f \,\mathrm{d}V.$$

The proposition applied to this fixed dominating measure shows that v, and hence u, has ordinary type φ. For the normalized equation apply it to the measure $c f \,\mathrm{d}V$. □

Part X
Problems and historical notes

Chapter 32
Problems in Hermitian pluripotential theory

Let X be a connected compact complex manifold of dimension $n \geq 1$.

A uniform Gauduchon trace bound makes a family of positive currents relatively compact, but its limits can lose the Monge–Ampère mass or singularity data needed for an equation. Even convergence of currents with integrable coefficients and convergence of their traces in total variation can coexist with infinite determinant mass, by Proposition 6.7.1 and Corollary 6.7.1. The envelope class can also contain infinite-mass potentials, by Proposition 15.3.3. Passing from these linear bounds to an exact equation therefore requires additional control at the nonlinear limit.

Strict positivity and preservation of singularities impose further requirements. A positive variational volume must produce one strict current to prove bigness. An envelope realizing divisorial bounds must avoid acquiring extra orders to recover the prescribed type. The examples below test these requirements separately: bounded upper mass, a positive scalar and an exact divisorial profile cannot replace one another.

32.1 Threshold equations and singular limits

The equation problem has two separate steps: construct a subsolution at the optimal coefficient, then pass to a limit without leaving excess Monge–Ampère mass. Fix a Hermitian form h and a smooth positive probability volume form $\mathrm{d}V$. We use the density threshold $b_h(f)$ and root space $\mathcal{D}_h$ of Theorem 13.24.1 and Proposition 13.26.2.

For semipositive big backgrounds, Theorem 30.1.1 solves both the positive-exponent and normalized equations under its model condition and positive lower mass. For arbitrary smooth big backgrounds, Theorems 31.7.1 and 31.7.2 give both equations under bounded mass and positive lower model-reference volume. The minimal ordinary-type L^p theorem in Section 13.15 needs neither metric mass property; under the second regime, Corollary 31.8.1 also recovers the prescribed ordinary model type.

Under bounded mass, Theorems 31.5.1 and 31.7.3 also permit the ordinary model references to converge in d_θ, with a common positive lower-volume bound. For densities converging in L^1 of a fixed non-pluripolar measure, the positive-exponent potentials converge in capacity and their measures in total variation. At exponent zero the scalars and measures converge, while the normalized potentials satisfy the limiting equation along each L^1 convergent subsequence. Identifying a single potential requires an additional uniqueness or selection argument.

Removing these mass hypotheses means replacing their uses in compactness. The semipositive proof constructs a common lower envelope using finite auxiliary contact measures, then obtains the moving-measure limit by capacity convergence and sublevel tails. The arbitrary-background proof uses finite upper relative volumes to construct a rooftop between classes and maximize the weighted measure. In either case the solution measure must be shown finite. The formal existence of the envelopes alone does not supply this step.

32.1.1 Individual density thresholds

The distinction between all measures and one fixed density is essential. Corollary 13.23.1 characterizes solvability for every finite non-pluripolar measure by $v_-(h) > 0$. When it vanishes, Theorem 13.23.1 gives an L^1 probability density, bounded below by a positive constant, which admits no positive scaled subsolution. Such obstructing densities are L^1 dense, although every smooth strictly positive density is solvable. Theorem 25.15.1 realizes this obstruction on a compact threefold.

For an individual nonnegative L^1 density, the bounded-truncation normalization constants decrease to $b_h(f)$. The upper-envelope and damped-continuity argument attains this coefficient with a subsolution, but does not prove an exact equation or finite mass. Smooth tests alone miss this threshold, by Theorem 13.24.2. The Gauduchon formula in Theorem 13.25.1 makes it upper semicontinuous in L^1. Its positive domain is described by density roots in $\mathcal{D}_h$, independently of the Hermitian metric. For $p \in \mathcal{D}_h$, the sum norm $\|p\|_{L^n} + \rho_h(p)$ makes this space Banach. Positive lower mass makes its domain all of L^n; finite upper mass is equivalent to completeness for ρ_h alone, by Propositions 13.25.1, 13.26.2 and 13.26.3.

Problem 32.1.1 Suppose $v_-(h) = 0$. For a nonzero $0 \le f \in L^1(\mathrm{d}V)$ with $b_h(f) > 0$, does there exist $u \in \mathrm{PSH}(X, h)$ such that

$$\mathrm{MA}_h^{\mathrm{ext}}(u) = b_h(f) f \, \mathrm{d}V?$$

Determine the singularity or envelope class in which such an equation should be posed.

When $v_-(h) > 0$, Corollary 30.7.1 gives an exact solution in $\mathcal{E}_{\mathrm{env}}(X, h, 0)$ for every nonzero nonnegative L^1 density. Finite-measure comparison identifies its scalar, and dominated continuity gives capacity convergence along every L^1-convergent

subsequence of normalized greatest exponential solutions. This includes the Hopf example without bounded mass. For L^p densities, $p > 1$, the continuous equation and Proposition 13.24.2 give the answer without positive lower mass.

For the data in Problem 32.1.1, Corollary 13.28.1 gives greatest exponential solutions, convergence of their scalars to $b_h(f)$ and total-variation convergence of their measures. Capacity convergence of the normalized potentials is the missing step. Under the root-tail condition of Proposition 13.26.4, Proposition 13.28.1 quantifies the scalar and measure errors; a rate linear in the exponent needs additional control. Positivity and integrability of f alone do not ensure a positive threshold. Exponential subsolutions when $b_h(f) = 0$, and an extension of the density criterion to arbitrary non-pluripolar measures, are separate questions.

32.1.2 Dual attainment and exact equations

A maximizing Gauduchon current can convert the threshold inequality into equality, provided its regularity or its weighted moments prevent loss in the limit. For density roots in the closed order ideal $\mathcal{D}_h^0$ of Proposition 13.26.4, Theorem 25.9.1 and Proposition 25.9.3 give attainment and represent all supporting probabilities. These probabilities are absolutely continuous, and the trace errors along smooth maximizing sequences tend to zero in total variation by Proposition 25.9.4. Uniform integrability along just one maximizing sequence already gives attainment and convergence of its determinant-weighted measures, by Proposition 25.9.2. That weaker hypothesis does not give the representation of all supporting probabilities.

In dimension at least two, a smooth strictly positive maximizer forces every threshold subsolution to be smooth and exact, with uniqueness modulo constants, and makes the density root smooth and nonnegative; see Corollary 25.9.1. For a closed background, Corollary 25.9.3 characterizes smooth dual attainment by smooth strict positivity of the density. The strict positivity conclusion fails for nonclosed backgrounds on every compact complex manifold of dimension at least two, by Example 25.9.1.

Theorem 25.9.2 also allows a smooth semipositive maximizer which is strictly positive outside a proper analytic set Z. It gives an exact global equation, smoothness off Z and uniqueness of all threshold subsolutions modulo constants. The proof takes the maximum of two subsolutions and uses a Gauduchon integral to eliminate their divisorial difference; no singularity type is prescribed. The positive pole-divisor classes are linearly independent by Corollary 25.9.6. Write $b = b_h(f)$ and, for the exponential parameter $\lambda > 0$, let u_λ be the greatest exponential solution and put $c_\lambda = \exp(\lambda \sup_X u_\lambda)$. In this regime Proposition 25.9.5 gives whole-family normalized L^1 convergence, a fixed lower barrier and, for a fixed finite constant C, the estimate $b \le c_\lambda \le b e^{C\lambda}$. A genuine smaller-background envelope of the threshold solution gives capacity convergence by Corollary 25.9.4; L^p data with $p > 1$ give a continuous solution and uniform convergence by Corollary 25.9.5. Example 25.9.2

has a smooth solution but a nondifferentiable density root on Z. For general L^1 data, automatic genuineness of that envelope and regularity across Z remain open here.

The moment criterion Theorem 25.9.3 uses no regularity of a maximizing current. Uniform first moments of normalized quasi-plurisubharmonic functions against one smooth maximizing sequence give a fixed lower barrier for the exponential solutions. Their normalized cluster limits lie below the greatest normalized threshold subsolution Φ_f. If $P_{\delta h}(\Phi_f)$ is genuine for some $0 < \delta < 1$, the lower-barrier criterion gives subsequential capacity convergence and an exact threshold equation, without either mass property. An entropy bound on the test probabilities supplies both the first moments and the selected-sequence uniform integrability above. By Proposition 14.6.1, such an entropy bound is equivalent to a uniform bound on these probabilities for H_0, the convex dual of the zero-reference normalization defined in Chapter 14. It supplies no smaller-background envelope.

Thus singular-current attainment alone does not settle the equation. At zero lower mass, it is still unknown here whether root tails suffice, or even whether $\mathcal{D}_h^0 = \mathcal{D}_h$. Proposition 25.9.1 characterizes the tail condition by continuity under uniformly bounded logarithmic perturbations tending to zero in measure. This is an order-interval statement, not unrestricted L^1 continuity of the threshold.

Capacity convergence is essential here. The finite atomic curve models of Theorem 15.4.1 converge in L^1 to a smooth positive-mass potential while their non-pluripolar measures vanish. The existence arguments obtain stronger convergence from their measure bounds; a carrier representation alone, even in a prescribed type, does not provide it.

32.1.3 Regularity of singular solutions

Regularity estimates supply the compactness needed to select a limit as the exponential parameter vanishes. For bounded solutions, Theorem 13.5.1 gives continuity wherever a fixed strict barrier is locally bounded. In an arbitrary smooth big background, Theorem 13.17.1 makes a locally Hölder solution with minimal singularities smooth where a strict analytic-singularity barrier and the positive density are smooth. Global L^p control supplies the Hölder premise by Theorem 13.16.1 and Corollary 13.17.1.

For a fixed ordinary type above the strict barrier, Corollary 13.14.1 gives compactness for uniform convergence of differences under common reference bounds and a uniform L^p density bound. The reference need not have minimal singularities, so this compactness does not by itself make the potentials regular. In the minimal type, the scalar and density convergence of Proposition 13.16.1 identifies each normalized exponential cluster equation.

For a compact Kähler manifold and a smooth positive global volume form, Theorem 27.10.1 proves smoothness of the full-mass solution on the ample locus of every big class, including non-nef classes. Smoothness of the density only on this open set does not imply global L^p integrability. On an arbitrary smooth-density open subset, it does not even imply continuity of a bounded minimal solution:

Theorem 13.19.2 gives a compact Kähler counterexample with constant density on a ball. Under local closedness, Theorem 13.19.1 gives a sufficient condition in terms of smooth pointwise boundary traces. The resolved logarithmic types of Theorem 13.20.1 admit Hölder remainders when their residual form is positive. For exceptional minimal poles, Theorems 13.21.1 and 13.21.2 instead give a Hölder remainder on the target and quantify the Jacobian loss in integrability. Neither result covers arbitrary prescribed types or derives a global Hölder bound from density information confined to the regular locus. At a density zero, even smooth data need not give a smooth residual: Proposition 27.11.3 and Example 27.11.3 give a necessary vanishing order and show that it is not sufficient. The corresponding questions beyond these resolved and explicit geometric cases remain open here.

32.1.4 Uniqueness and selection

Zeros of the density can separate regions where the linearized equation is elliptic. The issue is whether a difference of solutions can vary between these regions. Throughout this discussion ω is Hermitian, f is smooth and nonnegative, and potentials are normalized by their supremum.

The counterexample in Theorem 11.6.1 shows that a smooth nonnegative density can have two distinct smooth normalized solutions even on $\mathbb{P}^3$. The further construction in Theorem 11.7.1 has uncountably many smooth normalized solutions and a density positive off two real hypersurfaces. By modifying the background, Theorem 11.8.1 achieves exactly quadratic vanishing on those hypersurfaces. Thus neither zero smooth volume of the zero set nor finite-order vanishing suffices for uniqueness. All the backgrounds are strictly Hermitian and have bounded mass. The degeneration causing nonuniqueness lies in the density and the solution forms.

This places a necessary restriction on any geometric use of the equation. A construction that assigns a canonical potential to arbitrary nonnegative data must specify a selection procedure and prove that it is well defined. Existence in the ordinary minimal singularity type does not provide such a selection. The unique scalar, and the corresponding variational volume for strictly positive data, remain meaningful without uniqueness of every potential. On a compact Hermitian surface, Corollary 11.3.1 proves uniqueness for C^2 potentials and continuous density positive on a dense set, with no differential condition on the metric. In dimension three, Theorem 11.3.1 instead uses a sign condition on $\mathrm{dd^c}\omega$. The cofactor identity in (11.20) and the signed quadratic-difference argument provide two distinct integration mechanisms.

When $\{f > 0\}$ is connected and dense, Theorem 11.4.1 proves that two bounded normalized solutions with the same positive scalar coincide. The result applies to $c f \, \mathrm{d}V$ by absorbing the smooth positive ratio $\mathrm{d}V/\omega^n$ into the density. All the hypersurface counterexamples here have three positive-density components, including the analytic example in Theorem 11.9.1. They do not refute Conjecture 32.6.1, which concerns strictly positive densities e^F with F continuous.

Selection by the exponential equation is settled for every fixed nonzero L^p density, $p > 1$. The norming probabilities in Theorem 14.11.1 select the greatest critical subsolution satisfying their integral normalizations. In a Hermitian background the normalized solutions converge uniformly as the exponent tends to zero; in an arbitrary smooth big background convergence is in $L^1(X)$ and locally uniform off the strict barrier's pole set. The selected limit preserves the symmetries of the data by Proposition 13.18.1. For the projective example, Theorem 13.18.1 identifies it with the explicit symmetric $C^{1,1}$ solution of Proposition 11.6.1. Selection therefore coexists with nonuniqueness of the normalized critical equation. The remaining questions concern merely integrable data and more general singular reference types.

For a nonzero density at positive threshold in dimension at least two, Corollaries 25.9.1 and 25.9.2 give uniqueness of the normalized threshold solution and uniform convergence of the whole normalized positive-exponent family whenever a smooth strictly positive maximizing Gauduchon current exists. The construction in Example 25.9.1 satisfies this condition even though the density vanishes on an open set. Thus the size of the zero set alone does not determine uniqueness; the dual condition also involves the background.

32.2 Positive Gauduchon volume and bigness

The Gauduchon volume tests a background through smooth metrics, whereas bigness asks for one potential making that background strictly positive. Fix a smooth positive probability form μ and retain h. The question is whether positivity of the fixed-density volume of Definition 25.1.1 supplies such a potential at a nef boundary point.

Conjecture 32.2.1 Let θ be a nef smooth real $(1,1)$-background on a compact complex manifold. Then

$$\mathcal{V}_\mu(\theta) > 0 \quad \Longleftrightarrow \quad \theta \text{ is big.}$$

Here μ is any smooth strictly positive probability measure.

The choice of μ does not affect positivity, by Proposition 25.12.1. Taking an infimum over all smooth probability densities would change the question: that infimum vanishes for the big Hermitian backgrounds of Theorem 25.15.1. Bigness implies the stated positivity, since pseudoeffectivity of $\theta - ah$ gives $\lambda_\mu(\theta) \geq a\lambda_\mu(h) > 0$. In dimension one the converse follows from the integral formula. For a closed nef class on a compact Kähler manifold, $\mathcal{V}_\mu$ is its top self-intersection, and the equivalence is the nef-and-big criterion recalled in [Xia]. Theorem 25.11.1 treats a nonclosed endpoint on a compact Kähler manifold when it is a closed nef positive-volume core plus a semipositive form. The remaining implication concerns arbitrary nef smooth backgrounds without that structure.

32.2.1 Collapse of normalized Gauduchon tests

The root volume and the bigness threshold divide the same pairing by different quantities. Positive volume can fail to force bigness only if these denominators separate along tests approaching the threshold. The following calculation isolates that degeneration.

Let θ be a smooth pseudoeffective real $(1,1)$-form. For a Gauduchon metric γ, abbreviate the pairings by

$$A_\theta(\gamma) = \int_X \theta \wedge \gamma^{n-1}, \qquad A_h(\gamma) = \int_X h \wedge \gamma^{n-1}.$$

The threshold $\kappa_h(\theta)$ of Corollary 4.3.1 uses the quotient A_θ/A_h, whereas $\lambda_\mu(\theta)$ uses A_θ/D_μ with the determinant denominator of Definition 25.1.1.

Proposition 32.2.1 (1) *Suppose $n \geq 2$, $\lambda_\mu(\theta) > 0$, and θ is not big. Every sequence of Gauduchon metrics satisfying*

$$A_h(\gamma_j) = 1, \qquad A_\theta(\gamma_j) \longrightarrow 0$$

then satisfies $D_\mu(\gamma_j) \to 0$. Such sequences exist.

(2) *More generally, fix a real number $\delta > 0$. If a sequence with $A_h(\gamma_j) = 1$ satisfies*

$$A_\theta(\gamma_j) \longrightarrow \kappa_h(\theta), \qquad \liminf_j D_\mu(\gamma_j) \geq \delta > 0,$$

then

$$\kappa_h(\theta) \geq \delta\lambda_\mu(\theta).$$

In particular, positive root volume then implies bigness.

Proof If θ is not big, pseudoeffectivity and the threshold criterion give $\kappa_h(\theta) = 0$. Choose test metrics whose ratios approach zero. For $n \geq 2$, rescaling makes $A_h = 1$ without changing either quotient. The definition of the root volume gives

$$\lambda_\mu(\theta)D_\mu(\gamma) \leq A_\theta(\gamma). \tag{32.1}$$

Dividing by the positive root volume proves the first assertion. Applying the same inequality to the sequence in the second assertion and taking lower limits gives $\delta\lambda_\mu(\theta) \leq \kappa_h(\theta)$. If the root volume is positive, the threshold criterion gives bigness.□

There is always a uniform upper bound for D_μ/A_h. Indeed, the smooth equation for h, or (32.1) with $\theta = h$, gives

$$D_\mu(\gamma) \leq \lambda_\mu(h)^{-1}A_h(\gamma).$$

The missing estimate in the converse direction is a lower bound along suitable threshold-minimizing sequences. Compactness of normalized potentials does not provide compactness of these Gauduchon metrics.

In dimension one there is no such normalization issue. The root volume is $\int_X \theta$, and $\kappa_h(\theta) = \int_X \theta / \int_X h$. Positive volume and bigness are therefore equivalent by the same threshold criterion.

A Gauduchon equation can replace the volume density of a test while retaining its Aeppli class. This preserves the numerator when θ is closed. For nonclosed θ, integration by parts leaves derivatives of the background, so such a deformation must also control the change in the pairing. This is the additional estimate needed to use the smooth equation against collapse.

32.3 Closed positivity and geometric equations

A strict closed current places the manifold in Fujiki's class, whereas a strict nonclosed current preserves the torsion of its background. This distinction connects the volume-to-bigness question to geometric structure and limits what a scalar potential equation can prove. The external geometric inputs used below have their exact scopes in Section 33.18.

32.3.1 Nef classes, bigness and the Fujiki class

The existence of a closed Kähler current characterizes Fujiki's class $\mathcal{C}$. The transcendental Grauert–Riemenschneider conjecture asks whether a closed nef real $(1,1)$-class α with

$$\int_X \alpha^n > 0$$

is big. Under these hypotheses this is equivalent to $X \in \mathcal{C}$; the exact statements and sources are recorded in Section 33.18.

The mass obstruction is precise. Under the same nefness and positive top-intersection hypotheses, one has

$$\alpha \text{ is big} \quad \Longleftrightarrow \quad v_+(h) < \infty$$

for a Hermitian metric h. There is also a characterization of $\mathcal{C}$ by bounded mass and positivity of an upper class volume for some closed Bott–Chern class. That volume differs from $\mathcal{V}_\mu$; both results are specified in Section 33.18.

The Hopf threefold tests the mass hypothesis but does not refute this conjecture: a counterexample would also need a closed nef class with positive top self-intersection. A proof must explain why that closed positivity forces bounded mass in the geometry at hand.

The elliptic threefold of Theorem 25.15.1 supplies the complementary test: it has bounded mass, but no big closed class. Its entire nef cone is the nonnegative span of two smooth rank-one classes, and every top self-intersection is zero, by Proposition 27.9.1.

Hence this lower-mass example also leaves the geometric conjecture untouched. The exact boundary calculation in Proposition 27.9.4 describes its degeneration without identifying a nonclosed metric with a closed class.

The volume of this book separates two possible steps:

$$\int_X \alpha^n > 0 \quad \overset{?}{\Longrightarrow} \quad \mathcal{V}_\mu(\alpha) > 0 \quad \overset{?}{\Longrightarrow} \quad \alpha \text{ is big.} \tag{32.2}$$

Neither arrow is established here in the unrestricted compact complex setting. The second is the closed special case of Conjecture 32.2.1. On a compact Kähler manifold both follow from the usual nef volume formula and the classical nef-and-big criterion. Without closedness, positivity of $\mathcal{V}_\mu$ cannot characterize $\mathcal{C}$: every Hermitian metric h already satisfies $\mathcal{V}_\mu(h) > 0$, on every compact complex manifold.

Problem 32.3.1 Let α be closed and nef with $\int_X \alpha^n > 0$. Prove a positive lower bound for $\lambda_\mu(\alpha)$, and rule out the collapse in Proposition 32.2.1 for a sequence minimizing the bigness threshold of α.

The relevant test metrics can be normalized by $\int_X h \wedge \gamma^{n-1} = 1$. A positive lower bound for $D_\mu(\gamma)$ along one threshold-minimizing sequence would finish the second step. The Gauduchon equation can change the volume density while retaining an Aeppli class. Since α is closed, it also retains the numerator paired against α. The missing estimate is control of the trace normalization and determinant denominator after that change. Merely prescribing a volume density does not give this estimate.

32.3.2 Positive lower volume and concentration

Positive lower mass is an endpoint integrability property of the Gauduchon tests. It is independent of finite upper mass: the threefold in Theorem 25.15.1 has $v_-(h) = 0$ for every Hermitian metric and admits a balanced metric with finite upper mass, while the normalized Hopf metric has $v_- = 6$ and $v_+ = \infty$.

For $n \geq 2$, use the normalized current cone $\mathcal{G}_h$ and determinant measure of (25.4), writing

$$\Gamma_0(T) = k_T \, \mathrm{d}V, \qquad T \in \mathcal{G}_h.$$

By Theorem 25.8.2,

$$v_-(h) > 0 \quad \Longleftrightarrow \quad \sup_{T \in \mathcal{G}_h} \|k_T\|_{L^{n/(n-1)}(\mathrm{d}V)} < \infty. \tag{32.3}$$

The exact supremum is $v_-(h)^{-1/n}$, including the value $+\infty$. Uniform estimates for every smaller exponent $1 < q < n/(n-1)$ hold without either volume property. The lower-mass example violates the endpoint already for smooth strictly positive tests. Thus positivity, $\mathrm{dd^c}T = 0$ and normalized trace alone do not control concentration at that exponent.

Problem 32.3.2 Find geometric conditions beyond the metric identities in Corollary 25.14.1 that imply the endpoint estimate in (32.3), and identify how they exclude the lower-mass degeneration of Section 25.15.

Corollary 25.14.1 proves the endpoint bound when a metric has its first two powers dd^c-closed, including all compact Kähler manifolds and all compact complex surfaces. For a fixed smooth strictly positive density, Theorem 25.6.1 gives one smooth maximizing current with bounded determinant density. However, Proposition 25.14.1 produces maximizing sequences in fixed Kähler classes with divergent moments at every higher exponent. Exact maximizers and maximizing sequences therefore require different estimates.

The balanced condition and bounded upper mass are insufficient, even together. The explicit identity (25.25) relates the failing lower bound to a vertical Dirichlet integral on an elliptic fibration. Meanwhile the horizontal curvature forms α, β of Section 25.15 have a positive exact product $\alpha \wedge \beta$, which excludes pluriclosed metrics and closed Kähler currents. These calculations supply geometric information about this example without giving a classification of all lower-mass degeneration.

The equation consequence is exact for a Hermitian background: Corollary 13.23.1 characterizes positive lower mass by solvability for all finite non-pluripolar measures. When it vanishes, even a positive L^1 density bounded away from zero can admit no positive scaled subsolution. The remaining equation questions concern specified measures or additional geometric hypotheses, and separately the effect of the exponential weight.

32.3.3 Primitive elliptic bundles

On a primitive elliptic bundle, the variation of top-degree mass is a vertical Dirichlet integral. This gives an exact upper mass and a positive lower bound for separated potentials. The lower mass over all potentials remains a geometric question: the collapse construction over a product of curves does not transfer directly to a base with no curves.

Let (B, η) be a connected compact Kähler surface and let $L \to B$ be a holomorphic line bundle such that

$$\int_B c_1(L) \wedge \eta = 0.$$

Choose a smooth Hermitian metric on L with primitive curvature. To do so, start with any metric: the trace of its curvature has mean zero, and the scalar Poisson equation supplies a smooth change of metric cancelling that trace. Write $\pi : L \to B$ for the bundle projection and w for a nonzero fiber vector, whose norm is taken with this metric. Let c be the smooth closed curvature representative of $-c_1(L)$, so $\eta \wedge c = 0$. On the complement $L^\times$ of the zero section, put

$$t = \log |w|^2, \qquad dd^c t = \pi^* c.$$

We use the same symbol π for the induced quotient projection below. For $q > 1$ form the quotient

$$X = L^{\times}/\langle w \mapsto qw\rangle, \qquad \pi : X \longrightarrow B, \qquad T = 2\log q, \qquad A = \int_B \eta^2.$$

The action increases t by T. It is free and properly discontinuous, and the compact annulus $1 \le |w| \le q$ meets every orbit. Hence X is a connected compact complex threefold. For a trivializing open set $U \subset B$, a local holomorphic frame of L identifies the fibration with $U \times (\mathbb{C}^*/q^{\mathbb{Z}})$.

Both $\mathrm{d}t$ and $\mathrm{d}^{\mathrm{c}}t$ descend, and we set

$$D = \mathrm{d}t \wedge \mathrm{d}^{\mathrm{c}}t, \qquad \omega = \pi^*\eta + D.$$

The rank-one form D is semipositive and positive on the vertical line. Its kernel projects isomorphically onto the tangent space of B, so ω is Hermitian.

Proposition 32.3.1 *The metric ω is balanced and satisfies*

$$\mathrm{dd}^{\mathrm{c}}\omega = -\pi^*(c^2) \ge 0, \qquad \int_X \omega^3 = 3TA.$$

For every smooth real u,

$$\int_X (\omega + \mathrm{dd}^{\mathrm{c}}u)^3 = 3TA - 3\int_X \mathrm{d}u \wedge \mathrm{d}^{\mathrm{c}}u \wedge \big(-\pi^*(c^2)\big). \tag{32.4}$$

In particular $v_+(\omega) = 3TA$, and the canonical bundles satisfy $K_X \simeq \pi^*K_B$. If $c_1(L) \neq 0$ in real cohomology, X admits neither a pluriclosed metric nor a closed Kähler current. If $c_1(L) = 0$ in real cohomology, the chosen ω is Kähler.

Proof Write $a = \mathrm{d}t$ and $b = \mathrm{d}^{\mathrm{c}}t$. Then $\mathrm{d}a = 0$, $\mathrm{d}b = \pi^*c$, $\mathrm{d}^{\mathrm{c}}a = -\pi^*c$ and $\mathrm{d}^{\mathrm{c}}b = 0$. Consequently

$$D^2 = 0, \qquad \mathrm{d}D = -a \wedge \pi^*c, \qquad \mathrm{dd}^{\mathrm{c}}D = -\pi^*(c^2),$$

$$\mathrm{d}(\omega^2) = -2\pi^*(\eta \wedge c) \wedge a = 0.$$

The eigenvalues of c relative to η are λ and $-\lambda$, so $c^2 = -\lambda^2\eta^2$. This proves the asserted sign.

Fix an elliptic fiber $E = \pi^{-1}(b_0)$, where $b_0 \in B$. In a logarithmic fiber coordinate $x+\mathrm{i}y$, the real periods are $\log q$ and 2π, while $t = 2x+$ a constant and $D|_E = \pi^{-1}\mathrm{d}x\wedge\mathrm{d}y$. Thus $\int_E D = T$. Since $\omega^3 = 3\pi^*(\eta^2) \wedge D$, the volume formula follows.

To compute the mass variation, expand the cube in (32.4). The terms with one and three factors $\mathrm{dd}^{\mathrm{c}}u$ integrate to zero: ω^2 is closed, and $\mathrm{dd}^{\mathrm{c}}u$ is exact and closed. For the remaining term, put $S = \mathrm{dd}^{\mathrm{c}}\omega$. Integration by parts gives

$$\int_X (\mathrm{dd}^{\mathrm{c}}u)^2 \wedge \omega = \int_X u\,\mathrm{dd}^{\mathrm{c}}u \wedge S = -\int_X \mathrm{d}u \wedge \mathrm{d}^{\mathrm{c}}u \wedge S,$$

where the last equality uses $\mathrm{d}S = 0$. This proves the identity. Its last integral is nonnegative, and $u = 0$ attains the upper bound. The equality of smooth and bounded mass extrema in Theorem 25.12.1 proves the assertion about v_+.

If c is exact, Stokes gives $\int_B c^2 = 0$. The pointwise identity $c^2 = -\lambda^2\eta^2$ then makes $c = 0$, and ω is closed. Otherwise $S = -\pi^*(c^2)$ is nonzero, smooth and semipositive. For a pluriclosed metric γ, Stokes would give

$$0 < \int_X \gamma \wedge S = \int_X \omega \wedge \mathrm{dd^c}\gamma = 0.$$

For a closed positive current $R \geq \varepsilon\omega$, it would likewise give $0 < \int_X R \wedge S = \int_X \omega \wedge \mathrm{dd^c} R = 0$. All pairings are with smooth forms.

Finally, let w be a local nonzero holomorphic fiber coordinate and σ a local holomorphic two-form on B. The form $\pi^*\sigma \wedge \mathrm{d}w/w$ is invariant under the quotient action. Changing the line-bundle frame adds to $\mathrm{d}w/w$ a horizontal one-form, whose wedge with σ vanishes. These identifications have the transition functions of K_B and prove $K_X \simeq \pi^* K_B$. □

For separated potentials, positivity bounds both the amplitude and the downward slope of the vertical derivative. The sharp variance bound under these constraints determines how much mass such a potential can lose. For $M, T > 0$, set

$$F(M,T) = \begin{cases} T^2/12, & T \leq 2M, \\ M^2 - 4M^3/(3T), & T \geq 2M. \end{cases}$$

Lemma 32.3.1 *If a smooth real T-periodic function p has mean zero, $|p| < M$ and $p' > -1$, then*

$$\frac{1}{T}\int_0^T p^2 \,\mathrm{d}s < F(M,T).$$

The supremum of the left side over this class is $F(M,T)$.

Proof For $p = 0$ the inequality follows from $F(M,T) > 0$. Otherwise set $a = \max p > 0$, $-b = \min p < 0$ and $r = a + b$. On a forward interval I from a maximum to a subsequent minimum, integration of $-p' < 1$ gives $r < |I| \leq T$; also $r < 2M$. The function $G(s) = (a - s)(s + b)$ is nonnegative on the range of p. Since $-p' < 1$, integration on I gives

$$abT - \int_0^T p^2 \,\mathrm{d}s = \int_0^T G(p)\,\mathrm{d}s \geq -\int_I G(p)p' \,\mathrm{d}s = \frac{r^3}{6}.$$

Since $ab \leq r^2/4$, the average square is at most $r^2/4 - r^3/(6T)$. This expression increases strictly for $0 < r < T$. Its value at $\min\{2M, T\}$ is $F(M,T)$.

For sharpness when $T \geq 2M$, use equal constant intervals at M and $-M$, each of length $(T - 2M)/2$, joined by a descent of slope -1 and length $2M$. Close the period by an upward jump. When $T \leq 2M$, use $p_0(s) = T/2 - s$ for $0 \leq s < T$, with its upward jump at the join. Both profiles have mean zero, satisfy $|p_0| \leq M$ and $p_0' \geq -1$

distributionally, and have average square $F(M,T)$. Convolution with a nonnegative periodic approximate identity preserves the mean and the weak derivative bound. Multiplication by $1-\varepsilon$ makes both inequalities strict. As the convolution scale and ε tend to zero, the resulting functions converge to the profile in L^2. This also treats $T = 2M$. □

Proposition 32.3.2 *Suppose $c_1(L) \neq 0$ in real cohomology, and put*

$$\kappa = -\frac{\int_B c^2}{A} > 0.$$

Every smooth potential $u = f(t) + \pi^* v$ with f of period T, $v \in C^\infty(B,\mathbb{R})$ and $\omega + \mathrm{dd^c} u > 0$ satisfies

$$\int_X (\omega + \mathrm{dd^c} u)^3 > 3TA\big(1 - \kappa F(\kappa^{-1/2}, T)\big) > 0.$$

Let $\lambda \geq 0$ be the nonnegative eigenvalue function of c relative to η, so $c^2 = -\lambda^2\eta^2$, and put $M_0 = 1/\max_B \lambda$. For the subclass $v = 0$, the exact mass infimum is

$$3TA\big(1 - \kappa F(M_0, T)\big) > 0.$$

It is approached by smooth strictly admissible radial potentials and is not attained in that class.

Proof Put $p = f'$. The chain rule gives

$$\omega + \mathrm{dd^c} u = \pi^*(\eta + pc + \mathrm{dd^c} v) + (1 + p')D.$$

Positivity on the vertical line implies $p' > -1$. Every base point sees every value of t modulo T, so restriction to $\ker \partial t$ makes $\eta + pc + \mathrm{dd^c} v$ a Kähler form on B for each value of p. Its square has integral

$$\int_B (\eta + pc + \mathrm{dd^c} v)^2 = A(1 - \kappa p^2) > 0.$$

Here primitivity removes the mixed intersection and Stokes removes the exact terms. Hence $|p| < \kappa^{-1/2}$.

Only the vertical derivative enters (32.4). For a periodic function F_0, fiber integration gives $\int_E F_0(t)D = \int_0^T F_0(s)\,\mathrm{d}s$, and therefore

$$\int_X (\omega + \mathrm{dd^c} u)^3 = 3TA\left(1 - \frac{\kappa}{T}\int_0^T p^2\,\mathrm{d}s\right).$$

Apply Lemma 32.3.1 with $M = \kappa^{-1/2}$. The formula for F gives $F(M,T) < M^2$, proving the positive bound.

For $v = 0$, diagonalizing c against η shows that the horizontal positivity condition is exactly $|p| < M_0$. Together with $p' > -1$ it also suffices for positivity of the full

form. Moreover κ is the η^2-average of λ^2, so $\kappa \le M_0^{-2}$. The lemma proves the claimed positive infimum and nonattainment. Its smooth approximating functions have mean zero and hence periodic primitives f_ε. The potentials $f_\varepsilon(t)$ prove sharpness. □

The first bound is not asserted to be sharp for the separated class on an arbitrary base. Positivity of the class $[\eta] + p[c]$ alone does not supply one function v that makes all of its indicated representatives positive. In the earlier $\mathbf{P}^1 \times \mathbf{P}^1$ example, $A = 2$, $\kappa = 1$ and $T = 4\pi$; the bound is 8 and is sharp.

A base with no curves removes the horizontal curve averages used in the known collapse example. The following construction uses the K3 period theorem and Kähler-cone description [Huy16, Theorems 7.4.1 and 8.5.2 in the author draft].

Example 32.3.1 There is a K3 surface B with $\mathrm{Pic}(B) = \mathbb{Z}[L]$, $c_1(L)^2 = -4$, and a Kähler form η orthogonal to $c_1(L)$. It has no compact curves. The associated X has trivial canonical bundle, a balanced metric of finite upper mass, and no pluriclosed metric or closed Kähler current. It has no divisors, and every compact irreducible curve is an elliptic fiber.

Proof In a hyperbolic plane of the K3 lattice choose a basis e, f with $e^2 = f^2 = 0$ and $e \cdot f = 1$, and put $\ell = e - 2f$. This is primitive and has square -4. Choose a period in $\ell^\perp$ outside the hyperplanes perpendicular to integral vectors not in $\mathbb{Z}\ell$. Such a period exists by Baire's theorem: the period domain in $\ell^\perp$, of signature $(3, 18)$, is nonempty and each excluded hyperplane is proper. The period theorem and the integral $(1, 1)$-class description give the asserted Picard group. An irreducible curve has nonzero class by its positive Kähler area. Its possible square would be $-4m^2 < -2$, contradicting adjunction. Thus there are no curves.

The orthogonal complement of ℓ in $H^{1,1}(B, \mathbb{R})$ has signature $(1, 18)$. Choose a positive-square class there, with sign in the positive component containing Kähler classes. As there are no rational curves, the Kähler-cone theorem makes it Kähler. Choose η in that class. Proposition 32.3.1 gives the metric and canonical-bundle assertions.

If a divisor $Z \subset X$ dominated B, the projection of its integration current would be $\pi_*[Z] = m$ for a positive generic degree m. Stokes and the projection formula would then give

$$0 = \int_X [Z] \wedge \mathrm{dd^c}\omega = m \int_B (-c^2) > 0.$$

Thus no divisor dominates B. The proper image of any other irreducible divisor would be a curve or a point. The base has no curves, and a fiber has dimension one, so both alternatives are impossible. The proper image of an irreducible compact curve is likewise a point; the curve is therefore an entire elliptic fiber. □

Problem 32.3.3 Determine the lower Monge–Ampère mass of the primitive elliptic bundles above, in particular those of Example 32.3.1. Does it vanish whenever the real Chern class is nonzero?

The estimates for separated potentials do not answer this question. The known collapse on $\mathbf{P}^1 \times \mathbf{P}^1$ leaves that class of potentials. On the K3 example every compact

curve is vertical, so averaging such curves cannot supply the horizontal directions used in Lemma 25.15.2. A proof of collapse would need a different construction, while positive lower mass would require a bound for all admissible potentials or all normalized Gauduchon tests.

32.3.4 Null loci and the singularities of the Chern–Ricci flow

Return to the general manifold X of dimension n. At a finite maximal time τ of the Chern–Ricci flow, the limiting background is nef. To identify its regular region by a null locus, one first needs bigness. The dual currents detect failure of strict positivity, but their determinant degeneration does not by itself locate an analytic exceptional set. Write ω_0 for the initial metric and $\theta_\tau = \omega_0 - \tau \operatorname{Ric}(\omega_0)$, with the Ricci convention of Corollary 25.17.1.

For a big smooth background θ, the non-Hermitian locus $E_{\mathrm{nH}}(\theta)$ is the intersection of the singular sets of its strict currents with *almost analytic singularities*; one such current attains the intersection. Here the potential is locally a positive multiple of the logarithm of an ideal norm up to a bounded remainder, and that remainder becomes smooth after subtracting the divisorial logarithms on a suitable log resolution. These are the precise singularities used by the theorem recorded in Section 33.18.

For nef and big θ, define

$$\operatorname{Null}(\theta) = \bigcup_{\substack{V \subset X \text{ irreducible analytic} \\ \dim V > 0,\ \theta|_V \text{ not big}}} V.$$

Bigness on a singular subvariety has its intrinsic analytic meaning. The qualitative equality $E_{\mathrm{nH}}(\theta) = \operatorname{Null}(\theta)$ holds for nef and big smooth backgrounds on arbitrary compact complex manifolds. The numerical criterion in Section 33.18 has additional hypotheses: bounded mass on X and positive lower volume on its positive-dimensional submanifolds.

The known flow theorem recorded in Section 33.18 identifies finite-time Chern–Ricci singularities with the null locus under positive limiting lower volume, for arbitrary initial metrics in dimension two, and for Guan–Li initial metrics in higher dimensions. Here Guan–Li means $\mathrm{dd^c}\omega_0 = 0$ and $\mathrm{d}\omega_0 \wedge \mathrm{d^c}\omega_0 = 0$. These hypotheses must not be silently replaced by positivity of $\mathcal{V}_\mu$.

The current calculation in Corollary 25.17.1 applies to arbitrary initial metrics. The normalized annihilating currents of θ_τ satisfy

$$\int_X \theta_\tau \wedge T = 0.$$

If $\mathcal{V}_\mu(\theta_\tau) > 0$, their determinant-weighted measures vanish by Proposition 25.17.1. This is a precise degeneracy statement, but it does not imply that their traces are

concentrated on analytic sets. A smooth positive semidefinite matrix of deficient rank can be defined on an open set.

Problem 32.3.4 Suppose a Chern–Ricci flow has finite maximal time τ and $\mathcal{V}_\mu(\theta_\tau) > 0$. Can one prove that θ_τ is big and that the flow converges smoothly to a positive metric locally on $X \setminus E_{\mathrm{nH}}(\theta_\tau)$, without a bounded mass hypothesis? How do the annihilating Gauduchon currents describe the resulting degeneracy?

The first assertion is a case of the volume-to-bigness question. Once bigness of θ_τ is known, the parabolic theorem recorded in Section 33.18 supplies convergence as currents and smooth local convergence off $E_{\mathrm{nH}}(\theta_\tau)$, by parabolic estimates. Combined with nefness and the qualitative null-locus theorem, this identifies the analytic exceptional set. Thus the missing step in this route is the proposed numerical criterion for bigness. The local envelope continuity and homogeneous replacement arguments of Theorem 13.7.1 supply an elliptic foundation away from a strict barrier's poles; they are not a reconstruction of those parabolic estimates.

Completeness requires an additional metric estimate. In Theorem 27.14.1, the unique finite-energy solution of a divisorial exponential equation in a big closed class is incomplete on every nonempty connected open subset of its ample locus. Its minimal divisorial singularity cancels the prescribed pole. This resolves the unrestricted big-class completeness question negatively; it does not decide completeness under nefness or under the compatibility conditions of the geometric smoothing problem discussed in Section 33.15. In particular it does not settle Problem 32.3.4, whose limiting background is nef.

For integrable poles weakened by logarithmic factors, Corollary 27.15.1 gives a sufficient condition in terms of the density, the potential and the tangential determinant. The proof controls normal length and applies to arbitrary Hermitian backgrounds. The torus models in Examples 27.15.1 and 27.15.2 attain its critical exponent and show its sharpness, even with a Kähler background and full mass. Their pole coefficients need not extend smoothly and positively across the divisor; this distinction preserves the separate scope of the geometric smoothing question.

The regularity of the residual also matters. By Theorem 1.18.1, a Hölder residual on a compact manifold gives finite distance to its analytic singular set wherever it defines a smooth positive metric off that set. Continuity alone does not suffice: the prescribed radial densities in Theorem 27.12.2 have bounded continuous minimal remainders and complete regular ends when $n < d \leq 2n$. These examples concern their specified densities; they do not establish the flow compatibility in Problem 32.3.4.

32.3.5 Balanced and pluriclosed metrics

The balanced–pluriclosed conjecture asks whether the coexistence of these two metric structures forces a compact complex manifold to be Kähler. The balanced metric β and pluriclosed metric ω may differ; a single metric satisfying both conditions is already Kähler. The source and flow formulation are given in Section 33.18.

A pluriclosed manifold in C is Kähler, by the result recorded there. It would therefore suffice to construct a closed Kähler current from the two metrics. The scalar potential equation alone cannot produce the required closedness.

For every distributional potential u,

$$\mathrm{d}(\omega + \mathrm{dd}^{\mathrm{c}} u) = \mathrm{d}\omega.$$

Thus if the starting pluriclosed metric is nonclosed, no potential in its fixed background makes it Kähler. The finite-volume equation, including its smooth solution, does not change this differential datum. Similarly, a conformal Gauduchon representative need not be balanced. Positivity and a scalar normalization cannot be used to replace either missing closedness condition.

The balanced form β^{n-1} pairs trivially with exact real two-forms and can therefore give a conserved trace along a flow. Under $c_1(X) = 0$ in real cohomology, the flow conjecture in Section 33.18 proposes long-time existence and convergence of pluriclosed flow to a Kähler metric when a balanced metric exists. The invariant examples recorded there use the conserved trace. For general metrics, turning that integral into a pointwise bound remains the analytic difficulty.

Problem 32.3.5 Use the coexistence of a balanced metric and a pluriclosed metric to construct closed positive data, or to control a geometric flow that can change the torsion. Determine what estimate beyond the finite balanced trace prevents metric degeneration. Can determinant estimates for Gauduchon currents contribute to that estimate?

For families with a controlled balanced trace and determinant density, the remaining analytic target is a local ellipticity estimate. The scalar trace identity supplies no such estimate for arbitrary families.

32.3.6 Morse inequalities and the choice of volume

Subtracting a nef class has a controlled volume cost in the closed Kähler theory. For closed nef classes α, β on a compact Kähler manifold, the transcendental Morse inequality is

$$\mathrm{Vol}(\alpha - \beta) \geq \int_X \alpha^n - n \int_X \alpha^{n-1} \wedge \beta$$

where Vol is the usual class volume. Its source and the resulting differentiability and cone-duality theorems are recorded in Section 33.18; their proofs are not reconstructed here. To formulate an analogue on an arbitrary compact complex manifold, one must first choose the volume being estimated.

For nonclosed backgrounds, the displayed intersection expression is not generally invariant under changing representatives. Also, the upper class volume and $\mathcal{V}_\mu$ have different definitions and different possible finiteness behavior. A Hermitian analogue must first specify which invariant it estimates.

The first-variation formula gives a finite quantity to investigate. Fix smooth real $(1,1)$-forms θ, η and $t > 0$. Suppose $\theta_s = \theta - s\eta$ has a smooth positive representative for $0 \le s \le t$, and let γ_s be a minimizing Gauduchon metric for its smooth equation. By Proposition 25.2.1 and the fundamental theorem of calculus,

$$\lambda_\mu(\theta - t\eta) = \lambda_\mu(\theta) - \int_0^t \frac{\int_X \eta \wedge \gamma_s^{n-1}}{D_\mu(\gamma_s)}\, \mathrm{d}s. \tag{32.5}$$

The quotient is unchanged by rescaling the minimizer. This identity is valid on the stated positive segment and uses no upper Monge–Ampère mass bound. A uniform upper bound on its integrand gives a lower bound for the remaining root volume. Theorem 25.11.1 extends the identity to a genuine nef endpoint and bounds its improper integral when a closed nef positive-volume core is supplemented by a semipositive form. The fixed tests retain the prescribed density, and (25.13) isolates the correction involving $\mathrm{dd^c}B$. The endpoint is big by the closed core; Example 25.11.1 gives a nonclosed example with an explicit strict current.

Problem 32.3.6 Find geometric hypotheses controlling the cost in (32.5) up to a nef endpoint beyond the closed-core regime of Theorem 25.11.1. Determine when positive root volume forces bigness without that core. For closed backgrounds, compare the estimate with classical Morse inequalities; for nonclosed backgrounds, retain the differential data and the dependence on μ.

The singularity and divisorial constructions in the book become relevant at this stage: they record how much positivity can be spent on imposed poles, while the residual Gauduchon volume stays finite. They do not yet give the moving intersection product or the birational volume formula of the closed theory. Recovering such geometric information is a further objective, with the finite-divisor and Hopf calculations serving as explicit tests.

32.4 Closed and twisted classes

A volume attached to a closed Bott–Chern class must survive changes of representative. A volume equation for an LCK form must instead retain its Lee class and twisted differential. The examples below show where ordinary Hermitian perturbations fail to preserve these requirements. We use the Lee operator D_λ and the map j_X of (28.6) and Section 28.6.

The loci of Chapter 27 distinguish small negative errors for closed currents from strict closed positivity. Any proposed numerical restricted-volume construction in the general Hermitian setting must first pass Proposition 27.5.1: an unrestricted supremum over smooth Hermitian-error representatives is already infinite in the zero class on the Hopf threefold. The failure occurs at each fixed error size. Taking the error to zero afterwards cannot recover a class invariant from that expression.

The global subsolution threshold also constrains divisorial singularities. For a subsolution at this coefficient, Proposition 27.8.1 places every smooth maximizing Gauduchon sequence in the annihilator of each component class in the limit. If one such sequence has uniformly integrable determinant-weighted densities, Corollary 27.8.1 excludes classes with suitable smooth semipositive representatives, including the hyperplane divisor on the Hopf threefold. For a nef component class, Corollary 27.8.2 forces its fixed-density variational volume to vanish. Root tails suffice for both conclusions. Characterizing the remaining possible components requires a realization theorem. The smooth-test closure can be smaller than the full current set, as Example 27.8.1 shows.

For closed classes on manifolds in class C, Section 27.6 supplies a finite resolution-independent restricted invariant. Its positivity is equivalent to bigness on the resolved subvariety, and for a nef ambient class its zero-invariant union is the null locus. For an actual divisorial trace with bounded remainder, Section 27.7 computes the individual mass and turns positive residual mass into a strict current more singular than the trace. Preserving its bounded-difference type requires more: Example 27.7.2 shows the obstruction even on a projective surface. A corresponding construction for general nonclosed backgrounds, and its relation to intrinsic multiplier-ideal traces beyond this residual regime, remain open.

The LCK setting gives a separate set of equation questions. By Corollary 28.2.1, a compact strictly LCK manifold has no big ordinary Bott–Chern class. Its positive LCK form belongs instead to a twisted class, whose representatives are described using D_λ. The coupled system (28.9) identifies the additional conformal equation imposed by Gauduchon's condition. The uniqueness problem Problem 28.4.1 concerns that system with a fixed Lee class and twisted class; the ordinary-potential nonuniqueness examples do not decide it.

Two geometric questions use the Lee form in different ways. The pseudoeffectivity conjecture Conjecture 28.2.1 requires positivity against all Gauduchon tests, extending the positive pairing obtained from the LCK metric itself. The Calabi–Yau conjecture Conjecture 28.5.1 asks for a Vaisman metric on the manifold. The proved rigidity theorem Theorem 28.5.1 applies when one metric already satisfies both Chern–Ricci flatness and the Gauduchon condition. Arranging those conditions simultaneously is the remaining geometric existence issue.

The Lee–Gauduchon cone makes one part of this geometry invariant under bimeromorphic changes. Its dual is the cone of degree-one classes whose image under j_X is pseudoeffective; the strict inequalities exclude $\ker j_X$. The full duality and modification argument are proved in Section 28.6. A nonzero positive class in this image obstructs balanced metrics throughout the bimeromorphic class. The Hopf calculation gives a positive half-ray, despite unbounded Monge–Ampère masses. Thus this cohomological invariant addresses a geometric obstruction while leaving the nonlinear estimates for the scalar equation as a separate task. The converse to its balanced-metric consequence fails: Example 28.7.1 has zero Lee–Gauduchon cone and no strongly Gauduchon metric.

32.5 Degeneration orders and finite-distance boundaries

Approaching a nef background through positive ones gives two kinds of boundary data: the rate at which volume vanishes and the distance traveled in the Hessian metric of Section 25.5. They need not detect the same boundary points.

The orders of Proposition 25.16.1 are independent of the smooth probability measure and the Hermitian perturbation direction, but they need not coincide. On a fixed product of elliptic curves, Proposition 25.16.5 realizes every pair $(1+\alpha, 1+\beta)$ with $0 < \alpha < \beta < 1$, as well as irrational common orders. Proposition 25.16.4 gives zero volume with decay slower than every positive power. Equality and integrality therefore require additional geometry, as in the closed Kähler case governed by numerical dimension.

Under every modification both orders can only increase, by Proposition 25.16.6. A single blowup can increase them strictly or remove positive limiting volume while both remain zero. Positive-current subtraction gives the same one-sided comparison when the residual stays nef. Perturbative minimal multiplicities vanish for nef backgrounds, whereas the actual minimal boundary current in Corollary 25.16.1 has a nonzero divisor whose removal changes the order. The remaining questions in Problem 25.16.1 concern equality and geometric restrictions on these behaviors.

The background-class metric yields a different kind of boundary information. On a finite-dimensional cone it splits as a product of the logarithmic volume direction and a unit-volume hypersurface. Consequently a path whose volume tends to zero or infinity has infinite length. Yet the blowup example in Section 25.5 has a finite-distance boundary with positive nef volume. It follows that completing this metric cone cannot simply mean adjoining all nef classes, or declaring every nef boundary infinitely far away.

Problem 32.5.1 For a specified finite-dimensional cone of smooth background classes, describe the metric completion of its unit-volume hypersurface. Which boundary points admit a current representative, and what estimates on the minimizing Gauduchon metrics distinguish finite from infinite distance?

The cone must be specified in this question. Allowing additional directions can shorten path distances. The finite-dimensional metric calculations do not establish a distance on the entire infinite-dimensional quotient.

32.6 Weak Gauduchon adjoints

Differentiating the smooth equation produces a Gauduchon adjoint probability. At a continuous logarithmic density, supporting probabilities of the concave normalization functional provide a weak replacement. In the background and reference regimes of Chapter 14, the set $\mathcal{A}_\varphi(F)$ is nonempty, convex, weakly compact and absolutely continuous. Uniqueness of this probability would make the scalar functional differentiable even where a classical linearized equation is unavailable.

Conjecture 32.6.1 For a Hermitian background ω, the nonsingular reference $\varphi = 0$, and every real $F \in C(X)$, the set $\mathcal{A}_0(F)$ consists of a single probability measure.

Equivalently, the concave normalization functional is Gâteaux differentiable at every continuous logarithmic density in this regime. The conjecture holds for smooth F by the linearized equation, and in the closed Kähler setting by the explicit normalization. It would imply that smooth approximations of F have a unique weak limit of their Gauduchon adjoint probabilities, by Proposition 14.2.1.

A direct bounded-potential result is Proposition 4.6.1. If every smooth positive potential perturbation of ω is Gauduchon, normalized Monge–Ampère measures extend the smooth adjoint probabilities and converge under capacity convergence of uniformly bounded potentials. This applies on every Gauduchon surface. It does not define a weak linearized operator or settle approximation by unbounded potentials in a general background.

There is also a partial result from Baire category: Theorem 14.2.2 proves uniqueness on a dense G_δ of continuous logarithmic densities, in every fixed background and reference regime of Chapter 14. At those densities all choices of approximating supporting probabilities have the same weak limit. The assertion is generic in the uniform topology; it does not eliminate exceptional continuous densities.

For $n \geq 2$, the compact current formulation gives a more precise target. By Theorem 25.7.1, the supporting probabilities are exactly the normalized determinant-weighted measures of maximizing Gauduchon currents. Thus the conjecture asks whether these weighted measures agree for all maximizers. Corollary 25.8.1 already bounds their densities uniformly in every L^q, $1 < q < n/(n-1)$, on bounded sets of F; the endpoint is available if $\nu_-(\omega) > 0$. These estimates give compactness, but do not prove uniqueness. For a fixed smooth strictly positive density the maximizing current is already unique, smooth and strictly positive. The example in Proposition 25.14.1 shows that a sequence approaching its value can nevertheless have unbounded determinant moments above the critical exponent. Neither statement resolves uniqueness for merely continuous F. Another approach would show directly that the upper and lower directional derivatives of B_0 agree.

Relative inequalities between positive residual forms already give uniform convergence of the whole scalar functional, by Theorem 14.8.1. Under uniformly bounded L^1 changes of the tests, Corollary 14.8.1 gives uniformly integrable supports and weak L^1 cluster limits in the limiting supporting set. Smooth-test adjoints together with the scalar at zero determine the functional by (14.12). These statements leave the conjectured uniqueness at every continuous test unchanged.

For a singular reference or a degenerate background, the same uniqueness question remains meaningful with the hypotheses of Chapter 14. A further geometric interpretation requires a well-defined weak linearization. For a smooth background θ and $u \in \mathrm{PSH}(X, \theta)$, write $\theta_u = \theta + \mathrm{dd^c} u$, and let ρ be a measurable weight. The expression $\mathrm{dd^c}(\rho\theta_u^{n-1})$ is not a definition when multiplication of the product by ρ has not been justified.

32.7 Singularity topology

A topology for singularities must retain information that weak convergence of currents can lose. Upper-volume deficits do so when bounded mass and positive lower volume provide uniform control. On curves, polar curvature gives a complete description even at zero volume. The remaining problem is to extend that control to higher-dimensional degenerations and to backgrounds with unbounded mass.

32.7.1 Weak convergence and singularity data

The complete Gauduchon trace metric induces the L^1 topology on normalized potentials. The elementary ideal-envelope metric uses the same topology on canonical ideal representatives. Its completion can acquire points that are not representatives of their own ideal classes: on a curve, Theorem 15.4.1 approximates every normalized potential by ideal envelopes with finite logarithmic poles and zero non-pluripolar mass. In particular, these envelopes can converge to a smooth potential with strictly positive curvature. The convergence is not in capacity, since damped continuity would then preserve their zero measures.

A positive mass bound does not repair this failure. In Proposition 15.4.2, analytic ideal envelopes of one fixed positive mass converge in the elementary ideal metric to the smooth model 0, whose mass is larger. Nor can the normalization functional recover the lost data. On the closed model domain of Theorem 14.10.1, it records exactly the mass. The positive-residual examples in Examples 14.10.1 and 14.10.2 have identical functionals and supporting sets but different poles, multiplier ideals and normalized potential limits, including in genuinely nonclosed backgrounds.

An order bound gives more control than weak convergence alone. Under a one-sided ideal bound, Proposition 15.5.1 identifies the envelope limit from divisorial Lelong numbers. It gives the criterion for $P_\theta[u] = P_\theta[u]_{\mathcal{I}}$ along ordered analytic approximations, including dyadic regularization. Fixed-weight ideal inclusions as in Corollary 15.5.1 supply the required divisorial convergence. Without the order bound, convergence of every divisorial number together with L^1 convergence is insufficient, by Example 15.5.1. These tests still make sense when upper volume is infinite. In contrast, analytic classes are dense throughout the elementary ideal metric in a Hermitian background, by Corollary 15.5.3, so density for that metric cannot characterize envelope equality.

Finite damped measures control nonlinear limits throughout the potential space, including infinite-mass potentials. Their weak continuity already follows from capacity convergence. Requiring that continuity in addition to capacity convergence therefore imposes no further condition on multiplier ideals.

The uniform topology $\tau_{\omega,0}$ in Section 15.6 combines both envelope coordinates with controlled ideal inclusions and full divisorial variation on each fixed compact model. Its analytic closure and sequential analytic closure are exactly the ordinary classes satisfying $P_\omega[u] = P_\omega[u]_{\mathcal{I}}$, by Theorem 15.6.1. This answers the

analytic-characterization question for every Hermitian background, including classes represented by infinite-mass potentials. The topology is not T_0 on ordinary classes. Smooth-gauge transport is exact when the obstacle changes with the background. With zero obstacles fixed, it is discontinuous already on curves, even within a positive-mass cutoff, by Proposition 15.6.4. The all-obstacle uniformity of Section 15.6.4 is invariant under smooth background changes. Its analytic closure requires envelope equality at every smooth obstacle. On curves this is the same analytic locus, but Proposition 29.12.4 shows that the topology remains incomplete and strictly weaker than polar total variation. In higher dimension, membership in the relative envelope class of the ideal envelope suffices, by Proposition 15.6.6. Under bounded mass and positive lower ideal volume, Proposition 29.8.1 makes zero-obstacle equality sufficient. The implication beyond these regimes remains a separate question.

32.7.2 Volume comparison and analytic approximation

In this subsection, assume bounded mass and fix a smooth big background θ. The comparison d_θ of Section 29.6 measures upper-volume deficits between ordinary models. A common positive lower-volume bound converts a small deficit into a stretching estimate. The resulting quantitative L^1 bound and three-point modulus give separation and Cauchy completeness in Theorem 29.6.1; the cutoff must be closed, as Example 29.6.1 shows. Rooftop continuity supplies monotone lower bounds, which give capacity convergence by Proposition 29.7.1 and Theorem 29.7.1.

Analytic approximation then detects precisely the ideal models. Theorem 29.8.1 corrects analytic regularizations within the fixed background; their ordinary models converge to the ideal envelope, and their limiting comparison with the original potential equals its upper-volume excess. Conversely, Theorem 29.8.2 makes ideal models closed under this convergence at positive lower volume. Although d_θ fails the triangle inequality even on projective analytic models, Theorem 29.11.1 constructs a complete metric with the same convergent sequences. Its Cauchy criterion also controls reciprocal lower volumes, excluding approach to zero volume. The ideal models form a closed complete subspace and are the completion of the positive-volume analytic models, by Corollary 29.11.1.

Monotone limits explain where positivity enters. For decreasing ordinary models in backgrounds decreasing uniformly to a smooth big form, Theorem 29.5.2 gives lower-volume continuity even when the limiting volume is zero. Positive limiting lower volume also gives upper-volume and Lelong continuity. For increasing sequences in a fixed background, Theorem 29.5.1 gives ordinary-envelope and volume continuity under that positivity condition. For arbitrary decreasing potentials with positive-lower-volume limit, Proposition 29.5.1 identifies envelope convergence with upper-volume convergence. Likewise, Proposition 29.8.1 identifies ordinary and ideal envelope equality with upper-volume equality when the ideal envelope has positive lower volume.

The comparison is compatible with changes of background as long as these controls persist. Smooth gauges preserve it after ordinary projection, by Proposition 29.9.1. The fixed-potential limit in Corollary 29.5.1 retains lower volume and, at positive lower volume, the ordinary envelope. The uniform estimate (29.9) allows the curvature errors in analytic regularization to tend to zero, as in Theorem 29.9.1. For two-sided perturbations, convex correction gives admissible transported models and a uniform comparison independent of the chosen strict reference at the level of limiting convergence. This yields the analytic criterion of Theorems 29.10.1 and 29.10.2. These arguments do not cover loss of bigness or the higher-dimensional zero-volume boundary.

32.7.3 The curve boundary

On a curve, polar curvature determines the ordinary model and its atomic part determines the ideal model. Theorem 29.12.1 identifies d_θ with total variation of polar curvature, including at zero non-pluripolar mass. All ordinary models form a complete geodesic space, and the ideal models form the closed subspace obtained by completing positive-mass analytic models, by Corollary 29.12.1. On positive-mass ideal models, this agrees with the source topology described in Theorem 18.5.1 and Corollary 18.5.1. For the source metric, this space is not totally bounded. Thus its completeness has different content from the compact completion obtained from weak trace convergence.

Atomic extraction is a 1-Lipschitz retraction and selects the unique nearest ideal model. The distance to the analytic closure is exactly the diffuse polar mass, by Proposition 29.12.1. The zero-mass model of Example 29.12.2 has trivial multiplier ideals but positive distance from that closure. Total-variation convergence implies L^1 and capacity convergence even at zero mass, by Theorem 29.12.2; the moving pole of Example 29.12.1 shows that capacity convergence does not imply total-variation convergence.

Background degeneration is also explicit on curves. For smooth backgrounds of positive area, Theorem 29.13.1 gives compatible area-rescaling correspondences, including for zero-mass models. Their normalized representatives vary uniformly in L^1 and capacity under uniform convergence of the backgrounds, by Corollary 29.13.1. When the limiting area is zero, Theorem 1.8.1 makes every normalized potential converge to the unique limiting potential, uniformly over the admissible families. The unscaled model-space diameters tend to zero.

On ideal models, $\tau_{\theta,0}$ is the atomic total-variation topology and its canonical metric includes the zero-mass boundary, by Theorem 29.12.3. On general ordinary models it is weaker than polar total variation: diffuse polar curvature can concentrate into an atom in the envelope limit, producing the Cauchy sequence with no ordinary-type limit in Proposition 29.12.2. Thus analytic characterization on the whole ordinary space and a complete extension of the model topology are separate requirements.

Problem 32.7.1 Extend the complete metrizable topology of Theorem 29.11.1 on ordinary models beyond bounded mass and positive lower volume, retaining analytic closure exactly at the ideal models. Preserve the smooth-gauge and two-sided background compatibility of Sections 29.9 and 29.10, and treat background limits losing bigness or a uniform positive lower-volume bound. Recover the complete polar-variation topology of Corollary 29.12.1 on all curve models, and determine whether the higher-dimensional extension is completely metrizable.

The extension must retain the singularity information lost by weak limits. The class of the infinite-mass envelope-good potential in Proposition 15.3.3 is already covered by $\tau_{\omega,0}$, while its canonical envelopes have finite mass. The curve result shows that zero volume alone need not prevent completeness; in higher dimension the stretching argument still needs a positive lower-volume bound.

32.8 Trace realization and recovery

An envelope realizing divisorial lower bounds can have orders larger than those prescribed. For a quasi-plurisubharmonic function u on X and a connected compact submanifold $Y \subset X$ with $\nu(u, Y) = 0$, the trace profile and envelope of (18.1) and (18.5) therefore give a realization problem. Theorem 18.6.1 constructs the envelope in the original restricted background and identifies it as the L^1 limit of the restricted analytic envelopes. Theorem 18.6.2 gives exact pullback compatibility for the inherited data and convergence under vanishing Hermitian perturbations on any compact model. Exact realization asks for equality in the divisorial inequalities. When Y has bounded mass and positive lower mass, Corollary 29.5.2 proves this equality in sufficiently large backgrounds. The Kähler target case also follows directly from the closed theory in Theorem 18.6.3. Analytic ambient singularities have exact ordinary restrictions. Theorem 18.7.1 realizes all orders in higher dimension when the data reduce to an ambient curve problem, with an ordinary type independent of a sufficiently large background. In a larger Hermitian background, Theorem 18.6.4 supplies positive finite envelope mass for general data.

Problem 32.8.1 If Y fails at least one of the two mass properties, can the Hermitian background Ω in Theorem 18.6.4 be chosen so that

$$\nu\big(\mathcal{T}_\Omega(u; Y) \circ \pi, E\big) = \tau_{Y,u}(E)$$

for every prime divisor E on every smooth modification π of Y? If so, determine a background-compatible ordinary or ideal singularity class of this realization, and prove its addition and approximation properties.

The quantified equality is stronger than realizing the divisorial part on one model. Corollary 1.17.1 gives a canonical ordinary type for that part by combining its Gauduchon mass bound with a smooth current decomposition. It does not determine higher-codimensional trace data on further modifications. The exceptional-divisor

formula in Theorem 22.8.1 describes the discrepancy between the canonical types on two models, including a cocycle and approximation in linear mass. Example 22.8.1 shows that a nonzero exceptional term is compatible with an exact trace potential: the term records information absent from the original model's divisorial part. Increasing the background ensures a competitor with controlled curvature; it does not automatically exclude additional divisorial orders. The curve-fibration proof reduces every further-model order to a one-variable multiplicity. Under the two mass properties, Theorem 29.5.2 controls Lelong numbers through the upper-volume deficit and a positive lower volume. Extending that control beyond these mass conditions remains the missing step.

Even exact realization does not control the mass of all ideally equivalent representatives. In Proposition 18.8.2, a product with the Hopf threefold has one smooth positive extended measure and one infinite-mass representative with exactly the same ideals and trace numbers. A proposed volume or continuity characterization must specify which representative or mass domain it uses.

32.8.1 Recovery of compatible divisor systems

Prescribing finitely many poles leaves a residual smooth background. If that background has a positive representative, Theorem 25.18.1 solves the residual equation for a smooth strictly positive probability measure. Every threshold subsolution dominating the prescribed divisor differs from this solution by a constant. A solution in the prescribed ordinary type consequently has the same scalar and a smooth remainder. The difficulty in passing to compatible data on all modifications is to keep this positivity and exact singularity information uniformly.

Passing to all modifications introduces two further requirements. A smooth positive probability measure can pull back to a density that vanishes on the exceptional divisor. The smooth dual formula cannot be applied to that density without proving a degenerate extension. The residual classes can be compared explicitly: Corollary 23.2.1 subtracts the effective exceptional Bott–Chern class while retaining the prescribed exterior derivative. The coefficient approximants converge in linear trace mass, but no nonlinear volume transformation follows from this formula. Neither the degenerate equation issue nor the volume issue is resolved by assigning an intersection number to a nonclosed form.

The envelope-system construction in Chapter 24 gives an order adjunction and an exact criterion for recovery: the perturbation limit must retain the prescribed ideal type. Under bounded mass and bigness, Theorem 29.5.3 supplies a sufficient condition using only the initial slices: their lower singularity volumes must have a positive limit as the perturbation vanishes. Concavity then gives positive lower volume at every finite parameter, and stretching preserves all local divisorial orders. The ruled-surface example has zero residual volume and shows why unrestricted recovery fails.

Problem 32.8.2 For a smooth big background on a manifold without bounded mass, find a condition on a compatible Hermitian envelope system that forces recovery of its ideal type at every finite parameter. It should recover the positive limiting lower-volume criterion of Theorem 29.5.3 when bounded mass holds. Specify the invariant and uniform estimates that replace the finite upper-volume deficit, and their behavior under modifications.

For fixed finite divisor support, Proposition 25.18.2 gives a positive-cone case: if the coefficients converge and the limiting residual background still has a smooth positive representative, the normalized smooth remainders converge smoothly. The singular equation solutions converge in capacity, their measures converge in total variation, and their logarithmic coefficients remain exact on every modification. Multiplier ideals can nevertheless jump at an integer coefficient. For moving sections in fixed finite-dimensional spaces, positive limiting weights and residuals give the full-potential stability of Theorem 14.9.2, in its two stated regimes. Recovery beyond the bounded-mass positive-volume regime, degeneration of residual positivity, and estimates uniform over infinitely many models still require additional arguments.

32.9 Sections, kernels and the dimension of growth

A Hermitian form on the manifold does not become the curvature of a holomorphic line bundle unless it is closed. The algebraic part of the theory must therefore keep its closed curvature data and its analytic background separate. The weighted spaces of sections, their multiplicative variant, and their intrinsic flag bodies do so. Their growth dimension can be strictly less than the complex dimension of the manifold.

For partial kernels, the general upper envelope bound is proved. A matching lower bound is available in the explicit regimes below, while the general problem remains open. Theorem 26.6.1 gives the full-dimensional kernel measure limit on a manifold with a Kähler modification and a positive-mass closed-curvature potential. The proof retains the relative canonical bundle: its Jacobian factor corrects both ideal integrability and the Hilbert norm. This geometric assumption implies BMP; the birational transformation identities themselves do not require it. On a holomorphic submersion with descended data, the source equilibrium theorem on the base gives a different exact limit upstairs. Write m for the positive integer tensor power of the selecting line bundle on the Hopf n-fold. The Hopf example has growth degree $n-1$, and normalizing by m^n would erase it. Adding the vertical Hermitian form recovers a nonzero measure, with the factor computed in (26.3).

Isolated point poles which do not descend are treated in Theorem 26.8.1: their projected jet conditions differ from the base multiplier ideals by only finitely many orders. For a single point on the Hopf manifold, Theorem 26.9.1 proves the potential lower asymptotic as well as total-variation convergence of the kernel measures. Moving the pole along its fiber leaves every selected space unchanged. These limits therefore recover the projected equilibrium, while the original Hermitian multiplier-ideal type can be different.

Theorem 26.11.1 gives a further regime: a logarithmic pole along a section over a hyperplane, with arbitrary positive-dimensional compact connected fibers. The growth degree is the base dimension. Its partial potentials attain the domination envelope of the original upstairs singularity, and the kernel measures converge in total variation for every smooth positive norm volume. The base limit is fixed by the line-bundle metric; its lift retains precisely the conditional fiber probabilities.

Problem 32.9.1 Identify hypotheses on a weighted section series and its intrinsic growth dimension under which the partial Fubini–Study potentials attain the envelope lower bound and the normalized kernel measures converge. Relate the limiting measure to the Hermitian background while retaining the closed curvature and the growth dimension of the series.

Beyond the isolated-point, horizontal projective and descended regimes proved above, this question includes two tasks: producing enough sections with the required ideal constraints, and controlling the resulting measure. A Bergman upper bound supplies neither. In a non-Moishezon setting, ambient-degree normalization may already force every kernel measure to vanish, independently of singularity theory.

Locality of Hölder regularity and regularity of weighted extremals under a uniform density-in-capacity condition provide specific cases of the regularity problem; see Section 33.17. These hypotheses do not cover every singular envelope or settle the lower-asymptotic problem above.

The finite-codimension comparison suggests a more specific question on a fibration. It asks for an analytic description of the section constraints after descent.

Problem 32.9.2 Let $\pi : X \to B$ be a surjective holomorphic submersion of connected compact complex manifolds with connected fibers, where B is Kähler of dimension $q \geq 1$. Fix a Hermitian form ω on X, a holomorphic line bundle A with a smooth Hermitian metric h on B, and $\varphi \in \mathrm{PSH}(X, \omega)$. Set $\theta = c_1(A, h)$. For positive integers m, identify

$$V_m = H^0\big(X, \pi^* A^m \otimes \mathcal{I}(m\varphi)\big) \subseteq H^0(B, A^m)$$

by holomorphic descent. Find geometric or analytic conditions that produce $\psi \in \mathrm{PSH}(B, \theta)$ of positive non-pluripolar mass such that, for all sufficiently large m,

$$W_m = H^0\big(B, A^m \otimes \mathcal{I}(m\psi)\big) \subseteq V_m,$$
$$\dim V_m - \dim W_m = o(m^q).$$

Determine which features of the singularities of φ along the fibers suffice for this approximation.

For descended continuous weights and probability norms, Lemma 26.8.1 and the base equilibrium theorem would then give the kernel-measure and determinant-functional limits. They do not by themselves give convergence of the potentials. The finite isolated-point case in Theorem 26.8.1 supplies the stronger error $O(m^{q-1})$, under its stated hypothesis on the projected poles. Theorem 26.10.1 gives the same error for a logarithmic singularity along a section over a smooth base divisor, under

the curvature margin stated there. Its base reference is explicitly divisorial, and Example 26.10.1 proves sharpness of the error order. The general problem does not assert that every Hermitian-positive selecting weight admits such a base reference.

32.9.1 Growing vanishing orders on curves

Fix an integer $r \geq 1$ and nonnegative integers q_m, indexed by positive integers m. The model in Section 26.12 treats the smooth weight $|z|^{2r}$ and selecting ideals (z^{q_m}) on $\mathbb{C}$. When $q_m \to \infty$, the kernel vanishes at the natural rescaled limit, but has a Gaussian transition at the larger radius $(q_m/(rm))^{1/(2r)}$. For $q_m = \lfloor m\lambda \rfloor$ with a real $\lambda > 0$, the limiting kernel measure omits the pole atom of the limiting envelope current. For a compact curve with the curvature and local expansion in Theorem 26.13.1, that theorem transfers the Gaussian transition to global sections when $q_m = o(m^{1/(2r+1)})$. The next problem concerns larger growing orders.

Problem 32.9.3 Let C be a connected compact Riemann surface, let $x \in C$, and let (L, h) be a smooth Hermitian holomorphic line bundle whose curvature is semipositive and strictly positive on $C \setminus \{x\}$. Fix an integer $r \geq 1$. Suppose that in a local coordinate z centered at x and a holomorphic frame e one has

$$|e|_h^2 = \mathrm{e}^{-\phi}, \qquad \phi(z) = |z|^{2r} + O(|z|^{2r+1}).$$

Let ν be a smooth positive volume form with local expression $a(z)\,\mathrm{d}A$ and $a(0) = 1$, where $\mathrm{d}A$ is Euclidean area. For positive integers m, choose integers $q_m \to \infty$ with $q_m = o(m)$. Give

$$V_m = H^0(C, L^m \otimes \mathfrak{m}_x^{q_m})$$

the norm $\|s\|^2 = \int_C |s|_{h^m}^2 \,\mathrm{d}\nu$, where $\mathfrak{m}_x$ is the ideal sheaf of x. Write $B_m = \sum_j |s_{m,j}|_{h^m}^2$ for an orthonormal basis $(s_{m,j})$ of V_m, with value zero when $V_m = 0$. Determine the growth ranges of q_m for which, for each fixed real τ, every sequence of points x_m in the coordinate neighborhood satisfying

$$m|z(x_m)|^{2r} = q_m/r + \tau\sqrt{q_m/r}$$

obeys

$$\frac{\pi B_m(x_m)}{mr^2|z(x_m)|^{2r-2}} \longrightarrow \Phi(\tau) := \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\tau} \mathrm{e}^{-s^2/2}\,\mathrm{d}s.$$

When this transition law fails, determine how further terms of the local weight change its center and scale.

For the data of this problem, put $R_m = (q_m/(rm))^{1/(2r)}$. The range $q_m = o(m^{1/(2r+1)})$ is settled by Theorem 26.13.1: on every fixed multiple of the disc of radius R_m, the local weight error satisfies $mR_m^{2r+1} = q_m R_m/r = o(1)$. Its proof constructs global sections with the growing vanishing condition. Extending the range requires control of the weight remainder at the transition scale, rather than

only a fixed-order kernel limit. Higher-dimensional analogues and transfer through Hermitian fibrations remain separate questions.

32.10 Equations on singular spaces

On a resolution of a singular space, a pulled-back Hermitian form degenerates and the pulled-back density can vanish or become singular. Existence upstairs must therefore be combined with an estimate that permits descent. Chapter A provides potentials, envelopes, ideal equivalence and extended measures on reduced irreducible compact unibranch spaces, retaining the original nonclosed background. On projective cones, Theorems A.6.1 and A.6.2 now supply explicit equations in full ordinary logarithmic classes and in the closed bounded class, for the prescribed angular and radial data. The logarithmic construction includes determinant-preserving nonclosed deformations. Its full solution descends even when the smooth residual is nonconstant on the exceptional fiber. The continuity results discussed in Section 33.17 use additional geometric and integrability hypotheses, notably log terminal singularities.

A conformal Gauduchon metric, uniformly comparable to the original metric, is available on the regular locus of a compact reduced irreducible Hermitian space of dimension at least two; see Section 33.17. This existence theorem is not reconstructed in the appendix and does not by itself supply the nonlinear equations asked for below.

Problem 32.10.1 Beyond the projective-cone constructions of Section A.6, determine geometric and measure conditions on compact unibranch Hermitian spaces that give existence and uniqueness in specified singularity classes. Identify estimates ensuring finite mass, descent and continuity, and their behavior under common refinements.

In the cone examples, the pulled-back density vanishes on the exceptional divisor, and the logarithmic remainder solves an equation in a strictly positive background. The closed bounded endpoint instead uses an energy which is coercive off that divisor. More general spaces require a replacement for these two controls. The linear trace construction also needs a separate singular-space formulation; there is no smooth elliptic Green operator across the singular locus without additional analysis.

Chapter 33
Historical notes

The passage from Kähler to Hermitian pluripotential theory separates two changes. A closed class can be studied on a non-Kähler manifold, while a nonclosed background loses cohomological mass control even on a Kähler manifold. Local regularization and products persist through smooth corrections; global equations instead depend on Gauduchon duality, relative envelopes and the available mass bounds. These mechanisms connect the sources below. Their hypotheses also determine whether a solution retains the ordinary singularity type of its reference or belongs only to its relative envelope class.

33.1 Plurisubharmonic functions and regularization

Let X be a connected compact complex manifold of dimension $n \geq 1$.

Local compactness, integrability and regularization survive the passage to a nonclosed background because they apply to a psh function after a smooth local correction. The normalized L^1 estimate of Theorem 1.4.1, also recorded in [Din16, Proposition 6.2], follows by propagating local compactness over a connected coordinate cover. The exponential-integrability input recalled in [Zer01, Section 1] becomes uniform over normalized potentials through the weighted-sum contradiction of Lemma 1.5.1.

The comparison of convolutions in different coordinates in Lemma 1.9.1 and Corollary 1.9.1 is the bounded part of the argument in [BK07, §2, Lemmas 4–5]. The full regularization theorem of [BK07, Theorem 1] permits unbounded potentials and a positive continuous background. The proof of Theorem 1.11.1 retains the bounded case needed for the product constructions, including the decreasing diagonal and the rescaling that preserves a Hermitian background.

For $p \in \mathrm{PSH}(X, \theta)$ with θ a smooth real $(1,1)$-form, the Chern-connection expansion of Demailly [Dem94, Theorem 4.1] applies to the closed current $\mathrm{dd^c}p$ with an arbitrary smooth lower bound $-\theta$. In particular, the lower bound need not be closed. The monotonicity and logarithmic convexity of the regularization enter (1.27);

its linear error rate follows from the curvature inequality in [Dem94, Lemma 4.4] and the expansion in [Dem94, Proposition 3.8, equations (3.11), (4.3), and (4.5)]. Here the curvature inequality requires no small ϵ term or M_ϵ correction, and local trace bounds make the constants uniform in the center and radius.

The same expansion underlies the truncated transform of Lemma 1.21.1; see [Dem94, Theorem 4.1 and Remark 4.7] and [KN19, Lemma 4.1]. In Lemma 1.21.3, the full base–fiber error is retained as a modulus tending to zero, with a linear-radius term absorbing the mixed and vertical errors as in [Dem94, Remark 4.7]. This formulation accommodates the unbounded, finite-L^1 potentials in Theorem 13.7.1. Localizing the averaging radius then permits the minimum-principle argument off the singular barrier.

The resolvable regularization of Theorem 1.15.2 is [DP04, Theorem 3.2]. It allows an arbitrary smooth lower bound on a Hermitian manifold and supplies a smooth remainder after resolution. This is a separate regularizing sequence from the prescribed multiplier-ideal sequence in Theorem 1.15.1; simultaneous possession of their additional properties is not assumed.

The analyticity of positive Lelong level sets recalled in Section 1.13.1 and used in Lemma 25.15.1 is Siu's theorem [Siu74, Main Theorem, p. 53]. Smooth local corrections allow its application to quasi-plurisubharmonic functions.

The fixed non-pluripolar measure convergence in Lemma 1.23.1 uses Cegrell's classical fixed-measure argument, also written in extraction form in [KN25, Lemma 2.1 and Corollary 2.2]. Applying the extraction statement to each subsequence gives convergence of the whole bounded sequence.

33.2 Mass and energy

On a nonclosed background, variation of the Monge–Ampère mass and failure of the usual energy primitive are related manifestations of torsion. Chiose's polarization argument [Chi24, Theorems 1–2] gives the mass-invariance characterization in Theorem 2.2.2. Yi Li constructs torsion-corrected energies in [Li10, Equation (1.18) and Theorem 1.2]. The correction changes the one-form being integrated, consistently with the obstruction for the raw one-form in Theorem 3.1.1; Proposition 3.3.1 makes the difference explicit by first variation.

The failure of bounded mass on a Hopf threefold is Xia–Zhang's counterexample [XZ26, Theorem 1.1]. Its Gauduchon background converts vertical Dirichlet energy into Monge–Ampère mass, as developed in Chapter 8. Combining it with Theorem 6.6.1 produces infinite extended mass with trivial multiplier ideals at every positive weight. The coefficient-integrability and convergence results of Section 6.7 describe the linear current carried by this construction. The source uses $\mathrm{i}\partial\bar{\partial}$ [XZ26, Section 1]; dividing its background forms by 2π and leaving its potentials unchanged preserves positivity and changes the coefficient of the mass identity.

Failure of positive lower mass is a different phenomenon. It occurs on the elliptic threefold of Section 25.15. Sufficient conditions for a lower bound instead come from

the sign of torsion: (25.16) is the dimension-three case of [AGL23, Theorem 3.2], where $\mathrm{dd^c}\omega \leq 0$ is called plurinegativity.

The principal elliptic bundles in Section 32.3.3 are a special case of the torus bundles of Goldstein–Prokushkin [GP04, Section 2]. Their metric calculation gives an upper bound and does not decide whether lower mass vanishes.

33.3 Gauduchon metrics and positivity

Gauduchon metrics replace cohomological mass conservation by a linear trace identity. In dimension $n \geq 2$, the conformal theorem recalled immediately before [TW10, Lemma 2.3] starts from a Hermitian form η and constructs a smooth real function f such that $\gamma = \mathrm{e}^f \eta$ is Gauduchon; the multiplier of Theorem 4.1.1 is $\rho = \mathrm{e}^{(n-1)f}$. The resulting trace comparison, Corollary 4.2.1, also underlies the first mixed-volume bounds of [GL22, Proposition 3.3].

The trace identity has a dual form: positivity can be tested against Gauduchon metrics. For an arbitrary smooth background, the criterion in Theorem 4.3.1 is recorded in [BGL25, Lemma 1.15]. Its proof combines separation with a scalar elliptic right inverse. The same linear control supplies the normalization bound (4.3) used in the weak existence arguments of [BGL25, Section 4].

33.4 Bounded products and extended measures

The bounded product is local even when its background is not closed. Expanding against smooth local corrections, as in [KN15, Section 1] and [BGL25, Section 2.1], gives Proposition 1.7.1. Restricted truncations then define the extended measure of Theorem 6.1.1, by the mechanism of [BGL25, Proposition 2.1]. Finiteness becomes decisive when the restrictions are removed in the category of currents: [BGL25, Theorem 2.2] uses a uniform variation bound. An infinite extended positive measure retains the local construction but is not a global current of order zero.

The metric independence and reduction to smooth potentials in Proposition 6.5.1 recover the comparison recorded after [BGL25, Definition 1.16]. Exact decreasing regularization supplies the passage from smooth to bounded potentials. The Gauduchon trace bound and cancellation of the exact top wedge used in Proposition 7.1.1 also appear in [GL22, Proposition 3.3(i), (vi)]. The range estimate retains their polynomial dependence on the oscillation.

The local mixed inequality in Theorem 1.6.2 is [Din09, Theorem 1.2], for locally bounded psh functions and nonnegative measurable smooth-volume densities. Its extension to smooth backgrounds, proved in Section 1.7, is used in Section 13.17.1. Freezing the two smooth backgrounds gives Lemma 1.7.3 and the trace inequalities (13.9) and Lemma 6.4.2; the latter retains the linear trace mass on the pole set. The

coefficient-freezing proof also recovers the smooth-volume inequality used in [Ngu16, Lemma 1.9].

33.5 Envelopes and comparison

Contact sets connect order of potentials to order of their measures. The semipositive big case of Theorem 12.1.2 is [ALS25, Theorem 2.2]; the bounded-to-unbounded passage here removes bigness and a uniform mass assumption. For a general big background, [BGL25, Theorem 2.9 and Proposition 2.11] proves contact concentration under bounded mass. Local replacement and a strict singular barrier give Theorem 13.7.2 without that assumption.

The fixed-point estimate Proposition 12.1.4 is [ALS25, Proposition 2.12]. For a smooth semipositive θ and an ordinary envelope fixed point $\varphi \in \mathrm{PSH}(X, \theta)$, the present proof uses $P_\theta(\varphi + 1, 0) = \varphi$ and contact with the first obstacle is pluripolar, so only contact with zero contributes. For bounded potentials, the envelope proof of domination in Theorem 12.2.1 is due to Guedj–Lu [GL22, Proposition 2.8].

Difference envelopes provide a relative class in which singular potentials can still be compared. The construction is recorded for semipositive big backgrounds in [ALS25, Proposition 2.4 and Lemma 2.5] and for arbitrary big backgrounds in [Li26, Section 2.1]. The finite-sublevel estimate turns nonempty competitors into a genuine envelope. The class in Definition 12.4.1 follows [ALS25, Definition 3.2]; its convex stretching property, Lemma 12.4.1, is the argument of [ALS25, Proposition 3.5].

The model condition in (12.1) and its individual positivity consequence appear in [ALS25, Proposition 2.10 and the discussion after Definition 2.13]. That paper does not assume bounded mass for its same-type equations with L^p data. Individual positivity must be distinguished from a positive infimum over the whole ordinary type. The lower-volume mechanism in Theorem 12.5.1 is that of [ALS25, Proposition 3.8].

Lower semicontinuous obstacles require an approximation of the contact argument. The proof of Proposition 12.1.2 follows [GL22, Theorem 2.3], using the increasing bounded convergence theorem [BT82, Theorem 7.4]. Smooth local corrections and expansion against the remaining background give the nonclosed application.

The modified comparison principle in Theorem 10.2.1 is due to Kołodziej–Nguyen [KN15, Theorem 0.2]. The proof follows the finite induction of Guedj–Lu [GL22, Theorem 1.5], with enlarged dimensional constants.

33.6 Smooth and continuous equations

The smooth Hermitian equation must determine a normalization constant together with the potential. For $n \geq 2$, the existence theorem Theorem 9.1.1 is [TW10, Corollary 1]. For a smooth logarithmic density F, their equation uses $\mathrm{i}\partial\bar{\partial}\varphi$; setting $u = 2\pi\varphi$

gives the present convention, while a prescribed smooth volume $\mathrm{d}V$ is absorbed into $F + \log(\mathrm{d}V/\omega^n)$. The Gauduchon concavity argument for the normalization constant in Theorem 9.7.1 also appears in [Col21, Theorem 2.13].

The weighted integration proof of the potential estimate in Chapter 9 follows Cherrier and Tosatti–Weinkove [Che87, TW10]. A finite recurrence absorbs torsion before Sobolev iteration. The torsion calculation for the second-order estimate in Chapter 9 goes back to Cherrier. Its completed-square form and the maximum argument follow [TW15, Sections 3 and 9].

Prescribing the volume of a Gauduchon metric leads to a different equation, with first-order torsion terms. The existence result Theorem 9.9.1 is [STW17, Theorem 1.4], again with the potential multiplied by 2π. The potential bound of Section 9.10 follows [Szé18, Propositions 10–11 and Remark 12], specialized to the Gauduchon determinant. The second-order argument in Section 9.13 follows [STW17, Section 3]; the identities (9.20) from [STW17, Section 4] explain the structure of the first-order perturbation.

The gradient argument combines rescaling and a Liouville theorem. In (9.38), the first-order term is retained before rescaling and vanishes at rate A_j^{-1}, as in [Szé18, Section 6]. The Euclidean argument in Section 9.14 specializes [Szé18, Theorem 20] to the Gauduchon cone. Once uniform ellipticity is available, Lemma 9.16.1 uses the variable-coefficient Evans–Krylov theorem [TWWY15, Definition 3.2 and Theorem 3.4].

For L^p densities, the continuous normalized and exponential equations are respectively [KN15, Theorem 0.1] and [Ngu16, Theorem 0.1]; see Theorem 10.1.1 and (10.1). Their operator is twice the present one. For a target exponent $\lambda > 0$, the conversion is $u = 2\varphi$ and the exponential reference parameter is 2λ. These results require neither bounded mass nor positive lower volume. Local capacity estimates supply the compactness: the deductions in [KN15, Section 5] and [Din16, Section 6] lead to Theorem 10.3.2, the statement of [KN15, Corollary 5.10]. A common density bound is essential here; normalized L^1 compactness alone does not suffice.

The exponential term supplies comparison even when the densities have no positive lower bound; Lemma 10.4.2 recovers [Ngu16, Lemma 2.3]. Letting the exponent tend to zero then identifies the normalization constant through (10.9), as in [Ngu16, Corollary 0.2]. This proves scalar uniqueness without proving uniqueness of the limiting potential. The weighted mean (10.8), from [Ngu16, Proposition 2.5], controls the normalization; here its lower bound follows directly from the sublevel estimate.

Prescribed singularities require relative-envelope arguments and distinct assumptions on the reference and the ambient metric. The results of [ALS25, Li26] are separated accordingly in Chapters 13, 30 and 31; their provenance is discussed in Section 33.9.

33.7 Uniqueness and selection

At exponent zero, uniqueness of the scalar leaves a separate question about the potential. Density degeneration affects comparison, while the positive-exponent family can impose a choice of critical solution. The restricted-density mass-concentration proof around (11.38) follows [LPT21, proof of Theorem 3.7 and Remark 3.8]. The exterior-ball proof of (11.19) instead uses maximum propagation, following [BG19, Proposition 2.4 and Theorem 2.5 in the cited preprint].

The bounded uniqueness theorem Theorem 11.4.1 uses level-energy estimates and auxiliary test potentials for a smooth nonnegative density with connected dense positive set. The mixed inequality and coefficient-density passage use the local calculus in Chapter 1. Bounded continuity is obtained internally by proper comparison in Corollary 10.4.1. The smooth local Dirichlet input for Proposition 11.4.3 is [Gua98, Theorem 1.1, pp. 687–688]; the quadratic strict subsolution verifies its hypothesis. The real contact-set estimate then gives a constant depending only on the fixed geometry and the L^2 bound, rather than on the height of the test density.

For measurable densities bounded above and bounded away from zero, uniqueness is proved in Corollary 11.4.1 from the level-energy argument; its classical counterpart is [KN19, Corollary 3.9 and Remark 3.13], with the locators referring to the author version specified in that entry.

33.8 Volumes under mass bounds

Upper and lower singularity volumes retain information that a single mass loses for a nonclosed background. Under bounded mass, their comparison controls envelopes and gives a topology at positive lower volume. On curves, polar curvature describes the boundary where that lower volume vanishes.

33.8.1 Singularity volumes and relative classes

The definitions used in Section 29.1 are [BGL25, Definition 3.1], expressed in potentials. For closed backgrounds they coincide by invariance of non-pluripolar mass under bounded changes of singularities, as recalled in [BGL25, Example 3.2]. In the big regime, monotonicity Theorem 29.1.1 is [BGL25, Proposition 3.7]; the lower-volume addition proof of Proposition 29.1.1 is [Li26, Proposition 2.5].

The relative full-mass class Definition 29.2.1 is [Li26, Definition 3.2]. The stretching criterion Lemma 29.2.1 reconstructs [Li26, Lemma 3.1], retaining the upper-volume deficit and choosing a representative close to its upper volume. Its zero-deficit case gives Theorem 29.2.1. Uniform Lelong bounds turn the convex comparisons into Proposition 29.2.1.

The domination theorem Theorem 29.3.1 reconstructs [Li26, Proposition 3.2]; its proof uses positive lower volume of the reference, without requiring it to be an envelope fixed point. The relative obstacle estimate Lemma 29.3.1 is [Li26, Proposition 3.3]. Together they give the upper-volume identity of [Li26, Lemma 3.2] and the lower-volume identity of [Li26, Corollary 3.3], collected in Theorem 29.3.2. For $u \in \mathrm{PSH}(X, \theta)$, the proof first checks that the increasing obstacle limit has the ordinary type of $P_\theta[u]$, so domination applies before volume equality is established. The resulting full-mass characterization gives the volume criterion in Proposition 29.8.1.

The small-stretch construction Lemma 29.4.1 is [Li26, Proposition 3.1]. Its mixed-mass bound gives the explicit uniform stretching scale. The additive lower-volume estimate in Proposition 29.4.2 is the smooth-perturbation case of [BGL25, Proposition 3.3(ii)], with the mixed bound of [BGL25, Lemma 2.4] written in potentials. The strict barrier gives the multiplicative comparison at positive lower volume, as in [Li26, Proposition 3.5(ii), Step 1]. Thus Corollary 29.5.1 gives lower-volume continuity for every fixed potential; upper-volume and ordinary-envelope continuity retain positive lower volume.

The increasing-limit theorem Theorem 29.5.1 reconstructs [Li26, Proposition 3.5(i)]. Contact identifies the envelope limit and its upper volume; the deficit estimate then proves positivity of the approximating lower volumes. The decreasing-model theorem Theorem 29.5.2 reconstructs [Li26, Proposition 3.5(ii)], including its zero-lower-volume conclusion. The uniform small stretch shows that a positive lower bound along the models forces positive lower volume of their limit. Strict corrections and tail envelopes then reduce background variation to the fixed-background argument. The source states perturbations $\theta + \varepsilon_j \omega$; the proof here treats arbitrary smooth backgrounds decreasing uniformly in form order. The positive-lower-volume criterion Proposition 29.5.1 reconstructs [Li26, Proposition 3.5(iii)]. The deficit bound gives uniform Lelong convergence on each fixed compact model, and restricted analytic envelopes yield the exact traces of Corollary 29.5.2.

The semipositive consequence Proposition 29.4.1 is the forward direction of [ALS25, Proposition 2.11], deduced from the strict barrier. This direction requires neither the ordinary envelope identity nor $\nu_-(\theta) > 0$; the latter assumption is used for the converse. Applying it to an ordinary envelope fixed point compares (29.4) with (12.1).

33.8.2 Model topology and analytic approximation

The comparison function in Definition 29.6.1 is introduced in [Li26, Section 4.2]. Its separation follows from the envelope characterization. The normalized stretching argument of [Li26, Lemma 4.2] gives the quantitative L^1 bound in Proposition 29.6.1, and the three-point estimate (29.6) passes convergence from a subsequence to the whole Cauchy sequence. The tail-maximum construction of Theorem 29.6.1 follows [Li26, Theorem 4.3], with the closed lower bound $\underline{\mathrm{vol}}_\theta(u) \geq \delta$. The strict cutoff $> \delta$ used in version 1 of that source is not preserved at a Cauchy limit, as Example 29.6.1

shows. The source explicitly allows failure of the triangle inequality; Example 29.11.1 gives an intersecting-divisor example of that failure.

The finite-rooftop criterion Lemma 29.7.1 reconstructs [Li26, Lemma 4.1]: representatives close to their upper volumes make collapse contradict the lower volume of the common upper bound. The softened-stretch argument of [Li26, Proposition 4.4] gives Proposition 29.7.1; only the limiting potential needs positive lower volume, and the same comparison gives lower-volume convergence of the roofs. The two stretching scales in [Li26, Theorem 4.4] produce the monotone bounds of Theorem 29.7.1. Finite rooftops remain normalized models, so decreasing-model continuity passes their volume bounds to the infinite tail. Monotone capacity convergence then applies, and the subsequence criterion gives convergence of the original sequence.

The analytic approximation results of Section 29.8 combine the preceding volume continuity with the dyadic ideal-envelope limit. Rational mixing with a strict analytic potential removes the curvature error in any smooth big background. Two background comparisons then give the exact upper-volume excess in (29.8). Conversely, stretching and concavity of the ideal envelope make its fixed points closed under upper-volume comparison at positive lower volume. This proves the analytic criterion Theorem 29.8.2 and the inherited Cauchy completeness there. The higher-dimensional zero-lower-volume boundary and ideal-theoretic positivity criteria in Problem 29.8.1 remain separate questions.

The background results in Section 29.9 follow from the same volume and envelope arguments. Smooth-gauge transport preserves the competitor currents, and ordinary projection corrects the zero obstacle. The explicit small-stretch scale makes the strict barrier uniform on positive lower-volume sectors. This yields (29.9) and extends the analytic criterion to vanishing positive background errors. These statements concern singularity volumes and canonical models; they do not identify the raw envelope mass with a gauge invariant or prescribe convergence of arbitrary representatives.

The two-sided transport in Section 29.10 mixes the original model with a strict analytic potential. A common upper background bounds both this transport and the maximum of two transported models. The resulting volume bounds prove uniform comparison, convergence of the transported reference, and equivalence of the limiting comparisons for different strict corrections. The nearby-background estimate and the three-point modulus then reduce its analytic criterion to Theorem 29.9.1.

The complete metric in Section 29.11 is obtained by the standard dyadic finite-chain construction, proved directly from the three-point modulus. Stretching first gives lower-volume continuity, and reciprocal lower volume controls the zero-volume boundary. The analytic completion statement then follows from the preceding approximation criterion. This metrization concerns the topology of the upper-volume comparison; it does not turn that formula into a metric.

33.8.3 Curves and zero-volume limits

The planar inverse maximum principle in Lemma 1.6.3 is the two-dimensional form of the result recorded in [BP04, Theorem 3.1], where it is attributed to Dupaigne and Ponce. The proof here uses harmonic barriers on shrinking neighborhoods of a polar compact set. Applied to differences, it gives order of the polar curvature measures; subtracting a common Green potential gives the maximum formula (1.8). These general local facts are placed in Chapter 1.

The curve classification in Section 29.12 combines this polar order with the Green representation from Section 18.3. A prescribed polar measure gives an ordinary model equivalent to its Green potential. The positive-mass envelope characterization and harmonic uniqueness at zero mass identify every ordinary envelope with such a model. The maximum formula then turns the upper-volume comparison into total variation. Completeness, geodesics and the nearest ideal model follow from the corresponding measure operations. The exact analytic approximation defect is the nonatomic polar mass, which equals the full curvature mass for the Cantor example.

The curve continuity in Theorem 29.12.2 uses the positive-volume capacity theorem in the interior. At zero mass, polar curvature exhausts the full current, giving total-variation convergence of the currents. The local estimate Lemma 1.6.2 follows from the disc Green representation, the harmonic mean-value estimate and the weighted Chern–Levine–Nirenberg inequality. It gives capacity convergence at this boundary. Moving a pole separates this topology from total variation, while the finite-atomic approximation in Theorem 15.4.1 separates weak curvature convergence from capacity convergence.

The background correspondences in Section 29.13 follow from the same polar-measure classification. Smooth gauges extend it to every smooth curve background of positive area. Equal-area changes preserve total variation; area rescaling gives homotheties compatible with atomic extraction and hence with ideal projection. The scalar Poisson equation realizes them by a smooth gauge followed by ordinary projection. Comparing the new obstacle with its maximum and minimum gives a uniform bound, which yields L^1 and capacity continuity through the zero-mass boundary. Rationality is required only when asserting preservation of analytic singularities under area rescaling, because the book uses rational analytic coefficients. At zero area, positivity and Stokes leave a single normalized potential. The general convergence statement Theorem 1.8.1, placed in Chapter 1, follows from normalized compactness, harmonic uniqueness and the local variation-capacity estimate. It also gives the collapse of model representatives as the unscaled metric diameters tend to zero.

33.9 Equations with prescribed singularities

Relative envelopes allow an equation to retain prescribed singularities while its measure changes. The ordinary-type equations with L^p data, the general-measure

equations and the equations under metric mass bounds use this framework in different ways. The distinction concerns both the estimates available and the singularity type of the resulting solution.

33.9.1 Ordinary-type equations

The ordinary-type equations of Theorem 13.11.1 are [ALS25, Theorems 5.4 and 5.7]. Replacing the reference volume ω^n by $\mathrm{d}V$ multiplies the density by a smooth positive factor and preserves its L^p class. A positive constant change of the Hessian normalization is absorbed by rescaling both the potential and the exponential parameter.

The relative estimate Theorem 13.11.2 and rooftop supersolution Lemma 13.11.2 are the two constructions in [ALS25, Lemmas 5.5–5.6]. Together they control the solution within the ordinary type. For a big background, the bounded-subsolution argument of [GL23, Lemma 3.3 and Theorem 3.4] gives Lemma 13.2.1 and Theorem 13.2.1.

Strict singular barriers extend the comparison argument to minimal singularities. The development from Section 13.12 follows [BGL25, Section 4], where the bounded mass assumption from the earlier sections is dropped. Its minimal ordinary-type equations, culminating here in Theorem 13.15.2, are [BGL25, Theorems 4.6–4.7]. The solution has finite mass, even though the whole ordinary type is not assumed to have uniformly bounded mass.

33.9.2 General measures in semipositive backgrounds

The local decomposition used in Lemma 1.6.1 is [Ceg98, Theorem 6.3], on a bounded hyperconvex domain in dimension at least two. For a non-pluripolar measure, the pluripolar summand vanishes. The global constructions in Lemma 13.22.1 and Proposition 13.22.1 patch these local representations.

The bounded-subsolution theorem Theorem 13.9.1, from [ALS25, Theorem 5.1], solves the exponential equation for a bounded density relative to the measure of a bounded potential. Truncating the density in a measure representation brings this theorem into the general-measure construction. Domination then permits passage to the limit. The finite-measure domination theorem Theorem 30.2.1 in Chapter 30 reconstructs [ALS25, Theorem 3.14]. The fixed finite measure prevents collapse; the other members of the envelope class need not have finite mass. The proof of Theorem 30.2.2 also allows the compared measure to be infinite. Singular strict domination, Theorem 13.1.1, follows [ALS25, Theorem 3.11].

Passing to a measure limit requires noncollapse of lower envelopes. The fixed-dominating-measure theorem Theorem 30.3.1 reconstructs [ALS25, Lemma 4.2 and Theorem 4.1, condition (2)]. Positive lower mass gives noncollapse in this

case; bounded upper mass is not used. The common-envelope argument preceding Corollary 30.4.1 instead supplies the first alternative of [ALS25, Lemma 4.2]. For approximants of the reference's ordinary type, that alternative requires no positive lower-volume hypothesis.

A common ordinary-type barrier also gives Corollary 13.8.1, the corresponding case of [ALS25, Theorem 4.1]. The reference need not be an envelope fixed point: the smaller-background condition suffices. The moving bounded integrals are handled by the torsion calculation in Lemma 1.23.3, following [KN25, Lemma 2.3].

The resulting general-measure equations Theorem 30.1.1 are [ALS25, Theorem 5.8 and Corollary 5.9]. The minimal and prescribed-reference constructions appear in Sections 30.5 and 30.6; they use the density approximations of that proof and retain the smaller-background model condition and positive lower mass. The solutions have finite mass, but arbitrary measure data do not give bounded difference from the reference. For a Hermitian background and minimal reference, Corollary 30.7.1 identifies the equation scalar with the individual density threshold. The remaining question in Chapter 32 and Problem 32.1.1 concerns zero lower mass. The weighted addition theorem Theorem 13.10.1 reconstructs [ALS25, Theorem 5.2], localizing the passage to the equation.

33.9.3 Big backgrounds under mass bounds

Li's general-measure theory [Li26, Theorems 6.1–6.2 and Corollary 6.1] assumes both metric volume properties. It gives Theorem 31.1.1 and the model-reference constructions of Section 31.7. Under bounded mass, positivity of the model reference's lower volume implies the metric positive-lower-volume property. Expressing the hypothesis through the reference thus reorganizes this regime without enlarging it.

The relative foundations use the domination, stretching and obstacle estimates recorded in Section 33.8. Constructing the requisite rooftop precedes the application of Lemma 29.3.1 to relative projection in (31.9).

Moment and continuity estimates connect these envelope operations to existence. The first-moment compactness argument using (31.4) follows [Li26, Proposition 4.2], by choosing an equivalent reference potential whose mass lies sufficiently close to the upper volume; the upper volume need not be attained. Fixed-measure continuity is [Li26, Theorem 4.2]: finite rooftops use only the envelope class, while the mass interval prevents collapse and controls tails.

The varying-reference closure theorem Theorem 31.5.1 reconstructs [Li26, Corollary 4.1]. Projection to ordered lower models provides a fixed relative class for each tail, and an exponential contact estimate prevents collapse. The positive-exponent stability in [Li26, Corollary 6.2] assumes uniformly bounded L^p data and ordinary reference type. The theorem Theorem 31.7.3 instead treats the relative full-mass class: the models converge in the volume comparison with a common positive lower-volume bound, and the densities converge in L^1 against a fixed finite non-pluripolar measure. A summable extraction produces the dominator needed for closure. At

positive exponent, uniqueness gives convergence of the whole potential sequence in capacity. At exponent zero, the scalars and measures converge. Every L^1 cluster limit of the normalized potentials solves the limiting equation, and the corresponding subsequence converges in capacity. In both cases, identifying the right-hand sides in L^1 gives total-variation convergence of the measures.

For a minimal reference, the general-measure equations of Section 31.4.1 are [Li26, Theorems 5.2–5.3]. The higher-moment estimate in Section 31.3.1 follows the concave-weight argument of [Li26, Proposition 5.1 and Lemma 5.1], with integrability against the fixed singular measure stated explicitly. The measure smoothing used in the existence proof is collected in Lemmas 1.23.4 and 1.23.5.

The L^p ordinary-type conclusion of Corollary 31.8.1 is [Li26, Corollary 6.1]. The relative estimate Proposition 31.8.1 starts in the envelope class and uses a uniform moment of order $m > 2n+1$. The concave-weight method is related to [Li26, Theorem 5.1], whose stated hypothesis already assumes ordinary reference type and whose moment exponent is $m > n$. That statement is not used to infer the ordinary type of an arbitrary envelope-class solution.

Addition requires a common lower control during truncation. The weighted construction in Section 31.6.1 reconstructs [Li26, Proposition 6.1]. For unrestricted addition, (31.17) follows [Li26, Theorem 6.3]; an extra rooftop supplies the common lower bound needed to pass through the cutoff.

33.10 Normalization and adjoint measures

The normalization constant can be studied through its adjoint probability measures even when the potential itself is not unique. The smooth differential method appears in [Col21]; Theorem 14.1.1 replaces differentiation of smooth solutions by mixed products and domination.

For the mass quotient and closed examples in Section 14.10, the prescribed-type input, including uniqueness, is [DDNL18, Theorems 1.4(i) and 4.29, pp. 2052 and 2079–2080]; bounded-difference mass invariance is its [DDNL18, Proposition 2.1, p. 2055]. Its hypotheses are a compact Kähler manifold, a smooth closed big class, a model reference with small unbounded locus, and positive mass. The positive-residual factorization concerns actual ordinary divisorial types and does not assert a model-envelope identity for every real weighted sum.

33.11 Singularity classes and traces

A singularity theory for nonclosed backgrounds must retain torsion as well as divisorial information. The quotient of smooth real $(1,1)$-forms by smooth dd^c-terms in Chapter 23 is used in [BGL25, Section 1]; fixing the exterior derivative preserves that datum on each modification.

Resolution makes addition of traces accessible to multiplier-ideal approximation. The comparison in Proposition 18.1.1 is the argument underlying Cao's addition theorem [Cao15, proof of Theorem 1.4]; it does not require a global closed background. The recovery theorem Theorem 29.5.3 combines the decreasing-model and stretching arguments with the local ideal criterion. Uniform exponential integrability bounds each local divisorial order of the auxiliary potentials, so the proof does not require a local modification to extend over the compact manifold. The ruled-surface construction of Section 24.5, based on [DPS94, Example 1.7], exhibits the boundary behavior relevant to that recovery.

The closed model-envelope continuity used in Theorem 18.6.3 is [DX24, Proposition 2.1], also presented in [Xia, Proposition 3.1.12]. The Lelong-number argument uses small convex corrections from [DDNL21, Lemma 4.3]. Applying this continuity on the submanifold suffices here: the restricted approximants have a common curvature bound even when the ambient background is nonclosed.

33.12 Section growth and partial kernels

Section growth links singular potentials to algebraic dimension through a projective modification. The form of Moishezon's theorem used in Chapter 20 is recorded in [Moi67, Introduction, p. 1332] and retained as a classical geometric input. A Kähler modification also yields the bounded-mass conclusion of Lemma 20.4.1, by the upper-volume pullback argument of [GL22, Lemma 3.6].

For section rings constructed through algebraic reduction, see [Wu24]. The flag construction in Chapter 20 uses a specified compact flag on an auxiliary modification; it makes no assertion about centers of arbitrary meromorphic-field valuations.

33.13 Gauduchon variational volume

The Gauduchon variational volume extracts a homogeneous concave quantity from an elliptic normalization problem. Its method belongs to the construction of [Col21, Theorem 2.13]; Chapter 25 develops the adjoint formula and its dependence on the chosen probability measure.

For a closed Kähler class represented by θ, this construction meets the Hessian geometry of the Kähler cone. The metric of Section 25.3 is $1/n$ times the Hessian of $-\log \int_X \theta^n$ used in [Mag12, Section 1]. Completeness and positive-volume boundaries at finite distance are studied in [Mag12, Proposition 4.4 and Lemma 4.5]; Section 25.5 gives the homogeneous argument in the present normalization.

In Section 25.11, bigness of the closed nef positive-intersection core follows from [DP04, Theorem 0.5], the classical criterion recalled in [Xia]; no such criterion for arbitrary nonclosed nef backgrounds is used.

The determinant endpoint bound in Theorem 25.8.2 follows internally from the mass-invariance criterion and the lower-mass quotient. Related mass criteria are [GL22, Propositions 3.2 and 3.3(iv)]. The concentrating family in Section 25.14 distinguishes the unique smooth maximizer for one density from maximizing sequences with divergent higher moments.

The classical analytic input in Section 25.16.2 is Hironaka's elimination of indeterminacies: a modification is dominated by smooth-center blowups. The principalization and domination argument is explained in [AKMW02, Sections 1.2.3–1.2.4, pp. 538–539, and Section 5.4, p. 564], with the analytic case explicit. Exceptional-divisor curvature then supplies the uniform pairing bound used here; no factorization into a sequence of blowdowns is needed for this estimate.

The existence of a scalar conformal factor for a weak Monge–Ampère power is raised in [Lu22, Chapter 3, Section 5]. The finite-current representation of Theorem 25.7.1 supplies a supporting probability without defining that weak power or its scalar factor.

33.14 Closed classes and singular loci

For closed classes on a Hermitian manifold, an almost positive representative keeps cohomology separate from the auxiliary metric. The convention in Section 27.1 follows [Bou02, Section 2.1.2]; it does not assign a Bott–Chern class to a nonclosed Hermitian form.

The smooth and analytic regularizations in Section 1.15.1, used for Proposition 27.2.1, have the lower bounds in [Bou04, Theorem 2.1]; see also [Bou02, Theorems 1.2.9–1.2.10]. Smooth regularization loses a multiple of a continuous approximation of the Lelong numbers; analytic regularization has vanishing uniform curvature error and does not increase Lelong numbers.

Minimal multiplicities, Definition 27.2.1, are defined through these almost positive approximating families as in [Bou02, Definition 2.1.5]. The distinction from a single minimal current is essential: the nef ruled-surface example in [Bou02, Section 1.2.3] has a current with minimal singularities and nonzero Lelong numbers.

The non-Kähler locus in Definition 27.3.1 is defined in [Bou02, Definition 2.1.19]. The finite-intersection and regularized-maximum proof of Theorem 27.3.1 is the argument of [Bou02, Theorem 2.1.20].

The equivalence between a Kähler current and a Kähler modification, used in Chapter 27, is the classical criterion recalled in [Bou02, Theorem 1.2.11]. It is retained as an external geometric input. The equality of the non-Kähler and null loci for a nef and big class is the transcendental Nakamaye theorem of Collins–Tosatti [CT15], also valid on compact manifolds in Fujiki's class $\mathcal{C}$. The discussion in Chapter 27 recalls this external theorem.

Restricted volumes provide a numerical test for these loci. The numerical restricted volume in (27.1) follows Collins–Tosatti [CT22, Section 2]. The smooth remainder in the analytic singularities is part of that definition.

The existence of Kähler models for every subvariety uses Fujiki's closure of class $\mathcal{C}$ under closed subspaces [Fuj78, Lemma 4.6(2)], followed by the Kähler modification theorem of Varouchas in the form stated and explained by Barlet [Bar00, Theorem 10, pp. 31–32]. A common Kähler refinement follows by embedded resolution of the graph in a Kähler product, using [BM97, Theorem 1.6 and Remark 1.7(2)]. These are classical geometric inputs. Potential descent is already proved in Proposition 1.22.2, so the mass-invariance proof does not repeat it.

For Proposition 27.6.2, positive non-pluripolar mass implies bigness by [BEGZ10, Proposition 1.22]. The positive-product definition, its agreement with the cup product for nef classes, and attainment by minimal currents are in [BEGZ10, Definition 1.17, p. 219 and Proposition 1.20]. The normalized diagonal current in (27.7) uses the mass-concentration theorem [DP04, Proposition 2.6] and spells out the coefficient in the construction of [DP04, Theorem 2.12]. These Kähler results remain explicit classical inputs. The bounded-residual mass identity in Lemma 27.7.1 instead follows directly from the book's bounded mass formula by polynomial comparison. The nef comparison with numerical restricted volume is [CT22, Lemma 2.6]; the final equality in (27.9) uses the already recalled theorem [CT15, Theorem 1.1].

33.15 Closed classes and metric completeness

The incomplete example Theorem 27.14.1 tests the big-class completeness question raised in [DNGL26, Question 3.5]. It does not settle the flow problem of [DNGL26, Section 3.1], where the class, divisor and coefficients must come from specified flow data. Likewise, the normal-length examples have logarithmically weakened divisorial poles, not the nonintegrable pole with a smooth positive coefficient across the divisor required in that setting.

The energy argument in Section 27.14 uses the classical finite-energy class of [BEGZ10, Section 2], relative to a minimal-singularity reference. Finite mixed masses and relative energy are supplied by [BEGZ10, Proposition 1.6 and Section 2.2]. Uniqueness uses full-mass comparison [BEGZ10, Corollary 2.3]: the unweighted reference measure is infinite, but the two Monge–Ampère measures being compared are finite.

Hölder regularity imposes a separate metric restriction. For a positive residual current with a Hölder potential, smooth and strictly positive off a proper analytic set, Theorem 1.18.1 puts every point of that set at finite completion distance. In the compact setting, deleting a nonempty analytic set therefore gives an incomplete metric. Continuity alone permits complete ends: the prescribed radial densities of Theorem 27.12.2 have different logarithmic thresholds for boundedness of the minimal remainder and finite normal length. Their complete bounded examples retain the prescribed-density scope and do not supply the flow compatibility discussed above.

33.16 Locally conformally Kähler geometry

In locally conformally Kähler geometry, the Lee form constrains both metric rigidity and prescribed volume. The Lee-class pseudoeffectivity conjecture Conjecture 28.2.1 is formulated in terms of Gauduchon degrees in [OV25a, arXiv version, Conjecture 4.14 and Remark 4.16]; its balanced-metric consequence is Proposition 28.2.2. The Vaisman volume theorem of Ornea–Verbitsky [OV24a, arXiv version, Theorem 4.1 and Lemmas 4.2–4.4] requires invariance under both the Lee and anti-Lee vector fields. The proof of Theorem 28.4.1 uses Theorem 9.1.1: uniqueness gives a basic potential, transverse Stokes integration fixes the scalar, and the reconstructed Lee form is parallel because it is closed with Killing dual. The uniqueness argument retains the candidate metric's scale in (28.13) until integration determines it.

The proof of Theorem 28.5.1 specializes the argument of [BO25, Section 4] to the Chern–Ricci-flat case. The manifold-level conjecture Conjecture 28.5.1 is stated in [BO25, Introduction]. Vanishing first Bott–Chern class supplies a Chern–Ricci-flat conformal representative, which need not be the Gauduchon representative required by the rigidity argument.

The dual description and bimeromorphic invariance in Section 28.6 are due to Ornea–Verbitsky [OV25b, Theorems 3.11 and 4.5]. The distinction between vanishing Lee–Gauduchon class and the strongly Gauduchon condition in Section 28.7 follows [OV25b, Remarks 5.6–5.8]. The strongly Gauduchon existence criterion in Section 28.7 is due to Popovici [Pop13, Proposition 1.2].

33.17 Regularity and singular spaces

Regularity in a big class is first a local question on the locus where strict barriers are smooth. For a closed class on a compact Kähler manifold, existence and uniqueness in (27.16) are [BEGZ10, Theorem A, equivalently Theorem 3.1]; the minimal bounds (27.17) are [BEGZ10, Theorem B, quantitatively Theorem 4.1]. The same full-mass uniqueness theorem identifies the closed solutions in Sections 13.20, 13.21, 27.12 and 27.13. The nef case of ample-locus smoothness is [BEGZ10, Theorem C]; Theorem 27.10.1 gives smoothness in every closed big class for a smooth positive volume form.

For a possibly nonclosed big background, the local Hölder theorem Theorem 13.16.1 reconstructs [Dan26b, Theorem A and Theorem 3.1]. This supplies the local input for the ample-locus argument while retaining the unbounded global potential and nonclosed lower curvature bound. The formulation with a positive scalar and an equation only on the regular locus is equivalent, since the analytic complement carries neither smooth volume nor non-pluripolar mass.

The local regularity proof also needs a quantitative stability estimate. The strict-barrier estimate of Section 13.14 reconstructs [Dan26b, Theorem 3.3 and Remark 3.4] in the minimal case, with the comparison potential normalized explicitly. The auxiliary-subsolution construction of Section 13.3 first treats a semipositive big background.

The interior theorem Theorem 13.17.1 combines this Hölder control with smooth positive density on the entire regular locus of one fixed strict analytic barrier. That domain is part of the hypothesis: smoothness of the density on a smaller open set does not give the same theorem by restriction. The compact Pogorelov example of Example 27.11.2 has smooth positive density on an inner chart but loses density regularity along the divisor in the gluing region, so it does not meet that hypothesis.

Local smoothing uses the Dirichlet theorem of Guan–Li [GL10, Theorem 1.1, p. 1186]. In (13.46), the background is zero and the metric is $(\mathrm{i}/2)\partial\bar{\partial}|z|^2$; a large quadratic correction supplies a strict subsolution for density G. The same input constructs the approximants in Section 13.17.3 and starts the passage to continuous boundary values and zero density in Section 13.6. In dimension one it is replaced by the Poisson Dirichlet problem. Once uniform ellipticity holds in (13.14), the variable-coefficient theorem [TWWY15, Definition 3.2, Theorem 3.4, and the paragraph following it], followed by difference quotients and Schauder estimates, gives smoothness.

The local identification in Section 13.19 uses bounded Dirichlet uniqueness [BT82, Corollary 4.4, p. 17]. Degeneration of the density also matters for global stability: [KN21, Theorem 4.1] assumes $f \geq c_0 > 0$ in addition to its capacity hypotheses. The nonuniqueness densities of Chapter 11 lack this lower bound even when positive almost everywhere.

For compact sets in a Hermitian manifold, Ahn [Ahn26, Theorems 1.1–1.2] proves pointwise locality of Hölder continuity of the extremal function. The weighted conclusion assumes Hölder continuity of both the unweighted extremal and the weight, together with uniform density in capacity. The same density condition enters the equivalence with local and weak local Hölder properties. These are the regularity regimes referred to in Chapter 32.

On singular spaces, regularity upstairs must be combined with descent. Dang [Dan26b, Theorem B] proves continuity results for bounded solutions on resolutions under additional geometric and integrability hypotheses, notably the log terminal setting. This result does not cover arbitrary unibranch singularities in the problem posed in Chapter 32. Pan [Pan25, Theorems A and A'] constructs a Gauduchon metric on the regular locus of a compact reduced irreducible Hermitian space of dimension at least two. It is conformal and uniformly comparable to the given metric, and unique up to scale within that class. Control of conformal factors on a resolution is central to the construction. This external result motivates the singular trace questions in Chapter 32.

33.18 Geometric problems

A closed strict current has geometric consequences that a positive nonclosed background need not have. This distinction governs the passage from equation normalizations to the geometric criteria below. The Kähler-current characterization of Fujiki's class used in Section 32.3 is [DP04, Theorem 0.7]; the transcendental

Grauert–Riemenschneider conjecture is [DP04, Conjecture 0.8]. These published locators correspond to Theorem 0.6 and Conjecture 0.7 in the arXiv version.

Under nefness and positive top self-intersection, the equivalence between bigness and bounded mass cited in Section 32.3 is [GL22, Theorem 4.9]. The characterization of Fujiki's class by bounded mass and positive upper class volume is [BGL25, Theorem C]; that upper volume differs from the fixed-probability Gauduchon functional.

The closed Kähler transcendental Morse inequality, volume differentiability and cone-duality conclusions mentioned in Section 32.3 are [Tos26, Theorem 1.1 and Corollary 1.2, version 2], retained as literature results without reconstructing their proof. The qualitative criterion under bounded mass is [GL22, Theorem 4.12].

The non-Hermitian locus in Section 32.3 uses [Dan26a, Definition 2.12 and Theorem 2.14]. Its almost analytic singularities have the exact meaning of [Dan26a, Definition 2.8], including a smooth remainder on a suitable log resolution. Dang's qualitative null-locus equality for nef and big, possibly nonclosed backgrounds is [Dan26a, Theorem 1.1]. Theorem A of that paper gives a numerical criterion under bounded mass on the ambient manifold and positive lower volume on its positive-dimensional submanifolds. These are distinct scopes.

Theorem C of [Dan26a] identifies finite-time Chern–Ricci singularities with the null locus under positive limiting lower volume, for arbitrary initial metrics in dimension two and Guan–Li initial metrics in higher dimensions. Writing ω_0 for the initial metric, the latter satisfy $\mathrm{dd^c}\omega_0 = 0$ and $\mathrm{d}\omega_0 \wedge \mathrm{d^c}\omega_0 = 0$. Once the limiting background is big, [Dan26a, Theorem 4.3] gives current convergence and smooth local convergence off its non-Hermitian locus. This is the parabolic input distinguished from the numerical bigness question in Problem 32.3.4.

The maximal-time theorem Theorem 25.17.2 is [TW15, Theorem 1.2, (1.4), and Section 4, arXiv version]. Its proof here combines the positive affine path and time-weighted potential estimates with Lemma 9.4.1 and classical parabolic regularity. Potentials are multiplied by 2π; time is unchanged. The geometric smoothing conditions of [DNGL26, Section 3.1] retain the further compatibility discussed in Section 33.15.

The balanced–pluriclosed conjecture referred to in Section 32.3 is stated in [FV26, Conjecture 1.1]. The balanced and pluriclosed metrics are allowed to be different. The assertion that a pluriclosed manifold in Fujiki's class is Kähler is [Chi14, Theorem 0.2 in the author manuscript]. The locator refers to the author manuscript used in Section 32.3.

The proposed long-time existence and convergence of pluriclosed flow in the balanced setting with $c_1(X) = 0$ is [FV26, Conjecture 1.2]. The discussion in Section 32.3 distinguishes a conserved balanced trace from a pointwise metric bound.

Appendix A
Potentials and equations on unibranch spaces

The pullback of a smooth background has no vertical component. The maximum principle on connected compact fibers therefore makes potentials descend. This correspondence preserves envelopes and convergence, while exceptional sets of zero mass make the extended Monge–Ampère measure independent of the resolution.

Removing a logarithmic pole can leave a strictly positive residual background. On projective cones this gives uniqueness in the prescribed ordinary class, including certain nonclosed backgrounds. The bounded equation instead uses closedness and an energy identity. The full logarithmic potential descends even when its smooth residual varies along the contracted divisor.

Throughout this appendix X is a reduced irreducible compact unibranch complex space of pure dimension $n \geq 1$. Smooth functions and forms on X are understood through compatible local embeddings in complex Euclidean spaces. A Hermitian form is locally the restriction of a smooth positive ambient form. Fix a smooth real $(1,1)$-form θ on X.

Resolutions and common smooth refinements exist, and a proper bimeromorphic map to a unibranch space has connected fibers; see [Xia, Theorems B.1.1–B.1.3]. The Fornaess–Narasimhan criterion identifies plurisubharmonic functions by upper semicontinuity and subharmonicity on holomorphic disks [Xia, Theorem B.2.1]. The ramified disk lift used below is the argument in [Xia, proof of Theorem B.2.2].

A.1 Potentials and disk tests

The pullback of θ to the unit disk Δ has a smooth local potential. Disk tests therefore express the background inequality even when θ has no local potential on X.

Definition A.1.1 An upper semicontinuous function $u : X \to [-\infty, +\infty)$, not identically $-\infty$ on any nonempty open subset, is *θ-plurisubharmonic* if the following disk condition holds. For every holomorphic $f : \Delta \to X$, the function $u \circ f + a$ is subharmonic or identically $-\infty$ wherever a is a smooth local potential for $f^*\theta$. The set of these functions is denoted by $\mathrm{PSH}(X, \theta)$.

Two choices of a differ by a harmonic function, so the condition is independent of that choice. On every disk where the pullback is genuine, it gives $\mathrm{dd^c}(u \circ f) + f^*\theta \geq 0$ in distributions.

Lemma A.1.1 *Every member of* $\mathrm{PSH}(X, \theta)$ *is locally the sum of a plurisubharmonic function on X and a smooth function. If X is smooth, this definition agrees with Definition 1.1.1.*

Proof In a local embedding choose an ambient smooth strictly plurisubharmonic function r with $\mathrm{dd^c} r \geq \theta$ on a smaller neighborhood. For a disk f and a local potential a of $f^*\theta$, the function $r \circ f - a$ is subharmonic. Hence $(u + r) \circ f$ is subharmonic or identically $-\infty$. The Fornaess–Narasimhan criterion and the local nondegeneracy of u make $u + r$ plurisubharmonic. The smooth case now follows from Lemma 1.22.1.□

A.2 Descent through a resolution

The pullback of the background has no vertical component. A local correction therefore makes a potential plurisubharmonic near a fiber, and connectedness identifies the constants on its irreducible components.

Theorem A.2.1 *Let $\pi : Y \to X$ be a resolution. Pullback is an order-preserving bijection*

$$\pi^* : \mathrm{PSH}(X, \theta) \longrightarrow \mathrm{PSH}(Y, \pi^*\theta).$$

It preserves suprema of individual functions, boundedness, and ordinary singularity comparisons.

Proof Pullback preserves the disk inequalities. Every nonempty open subset of Y meets the isomorphism locus, so the pullback is locally genuine. Injectivity follows from surjectivity of π.

For the converse take $v \in \mathrm{PSH}(Y, \pi^*\theta)$. Near a fiber choose the local correction r on X used above. The function $v + r \circ \pi$ is plurisubharmonic on its inverse image, since

$$\mathrm{dd^c}(v + r \circ \pi) \geq \pi^*(\mathrm{dd^c} r - \theta) \geq 0.$$

Its restriction to each compact irreducible component of the fiber is constant, possibly identically $-\infty$, by the maximum principle on a resolution of that component. Connectedness of the fiber makes the constants equal: intersecting components have the same value, and the finite component intersection graph is connected. Since $r \circ \pi$ is constant on the fiber, so is v.

Thus $v = u \circ \pi$ for a unique function u on X. It is upper semicontinuous: for each real c, the closed set $\{v \geq c\}$ has closed image under the proper map, and that image is $\{u \geq c\}$. If u were identically $-\infty$ on a nonempty open set, the same would hold for v on its open inverse image. Thus u is locally genuine.

It remains to retain the exact background in the disk test. Let $f : \Delta \to X$ have a genuine pullback $u \circ f$. Near a chosen point, take a curve in $\Delta \times_X Y$ dominating

the disk. Its normalization gives, after shrinking, a finite cover $\rho(t) = t^k$ and a holomorphic lift $f' : \Delta \to Y$ with $\pi \circ f' = f \circ \rho$. Choose a smooth local function a on the original disk with $\mathrm{dd^c} a = f^*\theta$. The function

$$(u \circ f + a) \circ \rho = v \circ f' + a \circ \rho$$

is subharmonic by Lemma 1.22.1. Subharmonicity descends through ρ by the finite-disk argument following that lemma. Thus $u \circ f + a$ is subharmonic at every point, proving the bijection.

Surjectivity preserves suprema, uniform bounds and inequalities with constants, proving the remaining assertions. □

A.3 Envelopes and normalized compactness

Fix a resolution $\pi : Y \to X$. The envelope constructions of Section 1.19 and Definition 1.19.1 now apply on X. In the following obstacle statement the inequality against f is pointwise, and its family of minorants is assumed nonempty and locally uniformly bounded above. The descent bijection identifies the competitors; the remaining issue is whether upper semicontinuous regularization commutes with pullback.

Proposition A.3.1 *For such an extended-real obstacle $f : X \to [-\infty, +\infty]$, the envelope is genuine and*

$$P_{\pi^*\theta}(f \circ \pi) = P_\theta(f) \circ \pi.$$

It also holds for the ordinary model envelope $P_\theta[u]$, for every $u \in \mathrm{PSH}(X, \theta)$.

Proof The preceding bijection identifies the competitor families. Let s be their pointwise supremum on X; its pullback is the pointwise supremum upstairs. Its regularization on Y is a genuine $\pi^*\theta$-plurisubharmonic function, since there is a genuine lower competitor. It descends to a θ-plurisubharmonic function q by the theorem. Since $q \geq s$ and q is upper semicontinuous, $q \geq s^*$.

For the reverse inequality fix $x \in X$ and $y \in \pi^{-1}(x)$. For each neighborhood U of x, the unregularized supremum upstairs is bounded by $\sup_U s$ on the open neighborhood $\pi^{-1}(U)$ of y. Therefore $q(x) \leq \sup_U s$. Taking the infimum over U gives $q(x) \leq s^*(x)$. This proves both the identity and genuineness downstairs. The proof for $P_\theta[u]$ is identical, because ordinary singularity order and its constants are preserved by pullback. □

The normalized space $\{u \in \mathrm{PSH}(X, \theta) : \sup_X u = 0\}$ is compact in the L^1 topology of its pullbacks to Y, by Theorem 1.4.1 and descent. Smooth positive volume forms give equivalent norms on a fixed compact resolution. Under a further modification, pullback is continuous by Lemma 1.22.2 and bijective by descent. It is therefore a homeomorphism between the normalized compact spaces. A common resolution compares any two choices. This proves independence of the topology without asserting a comparison of the norms on different models.

A.4 Ideal envelopes defined on resolutions

Multiplier ideals on smooth models give a singularity relation on X. Call two potentials on X *resolution-ideal equivalent* if their pullbacks are $\mathcal{I}$-equivalent on one smooth resolution. The all-weight equivalence of Lemma 1.22.6, applied on a common refinement, makes this independent of the resolution. The smooth projection of Theorem 15.2.1 then defines an envelope in this class.

Proposition A.4.1 *For $u \in \mathrm{PSH}(X, \theta)$, there is a unique potential $P_\theta[u]^{\mathrm{res}}_{\mathcal{I}}$ on X satisfying*

$$P_\theta[u]^{\mathrm{res}}_{\mathcal{I}} \circ \pi = P_{\pi^*\theta}[u \circ \pi]_{\mathcal{I}}$$

on every resolution. It preserves the resolution-ideal class, is idempotent, has supremum zero, and is independent of the resolution.

Proof For a fixed resolution the right side exists by Theorem 15.2.1 and descends by Theorem A.2.1. Under a further smooth modification, the ideal-envelope competitor families correspond by the descent theorem and Lemma 1.22.6. The regularized suprema commute by the argument of Proposition A.3.1. Thus both resolutions give the same pullback on a common refinement and hence the same function on X. Projection, preservation of the class and normalization follow from the corresponding smooth assertions and surjectivity. □

An intrinsic restricted-ideal trace on a subspace contained in the singular locus requires regularization compatible with restriction there. The resolution-ideal envelope alone supplies neither this regularization nor the trace.

A.5 Extended measures and finite damping

Write X_{reg} and X_{sing} for the regular and singular loci, and choose a resolution $\pi : Y \to X$. Two resolutions are isomorphic over the complement of a proper analytic subset of X. Their extended measures agree there, while non-pluripolarity removes the omitted sets from the comparison, even when the total mass is infinite.

For $u \in \mathrm{PSH}(X, \theta)$ define

$$\mathrm{MA}^{\mathrm{ext}}_{X,\theta}(u) = \pi_* \mathrm{MA}^{\mathrm{ext}}_{Y,\pi^*\theta}(u \circ \pi), \tag{A.1}$$

where $\mathrm{MA}^{\mathrm{ext}}_{Y,\pi^*\theta}$ is the extended measure of Theorem 6.1.1 on Y.

Theorem A.5.1 *The right side is the same for every smooth resolution, including further modifications over regular points. It is a positive sigma-finite Borel measure, gives zero mass to X_{sing} and to locally pluripolar sets, and may have infinite total mass. For bounded u it is finite. Put $r_n = \max(n-2, 0)$. There are finite constants C and, for each $q > r_n$, C_q, depending only on the fixed background data and on q in the second case, such that for every nonpositive $u \in \mathrm{PSH}(X, \theta)$ and $p > 0$,*

$$\int_X \mathrm{e}^{pu}\, \mathrm{MA}^{\mathrm{ext}}_{X,\theta}(u) \le C(1 + p^{-1})^{r_n}, \qquad \int_X (1-u)^{-q}\, \mathrm{MA}^{\mathrm{ext}}_{X,\theta}(u) \le C_q.$$

The constants may be obtained on any one fixed resolution and are uniform over all such u.

Proof For two resolutions choose a proper analytic subset $S \subset X$ containing X_{sing} and both exceptional images. Each map is an isomorphism over $X \setminus S$. There the two measures are the same bounded local expansion followed by truncation and exhaustion. On each resolution, the inverse image of S is a proper analytic set and has zero extended mass by Theorem 6.1.1. Hence both pushforwards give S zero mass. This proves independence, including for resolutions with exceptional sets over regular points.

Positivity follows upstairs. On $X \setminus S$ either resolution is a Borel isomorphism, so a countable finite-measure exhaustion upstairs pushes forward to one downstairs; adjoining the null set S proves sigma-finiteness. A locally pluripolar set pulls back to a pluripolar set off the exceptional locus, and that locus itself has zero mass. This gives the asserted vanishing.

If u is bounded, its pullback is bounded on the compact smooth resolution. Local bounded-product estimates give finite total mass. For general nonpositive u, apply Theorem 7.6.1 on Y to the arbitrary smooth background $\pi^*\theta$. Pushforward and $u \circ \pi$ identify the weighted integrals exactly. The constants come from the fixed smooth background and Hermitian metric on Y. □

Define capacity convergence on X by capacity convergence of pullbacks to a resolution, using a smooth Hermitian metric there. The local capacity comparison of Section 1.8 makes the metric immaterial, and Lemma 1.22.3 applied on a common resolution makes the choice of model immaterial.

It follows from Theorem 7.6.1 and proper pushforward that both damped measures in the theorem are weakly continuous under this capacity convergence, for nonpositive potentials. For a sequence of nonpositive potentials $u_j \in \mathrm{PSH}(X, \theta)$ converging in capacity, the undamped measure is weakly continuous if, in addition, its total masses are uniformly bounded and its masses on $\{u_j < -T\}$ tend uniformly to zero as the real number T tends to infinity. Pullback preserves these mass and sublevel conditions.

General equations on singular spaces require estimates for the degeneracy of the pulled-back background and density; see Problem 32.10.1. For the cone geometry below, a radial construction gives solutions in a prescribed logarithmic class and in the closed bounded class.

A.6 Equations on projective cones

The residual construction of Section 13.20 has a singular-space counterpart on a projective cone. A logarithmic pole at the vertex becomes a divisor on the incidence resolution, and removing it leaves a positive residual background. The radial equation below determines both the solution and its normalization scalar. Bounded uniqueness applies because the lifted density is smooth with connected dense positive locus, even though it vanishes on the exceptional section.

Let $B \subset \mathbb{P}^N$ be a smooth connected, linearly nondegenerate, projectively normal variety of dimension $n-1$, where $N \geq n \geq 2$. Let $X \subset \mathbb{P}^{N+1}$ be its projective cone, with vertex $p = [0 : \cdots : 0 : 1]$, and put $\theta_0 = \omega_{\mathrm{FS}}|_X$. The incidence resolution $\pi : Y \to X$ is a smooth $\mathbb{P}^1$-bundle $f : Y \to B$; its section $E = \pi^{-1}(p)$ is contracted, and π is an isomorphism off E. These cones are reduced, irreducible, compact and normal, hence unibranch, and their vertices are singular.

The homogeneous coordinate ring of B is normal by projective normality. Adjoining W preserves normality, and its degree-zero projective charts are normal. Algebraic normality passes to the associated analytic space by the algebraic–analytic normality theorem. Smoothness and connectedness of B give irreducibility. In the chart $W = 1$, the ideal of the affine cone has no linear terms, by linear nondegeneracy. Its tangent dimension at the origin is $N + 1 > n$, proving that p is singular. The incidence construction is the bundle of lines in $\mathcal{O}_B \oplus \mathcal{O}_B(-1)$ and has the stated contraction and isomorphism properties.

Write

$$\beta = \pi^*\theta_0, \qquad \sigma = f^*\omega_B, \qquad \omega_B = \omega_{\mathrm{FS}}|_B,$$
$$d = \int_B \omega_B^{n-1}, \qquad x = \frac{\|Z\|^2}{\|Z\|^2 + |W|^2}.$$

The same symbol x denotes its pullback to Y. Choose $\Psi \in C^\infty(\mathbb{P}^N, \mathbb{R})$ with $\omega_{\mathrm{FS}} + \mathrm{dd^c}\Psi > 0$ and $k \in C^\infty([0,1])$ with $k > 0$. Put $\psi = \Psi|_B$ and define on Y

$$\begin{gathered} s = \frac{\mathrm{e}^\psi x}{1 - x + \mathrm{e}^\psi x}, \qquad \ell = \log(1 - x + \mathrm{e}^\psi x), \qquad \sigma_\psi = f^*(\omega_B + \mathrm{dd^c}\psi), \\ B_\psi = \beta + \mathrm{dd^c}\ell, \qquad \nu = k(s)B_\psi^n, \qquad \mu = \pi_*\nu, \\ I = \int_0^1 k(r)r^{n-1}\,\mathrm{d}r, \qquad M = ndI. \end{gathered} \tag{A.2}$$

These data prescribe the measure before the equation is solved. The local form and fiber integral below identify its degeneracy along E and its total mass M.

Choose a local holomorphic lift $b(z)$ of the embedding of B and write $S(z) = \sum |b_j(z)|^2$, $Z = tb(z)$. On this chart

$$\beta = \mathrm{dd^c}\log(1 + |t|^2 S), \qquad \sigma = \mathrm{dd^c}\log S.$$

On $0 < x < 1$ put $\rho = \log(|t|^2 S)$, $\rho_\psi = \rho + \psi$, $h(r) = \log(1+\mathrm{e}^r)$ and $\zeta_\psi = \mathrm{d}\rho_\psi \wedge \mathrm{d}^c\rho_\psi$. Then $x = h'(\rho)$, $s = h'(\rho_\psi)$ and $\ell = h(\rho_\psi) - h(\rho)$. Differentiation gives

$$B_\psi = s\sigma_\psi + s(1-s)\zeta_\psi, \qquad \sigma_\psi^n = 0, \qquad \zeta_\psi^2 = 0. \tag{A.3}$$

Near E, a local potential is $\log(1 + \mathrm{e}^\psi |t|^2 S)$; at $t = 0$ its Hessian has rank one in the transverse direction. At the infinity section, with $v = 1/t$, a local potential is $\log(\mathrm{e}^\psi S + |v|^2)$; its tangential Hessian is σ_ψ and its transverse coefficient is positive. These computations show that B_ψ is smooth and semipositive, positive off E. Since $n \geq 2$, its top power vanishes on E.

Along a fiber, $\mathrm{d}^c\rho_\psi = \mathrm{d}(\arg t)/(2\pi)$ and $\mathrm{d}\rho_\psi = \mathrm{d}s/[s(1-s)]$. Base components disappear after wedging with σ_ψ^{n-1}, so (A.3) gives

$$f_*\nu = nI(\omega_B + \mathrm{dd}^c\psi)^{n-1}.$$

Smooth closed Stokes on B gives $\int_B (\omega_B + \mathrm{dd}^c\psi)^{n-1} = d$, hence $\nu(Y) = M$. Smooth measures give E zero mass. Therefore $\mu(X) = M$ and $\mu(\{p\}) = 0$. Taking $\psi = 0$ and $k = 1$ gives $\int_Y \beta^n = d$.

A.6.1 Logarithmic poles at the vertex

The logarithmic pole supplies the tangential positivity missing from β on E. Fix $0 < a < 1$ and put $\varphi = a \log x$, with value $-\infty$ at p. Set

$$\begin{gathered}
\beta_a = (1-a)\beta + a\sigma, \qquad V = d(1-a^n), \qquad c_a = V/M, \\
P_a(s) = \left(a^n + c_a n \int_0^s k(r) r^{n-1}\,\mathrm{d}r\right)^{1/n}, \\
Q_a'(s) = \frac{P_a(s) - a - (1-a)s}{s(1-s)}, \qquad Q_a(1) = 0, \\
q_a = a\psi + (1-a)\ell + Q_a(s), \qquad A_a = \beta_a + \mathrm{dd}^c q_a.
\end{gathered} \tag{A.4}$$

The endpoint values in Q_a' are removable. The form β_a is strictly positive everywhere, whereas A_a is semipositive and strictly positive off E. On that complement one may view A_a as a form on $X \setminus \{p\}$.

Choose a relatively compact coordinate ball $O \subset X \setminus \{p\}$ and a smooth compactly supported endomorphism H of $T^{1,0}O$ which is A_a-self-adjoint and trace-free. On O define

$$A_H(v, \bar{w}) = A_a(\exp(H)v, \bar{w}), \qquad \eta_H = A_H - A_a, \qquad \theta_H = \theta_0 + \eta_H, \tag{A.5}$$

extending η_H by zero. We call H admissible when

$$\theta_H > 0, \qquad \beta_{a,H} := \beta_a + \pi^*\eta_H > 0. \tag{A.6}$$

These are explicit open inequalities in the prescribed data. Sufficiently small H satisfies them.

Theorem A.6.1 *For every admissible H, the ordinary class*

$$C_{a,H} = \{w \in \mathrm{PSH}(X, \theta_H) : \ \varphi - C \le w \le \varphi + C \text{ for some } C < \infty\}$$

contains a unique pair $c > 0$, u with $\sup_X u = 0$ such that

$$\mathrm{MA}^{\mathrm{ext}}_{X,\theta_H}(u) = c\mu.$$

It is given by $c = c_a$ and

$$u \circ \pi = \varphi \circ \pi + q_a - K, \qquad K = \sup_Y(\varphi \circ \pi + q_a).$$

Every class member has finite extended mass. The solution has mass V and is the same for every admissible H. For $H = 0$, every class member has mass V.

The solution is continuous with values in the extended real line, is smooth off p, and tends uniformly to $-\infty$ at p. Its smooth residual satisfies

$$(q_a - K)|_E = a\psi + Q_a(0) - K.$$

If ψ is nonconstant, this residual is not the pullback of any function on X. Arbitrarily small admissible choices of H give genuinely nonclosed θ_H.

For every $w \in C_{a,H}$ and every smooth resolution $\tau : Z \to X$, including further modifications over regular points,

$$\tau_* \left\langle \left(\tau^*\theta_H + \mathrm{dd^c}(w \circ \tau)\right)^n \right\rangle = \mathrm{MA}^{\mathrm{ext}}_{X,\theta_H}(w). \tag{A.7}$$

The upstairs measure is finite. Thus the measure, normalized descended solution and scalar are independent of the resolution and of every common refinement.

Proof The divisor formula is

$$\beta + \mathrm{dd^c}(\varphi \circ \pi) = a[E] + \beta_a. \tag{A.8}$$

Indeed, $\log x = \log |t|^2 + \log S - \log(1 + |t|^2 S)$ and $\mathrm{dd^c} \log |t|^2 = [t = 0]$. At E the kernels of β and σ are complementary. Off E, $\beta > 0$. Thus β_a is a smooth Kähler form. Fiber integration also gives

$$\int_Y \beta_a^n = nd \int_0^1 [a + (1-a)x]^{n-1}(1-a)\,\mathrm{d}x = d(1 - a^n) = V. \tag{A.9}$$

From (A.4), $P_a(0) = a$, $P_a(1) = 1$ and

$$P_a'(s) = \frac{c_a k(s) s^{n-1}}{P_a(s)^{n-1}}.$$

The numerator defining Q_a' vanishes at both ends, proving smoothness of Q_a. On $0 < x < 1$, a local potential for the full positive current is $F_a(\rho_\psi)$, where

$$F_a(r) = ar + (1-a)h(r) + Q_a(h'(r)).$$

Direct differentiation gives $F_a' = P_a(s)$ and $F_a'' = c_a k(s) s^n (1-s)/P_a(s)^{n-1}$. Consequently

$$A_a = P_a(s)\sigma_\psi + \frac{c_a k(s) s^{n-1}}{P_a(s)^{n-1}}(B_\psi - s\sigma_\psi), \qquad A_a^n = c_a \nu. \tag{A.10}$$

The coefficients are smooth since $P_a \geq a > 0$. On E the restriction is $a\sigma_\psi$; at infinity the horizontal coefficient is one and the transverse coefficient relative to $B_\psi - \sigma_\psi$ is $c_a k(1) > 0$. This proves all the asserted positivity properties, including at both sections.

The functions $\varphi \circ \pi$ and $\varphi \circ \pi + q_a$ are quasi-plurisubharmonic on Y. By (A.8), their exact β-currents are $a[E] + \beta_a$ and $a[E] + A_a$, respectively, so both are positive. The descent theorem Theorem A.2.1 gives $\varphi \in \mathrm{PSH}(X, \theta_0)$ and a potential u_0 with $u_0 \circ \pi = \varphi \circ \pi + q_a$. The bounded residual q_a puts u_0 in the ordinary class of φ and gives the uniform logarithmic asymptotic at p. Its finite supremum is K in Theorem A.6.1.

For an admissible H, self-adjointness makes $\exp H$ positive and $\det \exp H = \exp(\operatorname{tr} H) = 1$. Hence

$$(A_a + \pi^*\eta_H)^n = A_a^n = c_a \nu, \qquad \beta_{a,H} + \mathrm{dd^c}(q_a - K) = A_a + \pi^*\eta_H. \tag{A.11}$$

The full function $u = u_0 - K$ is unchanged. On the regular locus its exact θ_H-form is the positive form in (A.11); near p the background and potential are unchanged. Thus u is θ_H-plurisubharmonic. Formula (A.8) with the exact deformation gives $\pi^*\theta_H + \mathrm{dd^c}(u \circ \pi) = a[E] + A_a + \pi^*\eta_H$, proving its equation after zero extension. Its mass is $c_a M = V$. This calculation uses determinant preservation, rather than a class-mass identity for a nonclosed form.

Bounded residuals and uniqueness.

Let $w \in C_{a,H}$. On $Y \setminus E$, its residual $t = w \circ \pi - \varphi \circ \pi$ is bounded and $\beta_{a,H}$-plurisubharmonic. By Lemma 1.3.7, it extends with that exact background across E. Conversely, for any bounded $\beta_{a,H}$-plurisubharmonic t, the function $\varphi \circ \pi + t$ has positive current $a[E] + \beta_{a,H} + \mathrm{dd^c} t$ in the pulled-back background. It descends by Theorem A.2.1, and its uniform bounds relative to $\varphi \circ \pi$ put the descended potential in $C_{a,H}$. Thus bounded residuals correspond exactly to the prescribed ordinary class.

The bounded product is finite on compact Y and gives E zero mass, by Theorem 1.6.1 and Proposition 1.7.1. Thus

$$\mathrm{MA}^{\mathrm{ext}}_{X,\theta_H}(w) = \pi_*\big(\beta_{a,H} + \mathrm{dd^c} t\big)^n. \tag{A.12}$$

For $H = 0$, closedness makes every positive-degree exact term in the bounded expansion integrate to zero. Thus (A.9) gives mass V for every competitor. The nonclosed case uses only finiteness here.

If w solves with scalar $c' > 0$, its lifted measure is $c'\nu$, since both sides give E zero mass. Both residual solutions are continuous by Corollary 10.4.1; scalar uniqueness in Proposition 10.5.1 gives $c' = c_a$. The density $\nu/\beta^n_{a,H}$ is smooth and nonnegative with positive locus exactly $Y \setminus E$. This locus is connected and dense: it is a bundle with connected fibers $\mathbb{P}^1$ minus one point over connected B. Theorem 11.4.1 gives $t - (q_a - K)$ constant. The normalization fixes the constant, and both full functions equal $-\infty$ at p. This proves uniqueness against the ordinary class.

The uniform logarithmic asymptotic proves extended-real continuity of the full solution. Its residual restriction is $a\psi + Q_a(0) - K$. If ψ is nonconstant, it cannot be a pullback from X, since a pullback is constant on E. This concerns the residual; the full potential is $-\infty$ on the whole fiber and does descend.

Nonclosed deformations.

Fix $z_0 \in O$ and a smooth A_a-unitary frame on a smaller ball. Let $K_0 = \operatorname{diag}(1, -1, 0, \dots, 0)$ in this frame. Choose a real compactly supported χ with $\chi(z_0) = 0$ and $\mathrm{d}\chi(z_0)$ a nonzero covector in the first two complex directions, and put $H = \epsilon\chi K_0$. At z_0,

$$\mathrm{d}\eta_H = \epsilon\,\mathrm{d}\chi \wedge \gamma, \qquad \gamma = A_a(K_0\,\cdot, \cdot).$$

Terms differentiating the frame and A_a vanish because $\chi(z_0) = 0$. The restriction of γ to these four real directions is nondegenerate, so wedge multiplication by γ sends a nonzero one-form to a nonzero three-form. Thus $\mathrm{d}\theta_H(z_0) \neq 0$ for $\epsilon \neq 0$. On the fixed compact support, both strict inequalities (A.6) persist for sufficiently small $|\epsilon|$, and $H \to 0$ smoothly. This proves existence of arbitrarily small nonclosed choices; admissibility itself does not impose nonclosedness.

For any smooth resolution $\tau : Z \to X$, Theorem A.5.1 identifies its pushforward product with (A.12), proving (A.7). Its total mass equals the finite mass downstairs. Since the ordinary class, equation and normalization are intrinsic, uniqueness identifies the same solution and scalar on every resolution and common refinement. □

A.6.2 The bounded equation

Without a logarithmic pole, the residual background is β, which degenerates along E. The radial construction still gives a bounded solution, and closedness supplies an energy argument for its uniqueness. Put

$$c_b = \frac{d}{M} = \frac{1}{nI}, \qquad R(s) = \left(c_b n \int_0^1 k(sr) r^{n-1}\,\mathrm{d}r\right)^{1/n}, \qquad Q'(s) = \frac{R(s) - 1}{1 - s}, \qquad Q(1) = 0. \tag{A.13}$$

Here R is smooth and positive, $R(1) = 1$, and Q is smooth at both endpoints.

Theorem A.6.2 *For the same cone and prescribed measure, the bounded θ_0-plurisubharmonic class contains a unique normalized pair $c > 0$, u solving $\mathrm{MA}^{\mathrm{ext}}_{X,\theta_0}(u) = c\mu$. The scalar is c_b. The smooth function $\ell + Q(s)$ on Y is constant on E and descends to a finite continuous potential on X; its supremum-zero normalization is u.*

Every bounded competitor has mass d. Formula (A.7), with θ_0 in place of θ_H, holds on every resolution and common refinement. If $\Psi = 0$ and k is nonconstant, or if ψ is nonconstant, the solution is nonconstant.

Proof Set $P(s) = sR(s)$ with the data in (A.13). Then

$$P(s)^n = c_b n \int_0^s k(r) r^{n-1}\,\mathrm{d}r, \qquad P(0) = 0, \qquad P(1) = 1. \tag{A.14}$$

Smoothness and positivity of R follow from its integral formula. Since $R(1) = 1$, Q is smooth at $s = 1$, and its derivative is smooth at zero as well. Put $U = \ell + Q(s)$. It is smooth on Y and equals $Q(0)$ on E. On $0 < x < 1$, a local potential for $\beta + \mathrm{dd^c}U$ is $F(\rho_\psi)$, where $F(r) = h(r) + Q(h'(r))$. Differentiation gives $F' = P(s)$ and $F'' = c_b k(s) s^n (1-s)/P(s)^{n-1}$.

For $A = \beta + \mathrm{dd^c}U$,

$$A = P(s)\sigma_\psi + \frac{c_b k(s)}{R(s)^{n-1}}(B_\psi - s\sigma_\psi), \qquad A^n = c_b \nu. \tag{A.15}$$

The formula shows smoothness at zero directly. At infinity the horizontal coefficient is one and the transverse coefficient is positive. Thus A is semipositive on Y and strictly positive on $Y \setminus E$. By Theorem A.2.1, $U = u_0 \circ \pi$ for $u_0 \in \mathrm{PSH}(X, \theta_0)$. The proper surjection π is a quotient map, so smoothness of U makes u_0 continuous. Subtracting $K_b = \sup_X u_0$ gives the required solution.

For any bounded θ_0-plurisubharmonic w, the pullback $t = w \circ \pi$ is bounded β-plurisubharmonic. Its product is analytic-null and pushes forward to $\mathrm{MA}^{\mathrm{ext}}_{X,\theta_0}(w)$. Integrating the bounded closed expansion gives $\int_Y (\beta + \mathrm{dd^c}t)^n = \int_Y \beta^n = d$, fixing the scalar of any solution to be $c_b = d/M$. The resolution identity follows from Theorem A.5.1. If w solves, put $q = U - K_b$, $D = \beta + \mathrm{dd^c}t$ and $v = t - q$. The bounded integration-by-parts calculus of Section 1.6 gives

$$0 = \int_Y v(D^n - A^n) = -\sum_{j=0}^{n-1} \int_Y \mathrm{d}v \wedge \mathrm{d^c}v \wedge D^j \wedge A^{n-1-j}.$$

Every energy is nonnegative, so the term with the smooth factor A^{n-1} vanishes. By the weak-gradient interpretation, strict positivity of A on each relatively compact chart of $Y \setminus E$ forces $\mathrm{d}v = 0$ there. Connectedness makes v constant almost everywhere on that complement. The analytic divisor has ordinary volume zero, so Lemma 1.3.3 gives equality everywhere on Y. Surjectivity and normalization give $w = u$ on X.

If $\Psi = 0$ and the solution is constant, then $U = Q(x)$ is constant, so $Q' = 0$ and $R = 1$. Differentiating (A.14) gives $c_b k = 1$, forcing k constant. If ψ is nonconstant, the restriction of U to the infinity section is ψ, so the solution is nonconstant for every allowed k.

Example A.6.1 Nonconstant k alone does not imply nonconstant solution for unrestricted angular data. Take $\Psi = \log 2$ and $k(s) = 2/(2-s)^{n+1}$. Then $s = 2x/(1+x)$, $\ell = \log(1+x)$ and

$$\frac{B_\psi^n}{\beta^n} = \frac{s^n(1-s)}{x^n(1-x)} = \frac{(2-s)^{n+1}}{2}.$$

Thus $\nu = \beta^n$. Since $\frac{\mathrm{d}}{\mathrm{d}s}(s/(2-s))^n = nk(s)s^{n-1}$, one has $I = 1/n$, $M = d$ and $c_b = 1$. The endpoint formulas give $R = 1/(2-s)$ and $Q(s) = \log(2-s)$, hence $U = \log 2$. The normalized solution is zero, although k is nonconstant.

A.6.3 The quadric cone

Let $X = \{Z_0 Z_1 = Z_2^2\} \subset \mathbb{P}^3$. Its base conic is the quadratic Veronese image of $\mathbb{P}^1$, so $n = d = 2$. Its homogeneous ring $\mathbb{C}[\xi^2, \xi\eta, \eta^2]$ is the invariant ring of the sign involution on the normal polynomial ring, and is normal; thus the base is projectively normal. In the chart

$$\pi(z,t) = (t, tz^2, tz), \qquad S = 1 + |z|^2 + |z|^4, \qquad L = 1 + |t|^2 S,$$

one has $\beta = \mathrm{dd^c} \log L$. Differentiation gives

$$\det((\log L)_{j\bar{k}}) = \frac{|t|^2(SS_{z\bar{z}} - |S_z|^2)}{L^3} = \frac{|t|^2(1 + 4|z|^2 + |z|^4)}{L^3}.$$

Writing λ_4 for Euclidean volume gives

$$\frac{\beta^2}{2} = \frac{|t|^2(1 + 4|z|^2 + |z|^4)}{\pi^2(1 + |t|^2(1 + |z|^2 + |z|^4))^3}\,\mathrm{d}\lambda_4. \tag{A.16}$$

The transverse vanishing order on $E = \{t = 0\}$ is exactly two. For $\Psi = 0$, $k = 1/2$, one has $I = 1/4$ and $M = 1$, so $\mu_0 = \theta_0^2/2$, extended by zero at p, is an ambient-volume probability. The logarithmic scalar and solution mass are $c_a = 2(1 - a^2)$ for every admissible deformation, including the genuinely nonclosed choices. This exact normalization follows from determinant preservation.

For $\Psi = 0$, $k(s) = 1 + s$, one has $I = 5/6$, $M = 10/3$ and $c_b = 3/5$. Put $R(x) = \sqrt{(3 + 2x)/5}$ and $r_0 = \sqrt{3/5}$. The normalized bounded solution is

$$u(x) = -2(R(x) - r_0) + 2\log\frac{1 + R(x)}{1 + r_0}. \tag{A.17}$$

Indeed, $u' = -2/[5(1 + R)] = (R - 1)/(1 - x)$ with its removable value at one. It is $Q(x) - Q(0)$, strictly decreasing in x, with $u(p) = \sup_X u = 0$. The function of x is smooth on a neighborhood of $[0, 1]$, and x is ambient smooth, so u is a smooth ambient restriction. Its prescribed measure is $\mu_1 = (1 + x)\theta_0^2$, of mass $10/3$, and $\mathrm{MA}^{\mathrm{ext}}_{X,\theta_0}(u) = (3/5)\mu_1$, of mass two. Formula (A.16) also verifies the exceptional vanishing of its lift.

$\Psi = \epsilon|Z_0|^2/(|Z_0|^2 + |Z_1|^2 + |Z_2|^2)$ is smooth on $\mathbb{P}^2$ and preserves $\omega_{\mathrm{FS}} + \mathrm{dd^c}\Psi > 0$ for sufficiently small $|\epsilon|$. Its restriction to the conic takes values ϵ and zero at $[1 : 0 : 0]$ and $[0 : 1 : 0]$. For small nonzero ϵ, the logarithmic residual is nonconstant on E and fails to descend. Supported nonclosed deformations leave that residual, and the full descended solution, unchanged.

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